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Preface

Structure of Manual

This document defines the PSS®E Application Program Interface (API) to the various engineering functions in PSS®E.

How API Routines are Described

The description for each API routine starts with a brief explanation of its function, followed by one or more language specific syntaxes for calling the routine, followed by a detailed description of the arguments used in those calling sequences. The nomenclature used in the argument descriptions is most consistent with the Fortran programming language.

The syntaxes for calling the API routines will include at least one of the PSS®E automation file languages, and will usually include a syntax for use from a Fortran program. The set of syntaxes will include all the PSS®E automation file languages where the API routine can be used. Some general comments about those syntaxes may be helpful, as follows:

PSS®E Automation File Languages

Batch commands:this is a free-format input syntax. Items can be separated by a comma or spaces, strings need only be quoted if they contain spaces or commas, omitted items can be indicated by consecutive commas, arrays are entered as a sequence of scalar values, and a semi-colon can be used to terminate the command and return the remaining items to default values (it must be a separate data item, i.e., it must be separated from the previous data item by a space or comma). Except for unquoted filenames, a forward slash will terminate a line. Commands may be continued over multiple lines. The sequence @! at the beginning of the line will cause that line to be ignored.

Python commands: Python is an open source language; its syntax is well documented. Some of are arguments are defined as arrays (in keeping with Fortran nomenclature, as mentioned above). Python does not have arrays, and in general you should interpret arrays to mean lists (although tuples will usually work on input). Arrays that have some elements as input and some as output in the Fortran syntax are represented in the Python syntax as two separate arrays (i.e. lists). Keywords can be used for arguments, and any argument can be omitted. Additionally, any element of an input array can be specified as a keyword, e.g. myarray06=2 specifies that the 6th element of the array myarray has the value 2. If an alias is documented for a specific array element, it can also be used as a keyword for that element.

Fortran call statements: Fortran is a well-documented language (where it matters, we will use the Fortran 2008 standard). We will not use features that require explicit interfaces without stating so clearly in the description. Unless specifically stated, arguments cannot be omitted.

IPLAN commands:the IPLAN language is described in the IPLAN manual. For purposes of the CALL statement it resembles Fortran. Unless specifically stated, arguments cannot be omitted.

Return Codes / Error Codes

Most API routines return a integer return code, where zero indicates success. In Python, where multiple values are returned (a tuple), the return code will always be the first value. When non-successful (usually non-zero)

error codes are returned, other returned values are set to None in Python, and are either uninitialized or unchanged (for input/output values) in other syntaxes.

The Python interface can be made to throw exceptions instead of returning error codes values (the default). Enter:

```
psspy.setThrowPsseExceptions(True)
```

or

```
psspy.param.throwPsseExceptions = True
```

To see if the Python interface is set to throw exceptions instead of returning error code values:

```
val = psspy.getThrowPsseExceptions()
```

Default Values

When values are omitted, special “no input” values are supplied. These special values – for each variable type – are meant to indicate that “no input” was supplied. Many of the API routines permit this and will use documented default values when “no input” values are supplied.

The actual “no input” values can be retrieved from the API routines GETDEFAULTINT, GETDEFAULTREAL, and GETDEFAULTCHAR (capitalization depends on syntax). Users should never depend on the actual values returned, by these functions. They should be used only for supplying values to the PSS®E API routines.

When automation files are recorded, items that have the default value are omitted in batch commands (i.e. recorded as ",,") and recorded as _i, _f, _s, or _o (for integer, real/float, character/string, and filename, respectively). In the PSS®E embedded Python environment there will be local variables _i, _f, and _s set to the values returned from GETDEFAULTxxx routines mentioned above (_o will be treated as blanks). Outside of PSS®E it is the user’s responsibility to set those variables.

Recording

Two recording formats are available: batch commands and Python.

Not all API routines are recordable. Routines that only retrieve data do not record, for example. If an API routine is recordable, it will have a batch command format specified; routines without a batch command format specified are not recordable.

Python Modules

Most of the PSS®E API is defined in the module psspy. Later chapters discuss routines defined in other modules. In keeping with normal Python documentation the namespace is not included as part of the syntax for the individual routines. This may seem confusing to newcomers to Python. For those whose first exposure to Python is inside of PSS®E, be aware that, normally, you will use the, say, case API routine by entering:

psspy.case(filename)

not

case(filename)

Chapter 1
Power Flow Operation

This chapter describes the API routines corresponding to the operational functions of the PSS®E Power Flow and Fault Analysis. Also included are those routines that control basic program operations and the Open Access option.

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1.1. ACCC

This API routine is obsolete. It has been replaced by the API routine accc_with_dsp_2 .

Use this API to apply the AC contingency calculation function (activity ACCC).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ACCC  TOL  OPTIONS(1)..OPTIONS(7)  DFXFILE  ACCFILE  THRFILE
```

Python command syntax:

```
ierr = accc(tol, options, dfxfile, accfile, thrfile)
```

Fortran command syntax:

```
CALL ACCCAPI(TOL, OPTIONS, DFXFILE, ACCFILE, THRFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTIONS (7)	array of 7 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable

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	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	solution method flag (0 by default)
	OPTIONS(6) = 0	FDNS
	OPTIONS(6) = 1	FNSL
	OPTIONS(6) = 2	optimized FDNS
	OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTIONS(7) = 0	disable
	OPTIONS(7) = 1	enable
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output file (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	no Distribution Factor Data File specified
	IERR = 7	no Contingency Solution Output File specified
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	bus type code and series element status inconsistencies
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	prerequisite requirements for API are not met

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1.2. ACCC_2

This API routine is obsolete. It has been replaced by the API routine accc_with_dsp_3 .

Use this API to apply the AC contingency calculation function (activity ACCC).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ACCC_2  TOL  OPTIONS(1)..OPTIONS(7)  DFXFILE  ACCFILE  THRFILE
```

Python command syntax:

```
ierr = accc_2(tol, options, dfxfile, accfile, thrfile)
```

Fortran command syntax:

```
CALL ACCC_2(TOL, OPTIONS, DFXFILE, ACCFILE, THRFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTIONS (9)	array of 9 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(6)	solution method flag (0 by default)
	OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTIONS(8)	induction motor treatment flag (0 by default)
	OPTIONS(9)	induction machine failure flag (0 by default)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output file (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred.
	IERR > 0	as for ACCC_WITH_DSP_3.

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1.3. ACCC_MULTIPLE_RUN_REPORT

This API routine is obsolete. It has been replaced by the API routine accc_multiple_run_report_2 .

Use this API to report the results of up to nine executions of the AC Contingency Calculation function.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ACCC_MULTIPLE_RUN_REPORT  OPTIONS(1)..OPTIONS(14)
 VALUES(1)..VALUES(7)  NFILES  ACFILES(1)..ACFILES(NFILES)
```

Python command syntax:

```
ierr = accc_multiple_run_report(options, values, nfiles, acfiles)
```

Fortran command syntax:

```
CALL ACCC_MULTIPLE_RUN_REPORT(OPTIONS, VALUES, NFILES, ACFILES, IERR)
```

Where:

Integer OPTIONS (14)	array of 14 elements specifying defining reporting options(input).	
	OPTIONS(1)	column headings code (1 by default)
	OPTIONS(1) = 1	Contingency Solution Output File names
	OPTIONS(1) = 2	Saved Case File names as contained in the Contingency Solution Output Files
	OPTIONS(2)	base case rating set (rating set program option setting by default)
	OPTIONS(2) = 1	rate A
	OPTIONS(2) = 2	rate B
	OPTIONS(2) = 3	rate C
	OPTIONS(3)	contingency case rating set (rating set program option setting by default)
	OPTIONS(3) = 1	rate A
	OPTIONS(3) = 2	rate B
	OPTIONS(3) = 3	rate C
	OPTIONS(4)	print monitored elements summary report (0 by default)
	OPTIONS(4) = 0	no
	OPTIONS(4) = 1	yes
	OPTIONS(5)	print missing monitored elements report (0 by default)
	OPTIONS(5) = 0	no
	OPTIONS(5) = 1	yes
	OPTIONS(6)	print missing monitored voltage buses report (0 by default)
	OPTIONS(6) = 0	no
	OPTIONS(6) = 1	yes

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OPTIONS(7)	print contingency legend (2 by default)
OPTIONS(7) = 0	no
OPTIONS(7) = 1	complete master legend report only
OPTIONS(7) = 2	reduced legend with each table
OPTIONS(7) = 3	both master legend report and legend with each table
OPTIONS(8)	print missing contingencies report (0 by default)
OPTIONS(8) = 0	no
OPTIONS(8) = 1	yes
OPTIONS(9)	print non-converged contingencies report (0 by default)
OPTIONS(9) = 0	no
OPTIONS(9) = 1	yes
OPTIONS(10)	print loading violation reports (0 by default)
OPTIONS(10) = 0	no
OPTIONS(10) = 1	base case and worst case contingency violations only
OPTIONS(10) = 2	base case and all contingency violations only
OPTIONS(10) = 3	base case, worst case contingency, and all contingency violations
OPTIONS(11)	print voltage violation reports (0 by default)
OPTIONS(11) = 0	no
OPTIONS(11) = 1	base case and worst case contingency violations only
OPTIONS(11) = 2	base case and all contingency violations only
OPTIONS(11) = 3	base case, worst case contingency, and all contingency violations
OPTIONS(12)	option for processing interfaces in loading violation reports (1 by default)
OPTIONS(12) = 0	exclude interfaces from loading violation checking and reporting
OPTIONS(12) = 1	check and report interface loading violations
OPTIONS(13)	in loading violation reports, exclude monitored branches and interfaces that show loading violations in the base case from being checked and reported in contingency case reports (0 by default)
OPTIONS(13) = 0	no
OPTIONS(13) = 1	yes
OPTIONS(14)	in voltage range violation reports, exclude monitored buses that show voltage range violations in the base case from the corresponding check in contingency case reports (0 by default)
OPTIONS(14) = 0	no

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	OPTIONS(14) = 1	yes
Real VALUES (7)	array of 7 elements specifying reporting parameters(input).	
	VALUES(1)	bus mismatch converged tolerance (MW or Mvar) (0.5 by default)
	VALUES(2)	system mismatch converged tolerance (MVA) (5.0 by default)
	VALUES(3)	in loading violation reports, percent of flow rating for reporting (100.0 by default)
	VALUES(4)	in the worst case contingency loading violation reports, percent of flow rating for counting (> VALUES(3)) (100.0 by default)
	VALUES(5)	in loading violation reports, minimum contingency case flow change from base case value for reporting and counting (0.0 by default)
	VALUES(6)	in loading violation reports, minimum contingency case percent loading increase from base case value for reporting and counting (0.0 by default)
	VALUES(7)	minimum contingency case voltage change for reporting and counting in voltage range violations (0.0 by default)
Integer NFILES	number of Contingency Solution Output Files to be processed (1 through 22) (input; no default allowed).	
Character*260 ACFILES (NFILES)	NFILES names of Contingency Solution Output Files (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	invalid NFILES value
	IERR = 4	the percent for counting is less than the percent for reporting
	IERR = 5	file is not in the form of a current Contingency Solution Output File
	IERR = 6	error opening a Contingency Solution Output File
	IERR = 7	error reading a Contingency Solution Output File
	IERR = 8	error opening a temporary file
	IERR = 9	error opening a subfile in the temporary file
	IERR = 10	error reading or writing the temporary file
	IERR = 11	error closing a subfile in the temporary file
	IERR = 12	prerequisite requirements for API are not met

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1.4. ACCC_MULTIPLE_RUN_REPORT_2

Use this API to report the results of up to twenty two executions of the AC Contingency Calculation function.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_ACCC_MULTIPLE_RUN_REPORT_2  OPTIONS(1)..OPTIONS(16)
 VALUES(1)..VALUES(7)  NFILES  ACFILES(1)..ACFILES(NFILES)
```

Python command syntax:

```
ierr = accc_multiple_run_report_2(options, values, nfiles, acfiles)
```

Fortran command syntax:

```
CALL ACCC_MULTIPLE_RUN_REPORT_2(OPTIONS, VALUES, NFILES, ACFILES, IERR)
```

Where:

Integer OPTIONS (16)	array of 16 elements specifying defining reporting options(input).	
	OPTIONS(1)	column headings code (1 by default)
	OPTIONS(1) = 1	Contingency Solution Output File names
	OPTIONS(1) = 2	Saved Case File names as contained in the Contingency Solution Output Files
	OPTIONS(2)	base case rating set (rating set program option setting by default)
	OPTIONS(2) = 1	rate A
	OPTIONS(2) = 2	rate B
	OPTIONS(2) = 3	rate C
	OPTIONS(3)	contingency case rating set (rating set program option setting by default)
	OPTIONS(3) = 1	rate A
	OPTIONS(3) = 2	rate B
	OPTIONS(3) = 3	rate C
	OPTIONS(4)	base case voltage limit when STATUS(1) is 0, 1, 3 or 4
	OPTIONS(4) = 1	normal limit
	OPTIONS(4) = 2	emergency limit
	OPTIONS(5)	contingency case voltage limit when STATUS(1) is 0, 1, 3 or 4
	OPTIONS(5) = 1	normal limit
	OPTIONS(5) = 2	emergency limit
	OPTIONS(6)	print monitored elements summary report (0 by default)
	OPTIONS(6) = 0	no
	OPTIONS(6) = 1	yes

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OPTIONS(7)	print missing monitored elements report (0 by default)
OPTIONS(7) = 0	no
OPTIONS(7) = 1	yes
OPTIONS(8)	print missing monitored voltage buses report (0 by default)
OPTIONS(8) = 0	no
OPTIONS(8) = 1	yes
OPTIONS(9)	print contingency legend (2 by default)
OPTIONS(9) = 0	no
OPTIONS(9) = 1	complete master legend report only
OPTIONS(9) = 2	reduced legend with each table
OPTIONS(9) = 3	both master legend report and legend with each table
OPTIONS(10)	print missing contingencies report (0 by default)
OPTIONS(10) = 0	no
OPTIONS(10) = 1	yes
OPTIONS(11)	print non-converged contingencies report (0 by default)
OPTIONS(11) = 0	no
OPTIONS(11) = 1	yes
OPTIONS(12)	print loading violation reports (0 by default)
OPTIONS(12) = 0	no
OPTIONS(12) = 1	base case and worst case contingency violations only
OPTIONS(12) = 2	base case and all contingency violations only
OPTIONS(12) = 3	base case, worst case contingency, and all contingency violations
OPTIONS(13)	print voltage violation reports (0 by default)
OPTIONS(13) = 0	no
OPTIONS(13) = 1	base case and worst case contingency violations only
OPTIONS(13) = 2	base case and all contingency violations only
OPTIONS(13) = 3	base case, worst case contingency, and all contingency violations
OPTIONS(14)	option for processing interfaces in loading violation reports (1 by default)
OPTIONS(14) = 0	exclude interfaces from loading violation checking and reporting
OPTIONS(14) = 1	check and report interface loading violations
OPTIONS(15)	in loading violation reports, exclude monitored branches and interfaces that show loading violations in the base case from be-

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		ing checked and reported in contingency case reports (0 by default)
	OPTIONS(15) = 0	no
	OPTIONS(15) = 1	yes
	OPTIONS(16)	in voltage range violation reports, exclude monitored buses that show voltage range violations in the base case from the corresponding check in contingency case reports (0 by default)
	OPTIONS(16) = 0	no
	OPTIONS(16) = 1	yes
Real VALUES (7)	array of 7 elements specifying reporting parameters(input).	
	VALUES(1)	bus mismatch converged tolerance (MW or Mvar) (0.5 by default)
	VALUES(2)	system mismatch converged tolerance (MVA) (5.0 by default)
	VALUES(3)	in loading violation reports, percent of flow rating for reporting (100.0 by default)
	VALUES(4)	in the worst case contingency loading violation reports, percent of flow rating for counting (> VALUES(3)) (100.0 by default)
	VALUES(5)	in loading violation reports, minimum contingency case flow change from base case value for reporting and counting (0.0 by default)
	VALUES(6)	in loading violation reports, minimum contingency case percent loading increase from base case value for reporting and counting (0.0 by default)
	VALUES(7)	minimum contingency case voltage change for reporting and counting in voltage range violations (0.0 by default)
Integer NFILES	number of Contingency Solution Output Files to be processed (1 through 9) (input; no default allowed).	
Character*260 ACFILES (NFILES)	NFILES names of Contingency Solution Output Files (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	invalid NFILES value
	IERR = 4	the percent for counting is less than the percent for reporting
	IERR = 5	file is not in the form of a current Contingency Solution Output File

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IERR = 6	error opening a Contingency Solution Output File
IERR = 7	error reading a Contingency Solution Output File
IERR = 8	error opening a temporary file
IERR = 9	error opening a subfile in the temporary file
IERR = 10	error reading or writing the temporary file
IERR = 11	error closing a subfile in the temporary file
IERR = 12	prerequisite requirements for API are not met

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1.5. ACCC_PARALLEL

This API routine is obsolete. It has been replaced by the API routine accc_parallel_2 .

Use this API routine to run the initial release of the parallel implementation of the AC contingency calculation function (use ACCC_WITH_DSP_3 to run AC contingency calculation function serially). The API routine distributes contingency cases among multiple processors evenly, and tests the contingency cases assigned to each processor in parallel.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_ACCC_PARALLEL  TOL OPTACC(1)..OPTACC(10) LABEL DFXFILE ACCFILE THRFILE
 INLFILE
```

Python command syntax:

```
ierr = accc_parallel(tol, optacc, label, dfxfile, accfile, thrfile,
 inlfile)
```

Fortran command syntax:

```
CALL ACCC_PARALLEL(TOL, OPTACC, LABEL, DFXFILE, ACCFILE, THRFILE, INLFILE,
 IERR)
```

Where:

Real TOL	Is the mismatch tolerance (input; Newton solution convergence tolerance, TOLN, by default).	
Integer OPTACC (10)	array of elements specifying solution options. The values are as follows:(input).	
	OPTACC(1)	tap adjustment flag. (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag. (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag. (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag. (dc tap adjustment option setting by default)

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OPTACC(4) = 0	disable
OPTACC(4) = 1	enable
OPTACC(5)	switched shunt adjustment flag. (switched shunt adjustment option setting by default)
OPTACC(5) = 0	disable
OPTACC(5) = 1	enable
OPTACC(5) = 2	enable continuous mode, disable discrete mode
OPTACC(6)	solution method flag. (0 by default)
OPTACC(6) = 0	FDNS
OPTACC(6) = 1	FNSL
OPTACC(6) = 2	optimized FDNS
OPTACC(7)	non-divergent solution flag. (non-divergent solution option setting by default)
OPTACC(7) = 0	disable
OPTACC(7) = 1	enable
OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage) (0 by default)
OPTACC(8) = 0	stall
OPTACC(8) = 1	trip
OPTACC(9)	induction machine failure flag (0 by default)
OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the stalled or tripped state
OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the stalled or tripped state.
OPTACC(10)	dispatch mode (0 by default)
OPTACC(10) = 0	disable
OPTACC(10) = 1	subsystem machines (reserve)
OPTACC(10) = 2	subsystem machines (pmax)
OPTACC(10) = 3	subsystem machines (inertia)
OPTACC(10) = 4	subsystem machines (governor droop)

Character*32 LABEL	Is the name of the generation dispatch subsystem (blank by default, no default allowed if OPTIONS(10) is not 0). (input).
Character*260 DFXFILE	Is the name of the Distribution Factor Data file. (input; no default allowed).
Character*260 ACCFILE	Is the name of the Contingency Solution Output file. (input; no default allowed).
Character*260 THRFILE	Is the name of the Load Throwover Data file. (input; blank by default).
Character*260 INLFILE	Is the name of the Unit Inertia and Governor Data File. (input; blank by default).

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Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation dispatch subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	no Distribution Factor Data File specified
	IERR = 9	no Contingency Solution Output File specified
	IERR = 10	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 11	buses with bus type code and series element status inconsistencies
	IERR = 12	no ZIP Archive Output File specified
	IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	error opening Unit Inertia and Governor Data File
	IERR = 55	error opening ZIP Archive Output File
	IERR = 56	prerequisite requirements for API are not met

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1.6. ACCC_PARALLEL_2

Use this API routine to run the second release of the parallel implementation of the AC contingency calculation function (use ACCC_WITH_DSP_3 to run AC contingency calculation function serially). The API routine distributes contingency cases among multiple processors evenly, and tests the contingency cases assigned to each processor in parallel.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_ACCC_PARALLEL_2  TOL OPTACC(1)..OPTACC(11) LABEL DFXFILE ACCFILE
 THRFILE INLFILE ZIPFILE
```

Python command syntax:

```
ierr = accc_parallel_2(tol, optacc, label, dfxfile, accfile, thrfile,
 inlfile, zipfile)
```

Fortran command syntax:

```
CALL ACCC_PARALLEL_2(TOL, OPTACC, LABEL, DFXFILE, ACCFILE, THRFILE,
 INLFILE, ZIPFILE, IERR)
```

Where:

Real TOL	Is the mismatch tolerance (input; Newton solution convergence tolerance, TOLN, by default).	
Integer OPTACC (11)	array of elements specifying solution options. The values are as follows:(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable

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OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
OPTACC(5) = 0	disable
OPTACC(5) = 1	enable
OPTACC(5) = 2	enable continuous mode, disable discrete mode
OPTACC(6)	solution method flag (0 by default)
OPTACC(6) = 0	FDNS
OPTACC(6) = 1	FNSL
OPTACC(6) = 2	optimized FDNS
OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
OPTACC(7) = 0	disable
OPTACC(7) = 1	enable
OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage) (0 by default)
OPTACC(8) = 0	stall
OPTACC(8) = 1	trip
OPTACC(9)	induction machine failure flag (0 by default)
OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
OPTACC(10)	dispatch mode (0 by default)
OPTACC(10) = 0	disable
OPTACC(10) = 1	subsystem machines (reserve)
OPTACC(10) = 2	subsystem machines (pmax)
OPTACC(10) = 3	subsystem machines (inertia)
OPTACC(10) = 4	subsystem machines (governor droop)
OPTACC(11)	ZIP archive flag (0 by default)
OPTACC(11) = 0	do not write a ZIP archive file
OPTACC(11) = 1	write a ZIP archive using the file specified as ZIPFILE

Character*32 LABEL	Is the name of the generation dispatch subsystem (blank by default, no default allowed if OPTIONS(10) is not 0) (input).
Character*260 DFXFILE	Is the name of the Distribution Factor Data file (input; no default allowed).
Character*260 ACCFILE	Is the name of the Contingency Solution Output file (input; no default allowed).
Character*260 THRFILE	Is the name of the Load Throwover Data file (input; blank by default).

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Character*260 INLFILE	Is the name of the Unit Inertia and Governor Data File (input; blank by default).
Character*260 ZIPFILE	Is the name of the ZIP Archive Output File (input; blank by default).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	invalid TOL value
IERR = 2	invalid OPTIONS value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation dispatch subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	no Distribution Factor Data File specified
IERR = 9	no Contingency Solution Output File specified
IERR = 10	in-service induction machines are in the "stalled" or "tripped" state
IERR = 11	buses with bus type code and series element status inconsistencies
IERR = 12	no ZIP Archive Output File specified
IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	error opening Unit Inertia and Governor Data File
IERR = 55	error opening ZIP Archive Output File
IERR = 56	prerequisite requirements for API are not met.

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1.7. ACCC_SINGLE_RUN_REPORT

This API routine is obsolete. It has been replaced by the API routine accc_single_run_report_2 .

Use this API, the AC Contingency Report function, to report the results of the AC Contingency Calculation function.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ACCC_SINGLE_RUN_REPORT  STATUS(1)..STATUS(8)  INTVAL(1)..INTVAL(5)
 REALVAL(1)..REALVAL(7)  RFILE
```

Python command syntax:

```
ierr = accc_single_run_report(status, intval, realval, rfile)
```

Fortran command syntax:

```
CALL ACCC_SINGLE_RUN_REPORT(STATUS, INTVAL, REALVAL, RFILE, IERR)
```

Where:

Integer STATUS (8)	array of 8 elements specifying (input).	
	STATUS(1)	report format (3 by default)
	STATUS(1) = 0	spreadsheet overload report
	STATUS(1) = 1	spreadsheet loading table
	STATUS(1) = 2	available capacity table
	STATUS(1) = 3	non-spreadsheet overload report
	STATUS(1) = 4	non-spreadsheet loading table
	STATUS(1) = 5	non-converged networks report
	STATUS(2)	base case rating set; used only when STATUS(1) is 0, 1, 3 or 4 (rating set program option setting by default)
	STATUS(2) = 1	rate A
	STATUS(2) = 2	rate B
	STATUS(2) = 3	rate C
	STATUS(3)	contingency case rating set when STATUS(1) is 0, 1, 3 or 4; base case and contingency case rating set when STATUS(1) is 2 (rating set program option setting by default)
	STATUS(3) = 1	rate A
	STATUS(3) = 2	rate B
	STATUS(3) = 3	rate C
	STATUS(4)	exclude interfaces from report; used only when STATUS(1) is 0, 1, 2, 3 or 4 (0 by default)
	STATUS(4) = 0	no
	STATUS(4) = 1	yes

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	STATUS(5)	run voltage limit check; used only when STATUS(1) is 0, 1, 3 or 4 (0 by default)
	STATUS(5) = 0	no
	STATUS(5) = 1	yes
	STATUS(6)	in overload reports, exclude monitored branches and interfaces that show loading violations in the base case from being checked and reported in contingency cases; used only when STATUS(1) is 0 or 3 (0 by default)
	STATUS(6) = 0	no
	STATUS(6) = 1	yes
	STATUS(7)	in voltage range violation reports, exclude monitored buses that show voltage range violations in the base case from the corresponding check in contingency case reports; used only when STATUS(1) is 0 or 3 (0 by default)
	STATUS(7) = 0	no
	STATUS(7) = 1	yes
	STATUS(8)	exclude cases with no overloads from non-spreadsheet overload report; used only when STATUS(1) is 3 (0 by default)
	STATUS(8) = 0	no
	STATUS(8) = 1	yes
Integer INTVAL (5)	array of 5 elements specifying (input).	
	INTVAL(1)	number of low voltage range violations filtering criterion (0 by default)
	INTVAL(2)	number of high voltage range violations filtering criterion (0 by default)
	INTVAL(3)	number of voltage deviation violations filtering criterion; not applied to base case (0 by default)
	INTVAL(4)	number of buses in the largest disconnected island filtering criterion; not applied to base case (0 by default)
	INTVAL(5)	maximum number of elements in the available capacity table (no limit by default)
Real REALVAL (7)	array of 7 elements specifying (input).	
	REALVAL(1)	bus mismatch converged tolerance (MW or Mvar) (0.5 by default)
	REALVAL(2)	system mismatch converged tolerance (MVA) (5.0 by default)
	REALVAL(3)	percent of flow rating; used only when STATUS(1) is 0, 3 or 4 (100.0 by default)

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	REALVAL(4)	in overload reports, minimum contingency case flow change from base case value; used only when STATUS(1) is 0 or 3 (0.0 by default)
	REALVAL(5)	in overload reports, minimum contingency case percent loading increase from base case value; used only when STATUS(1) is 0 or 3 (0.0 by default)
	REALVAL(6)	in voltage range violation reports, minimum contingency case voltage change from base case value; used only when STATUS(1) is 0, 1, 3 or 4 (0.0 by default)
	REALVAL(7)	cutoff threshold for available capacity table; used only when STATUS(1) is 2 (99999.0 by default)
Character*260 RFILE	Contingency Solution Output File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid STATUS value
	IERR = 2	invalid INTVAL value
	IERR = 3	invalid REALVAL value
	IERR = 4	error opening RFILE
	IERR = 5	error reading RFILE
	IERR = 6	prerequisite requirements for API are not met

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1.8. ACCC_SINGLE_RUN_REPORT_2

This API routine is obsolete. It has been replaced by the API routine accc_single_run_report_3 .

This API is the second release of the AC Contingency Report function. Use this API to report the results of the AC Contingency Calculation function.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_ACCC_SINGLE_RUN_REPORT_2  STATUS(1)..STATUS(9)  INTVAL(1)..INTVAL(5)
 REALVAL(1)..REALVAL(7)  RFILE
```

Python command syntax:

```
ierr = accc_single_run_report_2(status, intval, realval, rfile)
```

Fortran command syntax:

```
CALL ACCC_SINGLE_RUN_REPORT_2(STATUS, INTVAL, REALVAL, RFILE, IERR)
```

Where:

Integer STATUS (9)	array of 9 elements specifying (input).	
	STATUS(1)	report format (3 by default)
	STATUS(1) = 0	spreadsheet overload report
	STATUS(1) = 1	spreadsheet loading table
	STATUS(1) = 2	available capacity table
	STATUS(1) = 3	non-spreadsheet overload report
	STATUS(1) = 4	non-spreadsheet loading table
	STATUS(1) = 5	non-converged networks report
	STATUS(1) = 6	non-spreadsheet corrective actions report
	STATUS(2)	base case rating set; used only when STATUS(1) is 0, 1, 3 or 4 (rating set program option setting by default)
	STATUS(2) = 1	rate A
	STATUS(2) = 2	rate B
	STATUS(2) = 3	rate C
	STATUS(3)	contingency case rating set when STATUS(1) is 0, 1, 3 or 4; base case and contingency case rating set when STATUS(1) is 2 (rating set program option setting by default)
	STATUS(3) = 1	rate A
	STATUS(3) = 2	rate B
	STATUS(3) = 3	rate C
	STATUS(4)	exclude interfaces from report; used only when STATUS(1) is 0, 1, 2, 3, 4 (0 by default)
	STATUS(4) = 0	no
	STATUS(4) = 1	yes

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	STATUS(5)	run voltage limit check; used only when STATUS(1) is 0, 1, 3, 4 (0 by default)
	STATUS(5) = 0	no
	STATUS(5) = 1	yes
	STATUS(6)	in overload reports, exclude monitored branches and interfaces that show loading violations in the base case from being checked and reported in contingency cases; used only when STATUS(1) is 0 or 3 (0 by default)
	STATUS(6) = 0	no
	STATUS(6) = 1	yes
	STATUS(7)	in voltage range violation reports, exclude monitored buses that show voltage range violations in the base case from the corresponding check in contingency case reports; used only when STATUS(1) is 0 or 3 (0 by default)
	STATUS(7) = 0	no
	STATUS(7) = 1	yes
	STATUS(8)	exclude cases with no overloads from non-spreadsheet overload report or non-spreadsheet corrective actions report; used only when STATUS(1) is 3 (0 by default)
	STATUS(8) = 0	no
	STATUS(8) = 1	yes
	STATUS(9)	report post-tripping action solutions (0 by default)
	STATUS(9) = 0	no
	STATUS(9) = 1	yes
Integer INTVAL (5)	array of 5 elements specifying (input).	
	INTVAL(1)	number of low voltage range violations filtering criterion (0 by default)
	INTVAL(2)	number of high voltage range violations filtering criterion (0 by default)
	INTVAL(3)	number of voltage deviation violations filtering criterion; not applied to base case (0 by default)
	INTVAL(4)	number of buses in the largest disconnected island filtering criterion; not applied to base case (0 by default)
	INTVAL(5)	maximum number of elements in the available capacity table (no limit by default)
Real REALVAL (7)	array of 7 elements specifying (input).	
	REALVAL(1)	bus mismatch converged tolerance (MW or Mvar) (0.5 by default)

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	REALVAL(2)	system mismatch converged tolerance (MVA) (5.0 by default)
	REALVAL(3)	percent of flow rating; used only when STATUS(1) is 0, 3 or 4 (100.0 by default)
	REALVAL(4)	in overload reports, minimum contingency case flow change from base case value; used only when STATUS(1) is 0 or 3 (0.0 by default)
	REALVAL(5)	in overload reports, minimum contingency case percent loading increase from base case value; used only when STATUS(1) is 0 or 3 (0.0 by default)
	REALVAL(6)	in voltage range violation reports, minimum contingency case voltage change from base case value; used only when STATUS(1) is 0, 1, 3 or 4 (0.0 by default)
	REALVAL(7)	cutoff threshold for available capacity table; used only when STATUS(1) is 2 (99999.0 by default)
Character*260 RFILE	Contingency Solution Output File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid STATUS value
	IERR = 2	invalid INTVAL value
	IERR = 3	invalid REALVAL value
	IERR = 4	error opening RFILE
	IERR = 5	error reading RFILE
	IERR = 6	prerequisite requirements for API are not met

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1.9. ACCC_SINGLE_RUN_REPORT_3

This API routine is obsolete. It has been replaced by the API routine accc_single_run_report_4 .

This API is the third release of the AC Contingency Report function. Use this API to report the results of the AC Contingency Calculation function.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ACCC_SINGLE_RUN_REPORT_3  STATUS(1)..STATUS(10)  INTVAL(1)..INTVAL(5)
 REALVAL(1)..REALVAL(7)  RFILE
```

Python command syntax:

```
ierr = accc_single_run_report_3(status, intval, realval, rfile)
```

Fortran command syntax:

```
CALL ACCC_SINGLE_RUN_REPORT_3(STATUS, INTVAL, REALVAL, RFILE, IERR)
```

Where:

Integer STATUS (10)	array of 10 elements specifying (input).	
	STATUS(1)	report format (3 by default)
	STATUS(1) = 0	spreadsheet overload report
	STATUS(1) = 1	spreadsheet loading table
	STATUS(1) = 2	available capacity table
	STATUS(1) = 3	non-spreadsheet overload report
	STATUS(1) = 4	non-spreadsheet loading table
	STATUS(1) = 5	non-converged networks report
	STATUS(1) = 6	non-spreadsheet corrective actions report
	STATUS(2)	base case rating set; used only when STATUS(1) is 0, 1, 3 or 4 (rating set program option setting by default)
	STATUS(2) = 1	rate A
	STATUS(2) = 2	rate B
	STATUS(2) = 3	rate C
	STATUS(3)	contingency case rating set when STATUS(1) is 0, 1, 3 or 4; base case and contingency case rating set when STATUS(1) is 2 (rating set program option setting by default)
	STATUS(3) = 1	rate A
	STATUS(3) = 2	rate B
	STATUS(3) = 3	rate C
	STATUS(4)	exclude interfaces from report; used only when STATUS(1) is 0, 1, 2, 3, 4 (0 by default)
	STATUS(4) = 0	no
	STATUS(4) = 1	yes

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	STATUS(5)	run voltage limit check; used only when STATUS(1) is 0, 1, 3, 4 (0 by default)
	STATUS(5) = 0	no
	STATUS(5) = 1	yes
	STATUS(6)	in overload reports, exclude monitored branches and interfaces that show loading violations in the base case from being checked and reported in contingency cases; used only when STATUS(1) is 0 or 3 (0 by default)
	STATUS(6) = 0	no
	STATUS(6) = 1	yes
	STATUS(7)	in voltage range violation reports, exclude monitored buses that show voltage range violations in the base case from the corresponding check in contingency case reports; used only when STATUS(1) is 0 or 3 (0 by default)
	STATUS(7) = 0	no
	STATUS(7) = 1	yes
	STATUS(8)	exclude cases with no overloads from non-spreadsheet overload report or non-spreadsheet corrective actions report; used only when STATUS(1) is 3 (0 by default)
	STATUS(8) = 0	no
	STATUS(8) = 1	yes
	STATUS(9)	report post-tripping action solutions (0 by default)
	STATUS(9) = 0	no
	STATUS(9) = 1	yes
	STATUS(10)	report loss of loads (0 by default)
	STATUS(10) = 0	no
	STATUS(10) = 1	yes
Integer INTVAL (5)	array of 5 elements specifying (input).	
	INTVAL(1)	number of low voltage range violations filtering criterion (0 by default)
	INTVAL(2)	number of high voltage range violations filtering criterion (0 by default)
	INTVAL(3)	number of voltage deviation violations filtering criterion; not applied to base case (0 by default)
	INTVAL(4)	number of buses in the largest disconnected island filtering criterion; not applied to base case (0 by default)
	INTVAL(5)	maximum number of elements in the available capacity table (no limit by default)

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Real REALVAL (7)	array of 7 elements specifying (input).	
	REALVAL(1)	bus mismatch converged tolerance (MW or Mvar) (0.5 by default)
	REALVAL(2)	system mismatch converged tolerance (MVA) (5.0 by default)
	REALVAL(3)	percent of flow rating; used only when STATUS(1) is 0, 3 or 4 (100.0 by default)
	REALVAL(4)	in overload reports, minimum contingency case flow change from base case value; used only when STATUS(1) is 0 or 3 (0.0 by default)
	REALVAL(5)	in overload reports, minimum contingency case percent loading increase from base case value; used only when STATUS(1) is 0 or 3 (0.0 by default)
	REALVAL(6)	in voltage range violation reports, minimum contingency case voltage change from base case value; used only when STATUS(1) is 0, 1, 3 or 4 (0.0 by default)
	REALVAL(7)	cutoff threshold for available capacity table; used only when STATUS(1) is 2 (99999.0 by default)
Character*260 RFILE	Contingency Solution Output File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid STATUS value
	IERR = 2	invalid INTVAL value
	IERR = 3	invalid REALVAL value
	IERR = 4	error opening RFILE
	IERR = 5	error reading RFILE
	IERR = 6	prerequisite requirements for API are not met

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1.10. ACCC_SINGLE_RUN_REPORT_4

This API is the fourth release of the AC Contingency Report function. Use this API to report the results of the AC Contingency Calculation function.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_ACCC_SINGLE_RUN_REPORT_4  STATUS(1)..STATUS(12)  INTVAL(1)..INTVAL(5)
 REALVAL(1)..REALVAL(7)  RFILE
```

Python command syntax:

```
ierr = accc_single_run_report_4(status, intval, realval, rfile)
```

Fortran command syntax:

```
CALL ACCC_SINGLE_RUN_REPORT_4(STATUS, INTVAL, REALVAL, RFILE, IERR)
```

Where:

Integer STATUS (12)	array of 12 elements specifying (input).	
	STATUS(1)	report format (3 by default)
	STATUS(1) = 0	spreadsheet overload report
	STATUS(1) = 1	spreadsheet loading table
	STATUS(1) = 2	available capacity table
	STATUS(1) = 3	non-spreadsheet overload report
	STATUS(1) = 4	non-spreadsheet loading table
	STATUS(1) = 5	non-converged networks report
	STATUS(1) = 6	non-spreadsheet corrective actions report
	STATUS(2)	base case rating set; used only when STATUS(1) is 0, 1, 3 or 4 (rating set program option setting by default)
	STATUS(2) = 1	rate A
	STATUS(2) = 2	rate B
	STATUS(2) = 3	rate C
	STATUS(3)	contingency case rating set when STATUS(1) is 0, 1, 3 or 4; base case and contingency case rating set when STATUS(1) is 2 (rating set program option setting by default)
	STATUS(3) = 1	rate A
	STATUS(3) = 2	rate B
	STATUS(3) = 3	rate C
	STATUS(4)	base case voltage limit when STATUS(1) is 0, 1, 3 or 4
	STATUS(4) = 1	normal limit
	STATUS(4) = 2	emergency limit
	STATUS(5)	contingency case voltage limit when STATUS(1) is 0, 1, 3 or 4

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	STATUS(5) = 1	normal limit
	STATUS(5) = 2	emergency limit
	STATUS(6)	exclude interfaces from report; used only when STATUS(1) is 0, 1, 2, 3, 4 (0 by default)
	STATUS(6) = 0	no
	STATUS(6) = 1	yes
	STATUS(7)	run voltage limit check; used only when STATUS(1) is 0, 1, 3, 4 (0 by default)
	STATUS(7) = 0	no
	STATUS(7) = 1	yes
	STATUS(8)	in overload reports, exclude monitored branches and interfaces that show loading violations in the base case from being checked and reported in contingency cases; used only when STATUS(1) is 0 or 3 (0 by default)
	STATUS(8) = 0	no
	STATUS(8) = 1	yes
	STATUS(9)	in voltage range violation reports, exclude monitored buses that show voltage range violations in the base case from the corresponding check in contingency case reports; used only when STATUS(1) is 0 or 3 (0 by default)
	STATUS(9) = 0	no
	STATUS(9) = 1	yes
	STATUS(10)	exclude cases with no overloads from non-spreadsheet overload report or non-spreadsheet corrective actions report; used only when STATUS(1) is 3 (0 by default)
	STATUS(10) = 0	no
	STATUS(10) = 1	yes
	STATUS(11)	report post-tripping action solutions (0 by default)
	STATUS(11) = 0	no
	STATUS(11) = 1	yes
	STATUS(12)	report loss of loads (0 by default)
	STATUS(12) = 0	no
	STATUS(12) = 1	yes
Integer INTVAL (5)	array of 5 elements specifying (input).	
	INTVAL(1)	number of low voltage range violations filtering criterion (0 by default)
	INTVAL(2)	number of high voltage range violations filtering criterion (0 by default)

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	INTVAL(3)	number of voltage deviation violations filtering criterion; not applied to base case (0 by default)
	INTVAL(4)	number of buses in the largest disconnected island filtering criterion; not applied to base case (0 by default)
	INTVAL(5)	maximum number of elements in the available capacity table (no limit by default)
Real REALVAL (7)	array of 7 elements specifying (input).	
	REALVAL(1)	bus mismatch converged tolerance (MW or Mvar) (0.5 by default)
	REALVAL(2)	system mismatch converged tolerance (MVA) (5.0 by default)
	REALVAL(3)	percent of flow rating; used only when STATUS(1) is 0, 3 or 4 (100.0 by default)
	REALVAL(4)	in overload reports, minimum contingency case flow change from base case value; used only when STATUS(1) is 0 or 3 (0.0 by default)
	REALVAL(5)	in overload reports, minimum contingency case percent loading increase from base case value; used only when STATUS(1) is 0 or 3 (0.0 by default)
	REALVAL(6)	in voltage range violation reports, minimum contingency case voltage change from base case value; used only when STATUS(1) is 0, 1, 3 or 4 (0.0 by default)
	REALVAL(7)	cutoff threshold for available capacity table; used only when STATUS(1) is 2 (99999.0 by default)
Character*260 RFILE	Contingency Solution Output File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid STATUS value
	IERR = 2	invalid INTVAL value
	IERR = 3	invalid REALVAL value
	IERR = 4	error opening RFILE
	IERR = 5	error reading RFILE
	IERR = 6	prerequisite requirements for API are not met

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1.11. ACCC_TRIP_COR

This API routine is obsolete. It has been replaced by the API routine accc_trip_cor_2 .

Use this API to apply the AC contingency calculation with tripping simulations and corrective actions. When there are violations in the post-tripping simulation solutions, the corrections actions analysis is launched to eliminate the violations, with the objective of minimizing the controls adjustments, including load shedding, MW generation dispatch and phase shifter angle adjustments. Rating set program option setting and 100.0 percent of rating set are used to check branch and interface flow overload violations.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_ACCC_TRIP_COR  TOL  OPTACC(1)..OPTACC(8) OPTTRP(1)..OPTTRP(3)
 OPTCOR(1)..OPTCOR(5)  VALUES(1)..VALUES(5)  LABELS(1)..LABELS(4)  DFXFILE
  ACCFILE  THRFILE  INLFILE  TRPFILE
```

Python command syntax:

```
ierr = accc_trip_cor(tol, optacc, opttrp, optcor, values, labels, dfxfile,
 accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL ACCC_TRIP_COR(TOL, OPTACC, OPTTRP, OPTCOR, VALUES, LABELS, DFXFILE,
 ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)

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	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (reserve)
	OPTACC(8) = 2	subsystem machines (pmax)
	OPTACC(8) = 3	subsystem machines (inertia)
	OPTACC(8) = 4	subsystem machines (governor droop)
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable tripping simulation
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Integer OPTCOR (5)	array of 5 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective actions (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number of power flow solutions (from 0 to 10, 1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable

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	OPTCOR(5) = 1	enable
Real VALUES (5)	array of 5 elements specifying corrective actions parameters(input).	
	VALUES(1)	bus voltage violation tolerance (0.1 by default)
	VALUES(2)	branch flow overload tolerance (0.1 by default)
	VALUES(3)	generator control weighting factor (1.0 by default)
	VALUES(4)	load control weighting factor (1.0 by default)
	VALUES(5)	phase shifter control weighting factor (1.0 by default)
Character*32 LABELS (4)	array of 4 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of generator control subsystem (blank by default, no default allowed if OPTCOR(3) is 1)
	LABELS(3)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(4)	label of phase shifter control subsystem (blank by default, no default allowed if OPTCOR(5) is 1)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation dispatch subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	invalid OPTTRP value
	IERR = 9	invalid OPTCOR value

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IERR = 10	invalid VALUES value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 49	error opening Tripping Element Data File
IERR = 50	error opening Unit Inertia and Governor Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for API are not met

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1.12. ACCC_TRIP_COR_2

This API routine is obsolete. It has been replaced by the API routine accc_trip_cor_3 .

This API is the second release of the function to run AC contingency analysis with tripping simulation and corrective actions. Following a post contingency solution, trip events are simulated if their tripping conditions are satisfied. Corrective actions are then specified automatically to eliminate the violations in each post tripping solution with the objective of minimizing the control adjustments including load shedding, MW generation dispatch, phase shifter angle adjustments, off-line generator controls, tap setting adjustments and switched shunt controls. Please note that flows are checked against rating set program option setting and 100.0 percent of the rating to determine flow overload violations.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_ACCC_TRIP_COR_2  TOL OPTACC(1)..OPTACC(8) OPTTRP(1)..OPTTRP(3)
 OPTCOR(1)..OPTCOR(8) VALUES(1)..VALUES(8) LABELS(1)..LABELS(7) DFXFILE
 ACCFILE THRFILE INLFILE TRPFILE
```

Python command syntax:

```
ierr = accc_trip_cor_2(tol, optacc, opttrp, optcor, values, labels,
 dfxfile, accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL ACCC_TRIP_COR_2(TOL, OPTACC, OPTTRP, OPTCOR, VALUES, LABELS, DFXFILE,
 ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable

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	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous, disable discrete
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (RESERVE)
	OPTACC(8) = 2	subsystem machines (PMAX)
	OPTACC(8) = 3	subsystem machines (INERTIA)
	OPTACC(8) = 4	subsystem machines (GOVERNOR DROOP)
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable tripping simulation
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Integer OPTCOR (8)	array of 8 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective action (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number of power flow solutions (1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)

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	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
	OPTCOR(6)	off-line generator control flag (0 by default)
	OPTCOR(6) = 0	disable
	OPTCOR(6) = 1	enable
	OPTCOR(7)	tap setting adjustment flag (0 by default)
	OPTCOR(7) = 0	disable
	OPTCOR(7) = 1	enable
	OPTCOR(8)	switched shunt control flag (0 by default)
	OPTCOR(8) = 0	disable
	OPTCOR(8) = 1	enable
Real VALUES (8)	array of 8 elements specifying corrective actions parameters(input).	
	VALUES(1)	bus voltage violation tolerance (0.1 by default)
	VALUES(2)	branch overload violation tolerance (0.1 by default)
	VALUES(3)	generator control weighting factor (1.0 by default)
	VALUES(4)	load control weighting factor (1.0 by default)
	VALUES(5)	phase shifter control weighting factor (1.0 by default)
	VALUES(6)	off-line generator control weighting factor (1.0 by default)
	VALUES(7)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(8)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (7)	array of 7 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of generator control subsystem (blank by default, no default allowed if OPTCOR(3) is 1)
	LABELS(3)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(4)	label of phase shifter control subsystem (blank by default, no default allowed if OPTCOR(5) is 1)
	LABELS(5)	off-line generator control subsystem (blank by default; no default allowed if OPTCOR(6) is 1)
	LABELS(6)	tap setting adjustment subsystem (blank by default; no default allowed if OPTCOR(7) is 1)

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	LABELS(7)	switched shunt control subsystem (blank by default; no default allowed if OPTCOR(8) is 1)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of Unit Inertia Data File (input; blank for none).	
Character*260 TRPFILE	name of Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	invalid OPTTRP value
	IERR = 9	invalid OPTCOR value
	IERR = 10	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multisection line members
	IERR = 49	error opening Tripping Element Data File
	IERR = 50	error opening Unit Inertia Data File
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	prerequisite requirements for function not met

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1.13. ACCC_TRIP_COR_3

This API is the third release of the function to run AC contingency analysis with tripping simulation and corrective actions. Following a post contingency solution, trip events are simulated if their tripping conditions are satisfied. Corrective actions are then specified automatically to eliminate the violations in each post tripping solution with the objective of minimizing the control adjustments including load shedding, MW generation dispatch, phase shifter angle adjustments, off-line generator controls, tap setting adjustments and switched shunt controls. Please note that flows are checked against rating set program option setting and 100.0 percent of the rating to determine flow overload violations.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ACCC_TRIP_COR_3  TOL OPTACC(1)..OPTACC(10) OPTTRP(1)..OPTTRP(3)
 OPTCOR(1)..OPTCOR(8) VALUES(1)..VALUES(8) LABELS(1)..LABELS(7) DFXFILE
 ACCFILE THRFILE INLFILE TRPFILE
```

Python command syntax:

```
ierr = accc_trip_cor_3(tol, optacc, opttrp, optcor, values, labels,
 dfxfile, accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL ACCC_TRIP_COR_3(TOL, OPTACC, OPTTRP, OPTCOR, VALUES, LABELS, DFXFILE,
 ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (10)	array of 10 elements specifying power flow solution(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)

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	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous, disable discrete
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTACC(8) = 0	stall
	OPTACC(8) = 1	trip
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable
	OPTACC(10) = 1	subsystem machines (RESERVE)
	OPTACC(10) = 2	subsystem machines (PMAX)
	OPTACC(10) = 3	subsystem machines (INERTIA)
	OPTACC(10) = 4	subsystem machines (GOVERNOR DROOP)
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable tripping simulation
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Integer OPTCOR (8)	array of 8 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective action (0 by default)
	OPTCOR(1) = 0	disable

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	OPTCOR(1) = 1	enable
	OPTCOR(2)	number of power flow solutions (1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
	OPTCOR(6)	off-line generator control flag (0 by default)
	OPTCOR(6) = 0	disable
	OPTCOR(6) = 1	enable
	OPTCOR(7)	tap setting adjustment flag (0 by default)
	OPTCOR(7) = 0	disable
	OPTCOR(7) = 1	enable
	OPTCOR(8)	switched shunt control flag (0 by default)
	OPTCOR(8) = 0	disable
	OPTCOR(8) = 1	enable
Real VALUES (8)	array of 8 elements specifying corrective actions parameters(input).	
	VALUES(1)	bus voltage violation tolerance (0.1 by default)
	VALUES(2)	branch overload violation tolerance (0.1 by default)
	VALUES(3)	generator control weighting factor (1.0 by default)
	VALUES(4)	load control weighting factor (1.0 by default)
	VALUES(5)	phase shifter control weighting factor (1.0 by default)
	VALUES(6)	off-line generator control weighting factor (1.0 by default)
	VALUES(7)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(8)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (7)	array of 7 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of generator control subsystem (blank by default, no default allowed if OPTCOR(3) is 1)

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	LABELS(3)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(4)	label of phase shifter control subsystem (blank by default, no default allowed if OPTCOR(5) is 1)
	LABELS(5)	off-line generator control subsystem (blank by default; no default allowed if OPTCOR(6) is 1)
	LABELS(6)	tap setting adjustment subsystem (blank by default; no default allowed if OPTCOR(7) is 1)
	LABELS(7)	switched shunt control subsystem (blank by default; no default allowed if OPTCOR(8) is 1)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of Unit Inertia Data File (input; blank for none).	
Character*260 TRPFILE	name of Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	invalid OPTTRP value
	IERR = 11	invalid OPTCOR value
	IERR = 12	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multisection line members
	IERR = 49	error opening Tripping Element Data File
	IERR = 50	error opening Unit Inertia Data File
	IERR = 51	error opening Contingency Solution Output File

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IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.14. ACCC_WITH_COR

This API routine is obsolete. It has been replaced by the API routine accc_with_cor_2 .

Use this API to apply the AC contingency calculation with corrective actions. When there are violations in the post-contingency solutions, the corrections actions analysis is launched to eliminate the violations, with the objective of minimizing the controls adjustments, including load shedding, MW generation dispatch, and phase shifter angle adjustments. Rating set program option setting and 100.0 percent of the rating set are used to check branch and interface flow overload violations.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_ACCC_WITH_COR  TOL  OPTACC(1)..OPTACC(8)  OPTCOR(1)..OPTCOR(5)
 VALUES(1)..VALUES(5)  LABELS(1)..LABELS(4)  DFXFILE  ACCFILE  THRFILE
 INLFILE
```

Python command syntax:

```
ierr = accc_with_cor(tol, optacc, optcor, values, labels, dfxfile,
 accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_COR(TOL, OPTACC, OPTCOR, VALUES, LABELS, DFXFILE, ACCFILE,
 THRFILE, INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)

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	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (reserve)
	OPTACC(8) = 2	subsystem machines (pmax)
	OPTACC(8) = 3	subsystem machines (inertia)
	OPTACC(8) = 4	subsystem machines (governor droop)
Integer OPTCOR (5)	array of 5 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective actions (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number of power flow solutions (from 0 to 10, 1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load curtailment flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
Real VALUES (5)	array of 5 elements specifying corrective actions parameters(input).	
	VALUES(1)	bus voltage violation tolerance (0.1 by default)
	VALUES(2)	branch flow overload tolerance (0.1 by default)
	VALUES(3)	generator control weighting factor (1.0 by default)

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	VALUES(4)	load control weighting factor (1.0 by default)
	VALUES(5)	phase shifter control weighting factor (1.0 by default)
Character*32 LABELS (4)	array of 4 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of generator control subsystem (blank by default, no default allowed if OPTCOR(3) is 1)
	LABELS(3)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(4)	label of phase shifter control subsystem (blank by default, no default allowed if OPTCOR(5) is 1)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation dispatch subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	invalid OPTCOR value
	IERR = 9	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 50	error opening Unit Inertia and Governor Data File
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File

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IERR = 54	prerequisite requirements for API are not met

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1.15. ACCC_WITH_COR_2

This API routine is obsolete. It has been replaced by the API routine accc_with_cor_3 .

This API is the second release of function to run AC contingency analysis with corrective actions. Corrective actions are specified automatically to eliminate the violations in each post contingency solution with the objective of minimizing the controls adjustments including load shedding, MW generation dispatch, phase shifter angle adjustments, off-line generator controls, tap setting adjustments and switched shunt controls. Please note that flows are checked against rating set program option setting and 100.0 percent of the rating to determine flow overload violations.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_ACCC_WITH_COR_2  TOL OPTACC(1)..OPTACC(8) OPTCOR(1)..OPTCOR(8)
 VALUES(1)..VALUES(8) LABELS(1)..LABELS(7)  DFXFILE ACCFILE THRFILE
 INLFILE
```

Python command syntax:

```
ierr = accc_with_cor_2(tol, optacc, optcor, values, labels,  dfxfile,
 accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_COR_2(TOL, OPTACC, OPTCOR, VALUES, LABELS,  DFXFILE,
 ACCFILE, THRFILE, INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable

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	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous, disable discrete
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (RESERVE)
	OPTACC(8) = 2	subsystem machines (PMAX)
	OPTACC(8) = 3	subsystem machines (INERTIA)
	OPTACC(8) = 4	subsystem machines (GOVERNOR DROOP)
Integer OPTCOR (8)	array of 8 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective actions (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number of power flows (1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
	OPTCOR(6)	off-line generator control flag (0 by default)
	OPTCOR(6) = 0	disable
	OPTCOR(6) = 1	enable
	OPTCOR(7)	tap setting adjustment flag (0 by default)
	OPTCOR(7) = 0	disable
	OPTCOR(7) = 1	enable

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	OPTCOR(8)	switched shunt control flag (0 by default)
	OPTCOR(8) = 0	disable
	OPTCOR(8) = 1	enable
Real VALUES (8)	array of 8 elements specifying corrective actions parameters(input).	
	VALUES(1)	bus voltage violation tolerance (0.1 by default)
	VALUES(2)	branch overload violation tolerance (0.1 by default)
	VALUES(3)	generator control weighting factor (1.0 by default)
	VALUES(4)	load control weighting factor (1.0 by default)
	VALUES(5)	phase shifter control weighting factor (1.0 by default)
	VALUES(6)	off-line generator control weighting factor (1.0 by default)
	VALUES(7)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(8)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (7)	array of 7 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of generator control subsystem (blank by default, no default allowed if OPTCOR(3) is 1)
	LABELS(3)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(4)	label of phase shifter control subsystem (blank by default, no default allowed if OPTCOR(5) is 1)
	LABELS(5)	off-line generator control subsystem (blank by default; no default allowed if OPTCOR(6) is 1)
	LABELS(6)	tap setting adjustment subsystem (blank by default; no default allowed if OPTCOR(7) is 1)
	LABELS(7)	switched shunt control subsystem (blank by default; no default allowed if OPTCOR(8) is 1)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of Unit Inertia Data File (input; blank for none).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred

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IERR = 1	invalid TOL value
IERR = 2	invalid OPTACC value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	invalid OPTCOR value
IERR = 9	invalid VALUES value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multisection line members
IERR = 50	error opening Unit Inertia Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.16. ACCC_WITH_COR_3

This API is the third release of function to run AC contingency analysis with corrective actions. Corrective actions are specified automatically to eliminate the violations in each post contingency solution with the objective of minimizing the controls adjustments including load shedding, MW generation dispatch, phase shifter angle adjustments, off-line generator controls, tap setting adjustments and switched shunt controls. Please note that flows are checked against rating set program option setting and 100.0 percent of the rating to determine flow overload violations.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ACCC_WITH_COR_3  TOL OPTACC(1)..OPTACC(10) OPTCOR(1)..OPTCOR(8)
 VALUES(1)..VALUES(8) LABELS(1)..LABELS(7)  DFXFILE ACCFILE THRFILE
 INLFILE
```

Python command syntax:

```
ierr = accc_with_cor_3(tol, optacc, optcor, values, labels,  dfxfile,
 accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_COR_3(TOL, OPTACC, OPTCOR, VALUES, LABELS,  DFXFILE,
 ACCFILE, THRFILE, INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN, by default).	
Integer OPTACC (10)	array of 10 elements specifying power flow solution(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable

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	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous, disable discrete
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTACC(8) = 0	stall
	OPTACC(8) = 1	trip
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable
	OPTACC(10) = 1	subsystem machines (RESERVE)
	OPTACC(10) = 2	subsystem machines (PMAX)
	OPTACC(10) = 3	subsystem machines (INERTIA)
	OPTACC(10) = 4	subsystem machines (GOVERNOR DROOP)
Integer OPTCOR (8)	array of 8 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective actions (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number of power flows (1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)

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	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
	OPTCOR(6)	off-line generator control flag (0 by default)
	OPTCOR(6) = 0	disable
	OPTCOR(6) = 1	enable
	OPTCOR(7)	tap setting adjustment flag (0 by default)
	OPTCOR(7) = 0	disable
	OPTCOR(7) = 1	enable
	OPTCOR(8)	switched shunt control flag (0 by default)
	OPTCOR(8) = 0	disable
	OPTCOR(8) = 1	enable
Real VALUES (8)	array of 8 elements specifying corrective actions parameters(input).	
	VALUES(1)	bus voltage violation tolerance (0.1 by default)
	VALUES(2)	branch overload violation tolerance (0.1 by default)
	VALUES(3)	generator control weighting factor (1.0 by default)
	VALUES(4)	load control weighting factor (1.0 by default)
	VALUES(5)	phase shifter control weighting factor (1.0 by default)
	VALUES(6)	off-line generator control weighting factor (1.0 by default)
	VALUES(7)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(8)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (7)	array of 7 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of generator control subsystem (blank by default, no default allowed if OPTCOR(3) is 1)
	LABELS(3)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(4)	label of phase shifter control subsystem (blank by default, no default allowed if OPTCOR(5) is 1)
	LABELS(5)	off-line generator control subsystem (blank by default; no default allowed if OPTCOR(6) is 1)
	LABELS(6)	tap setting adjustment subsystem (blank by default; no default allowed if OPTCOR(7) is 1)

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	LABELS(7)	switched shunt control subsystem (blank by default; no default allowed if OPTCOR(8) is 1)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of Unit Inertia Data File (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	invalid OPTCOR value
	IERR = 11	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multisection line members
	IERR = 50	error opening Unit Inertia Data File
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	prerequisite requirements for function not met

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1.17. ACCC_WITH_DSP

This API routine is obsolete. It has been replaced by the API routine accc_with_dsp_2 .

Use this API to apply the AC contingency calculation function with a generation dispatch function to handle imbalances in power resources and demand due to contingencies.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ACCC_WITH_DSP  TOL  OPTACC(1)..OPTACC(8)  LABEL  DFXFILE  ACCFILE
 THRFILE  INLFILE
```

Python command syntax:

```
ierr = accc_with_dsp(tol, optacc, label, dfxfile, accfile, thrfile,
 inlfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_DSP(TOL, OPTACC, LABEL, DFXFILE, ACCFILE, THRFILE, INLFILE,
 IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable

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	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (reserve)
	OPTACC(8) = 2	subsystem machines (pmax)
	OPTACC(8) = 3	subsystem machines (inertia)
	OPTACC(8) = 4	subsystem machines (governor droop)
Character*32 LABEL	name of the generation dispatch subsystem (input; blank by default, no default allowed if OPTACC(8) is not 0).	
Character*260 DFXFILE	name of the Distribution Factor Data file (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output file (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data file (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation dispatch subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	no Distribution Factor Data File specified
	IERR = 9	no Contingency Solution Output File specified
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX

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IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	error opening Unit Inertia and Governor Data File
IERR = 55	prerequisite requirements for API are not met

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1.18. ACCC_WITH_DSP_2

This API routine is obsolete. It has been replaced by the API routine accc_with_dsp_3 .

Use this API to run the second release of AC contingency calculation function (ACCC, ACCC_WITH_DSP). The API provides the options to handle induction machines in a contingency analysis.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ACCC_WITH_DSP_2  TOL  OPTIONS(1)..OPTIONS(10)  LABEL  DFXFILE  ACCFILE
  THRFILE  INLFILE
```

Python command syntax:

```
ierr = accc_with_dsp_2(tol, options, label, dfxfile, accfile, thrfile,
 inlfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_DSP_2(TOL, OPTIONS, LABEL, DFXFILE, ACCFILE, THRFILE,
 INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTIONS (10)	array of 10 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)

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	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	solution method flag (0 by default)
	OPTIONS(6) = 0	FDNS
	OPTIONS(6) = 1	FNSL
	OPTIONS(6) = 2	optimized FDNS
	OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTIONS(7) = 0	disable
	OPTIONS(7) = 1	enable
	OPTIONS(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTIONS(8) = 0	stall
	OPTIONS(8) = 1	trip
	OPTIONS(9)	induction machine failure flag (0 by default)
	OPTIONS(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTIONS(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTIONS(10)	dispatch mode (0 by default)
	OPTIONS(10) = 0	disable
	OPTIONS(10) = 1	subsystem machines (reserve)
	OPTIONS(10) = 2	subsystem machines (pmax)
	OPTIONS(10) = 3	subsystem machines (inertia)
	OPTIONS(10) = 4	subsystem machines (governor droop)
Character*32 LABEL	name of the generation dispatch subsystem (input; blank by default, no default allowed if OPTIONS(10) is not 0).	
Character*260 DFXFILE	name of the Distribution Factor Data file (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output file (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data file (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted

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IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation dispatch subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	no Distribution Factor Data File specified
IERR = 9	no Contingency Solution Output File specified
IERR = 10	in-service induction machines are in the "stalled" or "tripped" state
IERR = 11	buses with bus type code and series element status inconsistencies
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	error opening Unit Inertia and Governor Data File
IERR = 55	prerequisite requirements for API are not met

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1.19. ACCC_WITH_DSP_3

Use this API to run the second release of AC contingency calculation function (ACCC, ACCC_WITH_DSP). The API provides the options to handle induction machines in a contingency analysis.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_ACCC_WITH_DSP_3  TOL OPTACC(1)..OPTACC(11) LABEL DFXFILE ACCFILE
 THRFILE INLFILE ZIPFILE
```

Python command syntax:

```
ierr = accc_with_dsp_3(tol, optacc, label, dfxfile, accfile, thrfile,
 inlfile, zipfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_DSP_3(TOL, OPTACC, LABEL, DFXFILE, ACCFILE, THRFILE,
 INLFILE, ZIPFILE, IERR)
```

Where:

Real TOL	Is the mismatch tolerance. (input; Newton solution convergence tolerance, TOLN, by default).	
Integer OPTACC (11)	array of elements specifying solution options. The values are as follows:(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable

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OPTACC(5) = 1	enable
OPTACC(5) = 2	enable continuous mode, disable discrete mode
OPTACC(6)	solution method flag (0 by default)
OPTACC(6) = 0	FDNS
OPTACC(6) = 1	FNSL
OPTACC(6) = 2	optimized FDNS
OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
OPTACC(7) = 0	disable
OPTACC(7) = 1	enable
OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage)
OPTACC(8) = 0	stall
OPTACC(8) = 1	trip
OPTACC(9)	induction machine failure flag (0 by default)
OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
OPTACC(10)	dispatch mode (0 by default)
OPTACC(10) = 0	disable
OPTACC(10) = 1	subsystem machines (reserve)
OPTACC(10) = 2	subsystem machines (pmax)
OPTACC(10) = 3	subsystem machines (inertia)
OPTACC(10) = 4	subsystem machines (governor droop)
OPTACC(11)	ZIP archive flag (0 by default)
OPTACC(11) = 0	do not write a ZIP archive file
OPTACC(11) = 1	write a ZIP archive using the file specified as ZIPFILE

Character*32 LABEL	Is the name of the generation dispatch subsystem (blank by default, no default allowed if OPTIONS(10) is not 0). (input).
Character*260 DFXFILE	Is the name of the Distribution Factor Data file (input; no default allowed).
Character*260 ACCFILE	Is the name of the Contingency Solution Output file (input; no default allowed).
Character*260 THRFILE	Is the name of the Load Throwover Data file (input; blank by default).
Character*260 INLFILE	Is the name of the Unit Inertia and Governor Data File (input; blank by default).
Character*260 ZIPFILE	Is the name of the ZIP Archive Output File (input; blank by default).
Integer IERR	error code (output).

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IERR = 0	no error occurred
IERR = 1	invalid TOL value
IERR = 2	invalid OPTIONS value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation dispatch subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	no Distribution Factor Data File specified
IERR = 9	no AC Contingency Solution Output File specified
IERR = 10	in-service induction machines are in the "stalled" or "tripped" state
IERR = 11	buses with bus type code and series element status inconsistencies
IERR = 12	no ZIP Archive Output File specified
IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	error opening Unit Inertia and Governor Data File
IERR = 55	error opening ZIP Archive Output File
IERR = 56	prerequisite requirements for API are not met.

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1.20. ACCC_WITH_TRIP

This API routine is obsolete. It has been replaced by the API routine accc_with_trip_2 .

Use this API to apply the AC contingency calculation function with a post-contingency tripping function. This may be used, for example, to simulate switchings that occur when specified monitored quantities are beyond threshold values.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ACCC_WITH_TRIP  TOL  OPTACC(1)..OPTACC(8)  OPTTRP(1)..OPTTRP(3)  LABEL
   DFXFILE   ACCFILE   THRFILE  INLFILE  TRPFILE
```

Python command syntax:

```
ierr = accc_with_trip(tol, optacc, opttrp, label, dfxfile, accfile,
 thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_TRIP(TOL, OPTACC, OPTTRP, LABEL, DFXFILE, ACCFILE, THRFILE,
 INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable

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	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (reserve)
	OPTACC(8) = 2	subsystem machines (pmax)
	OPTACC(8) = 3	subsystem machines (inertia)
	OPTACC(8) = 4	subsystem machines (governor droop)
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable tripping simulation
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Character*32 LABEL	name of the generation dispatch subsystem (input; blank by default; no default allowed if OPTACC(8) is not 0).	
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted

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IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	Invalid OPTTRP value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 49	error opening Tripping Element Data File
IERR = 50	error opening Unit Inertia and Governor Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.21. ACCC_WITH_TRIP_2

Use this API to run the second version of AC contingency calculation function with a post-contingency tripping function. This may be used, for example, to simulate switchings that occur when specified monitored quantities are beyond threshold values.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ACCC_WITH_TRIP_2  TOL  OPTACC(1)..OPTACC(10)  OPTTRP(1)..OPTTRP(3)
 LABEL   DFXFILE   ACCFILE   THRFILE  INLFILE  TRPFILE
```

Python command syntax:

```
ierr = accc_with_trip_2(tol, optacc, opttrp, label, dfxfile, accfile,
 thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_TRIP_2(TOL, OPTACC, OPTTRP, LABEL, DFXFILE, ACCFILE,
 THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (10)	array of 10 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable

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	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTACC(8) = 0	stall
	OPTACC(8) = 1	trip
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable
	OPTACC(10) = 1	subsystem machines (reserve)
	OPTACC(10) = 2	subsystem machines (pmax)
	OPTACC(10) = 3	subsystem machines (inertia)
	OPTACC(10) = 4	subsystem machines (governor droop)
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable tripping simulation
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Character*32 LABEL	name of the generation dispatch subsystem (input; blank by default; no default allowed if OPTACC(8) is not 0).	
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	

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Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).
Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	invalid TOL value
IERR = 2	invalid OPTACC value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
IERR = 9	buses with bus type code and series element status inconsistencies
IERR = 10	Invalid OPTTRP value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-sec-tion line members
IERR = 49	error opening Tripping Element Data File
IERR = 50	error opening Unit Inertia and Governor Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.22. ACCC_WITH_TRIP_PARALLEL

Use this API to run the parallel version of AC contingency calculation function with a post-contingency tripping function (use ACCC_WITH_TRIP_2 to run the function sequentially). The API distributes contingency cases among multiple processors evenly, and tests the contingency cases and following tripping actions assigned to each processor in parallel.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_ACCC_WITH_TRIP_PARALLEL  TOL OPTACC(1)..OPTACC(10)
 OPTTRP(1)..OPTTRP(3) LABEL DFXFILE ACCFILE THRFILE INLFILE TRPFILE
```

Python command syntax:

```
ierr = accc_with_trip_parallel(tol, optacc, opttrp, label, dfxfile,
 accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL ACCC_WITH_TRIP_PARALLEL(TOL, OPTACC, OPTTRP, LABEL, DFXFILE, ACCFILE,
 THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	Is the mismatch tolerance (input; Newton solution convergence tolerance, TOLN, by default).	
Integer OPTACC (10)	array of elements specifying power flow solution options. The values are as follows:(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable.
	OPTACC(1) = 1	enable stepping adjustment.
	OPTACC(1) = 2	enable direct adjustment.
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable.
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange.
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange.
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable.
	OPTACC(3) = 1	enable.
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable.
	OPTACC(4) = 1	enable.

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	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable.
	OPTACC(5) = 1	enable.
	OPTACC(5) = 2	enable continuous mode, disable discrete mode.
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS.
	OPTACC(6) = 1	FNSL.
	OPTACC(6) = 2	optimized FDNS.
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable.
	OPTACC(7) = 1	enable.
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage). (0 by default)
	OPTACC(8) = 0	stall.
	OPTACC(8) = 1	trip.
	OPTACC(9)	induction machine failure flag. (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state.
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state.
	OPTACC(10)	dispatch mode. (0 by default)
	OPTACC(10) = 0	disable.
	OPTACC(10) = 1	subsystem machines (reserve).
	OPTACC(10) = 2	subsystem machines (pmax).
	OPTACC(10) = 3	subsystem machines (inertia).
	OPTACC(10) = 4	subsystem machines (governor droop).
Integer OPTTRP (3)	array of elements specifying tripping simulation options. The values are as follows:(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable.
	OPTTRP(1) = 1	enable tripping simulation.
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Character*32 LABEL	Is the name of the generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0). (input).	

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Character*260 DFXFILE	Is the name of the Distribution Factor Data File. (input; no default allowed).
Character*260 ACCFILE	Is the name of the Contingency Solution Output File. (input; no default allowed).
Character*260 THRFILE	Is the name of the Load Throwover Data File. (input; blank by default).
Character*260 INLFILE	Is the name of the Unit Inertia and Governor Data File. (input; blank by default).
Character*260 TRPFILE	Is the name of the Tripping Element Data File (input, blank for none; no default allowed if OPTTRP(1) is 1). (input).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	invalid TOL value
IERR = 2	invalid OPTACC value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation dispatch subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
IERR = 9	buses with bus type code and series element status inconsistencies
IERR = 10	Invalid OPTTRP value.
IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 49	error opening Tripping Element Data File.
IERR = 50	error opening Unit Inertia and Governor Data File.
IERR = 51	error opening Contingency Solution Output File.
IERR = 52	error opening Distribution Factor Data File.
IERR = 53	error opening Load Throwover Data File.
IERR = 54	prerequisite requirements for API are not met.

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1.23. ACCOR

This API routine is obsolete. It has been replaced by the API routine accor_2 .

Use this API to apply corrective actions to the base case using successive linear programming methods. The corrective actions are to eliminate flow overloads and voltage violations with the objective of minimizing the control adjustments, including load shedding, MW generations dispatch, and phase shifter angle adjustments.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_ACCOR  OPTIONS(1)..OPTIONS(10)  VALUES(1)..VALUES(7)
 LABELS(1)..LABELS(3)  DFXFILE
```

Python command syntax:

```
ierr = accor(options, values, labels, dfxfile)
```

Fortran command syntax:

```
CALL ACCORAPI(OPTIONS, VALUES, LABELS, DFXFILE, IERR)
```

Where:

Integer OPTIONS (10)	array of 10 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable

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	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	rating set (rating set program option setting by default)
	OPTIONS(6) = 1	rating A
	OPTIONS(6) = 2	rating B
	OPTIONS(6) = 3	rating C
	OPTIONS(7)	number of power flow solutions (from 1 to 10; 1 by default)
	OPTIONS(8)	generator control flag (1 by default)
	OPTIONS(8) = 0	disable
	OPTIONS(8) = 1	enable
	OPTIONS(9)	load control flag (0 by default)
	OPTIONS(9) = 0	disable
	OPTIONS(9) = 1	enable
	OPTIONS(10)	phase shifter control flag (1 by default)
	OPTIONS(10) = 0	disable
	OPTIONS(10) = 1	enable
Real VALUES (7)	array of 7 elements specifying AC corrective actions parameters(input).	
	VALUES(1)	MW mismatch tolerance (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	percent of rating set (100.0 by default)
	VALUES(3)	bus voltage violation tolerance (0.1 by default)
	VALUES(4)	branch flow overload tolerance (0.1 by default)
	VALUES(5)	generator control weighting factor (1.0 by default)
	VALUES(6)	load control weighting factor (1.0 by default)
	VALUES(7)	phase shifter control weighting factor (1.0 by default)
Character*32 LABELS (3)	array of 3 elements specifying subsystem labels(input).	
	LABELS(1)	label of generator control subsystem (blank by default; no default allowed if OPTIONS(8) is 1)
	LABELS(2)	label of load control subsystem (blank by default; no default allowed if OPTIONS(9) is 1)
	LABELS(3)	label of phase shifter control subsystem (blank by default; no default allowed if OPTIONS(10) is 1)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred

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IERR = 1	invalid OPTIONS value
IERR = 2	invalid VALUES value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	No controls
IERR = 6	control subsystem not defined
IERR = 7	Distribution Factor Data File not specified
IERR = 8	largest mismatch exceeds tolerance
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening Distribution Factor Data File
IERR = 52	prerequisite requirements for function not met

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1.24. ACCOR_2

This API routine is obsolete. It has been replaced by the API routine accor_3 .

This API is the second release of corrective action function. Use this API to augment control strategies with off-line generator control, tap setting adjustments and switched shunt controls.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_ACCOR_2  OPTIONS(1)..OPTIONS(13)  VALUES(1)..VALUES(10)
 LABELS(1)..LABELS(6)  DFXFILE
```

Python command syntax:

```
ierr = accor_2(options, values, labels, dfxfile)
```

Fortran command syntax:

```
CALL ACCORAPI_2(OPTIONS, VALUES, LABELS, DFXFILE, IERR)
```

Where:

Integer OPTIONS (13)	array of 13 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable

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	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	rating set (rating set program option setting by default)
	OPTIONS(6) = 1	rating A
	OPTIONS(6) = 2	rating B
	OPTIONS(6) = 3	rating C
	OPTIONS(7)	number of power flow solutions (from 1 to 10; 1 by default)
	OPTIONS(8)	generator control flag (1 by default)
	OPTIONS(8) = 0	disable
	OPTIONS(8) = 1	enable
	OPTIONS(9)	load control flag (0 by default)
	OPTIONS(9) = 0	disable
	OPTIONS(9) = 1	enable
	OPTIONS(10)	phase shifter control flag (1 by default)
	OPTIONS(10) = 0	disable
	OPTIONS(10) = 1	enable
	OPTIONS(11)	off-line generator control flag (0 by default)
	OPTIONS(11) = 0	disable
	OPTIONS(11) = 1	enable
	OPTIONS(12)	tap setting adjustment flag (0 by default)
	OPTIONS(12) = 0	disable
	OPTIONS(12) = 1	enable
	OPTIONS(13)	switched shunt control flag (0 by default)
	OPTIONS(13) = 0	disable
	OPTIONS(13) = 1	enable
Real VALUES (10)	array of 10 elements specifying AC corrective actions parameters(input).	
	VALUES(1)	MW mismatch tolerance (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	percent of rating set (100.0 by default)
	VALUES(3)	bus voltage violation tolerance (0.1 by default)
	VALUES(4)	branch flow overload tolerance (0.1 by default)
	VALUES(5)	generator control weighting factor (1.0 by default)
	VALUES(6)	load control weighting factor (1.0 by default)
	VALUES(7)	phase shifter control weighting factor (1.0 by default)
	VALUES(8)	off-line generator control weighting factor (1.0 by default)
	VALUES(9)	tap setting adjustment weighting factor (1.0 by default)

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	VALUES(10)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (6)	array of 6 elements specifying subsystem labels(input).	
	LABELS(1)	label of generator control subsystem (blank by default; no default allowed if OPTIONS(8) is 1)
	LABELS(2)	label of load control subsystem (blank by default; no default allowed if OPTIONS(9) is 1)
	LABELS(3)	label of phase shifter control subsystem (blank by default; no default allowed if OPTIONS(10) is 1)
	LABELS(4)	off-line generator control subsystem (blank by default; no default allowed if OPTIONS(11) is 1)
	LABELS(5)	tap setting adjustment subsystem (blank by default; no default allowed if OPTIONS(12) is 1)
	LABELS(6)	switched shunt control subsystem (blank by default; no default allowed if OPTIONS(13) is 1)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus, use activity TREE
	IERR = 5	No controls
	IERR = 6	control subsystem not defined
	IERR = 7	Distribution Factor Data File not specified
	IERR = 8	largest mismatch exceeds tolerance
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file, run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 51	error opening Distribution Factor Data File
	IERR = 52	prerequisite requirements for function not met

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1.25. ACCOR_3

This API is the third release of corrective action function. Use this API to specify voltage limits in determining voltage violations.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_ACCOR_3  OPTIONS(1)..OPTIONS(14) VALUES(1)..VALUES(10)
 LABELS(1)..LABELS(6) DFXFILE
```

Python command syntax:

```
ierr = accor_3(options, values, labels, dfxfile)
```

Fortran command syntax:

```
CALL ACCORAPI_3(OPTIONS, VALUES, LABELS, DFXFILE, IERR)
```

Where:

Integer OPTIONS (14)	array of elements specifying solution options. The values are as follows:(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable.
	OPTIONS(1) = 1	enable stepping adjustment.
	OPTIONS(1) = 2	enable direct adjustment.
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable.
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange.
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange.
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable.
	OPTIONS(3) = 1	enable.
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable.
	OPTIONS(4) = 1	enable.
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable.
	OPTIONS(5) = 1	enable.
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode.

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	OPTIONS(6)	rating set (rating set program option setting by default)
	OPTIONS(6) = 1	rating A
	OPTIONS(6) = 2	rating B
	OPTIONS(6) = 3	rating C
	OPTIONS(7)	voltage limit
	OPTIONS(7) = 1	normal
	OPTIONS(7) = 2	emergency
	OPTIONS(8)	number of power flow solutions (from 1 to 10; 1 by default)
	OPTIONS(9)	generator control flag (1 by default)
	OPTIONS(9) = 0	disable.
	OPTIONS(9) = 1	enable.
	OPTIONS(10)	load control flag. (0 by default)
	OPTIONS(10) = 0	disable.
	OPTIONS(10) = 1	enable.
	OPTIONS(11)	phase shifter control flag (1 by default)
	OPTIONS(11) = 0	disable.
	OPTIONS(11) = 1	enable.
	OPTIONS(12)	off-line generator control flag (0 by default)
	OPTIONS(12) = 0	disable.
	OPTIONS(12) = 1	enable.
	OPTIONS(13)	tap setting adjustment flag (0 by default)
	OPTIONS(13) = 0	disable.
	OPTIONS(13) = 1	enable.
	OPTIONS(14)	switched shunt control flag (0 by default)
	OPTIONS(14) = 0	disable.
	OPTIONS(14) = 1	enable.
Real VALUES (10)	array of elements specifying AC corrective actions parameters. The values are as follows:(input).	
	VALUES(1)	MW mismatch tolerance (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	percent of rating set (100.0 by default)
	VALUES(3)	bus voltage violation tolerance (0.1 by default)
	VALUES(4)	branch flow overload tolerance (0.1 by default)
	VALUES(5)	generator control weighting factor (1.0 by default)
	VALUES(6)	load control weighting factor (1.0 by default)
	VALUES(7)	phase shifter control weighting factor (1.0 by default)
	VALUES(8)	off-line generator control weighting factor (1.0 by default)

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	VALUES(9)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(10)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (6)	array of elements specifying subsystem labels (input). The values are as follows:(input).	
	LABELS(1)	label of generator control subsystem (blank by default; no default allowed if OPTIONS(9) is 1).
	LABELS(2)	label of load control subsystem (blank by default; no default allowed if OPTIONS(10) is 1).
	LABELS(3)	label of phase shifter control subsystem (blank by default; no default allowed if OPTIONS(11) is 1).
	LABELS(4)	off-line generator control subsystem (blank by default; no default allowed if OPTIONS(12) is 1).
	LABELS(5)	tap setting adjustment subsystem (blank by default; no default allowed if OPTIONS(13) is 1).
	LABELS(6)	switched shunt control subsystem (blank by default; no default allowed if OPTIONS(14) is 1).
Character*260 DFXFILE	Is the name of the Distribution Factor Data File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	No controls.
	IERR = 6	control subsystem not defined.
	IERR = 7	Distribution Factor Data File not specified.
	IERR = 8	largest mismatch exceeds tolerance.
	IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX.
	IERR = 22	monitored elements exceed limit when adding multi-section line members.
	IERR = 51	error opening Distribution Factor Data File.
	IERR = 52	prerequisite requirements for API are not met.

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1.26. ALERT_OUTPUT

Use this API to specify the alert device.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_ALERT_OUTPUT  ISLCT  FILARG  OPTIONS(1)  OPTIONS(2)
```

Python command syntax:

```
ierr = alert_output(islct, filarg, options)
```

Fortran command syntax:

```
CALL OPENALERTDEVICE(ISLCT, FILARG, OPTIONS, IERR)
```

Where:

Integer ISLCT	virtual device selector (input).	
	ISLCT = 1	standard destination. In the PSSE GUI standard output for alerts is a message box and a copy to the Alerts/Warnings tab of the output bar.
	ISLCT = 2	direct output to a file.
	ISLCT = 3	direct output to a printer.
	ISLCT = 4	direct output to progress device.
	ISLCT = 5	direct output to report device.
	ISLCT = 6	no output.
Character*260 FILARG	printer name or filename; ignored if ISLCT is not 2 or 3 (input).	
Integer OPTIONS (2)	array of 2 elements specifying open and printing options; ignored if ISLCT is not 2 or 3(input).	
	OPTIONS(1)	file and printer open options: The OPTIONS(1) values may be summed to select multiple options
	OPTIONS(1) = 0	open with carriage control format and, for files, for overwrite of existing files
	OPTIONS(1) = 1	open with list format
	OPTIONS(1) = 2	open file for append (file output only)
	OPTIONS(1) = 2	set printer orientation to portrait mode (printer output only)
	OPTIONS(1) = 4	open printer in immediate print mode (printer output only)
	OPTIONS(2)	number of copies to print (printer only)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = 2	failed to open device

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1.27. ALPH

Use this API to print an alphabetically sorted table of all buses in a specified subsystem of the working case (activity ALPH).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ALPH  SID  ALL
```

Python command syntax:

```
ierr = alph(sid, all)
```

Fortran command syntax:

```
CALL ALPHNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.28. ANSI

This API routine is obsolete. It has been replaced by the API routine ansi_2 .

Use this API to calculate fault currents according to the ANSI standard (activity ANSI). To designate fault specification data as part of the API's calling sequence, INARG must be positive; to designate fault specification data in an ANSI Fault Specification Data input file, INARG must be zero and the filename specified as FILARG.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ANSI  OPTIONS(1)..OPTIONS(3)  VALUES(1)..VALUES(4)
 INARG  BUSARY(1)..BUSARY(INARG)  VLTARY(1)..VLTARY(INARG)
 CPTARY(1)..CPTARY(INARG)  FILARG
```

Python command syntax:

```
ierr = ansi(options, values, inarg, busary, vltary, cptary, filarg)
```

Fortran command syntax:

```
CALL ANSIAPI(OPTIONS, VALUES, INARG, BUSARY, VLTARY, CPTARY, FILARG, IERR)
```

Where:

Integer OPTIONS (3)	array of 3 elements specifying solution and reporting options(input).	
	OPTIONS(1)	transformer impedance correction option (1 by default)
	OPTIONS(1) = 0	do not apply to zero sequence
	OPTIONS(1) = 1	apply to zero sequence
	OPTIONS(2)	set of ANSI decrement factors to use (1 by default)
	OPTIONS(2) = 0	account for dc decrement only (use ANSI table 3)
	OPTIONS(2) = 1	account for ac and dc decrements (use ANSI tables 1 and 2)
	OPTIONS(3)	option (0 by default)
	OPTIONS(3) = 0	summary output using ANSI X only, report format (blank delimited output fields)
	OPTIONS(3) = 1	summary output using ANSI X only, spreadsheet input format (comma delimited output fields)
	OPTIONS(3) = 2	detailed output using ANSI X only
	OPTIONS(3) = 3	detailed output using ANSI R and X
	OPTIONS(3) = 4	summary output using ANSI R and X, report format (blank delimited output fields)

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	OPTIONS(3) = 5	summary output using ANSI R and X, spreadsheet input format (comma delimited output fields)
Real VALUES (4)	array of 4 elements specifying divisors for calculating resistances of network elements where resistance is zero in the working case(input).	
	VALUES(1)	used for branches in the positive sequence network (40.0 by default)
	VALUES(2)	used for machines in the positive sequence network (80.0 by default)
	VALUES(3)	used for branches in the zero sequence network (40.0 by default)
	VALUES(4)	used for machines in the zero sequence network (80.0 by default)
Integer INARG	fault specification option (input; 0 by default).	
	INARG = 0	fault specifications specified in input file.
	INARG > 0	fault specifications specified in calling sequence.
Integer BUSARY (INARG)	a list of faulted buses; ignored if INARG is 0 (if INARG > 0) (input; no default allowed).	
Real VLTARY (INARG)	a list of maximum operating voltages in pu; ignored if INARG is 0 (input; 1.0).	
Real CPTARY (INARG)	a list of contact parting times in seconds; ignored if INARG is 0 (input; 1 cycle).	
Character*260 FILARG	filename of ANSI Fault Specification Data File; ignored if INARG > 0 (if INARG is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	invalid INARG value
	IERR = 4	fault analysis is invalid due to the presence of in-service induction machines that are in the "stalled" or "tripped" state
	IERR = 5	bus type code and series element status inconsistencies
	IERR = 6	error opening file FILARG
	IERR = 7	prerequisite requirements for API are not met

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1.29. ANSI_2

This API routine executes the second release of the ANSI fault calculation function. Use this API routine to calculate fault currents according to the ANSI standard (activity ANSI). To designate fault specification data as part of the API's calling sequence, INARG must be positive; to designate fault specification data in an ANSI Fault Specification Data input file, INARG must be zero and the filename specified as FILARG.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ANSI_2  OPTIONS(1)..OPTIONS(5)  VALUES(1)..VALUES(4)
 INARG  BUSARY(1)..BUSARY(INARG)  VLTARY(1)..VLTARY(INARG)
 CPTARY(1)..CPTARY(INARG)  FILARG
```

Python command syntax:

```
ierr = ansi_2(options, values, inarg, busary, vltary, cptary, filarg)
```

Fortran command syntax:

```
CALL ANSI_2(OPTIONS,VALUES,INARG,BUSARY,VLTARY,CPTARY,FILARG,IERR)
```

Where:

Integer OPTIONS (5)	array of 5 elements specifying solution and reporting options(input).	
	OPTIONS(1)	fault analysis generator reactance option (0 by default)
	OPTIONS(1) = 0	use subtransient reactance
	OPTIONS(1) = 1	use transient reactance
	OPTIONS(1) = 2	use synchronous reactance
	OPTIONS(2)	transformer impedance correction option (1 by default)
	OPTIONS(2) = 0	do not apply to zero sequence
	OPTIONS(2) = 1	apply to zero sequence
	OPTIONS(3)	set of ANSI decrement factors to use (1 by default)
	OPTIONS(3) = 0	account for dc decrement only (use ANSI table 3)
	OPTIONS(3) = 1	account for ac and dc decrements (use ANSI tables 1 and 2)
	OPTIONS(4)	option (0 by default)
	OPTIONS(4) = 0	summary output using ANSI X only, report format (blank delimited output fields)
	OPTIONS(4) = 1	summary output using ANSI X only, spreadsheet input format (comma delimited output fields)
	OPTIONS(4) = 2	detailed output using ANSI X only
	OPTIONS(4) = 3	detailed output using ANSI R and X

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	OPTIONS(4) = 4	summary output using ANSI R and X, report format (blank delimited output fields)
	OPTIONS(4) = 5	summary output using ANSI R and X, spreadsheet input format (comma delimited output fields)
	OPTIONS(5)	breaker rating option (0 by default)
	OPTIONS(5) = 0	IEEE total current basis
	OPTIONS(5) = 1	IEEE symmetrical current basis
Real VALUES (4)	array of 4 elements specifying divisors for calculating resistances of network elements where resistance is zero in the working case(input).	
	VALUES(1)	used for branches in the positive sequence network (40.0 by default)
	VALUES(2)	used for machines in the positive sequence network (80.0 by default)
	VALUES(3)	used for branches in the zero sequence network (40.0 by default)
	VALUES(4)	used for machines in the zero sequence network (80.0 by default)
Integer INARG	fault specification option (input; 0 by default).	
	INARG = 0	fault specifications specified in input file.
	INARG > 0	fault specifications specified in calling sequence.
Integer BUSARY (INARG)	a list of faulted buses; ignored if INARG is 0 (if INARG > 0) (input; no default allowed).	
Real VLTARY (INARG)	a list of maximum operating voltages in pu; ignored if INARG is 0 (input; 1.0).	
Real CPTARY (INARG)	a list of contact parting times in seconds; ignored if INARG is 0 (input; 1 cycle).	
Character*260 FILARG	filename of ANSI Fault Specification Data File; ignored if INARG > 0 (if INARG is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	invalid INARG value
	IERR = 4	fault analysis is invalid due to the presence of in-service induction machines that are in the "stalled" or "tripped" state
	IERR = 5	bus type code and series element status inconsistencies
	IERR = 6	error opening file FILARG
	IERR = 7	prerequisite requirements for API are not met

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1.30. APPEND_ACCC

Use this API to replicate system conditions of a contingency case solution, as contained in a designated Saved Case File, in the form of a Contingency Solution Output File. Results may be appended to an existing Contingency Solution Output File, or an new Contingency Solution Output File may be created.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_APPEND_ACCC  OPTION  DFXFILE  BASFILE  CNTFILE  ACCFILE  LABEL
 NUMDESC  DESCRPT(1)..DESCRPT(NUMDESC)
```

Python command syntax:

```
ierr = append_accc(option, dfxfile, basfile, cntfile, accfile, label,
 numdesc, descrpt)
```

Fortran command syntax:

```
CALL APPEND_ACCC(OPTION, DFXFILE, BASFILE, CNTFILE, ACCFILE, LABEL,
 NUMDESC, DESCRPT, IERR)
```

Where:

Integer OPTION	append flag (input; 1 by default). OPTION = 0 create a new Contingency Solution Output File in the file specified in ACCFILE. OPTION = 1 append results to the existing Contingency Solution Output File specified in ACCFILE
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).
Character*260 BASFILE	name of the Saved Case File containing the pre-contingency power flow working case. This is the case that was present when the Distribution Factor Data File specified in DFXFILE was constructed; it is also the case from which the contingency case was formed. Its solution is in the Saved Case File specified in CNTFILE (input; no default allowed).
Character*260 CNTFILE	name of the Saved Case File containing the contingency case solution for which results are to be placed in the Contingency Solution Output File specified in ACCFILE. This case must be formed by imposing status and/or boundary condition changes on the pre-contingency case specified in BASFILE, and then solving using one or more of the power flow solution activities (input; no default allowed).
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).
Character*32 LABEL	12 character contingency label (input; blank by default).
Integer NUMDESC	number of lines of text used to describe the contingency case; must be between 1 and 16 (input; no default allowed).
Character*160 DESCRPT (NUMDESC)	Are NUMDESC lines of text describing the contingency case (input; blank by default).

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Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTION value
	IERR = 2	invalid NUMDESC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	no base case Saved Case File specified
	IERR = 6	no contingency case Saved Case File specified
	IERR = 7	Contingency Solution Output File not found
	IERR = 8	DFAX File and Contingency Solution Output File are not compatible
	IERR = 9	no Distribution Factor Data File specified
	IERR = 10	no Contingency Solution Output File specified
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening pre-contingency Saved Case File
	IERR = 54	error opening contingency case Saved Case File
	IERR = 55	error reading Contingency Solution Output File
	IERR = 56	prerequisite requirements for API are not met

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1.31. AREA

This API routine is obsolete. It has been replaced by the API routine area_2 .

Use this API to tabulate area totals by area, as well as the desired area net interchange (activity AREA).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_AREA  SID  ALL
```

Python command syntax:

```
ierr = area(sid, all)
```

Fortran command syntax:

```
CALL AREANW(SID, ALL, IERR)
```

Where:

Integer SID	a valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Area subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all areas or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all areas.
	ALL = 0	process only areas in area subsystem SID.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 4	prerequisite requirements for API are not met

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1.32. AREA_2

Use this API to tabulate area totals by area, as well as the desired area net interchange (activity AREA).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_AREA_2  SID  ALL  OPT
```

Python command syntax:

```
ierr = area_2(sid, all, opt)
```

Fortran command syntax:

```
CALL AREA_2(SID, ALL, OPT, IERR)
```

Where:

Integer SID	a valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Area subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all areas or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all areas.
	ALL = 0	process only areas in area subsystem SID.
Integer OPT	load and induction machine reporting option (input; 2 by default).	
	OPT = 1	tabulate loads, induction generators and induction motors connected to area buses
	OPT = 2	tabulate loads, induction generators and induction motors assigned to the area
	OPT = 3	tabulate three load and induction machine subtotals for each area: -area load and induction machines connected to area buses -area load and induction machines connected to non-area buses -non-area load and induction machines connected to area buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	Invalid OPT value
	IERR = 4	prerequisite requirements for API are not met

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1.33. AREA_ZONE

Use this API to tabulate area totals by area, along with subtotals by zone. The desired net interchange of each area reported is also tabulated.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_AREA_ZONE  SID  ALL
```

Python command syntax:

```
ierr = area_zone(sid, all)
```

Fortran command syntax:

```
CALL AREA_ZONE(SID, ALL, IERR)
```

Where:

Integer SID	a valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Area subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all areas or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all areas.
	ALL = 0	process only areas in area subsystem SID.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.34. ARNM

This API routine is obsolete. It has been replaced by the API routine arnm_2 .

Use this API to reassign the buses and loads in a specified subsystem of the working case from their original area to a designated area (activity ARNM).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ARNM  SID  ALL  STATUS(1)  STATUS(2)  IANEW
```

Python command syntax:

```
ierr = arnm(sid, all, status, ianew)
```

Fortran command syntax:

```
CALL ARNMNW(SID, ALL, STATUS, IANEW, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer STATUS (2)	array of 2 elements specifying (input).	
	STATUS(1)	Area assignments of buses to IANEW; (0 by default)
	STATUS(1) = 1	Change area assignments of buses to IANEW
	STATUS(1) = 0	Leave area assignments of buses unchanged
	STATUS(2)	Area assignments of loads to IANEW; (0 by default)
	STATUS(2) = 1	Change area assignments of loads to IANEW
	STATUS(2) = 0	Leave area assignments of loads unchanged
Integer IANEW	new area to be assigned to specified buses and/or loads. IANEW must be between 1 and 9999 (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid new area number
	IERR = 5	area table is full; cannot add area to the working case

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IERR = 6	prerequisite requirements for API are not met

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1.35. ARNM_2

This API routine is the second release of the area renumbering function. Use this API routine to reassign the buses, loads and/or induction machines in a specified subsystem of the working case from their original area to a designated area (activity ARNM).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ARNM_2  SID  ALL  STATUS(1)  STATUS(2)  STATUS(3)  IANEW
```

Python command syntax:

```
ierr = arnm_2(sid, all, status, ianew)
```

Fortran command syntax:

```
CALL ARNM_2(SID, ALL, STATUS, IANEW, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer STATUS (3)	array of 3 elements specifying (input).	
	STATUS(1)	Area assignments of buses to IANEW; (0 by default)
	STATUS(1) = 1	Change area assignments of buses to IANEW
	STATUS(1) = 0	Leave area assignments of buses unchanged
	STATUS(2)	Area assignments of loads to IANEW; (0 by default)
	STATUS(2) = 1	Change area assignments of loads to IANEW
	STATUS(2) = 0	Leave area assignments of loads unchanged
	STATUS(3)	Area assignments of induction machines to IANEW; (0 by default)
	STATUS(3) = 1	Change area assignments of induction machines to IANEW
	STATUS(3) = 0	Leave area assignments of induction machines unchanged
Integer IANEW	new area to be assigned to specified buses, load, and/or induction machines. IANEW must be between 1 and 9999 (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined

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IERR = 2	invalid ALL value
IERR = 3	invalid STATUS value
IERR = 4	invalid new area number
IERR = 5	area table is full; cannot add area to the working case
IERR = 6	prerequisite requirements for API are not met

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1.36. ASCC

This API routine is obsolete. It has been replaced by the API routine ascc_1a .

There is a new release of this API, ASCC_2. This API is kept to be able to produce ASCC text output in the old format. Use this API to apply a series of faults at various locations in the working case (activity ASCC).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ASCC  SID  ALL  STATUS(1)..STATUS(10)  RELFILE  FCDFILE
```

Python command syntax:

```
ierr = ascc(sid, all, status, relfile, fcdfile)
```

Fortran command syntax:

```
CALL ASCCNW(SID, ALL, STATUS, RELFILE, FCDFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (10)	array of 10 elements specifying integer options(input).	
	STATUS(1)	Three phase and line-to-ground fault (default alias is flt3phlg) (0 by default)
	STATUS(1) = 0	three phase faults only
	STATUS(1) = 1	three phase and line-to-ground faults
	STATUS(2)	line-out faults (default alias is linout) (0 by default)
	STATUS(2) = 0	omit
	STATUS(2) = 1	include
	STATUS(3)	line-end faults (default alias is linend) (0 by default)
	STATUS(3) = 0	omit
	STATUS(3) = 1	include
	STATUS(4)	flat conditions option (default alias is flatop) (0 by default)
	STATUS(4) = 0	pre-fault conditions taken from working case
	STATUS(4) = 1	impose flat conditions as pre-fault condition
	STATUS(5)	double-precision fault calculation option (default alias is nouse1) (ignored, but must be

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		specified as 0 or 1; 0 by default; double precision always used)
	STATUS(6)	report option (default alias is rptop) (1 by default)
	STATUS(6) = 1	full output at home bus and N levels away
	STATUS(6) = 2	full output at home bus, summary N levels away
	STATUS(6) = 3	fault current summary table
	STATUS(7)	number of levels back for contributions if STATUS(6) = 1 or 2. STATUS(7) > 0 (default alias is rptlvl) (0 by default)
	STATUS(8)	branch quantity code for line to ground faults (default alias is brnop) (0 by default)
	STATUS(8) = 0	A phase
	STATUS(8) = 1	3 * IO
	STATUS(8) = 2	both
	STATUS(9)	dc line and FACTS device option (default alias is dcload) (0 by default)
	STATUS(9) = 0	blocked
	STATUS(9) = 1	represent as load
	STATUS(10)	transformer impedance correction option (default alias is zcorec) (1 by default)
	STATUS(10) = 0	do not apply to zero sequence
	STATUS(10) = 1	apply to zero sequence
Character*260 RELFILE	Relay Data (.rel) file (input; blank by default).	
Character*260 FCDFILE	Fault Control (.fcd) file (input; blank by default).	
Integer IERR	error code (output).	

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1.37. ASCC_1A

This API routine has been replaced by the API routine ASCC_2. This API routine is retained so that fault output in the form of the original ASCC may be obtained.
Use this API routine to apply a series of faults at various locations in the working case (activity ASCC).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ASCC_1A  SID  ALL  STATUS(1)..STATUS(10)  RELFILE  FCDFILE
```

Python command syntax:

```
ierr = ascc_1a(sid, all, status, relfile, fcdfile)
```

Fortran command syntax:

```
CALL ASCC_1A(SID, ALL, STATUS, RELFILE, FCDFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (10)	array of 10 elements specifying integer options(input).	
	STATUS(1)	Three phase and line-to-ground fault (default alias is flt3phlg) (0 by default)
	STATUS(1) = 0	three phase faults only
	STATUS(1) = 1	three phase and line-to-ground faults
	STATUS(2)	line-out faults (default alias is linout) (0 by default)
	STATUS(2) = 0	omit
	STATUS(2) = 1	include
	STATUS(3)	line-end faults (default alias is linend) (0 by default)
	STATUS(3) = 0	omit
	STATUS(3) = 1	include
	STATUS(4)	flat conditions option (default alias is flatop) (0 by default)
	STATUS(4) = 0	pre-fault conditions taken from working case
	STATUS(4) = 1	impose flat conditions as pre-fault condition
	STATUS(5)	fault analysis generator reactance option; ignored if generators are already "converted" (default alias is genxop) (0 by default)

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	STATUS(5) = 0	use subtransient reactance
	STATUS(5) = 1	use transient reactance
	STATUS(5) = 2	use synchronous reactance
	STATUS(6)	option (default alias is rptop) (1 by default)
	STATUS(6) = 1	full output at home bus and N levels away
	STATUS(6) = 2	full output at home bus, summary N levels away
	STATUS(6) = 3	fault current summary table
	STATUS(7)	number of levels back for contributions if STATUS(6) = 1 or 2. STATUS(7) > 0 (default alias is rptlvl) (0 by default)
	STATUS(8)	branch quantity code for line to ground faults (default alias is brnop) (0 by default)
	STATUS(8) = 0	A phase
	STATUS(8) = 1	3 * IO
	STATUS(8) = 2	both
	STATUS(9)	dc line and FACTS device option (default alias is dcload) (0 by default)
	STATUS(9) = 0	blocked
	STATUS(9) = 1	represent as load
	STATUS(10)	transformer impedance correction option (default alias is zcorec) (1 by default)
	STATUS(10) = 0	do not apply to zero sequence
	STATUS(10) = 1	apply to zero sequence
Character*260 RELFILE	Relay Data (.rel) file (input; blank by default).	
Character*260 FCDFILE	Fault Control (.fcd) file (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	line end fault is invalid; bus table is full
	IERR = 5	fault analysis is invalid due to the presence of in-service induction machines that are in the "stalled" or "tripped" state
	IERR = 6	bus type code and series element status inconsistencies
	IERR = 7	error opening RELFILE
	IERR = 8	error opening FCDFILE
	IERR = 9	prerequisite requirements for API are not met

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1.38. ASCC_2

This API routine is obsolete. It has been replaced by the API routine ascc_3 .

This API executes the second release of the activity ASCC. This API is used to apply a series of faults at various locations in the working case (activity ASCC_2).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_ASCC_2  SID ALL STATUS(1)..STATUS(13) RELFILE FCDFILE SCFILE
```

Python command syntax:

```
ierr = ascc_2(sid, all, status, relfile, fcdfile, scfile)
```

Fortran command syntax:

```
CALL ASCC2(SID, ALL, STATUS, RELFILE, FCDFILE, SCFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or selected subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (13)	array of 13 elements specifying integer options(input).	
	STATUS(1)	line-to-ground faults (default alias is fltlg) (0 by default)
	STATUS(1) = 0	omit
	STATUS(1) = 1	include
	STATUS(2)	line-out faults (default alias is linout) (0 by default)
	STATUS(2) = 0	omit
	STATUS(2) = 1	include
	STATUS(3)	line-end faults (default alias is linend) (0 by default)
	STATUS(3) = 0	omit
	STATUS(3) = 1	include
	STATUS(4)	flat conditions option (default alias is flatop) (0 by default)
	STATUS(4) = 0	pre-fault conditions taken from working case
	STATUS(4) = 1	impose flat conditions as pre-fault condition
	STATUS(5)	double-precision fault calculation option (default alias is nouse1) (ignored, but must be

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		specified as 0 or 1; 0 by default; double precision always used)
	STATUS(6)	report option (default alias is rptop) (1 by default)
	STATUS(6) = 1	total fault currents
	STATUS(6) = 2	fault contributions to N levels away
	STATUS(6) = 3	total fault currents and fault contributions to N levels away
	STATUS(7)	number of levels back for contributions (default alias is rptlvl) (0 by default)
	STATUS(7) = 0	used if STATUS(6) = 2 or 3
	STATUS(8)	branch quantity code (default alias is nouse2) (ignored, but must be specified as 0 or 1; 0 by default)
	STATUS(9)	dc line and FACTS device option (default alias is dcload) (0 by default)
	STATUS(9) = 0	blocked
	STATUS(9) = 1	represent as load
	STATUS(10)	zero sequence transformer impedance correction option (default alias is zcorec) (0 by default)
	STATUS(10) = 0	ignore
	STATUS(10) = 1	apply
	STATUS(11)	three phase faults (default alias is flt3ph) (0 by default)
	STATUS(11) = 0	omit
	STATUS(11) = 1	include
	STATUS(12)	line-line-to-ground faults (default alias is fltllg) (0 by default)
	STATUS(12) = 0	omit
	STATUS(12) = 1	include
	STATUS(13)	line-to-line faults (default alias is fltll) (0 by default)
	STATUS(13) = 0	omit
	STATUS(13) = 1	include
Character*260 RELFILE	Relay Data (.rel) file (input; blank by default).	
Character*260 FCDFILE	Fault Control (.fcd) file (input; blank by default).	
Character*260 SCFILE	ASCC_2 Results (.sc) file (input; blank by default).	
	scfile =	temporary file created
	scfile = nooutput	no file created
Integer IERR	error code (output).	

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1.39. ASCC_3

This API executes the third release of the activity ASCC. This API is used to apply a series of faults at various locations in the working case (activity ASCC_3).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ASCC_3  SID ALL STATUS(1)..STATUS(17) VALUES(1) RELFILE FCDFILE SCFILE
```

Python command syntax:

```
ierr = ascc_3(sid, all, status, values, relfile, fcdfile, scfile)
```

Fortran command syntax:

```
CALL ASCC3(SID, ALL, STATUS, VALUES, RELFILE, FCDFILE, SCFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or selected subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (17)	array of 17 elements specifying integer options(input).	
	STATUS(1)	line-to-ground faults (default alias is fltlg) (0 by default)
	STATUS(1) = 0	omit
	STATUS(1) = 1	include
	STATUS(2)	line-out faults (default alias is linout) (0 by default)
	STATUS(2) = 0	omit
	STATUS(2) = 1	include
	STATUS(3)	line-end faults (default alias is linend) (0 by default)
	STATUS(3) = 0	omit
	STATUS(3) = 1	include
	STATUS(4)	bus voltage option (default alias is voltop) (0 by default)
	STATUS(4) = 0	use bus voltages from power flow solution
	STATUS(4) = 1	set all bus voltages at specified value and at 0 deg
	STATUS(4) = 2	set faulted bus voltage at specified value and at 0 deg
	STATUS(5)	fault analysis generator reactance option; ignored if generators are already

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	"converted" (default alias is genxop) (0 by default)
STATUS(5) = 0	use subtransient reactance
STATUS(5) = 1	use transient reactance
STATUS(5) = 2	use synchronous reactance
STATUS(6)	report option (default alias is rptop) (1 by default)
STATUS(6) = 0	fault current summary table
STATUS(6) = 1	total fault currents
STATUS(6) = 2	fault contributions to N levels away
STATUS(6) = 3	total fault currents and fault contributions to N levels away
STATUS(7)	number of levels back for contributions (default alias is rptlvl) (0 by default)
STATUS(7) = 0	used if STATUS(6) = 2 or 3
STATUS(8)	transformer tap ratios and phase shift angles option (default alias is tpunty) (0 by default)
STATUS(8) = 0	leave tap ratios and phase shift angles unchanged
STATUS(8) = 1	set tap ratios to 1.0 pu and phase shift angles to 0
STATUS(8) = 2	set tap ratios to 1.0 pu and phase shift angles unchanged
STATUS(8) = 3	set tap ratios unchanged and phase shift angles to 0
STATUS(9)	dc line and FACTS device option (default alias is dcload) (0 by default)
STATUS(9) = 0	blocked
STATUS(9) = 1	represent as load
STATUS(10)	zero sequence transformer impedance correction option (default alias is zcorec) (1 by default)
STATUS(10) = 0	ignore
STATUS(10) = 1	apply
STATUS(11)	three phase faults (default alias is flt3ph) (0 by default)
STATUS(11) = 0	omit
STATUS(11) = 1	include
STATUS(12)	line-line-to-ground faults (default alias is fltllg) (0 by default)
STATUS(12) = 0	omit
STATUS(12) = 1	include
STATUS(13)	line-to-line faults (default alias is fltll) (0 by default)
STATUS(13) = 0	omit

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	STATUS(13) = 1	include
	STATUS(14)	line charging option (default alias is lnchrg) (0 by default)
	STATUS(14) = 0	leave unchanged
	STATUS(14) = 1	set to 0.0 in the positive and negative sequences
	STATUS(14) = 2	set to 0.0 in all sequences
	STATUS(15)	line, fixed and switched shunts, and magnetizing admittance option (default alias is shntop) (0 by default)
	STATUS(15) = 0	leave unchanged
	STATUS(15) = 1	set to 0.0 in the positive and negative sequences
	STATUS(15) = 2	set to 0.0 in all sequences
	STATUS(16)	load option (default alias is loadop) (0 by default)
	STATUS(16) = 0	leave unchanged
	STATUS(16) = 1	set to 0.0 in the positive and negative sequences
	STATUS(16) = 2	set to 0.0 in all sequences
	STATUS(17)	synchronous and asynchronous machines power option (default alias is machpq) (0 by default)
	STATUS(17) = 0	use real and reactive power outputs from power flow solution
	STATUS(17) = 1	set real and reactive power outputs to 0.0
Real VALUES (1)	array of 1 elements specifying real values(input).	
	VALUES(1)	User specified pu bus voltage value (Used when STATUS (4) = 1 or 2) (default alias is volts)
Character*260 RELFILE	Relay Data (.rel) file (input; blank by default).	
Character*260 FCDFILE	Fault Control (.fcd) file (input; blank by default).	
Character*260 SCFILE	Results (.sc) file (input; blank by default).	
	scfile =	temporary file created
	scfile = nooutput	no file created
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	line end fault invalid; bus table is full
	IERR = 5	error opening RELFILE
	IERR = 6	error opening FCDFILE

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IERR = 7	prerequisite requirements for function not met
IERR = 8	no faults specified
IERR = 9	error opening SCFILE
IERR = 10	SCFILE error
IERR = 11	bus SC results not available
IERR = 12	branch SC results not available
IERR = 13	three winding transformer SC results not available
IERR = 14	ncorrect SCFILE
IERR = 15	SCFILE not allowed

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1.40. ASCC_SCFILE

This API reads ASCC short circuit results from SCFILE and produces same ASCC output report that was generated when SCFILE was created. SCFILE is read into PSSE memory to enable displaying ASCC results onto slider diagrams. For this API to work properly, it is required that working case in PSSE memory should be the same Saved case that was used when creating the SCFILE.

This API routine was first introduced in release 32.0.2.

Batch command syntax:

```
BAT_ASCC_SCFILE  SCFILE
```

Python command syntax:

```
ierr = ascc_scfile(scfile)
```

Fortran command syntax:

```
CALL ASCC_SCFILE(SCFILE,IERR)
```

Where:

Character*260 SCFILE	Activity ASCC Results (.sc) file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 7	prerequisite requirements for function not met
	IERR = 9	error opening SCFILE
	IERR = 10	SCFILE error
	IERR = 11	bus SC results not available
	IERR = 12	branch SC results not available
	IERR = 13	three winding transformer SC results not available
	IERR = 14	incorrect SCFILE
	IERR = 15	SCFILE not allowed

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1.41. BASE_FREQUENCY

Use this API to specify or return the value of the base frequency option setting.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_BASE_FREQUENCY  RVAL
```

Python command syntax:

```
ierr = base_frequency(rval)

ierr, rval = base_frequency()
```

Fortran command syntax:

```
CALL BASE_FREQUENCY_API(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	base frequency (usually 50 or 60) (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 60.0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.42. BGEN

Use this API to convert the mismatch at boundary buses to equivalent load and/or generation (activity BGEN). It is intended to be used following the removal of a subsystem from a solved working case, with the flows to the deleted subsystem replaced by load and/or generation at those retained buses that were directly connected to the removed subsystem.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BGEN  SID  ALL  STATUS
```

Python command syntax:

```
ierr = bgen(sid, all, status)
```

Fortran command syntax:

```
CALL BGENNW(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer STATUS	type of boundary condition modification to be used (input; 0 by default).	
	STATUS = 0	generator for inflow; load for outflow.
	STATUS = 1	all equivalent generators.
	STATUS = 2	all equivalent loads.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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1.43. BKDY

Use this API to calculate and report circuit breaker interrupting duty for 3-phase faults at all buses in a specified subsystem of the working case (activity BKDY).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BKDY  SID  ALL  APIOPT  LVLBAK  FLTTIM  BFILE  FFILE
```

Python command syntax:

```
ierr = bkdy(sid, all, apiopt, lvlbak, flttim, bfile, ffile)
```

Fortran command syntax:

```
CALL BKDYNW(SID, ALL, APIOPT, LVLBAK, FLTTIM, BFILE, FFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer APIOPT	mode of operation in the API (input; 1 by default).	
	APIOPT = 1	fault applications specified in subsystem.
	APIOPT = 2	fault applications specified in FFILE.
Integer LVLBAK	number of levels back for contributions (input; 0 by default).	
	LVLBAK < 0	fault currents only.
	LVLBAK = 0	fault currents and local contributions.
	LVLBAK > 0	fault currents and local contributions, plus contributions LVLBAK levels back.
Real FLTTIM	default fault duty time. FLTTIM >= 0.0 (input; 0.0).	
Character*260 BFILE	Breaker Duty Data file (input; no default allowed).	
Character*260 FFILE	Fault Specification Data file (if APIOPT is 2) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid APIOPT value
	IERR = 4	invalid FLTTIM value
	IERR = 5	BFILE is blank
	IERR = 6	FFILE is blank

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IERR = 7	generators are not converted
IERR = 8	fixed bus shunt table is full
IERR = 9	error opening BFILE or FFILE
IERR = 10	prerequisite requirements for API are not met

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1.44. BRCH

This API routine is obsolete. It has been replaced by the API routine brch_2 .

Use this API to tabulate those branches where impedances or other characteristics are such that they may be detrimental to the rate of convergence of one or more of the power flow solution activities.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BRCH  SID  ALL  DOTEST(1)..DOTEST(10)  TOLVAL(1)..TOLVAL(7)
```

Python command syntax:

```
ierr = brch(sid, all, dotest, tolval)
```

Fortran command syntax:

```
CALL BRCHNW(SID, ALL, DOTEST, TOLVAL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).
Integer ALL	all buses or specified subsystem flag (input; 1 by default). ALL = 0 process only buses in subsystem SID ALL = 1 process all buses
Integer DOTEST (10)	array of 10 elements specifying options. Each element determines whether a specific branch check should be applied. For DOTEST(I) = 1, the corresponding branch check is applied; for DOTEST(I) = 0, the check is bypassed(input). DOTEST(1) check for small reactances DOTEST(2) check for large reactances DOTEST(3) check for large resistance to reactance ratios (R/X) DOTEST(4) check for negative reactances DOTEST(5) check for large ratio of largest to smallest reactance at each bus DOTEST(6) check for high or negative charging DOTEST(7) check data of parallel transformers DOTEST(8) check for high tap ratios DOTEST(9) check for low tap ratios DOTEST(10) check for zero sequence non-transformer branch impedances of 0.0
Real TOLVAL (7)	array of 7 elements specifying options. Each element contains a limit or threshold value used in one of the branch checks(input).

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	TOLVAL(1)	>0.0 reactance lower limit (0.0005 by default)
	TOLVAL(2)	>0.0 reactance upper limit (1.0 by default)
	TOLVAL(3)	>0.0 R/X threshold (0.66667 by default)
	TOLVAL(4)	>=10.0 Xmax/Xmin threshold (500.0 by default)
	TOLVAL(5)	>=0.0 charging upper limit (5.0 by default)
	TOLVAL(6)	>0.8 high tap ratio limit (1.1 by default)
	TOLVAL(7)	>0.0 and < 1.2 low tap ratio limit (0.9 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid DOTEST value
	IERR = 4	invalid TOLVAL value
	IERR = 5	prerequisite requirements for API are not met

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1.45. BRCH_2

Use this API to tabulate those branches where impedances or other characteristics are such that they may be detrimental to the rate of convergence of one or more of the power flow solution activities.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_BRCH_2  SID ALL DOTEST(1)..DOTEST(11) TOLVAL(1)..TOLVAL(7)
```

Python command syntax:

```
ierr = brch_2(sid, all, dotest, tolval)
```

Fortran command syntax:

```
CALL BRCH_2(SID, ALL, DOTEST, TOLVAL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer DOTEST (11)	array of elements specifying options. Each element determines whether a specific branch check should be applied. For DOTEST(I) = 1, the corresponding branch check is applied; for DOTEST(I) = 0, the check is bypassed(input).	
	DOTEST(1)	check for small reactances.
	DOTEST(2)	check for large reactances.
	DOTEST(3)	check for large resistance to reactance ratios (R/X).
	DOTEST(4)	check for negative reactances.
	DOTEST(5)	check for large ratio of largest to smallest reactance at each bus.
	DOTEST(6)	check for high or negative charging.
	DOTEST(7)	check data of parallel transformers.
	DOTEST(8)	check for high tap ratios.
	DOTEST(9)	check for low tap ratios.
	DOTEST(10)	check for inconsistent transformer loss data values (copper loss and no-load loss).
	DOTEST(11)	check for zero sequence non-transformer branch impedances of 0.0.
Real TOLVAL (7)	array of elements specifying options. Each element contains a limit or threshold value used in one of the branch checks (input).	
	TOLVAL(1)	>0.0. reactance lower limit (0.0005 by default).

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	TOLVAL(2)	>0.0. reactance upper limit (1.0 by default).
	TOLVAL(3)	>0.0. R/X threshold (0.66667 by default).
	TOLVAL(4)	>=10.0. Xmax/Xmin threshold (500.0 by default).
	TOLVAL(5)	>=0.0. charging upper limit (5.0 by default).
	TOLVAL(6)	>0.8. high tap ratio limit (1.1 by default).
	TOLVAL(7)	> 0.0 and < 1.2. low tap ratio limit (0.9 by default).
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid DOTEST value
	IERR = 4	invalid TOLVAL value
	IERR = 5	prerequisite requirements for API are not met

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1.46. BSNM

Use this API to change the bus numbers of specified network buses in the working case and retain a tabulation, in file form, of bus number changes made (activity BSNM). This API must be referenced with one OPT 0 reference followed by one of the following:
- one OPT 5, 11 or 12 reference.
- one or more OPT 1 references, followed by one OPT 7 reference.
- one or more OPT 2 references, followed by one OPT 7 reference.
- one or more OPT 3 references, followed by one OPT 7 reference.
- one or more OPT 4 references, followed by one OPT 7 reference.
- one OPT 15 reference for each area in the case containing at least one bus, in ascending area number order, followed by one OPT 25 reference.
- zero or more OPT 15 references in the same order as above, followed by one OPT 35 reference.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BSNM  SID  ALL  OPT  STATUS  BUSRNG(1)  BUSRNG(2)  OLDNAM  TFILE  OUT
 OFILE
```

Python command syntax:

```
ierr = bsnm(sid, all, opt, status, busrng, oldnam, tfile, out, ofile)
```

Fortran command syntax:

```
CALL BSNMNW(SID, ALL, OPT, STATUS, BUSRNG, OLDNAM, TFILE, OUT, OFILE,
 IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer OPT	method used to renumber the buses (input; no default allowed).	
	OPT = 0	initialize for renumbering.
	OPT = 1	renumber buses based on name/new number pairs.
	OPT = 11	renumber using name translation file.
	OPT = 2	renumber buses based on old number/new number pairs.
	OPT = 12	renumber buses using number translation file.
	OPT = 3	pack all buses in specified low/high pairs.
	OPT = 4	renumber buses based on the subsystem specified in the OPT 0 reference.
	OPT = 5	renumber all buses.

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	OPT = 15	renumber all buses blocking by area.
	OPT = 25	apply blocking by area changes.
	OPT = 35	cancel blocking by area changes.
	OPT = 7	cleanup from renumbering.
Integer STATUS	If OPT=15, is the number of the area where the bus number range is specified in BUSRNG. STATUS must be between 1 and AREAS (the highest area number allowed at the present size level of PSSE). If OPT=4: STATUS = 0 indicates a new bus number range is provided. STATUS = 1 indicates a bus number offset is provided. Otherwise STATUS is ignored (input).	
Integer BUSRNG (2)	If OPT = 1: BUSRNG(1) is the new bus number. If OPT = 2: BUSRNG(1) is the old bus number. BUSRNG(2) is the new bus number. If OPT = 3: BUSRNG(1) is the bus number at which to start packing sequentially. BUSRNG(2) is the bus number at which to end packing sequentially. If OPT = 4 and STATUS = 0: BUSRNG(1) is the starting bus number for the range into which the subsystem is to be renumbered. BUSRNG(2) is the ending bus number for the range into which the subsystem is to be renumbered. If OPT = 4 and STATUS = 1: BUSRNG(1) is amount by which all buses in the specified subsystem are offset. If OPT = 5: BUSRNG(1) bus number at which to start renumbering all buses in the working case. Buses are re-sequenced in increments of 1. If OPT = 15: BUSRNG(1) is the starting bus number for the range into which the area is to be renumbered. BUSRNG(2) is the ending bus number for the range into which the area is to be renumbered (input).	
Character*18 OLDNAM	extended bus name of the bus to be renumbered when OPT is 1 (when OPT is 1) (input; no default allowed).	
Character*260 TFILE	bus name/number translation file. This file is only used if OPT = 11 or 12. Data in the translation file must be in the form specified in the PSSE Program Operation Manual. (when OPT is 11 or 12) (input; no default allowed).	
Integer OUT	option selection when OPT is 0 (input; 1 by default).	
	OUT = 0	direct output to a data file specified by OFILE.
	OUT = 1	direct output to the report device.
	OUT = 2	no output.
Character*260 OFILE	Bus Renumbering Translation file for the records of activity BSNM (when OPT is 0 and OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OUT value
	IERR = 4	invalid OPT 0 call; bus renumbering already initialized
	IERR = 5	prior OPT is not 0 (OPT=1,2,11,12,5,25)
	IERR = 6	bus not found (OPT=1,2,11,12)

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IERR = 7	invalid new bus number (OPT=1,2,11,12,4)
IERR = 8	bus number is already assigned to another bus (OPT=1,2,11,12,4)
IERR = 9	prior OPT not 0 or 3 (OPT=3,4,15,35)
IERR = 10	invalid starting bus number (OPT=3,4,5,15)
IERR = 11	invalid ending bus number (OPT=3,4,15)
IERR = 12	starting bus number is greater than ending bus number (OPT=3,4,15)
IERR = 13	invalid STATUS value (OPT=4)
IERR = 14	invalid offset value (OPT=4)
IERR = 15	not enough unused bus numbers in specified bus number range (OPT=4)
IERR = 16	not enough numbers between BUSRNG(1) and 999997 (OPT=5)
IERR = 17	area specified but no more areas left to process (OPT=15)
IERR = 18	area out of sequence (OPT=15)
IERR = 19	bus number range of an area overlaps that of another area (OPT=15)
IERR = 20	bus number range of an area is not large enough (OPT=15)
IERR = 21	not enough OPT 15 calls (OPT=25)
IERR = 22	invalid OPT 7 call; bus renumbering is not initialized (OPT=7)
IERR = 23	invalid OPT value
IERR = 24	error opening TFILE (OPT=11,12)
IERR = 25	error opening OFILE
IERR = 26	prerequisite requirements for API are not met

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1.47. BUSN

Use this API to tabulate unused bus numbers within a specified bus number range (activity BUSN).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BUSN  BUSLO  BUSHI
```

Python command syntax:

```
ierr = busn(buslo, bushi)
```

Fortran command syntax:

```
CALL BUSNAPI(BUSLO, BUSHI, IERR)
```

Where:

Integer BUSLO	low limit of bus number range (input; 1 by default).	
Integer BUSHI	high limit of bus number range (input; 999,997).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid starting bus number
	IERR = 2	starting bus number is greater than ending bus number
	IERR = 3	prerequisite requirements for API are not met

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1.48. BUS_INPUT

Use this API to specify or return the bus input option setting, for either numbers or names.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_BUS_INPUT  IVAL
```

Python command syntax:

```
ierr = bus_input(ival)

ierr, ival = bus_input()
```

Fortran command syntax:

```
CALL BUS_INPUT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	bus numbers.
	IVAL = 1	extended bus names.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.49. BUS_OUTPUT

Use this API to specify or return the bus output option setting, for either numbers or names.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_BUS_OUTPUT  IVAL
```

Python command syntax:

```
ierr = bus_output(ival)

ierr, ival = bus_output()
```

Fortran command syntax:

```
CALL BUS_OUTPUT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	bus numbers.
	IVAL = 1	extended bus names.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.50. BUS_SIZE_LEVEL

Use this API to specify or return the value of the PSSE size level option setting to a multiple of 1,000 between 1,000 and 200,000.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_BUS_SIZE_LEVEL  IVAL
```

Python command syntax:

```
ierr = bus_size_level(ival)

ierr, ival = bus_size_level()
```

Fortran command syntax:

```
CALL BUS_SIZE_LEVEL_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1000 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.51. CASE

Use this API to open a PSSE Saved Case file and transfers its data into the PSSE working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CASE  SFILE
```

Python command syntax:

```
ierr = case(sfile)
```

Fortran command syntax:

```
CALL CASENW(SFILE, IERR)
```

Where:

Character*260 SFILE	name of the PSSE saved case file If SFILE is "*", CASENW attempts to reopen the most recently accessed Saved Case File. If no Saved Case File was accessed in the current execution of PSSE, CASENW returns an error code (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SFILE is blank
	IERR = 2	error reading from SFILE
	IERR = 3	error opening SFILE
	IERR = 4	prerequisite requirements for API are not met

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1.52. CHECKVOLTAGELIMITS

Use this API to tabulate those buses where voltage magnitude is beyond their normal or emergency voltage limits. This check may be performed for all buses in the working case or for all buses in a specified subsystem.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_CHECKVOLTAGELIMITS  SID ALL OPT THRSH
```

Python command syntax:

```
ierr = checkvoltagelimits(sid, all, opt, thrsh)
```

Fortran command syntax:

```
CALL CHECKVOLTAGELIMITS(SID, ALL, OPT, THRSH, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer OPT	set of voltage limits to be used (input; 1 by default).	
	OPT = 1	normal limits
	OPT = 2	emergency limits
Real THRSH	voltage limit threshold (-0.05 < THRSH < 0.05) (input0.0).	
	THRSH = 0.0	log voltages above upper limit or below lower limit
	THRSH > 0.0	log voltages beyond or within THRSH of a limit
	THRSH < 0.0	log voltage outside a limit by more than (-THRSH)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	invalid THRSH value
	IERR = 5	prerequisite requirements for API are not met

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1.53. CHECK_POWERFLOW_DATA

Use this API routine to perform data checks on the selected categories of powerflow data for all buses in the working case or for all buses in a specified subsystem.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_CHECK_POWERFLOW_DATA  SID ALL OPT
```

Python command syntax:

```
ierr = check_powerflow_data(sid, all, opt)
```

Fortran command syntax:

```
CALL CHECK_POWERFLOW_DATA(SID, ALL, OPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer OPT	powerflow data category to be checked (input; 0 by default).	
	OPT = 0	all data categories
	OPT = 1	bus data
	OPT = 2	load data
	OPT = 3	plant data
	OPT = 4	generator unit data
	OPT = 5	induction machine data
	OPT = 6	fixed bus shunt data
	OPT = 7	switched shunt data
	OPT = 8	non-transformer branch data
	OPT = 9	two-winding transformer data
	OPT = 10	three-winding transformer data
	OPT = 11	transformer impedance table correction data
	OPT = 12	multi-section line data
	OPT = 13	two-terminal dc line data
	OPT = 14	multi-terminal dc line data
	OPT = 15	VSC dc line data
	OPT = 16	FACTS device data
	OPT = 17	GNE device data
	OPT = 18	area interchange data
	OPT = 19	owner data

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	OPT = 20	zone data
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	prerequisite requirements for API are not met

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1.54. CHECK_SEQUENCE_DATA

Use this API routine to perform data checks on the selected categories of sequence data for all buses in the working case or for all buses in a specified subsystem.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_CHECK_SEQUENCE_DATA  SID ALL OPT
```

Python command syntax:

```
ierr = check_sequence_data(sid, all, opt)
```

Fortran command syntax:

```
CALL CHECK_SEQUENCE_DATA(SID, ALL, OPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer OPT	sequence data category to be checked (input; 0 by default).	
	OPT = 0	all data categories
	OPT = 1	load data
	OPT = 2	generator unit data
	OPT = 3	induction machine data
	OPT = 4	fixed bus shunt data
	OPT = 5	switched shunt data
	OPT = 6	non-transformer branch data
	OPT = 7	two-winding transformer data
	OPT = 8	three-winding transformer data
	OPT = 9	zero sequence mutual data
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	prerequisite requirements for API are not met

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1.55. CLOSE_POWERFLOW

Removes the current powerflow working case from PSSE's working memory. Thus, the PSSE powerflow arrays are in the same state as they are immediately following PSSE startup. The default PSSE program option settings are unchanged.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CLOSE_POWERFLOW
```

Python command syntax:

```
ierr = close_powerflow()
```

Fortran command syntax:

```
CALL CLOSE_POWERFLOW(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.56. CLOSE_REPORT

Sets the report output device to the standard output and sets the command line to request device selection for individual reporting activities (Activity CLOS).

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_CLOSE_REPORT
```

Python command syntax:

```
ierr = close_report()
```

Fortran command syntax:

```
CALL CLOSEREPORTDEVICE(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.57. CMPR

Use this API to tabulate certain case totals, as contained in the working case, with those of a designated Saved Case (activity CMPR). Either system totals, area totals, owner totals, or zone totals may be compared. It must be referenced using one of the following procedures:
- One reference with APIOPT=1, followed by any number of references with APIOPT=2, followed by one reference with APIOPT=3.
- One reference with APIOPT=0. This automatically combines the processing of one APIOPT 1 reference, followed by one APIOPT 2 reference, followed by one APIOPT 3 reference.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CMPR  SID  APIOPT  STATUS(1)..STATUS(4)  THRSH  CFILE
```

Python command syntax:

```
ierr = cmpr(sid, apiopt, status, thrsh, cfile)
```

Fortran command syntax:

```
CALL CMPRNW(SID, APIOPT, STATUS, THRSH, CFILE, IERR)
```

Where:

Integer SID	a valid area, zone or owner subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; used only when APIOPT is 0 or 1; 0).	
Integer APIOPT	mode of operation in the API (input; 0 by default).	
	APIOPT = 0	initialize for case totals comparison, run the comparison specified by STATUS(3), and run post-processing clean up.
	APIOPT = 1	initialize system for system, area, owner, zone totals.
	APIOPT = 2	calculate and report system, area, owner, zone totals.
	APIOPT = 3	postprocessing housekeeping.
Integer STATUS (4)	array of 4 elements specifying The value of each element is as follows(input).	
	STATUS(1)	tabulation method (used only when APIOPT is 0 or 1; 0 by default)
	STATUS(1) = 0	tabulate system totals
	STATUS(1) = 1	tabulate area totals for area subsystem SID
	STATUS(1) = 2	tabulate owner totals for owner subsystem SID
	STATUS(1) = 3	tabulate zone totals for zone subsystem SID
	STATUS(2)	units for the difference threshold (used only when APIOPT is 0 or 2; 0 by default)
	STATUS(2) = 0	threshold in engineering units

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	STATUS(2) = 1	threshold in percent
	STATUS(3)	quantity to be compared (used only when APIOPT is 0 or 2; 1 by default)
	STATUS(3) = 1	compare generation
	STATUS(3) = 2	compare load
	STATUS(3) = 3	compare losses
	STATUS(3) = 4	compare mismatch
	STATUS(3) = 5	compare interchange; not valid for STATUS(1) = 2
	STATUS(4)	file to restore into the working case upon completion (used only when APIOPT is 0 or 3; 0 by default)
	STATUS(4) = 0	restore the working case
	STATUS(4) = 1	restore the comparison file to the working case
Real THRSH	difference threshold, in either engineering units or a percent, based on the value of STATUS(2) (input; used only when APIOPT is 0 or 2; 0.0).	
Character*260 CFILE	Saved Case file to be compared against the working case (when APIOPT is 0 or 1) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid APIOPT value
	IERR = 3	invalid STATUS value
	IERR = 4	owner (STATUS(1)=2) interchange (STATUS(3)=5) combination is invalid
	IERR = 5	invalid THRSH value
	IERR = 6	CFILE is blank
	IERR = 7	error writing the working case to a temporary file
	IERR = 8	unexpected APIOPT value
	IERR = 9	error reading the comparison case
	IERR = 10	error opening the temporary file
	IERR = 11	prerequisite requirements for API are not met

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1.58. CNTB

Use this API to tabulate the voltage setpoints and desired voltage bands of voltage-controlling equipment in the working case (activity CNTB). Optionally, new scheduled voltages may be specified.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CNTB  SID  ALL  APIOPT  STATUS(1)..STATUS(3)  V(1)  V(2)
```

Python command syntax:

```
ierr, next, bus, ckt, v = cntb(sid, all, apiopt, status, v)
```

Fortran command syntax:

```
CALL CNTBNW(SID, ALL, APIOPT, STATUS, NEXT, BUS, CKT, V, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (used only when APIOPT is 2) (input; 0 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer APIOPT	mode of operation in the API (input).	
	APIOPT = 1	initialize.
	APIOPT = 2	specify voltage setpoint for controlled bus.
	APIOPT = 3	specify voltage band for load drop compensating transformer.
Integer STATUS (3)	array of 3 elements specifying used to control reporting and processing options(input).	
	STATUS(1)	varies by APIOPT
	STATUS(1) = 0	for APIOPT=1, list subsystem controlled buses with conflicts. for APIOPT=2 or 3, no change
	STATUS(1) = 1	for APIOPT=1, list all subsystem controlled buses. for APIOPT=2, change voltage setpoints to new setpoint in V(1). for APIOPT=3, change voltage band to new voltage band in V(1) and V(2)
	STATUS(2)	varies by APIOPT
	STATUS(2) = 0	for APIOPT=1, report at OPENed output device. for APIOPT=2 or 3, not used
	STATUS(2) = 1	for APIOPT=1, list in progress area and allow voltage schedule change. for APIOPT=2 or 3, not used
	STATUS(3)	varies by APIOPT

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	STATUS(3) = 0	for APIOPT=1, process all controlling elements and controlled buses. for APIOPT=2 or 3, not used
	STATUS(3) = 1	for APIOPT=1, process active controlling elements and controlled buses. for APIOPT=2 or 3, not used
Integer NEXT	returned as the next APIOPT value expected by the API (output).	
	NEXT = 1	done with this execution of CNTBNW.
	NEXT = 2	change voltage setpoint/schedule.
	NEXT = 3	change VMIN, VMAX of load drop compensating transformer.
Integer BUS (3)	array of 3 elements specifying a returned three bus values(output).	
	BUS(1)	Returned as controlled bus for which voltage schedule may be changed when Next=2.\n Returned as controlled bus for which voltage schedule may be changed when Next=3
	BUS(2)	Returned as the from bus of the transformer when Next=3
	BUS(3)	Returned as the to bus of the transformer when Next=3
Character*2 CKT	returned as the circuit ID of the transformer returned in BUS(2) and BUS(3) when NEXT returned as 3 (output).	
Real V (2)	array of 2 elements specifying Voltage values(input).	
	V(1)	New setpoint when APIOPT=2, new VMIN when APIOPT=3. Returned as present setpoint when NEXT=2, returned as present VMIN when NEXT=3
	V(2)	New VMAX when APIOPT=3. Returned as present VMAX when NEXT=3
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid new voltage setpoint value (V(1) <= 0.5 or V(1) >= 1.5)
	IERR = 5	VMIN is not less than VMAX
	IERR = 6	invalid APIOPT value
	IERR = 7	unexpected APIOPT value
	IERR = 8	prerequisite requirements for API are not met

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1.59. CONG

Use this API to convert generators from their power flow representation in preparation for switching studies and dynamic simulations (activity CONG).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CONG  OPT
```

Python command syntax:

```
ierr = cong(opt)
```

Fortran command syntax:

```
CALL CONGAPI(OPT, IERR)
```

Where:

Integer OPT	machine impedance option (input; 0 by default).	
	OPT = 0	use ZSORCE and IMZSRC.
	OPT = 1	use fault analysis subtransient impedance and IMZPOS.
	OPT = 2	use fault analysis transient impedance and IMZPOS.
	OPT = 3	use fault analysis synchronous impedance and IMZPOS.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	generators are already converted
	IERR = 3	generator conversion failed due to a machine impedance of zero
	IERR = 4	generator conversion failed due to the presence of in-service induction machines that are in the "stalled" or "tripped" state
	IERR = 5	prerequisite requirements for API are not met

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1.60. CONL

Use this API to convert the constant MVA load for a specified grouping of network loads to a specified mixture of the constant MVA, constant current, and constant admittance load characteristics (activity CONL).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CONL  SID  ALL  APIOPT  STATUS(1)  STATUS(2)  LOADIN(1)..  LOADIN(2)
 LOADIN(3)  LOADIN(4)
```

Python command syntax:

```
ierr, rlods = conl(sid, all, apiopt, status, loadin)
```

Fortran command syntax:

```
CALL CONLNW(SID, ALL, APIOPT, STATUS, LOADIN, RLODS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; used only when APIOPT is 2; 0).	
Integer ALL	all buses or specified subsystem flag (input; used only when APIOPT is 2; 0).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer APIOPT	mode of operation in the API (input; no default allowed).	
	APIOPT = 1	initialize for load conversion.
	APIOPT = 2	convert loads.
	APIOPT = 3	postprocessing housekeeping.
Integer STATUS (2)	array of 2 elements specifying The value of each element is as follows(input).	
	STATUS(1)	method of conversion (used only when APIOPT is 1; 0 by default)
	STATUS(1) = 0	convert existing constant MVA load
	STATUS(1) = 1	reconstruct loads using present voltage, then convert
	STATUS(1) = 2	reconstruct loads using unity voltage, then convert
	STATUS(2)	a special OPF bus type code flag (used only when APIOPT is 2; 0 by default)
	STATUS(2) = 0	all buses in subsystem
	STATUS(2) = 1	only Type 1 buses in subsystem
	STATUS(2) = 2	only Type 2 and 3 buses in subsystem
Real LOADIN (4)	array of 4 elements specifying (input).	

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	LOADIN(1)	percent of active power load to be converted to the constant current characteristic
	LOADIN(2)	percent of active power load to be converted to the constant admittance characteristic
	LOADIN(3)	percent of reactive power load to be converted to the constant current characteristic
	LOADIN(4)	percent of reactive power load to be converted to the constant admittance characteristic
Integer RLODS	returned as the number of unconverted loads remaining in the working case. If the value is nonzero after a call with APIOPT = 2, and the remaining loads are to be converted, the API is called again with ALL = 1 and APIOPT = 2 (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid APIOPT value
	IERR = 4	invalid STATUS value
	IERR = 5	unexpected APIOPT value
	IERR = 6	prerequisite requirements for API are not met

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1.61. CONNECTIVITY_CHECK

Use this API to specify or return the option to enable or disable the solution connectivity checking option setting.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_CONNECTIVITY_CHECK  IVAL
```

Python command syntax:

```
ierr = connectivity_check(ival)

ierr, ival = connectivity_check()
```

Fortran command syntax:

```
CALL CONNECTIVITY_CHECK_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1 by default; else output).	
	IVAL = 0	no solution connectivity check before power flow solutions.
	IVAL = 1	run solution connectivity check before each power flow solution.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.62. CONTROL_AREA_INTERCHANGE

Use this API to to specify or return the area interchange control option setting to disabled, or enabled with tie lines only, or enabled with tie lines and loads.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_CONTROL_AREA_INTERCHANGE  IVAL
```

Python command syntax:

```
ierr = control_area_interchange(ival)

ierr, ival = control_area_interchange()
```

Fortran command syntax:

```
CALL CONTROL_AREA_INTERCHANGE_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	disabled.
	IVAL = 1	enabled with interchange determined from tie lines only.
	IVAL = 2	enabled with interchange determined from tie lines and loads.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.63. DCCC

Use this API to run contingency case solutions using a linear network (dc) model (activity DCCC).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_DCCC  OPTIONS(1)..OPTIONS(6)  VALUES(1)..VALUES(3)  DFXFILE
```

Python command syntax:

```
ierr = dccc(options, values, dfxfile)
```

Fortran command syntax:

```
CALL DCCCAPI(OPTIONS, VALUES, DFXFILE, IERR)
```

Where:

Integer OPTIONS (6)	array of 6 elements specifying calculation and reporting options. The value of each element is as follows(input).	
	OPTIONS(1)	contingency case rating set (rating set program option setting by default)
	OPTIONS(1) = 1	RATEA
	OPTIONS(1) = 2	RATEB
	OPTIONS(1) = 3	RATEC
	OPTIONS(2)	base case line flow code (0 by default)
	OPTIONS(2) = 0	dc base case
	OPTIONS(2) = 1	ac base case
	OPTIONS(3)	format code (0 by default)
	OPTIONS(3) = 0	overload output
	OPTIONS(3) = 1	full output
	OPTIONS(4)	exclude cases with no overloads from the overload report (0 by default)
	OPTIONS(4) = 0	no
	OPTIONS(4) = 1	yes
	OPTIONS(5)	convert MVA ratings to estimated MW ratings (0 by default)
	OPTIONS(5) = 0	no
	OPTIONS(5) = 1	yes
	OPTIONS(6)	code to discontinue activity if MW mismatch exceeds TOLS(1) (0 by default)
	OPTIONS(6) = 0	discontinue
	OPTIONS(6) = 1	continue
Real VALUES (3)	array of 3 elements specifying tolerances and thresholds. The value of each element is as follows(input).	
	VALUES(1)	required MW mismatch tolerance for continuing (Newton solution convergence tolerance, TOLN by default)

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	VALUES(2)	percent of rating for reporting an element as overloaded (100.0 by default)
	VALUES(3)	minimum contingency case flow change for including an element in the overload report (0.0 by default)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	no Distribution Factor Data File specified
	IERR = 4	generators are converted
	IERR = 5	buses in island(s) without a swing bus; use activity TREE
	IERR = 6	largest mismatch exceeds mismatch tolerance
	IERR = 7	error opening file DFXFILE
	IERR = 8	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 9	file DFXFILE does not contain distribution factors; run DFAX with distribution factor option enabled
	IERR = 10	no monitored branches or interfaces defined
	IERR = 11	prerequisite requirements for API are not met

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1.64. DCCOR

Use this API to apply corrective actions to the base case using linear programming methods. The corrective actions are to eliminate flow overloads with the objective of minimizing the control adjustments, including load shedding, MW generations dispatch, and phase shifter angle adjustments.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_DCCOR  OPTIONS(1)..OPTIONS(7)  VALUES(1)..VALUES(6)
 LABELS(1)..LABELS(3)  DFXFILE
```

Python command syntax:

```
ierr = dccor(options, values, labels, dfxfile)
```

Fortran command syntax:

```
CALL DCCORAPI(OPTIONS, VALUES, LABELS, DFXFILE, IERR)
```

Where:

Integer OPTIONS (7)	array of 7 elements specifying DC corrective actions solution options(input).	
	OPTIONS(1)	rating set (rating set program option setting by default)
	OPTIONS(1) = 1	rating A
	OPTIONS(1) = 2	rating B
	OPTIONS(1) = 3	rating C
	OPTIONS(2)	generator control flag (1 by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable
	OPTIONS(3)	load control flag (0 by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	phase shifter control flag (1 by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	convert MVA ratings to estimated MW ratings (0 by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(6)	code to discontinue activity if MW mismatch exceeds tolerance (VALUES(1) (0 by default)
	OPTIONS(6) = 0	disable (i.e. continue, do not discontinue activity)
	OPTIONS(6) = 1	enable (i.e. discontinue)
	OPTIONS(7)	modify network data with corrective actions solutions (1 by default)

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	OPTIONS(7) = 0	disable (i.e. do not modify network data)
	OPTIONS(7) = 1	enable (i.e. allow network data to be modified)
Real VALUES (6)	array of 6 elements specifying DC corrective actions parameters(input).	
	VALUES(1)	MW mismatch tolerance for continuing (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	percent of rating set (100.0 by default)
	VALUES(3)	branch flow overload tolerance (0.1 by default)
	VALUES(4)	generator control weighting factor (1.0 by default)
	VALUES(5)	load control weighting factor (1.0 by default)
	VALUES(6)	phase shifter control weighting factor (1.0 by default)
Character*32 LABELS (3)	array of 3 elements specifying subsystem labels(input).	
	LABELS(1)	label of generator control subsystem (blank by default; no default allowed if OPTIONS(2) is 1)
	LABELS(2)	label of load control subsystem (blank by default; no default allowed if OPTIONS(3) is 1)
	LABELS(3)	label of phase shifter control subsystem (blank by default; no default allowed if OPTIONS(4) is 1)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	No controls
	IERR = 6	control subsystem not defined
	IERR = 7	Distribution Factor Data File not specified
	IERR = 8	largest mismatch exceeds tolerance
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 51	error opening Distribution Factor Data File
	IERR = 52	prerequisite requirements for function not met

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1.65. DCLF

Use this API to apply the dc analogy network solution algorithm to the network modeled in the working case (activity DCLF).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_DCLF  SID  ALL  STATUS(1)..STATUS(4)  BRNCH(1)..BRNCH(3) CKT
```

Python command syntax:

```
ierr = dclf(sid, all, status, brnch, ckt)
```

Fortran command syntax:

```
CALL DCLFAPI(SID, ALL, STATUS, BRNCH, CKT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID.
	ALL = 1	process all buses.
Integer STATUS (4)	array of 4 elements specifying The value of each element is as follows(input).	
	STATUS(1)	rating used in tabulating the report (default rating set program option setting)
	STATUS(1) = 1	RATEA
	STATUS(1) = 2	RATEB
	STATUS(1) = 3	RATEC
	STATUS(2)	voltage that should be saved (0 by default)
	STATUS(2) = 0	original voltages
	STATUS(2) = 1	base case dc power flow voltages
	STATUS(2) = 2	change case dc power flow voltages
	STATUS(3)	status of the changed branch upon completion (0 by default)
	STATUS(3) = 0	leave unchanged
	STATUS(3) = 1	save changed status
	STATUS(4)	action code if the MW mismatch exceeds 0.5 MW (1 by default)
	STATUS(4) = 0	discontinue
	STATUS(4) = 1	continue
Integer BRNCH (3)	array of 3 elements specifying The use of each element is as follows. The values can all be zero or defaulted if no branch is to be changed for the solution(input).	

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	BRNCH(1)	from bus of the branch for which status is to be changed for the solution (0 by default)
	BRNCH(2)	to bus of the branch for which status is to be changed for the solution (0 by default)
	BRNCH(3)	third bus connected by a three-winding transformer, or zero for a nontransformer branch or a two-winding transformer (0 by default)
Character*2 CKT	circuit ID of the branch for which status is to be changed for the solution (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	bus not found
	IERR = 5	branch not found
	IERR = 6	three-winding transformer not found
	IERR = 7	multi-section line specified but multi-section line reporting option is disabled
	IERR = 8	closing a multi-section line is invalid in DCLF
	IERR = 9	returning a Type 4 bus to service is not allowed in DCLF
	IERR = 10	closing a three-winding transformer is not allowed in DCLF
	IERR = 11	generators are converted
	IERR = 12	buses in island(s) without a swing bus; use activity TREE
	IERR = 13	largest mismatch exceeds 0.5 MW
	IERR = 14	prerequisite requirements for API are not met

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1.66. DC_TAP_ADJUSTMENT

Use this API to specify or return the option to enable or disable the dc tap adjustment option setting.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_DC_TAP_ADJUSTMENT  IVAL
```

Python command syntax:

```
ierr = dc_tap_adjustment(ival)

ierr, ival = dc_tap_adjustment()
```

Fortran command syntax:

```
CALL DC_TAP_ADJUSTMENT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	disabled.
	IVAL = 1	enabled.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.67. DELTMPFILES

Delete closed temporary files. PSSE creates a subdirectory within the user's "temporary" directory; only files there are removed.

This API routine was first introduced in release 32.2.0.

Batch command syntax:

```
BAT_DELTMPFILES
```

Python command syntax:

```
ierr = deltmpfiles()
```

Fortran command syntax:

```
CALL DELTMPFILES(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 1	prerequisite requirements for API are not met.

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1.68. DFAX

This API routine is obsolete. It has been replaced by the API routine dfax_2 .

Use this API to construct a Distribution Factor Data File (activity DFAX).

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_DFAX  OPTIONS(1)  OPTIONS(2)  SUBFILE  MONFILE  CONFILE  DFXFILE
```

Python command syntax:

```
ierr = dfax(options, subfile, monfile, confile, dfxfile)
```

Fortran command syntax:

```
CALL DFAXAPI(OPTIONS, SUBFILE, MONFILE, CONFILE, DFXFILE, IERR)
```

Where:

Integer OPTIONS (2)	array of 2 elements specifying calculation options. The value of each element is as follows(input).
	OPTIONS(1) — distribution factor option flag (1 by default)
	OPTIONS(1) = 0 — do not calculate distribution factors (i.e., DFAX,AC)
	OPTIONS(1) = 1 — calculate distribution factors
	OPTIONS(2) — monitored element sorting flag (0 by default)
	OPTIONS(2) = 0 — do not sort (i.e., leave in Monitored Element Description File order)
	OPTIONS(2) = 1 — sort
Character*260 SUBFILE	name of the Subsystem Description File; blank for none (input; blank by default).
Character*260 MONFILE	name of Monitored Element Description File (input; no default allowed).
Character*260 CONFILE	name of Contingency Description Data File (input; no default allowed).
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).
Integer IERR	error code (output).
	IERR = 0 — no error occurred
	IERR = 1 — invalid OPTIONS value
	IERR = 2 — generators are converted
	IERR = 3 — buses in island(s) without a swing bus; use activity TREE
	IERR = 4 — no Distribution Factor Data File specified
	IERR = 5 — no Monitored Element Data input file specified

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IERR = 6	no Contingency Description Data file specified
IERR = 7	fatal error reading input file
IERR = 8	error opening output file DFXFILE
IERR = 9	error opening input file SUBFILE
IERR = 10	error opening input file MONFILE
IERR = 11	error opening input file CONFILE
IERR = 12	prerequisite requirements for API are not met

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1.69. DFAX_2

Use this API to construct a Distribution Factor Data File (activity DFAX).

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_DFAX_2  OPTIONS(1)..OPTIONS(3) SUBFILE MONFILE CONFILE DFXFILE
```

Python command syntax:

```
ierr = dfax_2(options, subfile, monfile, confile, dfxfile)
```

Fortran command syntax:

```
CALL DFAXAPI_2(OPTIONS, SUBFILE, MONFILE, CONFILE, DFXFILE, IERR)
```

Where:

Integer OPTIONS (3)	array of elements specifying calculation options. The value of each element is as follows.(input).	
	OPTIONS(1)	distribution factor option flag. (1 by default)
	OPTIONS(1) = 0	do not calculate distribution factors (i.e., DFAX,AC).
	OPTIONS(1) = 1	calculate distribution factors.
	OPTIONS(2)	monitored element sorting flag (0 by default)
	OPTIONS(2) = 0	do not sort (i.e., leave in Monitored Element Description File order).
	OPTIONS(2) = 1	sort.
	OPTIONS(3)	out-of-service monitored branch flag (0 by default)
	OPTIONS(3) = 0	eliminate out-of-service branch from monitored branches.
	OPTIONS(3) = 1	keep out-of-service branch in monitored branches.
Character*260 SUBFILE	Is the name of the Subsystem Description File (input; blank by default).	
Character*260 MONFILE	Is the name of Monitored Element Description File (input; no default allowed).	
Character*260 CONFILE	Is the name of Contingency Description Data File (input; no default allowed).	
Character*260 DFXFILE	Is the name of Distribution Factor Data File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred.
	IERR = 1	invalid OPTIONS value
	IERR = 2	generators are converted.

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IERR = 3	buses in island(s) without a swing bus; use activity TREE
IERR = 4	no Distribution Factor Data File specified
IERR = 5	no Monitored Element Data input file specified.
IERR = 6	no Contingency Description Data file specified.
IERR = 7	fatal error reading input file.
IERR = 8	error opening output file DFXFILE.
IERR = 9	error opening input file SUBFILE.
IERR = 10	error opening input file MONFILE.
IERR = 11	error opening input file CONFILE.
IERR = 12	prerequisite requirements for API are not met.

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1.70. DFAX_CONTINGENCY

Use this API to combine one contingency in the first Distribution Factor file with one contingency in the second Distribution Factor file and so on, till specified contingency level is reached, to create multiple event contingencies and then export resulting contingencies to user defined report device.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr = dfax_contingency(option, confile, dfxfile)
```

Fortran command syntax:

```
CALL DFAX_CONTINGENCY(OPTION, CONFILE, DFXFILE, IERR)
```

Where:

Integer OPTION	level of multiple contingencies (input; from 1 to 3; 1).	
Character*260 CONFILE	name of Contingency Description Data File (input; blank by default).	
Character*260 DFXFILE (OPTION)	OPTION names of Distribution Factor Data Files where contingencies are combined each other to build OPTION level contingencies (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTION value
	IERR = 2	Input file is not a Distribution Factor Data File
	IERR = 3	error reading Distribution Factor Data File
	IERR = 51	error opening Contingency Description Data File
	IERR = 52	error opening Distribution Factor Data File

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1.71. DFTI

Use this API to compare tie lines, as contained in the working case, with those of a designated Saved Case (activity DFTI). It must be referenced using one of the following procedures:
- One reference with APIOPT=1, followed by any number of references with APIOPT=2, followed by one reference with APIOPT=3.
- One reference with APIOPT=0. This automatically combines the processing of one APIOPT 1 reference, followed by one APIOPT 2 reference, followed by one APIOPT 3 reference.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_DFTI  SID  APIOPT  STATUS(1)..STATUS(5)  THRSH  CFILE
```

Python command syntax:

```
ierr, seqflg = dfti(sid, apiopt, status, thrsh, cfile)
```

Fortran command syntax:

```
CALL DFTIAPI(SID, APIOPT, STATUS, THRSH, CFILE, SEQFLG, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; used only when APIOPT is 0 or 1; 0).	
Integer APIOPT	mode of operation of DFTIAPI (input; 0 by default).	
	APIOPT = 0	initialize for case comparison, run the comparison specified by STATUS(4), and run post-processing clean up.
	APIOPT = 1	initialize for case comparison.
	APIOPT = 2	run a comparison.
	APIOPT = 3	post-processing clean up.
Integer STATUS (5)	array of 5 elements specifying comparison options(input).	
	STATUS(1)	defines the working case content following an APIOPT=0 or 3 reference (used only when APIOPT is 0, 1 or 3; 0 by default)
	STATUS(1) = 0	original working case
	STATUS(1) = 1	Saved Case specified in CFILE
	STATUS(2)	flag for method of considering buses in the two cases to be the same bus (used only when APIOPT is 0 or 1; 0 by default)
	STATUS(2) = 0	same bus numbers
	STATUS(2) = 1	same extended bus names
	STATUS(2) = 2	same bus numbers and extended bus names
	STATUS(3)	flag for specifying THRESH (used only when APIOPT is 0 or 2; 0 by default)
	STATUS(3) = 0	engineering units

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	STATUS(3) = 1	percent
	STATUS(4)	code for the comparison to be run (used only when APIOPT is 0 or 2; 1 by default)
	STATUS(4) = 1	branch status
	STATUS(4) = 2	line R, X, B
	STATUS(4) = 3	line shunts
	STATUS(4) = 4	line ratings
	STATUS(4) = 5	metered end
	STATUS(4) = 6	line length
	STATUS(4) = 7	transformers
	STATUS(4) = 8	flow MW or MVAR (from bus)
	STATUS(4) = 9	flow MW or MVAR (from & to)
	STATUS(4) = 10	line MW or MVAR losses
	STATUS(4) = 11	zero sequence R, X, B
	STATUS(4) = 12	zero sequence line shunts
	STATUS(4) = 13	transformer connection codes
	STATUS(4) = 14	flow MW (from bus)
	STATUS(4) = 15	flow MVAR (from bus)
	STATUS(4) = 16	flow MW (from and to buses)
	STATUS(4) = 17	flow MVAR (from and to buses)
	STATUS(4) = 18	line MW losses
	STATUS(4) = 19	line MVAR losses
	STATUS(5)	line rating to be compared when STATUS(4) is 4 (used only when APIOPT is 0 or 2; 0 by default)
	STATUS(5) = 0	all ratings
	STATUS(5) = 1	RATEA
	STATUS(5) = 2	RATEB
	STATUS(5) = 3	RATEC
Real THRSH	difference threshold (input; used only when APIOPT is 0 or 2; 0.0).	
Character*260 CFILE	name of Saved Case File in which tie lines are to be compared to those of the working case (when APIOPT is 0 or 1) (input; no default allowed).	
Logical SEQFLG (2)	array of 2 elements specifying returned when APIOPT=0 or 1(output).	
	SEQFLG(1)	true if there is sequence data in the working case
	SEQFLG(2)	true if there is sequence data in the Saved Case in file CFILE
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid STATUS value
	IERR = 3	CFILE is blank

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IERR = 4	no ties to compare
IERR = 5	invalid APIOPT value
IERR = 6	unexpected APIOPT value
IERR = 7	invalid THRSH value
IERR = 8	error reading the comparison case
IERR = 9	prerequisite requirements for API are not met

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1.72. DIFF

Use this API to compare specified power flow data and solution results, as contained in the working case, with those of a designated Saved Case (activity DIFF). It must be referenced first with APIOPT=1, followed by any number of references with APIOPT=2, followed by one reference with APIOPT=3:

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_DIFF  SID  ALL  APIOPT  STATUS(1)..STATUS(4)  THRSH(1)..THRSH(3)
 CFILE
```

Python command syntax:

```
ierr, seqflg = diff(sid, all, apiopt, status, thrsh, cfile)
```

Fortran command syntax:

```
CALL DIFFNW(SID, ALL, APIOPT, STATUS, THRSH, CFILE, SEQFLG, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer APIOPT	mode of operation in the API (input).	
	APIOPT = 1	initialize for case comparison.
	APIOPT = 2	run case comparison for specified data category.
	APIOPT = 3	postprocessing housekeeping.
Integer STATUS (4)	STATUS has different values/meanings depending on the value of APIOPT. If APIOPT = 1, STATUS(1) is the case to leave in the working case (input; 0 by default). STATUS(1) = 0 original working case. STATUS(1) = 1 saved case specified in CFILE. If APIOPT = 1, STATUS(2) is the method used to consider two buses the same for comparison (0 by default). STATUS(2) = 0 bus numbers must be the same. STATUS(2) = 1 bus names must be the same. STATUS(2) = 2 bus names and numbers must be the same. If APIOPT = 2, STATUS(1) is the method used for specifying difference thresholds (0 by default). STATUS(1) = 0 threshold in engineering units. STATUS(1) = 1 threshold as a percent. If APIOPT = 2, STATUS(2) is the comparison to be run (1 by default). = 1, bus identifiers = 2, bus type codes = 3, machine status = 4, generator MW = 5, generator MW or MVAR = 6, bus loads = 7, bus shunts = 8, switched shunts = 9, voltage = 10, voltage and angle = 11, Mbase and Zsorce = 12, Mbase, Rpos and X" = 13, Mbase and Zneg = 14, Mbase and Zzero = 15, negative sequence loads = 16, zero sequence loads = 17, branch status = 18, line R, X, B = 19, line shunts = 20, line ratings = 21, metered end = 22, transformers = 23,	

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flow MW or MVAR (from bus) = 24, flow MW or MVAR (from & to) = 25, line MW or MVAR losses= 26, zero sequence R, X, B = 27, zero sequence line shunts = 28, connection codes = 29, zero sequence mutuals = 30, multi-section lines = 31, multi-section metered end = 32, load status = 33, line lengths = 34, generator MVAR = 35, flow MW (from bus) = 36, flow MVAR (from bus) = 37, flow MW (from and to) = 38, flow MVAR (from and to) = 39, line MW losses = 40, line MVAR losses = 41, fixed bus shunt status = 42, switched shunt status = 43, scalable load flag = 44, Mbase, Rpos and X' = 45, Mbase, Rpos and Xs If APIOPT = 2, STATUS(3) is the load characteristic to be compared (0 by default). STATUS(3) = 0 total nominal load. STATUS(3) = 1 constant MVA. STATUS(3) = 2 constant current. STATUS(3) = 3 constant admittance. If APIOPT = 2, STATUS(4) is the line rating set to be compared (0 by default). STATUS(4) = 0 all ratings. STATUS(4) = 1 RATEA. STATUS(4) = 2 RATEB. STATUS(4) = 3 RATEC. If APIOPT = 3, STATUS(1) is the case to leave in the working case (0 by default). STATUS(1) = 0 original working case. STATUS(1) = 1 saved case specified in CFILE).

Real THRSH (3) — array of 3 elements specifying (used only when APIOPT is 2)(input).

THRSH(1)	difference threshold
THRSH(2)	voltage or tap ratio threshold
THRSH(3)	le threshold

Character*260 CFILE — saved case to be compared against the data currently in the working case (when APIOPT is 1) (input; no default allowed).

Logical SEQFLG (2) — array of 2 elements specifying returned when APIOPT = 1(output).

SEQFLG(1)	true if there is sequence data in the working case
SEQFLG(2)	true if there is sequence data in the Saved Case in file CFILE

Integer IERR — error code (output).

IERR = 0	no error occurred
IERR = 1	invalid SID value or subsystem SID is not defined
IERR = 2	invalid ALL value
IERR = 3	invalid APIOPT value
IERR = 4	invalid STATUS value
IERR = 5	invalid THRSH value
IERR = 6	CFILE is blank
IERR = 7	unexpected APIOPT value
IERR = 8	error building the case to case translation files
IERR = 9	error reading the comparison case
IERR = 10	prerequisite requirements for API are not met

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1.73. DSCN

Use this API to electrically disconnect a bus (activity DSCN).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_DSCN  BUS
```

Python command syntax:

```
ierr = dscn(bus)
```

Fortran command syntax:

```
CALL DSCNAPI(BUS, IERR)
```

Where:

Integer BUS	number of the bus to be disconnected (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus BUS not found
	IERR = 2	prerequisite requirements for API are not met

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1.74. ECDI

Use this API to place machines in a specified subsystem on- or off-line to satisfy a given subsystem minimum capacity; the in-service machines in the subsystem are then dispatched on the basis of incremental cost to meet a specified total subsystem generation (activity ECDI).
This API must be referenced using one of the following procedures:
- One reference with APIOPT=1, followed by one reference with APIOPT=2, followed by one or more references with APIOPT=3, followed by one reference with APIOPT=4.
- One reference with APIOPT=0. This automatically combines the processing of one APIOPT 1 reference, followed by one APIOPT 2 reference, followed by one APIOPT 3 reference, followed by one APIOPT 4 reference.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ECDI  SID  ALL  APIOPT  CFILE  STATUS  VALUES(1)  VALUES(2)
```

Python command syntax:

```
ierr = ecdi(sid, all, apiopt, cfile, status, values)
```

Fortran command syntax:

```
CALL ECDIAPI(SID, ALL, APIOPT, CFILE, STATUS, VALUES, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; used only when APIOPT is 0 or 2; 0).	
Integer ALL	all buses or specified subsystem flag (input; used only when APIOPT is 0 or 2; 1).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer APIOPT	mode of operation of the API (input; 0 by default).	
	APIOPT = 0	initialize, specify subsystem, process, and clean up.
	APIOPT = 1	initialize for dispatch.
	APIOPT = 2	specify subsystem.
	APIOPT = 3	apply calculation.
	APIOPT = 4	postprocessing housekeeping.
Character*260 CFILE	name of the Economic Dispatch Data File (when APIOPT is 0 or 1) (input; no default allowed).	
Integer STATUS	unit commitment option (input; used only when APIOPT is 0 or 2; 0).	
	STATUS = 0	start from current commitment profile.
	STATUS = 1	new commitment profile (start with all units off-line).
	STATUS = 2	unit commitment disabled.

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Real VALUES (2)	array of 2 elements specifying (used only when APIOPT is 0 or 3)(input).	
	VALUES(1)	desired loading (present loading by default)
	VALUES(2)	desired minimum capacity of units being dispatched (0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid APIOPT value
	IERR = 3	invalid ALL value
	IERR = 4	invalid STATUS value
	IERR = 5	dispatch unable to meet desired power and capacity
	IERR = 6	no dispatch data for any subsystem machines
	IERR = 7	negative VALUES value
	IERR = 8	unexpected APIOPT value
	IERR = 9	error opening CFILE
	IERR = 10	prerequisite requirements for API are not met

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1.75. ECHO

Use this API to enable or disable response echoing (activity ECHO).

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_ECHO  IDVFIL
```

Python command syntax:

```
ierr = echo(idvfil)
```

Fortran command syntax:

```
CALL ECHOAPI(IDVFIL, IERR)
```

Where:

Character*260 IDVFIL	name of Response File to be constructed (blank to disable) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	error opening IDVFIL
	IERR = 2	prerequisite requirements for API are not met

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1.76. EEQV

Use this API to construct an electrical equivalent of a specified subsystem of the working case (activity EEQV).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_EEQV  SID  ALL  STATUS(1)..STATUS(6)  DVAL(1)  DVAL(2)
```

Python command syntax:

```
ierr = eeqv(sid, all, status, dval)
```

Fortran command syntax:

```
CALL EEQVNW(SID, ALL, STATUS, DVAL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer STATUS (6)	array of 6 elements specifying They are used to control equivalencing options. A value of 0 means the option is not enabled. A value of 1 means the option is enabled. They are as follows(input).	
	STATUS(1)	retain area boundary buses (0 by default)
	STATUS(2)	retain zone boundary buses (0 by default)
	STATUS(3)	suppress the equivalencing option for phase shifters (0 by default)
	STATUS(4)	retain controlled buses (0 by default)
	STATUS(5)	retain existing branches between retained buses (0 by default)
	STATUS(6)	net load and shunt at retained buses (0 by default)
Real DVAL (2)	array of 2 elements specifying They are as follows(input).	
	DVAL(1)	minimum active or reactive power generation for retaining generator buses (0.0 by default)
	DVAL(2)	branch threshold tolerance (10.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value

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IERR = 3	invalid STATUS value
IERR = 4	invalid DVAL value
IERR = 5	prerequisite requirements for API are not met

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1.77. EQRD

Use this API to build an electrical equivalent of radial and, optionally, two-point Type 1 buses in a specified subsystem of the working case (activity EQRD).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_EQRD  SID  ALL  STATUS(1)..STATUS(8)
```

Python command syntax:

```
ierr = eqrd(sid, all, status)
```

Fortran command syntax:

```
CALL EQRDNW(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer STATUS (8)	array of 8 elements specifying They are used to control equivalencing options. A value of 0 means the option is not enabled. A value of 1 means the option is enabled. They are as follows(input).	
	STATUS(1)	build an electrical equivalent of radial buses only (0 by default)
	STATUS(2)	suppress the equivalencing option for transformers (0 by default)
	STATUS(3)	suppress the equivalencing option for zero impedance lines (0 by default)
	STATUS(4)	suppress the equivalencing option for controlled buses (0 by default)
	STATUS(5)	suppress the equivalencing option for mutually coupled branches (0 by default)
	STATUS(6)	suppress the equivalencing option for area boundary buses (0 by default)
	STATUS(7)	suppress the equivalencing option for zone boundary buses (0 by default)
	STATUS(8)	apply transformer impedance correction to zero sequence network (1 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined

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IERR = 2	invalid ALL value
IERR = 3	invalid STATUS value
IERR = 4	error writing the working case to a temporary file
IERR = 5	load table is full; restoring original working case
IERR = 6	fixed bus shunt table is full; restoring original working case
IERR = 7	error opening temporary file
IERR = 8	prerequisite requirements for API are not met

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1.78. EXAM

Use this API to tabulate all power flow data pertaining to a specified bus (activity EXAM).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_EXAM  SID  ALL
```

Python command syntax:

```
ierr = exam(sid, all)
```

Fortran command syntax:

```
CALL EXAMNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.79. EXTR

Use this API to purge specified buses and all branches connected to them from the working case (activity EXTR).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_EXTR  SID  ALL  STATUS(1)  STATUS(2)
```

Python command syntax:

```
ierr = extr(sid, all, status)
```

Fortran command syntax:

```
CALL EXTRNW(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (when ALL is 0) (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; 0 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer STATUS (2)	array of 2 elements specifying They are used to control subsystem removal options. A value of 0 means the option is not enabled. A value of 1 means the option is enabled. They are as follows(input).	
	STATUS(1)	remove generator entries for out-of-service plants (0 by default)
	STATUS(2)	change codes of boundary buses (0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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1.80. FACT

This API routine is obsolete. It has been replaced by the API routine facts_chng_2 .

Use this API to factorize the network admittance matrix in preparation for switching studies and dynamic simulations (activity FACT).

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_FACT
```

Python command syntax:

```
ierr = fact()
```

Fortran command syntax:

```
CALL FACTAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	generators are not converted
	IERR = 2	prerequisite requirements for API are not met

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1.81. FDNS

Use this API to apply the fixed slope decoupled Newton-Raphson power flow calculation (activity FDNS).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_FDNS  OPTIONS(1)..OPTIONS(8)
```

Python command syntax:

```
ierr = fdns(options)
```

Fortran command syntax:

```
CALL FDNSAPI(OPTIONS, IERR)
```

Where:

Integer OPTIONS (8)	array of 8 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (use tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (use area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (use phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (use dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (use switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode

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	OPTIONS(6)	flat start flag (0 by default)
	OPTIONS(6) = 0	do not flat start
	OPTIONS(6) = 1	flat start
	OPTIONS(6) = 2	flat start, then estimate voltage magnitudes
	OPTIONS(6) = 3	flat start, then estimate voltage phase angles
	OPTIONS(6) = 4	flat start, then estimate voltage magnitudes and phase angles
	OPTIONS(7)	var limit flag (99 by default)
	OPTIONS(7) = 0	apply var limits immediately
	OPTIONS(7) = 0	apply var limits on iteration n (or sooner if mismatch gets small)
	OPTIONS(7) = -1	ignore var limits
	OPTIONS(8)	non-divergent solution flag (use non-divergent solution option setting by default)
	OPTIONS(8) = 0	disable
	OPTIONS(8) = 1	enable
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	generators are converted
	IERR = 3	buses in island(s) without a swing bus; use activity TREE
	IERR = 4	bus type code and series element status inconsistencies
	IERR = 5	prerequisite requirements for API are not met

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1.82. FILE_OVERWRITE

Use this API to specify or return the option to set the file overwrite option setting to either ask first or overwrite.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_FILE_OVERWRITE  IVAL
```

Python command syntax:

```
ierr = file_overwrite(ival)

ierr, ival = file_overwrite()
```

Fortran command syntax:

```
CALL FILE_OVERWRITE_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1 by default; else output).	
	IVAL = 0	ask first.
	IVAL = 1	overwrite.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.83. FIND

Use this API to tabulate a list of buses matching a partial extended bus name (activity FIND).

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_FIND  NAME BSKV
```

Python command syntax:

```
ierr = find(name, bskv)
```

Fortran command syntax:

```
CALL FINDAPI(NAME, BSKV, IERR)
```

Where:

Character*24 NAME	bus name, including imbedded asterisks that are treated as wildcard characters (input; blank by default).	
Character*12 BSKV	bus base voltage (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	error decoding BSKV
	IERR = 2	prerequisite requirements for API are not met

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1.84. FLAT

This API routine is obsolete. It has been replaced by the API routine flat_2 .

Use this API to specify various levels of flat conditions in preparation for fault analysis calculations (activity FLAT).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_FLAT  OPTIONS(1)..OPTIONS(4)  VALUES(1)  VALUES(2)
```

Python command syntax:

```
ierr = flat(options, values)
```

Fortran command syntax:

```
CALL FLATAPI(OPTIONS, VALUES, IERR)
```

Where:

Integer OPTIONS (4)	array of 4 elements specifying boundary condition options(input).	
	OPTIONS(1)	flat option (0 by default)
	OPTIONS(1) = 0	only change voltages (activity FLAT)
	OPTIONS(1) = 1	set classical fault analysis conditions (activity FLAT,CL)
	OPTIONS(1) = 2	set conditions in preparation for IEC standard 909 calculations (activity FLAT,IEC)
	OPTIONS(2)	tap ratio treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(2) = 0	leave tap ratios unchanged
	OPTIONS(2) = 1	set tap ratios to 1.0
	OPTIONS(3)	charging treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(3) = 0	leave charging unchanged
	OPTIONS(3) = 1	set charging to 0.0
	OPTIONS(4)	shunt treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(4) = 0	leave shunts unchanged
	OPTIONS(4) = 1	set line shunts to 0.0 in the positive (and hence negative) sequence, and fixed and switched shunts to 0.0 in all three sequence networks
	OPTIONS(4) = 2	set line shunts, fixed shunts, and switched shunts to 0.0 in all sequence networks, and negative and zero sequence shunt loads to 0.0

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Real VALUES (2)	array of 2 elements specifying boundary condition values (ignored unless OPTIONS(1) is 2)(input).	
	VALUES(1)	voltage magnitude for all buses, or 0.0 to use present voltage magnitudes (0.0 by default)
	VALUES(2)	generator power factor, or 0.0 to leave generator Mvar unchanged (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	prerequisite requirements for API are not met

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1.85. FLAT_2

This API routine is the second release of the flat conditions function. Use this API routine to specify various levels of flat conditions in preparation for fault analysis calculations (activity FLAT)

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_FLAT_2  OPTIONS(1)..OPTIONS(8)  VALUES(1)  VALUES(2)
```

Python command syntax:

```
ierr = flat_2(options, values)
```

Fortran command syntax:

```
CALL FLAT_2(OPTIONS, VALUES, IERR)
```

Where:

Integer OPTIONS (8)	array of 8 elements specifying boundary condition options(input).	
	OPTIONS(1)	flat option (0 by default)
	OPTIONS(1) = 0	only change voltages (activity FLAT)
	OPTIONS(1) = 1	set classical fault analysis conditions (activity FLAT,CL)
	OPTIONS(1) = 2	set conditions in preparation for IEC standard 909 calculations (activity FLAT,IEC)
	OPTIONS(2)	tap ratio treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(2) = 0	leave tap ratios unchanged
	OPTIONS(2) = 1	set tap ratios to 1.0
	OPTIONS(3)	charging treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(3) = 0	leave charging unchanged
	OPTIONS(3) = 1	set charging to 0.0
	OPTIONS(4)	fixed bus shunt treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(4) = 0	leave fixed bus shunts unchanged
	OPTIONS(4) = 1	set fixed bus shunts to 0.0 in the positive (and hence negative) sequence
	OPTIONS(4) = 2	set fixed bus shunts to 0.0 in all sequence networks
	OPTIONS(5)	switched shunt treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(5) = 0	leave switched shunts unchanged
	OPTIONS(5) = 1	set switched shunts to 0.0 in all sequence networks
	OPTIONS(6)	line shunt treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(6) = 0	leave line shunts unchanged

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	OPTIONS(6) = 1	set line shunts to 0.0 in the positive (and hence negative) sequence
	OPTIONS(6) = 2	set line shunts to 0.0 in all sequence networks
	OPTIONS(7)	transformer magnetizing admittance treatment flag (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(7) = 0	leave transformer magnetizing admittances unchanged
	OPTIONS(7) = 1	set transformer magnetizing admittances to 0.0 in the positive (and hence negative) sequence
	OPTIONS(8)	load treatment flag; this option applies to the constant power, current, and admittance load characteristic data (specified on Load Data records in the Power Flow Raw Data File), and to the exceptional negative and zero sequence load data (specified on Load Data records in the Sequence Data File) (ignored if OPTIONS(1) is 0) (0 by default)
	OPTIONS(8) = 0	leave loads unchanged
	OPTIONS(8) = 1	set the constant power, current and admittance load to 0.0 in all sequence networks; model (GNEG + j BNEG) in the negative sequence, and (GZERO + j BZERO) in the zero sequence
	OPTIONS(8) = 2	set the constant power, current and admittance load to 0.0 in all sequence networks, and (GNEG + j BNEG) to 0.0 in the negative sequence; model (GZERO + j BZERO) in the zero sequence
	OPTIONS(8) = 3	set the constant power, current and admittance load to 0.0 in all sequence networks, (GNEG + j BNEG) to 0.0 in the negative sequence, and (GZERO + j BZERO) to 0.0 in the zero sequence
Real VALUES (2)	array of 2 elements specifying boundary condition values (ignored unless OPTIONS(1) is 2)(input).	
	VALUES(1)	voltage magnitude for all buses, or 0.0 to use present voltage magnitudes (0.0 by default)
	VALUES(2)	generator power factor, or 0.0 to leave generator Mvar unchanged (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	prerequisite requirements for API are not met

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1.86. FNSL

Use this API to apply the Newton-Raphson power flow calculation (activity FNSL).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_FNSL  OPTIONS(1)..OPTIONS(8)
```

Python command syntax:

```
ierr = fnsl(options)
```

Fortran command syntax:

```
CALL FNSLAPI(OPTIONS, IERR)
```

Where:

Integer OPTIONS (8)	array of 8 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (use tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (use area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (use phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (use dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (use switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode

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	OPTIONS(6)	flat start flag (0 by default)
	OPTIONS(6) = 0	do not flat start
	OPTIONS(6) = 1	flat start
	OPTIONS(6) = 2	flat start, then estimate voltage magnitudes
	OPTIONS(6) = 3	flat start, then estimate voltage phase angles
	OPTIONS(6) = 4	flat start, then estimate voltage magnitudes and phase angles
	OPTIONS(7)	var limit flag (99 by default)
	OPTIONS(7) = 0	apply var limits immediately
	OPTIONS(7) = 0	apply var limits on iteration n (or sooner if mismatch gets small)
	OPTIONS(7) = -1	ignore var limits
	OPTIONS(8)	non-divergent solution flag (use non-divergent solution option setting by default)
	OPTIONS(8) = 0	disable
	OPTIONS(8) = 1	enable
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	generators are converted
	IERR = 3	buses in island(s) without a swing bus; use activity TREE
	IERR = 4	bus type code and series element status inconsistencies
	IERR = 5	prerequisite requirements for API are not met

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1.87. GCAP

Use this API to read a data file containing capability curve data and print a report of machine loading and limit data (activity GCAP). Optionally, machine reactive power limits in the working case may be updated.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_GCAP  SID  ALL  OPT  GFILE
```

Python command syntax:

```
ierr = gcap(sid, all, opt, gfile)
```

Fortran command syntax:

```
CALL GCAPNW(SID, ALL, OPT, GFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer OPT	update var limits option (input; 0 by default).	
	OPT = 0	leave var limits unchanged.
	OPT = 1	update var limits in the working case.
Character*260 GFILE	Machine Capability Data File (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	error opening GFILE
	IERR = 5	prerequisite requirements for API are not met

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1.88. GDIF

Use this API to calculate differences between the working case and a designated Saved Case (activity GDIF). Results may be programmatically accessed via the internally accessible fetch routines DIFTXT, DIFBUS, DIFBRN, DIF3WN, DIF2DC, DIFMDC and DIFFCT.
Usage is: call to GDIFAPI with APIOPT = 0;
multiple calls to fetch routines;
optionally, repeat of above steps for different Saved Case and/or subsystem;
final call to GDIFAPI with APIOPT = 1.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_GDIF  SID  ALL  APIOPT  NAMARG  FILARG
```

Python command syntax:

```
ierr = gdif(sid, all, apiopt, namarg, filarg)
```

Fortran command syntax:

```
CALL GDIFAPI(SID, ALL, APIOPT, NAMARG, FILARG, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; used only when APIOPT is 0; 0).	
Integer ALL	all buses or specified subsystem flag (input; used only when APIOPT is 0; 1).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer APIOPT	mode of operation of GDIFAPI (input; no default allowed).	
	APIOPT = 0	apply difference calculations and fill arrays used by fetch routines.
	APIOPT = 1	deallocate arrays used by fetch routines.
Integer NAMARG	flag indicating how to match up buses in the two cases (input; used only when APIOPT is 0; 0).	
	NAMARG = 0	by bus number only.
	NAMARG = 1	by extended bus name only.
	NAMARG = 2	by both number and name.
Character*260 FILARG	name of Saved Case File to be compared to the working case (when APIOPT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value

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IERR = 3	invalid APIOPT value
IERR = 4	invalid NAMARG value
IERR = 5	FILARG is blank
IERR = 6	error reading the comparison case
IERR = 7	prerequisite requirements for API are not met

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1.89. GENDSP

Use this API to impose a contingency specified in the Distribution Factor Data file and apply the generation dispatch algorithm used in contingency analysis on the working case. The network can then be solved by power flow solutions.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_GENDSP  OPTIONS(1)  OPTIONS(2)  LABELS(1)  LABELS(2)  DFXFILE  THRFILE
  INLFILE
```

Python command syntax:

```
ierr = gendsp(options, labels, dfxfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL GENDSPAPI(OPTIONS, LABELS, DFXFILE, THRFILE, INLFILE, IERR)
```

Where:

Integer OPTIONS (2)	array of 2 elements specifying options(input).	
	OPTIONS(1)	dispatch mode (1 by default)
	OPTIONS(1) = 0	disable dispatch
	OPTIONS(1) = 1	subsystem machines (RESERVE)
	OPTIONS(1) = 2	subsystem machines (PMAX)
	OPTIONS(1) = 3	subsystem machines (INERTIA)
	OPTIONS(1) = 4	subsystem machines (GOVERNOR DROOP)
	OPTIONS(2)	study case (1 by default)
	OPTIONS(2) = 0	impose a contingency case on the working case
	OPTIONS(2) = 1	working case
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTIONS(1) is from 1 to 4)
	LABELS(2)	contingency label (blank by default, no default allowed if OPTIONS(2) is 0)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPTIONS value
	IERR = 2	no dispatch subsystem specified

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IERR = 3	no contingency specified
IERR = 4	no Distribution Factor Data File specified
IERR = 5	generators are converted
IERR = 6	too many islands in base case
IERR = 7	contingency not found in Distribution Factor Data File
IERR = 21	file is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 23	no subsystem defined in Distribution Factor Data file
IERR = 24	subsystem not found
IERR = 51	error opening Distribution Factor Data File
IERR = 52	error opening Load Throwover Data File
IERR = 53	error opening Unit Inertia and Governor Data File
IERR = 54	prerequisite requirements for function not met

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1.90. GENS

Use this API to tabulate the loading and voltage conditions at plant buses (activity GENS).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_GENS  SID  ALL  OPT  VREV
```

Python command syntax:

```
ierr = gens(sid, all, opt, vrev)
```

Fortran command syntax:

```
CALL GENSNW(SID, ALL, OPT, VREV, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer OPT	option (input; 4 by default).	
	OPT = 1	tabulate var limited plants with unequal var limits.
	OPT = 2	tabulate all var limited plants.
	OPT = 3	tabulate on-line plants.
	OPT = 4	tabulate all plants.
Integer VREV	option to select the units in which voltages are tabulated (input; 0 by default).	
	VREV = 0	use default voltage units.
	VREV = 1	use alternate voltage units.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	invalid VREV value
	IERR = 5	prerequisite requirements for API are not met

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1.91. GEOL

Use this API to tabulate the loading and voltage conditions at the generator terminals for online machines at Type 2 and 3 buses in the working case (activity GEOL).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_GEOL  SID  ALL  OPT
```

Python command syntax:

```
ierr = geol(sid, all, opt)
```

Fortran command syntax:

```
CALL GEOLNW(SID, ALL, OPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer OPT	option (input; 1 by default).	
	OPT = 1	tabulate overloaded machines only.
	OPT = 2	tabulate all machines.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	prerequisite requirements for API are not met

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1.92. GETCONTINGENCYSAVEDCASE

Use this API routine to place the working case in the form of a specified system condition as calculated during a previous run of one of the members of the the AC contingency calculation family. This function retrieves data from a ZIP Archive Ouptut File that was created during the contingency calculation.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_GETCONTINGENCYSAVEDCASE  PATHZIP ISVFILE
```

Python command syntax:

```
ierr = getcontingencysavedcase(pathzip, isvfile)
```

Fortran command syntax:

```
CALL GETCONTINGENCYSAVEDCASE(PATHZIP, ISVFILE, IERR)
```

Where:

Character*260 PATHZIP	Is the name of the ZIP Archive Output File (input; no default allowed).	
Character*260 ISVFILE	Is the name of an Incremental Saved Case File (.isv file) contained in PATHZIP (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred.
	IERR = 1	unable to pick up the base Saved Case File from the ZIP Archive File.
	IERR = 2	error opening the Incremental Saved Case File.
	IERR = 3	error closing the Incremental Saved Case File.
	IERR = 4	prerequisite requirements for API are not met.

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1.93. GIC

This API routine is obsolete. It has been replaced by the API routine gic_2 .

Use this API to calculate Geomagnetically Induced Currents (GIC) in electric power system network as a result of Geomagnetic Disturbance (GMD).

This API routine was first introduced in release 32.2 and 33.3.

Batch command syntax:

```
BAT_GIC  SID ALL APIOPTNS(1)..APIOPTNS(4) VALUES(1)..VALUES(5)
 RPTOPTNS(1)..RPTOPTNS(7) GICDATAFIL GICADDFIL GICPURGFIL GICRNWKFIL
```

Python command syntax:

```
ierr = gic(sid, all, apioptns, values, rptoptns, gicdatafil, gicaddfil,
 gicpurgfil, gicrnwkfil)
```

Fortran command syntax:

```
CALL GIC(SID, ALL, APIOPTNS, VALUES, RPTOPTNS, GICDATAFIL, GICADDFIL,
 GICPURGFIL, GICRNWKFIL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID.
	ALL = 1	process all buses.
Integer APIOPTNS (4)	array of 4 elements specifying Integer Options(input).	
	APIOPTNS(1)	Electric Field Type (default alias is efield_type) (0 by default)
	APIOPTNS(1) = 0	for uniform geoelectric field
	APIOPTNS(2)	Units of Electric Field Magnitude (default alias is efield_unit) (0 by default)
	APIOPTNS(2) = 0	for volts/km
	APIOPTNS(2) = 1	for volts/mile
	APIOPTNS(3)	Option to add GIC updates to base case (default alias is addfile_optn) (1 by default)
	APIOPTNS(3) = 0	add updates to Saved Case
	APIOPTNS(3) = 1	create RDCH raw file of GIC updates
	APIOPTNS(4)	Number of levels of inter-tie buses to add to study subsystem (default alias is tielevels) (0 by default)
	APIOPTNS(4) = 0	consider only study subsystem buses, no buses from inter-ties

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	APIOPTNS(4) = 0	consider only study subsystem buses, no buses from inter-ties
Real VALUES (5)	array of 5 elements specifying Real Options(input).	
	VALUES(1)	electric field magnitude in units defined by charoptns(2), not used when charoptns(1)=nonuniform (default alias is efield_mag) (0 by default)
	VALUES(1) = 0	when charoptns(1)=uniform, no default allowed
	VALUES(1) = 0	when charoptns(1)=benchmark, when specified as 0.0, it will be set to 8.0 V/km (default benchmark event strength)
	VALUES(1) = 0	for charoptns(1)=benchmark, used as specified
	VALUES(2)	electric field direction in degrees, range 0 to 360 degrees (default alias is efield_deg) (0 by default)
	VALUES(2) = 0	when charoptns(1)=uniform or benchmark, not used when charoptns(1)=nonuniform
	VALUES(3)	substation grounding dc resistance in ohms (default alias is substation_r) (0.1 by default)
	VALUES(3) = 0	Substation is ungrounded
	VALUES(3) = 0	Use specified grounding dc resistance
	VALUES(4)	transmission line X/R ratio, must be >0, used to calculate branch DC resistance if R=0.0 in network data (default alias is branch_xbyr) (30 by default)
	VALUES(4) = 0	Use default
	VALUES(4) = 0	Use specified
	VALUES(5)	transformer winding X/R ratio, must be >0, used to calculate winding DC resistance if R=0.0 in network data (default alias is transformer_xbyr) (30 by default)
	VALUES(5) = 0	Use default
	VALUES(5) = 0	Use specified
Integer RPTOPTNS (7)	array of 7 elements specifying Report Options(input).	
	RPTOPTNS(1)	what to report (default alias is rptoptn) (-1 by default)
	RPTOPTNS(1) = -1	report all
	RPTOPTNS(1) = 0	no report
	RPTOPTNS(1) = 0	Report these many maximum values in each category
	RPTOPTNS(2)	report induced branch voltages (default alias is rptbrn_indv) (0 by default)
	RPTOPTNS(2) = 0	No
	RPTOPTNS(2) = 1	Yes

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	RPTOPTNS(3)	report DC bus voltages (default alias is rptdc_busv) (0 by default)
	RPTOPTNS(3) = 0	No
	RPTOPTNS(3) = 1	Yes
	RPTOPTNS(4)	report branch GIC flows (default alias is rptbrn_gic) (0 by default)
	RPTOPTNS(4) = 0	No
	RPTOPTNS(4) = 1	Yes
	RPTOPTNS(5)	report transformer GIC flows (default alias is rpttrn_gic) (1 by default)
	RPTOPTNS(5) = 0	No
	RPTOPTNS(5) = 1	Yes
	RPTOPTNS(6)	report substation GIC flows (default alias is rptstn_gic) (0 by default)
	RPTOPTNS(6) = 0	No
	RPTOPTNS(6) = 1	Yes
	RPTOPTNS(7)	report transformer losses (default alias is rpttrn_q) (1 by default)
	RPTOPTNS(7) = 0	No
	RPTOPTNS(7) = 1	Yes
Character*260 GICDATAFIL	GIC data file name (input). (input; no default allowed).	
Character*260 GICADDFIL	GIC updates to Base Case file name (output). (input; blank by default).	
Character*260 GICPURGFIL	RDCH file to remove GIC updates from GIC updated case in working memory to set it back to Base Case network condition (output). (input; blank by default).	
Character*260 GICRNWKFIL	GIC dc resistive network raw file. This represents the dc network used to calculate GIC flow (output). (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 1	DLL not loaded or entry point not found.

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1.94. GIC_2

This API routine is obsolete. It has been replaced by the API routine gic_3 .

Use this API to calculate Geomagnetically Induced Currents (GIC) in electric power system network as a result of Geomagnetic Disturbance (GMD).

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_GIC_2  SID ALL INTGOPTNS(1)..INTGOPTNS(4) REALOPTNS(1)..REALOPTNS(7)
 CHAROPTNS(1)..CHAROPTNS(7) EJETOPTNS(1)..EJETOPTNS(5)
 FILEOPTNS(1)..FILEOPTNS(4) RPTOPTNS(1)..RPTOPTNS(7)
```

Python command syntax:

```
ierr = gic_2(sid, all, intgoptns, realoptns, charoptns, ejetoptns,
 fileoptns, rptoptns)
```

Fortran command syntax:

```
CALL GIC_2(SID, ALL, INTGOPTNS, REALOPTNS, CHAROPTNS, EJETOPTNS,
 FILEOPTNS, RPTOPTNS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID.
	ALL = 1	process all buses.
Integer INTGOPTNS (4)	array of 4 elements specifying Integer Options(input).	
	INTGOPTNS(1)	Number of levels of inter-tie buses to add to study subsystem (default alias is tielevels) (0 by default)
	INTGOPTNS(1) = 0	consider only study subsystem buses, no buses from inter-ties
	INTGOPTNS(1) = 0	consider only study subsystem buses, no buses from inter-ties
	INTGOPTNS(2)	Year number to scale benchmark event GMD storm. These scaling factors account in the influence of geomagnetic latitude and earth model on the estimated geoelectric field magnitude and are provided in NERC TPL-007-1. (default alias is study_year) (0 by default)
	INTGOPTNS(2) = 0	current year, used in benchmark geoelectric field calculations only.
	INTGOPTNS(2) = 0	specific year

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	INTGOPTNS(3)	Subsystem sid for local GMD hot spots (default alias is sid_local) (0 by default)
	INTGOPTNS(3) = 0	No local GMD hot spots
	INTGOPTNS(3) = 0	subsystem number (1 - 11). Buses and branches in this subsystem will have Efield due to local GMD hot spots.
	INTGOPTNS(4)	Option for Transformer Thermal Analysis (not implemented yet) (default alias is thermal_ana_optn) (0 by default)
	INTGOPTNS(4) = -1	perform on all transformers
	INTGOPTNS(4) = 0	do not perform
	INTGOPTNS(4) = 0	perform on these many top transformers ordered by effective GIC flow
Real REALOPTNS (7)	array of 7 elements specifying Real Options(input).	
	REALOPTNS(1)	electric field magnitude in units defined by charoptns(2), not used when charoptns(1)=nonuniform (default alias is efield_mag) (0 by default)
	REALOPTNS(1) = 0	when charoptns(1)=uniform, no default allowed
	REALOPTNS(1) = 0	when charoptns(1)=benchmark, when specified as 0.0, it will be set to 8.0 V/km (default benchmark event strength)
	REALOPTNS(1) = 0	for charoptns(1)=benchmark, used as specified
	REALOPTNS(2)	electric field direction in degrees, range 0 to 360 degrees (default alias is efield_deg) (0 by default)
	REALOPTNS(2) = 0	when charoptns(1)=uniform or benchmark, not used when charoptns(1)=nonuniform
	REALOPTNS(3)	substation grounding dc resistance in ohms (default alias is substation_r) (0.1 by default)
	REALOPTNS(3) = 0	Substation is ungrounded
	REALOPTNS(3) = 0	Use specified grounding dc resistance
	REALOPTNS(4)	transmission line X/R ratio, must be >0, used to calculate branch DC resistance if R=0.0 in network data (default alias is branch_xbyr) (30 by default)
	REALOPTNS(4) = 0	Use default
	REALOPTNS(4) = 0	Use specified
	REALOPTNS(5)	transformer winding X/R ratio, must be >0, used to calculate winding DC resistance if R=0.0 in network data (default alias is transformer_xbyr) (30 by default)
	REALOPTNS(5) = 0	Use default
	REALOPTNS(5) = 0	Use specified

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	REALOPTNS(6)	local GMD hot spots electric field magnitude in units defined by charoptns(2), not used when charoptns(1)=nonuniform (default alias is efield_mag_local) (realoptns(1) by default)
	REALOPTNS(6) = 0	when charoptns(1)=uniform or benchmark, not used when charoptns(1)=nonuniform
	REALOPTNS(7)	local GMD hot spots electric field direction in degrees, range 0 to 360 degrees (default alias is efield_deg_local) (realoptns(2) by default)
	REALOPTNS(7) = 0	when charoptns(1)=uniform or benchmark, not used when charoptns(1)=nonuniform
Character*12 CHAROPTNS (7)	array of 7 elements specifying Character Options(input).	
	CHAROPTNS(1)	Electric Field Type (default alias is efield_type) (uniform by default)
	CHAROPTNS(1) = uniform	for uniform geoelectric field
	CHAROPTNS(1) = benchmark	for benchmark event geoelectric field
	CHAROPTNS(1) = nonuniform	for nonuniform geoelectric field
	CHAROPTNS(2)	Units of Electric Field Magnitude (default alias is efield_unit) (v/km by default)
	CHAROPTNS(2) = v/km	for volts/km
	CHAROPTNS(2) = v/mi	for volts/mile
	CHAROPTNS(3)	Option to add GIC updates to base case (default alias is addfile_optn) (rdch by default)
	CHAROPTNS(3) = sav	add updates to Saved Case
	CHAROPTNS(3) = rdch	create RDCH raw file of GIC updates
	CHAROPTNS(4)	Option to select method for GIC to Mvar Calculation (default alias is gic2mvar_optn) (kfactors by default)
	CHAROPTNS(4) = kfactors	for GIC to Mvar scaling factors from GIC data file when provided, otherwise default scaling factors
	CHAROPTNS(4) = capchar	for GIC to Mvar scaling factors from GIC data file when provided, otherwise scaling factors from default transformer Mvar and GIC characteristics
	CHAROPTNS(5)	Earth Model Name. A Standard or User defined model name must be provided when Benchmark Event or Non-uniform electric field is to be modeled or Transformer Thermal Analysis is to be performed. (default alias is earth_model_name) (by default)

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	CHAROPTNS(6) Option to scan storm event scenarios (default alias is scan_storm_event) (by default)
	CHAROPTNS(6) = no storm orientation and/or magnitude scan
	CHAROPTNS(6) = scan_deg perform storm orientation scan that give maximum Var losses when charoptns(1) is uniform or benchmark, create report for only orientation that gives maximum VAR losses
	CHAROPTNS(6) = scan_deg_rpt perform storm orientation scan that give maximum Var losses when charoptns(1) is uniform or benchmark, create report for each orientation scanned
	CHAROPTNS(6) = scan_mag perform storm electric field magnitude that yield converged power flow solution when charoptns(1) is uniform or benchmark and charoptns(7) specifies PF method. It uses orienation specified and creates report for only final electric field magnitude.
	CHAROPTNS(6) = scan_mag_rpt perform storm electric field magnitude that yield converged power flow solution when charoptns(1) is uniform or benchmark and charoptns(7) specifies PF method. It uses orienation specified and creates report for each scanned electric field magnitude.
	CHAROPTNS(6) = scan_d_m perform storm orientation scan that give maximum Var losses when charoptns(1) is uniform or benchmark, then perform magnitude scans tiil they yield converged power flow solution, create report for only orientation that gives maximum VAR losses and final electric field magnitude
	CHAROPTNS(6) = scan_d_m_rpt perform storm orientation scan that give maximum Var losses when charoptns(1) is uniform or benchmark, then perform magnitude scans till they yield converged power flow solution, create report for all orientation scans and mangitude scans performed
	CHAROPTNS(7) Option to solve Power Flow with GIC losses added to the base case (default alias is power_flow_optn) (by default)
	CHAROPTNS(7) = do not perform power flow solution
	CHAROPTNS(7) = fdns use fixed slope decoupled Newton-Raphson method
	CHAROPTNS(7) = fnsl use Full Newton-Raphson method
	CHAROPTNS(7) = nsol use Decoupled Newton-Raphson method
Real EJETOPTNS (5)	array of 5 elements specifying Electrojet Characteristics, used only if charoptns(1)=nonuniform(input).

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	EJETOPTNS(1)	eletrojet current in million amperes, must be >0 (default alias is ejet_million_amps) (no default allowed)
	EJETOPTNS(2)	Cauchy distribution half-width in km, must be >0 (default alias is ejet_halfwidth_km) (no default allowed)
	EJETOPTNS(3)	period of variation in minutes, must be >0 (default alias is ejet_period_min) (no default allowed)
	EJETOPTNS(4)	height of current in km, must be >0 (default alias is ejet_height_km) (no default allowed)
	EJETOPTNS(5)	latitude of center of electrojet in degrees (default alias is ejet_center_deg) (no default allowed)
Character*260 FILEOPTNS (4)	array of 4 elements specifying File Used(input).	
	FILEOPTNS(1)	GIC data file name (input). (default alias is gicfile) (no default allowed)
	FILEOPTNS(2)	GIC updates to Base Case file name (output). (default alias is addfile) (by default)
	FILEOPTNS(2) =	no file created
	FILEOPTNS(3)	RDCH file to remove GIC updates from GIC updated case in working memory to set it back to Base Case network condition (output). (default alias is purgfile) (by default)
	FILEOPTNS(3) =	no file created
	FILEOPTNS(4)	GIC dc resistive network raw file. This represents the dc network used to calculate GIC flow. (output) (default alias is rnwkfile) (by default)
	FILEOPTNS(4) =	no file created
Integer RPTOPTNS (7)	array of 7 elements specifying Report Options(input).	
	RPTOPTNS(1)	what to report (default alias is rptoptn) (-1 by default)
	RPTOPTNS(1) = -1	report all
	RPTOPTNS(1) = 0	no report
	RPTOPTNS(1) = 0	Report these many maximum values in each category
	RPTOPTNS(2)	report induced branch voltages (default alias is rptbrn_indv) (0 by default)
	RPTOPTNS(2) = 0	No
	RPTOPTNS(2) = 1	Yes
	RPTOPTNS(3)	report DC bus voltages (default alias is rptdc_busv) (0 by default)
	RPTOPTNS(3) = 0	No
	RPTOPTNS(3) = 1	Yes

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	RPTOPTNS(4)	report branch GIC flows (default alias is rptbrn_gic) (0 by default)
	RPTOPTNS(4) = 0	No
	RPTOPTNS(4) = 1	Yes
	RPTOPTNS(5)	report transformer GIC flows (default alias is rpttrn_gic) (1 by default)
	RPTOPTNS(5) = 0	No
	RPTOPTNS(5) = 1	Yes
	RPTOPTNS(6)	report substation GIC flows (default alias is rptstn_gic) (0 by default)
	RPTOPTNS(6) = 0	No
	RPTOPTNS(6) = 1	Yes
	RPTOPTNS(7)	report transformer losses (default alias is rpttrn_q) (1 by default)
	RPTOPTNS(7) = 0	No
	RPTOPTNS(7) = 1	Yes
Integer IERR	error code (output).	
	IERR = 1	DLL not loaded or entry point not found.

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1.95. GIC_3

This API routine is obsolete. It has been replaced by the API routine gic_4 .

Use this API to calculate Geomagnetically Induced Currents (GIC) in electric power system network as a result of Geomagnetic Disturbance (GMD).

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_GIC_3  SID ALL INTGOPTNS(1)..INTGOPTNS(4) REALOPTNS(1)..REALOPTNS(7)
 CHAROPTNS(1)..CHAROPTNS(7) EJETOPTNS(1)..EJETOPTNS(5)
 FILEOPTNS(1)..FILEOPTNS(5) RPTOPTNS(1)..RPTOPTNS(7)
```

Python command syntax:

```
ierr = gic_3(sid, all, intgoptns, realoptns, charoptns, ejetoptns,
 fileoptns, rptoptns)
```

Fortran command syntax:

```
CALL GIC_3(SID, ALL, INTGOPTNS, REALOPTNS, CHAROPTNS, EJETOPTNS,
 FILEOPTNS, RPTOPTNS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID.
	ALL = 1	process all buses.
Integer INTGOPTNS (4)	array of 4 elements specifying Integer Options(input).	
	INTGOPTNS(1)	Number of levels of inter-tie buses to add to study subsystem (default alias is tielevels) (0 by default)
	INTGOPTNS(1) = 0	consider only study subsystem buses, no buses from inter-ties
	INTGOPTNS(1) = 0	consider only study subsystem buses, no buses from inter-ties
	INTGOPTNS(2)	Year number to scale benchmark event GMD storm. These scaling factors account in the influence of geomagnetic latitude and earth model on the estimated geoelectric field magnitude and are provided in NERC TPL-007-1. (default alias is study_year) (0 by default)
	INTGOPTNS(2) = 0	current year, used in benchmark geoelectric field calculations only.
	INTGOPTNS(2) = 0	specific year

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	INTGOPTNS(3)	Subsystem sid for local GMD hot spots (default alias is sid_local) (0 by default)
	INTGOPTNS(3) = 0	No local GMD hot spots
	INTGOPTNS(3) = 0	subsystem number (1 - 11). Buses and branches in this subsystem will have Efield due to local GMD hot spots.
	INTGOPTNS(4)	Option for Transformer Thermal Analysis (not implemented yet) (default alias is thermal_ana_optn) (0 by default)
	INTGOPTNS(4) = -1	perform on all transformers
	INTGOPTNS(4) = 0	do not perform
	INTGOPTNS(4) = 0	perform on these many top transformers ordered by effective GIC flow
Real REALOPTNS (7)	array of 7 elements specifying Real Options(input).	
	REALOPTNS(1)	electric field magnitude in units defined by charoptns(2), not used when charoptns(1)=nonuniform (default alias is efield_mag) (0 by default)
	REALOPTNS(1) = 0	when charoptns(1)=uniform, no default allowed
	REALOPTNS(1) = 0	when charoptns(1)=benchmark, when specified as 0.0, it will be set to 8.0 V/km (default benchmark event strength)
	REALOPTNS(1) = 0	for charoptns(1)=benchmark, used as specified
	REALOPTNS(2)	electric field direction in degrees, range 0 to 360 degrees (default alias is efield_deg) (0 by default)
	REALOPTNS(2) = 0	when charoptns(1)=uniform or benchmark, not used when charoptns(1)=nonuniform
	REALOPTNS(3)	substation grounding dc resistance in ohms (default alias is substation_r) (0.1 by default)
	REALOPTNS(3) = 0	Substation is ungrounded
	REALOPTNS(3) = 0	Use specified grounding dc resistance
	REALOPTNS(4)	transmission line X/R ratio, must be >0, used to calculate branch DC resistance if R=0.0 in network data (default alias is branch_xbyr) (30 by default)
	REALOPTNS(4) = 0	Use default
	REALOPTNS(4) = 0	Use specified
	REALOPTNS(5)	transformer winding X/R ratio, must be >0, used to calculate winding DC resistance if R=0.0 in network data (default alias is transformer_xbyr) (30 by default)
	REALOPTNS(5) = 0	Use default
	REALOPTNS(5) = 0	Use specified

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	REALOPTNS(6)	local GMD hot spots electric field magnitude in units defined by charoptns(2), not used when charoptns(1)=nonuniform (default alias is efield_mag_local) (realoptns(1) by default)
	REALOPTNS(6) = 0	when charoptns(1)=uniform or benchmark, not used when charoptns(1)=nonuniform
	REALOPTNS(7)	local GMD hot spots electric field direction in degrees, range 0 to 360 degrees (default alias is efield_deg_local) (realoptns(2) by default)
	REALOPTNS(7) = 0	when charoptns(1)=uniform or benchmark, not used when charoptns(1)=nonuniform
Character*12 CHAROPTNS (7)	array of 7 elements specifying Character Options(input).	
	CHAROPTNS(1)	Electric Field Type (default alias is efield_type) (uniform by default)
	CHAROPTNS(1) = uniform	for uniform geoelectric field
	CHAROPTNS(1) = benchmark	for benchmark event geoelectric field
	CHAROPTNS(1) = nonuniform	for nonuniform geoelectric field
	CHAROPTNS(2)	Units of Electric Field Magnitude (default alias is efield_unit) (v/km by default)
	CHAROPTNS(2) = v/km	for volts/km
	CHAROPTNS(2) = v/mi	for volts/mile
	CHAROPTNS(3)	Option to add GIC updates to base case (default alias is addfile_optn) (rdch by default)
	CHAROPTNS(3) = sav	add updates to Saved Case
	CHAROPTNS(3) = rdch	create RDCH raw file of GIC updates
	CHAROPTNS(4)	Option to select method for GIC to Mvar Calculation (default alias is gic2mvar_optn) (kfactors by default)
	CHAROPTNS(4) = kfactors	for GIC to Mvar scaling factors from GIC data file when provided, otherwise default scaling factors
	CHAROPTNS(4) = capchar	for GIC to Mvar scaling factors from GIC data file when provided, otherwise scaling factors from default transformer Mvar and GIC characteristics
	CHAROPTNS(5)	Earth Model Name. A Standard or User defined model name must be provided when Benchmark Event or Non-uniform electric field is to be modeled or Transformer Thermal Analysis is to be performed. (default alias is earth_model_name) (by default)

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	CHAROPTNS(6) Option to scan storm event scenarios (default alias is scan_storm_event) (by default)
	CHAROPTNS(6) = no storm orientation and/or magnitude scan
	CHAROPTNS(6) = scan_deg perform storm orientation scan that give maximum Var losses when charoptns(1) is uniform or benchmark, create report for only orientation that gives maximum VAR losses
	CHAROPTNS(6) = scan_deg_rpt perform storm orientation scan that give maximum Var losses when charoptns(1) is uniform or benchmark, create report for each orientation scanned
	CHAROPTNS(6) = scan_mag perform storm electric field magnitude that yield converged power flow solution when charoptns(1) is uniform or benchmark and charoptns(7) specifies PF method. It uses orienation specified and creates report for only final electric field magnitude.
	CHAROPTNS(6) = scan_mag_rpt perform storm electric field magnitude that yield converged power flow solution when charoptns(1) is uniform or benchmark and charoptns(7) specifies PF method. It uses orienation specified and creates report for each scanned electric field magnitude.
	CHAROPTNS(6) = scan_d_m perform storm orientation scan that give maximum Var losses when charoptns(1) is uniform or benchmark, then perform magnitude scans tiil they yield converged power flow solution, create report for only orientation that gives maximum VAR losses and final electric field magnitude
	CHAROPTNS(6) = scan_d_m_rpt perform storm orientation scan that give maximum Var losses when charoptns(1) is uniform or benchmark, then perform magnitude scans till they yield converged power flow solution, create report for all orientation scans and mangitude scans performed
	CHAROPTNS(7) Option to solve Power Flow with GIC losses added to the base case (default alias is power_flow_optn) (by default)
	CHAROPTNS(7) = do not perform power flow solution
	CHAROPTNS(7) = fdns use fixed slope decoupled Newton-Raphson method
	CHAROPTNS(7) = fnsl use Full Newton-Raphson method
	CHAROPTNS(7) = nsol use Decoupled Newton-Raphson method
Real EJETOPTNS (5)	array of 5 elements specifying Electrojet Characteristics, used only if charoptns(1)=nonuniform(input).

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	EJETOPTNS(1)	eletrojet current in million amperes, must be >0 (default alias is ejet_million_amps) (no default allowed)
	EJETOPTNS(2)	Cauchy distribution half-width in km, must be >0 (default alias is ejet_halfwidth_km) (no default allowed)
	EJETOPTNS(3)	period of variation in minutes, must be >0 (default alias is ejet_period_min) (no default allowed)
	EJETOPTNS(4)	height of current in km, must be >0 (default alias is ejet_height_km) (no default allowed)
	EJETOPTNS(5)	latitude of center of electrojet in degrees (default alias is ejet_center_deg) (no default allowed)
Character*260 FILEOPTNS (5)	array of 5 elements specifying File Used(input).	
	FILEOPTNS(1)	GIC data file name (input). (default alias is gicfile) (no default allowed)
	FILEOPTNS(2)	GIC updates to Base Case file name (output). (default alias is addfile) (by default)
	FILEOPTNS(2) =	no file created
	FILEOPTNS(3)	RDCH file to remove GIC updates from GIC updated case in working memory to set it back to Base Case network condition (output). (default alias is purgfile) (by default)
	FILEOPTNS(3) =	no file created
	FILEOPTNS(4)	GIC dc resistive network raw file. This represents the dc network used to calculate GIC flow. (output) (default alias is rnwkfile) (by default)
	FILEOPTNS(4) =	no file created
	FILEOPTNS(5)	GIC Results map data file for given Efield magnitude and degrees OR Efield magnitude and degrees scans which give maximum Var losses when scans are performed (output). This is used by GICMAPS to plot GIC results on network map. (default alias is pygicfile) (by default)
	FILEOPTNS(5) =	temporary file created
	FILEOPTNS(5) = nooutput	no file created
Integer RPTOPTNS (7)	array of 7 elements specifying Report Options(input).	
	RPTOPTNS(1)	what to report (default alias is rptoptn) (-1 by default)
	RPTOPTNS(1) = -1	report all
	RPTOPTNS(1) = 0	no report

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	RPTOPTNS(1) = 0	Report these many maximum values in each category
	RPTOPTNS(2)	report induced branch voltages (default alias is rptbrn_indv) (0 by default)
	RPTOPTNS(2) = 0	No
	RPTOPTNS(2) = 1	Yes
	RPTOPTNS(3)	report DC bus voltages (default alias is rptdc_busv) (0 by default)
	RPTOPTNS(3) = 0	No
	RPTOPTNS(3) = 1	Yes
	RPTOPTNS(4)	report branch GIC flows (default alias is rptbrn_gic) (0 by default)
	RPTOPTNS(4) = 0	No
	RPTOPTNS(4) = 1	Yes
	RPTOPTNS(5)	report transformer GIC flows (default alias is rpttrn_gic) (1 by default)
	RPTOPTNS(5) = 0	No
	RPTOPTNS(5) = 1	Yes
	RPTOPTNS(6)	report substation GIC flows (default alias is rptstn_gic) (0 by default)
	RPTOPTNS(6) = 0	No
	RPTOPTNS(6) = 1	Yes
	RPTOPTNS(7)	report transformer losses (default alias is rpttrn_q) (1 by default)
	RPTOPTNS(7) = 0	No
	RPTOPTNS(7) = 1	Yes
Integer IERR	error code (output).	
	IERR = 1	DLL not loaded or entry point not found.

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1.96. GIC_MVARLOSS_SCALING_FACTORS

Specify KFACTORS to determine transformer reactive power losses due to geomagnetic currents (GICs) flow.

This API routine was first introduced in release 32.2.1.

Batch command syntax:

```
BAT_GIC_MVARLOSS_SCALING_FACTORS  KFACTORS(1)..KFACTORS(9)
```

Python command syntax:

```
ierr = gic_mvarloss_scaling_factors(kfactors)
```

Fortran command syntax:

```
CALL GIC_MVARLOSS_SCALING_FACTORS(KFACTORS, IERR)
```

Where:

Real KFACTORS (9)	array of elements specifying Kfactor values. The values are as follows.(input).	
	KFACTORS(1)	Voltage level at which these kfactors are specified
	KFACTORS(2)	kfactor value for Shell form transformers
	KFACTORS(3)	kfactor value for core design - Single Phase (three separate cores) transformers
	KFACTORS(4)	kfactor value for core design - Three Phase 3 legged core form transformers
	KFACTORS(5)	kfactor value for core design - Three Phase 5 legged core form transformers
	KFACTORS(6)	kfactor value for core design - Three Phase 7 legged core form transformers
	KFACTORS(7)	kfactor value for unknown core, voltage level of <=200 kV
	KFACTORS(8)	kfactor value for unknown core, voltage level of <200 kV and <=400 kV
	KFACTORS(9)	kfactor value for unknown core, voltage level of >400 kV
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	DLL not loaded or entry point not found.

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1.97. GNESTATUS

Use this API to specify or return the status of a GNE device.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_GNESTATUS  'NAME'  IVAL
```

Python command syntax:

```
ierr = gnestatus(name, ival)

ierr, ival = gnestatus(name)
```

Fortran command syntax:

```
CALL GNESTATUS(NAME, IVAL, IOCODE, IERR)
```

Where:

Character*12 NAME	name of the GNE device for which the status is to be changed (input; no default allowed).	
Integer IVAL	value of the status of the specified GNE device (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, no default allowed; else output).	
	IVAL = 0	out-of-service.
	IVAL = 1	in-service.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the status to the value specified in IVAL.
	IOCODE = 1	return the current value of the status in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	GNE device not found
	IERR = 2	invalid GNE status value
	IERR = 3	invalid IOCODE value
	IERR = 4	prerequisite requirements for API are not met

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1.98. GNET

Use this API to change in-service generation to negative MVA load at all Type 2 and 3 buses in the subsystem specified by the user.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_GNET  SID  ALL
```

Python command syntax:

```
ierr = gnet(sid, all)
```

Fortran command syntax:

```
CALL GNETNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.99. IECS

This API routine is obsolete. It has been replaced by the API routine iecs_2 .

Use this API to apply a series of faults at various locations in the working case to compute short circuit currents according to IEC 60909 standard.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_IECS  SID ALL STATUS(1)..STATUS(8) BRKTIM IECFILE FCDFILE
```

Python command syntax:

```
ierr = iecs(sid, all, status, brktim, iecfile, fcdfile)
```

Fortran command syntax:

```
CALL IECSAPI(SID, ALL, STATUS, BRKTIM, IECFILE, FCDFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (8)	array of 8 elements specifying integer options(input).	
	STATUS(1)	three phase faults (default alias is flt3ph) (0 by default)
	STATUS(1) = 0	omit
	STATUS(1) = 1	include
	STATUS(2)	line-to-ground faults (default alias is fltlg) (0 by default)
	STATUS(2) = 0	omit
	STATUS(2) = 1	include
	STATUS(3)	line-line-to-ground faults (default alias is fltllg) (0 by default)
	STATUS(3) = 0	omit
	STATUS(3) = 1	include
	STATUS(4)	line-to-line faults (default alias is fltll) (0 by default)
	STATUS(4) = 0	omit
	STATUS(4) = 1	include
	STATUS(5)	report option (default alias is rptop) (1 by default)
	STATUS(5) = 1	total fault currents

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STATUS(5) = 2	fault contributions to N levels away
STATUS(5) = 3	total fault currents and fault contributions to N levels away
STATUS(6)	number of levels back for contributions (default alias is rptlvl) (0 by default)
STATUS(6) = 0	used if STATUS(5) = 2 or 3
STATUS(7)	location of fault (default alias is fltloc) (0 by default)
STATUS(7) = 1	fault at network bus
STATUS(7) = 2	fault at LV bus of Power Station Unit (PSU)
STATUS(7) = 3	fault at auxiliary transformer (connected to PSU) LV bus
STATUS(8)	report total fault currents in descending order of magnitude (AMPS) (0 by default)
STATUS(8) = 0	do not order
STATUS(8) = 1	descending order

Real BRKTIM	breaker contact parting time in seconds (input; 5/BaseFrequency).
Character*260 IECFILE	IEC Fault Calculation Data (.iec) file (input; blank by default).
Character*260 FCDFILE	Fault Control (.fcd) file (input; blank by default).
Integer IERR	error code (output).

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1.100. IECS_2

This API routine is obsolete. It has been replaced by the API routine iecs_3 .

This API executes the second release of the IEC short circuit calculations function. This API is used to apply a series of faults at various locations in the working case to compute short circuit currents according to IEC 60909 standard.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_IECS_2  SID ALL STATUS(1)..STATUS(14) BRKTIM IECFILE FCDFILE SCFILE
```

Python command syntax:

```
ierr = iecs_2(sid, all, status, brktim, iecfile, fcdfile, scfile)
```

Fortran command syntax:

```
CALL IECSAPI2(SID, ALL, STATUS, BRKTIM, IECFILE, FCDFILE, SCFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (14)	array of 14 elements specifying integer options(input).	
	STATUS(1)	three phase faults (default alias is flt3ph) (0 by default)
	STATUS(1) = 0	omit
	STATUS(1) = 1	include
	STATUS(2)	line-to-ground faults (default alias is fltlg) (0 by default)
	STATUS(2) = 0	omit
	STATUS(2) = 1	include
	STATUS(3)	line-line-to-ground faults (default alias is fltllg) (0 by default)
	STATUS(3) = 0	omit
	STATUS(3) = 1	include
	STATUS(4)	line-to-line faults (default alias is fltll) (0 by default)
	STATUS(4) = 0	omit
	STATUS(4) = 1	include
	STATUS(5)	report option (default alias is rptop) (1 by default)
	STATUS(5) = 1	total fault currents

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STATUS(5) = 2	fault contributions to N levels away
STATUS(5) = 3	total fault currents and fault contributions to N levels away
STATUS(6)	number of levels back for contributions (default alias is rptlvl) (0 by default)
STATUS(6) = 0	used if STATUS(5) = 2 or 3
STATUS(7)	location of fault (default alias is fltloc) (1 by default)
STATUS(7) = 1	fault at network bus
STATUS(7) = 2	fault at LV bus of Power Station Unit (PSU)
STATUS(7) = 3	fault at auxiliary transformer (connected to PSU) LV bus
STATUS(8)	line-out faults (default alias is linout) (0 by default)
STATUS(8) = 0	omit
STATUS(8) = 1	include
STATUS(9)	line-end faults (default alias is linend) (0 by default)
STATUS(9) = 0	omit
STATUS(9) = 1	include
STATUS(10)	transformer tap ratios and phase angles option (default alias is tpunty) (0 by default)
STATUS(10) = 0	leave tap ratios and phase angles unchanged
STATUS(10) = 1	set tap ratios 1.0 pu and phase angles to 0.0 deg
STATUS(11)	line charging option (default alias is lnchrg) (1 by default)
STATUS(11) = 1	set to 0.0 in positive and negative sequences
STATUS(11) = 2	set to 0.0 in all sequences
STATUS(11) = 3	leave line charging unchanged
STATUS(12)	loads, fixed and switched shunts and and magnetizing admittance option (default alias is shntop) (1 by default)
STATUS(12) = 1	set to 0.0 in the positive and negative sequences
STATUS(12) = 2	set to 0.0 in all sequences
STATUS(12) = 3	leave unchanged
STATUS(13)	dc line and FACTS device option (default alias is dcload) (0 by default)
STATUS(13) = 0	blocked
STATUS(13) = 1	represent as load
STATUS(14)	zero sequence transformer impedance correction option (default alias is zcorec) (0 by default)
STATUS(14) = 0	ignore

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	STATUS(14) = 1 apply
Real BRKTIM	breaker contact parting time in seconds (input; 5/BaseFrequency).
Character*260 IECFILE	IEC Fault Calculation Data (.iec) file (input; blank by default).
Character*260 FCDFILE	Fault Control (.fcd) file (input; blank by default).
Character*260 SCFILE	Results (.sc) file (input; blank by default). scfile = temporary file created scfile = nooutput no file created
Integer IERR	error code (output).

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1.101. IECS_3

This API routine is obsolete. It has been replaced by the API routine iecs_4 .

This API executes the third release of the IEC short circuit calculations function. This API is used to apply a series of faults at various locations in the working case to compute short circuit currents according to IEC 60909 standard.

This API routine was first introduced in release 32.0.5.

Batch command syntax:

```
BAT_IECS_3  SID ALL STATUS(1)..STATUS(15) VALUES(1) VALUES(2) IECFILE
 FCDFILE SCFILE
```

Python command syntax:

```
ierr = iecs_3(sid, all, status, values, iecfile, fcdfile, scfile)
```

Fortran command syntax:

```
CALL IECSAPI3(SID, ALL, STATUS, VALUES, IECFILE, FCDFILE, SCFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (15)	array of 15 elements specifying integer options(input).	
	STATUS(1)	three phase faults (default alias is flt3ph) (0 by default)
	STATUS(1) = 0	omit
	STATUS(1) = 1	include
	STATUS(2)	line-to-ground faults (default alias is fltlg) (0 by default)
	STATUS(2) = 0	omit
	STATUS(2) = 1	include
	STATUS(3)	line-line-to-ground faults (default alias is fltllg) (0 by default)
	STATUS(3) = 0	omit
	STATUS(3) = 1	include
	STATUS(4)	line-to-line faults (default alias is fltll) (0 by default)
	STATUS(4) = 0	omit
	STATUS(4) = 1	include
	STATUS(5)	report option (default alias is rptop) (1 by default)

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STATUS(5) = 1	total fault currents
STATUS(5) = 2	fault contributions to N levels away
STATUS(5) = 3	total fault currents and fault contributions to N levels away
STATUS(6)	number of levels back for contributions (default alias is rptlvl) (0 by default)
STATUS(6) = 0	used if STATUS(5) = 2 or 3
STATUS(7)	location of fault (default alias is fltloc) (1 by default)
STATUS(7) = 1	fault at network bus
STATUS(7) = 2	fault at LV bus of Power Station Unit (PSU)
STATUS(7) = 3	fault at auxiliary transformer (connected to PSU) LV bus
STATUS(8)	line-out faults (default alias is linout) (0 by default)
STATUS(8) = 0	omit
STATUS(8) = 1	include
STATUS(9)	line-end faults (default alias is linend) (0 by default)
STATUS(9) = 0	omit
STATUS(9) = 1	include
STATUS(10)	transformer tap ratios and phase angles option (default alias is tpunty) (0 by default)
STATUS(10) = 0	leave tap ratios and phase angles unchanged
STATUS(10) = 1	set tap ratios 1.0 pu and phase angles to 0.0 deg
STATUS(11)	line charging option (default alias is lnchrg) (1 by default)
STATUS(11) = 1	set to 0.0 in positive and negative sequences
STATUS(11) = 2	set to 0.0 in all sequences
STATUS(11) = 3	leave line charging unchanged
STATUS(12)	loads, fixed and switched shunts and and magnetizing admittance option (default alias is shntop) (1 by default)
STATUS(12) = 1	set to 0.0 in the positive and negative sequences
STATUS(12) = 2	set to 0.0 in all sequences
STATUS(12) = 3	leave unchanged
STATUS(13)	dc line and FACTS device option (default alias is dcload) (0 by default)
STATUS(13) = 0	blocked
STATUS(13) = 1	represent as load
STATUS(14)	zero sequence transformer impedance correction option (default alias is zcorec) (0 by default)

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	STATUS(14) = 0	ignore
	STATUS(14) = 1	apply
	STATUS(15)	option for Voltage Factor C (default alias is cfactor) (1 by default)
	STATUS(15) = 1	Voltage Factor C for maximum fault currents (default)
	STATUS(15) = 2	Voltage Factor C for minimum fault currents
	STATUS(15) = 3	Voltage Factor C as specified by 'vfactorc' value
Real VALUES (2)	array of 2 elements specifying real values(input).	
	VALUES(1)	breaker contact parting time in seconds (default alias is brktim) (5/base Frequency by default)
	VALUES(2)	user specified voltage factor c (default alias is ucfactor) (1.0 by default) This is used only when STATUS(15)=3)
Character*260 IECFILE	IEC Fault Calculation Data (.iec) file (input; blank by default).	
Character*260 FCDFILE	Fault Control (.fcd) file (input; blank by default).	
Character*260 SCFILE	Results (.sc) file (input; blank by default).	
	scfile =	temporary file created
	scfile = nooutput	no file created
Integer IERR	error code (output).	

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1.102. IECS_4

This API executes the fourth release of the IEC short circuit calculations function. This API is used to apply a series of faults at various locations in the working case to compute short circuit currents according to IEC 60909 standard.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_IECS_4  SID ALL STATUS(1)..STATUS(17) VALUES(1) VALUES(2) IECFILE
 FCDFILE SCFILE
```

Python command syntax:

```
ierr = iecs_4(sid, all, status, values, iecfile, fcdfile, scfile)
```

Fortran command syntax:

```
CALL IECSAPI4(SID, ALL, STATUS, VALUES, IECFILE, FCDFILE, SCFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (17)	array of 17 elements specifying integer options(input).	
	STATUS(1)	three phase faults (default alias is flt3ph) (0 by default)
	STATUS(1) = 0	omit
	STATUS(1) = 1	include
	STATUS(2)	line-to-ground faults (default alias is fltlg) (0 by default)
	STATUS(2) = 0	omit
	STATUS(2) = 1	include
	STATUS(3)	line-line-to-ground faults (default alias is fltllg) (0 by default)
	STATUS(3) = 0	omit
	STATUS(3) = 1	include
	STATUS(4)	line-to-line faults (default alias is fltll) (0 by default)
	STATUS(4) = 0	omit
	STATUS(4) = 1	include
	STATUS(5)	report option (default alias is rptop) (1 by default)
	STATUS(5) = 0	fault current summary table
	STATUS(5) = 1	total fault currents

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STATUS(5) = 2	fault contributions to N levels away
STATUS(5) = 3	total fault currents and fault contributions to N levels away
STATUS(6)	number of levels back for contributions (default alias is rptlvl) (0 by default)
STATUS(6) = 0	used if STATUS(5) = 2 or 3
STATUS(7)	location of fault (default alias is fltloc) (0 by default)
STATUS(7) = 0	fault at network bus
STATUS(7) = 1	fault at LV bus of Power Station Unit (PSU)
STATUS(7) = 2	fault at auxiliary transformer (connected to PSU) LV bus
STATUS(8)	line-out faults (default alias is linout) (0 by default)
STATUS(8) = 0	omit
STATUS(8) = 1	include
STATUS(9)	line-end faults (default alias is linend) (0 by default)
STATUS(9) = 0	omit
STATUS(9) = 1	include
STATUS(10)	transformer tap ratios and phase shift angles option (default alias is tpunty) (0 by default)
STATUS(10) = 0	leave tap ratios and phase shift angles unchanged
STATUS(10) = 1	set tap ratios to 1.0 pu and phase shift angles to 0 deg
STATUS(10) = 2	set tap ratios to 1.0 pu and phase shift angles unchanged
STATUS(10) = 3	set tap ratios unchanged and phase shift angles to 0 deg
STATUS(11)	line charging option (default alias is lnchrg) (1 by default)
STATUS(11) = 0	leave unchanged
STATUS(11) = 1	set to 0.0 in the positive and negative sequences
STATUS(11) = 2	set to 0.0 in all sequences
STATUS(12)	line, fixed and switched shunts, and magnetizing admittance option (default alias is shntop) (1 by default)
STATUS(12) = 0	leave unchanged
STATUS(12) = 1	set to 0.0 in the positive and negative sequences
STATUS(12) = 2	set to 0.0 in all sequences
STATUS(13)	dc line and FACTS device option (default alias is dcload) (0 by default)
STATUS(13) = 0	blocked

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	STATUS(13) = 1	represent as load
	STATUS(14)	zero sequence transformer impedance correction option (default alias is zcorec) (0 by default)
	STATUS(14) = 0	ignore
	STATUS(14) = 1	apply
	STATUS(15)	option for Voltage Factor C (default alias is cfactor) (0 by default)
	STATUS(15) = 0	Voltage Factor C for maximum fault currents
	STATUS(15) = 1	Voltage Factor C for minimum fault currents
	STATUS(15) = 2	Voltage Factor C as specified by 'vfactorc' value for maximum fault current calculations
	STATUS(15) = 3	Voltage Factor C as specified by 'vfactorc' value for minimum fault current calculations
	STATUS(16)	load option (default alias is loadop) (1 by default)
	STATUS(16) = 0	leave unchanged
	STATUS(16) = 1	set to 0.0 in the positive and negative sequences
	STATUS(16) = 2	set to 0.0 in all sequences
	STATUS(17)	fault analysis generator reactance option; ignored if generators are already "converted" (default alias is genxop) (0 by default)
	STATUS(17) = 0	use subtransient reactance
	STATUS(17) = 1	use transient reactance
	STATUS(17) = 2	use synchronous reactance
Real VALUES (2)	array of 2 elements specifying real values(input).	
	VALUES(1)	breaker contact parting time in seconds (default alias is brktim) (5/base Frequency by default)
	VALUES(2)	user specified voltage factor c (default alias is ucfactor) (1.0 by default) This is used only when STATUS(15)=2)
Character*260 IECFILE	IEC Fault Calculation Data (.iec) file (input; blank by default).	
Character*260 FCDFILE	Fault Control (.fcd) file (input; blank by default).	
Character*260 SCFILE	Results (.sc) file (input; blank by default).	
	scfile =	temporary file created
	scfile = nooutput	no file created
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value

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IERR = 3	invalid STATUS value
IERR = 4	invalid VALUES value
IERR = 5	no faults specified
IERR = 6	error opening IECFILE
IERR = 7	error opening FCDFILE
IERR = 8	prerequisite requirements for API are not met
IERR = 9	error opening SCFILE
IERR = 10	SCFILE error
IERR = 11	bus SC results not available
IERR = 12	branch SC results not available
IERR = 13	three winding transformer SC results not available
IERR = 14	incorrect SCFILE
IERR = 15	SCFILE not allowed

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1.103. IECS_SCFILE

This API reads IECS short circuit results from SCFILE and produces same IECS output report that was generated when SCFILE was created. SCFILE is read into PSSE memory to enable displaying IECS results onto slider diagrams. For this API to work properly, it is required that working case in PSSE memory should be the same Saved case that was used when creating the SCFILE.

This API routine was first introduced in release 32.0.3.

Batch command syntax:

```
BAT_IECS_SCFILE  SCFILE
```

Python command syntax:

```
ierr = iecs_scfile(scfile)
```

Fortran command syntax:

```
CALL IECS_SCFILE(SCFILE,IERR)
```

Where:

Character*260 SCFILE	IEC Results (.sc) file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 7	prerequisite requirements for function not met
	IERR = 9	error opening SCFILE
	IERR = 10	SCFILE error
	IERR = 11	bus SC results not available
	IERR = 12	branch SC results not available
	IERR = 13	three winding transformer SC results not available
	IERR = 14	incorrect SCFILE
	IERR = 15	SCFILE not allowed

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1.104. IMNET

Use this API routine to change the status of induction machines that are in-service to out-of-service, and to replace the power flowing from the network into the machine with constant power load. The routine can be directed to process either all in-service induction machines, all in-service induction machines within a specified subsystem, or all in-service induction machines outside of a specified subsystem.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_IMNET  SID  ALL  INOUT
```

Python command syntax:

```
ierr = imnet(sid, all, inout)
```

Fortran command syntax:

```
CALL IMNET(SID, ALL, INOUT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only induction machines within, or outside of, subsystem SID
	ALL = 1	process all induction machines
Integer INOUT	include or exclude induction machines in subsystem SID (input; 0 by default).	
	INOUT = 0	process only machines inside of subsystem SID
	INOUT = 1	process only machines outside of subsystem SID
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid INOUT value
	IERR = 4	prerequisite requirements for API are not met

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1.105. IMOUTAGE

Use this API routine to change the status of induction machines that are in-service, but that have been set to the "stalled" (for motors) or "tripped" (for generators) state by the power flow solution, to out-of-service.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_IMOUTAGE  SID  ALL  MOTOPT  GENOPT
```

Python command syntax:

```
ierr = imoutage(sid, all, motopt, genopt)
```

Fortran command syntax:

```
CALL IMOUTAGE(SID, ALL, MOTOPT, GENOPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only induction machines in subsystem SID
	ALL = 1	process all induction machines
Integer MOTOPT	"stalled" induction motor option (input; 0 by default).	
	MOTOPT = 0	set to out-of-service
	MOTOPT = 1	leave in-service
Integer GENOPT	"tripped" induction generator option (input; 0 by default).	
	GENOPT = 0	set to out-of-service
	GENOPT = 1	leave in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid MOTOPT value
	IERR = 4	invalid GENOPT value
	IERR = 5	prerequisite requirements for API are not met

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1.106. IMPC

Use this API to calculate the impact of transaction events on MW flows using a linear network (dc) model.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_IMPC  EVENT DFXFILE ATCFILE
```

Python command syntax:

```
ierr = impc(event, dfxfile, atcfile)
```

Fortran command syntax:

```
CALL IMPCAPI(EVENT, DFXFILE, ATCFILE, IERR)
```

Where:

Integer EVENT	transaction event number (input; no default allowed).	
Character*260 DFXFILE	name of DFAX data file (input; no default allowed).	
Character*260 ATCFILE	name of ATC updates file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no transactions defined
	IERR = 2	transaction event not defined
	IERR = 3	no name of DFAX data file specified
	IERR = 4	error opening ATC updates file
	IERR = 5	error opening file DFXFILE
	IERR = 6	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 8	no monitored branches or interfaces defined
	IERR = 9	prerequisite requirements for API are not met

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1.107. IMPLEMENT_TRANSFER

This API routine is obsolete. It has been replaced by the API routine implement_transfer_2 .

Use this API to apply a specified transfer using the same transfer dispatch methods that are available in the PV analysis calculation engine.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_IMPLEMENT_TRANSFER  OPTIONS(1)..OPTIONS(4)  RVAL  LABELS(1)  LABELS(2)
  DFXFILE  ECDFILE
```

Python command syntax:

```
ierr = implement_transfer(options, rval, labels, dfxfile, ecdfile)
```

Fortran command syntax:

```
CALL IMPLEMENT_TRANSFER(OPTIONS, RVAL, LABELS, DFXFILE, ECDFILE, IERR)
```

Where:

Integer OPTIONS (4)	array of 4 elements specifying transfer implementation options(input).	
	OPTIONS(1)	study (source) system transfer dispatch method (1 by default)
	OPTIONS(1) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
	OPTIONS(1) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
	OPTIONS(1) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(1) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(1) = 5	subsystem buses with positive MW machines in proportion to their MW output
	OPTIONS(1) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
	OPTIONS(1) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC)
	OPTIONS(1) = 8	subsystem buses with positive MW machines via ECDI
	OPTIONS(2)	opposing (sink) system transfer dispatch method (1 by default)
	OPTIONS(2) = 1	buses and distribution factors from DFAX file for buses with positive MW machines

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	OPTIONS(2) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
	OPTIONS(2) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(2) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(2) = 5	subsystem buses with positive MW machines in proportion to their MW output
	OPTIONS(2) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
	OPTIONS(2) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC)
	OPTIONS(2) = 8	subsystem buses with positive MW machines via ECDI
	OPTIONS(3)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
	OPTIONS(3) = 0	no limits
	OPTIONS(3) = 1	honor machine active power limits
	OPTIONS(4)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
	OPTIONS(4) = 0	no limits
	OPTIONS(4) = 1	enforce non-negative net active power constant MVA load
Real RVAL	transfer increment in MW (input; no default allowed).	
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which load is to be decreased (no default allowed)
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for which load is to be increased (no default allowed)
Character*260 DFXFILE	name of the DFAX data file (input; no default allowed).	
Character*260 ECDFILE	name of the Economic Dispatch Data file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid RVAL value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted

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IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	no DFAX data file specified
IERR = 6	no economic dispatch data file specified
IERR = 7	no participating dispatch buses in study system
IERR = 8	no participating dispatch buses in opposing system
IERR = 9	subsystem engine error
IERR = 10	unable to implement transfer in study system
IERR = 11	unable to implement transfer in opposing system
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 52	error opening DFAX data file
IERR = 54	prerequisite requirements for API are not met

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1.108. IMPLEMENT_TRANSFER_2

Use this API routine to apply a specified transfer using the same transfer dispatch methods that are available in the PV analysis calculation engine.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_IMPLEMENT_TRANSFER_2  OPTIONS(1)..OPTIONS(4) RVAL(1)..RVAL(2)
 LABELS(1)..LABELS(2) DFXFILE ECDFILE
```

Python command syntax:

```
ierr = implement_transfer_2(options, rval, labels, dfxfile, ecdfile)
```

Fortran command syntax:

```
CALL IMPLEMENT_TRANSFER_2(OPTIONS, RVAL, LABELS, DFXFILE, ECDFILE, IERR)
```

Where:

Integer OPTIONS (4)	array of elements specifying transfer implementation options (input). The values are as follows:(input).	
	OPTIONS(1)	study (source) system transfer dispatch method (1 by default)
	OPTIONS(1) = 1	buses and distribution factors from DFAX file for buses with positive MW machines.
	OPTIONS(1) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
	OPTIONS(1) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(1) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(1) = 5	subsystem buses with positive MW machines in proportion to their MW output.
	OPTIONS(1) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
	OPTIONS(1) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC).
	OPTIONS(1) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
	OPTIONS(1) = 9	subsystem buses with positive MW machines via ECDI with unit commitment enabled
	OPTIONS(2)	opposing (sink) system transfer dispatch method (1 by default)
	OPTIONS(2) = 1	buses and distribution factors from DFAX file for buses with positive MW machines.

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	OPTIONS(2) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
	OPTIONS(2) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(2) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(2) = 5	subsystem buses with positive MW machines in proportion to their MW output.
	OPTIONS(2) = 6	subsystem buses with positive MW machines in proportion to their MBASEs.
	OPTIONS(2) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC).
	OPTIONS(2) = 8	subsystem buses with positive MW machines via ECDI. with unit commitment disabled
	OPTIONS(2) = 9	subsystem buses with positive MW machines via ECDI with unit commitment enabled
	OPTIONS(3)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
	OPTIONS(3) = 0	no limits.
	OPTIONS(3) = 1	honor machine active power limits.
	OPTIONS(4)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
	OPTIONS(4) = 0	no limits.
	OPTIONS(4) = 1	enforce non-negative net active power constant MVA load.
Real RVAL (2)	array of elements specifying transfer implementation options. The values are as follows:(input).	
	RVAL(1)	the transfer increment in MW
	RVAL(2)	power factor for load increases in dispatch methods 2, 3 and 4 (0.0 by default). Specify RVAL(2) as < 0.0 to retain the original power factor at each load that is changed.
Character*32 LABELS (2)	array of elements specifying subsystem labels (input). The values are as follows:(input).	
	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which load is to be decreased
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for which load is to be increased
Character*260 DFXFILE	Is the name of the Distribution Factor Data file (input; no default allowed).	

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Character*260 ECDFILE	Is the name of the Economic Dispatch Data file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred.
	IERR = 1	invalid RVAL value.
	IERR = 2	invalid OPTIONS value.
	IERR = 3	generators are converted.
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	no Distribution Factor Data File specified.
	IERR = 6	no Economic Dispatch Data File specified.
	IERR = 7	no participating dispatch buses in study system.
	IERR = 8	no participating dispatch buses in opposing system
	IERR = 9	subsystem engine error.
	IERR = 10	Unable to implement transfer in study system.
	IERR = 11	Unable to implement transfer in opposing system
	IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 23	no subsystems defined.
	IERR = 24	subsystem not found.
	IERR = 25	specified subsystems overlap.
	IERR = 26	no in-service buses in subsystem WORLD.
	IERR = 50	error opening Economic Dispatch Data File.
	IERR = 52	error opening Distribution Factor Data File.
	IERR = 54	prerequisite requirements for API are not met.

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1.109. INLF

This API routine is obsolete. It has been replaced by the API routine inlf_2 .

Use this API to apply an inertial or governor response power flow calculation.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_INLF  OPTIONS(1)..OPTIONS(7)  IFILE
```

Python command syntax:

```
ierr = inlf(options, ifile)
```

Fortran command syntax:

```
CALL INLFAPI(OPTIONS, IFILE, IERR)
```

Where:

Integer OPTIONS (7)	array of 7 elements specifying solution options(input).	
	OPTIONS(1)	solution type (0 by default)
	OPTIONS(1) = 0	inertial
	OPTIONS(1) = 1	governor response
	OPTIONS(2)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(2) = -1	solution type default (inertial => disable; governor => stepping)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable stepping adjustment
	OPTIONS(2) = 2	enable direct adjustment
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = -1	solution type default (inertial => disable; governor => enable)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = -1	solution type default (inertial => disable; governor => enable)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = -1	solution type default (inertial => enable; governor => enable)

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	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	generator var limit flag (-1 (inertial) or 99 (governor) by default)
	OPTIONS(6) = 0	apply var limits immediately
	OPTIONS(6) = 0	apply var limits on iteration n (or sooner if mismatch gets small)
	OPTIONS(6) = -1	ignore var limits
	OPTIONS(7)	missing active power limits flag (only used in governor response power flow) (0 by default)
	OPTIONS(7) = 0	pre-PSSE-29 approach (0.0 to 1.0 pu for machines with no Unit Inertia and Governor Data File record read; working case values for machines with PMAX defaulted on ECDI data record)
	OPTIONS(7) = 1	0.0 to 1.0 pu for both categories of machines
	OPTIONS(7) = 2	working case values for both categories of machines
Character*260 IFILE	name of unit inertia and governor data file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	generators are converted
	IERR = 3	bus type code and series element status inconsistencies
	IERR = 4	error opening IFILE
	IERR = 5	prerequisite requirements for API are not met

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1.110. INLF_2

Use this API routine to run the second release of the inertial and governor response power flow calculation.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_INLF_2  OPTIONS(1)..OPTIONS(8)  IFILE
```

Python command syntax:

```
ierr = inlf_2(options, ifile)
```

Fortran command syntax:

```
CALL INLF_2(OPTIONS, IFILE, IERR)
```

Where:

Integer OPTIONS (8)	array of 8 elements specifying solution options(input).	
	OPTIONS(1)	solution type (0 by default)
	OPTIONS(1) = 0	inertial
	OPTIONS(1) = 1	governor response
	OPTIONS(2)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(2) = -1	solution type default (inertial => disable; governor => stepping)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable stepping adjustment
	OPTIONS(2) = 2	enable direct adjustment
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = -1	solution type default (inertial => disable; governor => enable)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = -1	solution type default (inertial => disable; governor => enable)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = -1	solution type default (inertial => enable; governor => enable)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable

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	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	generator var limit flag (-1 (inertial) or 99 (governor) by default)
	OPTIONS(6) = 0	apply var limits immediately
	OPTIONS(6) = 0	apply var limits on iteration n (or sooner if mismatch gets small)
	OPTIONS(6) = -1	ignore var limits
	OPTIONS(7)	induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
	OPTIONS(7) = 0	stall
	OPTIONS(7) = 1	trip
	OPTIONS(8)	missing active power limits flag (only used in governor response power flow) (0 by default)
	OPTIONS(8) = 0	pre-PSSE-29 approach (0.0 to 1.0 pu for machines with no Unit Inertia and Governor Data File record read; working case values for machines with PMAX defaulted on ECDI data record)
	OPTIONS(8) = 1	0.0 to 1.0 pu for both categories of machines
	OPTIONS(8) = 2	working case values for both categories of machines
Character*260 IFILE	name of unit inertia and governor data file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	generators are converted
	IERR = 3	bus type code and series element status inconsistencies
	IERR = 4	error opening IFILE
	IERR = 5	prerequisite requirements for API are not met

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1.111. INTA

Use this API to summarize tie flows between an interchange area and all other areas in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_INTA  SID  ALL
```

Python command syntax:

```
ierr = inta(sid, all)
```

Fortran command syntax:

```
CALL INTANW(SID, ALL, IERR)
```

Where:

Integer SID	a valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Area subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all areas or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only areas in area subsystem SID
	ALL = 1	process all areas
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.112. INTZ

Use this API to summarize tie flows between each zone and all other zones in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_INTZ  SID  ALL
```

Python command syntax:

```
ierr = intz(sid, all)
```

Fortran command syntax:

```
CALL INTZNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Zone subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all zones or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only zones in zone subsystem SID
	ALL = 1	process all zones
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.113. JOIN

The API combines two buses into a single bus.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_JOIN    IBUS    JBUS    STATUS
```

Python command syntax:

```
ierr = join(ibus, jbus, status)

ierr, idtrns = join(ibus, jbus, status, nid)
```

Fortran command syntax:

```
CALL JOINAPI(IBUS, JBUS, STATUS, NID, IDTRNS, IERR)
```

Where:

Integer IBUS	retained bus number (input).	
Integer JBUS	eliminated bus number (input).	
Integer STATUS	line shunt option (input; 1 by default).	
	STATUS = 0	ignore line shunts of deleted branches
	STATUS = 1	add line shunts of deleted branches to shunt of retained bus
Integer NID	number of elements in the IDTRNS array; 0 to suppress setting IDTRNS; 0 by default	
Character*18 IDTRNS (NID)	returned as the ID translation originally connected to bus JBUS. IDTRNS contents: Columns Content 1-2 LD for load. BR for non-transformer branch. IM for induction machine MC for machine. SH for fixed bus shunt T2 for two-winding transformer. T3 for three-winding transformer. 3-8 For BR and T2 records, bus number at the other end of the branch. For T3 records, bus number of one of the other buses connected by the transformer. 9-14 For T3 records, bus number of the other bus connected by the transformer. 15-16 Old identifier. 17-18 New identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid STATUS value
	IERR = 2	bus not found
	IERR = 3	invalid NID value
	IERR = 4	IDTRNS is not big enough
	IERR = 5	JOIN processing error
	IERR = 6	prerequisite requirements for API are not met

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1.114. LAMP

Use this API to print power flow solution output, including loadings in amps, in a traditional power flow report format. Bus related information is on the left side of the report and branch related information on the right side.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LAMP  SID  ALL
```

Python command syntax:

```
ierr = lamp(sid, all)
```

Fortran command syntax:

```
CALL LAMPNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.115. LINES_PER_PAGE

Use this API to specify or return the page length limits for the four output devices; also retrieve device names.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_LINES_PER_PAGE  IVAL(1)  IVAL(2)  IVAL(3)  IVAL(4)
```

Python command syntax:

```
ierr = lines_per_page(ival)

ierr, ival,devnam = lines_per_page()
```

Fortran command syntax:

```
CALL LINES_PER_PAGE_API(IVAL, DEVNAM, IOCODE, IERR)
```

Where:

Integer IVAL (4)	array of 4 elements specifying the page length limits(input).	
	IVAL(1)	lines per page for disk files (no change by default)
	IVAL(2)	lines per page for the report window (no change by default)
	IVAL(3)	lines per page for the first primary hard copy device (no change by default)
	IVAL(4)	lines per page for the second primary hard copy device (no change by default)
Character*40 DEVNAM (4)	array of 4 elements specifying the four devices (when IVAL is output)(output).	
	DEVNAM(1)	'FILE
	DEVNAM(2)	'REPORT WINDOW'
	DEVNAM(3)	name of first primary hard copy device
	DEVNAM(4)	name of second primary hard copy device
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option settings to the values specified in IVAL.
	IOCODE = 1	return the current values of the option settings in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	negative IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.116. LINES_PER_PAGE_ONE_DEVICE

Use this API to specify or return the page length limit and device name for one of the four output devices.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_LINES_PER_PAGE_ONE_DEVICE  DEVICE  IVAL
```

Python command syntax:

```
ierr = lines_per_page_one_device(device, ival)

ierr, ival,devnam = lines_per_page_one_device(device)
```

Fortran command syntax:

```
CALL LINES_PER_PAGE_ONE_DEVICE_API(DEVICE, IVAL, DEVNAM, IOCODE, IERR)
```

Where:

Integer DEVICE	Indicates which of the four devices is to be processed (input; no default allowed).	
	DEVICE = 1	for disk files.
	DEVICE = 2	for the report window.
	DEVICE = 3	for the first primary hard copy output device.
	DEVICE = 4	for the second primary hard copy output device.
Integer IVAL	value of the lines per page option setting of the device corresponding to DEVICE (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, no change by default; else output).	
Character*40 DEVNAM	returned as the device name of the device corresponding to DEVICE (when IVAL is output) (output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	negative IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	invalid DEVICE value
	IERR = 4	prerequisite requirements for API are not met

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1.117. LINE_SHUNT_REPORTING

Use this API to specify or return the option to enable or disable the line shunt reporting.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_LINE_SHUNT_REPORTING  IVAL
```

Python command syntax:

```
ierr = line_shunt_reporting(ival)

ierr, ival = line_shunt_reporting()
```

Fortran command syntax:

```
CALL LINE_SHUNT_REPORTING_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	no special reporting of line shunts in POUT, LOUT and LAMP.
	IVAL = 1	include special reporting of line shunts in POUT, LOUT and LAMP.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.118. LIST

Use this API routine to tabulate the power flow working case in a form suitable for problem data documentation.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LIST  SID  ALL  OPT  VREV
```

Python command syntax:

```
ierr = list(sid, all, opt, vrev)
```

Fortran command syntax:

```
CALL LISTNW(SID, ALL, OPT, VREV, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer OPT	power flow data category to be tabulated (input; 1 by default).	
	OPT = 1	case summary
	OPT = 2	bus data
	OPT = 3	switched shunt data
	OPT = 4	plant data
	OPT = 5	generator unit data
	OPT = 6	branch data (single entry)
	OPT = 7	branch data (double entry)
	OPT = 8	two-winding transformer impedance data
	OPT = 9	line shunt data
	OPT = 10	dc line data
	OPT = 11	area interchange data
	OPT = 12	full listing (single entry)
	OPT = 13	full listing (double entry)
	OPT = 14	impedance table correction data
	OPT = 15	multi-section line data
	OPT = 16	zone data
	OPT = 17	inter-area transfer data
	OPT = 18	load data
	OPT = 19	owner data
	OPT = 20	FACTS device sending end data

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	OPT = 21	two-winding transformer winding data
	OPT = 22	three-winding transformer impedance data
	OPT = 23	three-winding transformer winding data
	OPT = 24	three-winding transformer control data
	OPT = 25	three-winding transformer general data
	OPT = 26	two-winding transformer control data
	OPT = 27	FACTS device terminal end data
	OPT = 28	fixed bus shunt data
	OPT = 29	GNE device data
	OPT = 30	induction machine circuit data
	OPT = 31	induction machine miscellaneous data
Integer VREV	units used for displaying voltages (input; 0 by default).	
	VREV = 0	use default voltage units
	VREV = 1	use alternate voltage units
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	invalid VREV value
	IERR = 5	prerequisite requirements for API are not met

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1.119. LISTCONTINGENCYSAVEDCASES

Use this API routine to obtain a report listing some or all of the system conditions preserved in a ZIP Archive Output File that was created during a previous run of one of the members of the the AC contingency calculation family.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_LISTCONTINGENCYSAVEDCASES  PATHZIP LABEL
```

Python command syntax:

```
ierr = listcontingencysavedcases(pathzip, label)
```

Fortran command syntax:

```
CALL LISTCONTINGENCYSAVEDCASES(PATHZIP, LABEL, IERR)
```

Where:

Character*260 PATHZIP	Is the name of the ZIP Archive Output File (input; no default allowed).	
Character*32 LABEL	Is a label indicating the system condition(s) that are to be tabulated. blank: list all system conditions. contingency label: list all system conditions for the specified contingency label. Base Case: list all system conditions for the base case; use only for ZIP Archive Output Files written by the P-V and Q-V analysis functions. (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred.
	IERR = 1	error opening the file Names.phy in the ZIP Archive Output File.
	IERR = 2	error reading the file Names.phy in the ZIP Archive Output File.
	IERR = 3	error closing the file Names.phy in the ZIP Archive Output File.
	IERR = 4	internal error.
	IERR = 5	internal error.
	IERR = 6	prerequisite requirements for API are not met.

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1.120. LLRF

Use this API to apply the line loading relief calculation using a linear network (dc) model. It must first be called once with APIOPT set to 1, then once for each priority level for which the adjustment method is to be changed with APIOPT set to 2, and finally once with APIOPT set to 3.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_LLRF  APIOPT  OPTIONS(1)  OPTIONS(2)  RVAL  DFXFILE
```

Python command syntax:

```
ierr = llrf(apiopt, options, rval, dfxfile)
```

Fortran command syntax:

```
CALL LLRFAPI(APIOPT, OPTIONS, RVAL, DFXFILE, IERR)
```

Where:

Integer APIOPT	mode of operation (input).	
	APIOPT = 1	initialize
	APIOPT = 2	specify a priority level and the adjustment method to be applied; must be called once for each priority level, adjustment method pair to be specified
	APIOPT = 3	specify a line loading relief function, apply the required calculations, report results, and terminate
Integer OPTIONS (2)	array of 2 elements specifying options and values to be used in the current calculation phase of LLRFAPI; ignored when APIOPT is 1. When APIOPT is 2, the elements of OPTIONS define a priority level and the adjustment method to be used in curtailing or restoring transactions. When APIOPT is 3, the first element of OPTIONS defines the line loading relief function to be applied. For the curtail transactions and restore transactions functions, the second element of OPTIONS defines the monitored element to be used as the transaction adjustment target(input).	
	OPTIONS(1)	When APIOPT is 2, priority level When APIOPT is 3 (no default allowed)
	OPTIONS(1) = 0	curtail transactions
	OPTIONS(1) = 1	restore transactions
	OPTIONS(1) = 2	report distribution factor matrix
	OPTIONS(2)	When APIOPT is 3, monitored element list index (1 by default when OPTIONS(1) is 0 or 1).. When APIOPT is 2)
	OPTIONS(2) = 0	first in, last out (FILO)
	OPTIONS(2) = 1	decreasing order of distribution factor magnitude (DF-ORDER)

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	OPTIONS(2) = 2	distribution factor pro rata (DF-PRO RATA)
	OPTIONS(2) = 3	pro rata based on the product of distribution factor and transaction schedule (DF-SCHED-PRO RATA)
Real RVAL	used in the current calculation phase of LLRFAPI; ignored when APIOPT is 2. When APIOPT is 1, RVAL is the distribution factor tolerance used to preclude those transactions with negligible effect on the target monitored element from the transaction adjustment functions (input; 0.05 by default). When APIOPT is 3, RVAL is the target element adjustment MW (10.0 by default).	
Character*260 DFXFILE	name of DFAX data file when APIOPT is 1 (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no transactions defined
	IERR = 2	no DFAX data file specified
	IERR = 3	invalid OPTIONS value
	IERR = 4	invalid RVAL value
	IERR = 5	unexpected APIOPT value
	IERR = 6	invalid APIOPT value
	IERR = 7	error opening file DFXFILE
	IERR = 8	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 10	no monitored branches or interfaces defined
	IERR = 11	prerequisite requirements for API are not met

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1.121. LOAD_REDUCTION

Use this API to tabulate the amount of load reduction in a specified subsystem due to the voltage at the bus to which the load is connected being below PQBRAK (for constant MVA load) or 0.5 (for constant current load.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_LOAD_REDUCTION  SID  ALL  AOZOPT
```

Python command syntax:

```
ierr = load_reduction(sid, all, aozopt)
```

Fortran command syntax:

```
CALL LOAD_REDUCTION(SID, ALL, AOZOPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only loads in subsystem SID
	ALL = 1	process all loads
Integer AOZOPT	subsystem selection option when area, owner and/or zone are included in the subsystem selection criteria (input; 1 by default).	
	AOZOPT = 1	use the area, owner and/or zone assignments of the bus to which the load is connected
	AOZOPT = 2	use the area, owner and/or zone assignments of the load
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid AOZOPT value
	IERR = 4	prerequisite requirements for API are not met

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1.122. LOUT

Use this API to print the power flow solution results in a traditional power flow report format. Bus related information is on the left side of the report and branch related information on the right side.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LOUT  SID  ALL
```

Python command syntax:

```
ierr = lout(sid, all)
```

Fortran command syntax:

```
CALL LOUTNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.123. LTAP

Use this API to insert a bus at a designated location along a line.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LTAP  FRMBUS  TOBUS  'CKT'  FRACTION  NEWNUM  'NEWNAM'  NEWKV
```

Python command syntax:

```
ierr = ltap(frmbus, tobus, ckt, fraction, newnum, newnam, newkv)
```

Fortran command syntax:

```
CALL LTAPAPI(FRMBUS, TOBUS, CKT, FRACTION, NEWNUM, NEWNAM, NEWKV, IERR)
```

Where:

Integer FRMBUS	one of the buses connected by the branch to be tapped (input; no default allowed).	
Integer TOBUS	other bus connected by the branch to be tapped (input; no default allowed).	
Character*2 CKT	circuit identifier of the branch to be tapped (input; '1').	
Real FRACTION	new bus location relative to bus FRMBUS in per unit of total line length (input; 0.5).	
Integer NEWNUM	number to be assigned to the new bus (input; no default allowed).	
Character*12 NEWNAM	name to be assigned to the new bus (input; blank by default).	
Real NEWKV	base voltage to be assigned to the new bus (input; base voltage of bus FRMBUS).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	branch not found
	IERR = 3	branch is a transformer
	IERR = 4	invalid FRACTION value
	IERR = 5	invalid NEWNUM value
	IERR = 6	bus number is already assigned to another bus
	IERR = 7	invalid NEWKV value
	IERR = 8	bus, branch, or branch ownership tables full
	IERR = 9	prerequisite requirements for API are not met

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1.124. MACCC

This API routine is obsolete. It has been replaced by the API routine maccc_2 .

Use this API to run multiple level contingency analysis. User-specified and automatically-selected contingencies from built-in contingency ranking functions are evaluated individually and in combination with each other as overlapping outages up to the specified evaluation level.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MACCC  TOL  OPTACC(1)..OPTACC(8)  OPTMAC(1)..OPTMAC(41)
 OPTRNK(1)..OPTRNK(17)  VALUES(1)..VALUES(3)  LABELS(1)  LABELS(2) DFXFILE
  ACCFILE  THRFILE  INLFILE
```

Python command syntax:

```
ierr = maccc(tol, optacc, optmac, optrnk, values, labels, dfxfile,
 accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL MACCCAPI(TOL, OPTACC, OPTMAC, OPTRNK, VALUES, LABELS, DFXFILE,
 ACCFILE, THRFILE, INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable

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	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (reserve)
	OPTACC(8) = 2	subsystem machines (pmax)
	OPTACC(8) = 3	subsystem machines (inertia)
	OPTACC(8) = 4	subsystem machines (governor droop)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	run multiple contingency analysis (1 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(2)	contingency evaluation level (1<=OPTMAC (2)<=3; 1 by default)
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single machine and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations, otherwise
	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	
	OPTRNK(1)	include ties from contingency subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable

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OPTRNK(2)	contingency rating (rating set program option setting by default)
OPTRNK(2) = 1	rate A
OPTRNK(2) = 2	rate B
OPTRNK(2) = 3	rate C
OPTRNK(3)	convert ratings to MW (0 by default)
OPTRNK(3) = 0	disable
OPTRNK(3) = 1	enable
OPTRNK(4)	include machine contingencies from overload ranking (0 by default)
OPTRNK(4) = 0	disable
OPTRNK(4) = 1	enable
OPTRNK(5)	minimum of machine contingencies to evaluate (>=0; 1 by default)
OPTRNK(6)	maximum of machine contingencies to evaluate (>=0; 10 by default)
OPTRNK(7)	non-failure cutoff of machine contingencies (>=0; 5 by default)
OPTRNK(8)	include branch contingencies from overload ranking (0 by default)
OPTRNK(8) = 0	disable
OPTRNK(8) = 1	enable
OPTRNK(9)	minimum of branch contingencies from overload ranking to evaluate (>=0; 1 by default)
OPTRNK(10)	maximum of branch contingencies from overload ranking to evaluate (>=0; 10 by default)
OPTRNK(11)	non-failure cutoff of branch contingencies from overload ranking (>=0; 5 by default)
OPTRNK(12)	include branch contingencies from voltage depression ranking (0 by default)
OPTRNK(12) = 0	disable
OPTRNK(12) = 1	enable
OPTRNK(13)	minimum of branch contingencies from voltage depression ranking to evaluate (>=0; 1 by default)
OPTRNK(14)	maximum of branch contingencies from voltage depression ranking to evaluate (>=0; 10 by default)
OPTRNK(15)	non-failure cutoff of branch contingencies from voltage depression ranking (>=0; 5 by default)
OPTRNK(16)	include all branches in voltage depression ranking (0 by default)
OPTRNK(16) = 0	include only with non-zero rating

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	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Real VALUES (3)	array of 3 elements specifying ranking parameters(input).	
	VALUES(1)	small reactance threshold (0<= VALUES (1)<=0.005; 0.0005 by default)
	VALUES(2)	percent of rating set (0<= VALUES (1)<=200.0; 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (VALUES (3) >=0.0; 0.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of contingency subsystem (blank by default)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	invalid OPTMAC value
	IERR = 9	invalid OPTRNK value
	IERR = 10	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 50	error opening Unit Inertia and Governor Data File

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IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.125. MACCC_2

Use this API to run the second release of multiple level contingency analysis. User-specified and automatically-selected contingencies from built-in contingency ranking functions are evaluated individually and in combination with each other as overlapping outages up to the specified evaluation level.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MACCC_2  TOL  OPTACC(1)..OPTACC(10)  OPTMAC(1)..OPTMAC(41)
 OPTRNK(1)..OPTRNK(17)  VALUES(1)..VALUES(3)  LABELS(1)  LABELS(2) DFXFILE
  ACCFILE  THRFILE  INLFILE
```

Python command syntax:

```
ierr = maccc_2(tol, optacc, optmac, optrnk, values, labels, dfxfile,
 accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL MACCCAPI_2(TOL, OPTACC, OPTMAC, OPTRNK, VALUES, LABELS, DFXFILE,
 ACCFILE, THRFILE, INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (10)	array of 10 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)

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	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTACC(8) = 0	stall
	OPTACC(8) = 1	trip
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable
	OPTACC(10) = 1	subsystem machines (reserve)
	OPTACC(10) = 2	subsystem machines (pmax)
	OPTACC(10) = 3	subsystem machines (inertia)
	OPTACC(10) = 4	subsystem machines (governor droop)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	run multiple contingency analysis (1 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(2)	contingency evaluation level (1 by default). If it is 2 or 3, a newly ordered event list is built for the next level after solving AC power flow each case at current level; if it is -2 or -3, the event list at the first level is used for any higher levels)
	OPTMAC(2) = 1	N-1 contingency analysis
	OPTMAC(2) = 2	N-2 contingency analysis
	OPTMAC(2) = -2	N-2 contingency analysis

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	OPTMAC(2) = 3	N-3 contingency analysis
	OPTMAC(2) = -3	N-3 contingency analysis
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single machine and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations, otherwise
	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	
	OPTRNK(1)	include ties from contingency subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable
	OPTRNK(2)	contingency rating (rating set program option setting by default)
	OPTRNK(2) = 1	rate A
	OPTRNK(2) = 2	rate B
	OPTRNK(2) = 3	rate C
	OPTRNK(3)	convert ratings to MW (0 by default)
	OPTRNK(3) = 0	disable
	OPTRNK(3) = 1	enable
	OPTRNK(4)	include machine contingencies from overload ranking (0 by default)
	OPTRNK(4) = 0	disable
	OPTRNK(4) = 1	enable
	OPTRNK(5)	minimum of machine contingencies to evaluate (>=0; 1 by default)
	OPTRNK(6)	maximum of machine contingencies to evaluate (>=0; 10 by default)
	OPTRNK(7)	non-failure cutoff of machine contingencies (>=0; 5 by default)
	OPTRNK(8)	include branch contingencies from overload ranking (0 by default)
	OPTRNK(8) = 0	disable
	OPTRNK(8) = 1	enable
	OPTRNK(9)	minimum of branch contingencies from overload ranking to evaluate (>=0; 1 by default)

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	OPTRNK(10)	maximum of branch contingencies from overload ranking to evaluate (>=0; 10 by default)
	OPTRNK(11)	non-failure cutoff of branch contingencies from overload ranking (>=0; 5 by default)
	OPTRNK(12)	include branch contingencies from voltage depression ranking (0 by default)
	OPTRNK(12) = 0	disable
	OPTRNK(12) = 1	enable
	OPTRNK(13)	minimum of branch contingencies from voltage depression ranking to evaluate (>=0; 1 by default)
	OPTRNK(14)	maximum of branch contingencies from voltage depression ranking to eval-uate (>=0; 10 by default)
	OPTRNK(15)	non-failure cutoff of branch contingencies from voltage depression ranking (>=0; 5 by default)
	OPTRNK(16)	include all branches in voltage depression ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating
	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Real VALUES (3)	array of 3 elements specifying ranking parameters(input).	
	VALUES(1)	small reactance threshold (0<= VALUES (1)<=0.005; 0.0005 by default)
	VALUES(2)	percent of rating set (0<= VALUES (1)<=200.0; 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0.0; 0.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of contingency subsystem (blank by default)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	

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Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	invalid OPTMAC value
	IERR = 11	invalid OPTRNK value
	IERR = 12	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 50	error opening Unit Inertia and Governor Data File
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	prerequisite requirements for function not met

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1.126. MACCC_PARALLEL

Use this API to run the multiple level contingency analysis in parallel when the number of contingency processors in Program Settings is greater than 1. User-specified and automatically selected contingencies from built-in contingency ranking functions are evaluated individually and in combination with each other as overlapping outages up to the specified evaluation level.

This API routine was first introduced in release 33.3.0.

Batch command syntax:

```
BAT_MACCC_PARALLEL  TOL OPTACC(1)..OPTACC(10) OPTMAC(1)..OPTMAC(41)
 OPTRNK(1)..OPTRNK(17) VALUES(1)..VALUES(3) LABELS(1)..LABELS(2) DFXFILE
 ACCFILE THRFILE INLFILE
```

Python command syntax:

```
ierr = maccc_parallel(tol, optacc, optmac, optrnk, values, labels,
 dfxfile, accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL MACCC_PARALLEL(TOL, OPTACC, OPTMAC, OPTRNK, VALUES, LABELS, DFXFILE,
 ACCFILE, THRFILE, INLFILE, IERR)
```

Where:

Real TOL	Is the mismatch tolerance (input; Newton solution convergence tolerance, TOLN, by default).	
Integer OPTACC (10)	array of elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable.
	OPTACC(1) = 1	enable stepping adjustment.
	OPTACC(1) = 2	enable direct adjustment.
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable.
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange.
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange.
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable.
	OPTACC(3) = 1	enable.
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable.
	OPTACC(4) = 1	enable.

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	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable.
	OPTACC(5) = 1	enable.
	OPTACC(5) = 2	enable continuous mode, disable discrete mode.
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS.
	OPTACC(6) = 1	FNSL.
	OPTACC(6) = 2	optimized FDNS.
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable.
	OPTACC(7) = 1	enable.
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default) (applied when an induction motor fails to solve due to low by default)
	OPTACC(8) = 0	stall.
	OPTACC(8) = 1	trip.
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state.
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable.
	OPTACC(10) = 1	subsystem machines (reserve).
	OPTACC(10) = 2	subsystem machines (pmax).
	OPTACC(10) = 3	subsystem machines (inertia).
	OPTACC(10) = 4	subsystem machines (governor droop).
Integer OPTMAC (41)	array of elements specifying contingency evaluation level and combinations (input). The values are as follows:(input).	
	OPTMAC(1)	run multiple contingency analysis (1 by default)
	OPTMAC(1) = 0	disable.
	OPTMAC(1) = 1	enable.
	OPTMAC(2)	contingency evaluation level.
	OPTMAC(2) = -3	N-3 contingency analysis
	OPTMAC(2) = -2	N-2 contingency analysis
	OPTMAC(2) = 1	N-1 contingency analysis

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	OPTMAC(3)	each element represents a type of contingency combinations in the following order. (0 by default)
	OPTMAC(3) = 0	do not include the contigency combination
	OPTMAC(3) = 1	include the "S" contingency combination. This and the remaining elements of the array (i.e. 3-41) individually indicate whether particular contingency combinations are to be included. In the following S, U and B refer to user specified cases, single machine and single branch outages, respectively: {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}
Integer OPTRNK (17)	array of elements specifying ranking options. The values are as follows:(input).	
	OPTRNK(1)	include ties from contingency subsystem (0 by default)
	OPTRNK(1) = 0	disable.
	OPTRNK(1) = 1	enable.
	OPTRNK(2)	contingency rating (rating set program option setting by default)
	OPTRNK(2) = 1	rate A.
	OPTRNK(2) = 2	rate B.
	OPTRNK(2) = 3	rate C.
	OPTRNK(3)	convert ratings to MW (0 by default)
	OPTRNK(3) = 0	disable.
	OPTRNK(3) = 1	enable.
	OPTRNK(4)	include machine contingencies from overload ranking (0 by default)
	OPTRNK(4) = 0	disable.
	OPTRNK(4) = 1	enable.
	OPTRNK(5)	minimum of machine contingencies to evaluate (>=0; 1 by default)
	OPTRNK(6)	maximum of machine contingencies to evaluate (>=0; 10 by default)
	OPTRNK(7)	non-failure cutoff of machine contingencies (>=0; 5 by default)
	OPTRNK(8)	include branch contingencies from overload ranking (0 by default)
	OPTRNK(8) = 0	disable.
	OPTRNK(8) = 1	enable.
	OPTRNK(9)	minimum of branch contingencies from overload ranking to evaluate (>=0; 1 by default)

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	OPTRNK(10)	maximum of branch contingencies from overload ranking to evaluate (>=0; 10 by default)
	OPTRNK(11)	non-failure cutoff of branch contingencies from overload ranking (>=0; 5 by default)
	OPTRNK(12)	include branch contingencies from voltage depression ranking (0 by default)
	OPTRNK(12) = 0	disable.
	OPTRNK(12) = 1	enable.
	OPTRNK(13)	minimum of branch contingencies from voltage depression ranking to evaluate (>=0; 1 by default)
	OPTRNK(14)	maximum of branch contingencies from voltage depression ranking to evaluate (>=0; 10 by default)
	OPTRNK(15)	non-failure cutoff of branch contingencies from voltage depression ranking (>=0; 5 by default)
	OPTRNK(16)	include all branches in voltage depression ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating.
	OPTRNK(16) = 1	all.
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable.
	OPTRNK(17) = 1	enable.
Real VALUES (3)	array of elements specifying ranking parameters.The values are as follows: (input).	
	VALUES(1)	small reactance threshold (0<= VALUES(1)<=0.005; 0.0005 by default).
	VALUES(2)	percent of rating set (0<= VALUES(1)<=200.0; 100.0 by default).
	VALUES(3)	MW threshold for islanding contingencies (>=0.0; 0.0 by default).
Character*32 LABELS (2)	array of elements specifying subsystem labels. The values are as follows:(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0).
	LABELS(2)	label of contingency subsystem (blank by default)
Character*260 DFXFILE	Is the name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	Is the name of the Contingency Solution Output File (input; no default allowed).	

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Character*260 THRFILE	Is the name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	Is the name of the Unit Inertia and Governor Data File (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred.
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted.
	IERR = 4	buses in island(s) without a swing bus; use activity TREE.
	IERR = 5	largest mismatch exceeds mismatch tolerance.
	IERR = 6	generation subsystem is not defined.
	IERR = 7	too many islands in base case.
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	invalid OPTMAC value.
	IERR = 11	invalid OPTRNK value.
	IERR = 12	invalid VALUES value.
	IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 50	error opening Unit Inertia and Governor Data File
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	prerequisite requirements for API are not met

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1.127. MACCC_TRIP_COR

This API routine is obsolete. It has been replaced by the API routine maccc_trip_cor_2 .

Use this API to run multiple AC contingency analysis with tripping simulations and corrective actions. The user-specified and automatically-selected contingencies are evaluated individually and in combination with each other as overlapping contingencies. When there are violations in post-tripping simulation solutions, the corrective actions analysis will be launched to eliminate the violations, with the objective of minimizing the controls adjustments, including load shedding, MW generation dispatch and phase shifter angle adjustments.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_MACCC_TRIP_COR  TOL OPTACC(1)..OPTACC(8) OPTMAC(1)..OPTMAC(41)
 OPTRNK(1)..OPTRNK(17) OPTTRP(1)..OPTTRP(3) OPTCOR(1)..OPTCOR(5)
 VALUES(1)..VALUES(8)  LABELS(1)..LABELS(5)  DFXFILE  ACCFILE  THRFILE
 INLFILE TRPFILE
```

Python command syntax:

```
ierr = maccc_trip_cor(tol, optacc, optmac, optrnk, opttrp, optcor, values,
 labels, dfxfile, accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL MACCC_TRIP_COR(TOL, OPTACC, OPTMAC, OPTRNK, OPTTRP, OPTCOR, VALUES,
 LABELS, DFXFILE, ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable

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	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (reserve)
	OPTACC(8) = 2	subsystem machines (pmax)
	OPTACC(8) = 3	subsystem machines (inertia)
	OPTACC(8) = 4	subsystem machines (governor droop)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	run multiple contingency analysis (1 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(2)	contingency evaluation level (1<=OPTMAC (2)<=3; 1 by default)
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single machine and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations, otherwise
	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	

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OPTRNK(1)	include ties from contingency subsystem (0 by default)
OPTRNK(1) = 0	disable
OPTRNK(1) = 1	enable
OPTRNK(2)	contingency rating (rating set program option setting by default)
OPTRNK(2) = 1	rate A
OPTRNK(2) = 2	rate B
OPTRNK(2) = 3	rate C
OPTRNK(3)	convert ratings to MW (0 by default)
OPTRNK(3) = 0	disable
OPTRNK(3) = 1	enable
OPTRNK(4)	include machine contingencies from overload ranking (0 by default)
OPTRNK(4) = 0	disable
OPTRNK(4) = 1	enable
OPTRNK(5)	minimum of machine contingencies to evaluate (>=0; 1 by default)
OPTRNK(6)	maximum of machine contingencies to evaluate (>=0; 10 by default)
OPTRNK(7)	non-failure cutoff of machine contingencies (>=0; 5 by default)
OPTRNK(8)	include branch contingencies from overload ranking (0 by default)
OPTRNK(8) = 0	disable
OPTRNK(8) = 1	enable
OPTRNK(9)	minimum of branch contingencies from overload ranking to evaluate (>=0; 1 by default)
OPTRNK(10)	maximum of branch contingencies from overload ranking to evaluate (>=0; 10 by default)
OPTRNK(11)	non-failure cutoff of branch contingencies from overload ranking (>=0; 5 by default)
OPTRNK(12)	include branch contingencies from voltage depression ranking (0 by default)
OPTRNK(12) = 0	disable
OPTRNK(12) = 1	enable
OPTRNK(13)	minimum of branch contingencies from voltage depression ranking to evaluate (>=0; 1 by default)
OPTRNK(14)	maximum of branch contingencies from voltage depression ranking to evaluate (>=0; 10 by default)

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	OPTRNK(15)	non-failure cutoff of branch contingencies from voltage depression ranking (>=0; 5 by default)
	OPTRNK(16)	include all branches in voltage depression ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating
	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable tripping simulation
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Integer OPTCOR (5)	array of 5 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective actions (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number of power flow solutions (from 0 to 10; 1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
Real VALUES (8)	array of 8 elements specifying ranking and corrective action parameters(input).	
	VALUES(1)	small reactance threshold (0<=VALUES (1)<=0.005; 0.0005 by default)
	VALUES(2)	percent of rating (0<=VALUES (1)<=200.0; 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0.0; 0.0 by default)
	VALUES(4)	bus voltage violation tolerance (0.1 by default)

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	VALUES(5)	branch flow overload tolerance (0.1 by default)
	VALUES(6)	generator control weighting factor (1.0 by default)
	VALUES(7)	load control weighting factor (1.0 by default)
	VALUES(8)	phase shifter control weighting factor (1.0 by default)
Character*32 LABELS (5)	array of 5 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of contingency subsystem (blank by default)
	LABELS(3)	label of generator control subsystem (blank by default; no default allowed if OPTCOR(3) is 1)
	LABELS(4)	label of load control subsystem (blank by default; no default allowed if OPTCOR(4) is 1)
	LABELS(5)	label of phase shifter control subsystem (blank by default; no default allowed if OPTCOR(5) is 1)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation dispatch subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	invalid OPTMAC value
	IERR = 9	invalid OPTRNK value
	IERR = 10	invalid OPTTRP value
	IERR = 11	invalid OPTCOR value

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IERR = 12	invalid VALUES value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 49	error opening Tripping Element Data File
IERR = 50	error opening Unit Inertia and Governor Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for API are not met

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1.128. MACCC_TRIP_COR_2

This API routine is obsolete. It has been replaced by the API routine maccc_trip_cor_3 .

This API is the second release of function to run multiple AC contingency analysis with tripping simulations and corrective actions. The user-specified and automatically-selected contingencies are evaluated individually and in combination each other as overlapping contingencies. Following a post contingency solution, trip events are simulated if their tripping conditions are satisfied. Corrective actions are then specified automatically to eliminate the violations in each post tripping solution with the objective of minimizing the control adjustments including load shedding, MW generation dispatch, phase shifter angle adjustments, off-line generator controls, tap setting adjustments and switched shunt controls.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_MACCC_TRIP_COR_2  TOL  OPTACC(1)..OPTACC(8) OPTMAC(1)..OPTMAC(41)
 OPTRNK(1)..OPTRNK(17) OPTTRP(1)..OPTTRP(3) OPTCOR(1)..OPTCOR(8)
 VALUES(1)..VALUES(11)  LABELS(1)..LABELS(8) DFXFILE ACCFILE THRFILE
 INLFILE TRPFILE
```

Python command syntax:

```
ierr = maccc_trip_cor_2(tol, optacc, optmac, optrnk, opttrp, optcor,
 values, labels, dfxfile, accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL MACCC_TRIP_COR_2(TOL, OPTACC, OPTMAC, OPTRNK, OPTTRP, OPTCOR, VALUES,
 LABELS, DFXFILE, ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable

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	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous, disable discrete
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (RESERVE)
	OPTACC(8) = 2	subsystem machines (PMAX)
	OPTACC(8) = 3	subsystem machines (INERTIA)
	OPTACC(8) = 4	subsystem machines (GOVERNOR DROOP)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	allow multiple contingency analysis (0 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(1) = 2	enabled with failure cutoff disabled
	OPTMAC(2)	contingency evaluation level (1 by default)
	OPTMAC(2) = 1	N-1 contingency analysis
	OPTMAC(2) = 2	N-2 contingency analysis
	OPTMAC(2) = 3	N-3 contingency analysis
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single unit and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations

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	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	
	OPTRNK(1)	include ties from subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable
	OPTRNK(2)	contingency rating (rating set program option setting by default)
	OPTRNK(2) = 1	rate A
	OPTRNK(2) = 2	rate B
	OPTRNK(2) = 3	rate C
	OPTRNK(3)	convert ratings to MW (0 by default)
	OPTRNK(3) = 0	disable
	OPTRNK(3) = 1	enable
	OPTRNK(4)	machine overload contingencies (0 by default)
	OPTRNK(4) = 0	disable
	OPTRNK(4) = 1	enable
	OPTRNK(5)	minimum of machine overload contingencies to evaluate (>=0, 1 by default)
	OPTRNK(6)	maximum of machine overload contingencies to evaluate (>=0, 10 by default)
	OPTRNK(7)	cutoff of machine overload contingencies to evaluate (>=0, 5 by default)
	OPTRNK(8)	line overload contingencies (0 by default)
	OPTRNK(8) = 0	disable
	OPTRNK(8) = 1	enable
	OPTRNK(9)	minimum of line overload contingencies to evaluate (>=0, 1 by default)
	OPTRNK(10)	maximum of line overload contingencies to evaluate (>=0, 10 by default)
	OPTRNK(11)	cutoff of line overload contingencies (>=0, 5 by default)
	OPTRNK(12)	voltage collapse line contingencies (0 by default)
	OPTRNK(12) = 0	disable
	OPTRNK(12) = 1	enable
	OPTRNK(13)	minimum of voltage collapse line contingencies to evaluate (>=0, 1 by default)
	OPTRNK(14)	maximum of voltage collapse line contingencies to evaluate (>=0, 10 by default)
	OPTRNK(15)	cutoff of voltage collapse line contingencies (>=0, 5 be default)
	OPTRNK(16)	include all branches in voltage collapse ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating

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	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable tripping simulation
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Integer OPTCOR (8)	array of 8 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective action (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number power flows (1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
	OPTCOR(6)	off-line generator control flag (0 by default)
	OPTCOR(6) = 0	disable
	OPTCOR(6) = 1	enable
	OPTCOR(7)	tap setting adjustment flag (0 by default)
	OPTCOR(7) = 0	disable
	OPTCOR(7) = 1	enable
	OPTCOR(8)	switched shunt control flag (0 by default)
	OPTCOR(8) = 0	disable
	OPTCOR(8) = 1	enable
Real VALUES (11)	array of 11 elements specifying ranking parameters and corrective actions parameters(input).	
	VALUES(1)	small reactance threshold, (0<= VALUES (1)<=0.005, 0.0005 by default)
	VALUES(2)	percent of flow rating, (0<= VALUES (2)<=200.0, 100.0 by default)

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	VALUES(3)	MW threshold for islanding contingencies (>=0, 0.0 by default)
	VALUES(4)	bus voltage violation tolerance (0.1 by default)
	VALUES(5)	branch overload violation tolerance (0.1 by default)
	VALUES(6)	generator control weighting factor (1.0 by default)
	VALUES(7)	load control weighting factor (1.0 by default)
	VALUES(8)	phase shifter control weighting factor (1.0 by default)
	VALUES(9)	off-line generator control weighting factor (1.0 by default)
	VALUES(10)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(11)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (8)	array of 8 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of contingency subsystem (blank by default)
	LABELS(3)	label of generator control subsystem (blank by default for none, no default allowed if OPTCOR(3) is 1)
	LABELS(4)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(5)	label of phase shifter control subsystem (blank by default for none, no default allowed if OPTCOR(5) is 1)
	LABELS(6)	off-line generator control subsystem (blank by default; no default allowed if OPTCOR(6) is 1)
	LABELS(7)	tap setting adjustment subsystem (blank by default; no default allowed if OPTCOR(7) is 1)
	LABELS(8)	switched shunt control subsystem (blank by default; no default allowed if OPTCOR(8) is 1)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank for none).	

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Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	invalid OPTMAC value
	IERR = 9	invalid OPTRNK value
	IERR = 10	invalid OPTTRP value
	IERR = 11	invalid OPTCOR value
	IERR = 12	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multisection line members
	IERR = 49	error opening Tripping Element Data File
	IERR = 50	error opening Unit Inertia Data File
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	prerequisite requirements for function not met

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1.129. MACCC_TRIP_COR_3

This API is the third release of function to run multiple AC contingency analysis with tripping simulations and corrective actions. The user-specified and automatically-selected contingencies are evaluated individually and in combination each other as overlapping contingencies. Following a post contingency solution, trip events are simulated if their tripping conditions are satisfied. Corrective actions are then specified automatically to eliminate the violations in each post tripping solution with the objective of minimizing the control adjustments including load shedding, MW generation dispatch, phase shifter angle adjustments, off-line generator controls, tap setting adjustments and switched shunt controls.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MACCC_TRIP_COR_3  TOL  OPTACC(1)..OPTACC(10) OPTMAC(1)..OPTMAC(41)
 OPTRNK(1)..OPTRNK(17) OPTTRP(1)..OPTTRP(3) OPTCOR(1)..OPTCOR(8)
 VALUES(1)..VALUES(11)  LABELS(1)..LABELS(8) DFXFILE ACCFILE THRFILE
 INLFILE TRPFILE
```

Python command syntax:

```
ierr = maccc_trip_cor_3(tol, optacc, optmac, optrnk, opttrp, optcor,
 values, labels, dfxfile, accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL MACCC_TRIP_COR_3(TOL, OPTACC, OPTMAC, OPTRNK, OPTTRP, OPTCOR, VALUES,
 LABELS, DFXFILE, ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (10)	array of 10 elements specifying power flow solution(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable

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	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous, disable discrete
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTACC(8) = 0	stall
	OPTACC(8) = 1	trip
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable
	OPTACC(10) = 1	subsystem machines (RESERVE)
	OPTACC(10) = 2	subsystem machines (PMAX)
	OPTACC(10) = 3	subsystem machines (INERTIA)
	OPTACC(10) = 4	subsystem machines (GOVERNOR DROOP)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	allow multiple contingency analysis (0 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(1) = 2	enabled with failure cutoff disabled
	OPTMAC(2)	contingency evaluation level (1 by default). If it is 2 or 3, a newly ordered event list is built for the next level after solving AC power

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		flow each case at current level; if it is -2 or -3, the event list at the first level is used for any higher levels)
	OPTMAC(2) = 1	N-1 contingency analysis
	OPTMAC(2) = 2	N-2 contingency analysis
	OPTMAC(2) = -2	N-2 contingency analysis
	OPTMAC(2) = 3	N-3 contingency analysis
	OPTMAC(2) = -3	N-3 contingency analysis
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single unit and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations
	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	
	OPTRNK(1)	include ties from subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable
	OPTRNK(2)	contingency rating (rating set program option setting by default)
	OPTRNK(2) = 1	rate A
	OPTRNK(2) = 2	rate B
	OPTRNK(2) = 3	rate C
	OPTRNK(3)	convert ratings to MW (0 by default)
	OPTRNK(3) = 0	disable
	OPTRNK(3) = 1	enable
	OPTRNK(4)	machine overload contingencies (0 by default)
	OPTRNK(4) = 0	disable
	OPTRNK(4) = 1	enable
	OPTRNK(5)	minimum of machine overload contingencies to evaluate (>=0, 1 by default)
	OPTRNK(6)	maximum of machine overload contingencies to evaluate (>=0, 10 by default)
	OPTRNK(7)	cutoff of machine overload contingencies to evaluate (>=0, 5 by default)
	OPTRNK(8)	line overload contingencies (0 by default)
	OPTRNK(8) = 0	disable
	OPTRNK(8) = 1	enable

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	OPTRNK(9)	minimum of line overload contingencies to evaluate (>=0, 1 by default)
	OPTRNK(10)	maximum of line overload contingencies to evaluate (>=0, 10 by default)
	OPTRNK(11)	cutoff of line overload contingencies (>=0, 5 by default)
	OPTRNK(12)	voltage collapse line contingencies (0 by default)
	OPTRNK(12) = 0	disable
	OPTRNK(12) = 1	enable
	OPTRNK(13)	minimum of voltage collapse line contingencies to evaluate (>=0, 1 by default)
	OPTRNK(14)	maximum of voltage collapse line contingencies to evaluate (>=0, 10 by default)
	OPTRNK(15)	cutoff of voltage collapse line contingencies (>=0, 5 be default)
	OPTRNK(16)	include all branches in voltage collapse ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating
	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable tripping simulation
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Integer OPTCOR (8)	array of 8 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective action (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number power flows (1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)

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	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
	OPTCOR(6)	off-line generator control flag (0 by default)
	OPTCOR(6) = 0	disable
	OPTCOR(6) = 1	enable
	OPTCOR(7)	tap setting adjustment flag (0 by default)
	OPTCOR(7) = 0	disable
	OPTCOR(7) = 1	enable
	OPTCOR(8)	switched shunt control flag (0 by default)
	OPTCOR(8) = 0	disable
	OPTCOR(8) = 1	enable
Real VALUES (11)	array of 11 elements specifying ranking parameters and corrective actions parameters(input).	
	VALUES(1)	small reactance threshold, (0<= VALUES (1)<=0.005, 0.0005 by default)
	VALUES(2)	percent of flow rating, (0<= VALUES (2)<=200.0, 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0, 0.0 by default)
	VALUES(4)	bus voltage violation tolerance (0.1 by default)
	VALUES(5)	branch overload violation tolerance (0.1 by default)
	VALUES(6)	generator control weighting factor (1.0 by default)
	VALUES(7)	load control weighting factor (1.0 by default)
	VALUES(8)	phase shifter control weighting factor (1.0 by default)
	VALUES(9)	off-line generator control weighting factor (1.0 by default)
	VALUES(10)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(11)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (8)	array of 8 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of contingency subsystem (blank by default)
	LABELS(3)	label of generator control subsystem (blank by default for none, no default allowed if OPTCOR(3) is 1)
	LABELS(4)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)

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	LABELS(5)	label of phase shifter control subsystem (blank by default for none, no default allowed if OPTCOR(5) is 1)
	LABELS(6)	off-line generator control subsystem (blank by default; no default allowed if OPTCOR(6) is 1)
	LABELS(7)	tap setting adjustment subsystem (blank by default; no default allowed if OPTCOR(7) is 1)
	LABELS(8)	switched shunt control subsystem (blank by default; no default allowed if OPTCOR(8) is 1)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank for none).	
Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	invalid OPTMAC value
	IERR = 11	invalid OPTRNK value
	IERR = 12	invalid OPTTRP value
	IERR = 13	invalid OPTCOR value
	IERR = 14	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multisection line members
	IERR = 49	error opening Tripping Element Data File
	IERR = 50	error opening Unit Inertia Data File

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IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.130. MACCC_WITH_COR

This API routine is obsolete. It has been replaced by the API routine maccc_with_cor_2 .

Use this API to run multiple AC contingency analysis with corrective actions. The user-specified and automatically-selected contingencies are evaluated individually and in combination with each other as overlapping contingencies. The corrective actions analysis will be launched for the contingencies with flow and voltage violations to eliminate the violations with the objective of minimizing the controls adjustments including load shedding, MW generation dispatch and phase shifter angle adjustments.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_MACCC_WITH_COR  TOL  OPTACC(1)..OPTACC(8) OPTMAC(1)..OPTMAC(41)
 OPTRNK(1)..OPTRNK(17)  OPTCOR(1)..OPTCOR(5) VALUES(1)..VALUES(8)
 LABELS(1)..LABELS(5)  DFXFILE  ACCFILE  THRFILE  INLFILE
```

Python command syntax:

```
ierr = maccc_with_cor(tol, optacc, optmac, optrnk, optcor, values, labels,
 dfxfile, accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL MACCC_WITH_COR(TOL, OPTACC, OPTMAC, OPTRNK, OPTCOR, VALUES, LABELS,
 DFXFILE, ACCFILE, THRFILE, INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)

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	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (reserve)
	OPTACC(8) = 2	subsystem machines (pmax)
	OPTACC(8) = 3	subsystem machines (inertia)
	OPTACC(8) = 4	subsystem machines (governor droop)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	run multiple contingency analysis (1 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(2)	contingency evaluation level (1<=OPTMAC(2)<=3; 1 by default)
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single machine and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations
	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying contingency ranking options(input).	
	OPTRNK(1)	include ties from contingency subsystem (0 by default)
	OPTRNK(1) = 0	disable

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OPTRNK(1) = 1	enable
OPTRNK(2)	contingency rating (rating set program option setting by default)
OPTRNK(2) = 1	rate A
OPTRNK(2) = 2	rate B
OPTRNK(2) = 3	rate C
OPTRNK(3)	convert ratings to MW (0 by default)
OPTRNK(3) = 0	disable
OPTRNK(3) = 1	enable
OPTRNK(4)	include machine contingencies from overload ranking (0 by default)
OPTRNK(4) = 0	disable
OPTRNK(4) = 1	enable
OPTRNK(5)	minimum of machine contingencies to evaluate (>=0; 1 by default)
OPTRNK(6)	maximum of machine contingencies to evaluate (>=0; 10 by default)
OPTRNK(7)	non-failure cutoff of machine contingencies (>=0)
OPTRNK(8)	include branch contingencies from overload ranking (0 by default)
OPTRNK(8) = 0	disable
OPTRNK(8) = 1	enable
OPTRNK(9)	minimum of branch contingencies from overload ranking to evaluate (>=0; 1 by default)
OPTRNK(10)	maximum of branch contingencies from overload ranking to evaluate (>=0; 10 by default)
OPTRNK(11)	non-failure cutoff of branch contingencies from overload ranking (>=0; 5 by default)
OPTRNK(12)	include branch contingencies from voltage depression ranking (0 by default)
OPTRNK(12) = 0	disable
OPTRNK(12) = 1	enable
OPTRNK(13)	minimum of branch contingencies from voltage depression ranking to evaluate (>=0; 1 by default)
OPTRNK(14)	maximum of branch contingencies from voltage depression ranking to evaluate (>=0; 10 by default)
OPTRNK(15)	non-failure cutoff of branch contingencies from voltage depression ranking (>=0; 5 by default)
OPTRNK(16)	include all branches in voltage depression ranking (0 by default)

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	OPTRNK(16) = 0	include only with non-zero rating
	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTCOR (5)	array of 5 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective actions (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number of power flow solutions (from 0 to 10; 1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
Real VALUES (8)	array of 8 elements specifying ranking and corrective actions parameters(input).	
	VALUES(1)	small reactance threshold (0<=VALUES (1)<=0.005; 0.0005 by default)
	VALUES(2)	percent of rating set (0<=VALUES (1)<=200.0; 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0.0; 0.0 by default)
	VALUES(4)	bus voltage violation tolerance (0.1 by default)
	VALUES(5)	branch flow overload tolerance (0.1 by default)
	VALUES(6)	generator control weighting factor (1.0 by default)
	VALUES(7)	load control weighting factor (1.0 by default)
	VALUES(8)	phase shifter control weighting factor (1.0 by default)
Character*32 LABELS (5)	array of 5 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of contingency subsystem (blank by default)

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	LABELS(3)	label of generator control subsystem (blank by default; no default allowed if OPTCOR(3) is 1)
	LABELS(4)	label of load control subsystem (blank by default; no default allowed if OPTCOR(4) is 1)
	LABELS(5)	label of phase shifter control subsystem (blank by default; no default allowed if OPTCOR(5) is 1)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation dispatch subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	invalid OPTMAC value
	IERR = 9	invalid OPTRNK value
	IERR = 10	invalid OPTCOR value
	IERR = 11	invalid VALUES value
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 50	error opening Unit Inertia and Governor Data File
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	prerequisite requirements for API are not met

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1.131. MACCC_WITH_COR_2

This API routine is obsolete. It has been replaced by the API routine maccc_with_cor_3 .

This API is the second release of function to run multiple AC contingency analysis with corrective actions. The user-specified and automatically-selected contingencies are evaluated individually and in combination each other as overlapping contingencies. Corrective actions are specified automatically to eliminate the violations in each post contingency solution with the objective of minimizing the controls adjustments including load shedding, MW generation dispatch, phase shifter angle adjustments, off-line generator controls, tap setting adjustments and switched shunt controls.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_MACCC_WITH_COR_2  TOL  OPTACC(1)..OPTACC(8) OPTMAC(1).. OPTMAC(41)
 OPTRNK(1)..OPTRNK(17) OPTCOR(1)..OPTCOR(8) VALUES(1)..VALUES(11)
 LABELS(1)..LABELS(8) DFXFILE ACCFILE THRFILE INLFILE
```

Python command syntax:

```
ierr = maccc_with_cor_2(tol, optacc, optmac, optrnk, optcor, values,
 labels, dfxfile, accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL MACCC_WITH_COR_2(TOL, OPTACC, OPTMAC, OPTRNK, OPTCOR, VALUES,
 LABELS, DFXFILE, ACCFILE, THRFILE, INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable

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	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous, disable discrete
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (RESERVE)
	OPTACC(8) = 2	subsystem machines (PMAX)
	OPTACC(8) = 3	subsystem machines (INERTIA)
	OPTACC(8) = 4	subsystem machines (GOVERNOR DROOP)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	allow multiple contingency analysis (0 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(1) = 2	enabled with failure cutoff disabled
	OPTMAC(2)	contingency evaluation level (1 by default)
	OPTMAC(2) = 1	N-1 contingency analysis
	OPTMAC(2) = 2	N-2 contingency analysis
	OPTMAC(2) = 3	N-3 contingency analysis
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single unit and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations
	OPTMAC(3) = 0	(0 by default)

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Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	
	OPTRNK(1)	include ties from subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable
	OPTRNK(2)	contingency rating (rating set program option setting by default)
	OPTRNK(2) = 1	rate A
	OPTRNK(2) = 2	rate B
	OPTRNK(2) = 3	rate C
	OPTRNK(3)	convert ratings to MW (0 by default)
	OPTRNK(3) = 0	disable
	OPTRNK(3) = 1	enable
	OPTRNK(4)	machine overload contingencies (0 by default)
	OPTRNK(4) = 0	disable
	OPTRNK(4) = 1	enable
	OPTRNK(5)	minimum of machine overload contingencies to evaluate (>=0, 1 by default)
	OPTRNK(6)	maximum of machine overload contingencies to evaluate (>=0, 10 by default)
	OPTRNK(7)	cutoff of machine overload contingencies to evaluate (>=0, 5 by default)
	OPTRNK(8)	line overload contingencies (0 by default)
	OPTRNK(8) = 0	disable
	OPTRNK(8) = 1	enable
	OPTRNK(9)	minimum of line overload contingencies to evaluate (>=0, 1 by default)
	OPTRNK(10)	maximum of line overload contingencies to evaluate (>=0, 10 by default)
	OPTRNK(11)	cutoff of line overload contingencies (>=0, 5 by default)
	OPTRNK(12)	voltage collapse line contingencies (0 by default)
	OPTRNK(12) = 0	disable
	OPTRNK(12) = 1	enable
	OPTRNK(13)	minimum of voltage collapse line contingencies to evaluate (>=0, 1 by default)
	OPTRNK(14)	maximum of voltage collapse line contingencies to evaluate (>=0, 10 by default)
	OPTRNK(15)	cutoff of voltage collapse line contingencies (>=0, 5 be default)
	OPTRNK(16)	include all branches in voltage collapse ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating
	OPTRNK(16) = 1	all

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	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTCOR (8)	array of 8 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective action (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number power flows (1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
	OPTCOR(6)	off-line generator control flag (0 by default)
	OPTCOR(6) = 0	disable
	OPTCOR(6) = 1	enable
	OPTCOR(7)	tap setting adjustment flag (0 by default)
	OPTCOR(7) = 0	disable
	OPTCOR(7) = 1	enable
	OPTCOR(8)	switched shunt control flag (0 by default)
	OPTCOR(8) = 0	disable
	OPTCOR(8) = 1	enable
Real VALUES (11)	array of 11 elements specifying ranking parameters and corrective actions parameters(input).	
	VALUES(1)	small reactance threshold, (0<= VALUES (1)<=0.005, 0.0005 by default)
	VALUES(2)	percent of flow rating, (0<= VALUES (2)<=200.0, 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0, 0.0 by default)
	VALUES(4)	bus voltage violation tolerance (0.1 by default)
	VALUES(5)	branch overload violation tolerance (0.1 by default)
	VALUES(6)	generator control weighting factor (1.0 by default)
	VALUES(7)	load control weighting factor (1.0 by default)
	VALUES(8)	phase shifter control weighting factor (1.0 by default)

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	VALUES(9)	off-line generator control weighting factor (1.0 by default)
	VALUES(10)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(11)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (8)	array of 8 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is 1)
	LABELS(2)	label of contingency subsystem (blank by default)
	LABELS(3)	label of generator control subsystem (blank by default, no default allowed if OPTCOR(3) is 1)
	LABELS(4)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(5)	label of phase shifter control subsystem (blank by default, no default allowed if OPTCOR(5) is 1)
	LABELS(6)	off-line generator control subsystem (blank by default; no default allowed if OPTCOR(6) is 1)
	LABELS(7)	tap setting adjustment subsystem (blank by default; no default allowed if OPTCOR(7) is 1)
	LABELS(8)	switched shunt control subsystem (blank by default; no default allowed if OPTCOR(8) is 1)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of Unit Inertia and Governor Data File (input; blank for none).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	invalid OPTMAC value
	IERR = 9	invalid OPTRNK value

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IERR = 10	invalid OPTCOR value
IERR = 11	invalid VALUES value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multisection line members
IERR = 50	error opening Unit Inertia Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.132. MACCC_WITH_COR_3

This API is the third release of function to run multiple AC contingency analysis with corrective actions. The user-specified and automatically-selected contingencies are evaluated individually and in combination each other as overlapping contingencies. Corrective actions are specified automatically to eliminate the violations in each post contingency solution with the objective of minimizing the controls adjustments including load shedding, MW generation dispatch, phase shifter angle adjustments, off-line generator controls, tap setting adjustments and switched shunt controls.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MACCC_WITH_COR_3  TOL  OPTACC(1)..OPTACC(10) OPTMAC(1).. OPTMAC(41)
 OPTRNK(1)..OPTRNK(17) OPTCOR(1)..OPTCOR(8) VALUES(1)..VALUES(11)
 LABELS(1)..LABELS(8) DFXFILE ACCFILE THRFILE INLFILE
```

Python command syntax:

```
ierr = maccc_with_cor_3(tol, optacc, optmac, optrnk, optcor, values,
 labels, dfxfile, accfile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL MACCC_WITH_COR_3(TOL, OPTACC, OPTMAC, OPTRNK, OPTCOR, VALUES,
 LABELS, DFXFILE, ACCFILE, THRFILE, INLFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (10)	array of 10 elements specifying power flow solution(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable

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	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous, disable discrete
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTACC(8) = 0	stall
	OPTACC(8) = 1	trip
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable
	OPTACC(10) = 1	subsystem machines (RESERVE)
	OPTACC(10) = 2	subsystem machines (PMAX)
	OPTACC(10) = 3	subsystem machines (INERTIA)
	OPTACC(10) = 4	subsystem machines (GOVERNOR DROOP)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	allow multiple contingency analysis (0 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(1) = 2	enabled with failure cutoff disabled
	OPTMAC(2)	contingency evaluation level (1 by default). If it is 2 or 3, a newly ordered event list is built for the next level after solving AC power flow each case at current level; if it is -2 or -3, the event list at the first level is used for any higher levels)

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	OPTMAC(2) = 1	N-1 contingency analysis
	OPTMAC(2) = 2	N-2 contingency analysis
	OPTMAC(2) = -2	N-2 contingency analysis
	OPTMAC(2) = 3	N-3 contingency analysis
	OPTMAC(2) = -3	N-3 contingency analysis
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single unit and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations
	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	
	OPTRNK(1)	include ties from subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable
	OPTRNK(2)	contingency rating (rating set program option setting by default)
	OPTRNK(2) = 1	rate A
	OPTRNK(2) = 2	rate B
	OPTRNK(2) = 3	rate C
	OPTRNK(3)	convert ratings to MW (0 by default)
	OPTRNK(3) = 0	disable
	OPTRNK(3) = 1	enable
	OPTRNK(4)	machine overload contingencies (0 by default)
	OPTRNK(4) = 0	disable
	OPTRNK(4) = 1	enable
	OPTRNK(5)	minimum of machine overload contingencies to evaluate (>=0, 1 by default)
	OPTRNK(6)	maximum of machine overload contingencies to evaluate (>=0, 10 by default)
	OPTRNK(7)	cutoff of machine overload contingencies to evaluate (>=0, 5 by default)
	OPTRNK(8)	line overload contingencies (0 by default)
	OPTRNK(8) = 0	disable
	OPTRNK(8) = 1	enable
	OPTRNK(9)	minimum of line overload contingencies to evaluate (>=0, 1 by default)
	OPTRNK(10)	maximum of line overload contingencies to evaluate (>=0, 10 by default)

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	OPTRNK(11)	cutoff of line overload contingencies (>=0, 5 by default)
	OPTRNK(12)	voltage collapse line contingencies (0 by default)
	OPTRNK(12) = 0	disable
	OPTRNK(12) = 1	enable
	OPTRNK(13)	minimum of voltage collapse line contingencies to evaluate (>=0, 1 by default)
	OPTRNK(14)	maximum of voltage collapse line contingencies to evaluate (>=0, 10 by default)
	OPTRNK(15)	cutoff of voltage collapse line contingencies (>=0, 5 be default)
	OPTRNK(16)	include all branches in voltage collapse ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating
	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTCOR (8)	array of 8 elements specifying corrective actions options(input).	
	OPTCOR(1)	run corrective action (0 by default)
	OPTCOR(1) = 0	disable
	OPTCOR(1) = 1	enable
	OPTCOR(2)	number power flows (1 by default)
	OPTCOR(3)	generator control flag (1 by default)
	OPTCOR(3) = 0	disable
	OPTCOR(3) = 1	enable
	OPTCOR(4)	load control flag (0 by default)
	OPTCOR(4) = 0	disable
	OPTCOR(4) = 1	enable
	OPTCOR(5)	phase shifter control flag (1 by default)
	OPTCOR(5) = 0	disable
	OPTCOR(5) = 1	enable
	OPTCOR(6)	off-line generator control flag (0 by default)
	OPTCOR(6) = 0	disable
	OPTCOR(6) = 1	enable
	OPTCOR(7)	tap setting adjustment flag (0 by default)
	OPTCOR(7) = 0	disable
	OPTCOR(7) = 1	enable
	OPTCOR(8)	switched shunt control flag (0 by default)
	OPTCOR(8) = 0	disable
	OPTCOR(8) = 1	enable

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Real VALUES (11)	array of 11 elements specifying ranking parameters and corrective actions parameters(input).	
	VALUES(1)	small reactance threshold, (0<= VALUES (1)<=0.005, 0.0005 by default)
	VALUES(2)	percent of flow rating, (0<= VALUES (2)<=200.0, 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0, 0.0 by default)
	VALUES(4)	bus voltage violation tolerance (0.1 by default)
	VALUES(5)	branch overload violation tolerance (0.1 by default)
	VALUES(6)	generator control weighting factor (1.0 by default)
	VALUES(7)	load control weighting factor (1.0 by default)
	VALUES(8)	phase shifter control weighting factor (1.0 by default)
	VALUES(9)	off-line generator control weighting factor (1.0 by default)
	VALUES(10)	tap setting adjustment weighting factor (1.0 by default)
	VALUES(11)	switched shunt control weighting factor (1.0 by default)
Character*32 LABELS (8)	array of 8 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default, no default allowed if OPTACC(8) is 1)
	LABELS(2)	label of contingency subsystem (blank by default)
	LABELS(3)	label of generator control subsystem (blank by default, no default allowed if OPTCOR(3) is 1)
	LABELS(4)	label of load control subsystem (blank by default, no default allowed if OPTCOR(4) is 1)
	LABELS(5)	label of phase shifter control subsystem (blank by default, no default allowed if OPTCOR(5) is 1)
	LABELS(6)	off-line generator control subsystem (blank by default; no default allowed if OPTCOR(6) is 1)
	LABELS(7)	tap setting adjustment subsystem (blank by default; no default allowed if OPTCOR(7) is 1)
	LABELS(8)	switched shunt control subsystem (blank by default; no default allowed if OPTCOR(8) is 1)
Character*260 DFXFILE	name of Distribution Factor Data File (input; no default allowed).	

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Character*260 ACCFILE	name of Contingency Solution Output File (input; no default allowed).
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).
Character*260 INLFILE	name of Unit Inertia and Governor Data File (input; blank for none).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	invalid TOL value
IERR = 2	invalid OPTACC value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
IERR = 9	buses with bus type code and series element status inconsistencies
IERR = 10	invalid OPTMAC value
IERR = 11	invalid OPTRNK value
IERR = 12	invalid OPTCOR value
IERR = 13	invalid VALUES value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multisection line members
IERR = 50	error opening Unit Inertia Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.133. MACCC_WITH_TRIP

This API routine is obsolete. It has been replaced by the API routine maccc_with_trip_2 .

Use this API to run multiple level contingency analysis with tripping simulation.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MACCC_WITH_TRIP  TOL OPTACC(1)..OPTACC(8) OPTMAC(1).. OPTMAC(41)
 OPTRNK(1).. OPTRNK(17) OPTTRP(1)..OPTTRP(3) VALUES(1).. VALUES(3)
 LABELS(1) LABELS(2) DFXFILE ACCFILE THRFILE INLFILE TRPFILE
```

Python command syntax:

```
ierr = maccc_with_trip(tol, optacc, optmac, optrnk, opttrp, values,
 labels, dfxfile, accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL MACCC_WITH_TRIP(TOL, OPTACC, OPTMAC, OPTRNK, OPTTRP, VALUES,  LABELS,
 DFXFILE, ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (8)	array of 8 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)

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	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	dispatch mode (0 by default)
	OPTACC(8) = 0	disable
	OPTACC(8) = 1	subsystem machines (reserve)
	OPTACC(8) = 2	subsystem machines (pmax)
	OPTACC(8) = 3	subsystem machines (inertia)
	OPTACC(8) = 4	subsystem machines (governor droop)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	run multiple contingency analysis (1 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(2)	contingency evaluation level (1<=OPTMAC (2)<=3; 1 by default)
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single machine and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations, otherwise
	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	
	OPTRNK(1)	include ties from contingency subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable
	OPTRNK(2)	contingency rating (rating set program option setting by default)

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OPTRNK(2) = 1	rate A
OPTRNK(2) = 2	rate B
OPTRNK(2) = 3	rate C
OPTRNK(3)	convert ratings to MW (0 by default)
OPTRNK(3) = 0	disable
OPTRNK(3) = 1	enable
OPTRNK(4)	include machine contingencies from overload ranking (0 by default)
OPTRNK(4) = 0	disable
OPTRNK(4) = 1	enable
OPTRNK(5)	minimum of machine contingencies to evaluate (>=0; 1 by default)
OPTRNK(6)	maximum of machine contingencies to evaluate (>=0; 10 by default)
OPTRNK(7)	non-failure cutoff of machine contingencies (>=0; 5 by default)
OPTRNK(8)	include branch contingencies from overload ranking (0 by default)
OPTRNK(8) = 0	disable
OPTRNK(8) = 1	enable
OPTRNK(9)	minimum of branch contingencies from overload ranking to evaluate (>=0; 1 by default)
OPTRNK(10)	maximum of branch contingencies from overload ranking to evaluate (>=0; 10 by default)
OPTRNK(11)	non-failure cutoff of branch contingencies from overload ranking (>=0; 5 by default)
OPTRNK(12)	include branch contingencies from voltage depression ranking (0 by default)
OPTRNK(12) = 0	disable
OPTRNK(12) = 1	enable
OPTRNK(13)	minimum of branch contingencies from voltage depression ranking to evaluate (>=0; 1 by default)
OPTRNK(14)	maximum of branch contingencies from voltage depression ranking to evaluate (>=0; 10 by default)
OPTRNK(15)	non-failure cutoff of branch contingencies from voltage depression ranking (>=0; 5 by default)
OPTRNK(16)	include all branches in voltage depression ranking (0 by default)
OPTRNK(16) = 0	include only with non-zero rating
OPTRNK(16) = 1	all

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	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Real VALUES (3)	array of 3 elements specifying ranking parameters(input).	
	VALUES(1)	small reactance threshold (0<= VALUES (1)<=0.005; 0.0005 by default)
	VALUES(2)	percent of rating (0<= VALUES (1)<=200.0; 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0; 0.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of contingency subsystem (blank by default)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation subsystem is not defined

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IERR = 7	too many islands in base case
IERR = 8	invalid OPTMAC value
IERR = 9	invalid OPTRNK value
IERR = 10	invalid VALUES value
IERR = 11	invalid OPTTRP value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 49	error opening Tripping Element Data File
IERR = 50	error opening Unit Inertia and Governor Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.134. MACCC_WITH_TRIP_2

Use this API to run the second version of multiple level contingency analysis with tripping simulation.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MACCC_WITH_TRIP_2  TOL OPTACC(1)..OPTACC(10) OPTMAC(1).. OPTMAC(41)
 OPTRNK(1).. OPTRNK(17) OPTTRP(1)..OPTTRP(3) VALUES(1).. VALUES(3)
 LABELS(1) LABELS(2) DFXFILE ACCFILE THRFILE INLFILE TRPFILE
```

Python command syntax:

```
ierr = maccc_with_trip_2(tol, optacc, optmac, optrnk, opttrp, values,
 labels, dfxfile, accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL MACCC_WITH_TRIP_2(TOL, OPTACC, OPTMAC, OPTRNK, OPTTRP, VALUES,
 LABELS, DFXFILE, ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, TOLN).	
Integer OPTACC (10)	array of 10 elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable
	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable

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	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTACC(8) = 0	stall
	OPTACC(8) = 1	trip
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable
	OPTACC(10) = 1	subsystem machines (reserve)
	OPTACC(10) = 2	subsystem machines (pmax)
	OPTACC(10) = 3	subsystem machines (inertia)
	OPTACC(10) = 4	subsystem machines (governor droop)
Integer OPTMAC (41)	array of 41 elements specifying contingency evaluation level and combinations(input).	
	OPTMAC(1)	run multiple contingency analysis (1 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(2)	contingency evaluation level (1 by default). If it is 2 or 3, a newly ordered event list is built for the next level after solving AC power flow each case at current level; if it is -2 or -3, the event list at the first level is used for any higher levels)
	OPTMAC(2) = 1	N-1 contingency analysis
	OPTMAC(2) = 2	N-2 contingency analysis
	OPTMAC(2) = -2	N-2 contingency analysis
	OPTMAC(2) = 3	N-3 contingency analysis
	OPTMAC(2) = -3	N-3 contingency analysis

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	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single machine and single branch outages respectively
	OPTMAC(3) = 1	include the contingency combinations, otherwise
	OPTMAC(3) = 0	(0 by default)
Integer OPTRNK (17)	array of 17 elements specifying ranking options(input).	
	OPTRNK(1)	include ties from contingency subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable
	OPTRNK(2)	contingency rating (rating set program option setting by default)
	OPTRNK(2) = 1	rate A
	OPTRNK(2) = 2	rate B
	OPTRNK(2) = 3	rate C
	OPTRNK(3)	convert ratings to MW (0 by default)
	OPTRNK(3) = 0	disable
	OPTRNK(3) = 1	enable
	OPTRNK(4)	include machine contingencies from overload ranking (0 by default)
	OPTRNK(4) = 0	disable
	OPTRNK(4) = 1	enable
	OPTRNK(5)	minimum of machine contingencies to evaluate (>=0; 1 by default)
	OPTRNK(6)	maximum of machine contingencies to evaluate (>=0; 10 by default)
	OPTRNK(7)	non-failure cutoff of machine contingencies (>=0; 5 by default)
	OPTRNK(8)	include branch contingencies from overload ranking (0 by default)
	OPTRNK(8) = 0	disable
	OPTRNK(8) = 1	enable
	OPTRNK(9)	minimum of branch contingencies from overload ranking to evaluate (>=0; 1 by default)
	OPTRNK(10)	maximum of branch contingencies from overload ranking to evaluate (>=0; 10 by default)

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	OPTRNK(11)	non-failure cutoff of branch contingencies from overload ranking (>=0; 5 by default)
	OPTRNK(12)	include branch contingencies from voltage depression ranking (0 by default)
	OPTRNK(12) = 0	disable
	OPTRNK(12) = 1	enable
	OPTRNK(13)	minimum of branch contingencies from voltage depression ranking to evaluate (>=0; 1 by default)
	OPTRNK(14)	maximum of branch contingencies from voltage depression ranking to evaluate (>=0; 10 by default)
	OPTRNK(15)	non-failure cutoff of branch contingencies from voltage depression ranking (>=0; 5 by default)
	OPTRNK(16)	include all branches in voltage depression ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating
	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTTRP (3)	array of 3 elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Real VALUES (3)	array of 3 elements specifying ranking parameters(input).	
	VALUES(1)	small reactance threshold (0<= VALUES (1)<=0.005; 0.0005 by default)
	VALUES(2)	percent of rating (0<= VALUES (1)<=200.0; 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0; 0.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0)
	LABELS(2)	label of contingency subsystem (blank by default)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	

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Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).
Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	invalid TOL value
IERR = 2	invalid OPTACC value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	generation subsystem is not defined
IERR = 7	too many islands in base case
IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
IERR = 9	buses with bus type code and series element status inconsistencies
IERR = 10	invalid OPTMAC value
IERR = 11	invalid OPTRNK value
IERR = 12	invalid VALUES value
IERR = 13	invalid OPTTRP value
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-sec-tion line members
IERR = 49	error opening Tripping Element Data File
IERR = 50	error opening Unit Inertia and Governor Data File
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 54	prerequisite requirements for function not met

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1.135. MACCC_WITH_TRIP_PARALLEL

Use this API to run the multiple level contingency analysis with tripping simulation in parallel when the number of contingency processors in Program Settings is greater than 1. User-specified and automatically selected contingencies from built-in contingency ranking functions are evaluated individually and in combination with each other as overlapping outages up to the specified evaluation level.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_MACCC_WITH_TRIP_PARALLEL  TOL OPTACC(1)..OPTACC(10)
 OPTMAC(1)..OPTMAC(41) OPTRNK(1)..OPTRNK(17) OPTTRP(1)..OPTTRP(3)
 VALUES(1)..VALUES(3) LABELS(1)..LABELS(2) DFXFILE ACCFILE THRFILE INLFILE
 TRPFILE
```

Python command syntax:

```
ierr = maccc_with_trip_parallel(tol, optacc, optmac, optrnk, opttrp,
 values, labels, dfxfile, accfile, thrfile, inlfile, trpfile)
```

Fortran command syntax:

```
CALL MACCC_WITH_TRIP_PARALLEL(TOL, OPTACC, OPTMAC, OPTRNK, OPTTRP, VALUES,
 LABELS, DFXFILE, ACCFILE, THRFILE, INLFILE, TRPFILE, IERR)
```

Where:

Real TOL	mismatch tolerance (input; Newton solution convergence tolerance, by default).	
Integer OPTACC (10)	array of elements specifying power flow solution options(input).	
	OPTACC(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTACC(1) = 0	disable
	OPTACC(1) = 1	enable stepping adjustment
	OPTACC(1) = 2	enable direct adjustment
	OPTACC(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTACC(2) = 0	disable
	OPTACC(2) = 1	enable using tie line flows only in calculating area interchange
	OPTACC(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTACC(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTACC(3) = 0	disable
	OPTACC(3) = 1	enable
	OPTACC(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTACC(4) = 0	disable
	OPTACC(4) = 1	enable

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	OPTACC(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTACC(5) = 0	disable
	OPTACC(5) = 1	enable
	OPTACC(5) = 2	enable continuous mode, disable discrete mode
	OPTACC(6)	solution method flag (0 by default)
	OPTACC(6) = 0	FDNS
	OPTACC(6) = 1	FNSL
	OPTACC(6) = 2	optimized FDNS
	OPTACC(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTACC(7) = 0	disable
	OPTACC(7) = 1	enable
	OPTACC(8)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTACC(8) = 0	stall
	OPTACC(8) = 1	trip
	OPTACC(9)	induction machine failure flag (0 by default)
	OPTACC(9) = 0	treat contingency as non-converged if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(9) = 1	treat contingency as solved if it converges, even if any induction machines are placed in the "stalled" or "tripped" state
	OPTACC(10)	dispatch mode (0 by default)
	OPTACC(10) = 0	disable
	OPTACC(10) = 1	subsystem machines (reserve)
	OPTACC(10) = 2	subsystem machines (pmax)
	OPTACC(10) = 3	subsystem machines (inertia)
	OPTACC(10) = 4	subsystem machines (governor droop)
Integer OPTMAC (41)	array of elements specifying contingency evaluation level and combinations. (input).	
	OPTMAC(1)	run multiple contingency analysis (1 by default)
	OPTMAC(1) = 0	disable
	OPTMAC(1) = 1	enable
	OPTMAC(2)	contingency evaluation level (1 by default). If it is 2 or 3, a newly ordered event list is built for the next level after solving AC power flow each case at current level; if it is -2 or -3, the event list at the first level is used for any higher levels by default)
	OPTMAC(2) = -3	N-3 contingency analysis

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	OPTMAC(2) = -2	N-2 contingency analysis
	OPTMAC(2) = 1	N-1 contingency analysis
	OPTMAC(2) = 2	N-2 contingency analysis
	OPTMAC(2) = 3	N-3 contingency analysis
	OPTMAC(3)	each element represents a type of contingency combinations in the following order. {S, U, B, SS, SU, SB, US, UU, UB, BS, BU, BB, SSS, SSB, SSU, SBS, SBB, SBU, SUS, SUB, SUU, USS, USU, USB, UUS, UUB, UUU, UBS, UBU, UBB, BSS, BSU, BSB, SUS, SUB, SUU, BBS, BBU, BBB}, where S, U and B are user specified cases, single machine and single branch outages respectively. (0 by default)
	OPTMAC(3) = 0	not include the contingency combinations
	OPTMAC(3) = 1	include the contingency combinations
Integer OPTRNK (17)	array of elements specifying ranking options(input).	
	OPTRNK(1)	include ties from contingency subsystem (0 by default)
	OPTRNK(1) = 0	disable
	OPTRNK(1) = 1	enable
	OPTRNK(2)	contingency rating (rating set program option setting by default)
	OPTRNK(2) = 1	rate A
	OPTRNK(2) = 2	rate B
	OPTRNK(2) = 3	rate C
	OPTRNK(3)	convert ratings to MW (0 by default)
	OPTRNK(3) = 0	disable
	OPTRNK(3) = 1	enable
	OPTRNK(4)	include machine contingencies from overload ranking (0 by default)
	OPTRNK(4) = 0	disable
	OPTRNK(4) = 1	enable
	OPTRNK(5)	minimum of machine contingencies to evaluate (>=0; 1 by default)
	OPTRNK(6)	maximum of machine contingencies to evaluate (>=0; 10 by default)
	OPTRNK(7)	non-failure cutoff of machine contingencies ((>=0; 5 by default)
	OPTRNK(8)	include branch contingencies from overload ranking (0 by default)
	OPTRNK(8) = 0	disable
	OPTRNK(8) = 1	enable
	OPTRNK(9)	minimum of branch contingencies from overload ranking to evaluate (>=0; 1 by default)

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	OPTRNK(10)	maximum of branch contingencies from overload ranking to evaluate (>=0; 10 by default)
	OPTRNK(11)	non-failure cutoff of branch contingencies from overload ranking (>=0; 5 by default)
	OPTRNK(12)	include branch contingencies from voltage depression ranking (0 by default)
	OPTRNK(12) = 0	disable
	OPTRNK(12) = 1	enable
	OPTRNK(13)	minimum of branch contingencies from voltage depression ranking to evaluate (>=0; 1 by default)
	OPTRNK(14)	maximum of branch contingencies from voltage depression ranking to evaluate (>=0; 10 by default)
	OPTRNK(15)	non-failure cutoff of branch contingencies from voltage depression ranking (>=0; 5 by default)
	OPTRNK(16)	include all branches in voltage depression ranking (0 by default)
	OPTRNK(16) = 0	include only with non-zero rating
	OPTRNK(16) = 1	all
	OPTRNK(17)	include contingencies that create swingless island (0 by default)
	OPTRNK(17) = 0	disable
	OPTRNK(17) = 1	enable
Integer OPTTRP (3)	array of elements specifying tripping simulation options(input).	
	OPTTRP(1)	tripping simulation flag (0 by default)
	OPTTRP(1) = 0	disable
	OPTTRP(1) = 1	enable
	OPTTRP(2)	number of tripping events per power flow (1 by default)
	OPTTRP(3)	number of power flows per contingency (1 by default)
Real VALUES (3)	array of elements specifying ranking parameters(input).	
	VALUES(1)	small reactance threshold (0<= VALUES(1)<=0.005; 0.0005 by default)
	VALUES(2)	percent of rating (0<= VALUES (1)<=200.0; 100.0 by default)
	VALUES(3)	MW threshold for islanding contingencies (>=0; 0.0 by default)
Character*32 LABELS (2)	array of elements specifying subsystem labels(input).	
	LABELS(1)	label of generation dispatch subsystem (blank by default; no default allowed if OPTACC(8) is not 0 by default)

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	LABELS(2)	label of contingency subsystem (blank by default) (blank by default)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of the Contingency Solution Output File (input; no default allowed).	
Character*260 THRFILE	name of the Load Throwover Data File (input; blank by default).	
Character*260 INLFILE	name of the Unit Inertia and Governor Data File (input; blank by default).	
Character*260 TRPFILE	name of the Tripping Element Data File (input; blank for none; no default allowed if OPTTRP(1) is 1). (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid TOL value
	IERR = 2	invalid OPTACC value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation dispatch subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	invalid OPTMAC value
	IERR = 11	invalid OPTRNK value
	IERR = 12	invalid VALUES value
	IERR = 13	invalid OPTTRP value
	IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 49	error opening Tripping Element Data File
	IERR = 50	error opening Unit Inertia and Governor Data File
	IERR = 51	error opening Contingency Solution Output File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 54	prerequisite requirements for function not met.

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1.136. MATRIX_GROWTH_FACTOR

Use this API to specify or return the value of the matrix growth factor option setting.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_MATRIX_GROWTH_FACTOR  RVAL
```

Python command syntax:

```
ierr = matrix_growth_factor(rval)

ierr, rval = matrix_growth_factor()
```

Fortran command syntax:

```
CALL MATRIX_GROWTH_FACTOR_API(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 2.0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.137. MBID2DC

Use this API to change the name of the specified two-terminal dc line.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBID2DC  DC  DCNEW
```

Python command syntax:

```
ierr = mbid2dc(dc, dcnew)
```

Fortran command syntax:

```
CALL MBID2DC(DC, DCNEW, IERR)

CALL MBID2DC_II(NUM, NUMNEW, IERR)

CALL MBID2DC_IC(NUM, NAMENEW, IERR)

CALL MBID2DC_CI(NAME, NUMNEW, IERR)

CALL MBID2DC_CC(NAME, NAMENEW, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which the dc line name is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer or Character*12 DCNEW	quoted new name to be assigned to the two-terminal dc line; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which the dc line name is to be changed (input; no default allowed).	
Integer NUMNEW	new number to be assigned to the two-terminal dc line as a left-justified name (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line for which the dc line name is to be changed (input; no default allowed).	
Character*12 NAMENEW	new dc line name to be assigned to the two-terminal dc line (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	dc line DC not found
	IERR = 2	dc line DCNEW already exists
	IERR = 3	a non-blank new identifier must be specified
	IERR = 4	prerequisite requirements for API are not met

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1.138. MBID3WND

Use this API to change the identifier of the specified three-winding transformer.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBID3WND  IBUS  JBUS1  JBUS2  'ID'  'IDNEW'
```

Python command syntax:

```
ierr = mbid3wnd(ibus, jbus1, jbus2, id, idnew)
```

Fortran command syntax:

```
CALL MBID3WND(IBUS, JBUS1, JBUS2, ID, IDNEW, IERR)
```

Where:

Integer IBUS	one of the buses connected by the three-winding transformer (input; no default allowed).	
Integer JBUS1	other bus connected by the three-winding transformer (input; no default allowed).	
Integer JBUS2	other bus connected by the three-winding transformer (input; no default allowed).	
Character*2 ID	present circuit identifier of the three-winding transformer for which the identifier is to be changed (input; '1').	
Character*2 IDNEW	new identifier to be assigned to the three-winding transformer (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	circuit ID bus IBUS to JBUS1 to JBUS2 not found
	IERR = 3	circuit IDNEW bus IBUS to JBUS1 to JBUS2 already exists
	IERR = 4	invalid new identifier; cannot start with &, * or @
	IERR = 5	a non-blank new identifier must be specified
	IERR = 6	prerequisite requirements for API are not met

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1.139. MBIDATRN

Use this API to change the identifier of the specified inter-area transfer.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBIDATRN  FRMAREA  TOAREA  'ID'  'IDNEW'
```

Python command syntax:

```
ierr = mbidatrn(frmarea, toarea, id, idnew)
```

Fortran command syntax:

```
CALL MBIDATRN(FRMAREA, TOAREA, ID, IDNEW, IERR)
```

Where:

Integer FRMAREA	one of the areas involved in the transfer (input; no default allowed).	
Integer TOAREA	other area involved in the transfer (input; no default allowed).	
Character*1 ID	present inter-area transfer identifier (input; '1').	
Character*1 IDNEW	new identifier to be assigned to the inter-area transfer (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid area number
	IERR = 2	inter-area transfer ID from area FRMAREA to TOAREA not found
	IERR = 3	inter-area transfer ID from area FRMAREA to TOAREA already exists
	IERR = 4	invalid new identifier--must be 0 through 9 or A through Z
	IERR = 5	prerequisite requirements for API are not met

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1.140. MBIDBRN

Use this API to change the identifier of the specified non-transformer branch or two-winding transformer.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBIDBRN  IBUS  JBUS  'ID'  'IDNEW'
```

Python command syntax:

```
ierr = mbidbrn(ibus, jbus, id, idnew)
```

Fortran command syntax:

```
CALL MBIDBRN(IBUS, JBUS, ID, IDNEW, IERR)
```

Where:

Integer IBUS	one of the buses connected by the branch (input; no default allowed).	
Integer JBUS	other bus connected by the branch (input; no default allowed).	
Character*2 ID	present circuit identifier of the branch for which the identifier is to be changed (input; '1').	
Character*2 IDNEW	new identifier to be assigned to the branch (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	circuit ID from IBUS to JBUS not found
	IERR = 3	circuit ID from IBUS to JBUS already exists
	IERR = 4	invalid new identifier; cannot start with &, * or @
	IERR = 5	a non-blank new identifier must be specified
	IERR = 6	prerequisite requirements for API are not met

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1.141. MBIDFACTS

Use this API to change the name of the specified FACTS device.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBIDFACTS  FD  FDNEW
```

Python command syntax:

```
ierr = mbidfacts(fd, fdnew)
```

Fortran command syntax:

```
CALL MBIDFACTS(FD, FDNEW, IERR)

CALL MBIDFACTS_II(NUM, NUMNEW, IERR)

CALL MBIDFACTS_IC(NUM, NAMENEW, IERR)

CALL MBIDFACTS_CI(NAME, NUMNEW, IERR)

CALL MBIDFACTS_CC(NAME, NAMENEW, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device for which the FACTS device name is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer or Character*12 FDNEW	quoted new name to be assigned to the FACTS device; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device for which the FACTS device name is to be changed (input; no default allowed).	
Integer NUMNEW	new number to be assigned to the FACTS device as a left-justified name (input; no default allowed).	
Character*12 NAME	name of the FACTS device for which the FACTS device name is to be changed (input; no default allowed).	
Character*12 NAMENEW	new FACTS device name to be assigned to the FACTS device (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	FACTS device FD not found
	IERR = 2	FACTS device FDNEW already exists
	IERR = 3	a non-blank new identifier must be specified
	IERR = 4	prerequisite requirements for API are not met

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1.142. MBIDGNE

Use this API routine to change the GNE device name of the specified GNE device.

This API routine was first introduced in release 32.0.2.

Batch command syntax:

```
BAT_MBIDGNE  'NAME'  'NAMNEW'
```

Python command syntax:

```
ierr = mbidgne(name, namnew)
```

Fortran command syntax:

```
CALL MBIDGNE(NAME, NAMNEW, IERR)
```

Where:

Character*12 NAME	name of the GNE device for which the GNE device name is to be changed (input; no default allowed).	
Character*12 NAMNEW	new name to be assigned to the GNE device (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	GNE device NAME not found
	IERR = 2	GNE device NEWNAM already exists
	IERR = 3	a non-blank new identifier must be specified
	IERR = 4	prerequisite requirements for API are not met

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1.143. MBIDINDMAC

Use this API routine to change the identifier of the specified induction machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MBIDINDMAC  IBUS  'ID'  'IDNEW'
```

Python command syntax:

```
ierr = mbidindmac(ibus, id, idnew)
```

Fortran command syntax:

```
CALL MBIDINDMAC(IBUS, ID, IDNEW, IERR)
```

Where:

Integer IBUS	bus to which induction machine ID is connected (input; no default allowed).	
Character*2 ID	present induction machine identifier of the induction machine for which the identifier is to be changed (input; '1').	
Character*2 IDNEW	new identifier to be assigned to the induction machine (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	induction machine ID at bus IBUS not found
	IERR = 3	induction machine ID at bus IBUS already exists
	IERR = 4	a non-blank new identifier must be specified
	IERR = 5	prerequisite requirements for API are not met

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1.144. MBIDLOAD

Use this API to change the identifier of the specified load.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBIDLOAD  IBUS  'ID'  'IDNEW'
```

Python command syntax:

```
ierr = mbidload(ibus, id, idnew)
```

Fortran command syntax:

```
CALL MBIDLOAD(IBUS, ID, IDNEW, IERR)
```

Where:

Integer IBUS	bus to which the load ID is connected (input; no default allowed).	
Character*2 ID	present load identifier of the load for which the identifier is to be changed (input; '1').	
Character*2 IDNEW	new identifier to be assigned to the load (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	load ID at bus IBUS not found
	IERR = 3	load ID at bus IBUS already exists
	IERR = 4	a non-blank new identifier must be specified
	IERR = 5	prerequisite requirements for API are not met

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1.145. MBIDMAC

Use this API to change the identifier of the specified machine.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBIDMAC  IBUS  'ID'  'IDNEW'
```

Python command syntax:

```
ierr = mbidmac(ibus, id, idnew)
```

Fortran command syntax:

```
CALL MBIDMAC(IBUS, ID, IDNEW, IERR)
```

Where:

Integer IBUS	bus to which the machine ID is connected (input; no default allowed).	
Character*2 ID	present machine identifier of the machine for which the identifier is to be changed (input; '1').	
Character*2 IDNEW	new identifier to be assigned to the machine (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	machine ID at bus IBUS not found
	IERR = 3	machine ID at bus IBUS already exists
	IERR = 4	a non-blank new identifier must be specified
	IERR = 5	prerequisite requirements for API are not met

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1.146. MBIDMDC

Use this API to change the name of the specified multi-terminal dc line.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBIDMDC  DC  DCNEW
```

Python command syntax:

```
ierr = mbidmdc(dc, dcnew)
```

Fortran command syntax:

```
CALL MBIDMDC(DC, DCNEW, IERR)

CALL MBIDMDC_II(NUM, NUMNEW, IERR)

CALL MBIDMDC_IC(NUM, NAMENEW, IERR)

CALL MBIDMDC_CI(NAME, NUMNEW, IERR)

CALL MBIDMDC_CC(NAME, NAMENEW, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which the dc line name is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer or Character*12 DCNEW	quoted new name to be assigned to the multi-terminal dc line; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which the dc line name is to be changed (input; no default allowed).	
Integer NUMNEW	new number to be assigned to the multi-terminal dc line as a left-justified name (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line for which the dc line name is to be changed (input; no default allowed).	
Character*12 NAMENEW	new dc line name to be assigned to the multi-terminal dc line (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	dc line DC not found
	IERR = 2	dc line DCNEW already exists
	IERR = 3	a non-blank new identifier must be specified
	IERR = 4	prerequisite requirements for API are not met

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1.147. MBIDMSL

Use this API to change the identifier of the specified multi-section line.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MBIDMSL  IBUS  JBUS  'ID'  'IDNEW'
```

Python command syntax:

```
ierr = mbidmsl(ibus, jbus, id, idnew)
```

Fortran command syntax:

```
CALL MBIDMSL(IBUS, JBUS, ID, IDNEW, IERR)
```

Where:

Integer IBUS	one of the endpoint buses connected by the multi-section line (input; no default allowed).	
Integer JBUS	other endpoint bus connected by the multi-section line (input; no default allowed).	
Character*2 ID	present multi-section identifier of the multi-section line for which the identifier is to be changed (input; '&1').	
Character*2 IDNEW	new identifier to be assigned to the multi-section line (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	multi-section line ID from IBUS to JBUS not found
	IERR = 3	multi-section line ID from IBUS to JBUS already exists
	IERR = 4	invalid new identifier; must start with &
	IERR = 5	prerequisite requirements for API are not met

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1.148. MBIDSHUNT

Use this API to change the identifier of the specified fixed bus shunt.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_MBIDSHUNT  IBUS  'ID'  'IDNEW'
```

Python command syntax:

```
ierr = mbidshunt(ibus, id, idnew)
```

Fortran command syntax:

```
CALL MBIDSHUNT(IBUS, ID, IDNEW, IERR)
```

Where:

Integer IBUS	bus to which the shunt ID is connected (input; no default allowed).	
Character*2 ID	present shunt identifier of the shunt for which the identifier is to be changed (input; '1').	
Character*2 IDNEW	new identifier to be assigned to the shunt (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	shunt ID at bus IBUS not found
	IERR = 3	shunt ID at bus IBUS already exists
	IERR = 4	a non-blank new identifier must be specified
	IERR = 5	prerequisite requirements for API are not met

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1.149. MBIDVSC

Use this API to change the VSC dc line name of the specified VSC dc line.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_MBIDVSC  'NAME'  'NAMNEW'
```

Python command syntax:

```
ierr = mbidvsc(name, namnew)
```

Fortran command syntax:

```
CALL MBIDVSC(NAME, NAMNEW, IERR)
```

Where:

Character*12 NAME	name of the VSC dc line for which the VSC dc line name is to be changed (input; no default allowed).	
Character*12 NAMNEW	new name to be assigned to the VSC dc line (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	dc line NAME not found
	IERR = 2	dc line NEWNAM already exists
	IERR = 3	a non-blank new identifier must be specified
	IERR = 4	prerequisite requirements for API are not met

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1.150. MCRE

Use this API to read a Machine Impedance Data File and add the data specified in it to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MCRE  STATUS(1)  STATUS(2)  IFILE
```

Python command syntax:

```
ierr = mcre(status, ifile)
```

Fortran command syntax:

```
CALL MCREAPI(STATUS, IFILE, IERR)
```

Where:

Integer STATUS (2)	array of 2 elements specifying options(input).	
	STATUS(1)	new machine status option (0 by default)
	STATUS(1) = 0	set from STAT field on input record
	STATUS(1) = 1	set from status of other machines at the bus
	STATUS(2)	option for treatment of machines with no data record specified and other machines at the bus with record specified (0 by default)
	STATUS(2) = 0	place out-of-service
	STATUS(2) = 1	delete
Character*260 IFILE	name of machine impedance data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid STATUS value
	IERR = 2	generators are converted
	IERR = 3	error opening IFILE
	IERR = 4	prerequisite requirements for API are not met

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1.151. MODR

Use this API to uniformly increase or decrease the line resistances of in-service nontransformer branches. It must first be called once with APIOPT set to 1, then once for each subsystem to be processed with APIOPT set to 2, and finally once with APIOPT set to 3.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MODR  SID  ALL  APIOPT  IRAT  RLOAD  FMAX
```

Python command syntax:

```
ierr, nchng = modr(sid, all, apiopt, irat, rload, fmax)
```

Fortran command syntax:

```
CALL MODRNW(SID, ALL, APIOPT, IRAT, RLOAD, FMAX, NCHNG, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; used only when APIOPT is 2; 0).	
Integer ALL	all buses or specified subsystem flag (input; used only when APIOPT is 2; 1).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer APIOPT	mode of operation of the API (input; no default allowed).	
	APIOPT = 1	initialize system for line resistance modification
	APIOPT = 2	modify line resistances
	APIOPT = 3	postprocessing housekeeping
Integer IRAT	rating set on which the present values of branch resistances are based (input; used only when APIOPT is 1; default rating set program option setting).	
	IRAT = 1	RATEA
	IRAT = 2	RATEB
	IRAT = 3	RATEC
Real RLOAD	base line loading in percent. 0.0001 < RLOAD < 2000.0 (input; used only when APIOPT is 1; 100.0).	
Real FMAX	line resistance scaling factor in pu 0.0 < FMAX < 0.1 (input; used only when APIOPT is 1; 0.01).	
Integer NCHNG	returned as the number of branch resistances changed (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined

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IERR = 2	invalid ALL value
IERR = 3	invalid APIOPT value
IERR = 4	invalid IRAT value
IERR = 5	invalid RLOAD value
IERR = 6	invalid FMAX value
IERR = 7	unexpected APIOPT value
IERR = 8	prerequisite requirements for API are not met

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1.152. MOVE3WND

Use this API to disconnect the third bus of a specified three-winding transformer from the bus to which it is currently connected, and reconnect it to a designated bus.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MOVE3WND  IBUS  JBUS1  JBUS2  'CKT'  NEWTOBUS2  'NEWCKT'
```

Python command syntax:

```
ierr = move3wnd(ibus, jbus1, jbus2, ckt, newtobus2, newckt)
```

Fortran command syntax:

```
CALL MOVE3WND(IBUS, JBUS1, JBUS2, CKT, NEWTOBUS2, NEWCKT, IERR)
```

Where:

Integer IBUS	one of the buses connected by the three-winding transformer; the transformer will remain connected to this bus (input; no default allowed).	
Integer JBUS1	other bus connected by the three-winding transformer; the transformer will remain connected to this bus (input; no default allowed).	
Integer JBUS2	third bus connected by the three-winding transformer; this connection will be severed and the transformer connected to bus NEWTOBUS2 (input; no default allowed).	
Character*2 CKT	circuit identifier of the three-winding transformer to be rerouted (input; '1').	
Integer NEWTOBUS2	number of the bus to which the three-winding transformer is to be connected (input; no default allowed).	
Character*2 NEWCKT	new circuit identifier to be assigned to the rerouted three-winding transformer (input; CKT).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	three-winding transformer not found
	IERR = 3	new connection bus must not be the present from or to buses
	IERR = 4	circuit NEWCKT from IBUS to JBUS1 to NEWTOBUS2 already exists
	IERR = 5	invalid new identifier; cannot start with &, * or @
	IERR = 6	invalid new identifier; must not be blank
	IERR = 7	prerequisite requirements for API are not met

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1.153. MOVEBRN

Use this API to disconnect the to bus of a specified non-transformer branch or two-winding transformer from the bus to which it is currently connected, and reconnect it to a designated bus.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MOVEBRN  IBUS  JBUS  'CKT'  NEWTOBUS  'NEWCKT'
```

Python command syntax:

```
ierr = movebrn(ibus, jbus, ckt, newtobus, newckt)
```

Fortran command syntax:

```
CALL MOVEBRN(IBUS, JBUS, CKT, NEWTOBUS, NEWCKT, IERR)
```

Where:

Integer IBUS	one of the buses connected by the branch; the branch will remain connected to this bus (input; no default allowed).	
Integer JBUS	other bus connected by the branch; this connection will be severed and the branch connected to bus NEWTOBUS (input; no default allowed).	
Character*2 CKT	circuit identifier of the branch to be rerouted (input; '1').	
Integer NEWTOBUS	number of the bus to which the branch is to be connected (input; no default allowed).	
Character*2 NEWCKT	new circuit identifier to be assigned to the rerouted branch (input; CKT).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	circuit CKT from IBUS to JBUS not found
	IERR = 3	new connection bus must not be the present from or to bus
	IERR = 4	circuit NEWCKT from IBUS to NEWTOBUS already exists
	IERR = 5	invalid new identifier; cannot start with &, * or @
	IERR = 6	invalid new identifier; must not be blank
	IERR = 7	prerequisite requirements for API are not met

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1.154. MOVEINDMAC

Use this API routine to disconnect the specified induction machine from the bus to which it is currently connected, and reconnect it to a designated bus.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MOVEINDMAC  IBUS  'ID'  NEWBUS  'NEWID'
```

Python command syntax:

```
ierr = moveindmac(ibus, id, newbus, newid)
```

Fortran command syntax:

```
CALL MOVEINDMAC(IBUS, ID, NEWBUS, NEWID, IERR)
```

Where:

Integer IBUS	bus to which induction machine ID is connected (input; no default allowed).	
Character*2 ID	machine identifier of the induction machine to be moved (input; '1').	
Integer NEWBUS	number of the bus to which the induction machine is to be moved (input; ID).	
Character*2 NEWID	new induction machine identifier to be assigned to the relocated induction machine (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	induction machine not found
	IERR = 3	IBUS and NEWBUS are the same bus
	IERR = 4	induction machine NEWID at bus NEWBUS already exists
	IERR = 5	invalid new identifier; must not be blank
	IERR = 6	prerequisite requirements for API are not met

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1.155. MOVEINDMACS

Use this API routine to disconnect all of the induction machines at the specified bus, and reconnect them to a designated bus. Bus TOBUS must not have any induction machines connected to it at the time this API routine is called.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MOVEINDMACS  IBUS  NEWBUS
```

Python command syntax:

```
ierr = moveindmacs(ibus, newbus)
```

Fortran command syntax:

```
CALL MOVEINDMACS(IBUS, NEWBUS, IERR)
```

Where:

Integer IBUS	bus at which one or more induction machines currently exist (input; no default allowed).	
Integer NEWBUS	bus to which the induction machines on bus IBUS are to be moved (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no induction machines at bus IBUS
	IERR = 3	IBUS and NEWBUS are the same bus
	IERR = 4	NEWBUS already has induction machines (move individual machines)
	IERR = 5	prerequisite requirements for API are not met

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1.156. MOVELOAD

Use this API to disconnect the specified load from the bus to which it is currently connected, and reconnect it to a designated bus.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MOVELOAD  IBUS  'ID'  NEWBUS  'NEWID'
```

Python command syntax:

```
ierr = moveload(ibus, id, newbus, newid)
```

Fortran command syntax:

```
CALL MOVELOAD(IBUS, ID, NEWBUS, NEWID, IERR)
```

Where:

Integer IBUS	bus to which the load ID is connected (input; no default allowed).	
Character*2 ID	load identifier of the load to be moved (input; '1').	
Integer NEWBUS	number of the bus to which the load is to be moved (input; no default allowed).	
Character*2 NEWID	new load identifier to be assigned to the relocated load (input; ID).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	load not found
	IERR = 3	IBUS and NEWBUS are the same bus
	IERR = 4	load NEWID at bus NEWBUS already exists
	IERR = 5	invalid new identifier; must not be blank
	IERR = 6	prerequisite requirements for API are not met

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1.157. MOVELOADS

Use this API to disconnect all of the load from the specified bus, and reconnect it to a designated bus. Bus TOBUS must not have any load connected to it at the time this API is called.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MOVELOADS  IBUS  NEWBUS
```

Python command syntax:

```
ierr = moveloads(ibus, newbus)
```

Fortran command syntax:

```
CALL MOVELOADS(IBUS, NEWBUS, IERR)
```

Where:

Integer IBUS	bus on which one or more loads currently exist (input; no default allowed).	
Integer NEWBUS	bus to which the loads on bus IBUS are to be moved (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no loads at bus IBUS
	IERR = 3	IBUS and NEWBUS are the same bus
	IERR = 4	NEWBUS already has loads (move individual loads)
	IERR = 5	prerequisite requirements for API are not met

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1.158. MOVEMAC

Use this API to disconnect the specified machine from the bus to which it is currently connected, and re-connect it to a designated bus.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MOVEMAC  IBUS  'ID'  NEWBUS  'NEWID'
```

Python command syntax:

```
ierr = movemac(ibus, id, newbus, newid)
```

Fortran command syntax:

```
CALL MOVEMAC(IBUS, ID, NEWBUS, NEWID, IERR)
```

Where:

Integer IBUS	bus to which the machine ID is connected (input; no default allowed).	
Character*2 ID	machine identifier of the machine to be moved (input; '1').	
Integer NEWBUS	number of the bus to which the machine is to be moved (input; ID).	
Character*2 NEWID	new machine identifier to be assigned to the relocated machine (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	machine not found
	IERR = 3	IBUS and NEWBUS are the same bus
	IERR = 4	machine NEWID at bus NEWBUS already exists
	IERR = 5	plant table full (move machine to an existing plant)
	IERR = 6	invalid new identifier; must not be blank
	IERR = 7	prerequisite requirements for API are not met

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1.159. MOVEPLNT

Use this API to disconnect all of the machines from the specified bus, and reconnect it to a designated bus. Bus TOBUS must not have any plant data associated with it at the time this API is called.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MOVEPLNT  IBUS  NEWBUS
```

Python command syntax:

```
ierr = moveplnt(ibus, newbus)
```

Fortran command syntax:

```
CALL MOVEPLNT(IBUS, NEWBUS, IERR)
```

Where:

Integer IBUS	bus on which plant data currently exists (input; no default allowed).	
Integer NEWBUS	bus to which the plant on bus IBUS is to be moved (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no plant at bus IBUS
	IERR = 3	IBUS and NEWBUS are the same bus
	IERR = 4	NEWBUS already has plant data (move individual machines)
	IERR = 5	prerequisite requirements for API are not met

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1.160. MOVESHUNT

Use this API to disconnect the specified fixed shunt from the bus to which it is currently connected, and reconnect it to a designated bus.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_MOVESHUNT  IBUS  'ID'  NEWBUS  'NEWID'
```

Python command syntax:

```
ierr = moveshunt(ibus, id, newbus, newid)
```

Fortran command syntax:

```
CALL MOVESHUNT(IBUS, ID, NEWBUS, NEWID, IERR)
```

Where:

Integer IBUS	bus to which the shunt ID is connected (input; no default allowed).	
Character*2 ID	shunt identifier of the fixed shunt to be moved (input; '1').	
Integer NEWBUS	number of the bus to which the shunt is to be moved (input; no default allowed).	
Character*2 NEWID	new shunt identifier to be assigned to the relocated fixed shunt (input; ID).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	fixed shunt not found
	IERR = 3	IBUS and NEWBUS are the same bus
	IERR = 4	shunt NEWID at bus NEWBUS already exists
	IERR = 5	invalid new identifier; must not be blank
	IERR = 6	prerequisite requirements for API are not met

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1.161. MOVESHUNTS

Use this API to disconnect all of the fixed shunts from the specified bus and reconnect them to a designated bus. Bus TOBUS must not have any fixed shunt connected to it at the time this API is called.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_MOVESHUNTS  FRMBUS  NEWBUS
```

Python command syntax:

```
ierr = moveshunts(frmbus, newbus)
```

Fortran command syntax:

```
CALL MOVESHUNTS(FRMBUS, NEWBUS, IERR)
```

Where:

Integer FRMBUS	bus on which one or more fixed shunts currently exist (input; no default allowed).	
Integer NEWBUS	bus to which the fixed shunts on bus FRMBUS are to be moved (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no fixed shunts at bus FRMBUS
	IERR = 3	FRMBUS and NEWBUS are the same bus
	IERR = 4	NEWBUS already has fixed shunts (move individual shunts)
	IERR = 5	prerequisite requirements for API are not met

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1.162. MOVESWS

Use this API to disconnect the switched shunt load from the specified bus and reconnect it to a designated bus.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MOVESWS  IBUS  NEWBUS
```

Python command syntax:

```
ierr = movesws(ibus, newbus)
```

Fortran command syntax:

```
CALL MOVESWS(IBUS, NEWBUS, IERR)
```

Where:

Integer IBUS	bus on which a switched shunt currently exists (input; no default allowed).	
Integer NEWBUS	bus to which the switched shunt on bus IBUS is to be moved (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no switched shunt at bus IBUS
	IERR = 3	IBUS and NEWBUS are the same bus
	IERR = 4	prerequisite requirements for API are not met

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1.163. MSLV

Use this API to apply the modified Gauss-Seidel power flow calculation.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MSLV  OPTIONS(1)..OPTIONS(6)
```

Python command syntax:

```
ierr = mslv(options)
```

Fortran command syntax:

```
CALL MSLVAPI(OPTIONS, IERR)
```

Where:

Integer OPTIONS (6)	array of 6 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (use tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable
	OPTIONS(2)	area interchange adjustment flag (use area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	flat start flag (0 by default)
	OPTIONS(3) = 0	do not flat start
	OPTIONS(3) = 1	flat start
	OPTIONS(3) = 2	flat start, then estimate voltage magnitudes
	OPTIONS(3) = 3	flat start, then estimate voltage phase angles
	OPTIONS(3) = 4	flat start, then estimate voltage magnitudes and phase angles
	OPTIONS(4)	dc tap adjustment flag (use dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (use switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable

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	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	var limit flag (0 by default)
	OPTIONS(6) = 0	apply var limits
	OPTIONS(6) = 1	ignore var limits
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	generators are converted
	IERR = 3	buses in island(s) without a swing bus; use activity TREE
	IERR = 4	bus type code and series element status inconsistencies
	IERR = 5	prerequisite requirements for API are not met

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1.164. MSUM

Use this API to print a summary of mileage by owner.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MSUM
```

Python command syntax:

```
ierr = msum()
```

Fortran command syntax:

```
CALL MSUMAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 4	prerequisite requirements for API are not met

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1.165. MTDC

Use this API to produce a report of bus voltages and flows for each in-service multi-terminal dc line.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MTDC
```

Python command syntax:

```
ierr = mtdc()
```

Fortran command syntax:

```
CALL MTDCAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no unblocked multi-terminal dc lines in this case
	IERR = 2	prerequisite requirements for API are not met

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1.166. MULTISECTION_REPORTING

Use this API to specify or return the option to enable or disable multi-section line reporting.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MULTISECTION_REPORTING  IVAL
```

Python command syntax:

```
ierr = multisection_reporting(ival)

ierr, ival = multisection_reporting()
```

Fortran command syntax:

```
CALL MULTISECTION_LINE_REPORTING_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1 by default; else output).	
	IVAL = 0	off.
	IVAL = 1	on.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.167. MWMI

Use this API to apply the MAPP MW-mile calculation
.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MWMI  SID  ALL  STATUS(1)..STATUS(4)  VALUES(1)..VALUES(3)  INPFILE
 TRNFILE  OUTFILE
```

Python command syntax:

```
ierr = mwmi(sid, all, status, values, inpfile, trnfile, outfile)
```

Fortran command syntax:

```
CALL MWMIAPI(SID, ALL, STATUS, VALUES, INPFILE, TRNFILE, OUTFILE, IERR)
```

Where:

Integer SID	a valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Area subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all areas or specified subsystem flag (input; 1 by default).	
	ALL = 0	only areas in area subsystem SID are in the pool
	ALL = 1	all areas are in the pool
Integer STATUS (4)	array of 4 elements specifying solution options(input).	
	STATUS(1)	rating used (default rating set program option setting)
	STATUS(1) = 1	RATEA
	STATUS(1) = 2	RATEB
	STATUS(1) = 3	RATEC
	STATUS(2)	option to list data (0 by default)
	STATUS(2) = 0	no
	STATUS(2) = 1	yes
	STATUS(3)	option to include buyer and seller branches in the detailed branch output file (0 by default)
	STATUS(3) = 0	no
	STATUS(3) = 1	yes
	STATUS(4)	action code if the MW mismatch exceeds 0.5 MW (0 by default)
	STATUS(4) = 0	discontinue
	STATUS(4) = 1	continue
Real VALUES (3)	array of 3 elements specifying solution parameters(input).	

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	VALUES(1)	percent load reduction for calculating capacity for serving load (10.0 by default)
	VALUES(2)	MW for each transaction; used when no transactions data file is specified (10.0 by default)
	VALUES(3)	distribution factor threshold for detailed branch (0.0 by default)
Character*260 INPFILE	name of MWMI data file (input; no default allowed).	
Character*260 TRNFILE	name of the transactions raw data file; blank for none (input; blank by default).	
Character*260 OUTFILE	name of the detailed branch output file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid VALUES value
	IERR = 5	INPFILE is blank
	IERR = 6	generators are converted
	IERR = 7	buses in island(s) without a swing bus; use activity TREE
	IERR = 8	largest mismatch exceeds 0.5 MW
	IERR = 9	error opening INPFILE
	IERR = 10	error opening OUTFILE
	IERR = 11	error opening TRNFILE
	IERR = 12	prerequisite requirements for API are not met

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1.168. MWOMWMSF

Use this API to tabulate generation (MW) on MW-mile shift factors for a specified transaction event.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MWOMWMSF   EVENT
```

Python command syntax:

```
ierr = mwomwmsf(event)
```

Fortran command syntax:

```
CALL MWOMWMSFAPI(EVENT, IERR)
```

Where:

Integer EVENT	transaction event number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no transactions defined
	IERR = 2	invalid transaction event number
	IERR = 3	transaction event not defined
	IERR = 4	prerequisite requirements for API are not met

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1.169. MWOMWOSF

Use this API to tabulate generation (MW) on MW-ohm shift factors for a specified transaction event.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MWOMWOSF   EVENT
```

Python command syntax:

```
ierr = mwomwosf(event)
```

Fortran command syntax:

```
CALL MWOMWOSFAPI(EVENT, IERR)
```

Where:

Integer EVENT	transaction event number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no transactions defined
	IERR = 2	invalid transaction event number
	IERR = 3	transaction event not defined
	IERR = 4	prerequisite requirements for API are not met

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1.170. N11_ACCC

This API routine is obsolete. It has been replaced by the API routine n11_accc_2 .

This API is used to perform N-1-1 contingency analysis. Each pair of N-1-1 contingencies is simulated in the way that the first N-1 contingency is simulated by a full AC load flow solution with the power flow options for the first N-1 contingency; the post solution of the first N-1 contingency is used as the base case for the second N-1 contingency, the second N-1 contingency is imposed and solved by a full AC load flow solution with power flow options for the second N-1 contingency.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_N11_ACCC  OPTIONS(1)..OPTIONS(18) VALUES(1) VALUES(2) LABEL DFXFILE
 ACCFILE CONFILE INLFILE THRFILE
```

Python command syntax:

```
ierr = n11_accc(options,values,label,dfxfile,accfile,confile,inlfile,
 thrfile)
```

Fortran command syntax:

```
CALL N11_ACCC(OPTIONS,VALUES,LABEL,DFXFILE,ACCFILE,CONFILE,INLFILE,
 THRFILE,IERR)
```

Where:

Integer OPTIONS (18)	array of 18 elements specifying integer specifying power flow options(input).	
	OPTIONS(1)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag for the first N-1 contingency (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag for the first N-1 contingency (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable

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OPTIONS(4)	dc tap adjustment flag for the first N-1 contingency (dc tap adjustment option setting by default)
OPTIONS(4) = 0	disable
OPTIONS(4) = 1	enable
OPTIONS(5)	switched shunt adjustment flag for the first N-1 contingency (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable all
OPTIONS(5) = 2	enable continuous, disable discrete
OPTIONS(6)	induction motor treatment flag for the first N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(6) = 0	stall
OPTIONS(6) = 1	trip
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	Fixed slope decoupled Newton-Raphson method
OPTIONS(7) = 1	Full Newton-Raphson method
OPTIONS(7) = 2	Optimized fixed slope decoupled Newton-Raphson method
OPTIONS(8)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(8) = 0	Disable
OPTIONS(8) = 1	Enable
OPTIONS(9)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
OPTIONS(9) = 0	disable
OPTIONS(9) = 1	enable stepping adjustment
OPTIONS(9) = 2	enable direct adjustment
OPTIONS(10)	area interchange adjustment flag for the first N-1 contingency (area interchange adjustment option setting by default)
OPTIONS(10) = 0	disable
OPTIONS(10) = 1	enable using tie line flows only in calculating area interchange
OPTIONS(10) = 2	enable using tie line flows and loads in calculating area interchange
OPTIONS(11)	phase shift adjustment flag for the first N-1 contingency (phase shift adjustment option setting by default)
OPTIONS(11) = 0	disable
OPTIONS(11) = 1	enable

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	OPTIONS(12)	dc tap adjustment flag for the first N-1 contingency (dc tap adjustment option setting by default)
	OPTIONS(12) = 0	disable
	OPTIONS(12) = 1	enable
Real VALUES (2)	array of 2 elements specifying a real(input).	
	VALUES(1)	Mismatch tolerance (TOLN by default)
	VALUES(2)	Percent of rating for the first N-1 contingency and second N-1 contingency case (100.0 by default)
Character*32 LABEL	label of generation dispatch system, required if the dispatch mode of OPTIONS(15) is not zero. The dispatch mode and the system provide for a default dispatch method for contingencies causing unbalance between load demand and generation (input).	
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of contingency analysis output file (input; no default allowed).	
Character*260 CONFILE	name of Contingency Description Data File for the second N-1 contingencies (input; blank for none).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of Unit Inertia Data File (input; blank for none).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid OPTIONS value
	IERR = 2	Invalid VALUES value
	IERR = 3	Generators are converted
	IERR = 4	Buses in island(s) without a swing bus--use activity TREE
	IERR = 5	Largest mismatch exceeds tolerance
	IERR = 6	Generation subsystem not specified
	IERR = 7	Too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	No Distribution Factor Data input file specified
	IERR = 11	No Contingency Analysis Output File specified
	IERR = 12	Can not read the Contingency Description Data File for the second N-1 contingencies
	IERR = 13	No second N-1 contingencies are specified
	IERR = 14	Invalid base/contingency case flag combination
	IERR = 21	File is not in the form of a PSS/E-25 or later DFAX file--run DFAX

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IERR = 22	Monitored elements exceed limit when adding multi-section line members
IERR = 51	Error opening distribution factor data file
IERR = 52	Error opening load throwover data file
IERR = 53	Error opening inertia data file
IERR = 54	Prerequisite requirements for function not met

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1.171. N11_ACCC_2

This API is the second release of N-1-1 contingency analysis solution. This API is used to perform N-1-1 contingency analysis. Each pair of N-1-1 contingencies is simulated in the way that the first N-1 contingency is simulated by a full AC load flow solution with the power flow options for the first N-1 contingency; the post solution of the first N-1 contingency is used as the base case for the second N-1 contingency, the second N-1 contingency is imposed and solved by a full AC load flow solution with power flow options for the second N-1 contingency.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_N11_ACCC_2  OPTIONS(1)..OPTIONS(20) VALUES(1) VALUES(2) LABEL DFXFILE
 ACCFILE CONFILE INLFILE THRFILE
```

Python command syntax:

```
ierr =
 n11_accc_2(options,values,label,dfxfile,accfile,confile,inlfile,thrfile)
```

Fortran command syntax:

```
CALL
 N11_ACCC_2(OPTIONS,VALUES,LABEL,DFXFILE,ACCFILE,CONFILE,INLFILE,THRFILE,IERR)
```

Where:

Integer OPTIONS (20)	array of 20 elements specifying integer specifying power flow options(input).	
	OPTIONS(1)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag for the first N-1 contingency (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag for the first N-1 contingency (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag for the first N-1 contingency (dc tap adjustment option setting by default)

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OPTIONS(4) = 0	disable
OPTIONS(4) = 1	enable
OPTIONS(5)	switched shunt adjustment flag for the first N-1 contingency (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable all
OPTIONS(5) = 2	enable continuous, disable discrete
OPTIONS(6)	induction motor treatment flag for the first N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(6) = 0	stall
OPTIONS(6) = 1	trip
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	Fixed slope decoupled Newton-Raphson method
OPTIONS(7) = 1	Full Newton-Raphson method
OPTIONS(7) = 2	Optimized fixed slope decoupled Newton-Raphson method
OPTIONS(8)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(8) = 0	Disable
OPTIONS(8) = 1	Enable
OPTIONS(9)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
OPTIONS(9) = 0	disable
OPTIONS(9) = 1	enable stepping adjustment
OPTIONS(9) = 2	enable direct adjustment
OPTIONS(10)	area interchange adjustment flag for the first N-1 contingency (area interchange adjustment option setting by default)
OPTIONS(10) = 0	disable
OPTIONS(10) = 1	enable using tie line flows only in calculating area interchange
OPTIONS(10) = 2	enable using tie line flows and loads in calculating area interchange
OPTIONS(11)	phase shift adjustment flag for the first N-1 contingency (phase shift adjustment option setting by default)
OPTIONS(11) = 0	disable
OPTIONS(11) = 1	enable
OPTIONS(12)	dc tap adjustment flag for the first N-1 contingency (dc tap adjustment option setting by default)

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	OPTIONS(12) = 0	disable
	OPTIONS(12) = 1	enable
	OPTIONS(13)	switched shunt adjustment flag for the first N-1 contingency (switched shunt adjustment option setting by default)
	OPTIONS(13) = 0	disable
	OPTIONS(13) = 1	enable
	OPTIONS(13) = 2	enable continuous, disable discrete
	OPTIONS(14)	induction motor treatment flag for the first N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTIONS(14) = 0	stall
	OPTIONS(14) = 1	trip
	OPTIONS(15)	generation dispatch method
	OPTIONS(15) = 1	subsystem machines (PMAX)
	OPTIONS(15) = 2	subsystem machines (PMAX)
	OPTIONS(15) = 3	subsystem machines (INERTIA)
	OPTIONS(15) = 4	subsystem machines (GOVERNOR DROOP)
	OPTIONS(16)	flag to skip the second N-1 contingencies if the first N-1 contingency is not clean
	OPTIONS(16) = 0	No
	OPTIONS(16) = 1	Yes
	OPTIONS(17)	Rating set for the first N-1 contingency case
	OPTIONS(17) = 1	RATE A
	OPTIONS(17) = 2	RATE B
	OPTIONS(17) = 3	RATE C
	OPTIONS(18)	Rating set for the second N-1 contingency case
	OPTIONS(18) = 1	RATE A
	OPTIONS(18) = 2	RATE B
	OPTIONS(18) = 3	RATE C
	OPTIONS(19)	Voltage limit for the first N-1 contingency case
	OPTIONS(19) = 1	normal
	OPTIONS(19) = 2	emergency
	OPTIONS(20)	Voltage limit for the second N-1 contingency case
	OPTIONS(20) = 1	normal
	OPTIONS(20) = 2	emergency
Real VALUES (2)	array of 2 elements specifying a real(input).	
	VALUES(1)	Mismatch tolerance (TOLN by default)

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	VALUES(2)	Percent of rating for the first N-1 contingency and second N-1 contingency case (100.0 by default)
Character*32 LABEL	label of generation dispatch system, required if the dispatch mode of OPTIONS(15) is not zero. The dispatch mode and the system provide for a default dispatch method for contingencies causing unbalance between load demand and generation (input).	
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of contingency analysis output file (input; no default allowed).	
Character*260 CONFILE	name of Contingency Description Data File for the second N-1 contingencies (input; blank for none).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of Unit Inertia Data File (input; blank for none).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid OPTIONS value
	IERR = 2	Invalid VALUES value
	IERR = 3	Generators are converted
	IERR = 4	Buses in island(s) without a swing bus--use activity TREE
	IERR = 5	Largest mismatch exceeds tolerance
	IERR = 6	Generation subsystem not specified
	IERR = 7	Too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	No Distribution Factor Data input file specified
	IERR = 11	No Contingency Analysis Output File specified
	IERR = 12	Can not read the Contingency Description Data File for the second N-1 contingencies
	IERR = 13	No second N-1 contingencies are specified
	IERR = 14	Invalid base/contingency case flag combination
	IERR = 21	File is not in the form of a PSS/E-25 or later DFAX file--run DFAX
	IERR = 22	Monitored elements exceed limit when adding multi-section line members
	IERR = 51	Error opening distribution factor data file
	IERR = 52	Error opening load throwover data file
	IERR = 53	Error opening inertia data file
	IERR = 54	Prerequisite requirements for function not met

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1.172. N11_ACCC_PARALLEL

Use this API to run the N-1-1 contingency analysis in parallel when the number of contingency processors in Program Settings is greater than 1. The API distributes contingency cases among multiple processors evenly, and tests the contingency cases and following tripping actions assigned to each processor in parallel.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_N11_ACCC_PARALLEL  OPTIONS(1)..OPTIONS(20) VALUES(1)..VALUES(2) LABEL
 DFXFILE ACCFILE CONFILE THRFILE INLFILE
```

Python command syntax:

```
ierr = n11_accc_parallel(options, values, label, dfxfile, accfile,
 confile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL N11_ACCC_PARALLEL(OPTIONS, VALUES, LABEL, DFXFILE, ACCFILE, CONFILE,
 THRFILE, INLFILE, IERR)
```

Where:

Integer OPTIONS (20)	array of elements specifying power flow options(input).	
	OPTIONS(1)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag for the first N-1 contingency (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag for the first N-1 contingency (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag for the first N-1 contingency (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable

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OPTIONS(5)	switched shunt adjustment flag for the first N-1 contingency (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable all
OPTIONS(5) = 2	enable continuous, disable discrete.
OPTIONS(6)	induction motor treatment flag for the first N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(6) = 0	stall.
OPTIONS(6) = 1	trip.
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	Fixed slope decoupled Newton-Raphson method
OPTIONS(7) = 1	Full Newton-Raphson method
OPTIONS(7) = 2	Optimized fixed slope decoupled Newton-Raphson method
OPTIONS(8)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(8) = 0	Disable.
OPTIONS(8) = 1	Enable.
OPTIONS(9)	tap adjustment flag for the second N-1 contingency (tap adjustment option setting by default)
OPTIONS(9) = 0	disable.
OPTIONS(9) = 1	enable stepping adjustment.
OPTIONS(9) = 2	enable direct adjustment.
OPTIONS(10)	area interchange adjustment flag for the second N-1 contingency (area interchange adjustment option setting by default)
OPTIONS(10) = 0	disable.
OPTIONS(10) = 1	enable using tie line flows only in calculating area interchange.
OPTIONS(10) = 2	enable using tie line flows and loads in calculating area interchange.
OPTIONS(11)	phase shift adjustment flag for the second N-1 contingency (phase shift adjustment option setting by default)
OPTIONS(11) = 0	disable.
OPTIONS(11) = 1	enable.
OPTIONS(12)	dc tap adjustment flag for the second N-1 contingency (dc tap adjustment option setting by default)
OPTIONS(12) = 0	disable.
OPTIONS(12) = 1	enable.

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	OPTIONS(13)	switched shunt adjustment flag for the second N-1 contingency (switched shunt adjustment option setting by default)
	OPTIONS(13) = 0	disable
	OPTIONS(13) = 1	enable
	OPTIONS(13) = 2	enable continuous, disable discrete.
	OPTIONS(14)	induction motor treatment flag for the second N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
	OPTIONS(14) = 0	stall
	OPTIONS(14) = 1	trip
	OPTIONS(15)	generation dispatch method
	OPTIONS(15) = 0	disable.
	OPTIONS(15) = 1	subsystem machines (RESERVE).
	OPTIONS(15) = 2	subsystem machines (PMAX).
	OPTIONS(15) = 3	subsystem machines (INERTIA).
	OPTIONS(15) = 4	subsystem machines (GOVERNOR DROOP).
	OPTIONS(16)	flag to skip the second N-1 contingencies if the first N-1 contingency is not clean
	OPTIONS(16) = 0	No
	OPTIONS(16) = 1	Yes
	OPTIONS(17)	Rating set for the first N-1 contingency case
	OPTIONS(17) = 1	rate A
	OPTIONS(17) = 2	rate B
	OPTIONS(17) = 3	rate C
	OPTIONS(18)	Rating set for the second N-1 contingency case
	OPTIONS(18) = 1	rate A
	OPTIONS(18) = 2	rate B
	OPTIONS(18) = 3	rate C
	OPTIONS(19)	voltage limit for the first N-1 contingency case
	OPTIONS(19) = 1	normal
	OPTIONS(19) = 2	emergency
	OPTIONS(20)	voltage limit for the second N-1 contingency case
	OPTIONS(20) = 1	normal
	OPTIONS(20) = 2	emergency
Real VALUES (2)	array of 2 elements specifying power flow options(input).	
	VALUES(1)	Mismatch tolerance (TOLN by default)
	VALUES(2)	Percent of rating for the first N-1 contingency and second N-1 contingency case (100.0 by default)

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Character*32 LABEL	The label of generation dispatch system, required if the dispatch mode of OPTIONS(15) is not zero. The dispatch mode and the system provide for a default dispatch method for contingencies causing unbalance between load demand and generation output. (input).
Character*260 DFXFILE	Is the name of the Distribution Factor Data File (input; no default allowed).
Character*260 ACCFILE	Is the name of contingency analysis output file (input; no default allowed).
Character*260 CONFILE	Is the name of Contingency Description Data File for the second N-1 contingencies (input; blank by default).
Character*260 THRFILE	Is the name of Load Throwover Data File (input; blank by default).
Character*260 INLFILE	Is the name of Unit Inertia Data File (input; blank by default).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	Invalid OPTIONS value
IERR = 2	Invalid VALUES value
IERR = 3	Generators are converted
IERR = 4	Buses in island(s) without a swing bus--use activity TREE
IERR = 5	Largest mismatch exceeds tolerance
IERR = 6	Generation subsystem not specified
IERR = 7	Too many islands in base case.
IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
IERR = 9	buses with bus type code and series element status inconsistencies
IERR = 10	No Distribution Factor Data input file specified
IERR = 11	No Contingency Solution Output File specified
IERR = 12	Can not read the Contingency Description Data File for the second N-1 contingencies
IERR = 13	No second N-1 contingencies are specified
IERR = 14	Invalid base/contingency case flag combination.
IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening Distribution Factor Data File
IERR = 52	error opening Load Throwover Data File
IERR = 53	error opening Unit Inertia and Governor Data File
IERR = 54	prerequisite requirements for API are not met

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1.173. N11_ACCC_PSCOPF

This API routine is obsolete. It has been replaced by the API routine n11_accc_pscopf_2 .

The API is used to perform N-1-1 contingency analysis in either corrective mode or security constrained optimal power flow mode (SCOPF). In corrective mode, for each pair of N-1 contingencies a full AC load flow is performed for the first N-1 contingency; the post contingency solution is checked against the operation limits and corrective actions are specified to mitigate the limit violations; the post-corrective action solution is used as the base case for the secondary contingency and a full AC load flow is performed to solve the second N-1 contingency.
In the SCOPF mode, a primary contingency is imposed and solved with a full AC load flow solution. A secure control is conducted on the system condition existing after the primary contingency with the secondary contingencies, control adjustments are specified by security constrained optimal power flow to ensure security of the system after the primary contingency. The system condition after the primary contingency and adjustments is used as the base case for secondary contingency; a full AC load flow solution is then performed to solve a secondary N-1 contingency.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_N11_ACCC_PSCOPF  OPTIONS(1)..OPTIONS(18) OPTCOR(1)..OPTCOR(9)
 VALUES(1)..VALUES(10)  LABELS(1)..LABELS(7) DFXFILE ACCFILE  CONFILE
 INLFILE THRFILE
```

Python command syntax:

```
ierr = n11_accc_pscopf(options, optcor, values, labels, dfxfile, accfile,
 confile, inlfile, thrfile)
```

Fortran command syntax:

```
CALL N11_ACCC_PSCOPF(OPTIONS, OPTCOR, VALUES, LABELS, DFXFILE, ACCFILE,
 CONFILE, INLFILE, THRFILE, IERR)
```

Where:

Integer OPTIONS (18)	array of 18 elements specifying integer specifying power flow options(input).	
	OPTIONS(1)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag for the first N-1 contingency (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange

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OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
OPTIONS(3)	phase shift adjustment flag for the first N-1 contingency (phase shift adjustment option setting by default)
OPTIONS(3) = 0	disable
OPTIONS(3) = 1	enable
OPTIONS(4)	dc tap adjustment flag for the first N-1 contingency (dc tap adjustment option setting by default)
OPTIONS(4) = 0	disable
OPTIONS(4) = 1	enable
OPTIONS(5)	switched shunt adjustment flag for the first N-1 contingency (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable all
OPTIONS(5) = 2	enable continuous, disable discrete
OPTIONS(6)	induction motor treatment flag for the first N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(6) = 0	stall
OPTIONS(6) = 1	trip
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	Fixed slope decoupled Newton-Raphson method
OPTIONS(7) = 1	Full Newton-Raphson method
OPTIONS(7) = 2	Optimized fixed slope decoupled Newton-Raphson method
OPTIONS(8)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(8) = 0	Disable
OPTIONS(8) = 1	Enable
OPTIONS(9)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
OPTIONS(9) = 0	disable
OPTIONS(9) = 1	enable stepping adjustment
OPTIONS(9) = 2	enable direct adjustment
OPTIONS(10)	area interchange adjustment flag for the first N-1 contingency (area interchange adjustment option setting by default)
OPTIONS(10) = 0	disable
OPTIONS(10) = 1	enable using tie line flows only in calculating area interchange

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OPTIONS(10) = 2	enable using tie line flows and loads in calculating area interchange
OPTIONS(11)	phase shift adjustment flag for the first N-1 contingency (phase shift adjustment option setting by default)
OPTIONS(11) = 0	disable
OPTIONS(11) = 1	enable
OPTIONS(12)	dc tap adjustment flag for the first N-1 contingency (dc tap adjustment option setting by default)
OPTIONS(12) = 0	disable
OPTIONS(12) = 1	enable
OPTIONS(13)	switched shunt adjustment flag for the first N-1 contingency (switched shunt adjustment option setting by default)
OPTIONS(13) = 0	disable
OPTIONS(13) = 1	enable
OPTIONS(13) = 2	enable continuous, disable discrete
OPTIONS(14)	induction motor treatment flag for the first N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(14) = 0	stall
OPTIONS(14) = 1	trip
OPTIONS(15)	dispatch mode (0 by default)
OPTIONS(15) = 0	disable
OPTIONS(15) = 1	subsystem machines (RESERVE)
OPTIONS(15) = 2	subsystem machines (PMAX)
OPTIONS(15) = 3	subsystem machines (INERTIA)
OPTIONS(15) = 4	subsystem machines (GOVERNOR DROOP)
OPTIONS(16)	flag to skip the second N-1 contingencies if the first N-1 contingency is not clean
OPTIONS(16) = 0	No
OPTIONS(16) = 1	Yes
OPTIONS(17)	Rating set for the first N-1 contingency case
OPTIONS(17) = 1	RATE A
OPTIONS(17) = 2	RATE B
OPTIONS(17) = 3	RATE C
OPTIONS(18)	Rating set for the second N-1 contingency case
OPTIONS(18) = 1	RATE A
OPTIONS(18) = 2	RATE B
OPTIONS(18) = 3	RATE C

Integer OPTCOR (9)	array of 9 elements specifying OPTCOR Is an integer specifying control options(input).

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	OPTCOR(1)	System adjustment mode
	OPTCOR(1) = 0	local adjustment only
	OPTCOR(1) = 1	Corrective action mode
	OPTCOR(1) = 2	Preventive security constrained optimal power flow mode
	OPTCOR(2)	maximum iteration number of PSCOPF, required if OPTCOR(1) is 2
	OPTCOR(3)	maximum number of a contingency that does not cause limit violations and instability issue to be tested in the iterative process of PSCOPF, required if OPTCOR(1) is 2
	OPTCOR(4)	generator control flag (1 by default)
	OPTCOR(4) = 0	disable generation MW dispatch in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(4) = 1	enable
	OPTCOR(5)	load control flag (0 by default)
	OPTCOR(5) = 0	disable load shedding control in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(5) = 1	enable
	OPTCOR(6)	phase shifter control flag (1 by default)
	OPTCOR(6) = 0	disable phase shifter control in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(6) = 1	enable
	OPTCOR(7)	off-line generator control flag (0 by default)
	OPTCOR(7) = 0	disable generation MW dispatch of off-line generators in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(7) = 1	enable
	OPTCOR(8)	tap setting adjustment flag (0 by default)
	OPTCOR(8) = 0	disable tap setting adjustment in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(8) = 1	enable
	OPTCOR(9)	switched shunt control flag (0 by default)
	OPTCOR(9) = 0	disable switch shunt control in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(9) = 1	enable
Real VALUES (10)	array of 10 elements specifying VALUES Is a real(input).	
	VALUES(1)	Mismatch tolerance (TOLN by default)
	VALUES(2)	Percentage of rating for the base case (100.0 bye default)

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	VALUES(3)	Voltage tolerance in linear programming
	VALUES(4)	Flow tolerance in linear programming
	VALUES(5)	Weighting factor of generation MW dispatch (1.0 bye default)
	VALUES(6)	Weighting factor of load control (1.0 bye default)
	VALUES(7)	Weighting factor of phase shifter control (1.0 bye default)
	VALUES(8)	Weighting factor of generation MW dispatch of off-line generators (1.0 bye default)
	VALUES(9)	Weighting factor of tap setting adjustment (1.0 by default)
	VALUES(10)	Weighting factor of switched shunt control (1.0 by default)
Character*32 LABELS (7)	array of 7 elements specifying specifiying subsystem labels(input).	
	LABELS(1)	label of generation dispatch system, required if the dispatch mode of OPTIONS(15) is not zero. The dispatch mode and the system provide for a default dispatch method for contingencies causing unbalance between load demand and generation (blank for none by default)
	LABELS(2)	label of generation dispatch control system. MW injections of the generators within the system are available for dispatch in the corrective action mode or preventive security constrained optimal power flow mode (blank for none by default)
	LABELS(3)	label of load control system. Loads in the load control system are available for curtailment in the corrective action mode or preventive security constrained optimal power flow mode (blank for none by default)
	LABELS(4)	label of phase shifter adjustment system. The phase angles of phase shifters in the system are regulated either by corrective action or preventive security constrained optimal power flow (blank for none by default)
	LABELS(5)	label of generation MW control of off-line generator control system. MW injections of the off-line generators within the system are available for dispatch in the corrective action mode or preventive security constrained optimal power flow mode (blank for none by default)
	LABELS(6)	label of tap adjustment system. The tap positions of transformers in voltage control mode are regulated either by corrective action or

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		preventive security constrained optimal power flow (blank for none by default)
	LABELS(7)	label of switched shunt control system. The admittances of switched shunts in voltage control mode are regulated either by corrective action or preventive security constrained optimal power flow (blank for none by default)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	name of contingency analysis output file (input; no default allowed).	
Character*260 CONFILE	name of Contingency Description Data File for the second N-1 contingencies (input; blank for none).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Character*260 INLFILE	name of Unit Inertia Data File (input; blank for none).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid OPTIONS value
	IERR = 2	Invalid VALUES value
	IERR = 3	Generators are converted
	IERR = 4	Buses in island(s) without a swing bus--use activity TREE
	IERR = 5	Largest mismatch exceeds tolerance
	IERR = 6	Generation subsystem not specified
	IERR = 7	Too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	No Distribution Factor Data input file specified
	IERR = 11	No Contingency Analysis Output File specified
	IERR = 12	Can not read the Contingency Description Data File for the second N-1 contingencies
	IERR = 13	No second N-1 contingencies are specified
	IERR = 14	Invalid base/contingency case flag combination
	IERR = 15	Invalid OPTCOR value
	IERR = 21	File is not in the form of a PSS/E-25 or later DFAX file--run DFAX
	IERR = 22	Monitored elements exceed limit when adding multi-section line members
	IERR = 51	Error opening distribution factor data file
	IERR = 52	Error opening load throwover data file
	IERR = 53	Error opening inertia data file

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IERR = 54	Prerequisite requirements for function not met

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1.174. N11_ACCC_PSCOPF_2

The API is the second release of N-1-1 contingency analysis solution in either corrective mode or security constrained optimal power flow mode (SCOPF). In corrective mode, for each pair of N-1 contingencies a full AC load flow is performed for the first N-1 ontingency; the post contingency solution is checked against the operation limits and corrective actions are specified to mitigate the limit violations; the post-corrective action solution is used as the base case for the secondary contingency and a full AC load flow is performed to solve the second N-1 contingency.
In the PSCOPF mode, a primary contingency is imposed and solved with a full AC load flow solution. A secure control is conducted on the system condition existing after the primary contingency with the secondary contingencies, control adjustments are specified by security constrained optimal power flow to ensure security of the system after the primary contingency. The system condition after the primary contingency and adjustments is used as the base case for secondary contingency; a full AC load flow solution is then performed to solve a secondary N-1 contingency.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_N11_ACCC_PSCOPF_2  OPTIONS(1)..OPTIONS(20) OPTCOR(1)..OPTCOR(9)
 VALUES(1)..VALUES(10) LABELS(1)..LABELS(7) DFXFILE ACCFILE CONFILE
 THRFILE INLFILE
```

Python command syntax:

```
ierr = n11_accc_pscopf_2(options, optcor, values, labels, dfxfile,
 accfile, confile, thrfile, inlfile)
```

Fortran command syntax:

```
CALL N11_ACCC_PSCOPF_2(OPTIONS, OPTCOR, VALUES, LABELS, DFXFILE, ACCFILE,
 CONFILE, THRFILE, INLFILE, IERR)
```

Where:

Integer OPTIONS (20)	array of elements specifying power flow options(input).	
	OPTIONS(1)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag for the first N-1 contingency (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange

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OPTIONS(3)	phase shift adjustment flag for the first N-1 contingency (phase shift adjustment option setting by default)
OPTIONS(3) = 0	disable
OPTIONS(3) = 1	enable
OPTIONS(4)	dc tap adjustment flag for the first N-1 contingency (dc tap adjustment option setting by default)
OPTIONS(4) = 0	disable
OPTIONS(4) = 1	enable
OPTIONS(5)	switched shunt adjustment flag for the first N-1 contingency (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable all
OPTIONS(5) = 2	enable continuous, disable discrete.
OPTIONS(6)	induction motor treatment flag for the first N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(6) = 0	stall.
OPTIONS(6) = 1	trip.
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	Fixed slope decoupled Newton-Raphson method
OPTIONS(7) = 1	Full Newton-Raphson method
OPTIONS(7) = 2	Optimized fixed slope decoupled Newton-Raphson method
OPTIONS(8)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(8) = 0	Disable
OPTIONS(8) = 1	Enable
OPTIONS(9)	tap adjustment flag for the second N-1 contingency (tap adjustment option setting by default)
OPTIONS(9) = 0	disable.
OPTIONS(9) = 1	enable stepping adjustment.
OPTIONS(9) = 2	enable direct adjustment.
OPTIONS(10)	area interchange adjustment flag for the second N-1 contingency (area interchange adjustment option setting by default)
OPTIONS(10) = 0	disable.
OPTIONS(10) = 1	enable using tie line flows only in calculating area interchange.
OPTIONS(10) = 2	enable using tie line flows and loads in calculating area interchange.

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OPTIONS(11)	phase shift adjustment flag for the second N-1 contingency (phase shift adjustment option setting by default)
OPTIONS(11) = 0	disable.
OPTIONS(11) = 1	enable.
OPTIONS(12)	dc tap adjustment flag for the second N-1 contingency (dc tap adjustment option setting by default)
OPTIONS(12) = 0	disable.
OPTIONS(12) = 1	enable.
OPTIONS(13)	switched shunt adjustment flag for the second N-1 contingency (switched shunt adjustment option setting by default)
OPTIONS(13) = 0	disable.
OPTIONS(13) = 1	enable.
OPTIONS(13) = 2	enable continuous, disable discrete.
OPTIONS(14)	induction motor treatment flag for the second N-1 contingency (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(14) = 0	stall.
OPTIONS(14) = 1	trip.
OPTIONS(15)	dispatch mode (0 by default)
OPTIONS(15) = 0	disable.
OPTIONS(15) = 1	subsystem machines (RESERVE).
OPTIONS(15) = 2	subsystem machines (PMAX).
OPTIONS(15) = 3	subsystem machines (INERTIA).
OPTIONS(15) = 4	subsystem machines (GOVERNOR DROOP).
OPTIONS(16)	flag to skip the second N-1 contingencies if the first N-1 contingency is not clean
OPTIONS(16) = 0	No
OPTIONS(16) = 1	Yes
OPTIONS(17)	Rating set for the first N-1 contingency case
OPTIONS(17) = 1	rate A
OPTIONS(17) = 2	rate B
OPTIONS(17) = 3	rate C
OPTIONS(18)	Rating set for the second N-1 contingency case
OPTIONS(18) = 1	rate A
OPTIONS(18) = 2	rate B
OPTIONS(18) = 3	rate C
OPTIONS(19)	voltage limit for the first N-1 contingency case
OPTIONS(19) = 1	normal
OPTIONS(19) = 2	emergency

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	OPTIONS(20)	voltage limit for the second N-1 contingency case
	OPTIONS(20) = 1	normal
	OPTIONS(20) = 2	emergency
Integer OPTCOR (9)	array of elements specifying control options(input).	
	OPTCOR(1)	System adjustment mode
	OPTCOR(1) = 0	local adjustment only
	OPTCOR(1) = 1	Corrective action mode
	OPTCOR(1) = 2	Preventive security constrained optimal power flow mode
	OPTCOR(2)	maximum iteration number of PSCOPF, required if OPTCOR(1) is 2
	OPTCOR(3)	maximum number of a contingency that does not cause limit violations and instability issue to be tested in the iterative process of PSCOPF, required if OPTCOR(1) is 2.
	OPTCOR(4)	generator control flag (0 by default)
	OPTCOR(4) = 0	disable generation MW dispatch in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(4) = 1	enable
	OPTCOR(5)	load control flag (0 by default)
	OPTCOR(5) = 0	disable load shedding control in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(5) = 1	enable
	OPTCOR(6)	phase shifter control flag (1 by default)
	OPTCOR(6) = 0	disable phase shifter control in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(6) = 1	enable
	OPTCOR(7)	off-line generator control flag (0 by default)
	OPTCOR(7) = 0	disable generation MW dispatch of off-line generators in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(7) = 1	enable
	OPTCOR(8)	tap setting adjustment flag (0 by default)
	OPTCOR(8) = 0	disable tap setting adjustment in the corrective action mode or preventive security constrained optimal power flow mode
	OPTCOR(8) = 1	enable
	OPTCOR(9)	switched shunt control flag (0 by default)
	OPTCOR(9) = 0	disable switch shunt control in the corrective action mode or preventive security constrained optimal power flow

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	OPTCOR(9) = 1	enable
Real VALUES (10)	array of elements specifying power flow settings(input).	
	VALUES(1)	Mismatch tolerance (TOLN by default)
	VALUES(2)	Percentage of rating for the base case (100.0 by default)
	VALUES(3)	Voltage tolerance in linear programming
	VALUES(4)	Flow tolerance in linear programming
	VALUES(5)	Weighting factor of generation MW dispatch (1.0 by default)
	VALUES(6)	Weighting factor of load control (1.0 by default)
	VALUES(7)	Weighting factor of phase shifter control (1.0 by default)
	VALUES(8)	Weighting factor of generation MW dispatch of off-line generators (1.0 by default)
	VALUES(9)	Weighting factor of tap setting adjustment (1.0 by default)
	VALUES(10)	Weighting factor of switched shunt control (1.0 by default)
Character*32 LABELS (7)	array of elements specifying subsystem labels (input).	
	LABELS(1)	The label of generation dispatch system, required if the dispatch mode of OPTIONS(15) is not zero. The dispatch mode and the system provide for a default dispatch method for contingencies causing unbalance between load demand and generation output (blank for none).
	LABELS(2)	The label of generation dispatch control system. MW injections of the generators within the system are available for dispatch in the corrective action mode or preventive security constrained optimal power flow mode (blank for none).
	LABELS(3)	The label of load control system. Loads in the load control system are available for curtailment in the corrective action mode or preventive security constrained optimal power flow mode (blank for none).
	LABELS(4)	The label of phase shifter adjustment system. The phase angles of phase shifters in the system are regulated either by corrective action or preventive security constrained optimal power flow (blank for none).
	LABELS(5)	The label of generation MW control of off-line generator control system. MW injections of the off-line generators within the system are available for dispatch in the corrective ac-

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		tion mode or preventive security constrained optimal power flow mode (blank for none).
	LABELS(6)	The label of tap adjustment system. The tap positions of transformers in voltage control mode are regulated either by corrective action or preventive security constrained optimal power flow (blank for none).
	LABELS(7)	The label of switched shunt control system. The admittances of switched shunts in voltage control mode are regulated either by corrective action or preventive security constrained optimal power flow (blank for none).
Character*260 DFXFILE	Is the name of the Distribution Factor Data File (input; no default allowed).	
Character*260 ACCFILE	Is the name of contingency analysis output file (input; no default allowed).	
Character*260 CONFILE	Is the name of Contingency Description Data File for the second N-1 contingencies (blank for none) (input).	
Character*260 THRFILE	Is the name of Load Throwover Data File (blank for none). (input).	
Character*260 INLFILE	Is the name of Unit Inertia Data File (blank for none). (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	generation dispatch subsystem is not defined
	IERR = 7	too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 9	buses with bus type code and series element status inconsistencies
	IERR = 10	no Distribution Factor Data File specified
	IERR = 11	no Contingency Analysis Output File specified
	IERR = 12	Can not read the Contingency Description Data File for the second N-1 contingencies
	IERR = 13	No second N-1 contingencies are specified
	IERR = 14	Invalid base/contingency case flag combination.
	IERR = 15	Invalid OPTCOR value

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IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening Distribution Factor Data File
IERR = 52	error opening Load Throwover Data File
IERR = 53	error opening Unit Inertia and Governor Data File
IERR = 54	prerequisite requirements for API are not met

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1.175. NETG

Use this API to change the in-service generation to negative MVA load at all Type 2 and 3 buses except those in the subsystem specified by the user.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_NETG  SID  ALL
```

Python command syntax:

```
ierr = netg(sid, all)
```

Fortran command syntax:

```
CALL NETGNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses outside of subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.176. NEWCAS

This API routine is obsolete. It has been replaced by the API routine newcase_2 .

Use this API to initialize a new power flow case. All existing working case data will be lost upon execution of this API.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_NEWCAS  BASEMVA  TITL1  TITL2
```

Python command syntax:

```
ierr = newcas(basemva, titl1, titl2)
```

Fortran command syntax:

```
CALL NEWCASNW(BASEMVA, TITL1, TITL2, IERR)
```

Where:

Real BASEMVA	new system base MVA. Valid base MVA values range from 1.0 to 10,000.0 (input; 100.0).	
Character*60 TITL1	first line of the new case title (input; blank by default).	
Character*60 TITL2	second line of the new case title (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid BASEMVA value
	IERR = 2	prerequisite requirements for API are not met

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1.177. NEWCASE_2

This API is the second release of the function used to initialize a new power flow case. All existing working case data will be lost upon execution of this API.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_NEWCASE_2  OPTIONS(1)  OPTIONS(2)  BASEMVA  BASEFREQ  TITL1  TITL2
```

Python command syntax:

```
ierr = newcase_2(options, basemva, basefreq, titl1, titl2)
```

Fortran command syntax:

```
CALL NEWCASE_2(OPTIONS, BASEMVA, BASEFREQ, TITL1, TITL2, IERR)
```

Where:

Integer OPTIONS (2)	array of 2 elements specifying units for ratings (used to set the per-cent loading program option settings)(input). OPTIONS(1) units for transformer ratings OPTIONS(1) = 0 current expressed as MVA OPTIONS(1) = 0 MVA OPTIONS(2) units for ratings of non-transformer branches OPTIONS(2) = 0 current expressed as MVA OPTIONS(2) = 0 MVA
Real BASEMVA	system base MVA. Valid base MVA values range from 1.0 to 10,000.0 (input; 100.0).
Real BASEFREQ	system base frequency. Valid base frequency values range from greater than 0.0 to less than 100.0 (input; present base frequency option setting).
Character*60 TITL1	first line of the new case title (input; blank by default).
Character*60 TITL2	second line of the new case title (input; blank by default).
Integer IERR	error code (output). IERR = 0 no error occurred IERR = 1 invalid BASEMVA value IERR = 2 invalid BASEFREQ value IERR = 3 prerequisite requirements for API are not met

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1.178. NEWSEQ

Use this API to initialize the fault analysis data arrays, set them to default values, and set the flag indicating that sequence data is present in the working case.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_NEWSEQ
```

Python command syntax:

```
ierr = newseq()
```

Fortran command syntax:

```
CALL NEWSEQ(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

1.179. NEWTON_TOLERANCE

Use this API to specify or return the value of the default Newton-Raphson convergence tolerance.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_NEWTON_TOLERANCE  RVAL
```

Python command syntax:

```
ierr = newton_tolerance(rval)

ierr, rval = newton_tolerance()
```

Fortran command syntax:

```
CALL NEWTON_TOLERANCE_API(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0.1 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.180. NEW_DIMENSION

Use this API to reset PSSE to an increased bus size level.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_NEW_DIMENSION  BUSES
```

Python command syntax:

```
ierr = new_dimension(buses)
```

Fortran command syntax:

```
CALL NEW_DIMENSION(BUSES, IERR)
```

Where:

Integer BUSES	maximum number of buses allowed at the new size level If BUSES is less than the current maximum number of buses, PSSE's capacity limits are not changed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid BUSES value
	IERR = 2	prerequisite requirements for API are not met

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1.181. NON_DIVERGENT

Use this API to specify or return the option to specify or return the option to enable or disable the non-divergent Newton power flow solution.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_NON_DIVERGENT  IVAL
```

Python command syntax:

```
ierr = non_divergent(ival)

ierr, ival = non_divergent()
```

Fortran command syntax:

```
CALL NON_DIVERGENT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	disable non-divergent Newton solution option.
	IVAL = 1	enable non-divergent Newton solution option.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.182. NON_TRANS_PERCENT_UNITS

Use this API to set the non-transformer branch percent units option setting to either MVA or current expressed as MVA.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_NON_TRANS_PERCENT_UNITS  IVAL
```

Python command syntax:

```
ierr = non_trans_percent_units(ival)

ierr, ival = non_trans_percent_units()
```

Fortran command syntax:

```
CALL NON_TRANS_PERCENT_UNITS_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1 by default; else output).	
	IVAL = 0	MVA.
	IVAL = 1	current expressed as MVA.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.183. NSOL

Use this API to apply the decoupled Newton-Raphson power flow calculation.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_NSOL  OPTIONS(1)  OPTIONS(2)  OPTIONS(3)  OPTIONS(4)  OPTIONS(5)
 OPTIONS(6)  OPTIONS(7)
```

Python command syntax:

```
ierr = nsol(options)
```

Fortran command syntax:

```
CALL NSOLAPI(OPTIONS, IERR)
```

Where:

Integer OPTIONS (7)	array of 7 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (use tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (use area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (use phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (use dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (use switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode

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	OPTIONS(6)	flat start flag (0 by default)
	OPTIONS(6) = 0	do not flat start
	OPTIONS(6) = 1	flat start
	OPTIONS(6) = 2	flat start, then estimate voltage magnitudes
	OPTIONS(6) = 3	flat start, then estimate voltage phase angles
	OPTIONS(6) = 4	flat start, then estimate voltage magnitudes and phase angles
	OPTIONS(7)	var limit flag (99 by default)
	OPTIONS(7) = 0	apply var limits immediately
	OPTIONS(7) = 0	apply var limits on iteration n (or sooner if mismatch gets small)
	OPTIONS(7) = -1	ignore var limits
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	generators are converted
	IERR = 3	buses in island(s) without a swing bus; use activity TREE
	IERR = 4	bus type code and series element status inconsistencies
	IERR = 5	prerequisite requirements for API are not met

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1.184. NUMBER_THREADS

Use this API to specify or return the number of processors available for parallel operations.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_NUMBER_THREADS  IVAL
```

Python command syntax:

```
ierr = number_threads(ival)

ierr, ival = number_threads()
```

Fortran command syntax:

```
CALL NUMBER_THREADS_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting between 1 and the number of processors of the computer (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.185. ORDR

Use this API to calculate a sparsity preserving ordering of buses in preparation for the processing of network matrices.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ORDR  OPT
```

Python command syntax:

```
ierr = ordr(opt)
```

Fortran command syntax:

```
CALL ORDRAPI(OPT, IERR)
```

Where:

Integer OPT	flag for ordering option (input; 0 by default).	
	OPT = 0	assume all branches between in-service buses are in-service
	OPT = 1	ignore out-of-service branches in calculating the bus ordering
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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1.186. OTDF

Use this API to tabulate vectors of distribution factors using a linear network (dc) model.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OTDF  DFXFILE
```

Python command syntax:

```
ierr = otdf(dfxfile)
```

Fortran command syntax:

```
CALL OTDFAPI(DFXFILE, IERR)
```

Where:

Character*260 DFXFILE	name of DFAX data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no DFAX data file specified
	IERR = 2	generators are converted
	IERR = 3	buses in island(s) without a swing bus; use activity TREE
	IERR = 4	error opening file DFXFILE
	IERR = 5	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 6	file DFXFILE does not contain distribution factors; run DFAX with distribution factor option enabled
	IERR = 7	no monitored branches or interfaces defined
	IERR = 8	prerequisite requirements for API are not met

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1.187. OUTPUT_Y_MATRIX

Use this API to display the network admittance matrix for a subsystem of the working case in the form of a list of matrix terms.

This API routine was first introduced in release 30.1.0.

Batch command syntax:

```
BAT_OUTPUT_Y_MATRIX  SID  ALL  TIES  OUT  OFILE
```

Python command syntax:

```
ierr = output_y_matrix(sid, all, ties, out, ofile)
```

Fortran command syntax:

```
CALL OUTPUT_Y_MATRIX(SID, ALL, TIES, OUT, OFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only rows corresponding to buses in subsystem SID
	ALL = 1	process the complete case
Integer TIES	option for the omission or inclusion of off-diagonal terms corresponding to ties from the specified subsystem; ignored if ALL is 1 (input; 0 by default).	
	TIES = 0	omit ties
	TIES = 1	include ties
Integer OUT	destination selection (input; 1 by default).	
	OUT = 0	direct output to a data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the file to which data records are to be written (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid TIES value
	IERR = 4	invalid OUT value
	IERR = 5	error opening OFILE
	IERR = 6	prerequisite requirements for API are not met

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1.188. OUTS

Use this API to tabulate those components in the working case that are removed from service.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OUTS  SID  ALL
```

Python command syntax:

```
ierr = outs(sid, all)
```

Fortran command syntax:

```
CALL OUTSNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.189. OWNM

This API routine is obsolete. It has been replaced by the API routine ownm_2 .

Use this API routine to reassign the buses, load, machines, and branches in a specified subsystem of the working case from their original owner to a designated owner.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OWNM  SID  ALL  STATUS(1)..STATUS(7)  IONEW
```

Python command syntax:

```
ierr = ownm(sid, all, status, ionew)
```

Fortran command syntax:

```
CALL OWNMAPI(SID, ALL, STATUS, IONEW, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS (7)	array of 7 elements specifying They are used to control reassignment options(input).	
	STATUS(1)	bus owners to IONEW (0 by default)
	STATUS(1) = 0	leave bus ownership unchanged
	STATUS(1) = 1	Change bus owners to IONEW
	STATUS(2)	load owners to IONEW (0 by default)
	STATUS(2) = 0	leave load ownership unchanged
	STATUS(2) = 1	change load owners to IONEW
	STATUS(3)	machine owners to IONEW (0 by default)
	STATUS(3) = 0	leave machine ownership unchanged
	STATUS(3) = 1	change machine owners to IONEW
	STATUS(4)	branch owners to IONEW (0 by default)
	STATUS(4) = 0	leave branch ownership unchanged
	STATUS(4) = 1	change branch owners to IONEW
	STATUS(5)	FACTS device owners to IONEW (0 by default)
	STATUS(5) = 0	leave FACTS device ownership unchanged
	STATUS(5) = 1	change FACTS device owners to IONEW
	STATUS(6)	VSC dc line owners to IONEW (0 by default)
	STATUS(6) = 0	leave VSC dc line ownership unchanged

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	STATUS(6) = 1	change VSC dc line owners to IONEW
	STATUS(7)	branch, series FACTS device, VSC dc line option (1 by default)
	STATUS(7) = 1	subsystem branches only
	STATUS(7) = 2	subsystem ties only
	STATUS(7) = 3	both subsystem branches and ties
Integer IONEW	new owner to which the specified equipment items are to be assigned. IONEW must be between 1 and 9999 (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid new owner number
	IERR = 5	owner table is full; cannot add owner to the working case
	IERR = 6	prerequisite requirements for API are not met

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1.190. OWNM_2

This API routine is obsolete. It has been replaced by the API routine ownm_3 .

This API routine is the second release of the owner renumbering function. Use this API routine to reassign the buses, loads, machines, branches, FACTS devices, VSC dc lines, and/or GNE devices in a specified sub-system of the working case from their original owner to a designated owner.

This API routine was first introduced in release 32.0.2.

Batch command syntax:

```
BAT_OWNM_2  SID  ALL  STATUS(1)..STATUS(8)  IONEW
```

Python command syntax:

```
ierr = ownm_2(sid, all, status, ionew)
```

Fortran command syntax:

```
CALL OWNMAPI_2(SID, ALL, STATUS, IONEW, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS (8)	array of 8 elements specifying They are used to control reassignment options(input).	
	STATUS(1)	bus owners to IONEW (0 by default)
	STATUS(1) = 0	leave bus ownership unchanged
	STATUS(1) = 1	Change bus owners to IONEW
	STATUS(2)	load owners to IONEW (0 by default)
	STATUS(2) = 0	leave load ownership unchanged
	STATUS(2) = 1	change load owners to IONEW
	STATUS(3)	machine owners to IONEW (0 by default)
	STATUS(3) = 0	leave machine ownership unchanged
	STATUS(3) = 1	change machine owners to IONEW
	STATUS(4)	branch owners to IONEW (0 by default)
	STATUS(4) = 0	leave branch ownership unchanged
	STATUS(4) = 1	change branch owners to IONEW
	STATUS(5)	FACTS device owners to IONEW (0 by default)
	STATUS(5) = 0	leave FACTS device ownership unchanged
	STATUS(5) = 1	change FACTS device owners to IONEW
	STATUS(6)	VSC dc line owners to IONEW (0 by default)

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	STATUS(6) = 0	leave VSC dc line ownership unchanged
	STATUS(6) = 1	change VSC dc line owners to IONEW
	STATUS(7)	GNE device owners to IONEW (0 by default)
	STATUS(7) = 0	leave GNE device ownership unchanged
	STATUS(7) = 1	change GNE device owners to IONEW
	STATUS(8)	branch, series FACTS device, VSC dc line, and series GNE device option (1 by default)
	STATUS(8) = 1	subsystem branches only
	STATUS(8) = 2	subsystem ties only
	STATUS(8) = 3	both subsystem branches and ties
Integer IONEW	new owner to which the specified equipment items are to be assigned. IONEW must be between 1 and 9999 (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid new owner number
	IERR = 5	owner table is full; cannot add owner to the working case
	IERR = 6	prerequisite requirements for API are not met

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1.191. OWNM_3

This API routine is the third release of the owner renumbering function. Use this API routine to reassign the buses, loads, machines, induction machines, branches, FACTS devices, VSC dc lines, and/or GNE devices in a specified subsystem of the working case from their original owner to a designated owner.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_OWNM_3  SID  ALL  STATUS(1)..STATUS(9)  IONEW
```

Python command syntax:

```
ierr = ownm_3(sid, all, status, ionew)
```

Fortran command syntax:

```
CALL OWNM_3(SID, ALL, STATUS, IONEW, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS (9)	array of 9 elements specifying They are used to control reassignment options(input).	
	STATUS(1)	bus owners to IONEW (0 by default)
	STATUS(1) = 0	leave bus ownership unchanged
	STATUS(1) = 1	change bus owners to IONEW
	STATUS(2)	load owners to IONEW (0 by default)
	STATUS(2) = 0	leave load ownership unchanged
	STATUS(2) = 1	change load owners to IONEW
	STATUS(3)	machine owners to IONEW (0 by default)
	STATUS(3) = 0	leave machine ownership unchanged
	STATUS(3) = 1	change machine owners to IONEW
	STATUS(4)	induction machine owners to IONEW (0 by default)
	STATUS(4) = 0	leave induction machine ownership unchanged
	STATUS(4) = 1	change induction machine owners to IONEW
	STATUS(5)	branch owners to IONEW (0 by default)
	STATUS(5) = 0	leave branch ownership unchanged
	STATUS(5) = 1	change branch owners to IONEW
	STATUS(6)	FACTS device owners to IONEW (0 by default)
	STATUS(6) = 0	leave FACTS device ownership unchanged

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	STATUS(6) = 1	change FACTS device owners to IONEW
	STATUS(7)	VSC dc line owners to IONEW (0 by default)
	STATUS(7) = 0	leave VSC dc line ownership unchanged
	STATUS(7) = 1	change VSC dc line owners to IONEW
	STATUS(8)	GNE device owners to IONEW (0 by default)
	STATUS(8) = 0	leave GNE device ownership unchanged
	STATUS(8) = 1	change GNE device owners to IONEW
	STATUS(9)	branch, series FACTS device, VSC dc line, and series GNE device option (1 by default)
	STATUS(9) = 1	subsystem branches only
	STATUS(9) = 2	subsystem ties only
	STATUS(9) = 3	both subsystem branches and ties
Integer IONEW	new owner to which the specified equipment items are to be assigned. IONEW must be between 1 and 9999 (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid new owner number
	IERR = 5	owner table is full; cannot add owner to the working case
	IERR = 6	prerequisite requirements for API are not met

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1.192. OWNR

Use this API to tabulate owner totals by owner.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OWNR  SID  ALL
```

Python command syntax:

```
ierr = ownr(sid, all)
```

Fortran command syntax:

```
CALL OWNRNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid owner subsystem identifier. Valid subsystem identifiers range from 0 to 11. Owner subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all owners or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only owners in owner subsystem SID
	ALL = 1	process all owners
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.193. PATH

Use this API to specify a directory pathname.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_PATH  PATHNAME
```

Python command syntax:

```
ierr = path(pathname)
```

Fortran command syntax:

```
CALL PATHAPI(PATHNAME, IERR)
```

Where:

Character*260 PATHNAME	name of path to be used (blank to disable) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	error setting path PATHNAME
	IERR = 2	prerequisite requirements for API are not met

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1.194. PHASE_SHIFT_ADJUSTMENT

Use this API to specify or return the option to enable or disable the phase shift adjustment.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_PHASE_SHIFT_ADJUSTMENT  IVAL
```

Python command syntax:

```
ierr = phase_shift_adjustment(ival)

ierr, ival = phase_shift_adjustment()
```

Fortran command syntax:

```
CALL PHASE_SHIFT_ADJUSTMENT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	disabled.
	IVAL = 1	enabled.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.195. POLY

Use this API to calculate interchange limits of a study system against two opposing systems using a linear network (dc) model.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_POLY  OPTIONS(1)..OPTIONS(15)  VALUES(1)..VALUES(6)
 LABELS(1)..LABELS(3)  DFXFILE  RESFILE
```

Python command syntax:

```
ierr = poly(options, values, labels, dfxfile, resfile)
```

Fortran command syntax:

```
CALL POLYAPI(OPTIONS, VALUES, LABELS, DFXFILE, RESFILE, IERR)
```

Where:

Integer OPTIONS (15)	array of 15 elements specifying calculation and reporting options(input).	
	OPTIONS(1)	base case rating set (rating set program option setting by default)
	OPTIONS(1) = 1	RATEA
	OPTIONS(1) = 2	RATEB
	OPTIONS(1) = 3	RATEC
	OPTIONS(2)	contingency case rating set (rating set program option setting by default)
	OPTIONS(2) = 1	RATEA
	OPTIONS(2) = 2	RATEB
	OPTIONS(2) = 3	RATEC
	OPTIONS(3)	base case line flow code (0 by default)
	OPTIONS(3) = 0	dc base case
	OPTIONS(3) = 1	ac base case
	OPTIONS(4)	phase shifter code (0 by default)
	OPTIONS(4) = 0	locked in base case shift solutions
	OPTIONS(4) = 1	regulating in base case shift solutions
	OPTIONS(5)	code for the treatment of base case line flow constraints in contingency cases (0 by default)
	OPTIONS(5) = 0	ignore
	OPTIONS(5) = 1	include
	OPTIONS(6)	list study system buses (0 by default)
	OPTIONS(6) = 0	no
	OPTIONS(6) = 1	yes
	OPTIONS(7)	list opposing system buses (0 by default)
	OPTIONS(7) = 0	no

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	OPTIONS(7) = 1	yes
	OPTIONS(8)	list study system tie lines (0 by default)
	OPTIONS(8) = 0	no
	OPTIONS(8) = 1	yes
	OPTIONS(9)	add study system tie lines to monitored line list (0 by default)
	OPTIONS(9) = 0	no
	OPTIONS(9) = 1	yes
	OPTIONS(10)	format code (0 by default)
	OPTIONS(10) = 0	summary output
	OPTIONS(10) = 1	full output
	OPTIONS(11)	convert MVA ratings to estimated MW ratings (0 by default)
	OPTIONS(11) = 0	no
	OPTIONS(11) = 1	yes
	OPTIONS(12)	discontinue activity if MW mismatch exceeds VALUES(1) (0 by default)
	OPTIONS(12) = 0	discontinue
	OPTIONS(12) = 1	continue
	OPTIONS(13)	graphical option (0 by default)
	OPTIONS(13) = 0	plot combined case only
	OPTIONS(13) = 1	plot combined case only
	OPTIONS(13) = 2	plot for all system conditions
	OPTIONS(14)	option for plotting constraints (0 by default)
	OPTIONS(14) = 0	plot all constraints
	OPTIONS(14) = 1	plot only constraints forming the boundary of the feasible region
	OPTIONS(15)	graphical format (0 by default)
	OPTIONS(15) = 0	single page per case
	OPTIONS(15) = 1	two pages per case
Real VALUES (6)	array of 6 elements specifying tolerances and thresholds(input).	
	VALUES(1)	required MW mismatch tolerance for continuing (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	percent of rating for reporting an element as overloaded (100.0 by default)
	VALUES(3)	maximum opposing system generation shift (10000.0 by default)
	VALUES(4)	maximum opposing system delta interchange for plotting (1000.0 by default)
	VALUES(5)	minimum distribution factor magnitude (0.0 by default)
	VALUES(6)	study system generation shift (100.0 by default)

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Character*32 LABELS (3)	array of 3 elements specifying subsystem labels(input).	
	LABELS(1)	study system label (no default allowed)
	LABELS(2)	first opposing system label (no default allowed)
	LABELS(3)	second opposing system label (no default allowed)
Character*260 DFXFILE	name of DFAX data file (input; no default allowed).	
Character*260 RESFILE	name of POLY results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	no Distribution Factor Data File specified
	IERR = 4	same label specified for at least two subsystems
	IERR = 5	no POLY Results Output File specified
	IERR = 6	generators are converted
	IERR = 7	buses in island(s) without a swing bus; use activity TREE
	IERR = 8	largest mismatch exceeds mismatch tolerance
	IERR = 9	error opening file DFXFILE
	IERR = 10	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 11	file DFXFILE does not contain distribution factors; run DFAX with distribution factor option enabled
	IERR = 12	no monitored branches or interfaces defined
	IERR = 13	no subsystems defined
	IERR = 14	subsystem not found
	IERR = 15	specified subsystems overlap
	IERR = 16	no in-service buses in subsystem WORLD
	IERR = 17	no tie lines from study system
	IERR = 18	only one subsystem defined
	IERR = 19	prerequisite requirements for API are not met

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1.196. POUT

Use this API to print the power flow solution results with boundary condition and flow information on the left side of the report and other information on the right side.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_POUT  SID  ALL
```

Python command syntax:

```
ierr = pout(sid, all)
```

Fortran command syntax:

```
CALL POUTNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.197. POWER_OUTPUT

Use this API to specify or return the option to display power output in either MVA or kVA.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_POWER_OUTPUT  IVAL
```

Python command syntax:

```
ierr = power_output(ival)

ierr, ival = power_output()
```

Fortran command syntax:

```
CALL POWER_OUTPUT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1 by default; else output).	
	IVAL = 0	kVA.
	IVAL = 1	MVA.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.198. PP_ACCC

Use this API to report the results of the AC contingency calculation function.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PP_ACCC  OPTIONS(1)..OPTIONS(5)  INTVAL(1)..INTVAL(5)
 REALVAL(1)..REALVAL(6) ACCFILE
```

Python command syntax:

```
ierr = pp_accc(options, intval, realval, accfile)
```

Fortran command syntax:

```
CALL PP_ACCC(OPTIONS, INTVAL, REALVAL, ACCFILE, IERR)
```

Where:

Integer OPTIONS (5)	array of 5 elements specifying reporting options(input).	
	OPTIONS(1)	contingency case rating (rating set program option setting by default)
	OPTIONS(1) = 0	RATEA
	OPTIONS(1) = 1	RATEB
	OPTIONS(1) = 2	RATEC
	OPTIONS(2)	report format (3 by default)
	OPTIONS(2) = 0	spreadsheet overload report
	OPTIONS(2) = 1	spreadsheet loading table
	OPTIONS(2) = 2	available capacity table
	OPTIONS(2) = 3	non-spreadsheet overload report
	OPTIONS(2) = 4	non-spreadsheet loading table
	OPTIONS(2) = 5	non-converged networks report
	OPTIONS(3)	exclude interfaces from report (0 by default)
	OPTIONS(3) = 0	no
	OPTIONS(3) = 1	yes
	OPTIONS(4)	run voltage limit check (0 by default)
	OPTIONS(4) = 0	no
	OPTIONS(4) = 1	yes
	OPTIONS(5)	exclude cases with no overloads from non-spreadsheet overload report (0 by default)
	OPTIONS(5) = 0	no
	OPTIONS(5) = 1	yes
Integer INTVAL (5)	array of 5 elements specifying integer reporting parameters(input).	
	INTVAL(1)	number of low voltage range violations (0 by default)
	INTVAL(2)	number of high voltage range violations (0 by default)

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	INTVAL(3)	number of voltage deviation violations (0 by default)
	INTVAL(4)	number of buses in the largest disconnected island (0 by default)
	INTVAL(5)	maximum number of elements in available capacity table (no limit by default)
Real REALVAL (6)	array of 6 elements specifying real reporting parameters(input).	
	REALVAL(1)	percent of flow rating (100.0 by default)
	REALVAL(2)	minimum contingency case flow change for overload report (0.0 by default)
	REALVAL(3)	cutoff threshold for available capacity table (99999.0 by default)
	REALVAL(4)	minimum contingency case voltage change for range violations (0.0 by default)
	REALVAL(5)	bus mismatch converged tolerance (MW or Mvar) (0.5 by default)
	REALVAL(6)	system mismatch converged tolerance (MVA) (5.0 by default)
Character*260 ACCFILE	name of the contingency solution output file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid INTVAL value
	IERR = 3	invalid REALVAL value
	IERR = 4	error opening RFILE
	IERR = 5	error reading RFILE
	IERR = 6	prerequisite requirements for API are not met

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1.199. PP_ACCC_MULTI_CASE

This API routine is obsolete. It has been replaced by the API routine accc_multiple_run_report .

Use this API to report the results of up to nine executions of the AC contingency calculation function.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_PP_ACCC_MULTI_CASE  OPTIONS(1)..OPTIONS(11)  VALUES(1)..VALUES(6)
 NFILES  ACCFILES(1)..ACCFILES(NFILES)
```

Python command syntax:

```
ierr = pp_accc_multi_case(options, values, nfiles, accfiles)
```

Fortran command syntax:

```
CALL PP_ACCC_MULTI_CASE(OPTIONS, VALUES, NFILES, ACCFILES, IERR)
```

Where:

Integer OPTIONS (11)	array of 11 elements specifying reporting options(input).	
	OPTIONS(1)	column headings code (1 by default)
	OPTIONS(1) = 1	contingency solution output file names
	OPTIONS(1) = 2	saved case file names as contained in the contingency solution output files
	OPTIONS(2)	rating set (rating set program option setting by default)
	OPTIONS(2) = 1	RateA
	OPTIONS(2) = 2	RateB
	OPTIONS(2) = 3	RateC
	OPTIONS(3)	print monitored elements summary report (0 by default)
	OPTIONS(3) = 0	no
	OPTIONS(3) = 1	yes
	OPTIONS(4)	print missing monitored elements report (0 by default)
	OPTIONS(4) = 0	no
	OPTIONS(4) = 1	yes
	OPTIONS(5)	print missing monitored voltage buses report (0 by default)
	OPTIONS(5) = 0	no
	OPTIONS(5) = 1	yes
	OPTIONS(6)	print contingency legend (2 by default)
	OPTIONS(6) = 0	no
	OPTIONS(6) = 1	complete master legend report only
	OPTIONS(6) = 2	reduced legend with each table

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	OPTIONS(6) = 3	both master legend report and legend with each table
	OPTIONS(7)	print missing contingencies report (0 by default)
	OPTIONS(7) = 0	no
	OPTIONS(7) = 1	yes
	OPTIONS(8)	print non-converged contingencies report (0 by default)
	OPTIONS(8) = 0	no
	OPTIONS(8) = 1	yes
	OPTIONS(9)	print loading violation reports (0 by default)
	OPTIONS(9) = 0	no
	OPTIONS(9) = 1	base case and worst case contingency violations only
	OPTIONS(9) = 2	base case and all contingency violations only
	OPTIONS(9) = 3	base case, worst case contingency, and all contingency violations
	OPTIONS(10)	print voltage violation reports (0 by default)
	OPTIONS(10) = 0	no
	OPTIONS(10) = 1	base case and worst case contingency violations only
	OPTIONS(10) = 2	base case and all contingency violations only
	OPTIONS(10) = 3	base case, worst case contingency, and all contingency violations
	OPTIONS(11)	option for processing interfaces in loading violation reports (1 by default)
	OPTIONS(11) = 0	exclude interfaces from loading violation checking and reporting
	OPTIONS(11) = 1	check and report interface loading violations
Real VALUES (6)	array of 6 elements specifying integer reporting parameters(input).	
	VALUES(1)	percent of flow rating for reporting (100.0 by default)
	VALUES(2)	percent of flow rating for counting in worst case contingency violations report (> VALUES(1)) (100.0 by default)
	VALUES(3)	minimum contingency case flow change for reporting and counting (0.0 by default)
	VALUES(4)	minimum contingency case voltage change for reporting and counting in voltage range violations (0.0 by default)
	VALUES(5)	bus mismatch converged tolerance (MW or Mvar) (0.5 by default)
	VALUES(6)	system mismatch converged tolerance (MVA) (5.0 by default)

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Integer NFILES	number of contingency solution output files to be processed (1 through 9) (input; no default allowed).	
Character*260 ACCFILES (NFILES)	NFILES names of contingency solution output files (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	invalid NFILES value
	IERR = 4	the percent for counting is less than the percent for reporting
	IERR = 5	file is not in the form of a current contingency solution output file
	IERR = 6	error opening a contingency solution output file
	IERR = 7	error reading a contingency solution output file
	IERR = 8	error opening a temporary file
	IERR = 9	error opening a subfile in the temporary file
	IERR = 10	error reading or writing the temporary file
	IERR = 11	error closing a subfile in the temporary file
	IERR = 12	prerequisite requirements for API are not met

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1.200. PRINT_OUTAGED_BRANCHES

Use this API to specify or return the option to enable or disable whether to print out-of-service branches.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PRINT_OUTAGED_BRANCHES  IVAL
```

Python command syntax:

```
ierr = print_outaged_branches(ival)

ierr, ival = print_outaged_branches()
```

Fortran command syntax:

```
CALL PRINT_OUTAGED_BRANCHES_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	do not print out-of-service branches.
	IVAL = 1	print out-of-service branches.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.201. PRINT_WINDING_BUSES

Use this API to specify or return the option to enable or disable whether to print winding "to" buses of three-winding transformers.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PRINT_WINDING_BUSES  IVAL
```

Python command syntax:

```
ierr = print_winding_buses(ival)

ierr, ival = print_winding_buses()
```

Fortran command syntax:

```
CALL PRINT_WINDING_BUSES_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	do not print winding "to" buses.
	IVAL = 1	print winding "to" buses.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.202. PROGRESS_OUTPUT

Use this API to specify the progress output device.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PROGRESS_OUTPUT  ISLCT  FILARG  OPTIONS(1)  OPTIONS(2)
```

Python command syntax:

```
ierr = progress_output(islct, filarg, options)
```

Fortran command syntax:

```
CALL OPENPROGRESSDEVICE(ISLCT, FILARG, OPTIONS, IERR)
```

Where:

Integer ISLCT	virtual device selector (input).	
	ISLCT = 1	standard destination. In the PSSE GUI standard output for progress is the Progress tab of the output bar.
	ISLCT = 2	direct output to a file.
	ISLCT = 3	direct output to a printer.
	ISLCT = 4	direct output to progress device.
	ISLCT = 5	direct output to report device.
	ISLCT = 6	no output.
Character*260 FILARG	printer name or filename; ignored if ISLCT is not 2 or 3 (input).	
Integer OPTIONS (2)	array of 2 elements specifying open and printing options; ignored if ISLCT is not 2 or 3(input).	
	OPTIONS(1)	file and printer open options: The OPTIONS(1) values may be summed to select multiple options
	OPTIONS(1) = 0	open with carriage control format and, for files, for overwrite of existing files
	OPTIONS(1) = 1	open with list format
	OPTIONS(1) = 2	open file for append (file output only)
	OPTIONS(1) = 2	set printer orientation to portrait mode (printer output only)
	OPTIONS(1) = 4	open printer in immediate print mode (printer output only)
	OPTIONS(2)	number of copies to print (printer only)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = 2	failed to open device

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1.203. PROMPT_OUTPUT

Use this API to specify the prompt output device.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PROMPT_OUTPUT  ISLCT  FILARG  OPTIONS(1)  OPTIONS(2)
```

Python command syntax:

```
ierr = prompt_output(islct, filarg, options)
```

Fortran command syntax:

```
CALL OPENPROMPTDEVICE(ISLCT, FILARG, OPTIONS, IERR)
```

Where:

Integer ISLCT	virtual device selector (input).	
	ISLCT = 1	standard destination. In the PSSE GUI standard output for prompts is the input window generated by user input requests.
	ISLCT = 2	direct output to a file.
	ISLCT = 3	direct output to a printer.
	ISLCT = 4	direct output to progress device.
	ISLCT = 5	direct output to report device.
	ISLCT = 6	no output.
Character*260 FILARG	printer name or filename; ignored if ISLCT is not 2 or 3 (input).	
Integer OPTIONS (2)	array of 2 elements specifying open and printing options; ignored if ISLCT is not 2 or 3(input).	
	OPTIONS(1)	file and printer open options: The OPTIONS(1) values may be summed to select multiple options
	OPTIONS(1) = 0	open with carriage control format and, for files, for overwrite of existing files
	OPTIONS(1) = 1	open with list format
	OPTIONS(1) = 2	open file for append (file output only)
	OPTIONS(1) = 2	set printer orientation to portrait mode (printer output only)
	OPTIONS(1) = 4	open printer in immediate print mode (printer output only)
	OPTIONS(2)	number of copies to print (printer only)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = 2	failed to open device

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1.204. PRTI

Use this API to print the 16 line long title.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_PRTI
```

Python command syntax:

```
ierr = prti()
```

Fortran command syntax:

```
CALL PRTIAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

1.205. PSCOPF

This API routine is obsolete. It has been replaced by the API routine pscopf_2 .

This API is used to find the settings of controls for a system to comply with base case and contingency case operation limits. The API is implemented with preventive security constrained optimal power flow method (PSCOPF). The objective of the PSCOPF is to minimize control adjustments subject to operation limits constraints. The available controls are generation MW outputs of on-line and off-line generators, loads, phase shifters, tap settings and switched shunts.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_PSCOPF  OPTIONS(1)..OPTIONS(26) VALUES(1)..VALUES(11)
 LABELS(1)..LABELS(7) DFXFILE INLFILE THRFILE
```

Python command syntax:

```
ierr = pscopf(options,values,labels,dfxfile,inlfile,thrfile)
```

Fortran command syntax:

```
CALL PSCOPFAPI(OPTIONS,VALUES, LABELS,DFXFILE,INLFILE,THRFILE,IERR)
```

Where:

Integer OPTIONS (26)	array of 26 elements specifying integer specifying power flow options(input).	
	OPTIONS(1)	tap adjustment flag for the base case (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag for the base case (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag for the base case (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag for the base case (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable

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OPTIONS(5)	switched shunt adjustment flag for the base case (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable all
OPTIONS(5) = 2	enable continuous, disable discrete
OPTIONS(6)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(6) = 0	stall
OPTIONS(6) = 1	trip
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	Fixed slope decoupled Newton-Raphson method
OPTIONS(7) = 1	Full Newton-Raphson method
OPTIONS(7) = 2	Optimized fixed slope decoupled Newton-Raphson method
OPTIONS(8)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(8) = 0	Disable
OPTIONS(8) = 1	Enable
OPTIONS(9)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
OPTIONS(9) = 0	disable
OPTIONS(9) = 1	enable stepping adjustment
OPTIONS(9) = 2	enable direct adjustment
OPTIONS(10)	area interchange adjustment flag for the contingency case (area interchange adjustment option setting by default)
OPTIONS(10) = 0	disable
OPTIONS(10) = 1	enable using tie line flows only in calculating area interchange
OPTIONS(10) = 2	enable using tie line flows and loads in calculating area interchange
OPTIONS(11)	phase shift adjustment flag for the contingency case (phase shift adjustment option setting by default)
OPTIONS(11) = 0	disable
OPTIONS(11) = 1	enable
OPTIONS(12)	dc tap adjustment flag for the contingency case (dc tap adjustment option setting by default)
OPTIONS(12) = 0	disable
OPTIONS(12) = 1	enable

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OPTIONS(13)	switched shunt adjustment flag for the contingency case (switched shunt adjustment option setting by default)
OPTIONS(13) = 0	disable
OPTIONS(13) = 1	enable
OPTIONS(13) = 2	enable continuous, disable discrete
OPTIONS(14)	induction motor treatment flag for the contingency case (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(14) = 0	stall
OPTIONS(14) = 1	trip
OPTIONS(15)	dispatch mode (0 by default)
OPTIONS(15) = 0	disable
OPTIONS(15) = 1	subsystem machines (RESERVE)
OPTIONS(15) = 2	subsystem machines (PMAX)
OPTIONS(15) = 3	subsystem machines (INERTIA)
OPTIONS(15) = 4	subsystem machines (GOVERNOR DROOP)
OPTIONS(16)	Rating set for the base case
OPTIONS(16) = 1	RATE A
OPTIONS(16) = 2	RATE B
OPTIONS(16) = 3	RATE C
OPTIONS(17)	Rating set for the contingency case
OPTIONS(17) = 1	RATE A
OPTIONS(17) = 2	RATE B
OPTIONS(17) = 3	RATE C
OPTIONS(18)	maximum iteration number of PSCOPF
OPTIONS(18) = 1	RATE A
OPTIONS(18) = 2	RATE B
OPTIONS(18) = 3	RATE C
OPTIONS(19)	maximum number of a contingency that does not cause limit violations and instability issue to be tested in the iterative process of PSCOPF
OPTIONS(20)	Handle initial violations in the base case
OPTIONS(20) = 0	Ignore the initial violations in the base case from the PSCOPF analysis
OPTIONS(20) = 1	Clean the initial violations in the base case
OPTIONS(21)	generator control flag (1 by default)
OPTIONS(21) = 0	disable generation MW dispatch
OPTIONS(21) = 1	enable
OPTIONS(22)	load control flag (0 by default)
OPTIONS(22) = 0	disable load shedding control
OPTIONS(22) = 1	enable
OPTIONS(23)	phase shifter control flag (1 by default)

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	OPTIONS(23) = 0	disable phase shifter control
	OPTIONS(23) = 1	enable
	OPTIONS(24)	off-line generator control flag (0 by default)
	OPTIONS(24) = 0	disable generation MW dispatch of off-line generators
	OPTIONS(24) = 1	enable
	OPTIONS(25)	tap setting adjustment flag (0 by default)
	OPTIONS(25) = 0	disable tap setting adjustment
	OPTIONS(25) = 1	enable
	OPTIONS(26)	switched shunt control flag (0 by default)
	OPTIONS(26) = 0	disable switch shunt control
	OPTIONS(26) = 1	enable
Real VALUES (11)	array of 11 elements specifying VALUES Is a real(input).	
	VALUES(1)	Mismatch tolerance (TOLN by default)
	VALUES(2)	Percentage of rating for the base case (100.0 by default)
	VALUES(3)	Percentage of rating for the contingency cases (100.0 by default)
	VALUES(4)	Voltage tolerance in linear programming
	VALUES(5)	Flow tolerance in linear programming
	VALUES(6)	Weighting factor of generation MW dispatch (1.0 by default)
	VALUES(7)	Weighting factor of load control (1.0 by default)
	VALUES(8)	Weighting factor of phase shifter control (1.0 by default)
	VALUES(9)	Weighting factor of generation MW dispatch of off-line generators (1.0 by default)
	VALUES(10)	Weighting factor of tap setting adjustment (1.0 by default)
	VALUES(11)	Weighting factor of switched shunt control (1.0 by default)
Character*32 LABELS (7)	array of 7 elements specifying a character(input).	
	LABELS(1)	label of generation dispatch system, required if the dispatch mode of OPTIONS(15) is not zero. The dispatch mode and the system provide for a default dispatch method for contingencies causing unbalance between load demand and generation (blank for none by default)
	LABELS(2)	label of generation dispatch control system. MW injections of the generators within the system are available for dispatch in the corrective action mode or preventive security constrained optimal power flow mode (blank for none by default)

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	LABELS(3)	label of load control system. Loads in the load control system are available for curtailment in the corrective action mode or preventive security constrained optimal power flow mode (blank for none by default)
	LABELS(4)	label of phase shifter adjustment system. The phase angles of phase shifters in the system are regulated either by corrective action or preventive security constrained optimal power flow (blank for none by default)
	LABELS(5)	label of generation MW control of off-line generator control system. MW injections of the off-line generators within the system are available for dispatch in the corrective action mode or preventive security constrained optimal power flow mode (blank for none by default)
	LABELS(6)	label of tap adjustment system. The tap positions of transformers in voltage control mode are regulated either by corrective action or preventive security constrained optimal power flow (blank for none by default)
	LABELS(7)	label of switched shunt control system. The admittances of switched shunts in voltage control mode are regulated either by corrective action or preventive security constrained optimal power flow (blank for none by default)
Character*260 DFXFILE	name of the Distribution Factor Data File (input; no default allowed).	
Character*260 INLFILE	name of Unit Inertia Data File (input; blank for none).	
Character*260 THRFILE	name of Load Throwover Data File (input; blank for none).	
Integer IERR	error code (output).	
	IERR = -2	PSCOPF terminates with active constraints
	IERR = -1	PSCOPF reaches the maximum iteration
	IERR = 0	No error
	IERR = 1	Invalid OPTIONS value
	IERR = 2	Invalid VALUES value
	IERR = 3	Generators are converted
	IERR = 4	Buses in island(s) without a swing bus--use activity TREE
	IERR = 5	Largest mismatch exceeds tolerance
	IERR = 6	Generation subsystem not specified
	IERR = 7	Too many islands in base case
	IERR = 8	in-service induction machines are in the "stalled" or "tripped" state

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IERR = 9	buses with bus type code and series element status inconsistencies
IERR = 10	No Distribution Factor Data input file specified
IERR = 11	No controls are specified
IERR = 12	No constraints are specified
IERR = 13	Invalid base/contingency case flag combination
IERR = 21	File is not in the form of a PSS/E-25 or later DFAX file--run DFAX
IERR = 22	Monitored elements exceed limit when adding multi-section line members
IERR = 51	Error opening distribution factor data file
IERR = 52	Error opening load throwover data file
IERR = 53	Error opening inertia data file
IERR = 54	Prerequisite requirements for function not met

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1.206. PSCOPF_2

This API is the second release of Preventive Security Constrained Optimal Power Flow solution (PSCOPF). This API is used to find the settings of controls for a system to comply with base case and contingency case operation limits. The API is implemented with preventive security constrained optimal power flow method (PSCOPF). The objective of the PSCOPF is to minimize control adjustments subject to operation limits constraints. The available controls are generation MW outputs of on-line and off-line generators, loads, phase shifters, tap settings and switched shunts.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_PSCOPF_2  OPTIONS(1)..OPTIONS(28) VALUES(1)..VALUES(11)
 LABELS(1)..LABELS(7) DFXFILE INLFILE THRFILE
```

Python command syntax:

```
ierr = pscopf_2(options, values, labels, dfxfile, inlfile, thrfile)
```

Fortran command syntax:

```
CALL PSCOPFAPI_2(OPTIONS, VALUES, LABELS, DFXFILE, INLFILE, THRFILE, IERR)
```

Where:

Integer OPTIONS (28)	array of elements specifying power flow options(input).	
	OPTIONS(1)	tap adjustment flag for the base case (tap adjustment option by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag for the base case (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange.
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag for the base case (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag for the base case (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable

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OPTIONS(5)	switched shunt adjustment flag for the base case (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable all
OPTIONS(5) = 2	enable continuous, disable discrete.
OPTIONS(6)	induction motor treatment flag (applied when an induction motor fails to solve due to low terminal bus voltage, 0 by default)
OPTIONS(6) = 0	stall.
OPTIONS(6) = 1	trip.
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	Fixed slope decoupled Newton-Raphson method
OPTIONS(7) = 1	Full Newton-Raphson method
OPTIONS(7) = 2	Optimized fixed slope decoupled Newton-Raphson method
OPTIONS(8)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(8) = 0	Disable
OPTIONS(8) = 1	Enable
OPTIONS(9)	tap adjustment flag for the first N-1 contingency (tap adjustment option setting by default)
OPTIONS(9) = 0	disable.
OPTIONS(9) = 1	enable stepping adjustment.
OPTIONS(9) = 2	enable direct adjustment.
OPTIONS(10)	area interchange adjustment flag for the contingency case (area interchange adjustment option setting by default)
OPTIONS(10) = 0	disable.
OPTIONS(10) = 1	enable using tie line flows only in calculating area interchange.
OPTIONS(10) = 2	enable using tie line flows and loads in calculating area interchange.
OPTIONS(11)	phase shift adjustment flag for the contingency case (phase shift adjustment option setting by default)
OPTIONS(11) = 0	disable.
OPTIONS(11) = 1	enable.
OPTIONS(12)	dc tap adjustment flag for the contingency case (dc tap adjustment option setting by default)
OPTIONS(12) = 0	disable.
OPTIONS(12) = 1	enable.

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OPTIONS(13)	switched shunt adjustment flag for the contingency case (switched shunt adjustment option setting by default)
OPTIONS(13) = 0	disable.
OPTIONS(13) = 1	enable.
OPTIONS(13) = 2	enable continuous, disable discrete.
OPTIONS(14)	induction motor treatment flag for the contingency case(applied when an induction motor fails to solve due to low terminal bus voltage) (0 by default)
OPTIONS(14) = 0	stall.
OPTIONS(14) = 1	trip.
OPTIONS(15)	dispatch mode (0 by default)
OPTIONS(15) = 0	disable.
OPTIONS(15) = 1	subsystem machines (RESERVE).
OPTIONS(15) = 2	subsystem machines (PMAX).
OPTIONS(15) = 3	subsystem machines (INERTIA).
OPTIONS(15) = 4	subsystem machines (GOVERNOR DROOP).
OPTIONS(16)	Rating set for the base case
OPTIONS(16) = 1	rate A
OPTIONS(16) = 2	rate B
OPTIONS(16) = 3	rate C
OPTIONS(17)	Rating set for the contingency case
OPTIONS(17) = 1	rate A
OPTIONS(17) = 2	rate B
OPTIONS(17) = 3	rate C
OPTIONS(18)	base case voltage limit
OPTIONS(18) = 1	normal
OPTIONS(18) = 2	emergency
OPTIONS(19)	contingency case voltage limit
OPTIONS(19) = 1	normal
OPTIONS(19) = 2	emergency
OPTIONS(20)	maximum iteration number of PSCOPF
OPTIONS(21)	maximum number of a contingency that does not cause limit violations and instability issue to be tested in the iterative process of PSCOPF
OPTIONS(22)	Handle initial violations in the base case
OPTIONS(22) = 0	Ignore the initial violations in the base case from the PSCOPF analysis
OPTIONS(22) = 1	Clean the initial violations in the base case
OPTIONS(23)	generator control flag (1 by default)
OPTIONS(23) = 0	disable generation MW dispatch
OPTIONS(23) = 1	enable
OPTIONS(24)	load control flag (0 by default)

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	OPTIONS(24) = 0	disable load shedding control
	OPTIONS(24) = 1	enable
	OPTIONS(25)	phase shifter control flag (1 by default)
	OPTIONS(25) = 0	disable phase shifter control
	OPTIONS(25) = 1	enable
	OPTIONS(26)	off-line generator control flag (0 by default)
	OPTIONS(26) = 0	disable generation MW dispatch of off-line generators
	OPTIONS(26) = 1	enable
	OPTIONS(27)	tap setting adjustment flag (0 by default)
	OPTIONS(27) = 0	disable tap setting adjustment
	OPTIONS(27) = 1	enable
	OPTIONS(28)	switched shunt control flag (0 by default)
	OPTIONS(28) = 0	disable switch shunt control
	OPTIONS(28) = 1	enable
Real VALUES (11)	array of elements specifying power flow options(input).	
	VALUES(1)	Mismatch tolerance (TOLN by default)
	VALUES(2)	Percentage of rating for the base case (100.0 by default)
	VALUES(3)	Percentage of rating for the contingency cases (100.0 by default)
	VALUES(4)	Voltage tolerance in linear programming
	VALUES(5)	Flow tolerance in linear programming
	VALUES(6)	Weighting factor of generation MW dispatch (1.0 by default)
	VALUES(7)	Weighting factor of load control (1.0 by default)
	VALUES(8)	Weighting factor of phase shifter control (1.0 by default)
	VALUES(9)	Weighting factor of generation MW dispatch of off-line generators (1.0 by default)
	VALUES(10)	Weighting factor of tap setting adjustment (1.0 by default)
	VALUES(11)	Weighting factor of switched shunt control (1.0 by default)
Character*32 LABELS (7)	array of elements specifying labels(input).	
	LABELS(1)	The label of generation dispatch system, required if the dispatch mode of OPTIONS(15) is not zero. The dispatch mode and the system provide for a default dispatch method for contingencies causing unbalance between load demand and generation output (blank for none).
	LABELS(2)	The label of generation dispatch control system. MW injections of the generators within

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		the system are available for dispatch in the corrective action mode or preventive security constrained optimal power flow mode (blank for none).
	LABELS(3)	The label of load control system. Loads in the load control system are available for curtailment in the corrective action mode or preventive security constrained optimal power flow mode (blank for none).
	LABELS(4)	The label of phase shifter adjustment system. The phase angles of phase shifters in the system are regulated either by corrective action or preventive security constrained optimal power flow (blank for none).
	LABELS(5)	The label of generation MW control of off-line generator control system. MW injections of the off-line generators within the system are available for dispatch in the corrective action mode or preventive security constrained optimal power flow mode (blank for none).
	LABELS(6)	The label of tap adjustment system. The tap positions of transformers in voltage control mode are regulated either by corrective action or preventive security constrained optimal power flow (blank for none).
	LABELS(7)	The label of switched shunt control system. The admittances of switched shunts in voltage control mode are regulated either by corrective action or preventive security constrained optimal power flow (blank for none).
Character*260 DFXFILE	Is the name of the Distribution Factor Data File (input; no default allowed).	
Character*260 INLFILE	Is the name of Load Throwover Data File (input, blank for none). (input).	
Character*260 THRFILE	Is the name of Unit Inertia Data File (input, blank for none). (input).	
Integer IERR	error code (output).	
	IERR = -2	PSCOPF terminates with active constraints
	IERR = -1	PSCOPF reaches the maximum iteration
	IERR = 0	No error
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance

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IERR = 6	Generation subsystem not specified
IERR = 7	too many islands in base case
IERR = 8	in-service induction machines are in the "stalled" or "tripped" state
IERR = 9	buses with bus type code and series element status inconsistencies
IERR = 10	no Distribution Factor Data File specified
IERR = 11	No controls are specified
IERR = 12	No constraints are specified
IERR = 13	Invalid base/contingency case flag combination.
IERR = 21	file DFXFILE is not in the form of a PSSE 25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening Distribution Factor Data File
IERR = 52	error opening Load Throwover Data File
IERR = 53	error opening Unit Inertia and Governor Data File
IERR = 54	prerequisite requirements for API are not met

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1.207. PSEB

Use this API to convert a PSEB command file into a PSSE response file.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PSEB  INPFILE  OUTFILE
```

Python command syntax:

```
ierr, respfil = pseb(inpfile, outfile)
```

Fortran command syntax:

```
CALL PSEBAPI(INPFILE, OUTFILE, RESPFIL, IERR)
```

Where:

Character*260 INPFILE	name of the PSEB command file (input; no default allowed).	
Character*260 OUTFILE	name of response file; blank to have PSEB assign the filename (input; blank by default).	
Character*260 RESPFIL	pathname of the response file constructed by PSEB (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	error opening INPFILE
	IERR = 2	error opening OUTFILE
	IERR = 3	errors detected while reading the PSEB Command Data input file
	IERR = 4	prerequisite requirements for API are not met

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1.208. PSSEHALT

This API routine is obsolete. It has been replaced by the API routine pssehalt_2 .

Use this API to end the operation of PSSE, closes all associated files and returns to the calling application.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PSSEHALT  DELETE
```

Python command syntax:

```
ierr = pssehalt(delete)
```

Fortran command syntax:

```
CALL HALTAPI(DELETE, IERR)
```

Where:

Integer DELETE	delete working files flag (input).	
	DELETE = 1	delete working files
	DELETE != 1	keep working files
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.209. PSSEHALT_2

Use this API to end the operation of PSSE, closes all associated files and returns to the calling application.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_PSSEHALT_2
```

Python command syntax:

```
ierr = pssehalt_2()
```

Fortran command syntax:

```
CALL HALTAPI_2(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.210. PURG

Use this API routine to delete specified outaged equipment items from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURG  SID  ALL  OPT  STATUS
```

Python command syntax:

```
ierr = purg(sid, all, opt, status)
```

Fortran command syntax:

```
CALL PURGNW(SID, ALL, OPT, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID.
	ALL = 1	process all buses.
Integer OPT	equipment type to remove (input; 1 by default).	
	OPT = 1	ac lines
	OPT = 2	switched shunts
	OPT = 3	machines
	OPT = 4	plants
	OPT = 5	two-terminal dc lines
	OPT = 6	multi-terminal dc lines
	OPT = 7	loads
	OPT = 8	FACTS devices
	OPT = 9	three-winding transformers
	OPT = 10	VSC dc lines
	OPT = 11	fixed bus shunts
	OPT = 12	GNE devices
	OPT = 13	induction machines
Integer STATUS	option for the treatment of out-of-service ties to other systems (input; 0 by default).	
	STATUS = 0	keep out-of-service ties
	STATUS = 1	remove out-of-service ties
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined

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IERR = 2	invalid ALL value
IERR = 3	invalid OPT value
IERR = 4	invalid STATUS value
IERR = 5	prerequisite requirements for API are not met

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1.211. PURG2DC

Use this API to delete the specified two-terminal dc line from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURG2DC  DC
```

Python command syntax:

```
ierr = purg2dc(dc)
```

Fortran command syntax:

```
CALL PURG2DC(DC, IERR)

CALL PURG2DC_I(NUM, IERR)

CALL PURG2DC_C(NAME, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line to be removed; if an un-quoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line to be removed; no default allowed	
Character*12 NAME	name of the two-terminal dc line to be removed; no default allowed	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	dc line not found
	IERR = 2	prerequisite requirements for API are not met

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1.212. PURG3WND

Use this API to delete the specified three-winding transformer from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURG3WND  IBUS  JBUS  KBUS  'CKT'
```

Python command syntax:

```
ierr = purg3wnd(ibus, jbus, kbus, ckt)
```

Fortran command syntax:

```
CALL PURG3WND(IBUS, JBUS, KBUS, CKT, IERR)
```

Where:

Integer IBUS	one of the buses connected by the three-winding transformer (input; no default allowed).	
Integer JBUS	the bus connected by the three-winding transformer (input; no default allowed).	
Integer KBUS	other bus connected by the three-winding transformer (input; no default allowed).	
Character*2 CKT	circuit identifier of the three-winding transformer to be deleted (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	three-winding transformer not found
	IERR = 3	prerequisite requirements for API are not met

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1.213. PURGAREA

Use this API to delete areas with no equipment assigned to them from the working case.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_PURGAREA  SID  ALL
```

Python command syntax:

```
ierr = purgarea(sid, all)
```

Fortran command syntax:

```
CALL PURGAREA(SID, ALL, IERR)
```

Where:

Integer SID	a valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Area subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all areas or specified subsystem flag (input; 1 by default).	
	ALL = 0	delete only areas in area subsystem SID with no equipment assigned to them.
	ALL = 1	delete all areas with no equipment assigned to them.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.214. PURGATRN

Use this API to delete the specified inter-area transfer from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGATRN  FRMAREA  TOAREA  'ID'
```

Python command syntax:

```
ierr = purgatrn(frmarea, toarea, id)
```

Fortran command syntax:

```
CALL PURGATRN(FRMAREA, TOAREA, ID, IERR)
```

Where:

Integer FRMAREA	one of the areas involved in the transfer (input; no default allowed).	
Integer TOAREA	other area involved in the transfer (input; no default allowed).	
Character*1 ID	inter-area transfer identifier (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	area not found
	IERR = 2	inter-area transfer not found
	IERR = 3	prerequisite requirements for API are not met

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1.215. PURGBRN

Use this API to delete the specified non-transformer branch or two-winding transformer from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGBRN  IBUS  JBUS  'CKT'
```

Python command syntax:

```
ierr = purgbrn(ibus, jbus, ckt)
```

Fortran command syntax:

```
CALL PURGBRN(IBUS, JBUS, CKT, IERR)
```

Where:

Integer IBUS	one of the buses connected by the branch (input; no default allowed).	
Integer JBUS	other bus connected by the branch (input; no default allowed).	
Character*2 CKT	circuit identifier of the branch to be deleted (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	branch not found
	IERR = 3	prerequisite requirements for API are not met

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1.216. PURGE_MULTI_TERM_DC_BUS

Use this API to delete the specified dc bus from the specified multi-terminal dc line in the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_MULTI_TERM_DC_BUS  DC  IBUS
```

Python command syntax:

```
ierr = purge_multi_term_dc_bus(dc, ibus)
```

Fortran command syntax:

```
CALL PURGMDC_DC_BUS_API(DC, IBUS, IERR, NERR)

CALL PURGMDC_DC_BUS_API_I(NUM, IBUS, IERR)

CALL PURGMDC_DC_BUS_API_C(NAME, IBUS, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line from which a dc bus is to be removed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line from which a dc bus is to be removed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line from which a dc bus is to be removed (input; no default allowed).	
Integer IBUS	number of the dc bus to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	dc line not found
	IERR = 2	invalid dc bus number
	IERR = 3	dc bus not present in this multi-terminal dc line
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - dc bus no longer connected to converter as first dc bus. - dc bus no longer connected to converter as second dc bus. - dc link deleted
Integer NERR	number of error entries in PBUF02	

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1.217. PURGE_MULTI_TERM_DC_CONVR

Use this API to delete the specified converter from the specified multi-terminal dc line in the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_MULTI_TERM_DC_CONVR  DC  IBUS
```

Python command syntax:

```
ierr = purge_multi_term_dc_convr(dc, ibus)
```

Fortran command syntax:

```
CALL PURGMDC_CONVERTER_API(DC, IBUS, IERR)

CALL PURGMDC_CONVERTER_API_I(NUM, IBUS, IERR)

CALL PURGMDC_CONVERTER_API_C(NAME, IBUS, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line from which a converter is to be removed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line from which a converter is to be removed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line from which a converter is to be removed (input; no default allowed).	
Integer IBUS	number of the ac bus from which the converter is to be removed (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	dc line not found
	IERR = 2	converter bus not found
	IERR = 3	bus is not a converter bus of this multi-terminal dc line
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - this converter bus is no longer its pole's voltage controlling converter. - the voltage setpoint of the new voltage controlling converter is not positive. - the voltage setpoint of the new voltage controlling converter is less than the mode switch voltage

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1.218. PURGE_MULTI_TERM_DC_LINK

Use this API to delete the specified dc link from the specified multi-terminal dc line in the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_MULTI_TERM_DC_LINK  DC  IBUS  JBUS  'CKT'
```

Python command syntax:

```
ierr = purge_multi_term_dc_link(dc, ibus, jbus, ckt)
```

Fortran command syntax:

```
CALL PURGMDC_DC_LINK_API(DC, IBUS, JBUS, CKT, IERR)

CALL PURGMDC_DC_LINK_API_I(NUM, IBUS, JBUS, CKT, IERR)

CALL PURGMDC_DC_LINK_API_C(NAME, IBUS, JBUS, CKT, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line from which a dc link is to be removed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line from which a dc link is to be removed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line from which a dc link is to be removed (input; no default allowed).	
Integer IBUS	one of the dc buses connected by the dc link to be removed (input).	
Integer JBUS	other dc bus connected by the dc link (input).	
Character*1 CKT	circuit identifier of the dc link to be deleted (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	dc line not found
	IERR = 2	invalid dc bus number
	IERR = 3	dc bus not present in this multi-terminal dc line
	IERR = 4	circuit identifier is more than one character
	IERR = 5	dc link not found in this multi-terminal dc line
	IERR = 6	prerequisite requirements for API are not met

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1.219. PURGFACTS

Use this API to delete the specified FACTS device from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGFACTS  FD
```

Python command syntax:

```
ierr = purgfacts(fd)
```

Fortran command syntax:

```
CALL PURGFACTS(FD, IERR)

CALL PURGFACTS_I(NUM, IERR)

CALL PURGFACTS_C(NAME, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device to be removed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device to be removed (input; no default allowed).	
Character*12 NAME	name of the FACTS device to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	FACTS device not found
	IERR = 2	prerequisite requirements for API are not met

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1.220. PURGGNE

Use this API routine to delete the specified GNE device from the working case.

This API routine was first introduced in release 32.0.2.

Batch command syntax:

```
BAT_PURGGNE  'NAME'
```

Python command syntax:

```
ierr = purggne(name)
```

Fortran command syntax:

```
CALL PURGGNE(NAME, IERR)
```

Where:

Character*12 NAME	name of the GNE device to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	GNE device not found
	IERR = 2	prerequisite requirements for API are not met

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1.221. PURGINDMAC

Use this API routine to delete the specified induction machine from the working case.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_PURGINDMAC  IBUS  'ID'
```

Python command syntax:

```
ierr = purgindmac(ibus, id)
```

Fortran command syntax:

```
CALL PURGINDMAC(IBUS, ID, IERR)
```

Where:

Integer IBUS	bus to which induction machine ID is connected (input; no default allowed).	
Character*2 ID	identifier of the induction machine to be removed (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no induction machines at bus
	IERR = 3	induction machine not found
	IERR = 4	prerequisite requirements for API are not met

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1.222. PURGINDMACS

Use this API routine to delete all induction machines from the specified bus in the working case.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_PURGINDMACS  IBUS
```

Python command syntax:

```
ierr = purgindmacs(ibus)
```

Fortran command syntax:

```
CALL PURGINDMACS(IBUS, IERR)
```

Where:

Integer IBUS	bus from which all induction machines are to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no induction machines at bus
	IERR = 3	prerequisite requirements for API are not met

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1.223. PURGLOAD

Use this API to delete the specified load from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGLOAD  IBUS  'ID'
```

Python command syntax:

```
ierr = purgload(ibus, id)
```

Fortran command syntax:

```
CALL PURGLOAD(IBUS, ID, IERR)
```

Where:

Integer IBUS	bus to which the load ID is connected (input; no default allowed).	
Character*2 ID	identifier of the load to be removed (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no loads at bus
	IERR = 3	load not found
	IERR = 4	prerequisite requirements for API are not met

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1.224. PURGLOADS

Use this API to delete all loads from the specified bus in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGLOADS  IBUS
```

Python command syntax:

```
ierr = purgloads(ibus)
```

Fortran command syntax:

```
CALL PURGLOADS(IBUS, IERR)
```

Where:

Integer IBUS	bus from which all loads are to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no loads at bus
	IERR = 3	prerequisite requirements for API are not met

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1.225. PURGMAC

Use this API to delete the specified machine from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGMAC  IBUS  'ID'
```

Python command syntax:

```
ierr = purgmac(ibus, id)
```

Fortran command syntax:

```
CALL PURGMAC(IBUS, ID, IERR)
```

Where:

Integer IBUS	bus to which the machine ID is connected (input; no default allowed).	
Character*2 ID	identifier of the machine to be removed (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no machines at bus
	IERR = 3	machine not found
	IERR = 4	prerequisite requirements for API are not met

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1.226. PURGMDC

Use this API to delete the specified multi-terminal dc line from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGMDC  DC
```

Python command syntax:

```
ierr = purgmdc(dc)
```

Fortran command syntax:

```
CALL PURGMDC(DC, IERR)

CALL PURGMDC_I(NUM, IERR)

CALL PURGMDC_C(NAME, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line to be removed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line to be removed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	dc line not found
	IERR = 2	prerequisite requirements for API are not met

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1.227. PURGMSL

Use this API to delete the specified multi-section line grouping from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGMSL  IBUS  JBUS  'CKT'
```

Python command syntax:

```
ierr = purgmsl(ibus, jbus, ckt)
```

Fortran command syntax:

```
CALL PURGMSL(IBUS, JBUS, CKT, IERR)
```

Where:

Integer IBUS	one of the endpoint buses of the multi-section line (input; no default allowed).	
Integer JBUS	other endpoint bus of the multi-section line (input; no default allowed).	
Character*2 CKT	circuit identifier of the multi-section line to be deleted; the first character must be an ampersand (input; '&1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	multi-section line not found
	IERR = 3	invalid multi-section line identifier (first character not &)
	IERR = 4	prerequisite requirements for API are not met

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1.228. PURGMUT

Use this API to delete the specified zero sequence mutual coupling from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGMUT  IBUS1  JBUS1  'CKT1'  IBUS2  JBUS2  'CKT2'
```

Python command syntax:

```
ierr = purgmut(ibus1, jbus1, ckt1, ibus2, jbus2, ckt2)
```

Fortran command syntax:

```
CALL PURGMUT(IBUS1, JBUS1, CKT1, IBUS2, JBUS2, CKT2, IERR)
```

Where:

Integer IBUS1	one of the buses connected by the first branch of the mutual coupling (input; no default allowed).	
Integer JBUS1	other bus connected by the first branch of the mutual coupling (input; no default allowed).	
Character*2 CKT1	circuit identifier of the first branch of the mutual coupling (input; '1').	
Integer IBUS2	one of the buses connected by the second branch of the mutual coupling (input; no default allowed).	
Integer JBUS2	other bus connected by the second branch of the mutual coupling (input; no default allowed).	
Character*2 CKT2	circuit identifier of the second branch of the mutual coupling (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	branch not found
	IERR = 3	mutual coupling not found
	IERR = 4	prerequisite requirements for API are not met

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1.229. PURGOWNER

Use this API to delete owners with no equipment assigned to them from the working case.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_PURGOWNER  SID  ALL
```

Python command syntax:

```
ierr = purgowner(sid, all)
```

Fortran command syntax:

```
CALL PURGOWNER(SID, ALL, IERR)
```

Where:

Integer SID	a valid owner subsystem identifier. Valid subsystem identifiers range from 0 to 11. Owner subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all owners or specified subsystem flag (input; 1 by default).	
	ALL = 0	delete only owners in owner subsystem SID with no equipment assigned to them
	ALL = 1	delete all owners with no equipment assigned to them
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.230. PURGPLNT

Use this API to delete the plant and machine data at the specified bus from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGPLNT  IBUS
```

Python command syntax:

```
ierr = purgplnt(ibus)
```

Fortran command syntax:

```
CALL PURGPLNT(IBUS, IERR)
```

Where:

Integer IBUS	bus from which the plant is to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no machines at bus
	IERR = 3	prerequisite requirements for API are not met

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1.231. PURGSHUNT

Use this API to delete the specified fixed bus shunt from the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGSHUNT  IBUS  'ID'
```

Python command syntax:

```
ierr = purgshunt(ibus, id)
```

Fortran command syntax:

```
CALL PURGSHUNT(IBUS, ID, IERR)
```

Where:

Integer IBUS	bus to which the shunt is connected (input; no default allowed).	
Character*2 ID	identifier of the shunt to be removed (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no fixed shunts at bus
	IERR = 3	shunt not found
	IERR = 4	prerequisite requirements for API are not met

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1.232. PURGSHUNTS

Use this API to delete all fixed shunts from the specified bus in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGSHUNTS  IBUS
```

Python command syntax:

```
ierr = purgshunts(ibus)
```

Fortran command syntax:

```
CALL PURGSHUNTS(IBUS, IERR)
```

Where:

Integer IBUS	bus from which all fixed shunts are to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no fixed shunts at bus
	IERR = 3	prerequisite requirements for API are not met

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1.233. PURGSWS

Use this API to delete the specified switched shunt from the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PURGSWS  IBUS
```

Python command syntax:

```
ierr = purgsws(ibus)
```

Fortran command syntax:

```
CALL PURGSWS(IBUS, IERR)
```

Where:

Integer IBUS	bus from which the switched shunt is to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	no switched shunt at bus
	IERR = 3	prerequisite requirements for API are not met

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1.234. PURGVSC

Use this API to delete the specified VSC dc line from the working case.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_PURGVSC  'NAME'
```

Python command syntax:

```
ierr = purgvsc(name)
```

Fortran command syntax:

```
CALL PURGVSC(NAME, IERR)
```

Where:

Character*12 NAME	name of the VSC dc line to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	dc line not found
	IERR = 2	prerequisite requirements for API are not met

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1.235. PURGZONE

Use this API to delete zones with no equipment assigned to them from the working case.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_PURGZONE  SID  ALL
```

Python command syntax:

```
ierr = purgzone(sid, all)
```

Fortran command syntax:

```
CALL PURGZONE(SID, ALL, IERR)
```

Where:

Integer SID	a valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Zone subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all zones or specified subsystem flag (input; 1 by default).	
	ALL = 0	delete only zones in zone subsystem SID with no equipment assigned to them
	ALL = 1	delete all zones with no equipment assigned to them
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.236. PV_ENGINE

This API routine is obsolete. It has been replaced by the API routine pv_engine_1a .

Use this API routine to run the first release of the PV analysis calculation engine. The same set of automatic adjustments is applied in base case and contingency case transfer increment solutions.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PV_ENGINE  OPTIONS(1)..OPTIONS(16)  VALUES(1)..VALUES(6)  LABELS(1)
 LABELS(2)  DFXFILE  THRFILE  ECDFILE  ACCFILE
```

Python command syntax:

```
ierr = pv_engine(options, values, labels, dfxfile, thrfile, ecdfile,
 accfile)
```

Fortran command syntax:

```
CALL PV_ENGINE(OPTIONS, VALUES, LABELS, DFXFILE, THRFILE, ECDFILE,
 ACCFILE, IERR)
```

Where:

Integer OPTIONS (16)	array of 16 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable

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OPTIONS(5) = 2	enable continuous mode, disable discrete mode
OPTIONS(6)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(6) = 0	disable
OPTIONS(6) = 1	enable
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	FDNS
OPTIONS(7) = 1	FNSL
OPTIONS(7) = 2	optimized FDNS
OPTIONS(8)	var limit code for the contingency case initial power flow solution (0 by default)
OPTIONS(8) = 0	apply immediately
OPTIONS(8) = 1	initially ignore, then apply automatically
OPTIONS(9)	var limit code for the subsequent transfer increment cases (0 by default)
OPTIONS(9) = 0	apply immediately
OPTIONS(9) = 1	initially ignore, then apply automatically
OPTIONS(10)	rating set (rating set program option setting by default)
OPTIONS(10) = 1	RATEA
OPTIONS(10) = 2	RATEB
OPTIONS(10) = 3	RATEC
OPTIONS(11)	study (source) system transfer dispatch method (1 by default)
OPTIONS(11) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(11) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(11) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(11) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(11) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(11) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(11) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC)
OPTIONS(11) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(12)	opposing (sink) system transfer dispatch method (1 by default)

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	OPTIONS(12) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
	OPTIONS(12) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
	OPTIONS(12) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(12) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(12) = 5	subsystem buses with positive MW machines in proportion to their MW output
	OPTIONS(12) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
	OPTIONS(12) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC)
	OPTIONS(12) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
	OPTIONS(13)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
	OPTIONS(13) = 0	no limits
	OPTIONS(13) = 1	honor machine active power limits
	OPTIONS(14)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
	OPTIONS(14) = 0	no limits
	OPTIONS(14) = 1	enforce non-negative net active power constant MVA load
	OPTIONS(15)	flag to check for low voltage at any monitored bus (0 by default)
	OPTIONS(15) = 0	disable check
	OPTIONS(15) = 1	enable check
	OPTIONS(16)	flag to check for excessive loading on any monitored branch (0 by default)
	OPTIONS(16) = 0	disable check
	OPTIONS(16) = 1	enable check
Real VALUES (6)	array of 6 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial transfer increment in MW (100.0 by default)
	VALUES(3)	transfer increment tolerance in MW (10.0 by default)
	VALUES(4)	maximum incremental transfer in MW (1000.0 by default)

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	VALUES(5)	low voltage threshold in pu in the low voltage check (0.8 by default)
	VALUES(6)	percent of rating in the excessive branch loading check (100.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which the load is to be decreased (no default allowed)
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for which the load is to be increased (no default allowed)
Character*260 DFXFILE	name of distribution factor data file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 ECDFILE	name of economic dispatch data file; blank for none (input; blank by default).	
Character*260 ACCFILE	name of PV results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	initial transfer increment is less than transfer increment tolerance
	IERR = 7	no participating dispatch buses in study system
	IERR = 8	no participating dispatch buses in opposing system
	IERR = 9	subsystem engine error
	IERR = 10	ECDI engine in subsystem aaa
	IERR = 11	base case ECDI solution did not converge
	IERR = 12	no DFAX file specified
	IERR = 13	no PV results file specified
	IERR = 14	no economic dispatch data file specified
	IERR = 15	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 16	bus type code and series element status inconsistencies

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IERR = 21	DFXFILE file is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 51	error opening PV results file
IERR = 52	error opening DFAX file
IERR = 53	error opening load throwover data file
IERR = 54	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.237. PV_ENGINE_1A

This API routine is obsolete. It has been replaced by the API routine pv_engine_2 .

Use this API routine to run the PV analysis calculation engine. The same set of automatic adjustments and induction machine treatment is applied in base case and contingency case transfer increment solutions.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_PV_ENGINE_1A  OPTIONS(1)..OPTIONS(17)  VALUES(1)..VALUES(6)  LABELS(1)
  LABELS(2)  DFXFILE  THRFILE  ECDFILE  ACCFILE
```

Python command syntax:

```
ierr = pv_engine_1a(options, values, labels, dfxfile, thrfile, ecdfile,
 accfile)
```

Fortran command syntax:

```
CALL PV_ENGINE_1A(OPTIONS, VALUES, LABELS, DFXFILE, THRFILE, ECDFILE,
 ACCFILE, IERR)
```

Where:

Integer OPTIONS (17)	array of 17 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)

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OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable
OPTIONS(5) = 2	enable continuous mode, disable discrete mode
OPTIONS(6)	induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
OPTIONS(6) = 0	stall
OPTIONS(6) = 1	trip
OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(7) = 0	disable
OPTIONS(7) = 1	enable
OPTIONS(8)	solution method flag (0 by default)
OPTIONS(8) = 0	FDNS
OPTIONS(8) = 1	FNSL
OPTIONS(8) = 2	optimized FDNS
OPTIONS(9)	var limit code for the contingency case initial power flow solution (0 by default)
OPTIONS(9) = 0	apply immediately
OPTIONS(9) = 1	initially ignore, then apply automatically
OPTIONS(10)	var limit code for the subsequent transfer increment cases (0 by default)
OPTIONS(10) = 0	apply immediately
OPTIONS(10) = 1	initially ignore, then apply automatically
OPTIONS(11)	rating set (rating set program option setting by default)
OPTIONS(11) = 1	RATEA
OPTIONS(11) = 2	RATEB
OPTIONS(11) = 3	RATEC
OPTIONS(12)	study (source) system transfer dispatch method (1 by default)
OPTIONS(12) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(12) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(12) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(12) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(12) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(12) = 6	subsystem buses with positive MW machines in proportion to their MBASEs

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OPTIONS(12) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC)
OPTIONS(12) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(13)	opposing (sink) system transfer dispatch method (1 by default)
OPTIONS(13) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(13) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(13) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(13) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(13) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(13) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(13) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC)
OPTIONS(13) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(14)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
OPTIONS(14) = 0	no limits
OPTIONS(14) = 1	honor machine active power limits
OPTIONS(15)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
OPTIONS(15) = 0	no limits
OPTIONS(15) = 1	enforce non-negative net active power constant MVA load
OPTIONS(16)	flag to check for low voltage at any monitored bus (0 by default)
OPTIONS(16) = 0	disable check
OPTIONS(16) = 1	enable check
OPTIONS(17)	flag to check for excessive loading on any monitored branch (0 by default)
OPTIONS(17) = 0	disable check
OPTIONS(17) = 1	enable check
Real VALUES (6)	array of 6 elements specifying solution values(input).

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	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial transfer increment in MW (100.0 by default)
	VALUES(3)	transfer increment tolerance in MW (10.0 by default)
	VALUES(4)	maximum incremental transfer in MW (1000.0 by default)
	VALUES(5)	low voltage threshold in pu in the low voltage check (0.8 by default)
	VALUES(6)	percent of rating in the excessive branch loading check (100.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which the load is to be decreased (no default allowed)
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for which the load is to be increased (no default allowed)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 ECDFILE	name of cconomic dispatch data file; blank for none (input; blank by default).	
Character*260 ACCFILE	name of PV results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	initial transfer increment is less than transfer increment tolerance
	IERR = 7	no participating dispatch buses in study system
	IERR = 8	no participating dispatch buses in opposing system
	IERR = 9	subsystem engine error
	IERR = 10	ECDI engine error in subsystem aaa

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IERR = 11	base case ECDI solution did not converge
IERR = 12	no DFAX file specified
IERR = 13	no PV results file specified
IERR = 14	no economic dispatch data file specified
IERR = 15	in-service induction machines are in the "stalled" or "tripped" state
IERR = 16	bus type code and series element status in-consistencies
IERR = 21	file DFAX file is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 51	error opening PV results file
IERR = 52	error opening DFAX File
IERR = 53	error opening load throwover data file
IERR = 54	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.238. PV_ENGINE_2

This API routine is obsolete. It has been replaced by the API routine pv_engine_3 .

Use this API to run PV analysis calculation engine. It provides for a different set of automatic adjustments in contingency case solutions than is used in the base case transfer increment solutions.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_PV_ENGINE_2  OPTIONS(1)..OPTIONS(21)  VALUES(1)..VALUES(6)  LABELS(1)
 LABELS(2)  DFXFILE  THRFILE  ECDFILE  ACCFILE
```

Python command syntax:

```
ierr = pv_engine_2(options, values, labels, dfxfile, thrfile, ecdfile,
 accfile)
```

Fortran command syntax:

```
CALL PV_ENGINE_2(OPTIONS, VALUES, LABELS, DFXFILE, THRFILE, ECDFILE,
 ACCFILE, IERR)
```

Where:

Integer OPTIONS (21)	array of 21 elements specifying solution options(input).	
	OPTIONS(1)	base case tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	base case area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	base case phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	base case dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	base case switched shunt adjustment flag (switched shunt adjustment option setting by default)

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OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable
OPTIONS(5) = 2	enable continuous mode, disable discrete mode
OPTIONS(6)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(6) = 0	disable
OPTIONS(6) = 1	enable
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	FDNS
OPTIONS(7) = 1	FNSL
OPTIONS(7) = 2	optimized FDNS
OPTIONS(8)	var limit code for the contingency case power flow solutions (0 by default)
OPTIONS(8) = 0	apply immediately
OPTIONS(8) = 1	initially ignore, then apply automatically
OPTIONS(9)	var limit code for the base case transfer increment solutions (0 by default)
OPTIONS(9) = 0	apply immediately
OPTIONS(9) = 1	initially ignore, then apply automatically
OPTIONS(10)	rating set (rating set program option setting by default)
OPTIONS(10) = 1	RATEA
OPTIONS(10) = 2	RATEB
OPTIONS(10) = 3	RATEC
OPTIONS(11)	study (source) system transfer dispatch method (1 by default)
OPTIONS(11) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(11) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(11) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(11) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(11) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(11) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(11) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC)
OPTIONS(11) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled

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OPTIONS(12)	opposing (sink) system transfer dispatch method (1 by default)
OPTIONS(12) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(12) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(12) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(12) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(12) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(12) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(12) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC)
OPTIONS(12) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(13)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
OPTIONS(13) = 0	no limits
OPTIONS(13) = 1	honor machine active power limits
OPTIONS(14)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
OPTIONS(14) = 0	no limits
OPTIONS(14) = 1	enforce non-negative net active power constant MVA load
OPTIONS(15)	flag to check for low voltage at any monitored bus (0 by default)
OPTIONS(15) = 0	disable check
OPTIONS(15) = 1	enable check
OPTIONS(16)	flag to check for excessive loading on any monitored branch (0 by default)
OPTIONS(16) = 0	disable check
OPTIONS(16) = 1	enable check
OPTIONS(17)	contingency case tap adjustment flag (base case tap adjustment flag, OPTIONS (1) by default)
OPTIONS(17) = 0	disable
OPTIONS(17) = 1	enable stepping adjustment
OPTIONS(17) = 2	enable direct adjustment

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	OPTIONS(18)	contingency case area interchange adjustment flag (base case area interchange adjustment flag, OPTIONS (2) by default)
	OPTIONS(18) = 0	disable
	OPTIONS(18) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(18) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(19)	contingency case phase shift adjustment flag (base case phase shift adjustment flag, OPTIONS (3) by default)
	OPTIONS(19) = 0	disable
	OPTIONS(19) = 1	enable
	OPTIONS(20)	contingency case dc tap adjustment flag (base case dc tap adjustment flag, OPTIONS (4) by default)
	OPTIONS(20) = 0	disable
	OPTIONS(20) = 1	enable
	OPTIONS(21)	contingency case switched shunt adjustment flag (base case switched shunt adjustment flag, OPTIONS (5) by default)
	OPTIONS(21) = 0	disable
	OPTIONS(21) = 1	enable
	OPTIONS(21) = 2	enable continuous mode, disable discrete mode
Real VALUES (6)	array of 6 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial transfer increment in MW (100.0 by default)
	VALUES(3)	transfer increment tolerance in MW (10.0 by default)
	VALUES(4)	maximum incremental transfer in MW (1000.0 by default)
	VALUES(5)	low voltage threshold in pu in the low voltage check (0.8 by default)
	VALUES(6)	percent of rating in the excessive branch loading check (100.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which the load is to be decreased (no default allowed)
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for

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	which the load is to be increased (no default allowed)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).
Character*260 ECDFILE	name of economic dispatch data file; blank for none (input; blank by default).
Character*260 ACCFILE	name of PV results file (input; no default allowed).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	invalid VALUES value
IERR = 2	invalid OPTIONS value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	initial transfer increment is less than transfer increment tolerance
IERR = 7	no participating dispatch buses in study system
IERR = 8	no participating dispatch buses in opposing system
IERR = 9	subsystem engine error
IERR = 10	ECDI engine in subsystem aaa
IERR = 11	base case ECDI solution did not converge
IERR = 12	no DFAX file specified
IERR = 14	no economic dispatch data file specified
IERR = 15	in-service induction machines are in the "stalled" or "tripped" state
IERR = 16	bus type code and series element status inconsistencies
IERR = 21	DFXFILE file is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 51	error opening PV results file
IERR = 52	error opening DFAX file
IERR = 53	error opening load throwover data file

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IERR = 54	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.239. PV_ENGINE_3

This API routine is obsolete. It has been replaced by the API routine pv_engine_4 .

Use this API to run the PV analysis calculation engine. It provides for a different set of automatic adjustments in contingency case solutions than is used in the base case transfer increment solutions, and allows a negative minimum incremental transfer to be specified..

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_PV_ENGINE_3  OPTIONS(1)..OPTIONS(21)  VALUES(1)..VALUES(7)  LABELS(1)
 LABELS(2)  DFXFILE  THRFILE  ECDFILE  ACCFILE
```

Python command syntax:

```
ierr = pv_engine_3(options, values, labels, dfxfile, thrfile, ecdfile,
 accfile)
```

Fortran command syntax:

```
CALL PV_ENGINE_3(OPTIONS, VALUES, LABELS, DFXFILE, THRFILE, ECDFILE,
 ACCFILE, IERR)
```

Where:

Integer OPTIONS (21)	array of 21 elements specifying solution options(input).	
	OPTIONS(1)	base case tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	base case area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	base case phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	base case dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable

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OPTIONS(5)	base case switched shunt adjustment flag (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable
OPTIONS(5) = 2	enable continuous mode, disable discrete mode
OPTIONS(6)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(6) = 0	disable
OPTIONS(6) = 1	enable
OPTIONS(7)	solution method flag (0 by default)
OPTIONS(7) = 0	FDNS
OPTIONS(7) = 1	FNSL
OPTIONS(7) = 2	optimized FDNS
OPTIONS(8)	var limit code for the contingency case power flow solutions (0 by default)
OPTIONS(8) = 0	apply immediately
OPTIONS(8) = 1	initially ignore, then apply automatically
OPTIONS(9)	var limit code for the base case transfer increment solutions (0 by default)
OPTIONS(9) = 0	apply immediately
OPTIONS(9) = 1	initially ignore, then apply automatically
OPTIONS(10)	rating set (rating set program option setting by default)
OPTIONS(10) = 1	RATEA
OPTIONS(10) = 2	RATEB
OPTIONS(10) = 3	RATEC
OPTIONS(11)	study (source) system transfer dispatch method (1 by default)
OPTIONS(11) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(11) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(11) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(11) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(11) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(11) = 6	subsystem buses with positive MW machines in proportion to their MBASEs

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OPTIONS(11) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC)
OPTIONS(11) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(12)	opposing (sink) system transfer dispatch method (1 by default)
OPTIONS(12) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(12) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(12) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(12) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(12) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(12) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(12) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC)
OPTIONS(12) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(13)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
OPTIONS(13) = 0	no limits
OPTIONS(13) = 1	honor machine active power limits
OPTIONS(14)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
OPTIONS(14) = 0	no limits
OPTIONS(14) = 1	enforce non-negative net active power constant MVA load
OPTIONS(15)	flag to check for low voltage at any monitored bus (0 by default)
OPTIONS(15) = 0	disable check
OPTIONS(15) = 1	enable check
OPTIONS(16)	flag to check for excessive loading on any monitored branch (0 by default)
OPTIONS(16) = 0	disable check
OPTIONS(16) = 1	enable check
OPTIONS(17)	contingency case tap adjustment flag (base case tap adjustment flag, OPTIONS (1) by default)

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	OPTIONS(17) = 0	disable
	OPTIONS(17) = 1	enable stepping adjustment
	OPTIONS(17) = 2	enable direct adjustment
	OPTIONS(18)	contingency case area interchange adjustment flag (base case area interchange adjustment flag, OPTIONS (2) by default)
	OPTIONS(18) = 0	disable
	OPTIONS(18) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(18) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(19)	contingency case phase shift adjustment flag (base case phase shift adjustment flag, OPTIONS (3) by default)
	OPTIONS(19) = 0	disable
	OPTIONS(19) = 1	enable
	OPTIONS(20)	contingency case dc tap adjustment flag (base case dc tap adjustment flag, OPTIONS (4) by default)
	OPTIONS(20) = 0	disable
	OPTIONS(20) = 1	enable
	OPTIONS(21)	contingency case switched shunt adjustment flag (base case switched shunt adjustment flag, OPTIONS (5) by default)
	OPTIONS(21) = 0	disable
	OPTIONS(21) = 1	enable
	OPTIONS(21) = 2	enable continuous mode, disable discrete mode
Real VALUES (7)	array of 7 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial transfer increment in MW (100.0 by default)
	VALUES(3)	transfer increment tolerance in MW (10.0 by default)
	VALUES(4)	maximum incremental transfer in MW (1000.0 by default)
	VALUES(5)	low voltage threshold in pu in the low voltage check (0.8 by default)
	VALUES(6)	percent of rating in the excessive branch loading check (100.0 by default)
	VALUES(7)	minimum incremental transfer in MW (< 0.0) (0.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	

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	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which the load is to be decreased (no default allowed)
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for which the load is to be increased (no default allowed)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 ECDFILE	name of economic dispatch data file; blank for none (input; blank by default).	
Character*260 ACCFILE	name of PV results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	initial transfer increment is less than transfer increment tolerance
	IERR = 7	no participating dispatch buses in study system
	IERR = 8	no participating dispatch buses in opposing system
	IERR = 9	subsystem engine error
	IERR = 10	ECDI engine in subsystem aaa
	IERR = 11	base case ECDI solution did not converge
	IERR = 12	no DFAX File specified
	IERR = 13	no PV results file specified
	IERR = 14	no economic dispatch data file specified
	IERR = 15	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 16	invalid base/contingency case adjustment flag combination
	IERR = 17	bus type code and series element status inconsistencies
	IERR = 21	DFXFILE file is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members

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IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 51	error opening PV results file
IERR = 52	error opening DFAX file
IERR = 53	error opening load throwover data file
IERR = 54	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.240. PV_ENGINE_4

This API routine is obsolete. It has been replaced by the API routine pv_engine_5 .

Use this API routine to run the PV analysis calculation engine. It provides for a different set of automatic adjustments and induction machine treatment in contingency case solutions than is used in the base case transfer increment solutions, and allows a negative minimum incremental transfer to be specified..

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_PV_ENGINE_4  OPTIONS(1)..OPTIONS(23)  VALUES(1)..VALUES(7)  LABELS(1)
 LABELS(2)  DFXFILE  THRFILE  ECDFILE  ACCFILE
```

Python command syntax:

```
ierr = pv_engine_4(options, values, labels, dfxfile, thrfile, ecdfile,
 accfile)
```

Fortran command syntax:

```
CALL PV_ENGINE_4(OPTIONS, VALUES, LABELS, DFXFILE, THRFILE, ECDFILE,
 ACCFILE, IERR)
```

Where:

Integer OPTIONS (23)	array of 23 elements specifying solution options(input).	
	OPTIONS(1)	base case tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	base case area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	base case phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	base case dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable

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OPTIONS(5)	base case switched shunt adjustment flag (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable
OPTIONS(5) = 2	enable continuous mode, disable discrete mode
OPTIONS(6)	base case induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
OPTIONS(6) = 0	stall
OPTIONS(6) = 1	trip
OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(7) = 0	disable
OPTIONS(7) = 1	enable
OPTIONS(8)	solution method flag (0 by default)
OPTIONS(8) = 0	FDNS
OPTIONS(8) = 1	FNSL
OPTIONS(8) = 2	optimized FDNS
OPTIONS(9)	var limit code for the contingency case power flow solutions (0 by default)
OPTIONS(9) = 0	apply immediately
OPTIONS(9) = 1	initially ignore, then apply automatically
OPTIONS(10)	var limit code for the base case transfer increment solutions (0 by default)
OPTIONS(10) = 0	apply immediately
OPTIONS(10) = 1	initially ignore, then apply automatically
OPTIONS(11)	rating set (rating set program option setting by default)
OPTIONS(11) = 1	RATEA
OPTIONS(11) = 2	RATEB
OPTIONS(11) = 3	RATEC
OPTIONS(12)	study (source) system transfer dispatch method (1 by default)
OPTIONS(12) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(12) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(12) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(12) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load

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OPTIONS(12) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(12) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(12) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC)
OPTIONS(12) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(13)	opposing (sink) system transfer dispatch method (1 by default)
OPTIONS(13) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(13) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(13) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(13) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(13) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(13) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(13) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC)
OPTIONS(13) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(14)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
OPTIONS(14) = 0	no limits
OPTIONS(14) = 1	honor machine active power limits
OPTIONS(15)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
OPTIONS(15) = 0	no limits
OPTIONS(15) = 1	enforce non-negative net active power constant MVA load
OPTIONS(16)	flag to check for low voltage at any monitored bus (0 by default)
OPTIONS(16) = 0	disable check
OPTIONS(16) = 1	enable check
OPTIONS(17)	flag to check for excessive loading on any monitored branch (0 by default)
OPTIONS(17) = 0	disable check

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	OPTIONS(17) = 1	enable check
	OPTIONS(18)	contingency case tap adjustment flag (base case tap adjustment flag, OPTIONS (1) by default)
	OPTIONS(18) = 0	disable
	OPTIONS(18) = 1	enable stepping adjustment
	OPTIONS(18) = 2	enable direct adjustment
	OPTIONS(19)	contingency case area interchange adjustment flag (base case area interchange adjustment flag, OPTIONS (2) by default)
	OPTIONS(19) = 0	disable
	OPTIONS(19) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(19) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(20)	contingency case phase shift adjustment flag (base case phase shift adjustment flag, OPTIONS (3) by default)
	OPTIONS(20) = 0	disable
	OPTIONS(20) = 1	enable
	OPTIONS(21)	contingency case dc tap adjustment flag (base case dc tap adjustment flag, OPTIONS (4) by default)
	OPTIONS(21) = 0	disable
	OPTIONS(21) = 1	enable
	OPTIONS(22)	contingency case switched shunt adjustment flag (base case switched shunt adjustment flag, OPTIONS (5) by default)
	OPTIONS(22) = 0	disable
	OPTIONS(22) = 1	enable
	OPTIONS(22) = 2	enable continuous mode, disable discrete mode
	OPTIONS(23)	contingency case induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
	OPTIONS(23) = 0	stall
	OPTIONS(23) = 1	trip
Real VALUES (7)	array of 7 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial transfer increment in MW (100.0 by default)
	VALUES(3)	transfer increment tolerance in MW (10.0 by default)

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	VALUES(4)	maximum incremental transfer in MW (1000.0 by default)
	VALUES(5)	low voltage threshold in pu in the low voltage check (0.8 by default)
	VALUES(6)	percent of rating in the excessive branch loading check (100.0 by default)
	VALUES(7)	minimum incremental transfer in MW (< 0.0) (0.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which the load is to be decreased (no default allowed)
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for which the load is to be increased (no default allowed)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 ECDFILE	name of economic dispatch data file; blank for none (input; blank by default).	
Character*260 ACCFILE	name of PV results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	initial transfer increment is less than transfer increment tolerance
	IERR = 7	no participating dispatch buses in study system
	IERR = 8	no participating dispatch buses in opposing system
	IERR = 9	subsystem engine error
	IERR = 10	ECDI engine in subsystem aaa
	IERR = 11	base case ECDI solution did not converge
	IERR = 12	no DFAX file specified
	IERR = 13	no PV results file specified
	IERR = 14	no economic dispatch data file specified

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IERR = 15	in-service induction machines are in the "stalled" or "tripped" state
IERR = 16	bus type code and series element status inconsistencies
IERR = 17	invalid base/contingency case flag combination
IERR = 21	DFXFILE file is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 51	error opening PV results file
IERR = 52	error opening DFAX file
IERR = 53	error opening load throwover data file
IERR = 54	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.241. PV_ENGINE_5

This API routine is obsolete. It has been replaced by the API routine pv_engine_6 .

Use this API routine to run the PV analysis calculation engine. It provides for a different set of automatic adjustments, and for different treatment of induction motors that fail to solve due to low terminal voltage, in contingency case solutions than is used in the base case transfer increment solutions; it allows a negative minimum incremental transfer to be specified; for transfer increment methods involving load, it allows a power factor to be specified; and it provides for dispatching power unbalances resulting from the application of contingencies.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_PV_ENGINE_5  OPTIONS(1)..OPTIONS(24) VALUES(1)..VALUES(8)
 LABELS(1)..LABELS(3) DFXFILE THRFILE ECDFILE INLFILE ACCFILE
```

Python command syntax:

```
ierr = pv_engine_5(options, values, labels, dfxfile, thrfile, ecdfile,
 inlfile, accfile)
```

Fortran command syntax:

```
CALL PV_ENGINE_5(OPTIONS, VALUES, LABELS, DFXFILE, THRFILE, ECDFILE,
 INLFILE, ACCFILE, IERR)
```

Where:

Integer OPTIONS (24)	array of 24 elements specifying solution options(input).	
	OPTIONS(1)	base case tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	base case area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	base case phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	base case dc tap adjustment flag (dc tap adjustment option setting by default)

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OPTIONS(4) = 0	disable
OPTIONS(4) = 1	enable
OPTIONS(5)	base case switched shunt adjustment flag (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable
OPTIONS(5) = 2	enable continuous mode, disable discrete mode
OPTIONS(6)	base case induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
OPTIONS(6) = 0	stall
OPTIONS(6) = 1	trip
OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(7) = 0	disable
OPTIONS(7) = 1	enable
OPTIONS(8)	solution method flag (0 by default)
OPTIONS(8) = 0	FDNS
OPTIONS(8) = 1	FNSL
OPTIONS(8) = 2	optimized FDNS
OPTIONS(9)	var limit code for the contingency case power flow solutions (0 by default)
OPTIONS(9) = 0	apply immediately
OPTIONS(9) = 1	initially ignore, then apply automatically
OPTIONS(10)	var limit code for the base case transfer increment solutions (0 by default)
OPTIONS(10) = 0	apply immediately
OPTIONS(10) = 1	initially ignore, then apply automatically
OPTIONS(11)	rating set (rating set program option setting by default)
OPTIONS(11) = 1	RATEA
OPTIONS(11) = 2	RATEB
OPTIONS(11) = 3	RATEC
OPTIONS(12)	study (source) system transfer dispatch method (1 by default)
OPTIONS(12) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(12) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(12) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load

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OPTIONS(12) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(12) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(12) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(12) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC)
OPTIONS(12) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(13)	opposing (sink) system transfer dispatch method (1 by default)
OPTIONS(13) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(13) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(13) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(13) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(13) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(13) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(13) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC)
OPTIONS(13) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(14)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
OPTIONS(14) = 0	no limits
OPTIONS(14) = 1	honor machine active power limits
OPTIONS(15)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
OPTIONS(15) = 0	no limits
OPTIONS(15) = 1	enforce non-negative net active power constant MVA load
OPTIONS(16)	flag to check for low voltage at any monitored bus (0 by default)
OPTIONS(16) = 0	disable check
OPTIONS(16) = 1	enable check
OPTIONS(17)	flag to check for excessive loading on any monitored branch (0 by default)

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OPTIONS(17) = 0	disable check
OPTIONS(17) = 1	enable check
OPTIONS(18)	dispatch mode for power unbalances resulting from the application of contingencies (1 by default)
OPTIONS(18) = 0	disable
OPTIONS(18) = 1	in-service subsystem machines using reserve
OPTIONS(18) = 2	in-service subsystem machines using Pmax
OPTIONS(18) = 3	in-service subsystem machines using inertia
OPTIONS(18) = 4	in-service subsystem machines using governor droop
OPTIONS(19)	contingency case tap adjustment flag (base case tap adjustment flag, OPTIONS (1) by default)
OPTIONS(19) = 0	disable
OPTIONS(19) = 1	enable stepping adjustment
OPTIONS(19) = 2	enable direct adjustment
OPTIONS(20)	contingency case area interchange adjustment flag (base case area interchange adjustment flag, OPTIONS (2) by default)
OPTIONS(20) = 0	disable
OPTIONS(20) = 1	enable using tie line flows only in calculating area interchange
OPTIONS(20) = 2	enable using tie line flows and loads in calculating area interchange
OPTIONS(21)	contingency case phase shift adjustment flag (base case phase shift adjustment flag, OPTIONS (3) by default)
OPTIONS(21) = 0	disable
OPTIONS(21) = 1	enable
OPTIONS(22)	contingency case dc tap adjustment flag (base case dc tap adjustment flag, OPTIONS (4) by default)
OPTIONS(22) = 0	disable
OPTIONS(22) = 1	enable
OPTIONS(23)	contingency case switched shunt adjustment flag (base case switched shunt adjustment flag, OPTIONS (5) by default)
OPTIONS(23) = 0	disable
OPTIONS(23) = 1	enable
OPTIONS(23) = 2	enable continuous mode, disable discrete mode
OPTIONS(24)	contingency case induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)

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	OPTIONS(24) = 0	stall
	OPTIONS(24) = 1	trip
Real VALUES (8)	array of 8 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial transfer increment in MW (100.0 by default)
	VALUES(3)	transfer increment tolerance in MW (10.0 by default)
	VALUES(4)	maximum incremental transfer in MW (1000.0 by default)
	VALUES(5)	low voltage threshold in pu in the low voltage check (0.8 by default)
	VALUES(6)	percent of rating in the excessive branch loading check (100.0 by default)
	VALUES(7)	minimum incremental transfer in MW (< 0.0) (0.0 by default)
	VALUES(8)	power factor for load increases in dispatch methods 2, 3 and 4. Specify as <= 0.0 to retain the original power factor at each load that is changed (0.0 by default)
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which the load is to be decreased (no default allowed)
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for which the load is to be increased (no default allowed)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 ECDFILE	name of economic dispatch data file; blank for none (input; blank by default).	
Character*260 INLFILE	name of inertia and governor response file (input).	
Character*260 ACCFILE	name of PV results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE

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IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	initial transfer increment is less than transfer increment tolerance
IERR = 7	no participating dispatch buses in study system
IERR = 8	no participating dispatch buses in opposing system
IERR = 9	subsystem engine error
IERR = 10	ECDI engine in subsystem aaa
IERR = 11	base case ECDI solution did not converge
IERR = 12	no DFAX file specified
IERR = 13	no PV results file specified
IERR = 14	no economic dispatch data file specified
IERR = 15	in-service induction machines are in the "stalled" or "tripped" state
IERR = 16	bus type code and series element status inconsistencies
IERR = 17	invalid base/contingency case flag combination
IERR = 21	DFXFILE file is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 51	error opening PV results file
IERR = 52	error opening DFAX file
IERR = 53	error opening load throwover data file
IERR = 54	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.242. PV_ENGINE_6

Use this API routine to run the PV analysis calculation engine. It provides for a different set of automatic adjustments, and for different treatment of induction motors that fail to solve due to low terminal voltage, in contingency case solutions than is used in the base case transfer increment solutions; it allows a negative minimum incremental transfer to be specified; for transfer increment methods involving load, it allows a power factor to be specified; it provides for dispatching power unbalances resulting from the application of contingencies; and it allows for the preservation of milestone results in a ZIP Archive Output File.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_PV_ENGINE_6  OPTIONS(1)..OPTIONS(25) VALUES(1)..VALUES(8)
 LABELS(1)..LABELS(3) DFXFILE THRFILE ECDFILE INLFILE ACCFILE ZIPFILE
```

Python command syntax:

```
ierr = pv_engine_6(options, values, labels, dfxfile, thrfile, ecdfile,
 inlfile, accfile, zipfile)
```

Fortran command syntax:

```
CALL PV_ENGINE_6(OPTIONS, VALUES, LABELS, DFXFILE, THRFILE, ECDFILE,
 INLFILE, ACCFILE, ZIPFILE, IERR)
```

Where:

Integer OPTIONS (25)	array of 25 elements specifying solution options(input).	
	OPTIONS(1)	base case tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	base case area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	base case phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	base case dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable

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OPTIONS(5)	base case switched shunt adjustment flag (switched shunt adjustment option setting by default)
OPTIONS(5) = 0	disable
OPTIONS(5) = 1	enable
OPTIONS(5) = 2	enable continuous mode, disable discrete mode
OPTIONS(6)	base case induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
OPTIONS(6) = 0	stall
OPTIONS(6) = 1	trip
OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
OPTIONS(7) = 0	disable
OPTIONS(7) = 1	enable
OPTIONS(8)	solution method flag (0 by default)
OPTIONS(8) = 0	FDNS
OPTIONS(8) = 1	FNSL
OPTIONS(8) = 2	optimized FDNS
OPTIONS(9)	var limit code for the contingency case power flow solutions (0 by default)
OPTIONS(9) = 0	apply immediately
OPTIONS(9) = 1	initially ignore, then apply automatically
OPTIONS(10)	var limit code for the base case transfer increment solutions (0 by default)
OPTIONS(10) = 0	apply immediately
OPTIONS(10) = 1	initially ignore, then apply automatically
OPTIONS(11)	rating set (rating set program option setting by default)
OPTIONS(11) = 1	RATEA
OPTIONS(11) = 2	RATEB
OPTIONS(11) = 3	RATEC
OPTIONS(12)	study (source) system transfer dispatch method (1 by default)
OPTIONS(12) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(12) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(12) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(12) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load

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OPTIONS(12) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(12) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(12) = 7	subsystem buses with positive MW machines in proportion to their reserve (PMAXMC - PGENMC)
OPTIONS(12) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(13)	opposing (sink) system transfer dispatch method (1 by default)
OPTIONS(13) = 1	buses and distribution factors from DFAX file for buses with positive MW machines
OPTIONS(13) = 2	buses and distribution factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(13) = 3	buses and distribution factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(13) = 4	subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(13) = 5	subsystem buses with positive MW machines in proportion to their MW output
OPTIONS(13) = 6	subsystem buses with positive MW machines in proportion to their MBASEs
OPTIONS(13) = 7	subsystem buses with positive MW machines in proportion to their reserve (PGENMC - PMINMC)
OPTIONS(13) = 8	subsystem buses with positive MW machines via ECDI with unit commitment disabled
OPTIONS(14)	generation plant limits flag for transfer methods 1, 3, 5, 6 and 7 (0 by default)
OPTIONS(14) = 0	no limits
OPTIONS(14) = 1	honor machine active power limits
OPTIONS(15)	positive load flag for transfer methods 2, 3 and 4 (0 by default)
OPTIONS(15) = 0	no limits
OPTIONS(15) = 1	enforce non-negative net active power constant MVA load
OPTIONS(16)	flag to check for low voltage at any monitored bus (0 by default)
OPTIONS(16) = 0	disable check
OPTIONS(16) = 1	enable check
OPTIONS(17)	flag to check for excessive loading on any monitored branch (0 by default)
OPTIONS(17) = 0	disable check

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OPTIONS(17) = 1	enable check
OPTIONS(18)	dispatch mode for power unbalances resulting from the application of contingencies (1 by default)
OPTIONS(18) = 0	disable
OPTIONS(18) = 1	in-service subsystem machines using reserve
OPTIONS(18) = 2	in-service subsystem machines using Pmax
OPTIONS(18) = 3	in-service subsystem machines using inertia
OPTIONS(18) = 4	in-service subsystem machines using governor droop
OPTIONS(19)	write ZIP archive flag (0 by default)
OPTIONS(19) = 0	no ZIP archive
OPTIONS(19) = 1	Write ZIP Archive file ZIPFILE; preserve each system condition at its largest solved incremental transfer level.
OPTIONS(19) = 2	Write ZIP Archive file ZIPFILE; preserve each system condition at all of its solved incremental transfer levels.
OPTIONS(20)	contingency case tap adjustment flag (base case tap adjustment flag, OPTIONS (1) by default)
OPTIONS(20) = 0	disable
OPTIONS(20) = 1	enable stepping adjustment
OPTIONS(20) = 2	enable direct adjustment
OPTIONS(21)	contingency case area interchange adjustment flag (base case area interchange adjustment flag, OPTIONS (2) by default)
OPTIONS(21) = 0	disable
OPTIONS(21) = 1	enable using tie line flows only in calculating area interchange
OPTIONS(21) = 2	enable using tie line flows and loads in calculating area interchange
OPTIONS(22)	contingency case phase shift adjustment flag (base case phase shift adjustment flag, OPTIONS (3) by default)
OPTIONS(22) = 0	disable
OPTIONS(22) = 1	enable
OPTIONS(23)	contingency case dc tap adjustment flag (base case dc tap adjustment flag, OPTIONS (4) by default)
OPTIONS(23) = 0	disable
OPTIONS(23) = 1	enable
OPTIONS(24)	contingency case switched shunt adjustment flag (base case switched shunt adjustment flag, OPTIONS (5) by default)
OPTIONS(24) = 0	disable

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	OPTIONS(24) = 1	enable
	OPTIONS(24) = 2	enable continuous mode, disable discrete mode
	OPTIONS(25)	contingency case induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
	OPTIONS(25) = 0	stall
	OPTIONS(25) = 1	trip
Real VALUES (8)	array of 8 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial transfer increment in MW (100.0 by default)
	VALUES(3)	transfer increment tolerance in MW (10.0 by default)
	VALUES(4)	maximum incremental transfer in MW (1000.0 by default)
	VALUES(5)	low voltage threshold in pu in the low voltage check (0.8 by default)
	VALUES(6)	percent of rating in the excessive branch loading check (100.0 by default)
	VALUES(7)	minimum incremental transfer in MW (< 0.0) (0.0 by default)
	VALUES(8)	power factor for load increases in dispatch methods 2, 3 and 4. Specify as <= 0.0 to retain the original power factor at each load that is changed (0.0 by default)
Character*32 LABELS (3)	array of 3 elements specifying subsystem labels(input).	
	LABELS(1)	label of the study (source) system for which generation is to be increased and/or for which the load is to be decreased (no default allowed)
	LABELS(2)	label of the opposing (sink) system for which generation is to be decreased and/or for which the load is to be increased (no default allowed)
	LABELS(3)	label of the dispatch subsystem; used if OPTIONS(18) is 1 through 4 (blank by default)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 ECDFILE	name of economic dispatch data file; blank for none (input; blank by default).	

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Character*260 INLFILE	name of inertia and governor response file (input).
Character*260 ACCFILE	name of PV results file (input; no default allowed).
Character*260 ZIPFILE	name of ZIP archive output file (input).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	invalid VALUES value
IERR = 2	invalid OPTIONS value
IERR = 3	generators are converted
IERR = 4	buses in island(s) without a swing bus; use activity TREE
IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	initial transfer increment is less than transfer increment tolerance
IERR = 7	no participating dispatch buses in study system
IERR = 8	no participating dispatch buses in opposing system
IERR = 9	subsystem engine error
IERR = 10	ECDI engine in subsystem aaa
IERR = 11	base case ECDI solution did not converge
IERR = 12	no DFAX file specified
IERR = 13	no PV results file specified
IERR = 14	no economic dispatch data file specified
IERR = 15	in-service induction machines are in the "stalled" or "tripped" state
IERR = 16	bus type code and series element status inconsistencies
IERR = 17	invalid base/contingency case flag combination
IERR = 21	DFXFILE file is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 51	error opening PV results file
IERR = 52	error opening DFAX file
IERR = 53	error opening load throwover data file
IERR = 54	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.243. QV_ENGINE

This API routine is obsolete. It has been replaced by the API routine qv_engine_2 .

Use this API to run the QV analysis calculation engine.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_QV_ENGINE  OPTIONS(1)..OPTIONS(10)  VALUES(1)..VALUES(4)  DFXFILE
 THRFILE  ACCFILE
```

Python command syntax:

```
ierr = qv_engine(options, values, dfxfile, thrfile, accfile)
```

Fortran command syntax:

```
CALL QV_ENGINE(OPTIONS, VALUES, DFXFILE, THRFILE, ACCFILE, IERR)
```

Where:

Integer OPTIONS (10)	array of 10 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode

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	OPTIONS(6)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTIONS(6) = 0	disable
	OPTIONS(6) = 1	enable
	OPTIONS(7)	solution method flag (0 by default)
	OPTIONS(7) = 0	FDNS
	OPTIONS(7) = 1	FNSL
	OPTIONS(7) = 2	optimized FDNS
	OPTIONS(8)	var limit code for the VHI power flow solution (0 by default)
	OPTIONS(8) = 0	apply immediately
	OPTIONS(8) = 1	initially ignore, then apply automatically
	OPTIONS(9)	var limit code for the subsequent voltage decrement cases (0 by default)
	OPTIONS(9) = 0	apply immediately
	OPTIONS(9) = 1	initially ignore, then apply automatically
	OPTIONS(10)	study bus number (no default allowed)
Real VALUES (4)	array of 4 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial (maximum) pu voltage setpoint at the study bus (VHI) (1.1 by default)
	VALUES(3)	minimum pu voltage setpoint at the study bus (VLO) (0.9 by default)
	VALUES(4)	pu voltage setpoint decrement (positive) at the study bus (DLTAV) (0.01 by default)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 ACCFILE	name of QV results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	VHI is not greater than VLO
	IERR = 7	study bus not found
	IERR = 8	invalid study bus
	IERR = 9	error when adding a plant to the study bus

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IERR = 10	error when adding a machine to the study bus
IERR = 11	base case VHI solution did not converge
IERR = 12	no Distribution Factor Data File specified
IERR = 13	no Q-V Solution Output File specified
IERR = 14	in-service induction machines are in the "stalled" or "tripped" state
IERR = 15	bus type code and series element status in-consistencies
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening QV Results File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 55	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.244. QV_ENGINE_2

This API routine is obsolete. It has been replaced by the API routine qv_engine_3 .

Use this API routine to run the QV analysis calculation engine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_QV_ENGINE_2  OPTIONS(1)..OPTIONS(11)  VALUES(1)..VALUES(4)  DFXFILE
 THRFILE  ACCFILE
```

Python command syntax:

```
ierr = qv_engine_2(options, values, dfxfile, thrfile, accfile)
```

Fortran command syntax:

```
CALL QV_ENGINE_2(OPTIONS, VALUES, DFXFILE, THRFILE, ACCFILE, IERR)
```

Where:

Integer OPTIONS (11)	array of 11 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode

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	OPTIONS(6)	induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
	OPTIONS(6) = 0	stall
	OPTIONS(6) = 1	trip
	OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTIONS(7) = 0	disable
	OPTIONS(7) = 1	enable
	OPTIONS(8)	solution method flag (0 by default)
	OPTIONS(8) = 0	FDNS
	OPTIONS(8) = 1	FNSL
	OPTIONS(8) = 2	optimized FDNS
	OPTIONS(9)	var limit code for the VHI power flow solution (0 by default)
	OPTIONS(9) = 0	apply immediately
	OPTIONS(9) = 1	initially ignore, then apply automatically
	OPTIONS(10)	var limit code for the subsequent voltage decrement cases (0 by default)
	OPTIONS(10) = 0	apply immediately
	OPTIONS(10) = 1	initially ignore, then apply automatically
	OPTIONS(11)	study bus number (no default allowed)
Real VALUES (4)	array of 4 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial (maximum) pu voltage setpoint at the study bus (VHI) (1.1 by default)
	VALUES(3)	minimum pu voltage setpoint at the study bus (VLO) (0.9 by default)
	VALUES(4)	pu voltage setpoint decrement (positive) at the study bus (DLTAV) (0.01 by default)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 ACCFILE	name of QV results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE

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IERR = 5	largest mismatch exceeds mismatch tolerance
IERR = 6	VHI is not greater than VLO
IERR = 7	study bus not found
IERR = 8	invalid study bus
IERR = 9	error when adding a plant to the study bus
IERR = 10	error when adding a machine to the study bus
IERR = 11	base case VHI solution did not converge
IERR = 12	no Distribution Factor Data File specified
IERR = 13	no Q-V Solution Output File specified
IERR = 14	in-service induction machines are in the "stalled" or "tripped" state
IERR = 15	bus type code and series element status inconsistencies
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 51	error opening QV Results File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Load Throwover Data File
IERR = 55	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.245. QV_ENGINE_3

This API routine is obsolete. It has been replaced by the API routine qv_engine_4 .

Use this API routine to run the QV analysis calculation engine.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_QV_ENGINE_3  OPTIONS(1)..OPTIONS(12) VALUES(1)..VALUES(4) LABEL
 DFXFILE THRFILE INLFILE ACCFILE
```

Python command syntax:

```
ierr = qv_engine_3(options, values, label, dfxfile, thrfile, inlfile,
 accfile)
```

Fortran command syntax:

```
CALL QV_ENGINE_3(OPTIONS, VALUES, LABEL, DFXFILE, THRFILE, INLFILE,
 ACCFILE, IERR)
```

Where:

Integer OPTIONS (12)	array of 12 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable

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	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
	OPTIONS(6) = 0	stall
	OPTIONS(6) = 1	trip
	OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTIONS(7) = 0	disable
	OPTIONS(7) = 1	enable
	OPTIONS(8)	solution method flag (0 by default)
	OPTIONS(8) = 0	FDNS
	OPTIONS(8) = 1	FNSL
	OPTIONS(8) = 2	optimized FDNS
	OPTIONS(9)	var limit code for the VHI power flow solution (0 by default)
	OPTIONS(9) = 0	apply immediately
	OPTIONS(9) = 1	initially ignore, then apply automatically
	OPTIONS(10)	var limit code for the subsequent voltage decrement cases (0 by default)
	OPTIONS(10) = 0	apply immediately
	OPTIONS(10) = 1	initially ignore, then apply automatically
	OPTIONS(11)	study bus number (no default allowed)
	OPTIONS(12)	dispatch mode for power unbalances resulting from the application of contingencies (1 by default)
	OPTIONS(12) = 0	disable
	OPTIONS(12) = 1	in-service subsystem machines using reserve
	OPTIONS(12) = 2	in-service subsystem machines using Pmax
	OPTIONS(12) = 3	in-service subsystem machines using inertia
	OPTIONS(12) = 4	in-service subsystem machines using governor droop
Real VALUES (4)	array of 4 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial (maximum) pu voltage setpoint at the study bus (VHI) (1.1 by default)
	VALUES(3)	minimum pu voltage setpoint at the study bus (VLO) (0.9 by default)
	VALUES(4)	pu voltage setpoint decrement (positive) at the study bus (DLTAV) (0.01 by default)
Character*32 LABEL (1)	array of 1 elements specifying subsystem label(input).	

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	LABEL(1)	label of the dispatch subsystem; used if OPTIONS(12) is 1 through 4 (blank by default)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 INLFILE	name of inertia and governor response file; blank for none (input; blank by default).	
Character*260 ACCFILE	name of QV results file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	VHI is not greater than VLO
	IERR = 7	study bus not found
	IERR = 8	invalid study bus
	IERR = 9	error when adding a plant to the study bus
	IERR = 10	error when adding a machine to the study bus
	IERR = 11	base case VHI solution did not converge
	IERR = 12	no Distribution Factor Data File specified
	IERR = 13	no Q-V Solution Output File specified
	IERR = 14	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 15	bus type code and series element status inconsistencies
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 22	monitored elements exceed limit when adding multi-section line members
	IERR = 51	error opening QV Results File
	IERR = 52	error opening Distribution Factor Data File
	IERR = 53	error opening Load Throwover Data File
	IERR = 55	prerequisite requirements for API are not met
	IERR = 99	terminated by user interrupt or file error

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1.246. QV_ENGINE_4

Use this API routine to run the QV analysis calculation engine.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_QV_ENGINE_4  OPTIONS(1)..OPTIONS(13) VALUES(1)..VALUES(4) LABEL
 DFXFILE THRFILE INLFILE ACCFILE ZIPFILE
```

Python command syntax:

```
ierr = qv_engine_4(options, values, label, dfxfile, thrfile, inlfile,
 accfile, zipfile)
```

Fortran command syntax:

```
CALL QV_ENGINE_4(OPTIONS, VALUES, LABEL, DFXFILE, THRFILE, INLFILE,
 ACCFILE, ZIPFILE, IERR)
```

Where:

Integer OPTIONS (13)	array of 13 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable stepping adjustment
	OPTIONS(1) = 2	enable direct adjustment
	OPTIONS(2)	area interchange adjustment flag (area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	phase shift adjustment flag (phase shift adjustment option setting by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
	OPTIONS(4)	dc tap adjustment flag (dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(5) = 2	enable continuous mode, disable discrete mode

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	OPTIONS(6)	induction motor treatment flag; applied when an induction motor fails to solve due to low terminal voltage (0 by default)
	OPTIONS(6) = 0	stall
	OPTIONS(6) = 1	trip
	OPTIONS(7)	non-divergent solution flag (non-divergent solution option setting by default)
	OPTIONS(7) = 0	disable
	OPTIONS(7) = 1	enable
	OPTIONS(8)	solution method flag (0 by default)
	OPTIONS(8) = 0	FDNS
	OPTIONS(8) = 1	FNSL
	OPTIONS(8) = 2	optimized FDNS
	OPTIONS(9)	var limit code for the VHI power flow solution (0 by default)
	OPTIONS(9) = 0	apply immediately
	OPTIONS(9) = 1	initially ignore, then apply automatically
	OPTIONS(10)	var limit code for the subsequent voltage decrement cases (0 by default)
	OPTIONS(10) = 0	apply immediately
	OPTIONS(10) = 1	initially ignore, then apply automatically
	OPTIONS(11)	study bus number (no default allowed)
	OPTIONS(12)	dispatch mode for power unbalances resulting from the application of contingencies (1 by default)
	OPTIONS(12) = 0	disable
	OPTIONS(12) = 1	in-service subsystem machines using reserve
	OPTIONS(12) = 2	in-service subsystem machines using Pmax
	OPTIONS(12) = 3	in-service subsystem machines using inertia
	OPTIONS(12) = 4	in-service subsystem machines using governor droop
	OPTIONS(13)	write ZIP archive flag (0 by default)
	OPTIONS(13) = 0	no ZIP archive
	OPTIONS(13) = 1	Write ZIP Archive file ZIPFILE; preserve each system condition at its lowest solved study bus voltage level.
	OPTIONS(13) = 2	Write ZIP Archive file ZIPFILE; preserve each system condition at all solved study bus voltage levels.
Real VALUES (4)	array of 4 elements specifying solution values(input).	
	VALUES(1)	mismatch tolerance in MW and Mvar (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	initial (maximum) pu voltage setpoint at the study bus (VHI) (1.1 by default)

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	VALUES(3)	minimum pu voltage setpoint at the study bus (VLO) (0.9 by default)
	VALUES(4)	pu voltage setpoint decrement (positive) at the study bus (DLTAV) (0.01 by default)
Character*32 LABEL (1)	array of 1 elements specifying subsystem label(input).	
	LABEL(1)	label of the dispatch subsystem; used if OPTIONS(12) is 1 through 4 (blank by default)
Character*260 DFXFILE	name of DFAX file (input; no default allowed).	
Character*260 THRFILE	name of load throwover data file; blank for none (input; blank by default).	
Character*260 INLFILE	name of inertia and governor response file; blank for none (input; blank by default).	
Character*260 ACCFILE	name of QV results file (input; no default allowed).	
Character*260 ZIPFILE	name of ZIP archive output file (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VALUES value
	IERR = 2	invalid OPTIONS value
	IERR = 3	generators are converted
	IERR = 4	buses in island(s) without a swing bus; use activity TREE
	IERR = 5	largest mismatch exceeds mismatch tolerance
	IERR = 6	initial transfer increment is less than transfer increment tolerance
	IERR = 7	no participating dispatch buses in study system
	IERR = 8	no participating dispatch buses in opposing system
	IERR = 9	subsystem engine error
	IERR = 10	ECDI engine in subsystem aaa
	IERR = 11	base case ECDI solution did not converge
	IERR = 12	no DFAX file specified
	IERR = 13	no PV results file specified
	IERR = 14	no economic dispatch data file specified
	IERR = 15	in-service induction machines are in the "stalled" or "tripped" state
	IERR = 16	bus type code and series element status inconsistencies
	IERR = 17	invalid base/contingency case flag combination
	IERR = 21	DFXFILE file is not in the form of a PSSE-25 or later DFAX file; run DFAX

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IERR = 22	monitored elements exceed limit when adding multi-section line members
IERR = 23	no subsystems defined
IERR = 24	subsystem not found
IERR = 25	specified subsystems overlap
IERR = 26	no in-service buses in subsystem WORLD
IERR = 50	error opening economic dispatch data file
IERR = 51	error opening PV results file
IERR = 52	error opening DFAX file
IERR = 53	error opening load throwover data file
IERR = 54	error opening inertia and governor response data file
IERR = 55	error opening ZIP archive output file
IERR = 56	prerequisite requirements for API are not met
IERR = 99	terminated by user interrupt or file error

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1.247. RANK

Use this API to estimate the severity of designated single branch outage contingencies and builds a contingency solution output file with contingencies specified in decreasing order of their estimated severities.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RANK  SID  ALL  IVAL(1)..IVAL(7) RVAL(1)..RVAL(4)  SSFILE  MNFILE
 ACCFILE
```

Python command syntax:

```
ierr = rank(sid, all, ival, rval, ssfile, mnfile, accfile)
```

Fortran command syntax:

```
CALL RANKNW(SID, ALL, IVAL, RVAL, SSFILE, MNFILE, ACCFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IVAL (7)	array of 7 elements specifying ranking options(input).	
	IVAL(1)	allow ties from specified subsystem as contingencies (0 = no, 1 = yes; 0 by default)
	IVAL(2)	number of branch contingencies from overload ranking to include (>= 0) (0 by default)
	IVAL(3)	rating set (1=RATEA, 2=RATEB, 3=RATEC) (rating set program option setting by default)
	IVAL(4)	convert ratings to estimated MW ratings (0 = no, 1 = yes; 0 by default)
	IVAL(5)	number of voltage depression contingencies to include (>=0) (0 by default)
	IVAL(6)	code indicating the set of branches to include in the voltage depression performance index (0 by default)
	IVAL(6) = 0	include all branches
	IVAL(6) = 1	include only those with a nonzero rating
	IVAL(7)	include contingencies that create swingless islands (0 by default)
	IVAL(7) = 0	no
	IVAL(7) = 1	yes
Real RVAL (4)	array of 4 elements specifying ranking parameters(input).	

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	RVAL(1)	MW mismatch threshold (>= 0.0) (Newton solution convergence tolerance, TOLN by default)
	RVAL(2)	small reactance threshold (THRSHZ <= RVAL(2) <= 0.005) (0.0005 by default)
	RVAL(3)	percent of rating (0.0 <= RVAL(3) <= 200.0) (100.0 by default)
	RVAL(4)	MW threshold for islanding contingencies (>= 0.0) (100.0 by default)
Character*260 SSFILE	name of the subsystem description file, or blank for no subsystem (input; blank by default).	
Character*260 MNFILE	name of the monitored element description file (if IVAL(2) > 0 (i.e., if the branch outage overload ranker is enabled)) (input; no default allowed).	
Character*260 ACCFILE	name of the contingency solution output file, or blank for output to the progress window (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid IVAL value
	IERR = 4	invalid RVAL value
	IERR = 5	MNFILE is blank
	IERR = 6	no contingency types enabled
	IERR = 7	error opening SSFILE or MNFILE
	IERR = 8	generators are converted
	IERR = 9	buses in island(s) without a swing bus; use activity TREE
	IERR = 10	largest mismatch exceeds mismatch tolerance
	IERR = 11	prerequisite requirements for API are not met

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1.248. RANK_BRN_AND_MAC

Use this API to estimate the severity of designated single element outage contingencies and builds a contingency description data file with contingencies specified in decreasing order of their estimated severities.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_RANK_BRN_AND_MAC  SID  ALL  IVAL(1)..IVAL(8) RVAL(1)..RVAL(4)  SSFILE
 MNFILE  ACCFILE
```

Python command syntax:

```
ierr = rank_brn_and_mac(sid, all, ival, rval, ssfile, mnfile, accfile)
```

Fortran command syntax:

```
CALL RANKAPI(SID, ALL, IVAL, RVAL, SSFILE, MNFILE, ACCFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IVAL (8)	array of 8 elements specifying ranking options(input).	
	IVAL(1)	allow ties from specified subsystem as contingencies (0 by default)
	IVAL(1) = 0	no
	IVAL(1) = 1	yes
	IVAL(2)	number of branch contingencies from overload ranking to include (>= 0) (0 by default)
	IVAL(3)	rating set (1=RATEA, 2=RATEB, 3=RATEC) (rating set program option setting by default)
	IVAL(4)	convert ratings to estimated MW ratings (0 by default)
	IVAL(4) = 0	no
	IVAL(4) = 1	yes
	IVAL(5)	number of voltage depression contingencies to include (>=0) (0 by default)
	IVAL(6)	code indicating the set of branches to include in the voltage depression performance index (0 by default)
	IVAL(6) = 0	include all branches
	IVAL(6) = 1	include only those with a nonzero rating
	IVAL(7)	include contingencies that create swingless islands (0 by default)

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	IVAL(7) = 0	no
	IVAL(7) = 1	yes
	IVAL(8)	number of machine contingencies from overload ranking to include (>=0) (0 by default)
Real RVAL (4)	array of 4 elements specifying ranking parameters(input).	
	RVAL(1)	MW mismatch threshold (>= 0.0) (Newton solution convergence tolerance, TOLN by default)
	RVAL(2)	small reactance threshold (THRSHZ <= RVAL(2) <= 0.005) (0.0005 by default)
	RVAL(3)	percent of rating (0.0 <= RVAL(3) <= 200.0) (100.0 by default)
	RVAL(4)	MW threshold for islanding contingencies (>= 0.0) (100.0 by default)
Character*260 SSFILE	name of the subsystem description file, or blank for no subsystem (input; blank by default).	
Character*260 MNFILE	name of the monitored element description file (if IVAL(2) > 0 (i.e., if the branch outage overload ranker is enabled) or if IVAL(8) > 0 (i.e., if the machine outage overload ranker is enabled)) (input; no default allowed).	
Character*260 ACCFILE	name of the contingency description data file, or blank for output to the progress window (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid IVAL value
	IERR = 4	invalid RVAL value
	IERR = 5	MNFILE is blank
	IERR = 6	no contingency types enabled
	IERR = 7	error opening SSFILE or MNFILE
	IERR = 8	generators are converted
	IERR = 9	buses in island(s) without a swing bus; use activity TREE
	IERR = 10	largest mismatch exceeds mismatch tolerance
	IERR = 11	prerequisite requirements for API are not met

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1.249. RATE

This API routine is obsolete. It has been replaced by the API routine rate_2 .

Use this API to check branch loadings.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RATE  SID  ALL  LINFLG  RATING  FMAX
```

Python command syntax:

```
ierr = rate(sid, all, linflg, rating, fmax)
```

Fortran command syntax:

```
CALL RATENW(SID, ALL, LINFLG, RATING, FMAX, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only branches connected to buses in subsystem SID
	ALL = 1	process all branches
Integer LINFLG	branch check to be applied (input; 1 by default).	
	LINFLG = 1	check all lines
	LINFLG = 2	check non-transformer branches
	LINFLG = 3	check transformers
	LINFLG = 4	check all lines, all ratings
Integer RATING	rating to be used (input; default rating set program option setting).	
	RATING = 1	RATEA
	RATING = 2	RATEB
	RATING = 3	RATEC
Real FMAX	line loading limit, in percent; 0.0 =< FMAX =< 500.0 (input; 100.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid LINFLG value
	IERR = 4	invalid RATING value
	IERR = 5	invalid FMAX value

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IERR = 6	prerequisite requirements for API are not met

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1.250. RATE_2

Use this API to check branch loadings.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_RATE_2  SID  ALL  LINFLG  TRNFLG  BRKFLG  RATING  FMAX
```

Python command syntax:

```
ierr = rate_2(sid, all, linflg, trnflg, brkflg, rating, fmax)
```

Fortran command syntax:

```
CALL RATE_2(SID, ALL, LINFLG, TRNFLG, BRKFLG, RATING, FMAX, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only branches connected to buses in subsystem SID
	ALL = 1	process all branches
Integer LINFLG	include non-transformer branches that are not breakers or switches in the branch check to be applied (input; 1 by default).	
	LINFLG = 0	no
	LINFLG = 1	yes
Integer TRNFLG	include transformer branches in the branch check to be applied (input; 1 by default).	
	TRNFLG = 0	no
	TRNFLG = 1	yes
Integer BRKFLG	include branches that are breakers or switches in the branch check to be applied (input; 1 by default).	
	BRKFLG = 0	no
	BRKFLG = 1	yes
Integer RATING	rating to be used (input; default rating set program option setting).	
	RATING = 0	all rating sets
	RATING = 1	RATEA
	RATING = 2	RATEB
	RATING = 3	RATEC
Real FMAX	line loading limit, in percent; 0.0 =< FMAX =< 500.0 (input; 100.0).	
Integer IERR	error code (output).	
	IERR = 0	no error

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IERR = 1	invalid SID value or subsystem SID is not defined
IERR = 2	invalid ALL value
IERR = 3	invalid LINFLG value
IERR = 4	invalid TRNFLG value
IERR = 5	invalid BRKFLG value
IERR = 6	invalid RATING value
IERR = 7	invalid FMAX value
IERR = 8	prerequisite requirements for API are not met

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1.251. RATING_SET

Use this API to set the default rating set option setting to either RATEA, RATEB or RATEC (activity OPTN).

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_RATING_SET  IVAL
```

Python command syntax:

```
ierr = rating_set(ival)

ierr, ival = rating_set()
```

Fortran command syntax:

```
CALL RATING_SET_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command or for Python if specified as an argument, or for Fortran if IOCODE is 0; else 1 by default for input).	
	IVAL = 1	RATEA.
	IVAL = 2	RATEB.
	IVAL = 3	RATEC.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.252. RAWD

This API routine is obsolete. It has been replaced by the API routine rawd_2 .

Use this API to replicate the working case in the form of a power flow raw data file.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RAWD  SID  ALL  STATUS(1)..STATUS(6)  OUT  OFILE
```

Python command syntax:

```
ierr = rawd(sid, all, status, out, ofile)
```

Fortran command syntax:

```
CALL RAWDAPI(SID, ALL, STATUS, OUT, OFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses when STATUS(4) = 0; area ties only when STATUS(4) = 1
Integer STATUS (6)	array of 6 elements specifying options(input).	
	STATUS(1)	include records for Type 4 buses (1 by default)
	STATUS(1) = 0	no
	STATUS(1) = 1	yes
	STATUS(2)	include records for out-of-service branches (1 by default)
	STATUS(2) = 0	no
	STATUS(2) = 1	yes
	STATUS(3)	include records for equipment in the subsystem (1 by default)
	STATUS(3) = 0	no
	STATUS(3) = 1	yes
	STATUS(4)	include records for subsystem ties (0 by default)
	STATUS(4) = 0	no
	STATUS(4) = 1	yes
	STATUS(5)	use bus names as bus identifiers (0 by default)
	STATUS(5) = 0	no

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	STATUS(5) = 1	yes
	STATUS(6)	Raw Data File type (0 by default)
	STATUS(6) = 0	for activity READ with IC=0
	STATUS(6) = 1	for activity READ with IC=1
	STATUS(6) = 2	for activity RDCH
Integer OUT	destination selection (input; 1 by default).	
	OUT = 0	direct output to a data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the file to which data records are to be written (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid OUT value
	IERR = 5	error opening OFILE
	IERR = 6	prerequisite requirements for API are not met

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1.253. RAWD_2

Use this API to replicate the working case in the form of a power flow raw data file.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_RAWD_2  SID  ALL  STATUS(1)..STATUS(7)  OUT  OFILE
```

Python command syntax:

```
ierr = rawd_2(sid, all, status, out, ofile)
```

Fortran command syntax:

```
CALL RAWDAPI_2(SID, ALL, STATUS, OUT, OFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses when STATUS(4) = 0; only area tie branches when STATUS(4) = 1
Integer STATUS (7)	array of 7 elements specifying options(input).	
	STATUS(1)	include records for Type 4 buses (1 by default)
	STATUS(1) = 0	no
	STATUS(1) = 1	yes
	STATUS(2)	include records for out-of-service branches (1 by default)
	STATUS(2) = 0	no
	STATUS(2) = 1	yes
	STATUS(3)	include records for equipment in the subsystem (1 by default)
	STATUS(3) = 0	no
	STATUS(3) = 1	yes
	STATUS(4)	include records for subsystem tie branches (0 by default)
	STATUS(4) = 0	no
	STATUS(4) = 1	yes
	STATUS(5)	include load record option (honored when ALL=0) (0 by default)
	STATUS(5) = 0	for all loads at subsystem buses
	STATUS(5) = 1	for subsystem loads at all buses

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	STATUS(5) = 2	for all loads at subsystem buses and subsystem loads at non-subsystem buses
	STATUS(6)	use bus names as bus identifiers (0 by default)
	STATUS(6) = 0	no
	STATUS(6) = 1	yes
	STATUS(7)	Raw Data File type (0 by default)
	STATUS(7) = 0	for activity READ with IC=0
	STATUS(7) = 1	for activity READ with IC=1
	STATUS(7) = 2	for activity RDCH
Integer OUT	destination selection (input; 1 by default).	
	OUT = 0	direct output to a data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the file to which data records are to be written (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid OUT value
	IERR = 5	error opening OFILE
	IERR = 6	prerequisite requirements for API are not met

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1.254. RDCH

Use this API to read power flow change data into the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RDCH  NUMNAM  IFILE
```

Python command syntax:

```
ierr = rdch(numnam, ifile)
```

Fortran command syntax:

```
CALL RDCHAPI(NUMNAM, IFILE, IERR)
```

Where:

Integer NUMNAM	flag for bus number or name specification on records (input; 0 by default).	
	NUMNAM = 0	bus numbers
	NUMNAM = 1	bus names
Character*260 IFILE	name of power flow data file, Options (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid NUMNAM
	IERR = 10	error opening IFILE
	IERR = 11	prerequisite requirements for API are not met

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1.255. RDCHRAWVERSION

Use this API to read a Power Flow Change Raw Data File into the working case. Allowed raw file formats are PSSE-15 through current format.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_RDCHRAWVERSION  NUMNAM VERNUM IFILE
```

Python command syntax:

```
ierr = rdchrawversion(numnam, vernum, ifile)
```

Fortran command syntax:

```
CALL RDCHRAWVERSION(NUMNAM, VERNUM, IFILE, IERR)
```

Where:

Integer NUMNAM	flag for bus number or name specification on records (input; 0 by default).	
	NUMNAM = 0	bus numbers
	NUMNAM = 1	bus names
Character*14 VERNUM	version number corresponding to the format of IFILE VERNUM is in the format of a PSSE release number. Example: If IFILE format is of PSSE-29.5.1: VERNUM = '29' or = '29.5' or = '29.5.1' (input; no default allowed).	
Character*260 IFILE	name of power flow raw data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid NUMNAM value
	IERR = 2	invalid VERNUM value
	IERR = 3	unable to convert file
	IERR = 4	error opening temporary file
	IERR = 5	invalid revision number
	IERR = 10	error opening IFILE
	IERR = 11	prerequisite requirements for API are not met

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1.256. RDEQ

Use this API to build an electrical equivalent of radial and, optionally, two-point Type 1 buses outside of a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RDEQ  SID  ALL  STATUS(1)..STATUS(8)
```

Python command syntax:

```
ierr = rdeq(sid, all, status)
```

Fortran command syntax:

```
CALL RDEQNW(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses outside of subsystem SID
	ALL = 1	process all buses
Integer STATUS (8)	array of 8 elements specifying used to control equivalencing options. A value of 0 means the option is not enabled. A value of 1 means the option is enabled. They are as follows(input).	
	STATUS(1)	build an electrical equivalent of radial buses only (0 by default)
	STATUS(2)	suppress the equivalencing option for transformers (0 by default)
	STATUS(3)	suppress the equivalencing option for zero impedance lines (0 by default)
	STATUS(4)	suppress the equivalencing option for controlled buses (0 by default)
	STATUS(5)	suppress the equivalencing option for mutually coupled branches (0 by default)
	STATUS(6)	suppress the equivalencing option for area boundary buses (0 by default)
	STATUS(7)	suppress the equivalencing option for zone boundary buses (0 by default)
	STATUS(8)	apply transformer impedance correction to zero sequence network (1 by default)
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined

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IERR = 2	invalid ALL value
IERR = 3	invalid STATUS value
IERR = 4	error writing the working case to a temporary file
IERR = 5	load table is full; restoring original working case
IERR = 6	fixed bus shunt table is full; restoring original working case
IERR = 7	error opening temporary file
IERR = 8	prerequisite requirements for API are not met

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1.257. READ

Use this API to read a power flow raw data file and add all the data specified in it to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_READ  NUMNAM  IFILE
```

Python command syntax:

```
ierr = read(numnam, ifile)
```

Fortran command syntax:

```
CALL READAPI(NUMNAM, IFILE, IERR)
```

Where:

Integer NUMNAM	flag for bus number or name specification on records (input; 0 by default).	
	NUMNAM = 0	bus numbers
	NUMNAM = 1	bus names
Character*260 IFILE	filename of the power flow raw data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid NUMNAM value
	IERR = 2	invalid revision number
	IERR = 3	unable to convert file
	IERR = 4	error opening temporary file
	IERR = 10	error opening IFILE
	IERR = 11	prerequisite requirements for API are not met

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1.258. READRAWVERSION

Use this API to read a power flow raw data file and add all the data specified in it to the working case. Allowed raw file formats are PSSE-15 through current format.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_READRAWVERSION  NUMNAM VERNUM IFILE
```

Python command syntax:

```
ierr = readrawversion(numnam, vernum, ifile)
```

Fortran command syntax:

```
CALL READRAWVERSION(NUMNAM, VERNUM, IFILE, IERR)
```

Where:

Integer NUMNAM	flag for bus number or name specification on records (input; 0 by default).	
	NUMNAM = 0	bus numbers
	NUMNAM = 1	bus names
Character*14 VERNUM	version number corresponding to the format of IFILE. VERNUM is in the format of a PSSE release number. Example: If IFILE format is of PSSE-29.5.1: VERNUM = '29' or = '29.5' or = '29.5.1' (input; no default allowed).	
Character*260 IFILE	name of power flow raw data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid NUMNAM value
	IERR = 2	invalid VERNUM value
	IERR = 3	unable to convert file
	IERR = 4	error opening temporary file
	IERR = 5	invalid revision number
	IERR = 10	error opening IFILE
	IERR = 11	prerequisite requirements for API are not met

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1.259. READSUB

Use this API to read a power flow raw data file and add subsystem data specified in it to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_READSUB  NUMNAM  IFILE  STATUS(1)  STATUS(2)  STATUS(3)  USEKV
 BASEKV(1)  BASEKV(2)  SIDAREA  SIDOWNER  SIDZONE
```

Python command syntax:

```
ierr = readsub(numnam, ifile, status, usekv, basekv, sidarea, sidowner,
 sidzone)
```

Fortran command syntax:

```
CALL READSUB(NUMNAM, IFILE, STATUS, USEKV, BASEKV, SIDAREA, SIDOWNER,
 SIDZONE, IERR)
```

Where:

Integer NUMNAM	flag for bus number or name specification on records (input; 0 by default).	
	NUMNAM = 0	bus numbers
	NUMNAM = 1	bus names
Character*260 IFILE	name of the power flow raw data file (input; no default allowed).	
Integer STATUS (3)	array of 3 elements specifying options(input).	
	STATUS(1)	append flag (0 by default)
	STATUS(1) = 0	initialize working case before reading data
	STATUS(1) = 1	append data to that already in the working case
	STATUS(2)	boundary bus identification flag (0 by default)
	STATUS(2) = 0	do not change type codes of boundary buses
	STATUS(2) = 1	change type codes of boundary buses in preparation for the equivalencing calculation
	STATUS(3)	data option (1 by default)
	STATUS(3) = 1	add only data from within subsystem
	STATUS(3) = 2	add only ties from subsystem
	STATUS(3) = 3	add subsystem data plus ties
Integer USEKV	subsystem base voltage flag (input; 0 by default).	
	USEKV = 0	do not use base voltage as a subsystem selection criterion
	USEKV = 1	use base voltage as a subsystem selection criterion
Real BASEKV (2)	array of 2 elements specifying the base voltage range(input).	
	BASEKV(1)	minimum base voltage (0.0 by default)

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	BASEKV(2)	maximum base voltage (BASEKV (1) by default)
Integer SIDAREA	area subsystem identifier (0 through 11), or -1 to bypass subsystem specification by area (input; -1 by default).	
Integer SIDOWNER	owner subsystem identifier (0 through 11), or -1 to bypass subsystem specification by owner (input; -1 by default).	
Integer SIDZONE	zone subsystem identifier (0 through 11), or -1 to bypass subsystem specification by zone (input; -1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid NUMNAM value
	IERR = 2	invalid STATUS value
	IERR = 3	invalid USEKV value
	IERR = 4	invalid BASEKV value(s)
	IERR = 5	BASEKV(1) is greater than BASEKV(2)
	IERR = 6	invalid SIDAREA value or area subsystem SIDAREA is not defined
	IERR = 7	invalid SIDOWNER value or owner subsystem SIDOWNER is not defined
	IERR = 8	invalid SIDZONE value or zone subsystem SIDZONE is not defined
	IERR = 9	no subsystem specified
	IERR = 10	error opening IFILE
	IERR = 11	invalid revision number
	IERR = 12	prerequisite requirements for API are not met
	IERR = 14	unable to convert file
	IERR = 15	error opening temporary file

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1.260. READSUBRAWVERSION

Use this API to read a power flow raw data file and add subsystem data specified in it to the working case. Allowed raw file formats are PSSE-15 through current format.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_READSUBRAWVERSION  NUMNAM  VERNUM  IFILE  STATUS(1)  STATUS(2)
 STATUS(3)  USEKV  BASEKV(1)  BASEKV(2)  SIDAREA  SIDOWNER  SIDZONE
```

Python command syntax:

```
ierr = readsubrawversion(numnam, vernum, ifile, status, usekv, basekv,
 sidarea, sidowner, sidzone)
```

Fortran command syntax:

```
CALL READSUBRAWVERSION(NUMNAM, VERNUM, IFILE, STATUS, USEKV, BASEKV,
 SIDAREA, SIDOWNER, SIDZONE, IERR)
```

Where:

Integer NUMNAM	flag for bus number or name specification on records (input; 0 by default).	
	NUMNAM = 0	bus numbers
	NUMNAM = 1	bus names
Character*14 VERNUM	version number corresponding to the format of IFILE. VERNUM is in the format of a PSSE release number. Example: If IFILE format is of PSSE-29.5.1: VERNUM = '29' or = '29.5' or = '29.5.1' (input; no default allowed).	
Character*260 IFILE	filename of the power flow raw data file (input; no default allowed).	
Integer STATUS (3)	array of 3 elements specifying options(input).	
	STATUS(1)	append flag (0 by default)
	STATUS(1) = 0	initialize working case before reading data
	STATUS(1) = 1	append data to that already in the working case
	STATUS(2)	boundary bus identification flag (0 by default)
	STATUS(2) = 0	do not change type codes of boundary buses
	STATUS(2) = 1	change type codes of boundary buses in preparation for the equivalencing calculation
	STATUS(3)	data option (1 by default)
	STATUS(3) = 1	add only data from within subsystem
	STATUS(3) = 2	add only ties from subsystem
	STATUS(3) = 3	add subsystem data plus ties
Integer USEKV	subsystem base voltage flag (input; 0 by default).	

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	USEKV = 0	do not use base voltage as a subsystem selection criterion
	USEKV = 1	use base voltage as a subsystem selection criterion
Real BASEKV (2)	array of 2 elements specifying the base voltage range(input).	
	BASEKV(1)	minimum base voltage (0.0 by default)
	BASEKV(2)	maximum base voltage (BASEKV (1) by default)
Integer SIDAREA	area subsystem identifier (0 through 11), or -1 to bypass subsystem specification by area (input; -1 by default).	
Integer SIDOWNER	owner subsystem identifier (0 through 11), or -1 to bypass subsystem specification by owner (input; -1 by default).	
Integer SIDZONE	zone subsystem identifier (0 through 11), or -1 to bypass subsystem specification by zone (input; -1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid NUMNAM value
	IERR = 2	invalid STATUS value
	IERR = 3	invalid USEKV value
	IERR = 4	invalid BASEKV value(s)
	IERR = 5	BASEKV(1) is greater than BASEKV(2)
	IERR = 6	invalid SIDAREA value or area subsystem SIDAREA is not defined
	IERR = 7	invalid SIDOWNER value or owner subsystem SIDOWNER is not defined
	IERR = 8	invalid SIDZONE value or zone subsystem SIDZONE is not defined
	IERR = 9	no subsystem specified
	IERR = 10	error opening IFILE
	IERR = 11	invalid revision number
	IERR = 12	prerequisite requirements for API are not met
	IERR = 13	invalid VERNUM value
	IERR = 14	unable to convert file
	IERR = 15	error opening temporary file

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1.261. RECN

Use this API to electrically reconnect a bus.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RECN  IBUS
```

Python command syntax:

```
ierr = recn(ibus)
```

Fortran command syntax:

```
CALL RECNAPI(IBUS, IERR)
```

Where:

Integer IBUS	bus number of bus to be reconnected (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	prerequisite requirements for API are not met

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1.262. RECORDING_OPTIONS

Use this API to specify options that control the API recording process.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_RECORDING_OPTIONS  RECOPTS(1) RECOPTS(2)
```

Python command syntax:

```
ierr = recording_options(recopts)
```

Fortran command syntax:

```
CALL RECORDING_OPTIONS(RECOPTS, IERR)
```

Where:

Integer RECOPTS (2)	recording options. For future use (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid RECOPTS value
	IERR = 2	prerequisite requirements for API are not met

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1.263. REGB

Use this API to tabulate those buses where voltages are controlled by generation, switched shunts, and/or other voltage controlling equipment.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_REGB  SID  ALL
```

Python command syntax:

```
ierr = regb(sid, all)
```

Fortran command syntax:

```
CALL REGBNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.264. RELIND

This API routine is obsolete. It has been replaced by the API routine relind_2 .

Use this API to run probabilistic reliability assessment for transmission systems. Probabilistic indices on system problems are calculated in entire system or a given subsystem, these problems include flow overloading, voltage violation and loss of load, etc.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_RELIND  OPTIONS(1)..OPTIONS(7)  VALUES(1)..VALUES(3)  LABEL   ACCFILE
 DFXFILE  PRBFILE  STSFILE
```

Python command syntax:

```
ierr = relind(options, values, label, accfile, dfxfile, prbfile, stsfile)
```

Fortran command syntax:

```
CALL RELINDAPI(OPTIONS, VALUES, LABEL, ACCFILE, DFXFILE, PRBFILE, STSFILE,
 IERR)
```

Where:

Integer OPTIONS (7)	array of 7 elements specifying options(input).	
	OPTIONS(1)	report format (1 by default)
	OPTIONS(1) = 1	system problem summary report
	OPTIONS(1) = 2	system loss of load report
	OPTIONS(1) = 3	bus loss of load report
	OPTIONS(1) = 4	branch flow overloading report
	OPTIONS(1) = 5	bus voltage violation report
	OPTIONS(1) = 6	contingency summary
	OPTIONS(1) = 7	system problem indices
	OPTIONS(1) = 8	system loss of load indices
	OPTIONS(1) = 9	bus loss of load indices
	OPTIONS(1) = 10	branch overload problem indices
	OPTIONS(1) = 11	bus voltage problem indices
	OPTIONS(1) = 12	contingency summary with outage statistics
	OPTIONS(2)	mode (1 by default)
	OPTIONS(2) = 1	post contingency mode
	OPTIONS(2) = 2	post tripping mode
	OPTIONS(2) = 3	post corrective action mode
	OPTIONS(3)	rating set (rating set program option setting by default)
	OPTIONS(3) = 1	RATEA
	OPTIONS(3) = 2	RATEB
	OPTIONS(3) = 3	RATEC

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	OPTIONS(4)	include ties from study subsystem (0 by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	include interfaces (0 by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable
	OPTIONS(6)	Outage Statistics Data File (0 by default)
	OPTIONS(6) = 0	disable
	OPTIONS(6) = 1	enable
	OPTIONS(7)	normalize probability (0 by default)
	OPTIONS(7) = 0	disable
	OPTIONS(7) = 1	enable
Real VALUES (3)	array of 3 elements specifying reliability assessment parameters(input).	
	VALUES(1)	percent of rating set (>=0 and <=200.0, 100.0 by default)
	VALUES(2)	probability cutoff (>=0; 0.0001 by default)
	VALUES(3)	frequency cutoff (>=0; 0.0001 by default)
Character*32 LABEL	name of the study subsystem (input; blank by default, probabilistic reliability assessment is carried out in entire system).	
Character*260 ACCFILE	contingency solution output file (input; no default allowed).	
Character*260 DFXFILE	DFAX data file (input; no default allowed).	
Character*260 PRBFILE	outage statistics data file (input; blank for none if OPTIONS(1) is between 1 and 6, inclusive; no default allowed if OPTIONS(1) is greater).	
Character*260 STSFILE	output outage statistics data file (input; blank for none if OPTIONS(6) is zero or OPTIONS(1) is from 1 to 6; otherwise no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPTIONS value
	IERR = 2	no dispatch subsystem specified
	IERR = 3	no Contingency Solution Output File specified
	IERR = 4	no Distribution Factor Data File specified
	IERR = 5	no Outage Statistic Data File specified
	IERR = 6	no Output Outage Statistic Data File specified
	IERR = 7	generators are converted
	IERR = 8	buses in island(s) without a swing bus--use activity TREE
	IERR = 9	subsystem not found in Distribution Factor Data File
	IERR = 10	Contingency Solution Output File is not created by Multiple Level Contingency Analysis of PSSE 31 or later

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IERR = 11	case data is not the one used in contingency analysis
IERR = 12	Distribution Factor Data File is not the one used in contingency analysis
IERR = 13	error reading Contingency Solution Output file
IERR = 51	error opening Contingency Solution Output File
IERR = 52	error opening Distribution Factor Data File
IERR = 53	error opening Outage Statistic Data File
IERR = 54	error opening Output Outage Statistic Data File
IERR = 55	prerequisite requirements for function not met

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1.265. RELIND_2

Use this API to run probabilistic reliability assessment for transmission systems. Probabilistic indices on system problems are calculated in entire system or a given subsystem, these problems include flow overloading, voltage violation and loss of load, etc.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_RELIND_2  OPTREL(1)..OPTREL(8) RELREL(1)..RELREL(3) CHAARG ACCFILE
 DFXFILE PRBFILE STSFILE
```

Python command syntax:

```
ierr = relind_2(optrel, relrel, chaarg, accfile, dfxfile, prbfile,
 stsfile)
```

Fortran command syntax:

```
CALL RELINDAPI_2(OPTREL, RELREL, CHAARG, ACCFILE, DFXFILE, PRBFILE,
 STSFILE, IERR)
```

Where:

Integer OPTREL (8)	array of elements specifying output options(input).	
	OPTREL(1)	report format (1 by default)
	OPTREL(1) = 1	system problem summary report
	OPTREL(1) = 2	system loss of load report
	OPTREL(1) = 3	bus loss of load report
	OPTREL(1) = 4	branch flow overloading report
	OPTREL(1) = 5	bus voltage violation report
	OPTREL(1) = 6	contingency summary
	OPTREL(1) = 7	system problem indices
	OPTREL(1) = 8	system loss of load indices
	OPTREL(1) = 9	bus loss of load indices
	OPTREL(1) = 10	branch overload problem indices
	OPTREL(1) = 11	bus voltage problem indices
	OPTREL(1) = 12	contingency summary with outage statistics
	OPTREL(2)	output mode (1 by default)
	OPTREL(2) = 1	post contingency mode
	OPTREL(2) = 2	post tripping mode
	OPTREL(2) = 3	post corrective action mode
	OPTREL(3)	rating set (rating set program option setting by default)
	OPTREL(3) = 1	RATEA
	OPTREL(3) = 2	RATEB
	OPTREL(3) = 3	RATEC
	OPTREL(4)	voltage limit.
	OPTREL(4) = 1	normal

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	OPTREL(4) = 2	emergency
	OPTREL(5)	include ties from study subsystem (0 by default)
	OPTREL(5) = 0	disable
	OPTREL(5) = 1	enable
	OPTREL(6)	include interfaces (0 by default)
	OPTREL(6) = 0	disable
	OPTREL(6) = 1	enable
	OPTREL(7)	output Outage Statistics Data File (0 by default)
	OPTREL(7) = 0	disable
	OPTREL(7) = 1	enable
	OPTREL(8)	normalize probability (0 by default)
	OPTREL(8) = 0	disable
	OPTREL(8) = 1	enable
Real RELREL (3)	array of elements specifying reliability assessment parameters(input).	
	RELREL(1)	percent of rating set, (>=0 and <=200.0, 100.0 by default)
	RELREL(2)	probability cutoff (>=0; 0.0001 by default).
	RELREL(3)	frequency cutoff (>=0; 0.0001 by default).
Character*32 CHAARG	name of the study subsystem (blank by default, probabilistic reliability assessment is carried out in entire system). (input).	
Character*260 ACCFILE	contingency solution output file (input; no default allowed).	
Character*260 DFXFILE	DFAX data file (input; no default allowed).	
Character*260 PRBFILE	outage statistics data file (blank for none if OPTIONS(1) is between 1 and 6, inclusive; no default allowed if OPTIONS(1) is greater). (input).	
Character*260 STSFILE	output outage statistics data file (blank for none if OPTIONS(6) is zero or OPTIONS(1) is from 1 to 6; otherwise no default allowed). (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	no dispatch subsystem specified
	IERR = 3	no contingency solution output file specified
	IERR = 4	no DFAX data file specified
	IERR = 5	no outage statistic data file specified
	IERR = 6	no output outage statistic data file specified
	IERR = 7	generators are converted
	IERR = 8	buses in island(s) without a swing bus; use activity TREE
	IERR = 9	subsystem not found in DFAX data file

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IERR = 10	contingency solution output file is not created by Multiple Level Contingency Analysis of PSSE 31 or later
IERR = 11	case data is not the one used in contingency analysis
IERR = 12	DFAX data file is not the one used in contingency analysis
IERR = 13	error reading contingency solution output file
IERR = 51	error opening contingency solution output file
IERR = 52	error opening DFAX data file
IERR = 53	error opening outage statistic data file
IERR = 54	error opening output outage statistic data file
IERR = 55	prerequisite requirements for function not met

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1.266. REMM

Use this API to read transaction event data from a transactions raw data file and add it to PSSE working memory.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_REMM  OPT  IFILE
```

Python command syntax:

```
ierr = remm(opt, ifile)
```

Fortran command syntax:

```
CALL REMMAPI(OPT, IFILE, IERR)
```

Where:

Integer OPT	flag for new or change data records (input; 0 by default).	
	OPT = 0	new data; replaces existing transactions data.
	OPT = 1	change data; modifies existing transactions data.
Character*260 IFILE	name of transactions raw data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPT value
	IERR = 2	no transactions defined; change mode invalid
	IERR = 3	error opening IFILE
	IERR = 4	prerequisite requirements for API are not met

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1.267. REPORT_OUTPUT

Use this API to specify the report output device.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_REPORT_OUTPUT  ISLCT  FILARG  OPTIONS(1)  OPTIONS(2)
```

Python command syntax:

```
ierr = report_output(islct, filarg, options)
```

Fortran command syntax:

```
CALL OPENREPORTDEVICE(ISLCT, FILARG, OPTIONS, IERR)
```

Where:

Integer ISLCT	virtual device selector (input).	
	ISLCT = 1	standard destination. In the PSSE GUI standard output for reports is the Report tab of the output bar.
	ISLCT = 2	direct output to a file.
	ISLCT = 3	direct output to a printer.
	ISLCT = 4	direct output to progress device.
	ISLCT = 5	direct output to report device.
	ISLCT = 6	no output.
Character*260 FILARG	printer name or filename; ignored if ISLCT is not 2 or 3 (input).	
Integer OPTIONS (2)	array of 2 elements specifying open and printing options; ignored if ISLCT is not 2 or 3(input).	
	OPTIONS(1)	file and printer open options: The OPTIONS(1) values may be summed to select multiple options
	OPTIONS(1) = 0	open with carriage control format and, for files, for overwrite of existing files
	OPTIONS(1) = 1	open with list format
	OPTIONS(1) = 2	open file for append (file output only)
	OPTIONS(1) = 2	set printer orientation to portrait mode (printer output only)
	OPTIONS(1) = 4	open printer in immediate print mode (printer output only)
	OPTIONS(2)	number of copies to print (printer only)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = 2	failed to open device

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1.268. RESQ

Use this API to read sequence data from a sequence data file and add it to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RESQ  IFILE
```

Python command syntax:

```
ierr = resq(ifile)
```

Fortran command syntax:

```
CALL RESQAPI(IFILE, IERR)
```

Where:

Character*260 IFILE	name of the power flow sequence data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	error opening IFILE
	IERR = 2	prerequisite requirements for API are not met

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1.269. RESQVERSION

Use this API to read a network sequence data file and add all the data specified in it to the working case. Allowed seq file formats are PSSE-15 through the current format.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_RESQVERSION  VERNUM  IFILE
```

Python command syntax:

```
ierr = resqversion(vernum, ifile)
```

Fortran command syntax:

```
CALL RESQVERSION(VERNUM, IFILE, IERR)
```

Where:

Character*14 VERNUM	version number corresponding to the format of the IFILE. VERNUM is in the format of a PSSE release number. Example: If IFILE format is of PSSE-29.5.1: VERNUM = '29' or = '29.5' or = '29.5.1' (input; no default allowed).	
Character*260 IFILE	name of the power flow sequence data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 2	invalid VERNUM value
	IERR = 3	unable to convert file
	IERR = 4	error opening temporary file
	IERR = 5	invalid revision number
	IERR = 10	error opening IFILE
	IERR = 11	prerequisite requirements for API are not met

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1.270. RETI

Use this API to read the long title from an input file into the working case.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_RETI  IFILE
```

Python command syntax:

```
ierr = reti(ifile)
```

Fortran command syntax:

```
CALL RETIAPI(IFILE, IERR)
```

Where:

Character*260 IFILE	name of long title data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	error opening IFILE
	IERR = 2	prerequisite requirements for API are not met

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1.271. RETRY_PSSUSERPF

Use this API to allow the loading or use of the Powerflow Customization Interface (PCI) implementation module (pssuserpf) after a failure.
For efficiency reasons loading the PCI module is only attempted once. You may reset the flag controlling this with this API.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_RETRY_PSSUSERPF
```

Python command syntax:

```
ierr = retry_pssuserpf()
```

Fortran command syntax:

```
CALL RETRYPSSUSERPF_FC(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 1	prerequisite requirements for API are not met.

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1.272. REV29_NAMES

Use this API to set the extended bus name input format to either the PSSE-29 or PSSE-30 format.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_REV29_NAMES  IVAL
```

Python command syntax:

```
ierr = rev29_names(ival)
```

Fortran command syntax:

```
CALL REV29_NAMES_API(IVAL, IERR)
```

Where:

Integer IVAL	extended bus name option flag (input; 0 by default).	
	IVAL = 0	PSSE-30 format (12 characters plus base voltage)
	IVAL = 1	PSSE-29 or earlier format (8 characters plus base voltage)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	prerequisite requirements for API are not met

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1.273. RNFI

Use this API to reproduce the results of a working case bus renumbering operation in auxiliary data input files.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_RNFI  FTYPE  TRNFILE  INPFILE  OUTFILE
```

Python command syntax:

```
ierr = rnfi(ftype, trnfile, inpfile, outfile)
```

Fortran command syntax:

```
CALL RNFIAPI(FTYPE, TRNFILE, INPFILE, OUTFILE, IERR)
```

Where:

Integer FTYPE	type of data file to be processed (input; 1 by default).	
	FTYPE = 1	sequence data file
	FTYPE = 2	drawing coordinate data file
	FTYPE = 3	dynamics data file
	FTYPE = 4	machine impedance data file
	FTYPE = 5	economic dispatch data file
	FTYPE = 6	inertia and governor response data file
	FTYPE = 7	breaker duty data file
	FTYPE = 8	fault dspecification data file
	FTYPE = 9	fault control data file
	FTYPE = 10	machine capability Curve data file
	FTYPE = 11	load throwover data file
	FTYPE = 12	ANSI fault specification data file
	FTYPE = 13	subsystem description data file
	FTYPE = 14	monitored element data file
	FTYPE = 15	contingency description data file
	FTYPE = 16	tripping element data file
Character*260 TRNFILE	name of bus renumbering translation file (input; no default allowed).	
Character*260 INPFILE	name of data file to be renumbered (input; no default allowed).	
Character*260 OUTFILE	name of data file (blank for output to the report window) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid FTYPE value
	IERR = 2	error opening translation file
	IERR = 3	error opening input data file
	IERR = 4	error opening output data file

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IERR = 5	prerequisite requirements for API are not met

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1.274. RWCM

Use this API to replicate the working case in IEEE common tape format.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RWCM  OPT  OUT  OFILE
```

Python command syntax:

```
ierr = rwcm(opt, out, ofile)
```

Fortran command syntax:

```
CALL RWCMAPI(OPT, OUT, OFILE, IERR)
```

Where:

Integer OPT	dc line/FACTS/GNE device option (input; 0 by default).	
	OPT = 0	ignore dc lines, FACTS devices, and GNE devices
	OPT = 1	add powers to loads
Integer OUT	selection option (input; 1 by default).	
	OUT = 0	direct output to the data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the output file (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPT value
	IERR = 2	invalid OUT value
	IERR = 3	buses in case with 5 or 6 digit bus numbers
	IERR = 4	n buses in case but IEEE common format can handle only 9999
	IERR = 5	error opening OFILE
	IERR = 6	prerequisite requirements for API are not met

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1.275. RWMA

Use this API to replicate machine parametric data from the working case in the form of a machine impedance data file.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RWMA  SID  ALL  STATUS(1)  STATUS(2)  STATUS(3)  OUT  OFILE
```

Python command syntax:

```
ierr = rwma(sid, all, status, out, ofile)
```

Fortran command syntax:

```
CALL RWMANW(SID, ALL, STATUS, OUT, OFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS (3)	array of 3 elements specifying control options(input).	
	STATUS(1)	include records for out-of-service machines (0 by default)
	STATUS(1) = 0	exclude them
	STATUS(1) = 1	include them
	STATUS(2)	used when calculating P fractions (0 by default)
	STATUS(2) = 0	use PGEN
	STATUS(2) = 1	use MBASE
	STATUS(2) = 2	use PMAX
	STATUS(3)	used when calculating Q fractions (0 by default)
	STATUS(3) = 0	use QGEN
	STATUS(3) = 1	use MBASE
	STATUS(3) = 2	use QMAX
Integer OUT	option selection (input; 1 by default).	
	OUT = 0	direct output to a data file specified by OFILE.
	OUT = 1	direct output to the progress window.
Character*260 OFILE	output filename (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error

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IERR = 1	invalid SID value or subsystem SID is not defined
IERR = 2	invalid ALL value
IERR = 3	invalid STATUS value
IERR = 4	invalid OUT value
IERR = 5	error opening OFILE
IERR = 6	prerequisite requirements for API are not met

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1.276. RWMM

Use this API to replicate transaction event data in the form of a transactions data file.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RWMM  OUT  OFILE
```

Python command syntax:

```
ierr = rwmm(out, ofile)
```

Fortran command syntax:

```
CALL RWMMAPI(OUT, OFILE, IERR)
```

Where:

Integer OUT	selection option (input; 1 by default).	
	OUT = 0	direct output to the data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the file to which data records are to be written (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OUT value
	IERR = 2	error opening OFILE
	IERR = 3	prerequisite requirements for API are not met

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1.277. RWSQ

This API routine is obsolete. It has been replaced by the API routine rwsq_2 .

Use this API to replicate the sequence data contained in the working case in the form of a sequence data file.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RWSQ  SID  ALL  STATUS(1)..STATUS(4)  OUT  OFILE
```

Python command syntax:

```
ierr = rwsq(sid, all, status, out, ofile)
```

Fortran command syntax:

```
CALL RWSQNW(SID, ALL, STATUS, OUT, OFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses when STATUS(4) is 0; area ties only when STATUS(4) is 1
Integer STATUS (4)	array of 4 elements specifying options(input).	
	STATUS(1)	include records for Type 4 buses (1 by default)
	STATUS(1) = 0	no
	STATUS(1) = 1	yes
	STATUS(2)	include records for out-of-service branches (1 by default)
	STATUS(2) = 0	no
	STATUS(2) = 1	yes
	STATUS(3)	include records for equipment in the subsystem (1 by default)
	STATUS(3) = 0	no
	STATUS(3) = 1	yes
	STATUS(4)	include records for subsystem ties (0 by default)
	STATUS(4) = 0	no
	STATUS(4) = 1	yes
Integer OUT	option selection (input; 1 by default).	

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	OUT = 0	direct output to a data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the file to which data records are to be written (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid OUT value
	IERR = 5	error opening OFILE
	IERR = 6	prerequisite requirements for API are not met

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1.278. RWSQ_2

Use this API to replicate the sequence data contained in the working case in the form of a sequence data file.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_RWSQ_2  SID  ALL  STATUS(1)..STATUS(5)  OUT  OFILE
```

Python command syntax:

```
ierr = rwsq_2(sid, all, status, out, ofile)
```

Fortran command syntax:

```
CALL RWSQNW_2(SID, ALL, STATUS, OUT, OFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses when STATUS(4) is 0; area ties only when STATUS(4) is 1
Integer STATUS (5)	array of 5 elements specifying options(input).	
	STATUS(1)	include records for Type 4 buses (1 by default)
	STATUS(1) = 0	no
	STATUS(1) = 1	yes
	STATUS(2)	include records for out-of-service branches (1 by default)
	STATUS(2) = 0	no
	STATUS(2) = 1	yes
	STATUS(3)	include records for equipment in the subsystem (1 by default)
	STATUS(3) = 0	no
	STATUS(3) = 1	yes
	STATUS(4)	include records for subsystem ties (0 by default)
	STATUS(4) = 0	no
	STATUS(4) = 1	yes
	STATUS(5)	sequence data file type (0 by default)
	STATUS(5) = 0	for activity RESQ with IC=0
	STATUS(5) = 1	for activity RESQ with IC=1
Integer OUT	option selection (input; 1 by default).	

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	OUT = 0	direct output to a data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the file to which data records are to be written (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	invalid OUT value
	IERR = 5	error opening OFILE
	IERR = 6	prerequisite requirements for API are not met

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1.279. SAVE

Use this API to save the PSSE working case in a saved case file.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SAVE  SFILE
```

Python command syntax:

```
ierr = save(sfile)
```

Fortran command syntax:

```
CALL SAVENW(SFILE, IERR)
```

Where:

Character*260 SFILE	name of the PSSE saved case file If SFILE is "*", SAVENW attempts to use the most recently accessed Saved Case File. If no Saved Case File was accessed in the current execution of PSSE, SAVENW returns an error code (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SFILE is blank
	IERR = 2	error writing to SFILE
	IERR = 3	error opening SFILE
	IERR = 4	prerequisite requirements for API are not met

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1.280. SCAL

This API routine is obsolete. It has been replaced by the API routine scal_2 .

Use this API to uniformly increase or decrease any or all specified bus quantities for a specified group of buses (activity SCAL). It must be referenced using one of the following procedures:
- One reference with APIOPT=1, followed by one reference with APIOPT=2.
- One reference with APIOPT=0. This automatically combines the processing of one APIOPT 1 reference, followed by one APIOPT 2 reference.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SCAL  SID  ALL  APIOPT  STATUS(1)..STATUS(4) SCALVAL(1)..SCALVAL(7)
```

Python command syntax:

```
ierr, totals, moto = scal(sid, all, apiopt, status, scalval)
```

Fortran command syntax:

```
CALL SCALNW(SID, ALL, APIOPT, TOTALS, MOTO, STATUS, SCALVAL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer APIOPT	mode of operation in the API (input; 0 by default).	
	APIOPT = 0	initialize for scaling, then run the scaling and post-processing housekeeping.
	APIOPT = 1	initialize for scaling.
	APIOPT = 2	run the scaling and post-processing housekeeping.
Real TOTALS (11)	array of 11 elements specifying returned when APIOPT = 0 or 1. They are as follows(output).	
	TOTALS(1)	load Mvar total
	TOTALS(2)	load MW total
	TOTALS(3)	generation MW total
	TOTALS(4)	shunt MW total
	TOTALS(5)	reactor Mvar total
	TOTALS(6)	capacitor Mvar total
	TOTALS(7)	motor load MW total
	TOTALS(8)	generator PMAX
	TOTALS(9)	generator PMIN
	TOTALS(10)	motor load PMIN

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	TOTALS(11)	motor load PMAX
Integer MOTO	returned when APIOPT = 0 or 1 (output).	
	MOTO = 0	no motors in the specified subsystem.
	MOTO = 1	motors present in the specified subsystem.
Integer STATUS (4)	array of 4 elements specifying that are used to control scaling. The first three elements are checked and used when APIOPT = 0 or 2; the fourth element is checked and saved when APIOPT = 0 or 1. They are as follows(input).	
	STATUS(1)	method used to scale active power load, generation and motor data, and bus shunt data (0 by default)
	STATUS(1) = 0	no scaling
	STATUS(1) = 1	specify new total powers
	STATUS(1) = 2	specify percent changes
	STATUS(1) = 3	specify incremental powers
	STATUS(2)	flag to enforce machine power limits (0 by default)
	STATUS(2) = 0	ignore machine power limits
	STATUS(2) = 1	enforce machine power limits
	STATUS(3)	flag to specify the scaling rule to be enforced when changing the reactive power load (0 by default)
	STATUS(3) = 0	no change
	STATUS(3) = 1	constant P/Q ratio
	STATUS(3) = 2	new total Q load
	STATUS(3) = 3	percent change
	STATUS(3) = 4	new power factor
	STATUS(3) = 5	incremental Q load
	STATUS(4)	bus type code flag for load scaling; normally defaulted except when used by OPF (0 by default)
	STATUS(4) = 0	all buses in subsystem
	STATUS(4) = 1	only Type 1 buses in subsystem
	STATUS(4) = 2	only Type 2 and 3 buses in subsystem
Real SCALVAL (7)	array of 7 elements specifying used as scaling targets. Based on the value of STATUS(1), entries (1) through (6) are either total powers(input).	
	SCALVAL(1)	load MW total/percent/increment
	SCALVAL(2)	generation MW total/percent/increment
	SCALVAL(3)	shunt MW total/percent/increment
	SCALVAL(4)	reactor Mvar total/percent/increment
	SCALVAL(5)	capacitor Mvar total/percent/increment
	SCALVAL(6)	motor load MW total/percent/increment
	SCALVAL(7)	reactive load scaling parameter

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	If STATUS(3)	
	SCALVAL(7) = 2	new total Mvar load (present total Mvar load by default)."
	SCALVAL(7) = 3	percent change (0.0 by default)."
	SCALVAL(7) = 4	new power factor (1.0 by default)."
	SCALVAL(7) = 5	incremental Mvar load change (0.0 by default)."
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid APIOPT value
	IERR = 4	invalid STATUS value
	IERR = 5	unexpected APIOPT value
	IERR = 6	invalid power factor in SCALVAL(7)
	IERR = 7	generator or motor totals beyond limits
	IERR = 8	prerequisite requirements for API are not met

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1.281. SCAL_2

Use this API routine to uniformly increase or decrease any or all specified bus quantities for a specified group of buses. It must be referenced using one of the following procedures:
- One reference with APIOPT=1, followed by one reference with APIOPT=2.
- One reference with APIOPT=0. This automatically combines the processing of one APIOPT 1 reference, followed by one APIOPT 2 reference.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_SCAL_2  SID ALL APIOPT STATUS(1)..STATUS(5) SCALVAL(1)..SCALVAL(7)
```

Python command syntax:

```
ierr, totals, moto= scal_2(sid, all, apiopt, status, scalval)
```

Fortran command syntax:

```
CALL SCAL_2(SID, ALL, APIOPT, TOTALS, MOTO, STATUS, SCALVAL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID.
	ALL = 1	process all buses.
Integer APIOPT	mode of operation in the API (input; 0 by default).	
	APIOPT = 0	initialize for scaling, then run the scaling and post-processing housekeeping.
	APIOPT = 1	initialize for scaling.
	APIOPT = 2	run the scaling and post-processing housekeeping.
Real TOTALS (11)	array of elements specifying returned when APIOPT = 0 or 1. They are as follows.(output).	
	TOTALS(1)	load Mvar total.
	TOTALS(2)	load MW total.
	TOTALS(3)	generation MW total.
	TOTALS(4)	shunt MW total.
	TOTALS(5)	reactor Mvar total.
	TOTALS(6)	capacitor Mvar total.
	TOTALS(7)	motor load MW total.
	TOTALS(8)	generator PMAX.
	TOTALS(9)	generator PMIN.
	TOTALS(10)	motor load PMIN.
	TOTALS(11)	motor load PMAX.

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Integer MOTO	returned when APIOPT = 0 or 1. It indicates the presence of motors that are modeled as conventional generators with negative active power settings. (output).
	MOTO = 0 — no motors in the specified subsystem.
	MOTO = 1 — motors present in the specified subsystem.
Integer STATUS (5)	array of elements specifying control scaling. The first and fifth elements are checked and saved when APIOPT = 0 or 1; the second, third and fourth elements are checked and used when APIOPT = 0 or 2. They are as follows.(input).
	STATUS(1) — interruptible load scaling option (0 by default)
	STATUS(1) = 0 — scale both interruptible and non-interruptible scalable loads.
	STATUS(1) = 1 — scale only non-interruptible scalable loads.
	STATUS(1) = 2 — scale only interruptible scalable loads.
	STATUS(2) — method used to scale active power load, generation and motor data, and bus shunt data (0 by default)
	STATUS(2) = 0 — no scaling.
	STATUS(2) = 1 — specify new total powers.
	STATUS(2) = 2 — specify percent changes.
	STATUS(2) = 3 — specify incremental powers.
	STATUS(3) — flag to enforce machine power limits (0 by default)
	STATUS(3) = 0 — ignore machine power limits.
	STATUS(3) = 1 — enforce machine power limits.
	STATUS(4) — flag to specify the scaling rule to be enforced when changing the reactive power load (0 by default)
	STATUS(4) = 0 — no change.
	STATUS(4) = 1 — constant P/Q ratio.
	STATUS(4) = 2 — new total Q load.
	STATUS(4) = 3 — percent change.
	STATUS(4) = 4 — new power factor.
	STATUS(4) = 5 — incremental Q load.
	STATUS(5) — is the bus type code flag for load scaling; normally defaulted except when used by OPF (0 by default)
	STATUS(5) = 0 — all buses in subsystem.
	STATUS(5) = 1 — only Type 1 buses in subsystem.
	STATUS(5) = 2 — only Type 2 and 3 buses in subsystem.
Real SCALVAL (7)	array of elements specifying scaling targets. Based on the value of STATUS(1), entries (1) through (6) are either total powers (present total power by default), percent changes (0.0 by default), or incremental changes (0.0 by default). If STATUS(3) = 2, SCALVAL(7) = new total Mvar load (present total Mvar load by default). If STATUS(3) =

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	3, SCALVAL(7) = percent change (0.0 by default). If STATUS(3) = 4, SCALVAL(7) = new power factor (1.0 by default). If STATUS(3) = 5, SCALVAL(7) = incremental Mvar load change (0.0 by default).(input).
	SCALVAL(1) load MW total/percent/increment.
	SCALVAL(2) generation MW total/percent/increment.
	SCALVAL(3) shunt MW total/percent/increment.
	SCALVAL(4) reactor Mvar total/percent/increment.
	SCALVAL(5) capacitor Mvar total/percent/increment.
	SCALVAL(6) motor load MW total/percent/increment.
	SCALVAL(7) reactive load scaling parameter.
Integer IERR	error code (output).
	IERR = 0 no error occurred
	IERR = 1 invalid SID value or subsystem SID is not defined
	IERR = 2 invalid ALL value
	IERR = 3 invalid APIOPT value
	IERR = 4 invalid STATUS value
	IERR = 5 unexpected APIOPT value
	IERR = 6 invalid power factor in SCALVAL(7)
	IERR = 7 generator or motor totals beyond limits
	IERR = 8 prerequisite requirements for API are not met

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1.282. SCEQ

Use this API to construct network equivalents of the positive and zero sequence networks and calculates source impedances at equivalent source nodes for all three sequences in preparation for the unbalanced fault analysis activities of PSSE.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SCEQ  SID  ALL  STATUS  THRESH  RFILE  SFILE
```

Python command syntax:

```
ierr = sceq(sid, all, status, thresh, rfile, sfile)
```

Fortran command syntax:

```
CALL SCEQNW(SID, ALL, STATUS, THRESH, RFILE, SFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS	transformer impedance correction option (input; 1 by default).	
	STATUS = 0	do not apply to zero sequence
	STATUS = 1	apply to zero sequence
Real THRESH	equivalent branch threshold tolerance (input; 10.0).	
Character*260 RFILE	power flow raw data file; blank for none (input; blank by default).	
Character*260 SFILE	sequence data file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	case is not subject to classical assumptions
	IERR = 5	generators are not converted
	IERR = 6	error writing the working case to a temporary file
	IERR = 7	no buses being retained
	IERR = 8	no buses being processed for equivalence
	IERR = 9	too many branches in the equivalent
	IERR = 10	bus type codes other than 1, 2, 4, 5 and 6 present

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IERR = 11	substations present in subsystem to be equivalenced
IERR = 12	error opening RFILE or SFILE
IERR = 13	error opening temporary file
IERR = 14	prerequisite requirements for API are not met

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1.283. SCGR

This API routine is obsolete. It has been replaced by the API routine ascc_2 .

Use this API routine to calculate three phase and line-to-ground fault solutions. Results may be programmatically accessed via the internally accessible fetch routines SCGTXT, SCGBUS, SCGBRN, and SCG3WN. Usage is:
- invoke SCGR API with APIOPT = 0;
- multiple calls to fetch routines;
- optionally, repeat of above steps for different fault and/or different pre-fault conditions;
- final invocation of SCGR API with APIOPT = 1.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SCGR  SID  ALL  APIOPT  STATUS(1)..STATUS(8)
```

Python command syntax:

```
ierr = scgr(sid, all, apiopt, status)
```

Fortran command syntax:

```
CALL SCGRAPI(SID, ALL, APIOPT, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer APIOPT	mode of operation of SCGRAPI (input; 0 by default).	
	APIOPT = 0	apply fault calculations and fill arrays used by fetch routines
	APIOPT = 1	deallocate arrays used by fetch routines
Integer STATUS (8)	array of 8 elements specifying fault and options(input).	
	STATUS(1)	fault type (0 by default)
	STATUS(1) = 0	three-phase
	STATUS(1) = 1	line-to-ground
	STATUS(2)	L-G fault (ignored if STATUS(1) is 0) (0 by default)
	STATUS(2) = 0	A phase
	STATUS(2) = 1	3 * I0
	STATUS(3)	fault location (0 by default)
	STATUS(3) = 0	all buses in subsystem

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	STATUS(3) = 0	bus number of the subsystem bus to be faulted
	STATUS(4)	flat conditions option (0 by default)
	STATUS(4) = 0	working case conditions
	STATUS(4) = 1	flat conditions
	STATUS(5)	double-precision fault calculation option (ignored, but must be specified as 0 or 1; 0 by default; double precision always used)
	STATUS(6)	dc line and FACTS device option (ignored if STATUS(4) is 1) (0 by default)
	STATUS(6) = 0	block
	STATUS(6) = 1	represent as load
	STATUS(7)	transformer impedance correction option (ignored if STATUS(1) is 0 or STATUS(4) is 1 (1 by default)
	STATUS(7) = 0	do not apply to zero sequence
	STATUS(7) = 1	apply to zero sequence
	STATUS(8)	branch results option (0 by default)
	STATUS(8) = 0	currents
	STATUS(8) = 1	apparent impedances
Integer IERR	error code (output).	
	IERR = 1	SCGR is obsolete; use "Fault>Automatic sequencing fault calculation (ASCC).." (API routine ASCC_2)
	IERR = 8	prerequisite requirements for API are not met

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1.284. SCMU

Use this API to calculate simultaneous unbalances (activity SCMU). It must be called once with APIOPT set to 1, once for each of the simultaneous faults to be applied with APIOPT set to 2, and once with APIOPT set to 3.

This API routine was first introduced in release 29.0.0

Batch command syntax:

```
BAT_SCMU  APIOPT  OPTIONS(1)..OPTIONS(7)  RVAL(1)..RVAL(5)  CKTARG
```

Python command syntax:

```
ierr = scmu(apiopt, options, rval, cktarg)
```

Fortran command syntax:

```
CALL SCMUAPI(APIOPT, OPTIONS, RVAL, CKTARG, IERR)
```

Where:

Integer APIOPT	mode of operation (input).	
	APIOPT = 1	initialize; automatically runs activity SEQD if the sequence network setup option setting is enabled.
	APIOPT = 2	specify an unbalance to be applied; must be called once for each simultaneous unbalance to be imposed.
	APIOPT = 3	solve in the presence of the specified unbalances and terminate.
	APIOPT = 4	remove specified unbalances (cancels any unbalances specified by any preceding APIOPT 2 references, and leaves memory as it is immediately following an APIOPT 1 reference).
Integer OPTIONS (7)	Is an array of seven elements specifying options and input values to be used in the current calculation phase; ignored when APIOPT is 3 or 4 (input) When APIOPT is 1, the first three elements of OPTIONS define options to be used in the construction of the sequence admittance matrices. These values are used only when the sequence network setup option is enabled, and activity SEQD is to be automatically executed.	
	OPTIONS(1)	dc line and FACTS device option (0 by default).
	OPTIONS(1) = 0	block and ignore.
	OPTIONS(1) = 1	convert to constant admittance load
	OPTIONS(2)	transformer impedance correction option (1 by default)
	OPTIONS(2) = 0	do not apply to zero sequence

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	OPTIONS(2) = 1	apply to zero sequence
	OPTIONS(3)	fault analysis generator reactance option; ignored if generators are already "converted" (0 by default).
	OPTIONS(3) = 0	use subtransient reactance
	OPTIONS(3) = 1	use transient reactance
	OPTIONS(3) = 2	use synchronous reactance.
	When APIOPT is 2, the first element of OPTIONS defines the type of unbalance to be applied.	
	OPTIONS(1)	unbalance type (no default allowed).
	OPTIONS(1) = 1	first line-to-ground fault.
	OPTIONS(1) = 2	second line-to-ground fault.
	OPTIONS(1) = 3	first line-to-line or line-to-line-to-ground fault
	OPTIONS(1) = 4	second line-to-line or line-to-line-to-ground fault.
	OPTIONS(1) = 5	one phase closed
	OPTIONS(1) = 6	two phases closed.
	OPTIONS(1) = 7	three-phase fault (uses fault types 2 and 4)
	OPTIONS(1) = 8	branch unbalance with one end opened
	OPTIONS(1) = 9	branch unbalance with both ends closed
Real RVAL (5)	Is an array of five elements specifying input values used to define the current fault; ignored except when APIOPT is 2. The elements of RVAL are dependent on the value specified for OPTIONS(1) and are described at the end of this section (input).	
Character*2 CKTARG	circuit identifier of branch to be faulted; ignored except when APIOPT is 2 and OPTIONS(1) is 8 or 9 (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	unexpected APIOPT value
	IERR = 2	invalid APIOPT value
	IERR = 3	invalid unbalance type
	IERR = 4	bus not found
	IERR = 5	bus is out-of-service
	IERR = 6	invalid phase value
	IERR = 7	branch not found
	IERR = 8	three-winding transformer not found
	IERR = 9	invalid open end bus
	IERR = 10	invalid distance factor
	IERR = 11	terminated due to invalid fault specification or application
	IERR = 12	SEQD results are not compatible with the working case
	IERR = 13	fault analysis is invalid due to the presence of in-service induction machines that are in the "stalled" or "tripped" state

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	IERR = 14	bus type code and series element status inconsistencies
	IERR = 15	invalid OPTIONS value (from SEQD)
	IERR = 16	prerequisite requirements for API are not met

When APIOPT is 2 and OPTIONS(1) is 1 or 2, the following additional elements of OPTIONS and RVAL are used

Integer OPTIONS(7)	OPTIONS(2)	bus to be faulted (no default allowed).
	OPTIONS(3)	node number of a substation node connected to the specified bus; this defines the bus section to be faulted. If 0 is specified fora bus in a substation, the "parent' bus is faulted; 0 must be specified for a bus that is not in a substation (0 by default).
	OPTIONS(4)	faulted phase (1 by default)
	OPTIONS(4) = 1	phase A
	OPTIONS(4) = 2	phase B
	OPTIONS(4) = 3	phase C
Real RVAL(5)	RVAL(1)	resistance of the complex L-G fault impedance in pu (0.0 by default)
	RVAL(2)	reactance of the complex L-G fault impedance in pu (0.0 by default).

When APIOPT is 2 and OPTIONS(1) is 3 or 4, the following additional elements of OPTIONS and RVAL are used

Integer OPTIONS(7)	OPTIONS(2)	bus to be faulted (no default allowed).
	OPTIONS(3)	node number of a substation node connected to the specified bus; this defines the bus section to be faulted. If 0 is specified fora bus in a substation, the "parent' bus is faulted; 0 must be specified for a bus that is not in a substation (0 by default)
	OPTIONS(4)	excluded phase (1 by default)
	OPTIONS(4) = 1	phase A
	OPTIONS(4) = 2	phase B
	OPTIONS(4) = 3	phase C
Real RVAL(5)	RVAL(1)	resistance of the complex L-L fault impedance in pu (0.0 by default)
	RVAL(2)	reactance of the complex L-L fault impedance in pu (0.0 by default)
	RVAL(3)	resistance of the complex L-G fault impedance in pu (0.0 by default)
	RVAL(4)	reactance of the complex L-G fault impedance in pu (0.0 by default)

When APIOPT is 2 and OPTIONS(1) is 5, the following additional elements of OPTIONS and RVAL are used.

Integer OPTIONS(7)

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	OPTIONS(2)	one of the buses to be connected by the phase closed unbalance (no default allowed)
	OPTIONS(3)	node number of a substation node connected to the specified bus; this defines the bus section of the first bus. If 0 is specified for a bus in a substation, the "parent‟" bus is selected; 0 must be specified for a bus that is not in a substation (0 by default)
	OPTIONS(4)	the other bus to be connected by the phase closed unbalance (no default allowed)
	OPTIONS(5)	node number of a substation node connected to the specified bus; this defines the bus section of the second bus. If 0 is specified for a bus in a substation, the "parent" bus is selected; 0 must be specified for a bus that is not in a substation (0 by default)
	OPTIONS(6)	phase to be closed (1 by default)
	OPTIONS(4) = 1	phase A
	OPTIONS(4) = 2	phase B
	OPTIONS(4) = 3	phase C
Real RVAL(5)	RVAL(1)	resistance of the complex phase impedance in pu (0.0 by default).
	RVAL(2)	reactance of the complex phase impedance in pu (0.0 by default)

When APIOPT is 2 and OPTIONS(1) is 6, the following additional elements of OPTIONS and RVAL are used.

Integer OPTIONS(7)	OPTIONS(2)	one of the buses to be connected by the phase closed unbalance (no default allowed)
	OPTIONS(3)	node number of a substation node connected to the specified bus; this defines the bus section of the first bus. If 0 is specified for a bus in a substation, the "parent‟" bus is selected; 0 must be specified for a bus that is not in a substation (0 by default)
	OPTIONS(4)	the other bus to be connected by the phase closed unbalance (no default allowed)
	OPTIONS(5)	node number of a substation node connected to the specified bus; this defines the bus section of the second bus. If 0 is specified for a bus in a substation, the "parent" bus is selected; 0 must be specified for a bus that is not in a substation (0 by default)
	OPTIONS(6)	excluded phase (1 by default)
	OPTIONS(4) = 1	phase A
	OPTIONS(4) = 2	phase B
	OPTIONS(4) = 3	phase C
Real RVAL(5)		

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	RVAL(1)	resistance of the complex phase impedance in pu (0.0 by default).
	RVAL(2)	reactance of the complex phase impedance in pu (0.0 by default)

When APIOPT is 2 and OPTIONS(1) is 7, the following additional elements of OPTIONS and RVAL are used.

Integer OPTIONS(7)	OPTIONS(2)	bus to be faulted (no default allowed)
	OPTIONS(3)	node number of a substation node connected to the specified bus; this defines the bus section to be faulted. If 0 is specified for a bus in a substation, the "parent" bus is faulted; 0 must be specified fora bus that is not in a substation (0 by default)
Real RVAL(5)	None	

When APIOPT is 2 and OPTIONS(1) is 8, the following additional elements of OPTIONS and RVAL are used

Integer OPTIONS(7)	OPTIONS(2)	one of the buses connected by the branch to be faulted (no default allowed).
	OPTIONS(3)	another bus connected by the branch to be faulted (no default allowed)
	OPTIONS(4)	third bus connected by the three-winding transformer to be faulted, or zero for a twoterminal branch (0 by default)
	OPTIONS(5)	type of in-line fault to be applied (1 by default)
	OPTIONS(5) = 1	line-to-ground
	OPTIONS(5) = 2	line-to-line or line-to-line-to-ground fault
	OPTIONS(5) = 3	three-phase
	OPTIONS(6)	faulted (if OPTIONS(5) is 1) or excluded (if OPTIONS(5) is 2) phase (1 by default)
	OPTIONS(6) = 1	phase A
	OPTIONS(6) = 2	phase B
	OPTIONS(6) = 3	phase C
	OPTIONS(7)	opened end bus (must be identical to one of OPTIONS(2), OPTIONS(3), or, if it is nonzero, OPTIONS(4)) (OPTIONS(2) by default).
Real RVAL(5)	RVAL(1)	resistance of the complex L-G (if OPTIONS(5) is 1) or L-L (if OPTIONS(5) is 2) fault impedance in pu (0.0 by default).
	RVAL(2)	reactance of the complex L-G (if OPTIONS(5) is 1) or L-L (if OPTIONS(5) is 2) fault impedance in pu (0.0 by default).
	RVAL(3)	resistance of the complex L-G fault impedance in pu (if OPTIONS(5) is 2) (0.0 by default)

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	RVAL(4)	reactance of the complex L-G fault impedance in pu (if OPTIONS(5) is 2) (0.0 by default)
	RVAL(5)	fault location from closed end bus in pu of total line length; must be 1.0 if the branch is a transformer (1.0 by default)

When APIOPT is 2 and OPTIONS(1) is 9, the following additional elements of OPTIONS and RVAL are used.

Integer OPTIONS(7)	OPTIONS(2)	one of the buses connected by the branch to be faulted (no default allowed).
	OPTIONS(3)	the other bus connected by the branch to be faulted (no default allowed)
	OPTIONS(4)	type of in-line fault to be applied (1 by default)
	OPTIONS(5)	faulted (if OPTIONS(4) is 1) or excluded (if OPTIONS(4) is 2) phase (1 by default)
	OPTIONS(5) = 1	phase A
	OPTIONS(5) = 2	phase B
	OPTIONS(5) = 3	phase C
Real RVAL(5)	RVAL(1)	resistance of the complex L-G (if OPTIONS(4) is 1) or L-L (if OPTIONS(4) is 2) fault impedance in pu (0.0 by default)
	RVAL(2)	reactance of the complex L-G (if OPTIONS(4) is 1) or L-L (if OPTIONS(4) is 2) fault impedance in pu (0.0 by default)
	RVAL(3)	resistance of the complex L-G fault impedance in pu (if OPTIONS(4) is 2) (0.0 by default)
	RVAL(4)	reactance of the complex L-G fault impedance in pu (if OPTIONS(4) is 2) (0.0 by default)
	RVAL(5)	fault location from bus OPTIONS(2) in pu of total line length (0.5 by default)

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1.285. SCOP

Use this API to tabulate unbalanced fault solution output. It must be preceded by the execution of activity SCMU, either interactively, from a response file, or via Python. It must then be referenced using one of the following procedures:
- One reference with APIOPT=1, followed by any number of references with APIOPT=2, followed by one reference with APIOPT=3.
- One reference with APIOPT=0. This automatically combines the processing of one APIOPT 1 reference, followed by one APIOPT 2 reference, followed by one APIOPT 3 reference.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SCOP  SID  ALL  UNITS  APIOPT
```

Python command syntax:

```
ierr = scop(sid, all, units, apiopt)
```

Fortran command syntax:

```
CALL SCOPAPI(SID, ALL, UNITS, APIOPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier; used only when APIOPT is 0 or 2 and ALL is 0. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag; used only when APIOPT is 0 or 2 (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer UNITS	branch quantity flag; used only when APIOPT is 0 or 2 (input; 1 by default).	
	UNITS = 1	currents
	UNITS = 2	apparent impedances
	UNITS = 3	apparent admittances
Integer APIOPT	mode of operation (input; 0 by default).	
	APIOPT = 0	initialize for short circuit output, produce the output, and run post-processing clean up
	APIOPT = 1	initialize for short circuit output
	APIOPT = 2	produce output
	APIOPT = 3	post-processing clean up
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value

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IERR = 3	invalid UNITS value
IERR = 4	invalid APIOPT value
IERR = 5	unexpected APIOPT value
IERR = 6	SCMU results error; one of the following occurred: - No SCMU results file is open. - SCMU results are not compatible with the working case. - Error reading the SCMU results file
IERR = 7	prerequisite requirements for API are not met

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1.286. SENSITIVITY_FLOW

This API is used to calculate sensitivity factors of a branch flow to MW power at buses, MW power at generator buses, MW at load buses, phase angle of phase shifters, tap postion of tap changing transformers as well as addimittance of switched shunts:

This API routine was first introduced in release 32.1.0.

Batch command syntax:

```
BAT_SENSITIVITY_FLOW  FLWOPT(1) .. FLWOPT(5) OPTIONS(1) .. OPTIONS(9)
 VALUES(1) VALUES(2) BRNCKT LABELS(1) LABELS(2) DFXFILE
```

Python command syntax:

```
ierr = sensitivity_flow(flwopt,options,values,brnckt,labels,dfxfile)
```

Fortran command syntax:

```
CALL SENSITIVITY_FLOW(FLWOPT,OPTIONS,VALUES,BRNCKT,LABELS,DFXFILE,IERR)
```

Where:

Integer FLWOPT (5)	array of 5 elements specifying (input).	
	FLWOPT(1)	from bus of the branch for which sensitivity factors are calculated (no default is allowed)
	FLWOPT(2)	to bus of the branch (no default is allowed)
	FLWOPT(3)	third bus connected by a three-winding transformer, or zero for a non-transformer branch or a two-winding transformer (0 by default)
	FLWOPT(4)	network model on which sensitivity anlayiss is based
	FLWOPT(4) = 0	DC network model
	FLWOPT(4) = 1	AC network model
	FLWOPT(5)	branch flow type to which the sensitivity factors are calculated (If DC network is used, only the branch MW flow is valid for sensitivity analysis. 0 by default)
	FLWOPT(5) = 0	MW flow
	FLWOPT(5) = 1	MVar flow
	FLWOPT(5) = 2	MVA flow
	FLWOPT(5) = 3	Amps flow
Integer OPTIONS (9)	array of 9 elements specifying options for sensitvity analysis to a branch flow(input).	
	OPTIONS(1)	flag to calculate sensitivity factors at buses
	OPTIONS(1) = 0	Disable, do not calculate sensitivity factors at buses
	OPTIONS(1) = 1	Enable, calculate sensitivity factors at buses
	OPTIONS(2)	flag to calculate sensitivity factors at generator buses

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OPTIONS(2) = 0	Disable
OPTIONS(2) = 1	Enable
OPTIONS(3)	flag to calculate sensitivity factors at load buses
OPTIONS(3) = 0	Disable
OPTIONS(3) = 1	Enable
OPTIONS(4)	flag to calculate sensitivity factors of the branch flow to phase shifters
OPTIONS(4) = 0	Disable
OPTIONS(4) = 1	Enable
OPTIONS(5)	flag to calculate sensitivity factors of the branch flow to tap changing transformers (ignore if DC network model is used)
OPTIONS(5) = 0	Disable
OPTIONS(5) = 1	Enable
OPTIONS(6)	flag to calculate sensitivity factors of the branch flow to switched shunts (ignore if DC network model is used)
OPTIONS(6) = 0	Disable
OPTIONS(6) = 1	Enable
OPTIONS(7)	flag to specifiy the transfer from the components to which sensitivity factors of the branch flow are calculated to the opposing subsystem (the flag is ignored if none of the bus, generator, or load sensitivity calculation flags is selected)
OPTIONS(7) = 0	Import. The positive transfer is from the components to the opposing subsystem
OPTIONS(7) = 1	Export. The positive transfer is from the opposing subsystem to the components
OPTIONS(8)	opposing subsystem type
OPTIONS(8) = 0	Slack bus
OPTIONS(8) = 1	Single bus
OPTIONS(8) = 2	Subsystem
OPTIONS(9)	ignored if opposing subsystem type is slack bus. It is a bus number when opposing subsystem type is a single bus, or one of the following seven dispatch modes if the opposing subsystem type is a subsystem (1 by default)
OPTIONS(9) = 1	Buses & participating factors from DFAX file for buses with positive MW machines
OPTIONS(9) = 2	Buses & participating factors from DFAX file for buses with positive MW constant MVA load

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	OPTIONS(9) = 3	Buses & participating factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(9) = 4	Subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(9) = 5	Subsystem buses with positive MW machines in proportion to their MW output
	OPTIONS(9) = 6	Subsystem buses with positive MW machines in proportion to their MBASE
	OPTIONS(9) = 7	Subsystem buses with positive MW machines in proportion to their reserves (Pmaxmc-Pgenmc)
Real VALUES (2)	array of 2 elements specifying sensitivity analysis parameters(input).	
	VALUES(1)	mismatch tolerance for continuing sensitivity analysis in AC mode, ignored in DC mode
	VALUES(2)	sensitivity cutoff. Sensitivity factors less than the cutoff are not presented
Character*2 BRNCKT	branch circuit ID (input).	
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of the subsystem containing components to which the sensitivity factors of the branch flow are calcualted
	LABELS(2)	opposing subsystem label when the opposing subsystem type is a subsystem
Character*260 DFXFILE	DFAX file name (input).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid OPTIONS value
	IERR = 2	nvalid VALUES value
	IERR = 3	The branch can not be found
	IERR = 4	Generators are converted
	IERR = 5	Buses in island(s) without a swing bus--use activity TREE
	IERR = 6	Dispatch bus does not exist
	IERR = 7	No Distribution Factor Data input file specified
	IERR = 8	Subsystem containing components to which sensitivity factors are calculated is not specified
	IERR = 9	Opposing subsystem is not specified
	IERR = 10	Largest mismatch exceeds mismatch tolerance
	IERR = 11	The branch bus does not exist
	IERR = 12	The branch is out-of-service
	IERR = 13	The branch is a zero impedance line

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IERR = 14	The branch is a multi-section line but the Report Multi-Section line option is disabled
IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file, run DFAX
IERR = 22	Error linearizing Jacobian matrix in AC network model
IERR = 51	error opening Distribution Factor Data File
IERR = 52	prerequisite requirements for function not met

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1.287. SENSITIVITY_FLOWS

This API is used to calculate sensitivity factors of flows on the branches in a subsystem to MW power at buses, MW power at generator buses, MW power at load buses, phase angle of phase shifters, tap postion of tap changing transformers as well as addimittance of switched shunts.

This API routine was first introduced in release 32.1.0.

Batch command syntax:

```
BAT_SENSITIVITY_FLOWS  FLWOPT(1) FLWOPT(2) OPTIONS(1) .. OPTIONS(9)
 VALUES(1) VALUES(2) LABELS(1) .. LABELS(3) DFXFILE
```

Python command syntax:

```
ierr = sensitivity_flows(flwopt,options,values,labels,dfxfile)
```

Fortran command syntax:

```
CALL SENSITIVITY_FLOWS(FLWOPT,OPTIONS,VALUES,LABELS,DFXFILE,IERR)
```

Where:

Integer FLWOPT (2)	array of 2 elements specifying options for sensitivity analysis for branches in a subsystem(input).	
	FLWOPT(1)	network model on which sensitivity analysis is based
	FLWOPT(1) = 0	DC network model
	FLWOPT(1) = 1	AC network model
	FLWOPT(2)	branch flow type to which the sensitivity factors are calculated (when DC network is used, only the branch MW flow is valid for sensitivity analysis.)
	FLWOPT(2) = 0	MW flow
	FLWOPT(2) = 1	MVar flow
	FLWOPT(2) = 2	MVA flow
	FLWOPT(2) = 3	Amps flow
Integer OPTIONS (9)	array of 9 elements specifying options for sensitvity analysis to a branch flow(input).	
	OPTIONS(1)	flag to calculate sensitivity factors at buses
	OPTIONS(1) = 0	Disable, do not calculate sensitivity factors at buses
	OPTIONS(1) = 1	Enable, calculate sensitivity factors at buses
	OPTIONS(2)	flag to calculate sensitivity factors at generator buses
	OPTIONS(2) = 0	Disable
	OPTIONS(2) = 1	Enable
	OPTIONS(3)	flag to calculate sensitivity factors at load buses

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OPTIONS(3) = 0	Disable
OPTIONS(3) = 1	Enable
OPTIONS(4)	flag to calculate sensitivity factors of the branch flow to phase shifters
OPTIONS(4) = 0	Disable
OPTIONS(4) = 1	Enable
OPTIONS(5)	flag to calculate sensitivity factors of the branch flow to tap changing transformers (ignore if DC network model is used)
OPTIONS(5) = 0	Disable
OPTIONS(5) = 1	Enable
OPTIONS(6)	flag to calculate sensitivity factors of the branch flow to switched shunts (ignore if DC network model is used)
OPTIONS(6) = 0	Disable
OPTIONS(6) = 1	Enable
OPTIONS(7)	flag to specifiy the transfer from the components to which sensitivity factors of the branch flow are calculated to the opposing subsystem (the flag is ignored if none of the bus, generator, or load sensitivity calculation flags is selected)
OPTIONS(7) = 0	Import. The positive transfer is from the components to the opposing subsystem
OPTIONS(7) = 1	Export. The positive transfer is from the opposing subsystem to the components
OPTIONS(8)	opposing subsystem type
OPTIONS(8) = 0	Slack bus
OPTIONS(8) = 1	Single bus
OPTIONS(8) = 2	Subsystem
OPTIONS(9)	ignored if opposing subsystem type is slack bus. It is a bus number when opposing subsystem type is a single bus, or one of the following seven dispatch modes if the opposing subsystem type is a subsystem (1 by default)
OPTIONS(9) = 1	Buses & participating factors from DFAX file for buses with positive MW machines
OPTIONS(9) = 2	Buses & participating factors from DFAX file for buses with positive MW constant MVA load
OPTIONS(9) = 3	Buses & participating factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
OPTIONS(9) = 4	Subsystem buses with positive MW constant MVA load in proportion to their MW load
OPTIONS(9) = 5	Subsystem buses with positive MW machines in proportion to their MW output

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	OPTIONS(9) = 6	Subsystem buses with positive MW machines in proportion to their MBASE
	OPTIONS(9) = 7	Subsystem buses with positive MW machines in proportion to their reserves (Pmaxmc-Pgenmc)
Real VALUES (2)	array of 2 elements specifying sensitivity analysis parameters(input).	
	VALUES(1)	mismatch tolerance for continuing sensitivity analysis in AC mode, ignored in DC mode
	VALUES(2)	sensitivity cutoff. Sensitivity factors less than the cutoff are not presented
Character*32 LABELS (3)	array of 3 elements specifying subsystem labels(input).	
	LABELS(1)	label of the subsystem containing components to which the sensitivity factors of the branch flow are calcualted
	LABELS(2)	label of opposing subsystem when the opposing subsystem type is a subsystem
	LABELS(3)	label of the subsystem including branches for which sensitivity factors are calculated
Character*260 DFXFILE	DFAX file name (input).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid OPTIONS value
	IERR = 2	Invalid VALUES value
	IERR = 3	The branch can not be found
	IERR = 4	Generators are converted
	IERR = 5	Buses in island(s) without a swing bus--use activity TREE
	IERR = 6	Dispatch bus does not exist
	IERR = 7	No Distribution Factor Data input file specified
	IERR = 8	Subsystem containing components to which sensitivity factors are calculated is not specified
	IERR = 9	Opposing subsystem is not specified
	IERR = 10	Largest mismatch exceeds mismatch tolerance
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file, run DFAX
	IERR = 22	Error linearizing Jacobian matrix in AC network model
	IERR = 51	error opening Distribution Factor Data File
	IERR = 52	prerequisite requirements for function not met

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1.288. SENSITIVITY_INTERFACE

This API is used to calculate sensitivity factors of an interface flow to MW power at buses, MW power at generator buses, MW power at load buses, phase angle of phase shifters, tap position of tap changing transformers as well as admittance of switched shunts.

This API routine was first introduced in release 32.1.0.

Batch command syntax:

```
BAT_SENSITIVITY_INTERFACE  MODE OPTIONS(1) .. OPTIONS(9) VALUES(1)
 VALUES(2) LABELS(1) .. LABELS(3) DFXFILE
```

Python command syntax:

```
ierr = sensitivity_interface(mode,options,values,labels,dfxfile)
```

Fortran command syntax:

```
CALL SENSITIVITY_INTERFACE(MODE,OPTIONS,VALUES,LABELS,DFXFILE,IERR)
```

Where:

Integer MODE	network model on which sensitivity analysis is based (input; 0 by default).	
	MODE = 0	DC network model
	MODE = 1	AC network model
Integer OPTIONS (9)	array of 9 elements specifying options for sensitvity analysis to a branch flow(input).	
	OPTIONS(1)	flag to calculate sensitivity factors at buses
	OPTIONS(1) = 0	Disable, do not calculate sensitivity factors at buses
	OPTIONS(1) = 1	Enable, calculate sensitivity factors at buses
	OPTIONS(2)	flag to calculate sensitivity factors at generator buses
	OPTIONS(2) = 0	Disable
	OPTIONS(2) = 1	Enable
	OPTIONS(3)	flag to calculate sensitivity factors at load buses
	OPTIONS(3) = 0	Disable
	OPTIONS(3) = 1	Enable
	OPTIONS(4)	flag to calculate sensitivity factors of the branch flow to phase shifters
	OPTIONS(4) = 0	Disable
	OPTIONS(4) = 1	Enable
	OPTIONS(5)	flag to calculate sensitivity factors of the branch flow to tap changing transformers (ignore if DC network model is used)
	OPTIONS(5) = 0	Disable
	OPTIONS(5) = 1	Enable

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	OPTIONS(6)	flag to calculate sensitivity factors of the branch flow to switched shunts (ignore if DC network model is used)
	OPTIONS(6) = 0	Disable
	OPTIONS(6) = 1	Enable
	OPTIONS(7)	flag to specifiy the transfer from the components to which sensitivity factors of the branch flow are calculated to the opposing subsystem (the flag is ignored if none of the bus, generator, or load sensitivity calculation flags is selected)
	OPTIONS(7) = 0	Import. The positive transfer is from the components to the opposing subsystem
	OPTIONS(7) = 1	Export. The positive transfer is from the opposing subsystem to the components
	OPTIONS(8)	opposing subsystem type
	OPTIONS(8) = 0	Slack bus
	OPTIONS(8) = 1	Single bus
	OPTIONS(8) = 2	Subsystem
	OPTIONS(9)	ignored if opposing subsystem type is slack bus. It is a bus number when opposing subsystem type is a single bus, or one of the following seven dispatch modes if the opposing subsystem type is a subsystem (1 by default)
	OPTIONS(9) = 1	Buses & participating factors from DFAX file for buses with positive MW machines
	OPTIONS(9) = 2	Buses & participating factors from DFAX file for buses with positive MW constant MVA load
	OPTIONS(9) = 3	Buses & participating factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(9) = 4	Subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(9) = 5	Subsystem buses with positive MW machines in proportion to their MW output
	OPTIONS(9) = 6	Subsystem buses with positive MW machines in proportion to their MBASE
	OPTIONS(9) = 7	Subsystem buses with positive MW machines in proportion to their reserves (Pmaxmc-Pgenmc)
Real VALUES (2)	array of 2 elements specifying sensitivity analysis parameters(input).	
	VALUES(1)	mismatch tolerance for continuing sensitivity analysis in AC mode, ignored in DC mode
	VALUES(2)	sensitivity cutoff. Sensitivity factors less than the cutoff are not presented

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Character*32 LABELS (3)	array of 3 elements specifying subsystem/interface labels(input).	
	LABELS(1)	label of the subsystem containing components to which the sensitivity factors of the interface flow are calculated
	LABELS(2)	label of opposing subsystem when the opposing subsystem type is a subsystem
	LABELS(3)	label of the interface for which sensitivity factors are calculated
Character*260 DFXFILE	DFAX file name (input).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid OPTIONS value
	IERR = 2	Invalid VALUES value
	IERR = 3	Interface is not specified
	IERR = 4	Generators are converted
	IERR = 5	Buses in island(s) without a swing bus--use activity TREE
	IERR = 6	Dispatch bus does not exist
	IERR = 7	No Distribution Factor Data input file specified
	IERR = 8	Subsystem containing components to which sensitivity factors are calculated is not specified
	IERR = 9	Opposing subsystem is not specified
	IERR = 10	Largest mismatch exceeds mismatch tolerance
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file, run DFAX
	IERR = 22	Error in linearizing Jacobian matrix with AC network model
	IERR = 51	error opening Distribution Factor Data File
	IERR = 52	prerequisite requirements for function not met

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1.289. SENSITIVITY_VOLTAGE

This API is used to calculate sensitivity factors of a bus voltage to MW and MVar power at buses, MW power at generator buses, MW and MVar at load buses, phase angle of phase shifters, tap postion of tap changing transformers as well as addimittance of switched shunts.

This API routine was first introduced in release 32.1.0.

Batch command syntax:

```
BAT_SENSITIVITY_VOLTAGE  IBUS OPTIONS(1) .. OPTIONS(9) VALUES(1) VALUES(2)
 LABELS(1) LABELS(2) DFXFILE
```

Python command syntax:

```
ierr = sensitivity_voltage(ibus,options,values,labels,dfxfile)
```

Fortran command syntax:

```
CALL SENSITIVITY_VOLTAGE(IBUS,OPTIONS,VALUES,LABELS,DFXFILE,IERR)
```

Where:

Integer IBUS	bus number for which sensitivity factors are calculated (input).
Integer OPTIONS (9)	array of 9 elements specifying options for sensitvity analysis to a branch flow(input).

OPTIONS(1)	flag to calculate sensitivity factors at buses
OPTIONS(1) = 0	Disable, do not calculate sensitivity factors at buses
OPTIONS(1) = 1	Enable, calculate sensitivity factors at buses
OPTIONS(2)	flag to calculate sensitivity factors at generator buses
OPTIONS(2) = 0	Disable
OPTIONS(2) = 1	Enable
OPTIONS(3)	flag to calculate sensitivity factors at load buses
OPTIONS(3) = 0	Disable
OPTIONS(3) = 1	Enable
OPTIONS(4)	flag to calculate sensitivity factors of the branch flow to phase shifters
OPTIONS(4) = 0	Disable
OPTIONS(4) = 1	Enable
OPTIONS(5)	flag to calculate sensitivity factors of the branch flow to tap changing transformers (ignore if DC network model is used)
OPTIONS(5) = 0	Disable
OPTIONS(5) = 1	Enable
OPTIONS(6)	flag to calculate sensitivity factors of the branch flow to switched shunts (ignore if DC network model is used)
OPTIONS(6) = 0	Disable

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	OPTIONS(6) = 1	Enable
	OPTIONS(7)	flag to specifiy the transfer from the components to which sensitivity factors of the branch flow are calculated to the opposing subsystem (the flag is ignored if none of the bus, generator, or load sensitivity calculation flags is selected)
	OPTIONS(7) = 0	Import. The positive transfer is from the components to the opposing subsystem
	OPTIONS(7) = 1	Export. The positive transfer is from the opposing subsystem to the components
	OPTIONS(8)	opposing subsystem type
	OPTIONS(8) = 0	Slack bus
	OPTIONS(8) = 1	Single bus
	OPTIONS(8) = 2	Subsystem
	OPTIONS(9)	ignored if opposing subsystem type is slack bus. It is a bus number when opposing subsystem type is a single bus, or one of the following seven dispatch modes if the opposing subsystem type is a subsystem (1 by default)
	OPTIONS(9) = 1	Buses & participating factors from DFAX file for buses with positive MW machines
	OPTIONS(9) = 2	Buses & participating factors from DFAX file for buses with positive MW constant MVA load
	OPTIONS(9) = 3	Buses & participating factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(9) = 4	Subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(9) = 5	Subsystem buses with positive MW machines in proportion to their MW output
	OPTIONS(9) = 6	Subsystem buses with positive MW machines in proportion to their MBASE
	OPTIONS(9) = 7	Subsystem buses with positive MW machines in proportion to their reserves (Pmaxmc-Pgenmc)
Real VALUES (2)	array of 2 elements specifying sensitivity analysis parameters(input).	
	VALUES(1)	mismatch tolerance for continuing sensitivity analysis in AC mode, ignored in DC mode
	VALUES(2)	sensitivity cutoff. Sensitivity factors less than the cutoff are not presented
Character*32 LABELS (2)	array of 2 elements specifying subsystem labels(input).	
	LABELS(1)	label of the subsystem containing components to which the sensitivity factors of the bus voltage are calcualted

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	LABELS(2)	opposing subsystem label when the opposing subsystem type is a subsystem
Character*260 DFXFILE	DFAX file name (input).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid OPTIONS value
	IERR = 2	Invalid VALUES value
	IERR = 3	The bus does not exist
	IERR = 4	Generators are converted
	IERR = 5	Buses in island(s) without a swing bus--use activity TREE
	IERR = 6	Dispatch bus does not exist
	IERR = 7	No Distribution Factor Data input file specified
	IERR = 8	Subsystem containing components to which sensitivity factors are calculated is not specified
	IERR = 9	Opposing subsystem is not specified
	IERR = 10	Largest mismatch exceeds mismatch tolerance
	IERR = 11	The bus is a type 2 or type 3 bus
	IERR = 21	file DFXFILE is not in the form of a PSSE-25 or later DFAX file, run DFAX
	IERR = 22	Error linearizing Jacobian matrix in AC network model
	IERR = 51	error opening Distribution Factor Data File
	IERR = 52	prerequisite requirements for function not met

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1.290. SENSITIVITY_VOLTAGES

This API is used to calculate sensitivity factors of bus voltages in a subsystem to MW and MVar power at buses, MW power at generator buses, MW and MVar at load buses, phase angle of phase shifters, tap postion of tap changing transformers as well as addimittance of switched shunts.

This API routine was first introduced in release 32.1.0.

Batch command syntax:

```
BAT_SENSITIVITY_VOLTAGES  OPTIONS(1) .. OPTIONS(9) VALUES(1) VALUES(2)
 LABELS(1) .. LABELS(3) DFXFILE
```

Python command syntax:

```
ierr = sensitivity_voltages(options,values,labels,dfxfile)
```

Fortran command syntax:

```
CALL SENSITIVITY_VOLTAGES(OPTIONS,VALUES,LABELS,DFXFILE,IERR)
```

Where:

Integer OPTIONS (9)	array of 9 elements specifying options for sensitvity analysis to a branch flow(input).	
	OPTIONS(1)	flag to calculate sensitivity factors at buses
	OPTIONS(1) = 0	Disable, do not calculate sensitivity factors at buses
	OPTIONS(1) = 1	Enable, calculate sensitivity factors at buses
	OPTIONS(2)	flag to calculate sensitivity factors at generator buses
	OPTIONS(2) = 0	Disable
	OPTIONS(2) = 1	Enable
	OPTIONS(3)	flag to calculate sensitivity factors at load buses
	OPTIONS(3) = 0	Disable
	OPTIONS(3) = 1	Enable
	OPTIONS(4)	flag to calculate sensitivity factors of the branch flow to phase shifters
	OPTIONS(4) = 0	Disable
	OPTIONS(4) = 1	Enable
	OPTIONS(5)	flag to calculate sensitivity factors of the branch flow to tap changing transformers (ignore if DC network model is used)
	OPTIONS(5) = 0	Disable
	OPTIONS(5) = 1	Enable
	OPTIONS(6)	flag to calculate sensitivity factors of the branch flow to switched shunts (ignore if DC network model is used)
	OPTIONS(6) = 0	Disable
	OPTIONS(6) = 1	Enable

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	OPTIONS(7)	flag to specifiy the transfer from the components to which sensitivity factors of the branch flow are calculated to the opposing subsystem (the flag is ignored if none of the bus, generator, or load sensitivity calculation flags is selected)
	OPTIONS(7) = 0	Import. The positive transfer is from the components to the opposing subsystem
	OPTIONS(7) = 1	Export. The positive transfer is from the opposing subsystem to the components
	OPTIONS(8)	opposing subsystem type
	OPTIONS(8) = 0	Slack bus
	OPTIONS(8) = 1	Single bus
	OPTIONS(8) = 2	Subsystem
	OPTIONS(9)	ignored if opposing subsystem type is slack bus. It is a bus number when opposing subsystem type is a single bus, or one of the following seven dispatch modes if the opposing subsystem type is a subsystem (1 by default)
	OPTIONS(9) = 1	Buses & participating factors from DFAX file for buses with positive MW machines
	OPTIONS(9) = 2	Buses & participating factors from DFAX file for buses with positive MW constant MVA load
	OPTIONS(9) = 3	Buses & participating factors from DFAX file for buses with either positive MW machines or positive MW constant MVA load
	OPTIONS(9) = 4	Subsystem buses with positive MW constant MVA load in proportion to their MW load
	OPTIONS(9) = 5	Subsystem buses with positive MW machines in proportion to their MW output
	OPTIONS(9) = 6	Subsystem buses with positive MW machines in proportion to their MBASE
	OPTIONS(9) = 7	Subsystem buses with positive MW machines in proportion to their reserves (Pmaxmc-Pgenmc)
Real VALUES (2)	array of 2 elements specifying sensitivity analysis parameters(input).	
	VALUES(1)	mismatch tolerance for continuing sensitivity analysis in AC mode, ignored in DC mode
	VALUES(2)	sensitivity cutoff. Sensitivity factors less than the cutoff are not presented
Character*32 LABELS (3)	array of 3 elements specifying subsystem labels(input).	
	LABELS(1)	label of the subsystem containing components to which the sensitivity factors of the branch flow are calcualted

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	LABELS(2)	label of opposing subsystem when the opposing subsystem type is a subsystem
	LABELS(3)	label of the subsystem including branches for which sensitivity factors are calculated
Character*260 DFXFILE	DFAX file name (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid UNITS value
	IERR = 4	invalid APIOPT value
	IERR = 5	unexpected APIOPT value
	IERR = 6	SCMU results error; one of the following occurred: - No SCMU results file is open. - SCMU results are not compatible with the working case. - Error reading the SCMU results file
	IERR = 7	prerequisite requirements for API are not met

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1.291. SEQD

This API routine is obsolete. It has been replaced by the API routine seqd_2 .

Use this API to factorize the sequence matrices in preparation for unbalanced fault calculations.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SEQD  OPTIONS(1)  OPTIONS(2)
```

Python command syntax:

```
ierr = seqd(options)
```

Fortran command syntax:

```
CALL SEQDAPI(OPTIONS, IERR)
```

Where:

Integer OPTIONS (2)	array of 2 elements specifying options to be used in constructing the sequence matrices(input).	
	OPTIONS(1)	dc line and FACTS device option (0 by default)
	OPTIONS(1) = 0	block and ignore
	OPTIONS(1) = 1	convert to constant admittance load
	OPTIONS(2)	transformer impedance correction option (1 by default)
	OPTIONS(2) = 0	do not apply to zero sequence
	OPTIONS(2) = 1	apply to zero sequence
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	prerequisite requirements for API are not met

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1.292. SEQD_2

Use this API routine to factorize the sequence matrices in preparation for unbalanced fault calculations SCMU and SPCB.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SEQD_2  OPTIONS(1)  OPTIONS(2)  OPTIONS(3)
```

Python command syntax:

```
ierr = seqd_2(options)
```

Fortran command syntax:

```
CALL SEQD_2(OPTIONS, IERR)
```

Where:

Integer OPTIONS (3)	array of 3 elements specifying options to be used in constructing the sequence matrices(input).	
	OPTIONS(1)	dc line and FACTS device option (0 by default)
	OPTIONS(1) = 0	block and ignore
	OPTIONS(1) = 1	convert to constant admittance load
	OPTIONS(2)	transformer impedance correction option (1 by default)
	OPTIONS(2) = 0	do not apply to zero sequence
	OPTIONS(2) = 1	apply to zero sequence
	OPTIONS(3)	fault analysis generator reactance option; ignored if generators are already "converted" (0 by default)
	OPTIONS(3) = 0	use subtransient reactance
	OPTIONS(3) = 1	use transient reactance
	OPTIONS(3) = 2	use synchronous reactance
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	fault analysis is invalid due to the presence of in-service induction machines that are in the "stalled" or "tripped" state
	IERR = 3	bus type code and series element status inconsistencies
	IERR = 4	prerequisite requirements for API are not met

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1.293. SEQUENCE_NETWORK_SETUP

Use this API to specify or return the option to enable or disable automatic fault analysis sequence network setup.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SEQUENCE_NETWORK_SETUP  IVAL
```

Python command syntax:

```
ierr = sequence_network_setup(ival)

ierr, ival = sequence_network_setup()
```

Fortran command syntax:

```
CALL SEQUENCE_NETWORK_SETUP_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	SCMU assumes SEQD has already been executed.
	IVAL = 1	SCMU always executes activity SEQD.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.294. SET_INPUT_DEV

Use this API to set the terminal input device to a file.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SET_INPUT_DEV  IDVFIL
```

Python command syntax:

```
ierr = set_input_dev(idvfil)
```

Fortran command syntax:

```
CALL IDEVAPI(IDVFIL, IERR)
```

Where:

Character*260 IDVFIL	name of response file to be used for terminal (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	error opening IDVFIL
	IERR = 2	prerequisite requirements for API are not met

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1.295. SHNT

Use this API to tabulate fixed and/or switched bus shunts contained in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SHNT  SID  ALL  OPT
```

Python command syntax:

```
ierr = shnt(sid, all, opt)
```

Fortran command syntax:

```
CALL SHNTNW(SID, ALL, OPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer OPT	option to select the type of bus shunts tabulated (input; 1 by default).	
	OPT = 1	all shunts
	OPT = 2	fixed bus shunts
	OPT = 3	switched shunts
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	prerequisite requirements for API are not met

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1.296. SHORT_CIRCUIT_COORDINATES

Use this API to specify or return the option to set the fault analysis voltage and current output coordinates to either rectangular or polar coordinates.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SHORT_CIRCUIT_COORDINATES  IVAL
```

Python command syntax:

```
ierr = short_circuit_coordinates(ival)

ierr, ival = short_circuit_coordinates()
```

Fortran command syntax:

```
CALL SHORT_CIRCUIT_COORDINATES_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	rectangular coordinates.
	IVAL = 1	polar coordinates.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.297. SHORT_CIRCUIT_MODELING

Use this API to specify or return the option for the fault analysis modeling setting; either normal three-phase or center tapped two-phase.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SHORT_CIRCUIT_MODELING  IVAL
```

Python command syntax:

```
ierr = short_circuit_modeling(ival)

ierr, ival = short_circuit_modeling()
```

Fortran command syntax:

```
CALL SHORT_CIRCUIT_MODELING_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1 by default; else output).	
	IVAL = 0	center tapped two-phase.
	IVAL = 1	normal three-phase.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.298. SHORT_CIRCUIT_UNITS

Use this API to specify or return the option to set the fault analysis voltage and current output units to either per unit or physical units.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SHORT_CIRCUIT_UNITS  IVAL
```

Python command syntax:

```
ierr = short_circuit_units(ival)

ierr, ival = short_circuit_units()
```

Fortran command syntax:

```
CALL SHORT_CIRCUIT_UNITS_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	per unit.
	IVAL = 1	physical units.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.299. SHORT_CIRCUIT_WARNING

Use this API to specify or return the option to enable or disable the fault analysis warning option setting.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SHORT_CIRCUIT_WARNING  IVAL
```

Python command syntax:

```
ierr = short_circuit_warning(ival)

ierr, ival = short_circuit_warning()
```

Fortran command syntax:

```
CALL SHORT_CIRCUIT_WARNING_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 1 by default; else output).	
	IVAL = 0	disable printing of RESQ/TRSQ/solution warnings.
	IVAL = 1	print RESQ/TRSQ/solution warnings.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.300. SHOW

Use this API to tabulate summaries of Saved Case and/or Snapshot Files.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SHOW  OPTION  SFILE
```

Python command syntax:

```
ierr = show(option, sfile)
```

Fortran command syntax:

```
CALL SHOWAPI(OPTION, SFILE, IERR)
```

Where:

Integer OPTION	code for the type(s) of files to report (input; 1 by default).	
	OPTION = 0	file specified in SFILE
	OPTION = 1	all *.sav and *.snp files
	OPTION = 2	all *.sav files
	OPTION = 3	all *.snp files
Character*260 SFILE	name of saved case or snapshot file (when OPTION is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTION value
	IERR = 2	SFILE is blank when OPTION is 0
	IERR = 3	prerequisite requirements for API are not met

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1.301. SIZE

Use this API to obtain a summary of the number of components in the working case (activity SIZE).

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SIZE
```

Python command syntax:

```
ierr = size()
```

Fortran command syntax:

```
CALL SIZEAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.302. SOLV

Use this API to apply the Gauss-Seidel power flow calculation.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SOLV  OPTIONS(1)..OPTIONS(6)
```

Python command syntax:

```
ierr = solv(options)
```

Fortran command syntax:

```
CALL SOLVAPI(OPTIONS, IERR)
```

Where:

Integer OPTIONS (6)	array of 6 elements specifying solution options(input).	
	OPTIONS(1)	tap adjustment flag (use tap adjustment option setting by default)
	OPTIONS(1) = 0	disable
	OPTIONS(1) = 1	enable
	OPTIONS(2)	area interchange adjustment flag (use area interchange adjustment option setting by default)
	OPTIONS(2) = 0	disable
	OPTIONS(2) = 1	enable using tie line flows only in calculating area interchange
	OPTIONS(2) = 2	enable using tie line flows and loads in calculating area interchange
	OPTIONS(3)	flat start flag (0 by default)
	OPTIONS(3) = 0	do not flat start
	OPTIONS(3) = 1	flat start
	OPTIONS(3) = 2	flat start, then estimate voltage magnitudes
	OPTIONS(3) = 3	flat start, then estimate voltage phase angles
	OPTIONS(3) = 4	flat start, then estimate voltage magnitudes and phase angles
	OPTIONS(4)	dc tap adjustment flag (use dc tap adjustment option setting by default)
	OPTIONS(4) = 0	disable
	OPTIONS(4) = 1	enable
	OPTIONS(5)	switched shunt adjustment flag (use switched shunt adjustment option setting by default)
	OPTIONS(5) = 0	disable
	OPTIONS(5) = 1	enable

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	OPTIONS(5) = 2	enable continuous mode, disable discrete mode
	OPTIONS(6)	var limit flag (0 by default)
	OPTIONS(6) = 0	apply var limits
	OPTIONS(6) = 1	ignore var limits
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	generators are converted
	IERR = 3	buses in island(s) without a swing bus; use activity TREE
	IERR = 4	bus type code and series element status inconsistencies
	IERR = 5	prerequisite requirements for API are not met

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1.303. SPCB

Use this API to calculate positive sequence equivalents of branch unbalances.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SPCB  IBUS  JBUS  CKT  OPTIONS(1)..OPTIONS(4)  VALUES(1)..VALUES(5)
```

Python command syntax:

```
ierr = spcb(ibus, jbus, ckt, options, values)
```

Fortran command syntax:

```
CALL SPCBAPI(IBUS, JBUS, CKT, OPTIONS, VALUES, IERR)
```

Where:

Integer IBUS	from bus of branch subjected to unbalance (input; no default allowed).	
Integer JBUS	to bus of branch subjected to unbalance (input; no default allowed).	
Character*2 CKT	circuit identifier of branch subjected to unbalance (input; '1').	
Integer OPTIONS (4)	array of 4 elements specifying unbalance OPTIONS(input).	
	OPTIONS(1)	unbalance code (0 by default)
	OPTIONS(1) = 0	no unbalance; Thevenin impedances only
	OPTIONS(1) = 1	one phase open
	OPTIONS(1) = 2	two phases open
	OPTIONS(1) = 3	in-line fault
	OPTIONS(1) = 4	one breaker open
	OPTIONS(1) = 5	no unbalance
	OPTIONS(2)	path to ground code (used when OPTIONS(1) is 1 or 4) (0 by default)
	OPTIONS(2) = 0	no path to ground
	OPTIONS(2) = 1	include a path to ground
	OPTIONS(3)	type of in-line fault code (used when OPTIONS(1) is 3) (1 by default)
	OPTIONS(3) = 1	line-to-ground
	OPTIONS(3) = 2	line-to-line-to-ground
	OPTIONS(3) = 3	three-phase
	OPTIONS(4)	open end code (used when OPTIONS(1) is 4) (1 by default)
	OPTIONS(4) = 0	breaker at to bus is open
	OPTIONS(4) = 1	breaker at from bus is open
Real VALUES (3)	array of 3 elements specifying unbalance parameters(input).	
	VALUES(1)	fault location as fraction of line from from bus; 0.0 < VALUES(1) < 1.0 (used when OP-

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		TIONS(1) is 1 or 4 and OPTIONS(2) is 1; or when OPTIONS(1) is 3) (0.5 by default)
	VALUES(2)	& (3) Rl-g and Xl-g (used when OPTIONS(1) is 1 or 4 and OPTIONS(2) is 1; or when OPTIONS(1) is 3 and OPTIONS(3) is 1 or 2) (0.0 by default)
	VALUES(3)	& (5) Rl-l and Xl-l (used when OPTIONS(1) is 3 and OPTIONS(3) is 2) (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid with two phase modeling option enabled
	IERR = 2	bus not found
	IERR = 3	invalid OPTIONS value
	IERR = 4	invalid VALUES value
	IERR = 5	SEQD temporary file is not open
	IERR = 6	SEQD results are not compatible with the working case
	IERR = 7	terminated due to invalid fault specification or application
	IERR = 8	prerequisite requirements for API are not met

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1.304. SPIL

Use this API to calculate transmission interchange limits using a linear network (dc) model.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SPIL  OPTIONS(1)..OPTIONS(16)  VALUES(1)..VALUES(7)
 LABELS(1)..LABELS(8)  DFXFILE  PRTFILE
```

Python command syntax:

```
ierr = spil(options, values, labels, dfxfile, prtfile)
```

Fortran command syntax:

```
CALL SPILAPI(OPTIONS, VALUES, LABELS, DFXFILE, PRTFILE, IERR)
```

Where:

Integer OPTIONS (16) — array of 16 elements specifying calculation and reporting options(input).

OPTIONS(1)	base case rating set (rating set program option setting by default)
OPTIONS(1) = 1	RATEA
OPTIONS(1) = 2	RATEB
OPTIONS(1) = 3	RATEC
OPTIONS(2)	contingency case rating set (rating set program option setting by default)
OPTIONS(2) = 1	RATEA
OPTIONS(2) = 2	RATEB
OPTIONS(2) = 3	RATEC
OPTIONS(3)	base case line flow code (0 by default)
OPTIONS(3) = 0	dc base case
OPTIONS(3) = 1	ac base case
OPTIONS(4)	phase shifter code (0 by default)
OPTIONS(4) = 0	locked in base case shift solutions
OPTIONS(4) = 1	regulating in base case shift solutions
OPTIONS(5)	list study system buses (0 by default)
OPTIONS(5) = 0	no
OPTIONS(5) = 1	yes
OPTIONS(6)	list opposing system buses (0 by default)
OPTIONS(6) = 0	no
OPTIONS(6) = 1	yes
OPTIONS(7)	list study system tie lines (0 by default)
OPTIONS(7) = 0	no
OPTIONS(7) = 1	yes
OPTIONS(8)	add study system tie lines to monitored line list (0 by default)

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	OPTIONS(8) = 0	no
	OPTIONS(8) = 1	yes
	OPTIONS(9)	format code (0 by default)
	OPTIONS(9) = 0	summary output
	OPTIONS(9) = 1	full output
	OPTIONS(10)	interchange limit code (0 by default)
	OPTIONS(10) = 0	incremental transfer capability
	OPTIONS(10) = 1	total transfer capability
	OPTIONS(11)	apply summary minimum distribution factor, VALUES(4), to solution reports (0 by default)
	OPTIONS(11) = 0	no
	OPTIONS(11) = 1	yes
	OPTIONS(12)	convert MVA ratings to estimated MW ratings (0 by default)
	OPTIONS(12) = 0	no
	OPTIONS(12) = 1	yes
	OPTIONS(13)	code to discontinue activity if MW mismatch exceeds VALUES(1) (0 by default)
	OPTIONS(13) = 0	discontinue
	OPTIONS(13) = 1	continue
	OPTIONS(14)	summary table contingency descriptions (1 by default)
	OPTIONS(14) = 0	contingency labels
	OPTIONS(14) = 1	contingency events
	OPTIONS(14) = 2	both contingency labels and events
	OPTIONS(15)	number of elements to include in flow tables (60000 by default)
	OPTIONS(16)	summary table maximum times for reporting the same element (5 by default)
Real VALUES (7)	array of 7 elements specifying tolerances and thresholds(input).	
	VALUES(1)	required MW mismatch tolerance for continuing (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	percent of rating for reporting an element as overloaded (100.0 by default)
	VALUES(3)	summary table maximum import or export (>0.) (99999.0 by default)
	VALUES(4)	summary table minimum distribution factor magnitude (0.0 by default)
	VALUES(5)	minimum contingency case pre-shift flow change (0.0 by default)
	VALUES(6)	minimum contingency case distribution factor change (0.0 by default)
	VALUES(7)	study system generation shift (100.0 by default)

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Character*32 LABELS (8)	array of 8 elements specifying subsystem and interface labels(input).	
	LABELS(1)	study system label (no default allowed)
	LABELS(2)	opposing system label (no default allowed)
	LABELS(3)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(4)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(5)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(6)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(7)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(8)	interface label for optional interface limits summary table (blank for none) (blank by default)
Character*260 DFXFILE	name of DFAX data file (input; no default allowed).	
Character*260 PRTFILE	name of subsystem participation data file; blank for none (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	no DFAX data file specified
	IERR = 4	same label specified for study and opposing systems
	IERR = 5	generators are converted
	IERR = 6	buses in island(s) without a swing bus; use activity TREE
	IERR = 7	largest mismatch exceeds mismatch tolerance
	IERR = 8	error opening file DFXFILE
	IERR = 9	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 10	file DFXFILE does not contain distribution factors; run DFAX with distribution factor option enabled
	IERR = 11	no monitored branches or interfaces defined
	IERR = 12	no subsystems defined
	IERR = 13	subsystem not found

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IERR = 14	specified subsystems overlap
IERR = 15	no in-service buses in subsystem WORLD
IERR = 16	no tie lines from study system
IERR = 17	prerequisite requirements for API are not met

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1.305. SPLT

Use this API to add a bus to the working case and place a zero impedance line between bus BUS and the new bus. No other equipment is connected to the new bus by this API.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SPLT  BUS  NEWNUM  'NEWNAM'  NEWKV
```

Python command syntax:

```
ierr = splt(bus, newnum, newnam, newkv)
```

Fortran command syntax:

```
CALL SPLTAPI(BUS, NEWNUM, NEWNAM, NEWKV, IERR)
```

Where:

Integer BUS	number of the bus to be split (input; no default allowed).	
Integer NEWNUM	number to be assigned to the new bus (input; no default allowed).	
Character*12 NEWNAM	name to be assigned to the new bus (input; blank by default).	
Real NEWKV	base voltage to be assigned to the new bus (input; base voltage of bus BUS).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	invalid NEWNUM value
	IERR = 3	bus number is already assigned to another bus
	IERR = 4	invalid NEWKV value
	IERR = 5	bus, branch, or branch ownership tables are full
	IERR = 6	prerequisite requirements for API are not met

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1.306. SQEX

Use this API to tabulate all fault analysis data pertaining to a specified bus.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SQEX  SID  ALL
```

Python command syntax:

```
ierr = sqex(sid, all)
```

Fortran command syntax:

```
CALL SQEXNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.307. SQLI

Use this API to tabulate the sequence data in a form suitable for problem data documentation.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SQLI  SID  ALL  OPT
```

Python command syntax:

```
ierr = sqli(sid, all, opt)
```

Fortran command syntax:

```
CALL SQLINW(SID, ALL, OPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer OPT	category of sequence data to be tabulated (input; 1 by default).	
	OPT = 1	full listing
	OPT = 2	bus load data
	OPT = 3	generator data
	OPT = 4	branch data
	OPT = 5	two-winding positive sequence data
	OPT = 6	mutual coupling data
	OPT = 7	switched shunt data
	OPT = 8	three-winding general data
	OPT = 9	line shunt data
	OPT = 10	fixed shunt data
	OPT = 11	induction machine data
	OPT = 12	two-winding zero sequence data
	OPT = 13	three-winding positive sequence data
	OPT = 14	three-winding zero sequence data
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	prerequisite requirements for API are not met

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1.308. SRAIND

Use this API to run the substation reliability analysis. It applies failure modes and affects analysis of arbitrary bus-breaker configurations and measures the impact of substation equipment failures upon the power transfer capability of the substation.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SRAIND  OPTIONS(1)..OPTIONS(3) VALUES(1)..VALUES(5)
 LABELS(1)..LABELS(3) DFXFILE PRBFILE
```

Python command syntax:

```
ierr = sraind(options, values, labels, dfxfile, prbfile)
```

Fortran command syntax:

```
CALL SRAINDAPI(OPTIONS, VALUES, LABELS, DFXFILE, PRBFILE, IERR)
```

Where:

Integer OPTIONS (3)	array of 3 elements specifying options(input).	
	OPTIONS(1)	report format (1 by default)
	OPTIONS(1) = 1	contingency report
	OPTIONS(1) = 2	bus load curtailment report
	OPTIONS(1) = 3	substation load curtailment report
	OPTIONS(2)	rating set (rating set program option setting by default)
	OPTIONS(2) = 1	RATEA
	OPTIONS(2) = 2	RATEB
	OPTIONS(2) = 3	RATEC
	OPTIONS(3)	include system ties (0 by default)
	OPTIONS(3) = 0	disable
	OPTIONS(3) = 1	enable
Real VALUES (5)	array of 5 elements specifying substation reliability assessment parameters(input).	
	VALUES(1)	percent of flow rating (>=0, 100.0 by default)
	VALUES(2)	frequency cutoff (>=0, 0.0001 by default)
	VALUES(3)	percent of peak load (>=0, 100.0 by default)
	VALUES(4)	switching time in hours (>=0, 0.0 by default)
	VALUES(5)	breaker failure rate in hours per year (>=0, 0.0 by default)
Character*32 LABELS (3)	array of 3 elements specifying subsystem labels(input).	
	LABELS(1)	label of substation subsystem (no default is allowed)

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	LABELS(2)	label of source subsystem (no default is allowed)
	LABELS(3)	label of sink subsystem (no default is allowed)
Character*260 DFXFILE	name of DFAX data file (input; no default allowed).	
Character*260 PRBFILE	name of outage statistic data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	substation/source/sink subsytem not specified
	IERR = 4	no DFAX data file specified
	IERR = 5	no outage Statistic Data File specified
	IERR = 6	file is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 7	subsystem not found in DFAX data file
	IERR = 8	source/sink buses not in substation subsystem
	IERR = 9	source and sink systems overlap
	IERR = 51	error opening DFAX data file
	IERR = 52	error opening outage statistic data file
	IERR = 53	file system errors

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1.309. STOP

This API routine is obsolete. It has been replaced by the API routine stop_2 .

Use this API to end the operation of PSSE.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_STOP  DELETE
```

Python command syntax:

```
ierr = stop(delete)
```

Fortran command syntax:

```
CALL STOPAPI(DELETE, IERR)
```

Where:

Integer DELETE	delete working files (input; 0 by default).	
	DELETE = 1	delete working files
	DELETE != 1	keep working files
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.310. STOP_2

Use this API to end the operation of PSSE.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_STOP_2
```

Python command syntax:

```
ierr = stop_2()
```

Fortran command syntax:

```
CALL STOPAPI_2(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.311. SUBS

Use this API to summarize conditions in the working case by tabulating the conditions at each swing system bus, conditions at each area slack area bus, number of components, generation/ load/shunt totals, and loss/line shunt/charging totals by voltage levels.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SUBS  SID  ALL
```

Python command syntax:

```
ierr = subs(sid, all)
```

Fortran command syntax:

```
CALL SUBSNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.312. SWITCHED_SHUNT_ADJUSTMENT

Use this API to specify or return the option to enable or disable switched shunt adjustment.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_SWITCHED_SHUNT_ADJUSTMENT  IVAL
```

Python command syntax:

```
ierr = switched_shunt_adjustment(ival)

ierr, ival = switched_shunt_adjustment()
```

Fortran command syntax:

```
CALL SWITCHED_SHUNT_ADJUSTMENT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	disabled.
	IVAL = 1	enabled.
	IVAL = 2	continuous mode switched shunts enabled, discrete mode switched shunts locked.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.313. TAP_ADJUSTMENT

Use this API to specify or return the option for the default tap adjustment setting; either disabled, stepping or direct.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_TAP_ADJUSTMENT  IVAL
```

Python command syntax:

```
ierr = tap_adjustment(ival)

ierr, ival = tap_adjustment()
```

Fortran command syntax:

```
CALL TAP_ADJUSTMENT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	locked (i.e., tap adjustment is disabled).
	IVAL = 1	tap adjustment using the stepping method enabled.
	IVAL = 2	tap adjustment using the direct method enabled.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.314. TEXT

This API does nothing. Its purpose is to provide recordable documentation.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_TEXT  'CARG'
```

Python command syntax:

```
ierr = text(carg)
```

Fortran command syntax:

```
CALL TEXT_API(CARG, IERR)
```

Where:

Character*220 CARG	string of text (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.315. TFLG

Use this API to set or reset the adjustment control mode flags for all automatically adjustable transformers contained in the specified subsystem.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TFLG  SID  ALL  OPT
```

Python command syntax:

```
ierr = tflg(sid, all, opt)
```

Fortran command syntax:

```
CALL TFLGNW(SID, ALL, OPT, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer OPT	option to select the adjustment control mode flags of those transformers being processed (input; 1 by default).	
	OPT = 0	disable
	OPT = 1	enable
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	prerequisite requirements for API are not met

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1.316. TIES

Use this API to tabulate the flows on all area tie lines, with tie flows grouped by area.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TIES  SID  ALL
```

Python command syntax:

```
ierr = ties(sid, all)
```

Fortran command syntax:

```
CALL TIESNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Area subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all areas or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only areas in area subsystem SID
	ALL = 1	process all areas
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.317. TIEZ

Use this API to tabulate the flows on all zone tie lines, with tie flows grouped by zone.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TIEZ  SID  ALL
```

Python command syntax:

```
ierr = tiez(sid, all)
```

Fortran command syntax:

```
CALL TIEZNW(SID, ALL, IERR)
```

Where:

Integer SID	a valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Zone subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all zones or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only zones in zone subsystem SID
	ALL = 1	process all zones
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.318. TIME

Use this API to tabulate timing statistics.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_TIME  INIT
```

Python command syntax:

```
ierr = time(init)
```

Fortran command syntax:

```
CALL TIMEAPI(INIT, IERR)
```

Where:

Integer INIT	timing flag (input; positive).	
	INIT <= 0	do not initialize
	INIT > 0	initialize
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.319. TLST

Use this API to tabulate those transformers in the working case where off-nominal turns ratio or phase shift angle may be adjusted by the power flow solution activities.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TLST  SID  ALL  OPT  VREV
```

Python command syntax:

```
ierr = tlst(sid, all, opt, vrev)
```

Fortran command syntax:

```
CALL TLSTNW(SID, ALL, OPT, VREV, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer OPT	option to select the transformers tabulated (input; 0 by default).	
	OPT = 0	all
	OPT = 1	violations only
Integer VREV	option to change voltage units from the default units to the alternate units (input; 0 by default).	
	VREV = 0	use default voltage units
	VREV = 1	use alternate voltage units
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	invalid VREV value
	IERR = 5	prerequisite requirements for API are not met

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1.320. TLTG

Use this API to calculate transmission interchange limits using a linear network (dc) model.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_TLTG  OPTIONS(1)..OPTIONS(17)  VALUES(1)..VALUES(7)
 LABELS(1)..LABELS(8)  DFXFILE
```

Python command syntax:

```
ierr = tltg(options, values, labels, dfxfile)
```

Fortran command syntax:

```
CALL TLTGAPI(OPTIONS, VALUES, LABELS, DFXFILE, IERR)
```

Where:

Integer OPTIONS (17)	array of 17 elements specifying calculation and reporting options(input).	
	OPTIONS(1)	base case rating set (rating set program option setting by default)
	OPTIONS(1) = 1	RATEA
	OPTIONS(1) = 2	RATEB
	OPTIONS(1) = 3	RATEC
	OPTIONS(2)	contingency case rating set (rating set program option setting by default)
	OPTIONS(2) = 1	RATEA
	OPTIONS(2) = 2	RATEB
	OPTIONS(2) = 3	RATEC
	OPTIONS(3)	base case line flow code (0 by default)
	OPTIONS(3) = 0	dc base case
	OPTIONS(3) = 1	ac base case
	OPTIONS(4)	phase shifter code (0 by default)
	OPTIONS(4) = 0	locked in base case shift solutions
	OPTIONS(4) = 1	regulating in base case shift solutions
	OPTIONS(5)	code for the treatment of base case line flow constraints in contingency cases (0 by default)
	OPTIONS(5) = 0	ignore
	OPTIONS(5) = 1	include
	OPTIONS(6)	list study system buses (0 by default)
	OPTIONS(6) = 0	no
	OPTIONS(6) = 1	yes
	OPTIONS(7)	list opposing system buses (0 by default)
	OPTIONS(7) = 0	no
	OPTIONS(7) = 1	yes

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	OPTIONS(8)	list study system tie lines (0 by default)
	OPTIONS(8) = 0	no
	OPTIONS(8) = 1	yes
	OPTIONS(9)	add study system tie lines to monitored line list (0 by default)
	OPTIONS(9) = 0	no
	OPTIONS(9) = 1	yes
	OPTIONS(10)	format code (0 by default)
	OPTIONS(10) = 0	summary output
	OPTIONS(10) = 1	full output
	OPTIONS(11)	interchange limit code (0 by default)
	OPTIONS(11) = 0	incremental transfer capability
	OPTIONS(11) = 1	total transfer capability
	OPTIONS(12)	apply summary minimum distribution factor, VALUES(4), to solution reports (0 by default)
	OPTIONS(12) = 0	no
	OPTIONS(12) = 1	yes
	OPTIONS(13)	convert MVA ratings to estimated MW ratings (0 by default)
	OPTIONS(13) = 0	no
	OPTIONS(13) = 1	yes
	OPTIONS(14)	code to discontinue activity if MW mismatch exceeds VALUES(1) (0 by default)
	OPTIONS(14) = 0	discontinue
	OPTIONS(14) = 1	continue
	OPTIONS(15)	summary table contingency descriptions (1 by default)
	OPTIONS(15) = 0	contingency labels
	OPTIONS(15) = 1	contingency events
	OPTIONS(15) = 2	both contingency labels and events
	OPTIONS(16)	number of elements to include in flow tables (60000 by default)
	OPTIONS(17)	summary table maximum times for reporting the same element (5 by default)
Real VALUES (7)	array of 7 elements specifying tolerances and thresholds(input).	
	VALUES(1)	required MW mismatch tolerance for continuing (Newton solution convergence tolerance, TOLN by default)
	VALUES(2)	percent of rating for reporting an element as overloaded (100.0 by default)
	VALUES(3)	summary table maximum import or export (>0.) (99999.0 by default)
	VALUES(4)	summary table minimum distribution factor magnitude (0.0 by default)

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	VALUES(5)	minimum contingency case pre-shift flow change (0.0 by default)
	VALUES(6)	minimum contingency case distribution factor change (0.0 by default)
	VALUES(7)	study system generation shift (100.0 by default)
Character*32 LABELS (8)	array of 8 elements specifying subsystem and interface labels(input).	
	LABELS(1)	study system label (no default allowed)
	LABELS(2)	opposing system label (no default allowed)
	LABELS(3)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(4)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(5)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(6)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(7)	interface label for optional interface limits summary table (blank for none) (blank by default)
	LABELS(8)	interface label for optional interface limits summary table (blank for none) (blank by default)
Character*260 DFXFILE	name of DFAX data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid VALUES value
	IERR = 3	no DFAX data file specified
	IERR = 4	same label specified for study and opposing systems
	IERR = 5	generators are converted
	IERR = 6	buses in island(s) without a swing bus; use activity TREE
	IERR = 7	largest mismatch exceeds mismatch tolerance
	IERR = 8	error opening file DFXFILE
	IERR = 9	file DFXFILE is not in the form of a PSSE-25 or later DFAX file; run DFAX
	IERR = 10	file DFXFILE does not contain distribution factors; run DFAX with distribution factor option enabled

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IERR = 11	no monitored branches or interfaces defined
IERR = 12	no subsystems defined
IERR = 13	subsystem not found
IERR = 14	specified subsystems overlap
IERR = 15	no in-service buses in subsystem WORLD
IERR = 16	no tie lines from study system
IERR = 17	prerequisite requirements for API are not met

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1.321. TPCH

Use this API to check the adjustment data associated with voltage or flow controlling transformers. It must be referenced using one of the following procedures:
- One reference with APIOPT=1, followed by any number of either APIOPT=2 references and/or pairs of references with APIOPT=2 and then 3, followed by one reference with APIOPT=4.
- One reference with APIOPT=0. This automatically combines the processing of one APIOPT 1 reference, followed by one APIOPT 2 reference, followed by one APIOPT 3 reference if NTRN from the APIOPT 2 reference is greater than zero, followed by one APIOPT 4 reference.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TPCH  SID  ALL  APIOPT  OPT  STATUS(1)  STATUS(2)  THRSH(1)  THRSH(2)
```

Python command syntax:

```
ierr, ttyp, ntrn = tpch(sid, all, apiopt, opt, status, thrsh)
```

Fortran command syntax:

```
CALL TPCHNW(SID, ALL, APIOPT, OPT, STATUS, THRSH, TTYP, NTRN, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; used only when APIOPT is 0 or 1; 0).	
Integer ALL	all buses or specified subsystem flag (input; used only when APIOPT is 0 or 1; 1).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer APIOPT	mode of operation (input; 0 by default).	
	APIOPT = 0	initialize, check, modify adjustment data, and terminate
	APIOPT = 1	initialize for check and data modification
	APIOPT = 2	check transformers
	APIOPT = 3	modify adjustment data
	APIOPT = 4	terminate transformer checks
Integer OPT	option to select the type of adjustment (input; used only when APIOPT is 0 or 2; 1).	
	OPT = 1	tap step = 0. (V/Mvar)
	OPT = 2	0. < step < minimum (V/ Mvar)
	OPT = 3	tap step > maximum (V/ Mvar)
	OPT = 4	voltage band < 2*step (V)
	OPT = 5	voltage band < minimum (V)
	OPT = 6	voltage band > maximum (V)
	OPT = 8	flow band < minimum (MW/Mvar)
	OPT = 9	flow band > maximum (MW/Mvar)

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Integer STATUS (2)	array of 2 elements specifying data modification options (used only when APIOPT is 0 or 3)(input).
	STATUS(1) — modify steps or voltage bands, when OPT = 4 (1 by default)
	STATUS(1) = 1 — modify steps
	STATUS(1) = 2 — modify voltage bands
	STATUS(2) — used to indicate which transformers to change where OPTs 1, 2, 3, 8 or 9 (0 by default)
	STATUS(2) = 0 — change both types
	STATUS(2) = 1 — change voltage controlling transformers when OPT = 1, 2 or 3; change MW controlling transformers when OPT = 8 or 9
	STATUS(2) = 2 — change Mvar controlling transformers
Real THRSH (2)	array of 2 elements specifying data checking and modification parameters(input).
	THRSH(1) — threshold to use in initial test (used only when APIOPT is 0 or 2; defaults: 0.00625 when OPT is 2 or 3; 0.02 when OPT is 5 or 6; 5.0 when OPT is 8 or 9)
	THRSH(2) — new tap step, voltage band, or flow band (used only when APIOPT is 0 or 3)
	If OPT
	THRSH(2) = 1 — new tap step (0.00625 by default)""
	THRSH(2) = 2,3 — new tap step (THRSH(1) by default)""
	THRSH(2) = 5,6 — new voltage band (THRSH(1) by default)""
	THRSH(2) = 8,9 — new flow band (THRSH(1) by default)""
Logical TTYP	returned when APIOPT = 0, 2 or 3 and is TRUE. if both voltage (or MW if OPT = 8 or 9) and Mvar controlling transformers were found (output).
Integer NTRN	returned when APIOPT = 0, 2 or 3 and is the number of transformers that need adjustment when applying the adjustment type indicated by OPT (output).
Integer IERR	error code (output).
	IERR = 0 — no error occurred
	IERR = 1 — invalid SID value or subsystem SID is not defined
	IERR = 2 — invalid ALL value
	IERR = 3 — invalid APIOPT value
	IERR = 4 — invalid OPT value
	IERR = 5 — invalid STATUS value
	IERR = 6 — invalid THRSH value
	IERR = 7 — unexpected APIOPT value
	IERR = 8 — no controlling transformers in case

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IERR = 9	prerequisite requirements for API are not met

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1.322. TRANSACTION_DELETE

Use this API to delete a transaction event from PSSE working memory.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_TRANSACTION_DELETE  EVENT
```

Python command syntax:

```
ierr = transaction_delete(event)
```

Fortran command syntax:

```
CALL TRANSACTION_DELETE(EVENT, IERR, NERR)
```

Where:

Integer EVENT	transaction event number (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid transaction event number
	IERR = 2	prerequisite requirements for API are not met
Integer NERR	number of error entries in PBUF02 (output).	

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1.323. TRANSFORMER_PERCENT_UNITS

Use this API to specify or return the option to set the transformer percent units to either MVA or current expressed as MVA.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_TRANSFORMER_PERCENT_UNITS  IVAL
```

Python command syntax:

```
ierr = transformer_percent_units(ival)

ierr, ival = transformer_percent_units()
```

Fortran command syntax:

```
CALL TRANSFORMER_PERCENT_UNITS_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	MVA.
	IVAL = 1	current expressed as MVA.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.324. TRANSMISSION_LINE_UNITS

Use this API to to specify or return the option set the transmission line units to either per unit or ohms.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_TRANSMISSION_LINE_UNITS  IVAL
```

Python command syntax:

```
ierr = transmission_line_units(ival)

ierr, ival = transmission_line_units()
```

Fortran command syntax:

```
CALL TRANSMISSION_LINE_UNITS_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	per unit.
	IVAL = 1	ohms and microfarads.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.325. TREE

Use this API to check for the existence of in-service ac islands that do not contain a Type 3 (swing) bus. Following each successful call, it returns BUSES as the number of buses in a swingless island (0 for no more swingless islands). When a BUSES value of 0 is returned, no further calls are needed.
The API must be called once with APIOPT set to 1. If BUSES is returned as 0 (i.e., there are no swingless islands), no further calls are needed. Otherwise, if BUSES is greater than zero, it must be called one or more times with APIOPT set to 2 and OPTION set to indicate the disposition of the current swingless island. APIOPT 2 calls are required until either BUSES is returned as zero or an APIOPT 2 call is made with OPTION set to a negative value.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_TREE  APIOPT  OPTION
```

Python command syntax:

```
ierr, buses = tree(apiopt, option)
```

Fortran command syntax:

```
CALL TREEAPI(APIOPT, OPTION, BUSES, IERR)
```

Where:

Integer APIOPT	mode of operation (input; no default allowed).	
	APIOPT = 1	initialize and check for the presence of a swingless island
	APIOPT = 2	process previously detected island as dictated by OPTION; then check for the presence of another swingless island
Integer OPTION	option for the handling of previously detected swingless island (input; used only when APIOPT is 2; -1).	
	OPTION < 0	leave this island alone and terminate activity TREE
	OPTION = 0	leave this island alone and check for another swingless island
	OPTION > 0	disconnect this island, then check for another swingless island
Integer BUSES	returned as the number of buses in this swingless island; 0 if no more swingless islands (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid APIOPT value
	IERR = 2	unexpected APIOPT value
	IERR = 3	prerequisite requirements for API are not met

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1.326. TYSL

Use this API to run switching study network solutions.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TYSL  OPT
```

Python command syntax:

```
ierr = tysl(opt)
```

Fortran command syntax:

```
CALL TYSLAPI(OPT, IERR)
```

Where:

Integer OPT	flat start flag (input; 0 by default).	
	OPT = 0	use present voltage vector as starting point
	OPT = 1	flat start
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid OPT value
	IERR = 2	generators are not converted
	IERR = 3	network is not ordered
	IERR = 4	error reading the admittance matrix temporary file; one of the following occurred: - Admittance matrix temporary file is not open. - Reading matrix pointers: different sizes specified than were used in writing them. - Reading matrix terms: different sizes specified than were used in writing them
	IERR = 5	prerequisite requirements for API are not met

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1.327. USER

Use this API to run the user-written activity, subroutine USERAC.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_USER
```

Python command syntax:

```
ierr = user()
```

Fortran command syntax:

```
CALL USERAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.328. VAMM

Use this API to tabulate the vector absolute MW-mile report.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_VAMM
```

Python command syntax:

```
ierr = vamm()
```

Fortran command syntax:

```
CALL VAMMAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no transactions defined
	IERR = 4	prerequisite requirements for API are not met

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1.329. VAMO

Use this API to tabulate the vector absolute MW-ohm report.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_VAMO
```

Python command syntax:

```
ierr = vamo()
```

Fortran command syntax:

```
CALL VAMOAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no transactions defined
	IERR = 4	prerequisite requirements for API are not met

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1.330. VCHK

Use this API to tabulate those buses where voltage magnitude is outside a specified range.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_VCHK  SID  ALL  VLO  VHI
```

Python command syntax:

```
ierr = vchk(sid, all, vlo, vhi)
```

Fortran command syntax:

```
CALL VCHKNW(SID, ALL, VLO, VHI, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Real VLO	lower limit of the voltage band (input; 0.95).	
Real VHI	upper limit of the voltage band (input; 1.05).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.331. VOLTAGE_INPUT

Use this API to specify or return the voltage input option setting; either per unit or kV.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_VOLTAGE_INPUT  IVAL
```

Python command syntax:

```
ierr = voltage_input(ival)

ierr, ival = voltage_input()
```

Fortran command syntax:

```
CALL VOLTAGE_INPUT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	per unit.
	IVAL = 1	kV.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.332. VOLTAGE_OUTPUT

Use this API to specify or return the voltage output option setting; either per unit or kV.

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_VOLTAGE_OUTPUT  IVAL
```

Python command syntax:

```
ierr = voltage_output(ival)

ierr, ival = voltage_output()
```

Fortran command syntax:

```
CALL VOLTAGE_OUTPUT_API(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	value of the option setting (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, 0 by default; else output).	
	IVAL = 0	per unit.
	IVAL = 1	kV.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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1.333. VSMO

Use this API to tabulate the vector MW-ohm report.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_VSMO
```

Python command syntax:

```
ierr = vsmo()
```

Fortran command syntax:

```
CALL VSMOAPI(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	no transactions defined
	IERR = 4	prerequisite requirements for API are not met

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1.334. WRITERAWVERSION

Use this API to replicate the working case in the form of power flow raw data file compatible with PSSE 15 or later.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_WRITERAWVERSION  'VERNUM'  OUT  OFILE
```

Python command syntax:

```
ierr = writerawversion(vernum, out, ofile)
```

Fortran command syntax:

```
CALL WRITERAWVERSION(VERNUM, OUT, OFILE, IERR)
```

Where:

Integer or Character*14 VERNUM	value of OFILE version number VERNUM = '15' or later. VERNUM = '27' or later. Example: If OFILE format is to be PSSE 29.5.1: VERNUM = '29' or = '29.5' or = '29.5.1' (input; no default allowed).	
Integer OUT	output destination (input; 1 by default).	
	OUT = 0	to OFILE
	OUT = 1	to report window
Character*260 OFILE	name of the power flow raw data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VERNUM value
	IERR = 2	invalid OUT value
	IERR = 3	memory allocation Error
	IERR = 4	error reading data
	IERR = 5	error saving temporary 'sav' file
	IERR = 6	error opening temporary 'sav' file
	IERR = 7	error deleting temporary 'sav' file
	IERR = 8	rrror opening OFILE
	IERR = 9	prerequisite requirements for API are not met

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1.335. WRITESEQVERSION

This API routine is obsolete. It has been replaced by the API routine writeseqversion_2 .

Use this API to replicate the working case in the form of a network sequence data file compatible with PSSE 27 or later.

This API routine was first introduced in release 32.0.4.

Batch command syntax:

```
BAT_WRITESEQVERSION  'VERNUM'  OUT  OFILE
```

Python command syntax:

```
ierr = writeseqversion(vernum, out, ofile)
```

Fortran command syntax:

```
CALL WRITESEQVERSION(VERNUM, OUT, OFILE, IERR)
```

Where:

Integer or Character*14 VERNUM	value of OFILE version number VERNUM = '27' or later. Example: If OFILE format is to be PSSE 29.5.1: VERNUM = '29' or = '29.5' or = '29.5.1' (input; no default allowed).	
Integer OUT	output destination (input; 1 by default).	
	OUT = 0	to OFILE
	OUT = 1	to report window
Character*260 OFILE	name of the network sequence data file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VERNUM value
	IERR = 2	invalid OUT value
	IERR = 3	error opening OFILE
	IERR = 4	prerequisite requirements for API are not met

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1.336. WRITESEQVERSION_2

Use this API to replicate the working case in the form of a network sequence data file compatible with PSSE 27 or later.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_WRITESEQVERSION_2  'VERNUM'  OUT  OFILE GZOPTN
```

Python command syntax:

```
ierr = writeseqversion_2(vernum, out, ofile, gzoptn)
```

Fortran command syntax:

```
CALL WRITESEQVERSION_2(VERNUM, OUT, OFILE, GZOPTN, IERR)
```

Where:

Integer or Character*14 VERNUM	value of OFILE version number VERNUM = '27' or later. Example: If OFILE format is to be PSSE 29.5.1: VERNUM = '29' or = '29.5' or = '29.5.1' (input; no default allowed).	
Integer OUT	output destination (input; 1 by default).	
	OUT = 0	to OFILE
	OUT = 1	to REPORT WINDOW
Character*260 OFILE	name of the network sequence data file (input; no default allowed).	
Integer GZOPTN	generator reactance option (input; 0 by default).	
	GZOPTN = 0	save subtransient reactance
	GZOPTN = 1	save transient reactance
	GZOPTN = 2	save synchronous reactance
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VERNUM value
	IERR = 2	invalid OUT value
	IERR = 3	error opening OFILE
	IERR = 4	prerequisite requirements for API are not met

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1.337. WRITE_OPTIONS_FILE

Use this API to save the present values of the PSSE-25 program option settings to the file psse.opt.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_WRITE_OPTIONS_FILE
```

Python command syntax:

```
ierr = write_options_file()
```

Fortran command syntax:

```
CALL WRITE_OPTIONS_FILE_API(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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1.338. XEQV

Use this API to build an electrical equivalent of the portion of the working case outside of a specified subsystem of the working case.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_XEQV  SID  ALL  STATUS(1)..STATUS(6)  DVAL(1)  DVAL(2)
```

Python command syntax:

```
ierr = xeqv(sid, all, status, dval)
```

Fortran command syntax:

```
CALL XEQV(SID, ALL, STATUS, DVAL, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses outside of subsystem SID
	ALL = 1	process all buses
Integer STATUS (6)	array of 6 elements specifying They are used to control equivalencing options. A value of 0 means the option is not enabled. A value of 1 means the option is enabled. They are as follows(input).	
	STATUS(1)	retain area boundary buses (0 by default)
	STATUS(2)	retain zone boundary buses (0 by default)
	STATUS(3)	suppress the equivalencing option for phase shifters (0 by default)
	STATUS(4)	retain controlled buses (0 by default)
	STATUS(5)	retain existing branches between retained buses (0 by default)
	STATUS(6)	net load and shunt at retained buses (0 by default)
Real DVAL (2)	array of 2 elements specifying They are as follows(input).	
	DVAL(1)	minimum active or reactive power generation for retaining generator buses (0.0 by default)
	DVAL(2)	branch threshold tolerance (10.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value

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IERR = 3	invalid STATUS value
IERR = 4	invalid DVAL value
IERR = 5	prerequisite requirements for API are not met

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1.339. ZONE

This API routine is obsolete. It has been replaced by the API routine zone_2 .

Use this API to tabulate zone totals by zone.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ZONE  SID  ALL
```

Python command syntax:

```
ierr = zone(sid, all)
```

Fortran command syntax:

```
CALL ZONENW(SID, ALL, IERR)
```

Where:

Integer SID	a valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Zone subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all zones or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only zones in zone subsystem SID
	ALL = 1	process all zones
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 4	prerequisite requirements for API are not met

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1.340. ZONE_2

Use this API to tabulate zone totals by zone.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_ZONE_2  SID  ALL  OPT
```

Python command syntax:

```
ierr = zone_2(sid, all, opt)
```

Fortran command syntax:

```
CALL ZONE_2(SID, ALL, OPT, IERR)
```

Where:

Integer SID	a valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Zone subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all zones or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only zones in zone subsystem SID
	ALL = 1	process all zones
Integer OPT	load reporting option (input; 2 by default).	
	OPT = 1	tabulate load connected to zone buses
	OPT = 2	tabulate load assigned to the zone
	OPT = 3	tabulate three load subtotals for each zone: -zone load connected to zone buses -zone load connected to non-zone buses -non-zone load connected to zone buses
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	prerequisite requirements for API are not met

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1.341. ZONE_AREA

Use this API to tabulate zone totals by zone, along with subtotals by area.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ZONE_AREA  SID  ALL
```

Python command syntax:

```
ierr = zone_area(sid, all)
```

Fortran command syntax:

```
CALL ZONE_AREA(SID, ALL, IERR)
```

Where:

Integer SID	a valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Zone subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all zones or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only zones in zone subsystem SID
	ALL = 1	process all zones
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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1.342. ZONM

This API routine is obsolete. It has been replaced by the API routine zonm_2 .

Use this API to reassign the buses and loads in a specified subsystem of the working case from their original zone to a designated zone.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ZONM  SID  ALL  STATUS(1)  STATUS(2)  IZNEW
```

Python command syntax:

```
ierr = zonm(sid, all, status, iznew)
```

Fortran command syntax:

```
CALL ZONMNW(SID, ALL, STATUS, IZNEW, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS (2)	array of 2 elements specifying reporting options(input).	
	STATUS(1)	zone assignments of buses to IZNEW (0 by default)
	STATUS(1) = 0	don't change zone assignments of buses to IZNEW
	STATUS(1) = 1	change zone assignments of buses to IZNEW
	STATUS(2)	zone assignments of loads to IZNEW (0 by default)
	STATUS(2) = 0	Don't change zone assignments of loads to IZNEW
	STATUS(2) = 1	Change zone assignments of loads to IZNEW
Integer IZNEW	new zone to be assigned to specified buses and/or loads. IZNEW must be between 1 and 9999 (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value

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IERR = 4	invalid new zone number
IERR = 5	zone table is full; cannot add zone to the working case
IERR = 6	prerequisite requirements for API are not met

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1.343. ZONM_2

Use this API routine to reassign the buses, loads and/or induction machines in a specified subsystem of the working case from their original zone to a designated zone.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ZONM_2  SID  ALL  STATUS(1)  STATUS(2)  STATUS(3)  IZNEW
```

Python command syntax:

```
ierr = zonm_2(sid, all, status, iznew)
```

Fortran command syntax:

```
CALL ZONM_2(SID, ALL, STATUS, IZNEW, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS (3)	array of 3 elements specifying reporting options(input).	
	STATUS(1)	zone assignments of buses to IZNEW (0 by default)
	STATUS(1) = 0	leave zone assignments of buses unchanged
	STATUS(1) = 1	change zone assignments of buses to IZNEW
	STATUS(2)	zone assignments of loads to IZNEW (0 by default)
	STATUS(2) = 0	leave zone assignments of loads unchanged
	STATUS(2) = 1	change zone assignments of loads to IZNEW
	STATUS(3)	zone assignments of induction machines to IZNEW (0 by default)
	STATUS(3) = 0	leave zone assignments of induction machines unchanged
	STATUS(3) = 1	change zone assignments of induction machines to IZNEW
Integer IZNEW	new zone to be assigned to specified buses, load, and/or induction machines. IZNEW must be between 1 and 9999 (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value

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IERR = 3	invalid STATUS value
IERR = 4	invalid new zone number
IERR = 5	zone table is full; cannot add zone to the working case
IERR = 6	prerequisite requirements for API are not met

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Chapter 2
Power Flow Data

This chapter describes the API routines used to add or modify power flow data.

For most categories of power flow data, there are a pair of API routines for specifying data; one of these contains "_DATA" as part of the API name, and the other contains "_CHNG" in its API name. The "_DATA" API routines may be used to add a new network element to the working case, or to modify the data of an existing network element. The "_CHNG" API routines may be used only to modify the data of an existing network element; specifying a network element that does not already exist in the working case to one of the "_CHNG" API routines is treated as a fatal error by the API routine.

Generally, when an API is adding a new element to the working case, data items that are defaulted are set to the value listed in the data item's description. When an API is modifying the data of an existing element, defaulted data items are unchanged. Note that, except for the SEQ_MUTUAL_DATA API, the SEQ_ APIs always modify data of an existing element.

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2.1. AREA_DATA

Use this API to modify area data in the working case (see PSSE Program Operation Manual, Areas, Zones and Owners).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_AREA_DATA  IAREA   IBUS   REALAR(1)   REALAR(2)   'ARNAME'
```

Python command syntax:

```
ierr = area_data(iarea, ibus, realar, arname)
```

Fortran command syntax:

```
CALL AREA_API_2(IAREA, IBUS, REALAR, ARNAME, IERR)
```

Where:

Integer IAREA	area number between 1 and 9999 (input; no default allowed).	
Integer IBUS	area slack bus number (input; no default allowed).	
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	PDES, desired MW area interchange (0.0 by default)
	REALAR(2)	PTOL, desired MW interchange tolerance (0.1 * SBASE by default; for example, when system base MVA is 100.0, the default value of PTOL is 10 MW)
Character*12 ARNAME	area name (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid area number
	IERR = 2	area table is full
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Area slack bus not found. - Slack bus is not in this area. - No plant at slack bus. - Type code of slack bus is not 2 or 3. - A Type 3 bus is in the area but it is not the area slack bus. - No equipment is assigned to area

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2.2. BRANCH_CHNG

Use this API routine to modify the data of an existing non-transformer branch in the working case (see PSSE Program Operation Manual, Non-Transformer Branch Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_BRANCH_CHNG  IBUS  JBUS  'CKT'  INTGAR(1)..INTGAR(6)
 REALAR(1)..REALAR(15)
```

Python command syntax:

```
ierr = branch_chng(ibus, jbus, ckt, intgar, realar)
```

Fortran command syntax:

```
CALL BRANCH_CHNG(IBUS, JBUS, CKT, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Integer INTGAR (6)	array of 6 elements specifying (input).	
	INTGAR(1)	ST, branch status (default alias is ST) (1 by default)
	INTGAR(2)	METBUS, metered end bus number (IBUS or JBUS) (default alias is METBUS) (IBUS by default)
	INTGAR(3)	O1, first owner number (default alias is O1) (owner of bus IBUS by default)
	INTGAR(4)	O2, second owner number (default alias is O2) (0 by default)
	INTGAR(5)	O3, third owner number (default alias is O3) (0 by default)
	INTGAR(6)	O4, fourth owner number (default alias is O4) (0 by default)
Real REALAR (15)	array of 15 elements specifying (input).	
	REALAR(1)	R, nominal branch resistance (default alias is R) (0.0 by default)
	REALAR(2)	X, nominal branch reactance (default alias is X) (THRSHZ by default; 0.0001 if THRSHZ = 0.0)
	REALAR(3)	B, total line charging (default alias is B) (0.0 by default)
	REALAR(4)	RATEA, rating set A line rating (default alias is RATEA) (0.0 by default)

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	REALAR(5)	RATEB, rating set B line rating (default alias is RATEB) (0.0 by default)
	REALAR(6)	RATEC, rating set C line rating (default alias is RATEC) (0.0 by default)
	REALAR(7)	GI, real line shunt at bus IBUS end (default alias is GI) (0.0 by default)
	REALAR(8)	BI, reactive line shunt at bus IBUS end (default alias is BI) (0.0 by default)
	REALAR(9)	GJ, real line shunt at bus JBUS end (default alias is GJ) (0.0 by default)
	REALAR(10)	BJ, reactive line shunt at bus JBUS end (default alias is BJ) (0.0 by default)
	REALAR(11)	LEN, line length (default alias is LEN) (0.0 by default)
	REALAR(12)	F1, first owner fraction (default alias is F1) (1.0 by default)
	REALAR(13)	F2, second owner fraction (default alias is F2) (1.0 by default)
	REALAR(14)	F3, third owner fraction (default alias is F3) (1.0 by default)
	REALAR(15)	F4, fourth owner fraction (default alias is F4) (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	branch exists but it is a two-winding transformer
	IERR = 9	branch not found
	IERR = 10	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - branch reactance is 0.0 - line length is negative - zero impedance line table is full; branch is treated as a normal line - invalid metered end bus - invalid branch status - invalid owner number - invalid ownership fraction - no ownership data is specified - multi-section line grouping deleted - branch ownership table is full--only first n owners retained - branch is no longer treated as a zero impedance line - owner table is full

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2.3. BRANCH_DATA

Use this API to modify the data of an existing non-transformer branch in the working case or to add a new non-transformer branch to the working case (see PSSE Program Operation Manual, Non-Transformer Branch Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BRANCH_DATA  IBUS  JBUS  'CKT'  INTGAR(1)..INTGAR(6)
 REALAR(1)..REALAR(15)
```

Python command syntax:

```
ierr = branch_data(ibus, jbus, ckt, intgar, realar)
```

Fortran command syntax:

```
CALL BRANCH_API_2(IBUS, JBUS, CKT, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*12 CKT	circuit identifier (input; '1').	
Integer INTGAR (6)	array of 6 elements specifying (input).	
	INTGAR(1)	ST, branch status (default alias is ST) (1 by default)
	INTGAR(2)	METBUS, metered end bus number (IBUS or JBUS) (default alias is METBUS) (IBUS by default)
	INTGAR(3)	O1, first owner number (default alias is O1) (owner of bus IBUS by default)
	INTGAR(4)	O2, second owner number (default alias is O2) (0 by default)
	INTGAR(5)	O3, third owner number (default alias is O3) (0 by default)
	INTGAR(6)	O4, fourth owner number (default alias is O4) (0 by default)
Real REALAR (15)	array of 15 elements specifying (input).	
	REALAR(1)	R, nominal branch resistance (default alias is R) (0.0 by default)
	REALAR(2)	X, nominal branch reactance (default alias is X) (THRSHZ by default; 0.0001 if THRSHZ = 0.0)
	REALAR(3)	B, total line charging (default alias is B) (0.0 by default)
	REALAR(4)	RATEA, rating set A line rating (default alias is RATEA) (0.0 by default)

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	REALAR(5)	RATEB, rating set B line rating (default alias is RATEB) (0.0 by default)
	REALAR(6)	RATEC, rating set C line rating (default alias is RATEC) (0.0 by default)
	REALAR(7)	GI, real line shunt at bus IBUS end (default alias is GI) (0.0 by default)
	REALAR(8)	BI, reactive line shunt at bus IBUS end (default alias is BI) (0.0 by default)
	REALAR(9)	GJ, real line shunt at bus JBUS end (default alias is GJ) (0.0 by default)
	REALAR(10)	BJ, reactive line shunt at bus JBUS end (default alias is BJ) (0.0 by default)
	REALAR(11)	LEN, line length (default alias is LEN) (0.0 by default)
	REALAR(12)	F1, first owner fraction (default alias is F1) (1.0 by default)
	REALAR(13)	F2, second owner fraction (default alias is F2) (1.0 by default)
	REALAR(14)	F3, third owner fraction (default alias is F3) (1.0 by default)
	REALAR(15)	F4, fourth owner fraction (default alias is F4) (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	branch exists but it is a two-winding transformer
	IERR = 4	branch from a bus to itself
	IERR = 5	blank circuit identifier is invalid
	IERR = 6	circuit identifier may not start with &
	IERR = 7	branch table is full
	IERR = 8	branch ownership table is full
	IERR = 10	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - branch reactance is 0.0 - line length is negative - zero impedance line table is full; branch is treated as a normal line - invalid metered end bus - invalid branch status - invalid owner number - invalid ownership fraction - no ownership data is specified - multi-section line grouping deleted - branch ownership table is full--only first n owners retained - branch is no longer treated as a zero impedance line - owner table is full

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2.4. BUS_CHNG_3

Use this API routine to modify the data of an existing bus in the working case (see PSSE Program Operation Manual, Bus Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_BUS_CHNG_3  IBUS  INTGAR(1)..INTGAR(4)  REALAR(1)..REALAR(7)  'NAME'
```

Python command syntax:

```
ierr = bus_chng_3(ibus, intgar, realar, name)
```

Fortran command syntax:

```
CALL BUS_CHNG_3(IBUS, INTGAR, REALAR, NAME, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Integer INTGAR (4)	array of 4 elements specifying (input).	
	INTGAR(1)	IDE, bus type code (1 by default)
	INTGAR(2)	AREA, area number (1 by default)
	INTGAR(3)	ZONE, zone number (1 by default)
	INTGAR(4)	OWNER, owner number (1 by default)
Real REALAR (7)	array of 7 elements specifying (input).	
	REALAR(1)	BASKV, bus base voltage in kV (0.0 by default)
	REALAR(2)	VM, bus voltage magnitude in pu (1.0 by default)
	REALAR(3)	VA, bus voltage phase angle (0.0 by default)
	REALAR(4)	NMAXV, normal bus voltage magnitude high limit in pu (1.1 by default)
	REALAR(5)	NMINV, normal bus voltage magnitude low limit in pu (0.9 by default)
	REALAR(6)	EMAXV, emergency bus voltage magnitude high limit in pu (1.1 by default)
	REALAR(7)	EMINV, emergency bus voltage magnitude low limit in pu (0.9 by default)
Character*12 NAME	bus name (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid bus number
	IERR = 3	working case is not initialized
	IERR = 4	bus not found

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IERR = 5	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Invalid base voltage. - Invalid bus type code. - Invalid area, zone or owner number. - Bus name is more than 12 characters. - New area number but bus is the area swing for its old area. - Bus type code is 3 but is not the swing bus for its area. - Area table is full. - Zone table is full. - Owner table is full. - High voltage limit is not greater than the low voltage limit. - Normal voltage limit is less restrictive than the emergency limit

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2.5. BUS_DATA

This API routine is obsolete. It has been replaced by the API routine bus_data_2 .

Use this API to modify the data of an existing bus in the working case or to add a new bus to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BUS_DATA  IBUS  INTGAR(1)..INTGAR(4)  REALAR(1)..REALAR(5)  'NAME'
```

Python command syntax:

```
ierr = bus_data(ibus, intgar, realar, name)
```

Fortran command syntax:

```
CALL BUS_API(IBUS, INTGAR, REALAR, NAME, IERR, NERR)
```

Where:

Integer IBUS	bus number (input).	
Integer INTGAR (4)	array of 4 elements specifying (input).	
	INTGAR(1)	IDE, bus type code
	INTGAR(2)	AREA, area number
	INTGAR(3)	ZONE, zone number
	INTGAR(4)	OWNER, owner number
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	GL, fixed shunt admittance (conductance)
	REALAR(2)	BL, fixed shunt admittance (susceptance)
	REALAR(3)	BASKV, bus base voltage in kV
	REALAR(4)	VM, bus voltage magnitude in pu
	REALAR(5)	VA, bus voltage phase angle
Character*12 NAME	bus name (input).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid bus number
	IERR = 2	bus table is full
	IERR = 3	working case is not initialized
	IERR = 5	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Invalid base voltage. - Invalid bus type code. - Invalid area, zone or owner number. - Bus name is more than 12 characters. - New area number but bus is the

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	area swing for its old area. - Bus type code is 3 but is not the swing bus for its area. - Area table is full. - Zone table is full. - Owner table is full
Integer NERR	number of entries in PBUF02 containing error messages (output).

2.6. BUS_DATA_2

This API routine is obsolete. It has been replaced by the API routine bus_data_3 .

Use this API to modify the data of an existing bus in the working case or to add a new bus to the working case (see PSSE Program Operation Manual, Bus Data).

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_BUS_DATA_2  IBUS  INTGAR(1)..INTGAR(4)  REALAR(1)..REALAR(3)  'NAME'
```

Python command syntax:

```
ierr = bus_data_2(ibus, intgar, realar, name)
```

Fortran command syntax:

```
CALL BUS_API_2(IBUS, INTGAR, REALAR, NAME, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Integer INTGAR (4)	array of 4 elements specifying (input).	
	INTGAR(1)	IDE, bus type code (1 by default)
	INTGAR(2)	AREA, area number (1 by default)
	INTGAR(3)	ZONE, zone number (1 by default)
	INTGAR(4)	OWNER, owner number (1 by default)
Real REALAR (3)	array of 3 elements specifying (input).	
	REALAR(1)	BASKV, bus base voltage in kV (0.0 by default)
	REALAR(2)	VM, bus voltage magnitude in pu (1.0 by default)
	REALAR(3)	VA, bus voltage phase angle (0.0 by default)
Character*12 NAME	bus name (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid bus number
	IERR = 2	bus table is full
	IERR = 3	working case is not initialized
	IERR = 5	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Invalid base voltage. - Invalid bus type code. - Invalid area, zone or owner number. - Bus name is more than 12 characters. - New area number but bus is the

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area swing for its old area. - Bus type code is 3 but is not the swing bus for its area. - Area table is full. - Zone table is full. - Owner table is full

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2.7. BUS_DATA_3

Use this API routine to modify the data of an existing bus in the working case, or to add a new bus to the working case (see PSSE Program Operation Manual, Bus Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_BUS_DATA_3  IBUS  INTGAR(1)..INTGAR(4)  REALAR(1)..REALAR(7)  'NAME'
```

Python command syntax:

```
ierr = bus_data_3(ibus, intgar, realar, name)
```

Fortran command syntax:

```
CALL BUS_API_3(IBUS, INTGAR, REALAR, NAME, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Integer INTGAR (4)	array of 4 elements specifying (input).	
	INTGAR(1)	IDE, bus type code (1 by default)
	INTGAR(2)	AREA, area number (1 by default)
	INTGAR(3)	ZONE, zone number (1 by default)
	INTGAR(4)	OWNER, owner number (1 by default)
Real REALAR (7)	array of 7 elements specifying (input).	
	REALAR(1)	BASKV, bus base voltage in kV (0.0 by default)
	REALAR(2)	VM, bus voltage magnitude in pu (1.0 by default)
	REALAR(3)	VA, bus voltage phase angle (0.0 by default)
	REALAR(4)	NMAXV, normal bus voltage magnitude high limit in pu (1.1 by default)
	REALAR(5)	NMINV, normal bus voltage magnitude low limit in pu (0.9 by default)
	REALAR(6)	EMAXV, emergency bus voltage magnitude high limit in pu (1.1 by default)
	REALAR(7)	EMINV, emergency bus voltage magnitude low limit in pu (0.9 by default)
Character*12 NAME	bus name (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid bus number
	IERR = 2	bus table is full
	IERR = 3	working case is not initialized

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IERR = 5	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Invalid base voltage. - Invalid bus type code. - Invalid area, zone or owner number. - Bus name is more than 12 characters. - New area number but bus is the area swing for its old area. - Bus type code is 3 but is not the swing bus for its area. - Area table is full. - Zone table is full. - Owner table is full. - High voltage limit is not greater than the low voltage limit. - Normal voltage limit is less restrictive than the emergency limit

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2.8. BUS_NUMBER

Use this API to change the bus number of an existing bus in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BUS_NUMBER  IBUS  NEWBUS
```

Python command syntax:

```
ierr = bus_number(ibus, newbus)
```

Fortran command syntax:

```
CALL BUS_NUMBER_API_2(IBUS, NEWBUS, IERR)
```

Where:

Integer IBUS	current bus number of the bus for which the number is to be changed (input; no default allowed).	
Integer NEWBUS	new bus number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	invalid new bus number
	IERR = 3	bus number NEWBUS is already assigned to another bus
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - None

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2.9. CASE_TITLE_DATA

Use this API to change the two line case title in the working case (see PSSE Program Operation Manual, Case Identification Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CASE_TITLE_DATA  'LINE1'  'LINE2'
```

Python command syntax:

```
ierr = case_title_data(line1, line2)
```

Fortran command syntax:

```
CALL CASE_TITLE_API_2(LINE1, LINE2, IERR)
```

Where:

Character*60 LINE1	first title line (input; unchanged).	
Character*60 LINE2	second title line (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Title line exceeds 60 characters

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2.10. FACTS_CHNG_2

Use this API routine to modify the data of an existing FACTS device in the working case (see PSSE Program Operation Manual, FACTS Device Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_FACTS_CHNG_2  'NAME'  INTGAR(1)..INTGAR(6)  REALAR(1)..REALAR(13)
 'MNAME'
```

Python command syntax:

```
ierr, realaro = facts_chng_2(name, intgar, realari, mname)
```

Fortran command syntax:

```
CALL FACTS_CHNG_2(NAME, INTGAR, REALAR, MNAME, IERR)
```

Where:

Character*12 NAME	name of the FACTS device (input; no default allowed).	
Integer INTGAR (6)	array of 6 elements specifying (input).	
	INTGAR(1)	I, sending end bus number (no default allowed)
	INTGAR(2)	J, terminal end bus number (0 for STATCON) (0 by default)
	INTGAR(3)	MODE, control mode (1 by default)
	INTGAR(4)	OWNER, owner number (1 by default)
	INTGAR(5)	VSREF, series voltage reference code for modes 4 and 7 (0 for sending end voltage; 1 for series current) (0 by default)
	INTGAR(6)	REMOT, remote bus number (0 for sending end bus) (0 by default)
Real REALAR (19)	(first thirteen are values; last six are values) (input (output).).	
Real REALARI (13)	array of 13 elements specifying corresponding to values REALAR(1) through REALAR(13) as described above(input).	
	REALARI(1)	PDES, desired real power at terminal bus (0.0 by default)
	REALARI(2)	QDES, desired reactive power at terminal bus (0.0 by default)
	REALARI(3)	VSET, sending bus voltage setpoint (1.0 by default)
	REALARI(4)	SHMX, sending end maximum shunt current (MVA at rated voltage) (99.99 * SBASE by default)
	REALARI(5)	TRMX, maximum bridge real power transfer (99.99 * SBASE by default)
	REALARI(6)	VTMN, minimum terminal bus voltage magnitude (0.9 by default)

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	REALARI(7)	VTMX, maximum terminal bus voltage magnitude (1.1 by default)
	REALARI(8)	VSMX, maximum series voltage magnitude (1.0 by default)
	REALARI(9)	IMX, maximum series current (MVA at rated voltage) (0.0 by default)
	REALARI(10)	LINX, pu reactance of temporary series element (0.05 by default)
	REALARI(11)	RMPCT, percent of contributed reactive power (100.0 by default)
	REALARI(12)	SET1, real component of constant Z (MODE = 3), magnitude of constant V (MODE = 4), or VD (MODE = 7 or 8) (0.0 by default)
	REALARI(13)	SET2, imaginary component of constant Z (MODE = 3), angle of constant V (MODE = 4), or VQ (MODE = 7 or 8) (0.005 by default when MODE = 3; 0.0 otherwise)
Real REALARO (6)	array of 6 elements specifying corresponding to values REALAR(14) through REALAR(19) as described above(output).	
	REALARO(1)	real component of present effective series Z (returned)
	REALARO(2)	imaginary component of present effective series Z (returned)
	REALARO(3)	magnitude of present series voltage, FDVREF reference (returned)
	REALARO(4)	le of present series voltage, FDVREF reference (returned)
	REALARO(5)	real component of present series voltage, FDVREF reference (returned)
	REALARO(6)	imaginary component of present series voltage, FDVREF reference (returned)
Character*12 MNAME	name of the FACTS device that is the master device for this IPFC slave device (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 2	no sending end bus is specified
	IERR = 3	bus not found
	IERR = 4	the same bus is specified for the sending and terminal end buses
	IERR = 5	blank FACTS device name is invalid
	IERR = 6	FACTS device not found
	IERR = 7	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - Invalid FACTS device control mode. - Invalid owner number. - Invalid series voltage reference code. - Reg-

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ulated bus not found. - Shunt element specified as remotely regulating itself. - Type code of regulated bus is not 1 or 2. - Shunt element voltage setpoint must be positive. - Terminal bus voltage limit must be positive. - VTMX is less than VTMN. - Series voltage limit must be positive. - Shunt current limit must not be negative. - Bridge real power transfer limit must not be negative. - Series current limit must not be negative. - Series reactance must be positive. - Shunt current limit is not 0.0 for an IPFC device. - Bridge real power transfer limit is not 0.0 for an IPFC device. - Impedance setpoint has reactance of 0.0. - RMPCT is not positive. - Master IPFC FACTS device not found. - Bus is dummy bus of a multi-section line. - Multi-section line grouping deleted. - Owner table is full

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2.11. FACTS_DATA

This API routine is obsolete. It has been replaced by the API routine facts_data_2 .

Use this API to modify the data of an existing FACTS device in the working case or to add a new FACTS device to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_FACTS_DATA  FD  INTGAR(1)..INTGAR(5)  REALAR(1)..REALAR(13)
```

Python command syntax:

```
ierr, realaro = facts_data(fd, intgar, realari)
```

Fortran command syntax:

```
CALL FACTS_API(FD, INTGAR, REALAR, IERR, NERR)
```

Where:

Integer FD	number (which the API converts to a left-justified name) of the FACTS device for which data is to be specified (input; no default allowed).	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	I, sending end bus number (default alias is I)
	INTGAR(2)	J, terminal end bus number (0 for STATCON) (default alias is J)
	INTGAR(3)	MODE, control mode (default alias is MODE)
	INTGAR(4)	OWNER, owner number (default alias is OWNER)
	INTGAR(5)	VSREF, series voltage reference code for modes 4 and 7 (0 for sending end voltage; 1 for series current) (default alias is VSREF)
Real REALAR (19)	(input (output).).	
Real REALARI (13)	array of 13 elements specifying corresponding to values REALAR(1) through REALAR(13) as described above(input).	
	REALARI(1)	PDES, desired real power at terminal bus (default alias is PDES)
	REALARI(2)	QDES, desired reactive power at terminal bus (default alias is QDES)
	REALARI(3)	VSET, sending bus voltage setpoint (default alias is VSET)
	REALARI(4)	SHMX, sending end maximum shunt current (MVA at rated voltage) (default alias is SHMAX)
	REALARI(5)	TRMX, maximum bridge real power transfer (default alias is TRMX)

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	REALARI(6)	VTMN, minimum terminal bus voltage magnitude (default alias is VTMN)
	REALARI(7)	VTMX, maximum terminal bus voltage magnitude (default alias is VTMX)
	REALARI(8)	VSMX, maximum series voltage magnitude (default alias is VSMX)
	REALARI(9)	IMX, maximum series current (MVA at rated voltage) (default alias is IMX)
	REALARI(10)	LINX, pu reactance of temporary series element (default alias is LINX)
	REALARI(11)	RMPCT, percent of contributed reactive power (default alias is RMPCT)
	REALARI(12)	SET1, real component of constant Z (mode = 3), magnitude of constant V (mode = 4), or VD (mode = 7 or 8) (default alias is SET1)
	REALARI(13)	SET2, imaginary component of constant Z (mode = 3), angle of constant V (mode = 4), or VQ (mode = 7 or 8) (default alias is SET2)
Real REALARO (6)	array of 6 elements specifying corresponding to values REALAR(14) through REALAR(19) as described above(output).	
	REALARO(1)	real component of present effective series Z (returned)
	REALARO(2)	imaginary component of present effective series Z (returned)
	REALARO(3)	magnitude of present series voltage, FDVREF reference (returned)
	REALARO(4)	le of present series voltage, FDVREF reference (returned)
	REALARO(5)	real component of present series voltage, FDVREF reference (returned)
	REALARO(6)	imaginary component of present series voltage, FDVREF reference (returned)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	FACTS device table is full
	IERR = 2	no sending end bus is specified
	IERR = 3	bus not found
	IERR = 4	the same bus is specified for the sending and terminal end buses
	IERR = 6	prerequisite requirements for API are not met
	IERR = 7	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - Invalid FACTS device control model. - Invalid owner number. - Invalid series voltage reference code. - Shunt element voltage setpoint must be positive. -

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	Terminal bus voltage limit must be positive. - VTMX is less than VTMN. - Series voltage limit must be positive. - Shunt current limit must not be negative. - Bridge real power transfer limit must not be negative. - Series current limit must not be negative. - Series reactance must be positive. - Shunt current limit is not 0.0 for an IPFC device. - Bridge real power transfer limit is not 0.0 for an IPFC device. - Impedance setpoint has reactance of 0.0. - RMPCT is not positive. - Master IPFC FACTS device not found. - Bus is dummy bus of a multi-section line. - Multi-section line grouping deleted. - Owner table is full
Integer NERR	number of entries in PBUF02 containing error messages (output).

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2.12. FACTS_DATA_2

Use this API to modify the data of an existing FACTS device in the working case, or to add a new FACTS device to the working case (see PSSE Program Operation Manual, FACTS Device Data).

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_FACTS_DATA_2  'NAME'  INTGAR(1)..INTGAR(6)  REALAR(1)..REALAR(13)
 'MNAME'
```

Python command syntax:

```
ierr, realaro = facts_data_2(name, intgar, realari, mname)
```

Fortran command syntax:

```
CALL FACTS_API_2(NAME, INTGAR, REALAR, MNAME, IERR)
```

Where:

Character*12 NAME	name of the FACTS device (input; no default allowed).	
Integer INTGAR (6)	array of 6 elements specifying (input).	
	INTGAR(1)	I, sending end bus number (no default allowed)
	INTGAR(2)	J, terminal end bus number (0 for STATCON) (0 by default)
	INTGAR(3)	MODE, control mode (1 by default)
	INTGAR(4)	OWNER, owner number (1 by default)
	INTGAR(5)	VSREF, series voltage reference code for modes 4 and 7 (0 for sending end voltage; 1 for series current) (0 by default)
	INTGAR(6)	REMOT, remote bus number (0 for sending end bus) (0 by default)
Real REALAR (19)	(first thirteen are values; last six are values) (input (output).).	
Real REALARI (13)	array of 13 elements specifying corresponding to values REALAR(1) through REALAR(13) as described above(input).	
	REALARI(1)	PDES, desired real power at terminal bus (0.0 by default)
	REALARI(2)	QDES, desired reactive power at terminal bus (0.0 by default)
	REALARI(3)	VSET, sending bus voltage setpoint (1.0 by default)
	REALARI(4)	SHMX, sending end maximum shunt current (MVA at rated voltage) (99.99 * SBASE by default)
	REALARI(5)	TRMX, maximum bridge real power transfer (99.99 * SBASE by default)
	REALARI(6)	VTMN, minimum terminal bus voltage magnitude (0.9 by default)

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	REALARI(7)	VTMX, maximum terminal bus voltage magnitude (1.1 by default)
	REALARI(8)	VSMX, maximum series voltage magnitude (1.0 by default)
	REALARI(9)	IMX, maximum series current (MVA at rated voltage) (0.0 by default)
	REALARI(10)	LINX, pu reactance of temporary series element (0.05 by default)
	REALARI(11)	RMPCT, percent of contributed reactive power (100.0 by default)
	REALARI(12)	SET1, real component of constant Z (MODE = 3), magnitude of constant V (MODE = 4), or VD (MODE = 7 or 8) (0.0 by default)
	REALARI(13)	SET2, imaginary component of constant Z (MODE = 3), angle of constant V (MODE = 4), or VQ (MODE = 7 or 8) (0.005 by default when MODE = 3; 0.0 otherwise)
Real REALARO (6)	array of 6 elements specifying corresponding to values REALAR(14) through REALAR(19) as described above(output).	
	REALARO(1)	real component of present effective series Z (returned)
	REALARO(2)	imaginary component of present effective series Z (returned)
	REALARO(3)	magnitude of present series voltage, FDVREF reference (returned)
	REALARO(4)	le of present series voltage, FDVREF reference (returned)
	REALARO(5)	real component of present series voltage, FDVREF reference (returned)
	REALARO(6)	imaginary component of present series voltage, FDVREF reference (returned)
Character*12 MNAME	name of the FACTS device that is the master device for this IPFC slave device (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	FACTS device table is full
	IERR = 2	no sending end bus is specified
	IERR = 3	bus not found
	IERR = 4	the same bus is specified for the sending and terminal end buses
	IERR = 5	blank FACTS device name is invalid
	IERR = 6	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - Invalid FACTS device control mode. - Invalid owner number. - Invalid series voltage reference code. - Reg-

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ulated bus not found. - Shunt element specified as remotely regulating itself. - Type code of regulated bus is not 1 or 2. - Shunt element voltage setpoint must be positive. - Terminal bus voltage limit must be positive. - VTMX is less than VTMN. - Series voltage limit must be positive. - Shunt current limit must not be negative. - Bridge real power transfer limit must not be negative. - Series current limit must not be negative. - Series reactance must be positive. - Shunt current limit is not 0.0 for an IPFC device. - Bridge real power transfer limit is not 0.0 for an IPFC device. - Impedance setpoint has reactance of 0.0. - RMPCT is not positive. - Master IPFC FACTS device not found. - Bus is dummy bus of a multi-section line. - Multi-section line grouping deleted. - Owner table is full

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2.13. GNE_CHNG

Use this API routine to modify the data of an existing GNE device in the working case (see PSSE Program Operation Manual, GNE Device Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_GNE_CHNG  'NAME'  'MODEL'  NTERM  TRMNLS(1)..TRMNLS(NTERM)
 INTGAR(1)..INTGAR(3)  NRELCN  RELCON(1)..RELCON(NRELCN)  NINTCN
 INTCON(1)..INTCON(NINTCN)  NCHRCN  CHRCON(1)..CHRCON(NCHRCN)
```

Python command syntax:

```
ierr = gne_chng(name, model, nterm, trmnls, intgar, nrelcn, relcon,
 nintcn, intcon, nchrcn, chrcon)
```

Fortran command syntax:

```
CALL GNE_CHNG(NAME, MODEL, NTERM, TRMNLS, INTGAR, NRELCN, RELCON, NINTCN,
 INTCON, NCHRCN, CHRCON, IERR)
```

Where:

Character*12 NAME	name of the GNE device (input; no default allowed).
Character*16 MODEL	name of the BOSL model to be used by this GNE device. The BOSL model file "MODEL.mac" must be available (input; no default allowed).
Integer NTERM	number of terminals (i.e., buses) to which the model is connected. NTERM must be 1 for GNE-PQ and GNE-I models, and it must be either 1 or 2 for a GNE-Y model (input; no default allowed).
Integer TRMNLS (NTERM)	containing the bus numbers of the buses to which this model is connected (input; no default allowed).
Integer INTGAR (3)	array of 3 elements specifying (input). INTGAR(1) STATUS, model status (0 for out-of-service, 1 for in-service) (1 by default) INTGAR(2) OWNER, owner number (the owner to which bus TRMNLS (1) is assigned by default) INTGAR(3) NMETR, non-metered end bus number (TRMNLS (NTERM) by default)
Integer NRELCN	number of real constants required by the model (input; no default allowed).
Real RELCON (NRELCN)	real model constants (input; 0.0).
Integer NINTCN	number of integer constants required by the model (input; no default allowed).
Integer INTCON (NINTCN)	integer model constants (usually bus numbers) (TRMNLS) (input; 1 by default).

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Integer NCHRCN	number of two-character identifiers required required by the model (input; no default allowed).	
Character*12 CHRCON (NCHRCN)	containing character model constants (e.g., circuit identifiers) (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	blank GNE device name is invalid
	IERR = 2	invalid number of terminal buses for this BOSL model
	IERR = 3	bus not found
	IERR = 4	model is not a GNE model
	IERR = 5	number of constants specified differs from those required in the model's.mac file
	IERR = 7	blank model name is invalid for a new GNE device instance
	IERR = 8	GNE.mac file is not found or an error occurred while reading it
	IERR = 9	GNE device not found
	IERR = 10	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - Invalid GNE device status. - Invalid owner number. - Invalid non-metered end bus. - Bus specified in an integer constant is not found. - Owner table is full

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2.14. GNE_DATA

Use this API routine to run the GNE device data specification function. It modifies the data of an existing GNE device in the working case, or adds a new GNE device to the working case (see PSSE Program Operation Manual, GNE Device Data).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_GNE_DATA  'NAME'  'MODEL'  NTERM  TRMNLS(1)..TRMNLS(NTERM)
 INTGAR(1)..INTGAR(3)  NRELCN  RELCON(1)..RELCON(NRELCN)  NINTCN
 INTCON(1)..INTCON(NINTCN)  NCHRCN  CHRCON(1)..CHRCON(NCHRCN)
```

Python command syntax:

```
ierr = gne_data(name, model, nterm, trmnls, intgar, nrelcn, relcon,
 nintcn, intcon, nchrcn, chrcon)
```

Fortran command syntax:

```
CALL GNE_API(NAME, MODEL, NTERM, TRMNLS, INTGAR, NRELCN, RELCON, NINTCN,
 INTCON, NCHRCN, CHRCON, IERR)
```

Where:

Character*12 NAME	name of the GNE device (input; no default allowed).
Character*16 MODEL	name of the BOSL model to be used by this GNE device. The BOSL model file "MODEL.mac" must be available (input; no default allowed).
Integer NTERM	number of terminals (i.e., buses) to which the model is connected. NTERM must be 1 for GNE-PQ and GNE-I models, and it must be either 1 or 2 for a GNE-Y model (input; no default allowed).
Integer TRMNLS (NTERM)	containing the bus numbers of the buses to which this model is connected (input; no default allowed).
Integer INTGAR (3)	array of 3 elements specifying (input).
	INTGAR(1) STATUS, model status (0 for out-of-service, 1 for in-service) (1 by default)
	INTGAR(2) OWNER, owner number (the owner to which bus TRMNLS (1) is assigned by default)
	INTGAR(3) NMETR, non-metered end bus number (TRMNLS (NTERM) by default)
Integer NRELCN	number of real constants required by the model (input; no default allowed).
Real RELCON (NRELCN)	real model constants (input; 0.0).
Integer NINTCN	number of integer constants required by the model (input; no default allowed).
Integer INTCON (NINTCN)	integer model constants (usually bus numbers) (TRMNLS) (input; 1 by default).

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Integer NCHRCN	number of two-character identifiers required required by the model (input; no default allowed).	
Character*12 CHRCON (NCHRCN)	containing character model constants (e.g., circuit identifiers) (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	blank GNE device name is invalid
	IERR = 2	invalid number of terminal buses for this BOSL model
	IERR = 3	bus not found
	IERR = 4	model is not a GNE model
	IERR = 5	number of constants specified differs from those required in the model's.mac file
	IERR = 6	GNE device instance table is full
	IERR = 7	blank model name is invalid for a new GNE device instance
	IERR = 8	GNE.mac file is not found or an error occurred while reading it
	IERR = 10	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - Invalid GNE device status. - Invalid owner number. - Invalid non-metered end bus. - Bus specified in an integer constant is not found. - Owner table is full

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2.15. IMPEDANCE_CORRECTION_DATA

Use this API to modify the data of an existing transformer impedance correction table in the working case or to add a new transformer impedance correction table to the working case (see PSSE Program Operation Manual, Transformer Impedance Correction Table Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_IMPEDANCE_CORRECTION_DATA  I   REALAR(1,1)  REALAR(2,1)  REALAR(1,2)
 REALAR(2,2)..REALAR(1,11)  REALAR(2,11)
```

Python command syntax:

```
ierr, intgar = impedance_correction_data(i, realar)
```

Fortran command syntax:

```
CALL IMPEDANCE_CORRECTION_TAB_API_2(I, INTGAR, REALAR, IERR)
```

Where:

Integer I	table number (must be between 1 and the number of tables allowed at the present size level of PSSE) (input; no default allowed).	
Integer INTGAR (2)	array of 2 elements specifying (output).	
	INTGAR(1)	number of entries in table (returned)
	INTGAR(2)	table type flag (0=ratio, 1=angle) (returned)
Real REALAR (22)	22 values dimensioned (2,11), where REALAR(1,i) specifies Ti, and REALAR(2,i) specifies Fi. The values are specified such that the left subscript varies more rapidly: (1,1), (2,1), (1,2), (2,2), .. (1,11), (2,11). (REALAR(1,1) = REALAR (input; 2,1) =1.0 by default; all other values are 0.0 by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid transformer impedance correction table number
	IERR = 3	ABSCISSA values are not in ascending order
	IERR = 4	ABSCISSA value is outside of the range of -180 to +180
	IERR = 5	table has fewer than 2 points
	IERR = 6	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - point is beyond previous end of table - invalid scaling factor

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2.16. INDUCTION_MACHINE_CHNG

Use this API routine to modify the data of an existing induction machine in the working case (see PSSE Program Operation Manual, Induction Machine Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_INDUCTION_MACHINE_CHNG  IBUS  'ID'  INTGAR(1)..INTGAR(9)
 REALAR(1)..REALAR(23)
```

Python command syntax:

```
ierr = induction_machine_chng(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL INDUCTION_MACHINE_CHNG(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Integer INTGAR (9)	array of 9 elements specifying (input).	
	INTGAR(1)	STAT, induction machine status (1 by default)
	INTGAR(2)	SCODE, machine standard type (1 for NEMA, 2 for IEC) (1 by default)
	INTGAR(3)	DCODE, machine design code (0 for custom, 1 through 5 for MEMA designs A through E) (2 by default)
	INTGAR(4)	AREA, area number (area of bus IBUS by default)
	INTGAR(5)	ZONE, zone number (zone of bus IBUS by default)
	INTGAR(6)	OWNER, owner number (owner of bus IBUS by default)
	INTGAR(7)	TCODE, type of mechanical torque variation (1 for simple power law, 2 for WECC model) (1 by default)
	INTGAR(8)	BCODE, base machine power code (1 for mechanical power 2 for electrical power drawn) (1 by default)
	INTGAR(9)	PCODE, power setpoint code (1 for mechanical power 2 for electrical power drawn) (1 by default)
Real REALAR (23)	array of 23 elements specifying (input).	
	REALAR(1)	MBASE, machine base power (SBASE by default)

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	REALAR(2)	RATEKV, machine rated voltage (kV line to line); 0.0 for bus base voltage (0.0 by default)
	REALAR(3)	PSET, machine power setpoint in MW (positive for a motor) (0.0 by default)
	REALAR(4)	H, machine inertia in pu on machine base power (1.0 by default)
	REALAR(5)	A, load torque variation constant (1.0 by default)
	REALAR(6)	B, load torque variation constant (1.0 by default)
	REALAR(7)	D, load torque variation constant (1.0 by default)
	REALAR(8)	E, load torque variation constant (1.0 by default)
	REALAR(9)	RA, armature resistance (0.02494 by default)
	REALAR(10)	XA, armature leakage reactance (0.07482 by default)
	REALAR(11)	XM, magnetizing reactance (2.32786 by default)
	REALAR(12)	R1, first cage resistance (0.02078 by default)
	REALAR(13)	X1, first cage reactance (0.09145 by default)
	REALAR(14)	R2, second cage resistance (0.12471 by default)
	REALAR(15)	X2, second cage reactance (0.03326 by default)
	REALAR(16)	X3, third reactance (0.0 by default)
	REALAR(17)	E1, first voltage point on saturation curve (1.0 by default)
	REALAR(18)	SE1, saturation factor at E1 (0.04 by default)
	REALAR(19)	E2, second voltage point on saturation curve (1.2 by default)
	REALAR(20)	SE2, saturation factor at E2 (0.2 by default)
	REALAR(21)	IA1, starting stator current for applying stator leakage reactance saturation (1.0 by default)
	REALAR(22)	IA2, stator current at wiich the stator leakage reactance is fully saturated (5.0 by default)
	REALAR(23)	XAMULT, Xa multiplier at full leakage reactance saturation (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	machine identifier is more than two characters
	IERR = 5	induction machine not found

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IERR = 6	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - invalid machine status. - invalid machine standard code. - invalid machine design code. - invalid area, zone or owner number. - invalid mechanical torque variation method code. - invalid base power code. - invalid power setpoint code. - machine base power is not positive. - machine rated voltage is negative. - machine rated voltage is ignored because the bus has no base voltage specified for it. - bus base voltage and machine rated voltage differ by more than 20%. - power setpoint exceeds the threshold determined from the standard code and the power setpoint code. - invalid machine inertia. - magnitude of load torque variation constant D is not between 0.2 and 5.0. - the sign of load torque variation constant D is not consistent with that of the machine power setpoint. - machine parameter is negative. - XM is not positive. - Rn or Xn exceeds 999.0. - Rn and Xn are both 0.0. - one of R1 and Xn is 999.0 and the other is not. - E1 is greater than E2; interchanging E1 and E2. - E1 and E2 are equal. - SE1 is greater than SE2; interchanging SE1 and SE2. - IA1 is greater than IA2; interchanging IA1 and IA2. - IA1 and IA2 are equal. - XAMULT must be between 0.0 and 1.0. - area, zone or owner table is full

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2.17. INDUCTION_MACHINE_DATA

Use this API routine to modify the data of an existing induction machine in the working case, or to add a new induction machine to the working case (see PSSE Program Operation Manual, Induction Machine Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_INDUCTION_MACHINE_DATA  IBUS  'ID'  INTGAR(1)..INTGAR(9)
 REALAR(1)..REALAR(23)
```

Python command syntax:

```
ierr = induction_machine_data(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL INDUCTION_MACHINE_API(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Integer INTGAR (9)	array of 9 elements specifying (input).	
	INTGAR(1)	STAT, induction machine status (1 by default)
	INTGAR(2)	SCODE, machine standard type (1 for NEMA, 2 for IEC) (1 by default)
	INTGAR(3)	DCODE, machine design code (0 for custom, 1 through 5 for MEMA designs A through E) (2 by default)
	INTGAR(4)	AREA, area number (area of bus IBUS by default)
	INTGAR(5)	ZONE, zone number (zone of bus IBUS by default)
	INTGAR(6)	OWNER, owner number (owner of bus IBUS by default)
	INTGAR(7)	TCODE, type of mechanical torque variation (1 for simple power law, 2 for WECC model) (1 by default)
	INTGAR(8)	BCODE, base machine power code (1 for mechanical power 2 for electrical power drawn) (1 by default)
	INTGAR(9)	PCODE, power setpoint code (1 for mechanical power 2 for electrical power drawn) (1 by default)
Real REALAR (23)	array of 23 elements specifying (input).	
	REALAR(1)	MBASE, machine base power (SBASE by default)

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	REALAR(2)	RATEKV, machine rated voltage (kV line to line); 0.0 for bus base voltage (0.0 by default)
	REALAR(3)	PSET, machine power setpoint in MW (positive for a motor) (0.0 by default)
	REALAR(4)	H, machine inertia in pu on machine base power (1.0 by default)
	REALAR(5)	A, load torque variation constant (1.0 by default)
	REALAR(6)	B, load torque variation constant (1.0 by default)
	REALAR(7)	D, load torque variation constant (1.0 by default)
	REALAR(8)	E, load torque variation constant (1.0 by default)
	REALAR(9)	RA, armature resistance (0.02494 by default)
	REALAR(10)	XA, armature leakage reactance (0.07482 by default)
	REALAR(11)	XM, magnetizing reactance (2.32786 by default)
	REALAR(12)	R1, first cage resistance (0.02078 by default)
	REALAR(13)	X1, first cage reactance (0.09145 by default)
	REALAR(14)	R2, second cage resistance (0.12471 by default)
	REALAR(15)	X2, second cage reactance (0.03326 by default)
	REALAR(16)	X3, third reactance (0.0 by default)
	REALAR(17)	E1, first voltage point on saturation curve (1.0 by default)
	REALAR(18)	SE1, saturation factor at E1 (0.04 by default)
	REALAR(19)	E2, second voltage point on saturation curve (1.2 by default)
	REALAR(20)	SE2, saturation factor at E2 (0.2 by default)
	REALAR(21)	IA1, starting stator current for applying stator leakage reactance saturation (1.0 by default)
	REALAR(22)	IA2, stator current at wiich the stator leakage reactance is fully saturated (5.0 by default)
	REALAR(23)	XAMULT, Xa multiplier at full leakage reactance saturation (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	machine identifier is more than two characters
	IERR = 3	blank machine identifier is invalid

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IERR = 4	induction machine table is full
IERR = 6	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - invalid machine status. - invalid machine standard code. - invalid machine design code. - invalid area, zone or owner number. - invalid mechanical torque variation method code. - invalid base power code. - invalid power setpoint code. - machine base power is not positive. - machine rated voltage is negative. - machine rated voltage is ignored because the bus has no base voltage specified for it. - bus base voltage and machine rated voltage differ by more than 20%. - power setpoint exceeds the threshold determined from the standard code and the power setpoint code. - invalid machine inertia. - magnitude of load torque variation constant D is not between 0.2 and 5.0. - the sign of load torque variation constant D is not consistent with that of the machine power setpoint. - machine parameter is negative. - XM is not positive. - Rn or Xn exceeds 999.0. - Rn and Xn are both 0.0. - one of R1 and Xn is 999.0 and the other is not. - E1 is greater than E2; interchanging E1 and E2. - E1 and E2 are equal. - SE1 is greater than SE2; interchanging SE1 and SE2. - IA1 is greater than IA2; interchanging IA1 and IA2. - IA1 and IA2 are equal. - XAMULT must be between 0.0 and 1.0. - area, zone or owner table is full

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2.18. LOAD_CHNG_4

Use this API routine to modify the data of an existing load in the working case (see PSSE Program Operation Manual, Load Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_LOAD_CHNG_4  IBUS 'ID' INTGAR(1)..INTGAR(6) REALAR(1)..REALAR(6)
```

Python command syntax:

```
ierr = load_chng_4(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL LOAD_CHNG_4(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	load identifier (input; '1').	
Integer INTGAR (6)	array of 6 elements specifying (input).	
	INTGAR(1)	STATUS, load status (1 by default)
	INTGAR(2)	AREA, area number (area of bus IBUS by default)
	INTGAR(3)	ZONE, zone number (zone of bus IBUS by default)
	INTGAR(4)	OWNER, owner number (owner of bus IBUS by default)
	INTGAR(5)	SCALE, load scaling flag (0 = fixed, non-conforming; 1 = scalable, conforming) (1 by default)
	INTGAR(6)	INTRPT, interruptible load flag (0 = non-interruptible; 1 = interruptible) (0 by default)
Real REALAR (6)	array of 6 elements specifying (input).	
	REALAR(1)	PL, constant power active load (0.0 by default)
	REALAR(2)	QL, constant power reactive load (0.0 by default)
	REALAR(3)	IP, constant current active load (0.0 by default)
	REALAR(4)	IQ, constant current reactive load (0.0 by default)
	REALAR(5)	YP, constant admittance active load (0.0 by default)
	REALAR(6)	YQ, constant admittance reactive load (0.0 by default)

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Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	load identifier is more than two characters
	IERR = 5	load not found
	IERR = 6	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid load status. - invalid area, zone or owner number. - area table is full. - zone table is full. - owner table is full. - invalid load scaling flag. - invalid interruptible load flag

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2.19. LOAD_DATA

This API routine is obsolete. It has been replaced by the API routine load_data_3 .

Use this API to modify the data of an existing load in the working case or to add a new load to the working case.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_LOAD_DATA  IBUS  'ID'  INTGAR(1)..INTGAR(4)  REALAR(1)..REALAR(6)
```

Python command syntax:

```
ierr = load_data(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL LOAD_API_2(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	load identifier (input; '1').	
Integer INTGAR (4)	array of 4 elements specifying (input).	
	INTGAR(1)	STATUS, load status (1 by default)
	INTGAR(2)	AREA, area number (area of bus IBUS by default)
	INTGAR(3)	ZONE, zone number (zone of bus IBUS by default)
	INTGAR(4)	OWNER, owner number (owner of bus IBUS by default)
Real REALAR (6)	array of 6 elements specifying (input).	
	REALAR(1)	PL, constant power active load (0.0 by default)
	REALAR(2)	QL, constant power reactive load (0.0 by default)
	REALAR(3)	IP, constant current active load (0.0 by default)
	REALAR(4)	IQ, constant current reactive load (0.0 by default)
	REALAR(5)	YP, constant admittance active load (0.0 by default)
	REALAR(6)	YQ, constant admittance reactive load (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred

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IERR > 0	fatal error occurred
IERR = 1	bus not found
IERR = 2	load identifier is more than two characters
IERR = 3	blank load identifier is invalid
IERR = 4	load table is full
IERR = 6	prerequisite requirements for API are not me
IERR = -1	data error, warning and information messages; one or more of: - invalid load status. - invalid area, zone or owner number. - Area table is full. - Zone table is full. - Owner table is full

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2.20. LOAD_DATA_3

This API routine is obsolete. It has been replaced by the API routine load_data_4 .

Use this API to modify the data of an existing load in the working case or to add a new load to the working case.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_LOAD_DATA_3  IBUS 'ID' INTGAR(1)..INTGAR(5) REALAR(1)..REALAR(6)
```

Python command syntax:

```
ierr = load_data_3(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL LOAD_API_3(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	load identifier (input; '1').	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	STATUS, load status (1 by default)
	INTGAR(2)	AREA, area number (area of bus IBUS by default)
	INTGAR(3)	ZONE, zone number (zone of bus IBUS by default)
	INTGAR(4)	OWNER, owner number (owner of bus IBUS by default)
	INTGAR(5)	SCALE, load scaling flag (0 = fixed, non-conforming; 1 = scalable, conforming) (1 by default)
Real REALAR (6)	array of 6 elements specifying (input).	
	REALAR(1)	PL, constant power active load (0.0 by default)
	REALAR(2)	QL, constant power reactive load (0.0 by default)
	REALAR(3)	IP, constant current active load (0.0 by default)
	REALAR(4)	IQ, constant current reactive load (0.0 by default)
	REALAR(5)	YP, constant admittance active load (0.0 by default)
	REALAR(6)	YQ, constant admittance reactive load (0.0 by default)

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Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	load identifier is more than two characters
	IERR = 3	blank load identifier is invalid
	IERR = 4	load table is full
	IERR = 6	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid load status. - invalid area, zone or owner number. - area table is full. - zone table is full. - owner table is full. - invalid load scaling flag

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2.21. LOAD_DATA_4

Use this API routine to modify the data of an existing load in the working case, or to add a new load to the working case (see PSSE Program Operation Manual, Load Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_LOAD_DATA_4  IBUS 'ID' INTGAR(1)..INTGAR(6) REALAR(1)..REALAR(6)
```

Python command syntax:

```
ierr = load_data_4(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL LOAD_API_4(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	load identifier (input; '1').	
Integer INTGAR (6)	array of 6 elements specifying (input).	
	INTGAR(1)	STATUS, load status (1 by default)
	INTGAR(2)	AREA, area number (area of bus IBUS by default)
	INTGAR(3)	ZONE, zone number (zone of bus IBUS by default)
	INTGAR(4)	OWNER, owner number (owner of bus IBUS by default)
	INTGAR(5)	SCALE, load scaling flag (0 = fixed, non-conforming; 1 = scalable, conforming) (1 by default)
	INTGAR(6)	INTRPT, interruptible load flag (0 = non-interruptible; 1 = interruptible) (0 by default)
Real REALAR (6)	array of 6 elements specifying (input).	
	REALAR(1)	PL, constant power active load (0.0 by default)
	REALAR(2)	QL, constant power reactive load (0.0 by default)
	REALAR(3)	IP, constant current active load (0.0 by default)
	REALAR(4)	IQ, constant current reactive load (0.0 by default)
	REALAR(5)	YP, constant admittance active load (0.0 by default)
	REALAR(6)	YQ, constant admittance reactive load (0.0 by default)

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Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	load identifier is more than two characters
	IERR = 3	blank load identifier is invalid
	IERR = 4	load table is full
	IERR = 6	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid load status. - invalid area, zone or owner number. - area table is full. - zone table is full. - owner table is full. - invalid load scaling flag. - invalid interruptible load flag

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2.22. LONG_TITLE_DATA

Use this API to change the sixteen line long title in the working case (see PSSE Program Operation Manual, activity CHTI).

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_LONG_TITLE_DATA  'LNGLNS(1)'..'LNGLNS(16)'
```

Python command syntax:

```
ierr = long_title_data(lnglns)
```

Fortran command syntax:

```
CALL LONG_TITLE_API_2(LNGLNS, IERR)
```

Where:

Character*72 LNGLNS (16)	sixteen title lines (defaulted lines are unchanged) (input).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Title line exceeds 72 characters

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2.23. MACHINE_CHNG_2

Use this API routine to modify the data of an existing machine in the working case (see PSSE Program Operation Manual, Fixed Bus Shunt Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MACHINE_CHNG_2  IBUS  'ID'  INTGAR(1)..INTGAR(6)
 REALAR(1)..REALAR(17)
```

Python command syntax:

```
ierr = machine_chng_2(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL MACHINE_CHNG_2(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Integer INTGAR (6)	array of 6 elements specifying (input).	
	INTGAR(1)	STAT, machine status (1 by default)
	INTGAR(2)	O1, first owner number (owner of bus I by default)
	INTGAR(3)	O2, second owner number (0 by default)
	INTGAR(4)	O3, third owner number (0 by default)
	INTGAR(5)	O4, fourth owner number (0 by default)
	INTGAR(6)	WMOD, wind machine reactive power limits mode (0 if this machine is not a wind machine) (0 by default)
Real REALAR (17)	array of 17 elements specifying (input).	
	REALAR(1)	PG, machine active power (0.0 by default)
	REALAR(2)	QG, machine reactive power (0.0 by default)
	REALAR(3)	QT, machine reactive power upper limit (9999.0 by default)
	REALAR(4)	QB, machine reactive power lower limit (-9999.0 by default)
	REALAR(5)	PT, machine active power upper limit (9999.0 by default)
	REALAR(6)	PB, machine active power lower limit (-9999.0 by default)
	REALAR(7)	MBASE, machine MVA base (SBASE by default)
	REALAR(8)	ZR, machine resistance (0.0 by default)
	REALAR(9)	ZX, machine reactance (1.0 by default)

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	REALAR(10)	RT, step-up transformer resistance (0.0 by default)
	REALAR(11)	XT, step-up transformer reactance (0.0 by default)
	REALAR(12)	GTAP, step-up transformer tap ratio (1.0 by default)
	REALAR(13)	F1, first owner fraction (1.0 by default)
	REALAR(14)	F2, second owner fraction (1.0 by default)
	REALAR(15)	F3, third owner fraction (1.0 by default)
	REALAR(16)	F4, fourth owner fraction (1.0 by default)
	REALAR(17)	WPF, wind machine power factor (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	machine identifier is more than two characters
	IERR = 3	no plant data at this bus
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid machine status - invalid wind machine control mode - invalid wind machine power factor - invalid ZSORCE reactance - invalid owner number - invalid ownership fraction - no ownership data is specified - QMAX is less than QMIN - PMAX is less than PMIN - MBASE is not positive - wind machine has negative active power - machine is no longer treated as a wind machine - wind machine table is full--retained as a non-wind machine - machine ownership table is full--only first n owners retained - owner table is full

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2.24. MACHINE_DATA

This API routine is obsolete. It has been replaced by the API routine machine_data_2 .

Use this API to modify the data of an existing machine in the working case or to add a new machine to a plant bus in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MACHINE_DATA  IBUS  'ID'  INTGAR(1)..INTGAR(5)  REALAR(1)..REALAR(16)
```

Python command syntax:

```
ierr = machine_data(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL MACHINE_API(IBUS, ID, INTGAR, REALAR, IERR, NERR)
```

Where:

Integer IBUS	bus number (input).	
Character*2 ID	machine identifier (input).	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	STAT, machine status
	INTGAR(2)	O1, first owner number
	INTGAR(3)	O2, second owner number
	INTGAR(4)	O3, third owner number
	INTGAR(5)	O4, fourth owner number
Real REALAR (16)	array of 16 elements specifying (input).	
	REALAR(1)	PG, machine active power
	REALAR(2)	QG, machine reactive power
	REALAR(3)	QT, machine reactive power upper limit
	REALAR(4)	QB, machine reactive power lower limit
	REALAR(5)	PT, machine active power upper limit
	REALAR(6)	PB, machine active power lower limit
	REALAR(7)	MBASE, machine MVA base
	REALAR(8)	ZR, machine resistance
	REALAR(9)	ZX, machine reactance
	REALAR(10)	RT, step-up transformer resistance
	REALAR(11)	XT, step-up transformer reactance
	REALAR(12)	GTAP, step-up transformer tap ratio
	REALAR(13)	F1, first owner fraction
	REALAR(14)	F2, second owner fraction
	REALAR(15)	F3, third owner fraction

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	REALAR(16)	F4, fourth owner fraction
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	machine identifier is more than two characters
	IERR = 3	no plant data at bus
	IERR = 4	blank machine identifier is invalid
	IERR = 5	machine table full
	IERR = 6	machine ownership table full
	IERR = 7	prerequisite requirements for API are not met
	IERR = 8	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid machine status - invalid ZSORCE reactance - invalid ZSORCE: (0.0,0.0) - invalid owner number - invalid ownership fraction - no ownership data is specified - QMAX is less than QMIN - PMAX is less than PMIN - MBASE is not positive - machine ownership table is full--only first n owners retained - owner table is full
Integer NERR	number of entries in PBUF02 containing error messages (output).	

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2.25. MACHINE_DATA_2

Use this API to modify the data of an existing machine in the working case, or to add a new machine to a plant bus in the working case (see PSSE Program Operation Manual, Fixed Bus Shunt Data).

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_MACHINE_DATA_2  I  'ID'  INTGAR(1)..INTGAR(6)  REALAR(1)..REALAR(17)
```

Python command syntax:

```
ierr = machine_data_2(i, id, intgar, realar)
```

Fortran command syntax:

```
CALL MACHINE_API_2(I, ID, INTGAR, REALAR, IERR)
```

Where:

Integer I	bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Integer INTGAR (6)	array of 6 elements specifying (input).	
	INTGAR(1)	STAT, machine status (1 by default)
	INTGAR(2)	O1, first owner number (owner of bus I by default)
	INTGAR(3)	O2, second owner number (0 by default)
	INTGAR(4)	O3, third owner number (0 by default)
	INTGAR(5)	O4, fourth owner number (0 by default)
	INTGAR(6)	WMOD, wind machine reactive power limits mode (0 if this machine is not a wind machine) (0 by default)
Real REALAR (17)	array of 17 elements specifying (input).	
	REALAR(1)	PG, machine active power (0.0 by default)
	REALAR(2)	QG, machine reactive power (0.0 by default)
	REALAR(3)	QT, machine reactive power upper limit (9999.0 by default)
	REALAR(4)	QB, machine reactive power lower limit (-9999.0 by default)
	REALAR(5)	PT, machine active power upper limit (9999.0 by default)
	REALAR(6)	PB, machine active power lower limit (-9999.0 by default)
	REALAR(7)	MBASE, machine MVA base (SBASE by default)
	REALAR(8)	ZR, machine resistance (0.0 by default)
	REALAR(9)	ZX, machine reactance (1.0 by default)

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	REALAR(10)	RT, step-up transformer resistance (0.0 by default)
	REALAR(11)	XT, step-up transformer reactance (0.0 by default)
	REALAR(12)	GTAP, step-up transformer tap ratio (1.0 by default)
	REALAR(13)	F1, first owner fraction (1.0 by default)
	REALAR(14)	F2, second owner fraction (1.0 by default)
	REALAR(15)	F3, third owner fraction (1.0 by default)
	REALAR(16)	F4, fourth owner fraction (1.0 by default)
	REALAR(17)	WPF, wind machine power factor (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	machine identifier is more than two characters
	IERR = 3	no plant data at this bus
	IERR = 4	blank machine identifier is invalid
	IERR = 5	machine table is full
	IERR = 6	machine ownership table is full
	IERR = 8	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid machine status - invalid wind machine control mode - invalid wind machine power factor - invalid ZSORCE: reactance - invalid owner number - invalid ownership fraction - no ownership data is specified - QMAX is less than QMIN - PMAX is less than PMIN - MBASE is not positive - wind machine has negative active power - machine is no longer treated as a wind machine - wind machine table is full--retained as a non-wind machine - machine ownership table is full--only first n owners retained - owner table is full

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2.26. MULTI_SECTION_LINE_DATA

Use this API to modify the definition of an existing multi-section line grouping in the working case or to add a new multi-section line grouping to the working case (see PSSE Program Operation Manual, Multi-Section Line Grouping Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MULTI_SECTION_LINE_DATA  IBUS  JBUS  'ID'  INTGAR(1)..INTGAR(10)
```

Python command syntax:

```
ierr = multi_section_line_data(ibus, jbus, id, intgar)
```

Fortran command syntax:

```
CALL MULTI_SECTION_LINE_API_2(IBUS, JBUS, ID, INTGAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 ID	a multi-section line identifier; ID(1:1) must be '&' (input; '&1').	
Integer INTGAR (10)	array of 10 elements specifying (input).	
	INTGAR(1)	METBUS, metered end bus number (must be IBUS or JBUS) (IBUS by default)
	INTGAR(2)	DUM1 through DUM9 entries 2 through 10 are the bus numbers of the dummy buses encountered along this multi-section line, starting with the bus to which bus 'IBUS' is connected and ending with the bus to which bus 'JBUS' is connected; entries used are 2 through n+1, where n is the number of dummy buses; if n+1 is less than 10, entries n+2 through 10 must be set to 0 (No default is allowed for entry 2; entries 3 through 10 are 0)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	multi-section line identifier is more than two characters
	IERR = 3	multi-section line identifier does not start with &
	IERR = 4	no dummy buses specified
	IERR = 5	line section not found

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IERR = 6	circuit is already a member of another multi-section line
IERR = 7	bus appears more than once in the multi-section line path
IERR = 8	dummy bus does not have exactly two branches connected to it
IERR = 9	dummy bus is a converter bus of a dc line
IERR = 10	a FACTS device is connected to a dummy bus
IERR = 11	line section table is full
IERR = 12	multi-section line table is full
IERR = 13	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - invalid metered end bus - branch status and bus type code changes

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2.27. MULTI_SECTION_LINE_EDIT

Use this API to modify the status and/or metered end designation of an existing multi-section line grouping in the working case (see PSSE Program Operation Manual, Multi-Section Line Grouping Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MULTI_SECTION_LINE_EDIT  IBUS  JBUS  'ID'  INTGAR(1)  INTGAR(2)
```

Python command syntax:

```
ierr = multi_section_line_edit(ibus, jbus, id, intgar)
```

Fortran command syntax:

```
CALL MULTI_SECTION_LINE_EDIT_API_2(IBUS, JBUS, ID, INTGAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 ID	multi-section line identifier; ID(1:1) must be '&' (input; '&1').	
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	STAT, branch status (unchanged by default)
	INTGAR(1) = 1	for in-service
	INTGAR(1) = 0	for out-of-service
	INTGAR(2)	METBUS, metered end bus number (must be IBUS or JBUS) (unchanged by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	multi-section line identifier is more than two characters
	IERR = 3	multi-section line identifier does not start with &
	IERR = 4	multi-section line not found
	IERR = 5	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid metered end bus. - invalid branch status

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2.28. MULTI_TERM_DC_BUS_DATA

Use this API to modify the data of a dc bus of an existing multi-terminal dc line in the working case or to add a new dc bus to a multi-terminal dc line in the working case (see PSSE Program Operation Manual, Multi-Terminal DC Transmission Line Data). The dc line must have previously been specified via the MULTI_TERM_DC_LINE_DATA API.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MULTI_TERM_DC_BUS_DATA  DC  IDC  INTGAR(1)..INTGAR(5)  REALAR(1)
 'DCNAME'
```

Python command syntax:

```
ierr, realaro = multi_term_dc_bus_data(dc, idc, intgar, realari, dcname)
```

Fortran command syntax:

```
CALL MULTI_TERMINAL_DC_BUS_API(DC, IDC, INTGAR, REALAR, DCNAME, IERR)

CALL MULTI_TERMINAL_DC_BUS_API_I(NUM, IDC, INTGAR, REALAR, DCNAME, IERR)

CALL MULTI_TERMINAL_DC_BUS_API_C(NAME, IDC, INTGAR, REALAR, DCNAME, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which data for a dc bus is to be specified; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which data for a dc bus is to be specified; no default allowed	
Character*12 NAME	name of the multi-terminal dc line for which data for a dc bus is to be specified; no default allowed	
Integer IDC	dc bus number (must be between 1 and the number of dc buses in multi-terminal dc line DC) (input; no default allowed).	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	IB, converter bus number or zero (0 by default)
	INTGAR(2)	AREA, area number (1 by default)
	INTGAR(3)	ZONE, zone number (1 by default)
	INTGAR(4)	OWNER, owner number (1 by default)
	INTGAR(5)	IDC2, second dc bus number (0 by default)
Real REALAR (2)	(first is value; second is value) (input (output).).	
Real REALARI (1)	array of 1 elements specifying one element corresponding to values REALAR(1) as described above(input).	
	REALARI(1)	RGRND, resistance to ground (0.0 by default)

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Real REALARO (1)	array of 1 elements specifying one element corresponding to value REALAR(2) as described above(output). REALARO(1) dc voltage (kV) (returned)
Character*12 DCNAME	dc bus name (input; blank by default).
Integer IERR	error code (output). IERR = 0 no errors occurred IERR > 0 fatal error occurred IERR = 1 dc line not found IERR = 2 invalid dc bus number IERR = 3 prerequisite requirements for API are not met IERR = -1 data error, warning and information messages; one or more of: - data for lower numbered dc bus has not yet been entered. - converter bus not found. - bus is not a converter of this multi-terminal dc line. - this dc bus is not the first dc bus of any converter. - the first dc bus of some converter is set to this dc bus. - invalid second dc bus number. - this dc bus is being connected to a converter bus. - dc bus no longer connected to some other converter. - dc bus is being specified with no converter bus. - dc bus no longer connected to converter as its first dc bus. - dc bus no longer connected to converter as its second dc bus. - bus invalid as second dc bus because it is the first dc bus of another converter. - invalid area/zone/owner number. - area table is full. - zone table is full. - owner table is full

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2.29. MULTI_TERM_DC_CONVR_DATA

Use this API to modify the data of a dc converter of an existing multi-terminal dc line in the working case or to add a new converter to a multi-terminal dc line in the working case (see PSSE Program Operation Manual, Multi-Terminal DC Transmission Line Data). The dc line must have previously been specified via the MULTI_TERM_DC_LINE_DATA API.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MULTI_TERM_DC_CONVR_DATA  DC  INTGAR(1)..INTGAR(4)
 REALAR(1)..REALAR(13)
```

Python command syntax:

```
ierr, realaro = multi_term_dc_convr_data(dc, intgar, realari)
```

Fortran command syntax:

```
CALL MULTI_TERM_DC_CONVERTER_API(DC,INTGAR,REALAR,IERR)

CALL MULTI_TERM_DC_CONVERTER_API_I(NUM,INTGAR,REALAR,IERR)

CALL MULTI_TERM_DC_CONVERTER_API_C(NAME,INTGAR,REALAR,IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which data for a converter is to be specified; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which data for a converter is to be specified; no default allowed	
Character*12 NAME	name of the multi-terminal dc line for which data for a converter is to be specified; no default allowed	
Integer INTGAR (4)	array of 4 elements specifying (input).	
	INTGAR(1)	IB, converter bus number (no default allowed)
	INTGAR(2)	N, number of bridges in series (1 by default)
	INTGAR(3)	CNVCOD, pole code (0 by default)
	INTGAR(3) = 0	for a positive pole converter
	INTGAR(3) = 0	for a negative pole converter
	INTGAR(4)	VCCFLG, pole's voltage controlling converter flag (0 by default)
	INTGAR(4) = 0	no
	INTGAR(4) = 0	yes
Real REALAR (17)	(first thirteen are values; last four are values) (input (output).).	
Real REALARI (13)	array of 13 elements specifying corresponding to values REALAR(1) through REALAR(13) as described above(input).	

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	REALARI(1)	SETVL, scheduled current (amps), power (MW) or voltage (kv) (0.0 by default)
	REALARI(2)	MN, minimum firing angle (0.0 by default)
	REALARI(3)	MX, maximum firing angle (90.0 by default)
	REALARI(4)	TR, transformer ratio (1.0 by default)
	REALARI(5)	RC, commutating resistance (0.0 by default)
	REALARI(6)	XC, commutating reactance (0.0 by default)
	REALARI(7)	EBAS, primary base ac voltage (1.0 by default)
	REALARI(8)	TAP, tap setting (1.0 by default)
	REALARI(9)	TPMX, maximum tap setting (1.5 by default)
	REALARI(10)	TPMN, minimum tap setting (0.51 by default)
	REALARI(11)	TSTP, tap step (0.00625 by default)
	REALARI(12)	MARG, margin (pu) (0.0 by default)
	REALARI(13)	DCPF, converter participation factor (1.0 by default)
Real REALARO (4)	array of 4 elements specifying corresponding to values REALAR(14) through REALAR(17) as described above(output).	
	REALARO(1)	firing angle (returned)
	REALARO(2)	converter current (amps) (returned)
	REALARO(3)	active power into converter (MW or kW) (returned)
	REALARO(4)	reactive power into converter (MW or kW) (returned)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	dc line not found
	IERR = 2	no converter bus is specified
	IERR = 3	converter bus not found
	IERR = 4	too many converters specified
	IERR = 5	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - converter bus is a dummy bus of a multi-section line. - multi-section line grouping deleted. - invalid number of bridges. - invalid base ac voltage. - converter maximum angle is less than its minimum angle. - converter maximum tap is less than its minimum tap. - invalid tap step. - dc voltage setpoint is not positive. - scheduled voltage is less than mode switch voltage. - this converter bus is now on the other pole; it is no longer the voltage controlling converter of its previous pole. - there is no voltage

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controlling converter on a pole. - this converter replaces another converter as its pole's voltage controlling converter. - this converter must remain its pole's voltage controlling converter because it is the only converter on the pole

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2.30. MULTI_TERM_DC_LINE_CHNG

Use this API routine to modify the control mode and mode switch voltage of an existing multi-terminal dc line in the working case (see PSSE Program Operation Manual, Multi-Terminal DC Transmission Line Data). Converter data, dc bus data, and dc link data are specified via the MULTI_TERM_DC_CONVR_DATA, MULTI_TERM_DC_BUS_DATA, and MULTI_TERM_DC_LINK_DATA API routines respectively.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_MULTI_TERM_DC_LINE_CHNG  DC  INTGAR(1)  REALAR(1)
```

Python command syntax:

```
ierr, intgaro = multi_term_dc_line_chng(dc, intgari, realar)
```

Fortran command syntax:

```
CALL MULTI_TERMINAL_DC_LINE_CHNG(DC, INTGAR, REALAR, IERR)

CALL MULTI_TERMINAL_DC_LINE_CHNG_I(NUM, INTGAR, REALAR, IERR)

CALL MULTI_TERMINAL_DC_LINE_CHNG_C(NAME, INTGAR, REALAR, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which data is to be specified; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which data is to be specified; no default allowed	
Character*12 NAME	name of the multi-terminal dc line for which data is to be specified; no default allowed	
Integer INTGAR (8)	(first is value; last seven are values) (input (output).).	
Integer INTGARI (1)	array of 1 elements specifying one element corresponding to value INTGAR(1) as described above(input).	
	INTGARI(1)	MDC, control mode (0 by default)
Integer INTGARO (7)	array of 7 elements specifying corresponding to values INTGAR(2) through INTGAR(8) as described above(output).	
	INTGARO(1)	positive pole voltage controlling converter index (returned)
	INTGARO(2)	negative pole voltage controlling converter index, or zero if the negative pole is not being modeled (returned)
	INTGARO(3)	positive pole voltage controlling converter bus number (returned)
	INTGARO(4)	positive pole voltage controlling converter bus number (returned)

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	INTGARO(5)	number of converters (returned)
	INTGARO(6)	number of dc buses (returned)
	INTGARO(7)	number of dc links (returned)
Real REALAR (1)	array of 1 elements specifying (input).	
	REALAR(1)	VCMOD, mode switch dc voltage (kV) (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 2	blank dc line name is invalid
	IERR = 3	dc line not found
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid dc line control mode

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2.31. MULTI_TERM_DC_LINE_DATA

Use this API to modify the control mode and mode switch voltage of an existing multi-terminal dc line in the working case or to add a new multi-terminal dc line to the working case (see PSSE Program Operation Manual, Multi-Terminal DC Transmission Line Data). Converter data, dc bus data, and dc link data are specified via the MULTI_TERM_DC_CONVR_DATA, MULTI_TERM_DC_BUS_DATA, and MULTI_TERM_DC_LINK_DATA APIs respectively.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MULTI_TERM_DC_LINE_DATA  DC  INTGAR(1)  REALAR(1)
```

Python command syntax:

```
ierr, intgaro = multi_term_dc_line_data(dc, intgari, realar)
```

Fortran command syntax:

```
CALL MULTI_TERMINAL_DC_LINE_API(DC, INTGAR, REALAR, IERR)

CALL MULTI_TERMINAL_DC_LINE_API_I(NUM, INTGAR, REALAR, IERR)

CALL MULTI_TERMINAL_DC_LINE_API_C(NAME, INTGAR, REALAR, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which data is to be specified; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which data is to be specified; no default allowed
Character*12 NAME	name of the multi-terminal dc line for which data is to be specified; no default allowed
Integer INTGAR (8)	(first is value; last seven are values) (input (output).).
Integer INTGARI (1)	array of 1 elements specifying one element corresponding to value INTGAR(1) as described above(input).
	INTGARI(1) MDC, control mode (0 by default)
Integer INTGARO (7)	array of 7 elements specifying corresponding to values INTGAR(2) through INTGAR(8) as described above(output).
	INTGARO(1) positive pole voltage controlling converter index (returned)
	INTGARO(2) negative pole voltage controlling converter index, or zero if the negative pole is not being modeled (returned)
	INTGARO(3) positive pole voltage controlling converter bus number (returned)
	INTGARO(4) positive pole voltage controlling converter bus number (returned)

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	INTGARO(5)	number of converters (returned)
	INTGARO(6)	number of dc buses (returned)
	INTGARO(7)	number of dc links (returned)
Real REALAR (1)	array of 1 elements specifying (input).	
	REALAR(1)	VCMOD, mode switch dc voltage (kV) (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	multi-terminal dc line table is full
	IERR = 2	blank dc line name is invalid
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid dc line control mode

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2.32. MULTI_TERM_DC_LINK_DATA

Use this API to modify the data of a dc link of an existing multi-terminal dc line in the working case or to add a new dc link to a multi-terminal dc line in the working case (see PSSE Program Operation Manual, Multi-Terminal DC Transmission Line Data). The dc line must have previously been specified via the MULTI_TERM_DC_LINE_DATA API.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MULTI_TERM_DC_LINK_DATA  DC  IDC  JDC  'DCCKT'  INTGAR(1)  REALAR(1)
 REALAR(2)
```

Python command syntax:

```
ierr = multi_term_dc_link_data(dc,idc,jdc,dcckt,intgar,realar)
```

Fortran command syntax:

```
CALL MULTI_TERMINAL_DC_LINK_API(DC, IDC, JDC, DCCKT, INTGAR, REALAR, IERR)

CALL MULTI_TERMINAL_DC_LINK_API_I(NUM, IDC, JDC, DCCKT, INTGAR, REALAR,
 IERR)

CALL MULTI_TERMINAL_DC_LINK_API_C(NAME, IDC, JDC, DCCKT, INTGAR, REALAR,
 IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which data for a dc link is to be specified; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which data for a dc link is to be specified; no default allowed	
Character*12 NAME	name of the multi-terminal dc line for which data for a dc link is to be specified; no default allowed	
Integer IDC	dc bus number of from bus (must be between 1 and the number of dc buses in multi-terminal dc line DC) (input; no default allowed).	
Integer JDC	dc bus number of to bus (must be between 1 and the number of dc buses in multi-terminal dc line DC) (input; no default allowed).	
Character*1 DCCKT	circuit identifier (input; '1').	
Integer INTGAR (1)	array of 1 elements specifying (input).	
	INTGAR(1)	METBUS, metered end dc bus number (IDC by default)
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	RDC, dc link resistance (0.0 by default)
	REALAR(2)	LDC, dc link inductance (0.0 by default)

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Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	dc line not found
	IERR = 2	invalid dc bus number
	IERR = 3	circuit identifier is more than one character
	IERR = 4	dc bus not present in this multi-terminal dc line
	IERR = 5	branch from a bus to itself
	IERR = 6	too many dc links in this multi-terminal dc line
	IERR = 7	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid metered end bus. - invalid dc link resistance. - invalid dc link inductance

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2.33. OWNER_DATA

Use this API to modify owner data in the working case (see PSSE Program Operation Manual, Interarea Transfer Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OWNER_DATA  IOWNER  'OWNAME'
```

Python command syntax:

```
ierr = owner_data(iowner, owname)
```

Fortran command syntax:

```
CALL OWNER_API_2(IOWNER, OWNAME, IERR)
```

Where:

Integer IOWNER	owner number between 1 and 9999 (input; no default allowed).	
Character*12 OWNAME	owner name (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid owner number
	IERR = 2	owner table is full
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - no equipment is assigned to owner

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2.34. PBUS_ADD_MOD

Use this API to add or modify transaction event participating bus data.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PBUS_ADD_MOD  EVENT  IBUS  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = pbus_add_mod(event, ibus, realar)
```

Fortran command syntax:

```
CALL PBUS_ADD_MOD(EVENT, IBUS, REALAR, IERR, NERR)
```

Where:

Integer EVENT	transaction event number (input).	
Integer IBUS	number of the participating bus (input).	
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	participating bus load value
	REALAR(2)	participating bus generation value
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	error occurred
	IERR = 2	invalid transaction event number
	IERR = 3	transaction event not found
	IERR > 3	unable to add or modify participating bus data; error condition detected in data access routine
	IERR = 10	prerequisite requirements for API are not met
	IERR = -1	warnings: - None
Integer NERR	number of error entries in PBUF02 (output).	

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2.35. PBUS_DELETE

Use this API to remove a bus from the set of buses that are participating in a transaction event.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PBUS_DELETE  EVENT  IBUS
```

Python command syntax:

```
ierr = pbus_delete(event, ibus)
```

Fortran command syntax:

```
CALL PBUS_DELETE(EVENT, IBUS, IERR, NERR)
```

Where:

Integer EVENT	transaction event number (input).	
Integer IBUS	number of the participating bus (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid transaction event number
	IERR = 2	transaction event not found
	IERR = 3	prerequisite requirements for API are not met
Integer NERR	number of error entries in PBUF02 (output).	

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2.36. PLANT_CHNG

This API routine is obsolete. It has been replaced by the API routine plant_chng_3 .

Use this API routine to modify the data of an existing plant in the working case, if the plant bus is not a bus section within a substation. Use plant_chng_3 if data is to be modified at a plant associated with a bus section.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_PLANT_CHNG  IBUS  INTGAR(1)  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = plant_chng(ibus, intgar, realar)
```

Fortran command syntax:

```
CALL PLANT_CHNG(IBUS, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Integer INTGAR (1)	array of 1 elements specifying (input).	
	INTGAR(1)	IREG, remote bus number (0 to control voltage at bus IBUS) (0 by default)
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	VS, scheduled voltage magnitude (1.0 by default)
	REALAR(2)	RMPCT, percent of contributed reactive power (100.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 3	no plant at bus
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid scheduled voltage - regulated bus not found - plant specified as remotely regulating itself - type code of regulated bus is not 1 or 2 - RMPCT is not positive

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2.37. PLANT_DATA

Use this API to modify the data of an existing plant in the working case, or to add a new plant to the working case (see PSSE Program Operation Manual, Fixed Bus Shunt Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PLANT_DATA  IBUS  INTGAR(1)  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = plant_data(ibus, intgar, realar)
```

Fortran command syntax:

```
CALL PLANT_API_2(IBUS, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Integer INTGAR (1)	array of 1 elements specifying (input).	
	INTGAR(1)	IREG, remote bus number (0 to control voltage at bus IBUS) (0 by default)
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	VS, scheduled voltage magnitude (1.0 by default)
	REALAR(2)	RMPCT, percent of contributed reactive power (100.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	plant table is full
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid scheduled voltage - regulated bus not found - plant specified as remotely regulating itself - type code of regulated bus is not 1 or 2 - RMPCT is not positive

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2.38. SEQ_3_WIND_GROUNDING_DATA

Use this API to modify the zero sequence grounding data of an existing three-winding transformer in the working case (see PSSE Program Operation Manual, Zero Sequence Transformer Data).

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SEQ_3_WIND_GROUNDING_DATA  IBUS  JBUS  KBUS  'CKT'  INTGAR(1)
 REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = seq_3_wind_grounding_data(ibus, jbus, kbus, ckt, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_3_WINDING_GROUNDING_API_2(IBUS,JBUS,KBUS,CKT,INTGAR,REALAR,IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 CKT	transformer circuit identifier (input; '1').	
Integer INTGAR (1)	array of 1 elements specifying one element (defaulted data items are unchanged)(input).	
	INTGAR(1)	CC, connection code
Real REALAR (2)	array of 2 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	RG, zero sequence grounding resistance
	REALAR(2)	XG, zero sequence grounding reactance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	three-winding transformer not found
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Invalid winding connection code. - Connection code is such that more than one winding requires a grounding impedance

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2.39. SEQ_3_WIND_WINDING_DATA

Use this API to modify the zero sequence impedance data of one winding of an existing three-winding transformer in the working case (see PSSE Program Operation Manual, Zero Sequence Transformer Data).

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SEQ_3_WIND_WINDING_DATA  IBUS  JBUS  KBUS  'CKT'  WARG  REALAR(1)
 REALAR(2)
```

Python command syntax:

```
ierr = seq_3_wind_winding_data(ibus, jbus, kbus, ckt, warg, realar)
```

Fortran command syntax:

```
CALL SEQ_THREE_WINDING_WINDING_API_2(IBUS,JBUS,KBUS,CKT,WARG,REALAR,IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 CKT	transformer circuit identifier (input; '1').	
Integer WARG	winding number (1, 2 or 3) (input; no default allowed).	
Real REALAR (2)	array of 2 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	Ri, zero sequence winding resistance
	REALAR(2)	Xi, zero sequence winding reactance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	three-winding transformer not found
	IERR = 4	invalid winding number
	IERR = 5	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - None

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2.40. SEQ_BRANCH_DATA

This API routine is obsolete. It has been replaced by the API routine seq_branch_data_3 .

Use this API to modify the zero sequence data of an existing non-transformer branch in the working case (see PSSE Program Operation Manual, Zero Sequence Non-Transformer Branch Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SEQ_BRANCH_DATA  IBUS  JBUS  'CKT'  REALAR(1)..REALAR(7)
```

Python command syntax:

```
ierr = seq_branch_data(ibus, jbus, ckt, realar)
```

Fortran command syntax:

```
CALL SEQ_BRANCH_API_2(IBUS, JBUS, CKT, REALAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Real REALAR (7)	array of 7 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	RLINZ, zero sequence branch resistance
	REALAR(2)	XLINZ, zero sequence branch reactance
	REALAR(3)	BCHZ, zero sequence total line charging
	REALAR(4)	GI, zero sequence active line shunt at bus IBUS end
	REALAR(5)	BI, zero sequence reactive line shunt at bus IBUS end
	REALAR(6)	GJ, zero sequence active line shunt at bus JBUS end
	REALAR(7)	BJ, zero sequence reactive line shunt at bus JBUS end
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	circuit identifier may not start with &
	IERR = 4	branch not found
	IERR = 5	branch exists but it is a two-winding transformer

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IERR = 6	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - branch is treated as a zero impedance line but its zero sequence impedance is non-zero

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2.41. SEQ_BRANCH_DATA_3

Use this API to modify the zero sequence data of an existing non-transformer branch in the working case (see PSSE Program Operation Manual, Zero Sequence Non-Transformer Branch Data).

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_SEQ_BRANCH_DATA_3  IBUS JBUS CKT INTGAR REALAR(1)..REALAR(8)
```

Python command syntax:

```
ierr = seq_branch_data_3(ibus, jbus, ckt, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_BRANCH_API_3(IBUS, JBUS, CKT, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Integer INTGAR (1)	array of 1 elements specifying MOV-protected series capacitor flag(input).	
	INTGAR(1)	MOV-protected series capacitor flag
	INTGAR(1) = 0	not a MOV-protected branch
	INTGAR(1) = 1	MOV-protected series capacitor with MOV protection enabled
	INTGAR(1) = 2	MOV-protected series capacitor with MOV protection disabled
	INTGAR(1) = 3	spark gap protection enabled
Real REALAR (8)	array of 8 elements specifying (defaulted data items are un-changed)(input).	
	REALAR(1)	RLINZ, zero sequence branch resistance
	REALAR(2)	XLINZ, zero sequence branch reactance
	REALAR(3)	BCHZ, zero sequence total line charging
	REALAR(4)	GI, zero sequence active line shunt at bus IBUS end
	REALAR(5)	BI, zero sequence reactive line shunt at bus IBUS end
	REALAR(6)	GJ, zero sequence active line shunt at bus JBUS end
	REALAR(7)	BJ, zero sequence reactive line shunt at bus JBUS end
	REALAR(8)	MOVI, mov protected current level in kA
Integer IERR	error code (output).	

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IERR = 0	no errors occurred
IERR > 0	fatal error occurred
IERR = 1	bus not found
IERR = 2	circuit identifier is more than two characters
IERR = 3	circuit identifier may not start with &
IERR = 4	branch not found
IERR = 5	branch exists but it is a two-winding transformer
IERR = 6	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - branch is treated as a zero impedance line but its zero sequence impedance is non-zero

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2.42. SEQ_BUS_DATA

Bus load sequence data is now accessible as load sequence data.
Use this API routine to modify the sequence data of an existing bus in the working case (see PSSE Program Operation Manual, Negative Sequence Shunt Load Data and Zero Sequence Shunt Load Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SEQ_BUS_DATA  IBUS  REALAR(1)..REALAR(4)
```

Python command syntax:

```
ierr = seq_bus_data(ibus, realar)
```

Fortran command syntax:

```
CALL SEQ_BUS_API_2(IBUS, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Real REALAR (4)	array of 4 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	GNEG, negative sequence shunt admittance active component
	REALAR(2)	BNEG, negative sequence shunt admittance reactive component
	REALAR(3)	GZERO, zero sequence shunt admittance active component
	REALAR(4)	BZERO, zero sequence shunt admittance reactive component
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	the SEQ_BUS_DATA API routine is obsolete
	IERR = 2	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - None

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2.43. SEQ_FIXED_SHUNT_DATA

Use this API to modify the zero sequence data of an existing fixed bus shunt in the working case (see PSSE Program Operation Manual, Zero Sequence Fixed Shunt Data).

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SEQ_FIXED_SHUNT_DATA  IBUS  'ID'  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = seq_fixed_shunt_data(ibus, id, realar)
```

Fortran command syntax:

```
CALL SEQ_FIXED_SHUNT_API(IBUS, ID, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	shunt identifier (input; '1').	
Real REALAR (2)	array of 2 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	GSZERO, active power component of fixed shunt zero sequence admittance (conductance)
	REALAR(2)	BSZERO, reactive power component of fixed shunt zero sequence admittance (susceptance)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	shunt identifier is more than two characters
	IERR = 3	fixed shunt not found
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Admittance components with different signs in the positive and zero sequences

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2.44. SEQ_INDUCTION_MACH_DATA

Use this API routine to modify the sequence data of an existing induction machine in the working case (see PSSE Program Operation Manual, Induction Machine Impedance Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SEQ_INDUCTION_MACH_DATA  IBUS  'ID'  INTGAR(1)  INTGAR(2)
 REALAR(1)..REALAR(6)
```

Python command syntax:

```
ierr = seq_induction_mach_data(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_INDUCTION_MACH_API(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Integer INTGAR (2)	array of 2 elements specifying (defaulted data items are unchanged)(input).	
	INTGAR(1)	CZG, code indicating the units in which the zero sequence grounding impedance is specified
	INTGAR(1) = 1	for per unit
	INTGAR(1) = 2	for ohms
	INTGAR(2)	CGR, grounding code (0 = ungrounded; 1 = grounded) (0 by default)
Real REALAR (6)	array of 6 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	ILK2IR, ratio of positive sequence locked rotor current to rated current
	REALAR(2)	ROVERX, ratio of positive sequence resistance to reactance
	REALAR(3)	R0, machine zero sequence fault analysis resistance
	REALAR(4)	X0, machine zero sequence fault analysis reactance
	REALAR(5)	RG, machine zero sequence grounding resistance
	REALAR(6)	XG, machine zero sequence grounding reactance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred

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IERR > 0	fatal error occurred
IERR = 1	bus not found
IERR = 2	machine identifier is more than two characters
IERR = 3	induction machine not found
IERR = 4	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Invalid CZG data value. - invalid grounding code. - CZG is 2 but the bus has no base voltage specified

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2.45. SEQ_LOAD_DATA

Use this API routine to modify the sequence data of an existing load in the working case (see PSSE Program Operation Manual, Load Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SEQ_LOAD_DATA  IBUS  'ID'  INTGAR(1)  REALAR(1)..REALAR(4)
```

Python command syntax:

```
ierr = seq_load_data(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_LOAD_API(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	load identifier (input; '1').	
Integer INTGAR (1)	array of 1 elements specifying one element(input).	
	INTGAR(1)	CGR, load grounding code (0 = ungrounded; 1 = grounded) (0 by default)
Real REALAR (4)	array of 4 elements specifying (input).	
	REALAR(1)	GNEG, negative sequence active power load component; entered as admittance in MW at 1.0 per unit voltage (0.0 by default)
	REALAR(2)	BNEG, negative sequence reactive power load component; BNEG is negative for an inductive load and positive for a capacitive load; entered as admittance in MVar at 1.0 per unit voltage (0.0 by default)
	REALAR(3)	GZERO, zero sequence active power load component; entered as admittance in MW at 1.0 per unit voltage (0.0 by default)
	REALAR(4)	BZERO, zero sequence reactive power load component; BZERO is negative for an inductive load and positive for a capacitive load; entered as admittance in MVar at 1.0 per unit voltage (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	load identifier is more than two characters
	IERR = 3	load not found

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IERR = 4	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - invalid grounding code

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2.46. SEQ_MACHINE_DATA

This API routine is obsolete. It has been replaced by the API routine seq_machine_data_3 .

Use this API routine to modify the sequence data of an existing machine in the working case (see PSSE Program Operation Manual, Positive Sequence Generator Impedance Data, Negative Sequence Generator Impedance Data and Zero Sequence Generator Impedance Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SEQ_MACHINE_DATA  IBUS  'ID'  REALAR(1)..REALAR(6)
```

Python command syntax:

```
ierr = seq_machine_data(ibus, id, realar)
```

Fortran command syntax:

```
CALL SEQ_MACHINE_API_2(IBUS, ID, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Real REALAR (6)	array of 6 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	ZRPOS, machine positive sequence fault analysis resistance
	REALAR(2)	ZXPOS, machine positive sequence fault analysis reactance
	REALAR(3)	ZRNEG, machine negative sequence fault analysis resistance
	REALAR(4)	ZXNEG, machine negative sequence fault analysis reactance
	REALAR(5)	RZERO, machine zero sequence fault analysis resistance
	REALAR(6)	XZERO, machine zero sequence fault analysis reactance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	machine identifier is more than two characters
	IERR = 3	machine not found
	IERR = 4	prerequisite requirements for API are not met

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IERR = -1	data error, warning and information messages; one or more of: - Machine impedance is (0.0,0.0)

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2.47. SEQ_MACHINE_DATA_3

Use this API routine to modify the sequence data of an existing machine in the working case (see PSSE Program Operation Manual, Positive Sequence Generator Impedance Data, Negative Sequence Generator Impedance Data and Zero Sequence Generator Impedance Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SEQ_MACHINE_DATA_3  IBUS  'ID'  INTGAR(1)  REALAR(1)..REALAR(10)
```

Python command syntax:

```
ierr = seq_machine_data_3(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_MACHINE_API_3(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Integer INTGAR (1)	array of 1 elements specifying one element (defaulted data items are unchanged)(input).	
	INTGAR(1)	CZG, code indicating the units in which the zero sequence grounding impedance is specified
	INTGAR(1) = 1	for per unit
	INTGAR(1) = 2	for ohms
Real REALAR (10)	array of 10 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	RPOS, machine positive sequence fault analysis resistance
	REALAR(2)	X", machine positive sequence fault analysis subtransient reactance
	REALAR(3)	RNEG, machine negative sequence fault analysis resistance
	REALAR(4)	XNEG, machine negative sequence fault analysis reactance
	REALAR(5)	RZERO, machine zero sequence fault analysis resistance
	REALAR(6)	XZERO, machine zero sequence fault analysis reactance
	REALAR(7)	X', machine positive sequence fault analysis transient reactance
	REALAR(8)	Xs, machine positive sequence fault analysis synchronous reactance

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	REALAR(9)	RG, machine zero sequence grounding resistance
	REALAR(10)	XG, machine zero sequence grounding reactance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	machine identifier is more than two characters
	IERR = 3	machine not found
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Grounding impedance units code indicates input in ohms but no base voltage has been specified for the bus. - Invalid grounding impedance units code. - Invalid grounding impedance units code. - Invalid positive sequence reactance value. - Negative sequence impedance is (0.0,0.0). - Zero sequence impedance is (0.0,0.0)

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2.48. SEQ_MUTUAL_DATA

Use this API to modify existing zero sequence mutual impedance data in the working case or to add a new zero sequence mutual coupling to the working case (see PSSE Program Operation Manual, Zero Sequence Mutual Impedance Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SEQ_MUTUAL_DATA  IBUS  JBUS  'CKT1'  KBUS  LBUS  'CKT2'
 REALAR(1)..REALAR(6)
```

Python command syntax:

```
ierr = seq_mutual_data(ibus, jbus, ckt1, kbus, lbus, ckt2, realar)
```

Fortran command syntax:

```
CALL SEQ_MUTUAL_API_2(IBUS, JBUS, CKT1, KBUS, LBUS, CKT2, REALAR, IERR)
```

Where:

Integer IBUS	bus number of the from bus of the first branch (input; no default allowed).	
Integer JBUS	bus number of the to bus of the first branch (input; no default allowed).	
Character*2 CKT1	first branch circuit identifier (input; '1').	
Integer KBUS	bus number of the from bus of the second branch (input; no default allowed).	
Integer LBUS	bus number of the to bus of the second branch to bus (input; no default allowed).	
Character*2 CKT2	second branch circuit identifier (input; '1').	
Real REALAR (6)	array of 6 elements specifying (input).	
	REALAR(1)	RM, branch-to-branch mutual resistance (0.0 by default)
	REALAR(2)	XM, branch-to-branch mutual reactance (0.0 by default)
	REALAR(3)	BIJ1, B factor at IBUS end of the first branch (0.0 by default)
	REALAR(4)	BIJ2, B factor at JBUS end of the first branch (1.0 by default)
	REALAR(5)	BKL1, B factor at KBUS end of the second branch (0.0 by default)
	REALAR(6)	BKL2, B factor at LBUS end of the second branch (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred

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IERR = 1	bus not found
IERR = 2	circuit identifier is more than two characters
IERR = 3	circuit identifier may not start with &
IERR = 4	branch not found
IERR = 5	mutual couples a branch to itself
IERR = 6	branch is a transformer
IERR = 7	branch is a zero impedance line
IERR = 8	mutual coupling table is full
IERR = 9	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Invalid B factor (<0.0 or >1.0). - B1 is not less than B2

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2.49. SEQ_SWITCHED_SHUNT_DATA

Use this API to modify the zero sequence data of an existing switched shunt in the working case (see PSSE Program Operation Manual, Zero Sequence Switched Shunt Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SEQ_SWITCHED_SHUNT_DATA  IBUS  REALAR(1)..REALAR(8)
```

Python command syntax:

```
ierr, realaro = seq_switched_shunt_data(ibus, realari)
```

Fortran command syntax:

```
CALL SEQ_SWITCHED_SHUNT_API_2(IBUS, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Real REALAR (9)	(first eight are values; last one is value) (defaulted data items are unchanged) (input (output).).	
Real REALARI (8)	array of 8 elements specifying corresponding to values REALAR(1) through REALAR(8) as described above(input).	
	REALARI(1)	BZ1, zero sequence increment in MVAR per step for block 1
	REALARI(2)	BZ2, zero sequence increment in MVAR per step for block 2
	REALARI(3)	BZ3, zero sequence increment in MVAR per step for block 3
	REALARI(4)	BZ4, zero sequence increment in MVAR per step for block 4
	REALARI(5)	BZ5, zero sequence increment in MVAR per step for block 5
	REALARI(6)	BZ6, zero sequence increment in MVAR per step for block 6
	REALARI(7)	BZ7, zero sequence increment in MVAR per step for block 7
	REALARI(8)	BZ8, zero sequence increment in MVAR per step for block 8
Real REALARO (1)	array of 1 elements specifying one element corresponding to value REALAR(9) as described above(output).	
	REALARO(1)	present zero sequence switched shunt admittance in MVAR (returned)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred

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IERR = 1	bus not found
IERR = 2	no switched shunt at bus
IERR = 3	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Block has N zero and B non-zero. - Block has admittances with different signs in positive and zero sequences

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2.50. SEQ_THREE_WINDING_DATA

This API routine is obsolete. It has been replaced by the API routine seq_three_winding_data_3 .

Use this API to modify the zero sequence data of an existing three-winding transformer in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SEQ_THREE_WINDING_DATA  IBUS  JBUS  KBUS  'CKT'  INTGAR(1)
 REALAR(1)..REALAR(8)
```

Python command syntax:

```
ierr = seq_three_winding_data(ibus, jbus, kbus, ckt, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_THREE_WINDING_API(IBUS, JBUS, KBUS, CKT, INTGAR, REALAR, IERR,
 NERR)
```

Where:

Integer IBUS	bus number of one of the buses (input).	
Integer JBUS	bus number one of the other buses (input).	
Integer KBUS	bus number of the third bus (input).	
Character*2 CKT	transformer circuit identifier (input).	
Integer INTGAR (1)	array of 1 elements specifying one element(input).	
	INTGAR(1)	CC, connection code
Real REALAR (8)	array of 8 elements specifying (input).	
	REALAR(1)	RG, zero sequence grounding resistance
	REALAR(2)	XG, zero sequence grounding reactance
	REALAR(3)	R1, zero sequence winding one resistance
	REALAR(4)	X1, zero sequence winding one reactance
	REALAR(5)	R2, zero sequence winding two resistance
	REALAR(6)	X2, zero sequence winding two reactance
	REALAR(7)	R3, zero sequence winding three resistance
	REALAR(8)	X3, zero sequence winding three reactance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	three-winding transformer not found
	IERR = 4	prerequisite requirements for API are not met

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	IERR = -1	warnings - one or more of: - Invalid winding connection code. - Connection code is such that more than one winding requires a grounding impedance
Integer NERR	number of entries in PBUF02 containing error messages (output).	

2.51. SEQ_THREE_WINDING_DATA_3

Use this API to modify the zero sequence data of an existing three-winding transformer in the working case.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SEQ_THREE_WINDING_DATA_3  IBUS  JBUS  KBUS  'CKT'
 INTGAR(1)..INTGAR(3)  REALAR(1)..REALAR(14)
```

Python command syntax:

```
ierr = seq_three_winding_data_3(ibus, jbus, kbus, ckt, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_THREE_WINDING_API_3(IBUS, JBUS, KBUS, CKT, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input).	
Integer JBUS	bus number one of the other buses (input).	
Integer KBUS	bus number of the third bus (input).	
Character*2 CKT	transformer circuit identifier (input).	
Integer INTGAR (3)	array of 3 elements specifying (defaulted data items are unchanged)(input).	
	INTGAR(1)	CZ0, non-grounding impedance data I/O code
	INTGAR(2)	CZG, grounding impedance data I/O code
	INTGAR(3)	CC, connection code
Real REALAR (14)	array of 14 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	RG1, grounding resistance on winding 1 for an impedance grounded transformer
	REALAR(2)	XG1, grounding reactance on winding 1 for an impedance grounded transformer
	REALAR(3)	R01, winding 1, or winding 1 bus to winding 2 bus, leakage resistance
	REALAR(4)	X01, winding 1, or winding 1 bus to winding 2 bus, leakage reactance
	REALAR(5)	RG2, grounding resistance on winding 2 for an impedance grounded transformer
	REALAR(6)	XG2, grounding reactance on winding 2 for an impedance grounded transformer
	REALAR(7)	R02, winding 2, or winding 2 bus to winding 3 bus, leakage resistance

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	REALAR(8)	X02, winding 2, or winding 2 bus to winding 3 bus, leakage reactance
	REALAR(9)	RG3, grounding resistance on winding 3 for an impedance grounded transformer
	REALAR(10)	XG3, grounding reactance on winding 3 for an impedance grounded transformer
	REALAR(11)	R03, winding 3, or winding 3 bus to winding 1 bus, leakage resistance
	REALAR(12)	X03, winding 3, or winding 3 bus to winding 1 bus, leakage reactance
	REALAR(13)	RN, common neutral grounding resistance
	REALAR(14)	XN, common neutral grounding reactance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	three-winding transformer not found
	IERR = 4	CZG is 3 but at least one endpoint bus has no base voltage specified
	IERR = 5	invalid CZ0 or CZG value
	IERR = 6	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - Invalid winding connection code. - Invalid CZ0 or CZG data value. - CZ0 or CZG is not 1 with connection code 1 through 9 or a three-digit connection code. - CZG data value is 3 but at least one endpoint bus has no base voltage specified. - Non-zero neutral impedance specified but at least one endpoint bus has no base or nominal voltage specified. - A non-zero value is specified for an impedance that is not used with the specified connection code. - A value of zero is specified for an impedance that is used with the specified connection code. - Both ZGn and Z0n are used with the specified connection code and both are (0.0,0.0). - Connection code is 17 or 18 but at least one endpoint bus has no base or nominal voltage specified

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2.52. SEQ_TWO_WINDING_DATA

This API routine is obsolete. It has been replaced by the API routine seq_two_winding_data_3 .

Use this API to modify the zero sequence data of an existing two-winding transformer in the working case (see PSSE Program Operation Manual, Zero Sequence Transformer Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SEQ_TWO_WINDING_DATA  IBUS  JBUS  'CKT'  INTGAR(1)
 REALAR(1)..REALAR(6)
```

Python command syntax:

```
ierr = seq_two_winding_data(ibus, jbus, ckt, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_TWO_WINDING_API_2(IBUS, JBUS, CKT, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Integer INTGAR (1)	array of 1 elements specifying one element (defaulted data items are unchanged)(input).	
	INTGAR(1)	CC, connection code (1 through 8)
Real REALAR (6)	array of 6 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	RG, grounding resistance if connection code < 8; winding 1 side resistance to ground if CC = 8
	REALAR(2)	XG, grounding reactance if connection code < 8; winding 1 side reactance to ground if CC = 8
	REALAR(3)	R1, zero sequence transformer resistance
	REALAR(4)	X1, zero sequence transformer reactance
	REALAR(5)	RG2, winding 2 side resistance to ground if CC = 8; 0.0 otherwise
	REALAR(6)	XG2, winding 2 side reactance to ground if CC = 8; 0.0 otherwise
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found

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IERR = 2	circuit identifier is more than two characters
IERR = 3	circuit identifier may not start with &, * or @
IERR = 4	branch not found
IERR = 5	branch exists but it is a non-transformer branch
IERR = 6	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Invalid winding connection code. - Zero sequence impedance is (0.0,0.0). - Non-zero winding 2 impedance to ground not used with the specified connection code. - Winding n impedance to ground is (0.0,0.0) with connection code 8. - Grounding impedance is (0.0,0.0) with connection code 9. - A specified series impedance is (0.0,0.0) with connection code 9

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2.53. SEQ_TWO_WINDING_DATA_3

Use this API to modify the zero sequence data of an existing two-winding transformer in the working case (see PSSE Program Operation Manual, Zero Sequence Transformer Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SEQ_TWO_WINDING_DATA_3  IBUS  JBUS  'CKT'  INTGAR(1)..INTGAR(3)
 REALAR(1)..REALAR(10)
```

Python command syntax:

```
ierr = seq_two_winding_data_3(ibus, jbus, ckt, intgar, realar)
```

Fortran command syntax:

```
CALL SEQ_TWO_WINDING_API_3(IBUS, JBUS, CKT, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Integer INTGAR (3)	array of 3 elements specifying (defaulted data items are unchanged)(input).	
	INTGAR(1)	CC, connection code
	INTGAR(2)	CZ0, leakage impedance data I/O code
	INTGAR(3)	CZG, grounding impedance data I/O code
Real REALAR (10)	array of 10 elements specifying (defaulted data items are unchanged)(input).	
	REALAR(1)	RG1, grounding resistance at the winding 1 bus for an impedance grounded transformer
	REALAR(2)	XG1, grounding reactance at the winding 1 bus for an impedance grounded transformer
	REALAR(3)	R01, zero sequence leakage resistance connected to the winding 1 bus
	REALAR(4)	X01, zero sequence leakage reactance connected to the winding 1 bus
	REALAR(5)	RG2, grounding resistance at the winding 2 bus for an impedance grounded transformer
	REALAR(6)	XG2, grounding reactance at the winding 2 bus for an impedance grounded transformer
	REALAR(7)	R02, zero sequence leakage resistance connected to the winding 2 bus
	REALAR(8)	X02, zero sequence leakage reactance connected to the winding 2 bus
	REALAR(9)	RN, common neutral grounding resistance

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	REALAR(10)	XN, common neutral grounding reactance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	circuit identifier may not start with &, * or @
	IERR = 4	branch not found
	IERR = 5	branch exists but it is a non-transformer branch
	IERR = 6	CZG is 3 but at least one endpoint bus has no base voltage specified
	IERR = 7	invalid CZ0 or CZG value
	IERR = 8	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Invalid winding connection code. - Z01 is set to its positive sequence value. - A non-zero value is specified for an impedance that is not used with the specified connection code.. - A value of zero is specified for an impedance that is used with the specified connection code. - Invalid CZ0 or CZG data value. - CZ0 or CZG is not 1 with connection code 1 through 9. - CZG data value is 3 but at least one endpoint bus has no base voltage specified. - Non-zero neutral impedance specified but at least one endpoint bus has no base or nominal voltage specified. - Connection code is 18 through 22 but at least one endpoint bus has no base or nominal voltage specified

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2.54. SHUNT_CHNG

Use this API routine to modify the data of an existing fixed bus shunt in the working case (see PSSE Program Operation Manual, Fixed Bus Shunt Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SHUNT_CHNG  IBUS  'ID'  INTGAR(1)  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = shunt_chng(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL SHUNT_CHNG(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	shunt identifier (input; '1').	
Integer INTGAR (1)	See the SHUNT_DATA API routine for details (input).	
Real REALAR (2)	See the SHUNT_DATA API routine for details (input).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	shunt identifier is more than two characters
	IERR = 5	fixed bus shunt not found
	IERR = 6	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid shunt status

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2.55. SHUNT_DATA

Use this API to modify the data of an existing fixed bus shunt in the working case or to add a new fixed bus shunt to the working case (see PSSE Program Operation Manual, Fixed Bus Shunt Data).

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SHUNT_DATA  IBUS  'ID'  INTGAR(1)  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = shunt_data(ibus, id, intgar, realar)
```

Fortran command syntax:

```
CALL SHUNT_API(IBUS, ID, INTGAR, REALAR, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*2 ID	shunt identifier (input; '1').	
Integer INTGAR (1)	array of 1 elements specifying (input).	
	INTGAR(1)	STATUS, shunt status (1 by default)
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	GL, fixed shunt admittance (conductance) (0.0 by default)
	REALAR(2)	BL, fixed shunt admittance (susceptance) (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	shunt identifier is more than two characters
	IERR = 3	blank shunt identifier is invalid
	IERR = 4	fixed bus shunt table is full
	IERR = 6	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid shunt status

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2.56. SOLUTION_PARAMETERS

This API routine is obsolete. It has been replaced by the API routine solution_parameters_2 .

Use this API to modify the power flow solution parameters in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SOLUTION_PARAMETERS  INTGAR(1)..INTGAR(3)  REALAR(1)..REALAR(16)
```

Python command syntax:

```
ierr = solution_parameters(intgar, realar)
```

Fortran command syntax:

```
CALL SOLUTION_PARAMETER_API(INTGAR, REALAR, IERR, NERR)
```

Where:

Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	ITMX, Gauss-Seidel maximum number of iterations
	INTGAR(2)	ITMXN, Newton-Raphson maximum number of iterations
	INTGAR(3)	ITMXTY, TYSL maximum number of iterations
Real REALAR (16)	array of 16 elements specifying (input).	
	REALAR(1)	ACCP, Gauss-Seidel real component acceleration factor
	REALAR(2)	ACCQ, Gauss-Seidel imaginary component acceleration factor
	REALAR(3)	ACCM, special MSLV acceleration factor
	REALAR(4)	TOL, Gauss-Seidel convergence tolerance
	REALAR(5)	ACCN, Newton-Raphson acceleration factor
	REALAR(6)	TOLN, Newton-Raphson mismatch convergence tolerance
	REALAR(7)	ACCTY, TYSL acceleration factor
	REALAR(8)	TOLTY, TYSL convergence tolerance
	REALAR(9)	BLOWUP, blow-up threshold
	REALAR(10)	PQBRAK, constant power load characteristic breakpoint
	REALAR(11)	THRSHZ, zero impedance line threshold
	REALAR(12)	ADJTHR, automatic adjustment threshold tolerance
	REALAR(13)	ACCTAP, tap movement deceleration factor
	REALAR(14)	TAPLIM, maximum tap change per adjustment

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	REALAR(15)	DVLIM, largest | (delta v)/|v| | for Newton solutions
	REALAR(16)	NDVFCT, non-divergent required improvement factor
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = -1	warnings: - None
Integer NERR	number of entries in PBUF02 containing error messages (output).	

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2.57. SOLUTION_PARAMETERS_2

This API routine is obsolete. It has been replaced by the API routine solution_parameters_3 .

Use this API to modify the power flow solution parameters in the working case.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_SOLUTION_PARAMETERS_2  INTGAR(1)..INTGAR(3)  REALAR(1)..REALAR(18)
```

Python command syntax:

```
ierr = solution_parameters_2(intgar, realar)
```

Fortran command syntax:

```
CALL SOLUTION_PARAMETER_API_2(INTGAR, REALAR, IERR, NERR)
```

Where:

Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	ITMX, Gauss-Seidel maximum number of iterations
	INTGAR(2)	ITMXN, Newton-Raphson maximum number of iterations
	INTGAR(3)	ITMXTY, TYSL maximum number of iterations
Real REALAR (18)	array of 18 elements specifying (input).	
	REALAR(1)	ACCP, Gauss-Seidel real component acceleration factor
	REALAR(2)	ACCQ, Gauss-Seidel imaginary component acceleration factor
	REALAR(3)	ACCM, special MSLV acceleration factor
	REALAR(4)	TOL, Gauss-Seidel convergence tolerance
	REALAR(5)	ACCN, Newton-Raphson acceleration factor
	REALAR(6)	TOLN, Newton-Raphson mismatch convergence tolerance
	REALAR(7)	ACCTY, TYSL acceleration factor
	REALAR(8)	TOLTY, TYSL convergence tolerance
	REALAR(9)	BLOWUP, blow-up threshold
	REALAR(10)	PQBRAK, constant power load characteristic breakpoint
	REALAR(11)	THRSHZ, zero impedance line threshold
	REALAR(12)	ADJTHR, automatic adjustment threshold tolerance
	REALAR(13)	ACCTAP, tap movement deceleration factor
	REALAR(14)	TAPLIM, maximum tap change per adjustment

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	REALAR(15)	DVLIM, largest | (delta v)/|v| | for Newton solutions
	REALAR(16)	NDVFCT, non-divergent required improvement factor
	REALAR(17)	VCTOLQ, Newton-Raphson voltage controlled bus reactive power mismatch convergence tolerance
	REALAR(18)	VCTOLV, Newton-Raphson voltage controlled bus voltage error convergence tolerance
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = -1	warnings: - None
Integer NERR	number of entries in PBUF02 containing error messages (output).	

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2.58. SOLUTION_PARAMETERS_3

This API routine is obsolete. It has been replaced by the API routine solution_parameters_4 .

Use this API to modify the power flow solution parameters in the working case (see PSSE Program Operation Manual, Sections 6.3.13, 6.3.18, 6.5.1 and 11.7).

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SOLUTION_PARAMETERS_3  INTGAR(1)..INTGAR(3)  REALAR(1)..REALAR(19)
```

Python command syntax:

```
ierr = solution_parameters_3(intgar, realar)
```

Fortran command syntax:

```
CALL SOLUTION_PARAMETER_API_3(INTGAR, REALAR, IERR)
```

Where:

Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	ITMX, Gauss-Seidel maximum number of iterations (100 by default)
	INTGAR(2)	ITMXN, Newton-Raphson maximum number of iterations (20 by default)
	INTGAR(3)	ITMXTY, TYSL maximum number of iterations (20 by default)
Real REALAR (19)	array of 19 elements specifying (input).	
	REALAR(1)	ACCP, Gauss-Seidel real component acceleration factor (1.6 by default)
	REALAR(2)	ACCQ, Gauss-Seidel imaginary component acceleration factor (1.6 by default)
	REALAR(3)	ACCM, special MSLV acceleration factor (1.0 by default)
	REALAR(4)	TOL, Gauss-Seidel convergence tolerance (0.0001 by default)
	REALAR(5)	ACCN, Newton-Raphson acceleration factor (1.0 by default)
	REALAR(6)	TOLN, Newton-Raphson mismatch convergence tolerance (default Newton power flow solution tolerance option setting)
	REALAR(7)	ACCTY, TYSL acceleration factor (1.0 by default)
	REALAR(8)	TOLTY, TYSL convergence tolerance (0.00001 by default)
	REALAR(9)	BLOWUP, blow-up threshold (5.0 by default)

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	REALAR(10)	PQBRAK, constant power load characteristic breakpoint (0.7 by default)
	REALAR(11)	THRSHZ, zero impedance line threshold (0.0001 by default)
	REALAR(12)	ADJTHR, automatic adjustment threshold tolerance (0.005 by default)
	REALAR(13)	ACCTAP, tap movement deceleration factor (1.0 by default)
	REALAR(14)	TAPLIM, maximum tap change per adjustment (0.05 by default)
	REALAR(15)	DVLIM, largest | (delta v)/|v| | for Newton solutions (0.99 by default)
	REALAR(16)	NDVFCT, non-divergent required improvement factor (0.99 by default)
	REALAR(17)	VCTOLQ, Newton-Raphson voltage controlled bus reactive power mismatch convergence tolerance (default Newton power flow solution tolerance option setting)
	REALAR(18)	VCTOLV, Newton-Raphson voltage controlled bus voltage error convergence tolerance (0.00001 by default)
	REALAR(19)	SWVBND, percent of voltage controlling band mode switched shunts to be adjusted per power flow iteration (100.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Invalid percent value for switched shunt adjustment

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2.59. SOLUTION_PARAMETERS_4

Use this API to modify the power flow solution parameters in the working case (see PSSE Program Operation Manual, Sections 6.3.13, 6.3.18, 6.5.1 and 11.7).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SOLUTION_PARAMETERS_4  INTGAR(1)..INTGAR(5)  REALAR(1)..REALAR(19)
```

Python command syntax:

```
ierr = solution_parameters_4(intgar, realar)
```

Fortran command syntax:

```
CALL SOLUTION_PARAMETER_API_4(INTGAR, REALAR, IERR)
```

Where:

Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	ITMX, Gauss-Seidel maximum number of iterations (100 by default)
	INTGAR(2)	ITMXN, Newton-Raphson maximum number of iterations (20 by default)
	INTGAR(3)	ITMXTY, TYSL maximum number of iterations (20 by default)
	INTGAR(4)	MXTPSS, maximum number of times taps and/or switched shunts are adjusted during power flow solutions (100 by default)
	INTGAR(5)	MXSWIM, maximum number of times an induction machine can change state (e.g., from "running" to "stalled") during power flow solutions (10 by default)
Real REALAR (19)	array of 19 elements specifying (input).	
	REALAR(1)	ACCP, Gauss-Seidel real component acceleration factor (1.6 by default)
	REALAR(2)	ACCQ, Gauss-Seidel imaginary component acceleration factor (1.6 by default)
	REALAR(3)	ACCM, special MSLV acceleration factor (1.0 by default)
	REALAR(4)	TOL, Gauss-Seidel convergence tolerance (0.0001 by default)
	REALAR(5)	ACCN, Newton-Raphson acceleration factor (1.0 by default)
	REALAR(6)	TOLN, Newton-Raphson mismatch convergence tolerance (default Newton power flow solution tolerance option setting)
	REALAR(7)	ACCTY, TYSL acceleration factor (1.0 by default)

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	REALAR(8)	TOLTY, TYSL convergence tolerance (0.00001 by default)
	REALAR(9)	BLOWUP, blow-up threshold (5.0 by default)
	REALAR(10)	PQBRAK, constant power load characteristic breakpoint (0.7 by default)
	REALAR(11)	THRSHZ, zero impedance line threshold (0.0001 by default)
	REALAR(12)	ADJTHR, automatic adjustment threshold tolerance (0.005 by default)
	REALAR(13)	ACCTAP, tap movement deceleration factor (1.0 by default)
	REALAR(14)	TAPLIM, maximum tap change per adjustment (0.05 by default)
	REALAR(15)	DVLIM, largest | (delta v)/|v| | for Newton solutions (0.99 by default)
	REALAR(16)	NDVFCT, non-divergent required improvement factor (0.99 by default)
	REALAR(17)	VCTOLQ, Newton-Raphson voltage controlled bus reactive power mismatch convergence tolerance (default Newton power flow solution tolerance option setting)
	REALAR(18)	VCTOLV, Newton-Raphson voltage controlled bus voltage error convergence tolerance (0.00001 by default)
	REALAR(19)	SWVBND, percent of voltage controlling band mode switched shunts to be adjusted per power flow iteration (100.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Invalid percent value for switched shunt adjustment. - Invalid induction motor treatment code

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2.60. SWITCHED_SHUNT_CHNG_3

Use this API routine to modify the data of an existing switched shunt in the working case (see PSSE Program Operation Manual, Switched Shunt Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SWITCHED_SHUNT_CHNG_3  IBUS  INTGAR(1)..INTGAR(12)
 REALAR(1)..REALAR(12)  'RMIDNT'
```

Python command syntax:

```
ierr = switched_shunt_chng_3(ibus, intgar, realar, rmidnt)
```

Fortran command syntax:

```
CALL SWITCHED_SHUNT_CHNG_3(IBUS, INTGAR, REALAR, RMIDNT, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Integer INTGAR (12)	array of 12 elements specifying (input).	
	INTGAR(1)	N1, number of steps for block 1 (0 by default)
	INTGAR(2)	N2, number of steps for block 2 (0 by default)
	INTGAR(3)	N3, number of steps for block 3 (0 by default)
	INTGAR(4)	N4, number of steps for block 4 (0 by default)
	INTGAR(5)	N5, number of steps for block 5 (0 by default)
	INTGAR(6)	N6, number of steps for block 6 (0 by default)
	INTGAR(7)	N7, number of steps for block 7 (0 by default)
	INTGAR(8)	N8, number of steps for block 8 (0 by default)
	INTGAR(9)	MODSW, control mode (1 by default)
	INTGAR(10)	SWREM, remote bus number (0 by default)
	INTGAR(11)	STAT, status (1 by default)
	INTGAR(12)	ADJM, adjustment method (0 by default)
Real REALAR (12)	array of 12 elements specifying (input).	
	REALAR(1)	B1, admittance increment per step for block 1 (0.0 by default)
	REALAR(2)	B2, admittance increment per step for block 2 (0.0 by default)

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	REALAR(3)	B3, admittance increment per step for block 3 (0.0 by default)
	REALAR(4)	B4, admittance increment per step for block 4 (0.0 by default)
	REALAR(5)	B5, admittance increment per step for block 5 (0.0 by default)
	REALAR(6)	B6, admittance increment per step for block 6 (0.0 by default)
	REALAR(7)	B7, admittance increment per step for block 7 (0.0 by default)
	REALAR(8)	B8, admittance increment per step for block 8 (0.0 by default)
	REALAR(9)	VSWHI, desired voltage upper limit (1.0 by default)
	REALAR(10)	VSWLO, desired voltage lower limit (1.0 by default)
	REALAR(11)	BINIT, present switched shunt admittance (0.0 by default)
	REALAR(12)	RMPCT, percent of contributed reactive power (100.0 by default)
Character*12 RMIDNT	VSC dc line name (MODSW =4) or FACTS device name (MODSW = 6) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 3	switched shunt not found
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - invalid switched shunt control mode - control mode is 4 and a blank VSC dc line name is specified - VSC dc line not found - control mode is 6 and a blank FACTS device name is specified - FACTS device not found - non-blank device name specified with control mode 0, 1, 2, 3 or 5 - regulated bus not found - voltage controlling switched shunt specified as remotely regulating itself - type code of regulated bus is not 1 or 2 - VHIGH is less than VLOW for voltage controlling switched shunt - no regulated bus specified for plant, VSC converter, or switched shunt control - control mode is 3 and the regulated bus has no generation - control mode is 4 and the regulated bus is not a converter of the specified VSC dc line - control mode is 5 and the regulated bus has no switched shunt - regulated bus is not the sending end

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bus of the specified FACTS device - VLOW is not less than VHIGH for plant, VSC converter, switched shunt, or FACTS device control - VHIGH is greater than 1.0 for plant, VSC converter, switched shunt, or FACTS device control - VLOW is less than 0.0 for plant, VSC converter, switched shunt, or FACTS device control - RMPCT is not positive - block has N or B zero and the other non-zero - invalid number of steps in block - block with non-zero N is beyond a previous N=0 block - block 1 has no steps. - Reactor block is out of sequence. - Initial admittance is below minimum. - Initial admittance is above maximum. - Initial admittance is not on a step. - Block has admittances with different signs in positive and zero sequences. - Invalid switched shunt status. - Invalid switched shunt adjustment method

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2.61. SWITCHED_SHUNT_DATA

This API routine is obsolete. It has been replaced by the API routine switched_shunt_data_3 .

Use this API to modify the data of an existing switched shunt in the working case or to add a new switched shunt to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SWITCHED_SHUNT_DATA  IBUS  INTGAR(1)..INTGAR(10)
 REALAR(1)..REALAR(12)  'RMIDNT'
```

Python command syntax:

```
ierr = switched_shunt_data(ibus, intgar, realar, rmidnt)
```

Fortran command syntax:

```
CALL SWITCHED_SHUNT_API_2(IBUS, INTGAR, REALAR, RMIDNT, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Integer INTGAR (10)	array of 10 elements specifying (input).	
	INTGAR(1)	N1, number of steps for block 1 (0 by default)
	INTGAR(2)	N2, number of steps for block 2 (0 by default)
	INTGAR(3)	N3, number of steps for block 3 (0 by default)
	INTGAR(4)	N4, number of steps for block 4 (0 by default)
	INTGAR(5)	N5, number of steps for block 5 (0 by default)
	INTGAR(6)	N6, number of steps for block 6 (0 by default)
	INTGAR(7)	N7, number of steps for block 7 (0 by default)
	INTGAR(8)	N8, number of steps for block 8 (0 by default)
	INTGAR(9)	MODSW, control mode (1 by default)
	INTGAR(10)	SWREM, remote bus number (0 by default)
Real REALAR (12)	array of 12 elements specifying (input).	
	REALAR(1)	B1, admittance increment per step for block 1 (0.0 by default)
	REALAR(2)	B2, admittance increment per step for block 2 (0.0 by default)

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	REALAR(3)	B3, admittance increment per step for block 3 (0.0 by default)
	REALAR(4)	B4, admittance increment per step for block 4 (0.0 by default)
	REALAR(5)	B5, admittance increment per step for block 5 (0.0 by default)
	REALAR(6)	B6, admittance increment per step for block 6 (0.0 by default)
	REALAR(7)	B7, admittance increment per step for block 7 (0.0 by default)
	REALAR(8)	B8, admittance increment per step for block 8 (0.0 by default)
	REALAR(9)	VSWHI, desired voltage upper limit (1.0 by default)
	REALAR(10)	VSWLO, desired voltage lower limit (1.0 by default)
	REALAR(11)	BINIT, present switched shunt admittance (0.0 by default)
	REALAR(12)	RMPCT, percent of contributed reactive power (100.0 by default)
Character*12 RMIDNT	VSC dc line name (MODSW =4) or FACTS device name (MODSW = 6) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	switched shunt table is full
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - invalid switched shunt control mode - control mode is 4 and a blank VSC dc line name is specified - VSC dc line not found - control mode is 6 and a blank FACTS device name is specified - FACTS device not found - non-blank device name specified with control mode 0, 1, 2, 3 or 5 - regulated bus not found - voltage controlling switched shunt specified as remotely regulating itself - type code of regulated bus is not 1 or 2 - VHIGH is less than VLOW for voltage controlling switched shunt - no regulated bus specified for plant, VSC converter, or switched shunt control - control mode is 3 and the regulated bus has no generation - control mode is 4 and the regulated bus is not a converter of the specified VSC dc line - control mode is 5 and the regulated bus has no switched shunt - regulated bus is not the sending end

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bus of the specified FACTS device - VLOW is not less than VHIGH for plant, VSC converter, switched shunt, or FACTS device control - VHIGH is greater than 1.0 for plant, VSC converter, switched shunt, or FACTS device control - VLOW is less than 0.0 for plant, VSC converter, switched shunt, or FACTS device control - - RMPCT is not positive - block has N or B zero and the other non-zero - invalid number of steps in block - block with non-zero N is beyond a previous N=0 block - block 1 has no steps. - Reactor block is out of sequence. - Initial admittance is below minimum. - Initial admittance is above maximum. - Initial admittance is not on a step. - Block has admittances with different signs in positive and zero sequences

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2.62. SWITCHED_SHUNT_DATA_3

Use this API to modify the data of an existing switched shunt in the working case or to add a new switched shunt to the working case (see PSSE Program Operation Manual, Switched Shunt Data).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SWITCHED_SHUNT_DATA_3  IBUS  INTGAR(1)..INTGAR(12)
 REALAR(1)..REALAR(12)  'RMIDNT'
```

Python command syntax:

```
ierr = switched_shunt_data_3(ibus, intgar, realar, rmidnt)
```

Fortran command syntax:

```
CALL SWITCHED_SHUNT_API_3(IBUS, INTGAR, REALAR, RMIDNT, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Integer INTGAR (12)	array of 12 elements specifying (input).	
	INTGAR(1)	N1, number of steps for block 1 (0 by default)
	INTGAR(2)	N2, number of steps for block 2 (0 by default)
	INTGAR(3)	N3, number of steps for block 3 (0 by default)
	INTGAR(4)	N4, number of steps for block 4 (0 by default)
	INTGAR(5)	N5, number of steps for block 5 (0 by default)
	INTGAR(6)	N6, number of steps for block 6 (0 by default)
	INTGAR(7)	N7, number of steps for block 7 (0 by default)
	INTGAR(8)	N8, number of steps for block 8 (0 by default)
	INTGAR(9)	MODSW, control mode (1 by default)
	INTGAR(10)	SWREM, remote bus number (0 by default)
	INTGAR(11)	STAT, status (1 by default)
	INTGAR(12)	ADJM, adjustment method (0 by default)
Real REALAR (12)	array of 12 elements specifying (input).	
	REALAR(1)	B1, admittance increment per step for block 1 (0.0 by default)
	REALAR(2)	B2, admittance increment per step for block 2 (0.0 by default)

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	REALAR(3)	B3, admittance increment per step for block 3 (0.0 by default)
	REALAR(4)	B4, admittance increment per step for block 4 (0.0 by default)
	REALAR(5)	B5, admittance increment per step for block 5 (0.0 by default)
	REALAR(6)	B6, admittance increment per step for block 6 (0.0 by default)
	REALAR(7)	B7, admittance increment per step for block 7 (0.0 by default)
	REALAR(8)	B8, admittance increment per step for block 8 (0.0 by default)
	REALAR(9)	VSWHI, desired voltage upper limit (1.0 by default)
	REALAR(10)	VSWLO, desired voltage lower limit (1.0 by default)
	REALAR(11)	BINIT, present switched shunt admittance (0.0 by default)
	REALAR(12)	RMPCT, percent of contributed reactive power (100.0 by default)
Character*12 RMIDNT	VSC dc line name (MODSW =4) or FACTS device name (MODSW = 6) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	switched shunt table is full
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - invalid switched shunt control mode - control mode is 4 and a blank VSC dc line name is specified - VSC dc line not found - control mode is 6 and a blank FACTS device name is specified - FACTS device not found - non-blank device name specified with control mode 0, 1, 2, 3 or 5 - regulated bus not found - voltage controlling switched shunt specified as remotely regulating itself - type code of regulated bus is not 1 or 2 - VHIGH is less than VLOW for voltage controlling switched shunt - no regulated bus specified for plant, VSC converter, or switched shunt control - control mode is 3 and the regulated bus has no generation - control mode is 4 and the regulated bus is not a converter of the specified VSC dc line - control mode is 5 and the regulated bus has no switched shunt - regulated bus is not the sending end

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bus of the specified FACTS device - VLOW is not less than VHIGH for plant, VSC converter, switched shunt, or FACTS device control - VHIGH is greater than 1.0 for plant, VSC converter, switched shunt, or FACTS device control - VLOW is less than 0.0 for plant, VSC converter, switched shunt, or FACTS device control - RMPCT is not positive - block has N or B zero and the other non-zero - invalid number of steps in block - block with non-zero N is beyond a previous N=0 block - block 1 has no steps. - Reactor block is out of sequence. - Initial admittance is below minimum. - Initial admittance is above maximum. - Initial admittance is not on a step. - Block has admittances with different signs in positive and zero sequences. - Invalid switched shunt status. - Invalid switched shunt adjustment method

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2.63. THREE_WINDING_DATA

This API routine is obsolete. It has been replaced by the API routines three_wnd_impedance_data and three_wnd_winding_data .

Use this API to modify the data of an existing three-winding transformer in the working case or to add a new three-winding transformer to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_THREE_WINDING_DATA  IBUS  JBUS  KBUS  'CKT'  INTGAR(1)..INTGAR(17)
 REALAR(1)..REALAR(41)  'NAME'
```

Python command syntax:

```
ierr, realaro = three_winding_data(ibus, jbus, kbus, ckt, intgar, realari,
 name)
```

Fortran command syntax:

```
CALL THREE_WINDING_API(IBUS, JBUS, KBUS, CKT, INTGAR, REALAR, NAME, IERR,
 NERR)
```

Where:

Integer IBUS	bus number of one of the buses (input).	
Integer JBUS	bus number one of the other buses (input).	
Integer KBUS	bus number of the third bus (input).	
Character*2 CKT	transformer circuit identifier (input).	
Integer INTGAR (17)	array of 17 elements specifying (input).	
	INTGAR(1)	O1, first owner number
	INTGAR(2)	O2, second owner number
	INTGAR(3)	O3, third owner number
	INTGAR(4)	O4, fourth owner number
	INTGAR(5)	NTP1, number of tap positions
	INTGAR(6)	TAB1, impedance correction table number
	INTGAR(7)	CONT1, controlled bus number
	INTGAR(8)	SICOD1, negative for controlled bus on winding one side
	INTGAR(9)	COD1, adjustment control mode flag (-3 through +3)
	INTGAR(10)	CW, winding data I/O code
	INTGAR(11)	CZ, impedance data I/O code
	INTGAR(12)	CM, magnetizing admittance data I/O code
	INTGAR(13)	STAT, branch status
	INTGAR(14)	NMETBS, non-metered end bus number (IBUS, JBUS or KBUS)

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	INTGAR(15)	WN1BUS, winding one side bus number (IBUS, JBUS or KBUS)
	INTGAR(16)	WN2BUS, winding two side bus number (IBUS, JBUS or KBUS)
	INTGAR(17)	WN3BUS, winding three side bus number (IBUS, JBUS or KBUS)
Real REALAR (53)	array of elements specifying (input).(output).	
	REALAR(1)	R1-2, nominal bus one to two transformer resistance
	REALAR(2)	X1-2, nominal bus one to two transformer reactance
	REALAR(3)	R3-1, nominal bus three to one transformer resistance
	REALAR(4)	X3-1, nominal bus three to one transformer reactance
	REALAR(5)	R2-3, nominal bus two to three transformer resistance
	REALAR(6)	X2-3, nominal bus two to three transformer reactance
	REALAR(7)	SBS1-2, winding one to two base MVA
	REALAR(8)	SBS2-3, winding two to three base MVA
	REALAR(9)	SBS3-1, winding three to one base MVA
	REALAR(10)	MAG1, magnetizing conductance or no-load losses
	REALAR(11)	MAG2, magnetizing susceptance or exciting current
	REALAR(12)	WINDV1, winding one ratio or voltage
	REALAR(13)	NOMV1, winding one nominal voltage
	REALAR(14)	1, winding one phase shift angle
	REALAR(15)	RATA1, winding one rating set A line rating
	REALAR(16)	RATB1, winding one rating set B line rating
	REALAR(17)	RATC1, winding one rating set C line rating
	REALAR(18)	WINDV2, winding two ratio or voltage
	REALAR(19)	NOMV2, winding two nominal voltage
	REALAR(20)	2, winding two phase shift angle
	REALAR(21)	RATA2, winding two rating set A line rating
	REALAR(22)	RATB2, winding two rating set B line rating
	REALAR(23)	RATC2, winding two rating set C line rating
	REALAR(24)	WINDV3, winding three ratio or voltage
	REALAR(25)	NOMV3, winding three nominal voltage
	REALAR(26)	3, winding three phase shift angle
	REALAR(27)	RATA3, winding three rating set A line rating
	REALAR(28)	RATB3, winding three rating set B line rating
	REALAR(29)	RATC3, winding three rating set C line rating
	REALAR(30)	F1, first owner fraction

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	REALAR(31)	F2, second owner fraction
	REALAR(32)	F3, third owner fraction
	REALAR(33)	F4, fourth owner fraction
	REALAR(34)	RMA1, winding one ratio/angle high limit
	REALAR(35)	RMI1, winding one ratio/angle low limit
	REALAR(36)	VMA1, voltage or flow upper limit
	REALAR(37)	VMI1, voltage or flow lower limit
	REALAR(38)	CR1, load drop compensating resistance
	REALAR(39)	CX1, load drop compensating reactance
	REALAR(40)	VMSTAR, star bus voltage magnitude
	REALAR(41)	TAR, star bus voltage angle
	REALAR(42)	actual bus one to two resistance (returned)
	REALAR(43)	actual bus one to two reactance (returned)
	REALAR(44)	actual bus three to one resistance (returned)
	REALAR(45)	actual bus three to one reactance (returned)
	REALAR(46)	nominal winding one resistance (returned)
	REALAR(47)	nominal winding one reactance (returned)
	REALAR(48)	nominal winding two resistance (returned)
	REALAR(49)	nominal winding two reactance (returned)
	REALAR(50)	nominal winding three resistance (returned)
	REALAR(51)	nominal winding three reactance (returned)
	REALAR(52)	actual winding one resistance (returned)
	REALAR(53)	actual winding one reactance (returned)
Character*40 NAME	transformer name (input).	
Real REALARI	(input).	
Real REALARO	(output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	invalid CW, CZ and/or CM value
	IERR = 4	CW, CZ and/or CM >1 but no base voltage at one or more buses
	IERR = 5	three different buses must be specified
	IERR = 6	blank circuit identifier
	IERR = 7	circuit identifier may not start with '&'
	IERR = 8	three-winding transformer table full
	IERR = 9	bus table full
	IERR = 10	not enough branch table entries available
	IERR = 11	not enough room in two winding transformer table
	IERR = 12	not enough room in branch ownership table

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	IERR = 13	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - Invalid set of winding buses. - Invalid transformer status. - Invalid non-metered end bus. - Winding side buses interchanged. - Losses are too big relative to total impedance at winding base MVA. - Magnitude of winding impedance is less than 0.00001. - Controlled bus not found. - Invalid adjustment control mode flag. - Any two-winding transformer warnings
Integer NERR	number of entries in PBUF02 containing error messages (output).	

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2.64. THREE_WND_IMPEDANCE_DATA

This API routine is obsolete. It has been replaced by the API routine three_wnd_imped_data_3 .

Use this API routine to modify the impedance data of an existing three-winding transformer in the working case, or to add a new three-winding transformer to the working case (see PSSE Program Operation Manual, Non-Transformer Branch Data).

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_THREE_WND_IMPEDANCE_DATA  IBUS  JBUS  KBUS  'CKT'
 INTGAR(1)..INTGAR(12)  REALAR(1)..REALAR(17)  'NAME'
```

Python command syntax:

```
ierr, realaro = three_wnd_impedance_data(ibus, jbus, kbus, ckt, intgar,
 realari, name)
```

Fortran command syntax:

```
CALL THREE_WINDING_IMPEDANCE_API_2(IBUS, JBUS, KBUS, CKT, INTGAR, REALAR,
 NAME, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 CKT	transformer circuit identifier (input; '1').	
Integer INTGAR (12)	array of 12 elements specifying (input).	
	INTGAR(1)	O1, first owner number (owner of bus IBUS by default)
	INTGAR(2)	O2, second owner number (0 by default)
	INTGAR(3)	O3, third owner number (0 by default)
	INTGAR(4)	O4, fourth owner number (0 by default)
	INTGAR(5)	CW, winding data I/O code (1 by default)
	INTGAR(6)	CZ, impedance data I/O code (1 by default)
	INTGAR(7)	CM, magnetizing admittance data I/O code (1 by default)
	INTGAR(8)	STAT, branch status (1 by default)
	INTGAR(9)	NMETBS, non-metered end bus number (IBUS, JBUS or KBUS) (JBUS by default)
	INTGAR(10)	WN1BUS, winding one side bus number (IBUS, JBUS or KBUS) (IBUS by default)
	INTGAR(11)	WN2BUS, winding two side bus number (IBUS, JBUS or KBUS) (JBUS by default)
	INTGAR(12)	WN3BUS, winding three side bus number (IBUS, JBUS or KBUS) (KBUS by default)

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Real REALAR (23)	(first seventeen are values; last six are values) (input (output).).
Real REALARI (17)	array of 17 elements specifying corresponding to values REALAR(1) through REALAR(17) as described above(input).
REALARI(1)	R1-2, nominal bus one to two transformer resistance (0.0 by default)
REALARI(2)	X1-2, nominal bus one to two transformer reactance (0.0002 by default)
REALARI(3)	R2-3, nominal bus two to three transformer resistance (0.0 by default)
REALARI(4)	X2-3, nominal bus two to three transformer reactance (0.0002 by default)
REALARI(5)	R3-1, nominal bus three to one transformer resistance (0.0 by default)
REALARI(6)	X3-1, nominal bus three to one transformer reactance (0.0002 by default)
REALARI(7)	SBS1-2, winding one to two base MVA (SBASE by default)
REALARI(8)	SBS2-3, winding two to three base MVA (SBASE by default)
REALARI(9)	SBS3-1, winding three to one base MVA (SBASE by default)
REALARI(10)	MAG1, magnetizing conductance or no-load losses (0.0 by default)
REALARI(11)	MAG2, magnetizing susceptance or exciting current (0.0 by default)
REALARI(12)	F1, first owner fraction (1.0 by default)
REALARI(13)	F2, second owner fraction (0.0 by default)
REALARI(14)	F3, third owner fraction (0.0 by default)
REALARI(15)	F4, fourth owner fraction (0.0 by default)
REALARI(16)	VMSTAR, star bus voltage magnitude (1.0 by default)
REALARI(17)	TAR, star bus voltage angle (0.0 by default)
Real REALARO (6)	array of 6 elements specifying corresponding to values REALAR(18) through REALAR(23) as described above(output).
REALARO(1)	actual bus one to two resistance (returned)
REALARO(2)	actual bus one to two reactance (returned)
REALARO(3)	actual bus two to three resistance (returned)
REALARO(4)	actual bus two to three reactance (returned)
REALARO(5)	actual bus three to one resistance (returned)
REALARO(6)	actual bus three to one reactance (returned)
Character*40 NAME	transformer name (input; blank by default).
Integer IERR	error code (output).
IERR = 0	no error occurred
IERR > 0	fatal error occurred

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IERR = 1	bus not found
IERR = 2	circuit identifier is more than two characters
IERR = 3	invalid CW, CZ and/or CM value
IERR = 4	CW, CZ and/or CM >1 but at least one end-point bus has no base voltage
IERR = 5	three different buses must be specified
IERR = 6	blank circuit identifier is invalid
IERR = 7	transformer circuit identifier may not start with &, * or @
IERR = 8	three-winding transformer table is full
IERR = 9	bus table is full
IERR = 10	not enough branch table entries are available
IERR = 11	not enough two-winding transformer table entries are available
IERR = 12	not enough branch ownership table entries are available
IERR = 14	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Invalid set of winding buses. - Invalid CW, CZ and/or CM data value. - Invalid transformer status. - Invalid non-metered end bus. - Winding side buses interchanged. - Losses are too big relative to total impedance at winding base MVA. - Magnitude of winding impedance is less than 0.00001. - Any two-winding transformer warnings

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2.65. THREE_WND_IMPED_CHNG_3

Use this API routine to modify the impedance data of an existing three-winding transformer in the working case (see PSSE Program Operation Manual, Non-Transformer Branch Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_THREE_WND_IMPED_CHNG_3  IBUS  JBUS  KBUS  'CKT'  INTGAR(1)..INTGAR(12)
  REALAR(1)..REALAR(17)  'CHARAR(1)'  'CHARAR(2)'
```

Python command syntax:

```
ierr, realaro = three_wnd_imped_chng_3(ibus, jbus, kbus, ckt, intgar,
 realari, charar)
```

Fortran command syntax:

```
CALL THREE_WINDING_IMPEDANCE_CHNG_3(IBUS, JBUS, KBUS, CKT, INTGAR, REALAR,
 CHARAR, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 CKT	transformer circuit identifier (input; '1').	
Integer INTGAR (12)	array of 12 elements specifying (input).	
	INTGAR(1)	O1, first owner number (owner of bus IBUS by default)
	INTGAR(2)	O2, second owner number (0 by default)
	INTGAR(3)	O3, third owner number (0 by default)
	INTGAR(4)	O4, fourth owner number (0 by default)
	INTGAR(5)	CW, winding data I/O code (1 by default)
	INTGAR(6)	CZ, impedance data I/O code (1 by default)
	INTGAR(7)	CM, magnetizing admittance data I/O code (1 by default)
	INTGAR(8)	STAT, branch status (1 by default)
	INTGAR(9)	NMETBS, non-metered end bus number (IBUS, JBUS or KBUS) (JBUS by default)
	INTGAR(10)	WN1BUS, winding one side bus number (IBUS, JBUS or KBUS) (IBUS by default)
	INTGAR(11)	WN2BUS, winding two side bus number (IBUS, JBUS or KBUS) (JBUS by default)
	INTGAR(12)	WN3BUS, winding three side bus number (IBUS, JBUS or KBUS) (KBUS by default)
Real REALAR (23)	(first seventeen are values; last six are values) (input (output).).	
Real REALARI (17)	array of 17 elements specifying corresponding to values REALAR(1) through REALAR(17) as described above(input).	

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	REALARI(1)	R1-2, nominal bus one to two transformer resistance (0.0 by default)
	REALARI(2)	X1-2, nominal bus one to two transformer reactance (0.0002 by default)
	REALARI(3)	R2-3, nominal bus two to three transformer resistance (0.0 by default)
	REALARI(4)	X2-3, nominal bus two to three transformer reactance (0.0002 by default)
	REALARI(5)	R3-1, nominal bus three to one transformer resistance (0.0 by default)
	REALARI(6)	X3-1, nominal bus three to one transformer reactance (0.0002 by default)
	REALARI(7)	SBS1-2, winding one to two base MVA (SBASE by default)
	REALARI(8)	SBS2-3, winding two to three base MVA (SBASE by default)
	REALARI(9)	SBS3-1, winding three to one base MVA (SBASE by default)
	REALARI(10)	MAG1, magnetizing conductance or no-load losses (0.0 by default)
	REALARI(11)	MAG2, magnetizing susceptance or exciting current (0.0 by default)
	REALARI(12)	F1, first owner fraction (1.0 by default)
	REALARI(13)	F2, second owner fraction (0.0 by default)
	REALARI(14)	F3, third owner fraction (0.0 by default)
	REALARI(15)	F4, fourth owner fraction (0.0 by default)
	REALARI(16)	VMSTAR, star bus voltage magnitude (1.0 by default)
	REALARI(17)	TAR, star bus voltage angle (0.0 by default)
Real REALARO (6)	array of 6 elements specifying corresponding to values REALAR(18) through REALAR(23) as described above(output).	
	REALARO(1)	actual bus one to two resistance (returned)
	REALARO(2)	actual bus one to two reactance (returned)
	REALARO(3)	actual bus two to three resistance (returned)
	REALARO(4)	actual bus two to three reactance (returned)
	REALARO(5)	actual bus three to one resistance (returned)
	REALARO(6)	actual bus three to one reactance (returned)
Character*40 CHARAR (2)	array of 2 elements specifying (input).	
	CHARAR(1)	NAME, transformer name (blank by default)
	CHARAR(2)	VGRP, vector group name (blank by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found

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IERR = 2	circuit identifier is more than two characters
IERR = 3	invalid CW, CZ and/or CM value
IERR = 4	CW, CZ and/or CM >1 but at least one end-point bus has no base voltage
IERR = 13	three-winding transformer not found
IERR = 14	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Invalid set of winding buses. - Invalid CW, CZ and/or CM data value. - Invalid transformer status. - Invalid non-metered end bus. - Winding side buses interchanged. - Losses are too big relative to total impedance at winding base MVA. - Magnitude of winding impedance is less than 0.00001. - Vector group not found. - Any two-winding transformer warnings

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2.66. THREE_WND_IMPED_DATA_3

Use this API routine to modify the impedance data of an existing three-winding transformer in the working case, or to add a new three-winding transformer to the working case (see PSSE Program Operation Manual, Non-Transformer Branch Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_THREE_WND_IMPED_DATA_3  IBUS  JBUS  KBUS  'CKT'  INTGAR(1)..INTGAR(12)
  REALAR(1)..REALAR(17)  'CHARAR(1)'  'CHARAR(2)'
```

Python command syntax:

```
ierr, realaro = three_wnd_imped_data_3(ibus, jbus, kbus, ckt, intgar,
 realari, charar)
```

Fortran command syntax:

```
CALL THREE_WINDING_IMPEDANCE_API_3(IBUS, JBUS, KBUS, CKT, INTGAR, REALAR,
 CHARAR, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 CKT	transformer circuit identifier (input; '1').	
Integer INTGAR (12)	array of 12 elements specifying (input).	
	INTGAR(1)	O1, first owner number (owner of bus IBUS by default)
	INTGAR(2)	O2, second owner number (0 by default)
	INTGAR(3)	O3, third owner number (0 by default)
	INTGAR(4)	O4, fourth owner number (0 by default)
	INTGAR(5)	CW, winding data I/O code (1 by default)
	INTGAR(6)	CZ, impedance data I/O code (1 by default)
	INTGAR(7)	CM, magnetizing admittance data I/O code (1 by default)
	INTGAR(8)	STAT, branch status (1 by default)
	INTGAR(9)	NMETBS, non-metered end bus number (IBUS, JBUS or KBUS) (JBUS by default)
	INTGAR(10)	WN1BUS, winding one side bus number (IBUS, JBUS or KBUS) (IBUS by default)
	INTGAR(11)	WN2BUS, winding two side bus number (IBUS, JBUS or KBUS) (JBUS by default)
	INTGAR(12)	WN3BUS, winding three side bus number (IBUS, JBUS or KBUS) (KBUS by default)
Real REALAR (23)	(first seventeen are values; last six are values) (input (output).).	

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Real REALARI (17)	array of 17 elements specifying corresponding to values REALAR(1) through REALAR(17) as described above(input).	
	REALARI(1)	R1-2, nominal bus one to two transformer resistance (0.0 by default)
	REALARI(2)	X1-2, nominal bus one to two transformer reactance (0.0002 by default)
	REALARI(3)	R2-3, nominal bus two to three transformer resistance (0.0 by default)
	REALARI(4)	X2-3, nominal bus two to three transformer reactance (0.0002 by default)
	REALARI(5)	R3-1, nominal bus three to one transformer resistance (0.0 by default)
	REALARI(6)	X3-1, nominal bus three to one transformer reactance (0.0002 by default)
	REALARI(7)	SBS1-2, winding one to two base MVA (SBASE by default)
	REALARI(8)	SBS2-3, winding two to three base MVA (SBASE by default)
	REALARI(9)	SBS3-1, winding three to one base MVA (SBASE by default)
	REALARI(10)	MAG1, magnetizing conductance or no-load losses (0.0 by default)
	REALARI(11)	MAG2, magnetizing susceptance or exciting current (0.0 by default)
	REALARI(12)	F1, first owner fraction (1.0 by default)
	REALARI(13)	F2, second owner fraction (0.0 by default)
	REALARI(14)	F3, third owner fraction (0.0 by default)
	REALARI(15)	F4, fourth owner fraction (0.0 by default)
	REALARI(16)	VMSTAR, star bus voltage magnitude (1.0 by default)
	REALARI(17)	TAR, star bus voltage angle (0.0 by default)
Real REALARO (6)	array of 6 elements specifying corresponding to values REALAR(18) through REALAR(23) as described above(output).	
	REALARO(1)	actual bus one to two resistance (returned)
	REALARO(2)	actual bus one to two reactance (returned)
	REALARO(3)	actual bus two to three resistance (returned)
	REALARO(4)	actual bus two to three reactance (returned)
	REALARO(5)	actual bus three to one resistance (returned)
	REALARO(6)	actual bus three to one reactance (returned)
Character*40 CHARAR (2)	array of 2 elements specifying (input).	
	CHARAR(1)	NAME, transformer name (blank by default)
	CHARAR(2)	VGRP, vector group name (blank by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred

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IERR > 0	fatal error occurred
IERR = 1	bus not found
IERR = 2	circuit identifier is more than two characters
IERR = 3	invalid CW, CZ and/or CM value
IERR = 4	CW, CZ and/or CM >1 but at least one end-point bus has no base voltage
IERR = 5	three different buses must be specified
IERR = 6	blank circuit identifier is invalid
IERR = 7	transformer circuit identifier may not start with &, * or @
IERR = 8	three-winding transformer table is full
IERR = 9	bus table is full
IERR = 10	not enough branch table entries are available
IERR = 11	not enough two-winding transformer table entries are available
IERR = 12	not enough branch ownership table entries are available
IERR = 14	prerequisite requirements for API are not met
IERR = -1	data error, warning and information messages; one or more of: - Invalid set of winding buses. - Invalid CW, CZ and/or CM data value. - Invalid transformer status. - Invalid non-metered end bus. - Winding side buses interchanged. - Losses are too big relative to total impedance at winding base MVA. - Magnitude of winding impedance is less than 0.00001. - Vector group not found. - Any two-winding transformer warnings

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2.67. THREE_WND_WINDING_DATA

This API routine is obsolete. It has been replaced by the API routine three_wnd_winding_data_3 .

Use this API to modify the data of one winding of an existing three-winding transformer in the working case (see PSSE Program Operation Manual, Non-Transformer Branch Data).

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_THREE_WND_WINDING_DATA  IBUS  JBUS  KBUS  'CKT'  WARG
 INTGAR(1)..INTGAR(5)  REALAR(1)..REALAR(12)
```

Python command syntax:

```
ierr, realaro = three_wnd_winding_data(ibus, jbus, kbus, ckt, warg,
 intgar, realari)
```

Fortran command syntax:

```
CALL THREE_WINDING_WINDING_API_2(IBUS, JBUS, KBUS, CKT, WARG, INTGAR,
 REALAR, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 CKT	transformer circuit identifier (input; '1').	
Integer WARG	winding number (1, 2 or 3) (input; no default allowed).	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	NTPi, number of tap positions (33 by default)
	INTGAR(2)	TABi, impedance correction table number (0 by default)
	INTGAR(3)	CONTi, controlled bus number (0 by default)
	INTGAR(4)	SICODi, negative for controlled bus on winding bus side (1 by default)
	INTGAR(5)	CODi, adjustment control mode flag (-3 through +3) (0 by default)
Real REALAR (16)	(first twelve are input values; last four are output values) (input (output).).	
Real REALARI (12)	array of 12 elements specifying corresponding to values REALAR(1) through REALAR(12) as described above(input).	
	REALARI(1)	WINDVi, winding ratio or voltage (1.0 by default if CW of this transformer is 1 or 3; base voltage of the winding bus by default if CW is 2)

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	REALARI(2)	NOMVi, winding nominal voltage (0.0 by default)
	REALARI(3)	i, winding phase shift angle (0.0 by default)
	REALARI(4)	RATAi, winding rating set A line rating (0.0 by default)
	REALARI(5)	RATBi, winding rating set B line rating (0.0 by default)
	REALARI(6)	RATCi, winding rating set C line rating (0.0 by default)
	REALARI(7)	RMAi, winding ratio/angle high limit (1.1 by default)
	REALARI(8)	RMIi, winding ratio/angle low limit (0.9 by default)
	REALARI(9)	VMAi, winding voltage or flow upper limit (1.1 by default)
	REALARI(10)	VMIi, winding voltage or flow lower limit (0.9 by default)
	REALARI(11)	CRi, winding load drop compensating resistance (0.0 by default)
	REALARI(12)	CXi, winding load drop compensating reactance (0.0 by default)
Real REALARO (4)	array of 4 elements specifying corresponding to values REALAR(13) through REALAR(16) as described above(output).	
	REALARO(1)	winding nominal resistance (returned)
	REALARO(2)	winding nominal reactance (returned)
	REALARO(3)	winding actual resistance (returned)
	REALARO(4)	winding actual reactance (returned)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	invalid CW, CZ and/or CM value
	IERR = 4	CW and/or CZ >1 but no base voltage at winding bus
	IERR = 5	three winding transformer not found
	IERR = 6	invalid winding number
	IERR = 7	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Controlled bus not found. - Invalid adjustment control mode flag. - Any two-winding transformer warnings

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2.68. THREE_WND_WINDING_DATA_3

Use this API to modify the data of one winding of an existing three-winding transformer in the working case (see PSSE Program Operation Manual, Non-Transformer Branch Data).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_THREE_WND_WINDING_DATA_3  IBUS  JBUS  KBUS  'CKT'  WARG
 INTGAR(1)..INTGAR(5)  REALAR(1)..REALAR(12)
```

Python command syntax:

```
ierr, realaro = three_wnd_winding_data_3(ibus, jbus, kbus, ckt, warg,
 intgar, realari)
```

Fortran command syntax:

```
CALL THREE_WINDING_WINDING_API_3(IBUS, JBUS, KBUS, CKT, WARG, INTGAR,
 REALAR, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 CKT	transformer circuit identifier (input; '1').	
Integer WARG	winding number (1, 2 or 3) (input; no default allowed).	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	NTPi, number of tap positions (33 by default)
	INTGAR(2)	TABi, impedance correction table number (0 by default)
	INTGAR(3)	CONTi, controlled bus number (0 by default)
	INTGAR(4)	SICODi, negative for controlled bus on winding bus side (1 by default)
	INTGAR(5)	CODi, adjustment control mode flag (-3 through +3, -5 or 5) (0 by default)
Real REALAR (17)	(first thirteen are values; last four are values) (input (output).).	
Real REALARI (13)	array of 13 elements specifying corresponding to values REALAR(1) through REALAR(13) as described above(input).	
	REALARI(1)	WINDVi, winding ratio or voltage (1.0 by default if CW of this transformer is 1 or 3; base voltage of the winding bus by default if CW is 2)
	REALARI(2)	NOMVi, winding nominal voltage (0.0 by default)
	REALARI(3)	i, winding phase shift angle (0.0 by default)

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	REALARI(4)	RATAi, winding rating set A line rating (0.0 by default)
	REALARI(5)	RATBi, winding rating set B line rating (0.0 by default)
	REALARI(6)	RATCi, winding rating set C line rating (0.0 by default)
	REALARI(7)	RMAi, winding ratio/angle high limit (1.1 by default)
	REALARI(8)	RMIi, winding ratio/angle low limit (0.9 by default)
	REALARI(9)	VMAi, winding voltage or flow upper limit (1.1 by default)
	REALARI(10)	VMIi, winding voltage or flow lower limit (0.9 by default)
	REALARI(11)	CRi, winding load drop compensating resistance (0.0 by default)
	REALARI(12)	CXi, winding load drop compensating reactance (0.0 by default)
	REALARI(13)	CNXAi, winding connection angle (0.0 by default). Used with adjustment control mode 5 (unsymmetric phase shift control of active power) implemented in PSSE version 32.1)
Real REALARO (4)	array of 4 elements specifying corresponding to values REALAR(14) through REALAR(17) as described above(output).	
	REALARO(1)	winding nominal resistance (returned)
	REALARO(2)	winding nominal reactance (returned)
	REALARO(3)	winding actual resistance (returned)
	REALARO(4)	winding actual reactance (returned)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	invalid CW, CZ and/or CM value
	IERR = 4	CW and/or CZ >1 but no base voltage at winding bus
	IERR = 5	three winding transformer not found
	IERR = 6	invalid winding number
	IERR = 7	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Controlled bus not found. - Invalid adjustment control mode flag. - Any two-winding transformer warnings

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2.69. TRANSACTION_ADD_MOD

Use this API to add or modify transaction event data in PSSE working memory.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_TRANSACTION_ADD_MOD  EVENT  INTGAR(1)  INTGAR(2)  REALAR(1)  REALAR(2)
  'DSCRP'
```

Python command syntax:

```
ierr = transaction_add_mod(event, intgar, realar, dscrp)
```

Fortran command syntax:

```
CALL TRANSACTION_ADD_MOD(EVENT, INTGAR, REALAR, DSCRP, IERR, NERR)
```

Where:

Integer EVENT	transaction event number (input).	
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	transaction event status (0 = out-of-service, 1 = in-service)
	INTGAR(2)	transaction event priority
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	transaction event magnitude in MW
	REALAR(2)	transaction event curtailment in MW
Character*24 DSCRP	transaction event description (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	error occurred
	IERR = 2	invalid transaction event number
	IERR > 2	unable to add or modify transaction event data: error condition detected in data access routine
	IERR = 10	prerequisite requirements for API are not met
	IERR = -1	warnings: - None
Integer NERR	number of error entries in PBUF02 (output).	

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2.70. TRANSFER_CHNG

Use this API routine to modify the data of an inter-area transfer in the working case (see PSSE Program Operation Manual, Zone Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_TRANSFER_CHNG  APPLYF  ARFROM  ARTO  'TRID'  REALAR(1)
```

Python command syntax:

```
ierr = transfer_chng(applyf, arfrom, arto, trid, realar)
```

Fortran command syntax:

```
CALL TRANSFER_CHNG(APPLYF, ARFROM, ARTO, TRID, REALAR, IERR)
```

Where:

Integer APPLYF	APPLYF = 1 to modify from and to area desired net interchange entries (DESINT) (input).	
	APPLYF = 2	to leave DESINT entries unchanged. (Input; no default allowed.)
Integer ARFROM	from area number (must be an area already defined) (input; no default allowed).	
Integer ARTO	to area number (must be an area already defined) (input; no default allowed).	
Character*1 TRID	inter-area transfer identifier (input; '1').	
Real REALAR (1)	array of 1 elements specifying (input).	
	REALAR(1)	PTRAN, MW comprising this inter-area transfer (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid area number
	IERR = 2	inter-area transfer identifier is more than one character
	IERR = 3	invalid APPLYF value
	IERR = 8	area not found
	IERR = 9	inter-area transfer not found
	IERR = 10	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Changes to desired area net interchanges if APPLYF = 1

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2.71. TRANSFER_DATA

Use this API routine to modify the data of an inter-area transfer in the working case, or to add a new inter-area transfer to the working case (see PSSE Program Operation Manual, Zone Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TRANSFER_DATA  APPLYF  ARFROM  ARTO  'TRID'  REALAR(1)
```

Python command syntax:

```
ierr = transfer_data(applyf, arfrom, arto, trid, realar)
```

Fortran command syntax:

```
CALL TRANSFER_API_2(APPLYF, ARFROM, ARTO, TRID, REALAR, IERR)
```

Where:

Integer APPLYF	APPLYF = 1 to modify from and to area desired net interchange entries (DESINT) (input).	
	APPLYF = 2	to leave DESINT entries unchanged. (Input; no default allowed.)
Integer ARFROM	from area number (must be an area already defined) (input; no default allowed).	
Integer ARTO	to area number (must be an area already defined) (input; no default allowed).	
Character*1 TRID	inter-area transfer identifier (input; '1').	
Real REALAR (1)	array of 1 elements specifying (input).	
	REALAR(1)	PTRAN, MW comprising this inter-area transfer (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid area number
	IERR = 2	inter-area transfer identifier is more than one character
	IERR = 3	invalid APPLYF value
	IERR = 4	invalid inter-area transfer identifier
	IERR = 5	transfer from an area to itself
	IERR = 6	inter-area transfer table is full
	IERR = 7	inter-area transfer already exists
	IERR = 8	area not found
	IERR = 10	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Changes to desired area net interchanges if APPLYF = 1

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2.72. TWO_TERMINAL_DC_LINE_CHNG

Use this API routine to modify the link data of an existing two-terminal dc line in the working case (see PSSE Program Operation Manual, Area Interchange Data). Converter data for the rectifier and inverter are specified via the TWO_TERM_DC_CONVR_DATA API routine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_TWO_TERMINAL_DC_LINE_CHNG DC INTGAR(1) INTGAR(2) REALAR(1)..REALAR(8)
 'METER'
```

Python command syntax:

```
ierr, realaro = two_terminal_dc_line_chng(dc, intgar, realari, meter)
```

Fortran command syntax:

```
CALL TWO_TERMINAL_DC_LINE_CHNG(DC, INTGAR, REALAR, METER, IERR)

CALL TWO_TERMINAL_DC_LINE_CHNG_I(NUM, INTGAR, REALAR, METER, IERR)

CALL TWO_TERMINAL_DC_LINE_CHNG_C(NAME, INTGAR, REALAR, METER, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which data is to be specified; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which data is to be specified; no default allowed	
Character*12 NAME	name of the two-terminal dc line for which data is to be specified; no default allowed	
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	MDC, control mode (0, 1 or 2) (0 by default)
	INTGAR(2)	CCITMX, CCC Newton solution iteration limit (20 by default)
Real REALAR (9)	first eight are input values; last one is output value. (input (output).).	
Real REALARI (8)	array of 8 elements specifying values REALAR(1) through REALAR(8) as described above(input).	
	REALARI(1)	SETVL, scheduled current (amps) or power (MW) demand (0.0 by default)
	REALARI(2)	VSCHD, scheduled dc voltage (kV) (1.0 by default)
	REALARI(3)	VCMOD, mode switch dc voltage (kV) (0.0 by default)
	REALARI(4)	DELTI, margin (pu, > 0) (0.0 by default)
	REALARI(5)	RDC, dc line resistance (0.0 by default)

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	REALARI(6)	RCOMP, compounding resistance (0.0 by default)
	REALARI(7)	DCVMIN, minimum dc voltage (kV) (0.0 by default)
	REALARI(8)	CCCACC, CCC Newton solution acceleration (1.0 by default)
Real REALARO (1)	array of 1 elements specifying value REALAR(9) as described above(output).	
	REALARO(1)	dc line current (amps) (returned)
Character*1 METER	metered end flag: 'R' or 'I' (input; 'I').	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 2	metered end code is more than one character
	IERR = 3	blank dc line name is invalid
	IERR = 4	two-terminal dc line not found
	IERR = 5	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid dc line control mode - invalid dc line resistance - invalid compounding resistance - compounding resistance exceeds line resistance - scheduled voltage must exceed mode switch voltage - invalid metered end code

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2.73. TWO_TERMINAL_DC_LINE_DATA

Use this API routine to modify the link data of an existing two-terminal dc line in the working case, or to add a new two-terminal dc line and its link data to the working case (see PSSE Program Operation Manual, Area Interchange Data). Converter data for the rectifier and inverter are specified via the TWO_TERM_DC_CONVR_DATA API routine.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TWO_TERMINAL_DC_LINE_DATA DC INTGAR(1) INTGAR(2) REALAR(1)..REALAR(8)
 'METER'
```

Python command syntax:

```
ierr, realaro = two_terminal_dc_line_data(dc, intgar, realari, meter)
```

Fortran command syntax:

```
CALL TWO_TERMINAL_DC_LINE_API(DC, INTGAR, REALAR, METER, IERR)

CALL TWO_TERMINAL_DC_LINE_API_I(NUM, INTGAR, REALAR, METER, IERR)

CALL TWO_TERMINAL_DC_LINE_API_C(NAME, INTGAR, REALAR, METER, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which data is to be specified; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which data is to be specified; no default allowed	
Character*12 NAME	name of the two-terminal dc line for which data is to be specified; no default allowed	
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	MDC, control mode (0, 1 or 2) (0 by default)
	INTGAR(2)	CCITMX, CCC Newton solution iteration limit (20 by default)
Real REALAR (9)	first eight are input values; last one is output value. (input (output).).	
Real REALARI (8)	array of 8 elements specifying values REALAR(1) through REALAR(8) as described above(input).	
	REALARI(1)	SETVL, scheduled current (amps) or power (MW) demand (0.0 by default)
	REALARI(2)	VSCHD, scheduled dc voltage (kV) (1.0 by default)
	REALARI(3)	VCMOD, mode switch dc voltage (kV) (0.0 by default)
	REALARI(4)	DELTI, margin (pu, > 0) (0.0 by default)
	REALARI(5)	RDC, dc line resistance (0.0 by default)

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	REALARI(6)	RCOMP, compounding resistance (0.0 by default)
	REALARI(7)	DCVMIN, minimum dc voltage (kV) (0.0 by default)
	REALARI(8)	CCCACC, CCC Newton solution acceleration (1.0 by default)
Real REALARO (1)	array of 1 elements specifying value REALAR(9) as described above(output).	
	REALARO(1)	dc line current (amps) (returned)
Character*1 METER	metered end flag: 'R' or 'I' (input; 'I').	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	two-terminal dc line table is full
	IERR = 2	metered end code is more than one character
	IERR = 3	blank dc line name is invalid
	IERR = 5	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid dc line control mode - invalid dc line resistance - invalid compounding resistance - compounding resistance exceeds line resistance - scheduled voltage must exceed mode switch voltage - invalid metered end code

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2.74. TWO_TERM_DC_CONVR_DATA

Use this API to modify the data of a dc converter of an existing two-terminal dc line in the working case (see PSSE Program Operation Manual, Area Interchange Data). The dc line must have previously been specified via the TWO_TERMINAL_DC_LINE_DATA API.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_TWO_TERM_DC_CONVR_DATA  CNVFLG  DC  INTGAR(1)..INTGAR(5)
 REALAR(1)..REALAR(11)  'ID'
```

Python command syntax:

```
ierr, realaro = two_term_dc_convr_data(cnvflg, dc, intgar, realari, id)
```

Fortran command syntax:

```
CALL TWO_TERMINAL_DC_CONVERTER_API (CNVFLG, DC, INTGAR, REALAR, ID, IERR)

CALL TWO_TERMINAL_DC_CONVERTER_API_I (CNVFLG, NUM, INTGAR, REALAR, ID,
 IERR)

CALL TWO_TERMINAL_DC_CONVERTER_API_C (CNVFLG, NAME, INTGAR, REALAR, ID,
 IERR)
```

Where:

Integer CNVFLG	converter type (input; no default allowed).	
	CNVFLG = 1	for rectifier data.
	CNVFLG = 2	for inverter data.
Integer or Character*12 DC	quoted name of the two-terminal dc line for which data for one of its converters is to be specified; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which data for one of its converters is to be specified; no default allowed	
Character*12 NAME	name of the two-terminal dc line for which data for one of its converters is to be specified; no default allowed	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	IPR or IPI, converter bus number (no default allowed)
	INTGAR(2)	NBR or NBI, number of bridges in series (0 by default)
	INTGAR(3)	ICR or ICI, zero or bus number of firing angle bus (0 by default)
	INTGAR(4)	IFR or IFI, zero or bus number of winding one side of a two-winding ac transformer (0 by default)

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	INTGAR(5)	ITR or ITI, zero or bus number of winding two side of a two-winding ac transformer (0 by default)
Real REALAR (15)	(first eleven are values; last four are values) (input (output).).	
Real REALARI (11)	array of elements specifying values REALAR(1) through REALAR(11) as described above(input).	
	REALARI(1)	NR or ANMNI, minimum firing angle (90.0 by default)
	REALARI(2)	XR or ANMXI, maximum firing angle (0.0 by default)
	REALARI(3)	RCR or RCI, commutating resistance (0.0 by default)
	REALARI(4)	XCR or XCI, commutating reactance (0.0 by default)
	REALARI(5)	EBASR or EBASI, primary base ac voltage (0.0 by default)
	REALARI(6)	TRR or TRI, transformer ratio (1.0 by default)
	REALARI(7)	TAPR or TAPI, tap setting (1.0 by default)
	REALARI(8)	TMXR or TMXI, maximum tap setting (1.5 by default)
	REALARI(9)	TMNR or TMNI, minimum tap setting (0.51 by default)
	REALARI(10)	STPR or STPI, tap step (0.00625 by default)
	REALARI(11)	XCAPR or XCAPI, capacitive reactance of CCC type converter (0.0 by default)
Real REALARO (4)	array of elements specifying values REALAR(12) through REALAR(15) as described above(output).	
	REALARO(1)	firing angle (returned)
	REALARO(2)	dc voltage (kV) (returned)
	REALARO(3)	real power into converter (PWRCOD) (returned)
	REALARO(4)	reactive power into converter (PWRCOD) (returned)
Character*2 ID	ac transformer circuit identifier, IDR or IDI (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	dc line not found
	IERR = 2	converter type is not 1 or 2
	IERR = 3	transformer circuit identifier is more than two characters
	IERR = 4	no converter bus is specified
	IERR = 5	converter bus not found
	IERR = 6	prerequisite requirements for API are not met

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IERR = -1	data error, warning and information messages; one or more of: - Converter bus is dummy bus of a multi-section line. - Multi-section line grouping deleted. - Converter maximum angle is less than its minimum angle. - Converter maximum tap is less than its minimum tap. - Invalid tap step. - Commutating bus not found. - ac transformer bus not found. - Specified branch not found. - Specified branch is not a transformer. - From bus of specified branch is not winding one side bus

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2.75. TWO_WINDING_CHNG_4

Use this API routine to modify the data of an existing two-winding transformer in the working case (see PSSE Program Operation Manual, Owner Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_TWO_WINDING_CHNG_4  IBUS  JBUS  'CKT'  INTGAR(1)..INTGAR(15)
 REALAR(1)..REALAR(24)  'CHARAR(1)'  'CHARAR(2)'
```

Python command syntax:

```
ierr, realaro = two_winding_chng_4(ibus, jbus, ckt, intgar, realari,
 charar)
```

Fortran command syntax:

```
CALL TWO_WINDING_CHNG_4(IBUS, JBUS, CKT, INTGAR, REALAR, CHARAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Integer INTGAR (15)	array of 15 elements specifying (input).	
	INTGAR(1)	STAT, branch status (1 by default)
	INTGAR(2)	METBUS, metered end bus number (IBUS or JBUS) (IBUS by default)
	INTGAR(3)	O1, first owner number (owner of bus IBUS by default)
	INTGAR(4)	O2, second owner number (0 by default)
	INTGAR(5)	O3, third owner number (0 by default)
	INTGAR(6)	O4, fourth owner number (0 by default)
	INTGAR(7)	NTP1, number of tap positions (33 by default)
	INTGAR(8)	TAB1, impedance correction table number (0 by default)
	INTGAR(9)	WN1BUS, winding one side bus number (IBUS or JBUS) (IBUS by default)
	INTGAR(10)	CONT1, controlled bus number (0 by default)
	INTGAR(11)	SICOD1, negative for controlled bus on winding one side (1 by default)
	INTGAR(12)	COD1, adjustment control mode flag (-5 through +5) (0 by default)
	INTGAR(13)	CW, winding data I/O code (1 by default)
	INTGAR(14)	CZ, impedance data I/O code (1 by default)
	INTGAR(15)	CM, magnetizing admittance data I/O code (1 by default)

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Real REALAR (26) (first twenty-four are values; last two are values) (input (output).).

Real REALARI (24) array of 24 elements specifying corresponding to values REALAR(1) through REALAR(24) as described above(input).

REALARI(1)	R1-2, nominal transformer resistance (0.0 by default)
REALARI(2)	X1-2, nominal transformer reactance (THRSHZ by default; 0.0001 if THRSHZ = 0.0)
REALARI(3)	SBS1-2, winding base MVA (SBASE by default)
REALARI(4)	WINDV1, winding one ratio/voltage (1.0 by default if CW is 1 or 3; base voltage of bus IBUS by default if CW is 2)
REALARI(5)	NOMV1, winding one nominal voltage (0.0 by default)
REALARI(6)	1, winding one phase shift angle (0.0 by default)
REALARI(7)	WINDV2, winding two ratio/voltage (1.0 by default if CW is 1 or 3; base voltage of bus JBUS by default if CW is 2)
REALARI(8)	NOMV2, winding two nominal voltage (0.0 by default)
REALARI(9)	RATA1, rating set A line rating (0.0 by default)
REALARI(10)	RATB1, rating set B line rating (0.0 by default)
REALARI(11)	RATC1, rating set C line rating (0.0 by default)
REALARI(12)	F1, first owner fraction (1.0 by default)
REALARI(13)	F2, second owner fraction (1.0 by default)
REALARI(14)	F3, third owner fraction (1.0 by default)
REALARI(15)	F4, fourth owner fraction (1.0 by default)
REALARI(16)	MAG1, magnetizing conductance or no-load losses (0.0 by default)
REALARI(17)	MAG2, magnetizing susceptance or exciting current (0.0 by default)
REALARI(18)	RMA1, winding one ratio/angle high limit (1.1 by default)
REALARI(19)	RMI1, winding one ratio/angle low limit (0.9 by default)
REALARI(20)	VMA1, voltage or flow upper limit (1.1 by default)
REALARI(21)	VMI1, voltage or flow lower limit (0.9 by default)
REALARI(22)	CR1, load drop compensating resistance (0.0 by default)

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	REALARI(23)	CX1, load drop compensating reactance (0.0 by default)
	REALARI(24)	CNXA1, winding connection angle (0.0 by default). Used with adjustment control mode 5 (unsymmatric phase shift control of active power) implemented in PSSE version 32.1)
Real REALARO (2)	array of 2 elements specifying corresponding to values REALAR(25) through REALAR(26) as described above(output).	
	REALARO(1)	actual transformer resistance (returned)
	REALARO(2)	actual transformer reactance (returned)
Character*40 CHARAR (2)	array of 2 elements specifying (input).	
	CHARAR(1)	NAME, transformer name (blank by default)
	CHARAR(2)	VGRP, vector group name (blank by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	branch exists but it is a non-transformer branch
	IERR = 10	CW, CZ and/or CM >1 but at least one end-point bus has no base voltage
	IERR = 11	invalid CW, CZ and/or CM value
	IERR = 12	two-winding transformer not found
	IERR = 13	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid metered end bus - invalid transformer status - invalid owner number - invalid ownership fraction - no ownership data specified. - multi-section line grouping deleted - branch ownership table is full--only first n owners retained - owner table is full - invalid winding 1 side bus number - invalid CW, CZ and/or CM data value - controlled bus not found - invalid adjustment control mode flag - invalid impedance correction table number - invalid number of tap positions - winding nominal voltage is negative - winding bus base voltage and nominal voltage differ by more than 20% - invalid winding voltage - winding voltage and bus base voltage differ by more than 50% - RMAX is greater than 150% of bus base voltage - RMIN is less than 50% of bus base voltage - invalid tap ratio - tap ratio is outside the band 0.5 to 1.5 pu - RMAX is greater than 1.5 - RMIN is less than 0.5 - RMAX is less than

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RMIN - VMAX is less than VMIN - phase shift angle normalized between -180 and 180 - winding base MVA must be positive - MAG1 is too big relative to MAG2 at winding base MVA - magnetizing susceptance MAG2 is positive - losses are too big relative to total impedance at winding base MVA - branch reactance is 0.0 - winding 1 and winding 2 sides are interchanged (any two-terminal dc lines controlled by this transformer are logged). - winding connection angle normalized between -360 and 360 - adjustment control mode flag and the winding connection angle are inconsistent. - for asymmetric phase shifter, abnormally large or small tap setting results at RMA1 (or RMI1) and CNXA1. - for asymmetric phase shifter, the specified and derived winding tap settings differ. - Vector group not found

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2.76. TWO_WINDING_DATA

This API routine is obsolete. It has been replaced by the API routine two_winding_data_3 .

Use this API to modify the data of an existing two-winding transformer in the working case, or to add a new two-winding transformer to the working case (see PSSE Program Operation Manual, Owner Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_TWO_WINDING_DATA  IBUS  JBUS  'CKT'  INTGAR(1)..INTGAR(15)
 REALAR(1)..REALAR(23)  'NAME'
```

Python command syntax:

```
ierr, realaro = two_winding_data(ibus, jbus, ckt, intgar, realari, name)
```

Fortran command syntax:

```
CALL TWO_WINDING_API_2(IBUS, JBUS, CKT, INTGAR, REALAR, NAME, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Integer INTGAR (15)	array of 15 elements specifying (input).	
	INTGAR(1)	STAT, branch status (1 by default)
	INTGAR(2)	METBUS, metered end bus number (IBUS or JBUS) (IBUS by default)
	INTGAR(3)	O1, first owner number (owner of bus IBUS by default)
	INTGAR(4)	O2, second owner number (0 by default)
	INTGAR(5)	O3, third owner number (0 by default)
	INTGAR(6)	O4, fourth owner number (0 by default)
	INTGAR(7)	NTP1, number of tap positions (33 by default)
	INTGAR(8)	TAB1, impedance correction table number (0 by default)
	INTGAR(9)	WN1BUS, winding one side bus number (IBUS or JBUS) (IBUS by default)
	INTGAR(10)	CONT1, controlled bus number (0 by default)
	INTGAR(11)	SICOD1, negative for controlled bus on winding one side (1 by default)
	INTGAR(12)	COD1, adjustment control mode flag (-4 through +4) (0 by default)
	INTGAR(13)	CW, winding data I/O code (1 by default)
	INTGAR(14)	CZ, impedance data I/O code (1 by default)

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	INTGAR(15)	CM, magnetizing admittance data I/O code (1 by default)
Real REALAR (25)	(first twenty-three are values; last two are values) (input (output).).	
Real REALARI (23)	array of 23 elements specifying corresponding to values REALAR(1) through REALAR(23) as described above(input).	
	REALARI(1)	R1-2, nominal transformer resistance (0.0 by default)
	REALARI(2)	X1-2, nominal transformer reactance (THRSHZ by default; 0.0001 if THRSHZ = 0.0)
	REALARI(3)	SBS1-2, winding base MVA (SBASE by default)
	REALARI(4)	WINDV1, winding one ratio/voltage (1.0 by default if CW is 1 or 3; base voltage of bus IBUS by default if CW is 2)
	REALARI(5)	NOMV1, winding one nominal voltage (0.0 by default)
	REALARI(6)	1, winding one phase shift angle (0.0 by default)
	REALARI(7)	WINDV2, winding two ratio/voltage (1.0 by default if CW is 1 or 3; base voltage of bus JBUS by default if CW is 2)
	REALARI(8)	NOMV2, winding two nominal voltage (0.0 by default)
	REALARI(9)	RATA1, rating set A line rating (0.0 by default)
	REALARI(10)	RATB1, rating set B line rating (0.0 by default)
	REALARI(11)	RATC1, rating set C line rating (0.0 by default)
	REALARI(12)	F1, first owner fraction (1.0 by default)
	REALARI(13)	F2, second owner fraction (1.0 by default)
	REALARI(14)	F3, third owner fraction (1.0 by default)
	REALARI(15)	F4, fourth owner fraction (1.0 by default)
	REALARI(16)	MAG1, magnetizing conductance or no-load losses (0.0 by default)
	REALARI(17)	MAG2, magnetizing susceptance or exciting current (0.0 by default)
	REALARI(18)	RMA1, winding one ratio/angle high limit (1.1 by default)
	REALARI(19)	RMI1, winding one ratio/angle low limit (0.9 by default)
	REALARI(20)	VMA1, voltage or flow upper limit (1.1 by default)
	REALARI(21)	VMI1, voltage or flow lower limit (0.9 by default)

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	REALARI(22)	CR1, load drop compensating resistance (0.0 by default)
	REALARI(23)	CX1, load drop compensating reactance (0.0 by default)
Real REALARO (2)	array of 2 elements specifying corresponding to values REALAR(24) through REALAR(25) as described above(output).	
	REALARO(1)	actual transformer resistance (returned)
	REALARO(2)	actual transformer reactance (returned)
Character*12 NAME	transformer name (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	branch exists but it is a non-transformer branch
	IERR = 4	branch from a bus to itself
	IERR = 5	blank circuit identifier is invalid
	IERR = 6	circuit identifier may not start with &, * or @
	IERR = 7	branch table is full
	IERR = 8	branch ownership table is full
	IERR = 9	two-winding transformer table is full
	IERR = 10	CW, CZ and/or CM >1 but at least one end-point bus has no base voltage
	IERR = 11	invalid CW, CZ and/or CM value
	IERR = 13	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid metered end bus - invalid transformer status - invalid owner number - invalid ownership fraction - no ownership data specified. - multi-section line grouping deleted - branch ownership table is full--only first n owners retained - owner table is full - invalid winding 1 side bus number - invalid CW, CZ and/or CM data value - controlled bus not found - invalid adjustment control mode flag - invalid impedance correction table number - invalid number of tap positions - winding nominal voltage is negative - winding bus base voltage and nominal voltage differ by more than 20% - invalid winding voltage - winding voltage and bus base voltage differ by more than 50% - RMAX is greater than 150% of bus base voltage - RMIN is less than 50% of bus base voltage - invalid tap ratio - tap ratio is outside the band 0.5 to

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1.5 pu - RMAX is greater than 1.5 - RMIN is less than 0.5 - RMAX is less than RMIN - VMAX is less than VMIN - phase shift angle normalized between -180 and 180 - winding base MVA must be positive - MAG1 is too big relative to MAG2 at winding base MVA - losses are too big relative to total impedance at winding base MVA - branch reactance is 0.0 - winding 1 and winding 2 sides are interchanged (any two-terminal dc lines controlled by this transformer are logged)

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2.77. TWO_WINDING_DATA_3

This API routine is obsolete. It has been replaced by the API routine two_winding_data_4 .

Use this API to modify the data of an existing two-winding transformer in the working case, or to add a new two-winding transformer to the working case (see PSSE Program Operation Manual, Owner Data).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_TWO_WINDING_DATA_3  IBUS  JBUS  'CKT'  INTGAR(1)..INTGAR(15)
 REALAR(1)..REALAR(24)  'NAME'
```

Python command syntax:

```
ierr, realaro = two_winding_data_3(ibus, jbus, ckt, intgar, realari, name)
```

Fortran command syntax:

```
CALL TWO_WINDING_API_3(IBUS, JBUS, CKT, INTGAR, REALAR, NAME, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Integer INTGAR (15)	array of 15 elements specifying (input).	
	INTGAR(1)	STAT, branch status (1 by default)
	INTGAR(2)	METBUS, metered end bus number (IBUS or JBUS) (IBUS by default)
	INTGAR(3)	O1, first owner number (owner of bus IBUS by default)
	INTGAR(4)	O2, second owner number (0 by default)
	INTGAR(5)	O3, third owner number (0 by default)
	INTGAR(6)	O4, fourth owner number (0 by default)
	INTGAR(7)	NTP1, number of tap positions (33 by default)
	INTGAR(8)	TAB1, impedance correction table number (0 by default)
	INTGAR(9)	WN1BUS, winding one side bus number (IBUS or JBUS) (IBUS by default)
	INTGAR(10)	CONT1, controlled bus number (0 by default)
	INTGAR(11)	SICOD1, negative for controlled bus on winding one side (1 by default)
	INTGAR(12)	COD1, adjustment control mode flag (-5 through +5) (0 by default)
	INTGAR(13)	CW, winding data I/O code (1 by default)
	INTGAR(14)	CZ, impedance data I/O code (1 by default)

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	INTGAR(15)	CM, magnetizing admittance data I/O code (1 by default)
Real REALAR (26)	(first twenty-four are values; last two are values) (input (output).).	
Real REALARI (24)	array of 24 elements specifying corresponding to values REALAR(1) through REALAR(24) as described above(input).	
	REALARI(1)	R1-2, nominal transformer resistance (0.0 by default)
	REALARI(2)	X1-2, nominal transformer reactance (THRSHZ by default; 0.0001 if THRSHZ = 0.0)
	REALARI(3)	SBS1-2, winding base MVA (SBASE by default)
	REALARI(4)	WINDV1, winding one ratio/voltage (1.0 by default if CW is 1 or 3; base voltage of bus IBUS by default if CW is 2)
	REALARI(5)	NOMV1, winding one nominal voltage (0.0 by default)
	REALARI(6)	1, winding one phase shift angle (0.0 by default)
	REALARI(7)	WINDV2, winding two ratio/voltage (1.0 by default if CW is 1 or 3; base voltage of bus JBUS by default if CW is 2)
	REALARI(8)	NOMV2, winding two nominal voltage (0.0 by default)
	REALARI(9)	RATA1, rating set A line rating (0.0 by default)
	REALARI(10)	RATB1, rating set B line rating (0.0 by default)
	REALARI(11)	RATC1, rating set C line rating (0.0 by default)
	REALARI(12)	F1, first owner fraction (1.0 by default)
	REALARI(13)	F2, second owner fraction (1.0 by default)
	REALARI(14)	F3, third owner fraction (1.0 by default)
	REALARI(15)	F4, fourth owner fraction (1.0 by default)
	REALARI(16)	MAG1, magnetizing conductance or no-load losses (0.0 by default)
	REALARI(17)	MAG2, magnetizing susceptance or exciting current (0.0 by default)
	REALARI(18)	RMA1, winding one ratio/angle high limit (1.1 by default)
	REALARI(19)	RMI1, winding one ratio/angle low limit (0.9 by default)
	REALARI(20)	VMA1, voltage or flow upper limit (1.1 by default)
	REALARI(21)	VMI1, voltage or flow lower limit (0.9 by default)

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	REALARI(22)	CR1, load drop compensating resistance (0.0 by default)
	REALARI(23)	CX1, load drop compensating reactance (0.0 by default)
	REALARI(24)	CNXA1, winding connection angle (0.0 by default). Used with adjustment control mode 5 (unsymmatric phase shift control of active power) implemented in PSSE version 32.1)
Real REALARO (2)	array of 2 elements specifying corresponding to values REALAR(25) through REALAR(26) as described above(output).	
	REALARO(1)	actual transformer resistance (returned)
	REALARO(2)	actual transformer reactance (returned)
Character*12 NAME	transformer name (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	branch exists but it is a non-transformer branch
	IERR = 4	branch from a bus to itself
	IERR = 5	blank circuit identifier is invalid
	IERR = 6	circuit identifier may not start with &, * or @
	IERR = 7	branch table is full
	IERR = 8	branch ownership table is full
	IERR = 9	two-winding transformer table is full
	IERR = 10	CW, CZ and/or CM >1 but at least one end-point bus has no base voltage
	IERR = 11	invalid CW, CZ and/or CM value
	IERR = 13	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid metered end bus - invalid transformer status - invalid owner number - invalid ownership fraction - no ownership data specified. - multi-section line grouping deleted - branch ownership table is full--only first n owners retained - owner table is full - invalid winding 1 side bus number - invalid CW, CZ and/or CM data value - controlled bus not found - invalid adjustment control mode flag - invalid impedance correction table number - invalid number of tap positions - winding nominal voltage is negative - winding bus base voltage and nominal voltage differ by more than 20% - invalid winding voltage - winding voltage and bus base

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voltage differ by more than 50% - RMAX is greater than 150% of bus base voltage - RMIN is less than 50% of bus base voltage - invalid tap ratio - tap ratio is outside the band 0.5 to 1.5 pu - RMAX is greater than 1.5 - RMIN is less than 0.5 - RMAX is less than RMIN - VMAX is less than VMIN - phase shift angle normalized between -180 and 180 - winding base MVA must be positive - MAG1 is too big relative to MAG2 at winding base MVA - magnetizing susceptance MAG2 is positive - losses are too big relative to total impedance at winding base MVA - branch reactance is 0.0 - winding 1 and winding 2 sides are interchanged (any two-terminal dc lines controlled by this transformer are logged). - winding connection angle normalized between -360 and 360 - adjustment control mode flag and the winding connection angle are inconsistent. - for asymmetric phase shifter, abnormally large or small tap setting results at RMA1 (or RMI1) and CNXA1. - for asymmetric phase shifter, the specified and derived winding tap settings differ

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2.78. TWO_WINDING_DATA_4

Use this API to modify the data of an existing two-winding transformer in the working case, or to add a new two-winding transformer to the working case (see PSSE Program Operation Manual, Owner Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_TWO_WINDING_DATA_4  IBUS  JBUS  'CKT'  INTGAR(1)..INTGAR(15)
 REALAR(1)..REALAR(24)  'CHARAR(1)'  'CHARAR(2)'
```

Python command syntax:

```
ierr, realaro = two_winding_data_4(ibus, jbus, ckt, intgar, realari,
 charar)
```

Fortran command syntax:

```
CALL TWO_WINDING_API_4(IBUS, JBUS, CKT, INTGAR, REALAR, CHARAR, IERR)
```

Where:

Integer IBUS	bus number of from bus (input; no default allowed).	
Integer JBUS	bus number of to bus (input; no default allowed).	
Character*2 CKT	circuit identifier (input; '1').	
Integer INTGAR (15)	array of 15 elements specifying (input).	
	INTGAR(1)	STAT, branch status (1 by default)
	INTGAR(2)	METBUS, metered end bus number (IBUS or JBUS) (IBUS by default)
	INTGAR(3)	O1, first owner number (owner of bus IBUS by default)
	INTGAR(4)	O2, second owner number (0 by default)
	INTGAR(5)	O3, third owner number (0 by default)
	INTGAR(6)	O4, fourth owner number (0 by default)
	INTGAR(7)	NTP1, number of tap positions (33 by default)
	INTGAR(8)	TAB1, impedance correction table number (0 by default)
	INTGAR(9)	WN1BUS, winding one side bus number (IBUS or JBUS) (IBUS by default)
	INTGAR(10)	CONT1, controlled bus number (0 by default)
	INTGAR(11)	SICOD1, negative for controlled bus on winding one side (1 by default)
	INTGAR(12)	COD1, adjustment control mode flag (-5 through +5) (0 by default)
	INTGAR(13)	CW, winding data I/O code (1 by default)
	INTGAR(14)	CZ, impedance data I/O code (1 by default)
	INTGAR(15)	CM, magnetizing admittance data I/O code (1 by default)

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Real REALAR (26) (first twenty-four are values; last two are values) (input (output).).

Real REALARI (24) array of 24 elements specifying corresponding to values REALAR(1) through REALAR(24) as described above(input).

REALARI(1)	R1-2, nominal transformer resistance (0.0 by default)
REALARI(2)	X1-2, nominal transformer reactance (THRSHZ by default; 0.0001 if THRSHZ = 0.0)
REALARI(3)	SBS1-2, winding base MVA (SBASE by default)
REALARI(4)	WINDV1, winding one ratio/voltage (1.0 by default if CW is 1 or 3; base voltage of bus IBUS by default if CW is 2)
REALARI(5)	NOMV1, winding one nominal voltage (0.0 by default)
REALARI(6)	1, winding one phase shift angle (0.0 by default)
REALARI(7)	WINDV2, winding two ratio/voltage (1.0 by default if CW is 1 or 3; base voltage of bus JBUS by default if CW is 2)
REALARI(8)	NOMV2, winding two nominal voltage (0.0 by default)
REALARI(9)	RATA1, rating set A line rating (0.0 by default)
REALARI(10)	RATB1, rating set B line rating (0.0 by default)
REALARI(11)	RATC1, rating set C line rating (0.0 by default)
REALARI(12)	F1, first owner fraction (1.0 by default)
REALARI(13)	F2, second owner fraction (1.0 by default)
REALARI(14)	F3, third owner fraction (1.0 by default)
REALARI(15)	F4, fourth owner fraction (1.0 by default)
REALARI(16)	MAG1, magnetizing conductance or no-load losses (0.0 by default)
REALARI(17)	MAG2, magnetizing susceptance or exciting current (0.0 by default)
REALARI(18)	RMA1, winding one ratio/angle high limit (1.1 by default)
REALARI(19)	RMI1, winding one ratio/angle low limit (0.9 by default)
REALARI(20)	VMA1, voltage or flow upper limit (1.1 by default)
REALARI(21)	VMI1, voltage or flow lower limit (0.9 by default)
REALARI(22)	CR1, load drop compensating resistance (0.0 by default)

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	REALARI(23)	CX1, load drop compensating reactance (0.0 by default)
	REALARI(24)	CNXA1, winding connection angle (0.0 by default). Used with adjustment control mode 5 (unsymmatric phase shift control of active power) implemented in PSSE version 32.1)
Real REALARO (2)	array of 2 elements specifying corresponding to values REALAR(25) through REALAR(26) as described above(output).	
	REALARO(1)	actual transformer resistance (returned)
	REALARO(2)	actual transformer reactance (returned)
Character*40 CHARAR (2)	array of 2 elements specifying (input).	
	CHARAR(1)	NAME, transformer name (blank by default)
	CHARAR(2)	VGRP, vector group name (blank by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	bus not found
	IERR = 2	circuit identifier is more than two characters
	IERR = 3	branch exists but it is a non-transformer branch
	IERR = 4	branch from a bus to itself
	IERR = 5	blank circuit identifier is invalid
	IERR = 6	circuit identifier may not start with &, * or @
	IERR = 7	branch table is full
	IERR = 8	branch ownership table is full
	IERR = 9	two-winding transformer table is full
	IERR = 10	CW, CZ and/or CM >1 but at least one end-point bus has no base voltage
	IERR = 11	invalid CW, CZ and/or CM value
	IERR = 13	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid metered end bus - invalid transformer status - invalid owner number - invalid ownership fraction - no ownership data specified. - multi-section line grouping deleted - branch ownership table is full--only first n owners retained - owner table is full - invalid winding 1 side bus number - invalid CW, CZ and/or CM data value - controlled bus not found - invalid adjustment control mode flag - invalid impedance correction table number - invalid number of tap positions - winding nominal voltage is negative - winding bus base voltage and nominal voltage differ by more than 20% - invalid winding voltage - winding voltage and

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bus base voltage differ by more than 50% - RMAX is greater than 150% of bus base voltage - RMIN is less than 50% of bus base voltage - invalid tap ratio - tap ratio is outside the band 0.5 to 1.5 pu - RMAX is greater than 1.5 - RMIN is less than 0.5 - RMAX is less than RMIN - VMAX is less than VMIN - phase shift angle normalized between -180 and 180 - winding base MVA must be positive - MAG1 is too big relative to MAG2 at winding base MVA - magnetizing susceptance MAG2 is positive - losses are too big relative to total impedance at winding base MVA - branch reactance is 0.0 - winding 1 and winding 2 sides are interchanged (any two-terminal dc lines controlled by this transformer are logged). - winding connection angle normalized between -360 and 360 - adjustment control mode flag and the winding connection angle are inconsistent. - for asymmetric phase shifter, abnormally large or small tap setting results at RMA1 (or RMI1) and CNXA1. - for asymmetric phase shifter, the specified and derived winding tap settings differ. - vector group not found

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2.79. VSC_DC_CONVERTER_DATA

Use this API to modify the data of a dc converter of an existing VSC dc line in the working case (see PSSE Program Operation Manual, Voltage Source Converter (VSC) DC Transmission Line Data).

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_VSC_DC_CONVERTER_DATA  'NAME'  IC  INTGAR(1)..INTGAR(4)
 REALAR(1)..REALAR(11)
```

Python command syntax:

```
ierr = vsc_dc_converter_data(name, ic, intgar, realar)
```

Fortran command syntax:

```
CALL VSC_DC_LINE_CONVERTER_API_2(NAME, IC, INTGAR, REALAR, IERR)
```

Where:

Character*12 NAME	VSC dc line name (input; no default allowed).	
Integer IC	dc converter index (1 or 2) (input; no default allowed).	
Integer INTGAR (4)	array of 4 elements specifying (input).	
	INTGAR(1)	IBUS, converter bus number (no default allowed)
	INTGAR(2)	TYPE, dc control code (1 by default)
	INTGAR(2) = 0	out-of-service
	INTGAR(2) = 1	kV
	INTGAR(2) = 2	MW
	INTGAR(3)	MODE, ac control code (1 = voltage, 2 = power factor) (1 by default)
	INTGAR(4)	REMOT, remote bus number (0 for this converter to regulate voltage at bus IBUS) (0 by default)
Real REALAR (11)	array of 11 elements specifying (input).	
	REALAR(1)	DCSET, dc setpoint (kV or MW) (1.0 by default)
	REALAR(2)	ACSET, ac setpoint (pu voltage or power factor) (1.0 by default)
	REALAR(3)	ALOSS, A loss coefficient (kW) (0.0 by default)
	REALAR(4)	BLOSS, B loss coefficient (kW/amps) (0.0 by default)
	REALAR(5)	MINLOS, minimum converter losses (kW) (0.0 by default)
	REALAR(6)	SMAX, converter ac MVA rating (0.0 by default)

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	REALAR(7)	IMAX, converter ac current rating (amps) (0.0 by default)
	REALAR(8)	PWF, power weighting factor fraction (1.0 by default)
	REALAR(9)	MAXQ, reactive power upper limit (Mvar) (9999.0 by default)
	REALAR(10)	MINQ, reactive power lower limit (Mvar) (-9999.0 by default)
	REALAR(11)	RMPCT, percent of contributed reactive power (100.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid converter index
	IERR = 2	no converter bus is specified
	IERR = 3	dc line not found
	IERR = 4	converter bus not found
	IERR = 5	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - Converter bus is dummy bus of a multi-section line - Multi-section line grouping deleted. - Invalid dc control code - Invalid ac control mode - Regulated bus not found - Converter specified as remotely regulating itself. - Type code of regulated bus is not 1 or 2. - dc voltage setpoint is not positive - ac voltage setpoint is outside the band 0.75 to 1.25 pu - ac power factor setpoint is in error (magnitude exceeds 1.0) - Invalid power weighting factor fraction - MAXQ is less than MINQ. - RMPCT is not positive

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2.80. VSC_DC_LINE_CHNG

Use this API routine to modify the link data of an existing VSC dc line in the working case (see PSSE Program Operation Manual, Voltage Source Converter (VSC) DC Transmission Line Data).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_VSC_DC_LINE_CHNG  'NAME'  INTGAR(1)..INTGAR(5)  REALAR(1)..REALAR(5)
```

Python command syntax:

```
ierr = vsc_dc_line_chng(name, intgar, realar)
```

Fortran command syntax:

```
CALL VSC_DC_LINE_CHNG(NAME, INTGAR, REALAR, IERR)
```

Where:

Character*12 NAME	VSC dc line name (input; no default allowed).	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	MDC, control mode (0 or 1) (0 by default)
	INTGAR(2)	O1, first owner number (1 by default)
	INTGAR(3)	O2, second owner number (0 by default)
	INTGAR(4)	O3, third owner number (0 by default)
	INTGAR(5)	O4, fourth owner number (0 by default)
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	RDC, dc line resistance (0.0 by default)
	REALAR(2)	F1, first owner fraction (1.0 by default)
	REALAR(3)	F2, second owner fraction (1.0 by default)
	REALAR(4)	F3, third owner fraction (1.0 by default)
	REALAR(5)	F4, fourth owner fraction (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 2	blank dc line name is invalid
	IERR = 3	dc line not found
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid dc line control mode - invalid dc line resistance - invalid owner number - invalid ownership fraction - no ownership data specified - owner table is full

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2.81. VSC_DC_LINE_DATA

Use this API routine to modify the link data of an existing VSC dc line in the working case, or to add a new VSC dc line to the working case (see PSSE Program Operation Manual, Voltage Source Converter (VSC) DC Transmission Line Data).

This API routine was first introduced in release 29.0.0.

Batch command syntax:

```
BAT_VSC_DC_LINE_DATA  'NAME'  INTGAR(1)..INTGAR(5)  REALAR(1)..REALAR(5)
```

Python command syntax:

```
ierr = vsc_dc_line_data(name, intgar, realar)
```

Fortran command syntax:

```
CALL VSC_DC_LINE_API_2(NAME, INTGAR, REALAR, IERR)
```

Where:

Character*12 NAME	VSC dc line name (input; no default allowed).	
Integer INTGAR (5)	array of 5 elements specifying (input).	
	INTGAR(1)	MDC, control mode (0 or 1) (0 by default)
	INTGAR(2)	O1, first owner number (1 by default)
	INTGAR(3)	O2, second owner number (0 by default)
	INTGAR(4)	O3, third owner number (0 by default)
	INTGAR(5)	O4, fourth owner number (0 by default)
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	RDC, dc line resistance (0.0 by default)
	REALAR(2)	F1, first owner fraction (1.0 by default)
	REALAR(3)	F2, second owner fraction (1.0 by default)
	REALAR(4)	F3, third owner fraction (1.0 by default)
	REALAR(5)	F4, fourth owner fraction (1.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	VSC dc line table is full
	IERR = 2	blank dc line name is invalid
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - invalid dc line control mode - invalid dc line resistance - invalid owner number - invalid ownership fraction - no ownership data specified - owner table is full

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2.82. ZONE_DATA

Use this API to modify zone data in the working case (see PSSE Program Operation Manual, Zone Data).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ZONE_DATA  IZONE  'ZONAME'
```

Python command syntax:

```
ierr = zone_data(izone, zoname)
```

Fortran command syntax:

```
CALL ZONE_API_2(IZONE, ZONAME, IERR)
```

Where:

Integer IZONE	zone number between 1 and 9999 (input; no default allowed).	
Character*12 ZONAME	zone name (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	fatal error occurred
	IERR = 1	invalid zone number
	IERR = 2	zone table is full
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - No equipment is assigned to zone

Chapter 3
Optimal Power Flow

This chapter describes the API routines used in PSS®E Optimal Power Flow. Optimal Power Flow Program Operation APIs contains API routines corresponding to the operational functions of the PSS®E OPF. Additional information on the functions these commands execute is in the PSS®E Program Operation Manual, Chapter 14. Optimal Power Flow Data Changing APIs contains API routines specific to adding, modifying and deleting optimal power flow data.

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3.1. Optimal Power Flow Operation

3.1.1. ADD_DETAILS_TO_OPF_LOG

Use this API to specify or return the value of the OPF solution option to add details of the sensitivity values and Lagrange multipliers to the OPF Output Log file.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_ADD_DETAILS_TO_OPF_LOG  IVAL
```

Python command syntax:

```
ierr = add_details_to_opf_log(ival)

ierr, ival = add_details_to_opf_log()
```

Fortran command syntax:

```
CALL ADD_DETAILS_TO_OPF_LOG(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to append the OPF log file with information on sensitivity values and Lagrange multipliers (input; input if Batch command or for Python if specified as an argument, or for FORTRAN if IOCODE is 0; else output, 0 by default).	
	IVAL = 0	do not print additional details in log file.
	IVAL = 1	print additional details.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to the value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.2. APPLY_ALTERNATE_STEP_SIZE

Use this API to specify or return the value of the OPF solution option for applying an alternate lambda mismatch step size. This option works in conjunction with the OPF_BARRIER_STEP_LENGTH option setting (Section 3.1.25), which specifies the alternate barrier step length value.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_APPLY_ALTERNATE_STEP_SIZE    IVAL
```

Python command syntax:

```
ierr = apply_alternate_step_size(ival)

ierr, ival = apply_alternate_step_size()
```

Fortran command syntax:

```
CALL APPLY_ALTERNATE_STEP_SIZE(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to limit the step size taken during the OPF solution (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not limit the step size.
	IVAL = 1	limit the step size.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.3. APPLY_AUTOMATIC_SCALING

Use this API to specify or return the value of the OPF solution option to apply automatic scaling of the objective coefficients and derivatives to improve convergence of the OPF solution.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_APPLY_AUTOMATIC_SCALING   IVAL
```

Python command syntax:

```
ierr = apply_automatic_scaling(ival)

ierr, ival = apply_automatic_scaling()
```

Fortran command syntax:

```
CALL APPLY_AUTOMATIC_SCALING(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to apply automatic scaling of the objective during the solution (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not apply automatic scaling of the objective.
	IVAL = 1	apply automatic scaling of the objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.4. BAD_ITER_COARSE_LIMIT

Use this API to specify or return the number of bad iterations allowed when the barrier coefficient is greater than 0.011 (mu greater then -2) before the solution progress is terminated as being infeasible.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_BAD_ITER_COARSE_LIMIT   IVAL
```

Python command syntax:

```
ierr =  bad_iter_coarse_limit(ival)

ierr, ival = bad_iter_coarse_limit()
```

Fortran command syntax:

```
CALL BAD_ITER_COARSE_LIMIT(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Bad iteration coarse limit value (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.5. BAD_ITER_FINE_LIMIT

Use this API to specify or return the the bad iteration limit when the barrier coefficient is less than, or equal to 0.011 (mu less than or equal to -2) before the solution progress is terminated as being infeasible.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_BAD_ITER_FINE_LIMIT  IVAL
```

Python command syntax:

```
ierr =  bad_iter_fine_limit(ival)

ierr, ival = bad_iter_fine_limit()
```

Fortran command syntax:

```
CALL BAD_ITER_FINE_LIMIT(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Bad iteration fine limit value (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.6. CLAMP_NONOPTIMIZED_GENS

Use this API to specify or return the value of the OPF solution option to apply the clamp equation to non-optimized generators. The behavior of the clamp equation may be further customized through the OPF_INITIAL_CLAMP_TOL, OPF_FINAL_CLAMP_TOL and the OPF_CLAMP_DECAY_FACTOR option settings. For additional information on the clamp treatment of non-optimized generators, refer to Clamp Treatment in the PSSE Program Operation Manual.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_CLAMP_NONOPTIMIZED_GENS    IVAL
```

Python command syntax:

```
ierr = clamp_nonoptimized_gens(ival)

ierr, ival = clamp_nonoptimized_gens()
```

Fortran command syntax:

```
CALL CLAMP_NONOPTIMIZED_GENS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to apply the clamp equation to nonoptimized generators (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not clamp nonoptimized generators.
	IVAL = 1	clamp nonoptimized generators.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.7. CONSTRAIN_INTERFACE_FLOWS

Use this API to specify or return the value of the OPF option to constrain interface flows. The setting of this option affects recognition of any interface flow constraints and the behavior of the MINIMIZE_INTERFACE_FLOWS objective function option.
If the Constrain Interface Flows option is not enabled, the Minimize Interrface Flows objective and any interface flow constraints will be ignored and not used.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_CONSTRAIN_INTERFACE_FLOWS    IVAL
```

Python command syntax:

```
ierr = constrain_interface_flows(ival)

ierr, ival = constrain_interface_flows()
```

Fortran command syntax:

```
CALL CONSTRAIN_INTERFACE_FLOWS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the option to constrain interface flows (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not constrain interface flows.
	IVAL = 1	constrain interface flows.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.8. FINAL_OPF_BARRIER_COEFF

Use this API to specify or return the value of the final coefficient value that the barrier function is attempting to attain. It may be used in conjunction with the INITIAL_OPF_BARRIER_COEFF value.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_FINAL_OPF_BARRIER_COEFF    RVAL
```

Python command syntax:

```
ierr =  final_opf_barrier_coeff(rval)

ierr, rval = final_opf_barrier_coeff()
```

Fortran command syntax:

```
CALL FINAL_OPF_BARRIER_COEFF(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	final barrier coefficient value (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.9. IMPORT_ECDI

Use this API to import data from an Economics Dispatch Data File, as prepared for use in activity ECDI, into the OPF working data. This function removes all existing Optimal Power Flow Generator Dispatch Data, Active Power Dispatch Data, and Cost Curve Data, before adding new records from the ECD file.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_IMPORT_ECDI  FILE
```

Python command syntax:

```
ierr = import_ecdi(file)
```

Fortran command syntax:

```
CALL IMPORT_ECDI(FILE, IERR)
```

Where:

Character*260 FILE	name of the Economics Dispatch Data File (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	NEWOPF_API error occurred
	IERR = 2	ECDIAPI error occurred
	IERR = 3	CSTTBL_QUAD_API error occurred
	IERR = 4	OPFDSP_TBL_API error occurred
	IERR = 5	GENDSP_INDV_API error occurred
	IERR = 6	prerequisite requirements for API are not met

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3.1.10. INITIAL_OPF_BARRIER_COEFF

Use this API to specify or return the value of the initial OPF barrier coefficient. It may be used in conjunction with the FINAL_OPF_BARRIER_COEFF value.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_INITIAL_OPF_BARRIER_COEFF    RVAL
```

Python command syntax:

```
ierr =  initial_opf_barrier_coeff(rval)

ierr, rval = initial_opf_barrier_coeff()
```

Fortran command syntax:

```
CALL INITIAL_OPF_BARRIER_COEFF(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	initial barrier coefficient value (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.11. INTERFACE_FLOW_COST_COEFF

Use this API to specify or return the value of the coefficient for the interface flow objective. This API works in conjunction with the MINIMIZE_INTERFACE_FLOWS objective function option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_INTERFACE_FLOW_COST_COEFF    RVAL
```

Python command syntax:

```
ierr =  interface_flow_cost_coeff(rval)

ierr, rval = interface_flow_cost_coeff()
```

Fortran command syntax:

```
CALL INTERFACE_FLOW_COST_COEFF(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the interface flow cost coefficient (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.12. LSTO

Use this API to list OPF working case data in a form suitable for data documentation.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LSTO   SID  ALL  OPT  VREV
```

Python command syntax:

```
ierr = lsto(sid, all, opt, vrev)
```

Fortran command syntax:

```
CALL LSTONW(SID, ALL, OPT, VREV, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer OPT	number corresponding to the category of OPF data that is to be tabulated (input; no default allowed).	
	OPT = 1	case summary
	OPT = 2	bus attribute data
	OPT = 3	adjustable bus shunt data
	OPT = 4	bus load data
	OPT = 5	adjustable bus load table data
	OPT = 6	generator dispatch data
	OPT = 7	active power dispatch data
	OPT = 8	generation reserve data
	OPT = 9	generator reactive capability data
	OPT = 10	adjustable branch reactance data
	OPT = 11	linear cost table data
	OPT = 12	quadratic cost table data
	OPT = 13	polynomial and exponential cost table data
	OPT = 14	period reserve constraint data
	OPT = 15	branch flow constraint data
	OPT = 16	interface flow constraint data
	OPT = 17	linear constraint dependency data
	OPT = 21	full listing
Integer VREV	option to change voltage units from the default units to the alternate units (input; no default allowed).	
	VREV = 0	use default voltage units.

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	VREV = 1	use alternate voltage units.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid OPT value
	IERR = 4	invalid VREV value
	IERR = 5	prerequisite requirements for API are not met

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3.1.13. MINIMIZE_ADJ_BUS_SHUNTS

Use this API to specify or return the value of the OPF minimize adjustable bus shunts (var compensation) objective function option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_ADJ_BUS_SHUNTS    IVAL
```

Python command syntax:

```
ierr = minimize_adj_bus_shunts(ival)

ierr, ival = minimize_adj_bus_shunts()
```

Fortran command syntax:

```
CALL MINIMIZE_ADJ_BUS_SHUNTS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize adjustable bus shunts objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize adjustable bus shunts objective.
	IVAL = 1	enable the minimize adjustable bus shunts objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.14. MINIMIZE_FUEL_COST

Use this API to specify or return the value of the OPF minimize fuel cost objective option. The Minimize fuel cost objective and Minimize active power loss objective cannot be simultaneously employed due to the competing goals.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_FUEL_COST   IVAL
```

Python command syntax:

```
ierr = minimize_fuel_cost(ival)

ierr, ival = minimize_fuel_cost()
```

Fortran command syntax:

```
CALL MINIMIZE_FUEL_COST(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize fuel cost objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize fuel cost objective.
	IVAL = 1	enable the minimize fuel cost objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.15. MINIMIZE_INTERFACE_FLOWS

Use this API to specify or return the value of the OPF minimize interface flows objective option. Used in conjuction with INTERFACE_FLOW_COST_COEFFICIENT. A negative coefficient value will maximize interface flows.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_INTERFACE_FLOWS    IVAL
```

Python command syntax:

```
ierr = minimize_interface_flows(ival)

ierr, ival = minimize_interface_flows()
```

Fortran command syntax:

```
CALL MINIMIZE_INTERFACE_FLOWS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize interface flows objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize interface flows objective.
	IVAL = 1	enable the minimize interface flows objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.16. MINIMIZE_LOAD_ADJUSTMENTS

Use this API to specify or return the value of the OPF minimize adjustable bus load objective option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_LOAD_ADJUSTMENTS    IVAL
```

Python command syntax:

```
ierr = minimize_load_adjustments(ival)

ierr, ival = minimize_load_adjustments()
```

Fortran command syntax:

```
CALL MINIMIZE_LOAD_ADJUSTMENTS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize adjustable bus loads objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize adjustable bus loads objective.
	IVAL = 1	enable the minimize adjustable bus loads objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.17. MINIMIZE_P_LOSSES

Use this API to specify or return the value of the OPF minimize active power loss objective option. The Minimize active power loss objective and Minimize fuel cost objective cannot be simultaneously employed due to the competing goals

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_P_LOSSES    IVAL
```

Python command syntax:

```
ierr = minimize_p_losses(ival)

ierr, ival = minimize_p_losses()
```

Fortran command syntax:

```
CALL MINIMIZE_P_LOSSES(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize active power loss objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize active power loss objective.
	IVAL = 1	enable the minimize active power loss objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.1.18. MINIMIZE_P_SLACK

Use this API to specify or return the value of the OPF minimize active power slack objective option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_P_SLACK    IVAL
```

Python command syntax:

```
ierr = minimize_p_slack(ival)

ierr, ival = minimize_p_slack()
```

Fortran command syntax:

```
CALL MINIMIZE_P_SLACK(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize active power slack objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize active power slack objective.
	IVAL = 1	enable the minimize active power slack objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.19. MINIMIZE_Q_LOSSES

Use this API to specify or return the value of the OPF minimize reactive power loss objective option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_Q_LOSSES    IVAL
```

Python command syntax:

```
ierr = minimize_q_losses(ival)

ierr, ival = minimize_q_losses()
```

Fortran command syntax:

```
CALL MINIMIZE_Q_LOSSES(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize reactive power loss objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize reactive power loss objective.
	IVAL = 1	enable the minimize reactive power loss objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.20. MINIMIZE_Q_SLACK

Use this API to specify or return the value of the OPF minimize reactive power slack objective option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_Q_SLACK    IVAL
```

Python command syntax:

```
ierr = minimize_q_slack(ival)

ierr, ival = minimize_q_slack()
```

Fortran command syntax:

```
CALL MINIMIZE_Q_SLACK(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize reactive power slack objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize reactive power slack objective.
	IVAL = 1	enable the minimize reactive power slack objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.21. MINIMIZE_REACTIVE_RESERVE

Use this API to specify or return the value of the OPF minimize reactive reserve objective option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_REACTIVE_RESERVE    IVAL
```

Python command syntax:

```
ierr = minimize_reactive_reserve(ival)

ierr, ival = minimize_reactive_reserve()
```

Fortran command syntax:

```
CALL MINIMIZE_REACTIVE_RESERVE(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize reactive reserves objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize reactive reserves objective.
	IVAL = 1	enable the minimize reactive reserves objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.22. MINIMIZE_SERIES_COMP

Use this API to specify or return the value of the OPF minimize series compensation (adjustable branch reactance) objective option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_MINIMIZE_SERIES_COMP   IVAL
```

Python command syntax:

```
ierr = minimize_series_comp(ival)

ierr, ival = minimize_series_comp()
```

Fortran command syntax:

```
CALL MINIMIZE_SERIES_COMP(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the minimize series compensation objective (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not enable the minimize series compensation objective.
	IVAL = 1	enable the minimize series compensation objective.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.23. NOPF

Use this API to run the Optimal Power Flow solution. Subsystems that have been specified will be optimized to the objectives and tolerances specified, if a solution is achievable.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_NOPF   SID  ALL
```

Python command syntax:

```
ierr = nopf(sid, all)
```

Fortran command syntax:

```
CALL NOPF_SOLVE(SID, ALL, IERR)
```

Where:

Integer SID	a valid bus subsystem ID from 0 to 11. A bus subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	error initializing OPF data
	IERR = 4	generators are converted
	IERR = 5	Topology error: buses exist with bus type code and series element status inconsistencies
	IERR = 6	prerequisite requirements for API are not met

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3.1.24. OPEN_BUS_VOLTAGE_LIMITS

Use this API to specify or return the value of the OPF solution option to automatically adjust bus voltage limits, making them more open for initial feasibility.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPEN_BUS_VOLTAGE_LIMITS   IVAL
```

Python command syntax:

```
ierr =  open_bus_voltage_limits(ival)

ierr, ival = open_bus_voltage_limits()
```

Fortran command syntax:

```
CALL OPEN_BUS_VOLTAGE_LIMITS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to open bus voltage limits for initial feasibility (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not open up bus voltage limits.
	IVAL = 1	open bus voltage limits for initial feasibility.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.25. OPF_BARRIER_STEP_LENGTH

Use this API to specify or return the value of the barrier function step length setting.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_BARRIER_STEP_LENGTH    RVAL
```

Python command syntax:

```
ierr =  opf_barrier_step_length(rval)

ierr, rval = opf_barrier_step_length()
```

Fortran command syntax:

```
CALL OPF_BARRIER_STEP_LENGTH(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the barrier function step length (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.26. OPF_CLAMP_DECAY_FACTOR

Use this API to specify or return the value of the clamp equation decay factor. This API works in conjunction with the API option to CLAMP_NONOPTIMIZED_GENS, and is associated with the APIs for OPF_INITIAL_CLAMP_TOL and OPF_FINAL_CLAMP_TOL.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_CLAMP_DECAY_FACTOR    RVAL
```

Python command syntax:

```
ierr =  opf_clamp_decay_factor(rval)

ierr, rval = opf_clamp_decay_factor()
```

Fortran command syntax:

```
CALL OPF_CLAMP_DECAY_FACTOR(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the clamp equation decay factor (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.27. OPF_FINAL_CLAMP_TOL

Use this API to specify or return the value of the final clamp tolerance. This API works in conjunction with the API option to CLAMP_NONOPTIMIZED_GENS, and is associated with the APIs for OPF_INITIAL_CLAMP_TOL and the OPF_CLAMP_DECAY_FACTOR.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_FINAL_CLAMP_TOL    RVAL
```

Python command syntax:

```
ierr =  opf_final_clamp_tol(rval)

ierr, rval = opf_final_clamp_tol()
```

Fortran command syntax:

```
CALL OPF_FINAL_CLAMP_TOL(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the final clamp tolerance (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.28. OPF_FIXED_VOLTAGE_PENALTY

Use this API to specify or return the value of the penalty for fixed voltage excursions.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_FIXED_VOLTAGE_PENALTY    RVAL
```

Python command syntax:

```
ierr =  opf_fixed_voltage_penalty(rval)

ierr, rval = opf_fixed_voltage_penalty()
```

Fortran command syntax:

```
CALL OPF_FIXED_VOLTAGE_PENALTY(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the penalty for fixed voltage excursions (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.29. OPF_FIX_ALL_GENERATORS

Use this API to specify or return the value of the OPF option to treat all generators as nonoptimized (fixed).

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_FIX_ALL_GENERATORS    IVAL
```

Python command syntax:

```
ierr = opf_fix_all_generators(ival)

ierr, ival = opf_fix_all_generators()
```

Fortran command syntax:

```
CALL OPF_FIX_ALL_GENERATORS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the option to treat all generators as nonoptimized and fix their settings (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not treat all generators as nonoptimized.
	IVAL = 1	treat all generators as nonoptimized.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.1.30. OPF_FIX_PHASE_SHIFTERS

Use this API to specify or return the value of the OPF option to fix transformer phase shift angle settings.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_FIX_PHASE_SHIFTERS    IVAL
```

Python command syntax:

```
ierr = opf_fix_phase_shifters(ival)

ierr, ival = opf_fix_phase_shifters()
```

Fortran command syntax:

```
CALL OPF_FIX_PHASE_SHIFTERS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the option to fix transformer phase shift angles (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not fix transformer phase shift angle settings.
	IVAL = 1	fix transformer phase shift angle settings.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.31. OPF_FIX_SWITCHED_SHUNTS

Use this API to specify or return the value of the OPF option to fix switched shunt settings.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_FIX_SWITCHED_SHUNTS    IVAL
```

Python command syntax:

```
ierr = opf_fix_switched_shunts(ival)

ierr, ival = opf_fix_switched_shunts()
```

Fortran command syntax:

```
CALL OPF_FIX_SWITCHED_SHUNTS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the option to fix switched shunt settings (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not fix switched shunt settings.
	IVAL = 1	fix switched shunt settings.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.32. OPF_FIX_TAP_RATIOS

Use this API to specify or return the value of the OPF option to fix transformer tap ratio settings.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_FIX_TAP_RATIOS    IVAL
```

Python command syntax:

```
ierr = opf_fix_tap_ratios(ival)

ierr, ival = opf_fix_tap_ratios()
```

Fortran command syntax:

```
CALL OPF_FIX_TAP_RATIOS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the option to fix transformer tap ratio settings (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not fix transformer tap ratio settings.
	IVAL = 1	fix transformer tap ratio settings.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.33. OPF_INITIAL_CLAMP_TOL

Use this API to specify or return the value of the initial clamp tolerance. This API works in conjunction with the API option to CLAMP_NONOPTIMIZED_GENS, and is associated with the APIs for OPF_FINAL_CLAMP_TOL and the OPF_CLAMP_DECAY_FACTOR.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_INITIAL_CLAMP_TOL    RVAL
```

Python command syntax:

```
ierr =  opf_initial_clamp_tol(rval)

ierr, rval = opf_initial_clamp_tol()
```

Fortran command syntax:

```
CALL OPF_INITIAL_CLAMP_TOL(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the initial clamp tolerance (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.34. OPF_INTERIOR_SHIFT_FACTOR

Use this API to specify or return the value of the interior shift factor multiplier. This value indicates by how much a variable should be moved "more interior" of the upper and lower limits.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_OPF_INTERIOR_SHIFT_FACTOR   RVAL
```

Python command syntax:

```
ierr =  opf_interior_shift_factor(rval)

ierr, rval = opf_interior_shift_factor()
```

Fortran command syntax:

```
CALL OPF_INTERIOR_SHIFT_FACTOR(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	interior shift factor multiplier value (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.35. OPF_LAMBDA_TOLERANCE

Use this API to specify or return the value of the Lagrange multiplier blow-up tolerance during an optimal power flow solution.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_LAMBDA_TOLERANCE    RVAL
```

Python command syntax:

```
ierr =  opf_lambda_tolerance(rval)

ierr, rval = opf_lambda_tolerance()
```

Fortran command syntax:

```
CALL OPF_LAMBDA_TOLERANCE(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	lambda tolerance value (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.36. OPF_LF_CONTROL_PENALTY

Use this API to specify or return the value of the quadratic penalty for loadflow controls. This penalty is applied to optimized loadflow control variable excursions from their initial condition, in $/pu.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_LF_CONTROL_PENALTY    RVAL
```

Python command syntax:

```
ierr =  opf_lf_control_penalty(rval)

ierr, rval = opf_lf_control_penalty()
```

Fortran command syntax:

```
CALL OPF_LF_CONTROL_PENALTY(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the quadratic penalty for loadflow controls (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.37. OPF_MAX_TAP_RATIO_STEP

Use this API to specify or return the value of the maximum transformer tap ratio step.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_MAX_TAP_RATIO_STEP    RVAL
```

Python command syntax:

```
ierr =  opf_max_tap_ratio_step(rval)

ierr, rval = opf_max_tap_ratio_step()
```

Fortran command syntax:

```
CALL OPF_MAX_TAP_RATIO_STEP(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	maximum tap ratio step value (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.38. OPF_MIN_TAP_RATIO_STEP

Use this API to set or return the value of the minimum transformer tap ratio step.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_MIN_TAP_RATIO_STEP    RVAL
```

Python command syntax:

```
ierr =  opf_min_tap_ratio_step(rval)

ierr, rval = opf_min_tap_ratio_step()
```

Fortran command syntax:

```
CALL OPF_MIN_TAP_RATIO_STEP(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	minimum tap ratio step value (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.39. OPF_REGULATE_AREA_INT

Use this API to set or return the value of the OPF option to regulate area interchange.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_REGULATE_AREA_INT    IVAL
```

Python command syntax:

```
ierr = opf_regulate_area_int(ival)

ierr, ival = opf_regulate_area_int()
```

Fortran command syntax:

```
CALL OPF_REGULATE_AREA_INT(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to enable the option to regulate area interchange (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not regulate area interchange.
	IVAL = 1	regulate area interchange.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.40. OPF_ROUND_SWITCHED_SHUNTS

Use this API to set or return the value of the OPF solution option to discretize switched shunts.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_ROUND_SWITCHED_SHUNTS    IVAL
```

Python command syntax:

```
ierr = opf_round_switched_shunts(ival)

ierr, ival = opf_round_switched_shunts()
```

Fortran command syntax:

```
CALL OPF_ROUND_SWITCHED_SHUNTS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to discretize switched shunt settings after initial solution. (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not discretize switched shunt settings.
	IVAL = 1	discretize switched shunt settings.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.41. OPF_ROUND_TAP_RATIOS

Use this API to set or return the value of the OPF solution option to round transformer tap ratio settings.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_ROUND_TAP_RATIOS    IVAL
```

Python command syntax:

```
ierr = opf_round_tap_ratios(ival)

ierr, ival = opf_round_tap_ratios()
```

Fortran command syntax:

```
CALL OPF_ROUND_TAP_RATIOS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to round transformer tap ratio settings (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not round transformer tap ratio settings.
	IVAL = 1	round transformer tap ratio settings.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.42. OPF_SCALE_QGEN_LIMITS

Use this API to set or return the value of the OPF option to scale reactive generation limits in the clamped constraint equation.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_SCALE_QGEN_LIMITS   IVAL
```

Python command syntax:

```
ierr = opf_scale_qgen_limits(ival)

ierr, ival = opf_scale_qgen_limits()
```

Fortran command syntax:

```
CALL OPF_SCALE_QGEN_LIMITS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to scale reactive generation limits in the clamp equation (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not scale reactive generation limits.
	IVAL = 1	scale reactive generation limits.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.43. OPF_STEP_LENGTH_TOLERANCE

Use this API to set or return the value of the OPF minimum barrier step length tolerance.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_STEP_LENGTH_TOLERANCE    RVAL
```

Python command syntax:

```
ierr =  opf_step_length_tolerance(rval)

ierr, rval = opf_step_length_tolerance()
```

Fortran command syntax:

```
CALL OPF_STEP_LENGTH_TOLERANCE(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the minimum barrier step length tolerance (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.44. OPF_USE_GENERATOR_VSCHED

Use this API to set or return the value of the OPF option to employ the generator scheduled voltage.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPF_USE_GENERATOR_VSCHED    IVAL
```

Python command syntax:

```
ierr = opf_use_generator_vsched(ival)

ierr, ival = opf_use_generator_vsched()
```

Fortran command syntax:

```
CALL OPF_USE_GENERATOR_VSCHED(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to use generator scheduled voltage values (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not use generator scheduled voltages.
	IVAL = 1	use generator scheduled voltages.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.45. OPTO

This API routine is obsolete.

Please use the individual OPF option setting API's described in this section.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPTO   SID  ALL  IARY(1) .. IARY(36)  RARY(1) .. RARY(19)  OPFILE
 RCFILE
```

Python command syntax:

```
ierr = opto(sid, all, iary, rary, opfile, rcfile)
```

Fortran command syntax:

```
CALL OPTONW(SID, ALL, IARY, RARY, OPFILE, RCFILE, IERR)
```

Where:

Integer SID	a valid bus subsystem ID from 0 to 11. A bus subsystem must have been previously defined (input).	
Integer ALL	all buses or specified subsystem flag (input).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer IARY (36)	array of 36 elements specifying 36 integers, which are defined as follows(input).	
	IARY(1)	specifies the fuel cost objective
	IARY(1) = 0	do not employ minimize fuel cost objective
	IARY(1) = 1	employ the minimize fuel cost objective
	IARY(2)	specifies the active power slack bus generation objective
	IARY(2) = 0	do not employ minimize active power slack bus generation objective
	IARY(2) = 1	employ the minimize active power slack bus generation objective
	IARY(3)	specifies the reactive power slack generation objective
	IARY(3) = 0	do not employ the minimize reactive power slack generation objective
	IARY(3) = 1	employ the minimize reactive power slack generation objective
	IARY(4)	specifies the active power loss objective
	IARY(4) = 0	do not employ the minimize active power loss objective
	IARY(4) = 1	employ the minimize active power loss objective

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IARY(5)	specifies the reactive power loss objective
IARY(5) = 0	do not employ the minimize reactive power loss objective
IARY(5) = 1	employ the minimize reactive power loss objective
IARY(6)	specifies the adjustable branch reactance (series compensation) objective
IARY(6) = 0	do not employ the minimize adjustable branch reactance objective
IARY(6) = 1	employ the minimize adjustable branch reactance objective
IARY(7)	specifies the adjustable bus shunt (var compensation) objective
IARY(7) = 0	do not employ the minimize adjustable bus shunt objective
IARY(7) = 1	employ the minimize adjustable bus shunt objective
IARY(8)	specifies the adjustable bus load (load shedding/increase) objective
IARY(8) = 0	do not employ the adjustable bus load objective
IARY(8) = 1	employ the adjustable bus load objective
IARY(9)	specifies the interface flow objective
IARY(9) = 0	do not employ the interface flow objective
IARY(9) = 1	employ the interface flow objective
IARY(10)	specifies the reactive generation reserve objective
IARY(10) = 0	do not employ the reactive generation reserve objective
IARY(10) = 1	employ the reactive generation reserve objective
IARY(11)	specifies whether to apply the power flow mismatch tolerance to the dual variable problem
IARY(11) = 0	apply tolerance to the mismatch equation only
IARY(11) = 1	apply tolerance to the entire equation
IARY(12)	specifies whether to apply the clamp equation to nonoptimized generators
IARY(12) = 0	do not clamp nonoptimized generators
IARY(12) = 1	clamp nonoptimized generators
IARY(13)	specifies whether to apply the clamp equation to nonoptimized transformers. (Not currently functional)
IARY(13) = 0	do not clamp nonoptimized transformers
IARY(13) = 1	clamp nonoptimized transformers

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IARY(14)	specifies whether to constrain interface flows
IARY(14) = 0	do not constrain interface flows
IARY(14) = 1	constrain interface flows
IARY(15)	specifies whether to employ automatic scaling of the objective
IARY(15) = 0	do not employ automatic scaling
IARY(15) = 1	employ automatic scaling
IARY(16)	specifies whether to regulate area interchange
IARY(16) = 0	do not regulate area interchange
IARY(16) = 1	regulate area interchange
IARY(17)	specifies whether to fix transformer tap ratio settings
IARY(17) = 0	do not fix transformer tap ratio settings
IARY(17) = 1	fix transformer tap ratio settings
IARY(18)	specifies whether to fix transformer phase shift angles
IARY(18) = 0	do not fix transformer phase shift angles
IARY(18) = 1	fix transformer phase shift angles
IARY(19)	specifies whether to fix switched shunt settings
IARY(19) = 0	do not fix switched shunt settings
IARY(19) = 1	fix switched shunt settings
IARY(20)	specifies whether to treat all generators as nonoptimized
IARY(20) = 0	do not treat all generators as nonoptimized
IARY(20) = 1	treat all generators as nonoptimized
IARY(21)	specifies whether to employ the generator scheduled voltage
IARY(21) = 0	do not employ generator scheduled voltage
IARY(21) = 1	use generator scheduled voltage
IARY(22)	specifies whether to scale reactive generation limits on clamped constraint equations
IARY(22) = 0	do not scale reactive generation limits
IARY(22) = 1	scale reactive generation limits
IARY(23)	specifies whether to round transformer tap settings
IARY(23) = 0	do not round transformer tap settings
IARY(23) = 1	round transformer tap settings
IARY(24)	specifies whether to round switched shunt settings
IARY(24) = 0	do not round switched shunt settings
IARY(24) = 1	round switched shunt settings

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IARY(25)	specifies whether to limit the step size taken during solution
IARY(25) = 0	do not limit the step size
IARY(25) = 1	limit the step size taken
IARY(26)	specifies whether to produce an optimization log file
IARY(26) = 0	do not produce an optimization log file
IARY(26) = 1	produce an optimization log file
IARY(27)	specifies whether to include a detailed listing of Lagrange multipliers in the optimization log file. IARY(26) must be set to 1 for this to take effect
IARY(27) = 0	do not print details
IARY(27) = 1	print Lagrange multipliers
IARY(28)	specifies whether to produce a PSSE raw data change file. (Not currently functional)
IARY(28) = 0	do not produce an Read/Change File
IARY(28) = 1	produce an Read/Change File
IARY(29)	specifies whether to save the OPF solution option settings to the OPF Options File, PSSOPF.OPT
IARY(29) = 0	do not save settings
IARY(29) = 1	save settings to PSSOPF.OPT
IARY(30)	indicates whether the program should open voltage limits to ensure an initial feasible condition
IARY(30) = 0	do not have the program open the limits for an initial feasible condition
IARY(30) = 1	have the program open the limits for an initial feasible condition
IARY(31)	specifies whether to impose emergency bus voltage limits or normal bus voltage limits
IARY(31) = 0	impose normal voltage limits
IARY(31) = 1	impose emergency bus voltage limits
IARY(32)	specifies whether to impose emergency branch flow limits or normal branch flow limits
IARY(32) = 0	impose normal branch flow limits
IARY(32) = 1	impose emergency branch flow limits
IARY(33)	specifies whether to include transformer adjustment data in the Read/Change File (if one is produced according to the setting of IARY(28)). (Not currently functional)
IARY(33) = 0	do not include transformer adjustment data
IARY(33) = 1	include transformer adjustment data

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	IARY(34)	specifies the maximum iteration limit (corresponds to ITMXN of the power flow solution parameters)
	IARY(35)	specifies the bad iteration coarse limit
	IARY(36)	specifies the bad iteration fine limit
Real RARY (19)	array of 19 elements specifying which are defined as follows(input).	
	RARY(1)	specifies the initial barrier function coefficient
	RARY(2)	specifies the final barrier function coefficient
	RARY(3)	specifies the step length blow-up tolerance
	RARY(4)	specifies the final clamp equation tolerance value
	RARY(5)	specifies the initial clamp equation tolerance value
	RARY(6)	specifies the clamp equation decay factor
	RARY(7)	specifies the cost coefficient for the active power loss objective
	RARY(8)	specifies the cost coefficient for the reactive power loss objective
	RARY(9)	specifies the penalty value for power flow controls
	RARY(10)	specifies the cost coefficient for the interface flow objective
	RARY(11)	specifies the cost coefficient for the reactive generation reserve objective
	RARY(12)	specifies the penalty value for fixed voltage
	RARY(13)	specifies the lambda (Lagrange multiplier) blow-up tolerance
	RARY(14)	specifies the minimum amount to open up tap settings by
	RARY(15)	specifies the maximum amount to open up tap settings by
	RARY(16)	specifies the amount to shift the interior point by during solution
	RARY(17)	specifies the step length taken for the barrier method
	RARY(18)	specifies the tolerance value used to determine which bus shunts get added to the Read/Change File as switched shunts (if an RDCH file is produced according to IARY(28) and the minimize added vars objective is employed according to IARY(7)). (Not currently functional)
	RARY(19)	specifies the solution convergence tolerance, in MVA (corresponds to TOLN of the power flow solution parameters)

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Character*260 OPFILE	name of the OPF Optimization Log File that will be produced if IARY(26) is set to 1. It is a good idea to specify the filename in quotes (input).
Character*260 RCFILE	name of the PTI Read Change Data File that will be produced if IARY(28) is set to 1. It is a good idea to specify the filename in quotes. (Not currently functional) (input).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	invalid SID value or subsystem SID is not defined
IERR = 2	invalid ALL value
IERR = 3	invalid IARY value
IERR = 4	invalid RARY value
IERR = 5	error opening OFILE
IERR = 6	prerequisite requirements for API are not met

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3.1.46. PRODUCE_OPF_LOG_FILE

Use this API to set or return the value of the OPF solution option to produce an OPF solution log file, and if so, the name of the log file. This API can be used in conjunction with the ADD_DETAILS_TO_OPF_LOG API if additional details of the Lagrange multipliers are desired.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PRODUCE_OPF_LOG_FILE   IVAL  OFILE
```

Python command syntax:

```
ierr = produce_opf_log_file(ival, ofile)

ierr, ival,ofile = produce_opf_log_file()
```

Fortran command syntax:

```
CALL PRODUCE_OPF_LOG_FILE(IVAL, IOCODE, OFILE, IERR)
```

Where:

Integer IVAL	Indicates whether to produce an OPF Optimization Log File (input; if Batch command or for Python if specified as an argument, or for Fortran if IOCODE is 0; else 0 by default for input).	
	IVAL = 0	do not produce an OPF solution log file.
	IVAL = 1	produce an OPF solution log file.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Character*260 OFILE	OPF Optimization Log output filename (input; if Batch command or for Python if specified as an argument, or for FORTRAN if IOCODE is 0; else DETAIL.DAT by default for input).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.47. P_LOSSES_COST_COEFF

Use this API to set or return the value of the coefficient for the active power loss objective (OPF Solution Options). This API works in conjunction with the MINIMIZE_P_LOSSES objective function option.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_P_LOSSES_COST_COEFF    RVAL
```

Python command syntax:

```
ierr =  p_losses_cost_coeff(rval)

ierr, rval = p_losses_cost_coeff()
```

Fortran command syntax:

```
CALL P_LOSSES_COST_COEFF(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the active power loss cost coefficient (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.48. Q_LOSSES_COST_COEFF

Use this API to set or return the value of the coefficient for the reactive power loss objective. This API works in conjunction with the API option to MINIMIZE_Q_LOSSES.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_Q_LOSSES_COST_COEFF    RVAL
```

Python command syntax:

```
ierr =  q_losses_cost_coeff(rval)

ierr, rval = q_losses_cost_coeff()
```

Fortran command syntax:

```
CALL Q_LOSSES_COST_COEFF(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the reactive power loss cost coefficient (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.49. REACTIVE_RESV_COST_COEFF

Use this API to set or return the value of the coefficient for the reactive reserve objective. This API works in conjunction with the API option to MINIMIZE_REACTIVE_RESERVE.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_REACTIVE_RESV_COST_COEFF    RVAL
```

Python command syntax:

```
ierr =  reactive_resv_cost_coeff(rval)

ierr, rval = reactive_resv_cost_coeff()
```

Fortran command syntax:

```
CALL REACTIVE_RESV_COST_COEFF(RVAL, IOCODE, IERR)
```

Where:

Real RVAL	value of the reactive reserve cost coefficient (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in RVAL.
	IOCODE = 1	return the current value of the option setting in RVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.50. ROPF

Use this API to read an Optimal Power Flow Raw Data File.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ROPF   RFILE
```

Python command syntax:

```
ierr = ropf(rfile)
```

Fortran command syntax:

```
CALL ROPFNW(RFILE, IERR)
```

Where:

Character*260 RFILE	Optimal Power Flow Data File name, or '1' if is coming directly from the terminal or Response file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	error opening file
	IERR = 2	prerequisite requirements for API are not met

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3.1.51. RWOP

Use this API to replicate the OPF data contained in the working case in the form of an Optimal Power Flow Data File.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RWOP   SID  ALL  STATUS(1)..STATUS(4)  OUT  OFILE
```

Python command syntax:

```
ierr = rwop(sid, all, status, out, ofile)
```

Fortran command syntax:

```
CALL RWOPNW(SID, ALL, STATUS, OUT, OFILE, IERR)
```

Where:

Integer SID	a valid bus subsystem ID from 0 to 11. A bus subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified buses subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer STATUS (4)	array of 4 elements specifying Specifies whether to use various control options. A value of 0 means the option is not enabled. A value of 1 means the option is enabled.(input).	
	STATUS(1)	include isolated buses
	STATUS(2)	include out-of-service branches
	STATUS(3)	include subsystem data
	STATUS(4)	include subsystem tie lines
Integer OUT	option selection (input; no default allowed).	
	OUT = 0	direct output to a data file specified by OFILE.
	OUT = 1	direct output to the progress window.
Character*260 OFILE	output filename (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS(n) value
	IERR = 4	invalid OUT value
	IERR = 5	error opening OFILE
	IERR = 6	prerequisite requirements for API are not met

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3.1.52. SET_OPF_REPORT_SUBSYSTEM

Use this API to define the subsystem to be used when producing the opf output report. The records contained in the report will be limited to just the equipment residing within the defined subsystem. This includes bounded and infeasible reactive power generation, regulated buses, transformer tap ratios, transformer phase shift angles and bounded switched shunts.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SET_OPF_REPORT_SUBSYSTEM  OSID ALL
```

Python command syntax:

```
ierr = set_opf_report_subsystem(osid, all)
```

Fortran command syntax:

```
CALL SET_OPF_REPORT_SUBSYSTEM(OSID, ALL, IERR)
```

Where:

Integer OSID	Indicates the previously defined subsystem number to be used for the opf report (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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3.1.53. USE_DUAL_CRITERIA

Use this API to set or return the value of the OPF solution option to apply the power flow mismatch tolerance to the dual variable problem.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_USE_DUAL_CRITERIA    IVAL
```

Python command syntax:

```
ierr = use_dual_criteria(ival)

ierr, ival = use_dual_criteria()
```

Fortran command syntax:

```
CALL USE_DUAL_CRITERIA(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to apply the power flow mismatch tolerance to the dual variable problem (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	apply tolerance to the mismatch equation only.
	IVAL = 1	apply tolerance to the entire equation.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.54. USE_EMERGENCY_FLOW_LIMITS

Use this API to set or return the value of the OPF solution option to impose the emergency flow limits instead of the normal flow limits.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_USE_EMERGENCY_FLOW_LIMITS   IVAL
```

Python command syntax:

```
ierr =  use_emergency_flow_limits(ival)

ierr, ival = use_emergency_flow_limits()
```

Fortran command syntax:

```
CALL USE_EMERGENCY_FLOW_LIMITS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to use the emergency flow limits instead of the normal bus voltage limits (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not impose the emergency flow limits (normal flow limits will be imposed).
	IVAL = 1	impose emergency flow limits.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.55. USE_EMERGENCY_VOLT_LIMITS

Use this API to set or return the value of the OPF solution option to impose the emergency bus voltage limits instead of the normal bus voltage limits.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_USE_EMERGENCY_VOLT_LIMITS   IVAL
```

Python command syntax:

```
ierr =  use_emergency_volt_limits(ival)

ierr, ival = use_emergency_volt_limits()
```

Fortran command syntax:

```
CALL USE_EMERGENCY_VOLT_LIMITS(IVAL, IOCODE, IERR)
```

Where:

Integer IVAL	Indicates whether to use the emergency bus voltage limits instead of the normal bus voltage limits (input; if Batch command, or for Python if specified as an argument, or for Fortran if IOCODE is 0, then input of 0 by default; else output).	
	IVAL = 0	do not impose the emergency bus voltage limits (normal voltage limits will be imposed).
	IVAL = 1	impose emergency bus voltage limits.
Integer IOCODE	Indicates the operation mode of the API (no default allowed).	
	IOCODE = 0	set the option setting to value specified in IVAL.
	IOCODE = 1	return the current value of the option setting in IVAL.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IVAL value
	IERR = 2	invalid IOCODE value
	IERR = 3	prerequisite requirements for API are not met

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3.1.56. WRITE_OPF_OPTIONS_FILE

Use this API to save the PSSE OPF solution option settings to the PSSOPF.OPT file.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_WRITE_OPF_OPTIONS_FILE
```

Python command syntax:

```
ierr =  write_opf_options_file()
```

Fortran command syntax:

```
CALL WRITE_OPF_OPTIONS_FILE(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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3.2. Optimal Power Flow Data

3.2.1. NEWOPF

Use this API to initialize the working case with default OPF data for all bus oriented data records. It must be called prior to introducing any OPF data records. It has no effect if OPF data already exists in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_NEWOPF
```

Python command syntax:

```
ierr = newopf()
```

Fortran command syntax:

```
CALL NEWOPF_API(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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3.2.2. OPF_ADJBRX_INDV

Use this API to add an individual adjustable branch reactance data record to the working case. It may also be used to modify an existing adjustable branch reactance data record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_ADJBRX_INDV   EIBUS  EJBUS  CKTID  INTGAR(1)..INTGAR(3)
 REALAR(1)..REALAR(4)
```

Python command syntax:

```
ierr = opf_adjbrx_indv(eibus, ejbus, cktid, intgar, realar)
```

Fortran command syntax:

```
CALL OPF_ADJBRX_INDV(EIBUS, EJBUS, CKTID, INTGAR, REALAR, IERR)
```

Where:

Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	cost curve type. This value is not currently used by the program
	INTGAR(1) = 1	polynomial and exponential cost curve
	INTGAR(1) = 2	linear cost curve
	INTGAR(1) = 3	quadratic cost curve
	INTGAR(2)	adjustable branch reactance status (default of 1 if new record, current value otherwise)
	INTGAR(2) = 0	out of service
	INTGAR(2) = 1	in service
	INTGAR(3)	cost table cross reference number. This value is not currently used by the program
Real REALAR (4)	array of 4 elements specifying (input).	
	REALAR(1)	reactance multiplier in pu (default value of 1.0 if new record, current value otherwise)
	REALAR(2)	maximum reactance multiplier in pu (default value of 1.0 if new record, current value otherwise)
	REALAR(3)	minimum reactance multiplier in pu (default value of 1.0 if new record, current value otherwise)
	REALAR(4)	cost scale coefficient in $/pu (default value of 1.0 if new record, current value otherwise)
Integer IERR	error code (output).	

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IERR = -1	non-fatal error occurred. Warning messages are as follows: - Invalid cost curve type specified - Invalid branch reactance status specified. - Maximum branch reactance multiplier less than minimum multiplier - Branch reactance multiplier cannot be less than 0.1. - Branch reactance multiplier cannot be set to zero
IERR = 0	no error occurred
IERR = 1	invalid bus number
IERR = 2	specified branch does not exist
IERR = 3	branch reactance table full
IERR = 4	specified branch cannot be a transformer
IERR = 5	prerequisite requirements for API are not met

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3.2.3. OPF_ADJBRX_SUBSYS

Use this API to add or modify all OPF branch reactance data records in a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_ADJBRX_SUBSYS   SID  ALL  INTGAR(1)..INTGAR(3)
 REALAR(1)..REALAR(4)  USEVAL(1)..USEVAL(4)
```

Python command syntax:

```
ierr = opf_adjbrx_subsys(sid, all, intgar, realar, useval)
```

Fortran command syntax:

```
CALL OPF_ADJBRX_SUBSYS(SID, ALL, INTGAR, REALAR, USEVAL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	cost curve type. This value is not currently used by the program
	INTGAR(1) = 1	polynomial and exponential cost curve
	INTGAR(1) = 2	linear cost curve
	INTGAR(1) = 3	quadratic cost curve
	INTGAR(2)	adjustable branch reactance status (default value of 1 for new records, current values otherwise)
	INTGAR(2) = 0	out of service
	INTGAR(2) = 1	in service
	INTGAR(3)	cost table cross reference number. This value is not currently used by the program
Real REALAR (4)	array of 4 elements specifying (input).	
	REALAR(1)	reactance multiplier in pu (default value of 1.0 for new records, current value otherwise)
	REALAR(2)	maximum reactance multiplier in pu (default value of 1.0 for new records, current value otherwise)
	REALAR(3)	minimum reactance multiplier in pu (default value of 1.0 for new records, current value otherwise)

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	REALAR(4)	cost scale coefficient in $/pu (default value of 1.0 for new records, current value otherwise)
Integer USEVAL (4)	array of 4 elements specifying (default treatment same as value of 0) (input).	
	USEVAL(1)	indicates how the value of REALAR(1) should be used
	USEVAL(1) = 0	if the value of REALAR(1) is to be used as defined
	USEVAL(1) = 1	if the value of REALAR(1) is to be multiplied by the present value of the branch reactance multiplier
	USEVAL(1) = 2	if the value of REALAR(1) is to be added to the present value of the branch reactance multiplier
	USEVAL(2)	indicates how the value of REALAR(2) should be used
	USEVAL(2) = 0	if the value of REALAR(2) is to be used as defined
	USEVAL(2) = 1	if the value of REALAR(2) is to be multiplied by the present value of the maximum branch reactance multiplier
	USEVAL(2) = 2	if the value of REALAR(2) is to be added to the present value of the maximum branch reactance multiplier
	USEVAL(3)	indicates how the value of REALAR(3) should be used
	USEVAL(3) = 0	if the value of REALAR(3) is to be used as defined
	USEVAL(3) = 1	if the value of REALAR(3) is to be multiplied by the present value of the minimum branch reactance multiplier
	USEVAL(3) = 2	if the value of REALAR(3) is to be added to the present value of the minimum branch reactance multiplier
	USEVAL(4)	indicates how the value of REALAR(4) should be used
	USEVAL(4) = 0	if the value of REALAR(4) is to be used as defined
	USEVAL(4) = 1	if the value of REALAR(4) is to be multiplied by the present value of the cost scale coefficient
	USEVAL(4) = 2	if the value of REALAR(4) is to be added to the present value of the cost scale coefficient
Integer IERR	error code (output).	
	IERR = 0	no error occurred

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IERR = 1	invalid SID value or subsystem SID is not defined
IERR = 2	invalid ALL value
IERR = 3	adjustable branch reactance table is full
IERR = 4	prerequisite requirements for API are not met
IERR = -1	non-fatal error occurred. Warning messages are as follows: - Invalid cost curve type specified - Invalid branch reactance status specified. - Maximum branch reactance multiplier less than minimum multiplier - Branch reactance multiplier cannot be less than 0.1. - Branch reactance multiplier cannot be set to zero

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3.2.4. OPF_ADJLOAD_TBL

Use this API to add an individual adjustable bus load table record to the working case. It may also be used to modify an existing adjustable bus load table record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_ADJLOAD_TBL   TBL  INTGAR(1)..INTGAR(3)   REALAR(1)..REALAR(7)
```

Python command syntax:

```
ierr = opf_adjload_tbl(tbl, intgar, realar)
```

Fortran command syntax:

```
CALL OPF_ADJLOAD_TBL(TBL, INTGAR, REALAR, IERR)
```

Where:

Integer TBL	adjustable bus load table number (input; no default allowed).	
Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	cost curve type. This value is not currently used by the program
	INTGAR(1) = 1	polynomial and exponential cost curve
	INTGAR(1) = 2	linear cost curve
	INTGAR(1) = 3	quadratic cost curve
	INTGAR(2)	adjustable bus load status (default of 1 if new record, current value otherwise)
	INTGAR(2) = 0	out of service
	INTGAR(2) = 1	in service
	INTGAR(3)	cost table cross reference number. This value is not currently used by the program
Real REALAR (7)	array of 7 elements specifying (input).	
	REALAR(1)	load multiplier (default value of 1.0 if new record, current value otherwise)
	REALAR(2)	maximum load multiplier (default value of 1.0 if new record, current value otherwise)
	REALAR(3)	minimum load multiplier (default value of 1.0 if new record, current value otherwise)
	REALAR(4)	load ratio multiplier. This value is not currently used by the program
	REALAR(5)	maximum load ratio multiplier. This value is not currently used by the program
	REALAR(6)	minimum load ratio multiplier. This value is not currently used by the program
	REALAR(7)	load cost scale coefficient (default value of 1.0 if new record, current value otherwise)

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Integer IERR	error code (output).	
	IERR = -1	non-fatal error occurred. Warning messages are as follows: - invalid cost curve type specified - invalid load table status specified - Maximum load multiplier less than 0.0 - Minimum load multiplier less than 0.0 - maximum load multiplier less than minimum load multiplier - maximum load ratio multiplier less than minimum load ratio multiplier
	IERR = 0	no error occurred
	IERR = 1	adjustable bus load table is full
	IERR = 2	prerequisite requirements for API are not met

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3.2.5. OPF_ADJVAR_INDV

Use this API to add an individual OPF adjustable bus shunt data record to the working case. It may also be used to modify an existing adjustable bus shunt data record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_ADJVAR_INDV   EIBUS  SHNTID  INTGAR(1)..INTGAR(3)
 REALAR(1)..REALAR(4)
```

Python command syntax:

```
ierr = opf_adjvar_indv(eibus, shntid, intgar, realar)
```

Fortran command syntax:

```
CALL OPF_ADJVAR_INDV(EIBUS, SHNTID, INTGAR, REALAR, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 SHNTID	bus shunt identifier (input; no default allowed).	
Integer INTGAR (3)	array of 3 elements specifying The value of each is as follows(input).	
	INTGAR(1)	cost curve type. This value is not currently used by the program
	INTGAR(1) = 1	polynomial and exponential cost curve
	INTGAR(1) = 2	linear cost curve
	INTGAR(1) = 3	quadratic cost curve
	INTGAR(2)	status (default value of 1 if new record, current value otherwise)
	INTGAR(2) = 0	out of service
	INTGAR(2) = 1	in service
	INTGAR(3)	cost table cross reference number. This value is not currently used by the program
Real REALAR (4)	array of 4 elements specifying (input).	
	REALAR(1)	initial bus shunt susceptance in Mvar at nominal voltage (default value of 0.0 if new record, current value otherwise)
	REALAR(2)	maximum bus shunt susceptance in Mvar (default value of 0.0 if new record, current value otherwise)
	REALAR(3)	minimum bus shunt susceptance in Mvar (default value of 0.0 if new record, current value otherwise)
	REALAR(4)	cost scale coefficient in $/Mvar (default value of 1.0 if new record, current value otherwise)
Integer IERR	error code (output).	

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IERR = 0	no error occurred
IERR = 1	adjustable bus shunt identifier is more than two characters
IERR = 2	invalid bus number specified
IERR = 3	blank adjustable bus shunt identifier is invalid
IERR = 4	adjustable bus shunt table is full
IERR = 5	prerequisite requirements for API are not met
IERR = -1	non-fatal error occurred. Possible warning messages are as follows: - Invalid cost curve type specified. - Invalid adjustable bus shunt status specified. - Maximum shunt susceptance less than minimum shunt susceptance

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3.2.6. OPF_ADJVAR_SUBSYS

Use this API to add or modify all OPF adjustable bus shunt records in a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_ADJVAR_SUBSYS   SID  ALL  AUTOADD  SHNTID   INTGAR(1)..INTGAR(4)
 REALAR(1)..REALAR(4)   USEVAL(1)..USEVAL(4)
```

Python command syntax:

```
ierr = opf_adjvar_subsys(sid, all, autoadd, shntid, intgar, realar,
 useval)
```

Fortran command syntax:

```
CALL OPF_ADJVAR_SUBSYS(SID, ALL, AUTOADD, SHNTID, INTGAR, REALAR, USEVAL,
 IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer AUTOADD	Indicates whether new records can be added (default treatment same as value of 3) (input).	
	AUTOADD = 0	do not add a new var record if it does not already exist.
	AUTOADD = 1	add a new var record if it does not already exist.
	AUTOADD = 2	add a separate shunt record at the bus.
	AUTOADD = 3	only modify existing bus shunt records that correspond to the shunt identifier given in SHNTID.
Character*2 SHNTID	adjustable bus shunt identifier (input; '1').	
Integer INTGAR (4)	array of 4 elements specifying (input).	
	INTGAR(1)	cost curve type. This value is not currently used by the program
	INTGAR(1) = 1	polynomial and exponential cost curve
	INTGAR(1) = 2	piece-wise linear cost curve
	INTGAR(1) = 3	piece-wise quadratic cost curve
	INTGAR(2)	status (default value of 1 for new records, current value otherwise)
	INTGAR(2) = 0	out of service
	INTGAR(2) = 1	in service

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	INTGAR(3)	cost table cross reference number. This value is not currently used by the program
	INTGAR(4)	indicates which bus types should be processed (default treatment same as value of 0)
	INTGAR(4) = 0	process all buses in subsystem
	INTGAR(4) = 1	process only Type 1 buses in subsystem
	INTGAR(4) = 2	process only Type 2 and 3 buses in subsystem
Real REALAR (4)	array of 4 elements specifying (input).	
	REALAR(1)	bus shunt susceptance in Mvar at nominal voltage (default value of 0.0 for new records, current values used for existing ones)
	REALAR(2)	maximum bus shunt susceptance in Mvar (default value of 0.0 for new records, current value otherwise)
	REALAR(3)	minimum bus shunt susceptance in Mvar (default value of 0.0 for new records, current value otherwise)
	REALAR(4)	cost scale coefficient in $/Mvar (default value of 1.0 for new records, current value otherwise)
Integer USEVAL (4)	array of 4 elements specifying Used to indicate actions to be taken on the REALAR values. If USEVAL is 1 then value REALAR value is multiplied by the current value. If USEVAL is 2 then REALAR value is added to current real value. If USEVAL is 0 (default) then REALAR is used as provided.(input).	
	USEVAL(1)	indicates how the value of REALAR(1) should be adjusted
	USEVAL(1) = 0	if the value of REALAR(1) is to be used as defined
	USEVAL(1) = 1	if the value of REALAR(1) is to be multiplied by the present value of the bus shunt susceptance
	USEVAL(1) = 2	if the value of REALAR(1) is to be added to the present value of the bus shunt susceptance
	USEVAL(2)	indicates how the value of REALAR(2) should be used
	USEVAL(2) = 0	if the value of REALAR(2) is to be used as defined
	USEVAL(2) = 1	if the value of REALAR(2) is to be multiplied by the present value of the maximum bus shunt susceptance
	USEVAL(2) = 2	if the value of REALAR(2) is to be added to the present value of the maximum bus shunt susceptance

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	USEVAL(3)	indicates how the value of REALAR(3) should be used
	USEVAL(3) = 0	if the value of REALAR(3) is to be used as defined
	USEVAL(3) = 1	if the value of REALAR(3) is to be multiplied by the present value of the minimum bus shunt susceptance
	USEVAL(3) = 2	if the value of REALAR(3) is to be added to the present value of the minimum bus shunt susceptance
	USEVAL(4)	indicates how the value of REALAR(4) should be used
	USEVAL(4) = 0	if the value of REALAR(4) is to be used as defined
	USEVAL(4) = 1	if the value of REALAR(4) is to be multiplied by the present value of the cost scale coefficient
	USEVAL(4) = 2	if the value of REALAR(4) is to be added to the present value of the cost scale coefficient
Integer IERR	error code (output).	
	IERR = -1	non-fatal error occurred. Possible warning messages are as follows: - Invalid cost curve type specified. - Invalid adjustable bus shunt status specified. - Maximum shunt susceptance less than minimum shunt susceptance
	IERR = 0	no error occurred
	IERR = 1	adjustable bus shunt identifier is more than two characters
	IERR = 2	invalid SID value or subsystem SID is not defined
	IERR = 3	invalid ALL value
	IERR = 4	adjustable bus shunt table is full
	IERR = 5	prerequisite requirements for API are not met

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3.2.7. OPF_APDSP_TBL

Use this API to add an individual active power dispatch table record to the working case. It may also be used to modify an existing active power dispatch table record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_APDSP_TBL   TBL  INTGAR(1)..INTGAR(3)   REALAR(1)..REALAR(3)
```

Python command syntax:

```
ierr = opf_apdsp_tbl(tbl, intgar, realar)
```

Fortran command syntax:

```
CALL OPF_APDSP_TBL(TBL, INTGAR, REALAR, IERR)
```

Where:

Integer TBL	active power dispatch table number (input; no default allowed).	
Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	cost curve type (default value of 1 if new record, current value otherwise)
	INTGAR(1) = 1	polynomial and exponential cost curve
	INTGAR(1) = 2	linear cost curve
	INTGAR(1) = 3	quadratic cost curve
	INTGAR(2)	active power dispatch table status (default value of 1 if new record, current value otherwise)
	INTGAR(2) = 0	out of service
	INTGAR(2) = 1	in service
	INTGAR(3)	cost table cross reference number (default value of 0, current value otherwise)
Real REALAR (3)	array of 3 elements specifying (input).	
	REALAR(1)	maximum active power generation, in MW (default value of 9999.0, current value otherwise)
	REALAR(2)	minimum active power generation, in MW (default value of -9999.0, current value otherwise)
	REALAR(3)	fuel cost scale coefficient (default value of 1.0, current value otherwise)
Integer IERR	error code (output).	
	IERR = -1	non-fatal error occurred. Warning messages are as follows: - invalid cost curve type specified - invalid active power dispatch table status specified - maximum active power gener-

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	ation less than minimum active power generation
IERR = 0	no error occurred
IERR = 1	active power dispatch table is full
IERR = 2	prerequisite requirements for API are not met

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3.2.8. OPF_BRFLW_3WT_INDV

Use this API to add a flow constraint record for an individual winding of a three-winding transformer in the working case. It may also be used to modify an existing three-winding transformer flow constraint record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_BRFLW_3WT_INDV   EIBUS  EJBUS  EKBUS  CKTID  BFLWID  INTGAR(1)
 INTGAR(2)  REALAR(1)..REALAR(5)
```

Python command syntax:

```
ierr = opf_brflw_3wt_indv(eibus, ejbus, ekbus, cktid, bflwid, intgar,
 realar)
```

Fortran command syntax:

```
CALL OPF_BRFLW_3WT_INDV(EIBUS, EJBUS, EKBUS, CKTID, BFLWID, INTGAR,
 REALAR, IERR)
```

Where:

Integer EIBUS	from bus number The value of EIBUS indicates the from bus of the winding for which the flow constraint is being introduced (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Integer EKBUS	third bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Character*1 BFLWID	branch flow constraint identifier (input; no default allowed).	
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	branch flow type (default value of 4 if new record, current value otherwise)
	INTGAR(1) = 1	MW
	INTGAR(1) = 2	Mvar
	INTGAR(1) = 3	MVA
	INTGAR(1) = 4	Ampere
	INTGAR(2)	limit type imposed on the branch flow limits (default value of 1 if new record, current value otherwise)
	INTGAR(2) = 0	reporting only
	INTGAR(2) = 1	hard limit
	INTGAR(2) = 2	linear soft limit
	INTGAR(2) = 3	quadratic soft limit
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	maximum normal flow limit in physical units (default value of 0.0 for new record, current value otherwise)

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	REALAR(2)	minimum normal flow limit in physical units (default value of 0.0 for new record, current value otherwise)
	REALAR(3)	maximum emergency flow limit in physical units (default value of 0.0 for new record, current value otherwise)
	REALAR(4)	minimum emergency flow limit in physical units (default value of 0.0 for new record, current value otherwise)
	REALAR(5)	soft limit penalty weight (default value of 1.0 for new record, current value otherwise)
Integer IERR	error code (output).	
	IERR = -1	non-fatal error occurred. Possible warning messages are: - Invalid branch flow type specified. - Invalid branch flow limit type specified. - Maximum normal flow limit is less than the minimum normal flow limit. - Maximum emergency flow limit is less than the minimum emergency flow limit
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified three-winding transformer does not exist
	IERR = 3	specified winding of three-winding transformer not found
	IERR = 4	branch flow constraint table is full
	IERR = 5	prerequisite requirements for API are not met

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3.2.9. OPF_BRFLW_BRN_INDV

Use this API to add an individual branch or two-winding transformer flow constraint record to the working case. It may also be used to modify an existing branch or two-winding transformer flow constraint.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_BRFLW_BRN_INDV   EIBUS  EJBUS  CKTID  BFLWID  INTGAR(1)  INTGAR(2)
  REALAR(1)..REALAR(5)
```

Python command syntax:

```
ierr = opf_brflw_brn_indv(eibus, ejbus, cktid, bflwid, intgar, realar)
```

Fortran command syntax:

```
CALL OPF_BRFLW_BRN_INDV(EIBUS, EJBUS, CKTID, BFLWID, INTGAR, REALAR, IERR)
```

Where:

Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Character*1 BFLWID	branch flow constraint identifier (input; no default allowed).	
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	branch flow type (default value of 4 if new record, current value otherwise)
	INTGAR(1) = 1	MW
	INTGAR(1) = 2	Mvar
	INTGAR(1) = 3	MVA
	INTGAR(1) = 4	Ampere
	INTGAR(2)	limit type imposed on the branch flow limits (default value of 1 if new record, current value otherwise)
	INTGAR(2) = 0	reporting only
	INTGAR(2) = 1	hard limit
	INTGAR(2) = 2	linear soft limit
	INTGAR(2) = 3	quadratic soft limit
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	maximum normal flow limit in physical units (default value of 0.0 if new record, current value otherwise)
	REALAR(2)	minimum normal flow limit in physical units (default value of 0.0 if new record, current value otherwise)
	REALAR(3)	maximum emergency flow limit in physical units (default value of 0.0 if new record, current value otherwise)

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	REALAR(4)	minimum emergency flow limit in physical units (default value of 0.0 if new record, current value otherwise)
	REALAR(5)	soft limit penalty weight (default value of 1.0 if new record, current value otherwise)
Integer IERR	error code (output).	
	IERR = -1	non-fatal error occurred. Possible warning messages are: - Invalid branch flow type specified. - Invalid branch flow limit type specified. - Maximum normal flow limit is less than the minimum normal flow limit. - Maximum emergency flow limit is less than the minimum emergency flow limit
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified branch does not exist
	IERR = 3	branch flow constraint table is full
	IERR = 4	prerequisite requirements for API are not met

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3.2.10. OPF_BRFLW_SUBSYS

Use this API to add or modify all branch flow data records in a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_BRFLW_SUBSYS   SID  ALL  AUTOADD  BFLWID  INTGAR(1)..INTGAR(3)
 REALAR(1)..REALAR(5)  USEVAL(1)..USEVAL(5)
```

Python command syntax:

```
ierr = opf_brflw_subsys(sid, all, autoadd, bflwid, intgar, realar, useval)
```

Fortran command syntax:

```
CALL OPF_BRFLW_SUBSYS(SID, ALL, AUTOADD, BFLWID, INTGAR, REALAR, USEVAL,
 IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer AUTOADD	Indicates whether new records can be added, or are to be modified (input; no default allowed).	
	AUTOADD = 0	do not add a new branch flow record if it does not already exist.
	AUTOADD = 1	add a new branch flow record if it does not already exist.
	AUTOADD = 2	only modify existing branch flow records that correspond to the flow identifier given in BFLWID.
Character*1 BFLWID	branch flow identifier (input; no default allowed).	
Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	branch flow type (default value of 4 for new records, current value otherwise)
	INTGAR(1) = 1	MW
	INTGAR(1) = 2	Mvar
	INTGAR(1) = 3	MVA
	INTGAR(1) = 4	Ampere
	INTGAR(2)	limit type imposed on the branch flow limits (default value of 1 for new records, current value otherwise)
	INTGAR(2) = 0	reporting only
	INTGAR(2) = 1	hard limit
	INTGAR(2) = 2	linear soft limit

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	INTGAR(2) = 3	quadratic soft limit
	INTGAR(3)	default rate setting (default value of 0)
	INTGAR(3) = 0	use the limits that have been defined
	INTGAR(3) = 1	base flow limits on RATEA
	INTGAR(3) = 2	base flow limits on RATEB
	INTGAR(3) = 3	base flow limits on RATEC
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	maximum normal flow limit in physical units (default value of 0.0 for new records, current value otherwise)
	REALAR(2)	minimum normal flow limit in physical units (default value of 0.0 for new records, current value otherwise)
	REALAR(3)	maximum emergency flow limit in physical units (default value of 0.0 for new records, current value otherwise)
	REALAR(4)	minimum emergency flow limit in physical units (default value of 0.0 for new records, current value otherwise)
	REALAR(5)	soft limit penalty weight (default value of 1.0 for new records, current value otherwise)
Integer USEVAL (5)	array of 5 elements specifying Specifies whether the corresponding REALAR value should be applied as provided, or multiplied by the present value or added to the present value.(input).	
	USEVAL(1)	indicates how the value of REALAR(1) should be used
	USEVAL(1) = 0	if the value of REALAR(1) is to be used as defined
	USEVAL(1) = 1	if the value of REALAR(1) is to be multiplied by the present value of the maximum normal flow limit
	USEVAL(1) = 2	if the value of REALAR(1) is to be added to the present value of the maximum normal flow limit
	USEVAL(2)	indicates how the value of REALAR(2) should be used
	USEVAL(2) = 0	if the value of REALAR(2) is to be used as defined
	USEVAL(2) = 1	if the value of REALAR(2) is to be multiplied by the present value of the minimum normal flow limit
	USEVAL(2) = 2	if the value of REALAR(2) is to be added to the present value of the minimum normal flow limit
	USEVAL(3)	indicates how the value of REALAR(3) should be used

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	USEVAL(3) = 0	if the value of REALAR(3) is to be used as defined
	USEVAL(3) = 1	if the value of REALAR(3) is to be multiplied by the present value of the maximum emergency flow limit
	USEVAL(3) = 2	if the value of REALAR(3) is to be added to the present value of the maximum emergency flow limit
	USEVAL(4)	indicates how the value of REALAR(4) should be used
	USEVAL(4) = 0	if the value of REALAR(4) is to be used as defined
	USEVAL(4) = 1	if the value of REALAR(4) is to be multiplied by the present value of the minimum emergency flow limit
	USEVAL(4) = 2	if the value of REALAR(4) is to be added to the present value of the minimum emergency flow limit
	USEVAL(5)	indicates how the value of REALAR(5) should be used
	USEVAL(5) = 0	if the value of REALAR(5) is to be used as defined
	USEVAL(5) = 1	if the value of REALAR(5) is to be multiplied by the present value of the soft limit penalty weight
	USEVAL(5) = 2	if the value of REALAR(5) is to be added to the present value of the soft limit penalty weight
Integer IERR	error code (output).	
	IERR = -1	non-fatal error occurred. Possible warning messages are: - Invalid flow type specified. - Invalid branch flow limit type specified. - Maximum normal flow limit is less than the minimum normal flow limit. - Maximum emergency flow limit is less than the minimum emergency flow limit
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	branch flow constraint table is full
	IERR = 4	prerequisite requirements for API are not met

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3.2.11. OPF_BUS_INDV

Use this API to add OPF bus voltage magnitude data to the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_BUS_INDV   EIBUS  INTGAR(1)  INTGAR(2)  REALAR(1)..REALAR(5)
```

Python command syntax:

```
ierr = opf_bus_indv(eibus, intgar, realar)
```

Fortran command syntax:

```
CALL OPF_BUS_INDV(EIBUS, INTGAR, REALAR, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	limit type imposed on the bus voltage limits (default is the current value; initialized to 1)
	INTGAR(1) = 0	reporting only
	INTGAR(1) = 1	hard limit
	INTGAR(1) = 2	linear soft limit
	INTGAR(1) = 3	quadratic soft limit
	INTGAR(2)	indicates what to do with the defined voltage limits (default treatment same as value of 0)
	INTGAR(2) = 0	leave voltage limits as defined by REALAR(1) and REALAR(2)
	INTGAR(2) = 1	fix normal voltage limit at present voltage values, ignoring any values defined by REALAR(1) and REALAR(2)
	INTGAR(2) = 2	allow normal voltage limits to open up if they exceed the bounds defined by REALAR(1) or REALAR(2)
	INTGAR(2) = 3	set normal and emergency OPF voltage limits to the normal and emergency voltage limits as defined in the network bus data
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	normal maximum voltage limit in pu (default is the current value; initialized to 9999.0)
	REALAR(2)	normal minimum voltage limit in pu (default is the current value; initialized to -9999.0)
	REALAR(3)	emergency maximum voltage limit in pu (default is the current value; initialized to 9999.0)

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	REALAR(4)	emergency minimum voltage limit in pu (default is the current value; initialized to -9999.0)
	REALAR(5)	soft-limit penalty weight (default is the current value; initialized to 1.0)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	prerequisite requirements for API are not met
	IERR = -1	non-fatal error occurred. Warning messages are as follows: - Invalid limit type specified. - Maximum normal voltage limit less than minimum limit. - Maximum emergency voltage limit less than minimum limit

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3.2.12. OPF_BUS_SUBSYS

Use this API to add OPF bus attribute data for all records in a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_BUS_SUBSYS   SID  ALL  INTGAR(1)..INTGAR(3)  REALAR(1)..REALAR(5)
 USEVAL(1)..USEVAL(5)
```

Python command syntax:

```
ierr = opf_bus_subsys(sid, all, intgar, realar, useval)
```

Fortran command syntax:

```
CALL OPF_BUS_SUBSYS(SID, ALL, INTGAR, REALAR, USEVAL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses, use existing data values.
	ALL = 2	process all buses, use default data values.
Integer INTGAR (3)	array of 3 elements specifying (input).	
	INTGAR(1)	limit type imposed on the bus voltage limits (default is the current value; initialized to 1)
	INTGAR(1) = 0	reporting only
	INTGAR(1) = 1	hard limit
	INTGAR(1) = 2	linear soft limit
	INTGAR(1) = 3	quadratic soft limit
	INTGAR(2)	indicates what to do with the defined voltage limits (default treatment same as value of 0)
	INTGAR(2) = 0	leave voltage limits as defined by REALAR(1) and REALAR(2)
	INTGAR(2) = 1	fix normal voltage limit at present voltage values, ignoring any values defined by REALAR(1) and REALAR(2)
	INTGAR(2) = 2	allow normal voltage limits to open up if they exceed the bounds defined by REALAR(1) or REALAR(2)
	INTGAR(2) = 3	set normal and emergency OPF voltage limits to the normal and emergency voltage limits as defined in the network bus data
	INTGAR(3)	indicates which bus types should be processed (default treatment same as value of 0)
	INTGAR(3) = 0	all buses in subsystem

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	INTGAR(3) = 1	only Type 1 buses in subsystem
	INTGAR(3) = 2	only Type 2 and 3 buses in subsystem
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	normal maximum voltage limit in pu (default is the current value; initialized to 9999.0)
	REALAR(2)	normal minimum voltage limit in pu (default is the current value; initialized to -9999.0)
	REALAR(3)	emergency maximum voltage limit in pu (default is the current value; initialized to 9999.0)
	REALAR(4)	emergency minimum voltage limit in pu (default is the current value; initialized to -9999.0)
	REALAR(5)	soft-limit penalty weight (default is the current value; initialized to 1.0)
Integer USEVAL (5)	array of 5 elements specifying (default treatment same as value of 0) (input).	
	USEVAL(1)	indicates how the value of REALAR(1) should be used
	USEVAL(1) = 0	if the value of REALAR(1) is to be used as defined
	USEVAL(1) = 1	if the value of REALAR(1) is to be multiplied by the present value of the normal bus voltage minimum limit
	USEVAL(1) = 2	if the value of REALAR(1) is to be added to the present value of the normal bus voltage minimum limit
	USEVAL(2)	indicates how the value of REALAR(2) should be used
	USEVAL(2) = 0	if the value of REALAR(2) is to be used as defined
	USEVAL(2) = 1	if the value of REALAR(2) is to be multiplied by the present value of the normal bus voltage maximum limit
	USEVAL(2) = 2	if the value of REALAR(2) is to be added to the present value of the normal bus voltage maximum limit
	USEVAL(3)	indicates how the value of REALAR(3) should be used
	USEVAL(3) = 0	if the value of REALAR(3) is to be used as defined
	USEVAL(3) = 1	if the value of REALAR(3) is to be multiplied by the present value of the emergency bus voltage minimum limit

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	USEVAL(3) = 2	if the value of REALAR(3) is to be added to the present value of the emergency bus voltage minimum limit
	USEVAL(4)	indicates how the value of REALAR(4) should be used
	USEVAL(4) = 0	if the value of REALAR(4) is to be used as defined
	USEVAL(4) = 1	if the value of REALAR(4) is to be multiplied by the present value of the emergency bus voltage maximum limit
	USEVAL(4) = 2	if the value of REALAR(4) is to be added to the present value of the emergency bus voltage maximum limit
	USEVAL(5)	indicates how the value of PENWGT should be used
	USEVAL(5) = 0	if the value of PENWGT is to be used as defined
	USEVAL(5) = 1	if the value of PENWGT is to be multiplied by the present value of the soft limit penalty weight
	USEVAL(5) = 2	if the value of PENWGT is to be added to the present value of the soft limit penalty weight
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	non-fatal error occurred. Warning messages are as follows: - Invalid limit type specified. - Maximum normal voltage limit less than minimum limit. - Maximum emergency voltage limit less than minimum limit

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3.2.13. OPF_CSTTBL_LIN

Use this API to add a linear cost curve table record to the working case. It may also be used to modify an existing linear cost curve table record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_CSTTBL_LIN   TBL   LABL   NPRS
```

Python command syntax:

```
ierr = opf_csttbl_lin(tbl, labl, nprs, xy)
```

Fortran command syntax:

```
CALL OPF_CSTTBL_LIN(TBL, LABL, NPRS, XY, IERR)
```

Where:

Integer TBL	linear cost curve table number (input; no default allowed).	
Character*12 LABL	cost table description of, at most, 12 characters (input; ' ').	
Integer NPRS	number of (Xi, Yi) coordinate pairs being specified (input; no default allowed).	
Real XY (2*NPRS)	Coordinate pairs from 1 to NPRS. Xi is the control variable value. Yi is the total cost or energy consumption value. (input; Xi and Yi have defaults of 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	cost table data table is full
	IERR = 2	exceeded maximum number of allowable cost curve data pairs
	IERR = 3	number of data pairs must not be less than zero
	IERR = 4	incremental costs are not monotonically increasing
	IERR = 5	MW values are not monotonically increasing
	IERR = 6	prerequisite requirements for API are not met

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3.2.14. OPF_CSTTBL_POLY

Use this API to add a polynomial and exponential cost curve table record to the working case. It may also be used to modify an existing polynomial and exponential cost curve table record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_CSTTBL_POLY   TBL   LABL   COST   LINCST   QUACST   EXPCST   EXPON
```

Python command syntax:

```
ierr = opf_csttbl_poly(tbl, labl, cost, lincst, quacst, expcst, expon)
```

Fortran command syntax:

```
CALL OPF_CSTTBL_POLY(TBL, LABL, COST, LINCST, QUACST, EXPCST, EXPON, IERR)
```

Where:

Integer TBL	polynomial and exponential cost curve table number (input; no default allowed).	
Character*12 LABL	cost table description of, at most, 12 characters (input; ' ').	
Real COST	cost integration constant (input; 0.0).	
Real LINCST	linear cost coefficient value (input; 1.0).	
Real QUACST	quadratic cost coefficient value (input; 1.0).	
Real EXPCST	exponential cost coefficient value (input; 0.0).	
Real EXPON	exponent of the exponential cost coefficient (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	cost table data table full
	IERR = 2	exceeded maximum number of allowable cost curve data values
	IERR = 3	prerequisite requirements for API are not met

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3.2.15. OPF_CSTTBL_QUAD

Use this API to add a quadratic cost curve table record to the working case. It may also be used to modify an existing quadratic cost curve table record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_CSTTBL_QUAD   TBL   LABL   COST   NPRS
```

Python command syntax:

```
ierr = opf_csttbl_quad(tbl, labl, cost, nprs, xy)
```

Fortran command syntax:

```
CALL OPF_CSTTBL_QUAD(TBL, LABL, COST, NPRS, XY, IERR)
```

Where:

Integer TBL	quadratic cost curve table number (input; no default allowed).	
Character*12 LABL	cost table description of, at most, 12 characters (input; ' ').	
Real COST	cost integration constant (input; 0.0).	
Integer NPRS	number of (Xi, Yi) coordinate pairs to be specified (no default) (input).	
Real XY (2*NPRS)	Coordinate pairs from 1 to NPRS. Xi is the control variable value. Yi is the total cost or energy consumption value. (input; Xi and Yi have defaults of 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	cost table data table is full
	IERR = 2	exceeded maximum number of allowable cost curve data pairs
	IERR = 3	number of data pairs must not be negative
	IERR = 4	incremental costs are not monotonically increasing
	IERR = 5	MW values are not monotonically increasing
	IERR = 6	prerequisite requirements for API are not met

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3.2.16. OPF_GENDSP_INDV

Use this API to add generator dispatch data to an individual record in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_GENDSP_INDV   EIBUS  GENID  APDTBL  DSPTCH
```

Python command syntax:

```
ierr = opf_gendsp_indv(eibus, genid, apdtbl, dsptch)
```

Fortran command syntax:

```
CALL OPF_GENDSP_INDV(EIBUS, GENID, APDTBL, DSPTCH, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Integer APDTBL	active power dispatch table cross reference number (input; default of 0 if new record, current value otherwise).	
Real DSPTCH	dispatch fraction (default value of 1.0 if new record, current value otherwise) (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified machine does not exist
	IERR = 3	prerequisite requirements for API are not met

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3.2.17. OPF_GENDSP_SUBSYS

Use this API to add OPF generator dispatch data to all records in a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_GENDSP_SUBSYS   SID  ALL  APDTBL  DSPTCH  USEVAL
```

Python command syntax:

```
ierr = opf_gendsp_subsys(sid, all, apdtbl, dsptch, useval)
```

Fortran command syntax:

```
CALL OPF_GENDSP_SUBSYS(SID, ALL, APDTBL, DSPTCH, USEVAL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses, use existing data values.
	ALL = 2	process all buses, use default data values.
Integer APDTBL	active power dispatch table cross reference number (input; default of 0 for new record, current value otherwise).	
Real DSPTCH	dispatch fraction (input; default of 1.0 for new record, current value otherwise).	
Integer USEVAL	Indicates how the value of DSPTCH should be applied (default treatment same as value of 0) (input).	
	useval = 0	if the value of DSPTCH is to be used as defined.
	useval = 1	if the value of DSPTCH is to be multiplied by the present value of the dispatch fraction.
	useval = 2	if the value of DSPTCH is to be added to the present value of the dispatch fraction.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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3.2.18. OPF_GENRSV_INDV

Use this API to add an individual generator reserve data record to the working case. It may also be used to modify an existing generator reserve data record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_GENRSV_INDV   EIBUS  GENID  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = opf_genrsv_indv(eibus, genid, realar)
```

Fortran command syntax:

```
CALL OPF_GENRSV_INDV(EIBUS, GENID, REALAR, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	unit ramp rate in MW/minute (default value of 9999.0 if new record, current value otherwise)
	REALAR(2)	unit capability in MW (default value of 9999.0 if new record, current value otherwise)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified machine does not exist
	IERR = 3	generator reserve unit table full
	IERR = 4	prerequisite requirements for API are not met

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3.2.19. OPF_GENRSV_SUBSYS

Use this API to add or modify all OPF generator reserve records in a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_GENRSV_SUBSYS   SID  ALL  REALAR(1)  REALAR(2)  USEVAL(1)
 USEVAL(2)
```

Python command syntax:

```
ierr = opf_genrsv_subsys(sid, all, realar, useval)
```

Fortran command syntax:

```
CALL OPF_GENRSV_SUBSYS(SID, ALL, REALAR, USEVAL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	unit ramp rate in MW/minute (default value of 9999.0 for new records, current value used otherwise)
	REALAR(2)	unit capability in MW (default value of 9999.0 for new records, current value used otherwise)
Integer USEVAL (2)	array of 2 elements specifying (default treatment same as value of 0) (input).	
	USEVAL(1)	indicates how the value of REALAR(1) should be used
	USEVAL(1) = 0	if the value of REALAR(1) is to be used as defined
	USEVAL(1) = 1	if the value of REALAR(1) is to be multiplied by the present value of the unit ramp rate
	USEVAL(1) = 2	if the value of REALAR(1) is to be added to the present value of the unit ramp rate
	USEVAL(2)	indicates how the value of REALAR(2) should be used
	USEVAL(2) = 0	if the value of REALAR(2) is to be used as defined
	USEVAL(2) = 1	if the value of REALAR(2) is to be multiplied by the present value of the unit capability

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	USEVAL(2) = 2	if the value of REALAR(2) is to be added to the present value of the unit capability
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	generator reserve unit table is full
	IERR = 4	prerequisite requirements for API are not met

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3.2.20. OPF_GEN_RCAP_INDV

Use this API to add an individual generation reactive capability data record to the working case. It may also be used to modify an existing generation reactive capability data record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_GEN_RCAP_INDV   EIBUS  GENID  RCSTAT  REALAR(1)..REALAR(5)
```

Python command syntax:

```
ierr = opf_gen_rcap_indv(eibus, genid, rcstat, realar)
```

Fortran command syntax:

```
CALL OPF_GEN_RCAP_INDV(EIBUS, GENID, RCSTAT, REALAR, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; ' ').	
Integer RCSTAT	reactive capability limit status (input; 4 by default).	
	rcstat = 0	out of service.
	rcstat = 1	in service, no reactive generation limits.
	rcstat = 2	in service, inhibiting field increase.
	rcstat = 3	in service, inhibiting field decrease.
	rcstat = 4	in service, fixed Efd.
	rcstat = 5	in service, apply reactive generation limits.
Real REALAR (5)	array of 5 elements specifying (input).	
	REALAR(1)	direct-axis synchronous reactance in pu on Mbase (default of 1.0 if new record, current value otherwise)
	REALAR(2)	stator current limit in pu on Mbase (default of 1.0 if new record, current value otherwise)
	REALAR(3)	rated lagging power factor (default of 1.0 if new record, current value otherwise)
	REALAR(4)	rated leading power factor (default of 1.0 if new record, current value otherwise)
	REALAR(5)	maximum reactive absorption in pu on Mbase (default of 1.0 if new record, current value otherwise)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified machine does not exist
	IERR = 3	generator reactive capability data table is full

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IERR = 4	prerequisite requirements for API are not met
IERR = -1	non-fatal error occurred. Possible warning messages are: - Invalid reactive capability status specified

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3.2.21. OPF_GEN_RCAP_SUBSYS

Use this API to add or modify all OPF generator reactive capability records in a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_GEN_RCAP_SUBSYS   SID  ALL  RCSTAT  REALAR(1)..REALAR(5)
 USEVAL(1)..USEVAL(5)
```

Python command syntax:

```
ierr = opf_gen_rcap_subsys(sid, all, rcstat, realar, useval)
```

Fortran command syntax:

```
CALL OPF_GEN_RCAP_SUBSYS(SID, ALL, RCSTAT, REALAR, USEVAL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer RCSTAT	reactive capability limit status (default value of 4 for new records, current value otherwise) (input).	
	rcstat = 0	out of service.
	rcstat = 1	in service, no reactive generation limits.
	rcstat = 2	in service, inhibiting field increase.
	rcstat = 3	in service, inhibiting field decrease.
	rcstat = 4	in service, fixed Efd.
	rcstat = 5	in service, apply reactive generation limits.
Real REALAR (5)	array of 5 elements specifying The value of each is as follows(input).	
	REALAR(1)	direct-axis synchronous reactance in pu on Mbase (default value of 1.0 for new records, current value otherwise)
	REALAR(2)	stator current limit in pu on Mbase (default value of 1.0 for new records, current value otherwise)
	REALAR(3)	rated lagging power factor (default value of 1.0 for new records, current value otherwise)
	REALAR(4)	rated leading power factor (default value of 1.0 for new records, current value otherwise)
	REALAR(5)	maximum reactive absorption in pu on Mbase (default value of 1.0 for new records, current value otherwise)

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Integer USEVAL (5)	array of 5 elements specifying (default treatment same as value of 0) (input).
	USEVAL(1) indicates how the value of REALAR(1) should be used
	USEVAL(1) = 0 if the value of REALAR(1) is to be used as defined
	USEVAL(1) = 1 if the value of REALAR(1) is to be multiplied by the present value of the direct-axis synchronous reactance
	USEVAL(1) = 2 if the value of REALAR(1) is to be added to the present value of the direct-axis synchronous reactance
	USEVAL(2) indicates how the value of REALAR(2) should be used
	USEVAL(2) = 0 if the value of REALAR(2) is to be used as defined
	USEVAL(2) = 1 if the value of REALAR(2) is to be multiplied by the present value of the stator current limit
	USEVAL(2) = 2 if the value of REALAR(2) is to be added to the present value of the stator current limit
	USEVAL(3) indicates how the value of REALAR(3) should be used
	USEVAL(3) = 0 if the value of REALAR(3) is to be used as defined
	USEVAL(3) = 1 if the value of REALAR(3) is to be multiplied by the present value of the rated lagging power factor
	USEVAL(3) = 2 if the value of REALAR(3) is to be added to the present value of the rated lagging power factor
	USEVAL(4) indicates how the value of REALAR(4) should be used
	USEVAL(4) = 0 if the value of REALAR(4) is to be used as defined
	USEVAL(4) = 1 if the value of REALAR(4) is to be multiplied by the present value of the rated leading power factor
	USEVAL(4) = 2 if the value of REALAR(4) is to be added to the present value of the rated leading power factor
	USEVAL(5) indicates how the value of REALAR(5) should be used
	USEVAL(5) = 0 if the value of REALAR(5) is to be used as defined

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	USEVAL(5) = 1	if the value of REALAR(5) is to be multiplied by the present value of the maximum reactive absorption
	USEVAL(5) = 2	if the value of REALAR(5) is to be added to the present value of the maximum reactive absorption
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	generator reactive capability data table is full
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	non-fatal error occurred. Possible warning messages are: - Invalid reactive capability status specified

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3.2.22. OPF_INTFLW_3WT

Use this API to add a participating three-winding transformer winding to a specified interface flow constraint in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_INTFLW_3WT   IFID  EIBUS  EJBUS  EKBUS  CKTID
```

Python command syntax:

```
ierr = opf_intflw_3wt(ifid, eibus, ejbus, ekbus, cktid)
```

Fortran command syntax:

```
CALL OPF_INTFLW_3WT(IFID, EIBUS, EJBUS, EKBUS, CKTID, IERR)
```

Where:

Integer IFID	interface flow identifier (input; no default allowed).	
Integer EIBUS	from bus number The value of EIBUS indicates the from bus of the winding for which the flow constraint is being introduced (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Integer EKBUS	third bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	participating branch table is full
	IERR = 2	invalid interface flow identifier specified
	IERR = 3	invalid bus number specified
	IERR = 4	specified three-winding transformer does not exist
	IERR = 5	specified winding is already participating in the interface
	IERR = 6	prerequisite requirements for API are not met

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3.2.23. OPF_INTFLW_BRN

Use this API to add a participating branch to a specified interface flow constraint in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_INTFLW_BRN   IFID  EIBUS  EJBUS  CKTID
```

Python command syntax:

```
ierr = opf_intflw_brn(ifid, eibus, ejbus, cktid)
```

Fortran command syntax:

```
CALL OPF_INTFLW_BRN(IFID, EIBUS, EJBUS, CKTID, IERR)
```

Where:

Integer IFID	interface flow identifier (input; no default allowed).	
Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	participating branch table is full
	IERR = 2	invalid interface flow identifier specified
	IERR = 3	invalid bus number specified
	IERR = 4	specified branch does not exist
	IERR = 5	specified branch is already participating in the interface
	IERR = 6	prerequisite requirements for API are not met

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3.2.24. OPF_INTFLW_MAIN

Use this API to add an interface flow constraint record to the working case. It may also be used to modify an existing interface flow constraint record.
This API must be invoked prior to adding any participating branch or three-winding transformer flows to the interface flow constraint.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_INTFLW_MAIN   IFID  LABL  INTGAR(1)  INTGAR(2)
 REALAR(1)..REALAR(3)
```

Python command syntax:

```
ierr = opf_intflw_main(ifid, labl, intgar, realar)
```

Fortran command syntax:

```
CALL OPF_INTFLW_MAIN(IFID, LABL, INTGAR, REALAR, IERR)
```

Where:

Integer IFID	interface flow identifier (input; no default allowed).	
Character*32 LABL	interface flow description made up of, at most, 32 characters (input; ' ').	
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	interface flow type (default value of 1 if new record, current value otherwise)
	INTGAR(1) = 1	MW
	INTGAR(1) = 2	Mvar
	INTGAR(2)	interface flow limit type (default value of 1 if new record, current value otherwise)
	INTGAR(2) = 0	reporting only
	INTGAR(2) = 1	hard limit
	INTGAR(2) = 2	linear soft limit
	INTGAR(2) = 3	quadratic soft limit
Real REALAR (3)	array of 3 elements specifying (input).	
	REALAR(1)	maximum interface flow, in physical units (default value of 0.0 if new record, current value otherwise)
	REALAR(2)	minimum interface flow, in physical units (default value of 0.0 if new record, current value otherwise)
	REALAR(3)	soft limit penalty weight (default value of 1.0 if new record, current value otherwise)
Integer IERR	error code (output).	
	IERR = 0	no error occurred

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IERR = 1	interface flow constraint table is full
IERR = 2	prerequisite requirements for API are not met
IERR = -1	non-fatal error occurred. Possible warning messages are: - Invalid flow type specified. - Invalid interface flow limit type specified. - Maximum interface flow is less than the minimum interface flow

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3.2.25. OPF_INTFLW_RESET

Use this API to remove all interface branches associated with a particular interface flow constraint in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_INTFLW_RESET   IFID
```

Python command syntax:

```
ierr = opf_intflw_reset(ifid)
```

Fortran command syntax:

```
CALL OPF_INTFLW_RESET(IFID, IERR)
```

Where:

Integer IFID	interface flow identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid interface flow identifier specified
	IERR = 2	prerequisite requirements for API are not met

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3.2.26. OPF_LNCEQN_ADJLOAD

Use this API to add a participating adjustable bus load variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_ADJLOAD   IQID  ITBL  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_adjload(iqid, itbl, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_ADJLOAD(IQID, ITBL, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer ITBL	adjustable bus load table identifier (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable (input; default of 1.0 if new record, current value otherwise).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	specified adjustable bus load table does not exist
	IERR = 4	prerequisite requirements for API are not met

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3.2.27. OPF_LNCEQN_ADJVAR

Use this API to add a participating adjustable bus shunt variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_ADJVAR   IQID  EIBUS  SHNTID  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_adjvar(iqid, eibus, shntid, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_ADJVAR(IQID, EIBUS, SHNTID, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Character*2 SHNTID	OPF adjustable bus shunt identifier (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable (input; default of 1.0 if new record, current value otherwise).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	invalid bus number specified
	IERR = 4	specified adjustable bus shunt does not exist
	IERR = 5	prerequisite requirements for API are not met

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3.2.28. OPF_LNCEQN_BRFLOW

Use this API to add a participating branch flow variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_BRFLOW   IQID  EIBUS  EJBUS  EKBUS  CKTID  FLWID  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_brflow(iqid, eibus, ejbus, ekbus, cktid, flwid, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_BRFLOW(IQID, EIBUS, EJBUS, EKBUS, CKTID, FLWID, COEFF,
 IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Integer EKBUS	third bus number for a three-winding transformer and zero for a two-winding transformer (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Character*1 FLWID	branch flow identifier (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable. (input; Default value of 1.0 if new record, current value otherwise.).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	invalid bus number specified
	IERR = 4	specified branch/transformer does not exist
	IERR = 5	specified three-winding transformer does not exist
	IERR = 6	flow constraint does not exist at the specified branch
	IERR = 7	flow constraint does not exist at the specified three-winding transformer
	IERR = 8	prerequisite requirements for API are not met

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3.2.29. OPF_LNCEQN_INTFLOW

Use this API to add a participating interface flow constraint variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_INTFLOW   IQID  ITBL  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_intflow(iqid, itbl, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_INTFLOW(IQID, ITBL, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer ITBL	interface flow identifier (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable (input; Default value of 1.0 if new record, current value otherwise.).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	specified interface identifier does not exist
	IERR = 4	prerequisite requirements for API are not met

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3.2.30. OPF_LNCEQN_MAIN

Use this API to add a linear constraint dependency record to the working case. It may also be used to modify an existing linear constraint dependency record.
This API must be invoked prior to adding any participating linear constraint equation variable records to the specified linear constraint dependency equation.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_MAIN   IQID  LABL  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = opf_lnceqn_main(iqid, labl, realar)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_MAIN(IQID, LABL, REALAR, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Character*12 LABL	linear constraint description made up of, at most, 12 characters (input; ' ').	
Real REALAR (2)	array of 2 elements specifying (input).	
	REALAR(1)	maximum constraint slack value (Default value of 1.0 if new record, current value otherwise. by default)
	REALAR(2)	minimum constraint slack value (Default value of 1.0 if new record, current value otherwise. by default)
Integer IERR	error code (output).	
	IERR = -1	non-fatal error occurred. Possible warning messages are: - Maximum linear constraint slack is less than the minimum linear constraint slack
	IERR = 0	no error occurred
	IERR = 1	linear constraint table is full
	IERR = 2	prerequisite requirements for API are not met

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3.2.31. OPF_LNCEQN_PGEN

Use this API to add a participating active power dispatch variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_PGEN   IQID  ITBL  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_pgen(iqid, itbl, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_PGEN(IQID, ITBL, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer ITBL	active power dispatch table number (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable (input; Default value of 1.0 if new record, current value otherwise.).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	invalid active power dispatch table number specified
	IERR = 4	prerequisite requirements for API are not met

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3.2.32. OPF_LNCEQN_QGEN

Use this API to add a participating reactive power dispatch variable of a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_QGEN   IQID  EIBUS  GENID  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_qgen(iqid, eibus, genid, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_QGEN(IQID, EIBUS, GENID, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable (input; Default value of 1.0 if new record, current value otherwise.).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	specified bus does not exist
	IERR = 4	specified machine does not exist at bus
	IERR = 5	prerequisite requirements for API are not met

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3.2.33. OPF_LNCEQN_RESET

Use this API to remove all participating variables from a particular OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_RESET   IQID
```

Python command syntax:

```
ierr = opf_lnceqn_reset(iqid)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_RESET(IQID, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	Invalid linear constraint identifier specified
	IERR = 2	prerequisite requirements for API are not met

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3.2.34. OPF_LNCEQN_SWSHUNT

Use this API to add a participating switched shunt variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_SWSHUNT   IQID  EIBUS  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_swshunt(iqid, eibus, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_SWSHUNT(IQID, EIBUS, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable (input; Default of 1.0 if new record, current value otherwise).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	invalid bus number specified
	IERR = 4	specified bus does not have any switched shunt devices
	IERR = 5	prerequisite requirements for API are not met

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3.2.35. OPF_LNCEQN_TRAN

Use this API to add a participating transformer control variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_TRAN   IQID  EIBUS  EJBUS  EKBUS  CKTID  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_tran(iqid, eibus, ejbus, ekbus, cktid, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_TRAN(IQID, EIBUS, EJBUS, EKBUS, CKTID, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).
Integer EIBUS	from bus number (input; no default allowed).
Integer EJBUS	to bus number (input; no default allowed).
Integer EKBUS	third bus number if a three-winding transformer is specified and zero for a two-winding transformer (input; no default allowed).
Character*2 CKTID	circuit identifier (input; no default allowed).
Real COEFF	value of the coefficient associated with the equation variable (input; Default value of 1.0 if new record, current value otherwise).
Integer IERR	error code (output).

IERR = 0	no error occurred
IERR = 1	specified linear constraint equation does not exist
IERR = 2	participating equation variable table is full
IERR = 3	invalid bus number specified
IERR = 3	specified branch/transformer does not exist
IERR = 4	specified three-winding transformer does not exist
IERR = 5	specified branch is not a transformer
IERR = 6	specified branch is not a three-winding transformer
IERR = 8	prerequisite requirements for API are not met

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3.2.36. OPF_LNCEQN_VANG

Use this API to add a participating bus voltage angle variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_VANG   IQID  EIBUS  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_vang(iqid, eibus, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_VANG(IQID, EIBUS, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable (input; Default value of 1.0 if new record, current value otherwise).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	invalid bus number specified
	IERR = 4	prerequisite requirements for API are not met

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3.2.37. OPF_LNCEQN_VMAG

Use this API to add a participating bus voltage magnitude variable to a specified linear constraint dependency equation in the working case. It may also be used to modify an existing record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LNCEQN_VMAG   IQID  EIBUS  COEFF
```

Python command syntax:

```
ierr = opf_lnceqn_vmag(iqid, eibus, coeff)
```

Fortran command syntax:

```
CALL OPF_LNCEQN_VMAG(IQID, EIBUS, COEFF, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Real COEFF	value of the coefficient associated with the equation variable (input; Default value of 1.0 if new record, current value otherwise).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	participating equation variable table is full
	IERR = 3	invalid bus number specified
	IERR = 4	prerequisite requirements for API are not met

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3.2.38. OPF_LOAD_INDV

Use this API to add an individual OPF bus load data record to the working case. It may also be used to modify an existing bus load data record.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LOAD_INDV   EIBUS  LOADID  INTGAR
```

Python command syntax:

```
ierr = opf_load_indv(eibus, loadid, intgar)
```

Fortran command syntax:

```
CALL OPF_LOAD_INDV(EIBUS, LOADID, INTGAR, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 LOADID	bus load identifier (input; no default allowed).	
Integer INTGAR	adjustable bus load table cross reference number (input; Default value of 0 if new record, current value otherwise).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified load does not exist
	IERR = 3	prerequisite requirements for API are not met

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3.2.39. OPF_LOAD_SUBSYS

Use this API to add or modify all OPF bus load data records in a specified subsystem of the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_LOAD_SUBSYS   SID  ALL  INTGAR(1)  INTGAR(2)
```

Python command syntax:

```
ierr = opf_load_subsys(sid, all, intgar)
```

Fortran command syntax:

```
CALL OPF_LOAD_SUBSYS(SID, ALL, INTGAR, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses, use existing data values.
	ALL = 2	process all buses, use default data values.
Integer INTGAR (2)	array of 2 elements specifying (input).	
	INTGAR(1)	bus load table cross reference number (default value of 0, current value otherwise)
	INTGAR(2)	indicates which bus types should be processed (default treatment same as value of 0)
	INTGAR(2) = 0	all buses in subsystem
	INTGAR(2) = 1	only Type 1 buses in subsystem
	INTGAR(2) = 2	only Type 2 and 3 buses in subsystem
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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3.2.40. OPF_PERRSV_GEN

Use this API to add a participating generation reserve unit to a previously defined period reserve constraint in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_PERRSV_GEN   RSVID  EIBUS  GENID
```

Python command syntax:

```
ierr = opf_perrsv_gen(rsvid, eibus, genid)
```

Fortran command syntax:

```
CALL OPF_PERRSV_GEN(RSVID, EIBUS, GENID, IERR)
```

Where:

Integer RSVID	period reserve identifier (1 through 15) (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid period reserve identifier specified
	IERR = 2	invalid bus number specified
	IERR = 3	specified machine does not exist
	IERR = 4	specified unit does not have any generator reserve data defined
	IERR = 5	specified unit is already participating in the reserve constraint
	IERR = 6	generator reserve participating unit table full
	IERR = 7	prerequisite requirements for API are not met

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3.2.41. OPF_PERRSV_MAIN

Use this API to add a period reserve constraint record to the working case. It may also be used to modify an existing period reserve constraint record.
This API must be invoked prior to adding any participating generator reserve units (via OPF_PERRSV_GEN) to a specified period reserve constraint.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_PERRSV_MAIN   RSVID  STATUS  REALAR(1)  REALAR(2)
```

Python command syntax:

```
ierr = opf_perrsv_main(rsvid, status, realar)
```

Fortran command syntax:

```
CALL OPF_PERRSV_MAIN(RSVID, STATUS, REALAR, IERR)
```

Where:

Integer RSVID	a valid period reserve identifier (1 through 15) (input; no default allowed).	
Integer STATUS	status of the period reserve constraint (input; default value of 1 if new record, current value otherwise).	
	status = 0	out of service.
	status = 1	in service.
Real REALAR (2)	array of 2 elements specifying The value for each is as follows(input).	
	REALAR(1)	reserve limit, in MW (default value of 0.0 if new record, current value otherwise)
	REALAR(2)	period, in minutes (default value of 9999.0 if new record, current value otherwise)
Integer IERR	error code (output).	
	IERR = -1	non-fatal error occurred. Possible warning messages are: - Invalid period reserve status specified
	IERR = 0	no error occurred
	IERR = 1	invalid period reserve identifier specified
	IERR = 2	prerequisite requirements for API are not met

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3.2.42. OPF_PERRSV_RESET

Use this API to remove all generation reserve units associated with a particular period reserve constraint in the working case.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_OPF_PERRSV_RESET    RSVID
```

Python command syntax:

```
ierr = opf_perrsv_reset(rsvid)
```

Fortran command syntax:

```
CALL OPF_PERRSV_RESET(RSVID, IERR)
```

Where:

Integer RSVID	period reserve identifier (1 through 15) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid period reserve identifier specified
	IERR = 2	prerequisite requirements for API are not met

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3.2.43. PURGE_ALL_OPF_DATA

Use this API to purge and re-initialize all OPF data records in the working case. Unlike the NEWOPF_API call, any OPF data in the working case will be purged and re-initialized.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_ALL_OPF_DATA
```

Python command syntax:

```
ierr = purge_all_opf_data()
```

Fortran command syntax:

```
CALL PURGE_ALL_OPF_DATA(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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3.2.44. PURGE_OPF_ADJBRX_INDV

Use this API to purge an individual adjustable branch reactance data record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_ADJBRX_INDV   EIBUS  EJBUS  CKTID
```

Python command syntax:

```
ierr = purge_opf_adjbrx_indv(eibus, ejbus, cktid)
```

Fortran command syntax:

```
CALL PURGE_OPF_ADJBRX_INDV_2(EIBUS, EJBUS, CKTID, IERR)
```

Where:

Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified branch does not exist
	IERR = 3	prerequisite requirements for API are not met
	IERR = 4	specified branch cannot be a transformer

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3.2.45. PURGE_OPF_ADJBRX_SUBSYS

Use this API to purge all OPF branch reactance data records from a specified subsystem of the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_ADJBRX_SUBSYS   SID  ALL
```

Python command syntax:

```
ierr = purge_opf_adjbrx_subsys(sid, all)
```

Fortran command syntax:

```
CALL PURGE_OPF_ADJBRX_SUBSYS_2(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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3.2.46. PURGE_OPF_ADJLOAD_TBL

Use this API to purge an individual adjustable bus load table from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_ADJLOAD_TBL    TBL
```

Python command syntax:

```
ierr = purge_opf_adjload_tbl(tbl)
```

Fortran command syntax:

```
CALL PURGE_OPF_ADJLOAD_TBL_2(TBL, IERR)
```

Where:

Integer TBL	adjustable bus load table number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	adjustable bus load table does not exist
	IERR = 2	load table still referenced by bus load(s) -- unable to delete
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.47. PURGE_OPF_ADJVAR_INDV

Use this API to purge an individual OPF adjustable bus shunt data record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_ADJVAR_INDV   EIBUS   SHNTID
```

Python command syntax:

```
ierr = purge_opf_adjvar_indv(eibus, shntid)
```

Fortran command syntax:

```
CALL PURGE_OPF_ADJVAR_INDV_2(EIBUS, SHNTID, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 SHNTID	bus shunt identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	adjustable bus shunt identifier is more than two characters
	IERR = 2	invalid bus number specified
	IERR = 3	blank adjustable bus shunt identifier is invalid
	IERR = 4	unable to allocate sufficient memory
	IERR = 5	OPF adjustable bus shunt at bus not found
	IERR = 6	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.48. PURGE_OPF_ADJVAR_SUBSYS

Use this API to purge all OPF adjustable bus shunt records from a specified subsystem of the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_ADJVAR_SUBSYS   SID  ALL  SHNTID
```

Python command syntax:

```
ierr = purge_opf_adjvar_subsys(sid, all, shntid)
```

Fortran command syntax:

```
CALL PURGE_OPF_ADJVAR_SUBSYS_2(SID, ALL, SHNTID, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Character*2 SHNTID	adjustable bus shunt identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	adjustable bus shunt identifier is more than two characters
	IERR = 2	invalid SID value or subsystem SID is not defined
	IERR = 3	invalid ALL value
	IERR = 4	unable to allocate sufficient memory
	IERR = 5	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.49. PURGE_OPF_APDSP_TBL

Use this API to purge an individual active power dispatch table record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_APDSP_TBL    TBL
```

Python command syntax:

```
ierr = purge_opf_apdsp_tbl(tbl)
```

Fortran command syntax:

```
CALL PURGE_OPF_APDSP_TBL_2(TBL, IERR)
```

Where:

Integer TBL	active power dispatch table number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	active power dispatch table does not exist
	IERR = 2	active power dispatch table still referenced -- unable to delete
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.50. PURGE_OPF_BRFLW_3WT

Use this API to purge an individual flow constraint record of a three-winding transformer from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_BRFLW_3WT   EIBUS  EJBUS  EKBUS  CKTID  BFLWID
```

Python command syntax:

```
ierr = purge_opf_brflw_3wt(eibus, ejbus, ekbus, cktid, bflwid)
```

Fortran command syntax:

```
CALL PURGE_OPF_BRFLW_3WT_2(EIBUS, EJBUS, EKBUS, CKTID, BFLWID, IERR)
```

Where:

Integer EIBUS	from bus of the winding from which the flow constraint is being purged (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Integer EKBUS	third bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Character*1 BFLWID	branch flow constraint identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified three-winding transformer does not exist
	IERR = 3	specified winding of three-winding transformer not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.51. PURGE_OPF_BRFLW_BRN

Use this API to purge an individual branch or two-winding transformer flow constraint record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_BRFLW_BRN   EIBUS  EJBUS  CKTID  BFLWID
```

Python command syntax:

```
ierr = purge_opf_brflw_brn(eibus, ejbus, cktid, bflwid)
```

Fortran command syntax:

```
CALL PURGE_OPF_BRFLW_BRN_2(EIBUS, EJBUS, CKTID, BFLWID, IERR)
```

Where:

Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Character*1 BFLWID	branch flow constraint identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified branch does not exist
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.52. PURGE_OPF_BRFLW_SUBSYS

Use this API to purge all branch flow data records in a specified subsystem in the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_BRFLW_SUBSYS   SID  ALL  BFLWID
```

Python command syntax:

```
ierr = purge_opf_brflw_subsys(sid, all, bflwid)
```

Fortran command syntax:

```
CALL PURGE_OPF_BRFLW_SUBSYS_2(SID, ALL, BFLWID, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Character*1 BFLWID	branch flow identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.53. PURGE_OPF_BUS_INDV

Use this API to purge (re-initialize) OPF bus voltage magnitude data for an individual record in the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_BUS_INDV   EIBUS
```

Python command syntax:

```
ierr = purge_opf_bus_indv(eibus)
```

Fortran command syntax:

```
CALL PURGE_OPF_BUS_INDV_2(EIBUS, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.54. PURGE_OPF_BUS_SUBSYS

Use this API to purge (re-initialize) OPF bus attribute data for all records in a specified subsystem of the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_BUS_SUBSYS    SID   ALL
```

Python command syntax:

```
ierr = purge_opf_bus_subsys(sid, all)
```

Fortran command syntax:

```
CALL PURGE_OPF_BUS_SUBSYS_2(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses, use existing data values.
	ALL = 2	process all buses, use default data values.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.55. PURGE_OPF_CSTTBL_LIN

Use this API to purge an optimal power flow linear cost curve record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_CSTTBL_LIN   TBL
```

Python command syntax:

```
ierr = purge_opf_csttbl_lin(tbl)
```

Fortran command syntax:

```
CALL PURGE_OPF_CSTTBL_LIN_2(TBL, IERR)
```

Where:

Integer TBL	linear cost curve table number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear cost table does not exist
	IERR = 2	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.56. PURGE_OPF_CSTTBL_POLY

Use this API to purge an optimal power flow polynomial and exponential cost curve record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_CSTTBL_POLY    TBL
```

Python command syntax:

```
ierr = purge_opf_csttbl_poly(tbl)
```

Fortran command syntax:

```
CALL PURGE_OPF_CSTTBL_POLY_2(TBL, IERR)
```

Where:

Integer TBL	polynomial and exponential cost curve table number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified polynomial cost table does not exist
	IERR = 2	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.57. PURGE_OPF_CSTTBL_QUAD

Use this API to purge an optimal power flow quadratic cost curve record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_CSTTBL_QUAD    TBL
```

Python command syntax:

```
ierr = purge_opf_csttbl_quad(tbl)
```

Fortran command syntax:

```
CALL PURGE_OPF_CSTTBL_QUAD_2(TBL, IERR)
```

Where:

Integer TBL	quadratic cost curve table number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified quadratic cost table does not exist
	IERR = 2	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.58. PURGE_OPF_GENDSP_INDV

Use this API to purge generator dispatch data for an individual record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_GENDSP_INDV   EIBUS  GENID
```

Python command syntax:

```
ierr = purge_opf_gendsp_indv(eibus, genid)
```

Fortran command syntax:

```
CALL PURGE_OPF_GENDSP_INDV_2(EIBUS, GENID, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified machine does not exist
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.59. PURGE_OPF_GENDSP_SUBSYS

Use this API to purge generator dispatch data for all records from a specified subsystem of the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_GENDSP_SUBSYS   SID  ALL
```

Python command syntax:

```
ierr = purge_opf_gendsp_subsys(sid, all)
```

Fortran command syntax:

```
CALL PURGE_OPF_GENDSP_SUBSYS_2(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses, use existing data values.
	ALL = 2	process all buses, use default data values.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.60. PURGE_OPF_GENRSV_INDV

Use this API to purge an individual optimal power flow generator reserve data record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_GENRSV_INDV   EIBUS  GENID
```

Python command syntax:

```
ierr = purge_opf_genrsv_indv(eibus, genid)
```

Fortran command syntax:

```
CALL PURGE_OPF_GENRSV_INDV_2(EIBUS, GENID, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified machine does not exist
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.61. PURGE_OPF_GENRSV_SUBSYS

Use this API to purge all OPF generator reserve records in a specified subsystem from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_GENRSV_SUBSYS   SID  ALL
```

Python command syntax:

```
ierr = purge_opf_genrsv_subsys(sid, all)
```

Fortran command syntax:

```
CALL PURGE_OPF_GENRSV_SUBSYS_2(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.62. PURGE_OPF_GEN_RCAP_INDV

Use this API to purge an individual OPF generation reactive capability data record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_GEN_RCAP_INDV   EIBUS  GENID
```

Python command syntax:

```
ierr = purge_opf_gen_rcap_indv(eibus, genid)
```

Fortran command syntax:

```
CALL PURGE_OPF_GEN_RCAP_INDV_2(EIBUS, GENID, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified machine does not exist
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.63. PURGE_OPF_GEN_RCAP_SUBSYS

Use this API to purge all OPF generator reactive capability records in a specified subsystem from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_GEN_RCAP_SUBSYS   SID  ALL
```

Python command syntax:

```
ierr = purge_opf_gen_rcap_subsys(sid, all)
```

Fortran command syntax:

```
CALL PURGE_OPF_GEN_RCAP_SUBSYS_2(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.64. PURGE_OPF_INTFLW

Use this API to purge an OPF interface flow constraint record, including all of its participating branches, from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_INTFLW   IFID
```

Python command syntax:

```
ierr = purge_opf_intflw(ifid)
```

Fortran command syntax:

```
CALL PURGE_OPF_INTFLW_2(IFID, IERR)
```

Where:

Integer IFID	interface flow identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	interface flow identifier not found
	IERR = 2	unable to remove interface flow record due to participation in linear constraint record
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.65. PURGE_OPF_INTFLW_3WT

Use this API to remove a participating three-winding transformer winding from a specified OPF interface flow constraint in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_INTFLW_3WT   IFID  EIBUS  EJBUS  EKBUS  CKTID
```

Python command syntax:

```
ierr = purge_opf_intflw_3wt(ifid, eibus, ejbus, ekbus, cktid)
```

Fortran command syntax:

```
CALL PURGE_OPF_INTFLW_3WT(IFID, EIBUS, EJBUS, EKBUS, CKTID, IERR)
```

Where:

Integer IFID	interface flow identifier (input; no default allowed).	
Integer EIBUS	from bus number The value of EIBUS indicates the from bus of the winding to be removed from the flow constraint (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Integer EKBUS	third bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid interface flow identifier specified
	IERR = 2	invalid bus number specified
	IERR = 3	specified three-winding transformer does not exist
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.66. PURGE_OPF_INTFLW_BRN

Use this API to remove a participating branch from a specified OPF interface flow constraint in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_INTFLW_BRN   IFID  EIBUS  EJBUS  CKTID
```

Python command syntax:

```
ierr = purge_opf_intflw_brn(ifid, eibus, ejbus, cktid)
```

Fortran command syntax:

```
CALL PURGE_OPF_INTFLW_BRN(IFID, EIBUS, EJBUS, CKTID, IERR)
```

Where:

Integer IFID	interface flow identifier (input; no default allowed).	
Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid interface flow identifier specified
	IERR = 2	invalid bus number specified
	IERR = 3	specified branch does not exist
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.67. PURGE_OPF_LNCEQN

Use this API to purge an OPF linear constraint dependency record, including all of its participating variables, from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN   IQID
```

Python command syntax:

```
ierr = purge_opf_lnceqn(iqid)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_2(IQID, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	linear constraint equation number not found
	IERR = 2	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.68. PURGE_OPF_LNCEQN_ADJLOAD

Use this API to remove a participating adjustable bus load variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_ADJLOAD   IQID  ITBL
```

Python command syntax:

```
ierr = purge_opf_lnceqn_adjload(iqid, itbl)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_ADJLOAD(IQID, ITBL, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer ITBL	adjustable bus load table identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	specified adjustable bus load table does not exist
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.69. PURGE_OPF_LNCEQN_ADJVAR

Use this API to remove a participating adjustable bus shunt variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_ADJVAR   IQID  EIBUS  SHNTID
```

Python command syntax:

```
ierr = purge_opf_lnceqn_adjvar(iqid, eibus, shntid)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_ADJVAR(IQID, EIBUS, SHNTID, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Character*2 SHNTID	adjustable bus shunt identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	invalid bus number specified
	IERR = 3	specified adjustable bus shunt does not exist
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.70. PURGE_OPF_LNCEQN_BRFLOW

Use this API to remove a participating branch flow variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_BRFLOW   IQID  EIBUS  EJBUS  EKBUS  CKTID  FLWID
```

Python command syntax:

```
ierr = purge_opf_lnceqn_brflow(iqid, eibus, ejbus, ekbus, cktid, flwid)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_BRFLOW(IQID, EIBUS, EJBUS, EKBUS, CKTID, FLWID,
 IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Integer EKBUS	third bus number for a three-winding transformer and zero for a two-winding transformer (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Character*1 FLWID	branch flow identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	invalid bus number specified
	IERR = 3	specified branch/transformer does not exist
	IERR = 4	specified three-winding transformer does not exist
	IERR = 5	flow constraint does not exist at the specified branch
	IERR = 6	flow constraint does not exist at the specified three-winding transformer
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

3.2.71. PURGE_OPF_LNCEQN_INTFLOW

Use this API to remove a participating interface flow constraint variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_INTFLOW   IQID  ITBL
```

Python command syntax:

```
ierr = purge_opf_lnceqn_intflow(iqid, itbl)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_INTFLOW(IQID, ITBL, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer ITBL	interface flow identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	specified interface identifier does not exist
	IERR = 3	prerequisite requirements for API are not met

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3.2.72. PURGE_OPF_LNCEQN_PGEN

Use this API to remove a participating active power dispatch variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_PGEN   IQID  ITBL
```

Python command syntax:

```
ierr = purge_opf_lnceqn_pgen(iqid, itbl)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_PGEN(IQID, ITBL, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer ITBL	active power dispatch table number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	invalid active power dispatch table number specified
	IERR = 3	prerequisite requirements for API are not met

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3.2.73. PURGE_OPF_LNCEQN_QGEN

Use this API to remove a participating reactive power dispatch variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_QGEN   IQID  EIBUS  GENID
```

Python command syntax:

```
ierr = purge_opf_lnceqn_qgen(iqid, eibus, genid)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_QGEN(IQID, EIBUS, GENID, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	specified bus does not exist
	IERR = 3	specified machine does not exist at bus
	IERR = 4	prerequisite requirements for API are not met

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3.2.74. PURGE_OPF_LNCEQN_SWSHUNT

Use this API to remove a participating switched shunt variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_SWSHUNT   IQID  EIBUS
```

Python command syntax:

```
ierr = purge_opf_lnceqn_swshunt(iqid, eibus)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_SWSHUNT(IQID, EIBUS, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	invalid bus number specified
	IERR = 3	specified bus does not have any switched shunt devices
	IERR = 4	prerequisite requirements for API are not met

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3.2.75. PURGE_OPF_LNCEQN_TRAN

Use this API to remove a participating transformer control variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_TRAN   IQID  EIBUS  EJBUS  EKBUS  CKTID
```

Python command syntax:

```
ierr = purge_opf_lnceqn_tran(iqid, eibus, ejbus, ekbus, cktid)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_TRAN(IQID, EIBUS, EJBUS, EKBUS, CKTID, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	from bus number (input; no default allowed).	
Integer EJBUS	to bus number (input; no default allowed).	
Integer EKBUS	third bus number if a three-winding transformer is specified and zero for a two-winding transformer (input; no default allowed).	
Character*2 CKTID	circuit identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	invalid bus number specified
	IERR = 3	specified branch/transformer does not exist
	IERR = 4	specified three-winding transformer does not exist
	IERR = 5	specified branch is not a transformer
	IERR = 6	specified branch is not a three-winding transformer
	IERR = 7	prerequisite requirements for API are not met

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3.2.76. PURGE_OPF_LNCEQN_VANG

Use this API to remove a participating bus voltage angle variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_VANG   IQID  EIBUS
```

Python command syntax:

```
ierr = purge_opf_lnceqn_vang(iqid, eibus)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_VANG(IQID, EIBUS, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	invalid bus number specified
	IERR = 3	prerequisite requirements for API are not met

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3.2.77. PURGE_OPF_LNCEQN_VMAG

Use this API to remove a participating bus voltage magnitude variable from a specified OPF linear constraint dependency equation in the working case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LNCEQN_VMAG   IQID  EIBUS
```

Python command syntax:

```
ierr = purge_opf_lnceqn_vmag(iqid, eibus)
```

Fortran command syntax:

```
CALL PURGE_OPF_LNCEQN_VMAG(IQID, EIBUS, IERR)
```

Where:

Integer IQID	linear constraint equation identification number (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	specified linear constraint equation does not exist
	IERR = 2	invalid bus number specified
	IERR = 3	prerequisite requirements for API are not met

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3.2.78. PURGE_OPF_LOAD_INDV

Use this API to purge (re-initialize) an individual OPF bus load data record from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LOAD_INDV   EIBUS  LOADID
```

Python command syntax:

```
ierr = purge_opf_load_indv(eibus, loadid)
```

Fortran command syntax:

```
CALL PURGE_OPF_LOAD_INDV_2(EIBUS, LOADID, IERR)
```

Where:

Integer EIBUS	bus number (input; no default allowed).	
Character*2 LOADID	bus load identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid bus number specified
	IERR = 2	specified load does not exist
	IERR = 3	prerequisite requirements for API are not met

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3.2.79. PURGE_OPF_LOAD_SUBSYS

Use this API to purge (re-initialize) all OPF bus load data records in a specified subsystem of the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_LOAD_SUBSYS   SID  ALL
```

Python command syntax:

```
ierr = purge_opf_load_subsys(sid, all)
```

Fortran command syntax:

```
CALL PURGE_OPF_LOAD_SUBSYS_2(SID, ALL, IERR)
```

Where:

Integer SID	a valid subsystem ID from 0 to 11. A subsystem must have been previously defined (input; no default allowed).	
Integer ALL	all buses or specified subsystem flag (input; no default allowed).	
	ALL = 0	process only specified buses.
	ALL = 1	process all buses.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	prerequisite requirements for API are not met

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3.2.80. PURGE_OPF_PERRSV

Use this API to purge an OPF period reserve constraint record and its association with participating reserve units from the working case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_PERRSV    RSVID
```

Python command syntax:

```
ierr = purge_opf_perrsv(rsvid)
```

Fortran command syntax:

```
CALL PURGE_OPF_PERRSV_2(RSVID, IERR)
```

Where:

Integer RSVID	a valid period reserve identifier (1 through 15) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid period reserve identifier specified
	IERR = 2	prerequisite requirements for API are not met

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3.2.81. PURGE_OPF_PERRSV_GEN

Use this API to remove a participating period reserve unit from a specified OPF period reserve constraint.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PURGE_OPF_PERRSV_GEN   RSVID  EIBUS  GENID
```

Python command syntax:

```
ierr = purge_opf_perrsv_gen(rsvid, eibus, genid)
```

Fortran command syntax:

```
CALL PURGE_OPF_PERRSV_GEN(RSVID, EIBUS, GENID, IERR)
```

Where:

Integer RSVID	a valid period reserve identifier (1 through 15) (input; no default allowed).	
Integer EIBUS	bus number (input; no default allowed).	
Character*2 GENID	machine identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	period reserve constraint does not exist
	IERR = 2	bus not found
	IERR = 3	machine not found
	IERR = 4	machine does not have generator reserve data defined
	IERR = 5	prerequisite requirements for API are not met

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Chapter 4
Dynamics Simulation

This chapter describes the API routines used in PSS®E Dynamics Simulation. The first section, Dynamic Simulation Operation, contains API routines that handle the majority of dynamics simulation functions. The following sections contain API routines specific to the handling of plant related models, load related models, line relay models, dc line models, and FACTS device models, respectively. The section, Auxiliary Signal Model APIs, contains API routines specific to the handling of models that inject supplementary signals into dc lines and FACTS devices (auxiliary signal models). The last section, Set Disturbance, contains API routines for applying and removing disturbances during dynamics simulations.

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4.1. Dynamic Simulation Operation

4.1.1. ADDMODELLIBRARY

Use this API to add a library to the list to be searched for library models.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_ADDMODELLIBRARY  LIBRARYNAME
```

Python command syntax:

```
ierr = addmodellibrary(libraryname)
```

Fortran command syntax:

```
CALL ADDMODELLIBRARY(LIBRARYNAME, IERR)
```

Where:

Character*260 LIBRARYNAME	name of the library of model programs (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	library not found
	IERR = 2	library load failed
	IERR = 3	prerequisite requirements for API are not met

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4.1.2. ASTR

Use this API to calculate and replicates the state variable system matrices in the form required by the Linear Dynamic Analysis Program, LSYSAN (activity ASTR).

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ASTR  OPTION  MAXDERIV  PERTURB  NSTATE  MINSTATE(1)..MINSTATE(NSTATE)
   MAXSTATE(1)..MAXSTATE(NSTATE)  NOUTPUT  MINCHAN(1)..MINCHAN(NOUTPUT)
  MAXCHAN(1)..MAXCHAN(NOUTPUT)  NINPUT  INTYPE(1)..INTYPE(NINPUT)
 INPERT(1)..INPERT(NINPUT)  INBUS(1)..INBUS(NINPUT)
 INMCID(1)..INMCID(NINPUT)  INIDENT(1)..INIDENT(NINPUT)  OUTFILE
```

Python command syntax:

```
ierr = astr(option, maxderiv, perturb, nstate, minstate, maxstate,
 noutput, minchan, maxchan, ninput, intype, inpert, inbus, inmcid,
 inident, outfile)
```

Fortran command syntax:

```
CALL ASTRAPI(OPTION, MAXDERIV, PERTURB, NSTATE, MINSTATE, MAXSTATE,
 NOUTPUT, MINCHAN, MAXCHAN, NINPUT, INTYPE, INPERT, INBUS, INMCID,
 INIDENT, OUTFILE, IERR)
```

Where:

Integer OPTION	network solution convergence monitor option (input; 0 by default). OPTION = 1 automatically print the convergence monitor. OPTION = 0 print the convergence monitor only if it is enabled via the CM interrupt control code.
Real MAXDERIV	largest derivative threshold for considering that the initialization reflects a steady-state condition (input; 0.01).
Real PERTURB	amount by which the specified state variables are to be perturbed (input; 0.01).
Integer NSTATE	number of state variable ranges defining those state variables that are to be perturbed; 0 to include all STATEs (input; 0 by default).
Integer MINSTATE (NSTATE)	a list of starting STATE indices defining the ranges of state variables to be perturbed; ignored if NSTATE = 0 (input; no default allowed).
Integer MAXSTATE (NSTATE)	a list of ending STATE indices defining the ranges of state variables to be perturbed; ignored if NSTATE = 0 (input; no default allowed).
Integer NOUTPUT	number of channel ranges defining those output channels that are to be included in the system output vector; 0 to use the first 'NCHAN' or 50, whichever is less (input; 0 by default).
Integer MINCHAN (NOUTPUT)	a list of starting channel indices defining the ranges of output channels to be included in the system output vector; ignored if NOUTPUT = 0 (input; no default allowed).

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Integer MAXCHAN (NOUTPUT)	a list of ending channel indices defining the ranges of output channels to be included in the system output vector; ignored if NOUTPUT = 0 (input; no default allowed).
Integer NINPUT	number of system variables that are to be perturbed (input; 0 by default).
Integer INTYPE (NINPUT)	a list of variable type codes; ignored if NINPUT = 0 (input; no default allowed).
	INTYPE = 1 EFD, generator main field voltage (pu)
	INTYPE = 2 PMECH, turbine mechanical power (pu on MBASE)
	INTYPE = 3 VOTHSG, stabilizer output signal (pu)
	INTYPE = 4 VREF, voltage regulator voltage setpoint (pu)
	INTYPE = 5 VAR
Real INPERT (NINPUT)	a list of perturbation factors to be applied to those system variables that are to be perturbed; ignored if NINPUT = 0 (input; PERTURB).
Integer INBUS (NINPUT)	a list of buses (when the corresponding INTYPE is 1 through 4) and/or VAR indices (when the corresponding INTYPE is 5) used to define those system variables that are to be perturbed; ignored if NINPUT = 0 (input; no default allowed).
Character*2 INMCID (NINPUT)	a list of machine identifiers used to define those system variables that are to be perturbed (when the corresponding INTYPE is 1 through 4; ignored when the corresponding INTYPE is 5); ignored if NINPUT = 0 (input; '1').
Character*32 INIDENT (NINPUT)	a list of text strings used to identify those system variables that are to be perturbed; ignored if NINPUT = 0. If a blank identifier is specified, the ASTR API provides an appropriate identifier (input; blank by default).
Character*260 OUTFILE	name of the Matrix Output File into which the results of this API's calculation are to be placed; this file may then be used as to the LSYSAN program (input; no default allowed).
Integer IERR	error code (output).
	IERR = 0 no error occurred
	IERR = 1 generators are not converted
	IERR = 2 invalid OPTION value
	IERR = 3 invalid MAXDERIV value
	IERR = 4 invalid PERTURB value
	IERR = 5 derivative threshold exceeded--not in steady state
	IERR = 6 invalid code of input to be perturbed
	IERR = 7 invalid perturbation amount
	IERR = 8 machine is out-of-service
	IERR = 9 invalid VAR index
	IERR = 10 attempting to select more STATEs than are in the system model
	IERR = 11 too many channels selected

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IERR = 12	prior initialization modified the loads--pick up original converted case
IERR = 13	error opening OUTFILE
IERR = 14	prerequisite requirements for API are not met

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4.1.3. AULIST

Use this API to list auxiliary signal models that inject signals into dc lines or FACTS devices connected to subsystem buses, along with their storage locations in the dynamics data arrays.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_AULIST  SID  ALL  STATUS
```

Python command syntax:

```
ierr = aulist(sid, all, status)
```

Fortran command syntax:

```
CALL AULIST(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS	active/bypassed flag (input; 0 by default).	
	STATUS = 0	active and bypassed models
	STATUS = 1	active models only
	STATUS = 2	bypassed models only
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.1.4. CHANGE_CHANNEL_OUT_FILE

Use this API to modify the dynamic simulation channel output filename in dynamics working memory (see PSSE Program Operation Manual, activity ALTR).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_CHANNEL_OUT_FILE  OUTFILE
```

Python command syntax:

```
ierr = change_channel_out_file(outfile)
```

Fortran command syntax:

```
CALL CHANGE_CHANNEL_OUT_FILE(OUTFILE, IERR)
```

Where:

Character*260 OUTFILE	name of the Channel Output File (blank to bypass recording of the output channel values in a Channel Output File) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	error opening OUTFILE
	IERR = 2	prerequisite requirements for API are not met
	IERR = -1	warnings: - None

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4.1.5. CHANGE_CHRICN

Use this API to change the value of an element of the CHRICN array (a character ICON). The element is specified relative to the beginning of the CHRICN array.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CHANGE_CHRICN  NUMBER  'NEWVAL'
```

Python command syntax:

```
ierr = change_chricn(number, newval)
```

Fortran command syntax:

```
CALL CHANGE_CHRICN(NUMBER, NEWVAL, IERR)
```

Where:

Integer NUMBER	CHRICN number or array index (must be between 1 and the number of ICONs allowed at the present size level of PSSE) (input; no default allowed).	
Character*12 NEWVAL	new value to be assigned to CHRICN (NUMBER) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid CHRICN index
	IERR = 2	prerequisite requirements for API are not met

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4.1.6. CHANGE_CON

Use this API to change the value of an element of the CON array. The element is specified relative to the beginning of the CON array.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CHANGE_CON  NUMBER  NEWVAL
```

Python command syntax:

```
ierr = change_con(number, newval)
```

Fortran command syntax:

```
CALL CHANGE_CON(NUMBER, NEWVAL, IERR)
```

Where:

Integer NUMBER	CON number or array index (must be between 1 and the number of CONs allowed at the present size level of PSSE) (input; no default allowed).	
Real NEWVAL	new value to be assigned to CON (NUMBER) (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid CON index
	IERR = 2	prerequisite requirements for API are not met

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4.1.7. CHANGE_GREF

Use this API to change the value of the element of the GREF (governor reference) array associated with a specified machine.

This API routine was first introduced in release 31.0.2.

Batch command syntax:

```
BAT_CHANGE_GREF  IBUS  'ID'  NEWVAL
```

Python command syntax:

```
ierr = change_gref(ibus, id, newval)
```

Fortran command syntax:

```
CALL CHANGE_GREF(IBUS, ID, NEWVAL, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Real NEWVAL	new value to be assigned to the GREF array entry associated with machine ID at bus IBUS (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	machine not found
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - No turbine-governor model at this machine - Turbine-governor model at this machine is bypassed

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4.1.8. CHANGE_ICON

Use this API to change the value of an element of the ICON array (an integer ICON). The element is specified relative to the beginning of the ICON array.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CHANGE_ICON  NUMBER  NEWVAL
```

Python command syntax:

```
ierr = change_icon(number, newval)
```

Fortran command syntax:

```
CALL CHANGE_ICON(NUMBER, NEWVAL, IERR)
```

Where:

Integer NUMBER	ICON number or array index (must be between 1 and the number of ICONs allowed at the present size level of PSSE) (input; no default allowed).	
Integer NEWVAL	new value to be assigned to ICON (NUMBER) (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid ICON index
	IERR = 2	prerequisite requirements for API are not met

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4.1.9. CHANGE_STATE

Use this API to change the value of an element of the STATE array, as well as the values of the corresponding time derivative (DSTATE) and the associated memory cell used in the integration algorithm (STORE). The element is specified relative to the beginning of the STATE array.
Note: Although this API may be used to modify the value of a state variable (along with its time derivative and integration memory cell) used in state-space and response test simulations, this ability is rarely required in normal dynamic simulation work. Unless some very special purpose is being served, this API should not be used.
Note: This API should never be used during extended term simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_STATE  NUMBER  NEWVAL(1)..NEWVAL(3)
```

Python command syntax:

```
ierr = change_state(number, newval)
```

Fortran command syntax:

```
CALL CHANGE_STATE(NUMBER, NEWVAL, IERR)
```

Where:

Integer NUMBER	STATE number or array index (must be between 1 and the number of STATEs allowed at the present size level of PSSE) (input; no default allowed).	
Real NEWVAL (3)	array of 3 elements specifying (input).	
	NEWVAL(1)	new value to be assigned to STATE(NUMBER) (unchanged by default)
	NEWVAL(2)	new value to be assigned to DSTATE(NUMBER) (unchanged by default)
	NEWVAL(3)	new value to be assigned to STORE(NUMBER) (unchanged by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid STATE index
	IERR = 2	prerequisite requirements for API are not met

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4.1.10. CHANGE_SWSREF

Use this API to change the value of the element of the SWREF array associated with a specified switched shunt.

This API routine was first introduced in release 32.2.1.

Batch command syntax:

```
BAT_CHANGE_SWSREF  IBUS NEWVAL
```

Python command syntax:

```
ierr = change_swsref(ibus, newval)
```

Fortran command syntax:

```
CALL CHANGE_SWSREF(IBUS, NEWVAL, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input).	
Real NEWVAL	new value to be assigned to the SWSREF array entry associated with switched shunt at bus IBUS (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	switched shunt not found
	IERR = 2	prerequisite requirements for API are not met
	IERR = -1	data error, warning and information messages; one or more of: - no table mode for switched shunt at this bus. - switched shunt dynamic model at this switched shunt is bypassed.

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4.1.11. CHANGE_VAR

Use this API to change the value of an element of the VAR array. The element is specified relative to the beginning of the VAR array.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VAR  NUMBER  NEWVAL
```

Python command syntax:

```
ierr = change_var(number, newval)
```

Fortran command syntax:

```
CALL CHANGE_VAR(NUMBER, NEWVAL, IERR)
```

Where:

Integer NUMBER	VAR number or array index (must be between 1 and the number of VARs allowed at the present size level of PSSE) (input; no default allowed).	
Real NEWVAL	new value to be assigned to VAR (NUMBER) (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid VAR index
	IERR = 2	prerequisite requirements for API are not met

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4.1.12. CHANGE_VREF

Use this API to change the value of the element of the VREF array associated with a specified machine.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VREF  IBUS  'ID'  NEWVAL
```

Python command syntax:

```
ierr = change_vref(ibus, id, newval)
```

Fortran command syntax:

```
CALL CHANGE_VREF(IBUS, ID, NEWVAL, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Real NEWVAL	new value to be assigned to the VREF array entry associated with machine ID at bus IBUS (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	machine not found
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - No excitation system model at this machine - Excitation system model at this machine is bypassed

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4.1.13. DCLIST

Use this API to list dc line models connected to subsystem buses along with their storage locations in the dynamics data arrays.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DCLIST  SID  ALL  STATUS(1)  STATUS(2)
```

Python command syntax:

```
ierr = dclist(sid, all, status)
```

Fortran command syntax:

```
CALL DCLIST(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer STATUS (2)	array of 2 elements specifying a two-element array(input).	
	STATUS(1)	active/bypassed flag (0 by default)
	STATUS(1) = 0	list active and bypassed models
	STATUS(1) = 1	list active models only
	STATUS(1) = 2	list bypassed models only
	STATUS(2)	dc line model type (1 by default)
	STATUS(2) = 1	two-terminal dc line models
	STATUS(2) = 2	multi-terminal dc line models
	STATUS(2) = 3	VSC dc line models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.1.14. DLST

Use this API to tabulate specified portions of one or more of the PSSE dynamics data storage arrays.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DLST  NTYPE  ATYPE(1)..ATYPE(NTYPE)   FIRST(1)..FIRST(NTYPE)
 LAST(1)..LAST(NTYPE)
```

Python command syntax:

```
ierr = dlst(ntype, atype, first, last)
```

Fortran command syntax:

```
CALL DLSTAPI(NTYPE, ATYPE, FIRST, LAST, IERR)
```

Where:

Integer NTYPE	number of array portion specifications to be supplied; must be between 1 and 6, inclusive (input; no default allowed).	
Integer ATYPE (NTYPE)	a list of PSSE array type codes. No more than one occurrence of each of the valid ATYPE values is allowed in each use of this API (input; no default allowed).	
	ATYPE = 1	for CONs
	ATYPE = 2	for STATEs
	ATYPE = 3	for VARs
	ATYPE = 4	for ICONs and CHRICNs
	ATYPE = 5	for output channels
	ATYPE = 6	for CRT plot channels
Integer FIRST (NTYPE)	a list of starting indices in the data array(s) specified by the corresponding ATYPE entry (input; 1 by default).	
Integer LAST (NTYPE)	a list of ending indices in the data array(s) specified by the corresponding ATYPE entry (input; next available - 1 by default for ATYPE = 1 through 5; 6 by default for ATYPE = 6).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid NTYPE value
	IERR = 2	invalid ATYPE value (entry skipped)
	IERR = 3	ATYPE value previously specified (entry skipped)
	IERR = 4	prerequisite requirements for API are not met

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4.1.15. DOCU

Use this API to tabulate the data associated with equipment models referenced in the user's simulation set-up.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_DOCU  SID  ALL  STATUS(1)..STATUS(3)
```

Python command syntax:

```
ierr = docu(sid, all, status)
```

Fortran command syntax:

```
CALL DOCUNW(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (3)	array of 3 elements specifying reporting options(input).	
	STATUS(1)	processing mode (0 by default)
	STATUS(1) = 0	reporting mode
	STATUS(1) = 1	data checking mode
	STATUS(2)	models to process (3 by default)
	STATUS(2) = 0	process all CONEC models
	STATUS(2) = 1	process all CONET models
	STATUS(2) = 2	process all CHAN models
	STATUS(2) = 3	process models by subsystem
	STATUS(2) = 4	process Miscellaneous, "other," type models
	STATUS(3)	value depends on the value of STATUS(2). If STATUS(2) = 3 then STATUS(3) is the type of model for which data is to be processed (1 by default)
	If STATUS(2) = 4:	
	STATUS(3) = 0	Miscellaneous "Other" model. DOCU for every model instance
	STATUS(3) = 0	Miscellaneous "Other" model. DOCU for the specified model instance
	If STATUS(2) = 3:	
	STATUS(3) = 1	all models
	STATUS(3) = 2	all plant models
	STATUS(3) = 3	generator models

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	STATUS(3) = 4	compensator models
	STATUS(3) = 5	stabilizer models
	STATUS(3) = 6	minimum excitation limiter models
	STATUS(3) = 7	maximum excitation limiter models
	STATUS(3) = 8	excitation system models
	STATUS(3) = 9	turbine governor models
	STATUS(3) = 10	load characteristic models
	STATUS(3) = 11	load relay models
	STATUS(3) = 12	line relay models
	STATUS(3) = 13	auxiliary signal models
	STATUS(3) = 14	two-terminal dc line models
	STATUS(3) = 15	multi-terminal dc line models
	STATUS(3) = 16	VSC dc line models
	STATUS(3) = 17	FACTS device models
	STATUS(3) = 18	CONEC models
	STATUS(3) = 19	CONET models
	STATUS(3) = 20	CHAN models
	STATUS(3) = 21	turbine load controller
	STATUS(3) = 22	switched shunt models
	STATUS(3) = 23	all wind models
	STATUS(3) = 24	wind generator models
	STATUS(3) = 25	wind electrical control models
	STATUS(3) = 26	wind mechanical system models
	STATUS(3) = 27	wind pitch models
	STATUS(3) = 28	wind aerodynamic models
	STATUS(3) = 29	wind gust models
	STATUS(3) = 30	wind auxiliary signal models
	STATUS(3) = 31	induction machine models
	STATUS(3) = 32	branch device models
	STATUS(3) = 33	two-winding transformer device models
	STATUS(3) = 34	three-winding transformer models
	STATUS(3) = 35	protection models associated with machines
	STATUS(3) = 36	"other" models associated with machines
	STATUS(3) = 37	"other" models associated with bus
	STATUS(3) = 38	"other" models associated with two-terminal dc lines
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.1.16. DROPMODELLIBRARY

Use this API to remove a library from the list being searched for library models.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_DROPMODELLIBRARY  LIBRARYNAME
```

Python command syntax:

```
ierr = dropmodellibrary(libraryname)
```

Fortran command syntax:

```
CALL DROPMODELLIBRARY(LIBRARYNAME, IERR)
```

Where:

Character*260 LIBRARYNAME	name of the library of model programs (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	library not found
	IERR = 2	there are still models in use from this library; unload not done
	IERR = 3	library unload failed
	IERR = 4	prerequisite requirements for API are not met

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4.1.17. DROPMODELPROGRAM

Use this API to remove a model from the list of known library models; library models are added to this list the first time that they are called.

This API routine was first introduced in release 30.3.0.

Batch command syntax:

```
BAT_DROPMODELPROGRAM  PROGRAMNAME
```

Python command syntax:

```
ierr = dropmodelprogram(programname)
```

Fortran command syntax:

```
CALL DROPMODELPROGRAM(PROGRAMNAME, IERR)
```

Where:

Character*260 PROGRAMNAME	name of the model program (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	program not found
	IERR = 2	program unload failed
	IERR = 3	prerequisite requirements for API are not met

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4.1.18. DYDA

Use this API to replicate dynamics model data in the form of a Dynamics Data Input file.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_DYDA  SID  ALL  STATUS(1)..STATUS(3)  OUT  OFILE
```

Python command syntax:

```
ierr = dyda(sid, all, status, out, ofile)
```

Fortran command syntax:

```
CALL DYDANW(SID, ALL, STATUS, OUT, OFILE, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses.
	ALL = 0	process only buses in subsystem SID.
Integer STATUS (3)	array of 3 elements specifying a three-element array(input).	
	STATUS(1)	models to process (2 by default)
	STATUS(1) = 0	process all CONEC models
	STATUS(1) = 1	process all CONET models
	STATUS(1) = 2	process models by subsystem
	STATUS(1) = 3	process Miscellaneous other type models
	STATUS(2)	value depends on the value of STATUS(1). If STATUS(1) = 2 then STATUS(2) is the type of model for which data is to be processed (1 by default)
	If STATUS(1) = 3:	
	STATUS(2) = 0	Miscellaneous, "Other," model DYDA for every model instance.
	STATUS(2) = 0	Miscellaneous, "Other," model DYDA for the specificed model instance.
	If STATUS(1) = 2:	
	STATUS(2) = 1	all models
	STATUS(2) = 2	all plant models
	STATUS(2) = 3	generator models
	STATUS(2) = 4	compensator models
	STATUS(2) = 5	stabilizer models
	STATUS(2) = 6	minimum excitation limiter models
	STATUS(2) = 7	maximum excitation limiter models
	STATUS(2) = 8	excitation system models

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STATUS(2) = 9	turbine governor models
STATUS(2) = 10	load characteristic models
STATUS(2) = 11	load relay models
STATUS(2) = 12	line relay models
STATUS(2) = 13	auxiliary signal models
STATUS(2) = 14	two-terminal dc line models
STATUS(2) = 15	multi-terminal dc line models
STATUS(2) = 16	VSC dc line models
STATUS(2) = 17	FACTS device models
STATUS(2) = 18	CONEC models
STATUS(2) = 19	CONET models
STATUS(2) = 20	turbine load controller
STATUS(2) = 21	switched shunt models
STATUS(2) = 22	all wind models
STATUS(2) = 23	wind generator models
STATUS(2) = 24	wind electrical control models
STATUS(2) = 25	wind mechanical system models
STATUS(2) = 26	wind pitch models
STATUS(2) = 27	wind aerodynamic models
STATUS(2) = 28	wind gust models
STATUS(2) = 29	wind auxiliary signal models
STATUS(2) = 30	induction machine models
STATUS(2) = 31	branch device models
STATUS(2) = 32	two-winding transformer device models
STATUS(2) = 33	three-winding transformer models
STATUS(2) = 34	protection models associated with machines
STATUS(2) = 35	"other" models associated with machines
STATUS(2) = 36	"other" models associated with bus
STATUS(2) = 37	"other" models associated with two-terminal dc lines
STATUS(3)	load model option (0 by default)
STATUS(3) = 0	include all subsystem load models attached to loads in the specified subsystem
STATUS(3) = 1	exclude subsystem load models attached to loads in the specified subsystem except when subsystem number is explicitly in the list

Integer OUT — destination selection (input; 1 by default).

OUT = 0	direct output to a data file specified by OFILE.
OUT = 1	direct output to the progress window.

Character*260 OFILE — name of the file to which data records are to be written (when OUT is 0) (input; no default allowed).

Integer IERR — error code (output).

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IERR = 0	no error
IERR = 1	invalid SID value or subsystem SID is not defined
IERR = 2	invalid ALL value
IERR = 3	invalid STATUS value
IERR = 4	invalid OUT value
IERR = 5	error opening OFILE
IERR = 6	prerequisite requirements for API are not met

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4.1.19. DYNAMICSMODE

Use this API to return PSSE to dynamics mode following the use of the POWERFLOWMODE API while in dynamics mode. (The POWERFLOWMODE API would normally be followed by the use of one or more of the power flow data editor APIs).

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_DYNAMICSMODE  OPT
```

Python command syntax:

```
ierr = dynamicsmode(opt)
```

Fortran command syntax:

```
CALL DSMODE(OPT, IERR)
```

Where:

Logical OPT	factorize admittance matrix flag (input; FALSE).	
	OPT = TRUE	(or T or 1 in batch command and Python) to factorize the admittance matrix
	OPT = FALSE	(or F or 0 in batch command and Python) to omit the matrix factorization
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	PSSE was started up at its powerflow entry point
	IERR = 2	prerequisite requirements for API are not met

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4.1.20. DYNAMICS_SOLUTION_PARAMS

This API routine is obsolete. It has been replaced by the API routine dynamics_solution_param_2 .

Use this API to modify the dynamic simulation solution parameters in dynamics working memory.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DYNAMICS_SOLUTION_PARAMS  INTGAR(1)..INTGAR(8)  REALAR(1)..REALAR(8)
 OUTFILE
```

Python command syntax:

```
ierr = dynamics_solution_params(intgar, realar, outfile)
```

Fortran command syntax:

```
CALL DYNAMICS_SOLUTION_PARAMETERS(INTGAR, REALAR, OUTFILE, IERR)
```

Where:

Integer INTGAR (8)	array of 8 elements specifying (input).	
	INTGAR(1)	network solution maximum number of iterations
	INTGAR(2)	number of channels being monitored; must be less than or equal to the number of output channels defined
	INTGAR(3)	number of state variables in use in the simulation model; should not normally be changed by the user
	INTGAR(4)	next available location in the CON array
	INTGAR(5)	next available location in the STATE array
	INTGAR(6)	next available location in the VAR array
	INTGAR(7)	next available location in the ICON and CHRICN arrays
	INTGAR(8)	next available location in the channel arrays
Real REALAR (8)	array of 8 elements specifying (input).	
	REALAR(1)	acceleration sfactor used in the network solution
	REALAR(2)	convergence tolerance used in the network solution
	REALAR(3)	simulation time step (DELT)
	REALAR(4)	filter time constant used in calculating bus frequency deviations
	REALAR(5)	intermediate simulation mode time step threshold used in extended term simulations

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	REALAR(6)	large (island frequency) simulation mode time step threshold used in extended term simulations
	REALAR(7)	large (island frequency) simulation mode acceleration factor used in extended term simulations
	REALAR(8)	large (island frequency) simulation mode convergence tolerance used in extended term simulations
Character*260 OUTFILE	name of the Channel Output File (blank to bypass recording of the output channel values in a Channel Output File) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	error opening OUTFILE
	IERR = 2	prerequisite requirements for API are not met
	IERR = -1	warnings: - None

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4.1.21. DYNAMICS_SOLUTION_PARAM_2

Use this API to modify the dynamic simulation solution parameters (except the channel output filename) in dynamics working memory (see PSSE Program Operation Manual, activity ALTR).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_DYNAMICS_SOLUTION_PARAM_2  INTGAR(1)..INTGAR(8)  REALAR(1)..REALAR(8)
```

Python command syntax:

```
ierr = dynamics_solution_param_2(intgar, realar)
```

Fortran command syntax:

```
CALL DYNAMICS_SOLUTION_PARAMETERS_2(INTGAR, REALAR, IERR)
```

Where:

Integer INTGAR (8)	array of 8 elements specifying (input).	
	INTGAR(1)	network solution maximum number of iterations
	INTGAR(2)	number of channels being monitored; must be less than or equal to the number of output channels defined
	INTGAR(3)	number of state variables in use in the simulation model; should not normally be changed by the user
	INTGAR(4)	next available location in the CON array
	INTGAR(5)	next available location in the STATE array
	INTGAR(6)	next available location in the VAR array
	INTGAR(7)	next available location in the ICON and CHRICN arrays
	INTGAR(8)	next available location in the channel arrays
Real REALAR (8)	array of 8 elements specifying (input).	
	REALAR(1)	acceleration sfactor used in the network solution
	REALAR(2)	convergence tolerance used in the network solution
	REALAR(3)	simulation time step (DELT)
	REALAR(4)	filter time constant used in calculating bus frequency deviations
	REALAR(5)	intermediate simulation mode time step threshold used in extended term simulations
	REALAR(6)	large (island frequency) simulation mode time step threshold used in extended term simulations

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	REALAR(7)	large (island frequency) simulation mode acceleration factor used in extended term simulations
	REALAR(8)	large (island frequency) simulation mode convergence tolerance used in extended term simulations
Integer IERR	error code (output).	
	IERR = 0	no errors occurred
	IERR > 0	error occurred
	IERR = 1	prerequisite requirements for API are not met
	IERR = -1	warnings: - None

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4.1.22. DYNEXPORTCSV

Use this API to export dynamics engine and dynamics model messages that are displayed after running a dynamics simulation to CSV file format. If applicable the following data will be included in the specified output file:
type, severity, model, model type, bus number, bus name, base kv, area number, owner number, and engine/model messages

This API routine was first introduced in release 33.10.0 and 34.2.0.

Batch command syntax:

```
BAT_DYNEXPORTCSV CSVFILENAME
```

Python command syntax:

```
ierr = dynexportcsv(csvfilename)
```

Fortran command syntax:

```
CALL DYNEXPORTCSV(CSVFILENAME, IERR)
```

Where:

character*260 CSVFILENAME	name of the CSV file to hold exported data (input).	
Integer IERR	error code (output).	
	IERR = -1	memory allocation error
	IERR = 0	no error
	IERR = 1	error exporting to csv file

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4.1.23. DYRE_ADD

Use this API to read a Dynamics Model Raw Data File and append the model references specified in its data records to the simulation data already contained in dynamics working memory (activity DYRE,ADD).

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DYRE_ADD  STARTINDX(1)..STARTINDX(4)  DYREFILE  CONECFILE  CONETFILE
```

Python command syntax:

```
ierr = dyre_add(startindx, dyrefile, conecfile, conetfile)
```

Fortran command syntax:

```
CALL DYRE_ADD(STARTINDX, DYREFILE, CONECFILE, CONETFILE, IERR)
```

Where:

Integer STARTINDX (4)	array of 4 elements specifying starting locations in the dynamics data storage arrays(input).	
	STARTINDX(1)	starting CON index (the next available CON by default)
	STARTINDX(2)	starting STATE index (the next available STATE by default)
	STARTINDX(3)	starting VAR index (the next available VAR by default)
	STARTINDX(4)	starting ICON index (the next available ICON by default)
Character*260 DYREFILE	name of the Dynamics Model Raw Data File (input; no default allowed).	
Character*260 CONECFILE	name of the output file for CONEC additions; blank to route data to the progress area (input; blank by default).	
Character*260 CONETFILE	name of the output file for CONET additions; blank to route data to the progress area (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	generators are not converted
	IERR = 2	invalid STARTINDX value
	IERR = 3	machine model connection tables full--use pack plant model tables function
	IERR = 4	error opening output file DYREFILE
	IERR = 5	prerequisite requirements for API are not met

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4.1.24. DYRE_NEW

Use this API to clear dynamics working memory, read a Dynamics Data File, and place the model references specified on its data records into dynamics working memory. It optionally creates a command file for compiling the CONEC and CONET subroutines.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DYRE_NEW  STARTINDX(1)..STARTINDX(4)  DYREFILE  CONECFILE  CONETFILE
 COMPILFIL
```

Python command syntax:

```
ierr = dyre_new(startindx,dyrefile,conecfile,conetfile,compilfil)
```

Fortran command syntax:

```
CALL DYRE_NEW(STARTINDX, DYREFILE, CONECFILE, CONETFILE, COMPILFIL, IERR)
```

Where:

Integer STARTINDX (4)	array of 4 elements specifying starting locations in the dynamics data storage arrays(input).
	STARTINDX(1) starting CON index (1 by default)
	STARTINDX(2) starting STATE index (1 by default)
	STARTINDX(3) starting VAR index (1 by default)
	STARTINDX(4) starting ICON index (1 by default)
Character*260 DYREFILE	name of the Dynamics Model Raw Data File (input; no default allowed).
Character*260 CONECFILE	name of CONEC output file; blank for output to the progress area (input; blank by default).
Character*260 CONETFILE	name of CONET output file; blank for output to the progress area (input; blank by default).
Character*260 COMPILFIL	name of file containing commands to compile the CONEC and CONET output files; blank for none (input; blank by default).
Integer IERR	error code (output).
	IERR = 0 no error occurred
	IERR = 1 generators are not converted
	IERR = 2 invalid STARTINDX value
	IERR = 4 error opening output file DYREFILE
	IERR = 5 prerequisite requirements for API are not met

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4.1.25. ERUN

Use this API to calculate PSSE state-space dynamic simulations of excitation system response tests.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ERUN  TPAUSE  NPRT  NPLT  CRTPLT
```

Python command syntax:

```
ierr = erun(tpause, nprt, nplt, crtplt)
```

Fortran command syntax:

```
CALL ERUNAPI(TPAUSE, NPRT, NPLT, CRTPLT, IERR)
```

Where:

Real TPAUSE	value of simulated time at which the simulation should next pause (input; no default allowed).	
Integer NPRT	number of time steps between the printing of the channel values (input; unchanged).	
Integer NPLT	number of time steps between the writing of the output channel values to the current Channel Output File (input; unchanged).	
Integer CRTPLT	number of time steps between the plotting of those channel values that have been designated as CRT output channels (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	activity ESTR needs to be executed
	IERR = 2	error opening the current Channel Output File
	IERR = 3	prerequisite requirements for API are not met

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4.1.26. ESTR_OPEN_CIRCUIT_TEST

Use this API to initialize a PSSE dynamic simulation for excitation system open circuit response simulations (i.e., in preparation for activity ERUN) and to specify the Channel Output File into which the output channel values are to be recorded during the simulation.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ESTR_OPEN_CIRCUIT_TEST  OPTION  VDELTA  OUTFILE
```

Python command syntax:

```
ierr = estr_open_circuit_test(option, vdelta, outfile)
```

Fortran command syntax:

```
CALL ESTR_OPEN_CIRCUIT_TEST(OPTION, VDELTA, OUTFILE, IERR)
```

Where:

Integer OPTION	bus flag indicating those buses to be tested (input; 0 by default).	
	OPTION = 0	run test on all in-service machines with excitation system models.
	OPTION > 0	bus number for which machines are to be tested.
Real VDELTA	per unit voltage reference step applied at TIME = 0.0 (input; 0.05).	
Character*260 OUTFILE	name of the Channel Output File (blank to bypass recording of the output channel values in a Channel Output File) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	error opening OUTFILE
	IERR = 3	generators are not "converted"
	IERR = 4	prerequisite requirements for API are not met

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4.1.27. ESTR_RESPONSE_RATIO_TEST

Use this API to initialize a PSSE dynamic simulation for excitation system response ratio test simulations (i.e., in preparation for activity ERUN) and to specify the Channel Output File into which the output channel values are to be recorded during the simulation.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ESTR_RESPONSE_RATIO_TEST  OPTION  PFDEFALT  NMACHS
  BUSARY(1)..BUSARY(NMACHS)   IDARY(1)..IDARY(NMACHS)
 PFARY(1)..PFARY(NMACHS)  OUTFILE
```

Python command syntax:

```
ierr = estr_response_ratio_test(option, pfdefalt, nmachs, busary, idary,
 pfary, outfile)
```

Fortran command syntax:

```
CALL ESTR_RESPONSE_RATIO_TEST(OPTION, PFDEFALT, NMACHS, BUSARY, IDARY,
 PFARY, OUTFILE, IERR)
```

Where:

Integer OPTION	bus flag indicating those buses to be tested (input; 0 by default).	
	OPTION = 0	run test on all in-service machines with excitation system models.
	OPTION > 0	bus number for which machines are to be tested.
Real PFDEFALT	default power factor at which machines are initialized (input; 0.95).	
Integer NMACHS	number of machines to be initialized at other power factors (input; 0 by default).	
Integer BUSARY (NMACHS)	a list of bus numbers identifying machines to be initialized at other power factors; ignored if NMACHS = 0 (input; no default allowed).	
Character*2 IDARY (NMACHS)	a list of machine identifiers identifying machines to be initialized at other power factors; ignored if NMACHS = 0 (input; '1').	
Real PFARY (NMACHS)	a list of power factors; ignored if NMACHS = 0. Machine IDARY(n) at bus BUSARY(n) is initialized to the power factor specified in PFARY(n) (input; PFDEFALT).	
Character*260 OUTFILE	name of the Channel Output File (blank to bypass recording of the output channel values in a Channel Output File) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	invalid power factor
	IERR = 3	error opening OUTFILE

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IERR = 4	generators are not "converted"
IERR = 5	prerequisite requirements for API are not met

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4.1.28. FCLIST

Use this API to list FACTS device models connected to subsystem buses along with their storage locations in the dynamics data arrays.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_FCLIST  SID  ALL  STATUS
```

Python command syntax:

```
ierr = fclist(sid, all, status)
```

Fortran command syntax:

```
CALL FCLIST(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS	active/bypassed flag (input; 0 by default).	
	STATUS = 0	active and bypassed models
	STATUS = 1	active models only
	STATUS = 2	bypassed models only
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.1.29. GRUN

Use this API to calculate PSSE state-space dynamic simulations of turbine-governor response tests.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_GRUN  TPAUSE  NPRT  NPLT  CRTPLT
```

Python command syntax:

```
ierr = grun(tpause, nprt, nplt, crtplt)
```

Fortran command syntax:

```
CALL GRUNAPI(TPAUSE, NPRT, NPLT, CRTPLT, IERR)
```

Where:

Real TPAUSE	value of simulated time at which the simulation should next pause (input; no default allowed).	
Integer NPRT	number of time steps between the printing of the channel values (input; unchanged).	
Integer NPLT	number of time steps between the writing of the output channel values to the current Channel Output File (input; unchanged).	
Integer CRTPLT	number of time steps between the plotting of those channel values that have been designated as CRT output channels (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	activity GSTR needs to be executed
	IERR = 2	error opening the current Channel Output File
	IERR = 3	prerequisite requirements for API are not met

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4.1.30. GSTR

Use this API to initialize a PSSE dynamic simulation for governor response simulations and to specify the Channel Output File into which the output channel values are to be recorded during the simulation.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_GSTR  OPTION  PINITIAL  PDELTA  OUTFILE
```

Python command syntax:

```
ierr = gstr(option, pinitial, pdelta, outfile)
```

Fortran command syntax:

```
CALL GSTRAPI(OPTION, PINITIAL, PDELTA, OUTFILE, IERR)
```

Where:

Integer OPTION	bus flag indicating those buses to be tested (input; 0 by default).	
	OPTION = 0	run test on all in-service machines with turbine-governor models.
	OPTION = 0	bus number for which machines are to be tested.
Real PINITIAL	initial per unit loading of all machines being tested (input; 0.8).	
Real PDELTA	per unit loading step applied at TIME = 0.0 (input; 0.1).	
Character*260 OUTFILE	name of the Channel Output File (blank to bypass recording of the output channel values in a Channel Output File) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	bus not found
	IERR = 2	error opening OUTFILE
	IERR = 3	generators are not, "converted"
	IERR = 4	prerequisite requirements for API are not met

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4.1.31. INCREMENT_GREF

Use this API to add a specified value to the value of the element of the GREF (governor reference) array associated with a specified machine.

This API routine was first introduced in release 31.0.2.

Batch command syntax:

```
BAT_INCREMENT_GREF  IBUS  'ID'  NEWVAL
```

Python command syntax:

```
ierr = increment_gref(ibus, id, newval)
```

Fortran command syntax:

```
CALL INCREMENT_GREF(IBUS, ID, NEWVAL, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Real NEWVAL	new value to be added to the GREF array entry associated with machine ID at bus IBUS (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	machine not found
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - No turbine-governor model at this machine - Turbine-governor model at this machine is bypassed

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4.1.32. INCREMENT_SWSREF

Use this API to add a specified value to the value of the element of the SWREF (switched shunt reference) array associated with a specified switched shunt.

This API routine was first introduced in release 32.2.1.

Batch command syntax:

```
BAT_INCREMENT_SWSREF  IBUS NEWVAL
```

Python command syntax:

```
ierr = increment_swsref(ibus, newval)
```

Fortran command syntax:

```
CALL INCREMENT_SWSREF(IBUS, NEWVAL, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input).	
Real NEWVAL	new value to be added to the SWREF array entry associated with switched shunt at bus IBUS (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	switched shunt not found
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - no table model for switched shunt at this bus. - switched shunt dynamic model as this bus is bypassed.

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4.1.33. INCREMENT_VREF

Use this API to add a specified value to the value of the element of the VREF (voltage reference) array associated with a specified machine.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_INCREMENT_VREF  IBUS  'ID'  NEWVAL
```

Python command syntax:

```
ierr = increment_vref(ibus, id, newval)
```

Fortran command syntax:

```
CALL INCREMENT_VREF(IBUS, ID, NEWVAL, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Real NEWVAL	new value to be added to the VREF array entry associated with machine ID at bus IBUS (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	error occurred
	IERR = 1	bus not found
	IERR = 2	machine not found
	IERR = 3	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - no excitation system model at this machine - excitation system model at this machine is bypassed

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4.1.34. LDLIST

Use this API to list load models connected to subsystem loads along with their storage locations in the dynamics data arrays.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LDLIST  SID  ALL  STATUS(1)  STATUS(2)
```

Python command syntax:

```
ierr = ldlist(sid, all, status)
```

Fortran command syntax:

```
CALL LDLIST(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (2)	array of 2 elements specifying a two-element array(input).	
	STATUS(1)	active/bypassed flag (0 by default)
	STATUS(1) = 0	active and bypassed models
	STATUS(1) = 1	active models only
	STATUS(1) = 2	bypassed models only
	STATUS(2)	model connection report option (0 by default)
	STATUS(2) = 0	list models connected to loads
	STATUS(2) = 1	list loads connected to model
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.1.35. LOAD_ARRAY_CHANNEL

Use this API to add an output channel containing the active or reactive component of a designated load.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_LOAD_ARRAY_CHANNEL  STATUS(1)..STATUS(3)  'ID'  'IDENT'
```

Python command syntax:

```
ierr = load_array_channel(status, id, ident)
```

Fortran command syntax:

```
CALL LOAD_ARRAY_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (3)	array of 3 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	used to indicate the quantity to be placed in an channel (1 by default)
	STATUS(2) = 1	PLOAD
	STATUS(2) = 2	QLOAD
	STATUS(3)	number of the bus to which the load for which quantity is to be placed in an channel is connected (no default allowed)
Character*2 ID	load identifier (input; '1').	
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	bus not found
	IERR = 4	load not found
	IERR = 5	prerequisite requirements for API are not met

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4.1.36. MLST

Use this API to list plant models connected to subsystem machines along with their storage locations in the dynamics data arrays.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_MLST  SID  ALL  STATUS(1)..STATUS(3)
```

Python command syntax:

```
ierr = mlst(sid, all, status)
```

Fortran command syntax:

```
CALL MLSTNW(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (3)	array of 3 elements specifying a three-element array(input).	
	STATUS(1)	connected/unconnected flag (1 by default)
	STATUS(1) = 0	list all unconnected models
	STATUS(1) = 1	list connected subsystem models
	STATUS(2)	active/bypassed flag (0 by default)
	STATUS(2) = 0	active and bypassed models
	STATUS(2) = 1	active models only
	STATUS(2) = 2	bypassed models only
	STATUS(3)	plant model type (1 by default)
	STATUS(3) = 1	all plant models
	STATUS(3) = 2	generator models
	STATUS(3) = 3	compensator models
	STATUS(3) = 4	stabilizer models
	STATUS(3) = 5	minimum excitation models
	STATUS(3) = 6	maximum excitation models
	STATUS(3) = 7	excitation system models
	STATUS(3) = 8	governor models
	STATUS(3) = 9	turbine load controller models
Integer IERR	error code (output).	
	IERR = 0	no error

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IERR = 1	invalid SID value or subsystem SID is not defined
IERR = 2	invalid ALL value
IERR = 3	invalid STATUS value
IERR = 4	prerequisite requirements for API are not met

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4.1.37. MRUN

Use this API to calculate PSSE extended term dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MRUN  OPTION  TPAUSE  NPRT  NPLT  CRTPLT
```

Python command syntax:

```
ierr = mrun(option, tpause, nprt, nplt, crtplt)
```

Fortran command syntax:

```
CALL MRUNAPI(OPTION, TPAUSE, NPRT, NPLT, CRTPLT, IERR)
```

Where:

Integer OPTION	network solution convergence monitor option (input; 0 by default).	
	OPTION = 1	automatically print the convergence monitor.
	OPTION = 0	print the convergence monitor only if it is enabled via the CM interrupt control code.
Real TPAUSE	value of simulated time at which the simulation should next pause (input; no default allowed).	
Integer NPRT	number of time steps between the printing of the channel values (input; unchanged).	
Integer NPLT	number of time steps between the writing of the output channel values to the current Channel Output File (input; unchanged).	
Integer CRTPLT	number of time steps between the plotting of those channel values that have been designated as CRT output channels (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	long time step mode threshold is at or below intermediate mode threshold
	IERR = 2	activity MSTR needs to be executed
	IERR = 3	invalid OPTION value
	IERR = 4	generators are not converted
	IERR = 5	error opening the current Channel Output File
	IERR = 6	prerequisite requirements for API are not met

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4.1.38. MSTR

Use this API to initialize a PSSE dynamic simulation for extended term simulations and to specify the Channel Output File into which the output channel values are to be recorded during the dynamic simulation.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MSTR  OPTION  OUTFILE
```

Python command syntax:

```
ierr = mstr(option, outfile)
```

Fortran command syntax:

```
CALL MSTRAPI(OPTION, OUTFILE, IERR)
```

Where:

Integer OPTION	network solution convergence monitor option (input; 0 by default).	
	OPTION = 1	automatically print the convergence monitor.
	OPTION = 0	print the convergence monitor only if it is enabled via the CM interrupt control code.
Character*260 OUTFILE	name of the Channel Output File (blank to bypass recording of the output channel values in a Channel Output File) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	long time step mode threshold is at or below intermediate mode threshold
	IERR = 2	DELT is greater than intermediate time step mode threshold
	IERR = 3	generators are not converted
	IERR = 4	invalid OPTION value
	IERR = 5	Prior initialization modified the loads--pick up original converted case
	IERR = 6	error opening OUTFILE
	IERR = 7	prerequisite requirements for API are not met

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4.1.39. POWERFLOWMODE

Use this API to switch PSSE from dynamics mode to power flow mode. (This API would normally be followed by the use of one or more of the power flow data editor APIs, and then the DYNAMICSMODE API).
Batch command syntax.
BAT_POWERFLOWMODE

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_POWERFLOWMODE
```

Python command syntax:

```
ierr = powerflowmode()
```

Fortran command syntax:

```
CALL PFMODE(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

4.1.40. PSAS

Use this API to convert a PSAS Command File into a PSSE Response File.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PSAS  INPFILE  OUTFILE
```

Python command syntax:

```
ierr, respfil = psas(inpfile, outfile)
```

Fortran command syntax:

```
CALL PSASAPI(INPFILE, OUTFILE, RESPFIL, IERR)
```

Where:

Character*260 INPFILE	name of the PSAS command file (input; no default allowed).	
Character*260 OUTFILE	name of Response file; blank to have PSAS assign the filename (input; blank by default).	
Character*260 RESPFIL	pathname of the Response file constructed by PSAS (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	error opening INPFILE
	IERR = 2	error opening OUTFILE
	IERR = 3	errors detected while reading the PSAS command data input file
	IERR = 4	prerequisite requirements for API are not met

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4.1.41. RLLIST

Use this API to list line relay models connected to subsystem buses along with their storage locations in the dynamics data arrays.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RLLIST  SID  ALL  STATUS
```

Python command syntax:

```
ierr = rllist(sid, all, status)
```

Fortran command syntax:

```
CALL RLLIST(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS	active/bypassed flag (input; 0 by default).	
	STATUS = 0	active and bypassed models
	STATUS = 1	active models only
	STATUS = 2	bypassed models only
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.1.42. RSTR

Use this API to read a dynamics Snapshot File into PSSE working memory.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RSTR  SFILE
```

Python command syntax:

```
ierr = rstr(sfile)
```

Fortran command syntax:

```
CALL RSTRNW(SFILE, IERR)
```

Where:

Character*260 SFILE	snapshot file; '*' to restore the most recently accessed snapshot file in the current execution of PSSE (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	error opening SFILE
	IERR = 2	error reading from SFILE
	IERR = 3	prerequisite requirements for API are not met

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4.1.43. RUN

Use this API to calculate PSSE state-space dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_RUN  OPTION  TPAUSE  NPRT  NPLT  CRTPLT
```

Python command syntax:

```
ierr = run(option, tpause, nprt, nplt, crtplt)
```

Fortran command syntax:

```
CALL RUNAPI(OPTION, TPAUSE, NPRT, NPLT, CRTPLT, IERR)
```

Where:

Integer OPTION	network solution convergence monitor option (input; 0 by default).	
	OPTION = 1	automatically print the convergence monitor
	OPTION = 0	print the convergence monitor only if it is enabled via the CM interrupt control code
Real TPAUSE	value of simulated time at which the simulation should next pause (input; no default allowed).	
Integer NPRT	number of time steps between the printing of the channel values (input; unchanged).	
Integer NPLT	number of time steps between the writing of the output channel values to the current channel output file (input; unchanged).	
Integer CRTPLT	number of time steps between the plotting of those channel values that have been designated as CRT output channels (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	activity STRT needs to be executed
	IERR = 2	invalid OPTION value
	IERR = 3	generators are not converted
	IERR = 4	error opening the current channel output file
	IERR = 5	prerequisite requirements for API are not met

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4.1.44. RWDY

Use this API to replicate portions of dynamics model data in the form of either a breaker duty data file, unit inertia and governor data file, or PSSPLT relay characteristic data file.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_RWDY  OPTIONS(1)..OPTIONS(3)  OUT  OFILE
```

Python command syntax:

```
ierr = rwdy(options, out, ofile)
```

Fortran command syntax:

```
CALL RWDYAPI(OPTIONS, OUT, OFILE, IERR)
```

Where:

Integer OPTIONS (3)	array of 3 elements specifying (input).	
	OPTIONS(1)	type of file to be (1 by default)
	OPTIONS(1) = 1	breaker duty data file
	OPTIONS(1) = 2	unit inertia and governor data file
	OPTIONS(1) = 3	PSSPLT relay characteristic data file
	OPTIONS(2)	machine limits flag for activity INLF (ignored unless OPTIONS(1) = 2) (0 by default)
	OPTIONS(2) = 0	include machine limits
	OPTIONS(2) = 1	suppress machine limits
	OPTIONS(3)	flag indicating the treatment of machines without governor models in activity INLF (ignored unless OPTIONS(1) = 2) (0 by default)
	OPTIONS(3) = 0	use the working case limits
	OPTIONS(3) = 1	treat them as non-dispatchable
Integer OUT	destination selection (input; 1 by default).	
	OUT = 0	direct output to a data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the file to which data records are to be written (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPTIONS value
	IERR = 2	invalid OUT value
	IERR = 3	error opening OFILE
	IERR = 4	prerequisite requirements for API are not met

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4.1.45. SET_CHNFIL_TYPE

Use this API to set/Get the channel output file type (extended type or not).

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_SET_CHNFIL_TYPE  STATUS ICODE
```

Python command syntax:

```
ierr = set_chnfil_type(status)

ierr, status = set_chnfil_type()
```

Fortran command syntax:

```
CALL SET_CHNFIL_TYPE(STATUS, ICODE, IERR)
```

Where:

Integer STATUS	value of the option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	STATUS = 1	channel output file is of extended type
	STATUS != 1	channel output file is not of extended type
Integer ICODE	indicates the operation mode of the API (1 by default).	
	ICODE = 0	set the option setting to the value specified in STATUS
	ICODE > 0	return the current value of the option setting in STATUS
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.1.46. SET_GENANG

This API routine is obsolete. It has been replaced by the API routine set_genang_2 .

Use this API to enable or disable the simulation option setting that scans for generators for which the angle differs from the angular average by more than a specified threshold.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SET_GENANG  STATUS  ANGLE
```

Python command syntax:

```
ierr = set_genang(status, angle)

ierr, status,angle = set_genang()
```

Fortran command syntax:

```
CALL SET_GENANG(STATUS, ANGLE, ICODE, IERR)
```

Where:

Integer STATUS	option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	STATUS = 1	enable scanning for generators exceeding angle threshold.
	STATUS != 1	disable scanning option.
Real ANGLE	deviation threshold (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
Integer ICODE	operation mode of the API (input; no default allowed).	
	ICODE <= 0	set the option setting to the value specified in STATUS and the threshold to the value specified in ANGLE.
	ICODE > 0	return the current value of the option setting in STATUS and the threshold in ANGLE.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid angle threshold--must be >0.0 and <360.0
	IERR = 2	prerequisite requirements for API are not met

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4.1.47. SET_GENANG_2

Use this API to enable or disable the simulation option setting that scans for generators for which the angle differs from the angular average by more than a specified threshold. Scanning, if enabled, will be done for all machines whose machine MVA is greater than the specified value

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SET_GENANG_2  STATUS  ANGLE    MBASE
```

Python command syntax:

```
ierr = set_genang_2(status, angle, mbase)

ierr, status,angle,mbase = set_genang_2()
```

Fortran command syntax:

```
CALL SET_GENANG_2(STATUS, ANGLE, MBASE, ICODE, IERR)
```

Where:

Integer STATUS	option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	STATUS = 1	enable scanning for generators exceeding angle threshold.
	STATUS = 1	disable scanning option.
Real ANGLE	deviation threshold (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
Real MBASE	machine size (MVA) (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive). All machines whose MVAR ratings are above this threshold will be scanned).	
Integer ICODE	operation mode of the API (input; no default allowed).	
	ICODE <= 0	set the option setting to the value specified in STATUS and the threshold values specified in ANGLE and MBASE.
	ICODE > 0	return the current value of the option setting in STATUS and the threshold values in ANGLE and MBASE
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid angle threshold--must be >0.0 and <360.0
	IERR = 2	invalid angle threshold--must be >= 0.0
	IERR = 3	prerequisite requirements for API are not met

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4.1.48. SET_GENPWR

Use this API to enable or disable the simulation option setting that scans for generators for which the mechanical power differs from its electrical power by more than a specified threshold.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SET_GENPWR  STATUS  POWER
```

Python command syntax:

```
ierr = set_genpwr(status, power)

ierr, status,power = set_genpwr()
```

Fortran command syntax:

```
CALL SET_GENPWR(STATUS, POWER, ICODE, IERR)
```

Where:

Integer STATUS	option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	STATUS = 1	enable scanning for generators exceeding power unbalance threshold
	STATUS = 0	disable scanning option
Real POWER	power unbalance threshold (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
Integer ICODE	operation mode of the API (input; no default allowed).	
	ICODE <= 0	set the option setting to the value specified in STATUS and the threshold values specified in POWER
	ICODE > 0	return the current value of the option setting in STATUS and the threshold values in POWER
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid power unbalance threshold - must be >0.0 and <10.0
	IERR = 2	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.1.49. SET_NETFRQ

Use this API to enable or disable the simulation option setting that models the frequency dependence of network parameters.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SET_NETFRQ  STATUS
```

Python command syntax:

```
ierr = set_netfrq(status)

ierr, status = set_netfrq()
```

Fortran command syntax:

```
CALL SET_NETFRQ(STATUS, ICODE, IERR)
```

Where:

Integer STATUS	option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	STATUS = 0	disable network frequency dependence
	STATUS = 1	enable network frequency dependence
Integer ICODE	operation mode of the API (no default allowed).	
	ICODE = 0	set the option setting to the value specified in STATUS
	ICODE > 0	return the current value of the option setting in STATUS
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.1.50. SET_NEXT_CHANNEL

Use this API to assign the next available channel index value.

This API routine was first introduced in release 31.0.2.

Batch command syntax:

```
BAT_SET_NEXT_CHANNEL  IVAL
```

Python command syntax:

```
ierr = set_next_channel(ival)
```

Fortran command syntax:

```
CALL SET_NEXT_CHANNEL(IVAL, IERR)
```

Where:

Integer IVAL	value to assign as the next available channel index (input).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid channel index
	IERR = 2	index is beyond the maximum allowed
	IERR = 3	prerequisite requirements for API are not met

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4.1.51. SET_NEXT_ICON

Use this API to assign the next available ICON index value.

This API routine was first introduced in release 31.0.2.

Batch command syntax:

```
BAT_SET_NEXT_ICON  IVAL
```

Python command syntax:

```
ierr = set_next_icon(ival)
```

Fortran command syntax:

```
CALL SET_NEXT_ICON(IVAL, IERR)
```

Where:

Integer IVAL	value to assign as the next available ICON index (input).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid ICON index
	IERR = 2	index is beyond the maximum allowed
	IERR = 3	prerequisite requirements for API are not met

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4.1.52. SET_NEXT_VAR

Use this API to assign the next available VAR index value.

This API routine was first introduced in release 31.0.2.

Batch command syntax:

```
BAT_SET_NEXT_VAR  IVAL
```

Python command syntax:

```
ierr = set_next_var(ival)
```

Fortran command syntax:

```
CALL SET_NEXT_VAR(IVAL, IERR)
```

Where:

Integer IVAL	value to assign as the next available VAR index (input).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid VAR index
	IERR = 2	index is beyond the maximum allowed
	IERR = 3	prerequisite requirements for API are not met

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4.1.53. SET_OSSCAN

Use this API to enable or disable the simulation option setting that scans for out-of-step conditions using a generic relay.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SET_OSSCAN  STATUS  TRIP
```

Python command syntax:

```
ierr = set_osscan(status, trip)

ierr, status,trip = set_osscan()
```

Fortran command syntax:

```
CALL SET_OSSCAN(STATUS, TRIP, ICODE, IERR)
```

Where:

Integer STATUS	option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	STATUS = 0	disable out-of-step scanning
	STATUS = 1	enable out-of-step scanning
Integer TRIP	monitor trip flag (input; unchanged).	
	TRIP = 0	monitor only
	TRIP = 1	trip out-of-step circuits
Integer ICODE	operation mode of the API (input; no default allowed).	
	ICODE <= 0	set the option setting to the value specified in STATUS and the trip flag to the value specified in TRIP.
	ICODE > 0	return the current value of the option setting in STATUS and the trip flag in TRIP.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.1.54. SET_RELANG

Use this API to enable or disable the simulation option setting that expresses the ANGLE array relative to a designated reference angle.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SET_RELANG  SWITCH  IBUSEX  'ID'
```

Python command syntax:

```
ierr = set_relang(switch, ibusex, id)

ierr, switch,ibusex,id = set_relang()
```

Fortran command syntax:

```
CALL SET_RELANG(SWITCH, IBUSEX, ID, ICODE, IERR)
```

Where:

Integer SWITCH	option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	SWITCH = 0	disable relative angle calculation (use absolute angles)
	SWITCH = 1	enable relative angle calculation
Integer IBUSEX	relative angle calculation to be applied (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	IBUSEX > 0	use machine ID at bus IBUSEX as the reference machine
	IBUSEX = 0	use system average angle
	IBUSEX = -1	use system weighted average angle
Character*2 ID	machine identifier (input; if batch command or ICODE is 0 or negative; '1' by default) (if ICODE is positive).	
Integer ICODE	operation mode of the API (input; no default allowed).	
	ICODE < 0	set the option setting to the value specified in SWITCH and the relative angle parameters to the values specified in IBUSEX and ID.
	ICODE > 0	return the current value of the option setting in SWITCH and the relative angle parameters in IBUSEX and ID.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid IBUSEX value
	IERR = 2	bus not found
	IERR = 3	machine not found
	IERR = 4	prerequisite requirements for API are not met

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4.1.55. SET_RELSCN

Use this API to enable or disable the simulation option setting that scans branches using a generic branch relay.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SET_RELSCN  STATUS
```

Python command syntax:

```
ierr = set_relscn(status)

ierr, status = set_relscn()
```

Fortran command syntax:

```
CALL SET_RELSCN(STATUS, ICODE, IERR)
```

Where:

Integer STATUS	option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	STATUS = 0	disable relay scanning
	STATUS = 1	enable relay scanning
Integer ICODE	operation mode of the API (input; no default allowed).	
	ICODE < 0	set the option setting to the value specified in STATUS
	ICODE > 0	return the current value of the option setting in STATUS
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.1.56. SET_VLTSCN

Use this API to enable or disable the simulation option setting that scans buses for high or low values of voltage magnitude.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SET_VLTSCN  STATUS  VHI  VLO
```

Python command syntax:

```
ierr = set_vltscn(status, vhi, vlo)

ierr, status,vhi,vlo = set_vltscn()
```

Fortran command syntax:

```
CALL SET_VLTSCN(STATUS, VHI, VLO, ICODE, IERR)
```

Where:

Integer STATUS	option setting (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	STATUS = 1	enable voltage scanning
	STATUS = 1	disable voltage scanning
Real VHI	high voltage threshold (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
Real VLO	low voltage threshold (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
Integer ICODE	operation mode of the API (input; no default allowed).	
	ICODE < 0	set the option setting to the value specified in STATUS and the voltage limits to VHI and VLO.
	ICODE > 0	return the current value of the option setting in STATUS and the voltage limits in VHI and VLO.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid specification of VHI and VLO
	IERR = 2	prerequisite requirements for API are not met

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4.1.57. SET_VOLTAGE_DIP_CHECK

This API is used to enable or disable the simulation option setting which scans buses for voltage dip (back-swing).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SET_VOLTAGE_DIP_CHECK  VDPCHK   VDPTHR   VDPDUR
```

Python command syntax:

```
ierr = set_voltage_dip_check(vdpchk, vdpthr, vdpdur)
```

Fortran command syntax:

```
CALL SET_VOLTAGE_DIP_CHECK(VDPCHK, VDPTHR, VDPDUR, ICODE, IERR)
```

Where:

Integer VDPCHK	(input; flag to set voltage dip check: if command or ICODE is 0 or negative; unchanged).	
	VDPCHK = 0	disable voltage dip check
	VDPCHK = 1	enable voltage dip check
Real VDPTHR	(input; voltage dip threshold in per unit: if batch command or ICODE is 0 or negative; unchanged by default. if ICODE is positive. If the voltage swings-back below this threshold a timer is imitated to check for the dip duration).	
Real VDPDUR	(input; voltage dip duration in seconds: if batch command or ICODE is 0 or negative; unchanged).	
Integer ICODE	operation mode of API (input; no default allowed).	
	ICODE <= 0	set the option setting to the value specified in VDPCHK, VDPTHR, and VDPDUR.
	ICODE > 0	return the current value of the option setting in VDPCHK, VDPTHR, and VDPDUR.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API not met

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4.1.58. SET_VOLTAGE_REC_CHECK

This API is used to enable or disable the simulation option setting which scans buses for voltage recovery (primary and secondary recovery).

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SET_VOLTAGE_REC_CHECK  VPRCHK   VSRCHK   VPRTHR   VPRTIM   VSRTHR
 VSRTIM
```

Python command syntax:

```
ierr = set_voltage_rec_check(vprchk, vsrchk, vprthr, vprtim, vsrthr,
 vsrtim)
```

Fortran command syntax:

```
CALL SET_VOLTAGE_REC_CHECK(VPRCHK, VSRCHK, VPRTHR, VPRTIM, VSRTHR, VSRTIM,
 ICODE, IERR)
```

Where:

Integer VPRCHK	(input; flag to set voltage primary recovery check: if batch command or ICODE is 0 or negative; unchanged).	
	VPRCHK = 0	disable primary recovery check
	VPRCHK = 1	enable primary recovery check
Integer VSRCHK	(input; flag to set voltage secondary recovery check: if batch command or ICODE is 0 or negative; unchanged).	
	VSRCHK = 0	disable secondary recovery check
	VSRCHK = 1	enable secondary recovery check
Real VPRTHR	(input; voltage primary recovery threshold in per unit: if batch command or ICODE is 0 or negative; unchanged).	
Real VPRTIM	(input; voltage primary recovery time in seconds: if batch command or ICODE is 0 or negative; unchanged).	
Real VSRTHR	(input; voltage secondary recovery threshold in per unit: if batch command or ICODE is 0 or negative; unchanged).	
Real VSRTIM	(input; voltage secondary recovery time in seconds: if batch command or ICODE is 0 or negative; unchanged).	
Integer ICODE	operation mode of API (input; no default allowed).	
	ICODE <= 0	set the option setting to the value specified in VPRCHK, VSRCHK, VPRTHR, VPRTIM, VSRTHR and VSRTIM.
	ICODE > 0	return the current value of the option setting in VPRCHK, VSRCHK, VPRTHR, VPRTIM, VSRTHR and VSRTIM.
Integer IERR	error code (output).	
	IERR = 0	no error

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IERR = 1	invalid specification of VPRTHR and VSRTHR
IERR = 2	invalid specification of VPRTIM and VSRTIM
IERR = 3	prerequisite requirements for API not met

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4.1.59. SET_VOLT_VIOL_SUBSYS_FLAG

This API is used to set the flag to denote the bus subsystem to be used for voltage violation (voltage recovery and voltage dip) checks.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SET_VOLT_VIOL_SUBSYS_FLAG  ALL
```

Python command syntax:

```
ierr = set_volt_viol_subsys_flag(all)
```

Fortran command syntax:

```
CALL SET_VOLT_VIOL_SUBSYS_FLAG(ALL, ICODE, IERR)
```

Where:

Integer ALL	flag to set the subsystem definition for performing voltage violation check (input; if batch command or ICODE is 0 or negative; unchanged by default) (if ICODE is positive).	
	ALL = 0	Use existing bus subsystem definition
	ALL = 1	Do voltage violation check on all buses
Integer ICODE	Indicates the operation mode of API (input; no default allowed).	
	ICODE <= 0	set the option setting to the value specified in ALL
	ICODE > 0	return the current value of the option setting in ALL
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	Invalid ALL value
	IERR = 2	prerequisite requirements for API not met

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4.1.60. SIZE_DS

Use this API to obtain a summary of utilization in the general purpose dynamics data storage arrays as well as in the various model connection and allocation tables contained in dynamics working memory.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_SIZE_DS
```

Python command syntax:

```
ierr = size_ds()
```

Fortran command syntax:

```
CALL SIZE_DS(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.1.61. SNAP

Use this API to save PSSE dynamics working memory into a Snapshot file.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_SNAP  STATUS(1)..STATUS(5)  SFILE
```

Python command syntax:

```
ierr = snap(status, sfile)
```

Fortran command syntax:

```
CALL SNAPNW(STATUS, SFILE, IERR)
```

Where:

Integer STATUS (5)
: array of 5 elements specifying save options. For each entry, -1 may be specified to indicate that the number of elements of the corresponding array(s) to save in the Snapshot File is 1 through the next available - 1(input).

STATUS(1)	number of CONs to save (next available-1 by default)
STATUS(2)	number of STATEs to save (next available-1 by default)
STATUS(3)	number of VARs to save (next available-1 by default)
STATUS(4)	number of ICONs to save (next available-1 by default)
STATUS(5)	number of channels to save (next available-1 by default)

Character*260 SFILE
: snapshot file; '*' to use the most recently accessed snapshot file in the current execution of PSSE (input; no default allowed).

Integer IERR
: error code (output).

IERR = 0	no error occurred
IERR = 1	invalid STATUS value
IERR = 2	error opening SFILE
IERR = 3	error writing to SFILE
IERR = 4	prerequisite requirements for API are not met

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4.1.62. STRT

This API routine is obsolete. It has been replaced by the API routine strt_2 .

Use this API to initialize a PSSE dynamic simulation for state-space simulations (i.e., in preparation for activity RUN) and to specify the channel output file into which the output channel values are to be recorded during the dynamic simulation.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_STRT  OPTION  OUTFILE
```

Python command syntax:

```
ierr = strt(option, outfile)
```

Fortran command syntax:

```
CALL STRTAPI(OPTION, OUTFILE, IERR)
```

Where:

Integer OPTION	network solution convergence monitor option (input; 0 by default).	
	OPTION = 0	print the convergence monitor only if it is enabled via the CM interrupt control code.
	OPTION = 1	automatically print the convergence monitor.
Character*260 OUTFILE	name of the channel output file (blank to bypass recording of the output channel values in a Channel Output File) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	generators are not converted
	IERR = 2	invalid OPTION value
	IERR = 3	Prior initialization modified the loads--pick up original converted case
	IERR = 4	error opening OUTFILE
	IERR = 5	prerequisite requirements for API are not met

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4.1.63. STRT_2

This API routine is the second release of the dynamic simulation initialization function. Use this API routine to initialize a PSSE dynamic simulation for state-space simulations (i.e., in preparation for activity RUN) and to specify the channel output file into which the output channel values are to be recorded during the dynamic simulation.

This API routine was first introduced in release 33.9.0 and 34.2.0.

Batch command syntax:

```
BAT_STRT_2  OPTIONS  OUTFILE
```

Python command syntax:

```
ierr = strt_2(options, outfile)
```

Fortran command syntax:

```
CALL STRT_2(OPTIONS, OUTFILE, IERR)
```

Where:

Integer OPTIONS (2)	array of 2 elements specifying simulation initialization options(input).	
	OPTIONS(1)	Network solution convergence monitor option
	OPTIONS(1) = 0	bypass printing of the network solution convergence monitor
	OPTIONS(1) = 1	print the network solution convergence monitor
	OPTIONS(2)	Missing machine model netting option.
	OPTIONS(2) = 0	treat as a fatal error
	OPTIONS(2) = 1	net machine power
Character*260 OUTFILE	channel output file (blank to bypass recording of the output channel values in a channel output file) (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	generators are not converted
	IERR = 2	invalid OPTIONS value
	IERR = 3	prior initialization modified the loads - pick up original converted case
	IERR = 4	error opening OUTFILE
	IERR = 5	prerequisite requirements for API are not met

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4.1.64. SWSLIST

Use this API to list switched shunt models connected to subsystem buses along with their storage locations in the dynamics data arrays.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SWSLIST  SID  ALL  STATUS
```

Python command syntax:

```
ierr = swslist(sid, all, status)
```

Fortran command syntax:

```
CALL SWSLIST(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID
	ALL = 1	process all buses
Integer STATUS	active/bypassed flag (input; 0 by default).	
	STATUS = 0	active and bypassed models
	STATUS = 1	active models only
	STATUS = 2	bypassed models only
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.1.65. TRIG_VOLT_VIOLATION_CHECK

This API is used to trigger voltage violation (voltage recovery, voltage dip) checks.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_TRIG_VOLT_VIOLATION_CHECK  STATUS
```

Python command syntax:

```
ierr = trig_volt_violation_check(status)
```

Fortran command syntax:

```
CALL TRIG_VOLT_VIOLATION_CHECK(STATUS, ICODE, IERR)
```

Where:

Integer STATUS	option setting (input).	
	STATUS = 0	disable voltage violation check
	STATUS = 1	trigger voltage violation
Integer ICODE	operation mode of the API (input; no default allowed).	
	ICODE <= 0	set the option setting to the value specified in STATUS
	ICODE > 0	return the current value of the option setting in STATUS
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API not met

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4.1.66. WNLIST

Use this API to list wind models connected to subsystem buses along with their storage locations in the dynamics data arrays.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_WNLIST  SID  ALL  STATUS
```

Python command syntax:

```
ierr = wnlist(sid, all, status)
```

Fortran command syntax:

```
CALL WNLIST(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 0	process only buses in subsystem SID.
	ALL = 1	process all buses.
Integer STATUS (2)	array of 2 elements specifying API options(input).	
	STATUS(1)	active/bypassed flag (0 by default)
	STATUS(1) = 0	active and bypassed models
	STATUS(1) = 1	active models only
	STATUS(1) = 2	bypassed models only
	STATUS(2)	wind model type (1 by default)
	STATUS(2) = 1	all wind models
	STATUS(2) = 2	wind generator models
	STATUS(2) = 3	wind electrical models
	STATUS(2) = 4	wind mechanical models
	STATUS(2) = 5	wind pitch models
	STATUS(2) = 6	wind aerodynamic models
	STATUS(2) = 7	wind gust models
	STATUS(2) = 8	wind auxiliary signal models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.2. Output Chanel Operations

4.2.1. Dynamic Simulation Output Channel Operation

4.2.1.1. CHANGE_CHANNEL

Use this API to change the channel address and the channel identifier of a specified output channel.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_CHANGE_CHANNEL  CHNARG  ADRARG  IDNARG
```

Python command syntax:

```
ierr = change_channel(chnarg,adrarg, idnarg)
```

Fortran command syntax:

```
CALL CHANGE_CHANNEL(CHNARG, ADRARG, IDNARG, IERR)
```

Where:

Integer CHNARG	channel number for which the channel address and/or channel identifier has to be changed (input; no default allowed).	
Integer ADRARG	channel address (input; unchanged).	
Character*32 IDNARG	channel identifier (input; unchanged).	
Integer IERR	error code (output).	
	IERR = -1	warnings - one or more of: - channel number is larger than NCHAN
	IERR = 0	no error
	IERR = 1	invalid channel number
	IERR = 2	channel number is at or beyond the next available channel
	IERR = 3	invalid channel address
	IERR = 4	prerequisite requirements for API are not met

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4.2.1.2. CHSB

Use this API to specify, on a subsystem basis, the simulation variables to monitor during dynamic simulation runs (activity CHSB).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CHSB  SID  ALL  STATUS(1)..STATUS(6)
```

Python command syntax:

```
ierr = chsb(sid, all, status)
```

Fortran command syntax:

```
CALL CHSBNW(SID, ALL, STATUS, IERR)
```

Where:

Integer SID	valid area, bus, owner, or zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Area, bus, owner, or zone subsystem SID must have been previously defined (input; 0 by default).	
Integer ALL	all buses or specified subsystem flag (input; 1 by default).	
	ALL = 1	process all buses
	ALL = 0	process only buses in subsystem SID
Integer STATUS (6)	array of 6 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	used to indicate the type of quantities to be placed in channels (1 by default)
	STATUS(4) = 1	machine, bus, load, branch quantities by bus subsystem
	STATUS(4) = 2	tie quantity by area subsystem
	STATUS(4) = 3	tie quantity by zone subsystem
	STATUS(4) = 4	power totals for an area subsystem
	STATUS(4) = 5	power totals for a zone subsystem
	STATUS(4) = 6	power totals for an owner subsystem
	STATUS(4) = 7	power totals for all buses
	STATUS(4) = 8	machine angle statistics
	STATUS(5)	quantity code (1 by default). If STATUS(4) = 4, 5, 6, 7 or 8 then STATUS(5) is ignored)
	If STATUS(4) = 1:	
	STATUS(5) = 1	ANGLE, machine relative rotor angle (degrees).

STATUS(5) = 2	PELEC, machine electrical power (pu on SBASE).
STATUS(5) = 3	QELEC, machine reactive power.
STATUS(5) = 4	ETERM, machine terminal voltage (pu).
STATUS(5) = 5	EFD, generator main field voltage (pu).
STATUS(5) = 6	PMECH, turbine mechanical power (pu on MBASE).
STATUS(5) = 7	SPEED, machine speed deviation from nominal (pu).
STATUS(5) = 8	XADIFD, machine field current (pu).
STATUS(5) = 9	ECOMP, voltage regulator compensated voltage (pu).
STATUS(5) = 10	VOTHSG, stabilizer output signal (pu).
STATUS(5) = 11	VREF, voltage regulator voltage setpoint (pu).
STATUS(5) = 12	BSFREQ, bus pu frequency deviations.
STATUS(5) = 13	VOLT, bus pu voltages (complex).
STATUS(5) = 14	voltage and angle
STATUS(5) = 15	flow (P).
STATUS(5) = 16	flow (P and Q).
STATUS(5) = 17	flow (MVA).
STATUS(5) = 18	apparent impedance (R and X).
STATUS(5) = 21	ITERM.
STATUS(5) = 22	machine apparent impedance
STATUS(5) = 23	VUEL, minimum excitation limiter output signal (pu).
STATUS(5) = 24	VOEL, maximum excitation limiter output signal (pu).
STATUS(5) = 25	PLOAD.
STATUS(5) = 26	QLOAD.
STATUS(5) = 27	GREF, turbine governor reference.
STATUS(5) = 28	LCREF, turbine load control reference.
STATUS(5) = 29	WVLCTY, wind velocity (m/s).
STATUS(5) = 30	WTRBSP, wind turbine rotor speed deviation (pu).
STATUS(5) = 31	WPITCH, pitch angle (degrees).
STATUS(5) = 32	WAEROT, aerodynamic torque (pu on MBASE).
STATUS(5) = 33	WROTRV, rotor voltage (pu on MBASE).
STATUS(5) = 34	WROTRI, rotor current (pu on MBASE).
STATUS(5) = 35	WPCMND, active power command from wind control (pu on MBASE).
STATUS(5) = 36	WQCMND, reactive power command from wind control (pu on MBASE).

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	STATUS(5) = 37	WAUXSG, output of wind auxiliary control (pu on MBASE).
	STATUS(6)	out-of-service equipment option (0 by default)
	STATUS(6) = 0	omit out-of-service equipment
	STATUS(6) = 1	include out-of-service equipment
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid SID value or subsystem SID is not defined
	IERR = 2	invalid ALL value
	IERR = 3	invalid STATUS value
	IERR = 4	have just assigned channel n, which is the highest channel number
	IERR = 5	trying to assign locations beyond the end of the VAR or ICON array
	IERR = 6	machine angle statistics are already assigned to channels
	IERR = 7	n monitoring models specified--pointer table full
	IERR = 8	next channel is greater than the maximum allowed channel number
	IERR = 9	prerequisite requirements for API are not met

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4.2.1.3. CRT_PLOT_CHANNEL

Use this API to assign one of the main simulation output channels to a designated CRT plot channel.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CRT_PLOT_CHANNEL  CRTCHAN  CHANNEL  CMIN  CMAX
```

Python command syntax:

```
ierr = crt_plot_channel(crtchan, channel, cmin, cmax)
```

Fortran command syntax:

```
CALL CRT_PLOT_CHANNEL(CRTCHAN, CHANNEL, CMIN, CMAX, IERR)
```

Where:

Integer CRTCHAN	CRT plot channel number (1 through 6) (input; no default allowed).	
Integer CHANNEL	number of the main simulation channel to be assigned to CRT channel number CRTCHAN (must be between 1 and the number of output channels allowed at the present size level of PSSE), or 0 to disable CRT plot channel CRTCHAN (input; unchanged).	
Real CMIN	minimum plotting scale value for this CRT plot channel (input; unchanged).	
Real CMAX	maximum plotting scale value for this CRT plot channel (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	error occurred
	IERR = 1	invalid CRTCHAN value
	IERR = 2	invalid CHANNEL value
	IERR = 3	CMIN is not less than CMAX
	IERR = 4	prerequisite requirements for API are not met
	IERR = -1	warnings - one or more of: - CHANNEL is larger than NCHAN

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4.2.1.4. DELETE_ALL_PLOT_CHANNELS

Use this API to delete all the plot channels in the working case. Use of this API will reset the number of plot channels to zero and the next channel to one.

This API routine was first introduced in release 32.2.1.

Batch command syntax:

```
BAT_DELETE_ALL_PLOT_CHANNELS
```

Python command syntax:

```
ierr = delete_all_plot_channels()
```

Fortran command syntax:

```
CALL DELETE_ALL_PLOT_CHANNELS(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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4.2.1.5. DMPC

Use this API to replicate the output channel specifications from dynamics working memory in the form of a response file suitable for transferring them to another snapshot that models essentially the same system.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DMPC  OPTIONS(1)..OPTIONS(4)  OUT  OFILE
```

Python command syntax:

```
ierr = dmpc(options, out, ofile)
```

Fortran command syntax:

```
CALL DMPCAPI(OPTIONS, OUT, OFILE, IERR)
```

Where:

Integer OPTIONS (4)	array of 4 elements specifying (input).	
	OPTIONS(1)	channel selection code (1 by default)
	OPTIONS(1) = 0	skip channels containing STATEs or VARs not associated with channel monitoring models
	OPTIONS(1) = 1	include all channels
	OPTIONS(2)	starting channel code (-1 by default)
	OPTIONS(2) = 0	start at next available channel
	OPTIONS(2) = 0	keep existing channel numbers
	OPTIONS(2) = 0	starting channel number
	OPTIONS(3)	starting VAR code for channel monitoring models (-1 by default)
	OPTIONS(3) = 0	start at next available VAR
	OPTIONS(3) = 0	keep existing VAR indices
	OPTIONS(3) = 0	starting VAR index
	OPTIONS(4)	starting ICON code for channel monitoring models (-1 by default)
	OPTIONS(4) = 0	start at next available ICON
	OPTIONS(4) = 0	keep existing ICON indices
	OPTIONS(4) = 0	starting ICON index
Integer OUT	destination selection (input; 1 by default).	
	OUT = 0	direct output to a data file specified by OFILE
	OUT = 1	direct output to the progress window
Character*260 OFILE	name of the file to which data records are to be written (when OUT is 0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPTIONS(1) value

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IERR = 2	starting channel number is greater than the largest channel number allowed
IERR = 3	starting VAR index is greater than the largest VAR index allowed
IERR = 4	starting ICON index is greater than the largest ICON index allowed
IERR = 5	invalid OUT value
IERR = 6	error opening file OFILE
IERR = 7	prerequisite requirements for API are not met

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4.2.1.6. LIST_CHANNEL_MODELS

Use this API to tabulate the output channel monitoring models referenced in the dynamic model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_LIST_CHANNEL_MODELS  STATUS
```

Python command syntax:

```
ierr = list_channel_models(status)
```

Fortran command syntax:

```
CALL LIST_CHANNEL_MODELS(STATUS, IERR)
```

Where:

Integer STATUS	category of channel monitoring models to be listed (input; 0 by default).	
	STATUS = 0	all output channel monitoring models
	STATUS = 1	only models monitoring equipment that exists in the power flow working case
	STATUS = 2	only models monitoring equipment that is not present in the power flow working case
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	invalid STATUS value
	IERR = 2	prerequisite requirements for API are not met

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4.2.1.7. REMOVE_UNUSED_CHAN_MODELS

Use this API to delete from the output channel monitoring model table those model references indicating equipment that is not present in the power flow working case.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_REMOVE_UNUSED_CHAN_MODELS
```

Python command syntax:

```
ierr = remove_unused_chan_models()
```

Fortran command syntax:

```
CALL REMOVE_UNUSED_CHAN_MODELS(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	prerequisite requirements for API are not met

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4.2.2. Assign Machine Quantities to Output Channels

4.2.2.1. MACHINE_APP_R_X_CHANNEL

Use this API to add a pair of output channels containing the apparent impedance of the system, as seen from the terminals of a specified machine, along with a corresponding call to the GENTMZ monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MACHINE_APP_R_X_CHANNEL  STATUS(1)..STATUS(4)  'ID'  'IDENT(1)'
 'IDENT(2)'
```

Python command syntax:

```
ierr = machine_app_r_x_channel(status, id, ident)
```

Fortran command syntax:

```
CALL MACHINE_APP_R_X_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (4)	array of 4 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the bus to which the machine for which the apparent impedance is to be placed in channels is connected (no default allowed)
Character*2 ID	machine identifier (input; '1').	
Character*32 IDENT (2)	two identifiers to be assigned to the two channels. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed
	IERR = 4	starting ICON index is greater than the largest ICON index allowed

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IERR = 5	model GENTMZ needs 2 VARs but the last one exceeds the largest VAR index allowed
IERR = 6	model GENTMZ needs 2 ICONs but the last one exceeds the largest ICON index allowed
IERR = 7	the maximum number of channel monitoring models has already been specified
IERR = 8	model GENTMZ needs 2 channels but the last one exceeds the largest channel number allowed
IERR = 9	bus not found
IERR = 10	machine not found
IERR = 11	prerequisite requirements for API are not met

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4.2.2.2. MACHINE_ARRAY_CHANNEL

Use this API to add an output channel containing a plant related model variable of a designated type for a specified machine.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MACHINE_ARRAY_CHANNEL  STATUS(1)..STATUS(3)  'ID'  'IDENT'
```

Python command syntax:

```
ierr = machine_array_channel(status, id, ident)
```

Fortran command syntax:

```
CALL MACHINE_ARRAY_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (3)	array of 3 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	used to indicate the quantity to be placed in an channel (1 by default)
	STATUS(2) = 1	ANGLE, machine relative rotor angle (degrees)
	STATUS(2) = 2	PELEC, machine electrical power (pu on SBASE)
	STATUS(2) = 3	QELEC, machine reactive power
	STATUS(2) = 4	ETERM, machine terminal voltage (pu)
	STATUS(2) = 5	EFD, generator main field voltage (pu)
	STATUS(2) = 6	PMECH, turbine mechanical power (pu on MBASE)
	STATUS(2) = 7	SPEED, machine speed deviation from nominal (pu)
	STATUS(2) = 8	XADIFD, machine field current (pu)
	STATUS(2) = 9	ECOMP, voltage regulator compensated voltage (pu)
	STATUS(2) = 10	VOTHSG, stabilizer output signal (pu)
	STATUS(2) = 11	VREF, voltage regulator voltage setpoint (pu)
	STATUS(2) = 12	VUEL, minimum excitation limiter output signal (pu)
	STATUS(2) = 13	VOEL, maximum excitation limiter output signal (pu)
	STATUS(2) = 14	GREF, turbine governor reference
	STATUS(2) = 15	LCREF, turbine load control reference
	STATUS(2) = 16	WVLCTY, wind velocity (m/s)

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	STATUS(2) = 17	WTRBSP, wind turbine rotor speed deviation (pu)
	STATUS(2) = 18	WPITCH, pitch angle (degrees)
	STATUS(2) = 19	WAEROT, aerodynamic torque (pu on MBASE)
	STATUS(2) = 20	WROTRV, rotor voltage (pu on MBASE)
	STATUS(2) = 21	WROTRI, rotor current (pu on MBASE)
	STATUS(2) = 22	WPCMND, active power command from wind control (pu on MBASE)
	STATUS(2) = 23	WQCMND, reactive power command from wind control (pu on MBASE)
	STATUS(2) = 24	WAUXSG, output of wind auxiliary control (pu on MBASE)
	STATUS(3)	number of the bus to which the machine for which the quantity is to be placed in an channel is connected (no default allowed)
Character*2 ID	machine identifier (input; '1').	
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	bus not found
	IERR = 4	machine not found
	IERR = 5	prerequisite requirements for API are not met

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4.2.2.3. MACHINE_ITERM_CHANNEL

Use this API to add an output channel containing the terminal current of a specified machine, along with a corresponding call to the GENTMC monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MACHINE_ITERM_CHANNEL  STATUS(1)..STATUS(4)  'ID'  'IDENT'
```

Python command syntax:

```
ierr = machine_iterm_channel(status, id, ident)
```

Fortran command syntax:

```
CALL MACHINE_ITERM_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (4)	array of 4 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the bus to which the machine for which the terminal current is to be placed in an channel is connected (no default allowed)
Character*2 ID	machine identifier (input; '1').	
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed
	IERR = 4	starting ICON index is greater than the largest ICON index allowed
	IERR = 5	model GENTMC needs 2 VARs but the last one exceeds the largest VAR index allowed
	IERR = 6	model GENTMC needs 2 ICONs but the last one exceeds the largest ICON index allowed
	IERR = 7	the maximum number of channel monitoring models has already been specified

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IERR = 8	bus not found
IERR = 9	machine not found
IERR = 10	prerequisite requirements for API are not met

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4.2.3. Assign Branch Quantities to Output Channels

4.2.3.1. BRANCH_APP_R_X_CHANNEL

Use this API to add a pair of output channels containing the apparent impedance, as seen at the from bus of a specified branch, along with a corresponding call to the RELAY2 monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_BRANCH_APP_R_X_CHANNEL  STATUS(1)..STATUS(5)  'ID'  'IDENT(1)'
 'IDENT(2)'
```

Python command syntax:

```
ierr = branch_app_r_x_channel(status, id, ident)
```

Fortran command syntax:

```
CALL BRANCH_APP_R_X_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (5)	array of 5 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the from bus of the branch for which apparent impedance is to be placed in channels (no default allowed)
	STATUS(5)	number of the to bus of the branch for which apparent impedance is to be placed in channels (no default allowed)
Character*2 ID	circuit identifier (input; '1').	
Character*32 IDENT (2)	two identifiers to be assigned to the two channels. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed
	IERR = 4	starting ICON index is greater than the largest ICON index allowed

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IERR = 5	model RELAY2 needs 3 ICONs but the last one exceeds the largest ICON index allowed
IERR = 6	the maximum number of channel monitoring models has already been specified
IERR = 7	model RELAY2 needs 2 VARs but the last one exceeds the largest VAR index allowed
IERR = 8	model RELAY2 needs 2 channels but the last one exceeds the largest channel number allowed
IERR = 9	bus not found
IERR = 10	branch not found
IERR = 11	prerequisite requirements for API are not met

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4.2.3.2. BRANCH_MVA_CHANNEL

Use this API to add an output channel containing the MVA flow at the from bus of a specified branch, along with a corresponding call to the FLOW1 monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_BRANCH_MVA_CHANNEL  STATUS(1)..STATUS(5)  'ID'  'IDENT'
```

Python command syntax:

```
ierr = branch_mva_channel(status, id, ident)
```

Fortran command syntax:

```
CALL BRANCH_MVA_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (5)	array of 5 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the from bus of the branch for which MVA flow is to be placed in an channel (no default allowed)
	STATUS(5)	number of the to bus of the branch for which MVA is to be placed in an channel (no default allowed)
Character*2 ID	circuit identifier (input; '1').	
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed
	IERR = 4	starting ICON index is greater than the largest ICON index allowed
	IERR = 5	model FLOW1 needs 3 ICONs but the last one exceeds the largest ICON index allowed

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IERR = 6	the maximum number of channel monitoring models has already been specified
IERR = 7	bus not found
IERR = 8	branch not found
IERR = 9	prerequisite requirements for API are not met

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4.2.3.3. BRANCH_P_AND_Q_CHANNEL

Use this API to add a pair of output channels containing the active and reactive power flow at the from bus of a specified branch, along with a corresponding call to the FLOW1 monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_BRANCH_P_AND_Q_CHANNEL  STATUS(1)..STATUS(5)  'ID'  'IDENT(1)'
 'IDENT(2)'
```

Python command syntax:

```
ierr = branch_p_and_q_channel(status, id, ident)
```

Fortran command syntax:

```
CALL BRANCH_P_AND_Q_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (5)	array of 5 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the from bus of the branch for which active and reactive power flows are to be placed in channels (no default allowed)
	STATUS(5)	number of the to bus of the branch for which active and reactive power flows are to be placed in channels (no default allowed)
Character*2 ID	circuit identifier (input; '1').	
Character*32 IDENT (2)	two identifiers to be assigned to the two channels. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed
	IERR = 4	starting ICON index is greater than the largest ICON index allowed
	IERR = 5	model FLOW1 needs 3 ICONs but the last one exceeds the largest ICON index allowed

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IERR = 6	the maximum number of channel monitoring models has already been specified
IERR = 7	model FLOW1 needs 2 VARs but the last one exceeds the largest VAR index allowed
IERR = 8	model FLOW1 needs 2 channels but the last one exceeds the largest channel number allowed
IERR = 9	bus not found
IERR = 10	branch not found
IERR = 11	prerequisite requirements for API are not met

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4.2.3.4. BRANCH_P_CHANNEL

Use this API to add an output channel containing the active power flow at the from bus of a specified branch, along with a corresponding call to the FLOW1 monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_BRANCH_P_CHANNEL  STATUS(1)..STATUS(5)  'ID'  'IDENT'
```

Python command syntax:

```
ierr = branch_p_channel(status, id, ident)
```

Fortran command syntax:

```
CALL BRANCH_P_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (5)	array of 5 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the from bus of the branch for which active power flow is to be placed in an channel (no default allowed)
	STATUS(5)	number of the to bus of the branch for which active power flow is to be placed in an channel (no default allowed)
Character*2 ID	circuit identifier (input; '1').	
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed
	IERR = 4	starting ICON index is greater than the largest ICON index allowed
	IERR = 5	model FLOW1 needs 3 ICONs but the last one exceeds the largest ICON index allowed

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IERR = 6	the maximum number of channel monitoring models has already been specified
IERR = 7	bus not found
IERR = 8	branch not found
IERR = 9	prerequisite requirements for API are not met

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4.2.3.5. THREE_WND_APP_R_X_CHANNEL

Use this API to add a pair of output channels containing the apparent impedance as seen at the from bus of a specified three-winding transformer, along with a corresponding call to the RELAY3 monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_THREE_WND_APP_R_X_CHANNEL  STATUS(1)..STATUS(6)  'ID'  'IDENT(1)'
 'IDENT(2)'
```

Python command syntax:

```
ierr = three_wnd_app_r_x_channel(status, id, ident)
```

Fortran command syntax:

```
CALL THREE_WINDING_APP_R_X_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (6)	array of 6 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the from bus of the three-winding transformer for which apparent impedance is to be placed in channels (no default allowed)
	STATUS(5)	number of one of the to buses of the three-winding transformer for which apparent impedance is to be placed in channels (no default allowed)
	STATUS(6)	number of the third bus of the three-winding transformer for which apparent impedance is to be placed in channels (no default allowed)
Character*2 ID	circuit identifier (input; '1').	
Character*32 IDENT (2)	two identifiers to be assigned to the two channels. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed

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IERR = 4	starting ICON index is greater than the largest ICON index allowed
IERR = 5	model RELAY3 needs 4 ICONs but the last one exceeds the largest ICON index allowed
IERR = 6	the maximum number of channel monitoring models has already been specified
IERR = 7	model RELAY3 needs 2 VARs but the last one exceeds the largest VAR index allowed
IERR = 8	model RELAY3 needs 2 channels but the last one exceeds the largest channel number allowed
IERR = 9	bus not found
IERR = 10	branch not found
IERR = 11	prerequisite requirements for API are not met

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4.2.3.6. THREE_WND_MVA_CHANNEL

Use this API to add an output channel containing the MVA flow at the from bus of a specified three-winding transformer, along with a corresponding call to the FLOW3 monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_THREE_WND_MVA_CHANNEL  STATUS(1)..STATUS(6)  'ID'  'IDENT'
```

Python command syntax:

```
ierr = three_wnd_mva_channel(status, id, ident)
```

Fortran command syntax:

```
CALL THREE_WINDING_MVA_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (6)	array of 6 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the from bus of the three-winding transformer for which MVA flow is to be placed in an channel (no default allowed)
	STATUS(5)	number of one of the to buses of the three-winding transformer for which MVA flow is to be placed in an channel (no default allowed)
	STATUS(6)	number of the third bus of the three-winding transformer for which MVA flow is to be placed in an channel (no default allowed)
Character*2 ID	circuit identifier (input; '1').	
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed

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IERR = 4	starting ICON index is greater than the largest ICON index allowed
IERR = 5	model FLOW3 needs 4 ICONs but the last one exceeds the largest ICON index allowed
IERR = 6	the maximum number of channel monitoring models has already been specified
IERR = 7	bus not found
IERR = 8	branch not found
IERR = 9	prerequisite requirements for API are not met

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4.2.3.7. THREE_WND_P_AND_Q_CHANNEL

Use this API to add a pair of output channels containing the active and reactive power flow at the from bus of a specified three-winding transformer, along with a corresponding call to the FLOW3 monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_THREE_WND_P_AND_Q_CHANNEL  STATUS(1)..STATUS(6)  'ID'  'IDENT(1)'
 'IDENT(2)'
```

Python command syntax:

```
ierr = three_wnd_p_and_q_channel(status, id, ident)
```

Fortran command syntax:

```
CALL THREE_WINDING_P_AND_Q_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (6)	array of 6 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the from bus of the three-winding transformer for which active and reactive power flows are to be placed in channels (no default allowed)
	STATUS(5)	number of one of the to buses of the three-winding transformer for which active and reactive power flows are to be placed in channels (no default allowed)
	STATUS(6)	number of the third bus of the three-winding transformer for which active and reactive power flows are to be placed in channels (no default allowed)
Character*2 ID	circuit identifier (input; '1').	
Character*32 IDENT (2)	two identifiers to be assigned to the two channels. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed

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IERR = 3	starting VAR index is greater than the largest VAR index allowed
IERR = 4	starting ICON index is greater than the largest ICON index allowed
IERR = 5	model FLOW3 needs 4 ICONs but the last one exceeds the largest ICON index allowed
IERR = 6	the maximum number of channel monitoring models has already been specified
IERR = 7	model FLOW3 needs 2 VARs but the last one exceeds the largest VAR index allowed
IERR = 8	model FLOW3 needs 2 channels but the last one exceeds the largest channel number allowed
IERR = 9	bus not found
IERR = 10	branch not found
IERR = 11	prerequisite requirements for API are not met

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4.2.3.8. THREE_WND_P_CHANNEL

Use this API to add an output channel containing the active power flow at the from bus of a specified three-winding transformer, along with a corresponding call to the FLOW3 monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_THREE_WND_P_CHANNEL  STATUS(1)..STATUS(6)  'ID'  'IDENT'
```

Python command syntax:

```
ierr = three_wnd_p_channel(status, id, ident)
```

Fortran command syntax:

```
CALL THREE_WINDING_P_CHANNEL(STATUS, ID, IDENT, IERR)
```

Where:

Integer STATUS (6)	array of 6 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the from bus of the three-winding transformer for which active power flow is to be placed in an channel (no default allowed)
	STATUS(5)	number of one of the to buses of the three-winding transformer for which active power flow is to be placed in an channel (no default allowed)
	STATUS(6)	number of the third bus of the three-winding transformer for which active power flow is to be placed in an channel (no default allowed)
Character*2 ID	circuit identifier (input; '1').	
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed

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IERR = 4	starting ICON index is greater than the largest ICON index allowed
IERR = 5	model FLOW3 needs 4 ICONs but the last one exceeds the largest ICON index allowed
IERR = 6	the maximum number of channel monitoring models has already been specified
IERR = 7	bus not found
IERR = 8	branch not found
IERR = 9	prerequisite requirements for API are not met

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4.2.4. Assign Other Quantities to Output Channels

4.2.4.1. BUS_FREQUENCY_CHANNEL

Use this API to add an output channel containing the per unit frequency deviation at a specified bus.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_BUS_FREQUENCY_CHANNEL  STATUS(1)  STATUS(2)  'IDENT'
```

Python command syntax:

```
ierr = bus_frequency_channel(status, ident)
```

Fortran command syntax:

```
CALL BUS_FREQUENCY_CHANNEL(STATUS, IDENT, IERR)
```

Where:

Integer STATUS (2)	array of 2 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	number of the bus for which frequency deviation is to be placed in an channel is connected (no default allowed)
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	bus not found
	IERR = 4	prerequisite requirements for API are not met

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4.2.4.2. STATE_CHANNEL

Use this API to add an output channel containing the value of a specified STATE.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_STATE_CHANNEL  STATUS(1)  STATUS(2)  'IDENT'
```

Python command syntax:

```
ierr = state_channel(status, ident)
```

Fortran command syntax:

```
CALL STATE_CHANNEL(STATUS, IDENT, IERR)
```

Where:

Integer STATUS (2)	array of 2 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	index of the STATE for which the value is to be placed in an channel (no default allowed)
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	invalid STATE index
	IERR = 4	prerequisite requirements for API are not met

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4.2.4.3. VAR_CHANNEL

Use this API to add an output channel containing the value of a specified VAR.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_VAR_CHANNEL  STATUS(1)  STATUS(2)  'IDENT'
```

Python command syntax:

```
ierr = var_channel(status, ident)
```

Fortran command syntax:

```
CALL VAR_CHANNEL(STATUS, IDENT, IERR)
```

Where:

Integer STATUS (2)	array of 2 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	index of the VAR for which the value is to be placed in an channel (no default allowed)
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	invalid VAR index
	IERR = 4	prerequisite requirements for API are not met

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4.2.4.4. VOLTAGE_AND_ANGLE_CHANNEL

Use this API to add a pair of output channels containing the voltage magnitude in per unit and phase angle in degrees of a specified bus, along with a corresponding call to the VOLMAG monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_VOLTAGE_AND_ANGLE_CHANNEL  STATUS(1)..STATUS(4)  'IDENT(1)'
 'IDENT(2)'
```

Python command syntax:

```
ierr = voltage_and_angle_channel(status, ident)
```

Fortran command syntax:

```
CALL VOLTAGE_AND_ANGLE_CHANNEL(STATUS, IDENT, IERR)
```

Where:

Integer STATUS (4)	array of 4 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the bus for which voltage magnitude and phase angle are to be placed in channels (no default allowed)
Character*32 IDENT (2)	two identifiers to be assigned to the two channels. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed
	IERR = 4	starting ICON index is greater than the largest ICON index allowed
	IERR = 5	the maximum number of channel monitoring models has already been specified
	IERR = 6	bus not found
	IERR = 7	model VOLMAG needs 2 VARs but the last one exceeds the largest VAR index allowed

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IERR = 8	model VOLMAG needs 2 channels but the last one exceeds the largest channel number allowed
IERR = 9	prerequisite requirements for API are not met

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4.2.4.5. VOLTAGE_CHANNEL

Use this API to add an output channel containing the voltage magnitude in per unit of a specified bus, along with a corresponding call to the VOLMAG monitoring model.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_VOLTAGE_CHANNEL  STATUS(1)..STATUS(4)  'IDENT'
```

Python command syntax:

```
ierr = voltage_channel(status, ident)
```

Fortran command syntax:

```
CALL VOLTAGE_CHANNEL(STATUS, IDENT, IERR)
```

Where:

Integer STATUS (4)	array of 4 elements specifying (input).	
	STATUS(1)	starting channel index, or -1 for the next available (next available by default)
	STATUS(2)	starting VAR index, or -1 for the next available (next available by default)
	STATUS(3)	starting ICON index, or -1 for the next available (next available by default)
	STATUS(4)	number of the bus for which voltage magnitude is to be placed in an channel (no default allowed)
Character*32 IDENT	identifier to be assigned to the channel. If a blank channel identifier is specified, the API generates an appropriate channel identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid STATUS value
	IERR = 2	starting channel number is greater than the largest channel number allowed
	IERR = 3	starting VAR index is greater than the largest VAR index allowed
	IERR = 4	starting ICON index is greater than the largest ICON index allowed
	IERR = 5	the maximum number of channel monitoring models has already been specified
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

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4.3. Plant Related Models

4.3.1. ADD_PLANT_MODEL

Use this API to add a plant related model of a designated type to a specified machine. If a model of the designated type already exists at the machine, it is replaced.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ADD_PLANT_MODEL  IBUS  'ID'  MTYPE  'NAME'  IBUS2 'ID2'  NICN  IDATA
 CHDATA  NCON  RDATA
```

Python command syntax:

```
ierr = add_plant_model(ibus, id, mtype, name, ibus2, id2, nicn, idata,
 chdata, ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_PLANT_MODEL(IBUS, ID, MTYPE, NAME, IBUS2, ID2, NICN, IDATA,
 CHDATA, NCON, RDATA, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).
Character*2 ID	machine identifier (input; blank by default).
Integer MTYPE	model type (input; 1 by default). MTYPE = 1 generator model. MTYPE = 2 compensator model. MTYPE = 3 stabilizer model. MTYPE = 4 minimum excitation limiter model. MTYPE = 5 maximum excitation limiter model. MTYPE = 6 excitation system model. MTYPE = 7 turbine-governor model. MTYPE = 8 turbine-load controller model.
Character*16 NAME	model name (input; no default allowed).
Integer IBUS2	bus number of the second machine if this is a two machine model, or 0 for a single machine model (input; 0 by default).
Character*2 ID2	machine identifier of the second machine if this is a two machine model, or ' ' for a single machine model (input; blank by default).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended

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	for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	machine model connection tables full--use pack plant model tables function
	IERR = 2	plant model tables are corrupt
	IERR = 3	invalid MTYPE value
	IERR = 4	unknown model NAME
	IERR = 5	model is not consistent with MTYPE
	IERR = 6	# CONs specified is not the number expected
	IERR = 7	# ICONs specified is not the number expected
	IERR = 8	second machine not found where one is required
	IERR = 9	same machine specified as the first and second machines
	IERR = 10	maximum number of array elements exceeded
	IERR = 11	bus not found
	IERR = 12	machine not found
	IERR = 13	prerequisite requirements for API are not met

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4.3.2. CHANGE_PLMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of a designated plant related model at a specified machine. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_PLMOD_CHRICN  IBUS  'ID'  'NAME'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_plmod_chricn(ibus, id, name, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_PLMOD_CHRICN(IBUS, ID, NAME, J, CHDATA, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a plant-related model
	IERR = 3	no <model type> model at this machine
	IERR = 4	the <model type> model at this machine is aaa not NAME
	IERR = 5	no input ICONs used in model NAME
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	bus not found
	IERR = 8	machine not found
	IERR = 9	prerequisite requirements for API are not met

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4.3.3. CHANGE_PLMOD_CON

Use this API to change the value of a CON of a designated plant related model at a specified machine.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_PLMOD_CON  IBUS  'ID'  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_plmod_con(ibus, id, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_PLMOD_CON(IBUS, ID, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a plant-related model
	IERR = 3	no <model type> model at this machine
	IERR = 4	the <model type> model at this machine is aaa not NAME
	IERR = 5	no CONs used in model NAME
	IERR = 6	CON number outside of the CON range of this model
	IERR = 7	bus not found
	IERR = 8	machine not found
	IERR = 9	prerequisite requirements for API are not met

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4.3.4. CHANGE_PLMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_PLMOD_CON, CHANGE_PLMOD_ICON, and CHANGE_PLMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of a plant related model of a designated type at a specified machine.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CHANGE_PLMOD_DATA  IBUS  'ID'  MTYPE  'NAME'  J  RDATA  I  IDATA
 'CHDATA'
```

Python command syntax:

```
ierr = change_plmod_data(ibus,id,mtype,name,j,rdata,i,idata,chdata)
```

Fortran command syntax:

```
CALL CHANGE_PLMOD_DATA(IBUS,ID,MTYPE,NAME,J,RDATA,I,IDATA,CHDATA,IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).
Character*2 ID	machine identifier (input; '1').
Integer MTYPE	model type (input; 1 by default). MTYPE = 1 generator model. MTYPE = 2 compensator model. MTYPE = 3 stabilizer model. MTYPE = 4 minimum excitation limiter model. MTYPE = 5 maximum excitation limiter model. MTYPE = 6 excitation system model. MTYPE = 7 turbine-governor model. MTYPE = 8 turbine-load controller model.
Character*16 NAME	model name (input; no default allowed).
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).
Real RDATA	new CON value (input; unchanged).
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value,

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	the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer IERR	error code (output).

IERR = 0	no error
IERR = 1	invalid MTYPE value
IERR = 2	unknown model NAME
IERR = 3	no model of type MTYPE at this machine
IERR = 4	type MTYPE model at this machine is aaa not xxx
IERR = 5	CON number exceeds the number of CONs used by the model
IERR = 6	ICON number exceeds the number of ICONs used by the model
IERR = 7	neither the CON nor ICON number is positive
IERR = 8	bus not found
IERR = 9	machine not found
IERR = 10	prerequisite requirements for API are not met

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4.3.5. CHANGE_PLMOD_ICON

Use this API to change the value of an integer ICON of a designated plant related model at a specified machine. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_PLMOD_ICON  IBUS  'ID'  'NAME'  J  IDATA
```

Python command syntax:

```
ierr = change_plmod_icon(ibus, id, name, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_PLMOD_ICON(IBUS, ID, NAME, J, IDATA, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a plant-related model
	IERR = 3	no <model type> model at this machine
	IERR = 4	the <model type> model at this machine is aaa not NAME
	IERR = 5	no input ICONs used in model NAME
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	bus not found
	IERR = 8	machine not found
	IERR = 9	prerequisite requirements for API are not met

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4.3.6. CHANGE_PLMOD_VAR

Use this API to change the value of a VAR of a designated plant related model at a specified machine.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_PLMOD_VAR  IBUS  'ID'  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_plmod_var(ibus, id, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_PLMOD_VAR(IBUS, ID, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a plant-related model
	IERR = 3	no <model type> model at this machine
	IERR = 4	the <model type> model at this machine is aaa not NAME
	IERR = 5	no VARs used in model NAME
	IERR = 6	VAR number outside of the VAR range of this model
	IERR = 7	bus not found
	IERR = 8	machine not found
	IERR = 9	prerequisite requirements for API are not met

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4.3.7. GMB_ADD_PLANT_MODEL

Use this API to add a plant related GMB model to a specified machine. If a model of the designated type already exists at the machine, it is replaced.

This API routine was first introduced in release 32.0.2.

Batch command syntax:

```
BAT_GMB_ADD_PLANT_MODEL  IBUS  'ID'  'NAME'  IBUS2 'ID2'  NICN  IDATA
 CHDATA  NCON  RDATA
```

Python command syntax:

```
ierr = gmb_add_plant_model(ibus, id, name, ibus2, id2, nicn, idata,
 chdata, ncon, rdata)
```

Fortran command syntax:

```
CALL GMB_ADD_PLANT_MODEL(IBUS, ID, NAME, IBUS2, ID2, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer IBUS2	bus number of the second machine if this is a two machine model, or 0 for a single machine model (input; 0 by default).	
Character*2 ID2	machine identifier of the second machine if this is a two machine model, or ' ' for a single machine model (input; blank by default).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	machine model connection tables full--use pack plant model tables function
	IERR = 2	plant model tables are corrupt
	IERR = 3	unlicensed GMB model

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IERR = 4	error reading GMB model
IERR = 5	GMB model is not a plant type model
IERR = 6	# CONs specified is not the number expected
IERR = 7	# ICONs specified is not the number expected
IERR = 8	second machine not found where one is required
IERR = 9	same machine specified as the first and second machines
IERR = 10	maximum number of array elements exceeded
IERR = 11	number of user-defined models exceeds maximum allowed
IERR = 12	bus not found
IERR = 13	machine not found
IERR = 14	prerequisite requirements for API are not met

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4.3.8. PLMOD_CONSISTENCY

Use this API to check consistency among the plant related models referenced at each machine.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PLMOD_CONSISTENCY
```

Python command syntax:

```
ierr = plmod_consistency()
```

Fortran command syntax:

```
CALL PLMOD_CONSISTENCY(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.3.9. PLMOD_PACK

Use this API to remove entries that are marked as unused from the plant model connection tables and the plant model array allocation tables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PLMOD_PACK
```

Python command syntax:

```
ierr = plmod_pack()
```

Fortran command syntax:

```
CALL PLMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.3.10. PLMOD_REMOVE

Use this API to remove a plant related model of a designated type from a specified machine.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PLMOD_REMOVE  IBUS  'ID'  MTYPE
```

Python command syntax:

```
ierr = plmod_remove(ibus, id, mtype)
```

Fortran command syntax:

```
CALL PLMOD_REMOVE(IBUS, ID, MTYPE, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; blank by default).	
Integer MTYPE	model type (input; 1 by default).	
	MTYPE = 1	generator model
	MTYPE = 2	compensator model
	MTYPE = 3	stabilizer model
	MTYPE = 4	minimum excitation limiter model
	MTYPE = 5	maximum excitation limiter model
	MTYPE = 6	excitation system model
	MTYPE = 7	turbine-governor model
	MTYPE = 8	turbine-load controller model.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid MTYPE value
	IERR = 2	no model of type MTYPE at this machine
	IERR = 3	bus not found
	IERR = 4	machine not found
	IERR = 5	prerequisite requirements for API are not met

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4.3.11. PLMOD_STATUS

Use this API to change the status of a plant related model of a designated type at a specified machine.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PLMOD_STATUS  IBUS  'ID'  MTYPE  STATUS
```

Python command syntax:

```
ierr = plmod_status(ibus, id, mtype, status)
```

Fortran command syntax:

```
CALL PLMOD_STATUS(IBUS, ID, MTYPE, STATUS, IERR)
```

Where:

Integer IBUS	plant bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; blank by default).	
Integer MTYPE	model type (input; 1 by default).	
	MTYPE = 1	generator model
	MTYPE = 2	compensator model
	MTYPE = 3	stabilizer model
	MTYPE = 4	minimum excitation limiter model
	MTYPE = 5	maximum excitation limiter model
	MTYPE = 6	excitation system model
	MTYPE = 7	turbine-governor model
	MTYPE = 8	turbine-load controller model.
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid MTYPE value
	IERR = 2	no model of type MTYPE at this machine
	IERR = 3	invalid STATUS value
	IERR = 4	bus not found
	IERR = 5	machine not found
	IERR = 6	prerequisite requirements for API are not met

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4.3.12. PLMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those plant related models that are assigned to machines that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PLMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = plmod_unconnected(opt)
```

Fortran command syntax:

```
CALL PLMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected plant models
	OPT = 1	remove unconnected plant models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.3.13. PLMOD_USER

Use this API to list user-written plant model definitions or to remove user-written plant model definitions that are not assigned to any machines (unused) from the user model definition tables.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_PLMOD_USER  OPT
```

Python command syntax:

```
ierr = plmod_user(opt)
```

Fortran command syntax:

```
CALL PLMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written plant model definitions
	OPT = 1	remove unused user-written plant model definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.4. Wind Related Models

4.4.1. ADD_WIND_MODEL

Use this API to add a renewable related model of a designated type to a specified renewable machine. If a model of the designated type already exists at the machine, it is replaced.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_ADD_WIND_MODEL  IBUS  'ID'  MTYPE  'NAME'  NICN  IDATA  CHDATA  NCON
 RDATA
```

Python command syntax:

```
ierr = add_wind_model(ibus, id, mtype, name, nicn, idata, chdata, ncon,
 rdata)
```

Fortran command syntax:

```
CALL ADD_WIND_MODEL(IBUS, ID, MTYPE, NAME, NICN, IDATA, CHDATA, NCON,
 RDATA, IERR)
```

Where:

Integer IBUS	renewable machine bus number (input; no default allowed).
Character*2 ID	renewable machine identifier (input; blank by default).
Integer MTYPE	model type (input; 1 by default).MTYPE = 1 wind generator model). MTYPE = 2 renewable electrical control model MTYPE = 3 renewable mechanical system model MTYPE = 4 renewable pitch control model MTYPE = 5 renewable aerodynamic model MTYPE = 6 renewable gust model MTYPE = 7 renewable auxiliary control model
Character*16 NAME	model name (input; no default allowed).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer NCON	number of CONs required by the model (input; no default allowed).
Real RDATA (NCON)	CON data (input; 0.0).
Integer IERR	error code (output).

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IERR = 0	no error
IERR = 1	renewable machine model connection tables full--use pack renewable model tables function
IERR = 2	renewable model tables are corrupt
IERR = 3	invalid MTYPE value
IERR = 4	unknown model NAME
IERR = 5	model is not consistent with MTYPE
IERR = 6	# CONs specified is not the number expected
IERR = 7	# ICONs specified is not the number expected
IERR = 8	maximum number of array elements exceeded
IERR = 9	machine is not designated as a renewable machine in power flow
IERR = 10	bus not found
IERR = 11	machine not found
IERR = 12	prerequisite requirements for API are not met

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4.4.2. CHANGE_WNMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of a designated renewable related model at a specified renewable machine. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_CHANGE_WNMOD_CHRICN  IBUS  'ID'  'NAME'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_wnmod_chricn(ibus, id, name, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_WNMOD_CHRICN(IBUS, ID, NAME, J, CHDATA, IERR)
```

Where:

Integer IBUS	renewable machine bus number (input; no default allowed).	
Character*2 ID	renewable machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a renewable machine model
	IERR = 3	no <model type> model at this machine
	IERR = 4	the <model type> model at this machine is aaa not NAME
	IERR = 5	no input ICONs used in model NAME
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	machine is not designated as a renewable machine in power flow
	IERR = 8	bus not found
	IERR = 9	machine not found
	IERR = 10	prerequisite requirements for API are not met

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4.4.3. CHANGE_WNMOD_CON

Use this API to change the value of a CON of a designated renewable machine related model at a specified renewable machine.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_CHANGE_WNMOD_CON  IBUS  'ID'  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_wnmod_con(ibus, id, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_WNMOD_CON(IBUS, ID, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	renewable machine bus number (input; no default allowed).	
Character*2 ID	renewable machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a renewable machine model
	IERR = 3	no <model type> model at this machine
	IERR = 4	the <model type> model at this machine is aaa not NAME
	IERR = 5	no input CONs used in model NAME
	IERR = 6	CON number outside of the CON range of this model
	IERR = 7	machine is not designated as a renewable machine in power flow
	IERR = 8	bus not found
	IERR = 9	machine not found
	IERR = 10	prerequisite requirements for API are not met

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4.4.4. CHANGE_WNMOD_ICON

Use this API to change the value of an integer ICON of a designated renewable machine related model at a specified renewable machine. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_CHANGE_WNMOD_ICON  IBUS  'ID'  'NAME'  J  IDATA
```

Python command syntax:

```
ierr = change_wnmod_icon(ibus, id, name, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_WNMOD_ICON(IBUS, ID, NAME, J, IDATA, IERR)
```

Where:

Integer IBUS	renewable machine bus number (input; no default allowed).	
Character*2 ID	renewable machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a renewable machine model
	IERR = 3	no <model type> model at this machine
	IERR = 4	the <model type> model at this machine is aaa not NAME
	IERR = 5	no input ICONs used in model NAME
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	machine is not designated as a renewable machine in power flow
	IERR = 8	bus not found
	IERR = 9	machine not found
	IERR = 10	prerequisite requirements for API are not met

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4.4.5. CHANGE_WNMOD_VAR

Use this API to change the value of a VAR of a designated renewable machine related model at a specified renewable machine.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_WNMOD_VAR  IBUS  'ID'  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_wnmod_var(ibus, id, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_WNMOD_VAR(IBUS, ID, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	renewable machine bus number (input; no default allowed).	
Character*2 ID	renewable machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a renewable machine model
	IERR = 3	no <model type> model at this machine
	IERR = 4	the <model type> model at this machine is aaa not NAME
	IERR = 5	no input VARs used in model NAME
	IERR = 6	VAR number outside of the VAR range of this model
	IERR = 7	machine is not designated as a renewable machine in power flow
	IERR = 8	bus not found
	IERR = 9	machine not found
	IERR = 10	prerequisite requirements for API are not met

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4.4.6. WNMOD_CONSISTENCY

Use this API to check consistency among the renewable machine related models referenced at each machine.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_WNMOD_CONSISTENCY
```

Python command syntax:

```
ierr = wnmod_consistency()
```

Fortran command syntax:

```
CALL WNMOD_CONSISTENCY(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.4.7. WNMOD_PACK

Use this API to remove entries that are marked as unused from the wind model connection tables and the renewable machine model array allocation tables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_WNMOD_PACK
```

Python command syntax:

```
ierr = wnmod_pack()
```

Fortran command syntax:

```
CALL WNMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.4.8. WNMOD_REMOVE

Use this API to remove a renewable machine related model of a designated type from a specified renewable machine.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_WNMOD_REMOVE  IBUS  'ID'  MTYPE
```

Python command syntax:

```
ierr = wnmod_remove(ibus, id, mtype)
```

Fortran command syntax:

```
CALL WNMOD_REMOVE(IBUS, ID, MTYPE, IERR)
```

Where:

Integer IBUS	renewable machine bus number (input; no default allowed).	
Character*2 ID	renewable machine identifier (input; blank by default).	
Integer MTYPE	model type (input; 1 by default).	
	MTYPE = 1	renewable generator model
	MTYPE = 2	renewable electrical control model
	MTYPE = 3	renewable mechanical system model
	MTYPE = 4	renewable pitch control model
	MTYPE = 5	renewable aerodynamic model
	MTYPE = 6	renewable gust model
	MTYPE = 7	renewable auxiliary control model
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid MTYPE value
	IERR = 2	no model of type MTYPE at this machine
	IERR = 3	machine is not designated as a renewable machine in power flow
	IERR = 4	bus not found
	IERR = 5	machine not found
	IERR = 6	prerequisite requirements for API are not met

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4.4.9. WNMOD_STATUS

Use this API to change the status of a renewable machine related model of a designated type at a specified renewable machine.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_WNMOD_STATUS  IBUS  'ID'  MTYPE  STATUS
```

Python command syntax:

```
ierr = wnmod_status(ibus, id, mtype, status)
```

Fortran command syntax:

```
CALL WNMOD_STATUS(IBUS, ID, MTYPE, STATUS, IERR)
```

Where:

Integer IBUS	renewable machine bus number (input; no default allowed).	
Character*2 ID	renewable machine identifier (input; blank by default).	
Integer MTYPE	model type (input; 1 by default).	
	MTYPE = 1	renewable generator model
	MTYPE = 2	renewable electrical control model
	MTYPE = 3	renewable mechanical system model
	MTYPE = 4	renewable pitch control model
	MTYPE = 5	renewable aerodynamic model
	MTYPE = 6	renewable gust model
	MTYPE = 7	renewable auxiliary control model
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid MTYPE value
	IERR = 2	no model of type MTYPE at this machine
	IERR = 3	invalid STATUS value
	IERR = 4	machine is not designated as a renewable machine in power flow
	IERR = 5	bus not found
	IERR = 6	machine not found
	IERR = 7	prerequisite requirements for API are not met

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4.4.10. WNMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those renewable machine related models that are assigned to renewable machines that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_WNMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = wnmod_unconnected(opt)
```

Fortran command syntax:

```
CALL WNMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected renewable machine models
	OPT = 1	remove unconnected renewable machine models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.4.11. WNMOD_USER

Use this API to list user-written renewable machine model definitions or to remove user-written renewable machine model definitions that are not assigned to any renewable machines (unused) from the user model definition tables.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_WNMOD_USER  OPT
```

Python command syntax:

```
ierr = wnmod_user(opt)
```

Fortran command syntax:

```
CALL WNMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written renewable machine model definitions
	OPT = 1	remove unused user-written renewable machine model definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.5. Load Related Models

4.5.1. ADD_LOAD_MODEL

Use this API to add a load related model of a designated type to a specified load or subsystem. If a model of the designated type already exists at the load or subsystem, it is replaced.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ADD_LOAD_MODEL  IBUS  'ID'  MTYPE  LTYPE  'NAME'  NICN  IDATA  CHDATA
  NCON  RDATA
```

Python command syntax:

```
ierr = add_load_model(ibus, id, mtype, ltype, name, nicn, idata, chdata,
 ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_LOAD_MODEL(IBUS, ID, MTYPE, LTYPE, NAME, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)
```

Where:

Integer IBUS	bus, owner, zone or area number. Ignored when MTYPE is 4 (when MTYPE is less than 4) (input; no default allowed).
Character*2 ID	load identifier, or "*" to apply the model to all loads in the subsystem (input; blank by default).
Integer MTYPE	subsystem type (input; 0 by default). MTYPE = 0 bus model MTYPE = 1 owner model MTYPE = 2 zone model MTYPE = 3 area model MTYPE = 4 all model
Integer LTYPE	load model type (input; 1 by default). LTYPE = 1 load characteristic model LTYPE = 2 load relay model
Character*16 NAME	model name (input; no default allowed).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).

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Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	load model connection tables full--use pack load model tables function
	IERR = 2	invalid MTYPE value
	IERR = 3	invalid LTYPE value
	IERR = 4	unknown model NAME
	IERR = 5	model is not consistent with LTYPE
	IERR = 6	model is not consistent with MTYPE
	IERR = 7	# CONs specified is not the number expected
	IERR = 8	# ICONs specified is not the number expected
	IERR = 9	maximum number of array elements exceeded
	IERR = 10	type MTYPE load model table is full
	IERR = 11	no loads with the specified load identifier
	IERR = 12	invalid area/owner/zone number
	IERR = 13	no loads with the specified load identifier for specified subsystem
	IERR = 14	no loads at bus
	IERR = 15	bus not found
	IERR = 16	load not found
	IERR = 17	prerequisite requirements for API are not met

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4.5.2. CHANGE_LDMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of a load related model at a specified load or subsystem. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_LDMOD_CHRICN  IBUS  'ID'  'NAME'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_ldmod_chricn(ibus, id, name, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_LDMOD_CHRICN(IBUS, ID, NAME, J, CHDATA, IERR)
```

Where:

Integer IBUS	bus, owner, zone or area number. Ignored when an all type load model is specified (unless an all type load model is specified) (input; no default allowed).	
Character*2 ID	load identifier, or "*" if the model is applied to all loads in the subsystem (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a load-related model
	IERR = 3	no <subsystem type> type <load type> model for this id in tables for <subsystem type> IBUS
	IERR = 4	no all type <load type> model for this id in tables
	IERR = 5	<subsystem type> <load type> model for this id for <subsystem type> IBUS is aaa not xxx
	IERR = 6	all type <load type> model for this id is aaa not xxx
	IERR = 7	no input ICONs used in model NAME
	IERR = 8	ICON number outside of the ICON range of this model
	IERR = 9	prerequisite requirements for API are not met

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4.5.3. CHANGE_LDMOD_CON

Use this API to change the value of a CON of a load related model at a specified load or subsystem.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_LDMOD_CON  IBUS  'ID'  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_ldmod_con(ibus, id, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_LDMOD_CON(IBUS, ID, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	bus, owner, zone or area number. Ignored when an all type load model is specified (unless an all type load model is specified) (input; no default allowed).	
Character*2 ID	load identifier, or "*" if the model is applied to all loads in the subsystem (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a load-related model
	IERR = 3	no <subsystem type> type <load type> model for this id in tables for <subsystem type> IBUS
	IERR = 4	no all type <load type> model for this id in tables
	IERR = 5	<subsystem type> <load type> model for this id for <subsystem type> IBUS is aaa not xxx
	IERR = 6	all type <load type> model for this id is aaa not xxx
	IERR = 7	no CONs used in model NAME
	IERR = 8	CON number outside of the CON range of this model
	IERR = 9	prerequisite requirements for API are not met

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4.5.4. CHANGE_LDMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_LDMOD_CON, CHANGE_LDMOD_ICON, and CHANGE_LDMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of a load related model of a designated type at a specified load or subsystem.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CHANGE_LDMOD_DATA  IBUS  'ID'  MTYPE  LTYPE  'NAME'  J  RDATA  I
 IDATA  'CHDATA'
```

Python command syntax:

```
ierr = change_ldmod_data(ibus,id,mtype,ltype,name,j,rdata,i,idata,chdata)
```

Fortran command syntax:

```
CALL CHANGE_LDMOD_DATA(IBUS, ID, MTYPE, LTYPE, NAME, J, RDATA, I, IDATA,
 CHDATA, IERR)
```

Where:

Integer IBUS	bus, owner, zone or area number. Ignored when MTYPE is 4 (when MTYPE is less than 4) (input; no default allowed).
Character*2 ID	load identifier, or "*" if the model is applied to all loads in the subsystem (input; blank by default).
Integer MTYPE	subsystem type (input; 0 by default). MTYPE = 0 bus model MTYPE = 1 owner model MTYPE = 2 zone model MTYPE = 3 area model MTYPE = 4 all model
Integer LTYPE	load model type (input; 1 by default). LTYPE = 1 load characteristic model LTYPE = 2 load relay model
Character*16 NAME	model name (input; no default allowed).
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).
Real RDATA	new CON value (input; unchanged).
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).

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Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer IERR	error code (output).

IERR = 0	no error
IERR = 1	invalid MTYPE value
IERR = 2	invalid LTYPE value
IERR = 3	unknown model NAME
IERR = 4	no all type LTYPE model for this id in tables
IERR = 5	no STYPE type LTYPE model for this id in tables for MTYPE IBUS
IERR = 6	all type LTYPE model for this id is aaa not xxx
IERR = 7	MTYPE type LTYPE model for this id for MTYPE IBUS is aaa not xxx
IERR = 8	CON number exceeds the number of CONs used by the model
IERR = 9	ICON number exceeds the number of input ICONs used by the model
IERR = 10	neither the CON nor ICON number is positive
IERR = 11	prerequisite requirements for API are not met

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4.5.5. CHANGE_LDMOD_ICON

Use this API to change the value of an integer ICON of a load related model at a specified load or subsystem. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_LDMOD_ICON  IBUS  'ID'  'NAME'  J  IDATA
```

Python command syntax:

```
ierr = change_ldmod_icon(ibus, id, name, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_LDMOD_ICON(IBUS, ID, NAME, J, IDATA, IERR)
```

Where:

Integer IBUS	bus, owner, zone or area number. Ignored when an all type load model is specified (unless an all type load model is specified) (input; no default allowed).	
Character*2 ID	load identifier, or "*" if the model is applied to all loads in the subsystem (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a load-related model
	IERR = 3	no <subsystem type> type <load type> model for this id in tables for <subsystem type> IBUS
	IERR = 4	no all type <load type> model for this id in tables
	IERR = 5	<subsystem type> <load type> model for this id for <subsystem type> IBUS is aaa not xxx
	IERR = 6	all type <load type> model for this id is aaa not xxx
	IERR = 7	no input ICONs used in model NAME
	IERR = 8	ICON number outside of the ICON range of this model
	IERR = 9	prerequisite requirements for API are not met

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4.5.6. CHANGE_LDMOD_VAR

Use this API to change the value of a VAR of a load related model at a specified load or subsystem. This API can be used only for bus based load models. It cannot be used for changing VARs of owner, zone, area, and "all" type load models.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_LDMOD_VAR  IBUS  'ID'  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_ldmod_var(ibus, id, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_LDMOD_VAR(IBUS, ID, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	load bus number (input).	
Character*2 ID	load identifier, or "*" if the model is applied to all loads in the subsystem (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	model NAME is not a load-related model
	IERR = 3	no <subsystem type> type <load type> model for this id in tables for <subsystem type> IBUS
	IERR = 4	no "all" type <load type> model for this id in tables
	IERR = 5	<subsystem type> <load type> model for this id for <subsystem type> IBUS is aaa not xxx
	IERR = 6	"all" type <load type> model for this id is aaa not xxx
	IERR = 7	no VARs used in model NAME
	IERR = 8	VAR number outside of the VAR range of this model
	IERR = 9	VARs of only bus based (load type) models can be changed using this API
	IERR = 10	prerequisite requirements for API are not met

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4.5.7. LDMOD_PACK

Use this API to remove entries that are marked as unused from the load model connection tables and the load model array allocation tables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LDMOD_PACK
```

Python command syntax:

```
ierr = ldmod_pack()
```

Fortran command syntax:

```
CALL LDMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.5.8. LDMOD_REMOVE

Use this API to remove a load related model of a designated type from a specified load or subsystem.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LDMOD_REMOVE  MTYPE  IBUS  'ID'  LTYPE
```

Python command syntax:

```
ierr = ldmod_remove(mtype, ibus, id, ltype)
```

Fortran command syntax:

```
CALL LDMOD_REMOVE(MTYPE, IBUS, ID, LTYPE, IERR)
```

Where:

Integer MTYPE	subsystem type (input; 0 by default).	
	MTYPE = 0	bus model.
	MTYPE = 1	owner model.
	MTYPE = 2	zone model.
	MTYPE = 3	area model.
	MTYPE = 4	all model.
Integer IBUS	bus, owner, zone or area number. Ignored when MTYPE is 4 (when MTYPE is less than 4) (input; no default allowed).	
Character*2 ID	load identifier, or "*" if the model is applied to all loads in the subsystem (input; blank by default).	
Integer LTYPE	load model type (input; 1 by default).	
	LTYPE = 1	load characteristic model
	LTYPE = 2	load relay model
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid number specified for IBUS
	IERR = 2	invalid LTYPE value
	IERR = 3	invalid MTYPE value
	IERR = 4	no all type LTYPE model for this id in tables
	IERR = 5	no MTYPE type LTYPE model for this id in tables for MTYPE IBUS
	IERR = 6	prerequisite requirements for API are not met

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4.5.9. LDMOD_STATUS

Use this API to change the status of a load related model of a designated type for all subsystem loads at which it is applied.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LDMOD_STATUS  MTYPE  IBUS  'ID'  LTYPE  STATUS
```

Python command syntax:

```
ierr = ldmod_status(mtype, ibus, id, ltype, status)
```

Fortran command syntax:

```
CALL LDMOD_STATUS(MTYPE, IBUS, ID, LTYPE, STATUS, IERR)
```

Where:

Integer MTYPE	subsystem type (input; 0 by default).	
	MTYPE = 0	bus model
	MTYPE = 1	owner model
	MTYPE = 2	zone model
	MTYPE = 3	area model
	MTYPE = 4	all model
Integer IBUS	bus, owner, zone or area number. Ignored when MTYPE is 4 (when MTYPE is less than 4) (input; no default allowed).	
Character*2 ID	load identifier, or "*" if the model is applied to all loads in the subsystem (input; blank by default).	
Integer LTYPE	load model type (input; 1 by default).	
	LTYPE = 1	load characteristic model
	LTYPE = 2	load relay model
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid number specified for IBUS
	IERR = 2	invalid LTYPE value
	IERR = 3	invalid MTYPE value
	IERR = 4	no all type LTYPE model for this id in tables
	IERR = 5	no MTYPE type LTYPE model for this id in tables for MTYPE IBUS
	IERR = 6	invalid STATUS value
	IERR = 7	prerequisite requirements for API are not met

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4.5.10. LDMOD_STATUS2

Use this API to change the status of a load related model of a designated type at a specified load.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LDMOD_STATUS2  IBUS  'ID'  LTYPE  STATUS
```

Python command syntax:

```
ierr = ldmod_status2(ibus, id, ltype, status)
```

Fortran command syntax:

```
CALL LDMOD_STATUS2(IBUS, ID, LTYPE, STATUS, IERR)
```

Where:

Integer IBUS	load bus number (input; no default allowed).	
Character*2 ID	load identifier (input; '1').	
Integer LTYPE	load model type (input; 1 by default).	
	LTYPE = 1	load characteristic model
	LTYPE = 2	load relay model
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid LTYPE value
	IERR = 2	invalid STATUS value
	IERR = 3	no LTYPE model at this load
	IERR = 4	bus not found
	IERR = 5	load not found
	IERR = 6	prerequisite requirements for API are not met

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4.5.11. LDMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those load related models that are assigned to loads that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LDMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = ldmod_unconnected(opt)
```

Fortran command syntax:

```
CALL LDMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected load models
	OPT = 1	remove unconnected load models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.5.12. LDMOD_USER

Use this API to list user-written load model definitions or to remove user-written load model definitions that are not assigned to any loads (unused) from the user model definition tables.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_LDMOD_USER  OPT
```

Python command syntax:

```
ierr = ldmod_user(opt)
```

Fortran command syntax:

```
CALL LDMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written load model definitions
	OPT = 1	remove unused user-written load model definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.6. Line Relay Models

4.6.1. ADD_RELAY_MODEL

Use this API to add a line relay model in the designated relay slot of the from bus end of a specified branch. If a model already exists in the designated relay slot at the from bus end of the branch, it is replaced.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_ADD_RELAY_MODEL  IBUS  JBUS  'ID'  RS  'NAME'  NICN  IDATA  CHDATA
 NCON  RDATA
```

Python command syntax:

```
ierr = add_relay_model(ibus,jbus,id,rs,name,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL ADD_RELAY_MODEL(IBUS, JBUS, ID, RS, NAME, NICN, IDATA, CHDATA, NCON,
 RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; blank by default).	
Integer RS	relay slot (1 or 2) (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RS value
	IERR = 2	unknown model NAME
	IERR = 3	model is not a line relay model
	IERR = 4	# CONs specified is not the number expected
	IERR = 5	# ICONs specified is not the number expected

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IERR = 6	maximum number of array elements exceeded
IERR = 7	line relay model connection tables full--use pack line relay model tables function
IERR = 8	line relay model tables are corrupt
IERR = 9	bus not found
IERR = 10	branch not found
IERR = 11	prerequisite requirements for API are not met

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4.6.2. CHANGE_RLMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the line relay model in the designated relay slot of the from bus end of a specified branch. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_RLMOD_CHRICN  IBUS  JBUS  'ID'  RS  'NAME'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_rlmod_chricn(ibus, jbus, id, rs, name, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_RLMOD_CHRICN(IBUS,JBUS,ID,RS,NAME,J,CHDATA,IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; blank by default).	
Integer RS	relay slot (1 or 2) (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RS value
	IERR = 2	unknown model NAME
	IERR = 3	no relay model on this branch at the bus IBUS end
	IERR = 4	relay model at bus IBUS end is aaa not xxx
	IERR = 5	no input ICONs used in model NAME
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	bus not found
	IERR = 8	branch not found
	IERR = 9	prerequisite requirements for API are not met

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4.6.3. CHANGE_RLMOD_CON

Use this API to change the value of a CON of the line relay model in the designated relay slot of the from bus end of a specified branch.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_RLMOD_CON  IBUS  JBUS  'ID'  RS  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_rlmod_con(ibus, jbus, id, rs, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_RLMOD_CON(IBUS, JBUS, ID, RS, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; blank by default).	
Integer RS	relay slot (1 or 2) (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RS value
	IERR = 2	unknown model NAME
	IERR = 3	no relay model on this branch at the bus IBUS end
	IERR = 4	relay model at bus IBUS end is aaa not xxx
	IERR = 5	no CONs used in model NAME
	IERR = 6	CON number outside of the CON range of this model
	IERR = 7	bus not found
	IERR = 8	branch not found
	IERR = 9	prerequisite requirements for API are not met

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4.6.4. CHANGE_RLMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_RLMOD_CON, CHANGE_RLMOD_ICON, and CHANGE_RLMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of the line relay model in the designated relay slot of the from bus end of a specified branch.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_CHANGE_RLMOD_DATA  IBUS  JBUS  'ID'  RS  'NAME'  J  RDATA  I  IDATA
 'CHDATA'
```

Python command syntax:

```
ierr = change_rlmod_data(ibus,jbus,id,rs,name,j,rdata,i,idata,chdata)
```

Fortran command syntax:

```
CALL CHANGE_RLMOD_DATA(IBUS, JBUS, ID, RS, NAME, J, RDATA, I, IDATA,
 CHDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).
Integer JBUS	to bus number (input; no default allowed).
Character*2 ID	circuit identifier (input; blank by default).
Integer RS	relay slot (1 or 2) (input; 1 by default).
Character*16 NAME	model name (input; no default allowed).
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).
Real RDATA	new CON value (input; unchanged).
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer IERR	error code (output). IERR = 0 no error IERR = 1 invalid RS value

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IERR = 2	unknown model NAME
IERR = 3	no relay model on this branch at the bus IBUS end
IERR = 4	relay model at bus IBUS end is aaa not xxx
IERR = 5	CON number exceeds the number of CONs used by the model
IERR = 6	ICON number exceeds the number of ICONs used by the model
IERR = 7	neither the CON nor ICON number is positive
IERR = 8	bus not found
IERR = 9	branch not found
IERR = 10	prerequisite requirements for API are not met

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4.6.5. CHANGE_RLMOD_ICON

Use this API to change the value of an integer ICON of the line relay model in the designated relay slot of the from bus end of a specified branch. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_RLMOD_ICON  IBUS  JBUS  'ID'  RS  'NAME'  J  IDATA
```

Python command syntax:

```
ierr = change_rlmod_icon(ibus, jbus, id, rs, name, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_RLMOD_ICON(IBUS, JBUS, ID, RS, NAME, J, IDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; blank by default).	
Integer RS	relay slot (1 or 2) (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RS value
	IERR = 2	unknown model NAME
	IERR = 3	no relay model on this branch at the bus IBUS end
	IERR = 4	relay model at bus IBUS end is aaa not xxx
	IERR = 5	no input ICONs used in model NAME
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	bus not found
	IERR = 8	branch not found
	IERR = 9	prerequisite requirements for API are not met

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4.6.6. CHANGE_RLMOD_VAR

Use this API to change the value of a VAR of the line relay model in the designated relay slot of the from bus end of a specified branch.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_RLMOD_VAR  IBUS  JBUS  'ID'  RS  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_rlmod_var(ibus, jbus, id, rs, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_RLMOD_VAR(IBUS, JBUS, ID, RS, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; blank by default).	
Integer RS	relay slot (1 or 2) (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RS value
	IERR = 2	unknown model NAME
	IERR = 3	no relay model on this branch at the bus IBUS end
	IERR = 4	relay model at bus IBUS end is aaa not xxx
	IERR = 5	no VARs used in model NAME
	IERR = 6	VAR number outside of the VAR range of this model
	IERR = 7	bus not found
	IERR = 8	branch not found
	IERR = 9	prerequisite requirements for API are not met

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4.6.7. RLMOD_PACK

Use this API to remove entries that are marked as unused from the line relay model connection tables and the line relay model array allocation tables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RLMOD_PACK
```

Python command syntax:

```
ierr = rlmod_pack()
```

Fortran command syntax:

```
CALL RLMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.6.8. RLMOD_REMOVE

Use this API to remove the line relay model in the designated relay slot of the from bus end of a specified branch.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RLMOD_REMOVE  IBUS  JBUS 'ID'  RS
```

Python command syntax:

```
ierr = rlmod_remove(ibus, jbus, id, rs)
```

Fortran command syntax:

```
CALL RLMOD_REMOVE(IBUS, JBUS, ID, RS, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; '1').	
Integer RS	relay slot (1 or 2) (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RS value
	IERR = 2	no relay model on this branch at the bus IBUS end
	IERR = 3	bus not found
	IERR = 4	branch not found
	IERR = 5	prerequisite requirements for API are not met

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4.6.9. RLMOD_STATUS

Use this API to change the status of the line relay model in the designated relay slot of the from bus end of a specified branch.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RLMOD_STATUS  IBUS  JBUS  'ID'  RS  STATUS
```

Python command syntax:

```
ierr = rlmod_status(ibus, jbus, id, rs, status)
```

Fortran command syntax:

```
CALL RLMOD_STATUS(IBUS, JBUS, ID, RS, STATUS, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; '1').	
Integer RS	relay slot (1 or 2) (input; 1 by default).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid RS value
	IERR = 2	no relay model on this branch at the bus IBUS end
	IERR = 3	invalid STATUS value
	IERR = 4	bus not found
	IERR = 5	branch not found
	IERR = 6	prerequisite requirements for API are not met

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4.6.10. RLMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those line relay models that are assigned to branches that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RLMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = rlmod_unconnected(opt)
```

Fortran command syntax:

```
CALL RLMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected line relay models
	OPT = 1	remove unconnected line relay models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.6.11. RLMOD_USER

Use this API to list user-written line relay model definitions or to remove user-written line relay model definitions that are not assigned to any branches (unused) from the user model definition tables.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_RLMOD_USER  OPT
```

Python command syntax:

```
ierr = rlmod_user(opt)
```

Fortran command syntax:

```
CALL RLMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written line relay definitions
	OPT = 1	remove unused user-written line relay definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.7. DC Line Related Models

4.7.1. ADD_MTDCLINE_MODEL

Use this API to add a multi-terminal dc line model to the specified multi-terminal dc line. If a model already exists at the multi-terminal dc line, it is replaced.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ADD_MTDCLINE_MODEL  DC  'MODEL'  NICN  IDATA  CHDATA  NCON  RDATA
 IERR
```

Python command syntax:

```
ierr = add_mtdcline_model(dc,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL ADD_MTDCLINE_MODEL(DC,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)

CALL ADD_MTDCLINE_MODEL_I(NUM,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)

CALL ADD_MTDCLINE_MODEL_C(NAME,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line to which the model is to be added; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line to which the model is to be added (input; no default allowed).
Character*12 NAME	name of the multi-terminal dc line to which the model is to be added (input; no default allowed).
Character*16 MODEL	model name (input; no default allowed).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer NCON	number of CONs required by the model (input; no default allowed).
Real RDATA (NCON)	CON data (input; 0.0).
Integer IERR	error code (output).

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IERR = 0	no error
IERR = 1	dc line not found
IERR = 2	multi-terminal dc line model connection tables full--use pack dc line model tables function
IERR = 3	unknown model name
IERR = 4	model is not a multi-terminal dc line model
IERR = 5	# CONs specified is not the number expected
IERR = 6	# ICONs specified is not the number expected
IERR = 7	maximum number of array elements exceeded
IERR = 8	prerequisite requirements for API are not met

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4.7.2. ADD_TTDCLINE_MODEL

Use this API to add a two-terminal dc line model to the specified two-terminal dc line. If a model already exists at the two-terminal dc line, it is replaced.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ADD_TTDCLINE_MODEL  DC  'MODEL'  NICN  IDATA  CHDATA  NCON  RDATA
 IERR
```

Python command syntax:

```
ierr = add_ttdcline_model(dc,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL ADD_TTDCLINE_MODEL(DC,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)

CALL ADD_TTDCLINE_MODEL_I(NUM,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)

CALL ADD_TTDCLINE_MODEL_C(NAME,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line to which the model is to be added; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line to which the model is to be added (input; no default allowed).
Character*12 NAME	name of the two-terminal dc line to which the model is to be added (input; no default allowed).
Character*16 MODEL	model name (input; no default allowed).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer NCON	number of CONs required by the model (input; no default allowed).
Real RDATA (NCON)	CON data (input; 0.0).
Integer IERR	error code (output). IERR = 0 no error IERR = 1 dc line not found

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IERR = 2	two-terminal dc line model connection tables full--use pack dc line model tables function
IERR = 3	unknown model name
IERR = 4	model is not a two-terminal dc line model
IERR = 5	# CONs specified is not the number expected
IERR = 6	# ICONs specified is not the number expected
IERR = 7	maximum number of array elements exceeded
IERR = 8	prerequisite requirements for API are not met

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4.7.3. ADD_VSDCLINE_MODEL

Use this API to add a VSC dc line model to the specified VSC dc line. If a model already exists at the VSC dc line, it is replaced.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ADD_VSDCLINE_MODEL  'DCNAME'  'MODEL'  NICN  IDATA  CHDATA  NCON
 RDATA  IERR
```

Python command syntax:

```
ierr = add_vsdcline_model(dcname,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL ADD_VSDCLINE_MODEL(DCNAME,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)
```

Where:

Character*12 DCNAME	VSC dc line name (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	VSC dc line model connection tables full--use pack dc line model tables function
	IERR = 3	unknown model name
	IERR = 4	model is not a VSC dc line model
	IERR = 5	# CONs specified is not the number expected
	IERR = 6	# ICONs specified is not the number expected
	IERR = 7	maximum number of array elements exceeded
	IERR = 8	prerequisite requirements for API are not met

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4.7.4. CHANGE_MTDCLMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the multi-terminal dc line model of a specified multi-terminal dc line. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_MTDCLMOD_CHRICN  DC  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_mtdclmod_chricn(dc, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCLMOD_CHRICN(DC, MODEL, J, CHDATA, IERR)

CALL CHANGE_MTDCLMOD_CHRICN_I(NUM, MODEL, J, CHDATA, IERR)

CALL CHANGE_MTDCLMOD_CHRICN_C(NAME, MODEL, J, CHDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no input ICONs used in model
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.5. CHANGE_MTDCLMOD_CON

Use this API to change the value of a CON of the multi-terminal dc line model of a specified multi-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_MTDCLMOD_CON  DC  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_mtdclmod_con(dc, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCLMOD_CON(DC, MODEL, J, RDATA, IERR)

CALL CHANGE_MTDCLMOD_CON_I(NUM, MODEL, J, RDATA, IERR)

CALL CHANGE_MTDCLMOD_CON_C(NAME, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no CONs used in model
	IERR = 6	CON number outside of the CON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.6. CHANGE_MTDCLMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_MTDCLMOD_CON, CHANGE_MTDCLMOD_ICON, and CHANGE_MTDCLMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of the multi-terminal dc line model of a specified multi-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_MTDCLMOD_DATA  DC  'NAME'  J  RDATA  I  IDATA  'CHDATA'
```

Python command syntax:

```
ierr = change_mtdclmod_data(dc, name, j, rdata, i, idata, chdata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCLMOD_DATA(DC,NAME,J,RDATA,I,IDATA,CHDATA,IERR)
```

Where:

Integer DC	multi-terminal dc line number (input; no default allowed).
Character*16 NAME	model name (input; no default allowed).
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).
Real RDATA	new CON value (input; unchanged).
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer IERR	error code (output).
	IERR = 1 dc line not found
	IERR = 2 unknown model name
	IERR = 3 no model at this dc line
	IERR = 4 model at this dc line is aaa not xxx
	IERR = 5 no CONs used in model; or
	IERR = 5 no input ICONs used in model
	IERR = 6 CON number outside of the CON range of this model; or

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IERR = 6	ICON number outside of the ICON range of this model
IERR = 7	prerequisite requirements for API are not met

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4.7.7. CHANGE_MTDCLMOD_ICON

Use this API to change the value of an integer ICON of the multi-terminal dc line model of a specified multi-terminal dc line. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_MTDCLMOD_ICON  DC  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_mtdclmod_icon(dc, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCLMOD_ICON(DC, MODEL, J, IDATA, IERR)

CALL CHANGE_MTDCLMOD_ICON_I(NUM, MODEL, J, IDATA, IERR)

CALL CHANGE_MTDCLMOD_ICON_C(NAME, MODEL, J, IDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no input ICONs used in model
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.8. CHANGE_MTDCLMOD_VAR

Use this API to change the value of a VAR of the multi-terminal dc line model of a specified multi-terminal dc line.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_MTDCLMOD_VAR  DC  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_mtdclmod_var(dc, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCLMOD_VAR(DC, MODEL, J, RDATA, IERR)

CALL CHANGE_MTDCLMOD_VAR_I(NUM, MODEL, J, RDATA, IERR)

CALL CHANGE_MTDCLMOD_VAR_C(NAME, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no VARs used in model
	IERR = 6	VAR number outside of the VAR range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.9. CHANGE_TTDCLMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the two-terminal dc line model of a specified two-terminal dc line. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_TTDCLMOD_CHRICN  DC  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_ttdclmod_chricn(dc, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCLMOD_CHRICN(DC, MODEL, J, CHDATA, IERR)

CALL CHANGE_TTDCLMOD_CHRICN_I(NUM, MODEL, J, CHDATA, IERR)

CALL CHANGE_TTDCLMOD_CHRICN_C(NAME, MODEL, J, CHDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no input ICONs used in model
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.10. CHANGE_TTDCLMOD_CON

Use this API to change the value of a CON of the two-terminal dc line model of a specified two-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_TTDCLMOD_CON  DC  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_ttdclmod_con(dc, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCLMOD_CON(DC, MODEL, J, RDATA, IERR)

CALL CHANGE_TTDCLMOD_CON_I(NUM, MODEL, J, RDATA, IERR)

CALL CHANGE_TTDCLMOD_CON_C(NAME, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no CONs used in model
	IERR = 6	CON number outside of the CON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.11. CHANGE_TTDCLMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_TTDCLMOD_CON, CHANGE_TTDCLMOD_ICON, and CHANGE_TTDCLMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of the two-terminal dc line model of a specified two-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_TTDCLMOD_DATA  IDC  'NAME'  J  RDATA  I  IDATA  'CHDATA'
```

Python command syntax:

```
ierr = change_ttdclmod_data(idc, name, j, rdata, i, idata, chdata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCLMOD_DATA(IDC,NAME,J,RDATA,I,IDATA,CHDATA,IERR)
```

Where:

Integer IDC	two-terminal dc line number (input; no default allowed).
Character*16 NAME	model name (input; no default allowed).
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).
Real RDATA	new CON value (input; unchanged).
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer IERR	error code (output).
	IERR = 0 no error
	IERR = 1 dc line not found
	IERR = 2 unknown model name
	IERR = 3 no model at this dc line
	IERR = 4 model at this dc line is aaa not xxx
	IERR = 5 no CONs used in model; or
	IERR = 5 no input ICONs used in model

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IERR = 6	CON number outside of the CON range of this model; or
IERR = 6	ICON number outside of the ICON range of this model
IERR = 7	prerequisite requirements for API are not met

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4.7.12. CHANGE_TTDCLMOD_ICON

Use this API to change the value of an integer ICON of the two-terminal dc line model of a specified two-terminal dc line. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_TTDCLMOD_ICON  DC  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_ttdclmod_icon(dc, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCLMOD_ICON(DC, MODEL, J, IDATA, IERR)

CALL CHANGE_TTDCLMOD_ICON_I(NUM, MODEL, J, IDATA, IERR)

CALL CHANGE_TTDCLMOD_ICON_C(NAME, MODEL, J, IDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no input ICONs used in model
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.13. CHANGE_TTDCLMOD_VAR

Use this API to change the value of a VAR of the two-terminal dc line model of a specified two-terminal dc line.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_TTDCLMOD_VAR  DC  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_ttdclmod_var(dc, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCLMOD_VAR(DC, MODEL, J, RDATA, IERR)

CALL CHANGE_TTDCLMOD_VAR_I(NUM, MODEL, J, RDATA, IERR)

CALL CHANGE_TTDCLMOD_VAR_C(NAME, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no VARs used in model
	IERR = 6	VAR number outside of the VAR range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.14. CHANGE_VSDCLMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the VSC dc line model of a specified VSC dc line. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VSDCLMOD_CHRICN  'DCNAME'  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_vsdclmod_chricn(dcname, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCLMOD_CHRICN(DCNAME, MODEL, J, CHDATA, IERR)
```

Where:

Character*12 DCNAME	VSC dc line name (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no input ICONs used in model
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.15. CHANGE_VSDCLMOD_CON

Use this API to change the value of a CON of the VSC dc line model of a specified VSC dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VSDCLMOD_CON  'DCNAME'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_vsdclmod_con(dcname, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCLMOD_CON(DCNAME, MODEL, J, RDATA, IERR)
```

Where:

Character*12 DCNAME	VSC dc line name (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	VSC dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no CONs used in model
	IERR = 6	CON number outside of the CON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.16. CHANGE_VSDCLMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_VSDCLMOD_CON, CHANGE_VSDCLMOD_ICON, and CHANGE_VSDCLMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of the VSC dc line model of a specified VSC dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VSDCLMOD_DATA  'VSDCNM'  'NAME'  J  RDATA  I  IDATA  'CHDATA'
```

Python command syntax:

```
ierr = change_vsdclmod_data(vsdcnm,name,j,rdata,i,idata,chdata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCLMOD_DATA(VSDCNM,NAME,J,RDATA,I,IDATA,CHDATA,IERR)
```

Where:

Character*12 VSDCNM	VSC dc line name (input; no default allowed).
Character*16 NAME	model name (input; no default allowed).
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).
Real RDATA	new CON value (input; unchanged).
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer IERR	error code (output).
	IERR = 0 no error
	IERR = 1 dc line not found
	IERR = 2 unknown model name
	IERR = 3 no model at this dc line
	IERR = 4 model at this dc line is aaa not xxx
	IERR = 5 no CONs used in model; or
	IERR = 5 no input ICONs used in model
	IERR = 6 CON number outside of the CON range of this model; or

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IERR = 6	ICON number outside of the ICON range of this model
IERR = 7	prerequisite requirements for API are not met

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4.7.17. CHANGE_VSDCLMOD_ICON

Use this API to change the value of an integer ICON of the VSC dc line model of a specified VSC dc line. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VSDCLMOD_ICON  'DCNAME'  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_vsdclmod_icon(dcname, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCLMOD_ICON(DCNAME, MODEL, J, IDATA, IERR)
```

Where:

Character*12 DCNAME	VSC dc line name (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no input ICONs used in model
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.18. CHANGE_VSDCLMOD_VAR

Use this API to change the value of a VAR of the VSC dc line model of a specified VSC dc line.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_VSDCLMOD_VAR  'DCNAME'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_vsdclmod_var(dcname, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCLMOD_VAR(DCNAME, MODEL, J, RDATA, IERR)
```

Where:

Character*12 DCNAME	VSC dc line name (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	VSC dc line not found
	IERR = 2	unknown model name
	IERR = 3	no model at this dc line
	IERR = 4	model at this dc line is aaa not xxx
	IERR = 5	no VARs used in model
	IERR = 6	VAR number outside of the VAR range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.7.19. DCLMOD_PACK

Use this API to remove entries that are marked as unused from the dc line model connection tables and the dc line array allocation tables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DCLMOD_PACK
```

Python command syntax:

```
ierr = dclmod_pack()
```

Fortran command syntax:

```
CALL DCLMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.7.20. DCLMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those dc line models that are assigned to dc lines that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DCLMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = dclmod_unconnected(opt)
```

Fortran command syntax:

```
CALL DCLMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected dc line models
	OPT = 1	remove unconnected dc line models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.7.21. DCLMOD_USER

Use this API to list user-written dc line model definitions or to remove user-written dc line model definitions that are not assigned to any dc lines (unused) from the user model definition tables.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DCLMOD_USER  OPT
```

Python command syntax:

```
ierr = dclmod_user(opt)
```

Fortran command syntax:

```
CALL DCLMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written dc line model definitions.
	OPT = 1	remove unused user-written dc line model definitions.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.7.22. MTDCLMOD_REMOVE

Use this API to remove the multi-terminal dc line model from a specified multi-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MTDCLMOD_REMOVE  DC
```

Python command syntax:

```
ierr = mtdclmod_remove(dc)
```

Fortran command syntax:

```
CALL MTDCLMOD_REMOVE(DC, IERR)

CALL MTDCLMOD_REMOVE_I(NUM, IERR)

CALL MTDCLMOD_REMOVE_C(NAME, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which the model is to be removed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which the model is to be removed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line for which the model is to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	no model at this dc line
	IERR = 3	prerequisite requirements for API are not met

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4.7.23. MTDCLMOD_STATUS

Use this API to change the status of the multi-terminal dc line model at a specified multi-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MTDCLMOD_STATUS  DC  STATUS
```

Python command syntax:

```
ierr = mtdclmod_status(dc, status)
```

Fortran command syntax:

```
CALL MTDCLMOD_STATUS(DC, STATUS, IERR)

CALL MTDCLMOD_STATUS_I(NUM, STATUS, IERR)

CALL MTDCLMOD_STATUS_C(NAME, STATUS, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line for which the model status is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line for which the model status is to be changed (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line for which the model status is to be changed (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	no model at this dc line
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.7.24. TTDCLMOD_REMOVE

Use this API to remove the two-terminal dc line model from a specified two-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_TTDCLMOD_REMOVE  DC
```

Python command syntax:

```
ierr = ttdclmod_remove(dc)
```

Fortran command syntax:

```
CALL TTDCLMOD_REMOVE(DC, IERR)

CALL TTDCLMOD_REMOVE_I(NUM, IERR)

CALL TTDCLMOD_REMOVE_C(NAME, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which the model is to be removed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which the model is to be removed (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line for which the model is to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	no model at this dc line
	IERR = 3	prerequisite requirements for API are not met

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4.7.25. TTDCLMOD_STATUS

Use this API to change the status of the two-terminal dc line model at a specified two-terminal end line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_TTDCLMOD_STATUS  DC  STATUS
```

Python command syntax:

```
ierr = ttdclmod_status(dc, status)
```

Fortran command syntax:

```
CALL TTDCLMOD_STATUS(DC, STATUS, IERR)

CALL TTDCLMOD_STATUS_I(NUM, STATUS, IERR)

CALL TTDCLMOD_STATUS_C(NAME, STATUS, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line for which the model status is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line for which the model status is to be changed (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line for which the model status is to be changed (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	no model at this dc line
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.7.26. VSDCLMOD_REMOVE

Use this API to remove the VSC dc line model from a specified VSC dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_VSDCLMOD_REMOVE  'DCNAME'
```

Python command syntax:

```
ierr = vsdclmod_remove(dcname)
```

Fortran command syntax:

```
CALL VSDCLMOD_REMOVE(DCNAME, IERR)
```

Where:

Character*12 DCNAME	VSC dc line name (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	VSC dc line not found
	IERR = 2	no model at this VSC dc line
	IERR = 3	prerequisite requirements for API are not met

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4.7.27. VSDCLMOD_STATUS

Use this API to change the status of the VSC dc line model at a specified VSC dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_VSDCLMOD_STATUS  'DCNAME'  STATUS
```

Python command syntax:

```
ierr = vsdclmod_status(dcname, status)
```

Fortran command syntax:

```
CALL VSDCLMOD_STATUS(DCNAME, STATUS, IERR)
```

Where:

Character*12 DCNAME	VSC dc line name (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	VSC dc line not found
	IERR = 2	no model at this VSC dc line
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.8. FACTS Device Models

4.8.1. ADD_FACTS_DEVICE_MODEL

Use this API to add a FACTS device model to the specified FACTS device. If a model already exists at the FACTS device, it is replaced.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ADD_FACTS_DEVICE_MODEL FD 'MODEL' NICN IDATA CHDATA NCON RDATA
 IERR
```

Python command syntax:

```
ierr = add_facts_device_model(fd,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL ADD_FACTS_DEVICE_MODEL(FD,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)

CALL ADD_FACTS_DEVICE_MODEL_I(NUM,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)

CALL
 ADD_FACTS_DEVICE_MODEL_C(NAME,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device to which the model is to be added; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device to which the model is to be added (input; no default allowed).
Character*12 NAME	name of the FACTS device to which the model is to be added (input; no default allowed).
Character*16 MODEL	model name (input; no default allowed).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer NCON	number of CONs required by the model (input; no default allowed).
Real RDATA (NCON)	CON data (input; 0.0).
Integer IERR	error code (output).

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IERR = 0	no error
IERR = 1	FACTS device not found
IERR = 2	unknown model name
IERR = 3	model is not a FACTS device model
IERR = 4	# CONs specified is not the number expected
IERR = 5	# ICONs specified is not the number expected
IERR = 6	maximum number of array elements exceeded
IERR = 7	FACTS device model connection tables full; use pack FACTS device model tables function
IERR = 8	prerequisite requirements for API are not met

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4.8.2. CHANGE_FCTMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the FACTS device model of a specified FACTS device. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_FCTMOD_CHRICN  FD  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_fctmod_chricn(fd, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_FCTMOD_CHRICN(FD, MODEL, J, CHDATA, IERR)

CALL CHANGE_FCTMOD_CHRICN_I(NUM, MODEL, J, CHDATA, IERR)

CALL CHANGE_FCTMOD_CHRICN_C(NAME, MODEL, J, CHDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the FACTS device for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	unknown model name
	IERR = 3	no model at this FACTS device
	IERR = 4	model at this FACTS device is aaa not xxx
	IERR = 5	no input ICONs used in model
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.8.3. CHANGE_FCTMOD_CON

Use this API to change the value of a CON of the FACTS device model of a specified FACTS device.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_FCTMOD_CON  FD  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_fctmod_con(fd, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_FCTMOD_CON(FD, MODEL, J, RDATA, IERR)

CALL CHANGE_FCTMOD_CON_I(NUM, MODEL, J, RDATA, IERR)

CALL CHANGE_FCTMOD_CON_C(NAME, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the FACTS device for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	unknown model name
	IERR = 3	no model at this FACTS device
	IERR = 4	model at this FACTS device is aaa not xxx
	IERR = 5	no CONs used in model
	IERR = 6	CON number outside of the CON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.8.4. CHANGE_FCTMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_FCTMOD_CON, CHANGE_FCTMOD_ICON, and CHANGE_FCTMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of the FACTS device model of a specified FACTS device.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_FCTMOD_DATA  FD  'NAME'  J  RDATA  I  IDATA  'CHDATA'
```

Python command syntax:

```
ierr = change_fctmod_data(fd, name, j, rdata, i, idata, chdata)
```

Fortran command syntax:

```
CALL CHANGE_FCTMOD_DATA(FD,NAME,J,RDATA,I,IDATA,CHDATA,IERR)
```

Where:

Integer FD	FACTS device number (input; no default allowed).
Character*16 NAME	model name (input; no default allowed).
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).
Real RDATA	new CON value (input; unchanged).
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).
Integer IERR	error code (output).
	IERR = 0 no error
	IERR = 1 FACTS device not found
	IERR = 2 unknown model name
	IERR = 3 no model at this FACTS device
	IERR = 4 model at this FACTS device is aaa not xxx
	IERR = 5 no CONs used in model
	IERR = 5 no input ICONs used in model

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IERR = 6	CON number outside of the CON range of this model
IERR = 6	ICON number outside of the ICON range of this model
IERR = 7	prerequisite requirements for API are not met

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4.8.5. CHANGE_FCTMOD_ICON

Use this API to change the value of an integer ICON of the FACTS device model of a specified FACTS device. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_FCTMOD_ICON  FD  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_fctmod_icon(fd, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_FCTMOD_ICON(FD, MODEL, J, IDATA, IERR)

CALL CHANGE_FCTMOD_ICON_I(NUM, MODEL, J, IDATA, IERR)

CALL CHANGE_FCTMOD_ICON_C(NAME, MODEL, J, IDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the FACTS device for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	unknown model name
	IERR = 3	no model at this FACTS device
	IERR = 4	model at this FACTS device is aaa not xxx
	IERR = 5	no input ICONs used in model
	IERR = 6	ICON number outside of the ICON range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.8.6. CHANGE_FCTMOD_VAR

Use this API to change the value of a VAR of the FACTS device model of a specified FACTS device.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_FCTMOD_VAR  FD  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_fctmod_var(fd, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_FCTMOD_VAR(FD, MODEL, J, RDATA, IERR)

CALL CHANGE_FCTMOD_VAR_I(NUM, MODEL, J, RDATA, IERR)

CALL CHANGE_FCTMOD_VAR_C(NAME, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device for which model data is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device for which model data is to be changed (input; no default allowed).	
Character*12 NAME	name of the FACTS device for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	unknown model name
	IERR = 3	no model at this FACTS device
	IERR = 4	model at this FACTS device is aaa not xxx
	IERR = 5	no VARs used in model
	IERR = 6	VAR number outside of the VAR range of this model
	IERR = 7	prerequisite requirements for API are not met

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4.8.7. FCTMOD_PACK

Use this API to remove entries that are marked as unused from the FACTS device model connection tables and the FACTS device model array allocation tables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_FCTMOD_PACK
```

Python command syntax:

```
ierr = fctmod_pack()
```

Fortran command syntax:

```
CALL FCTMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.8.8. FCTMOD_REMOVE

Use this API to remove the FACTS device model from a specified FACTS device.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_FCTMOD_REMOVE  FD
```

Python command syntax:

```
ierr = fctmod_remove(fd)
```

Fortran command syntax:

```
CALL FCTMOD_REMOVE(FD, IERR)

CALL FCTMOD_REMOVE_I(NUM, IERR)

CALL FCTMOD_REMOVE_C(NAME, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device for which the model is to be removed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device for which the model is to be removed (input; no default allowed).	
Character*12 NAME	name of the FACTS device for which the model is to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	no model at this FACTS device
	IERR = 3	prerequisite requirements for API are not met

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4.8.9. FCTMOD_STATUS

Use this API to change the status of the FACTS device model at a specified FACTS device.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_FCTMOD_STATUS  FD  STATUS
```

Python command syntax:

```
ierr = fctmod_status(fd, status)
```

Fortran command syntax:

```
CALL FCTMOD_STATUS(FD, STATUS, IERR)

CALL FCTMOD_STATUS_I(NUM, STATUS, IERR)

CALL FCTMOD_STATUS_C(NAME, STATUS, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device for which the model status is to be changed; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device for which the model status is to be changed (input; no default allowed).	
Character*12 NAME	name of the FACTS device for which the model status is to be changed (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	no model at this FACTS device
	IERR = 3	invalid STATUS value
	IERR = 4	prerequisite requirements for API are not met

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4.8.10. FCTMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those FACTS device models that are assigned to FACTS device that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_FCTMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = fctmod_unconnected(opt)
```

Fortran command syntax:

```
CALL FCTMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected FACTS device models
	OPT = 1	remove unconnected FACTS device models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.8.11. FCTMOD_USER

Use this API to list user-written FACTS device model definitions or to remove user-written FACTS device model definitions that are not assigned to any FACTS devices (unused) from the user model definition tables.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_FCTMOD_USER  OPT
```

Python command syntax:

```
ierr = fctmod_user(opt)
```

Fortran command syntax:

```
CALL FCTMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written FACTS device model definitions
	OPT = 1	remove unused user-written FACTS device model definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.8.12. GMB_ADD_FACTS_DEV_MODEL

Use this API to add a GMB FACTS device model to the specified FACTS device. If a model already exists at the FACTS device, it is replaced.

This API routine was first introduced in release 32.0.2.

Batch command syntax:

```
BAT_GMB_ADD_FACTS_DEV_MODEL FD 'MODEL' NICN IDATA CHDATA NCON RDATA
  IERR
```

Python command syntax:

```
ierr = gmb_add_facts_dev_model(fd,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL GMB_ADD_FACTS_DEV_MODEL(FD,MODEL,NICN,IDATA,CHDATA,NCON,RDATA, IERR)
```

Where:

Character*12 FD	quoted name of the FACTS device to which the model is to be added; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device to which the model is to be added (input; no default allowed).	
Character*12 NAME	name of the FACTS device to which the model is to be added (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	error reading GMB model name
	IERR = 3	model is not a FACTS device model
	IERR = 4	# CONs specified is not the number expected
	IERR = 5	# ICONs specified is not the number expected

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IERR = 6	maximum number of array elements exceeded
IERR = 7	FACTS device model connection tables full; use pack FACTS device model tables function
IERR = 8	unlicensed GMB model name
IERR = 9	number of user-defined models exceeds maximum allowed
IERR = 10	prerequisite requirements for API are not met

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4.9. Switched Shunt Models

4.9.1. ADD_SWSHUNT_MODEL

Use this API to add a switched shunt model to the specified switched shunt device. If a model already exists at the switched shunt, it is replaced.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_ADD_SWSHUNT_MODEL  IBUS  'MODEL'  NICN  IDATA  CHDATA  NCON  RDATA
 IERR
```

Python command syntax:

```
ierr = add_swshunt_model(ibus,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL ADD_SWSHUNT_MODEL(IBUS,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a switched shunt model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	switched shunt model connection tables full--use pack switched shunt model tables function

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IERR = 7	bus not found
IERR = 8	switched shunt not found
IERR = 9	prerequisite requirements for API are not met

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4.9.2. CHANGE_SWSMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the switched shunt model of a specified switched shunt device. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_CHANGE_SWSMOD_CHRICN  IBUS  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_swsmod_chricn(ibus, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_SWSMOD_CHRICN(IBUS, MODEL, J, CHDATA, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no model at this switched shunt device
	IERR = 3	model at this switched shunt is aaa not xxx
	IERR = 4	no input ICONs used in model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	switched shunt not found
	IERR = 8	prerequisite requirements for API are not met

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4.9.3. CHANGE_SWSMOD_CON

Use this API to change the value of a CON of the switched shunt model of a specified switched shunt device.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_CHANGE_SWSMOD_CON  IBUS  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_swsmod_con(ibus, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_SWSMOD_CON(IBUS, MODEL, J, RDATA, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no model at this switched shunt device
	IERR = 3	model at this switched shunt is aaa not xxx
	IERR = 4	no input CONs used in model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	switched shunt not found
	IERR = 8	prerequisite requirements for API are not met

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4.9.4. CHANGE_SWSMOD_ICON

Use this API to change the value of an integer ICON (i.e., an element of the ICON array) of the switched shunt model of a specified switched shunt device. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_CHANGE_SWSMOD_ICON  IBUS  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_swsmod_icon(ibus, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_SWSMOD_ICON(IBUS, MODEL, J, IDATA, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no model at this switched shunt device
	IERR = 3	model at this switched shunt is aaa not xxx
	IERR = 4	no input ICONs used in model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	switched shunt not found
	IERR = 8	prerequisite requirements for API are not met

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4.9.5. CHANGE_SWSMOD_VAR

Use this API to change the value of a VAR of the switched shunt model of a specified switched shunt device.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_SWSMOD_VAR  IBUS  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_swsmod_var(ibus, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_SWSMOD_VAR(IBUS, MODEL, J, RDATA, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no model at this switched shunt device
	IERR = 3	model at this switched shunt is aaa not xxx
	IERR = 4	no input VARs used in model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	switched shunt not found
	IERR = 8	prerequisite requirements for API are not met

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4.9.6. GMB_ADD_SWSHUNT_MODEL

Use this API to add a GMB switched shunt model to the specified switched shunt device. If a model already exists at the switched shunt, it is replaced.

This API routine was first introduced in release 32.0.2.

Batch command syntax:

```
BAT_GMB_ADD_SWSHUNT_MODEL  IBUS  'MODEL'  NICN  IDATA  CHDATA  NCON  RDATA
  IERR
```

Python command syntax:

```
ierr = gmb_add_swshunt_model(ibus,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL GMB_ADD_SWSHUNT_MODEL(IBUS,MODEL,NICN,IDATA,CHDATA,NCON,RDATA,IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	error reading GMB model name
	IERR = 2	model is not a switched shunt model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	switched shunt model connection tables full--use pack switched shunt model tables function
	IERR = 7	unlicensed GMB model name
	IERR = 8	number of user-defined models exceeds maximum allowed

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IERR = 9	bus not found
IERR = 10	switched shunt not found
IERR = 11	prerequisite requirements for API are not met

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4.9.7. SWSMOD_PACK

Use this API to remove entries that are marked as unused from the switched shunt model connection tables and the switched shunt model array allocation tables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SWSMOD_PACK
```

Python command syntax:

```
ierr = swsmod_pack()
```

Fortran command syntax:

```
CALL SWSMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.9.8. SWSMOD_REMOVE

Use this API to remove the switched shunt model from a specified switched shunt.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SWSMOD_REMOVE  IBUS
```

Python command syntax:

```
ierr = swsmod_remove(ibus)
```

Fortran command syntax:

```
CALL SWSMOD_REMOVE(IBUS, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no model at this switched shunt device
	IERR = 2	bus not found
	IERR = 3	switched shunt not found
	IERR = 4	prerequisite requirements for API are not met

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4.9.9. SWSMOD_STATUS

Use this API to change the status of the switched shunt model at a specified switched shunt device.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SWSMOD_STATUS  IBUS  STATUS
```

Python command syntax:

```
ierr = swsmod_status(ibus, status)
```

Fortran command syntax:

```
CALL SWSMOD_STATUS(IBUS, STATUS, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service.
	STATUS = 1	in-service.
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no model at this switched shunt device
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	switched shunt not found
	IERR = 5	prerequisite requirements for API are not met

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4.9.10. SWSMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those switched shunt models that are assigned to switched shunt that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SWSMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = swsmod_unconnected(opt)
```

Fortran command syntax:

```
CALL SWSMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected switched shunt models
	OPT = 1	remove unconnected switched shunt models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.9.11. SWSMOD_USER

Use this API to list user-written switched shunt model definitions or to remove user-written switched shunt model definitions that are not assigned to any switched shunt devices (unused) from the user model definition tables.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SWSMOD_USER  OPT
```

Python command syntax:

```
ierr = swsmod_user(opt)
```

Fortran command syntax:

```
CALL SWSMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written switched shunt model definitions
	OPT = 1	remove unused user-written switched shunt model definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.10. Auxiliary Signal Models

4.10.1. ADD_FCTSAUXSIGNAL_MODEL

Use this API to add a FACTS device auxiliary signal model at the specified injection point of the designated FACTS device. If a model already exists at the specified injection point of the FACTS device, it is replaced.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ADD_FCTSAUXSIGNAL_MODEL  FD  ISGNDX  'MODEL'  NICN  IDATA  CHDATA
 NCON  RDATA  IERR
```

Python command syntax:

```
ierr = add_fctsauxsignal_model(fd, isgndx, model, nicn, idata, chdata,
 ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_FCTSAUXSIGNAL_MODEL(FD, ISGNDX, MODEL, NICN, IDATA, CHDATA, NCON,
 RDATA, IERR)

CALL ADD_FCTSAUXSIGNAL_MODEL_I(NUM, ISGNDX, MODEL, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)

CALL ADD_FCTSAUXSIGNAL_MODEL_C(NAME, ISGNDX, MODEL, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).
Character*12 NAME	name of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).
Character*16 MODEL	model name (input; no default allowed).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either

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	defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	invalid ISGNDX value for a FACTS device
	IERR = 3	unknown model name
	IERR = 4	model is not an auxiliary signal model
	IERR = 5	# CONs specified is not the number expected
	IERR = 6	# ICONs specified is not the number expected
	IERR = 7	maximum number of array elements exceeded
	IERR = 8	auxiliary signal model connection tables full--use pack auxiliary signal model tables function
	IERR = 9	this auxiliary signal may be used only with two-terminal dc lines
	IERR = 10	prerequisite requirements for API are not met

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4.10.2. ADD_MTDCAUXSIGNAL_MODEL

Use this API to add a multi-terminal dc line auxiliary signal model at the specified auxiliary signal index of the designated multi-terminal dc line. If a model already exists at the specified auxiliary signal index of the multi-terminal dc line, it is replaced.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ADD_MTDCAUXSIGNAL_MODEL  DC  ISGNDX  'MODEL'  NICN  IDATA  CHDATA
 NCON  RDATA  IERR
```

Python command syntax:

```
ierr = add_mtdcauxsignal_model(dc, isgndx, model, nicn, idata, chdata,
 ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_MTDCAUXSIGNAL_MODEL(DC, ISGNDX, MODEL, NICN, IDATA, CHDATA, NCON,
 RDATA, IERR)

CALL ADD_MTDCAUXSIGNAL_MODEL_I(NUM, ISGNDX, MODEL, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)

CALL ADD_MTDCAUXSIGNAL_MODEL_C(NAME, ISGNDX, MODEL, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).
Character*12 NAME	name of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).
Character*16 MODEL	model name (input; no default allowed).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).

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Integer NCON	number of CONs required by the model (input; no default allowed).
Real RDATA (NCON)	CON data (input; 0.0).
Integer IERR	error code (output).

IERR = 0	no error
IERR = 1	dc line not found
IERR = 2	invalid ISGNDX value for a multi-terminal dc line
IERR = 3	unknown model name
IERR = 4	model is not an auxiliary signal model
IERR = 5	# CONs specified is not the number expected
IERR = 6	# ICONs specified is not the number expected
IERR = 7	maximum number of array elements exceeded
IERR = 8	auxiliary signal model connection tables full--use pack auxiliary signal model tables function
IERR = 9	this auxiliary signal may be used only with two-terminal dc lines
IERR = 10	prerequisite requirements for API are not met

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4.10.3. ADD_TTDCAUXSIGNAL_MODEL

Use this API to add a two-terminal dc line auxiliary signal model at the specified injection point of the designated two-terminal dc line. If a model already exists at the specified injection point of the two-terminal dc line, it is replaced.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ADD_TTDCAUXSIGNAL_MODEL  DC  ISGNDX  'MODEL'  NICN  IDATA  CHDATA
 NCON  RDATA  IERR
```

Python command syntax:

```
ierr = add_ttdcauxsignal_model(dc, isgndx, model, nicn, idata, chdata,
 ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_TTDCAUXSIGNAL_MODEL(DC, ISGNDX, MODEL, NICN, IDATA, CHDATA, NCON,
 RDATA, IERR)

CALL ADD_TTDCAUXSIGNAL_MODEL_I(NUM, ISGNDX, MODEL, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)

CALL ADD_TTDCAUXSIGNAL_MODEL_C(NAME, ISGNDX, MODEL, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).
Character*12 NAME	name of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).
Character*16 MODEL	model name (input; no default allowed).
Integer NICN	number of ICONs required by the model (input; no default allowed).
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).

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Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a two-terminal dc line
	IERR = 3	unknown model name
	IERR = 4	model is not an auxiliary signal model
	IERR = 5	# CONs specified is not the number expected
	IERR = 6	# ICONs specified is not the number expected
	IERR = 7	maximum number of array elements exceeded
	IERR = 8	auxiliary signal model connection tables full--use pack auxiliary signal model tables function
	IERR = 10	prerequisite requirements for API are not met

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4.10.4. ADD_VSDCAUXSIGNAL_MODEL

Use this API to add a VSC dc line auxiliary signal model at the specified injection point of the designated VSC dc line. If a model already exists at the specified injection point of the VSC dc line, it is replaced.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_ADD_VSDCAUXSIGNAL_MODEL  'DCNAME'  ISGNDX  'MODEL'  NICN  IDATA
 CHDATA  NCON  RDATA  IERR
```

Python command syntax:

```
ierr = add_vsdcauxsignal_model(dcname, isgndx, model, nicn, idata, chdata,
 ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_VSDCAUXSIGNAL_MODEL(DCNAME, ISGNDX, MODEL, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)
```

Where:

Character*12 DCNAME	name of the VSC dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a VSC dc line
	IERR = 3	unknown model name
	IERR = 4	model is not an auxiliary signal model
	IERR = 5	# CONs specified is not the number expected
	IERR = 6	# ICONs specified is not the number expected

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IERR = 7	maximum number of array elements exceeded
IERR = 8	auxiliary signal model connection tables full--use pack auxiliary signal model tables function
IERR = 9	this auxiliary signal may be used only with two-terminal dc lines
IERR = 10	prerequisite requirements for API are not met

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4.10.5. AUXMOD_PACK

Use this API to remove entries that are marked as unused from the auxiliary signal model connection tables and the auxiliary signal array allocation tables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_AUXMOD_PACK
```

Python command syntax:

```
ierr = auxmod_pack()
```

Fortran command syntax:

```
CALL AUXMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.10.6. AUXMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those auxiliary signal models that are assigned to dc lines or FACTS devices that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_AUXMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = auxmod_unconnected(opt)
```

Fortran command syntax:

```
CALL AUXMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected auxiliary signal models
	OPT = 1	remove unconnected auxiliary signal models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.10.7. AUXMOD_USER

Use this API to list user-written auxiliary signal model definitions or to remove user-written auxiliary signal model definitions that are not assigned to any dc lines or FACTS devices (unused) from the user model definition tables.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_AUXMOD_USER  OPT
```

Python command syntax:

```
ierr = auxmod_user(opt)
```

Fortran command syntax:

```
CALL AUXMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written auxiliary signal model definitions
	OPT = 1	remove unused user-written auxiliary signal model definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.10.8. CHANGE_FCTSAUXMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of an auxiliary signal model of a specified FACTS device. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_FCTSAUXMOD_CHRICN  FD  ISGNDX  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_fctsauxmod_chricn(fd, isgndx, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_FCTSAUXMOD_CHRICN(FD, ISGNDX, MODEL, J, CHDATA, IERR)

CALL CHANGE_FCTSAUXMOD_CHRICN_I(NUM, ISGNDX, MODEL, J, CHDATA, IERR)

CALL CHANGE_FCTSAUXMOD_CHRICN_C(NAME, ISGNDX, MODEL, J, CHDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	invalid ISGNDX value for a FACTS device
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this FACTS device
	IERR = 5	model at this index for this FACTS device is aaa not xxx

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IERR = 6	no input ICONs used in model
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.9. CHANGE_FCTSAUXMOD_CON

Use this API to change the value of a CON of an auxiliary signal model of a specified FACTS device.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_FCTSAUXMOD_CON  FD  ISGNDX  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_fctsauxmod_con(fd, isgndx, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_FCTSAUXMOD_CON(FD, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_FCTSAUXMOD_CON_I(NUM, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_FCTSAUXMOD_CON_C(NAME, ISGNDX, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	invalid ISGNDX value for a FACTS device
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this FACTS device
	IERR = 5	model at this index for this FACTS device is aaa not xxx
	IERR = 6	no CONs used in model

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IERR = 7	CON number outside of the CON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.10. CHANGE_FCTSAUXMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_FCTSAUXMOD_CON, CHANGE_FCTSAUXMOD_ICON, and CHANGE_FCTSAUXMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of an auxiliary signal model of a specified FACTS device.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_FCTSAUXMOD_DATA  FD  ISGNDX  'NAME'  J  RDATA  I  IDATA
 'CHDATA'
```

Python command syntax:

```
ierr = change_fctsauxmod_data(fd,isgndx,name,j,rdata,i,idata,chdata)
```

Fortran command syntax:

```
CALL CHANGE_FCTSAUXMOD_DATA(FD,ISGNDX,NAME,J,RDATA,I,IDATA,CHDATA,IERR)
```

Where:

Integer FD	number of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).	
Real RDATA	new CON value (input; unchanged).	
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).	
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	invalid ISGNDX value for a FACTS device
	IERR = 3	unknown model name

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IERR = 4	no auxiliary signal model at this index for this FACTS device
IERR = 5	model at this index for this FACTS device is aaa not xxx
IERR = 6	no CONs used in model; or
IERR = 6	no input ICONs used in model; or
IERR = 7	CON number outside of the CON range of this model; or
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.11. CHANGE_FCTSAUXMOD_ICON

Use this API to change the value of an integer ICON of an auxiliary signal model of a specified FACTS device. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_FCTSAUXMOD_ICON  FD  ISGNDX  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_fctsauxmod_icon(fd, isgndx, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_FCTSAUXMOD_ICON(FD, ISGNDX, MODEL, J, IDATA, IERR)

CALL CHANGE_FCTSAUXMOD_ICON_I(NUM, ISGNDX, MODEL, J, IDATA, IERR)

CALL CHANGE_FCTSAUXMOD_ICON_C(NAME, ISGNDX, MODEL, J, IDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 2	FACTS device not found
	IERR = 3	invalid ISGNDX value for a FACTS device
	IERR = 4	unknown model name
	IERR = 5	no auxiliary signal model at this index for this FACTS device
	IERR = 6	model at this index for this FACTS device is aaa not xxx

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IERR = 7	no input ICONs used in model
IERR = 8	ICON number outside of the ICON range of this model
IERR = 9	prerequisite requirements for API are not met

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4.10.12. CHANGE_FCTSAUXMOD_VAR

Use this API to change the value of a VAR of an auxiliary signal model of a specified FACTS device.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_FCTSAUXMOD_VAR  FD  ISGNDX  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_fctsauxmod_var(fd, isgndx, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_FCTSAUXMOD_VAR(FD, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_FCTSAUXMOD_VAR_I(NUM, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_FCTSAUXMOD_VAR_C(NAME, ISGNDX, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	invalid ISGNDX value for a FACTS device
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this FACTS device
	IERR = 5	model at this index for this FACTS device is aaa not xxx
	IERR = 6	no VARs used in model

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IERR = 7	VAR number outside of the VAR range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.13. CHANGE_MTDCAUXMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of an auxiliary signal model of a specified multi-terminal dc line. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_MTDCAUXMOD_CHRICN  DC  ISGNDX  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_mtdcauxmod_chricn(dc, isgndx, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCAUXMOD_CHRICN(DC, ISGNDX, MODEL, J, CHDATA, IERR)

CALL CHANGE_MTDCAUXMOD_CHRICN_I(NUM, ISGNDX, MODEL, J, CHDATA, IERR)

CALL CHANGE_MTDCAUXMOD_CHRICN_C(NAME, ISGNDX, MODEL, J, CHDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).
Character*12 NAME	name of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).
Character*16 MODEL	model name (input; no default allowed).
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).
Character*12 CHDATA	new character ICON value (input; blank by default).
Integer IERR	error code (output).

IERR = 0	no error
IERR = 1	dc line not found
IERR = 2	invalid ISGNDX value for a multi-terminal dc line
IERR = 3	unknown model name
IERR = 4	no auxiliary signal model at this index for this multi-terminal dc line
IERR = 5	model at this index for this multi-terminal dc line is aaa not xxx

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IERR = 6	no input ICONs used in model
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.14. CHANGE_MTDCAUXMOD_CON

Use this API to change the value of a CON of an auxiliary signal model of a specified multi-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_MTDCAUXMOD_CON  DC  ISGNDX  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_mtdcauxmod_con(dc, isgndx, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCAUXMOD_CON(DC, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_MTDCAUXMOD_CON_I(NUM, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_MTDCAUXMOD_CON_C(NAME, ISGNDX, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a multi-terminal dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this multi-terminal dc line
	IERR = 5	model at this index for this multi-terminal dc line is aaa not xxx

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IERR = 6	no CONs used in model
IERR = 7	CON number outside of the CON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.15. CHANGE_MTDCAUXMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_MTDCAUXMOD_CON, CHANGE_MTDCAUXMOD_ICON, and CHANGE_MTDCAUXMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of an auxiliary signal model of a specified multi-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_MTDCAUXMOD_DATA  DC  ISGNDX  'NAME'  J  RDATA  I  IDATA
 'CHDATA'
```

Python command syntax:

```
ierr = change_mtdcauxmod_data(dc,isgndx,name,j,rdata,i,idata,chdata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCAUXMOD_DATA(DC,ISGNDX,NAME,J,RDATA,I,IDATA,CHDATA,IERR)
```

Where:

Integer DC	number of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).	
Real RDATA	new CON value (input; unchanged).	
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).	
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a multi-terminal dc line

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IERR = 3	unknown model name
IERR = 4	no auxiliary signal model at this index for this multi-terminal dc line
IERR = 5	model at this index for this multi-terminal dc line is aaa not xxx
IERR = 6	no CONs used in model; or
IERR = 6	no input ICONs used in model; or
IERR = 7	CON number outside of the CON range of this model; or
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.16. CHANGE_MTDCAUXMOD_ICON

Use this API to change the value of an integer ICON of an auxiliary signal model of a specified multi-terminal dc line. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_MTDCAUXMOD_ICON  DC  ISGNDX  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_mtdcauxmod_icon(dc, isgndx, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCAUXMOD_ICON(DC, ISGNDX, MODEL, J, IDATA, IERR)

CALL CHANGE_MTDCAUXMOD_ICON_I(NUM, ISGNDX, MODEL, J, IDATA, IERR)

CALL CHANGE_MTDCAUXMOD_ICON_C(NAME, ISGNDX, MODEL, J, IDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a multi-terminal dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this multi-terminal dc line
	IERR = 5	model at this index for this multi-terminal dc line is aaa not xxx

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IERR = 6	no input ICONs used in model
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.17. CHANGE_MTDCAUXMOD_VAR

Use this API to change the value of a VAR of an auxiliary signal model of a specified multi-terminal dc line.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_MTDCAUXMOD_VAR  DC  ISGNDX  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_mtdcauxmod_var(dc, isgndx, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_MTDCAUXMOD_VAR(DC, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_MTDCAUXMOD_VAR_I(NUM, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_MTDCAUXMOD_VAR_C(NAME, ISGNDX, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a multi-terminal dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this multi-terminal dc line
	IERR = 5	model at this index for this multi-terminal dc line is aaa not xxx

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IERR = 6	no VARs used in model
IERR = 7	VAR number outside of the VAR range of this model
IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.10.18. CHANGE_TTDCAUXMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of an auxiliary signal model of a specified two-terminal dc line. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_TTDCAUXMOD_CHRICN  DC  ISGNDX  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_ttdcauxmod_chricn(dc, isgndx, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCAUXMOD_CHRICN(DC, ISGNDX, MODEL, J, CHDATA, IERR)

CALL CHANGE_TTDCAUXMOD_CHRICN_I(NUM, ISGNDX, MODEL, J, CHDATA, IERR)

CALL CHANGE_TTDCAUXMOD_CHRICN_C(NAME, ISGNDX, MODEL, J, CHDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a two-terminal dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this two-terminal dc line
	IERR = 5	model at this index for this two-terminal dc line is aaa not xxx

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

IERR = 6	no input ICONs used in model
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.10.19. CHANGE_TTDCAUXMOD_CON

Use this API to change the value of a CON of an auxiliary signal model of a specified two-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_TTDCAUXMOD_CON  DC  ISGNDX  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_ttdcauxmod_con(dc, isgndx, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCAUXMOD_CON(DC, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_TTDCAUXMOD_CON_I(NUM, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_TTDCAUXMOD_CON_C(NAME, ISGNDX, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a two-terminal dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this two-terminal dc line
	IERR = 5	model at this index for this two-terminal dc line is aaa not xxx

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IERR = 6	no CONs used in model
IERR = 7	CON number outside of the CON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.20. CHANGE_TTDCAUXMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_TTDCAUXMOD_CON, CHANGE_TTDCAUXMOD_ICON, and CHANGE_TTDCAUXMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of an auxiliary signal model of a specified two-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_TTDCAUXMOD_DATA  DC  ISGNDX  'NAME'  J  RDATA  I  IDATA
 'CHDATA'
```

Python command syntax:

```
ierr = change_ttdcauxmod_data(dc,isgndx,name,j,rdata,i,idata,chdata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCAUXMOD_DATA(DC,ISGNDX,NAME,J,RDATA,I,IDATA,CHDATA,IERR)
```

Where:

Integer DC	number of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).	
Real RDATA	new CON value (input; unchanged).	
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).	
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a two-terminal dc line

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IERR = 3	unknown model name
IERR = 4	no auxiliary signal model at this index for this two-terminal dc line
IERR = 5	model at this index for this two-terminal dc line is aaa not xxx
IERR = 6	no CONs used in model
IERR = 6	no input ICONs used in model
IERR = 7	CON number outside of the CON range of this model
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.21. CHANGE_TTDCAUXMOD_ICON

Use this API to change the value of an integer ICON of an auxiliary signal model of a specified two-terminal dc line. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_TTDCAUXMOD_ICON  DC  ISGNDX  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_ttdcauxmod_icon(dc, isgndx, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCAUXMOD_ICON(DC, ISGNDX, MODEL, J, IDATA, IERR)

CALL CHANGE_TTDCAUXMOD_ICON_I(NUM, ISGNDX, MODEL, J, IDATA, IERR)

CALL CHANGE_TTDCAUXMOD_ICON_C(NAME, ISGNDX, MODEL, J, IDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a two-terminal dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this two-terminal dc line
	IERR = 5	model at this index for this two-terminal dc line is aaa not xxx

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IERR = 6	no input ICONs used in model
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.10.22. CHANGE_TTDCAUXMOD_VAR

Use this API to change the value of a VAR of an auxiliary signal model of a specified two-terminal dc line.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_TTDCAUXMOD_VAR  DC  ISGNDX  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_ttdcauxmod_var(dc, isgndx, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_TTDCAUXMOD_VAR(DC, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_TTDCAUXMOD_VAR_I(NUM, ISGNDX, MODEL, J, RDATA, IERR)

CALL CHANGE_TTDCAUXMOD_VAR_C(NAME, ISGNDX, MODEL, J, RDATA, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a two-terminal dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this two-terminal dc line
	IERR = 5	model at this index for this two-terminal dc line is aaa not xxx

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IERR = 6	no VARs used in model
IERR = 7	VAR number outside of the VAR range of this model
IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.10.23. CHANGE_VSDCAUXMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of an auxiliary signal model of a specified VSC dc line. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VSDCAUXMOD_CHRICN  'DCNAME'  ISGNDX  'MODEL'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_vsdcauxmod_chricn(dcname, isgndx, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCAUXMOD_CHRICN(DCNAME,ISGNDX,MODEL,J,CHDATA,IERR)
```

Where:

Character*12 DCNAME	name of the VSC dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a VSC dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this VSC dc line
	IERR = 5	model at this index for this VSC dc line is aaa not xxx
	IERR = 6	no input ICONs used in model
	IERR = 7	ICON number outside of the ICON range of this model
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.10.24. CHANGE_VSDCAUXMOD_CON

Use this API to change the value of a CON of an auxiliary signal model of a specified VSC dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VSDCAUXMOD_CON  'DCNAME'  ISGNDX  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_vsdcauxmod_con(dcname, isgndx, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCAUXMOD_CON(DCNAME, ISGNDX, MODEL, J, RDATA, IERR)
```

Where:

Character*12 DCNAME	name of the VSC dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a VSC dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this VSC dc line
	IERR = 5	model at this index for this VSC dc line is aaa not xxx
	IERR = 6	no CONs used in model
	IERR = 7	CON number outside of the CON range of this model
	IERR = 8	prerequisite requirements for API are not met

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4.10.25. CHANGE_VSDCAUXMOD_DATA

This API routine is obsolete. It has been replaced by the API routines CHANGE_VSDCAUXMOD_CON, CHANGE_VSDCAUXMOD_ICON, and CHANGE_VSDCAUXMOD_CHRICN .

Use this API to change the value of an ICON and/or CON of an auxiliary signal model of a specified VSC dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VSDCAUXMOD_DATA 'DCNAME' ISGNDX 'NAME' J RDATA I IDATA
 'CHDATA'
```

Python command syntax:

```
ierr = change_vsdcauxmod_data(dcname,isgndx,name,j,rdata,i,idata,chdata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCAUXMOD_DATA(DCNAME, ISGNDX, NAME, J, RDATA, I, IDATA,
 CHDATA, IERR)
```

Where:

Character*12 DCNAME	name of the VSC dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model, or 0 for no CON change (1 for CON(J), 2 for CON(J+1), etc.) (input; 0 by default).	
Real RDATA	new CON value (input; unchanged).	
Integer I	ICON number to change relative to the start of the block of ICONs assigned to this model, or 0 for no ICON change (1 for ICON(I), 2 for ICON(I+1), etc.) (input; 0 by default).	
Integer IDATA	new integer ICON value (input; unchanged by default). When a character value is intended for an ICON value, the IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA	new character ICON value (no default allowed if the character ICON is to be changed). When an integer value is intended for an ICON value, the CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a VSC dc line

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IERR = 3	unknown model name
IERR = 4	no auxiliary signal model at this index for this VSC dc line
IERR = 5	model at this index for this VSC dc line is aaa not xxx
IERR = 6	no CONs used in model
IERR = 6	no input ICONs used in model
IERR = 7	CON number outside of the CON range of this model
IERR = 7	ICON number outside of the ICON range of this model
IERR = 8	prerequisite requirements for API are not met

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4.10.26. CHANGE_VSDCAUXMOD_ICON

Use this API to change the value of an integer ICON of an auxiliary signal model of a specified VSC dc line. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CHANGE_VSDCAUXMOD_ICON  'DCNAME'  ISGNDX  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_vsdcauxmod_icon(dcname, isgndx, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCAUXMOD_ICON(DCNAME, ISGNDX, MODEL, J, IDATA, IERR)
```

Where:

Character*12 DCNAME	name of the VSC dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a VSC dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this VSC dc line
	IERR = 5	model at this index for this VSC dc line is aaa not xxx
	IERR = 6	no input ICONs used in model
	IERR = 7	ICON number outside of the ICON range of this model
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.10.27. CHANGE_VSDCAUXMOD_VAR

Use this API to change the value of a VAR of an auxiliary signal model of a specified VSC dc line.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CHANGE_VSDCAUXMOD_VAR  'DCNAME'  ISGNDX  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_vsdcauxmod_var(dcname, isgndx, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_VSDCAUXMOD_VAR(DCNAME, ISGNDX, MODEL, J, RDATA, IERR)
```

Where:

Character*12 DCNAME	name of the VSC dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a VSC dc line
	IERR = 3	unknown model name
	IERR = 4	no auxiliary signal model at this index for this VSC dc line
	IERR = 5	model at this index for this VSC dc line is aaa not xxx
	IERR = 6	no VARs used in model
	IERR = 7	VAR number outside of the VAR range of this model
	IERR = 8	prerequisite requirements for API are not met

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4.10.28. FCTSAUXMOD_REMOVE

Use this API to remove an auxiliary signal model from the specified injection point of a designated FACTS device.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_FCTSAUXMOD_REMOVE  FD  ISGNDX
```

Python command syntax:

```
ierr = fctsauxmod_remove(fd, isgndx)
```

Fortran command syntax:

```
CALL FCTSAUXMOD_REMOVE(FD, ISGNDX, IERR)

CALL FCTSAUXMOD_REMOVE_I(NUM, ISGNDX, IERR)

CALL FCTSAUXMOD_REMOVE_C(NAME, ISGNDX, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	invalid ISGNDX value for a FACTS device
	IERR = 3	no auxiliary signal model at this index for this FACTS device
	IERR = 4	prerequisite requirements for API are not met

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4.10.29. FCTSAUXMOD_STATUS

Use this API to change the status of the auxiliary signal model at the specified injection point of a designated FACTS device.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_FCTSAUXMOD_STATUS  FD  ISGNDX  STATUS
```

Python command syntax:

```
ierr = fctsauxmod_status(fd, isgndx, status)
```

Fortran command syntax:

```
CALL FCTSAUXMOD_STATUS(FD, ISGNDX, STATUS, IERR)

CALL FCTSAUXMOD_STATUS_I(NUM, ISGNDX, STATUS, IERR)

CALL FCTSAUXMOD_STATUS_C(NAME, ISGNDX, STATUS, IERR)
```

Where:

Integer or Character*12 FD	quoted name of the FACTS device receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the FACTS device receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	FACTS device not found
	IERR = 2	invalid ISGNDX value for a FACTS device
	IERR = 3	no auxiliary signal model at this index for this FACTS device
	IERR = 4	invalid STATUS value
	IERR = 5	prerequisite requirements for API are not met

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4.10.30. MTDCAUXMOD_REMOVE

Use this API to remove an auxiliary signal model from the specified auxiliary signal index of the designated multi-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MTDCAUXMOD_REMOVE  DC  ISGNDX
```

Python command syntax:

```
ierr = mtdcauxmod_remove(dc, isgndx)
```

Fortran command syntax:

```
CALL MTDCAUXMOD_REMOVE(DC, ISGNDX, IERR)

CALL MTDCAUXMOD_REMOVE_I(NUM, ISGNDX, IERR)

CALL MTDCAUXMOD_REMOVE_C(NAME, ISGNDX, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a multi-terminal dc line
	IERR = 3	no auxiliary signal model at this index for this multi-terminal dc line
	IERR = 4	prerequisite requirements for API are not met

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4.10.31. MTDCAUXMOD_STATUS

Use this API to change the status of the auxiliary signal model at the specified auxiliary signal index of the designated multi-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_MTDCAUXMOD_STATUS  DC  ISGNDX  STATUS
```

Python command syntax:

```
ierr = mtdcauxmod_status(dc, isgndx, status)
```

Fortran command syntax:

```
CALL MTDCAUXMOD_STATUS(DC, ISGNDX, STATUS, IERR)

CALL MTDCAUXMOD_STATUS_I(NUM, ISGNDX, STATUS, IERR)

CALL MTDCAUXMOD_STATUS_C(NAME, ISGNDX, STATUS, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the multi-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the multi-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a multi-terminal dc line
	IERR = 3	no auxiliary signal model at this index for this multi-terminal dc line
	IERR = 4	invalid STATUS value
	IERR = 5	prerequisite requirements for API are not met

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4.10.32. TTDCAUXMOD_REMOVE

Use this API to remove an auxiliary signal model from the specified injection point of the designated two-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_TTDCAUXMOD_REMOVE  DC  ISGNDX
```

Python command syntax:

```
ierr = ttdcauxmod_remove(dc, isgndx)
```

Fortran command syntax:

```
CALL TTDCAUXMOD_REMOVE(DC, ISGNDX, IERR)

CALL TTDCAUXMOD_REMOVE_I(NUM, ISGNDX, IERR)

CALL TTDCAUXMOD_REMOVE_C(NAME, ISGNDX, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a two-terminal dc line
	IERR = 3	no auxiliary signal model at this index for this two-terminal dc line
	IERR = 4	prerequisite requirements for API are not met

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4.10.33. TTDCAUXMOD_STATUS

Use this API to change the status of the auxiliary signal model at the specified injection point of the designated two-terminal dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_TTDCAUXMOD_STATUS  DC  ISGNDX  STATUS
```

Python command syntax:

```
ierr = ttdcauxmod_status(dc, isgndx, status)
```

Fortran command syntax:

```
CALL TTDCAUXMOD_STATUS(DC, ISGNDX, STATUS, IERR)

CALL TTDCAUXMOD_STATUS_I(NUM, ISGNDX, STATUS, IERR)

CALL TTDCAUXMOD_STATUS_C(NAME, ISGNDX, STATUS, IERR)
```

Where:

Integer or Character*12 DC	quoted name of the two-terminal dc line receiving the of the auxiliary signal model; if an unquoted integer value is specified, the API converts it to a left-justified name (input; no default allowed).	
Integer NUM	number (which the API converts to a left-justified name) of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Character*12 NAME	name of the two-terminal dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 2	dc line not found
	IERR = 3	invalid ISGNDX value for a two-terminal dc line
	IERR = 4	no auxiliary signal model at this index for this two-terminal dc line
	IERR = 5	invalid STATUS value
	IERR = 6	prerequisite requirements for API are not met

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4.10.34. VSDCAUXMOD_REMOVE

Use this API to remove an auxiliary signal model from the specified injection point of the designated VSC dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_VSDCAUXMOD_REMOVE  'DCNAME'  ISGNDX
```

Python command syntax:

```
ierr = vsdcauxmod_remove(dcname, isgndx)
```

Fortran command syntax:

```
CALL VSDCAUXMOD_REMOVE(DCNAME, ISGNDX, IERR)
```

Where:

Character*12 DCNAME	name of the VSC dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	dc line not found
	IERR = 2	invalid ISGNDX value for a VSC dc line
	IERR = 3	no auxiliary signal model at this index for this VSC dc line
	IERR = 4	prerequisite requirements for API are not met

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4.10.35. VSDCAUXMOD_STATUS

Use this API to change the status of the auxiliary signal model at the specified injection point of the designated VSC dc line.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_VSDCAUXMOD_STATUS  'DCNAME'  ISGNDX  STATUS
```

Python command syntax:

```
ierr = vsdcauxmod_status(dcname, isgndx, status)
```

Fortran command syntax:

```
CALL VSDCAUXMOD_STATUS(DCNAME, ISGNDX, STATUS, IERR)
```

Where:

Character*12 DCNAME	name of the VSC dc line receiving the of the auxiliary signal model (input; no default allowed).	
Integer ISGNDX	auxiliary signal injection point number, or the auxiliary signal index (input; 1 by default).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	VSC dc line not found
	IERR = 2	invalid ISGNDX value for a VSC dc line
	IERR = 3	no auxiliary signal model at this index for this VSC dc line
	IERR = 4	invalid STATUS value
	IERR = 5	prerequisite requirements for API are not met

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4.11. Induction Machine Models

4.11.1. ADD_INDMAC_MODEL

Use this API to add an induction machine model to a specified machine. If a model of the designated type already exists at the machine, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_INDMAC_MODEL  IBUS  'ID'  'NAME'  NICN  IDATA  CHDATA  NCON  RDATA
```

Python command syntax:

```
ierr = add_indmac_model(ibus, id, name, nicn, idata, chdata, ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_INDMAC_MODEL(IBUS, ID, NAME, NICN, IDATA, CHDATA, NCON, RDATA,
 IERR)
```

Where:

Integer IBUS	induction machine bus number (input; no default allowed).	
Character*2 ID	induction machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	Induction machine model connection tables full--use pack induction machine model tables function
	IERR = 2	Induction machine model tables are corrupt
	IERR = 3	unknown model NAME
	IERR = 4	model is not an induction machine model
	IERR = 5	# CONs specified is not the number expected
	IERR = 6	# ICONs specified is not the number expected

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IERR = 7	maximum number of array elements exceeded
IERR = 8	bus not found
IERR = 9	machine not found
IERR = 10	prerequisite requirements for API are not met

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4.11.2. CHANGE_IMMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of an induction machine model at a specified machine. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_IMMOD_CHRICN  IBUS  'ID'  'NAME'  J  'CHDATA'
```

Python command syntax:

```
ierr = change_immod_chricn(ibus, id, name, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_IMMOD_CHRICN(IBUS, ID, NAME, J, CHDATA, IERR)
```

Where:

Integer IBUS	induction machine bus number (input; no default allowed).	
Character*2 ID	induction machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	No model at this induction machine
	IERR = 3	model at this machine is aaa not NAME
	IERR = 4	no input ICONs used in model NAME
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

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4.11.3. CHANGE_IMMOD_CON

Use this API to change the value of a CON of a designated induction machine model at a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_IMMOD_CON  IBUS  'ID'  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_immod_con(ibus, id, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_IMMOD_CON(IBUS, ID, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	induction machine bus number (input; no default allowed).	
Character*2 ID	induction machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	CON number to change relative to the start of the block of CONs assigned to this model (1 for CON(J), 2 for CON(J+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	No model at this induction machine
	IERR = 3	model at this machine is aaa not NAME
	IERR = 4	no input ICONs used in model NAME
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

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4.11.4. CHANGE_IMMOD_ICON

Use this API to change the value of an integer ICON of a designated induction machine model at a specified machine. The corresponding character CHRICN is set to '$$'.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_IMMOD_ICON  IBUS  'ID'  'NAME'  J  IDATA
```

Python command syntax:

```
ierr = change_immod_icon(ibus, id, name, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_IMMOD_ICON(IBUS, ID, NAME, J, IDATA, IERR)
```

Where:

Integer IBUS	induction machine bus number (input; no default allowed).	
Character*2 ID	induction machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	ICON number to change relative to the start of the block of ICONs assigned to this model (1 for ICON(I), 2 for ICON(I+1), etc.) (input; no default allowed).	
Integer IDATA	new integer ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	No model at this induction machine
	IERR = 3	model at this machine is aaa not NAME
	IERR = 4	no input CONs used in model NAME
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

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4.11.5. CHANGE_IMMOD_VAR

Use this API to change the value of a VAR of a designated induction machine model at a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_IMMOD_VAR  IBUS  'ID'  'NAME'  J  RDATA
```

Python command syntax:

```
ierr = change_immod_var(ibus, id, name, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_IMMOD_VAR(IBUS, ID, NAME, J, RDATA, IERR)
```

Where:

Integer IBUS	induction machine bus number (input; no default allowed).	
Character*2 ID	induction machine identifier (input; blank by default).	
Character*16 NAME	model name (input; no default allowed).	
Integer J	VAR number to change relative to the start of the block of VARs assigned to this model (1 for VAR(J), 2 for VAR(J+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model NAME
	IERR = 2	No model at this induction machine
	IERR = 3	model at this machine is aaa not NAME
	IERR = 4	no input VARs used in model NAME
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

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4.11.6. IMMOD_PACK

Use this API to remove entries that are marked as unused from the induction machine model contables. It packs the entries remaining in these tables at the front of the various arrays.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_IMMOD_PACK
```

Python command syntax:

```
ierr = immod_pack()
```

Fortran command syntax:

```
CALL IMMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

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4.11.7. IMMOD_REMOVE

Use this API to remove an induction machine model of a designated type from a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_IMMOD_REMOVE  IBUS  'ID'
```

Python command syntax:

```
ierr = immod_remove(ibus, id)
```

Fortran command syntax:

```
CALL IMMOD_REMOVE(IBUS, ID, IERR)
```

Where:

Integer IBUS	induction machine bus number (input; no default allowed).	
Character*2 ID	induction machine identifier (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no induction machine model at this machine
	IERR = 2	bus not found
	IERR = 3	machine not found
	IERR = 4	prerequisite requirements for API are not met

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4.11.8. IMMOD_STATUS

Use this API to change the status of an induction machine model at a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_IMMOD_STATUS  IBUS  'ID'  STATUS
```

Python command syntax:

```
ierr = immod_status(ibus, id, status)
```

Fortran command syntax:

```
CALL IMMOD_STATUS(IBUS, ID, STATUS, IERR)
```

Where:

Integer IBUS	induction machine bus number (input; no default allowed).	
Character*2 ID	induction machine identifier (input; blank by default).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no induction machine model at this machine
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	machine not found
	IERR = 5	prerequisite requirements for API are not met

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4.11.9. IMMOD_UNCONNECTED

Use this API to list or remove from dynamics working memory those induction machine models that are assigned to machines that are not present in the current power flow working case (unconnected).

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_IMMOD_UNCONNECTED  OPT
```

Python command syntax:

```
ierr = immod_unconnected(opt)
```

Fortran command syntax:

```
CALL IMMOD_UNCONNECTED(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list unconnected induction machine models
	OPT = 1	remove unconnected induction machine models
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.11.10. IMMOD_USER

Use this API to list user-written induction machine model definitions or to remove user-written induction machine model definitions that are not assigned to any machines (unused) from the user model definition tables.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_IMMOD_USER  OPT
```

Python command syntax:

```
ierr = immod_user(opt)
```

Fortran command syntax:

```
CALL IMMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written induction machine model definitions
	OPT = 1	remove unused user-written induction machine model definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

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4.12. CCT Models

4.12.1. ADD_CCT2DCO_MODEL

Use this API to add an "other" type model attached to a 2-terminal dc line. If another model exists, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_CCT2DCO_MODEL  DC  'MODEL'  NICN  IDATA  CHDATA  NCON  RDATA
```

Python command syntax:

```
ierr = add_cct2dco_model(dc, model, nicn, idata, chdata, ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_CCT2DCO_MODEL(DC, MODEL, NICN, IDATA, CHDATA, NCON, RDATA, IERR)
```

Where:

Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a 2-terminal dc line "other" type model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function

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IERR = 7	dc line not found
IERR = 8	prerequisite requirements for API are not met

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4.12.2. ADD_CCT2WTD_MODEL

Use this API to add a 2-winding transformer device model to the specified 2-winding transformer. If a model already exists at the 2-winding transformer, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_CCT2WTD_MODEL  IBUS  JBUS  'ID'  'MODEL'  NICN  IDATA  CHDATA
 NCON  RDATA
```

Python command syntax:

```
ierr = add_cct2wtd_model(ibus,jbus,id,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL ADD_CCT2WTD_MODEL(IBUS, JBUS, ID, MODEL, NICN, IDATA, CHDATA, NCON,
 RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a 2-winding transformer device model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function

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IERR = 7	branch is not a transformer
IERR = 8	bus not found
IERR = 9	2-winding transformer not found
IERR = 10	prerequisite requirements for API are not met

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4.12.3. ADD_CCT3WTD_MODEL

Use this API to add a 3-winding transformer device model to the specified 3-winding transformer. If a model already exists at the 3-winding transformer, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_CCT3WTD_MODEL  IBUS JBUS KBUS ID MODEL NICN IDATA CHDATA NCON
 RDATA
```

Python command syntax:

```
ierr = add_cct3wtd_model(ibus, jbus, kbus, id, model, nicn, idata, chdata,
 ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_CCT3WTD_MODEL(IBUS, JBUS, KBUS, ID, MODEL, NICN, IDATA, CHDATA,
 NCON, RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Integer KBUS	third bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a 3-winding transformer device model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded

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IERR = 6	CCT model connection tables full-use pack CCT model tables function
IERR = 7	branch is not a transformer
IERR = 8	bus not found
IERR = 9	3-winding transformer not found
IERR = 10	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.4. ADD_CCTBRND_MODEL

Use this API to add a branch device model to the specified branch. If a model already exists at the branch, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_CCTBRND_MODEL  IBUS  JBUS  'ID'  'MODEL'  NICN  IDATA  CHDATA
 NCON  RDATA
```

Python command syntax:

```
ierr = add_cctbrnd_model(ibus,jbus,id,model,nicn,idata,chdata,ncon,rdata)
```

Fortran command syntax:

```
CALL ADD_CCTBRND_MODEL(IBUS, JBUS, ID, MODEL, NICN, IDATA, CHDATA, NCON,
 RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a branch device model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function

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IERR = 7	branch is a transformer
IERR = 8	bus not found
IERR = 9	branch not found
IERR = 10	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.5. ADD_CCTBRNO_MODEL

Use this API to add a branch "other" model to the specified branch. If a model already exists at the branch, it is replaced.

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_ADD_CCTBRNO_MODEL  IBUS JBUS ID MODEL NICN IDATA(1)..IDATA(NICN)
 CHDATA(1)..CHDATA(NICN) NCON RDATA(1)..RDATA(NCON)
```

Python command syntax:

```
ierr = add_cctbrno_model(ibus, jbus, id, model, nicn, idata, chdata, ncon,
 rdata)
```

Fortran command syntax:

```
CALL ADD_CCTBRNO_MODEL(IBUS, JBUS, ID, MODEL, NICN, IDATA, CHDATA, NCON,
 RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a branch device model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function

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IERR = 7	branch is a transformer
IERR = 8	bus not found
IERR = 9	branch not found
IERR = 10	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.6. ADD_CCTBUSO_MODEL

Use this API to add an "other" type model attached to a bus. If another model exists, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_CCTBUSO_MODEL  IBUS  'MODEL'  NICN  IDATA  CHDATA  NCON  RDATA
```

Python command syntax:

```
ierr = add_cctbuso_model(ibus, model, nicn, idata, chdata, ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_CCTBUSO_MODEL(IBUS, MODEL, NICN, IDATA, CHDATA, NCON, RDATA,
 IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a bus "other" type model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function
	IERR = 7	bus not found
	IERR = 8	prerequisite requirements for API are not met

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4.12.7. ADD_CCTMCNO_MODEL

Use this API to add a machine other model to the specified machine. If another model exists, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_CCTMCNO_MODEL  IBUS  'ID'  'MODEL'  NICN  IDATA  CHDATA  NCON
 RDATA
```

Python command syntax:

```
ierr = add_cctmcno_model(ibus, id, model, nicn, idata, chdata, ncon,
 rdata)
```

Fortran command syntax:

```
CALL ADD_CCTMCNO_MODEL(IBUS, ID, MODEL, NICN, IDATA, CHDATA, NCON, RDATA,
 IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a machine protection model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function
	IERR = 7	bus not found

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IERR = 8	machine not found
IERR = 9	prerequisite requirements for API are not met

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4.12.8. ADD_CCTMCNP_MODEL

Use this API to add a machine protection model to the specified machine. If another model exists, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_CCTMCNP_MODEL  IBUS  'ID'  'MODEL'  NICN  IDATA  CHDATA  NCON
 RDATA
```

Python command syntax:

```
ierr = add_cctmcnp_model(ibus, id, model, nicn, idata, chdata, ncon,
 rdata)
```

Fortran command syntax:

```
CALL ADD_CCTMCNP_MODEL(IBUS, ID, MODEL, NICN, IDATA, CHDATA, NCON, RDATA,
 IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a machine protection model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function
	IERR = 7	bus not found

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IERR = 8	machine not found
IERR = 9	prerequisite requirements for API are not met

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4.12.9. ADD_CCTMSCO_MODEL

Use this API to add a miscellaneous "other" type model with the specified model instance. If another model exists with the same model instance, it is replaced.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_ADD_CCTMSCO_MODEL  'MODEL'  MINS NICN  IDATA  CHDATA  NCON  RDATA
```

Python command syntax:

```
ierr = add_cctmsco_model(model, mins,  nicn, idata, chdata, ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_CCTMSCO_MODEL(MODEL, MINS, NICN, IDATA, CHDATA, NCON, RDATA,
 IERR)
```

Where:

Character*16 MODEL	model name (input; no default allowed).	
Integer MINS	model instance (>0) (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a bus "other" type model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function
	IERR = 7	invalid model instance
	IERR = 8	prerequisite requirements for API are not met

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4.12.10. ADD_CCTSWSO_MODEL

Use this API to add a switched shunt "other" model to the specified bus. If a model already exists at the bus, it is replaced.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_ADD_CCTSWSO_MODEL  IBUS MODEL NICN IDATA(1)..IDATA(NICN)
 CHDATA(1)..CHDATA(NICN) NCON RDATA(1)..RDATA(NCON)
```

Python command syntax:

```
ierr = add_cctswso_model(ibus, model, nicn, idata, chdata, ncon, rdata)
```

Fortran command syntax:

```
CALL ADD_CCTSWSO_MODEL(IBUS, MODEL, NICN, IDATA, CHDATA, NCON, RDATA,
 IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer NICN	number of ICONs required by the model (input; no default allowed).	
Integer IDATA (NICN)	ICON data (input; 0 by default). When a character value is intended for an ICON value, the corresponding IDATA entry is ignored and the integer ICON is set to 0).	
Character*12 CHDATA (NICN)	character ICON data (no default allowed for an ICON for which a character value is to be specified). When an integer value is intended for an ICON value, the corresponding CHDATA entry must be either defaulted or specified as either '$$' or ' '; the character ICON is set to '$$' (input; blank by default).	
Integer NCON	number of CONs required by the model (input; no default allowed).	
Real RDATA (NCON)	CON data (input; 0.0).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	model is not a switched shunt other model
	IERR = 3	# CONs specified is not the number expected
	IERR = 4	# ICONs specified is not the number expected
	IERR = 5	maximum number of array elements exceeded
	IERR = 6	CCT model connection tables full-use pack CCT model tables function
	IERR = 7	bus not found
	IERR = 8	switched shunt not found
	IERR = 9	prerequisite requirements for API are not met

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4.12.11. CCT2DCOMOD_REMOVE

Use this API to remove an "other" type model attached to a 2-terminal dc line.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCT2DCOMOD_REMOVE  DC  MODEL
```

Python command syntax:

```
ierr = cct2dcomod_remove(dc, model)
```

Fortran command syntax:

```
CALL CCT2DCOMOD_REMOVE(DC, MODEL, IERR)
```

Where:

Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	name of the model to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no "other" model at this dc line
	IERR = 2	dc line not found
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.12. CCT2DCOMOD_STATUS

Use this API to change status of an "other" type model attached to a 2-terminal dc line.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCT2DCOMOD_STATUS  DC  MODEL  STATUS
```

Python command syntax:

```
ierr = cct2dcomod_status(dc, model, status)
```

Fortran command syntax:

```
CALL CCT2DCOMOD_STATUS(DC, MODEL, STATUS, IERR)
```

Where:

Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	name of the model whose status is to be changed (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no "other" model at this dc line
	IERR = 2	invalid STATUS value
	IERR = 3	dc line not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.13. CCT2WTDMOD_REMOVE

Use this API to remove a 2-winding transformer device model attached to the specified 2-winding transformer.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCT2WTDMOD_REMOVE   IBUS  JBUS  'ID'
```

Python command syntax:

```
ierr = cct2wtdmod_remove(ibus, jbus, id)
```

Fortran command syntax:

```
CALL CCT2WTDMOD_REMOVE(IBUS, JBUS, ID, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no 2-winding transformer device model on this branch
	IERR = 2	bus not found
	IERR = 3	2-winding transformer not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.14. CCT2WTDMOD_STATUS

Use this API to change the status of a 2-winding transformer device model attached to the specified 2-winding transformer.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCT2WTDMOD_STATUS   IBUS  JBUS  'ID'  STATUS
```

Python command syntax:

```
ierr = cct2wtdmod_status(ibus, jbus, id, status)
```

Fortran command syntax:

```
CALL CCT2WTDMOD_STATUS(IBUS, JBUS, ID, STATUS, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no 2-winding transformer device model on this branch
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	2-winding transformer not found
	IERR = 5	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.15. CCT3WTDMOD_REMOVE

Use this API to remove a 3-winding transformer device model attached to the specified 3-winding transformer.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCT3WTDMOD_REMOVE  IBUS  JBUS  KBUS 'ID'
```

Python command syntax:

```
ierr = cct3wtdmod_remove(ibus, jbus, kbus, id)
```

Fortran command syntax:

```
CALL CCT3WTDMOD_REMOVE(IBUS, JBUS, KBUS, ID, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Integer KBUS	third bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no 3-winding transformer device model on this branch
	IERR = 2	bus not found
	IERR = 3	3-winding transformer not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.16. CCT3WTDMOD_STATUS

Use this API to change the status of a 3-winding transformer device model attached to the specified 3-winding transformer.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCT3WTDMOD_STATUS  IBUS  JBUS  KBUS 'ID'  STATUS
```

Python command syntax:

```
ierr = cct3wtdmod_status(ibus, jbus, kbus, id, status)
```

Fortran command syntax:

```
CALL CCT3WTDMOD_STATUS(IBUS, JBUS, KBUS, ID, STATUS, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Integer KBUS	third bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no 3-winding transformer device model on this branch
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	3-winding transformer not found
	IERR = 5	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.17. CCTBRNDMOD_REMOVE

Use this API to remove a branch device model attached to the specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTBRNMOD_REMOVE  IBUS JBUS ID
```

Python command syntax:

```
ierr = cctbrndmod_remove(ibus, jbus, id)
```

Fortran command syntax:

```
CALL CCTBRNDMOD_REMOVE(IBUS, JBUS, ID, IERR)
```

Where:

Integer IBUS	from bus number (input; Not Allowed).	
Integer JBUS	to bus number (input; Not Allowed).	
Character*2 ID	circuit identifier (input; Not Allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no branch device model on the branch
	IERR = 2	bus not found
	IERR = 3	branch not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.18. CCTBRNDMOD_STATUS

Use this API to change the status of a branch device model attached to the specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTBRNDMOD_STATUS  IBUS  JBUS  'ID'  STATUS
```

Python command syntax:

```
ierr = cctbrndmod_status(ibus, jbus, id, status)
```

Fortran command syntax:

```
CALL CCTBRNDMOD_STATUS(IBUS, JBUS, ID, STATUS, IERR)
```

Where:

Integer IBUS	from bus number (input).	
Integer JBUS	to bus number (input; Not Allowed).	
Character*2 ID	circuit identifier (input).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no branch device model on this branch
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	branch not found
	IERR = 5	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.19. CCTBRNOMOD_REMOVE

Use this API to remove a branch "other" model attached to the specified branch.

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_CCTBRNOMOD_REMOVE  IBUS JBUS ID MODEL
```

Python command syntax:

```
ierr = cctbrnomod_remove(ibus, jbus, id, model)
```

Fortran command syntax:

```
CALL CCTBRNOMOD_REMOVE(IBUS, JBUS, ID, MODEL, IERR)
```

Where:

Integer IBUS	from bus number (input; Not Allowed).	
Integer JBUS	to bus number (input; Not Allowed).	
Character*2 ID	circuit identifier (input; Not Allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no branch other model on the branch
	IERR = 2	bus not found
	IERR = 3	branch not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.20. CCTBRNOMOD_STATUS

Use this API to change the status of a branch "other" model attached to the specified branch.

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_CCTBRNOMOD_STATUS  IBUS JBUS ID MODEL STATUS
```

Python command syntax:

```
ierr = cctbrnomod_status(ibus, jbus, id, model, status)
```

Fortran command syntax:

```
CALL CCTBRNOMOD_STATUS(IBUS, JBUS, ID, MODEL, STATUS, IERR)
```

Where:

Integer IBUS	from bus number (input).	
Integer JBUS	to bus number (input; Not Allowed).	
Character*2 ID	circuit identifier (input).	
Character*16 MODEL	model name (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no branch other model on this branch
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	branch not found
	IERR = 5	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.21. CCTBUSOMOD_REMOVE

Use this API to remove an "other" type model attached to a bus.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTBUSOMOD_REMOVE  IBUS  MODEL
```

Python command syntax:

```
ierr = cctbusomod_remove(ibus, model)
```

Fortran command syntax:

```
CALL CCTBUSOMOD_REMOVE(IBUS, MODEL, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*16 MODEL	name of the model to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no bus other model
	IERR = 2	bus not found
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.22. CCTBUSOMOD_STATUS

Use this API to change status of an "other" type model attached to a bus.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTBUSOMOD_STATUS  IBUS  MODEL  STATUS
```

Python command syntax:

```
ierr = cctbusomod_status(ibus, model, status)
```

Fortran command syntax:

```
CALL CCTBUSOMOD_STATUS(IBUS, MODEL, STATUS, IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*16 MODEL	name of the model whose status is to be changed (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no bus other model at this bus
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.23. CCTMCNOMOD_REMOVE

Use this API to remove a machine "other" model attached to the specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTMCNOMOD_REMOVE  IBUS  'ID'  'MODEL'
```

Python command syntax:

```
ierr = cctmcnomod_remove(ibus, id, model)
```

Fortran command syntax:

```
CALL CCTMCNOMOD_REMOVE(IBUS, ID, MODEL, IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	name of the model to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no machine protection model at this machine
	IERR = 2	bus not found
	IERR = 3	machine not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.24. CCTMCNOMOD_STATUS

Use this API to change status of a machine "other" model attached to the specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTMCNOMOD_STATUS  IBUS  'ID'  MODEL  STATUS
```

Python command syntax:

```
ierr = cctmcnomod_status(ibus, id, model, status)
```

Fortran command syntax:

```
CALL CCTMCNOMOD_STATUS(IBUS, ID, MODEL, STATUS, IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	name of the model whose status is to be changed (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no machine other model at this machine
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	machine not found
	IERR = 5	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.25. CCTMCNPMOD_REMOVE

Use this API to remove a machine protection model attached to the specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTMCNPMOD_REMOVE  IBUS  'ID'  'MODEL'
```

Python command syntax:

```
ierr = cctmcnpmod_remove(ibus, id, model)
```

Fortran command syntax:

```
CALL CCTMCNPMOD_REMOVE(IBUS, ID, MODEL, IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	name of the model to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no machine protection model at this machine
	IERR = 2	bus not found
	IERR = 3	machine not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.26. CCTMCNPMOD_STATUS

Use this API to change status of a machine protection model attached to the specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTMCNPMOD_STATUS  IBUS  'ID'  MODEL  STATUS
```

Python command syntax:

```
ierr = cctmcnpmod_status(ibus, id, model, status)
```

Fortran command syntax:

```
CALL CCTMCNPMOD_STATUS(IBUS, ID, MODEL, STATUS, IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	name of the model whose status is to be changed (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no machine protection model at this machine
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	machine not found
	IERR = 5	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.27. CCTMOD_PACK

Use this API to pack CCT model definition tables.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTMOD_PACK
```

Python command syntax:

```
ierr = cctmod_pack()
```

Fortran command syntax:

```
CALL CCTMOD_PACK(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.28. CCTMOD_USER

Use this API to list or remove user-written CCT model definitions.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTMOD_USER    OPT
```

Python command syntax:

```
ierr = cctmod_user(opt)
```

Fortran command syntax:

```
CALL CCTMOD_USER(OPT, IERR)
```

Where:

Integer OPT	option code (input; 0 by default).	
	OPT = 0	list user-written model definitions
	OPT = 1	remove unused user-written model definitions
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	invalid OPT value
	IERR = 2	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.29. CCTMSCOMOD_REMOVE

Use this API to remove a miscellaneous "other" type model with the specified model instance.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTMSCOMOD_REMOVE  MINS
```

Python command syntax:

```
ierr = cctmscomod_remove(mins)
```

Fortran command syntax:

```
CALL CCTMSCOMOD_REMOVE(MINS, IERR)
```

Where:

Integer MINS	model instance (>0) (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no bus other model with this model instance at this bus
	IERR = 2	invalid model instance
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.30. CCTMSCOMOD_STATUS

Use this API to change status of a miscellaneous "other" type model with the specified model instance.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CCTMSCOMOD_STATUS  MINS  STATUS
```

Python command syntax:

```
ierr = cctmscomod_status(mins, status)
```

Fortran command syntax:

```
CALL CCTMSCOMOD_STATUS(MINS, STATUS, IERR)
```

Where:

Integer MINS	model instance (>0) (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no miscellaneous other model with this model instance
	IERR = 2	invalid STATUS value
	IERR = 3	invalid model instance
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.31. CCTSWSOMOD_REMOVE

Use this API to remove an "other" type model attached to a switched shunt.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_CCTSWSOMOD_REMOVE  IBUS MODEL
```

Python command syntax:

```
ierr = cctswsomod_remove(ibus, model)
```

Fortran command syntax:

```
CALL CCTSWSOMOD_REMOVE(IBUS, MODEL, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	name of the model to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no bus other model
	IERR = 2	bus not found
	IERR = 3	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.32. CCTSWSOMOD_STATUS

Use this API to change status of an "other" type model attached to a switched shunt.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_CCTSWSOMOD_STATUS  IBUS MODEL STATUS
```

Python command syntax:

```
ierr = cctswsomod_status(ibus, model, status)
```

Fortran command syntax:

```
CALL CCTSWSOMOD_STATUS(IBUS, MODEL, STATUS, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	name of the model whose status is to be changed (input; no default allowed).	
Integer STATUS	new model status (input; 1 by default).	
	STATUS = 0	out-of-service
	STATUS = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	no bus other model at this bus
	IERR = 2	invalid STATUS value
	IERR = 3	bus not found
	IERR = 4	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.33. CHANGE_CCT2DCOMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the "other" type model attached to a specified 2-terminal dc line. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT2DCOMOD_CHRICN  DC  'MODEL' J  CHDATA
```

Python command syntax:

```
ierr = change_cct2dcomod_chricn(dc, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT2DCOMOD_CHRICN(DC, MODEL, J, CHDATA,  IERR)
```

Where:

Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	dc line not found
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.34. CHANGE_CCT2DCOMOD_CON

Use this API to change the value of a real constant (CON) of the "other" type model attached to a specified 2-terminal dc line.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT2DCOMOD_CON  DC  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cct2dcomod_con(dc, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT2DCOMOD_CON(DC, MODEL, J, RDATA,  IERR)
```

Where:

Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	dc line not found
	IERR = 6	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.35. CHANGE_CCT2DCOMOD_ICON

Use this API to change the value of an integer ICON of the "other" type model attached to a specified 2-terminal dc line.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT2DCOMOD_ICON  DC  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_cct2dcomod_icon(dc, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCT2DCOMOD_ICON(DC, MODEL, J, IDATA,  IERR)
```

Where:

Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	dc line not found
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.36. CHANGE_CCT2DCOMOD_VAR

Use this API to change the value of VAR of the "other" type model attached to a specified 2-terminal dc line.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT2DCOMOD_VAR  DC  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cct2dcomod_var(dc, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT2DCOMOD_VAR(DC, MODEL, J, RDATA,  IERR)
```

Where:

Character*12 DC	quoted name of the two-terminal dc line for which model data is to be changed (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	dc line not found
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.37. CHANGE_CCT2WTDMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the 2-winding transformer device model of a specified branch. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT2WTDMOD_CHRICN  IBUS  JBUS  'ID'  'MODEL'  J  CHDATA
```

Python command syntax:

```
ierr = change_cct2wtdmod_chricn(ibus, jbus, id, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT2WTDMOD_CHRICN(IBUS, JBUS, ID, MODEL, J, CHDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.38. CHANGE_CCT2WTDMOD_CON

Use this API to change the value of a real constant (CON) of the 2-winding transformer device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT2WTDMOD_CON  IBUS  JBUS  'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cct2wtdmod_con(ibus, jbus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT2WTDMOD_CON(IBUS, JBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.39. CHANGE_CCT2WTDMOD_ICON

Use this API to change the value of an integer ICON of the 2-winding transformer device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT2WTDMOD_ICON  IBUS  JBUS  'ID'  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_cct2wtdmod_icon(ibus, jbus, id, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCT2WTDMOD_ICON(IBUS, JBUS, ID, MODEL, J, IDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.40. CHANGE_CCT2WTDMOD_VAR

Use this API to change the value of VAR of the 2-winding transformer device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT2WTDMOD_VAR  IBUS  JBUS  'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cct2wtdmod_var(ibus, jbus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT2WTDMOD_VAR(IBUS, JBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.41. CHANGE_CCT3WTDMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the 3-winding transformer device model of a specified branch. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT3WTDMOD_CHRICN  IBUS  JBUS  KBUS 'ID'  'MODEL'  J  CHDATA
```

Python command syntax:

```
ierr = change_cct3wtdmod_chricn(ibus, jbus, kbus, id, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT3WTDMOD_CHRICN(IBUS, JBUS, KBUS, ID, MODEL, J, CHDATA,
 IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Integer KBUS	third bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.42. CHANGE_CCT3WTDMOD_CON

Use this API to change the value of a real constant (CON) of the 3-winding transformer device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT3WTDMOD_CON  IBUS  JBUS  KBUS 'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cct3wtdmod_con(ibus, jbus, kbus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT3WTDMOD_CON(IBUS, JBUS, KBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Integer KBUS	third bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.43. CHANGE_CCT3WTDMOD_ICON

Use this API to change the value of an integer ICON of the 3-winding transformer device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT3WTDMOD_ICON  IBUS  JBUS  KBUS 'ID'  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_cct3wtdmod_icon(ibus, jbus, kbus, id, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCT3WTDMOD_ICON(IBUS, JBUS, KBUS, ID, MODEL, J, IDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Integer KBUS	third bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.44. CHANGE_CCT3WTDMOD_VAR

Use this API to change the value of VAR of the 3-winding transformer device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCT3WTDMOD_VAR  IBUS  JBUS  KBUS 'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cct3wtdmod_var(ibus, jbus, kbus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCT3WTDMOD_VAR(IBUS, JBUS, KBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Integer KBUS	third bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.45. CHANGE_CCTBRNDMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the branch device model of a specified branch. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTBRNDMOD_CHRICN  IBUS  JBUS  'ID'  'MODEL'  J  CHDATA
```

Python command syntax:

```
ierr = change_cctbrndmod_chricn(ibus, jbus, id, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBRNDMOD_CHRICN(IBUS, JBUS, ID, MODEL, J, CHDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.46. CHANGE_CCTBRNDMOD_CON

Use this API to change the value of a real constant (CON) of the branch device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTBRNDMOD_CON  IBUS  JBUS  'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cctbrndmod_con(ibus, jbus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBRNDMOD_CON(IBUS, JBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.47. CHANGE_CCTBRNDMOD_ICON

Use this API to change the value of an integer ICON of the branch device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTBRNDMOD_ICON  IBUS  JBUS  'ID'  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_cctbrndmod_icon(ibus, jbus, id, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBRNDMOD_ICON(IBUS, JBUS, ID, MODEL, J, IDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.48. CHANGE_CCTBRNDMOD_VAR

Use this API to change the value of VAR of the branch device model of a specified branch.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTBRNDMOD_VAR  IBUS  JBUS  'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cctbrndmod_var(ibus, jbus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBRNDMOD_VAR(IBUS, JBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.49. CHANGE_CCTBRNOMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the branch "other" model of a specified branch. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_CHANGE_CCTBRNOMOD_CHRICN  IBUS JBUS ID MODEL J CHDATA
```

Python command syntax:

```
ierr = change_cctbrnomod_chricn(ibus, jbus, id, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBRNOMOD_CHRICN(IBUS, JBUS, ID, MODEL, J, CHDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.50. CHANGE_CCTBRNOMOD_CON

Use this API to change the value of a real constant (CON) of the branch "other" model of a specified branch.

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_CHANGE_CCTBRNOMOD_CON  IBUS JBUS ID MODEL J RDATA
```

Python command syntax:

```
ierr = change_cctbrnomod_con(ibus, jbus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBRNOMOD_CON(IBUS, JBUS, ID, MODEL, J, RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.51. CHANGE_CCTBRNOMOD_ICON

Use this API to change the value of an integer ICON of the branch "other" model of a specified branch.

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_CHANGE_CCTBRNOMOD_ICON  IBUS JBUS ID MODEL J IDATA
```

Python command syntax:

```
ierr = change_cctbrnomod_icon(ibus, jbus, id, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBRNOMOD_ICON(IBUS, JBUS, ID, MODEL, J, IDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.52. CHANGE_CCTBRNOMOD_VAR

Use this API to change the value of VAR of the branch "other" model of a specified branch.

This API routine was first introduced in release 33.6.0.

Batch command syntax:

```
BAT_CHANGE_CCTBRNOMOD_VAR  IBUS JBUS ID MODEL J RDATA
```

Python command syntax:

```
ierr = change_cctbrnomod_var(ibus, jbus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBRNOMOD_VAR(IBUS, JBUS, ID, MODEL, J, RDATA, IERR)
```

Where:

Integer IBUS	from bus number (input; no default allowed).	
Integer JBUS	to bus number (input; no default allowed).	
Character*2 ID	circuit identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	branch not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.53. CHANGE_CCTBUSOMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the "other" type model attached to a specified bus. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTBUSOMOD_CHRICN  IBUS  'MODEL' J  CHDATA
```

Python command syntax:

```
ierr = change_cctbusomod_chricn(ibus, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBUSOMOD_CHRICN(IBUS, MODEL, J, CHDATA,  IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.54. CHANGE_CCTBUSOMOD_CON

Use this API to change the value of a real constant (CON) of the "other" type model attached to a specified bus.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTBUSOMOD_CON  IBUS  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cctbusomod_con(ibus, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBUSOMOD_CON(IBUS, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.55. CHANGE_CCTBUSOMOD_ICON

Use this API to change the value of an integer ICON of the "other" type model attached to a specified bus.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTBUSOMOD_ICON  IBUS  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_cctbusomod_icon(ibus, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBUSOMOD_ICON(IBUS, MODEL, J, IDATA,  IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.56. CHANGE_CCTBUSOMOD_VAR

Use this API to change the value of VAR of the "other" type model attached to a specified bus.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTBUSOMOD_VAR  IBUS  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cctbusomod_var(ibus, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTBUSOMOD_VAR(IBUS, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

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4.12.57. CHANGE_CCTMCNOMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the machine "other" model for a specified machine. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMCNOMOD_CHRICN  IBUS  'ID'  'MODEL' J  CHDATA
```

Python command syntax:

```
ierr = change_cctmcnomod_chricn(ibus, id, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMCNOMOD_CHRICN(IBUS, ID, MODEL, J, CHDATA,  IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.58. CHANGE_CCTMCNOMOD_CON

Use this API to change the value of a real constant (CON) of the machine "other" model for a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMCNOMOD_CON  IBUS  'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cctmcnomod_con(ibus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMCNOMOD_CON(IBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.59. CHANGE_CCTMCNOMOD_ICON

Use this API to change the value of an integer ICON of the machine "other" model for a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMCNOMOD_ICON  IBUS  'ID'  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_cctmcnomod_icon(ibus, id, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMCNOMOD_ICON(IBUS, ID, MODEL, J, IDATA,  IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.60. CHANGE_CCTMCNOMOD_VAR

Use this API to change the value of VAR of the machine "other" model for a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMCNOMOD_VAR  IBUS  'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cctmcnomod_var(ibus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMCNOMOD_VAR(IBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.61. CHANGE_CCTMCNPMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the machine protection model of a specified machine. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMCNPMOD_CHRICN  IBUS  'ID'  'MODEL' J  CHDATA
```

Python command syntax:

```
ierr = change_cctmcnpmod_chricn(ibus, id, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMCNPMOD_CHRICN(IBUS, ID, MODEL, J, CHDATA,  IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

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4.12.62. CHANGE_CCTMCNPMOD_CON

Use this API to change the value of a real constant (CON) of the machine protection model of a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMCNPMOD_CON  IBUS  'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cctmcnpmod_con(ibus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMCNPMOD_CON(IBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.63. CHANGE_CCTMCNPMOD_ICON

Use this API to change the value of an integer ICON of the machine protection model of a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMCNPMOD_ICON  IBUS  'ID'  'MODEL'  J  IDATA
```

Python command syntax:

```
ierr = change_cctmcnpmod_icon(ibus, id, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMCNPMOD_ICON(IBUS, ID, MODEL, J, IDATA,  IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.64. CHANGE_CCTMCNPMOD_VAR

Use this API to change the value of VAR of the machine protection model of a specified machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMCNPMOD_VAR  IBUS  'ID'  'MODEL'  J  RDATA
```

Python command syntax:

```
ierr = change_cctmcnpmod_var(ibus, id, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMCNPMOD_VAR(IBUS, ID, MODEL, J, RDATA,  IERR)
```

Where:

Integer IBUS	machine bus number (input; no default allowed).	
Character*2 ID	machine identifier (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	machine not found
	IERR = 8	prerequisite requirements for API are not met

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4.12.65. CHANGE_CCTMSCOMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the miscellaneous "other" type model with the specified model instance. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMSCOMOD_CHRICN  MODEL MINS J CHDATA
```

Python command syntax:

```
ierr = change_cctmscomod_chricn(model, mins, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMSCOMOD_CHRICN(MODEL, MINS, J, CHDATA,  IERR)
```

Where:

Character*16 MODEL	model name (input; no default allowed).	
Integer MINS	model instance (>0) (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model aaa with model instance xxx
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	invalid model instance value
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.66. CHANGE_CCTMSCOMOD_CON

Use this API to change the value of a real constant (CON) of the miscellaneous "other" type model with the specified model instance.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMSCOMOD_CON  'MODEL'  MINS J  RDATA
```

Python command syntax:

```
ierr = change_cctmscomod_con(model, mins, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMSCOMOD_CON(MODEL, MINS, J, RDATA,  IERR)
```

Where:

Character*16 MODEL	model name (input; no default allowed).	
Integer MINS	model instance (>0) (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model aaa with model instance xxx
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	invalid model instance value
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.67. CHANGE_CCTMSCOMOD_ICON

Use this API to change the value of an integer ICON of the miscellaneous "other" type model with the specified model instance.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMSCOMOD_ICON  'MODEL'  MINS J  IDATA
```

Python command syntax:

```
ierr = change_cctmscomod_icon(model, mins, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMSCOMOD_ICON(MODEL, MINS, J, IDATA,  IERR)
```

Where:

Character*16 MODEL	model name (input; no default allowed).	
Integer MINS	model instance (>0) (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model aaa with model instance xxx
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	invalid model instance value
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.68. CHANGE_CCTMSCOMOD_VAR

Use this API to change the value of VAR of the miscellaneous "other" type model with the specified model instance.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_CHANGE_CCTMSCOMOD_VAR  'MODEL'  MINS J  RDATA
```

Python command syntax:

```
ierr = change_cctmscomod_var(model, mins, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTMSCOMOD_VAR(MODEL, MINS, J, RDATA,  IERR)
```

Where:

Character*16 MODEL	model name (input; no default allowed).	
Integer MINS	model instance (>0) (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model aaa with model instance xxx
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	invalid model instance value
	IERR = 7	prerequisite requirements for API are not met

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4.12.69. CHANGE_CCTSWSOMOD_CHRICN

Use this API to change the value of a character ICON (i.e., an element of the CHRICN array) of the "other" type model attached to a specified switched shunt. The corresponding integer ICON is set to 0.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_CHANGE_CCTSWSOMOD_CHRICN  IBUS MODEL J CHDATA
```

Python command syntax:

```
ierr = change_cctswsomod_chricn(ibus, model, j, chdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTSWSOMOD_CHRICN(IBUS, MODEL, J, CHDATA, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Character*12 CHDATA	new character ICON value (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

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4.12.70. CHANGE_CCTSWSOMOD_CON

Use this API to change the value of a real constant (CON) of the "other" type model attached to a specified switched shunt.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_CHANGE_CCTSWSOMOD_CON  IBUS MODEL J RDATA
```

Python command syntax:

```
ierr = change_cctswsomod_con(ibus, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTSWSOMOD_CON(IBUS, MODEL, J, RDATA, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	CON number to change (J is relative to the start of the block of CONs assigned to this model, e,g., 1 for first CON(K), 2 for CON(K+1), etc.) (input; no default allowed).	
Real RDATA	new CON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no CONs used in the model
	IERR = 5	CON number outside of the CON range of this model
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.71. CHANGE_CCTSWSOMOD_ICON

Use this API to change the value of an integer ICON of the "other" type model attached to a specified switched shunt.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_CHANGE_CCTSWSOMOD_ICON  IBUS MODEL J IDATA
```

Python command syntax:

```
ierr = change_cctswsomod_icon(ibus, model, j, idata)
```

Fortran command syntax:

```
CALL CHANGE_CCTSWSOMOD_ICON(IBUS, MODEL, J, IDATA, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	ICON number to change (J is relative to the start of the block of ICONs assigned to this model, e,g., 1 for first ICON(M), 2 for ICON(M +1), etc.) (input; no default allowed).	
Integer IDATA	new ICON value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no ICONs used in the model
	IERR = 5	ICON number outside of the ICON range of this model
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.12.72. CHANGE_CCTSWSOMOD_VAR

Use this API to change the value of VAR of the "other" type model attached to a specified switched shunt.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_CHANGE_CCTSWSOMOD_VAR  IBUS MODEL J RDATA
```

Python command syntax:

```
ierr = change_cctswsomod_var(ibus, model, j, rdata)
```

Fortran command syntax:

```
CALL CHANGE_CCTSWSOMOD_VAR(IBUS, MODEL, J, RDATA, IERR)
```

Where:

Integer IBUS	switched shunt bus number (input; no default allowed).	
Character*16 MODEL	model name (input; no default allowed).	
Integer J	VAR number to change (J is relative to the start of the block of VARs assigned to this model, e,g., 1 for first VAR(L), 2 for VAR(L+1), etc.) (input; no default allowed).	
Real RDATA	new VAR value (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error
	IERR = 1	unknown model name
	IERR = 2	no CCT table entry for the model
	IERR = 3	model names do not match - model is aaa not xxx
	IERR = 4	no VARs used in the model
	IERR = 5	VAR number outside of the VAR range of this model
	IERR = 6	bus not found
	IERR = 7	prerequisite requirements for API are not met

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4.13. Set Disturbance

4.13.1. DIST_3WIND_CLOSE

Use this API to set a three-winding transformer to in-service during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_3WIND_CLOSE  IBUS  JBUS  KBUS  'ID'
```

Python command syntax:

```
ierr = dist_3wind_close(ibus, jbus, kbus, id)
```

Fortran command syntax:

```
CALL DIST_3WIND_CLOSEAPI(IBUS, JBUS, KBUS, ID, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number of one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 ID	circuit identifier (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	three-winding transformer not found
	IERR = 4	three-winding transformer is already in-service
	IERR = 5	bus is out-of-service
	IERR = 6	prerequisite requirements for API are not met

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

4.13.2. DIST_3WIND_FAULT

Use this API to apply a fault at the IBUS end of a three-winding transformer during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_3WIND_FAULT  IBUS  JBUS  KBUS  'ID'  UNITS  BASEKV  VALUES(1)
 VALUES(2)
```

Python command syntax:

```
ierr = dist_3wind_fault(ibus,jbus,kbus,id,units,basekv,values)
```

Fortran command syntax:

```
CALL DIST_3WIND_FAULTAPI(IBUS,JBUS,KBUS,ID,UNITS,BASEKV,VALUES,IERR)
```

Where:

Integer IBUS	bus number of the bus at which the fault is to be placed (input; no default allowed).	
Integer JBUS	bus number of one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 ID	circuit identifier (input; '1').	
Integer UNITS	units in which fault admittance or impedance is specified in VALUES (input; 1 by default).	
	UNITS = 1	admittance in MVA
	UNITS = 2	admittance in mhos
	UNITS = 3	impedance in ohms
Real BASEKV	base voltage in kV used to calculate the per unit fault admittance if UNITS is 2 or 3; ignored if UNITS is 1. If BASEKV is specified as 0.0, the base voltage of bus IBUS is used (input; 0.0).	
Real VALUES (2)	array of 2 elements specifying (input).	
	VALUES(1)	real component of the complex fault admittance or impedance according to the value specified for UNITS (0.0 by default)
	VALUES(2)	reactive component of the complex fault admittance or impedance according to the value specified for UNITS (-2.0E11 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	three-winding transformer not found
	IERR = 4	three-winding transformer winding is out-of-service

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IERR = 5	invalid UNITS value
IERR = 6	invalid BASEKV value (<0.0)
IERR = 7	both BASEKV and the base voltage of bus IBUS are 0.0
IERR = 8	prerequisite requirements for API are not met

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4.13.3. DIST_3WIND_TRIP

Use this API to set a three-winding transformer to out-of-service during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_3WIND_TRIP  IBUS  JBUS  KBUS  'ID'
```

Python command syntax:

```
ierr = dist_3wind_trip(ibus, jbus, kbus, id)
```

Fortran command syntax:

```
CALL DIST_3WIND_TRIPAPI(IBUS, JBUS, KBUS, ID, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number of one of the other buses (input; no default allowed).	
Integer KBUS	bus number of the third bus (input; no default allowed).	
Character*2 ID	circuit identifier (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	three-winding transformer not found
	IERR = 4	three-winding transformer is already out-of-service
	IERR = 5	prerequisite requirements for API are not met

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4.13.4. DIST_BRANCH_CLOSE

Use this API to set a non-transformer branch or a two-winding transformer to in-service during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_BRANCH_CLOSE  IBUS  JBUS  'ID'
```

Python command syntax:

```
ierr = dist_branch_close(ibus, jbus, id)
```

Fortran command syntax:

```
CALL DIST_BRANCH_CLOSEAPI(IBUS, JBUS, ID, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number of the other bus (input; no default allowed).	
Character*2 ID	circuit identifier (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	branch not found
	IERR = 4	branch is already in-service
	IERR = 5	bus is out-of-service
	IERR = 6	prerequisite requirements for API are not met

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4.13.5. DIST_BRANCH_FAULT

Use this API to apply a fault at the IBUS end of a non-transformer branch or a two-winding transformer during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_BRANCH_FAULT  IBUS  JBUS  'ID'  UNITS  BASEKV  VALUES(1)
 VALUES(2)
```

Python command syntax:

```
ierr = dist_branch_fault(ibus, jbus, id, units, basekv, values)
```

Fortran command syntax:

```
CALL DIST_BRANCH_FAULTAPI(IBUS,JBUS,ID,UNITS,BASEKV,VALUES,IERR)
```

Where:

Integer IBUS	bus number of the bus at which the fault is to be placed (input; no default allowed).	
Integer JBUS	bus number of the other bus (input; no default allowed).	
Character*2 ID	circuit identifier (input; '1').	
Integer UNITS	units in which fault admittance or impedance is specified in VALUES (input; 1 by default).	
	UNITS = 1	admittance in MVA
	UNITS = 2	admittance in mhos
	UNITS = 3	impedance in ohms
Real BASEKV	base voltage in kV used to calculate the per unit fault admittance if UNITS is 2 or 3; ignored if UNITS is 1. If BASEKV is specified as 0.0, the base voltage of bus IBUS is used (input; 0.0).	
Real VALUES (2)	array of 2 elements specifying (input).	
	VALUES(1)	real component of the complex fault admittance or impedance according to the value specified for UNITS (0.0 by default)
	VALUES(2)	reactive component of the complex fault admittance or impedance according to the value specified for UNITS (-2.0E11 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	branch not found
	IERR = 4	branch is out-of-service
	IERR = 5	invalid UNITS value

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IERR = 6	invalid BASEKV value (<0.0)
IERR = 7	both BASEKV and the base voltage of bus IBUS are 0.0
IERR = 8	prerequisite requirements for API are not met

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4.13.6. DIST_BRANCH_TRIP

Use this API to set a non-transformer branch or a two-winding transformer to out-of-service during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_BRANCH_TRIP  IBUS  JBUS  'ID'
```

Python command syntax:

```
ierr = dist_branch_trip(ibus, jbus, id)
```

Fortran command syntax:

```
CALL DIST_BRANCH_TRIPAPI(IBUS, JBUS, ID, IERR)
```

Where:

Integer IBUS	bus number of one of the buses (input; no default allowed).	
Integer JBUS	bus number of the other bus (input; no default allowed).	
Character*2 ID	circuit identifier (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	branch not found
	IERR = 4	branch is already out-of-service
	IERR = 5	prerequisite requirements for API are not met

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4.13.7. DIST_BUS_FAULT

Use this API to apply a fault at a bus during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_BUS_FAULT  IBUS  UNITS  BASEKV  VALUES(1)  VALUES(2)
```

Python command syntax:

```
ierr = dist_bus_fault(ibus, units, basekv, values)
```

Fortran command syntax:

```
CALL DIST_BUS_FAULTAPI(IBUS, UNITS, BASEKV, VALUES, IERR)
```

Where:

Integer IBUS	bus number of the bus at which the fault is to be placed (input; no default allowed).	
Integer UNITS	units in which fault admittance or impedance is specified in VALUES (input; 1 by default).	
	UNITS = 1	admittance in MVA
	UNITS = 2	admittance in mhos
	UNITS = 3	impedance in ohms
Real BASEKV	base voltage in kV used to calculate the per unit fault admittance if UNITS is 2 or 3; ignored if UNITS is 1. If BASEKV is specified as 0.0, the base voltage of bus IBUS is used (input; 0.0).	
Real VALUES (2)	array of 2 elements specifying (input).	
	VALUES(1)	real component of the complex fault admittance or impedance according to the value specified for UNITS (0.0 by default)
	VALUES(2)	reactive component of the complex fault admittance or impedance according to the value specified for UNITS (-2.0E11 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	bus is out-of-service
	IERR = 4	invalid UNITS value
	IERR = 5	invalid BASEKV value (<0.0)
	IERR = 6	both BASEKV and the base voltage of bus IBUS are 0.0
	IERR = 7	fixed bus shunt table is full
	IERR = 8	prerequisite requirements for API are not met

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4.13.8. DIST_BUS_TRIP

Use this API to disconnect a bus during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_BUS_TRIP  IBUS
```

Python command syntax:

```
ierr = dist_bus_trip(ibus)
```

Fortran command syntax:

```
CALL DIST_BUS_TRIPAPI(IBUS, IERR)
```

Where:

Integer IBUS	bus number of the bus to be disconnected (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR not successfully executed
	IERR = 2	bus not found
	IERR = 3	bus is already disconnected
	IERR = 4	prerequisite requirements for API are not met

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4.13.9. DIST_CLEAR_FAULT

Use this API to clear a fault during dynamic simulations. The fault must have previously been applied using one of the following APIs:
- DIST_3WIND_FAULT
- DIST_BRANCH_FAULT
- DIST_BUS_FAULT
- DIST_SCMU_FAULT
- DIST_SPCB_FAULT

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_CLEAR_FAULT  FAULT
```

Python command syntax:

```
ierr = dist_clear_fault(fault)
```

Fortran command syntax:

```
CALL DIST_CLEAR_FAULTAPI(FAULT, IERR)
```

Where:

Integer FAULT	index in the fault memory tables of the fault to be cleared (input; 1 by default). Faults are stored in the fault memory tables in the order in which they are applied; the index assigned to a fault is the next available location in the tables. Each time a fault is cleared using this API, the fault memory tables are compressed. Thus, if there are three faults active, the most recently applied fault will have index number 3).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR not successfully executed
	IERR = 2	bus not found
	IERR = 3	branch not found
	IERR = 4	three-winding transformer not found
	IERR = 5	fixed bus shunt not found
	IERR = 6	no faults in the fault memory tables
	IERR = 7	invalid FAULT value
	IERR = 8	prerequisite requirements for API are not met

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4.13.10. DIST_MACHINE_TRIP

Use this API to set a machine to out-of-service during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_MACHINE_TRIP  IBUS  'ID'
```

Python command syntax:

```
ierr = dist_machine_trip(ibus,id)
```

Fortran command syntax:

```
CALL DIST_MACHINE_TRIPAPI(IBUS, ID, IERR)
```

Where:

Integer IBUS	bus number of machine to be tripped (input; no default allowed).	
Character*2 ID	machine identifier (input; '1').	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR not successfully executed
	IERR = 2	bus not found
	IERR = 3	machine not found
	IERR = 4	machine is already out-of-service
	IERR = 5	prerequisite requirements for API are not met

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4.13.11. DIST_SCMU_FAULT

Use this API to calculate an unbalanced fault and apply the equivalent positive sequence fault admittance at a bus during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_SCMU_FAULT  OPTIONS(1)..OPTIONS(4)  VALUES(1)..VALUES(4)
```

Python command syntax:

```
ierr = dist_scmu_fault(options, values)
```

Fortran command syntax:

```
CALL DIST_SCMU_FAULTAPI(OPTIONS, VALUES, IERR)
```

Where:

Integer OPTIONS (4)	array of 4 elements specifying options and values. The first two elements of OPTIONS define options to be used in the construction of the sequence admittance matrices; the remaining entries define the fault to be applied(input).
OPTIONS(1)	dc line and FACTS device option (0 by default)
OPTIONS(1) = 0	block and ignore
OPTIONS(1) = 1	convert to constant admittance load
OPTIONS(2)	transformer impedance correction option (1 by default)
OPTIONS(2) = 0	do not apply to zero sequence
OPTIONS(2) = 1	apply to zero sequence
OPTIONS(3)	unbalanced type (1 by default)
OPTIONS(3) = 1	line-to-ground fault
OPTIONS(3) = 2	line-to-line or line-to-line-to-ground fault
OPTIONS(4)	bus to be faulted (no default allowed)
Real VALUES (4)	array of 4 elements specifying values used to define the fault. They are dependent on the value specified for OPTIONS(3)(input).
VALUES(1)	resistance of the complex L-G (if OPTIONS(3) is 1) or L-L (if OPTIONS(3) is 2) fault impedance in pu (0.0 by default)
VALUES(2)	reactance of the complex L-G (if OPTIONS(3) is 1) or L-L (if OPTIONS(3) is 2) fault impedance in pu (0.0 by default)
VALUES(3)	resistance of the complex L-G fault impedance in pu (if OPTIONS(3) is 2) (0.0 by default)
VALUES(4)	reactance of the complex L-G fault impedance in pu (if OPTIONS(3) is 2) (0.0 by default)

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Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR not successfully executed
	IERR = 2	fixed bus shunt table is full
	IERR = 3	error encountered in executing SCMUAPI
	IERR = 4	prerequisite requirements for API are not met

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4.13.12. DIST_SPCB_FAULT

This API routine is obsolete. It has been replaced by the API routine dist_spcb_fault_2 .

Use this API to calculate a branch unbalance and apply the equivalent positive sequence pi model at a branch during dynamic simulations.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_DIST_SPCB_FAULT  IBUS  JBUS  'ID'  OPTIONS(1)..OPTIONS(6)
 VALUES(1)..VALUES(5)
```

Python command syntax:

```
ierr = dist_spcb_fault(ibus, jbus, id, options, values)
```

Fortran command syntax:

```
CALL DIST_SPCB_FAULTAPI(IBUS, JBUS, ID, OPTIONS, VALUES, IERR)
```

Where:

Integer IBUS	from bus of branch to be subjected to unbalance (input; no default allowed).	
Integer JBUS	to bus of branch to be subjected to unbalance (input; no default allowed).	
Character*2 ID	circuit identifier of branch to be subjected to unbalance (input; '1').	
Integer OPTIONS (6)	array of 6 elements specifying unbalance options(input).	
	OPTIONS(1)	unbalance code (1 by default)
	OPTIONS(1) = 1	one phase open
	OPTIONS(1) = 2	two phases open
	OPTIONS(1) = 3	in-line fault
	OPTIONS(1) = 4	one breaker open
	OPTIONS(2)	path to ground code (used when OPTIONS(1) is 1 or 4) (0 by default)
	OPTIONS(2) = 0	no path to ground
	OPTIONS(2) = 1	include a path to ground
	OPTIONS(3)	type of in-line fault code (used when OPTIONS(1) is 3) (1 by default)
	OPTIONS(3) = 1	line-to-ground fault
	OPTIONS(3) = 2	line-to-line-to-ground fault
	OPTIONS(3) = 3	three phase fault
	OPTIONS(4)	open end code (used when OPTIONS(1) is 4) (1 by default)
	OPTIONS(4) = 0	breaker at bus JBUS is open
	OPTIONS(4) = 1	breaker at bus IBUS is open

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	OPTIONS(5)	dc line and FACTS device option (0 by default)
	OPTIONS(5) = 0	block and ignore
	OPTIONS(5) = 1	convert to constant admittance load
	OPTIONS(6)	transformer impedance correction option (1 by default)
	OPTIONS(6) = 0	do not apply to zero sequence
	OPTIONS(6) = 1	apply to zero sequence
Real VALUES (5)	array of 5 elements specifying unbalance parameters(input).	
	VALUES(1)	fault location as fraction of line from bus IBUS; 0.0 < VALUES(1) < 1.0 (used when OPTIONS(1) is 1 or 4 and OPTIONS(2) is 1; or when OPTIONS(1) is 3) (0.5 by default)
	VALUES(2)	Rl-g (used when OPTIONS(1) is 1 or 4 and OPTIONS(2) is 1; or when OPTIONS(1) is 3 and OPTIONS(3) is 1 or 2) (0.0 by default)
	VALUES(3)	Xl-g (used when OPTIONS(1) is 1 or 4 and OPTIONS(2) is 1; or when OPTIONS(1) is 3 and OPTIONS(3) is 1 or 2) (0.0 by default)
	VALUES(4)	Rl-l used when OPTIONS(1) is 3 and OPTIONS(3) is 2) (0.0 by default)
	VALUES(5)	Xl-l (used when OPTIONS(1) is 3 and OPTIONS(3) is 2) (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	branch not found
	IERR = 4	branch is out-of-service
	IERR = 5	error encountered in executing SPCBAPI
	IERR = 6	prerequisite requirements for API are not met

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4.13.13. DIST_SPCB_FAULT_2

Use this API routine to run the second release of the dynamics branch unbalance function. Use this API routine to calculate a branch unbalance and apply the equivalent positive sequence pi model at a branch during dynamic simulations.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_DIST_SPCB_FAULT_2  IBUS  JBUS  'ID'  OPTIONS(1)..OPTIONS(7)
 VALUES(1)..VALUES(5)
```

Python command syntax:

```
ierr = dist_spcb_fault_2(ibus, jbus, id, options, values)
```

Fortran command syntax:

```
CALL DIST_SPCB_FAULT_2(IBUS, JBUS, ID, OPTIONS, VALUES, IERR)
```

Where:

Integer IBUS	from bus of branch to be subjected to unbalance (input; no default allowed).	
Integer JBUS	to bus of branch to be subjected to unbalance (input; no default allowed).	
Character*2 ID	circuit identifier of branch to be subjected to unbalance (input; '1').	
Integer OPTIONS (7)	array of 7 elements specifying unbalance options(input).	
	OPTIONS(1)	unbalance code (1 by default)
	OPTIONS(1) = 1	one phase open
	OPTIONS(1) = 2	two phases open
	OPTIONS(1) = 3	in-line fault
	OPTIONS(1) = 4	one breaker open
	OPTIONS(2)	path to ground code (used when OPTIONS(1) is 1 or 4) (0 by default)
	OPTIONS(2) = 0	no path to ground
	OPTIONS(2) = 1	include a path to ground
	OPTIONS(3)	type of in-line fault code (used when OPTIONS(1) is 3) (1 by default)
	OPTIONS(3) = 1	line-to-ground fault
	OPTIONS(3) = 2	line-to-line-to-ground fault
	OPTIONS(3) = 3	three phase fault
	OPTIONS(4)	open end code (used when OPTIONS(1) is 4) (1 by default)
	OPTIONS(4) = 0	breaker at bus JBUS is open
	OPTIONS(4) = 1	breaker at bus IBUS is open
	OPTIONS(5)	dc line and FACTS device option (0 by default)

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	OPTIONS(5) = 0	block and ignore
	OPTIONS(5) = 1	convert to constant admittance load
	OPTIONS(6)	transformer impedance correction option (1 by default)
	OPTIONS(6) = 0	do not apply to zero sequence
	OPTIONS(6) = 1	apply to zero sequence
	OPTIONS(7)	fault analysis generator reactance option (0 by default)
	OPTIONS(7) = 0	use subtransient reactance
	OPTIONS(7) = 1	use transient reactance
	OPTIONS(7) = 2	use synchronous reactance
Real VALUES (5)	array of 5 elements specifying unbalance parameters(input).	
	VALUES(1)	fault location as fraction of line from bus IBUS; 0.0 < VALUES(1) < 1.0 (used when OPTIONS(1) is 1 or 4 and OPTIONS(2) is 1; or when OPTIONS(1) is 3) (0.5 by default)
	VALUES(2)	Rl-g (used when OPTIONS(1) is 1 or 4 and OPTIONS(2) is 1; or when OPTIONS(1) is 3 and OPTIONS(3) is 1 or 2) (0.0 by default)
	VALUES(3)	Xl-g (used when OPTIONS(1) is 1 or 4 and OPTIONS(2) is 1; or when OPTIONS(1) is 3 and OPTIONS(3) is 1 or 2) (0.0 by default)
	VALUES(4)	Rl-l used when OPTIONS(1) is 3 and OPTIONS(3) is 2) (0.0 by default)
	VALUES(5)	Xl-l (used when OPTIONS(1) is 3 and OPTIONS(3) is 2) (0.0 by default)
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	STRT or MSTR has not been successfully executed
	IERR = 2	bus not found
	IERR = 3	branch not found
	IERR = 4	branch is out-of-service
	IERR = 5	error encountered in executing SPCBAPI
	IERR = 6	prerequisite requirements for API are not met

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Chapter 5
Subsystem Definition

This chapter defines API routines that control subsystems. There are four kinds of subsystems in PSS®E: area subsystems, bus subsystems, owner subsystems, and zone subsystems. Area subsystems are a set of areas, bus subsystems are a set of buses, etc.

Many PSS®E API routines accept a subsystem definition to limit the scope of operation of the routine. Most such use bus subsystems. Bus subsystems can be defined in terms of areas, zones, and/or owners; but a bus subsystem defined by areas, for example, is not an area subsystem, which is a set of areas, not a set of buses in an area.

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5.1. Area Subsystems

5.1.1. ASYS

Use this API to define an area subsystem.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_ASYS  SID  NUM  AREAS(1) .. AREAS(NUM)
```

Python command syntax:

```
ierr = asys(sid, num, areas)
```

Fortran command syntax:

```
CALL SBSASET(SID, NUM, AREAS, IERR)
```

Where:

Integer SID	valid area subsystem ID. Valid area subsystem IDs range from 0 to 11 (input).	
Integer NUM	number of areas to set (input).	
Integer AREAS (NUM)	array that contains the areas to set (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid

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5.1.2. ASYSDEF

Use this API to set the definition of an area subsystem. This can be used to toggle a particular area subsystem on or off.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_ASYSDEF  SID  DEFINED
```

Python command syntax:

```
asysdef(sid, defined)
```

Fortran command syntax:

```
CALL SBSASETDEF(SID, DEFINED)
```

Where:

Integer SID	valid area subsystem ID. Valid area subsystem IDs range from 0 to 11 (input).
Integer DEFINED	flag used to mark the area subsystem as undefined/defined (input). DEFINED = 0 mark the area subsystem as undefined DEFINED = 1 mark the area subsystem as defined

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5.1.3. ASYSINIT

Use this API to initialize or re-initialize an area subsystem.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_ASYSINIT  SID
```

Python command syntax:

```
asysinit(sid)
```

Fortran command syntax:

```
CALL SBSACREATE(SID, IERR)
```

Where:

Integer SID	valid area subsystem ID. Valid area subsystem IDs range from 0 to 11 (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid

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5.2. Bus Subsystems

5.2.1. BSYS

Use this API to define a bus subsystem.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_BSYS  SID  USEKV BASEKV(1)  BASEKV(2) NUMAREA
 AREAS(1)..AREAS(NUMAREA)  NUMBUS  BUSNUM(1)..BUSNUM(NUMBUS) NUMOWNER
 OWNERS(1)..OWNERS(NUMOWNER)  NUMZONE  ZONES(1)..ZONES(NUMZONE)

BAT_BSYS  SID  USEKV BASEKV(1)  BASEKV(2) NUMAREA
 AREAS(1)..AREAS(NUMAREA)  NUMBUS  BUSNAM(1)..BUSNAM(NUMBUS) NUMOWNER
 OWNERS(1)..OWNERS(NUMOWNER)  NUMZONE  ZONES(1)..ZONES(NUMZONE)
```

Python command syntax:

```
ierr = bsys(sid, usekv, basekv, numarea, areas, numbus, buses, numowner,
 owners, numzone, zones)
```

Fortran command syntax:

```
CALL SBSBSET(SID, USEKV, BASEKV, NUMAREA, AREAS, NUMBUS, BUSES, NUMOWNER,
 OWNERS, NUMZONE, ZONES, IERR)

CALL SBSBSETA(SID, USEKV, BASEKV, NUMAREA, AREAS, NUMBUS, BUSNAM,
 NUMOWNER, OWNERS, NUMZONE, ZONES, IERR)
```

Where:

Integer SID	valid bus subsystem ID. Valid bus subsystem IDs range from 0 to 11 (input).
Integer USEKV	flag to determine whether to use BASEKV values in setting the subsystem (input). USEKV = 0 don't use BASEKV values USEKV = 1 use BASEKV values
Real BASEKV (2)	array of 2 elements specifying (input). BASEKV(1) minimum basekV limit BASEKV(2) maximum basekV limit
Integer NUMAREA	number of areas to set (input).
Integer AREAS (NUMAREA)	array that contains the areas to set (input).
Integer NUMBUS	number of buses to set (input).
Integer or Character*18 BUSES (NUMBUS)	array that contains the buses to set. Can be either BUSNUM or BUSNAM as defined above (input).
Integer BUSNUM (NUMBUS)	array that contains the buses to set
Character*12 BUSNAM (NUMBUS)	array that contains the extended bus names to set

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Integer NUMOWNER	number of owners to set (input).
Integer OWNERS (NUMOWNER)	array that contains the owners to set (input).
Integer NUMZONE	number of zones to set (input).
Integer ZONES (NUMZONE)	array that contains the zones to set (input).
Integer IERR	error code (output).
	IERR = 0 no error occurred
	IERR = 1 SID is not valid

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5.2.2. BSYSADD

Use this API to add elements to an existing bus subsystem.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_BSYSADD  SID  USEKV  BASEKV(1)  BASEKV(2)  NUMAREA
 AREAS(1)..AREAS(NUMAREA)  NUMBUS  BUSES(1)..BUSES(NUMBUS)  NUMOWNER
 OWNERS(1)..OWNERS(NUMOWNER)  NUMZONE  ZONES(1)..ZONES(NUMZONE)
```

Python command syntax:

```
ierr = bsysadd(sid, usekv, basekv, numarea, areas, numbus, buses,
 numowner, owners, numzone, zones)
```

Fortran command syntax:

```
CALL SBSBADD(SID, USEKV, BASEKV, NUMAREA, AREAS, NUMBUS, BUSES, NUMOWNER,
 OWNERS, NUMZONE, ZONES, IERR)
```

Where:

Integer SID	valid bus subsystem ID. Valid bus subsystem IDs range from 0 to 11 (input).	
Integer USEKV	flag to determine whether to use BASEKV values in setting the sub-system (input).	
	USEKV = 0	don't use BASEKV values
	USEKV = 1	use BASEKV values
Real BASEKV (2)	array of 2 elements specifying (input).	
	BASEKV(1)	minimum basekV limit
	BASEKV(2)	maximum basekV limit
Integer NUMAREA	number of areas to set (input).	
Integer AREAS (NUMAREA)	array that contains the areas to set (input).	
Integer NUMBUS	number of buses to set (input).	
Integer BUSES (NUMBUS)	array that contains the buses to set (input).	
Integer NUMOWNER	number of owners to set (input).	
Integer OWNERS (NUMOWNER)	array that contains the owners to set (input).	
Integer NUMZONE	number of zones to set (input).	
Integer ZONES (NUMZONE)	array that contains the zones to set (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid

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5.2.3. BSYSDEF

Use this API to set the definition of a bus subsystem. This can be used to toggle a particular bus subsystem on or off.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_BSYSDEF  SID  DEFINED
```

Python command syntax:

```
bsysdef(sid, defined)
```

Fortran command syntax:

```
CALL SBSBSETDEF(SID, DEFINED)
```

Where:

Integer SID	valid bus subsystem ID. Valid bus subsystem IDs range from 0 to 11 (input).
Integer DEFINED	flag used to mark the bus subsystem as undefined/defined (input). DEFINED = 0 mark the bus subsystem as undefined DEFINED = 1 mark the bus subsystem as defined

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5.2.4. BSYSDELO

Use this API to delete a single bus from a bus subsystem.

This API routine was first introduced in release 33.4.0.

Batch command syntax:

```
BAT_BSYSDELO  SID BUSNUM
```

Python command syntax:

```
ierr = bsysdelo(sid, busnum)
```

Fortran command syntax:

```
CALL SBSBDELO(SID, BUSNUM, IERR)
```

Where:

Integer SID	valid bus subsystem ID. Valid bus subsystem IDs range from 0 to 11 (input).	
Integer BUSNUM	bus number to delete from the subsystem (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid
	IERR = 2	bus subsystem is not defined
	IERR = 3	no buses defined in subsystem
	IERR = 4	BUSNUM doesn't exist in subsystem

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5.2.5. BSYSINIT

Use this API to initialize or re-initialize a bus subsystem.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_BSYSINIT  SID
```

Python command syntax:

```
bsysinit(sid)
```

Fortran command syntax:

```
CALL SBSBCREATE(SID, IERR)
```

Where:

Integer SID	valid bus subsystem ID. Valid bus subsystem IDs range from 0 to 11 (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid

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5.2.6. BSYSMEM

Use this API to replicate the specified bus subsystem in a bus subsystem file. This bus subsystem file may then be recalled at a later time to restore the specified bus subsystem.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BSYSMEM  SID  SFILE
```

Python command syntax:

```
ierr = bsysmem(sid, sfile)
```

Fortran command syntax:

```
CALL SBSBSMEM(SID, SFILE, IERR)
```

Where:

Integer SID	a valid bus subsystem id. Valid bus subsystem ids range from 0 to 11 (input).	
Character*260 SFILE	bus subsystem filename (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid
	IERR = 2	error opening SFILE

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5.2.7. BSYSO

Use this API to build a bus subsystem one bus at a time.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_BSYSO  SID BUSNUM
```

Python command syntax:

```
ierr = bsyso(sid, busnum)
```

Fortran command syntax:

```
CALL SBSBSETO(SID, BUSNUM, IERR)
```

Where:

Integer SID	valid bus subsystem ID. Valid bus subsystem IDs range from 0 to 11 (input).	
Integer BUSNUM	bus number to add to subsystem (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid
	IERR = 2	Bus subsystem contains other elements, not just buses

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5.2.8. BSYSRCL

Use this API to recall a specified bus subsystem saved in a bus subsystem file.

This API routine was first introduced in release 28.0.0.

Batch command syntax:

```
BAT_BSYSRCL  SID  SFILE
```

Python command syntax:

```
ierr = bsysrcl(sid, sfile)
```

Fortran command syntax:

```
CALL SBSBSRCL(SID, SFILE, IERR)
```

Where:

Integer SID	a valid bus subsystem id. Valid bus subsystem ids range from 0 to 11 (input).	
Character*260 SFILE	bus subsystem filename (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid
	IERR = 2	error opening SFILE
	IERR = 3	not a valid bus subsystem file
	IERR = 4	items in bus subsystem file not found in working case

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5.3. Owner Subsystems

5.3.1. OSYS

Use this API to define an owner subsystem.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OSYS  SID  NUM  OWNERS(1) .. OWNERS(NUM)
```

Python command syntax:

```
ierr = osys(sid, num, owners)
```

Fortran command syntax:

```
CALL SBSOSET(SID, NUM, OWNERS, IERR)
```

Where:

Integer SID	valid owner subsystem ID. Valid owner subsystem IDs range from 0 to 11 (input).	
Integer NUM	number of owners to set (input).	
Integer OWNERS (NUM)	array that contains the owners to set (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid

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5.3.2. OSYSDEF

Use this API to set the definition of an owner subsystem. This can be used to toggle a particular owner subsystem on or off.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_OSYSDEF  SID  DEFINED
```

Python command syntax:

```
osysdef(sid, defined)
```

Fortran command syntax:

```
CALL SBSOSETDEF(SID, DEFINED)
```

Where:

Integer SID	a valid owner subsystem ID. Valid owner subsystem IDs range from 0 to 11 (input).
Integer DEFINED	flag to mark the owner subsystem as undefined/defined (input). DEFINED = 0 mark the owner subsystem as undefined DEFINED = 1 mark the owner subsystem as defined

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5.3.3. OSYSINIT

Use this API to initialize or re-initialize an owner subsystem.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_OSYSINIT  SID
```

Python command syntax:

```
osysinit(sid)
```

Fortran command syntax:

```
CALL SBSOCREATE(SID, IERR)
```

Where:

Integer SID	a valid owner subsystem ID. Valid owner subsystem IDs range from 0 to 11 (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid

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5.4. Zone Subsystems

5.4.1. ZSYS

Use this API to define a zone subsystem.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_ZSYS  SID  NUM  ZONES(1) .. ZONES(NUM)
```

Python command syntax:

```
ierr = zsys(sid, num, zones)
```

Fortran command syntax:

```
CALL SBSZSET(SID, NUM, ZONES, IERR)
```

Where:

Integer SID	valid zone subsystem ID. Valid zone subsystem IDs range from 0 to 11 (input).	
Integer NUM	number of zones to set (input).	
Integer ZONES (NUM)	array that contains the zones to set (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid

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5.4.2. ZSYSDEF

Use this API to set the definition of an zone subsystem. This can be used to toggle a particular zone subsystem on or off.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_ZSYSDEF  SID  DEFINED
```

Python command syntax:

```
zsysdef(sid, defined)
```

Fortran command syntax:

```
CALL SBSZSETDEF(SID, DEFINED)
```

Where:

Integer SID	a valid zone subsystem ID. Valid zone subsystem IDs range from 0 to 11 (input).
Integer DEFINED	flag to mark the zone subsystem as undefined/defined (input). DEFINED = 0 mark the zone subsystem as undefined DEFINED = 1 mark the zone subsystem as defined

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5.4.3. ZSYSINIT

Use this API to initialize or re-initialize a zone subsystem.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_ZSYSINIT  SID
```

Python command syntax:

```
zsysinit(sid)
```

Fortran command syntax:

```
CALL SBSZCREATE(SID, IERR)
```

Where:

Integer SID	a valid zone subsystem ID. Valid zone subsystem IDs range from 0 to 11 (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR = 1	SID is not valid

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Chapter 6
Diagram View

This chapter defines API routines that are used to manipulate Slider Diagrams in the interface. They will elicit no action if a GUI is not present when invoked.

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6.1. CHANGEDIAGCONTOUR

Use this API to change the contour parameters for the contour in the active Slider diagram in the GUI. Requires a current network case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_CHANGEDIAGCONTOUR  ELEMENT QUANTITY METHOD COLOR ENABLE RES INFLUENCE
 BOUND MAX MIN
```

Python command syntax:

```
changediagcontour(element, quantity, method, color, enable, res,
 influence, bound, max, min)
```

Fortran command syntax:

```
CALL CHANGEDIAGCONTOUR(ELEMENT, QUANTITY, METHOD, COLOR, ENABLE, RES,
 INFLUENCE, BOUND, MAX, MIN)
```

Where:

Integer ELEMENT	item to be contoured e.g. Buses, branches, machines (input).	
	ELEMENT = 1	bus
	ELEMENT = 2	branch
	ELEMENT = 3	machine
Integer QUANTITY	element quantity to generate the contour on (input).	
Integer METHOD	algorithm used to generate the contour (input).	
	METHOD = 1	linear method
	METHOD = 2	linear approximation method
	METHOD = 3	quad tree method
Integer COLOR	color gradiant to use for the contour (input).	
	COLOR = 0	red scale
	COLOR = 1	green scale
	COLOR = 2	blue scale
	COLOR = 3	gray scale
	COLOR = 4	blue/low to red/high
	COLOR = 5	red/low to blue/high
Integer ENABLE	"switch" used to toggle the contour on/off (input).	
	ENABLE = 0	off
	ENABLE = 1	on
Real RES	resolution of the contour (10 to 2000). A higher resolution will result in greater computation times. (input).	
Real INFLUENCE	influence range (1 to 100). A larger range will result in greater blending. (input).	

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Real BOUND	value that controls how far beyond the outermost ELEMENTs on the diagram the contour extends (input).
Real MAX	maximum "quantity" value associated with the "high" color (input).
Real MIN	minimum "quantity" value associated with the "low" color (input).

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6.2. CHECKDIAGFILE

Use this API to check the active Slider Diagram in the GUI against the current network case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_CHECKDIAGFILE
```

Python command syntax:

```
checkdiagfile()
```

Fortran command syntax:

```
CALL CHECKDIAGFILE
```

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6.3. CLOSEDIAGFILE

Use this API to close the active Slider Diagram in the GUI without prompting to save the Diagram.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_CLOSEDIAGFILE
```

Python command syntax:

```
closediagfile()
```

Fortran command syntax:

```
CALL CLOSEDIAGFILE
```

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6.4. DELETEDIAGCONTOUR

Use this API to delete the contour from the active Slider Diagram in the GUI.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_DELETEDIAGCONTOUR
```

Python command syntax:

```
deletediagcontour()
```

Fortran command syntax:

```
CALL DELETEDIAGCONTOUR
```

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6.5. ENABLEDIAGCONTOUR

Use this API to specify the contour parameters and display a contour in the active Slider Diagram in the GUI. Requires a current network case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_ENABLEDIAGCONTOUR  ELEMENT QUANTITY METHOD COLOR ENABLE RES INFLUENCE
 BOUND MAX MIN
```

Python command syntax:

```
enablediagcontour(element, quantity, method, color, enable, res,
 influence, bound, max, min)
```

Fortran command syntax:

```
CALL ENABLEDIAGCONTOUR(ELEMENT, QUANTITY, METHOD, COLOR, ENABLE, RES,
 INFLUENCE, BOUND, MAX, MIN)
```

Where:

Integer ELEMENT	item to be contoured e.g. Buses, branches, machines (input).	
	ELEMENT = 1	Bus
	ELEMENT = 2	Branch
	ELEMENT = 3	Machine
Integer QUANTITY	element quantity to generate the contour on (input).	
Integer METHOD	algorithm used to generate the contour (input).	
	METHOD = 1	Linear method
	METHOD = 2	Linear approximation method
	METHOD = 3	Quad tree method
Integer COLOR	color gradiant to use for the contour (input).	
	COLOR = 0	Red scale
	COLOR = 1	Green scale
	COLOR = 2	Blue scale
	COLOR = 3	Gray scale
	COLOR = 4	Blue/low to Red/high
	COLOR = 5	Red/low to Blue/high
Integer ENABLE	"switch" used to toggle the contour on/off (input).	
	ENABLE = 0	Off
	ENABLE = 1	On
Real RES	is the resolution of the contour (10 to 2000). A higher resolution will result in greater computation times. (input).	
Real INFLUENCE	influence range (1 to 100). A larger range will result in greater blending. (input).	

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Real BOUND	a value that controls how far beyond the outermost ELEMENTs on the diagram the contour extends (input).
Real MAX	maximum "quantity" value associated with the "high" color (input).
Real MIN	minimum "quantity" value associated with the "low" color (input).

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6.6. EXPORTBUSLOCFILE

Use this API to export the bus location from the active Slider Diagram in the GUI to a file.

This API routine was first introduced in release 30.2.0.

Batch command syntax:

```
BAT_EXPORTBUSLOCFILE  BUSLOCFILE
```

Python command syntax:

```
exportbuslocfile(buslocfile)
```

Fortran command syntax:

```
CALL EXPORTBUSLOCFILE(BUSLOCFILE)
```

Where:

Character*260 BUSLOCFILE	name of the Bus Location Data File to be written (input).

6.7. EXPORTIMAGEFILE

Use this API to export an image of the active Slider Diagram in the GUI to a file.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_EXPORTIMAGEFILE  TYPE IMAGEFILE QUALITY
```

Python command syntax:

```
exportimagefile(type, imagefile, quality)
```

Fortran command syntax:

```
CALL EXPORTIMAGEFILE(TYPE, IMAGEFILE, QUALITY)
```

Where:

Integer TYPE	type of image to export (input).	
	Type = 1	Bitmap
	Type = 2	JPEG
	Type = 3	PNG
	Type = 4	GIF
	Type = 5	TIF
Character*260 IMAGEFILE	name of the image file to be written (input).	
Integer QUALITY	a value, between 1 and 100, used to control the quality of the exported image if TYPE = 2 (input).	

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6.8. GEXMBUS

Use this API to create a GEXM/GOUT display of a bus in an active Slider diagram in the GUI. Requires a current network case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_GEXMBUS   BUS
```

Python command syntax:

```
gexmbus(bus)
```

Fortran command syntax:

```
CALL GEXMBUS(BUS)
```

Where:

Integer BUS	bus number of the bus for which to create the GEXM display (input).

6.9. GROWBUS

Use this API to GROW a bus in an active Slider diagram in the GUI. Requires a current network case.

This API routine was first introduced in release 30.1.0.

Batch command syntax:

```
BAT_GROWBUS  BUSNUMBER XPOSITION YPOSITION
```

Python command syntax:

```
growbus(busnumber, xposition, yposition)
```

Fortran command syntax:

```
CALL GROWBUS(BUSNUMBER, XPOSITION, YPOSITION)
```

Where:

Integer BUSNUMBER	bus number of the bus to GROW (no default) (input).
Real XPOSITION	X coordinate of the bus on the Slider diagram (input).
Real YPOSITION	Y coordinate of the bus on the Slider diagram (input).

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6.10. GROWBUSLEVELS

Use this API to GROW a bus and all the connected buses up to the specified number of levels away in an active Slider diagram in the GUI. Requires a current network case.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_GROWBUSLEVELS  BUSNUMBER XPOSITION YPOSITION NUMLEVELS
```

Python command syntax:

```
growbuslevels(busnumber, xposition, yposition, numlevels)
```

Fortran command syntax:

```
CALL GROWBUSLEVELS(BUSNUMBER, XPOSITION, YPOSITION, NUMLEVELS)
```

Where:

Integer BUSNUMBER	bus number of the bus to GROW (no default) (input).
Real XPOSITION	X coordinate of the bus on the Slider diagram (input).
Real YPOSITION	Y coordinate of the bus on the Slider diagram (input).
Integer NUMLEVELS	number of levels; include all buses up to this number of connections from BUSNUMBER in the GROW operation (no default) (input).

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6.11. GROWDIAGRAM

This API routine is obsolete. It has been replaced by the API routine growdiagram_2 .

Use this API to GROW a bus and all the connected buses in an active Slider diagram in the GUI. Requires a current network case.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_GROWDIAGRAM  NUMLEVELS ITEMCOUNT ITEMLIST XPOSITION YPOSITION
```

Python command syntax:

```
growdiagram(numlevels, itemcount, itemlist, xposition, yposition)
```

Fortran command syntax:

```
CALL GROWDIAGRAM(NUMLEVELS, ITEMCOUNT, ITEMLIST, XPOSITION, YPOSITION)
```

Where:

Integer NUMLEVELS	number of levels; include all buses up to this number of connections from each item in ITEMLIST in the GROW operation. A value of -1 can be used to grow all buses (subsystem and in-service options still apply), however attempting to lay out several hundred buses or more will take a very large amount of time to complete as PSSE attempts to calculate best positions for each bus. It is still recommended that the user only grow several levels at once (input).
Integer ITEMCOUNT	number of items in ITEMLIST. ITEMCOUNT can be 0 if NUMLEVELS is -1. Under this condition diagram growth will start at an arbitrary bus (input; 0 by default).
Character*40 ITEMLIST (ITEMCOUNT)	list of items to grow. Each item is given by its generated map string (input).
Real XPOSITION	X coordinate of the first bus on the Slider diagram (input).
Real YPOSITION	X coordinate of the first bus on the Slider diagram (input).

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6.12. GRPG

Use this API to produce a GRPG report in the active Slider diagram in the GUI.

This API routine was first introduced in release 30.3.3.

Batch command syntax:

```
BAT_GRPG IMAGEFILE DUMFILE
```

Python command syntax:

```
ierr = grpg(imagefile, dumfile)
```

Fortran command syntax:

```
CALL GRPGAPI(IMAGEFILE, DUMFILE, IERR)
```

Where:

Character*260 IMAGEFILE	name of the GRPG file to be used (input).
Character*260 DUMFILE	dummy filename; this should be initialized to " " (input).
Integer IERR	error code (output).

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6.13. IMPORTDRAWFILE

Use this API to import a Draw Data File into an active Slider diagram in the GUI.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_IMPORTDRAWFILE  DRAWFILE
```

Python command syntax:

```
importdrawfile(drawfile)
```

Fortran command syntax:

```
CALL IMPORTDRAWFILE(DRAWFILE)
```

Where:

Character*260 DRAWFILE	name of an existing Draw Data File, *.drw (input).

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6.14. IMPORTIMAGEFILE

Use this API to import an image into the background layer of the active Slider diagram in the GUI.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_IMPORTIMAGEFILE  IMAGEFILE
```

Python command syntax:

```
importimagefile(imagefile)
```

Fortran command syntax:

```
CALL IMPORTIMAGEFILE(IMAGEFILE)
```

Where:

Character*260 IMAGEFILE	name of the image file to be imported (input).

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6.15. NEWDIAGFILE

Use this API to open a new Slider diagram in the GUI.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_NEWDIAGFILE
```

Python command syntax:

```
newdiagfile()
```

Fortran command syntax:

```
CALL NEWDIAGFILE
```

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6.16. NEXTDIAGFILEDATASET

This API is used to display the next element in the active data set on the active Slider diagram. Currently, this API is only used with ACCC results.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_NEXTDIAGFILEDATASET
```

Python command syntax:

```
nextdiagfileset()
```

Fortran command syntax:

```
CALL NEXTDIAGFILEDATASET
```

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6.17. OPENACCFILE

This API routine is obsolete. It has been replaced by the API routine openaccfile_2 .

This API is used to open an existing ACC results file for displaying results on diagram.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_OPENACCFILE  NBASERATE NCNTGRATE FPERRATE SZFILENAME
```

Python command syntax:

```
openaccfile(nbaserate, ncntgrate, fperrate, szfilename)
```

Fortran command syntax:

```
CALL OPENACCFILE(NBASERATE, NCNTGRATE, FPERRATE, SZFILENAME)
```

Where:

Integer NBASERATE	base case rating to use for results display (input). nBaseRate = 1 RateA nBaseRate = 2 RateB nBaseRate = 3 RateC
Integer NCNTGRATE	contingency case rating to use for results display (input). nCntgRate = 1 RateA nCntgRate = 2 RateB nCntgRate = 3 RateC
Real FPERRATE	loading percentage to use for results display (input).
Character*260 SZFILENAME	ACC file to open (input).

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6.18. OPENACCFILE_2

This API is used to open an existing ACC results file for displaying results on diagram.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_OPENACCFILE_2  NBASERATE NCNTGRATE NBASEVOLTLIM NCNTGVOLTLIM FPERRATE
 SZFILENAME
```

Python command syntax:

```
openaccfile_2(nbaserate, ncntgrate, nbasevoltlim, ncntgvoltlim, fperrate,
 szfilename)
```

Fortran command syntax:

```
CALL OPENACCFILE_2(NBASERATE, NCNTGRATE, NBASEVOLTLIM, NCNTGVOLTLIM,
 FPERRATE, SZFILENAME)
```

Where:

Integer NBASERATE	base case rating to use for results display (input).	
	nBaseRate = 1	RateA
	nBaseRate = 2	RateB
	nBaseRate = 3	RateC
Integer NCNTGRATE	contingency case rating to use for results display (input).	
	nCntgRate = 1	RateA
	nCntgRate = 2	RateB
	nCntgRate = 3	RateC
Integer NBASEVOLTLIM	base case voltage limit for results display (input).	
	nBaseVoltLim = 1	Don't use the limit
	nBaseVoltLim = 2	Use the limit
Integer NCNTGVOLTLIM	contingencyt case voltage limit for results display (input).	
	nCntgVoltLim = 1	don't use the limit
	nCntgVoltLim = 2	use the limit
Real FPERRATE	loading percentage to use for results display (input).	
Character*260 SZFILENAME	ACC file to open (input).	

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6.19. OPENBUSLOCFILE

Use this API to open an existing bus location file in the GUI. The bus location file can be used to control the placement of buses on a diagram during auto-draw and grow operations

This API routine was first introduced in release 30.0.1.

Batch command syntax:

```
BAT_OPENBUSLOCFILE  BUSLOCFILE
```

Python command syntax:

```
openbuslocfile(buslocfile)
```

Fortran command syntax:

```
CALL OPENBUSLOCFILE(BUSLOCFILE)
```

Where:

Character*260 BUSLOCFILE	name of existing Bus Location Data File (input).

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6.20. OPENDIAGFILE

Use this API to open an existing Slider diagram in the GUI.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_OPENDIAGFILE  SFILE
```

Python command syntax:

```
opendiagfile(sfile)
```

Fortran command syntax:

```
CALL OPENDIAGFILE(SFILE)
```

Where:

Character*260 SFILE	error code (output).	
	IERR = 0	no error occured
	IERR = 1	error opening the diagram file

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6.21. PREVDIAGFILEDATASET

This API is used to display the previous element in the active data set on the active Slider diagram. Currently, this API is only used with ACCC results.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_PREVDIAGFILEDATASET
```

Python command syntax:

```
prevdiagfiledataset()
```

Fortran command syntax:

```
CALL PREVDIAGFILEDATASET
```

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6.22. PRINTDIAGFILE

Use this API to print the active Slider diagram in the GUI.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PRINTDIAGFILE  SDEV COPIES ORIENT
```

Python command syntax:

```
printdiagfile(sdev, copies, orient)
```

Fortran command syntax:

```
CALL PRINTDIAGFILE(SDEV, COPIES, ORIENT)
```

Where:

Character*260 SDEV	name of valid attached Windows printer (input).	
Integer COPIES	number of copies to print (input).	
Integer ORIENT	orientation (input).	
	orient = 1	Portrait
	orient = 2	Landscape

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6.23. REFRESHDIAGCONTOUR

Use this API to refresh the contour in the active Slider diagram in the GUI. Requires a current network case. This API would be used in an automation file to force a contour refresh that wouldn't normally take place until the automation file completed.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_REFRESHDIAGCONTOUR
```

Python command syntax:

```
refreshdiagcontour()
```

Fortran command syntax:

```
CALL REFRESHDIAGCONTOUR
```

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6.24. REFRESHDIAGFILE

Use this API to refresh the results and styles in the active Slider diagram in the GUI. It requires a current network case. This API would be used in an automation file to force a results refresh that wouldn't normally take place until the automation file completed.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_REFRESHDIAGFILE
```

Python command syntax:

```
refreshdiagfile()
```

Fortran command syntax:

```
CALL REFRESHDIAGFILE
```

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6.25. RENUMDIAGFILE

Use this API to renumber buses in the active Slider diagram in the GUI.

This API routine was first introduced in release 30.2.1.

Batch command syntax:

```
BAT_RENUMDIAGFILE  TRNFILE
```

Python command syntax:

```
renumdiagfile(trnfile)
```

Fortran command syntax:

```
CALL RENUMDIAGFILE(TRNFILE)
```

Where:

Character*260 TRNFILE	name of Bus Number Translation File (input).

6.26. SAVEDIAGFILE

Use this API to save the active Slider diagram in the GUI.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SAVEDIAGFILE  SFILE
```

Python command syntax:

```
savediagfile(sfile)
```

Fortran command syntax:

```
CALL SAVEDIAGFILE(SFILE)
```

Where:

Character*260 SFILE	name of file to save the Slider diagram in (input).

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6.27. SETDIAGAUTOFILE

Use this API to specify an automation file that is run each time results are updated on the active Slider diagram.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SETDIAGAUTOFILE  AUTONAME
```

Python command syntax:

```
setdiagautofile(autoname)
```

Fortran command syntax:

```
CALL SETDIAGAUTOFILE(AUTONAME)
```

Where:

Character*260 AUTONAME	name of the automation file (input).

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6.28. SETDIAGFONT

Use this API to specify the default text font in the active Slider diagram.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGFONT  FNTSIZ FNTWGT FNTNAM
```

Python command syntax:

```
setdiagfont(fntsiz, fntwgt, fntnam)
```

Fortran command syntax:

```
CALL SETDIAGFONT(FNTSIZ, FNTWGT, FNTNAM)
```

Where:

Integer FNTSIZ	font point size (input).
Integer FNTWGT	font weight (input).
character*260 FNTNAM	name of the font family (input).

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6.29. SETDIAGPREC

Use this API to specify the precision of results displayed in Slider diagrams.

This API routine was first introduced in release 30.1.0.

Batch command syntax:

```
BAT_SETDIAGPREC  DIAGPREC
```

Python command syntax:

```
setdiagprec(diagprec)
```

Fortran command syntax:

```
CALL SETDIAGPREC(DIAGPREC)
```

Where:

Integer DIAGPREC (8)	number of digits after the decimal point for various result types. Currently, only DIAGPREC(1) is used for power flow results; valid values are 1 through 4 (input).

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6.30. SETDIAGPROP

This API routine is obsolete. It has been replaced by the API routine setdiagprop_2 .

Use this API to specify basic properties in the active Slider diagram.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGPROP  UNBNDCLR(1)..UNBNDCLR(3) BNDCLR(1)..BNDCLR(3)
 BKGCLR(1)..BKGCLR(3) PRTCLR(1)..PRTCLR(3) GRDCLR(1)..GRDCLR(3)
 UPARWCLR(1)..UPARWCLR(3) DWNARWCLR(1)..DWNARWCLR(3) LABCLR(1)..LABCLR(3)
 GRDSTYL GRDPTRN FLTVAL(1) FLTVAL(2) DBLVAL(1)..DBLVAL(10)
 LOGVAL(1)..LOGVAL(6) BARCHRT LABLHIDE PRNTMODE IMGDIR
```

Python command syntax:

```
setdiagprop(unbndclr, bndclr, bkgclr, prtclr, grdclr, uparwclr, dwnarwclr,
 labclr, grdstyl, grdptrn, fltval, dblval, logval, barchrt, lablhide,
 prntmode, imgdir)
```

Fortran command syntax:

```
CALL SETDIAGPROP(UNBNDCLR, BNDCLR, BKGCLR, PRTCLR, GRDCLR, UPARWCLR,
 DWNARWCLR, LABCLR, GRDSTYL, GRDPTRN, FLTVAL, DBLVAL, LOGVAL, BARCHRT,
 LABLHIDE, PRNTMODE, IMGDIR)
```

Where:

Integer UNBNDCLR (3)	array of 3 elements specifying unbound item color(input).	
	UNBNDCLR(1)	0->255 Red component
	UNBNDCLR(2)	0->255 Blue component
	UNBNDCLR(3)	0->255 Green component
Integer BNDCLR (3)	array of 3 elements specifying bound item color(input).	
	BNDCLR(1)	0->255 Red component
	BNDCLR(2)	0->255 Blue component
	BNDCLR(3)	0->255 Green component
Integer BKGCLR (3)	array of 3 elements specifying background color(input).	
	BKGCLR(1)	0->255 Red component
	BKGCLR(2)	0->255 Blue component
	BKGCLR(3)	0->255 Green component
Integer PRTCLR (3)	array of 3 elements specifying item connection port color(input).	
	PRTCLR(1)	0->255 Red component
	PRTCLR(2)	0->255 Blue component
	PRTCLR(3)	0->255 Green component
Integer GRDCLR (3)	array of 3 elements specifying on-diagram layout grid color(input).	
	GRDCLR(1)	0->255 Red component
	GRDCLR(2)	0->255 Blue component

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	GRDCLR(3)	0->255 Green component
Integer UPARWCLR (3)	array of 3 elements specifying active power animation arrow color(input).	
	UPARWCLR(1)	0->255 Red component
	UPARWCLR(2)	0->255 Blue component
	UPARWCLR(3)	0->255 Green component
Integer DWNARWCLR (3)	array of 3 elements specifying reactive power animation arrow color(input).	
	DWNARWCLR(1)	0->255 Red component
	DWNARWCLR(2)	0->255 Blue component
	DWNARWCLR(3)	0->255 Green component
Integer LABCLR (3)	array of 3 elements specifying label color(input).	
	LABCLR(1)	0->255 Red component
	LABCLR(2)	0->255 Blue component
	LABCLR(3)	0->255 Green component
Integer GRDSTYL	grid style (input).	
	GRDSTYL = 0	Dotted
	GRDSTYL = 1	Lined
Integer GRDPTRN	grid pattern (input).	
Real FLTVAL (2)	array of 2 elements specifying (input).	
	FLTVAL(1)	grid point size
	FLTVAL(2)	default line width in pixels
Real DBLVAL (10)	array of 10 elements specifying (input).	
	DBLVAL(1)	maximum zoom factor x 100, i.e. 10=1000% zoom
	DBLVAL(2)	minimum zoom factor
	DBLVAL(3)	zoom increment
	DBLVAL(4)	zoom at which to hide selection handles
	DBLVAL(5)	zoom at which to hide text
	DBLVAL(6)	zoom at which to hide grid
	DBLVAL(7)	multi-page print scaling
	DBLVAL(8)	pan distance
	DBLVAL(9)	grid spacing
	DBLVAL(10)	grid snap distance
Integer LOGVAL (6)	array of 6 elements specifying Each element has only two possible values. 0 = FALSE and 1 = TRUE. The elements are as follows(input).	
	LOGVAL(1)	no upside down text
	LOGVAL(2)	high quality printing
	LOGVAL(3)	display tooltips
	LOGVAL(4)	stipple grid
	LOGVAL(5)	hide unbound network items
	LOGVAL(6)	display filled busbars

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Integer BARCHRT	Two-winding transformer bar chart display side (input). BARCHRT = 0 high current loading side BARCHRT = 1 low current loading side BARCHRT = 2 both sides
Integer LABLHIDE	Label hide mode (input). LABLHIDE = 0 show all labels and results LABLHIDE = 1 hide results LABLHIDE = 2 hide all labels and results
Integer PRNTMODE	Printing mode (input). PRNTMODE = 0 as seen on the screen, WYSIWYG PRNTMODE = 1 fit the whole diagram on the page PRNTMODE = 2 multi-page
Character*260 IMGDIR	name of directory where image files for this diagram are stored (input).

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6.31. SETDIAGPROP_2

This API routine is obsolete. It has been replaced by the API routine setdiagprop_3 .

Use this API to specify basic properties in the active Slider diagram.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SETDIAGPROP_2  BKGCLR(1)..BKGCLR(3) PRTCLR(1)..PRTCLR(3)
 GRDCLR(1)..GRDCLR(3) UPARWCLR(1)..UPARWCLR(3) DWNARWCLR(1)..DWNARWCLR(3)
 LABCLR(1)..LABCLR(3) GRDSTYL GRDPTRN FLTVAL(1) FLTVAL(2)
 DBLVAL(1)..DBLVAL(10) LOGVAL(1)..LOGVAL(6) LABLHIDE PRNTMODE IMGDIR
```

Python command syntax:

```
setdiagprop_2(bkgclr, prtclr, grdclr, uparwclr, dwnarwclr, labclr,
 grdstyl, grdptrn, fltval, dblval, logval, lablhide, prntmode, imgdir)
```

Fortran command syntax:

```
CALL SETDIAGPROP_2(BKGCLR, PRTCLR, GRDCLR, UPARWCLR, DWNARWCLR, LABCLR,
 GRDSTYL, GRDPTRN, FLTVAL, DBLVAL, LOGVAL, LABLHIDE, PRNTMODE, IMGDIR)
```

Where:

Integer BKGCLR (3)	array of 3 elements specifying background color(input).	
	BKGCLR(1)	0->255 Red component
	BKGCLR(2)	0->255 Blue component
	BKGCLR(3)	0->255 Green component
Integer PRTCLR (3)	array of 3 elements specifying item connection port color(input).	
	PRTCLR(1)	0->255 Red component
	PRTCLR(2)	0->255 Blue component
	PRTCLR(3)	0->255 Green component
Integer GRDCLR (3)	array of 3 elements specifying on-diagram layout grid color(input).	
	GRDCLR(1)	0->255 Red component
	GRDCLR(2)	0->255 Blue component
	GRDCLR(3)	0->255 Green component
Integer UPARWCLR (3)	array of 3 elements specifying active power animation arrow color(input).	
	UPARWCLR(1)	0->255 Red component
	UPARWCLR(2)	0->255 Blue component
	UPARWCLR(3)	0->255 Green component
Integer DWNARWCLR (3)	array of 3 elements specifying reactive power animation arrow color(input).	
	DWNARWCLR(1)	0->255 Red component
	DWNARWCLR(2)	0->255 Blue component

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	DWNARWCLR(3)	0->255 Green component
Integer LABCLR (3)	array of 3 elements specifying label color(input).	
	LABCLR(1)	0->255 Red component
	LABCLR(2)	0->255 Blue component
	LABCLR(3)	0->255 Green component
Integer GRDSTYL	grid style (input).	
	GRDSTYL = 0	Dotted
	GRDSTYL = 1	Lined
Integer GRDPTRN	grid pattern (input).	
Real FLTVAL (2)	array of 2 elements specifying (input).	
	FLTVAL(1)	grid point size
	FLTVAL(2)	default line width in pixels
Real DBLVAL (10)	array of 10 elements specifying (input).	
	DBLVAL(1)	maximum zoom factor x 100, i.e. 10=1000% zoom
	DBLVAL(2)	minimum zoom factor
	DBLVAL(3)	zoom increment
	DBLVAL(4)	zoom at which to hide selection handles
	DBLVAL(5)	zoom at which to hide text
	DBLVAL(6)	zoom at which to hide grid
	DBLVAL(7)	multi-page print scaling
	DBLVAL(8)	pan distance
	DBLVAL(9)	grid spacing
	DBLVAL(10)	grid snap distance
Integer LOGVAL (6)	array of 6 elements specifying Each element has only two possible values. 0 = FALSE and 1 = TRUE. The elements are as follows(input).	
	LOGVAL(1)	no upside down text
	LOGVAL(2)	high quality printing
	LOGVAL(3)	display tooltips
	LOGVAL(4)	stipple grid
	LOGVAL(5)	hide unbound network items
	LOGVAL(6)	display filled busbars
Integer LABLHIDE	Label hide mode (input).	
	LABLHIDE = 0	show all labels and results
	LABLHIDE = 1	hide results
	LABLHIDE = 2	hide all labels and results
Integer PRNTMODE	Printing mode (input).	
	PRNTMODE = 0	as seen on the screen, WYSIWYG
	PRNTMODE = 1	fit the whole diagram on the page
	PRNTMODE = 2	multi-page
Character*260 IMGDIR	name of the directory where image files for this diagram are stored (input).	

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6.32. SETDIAGPROP_3

This API routine is obsolete. It has been replaced by the API routine setdiagprop_4 .

Use this API to specify basic properties in the active Slider diagram.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_SETDIAGPROP_3  BKGCLR(1)..BKGCLR(3) PRTCLR(1)..PRTCLR(3)
 GRDCLR(1)..GRDCLR(3) UPARWCLR(1)..UPARWCLR(3) DWNARWCLR(1)..DWNARWCLR(3)
 LABCLR(1)..LABCLR(3) GRDSTYL GRDPTRN FLTVAL(1)..FLTVAL(2)
 DBLVAL(1)..DBLVAL(10) LOGVAL(1)..LOGVAL(7) LABLHIDE PRNTMODE IMGDIR
```

Python command syntax:

```
setdiagprop_3(bkgclr, prtclr, grdclr, uparwclr, dwnarwclr, labclr,
 grdstyl, grdptrn, fltval, dblval, logval, lablhide, prntmode, imgdir)
```

Fortran command syntax:

```
CALL SETDIAGPROP_3(BKGCLR, PRTCLR, GRDCLR, UPARWCLR, DWNARWCLR, LABCLR,
 GRDSTYL, GRDPTRN, FLTVAL, DBLVAL, LOGVAL, LABLHIDE, PRNTMODE, IMGDIR)
```

Where:

Integer BKGCLR (3)	array of 3 elements specifying background color(input).	
	BKGCLR(1)	0->255 Red component
	BKGCLR(2)	0->255 Blue component
	BKGCLR(3)	0->255 Green component
Integer PRTCLR (3)	array of 3 elements specifying item connection port color(input).	
	PRTCLR(1)	0->255 Red component
	PRTCLR(2)	0->255 Blue component
	PRTCLR(3)	0->255 Green component
Integer GRDCLR (3)	array of 3 elements specifying on-diagram layout grid color(input).	
	GRDCLR(1)	0->255 Red component
	GRDCLR(2)	0->255 Blue component
	GRDCLR(3)	0->255 Green component
Integer UPARWCLR (3)	array of 3 elements specifying active power animation arrow color(input).	
	UPARWCLR(1)	0->255 Red component
	UPARWCLR(2)	0->255 Blue component
	UPARWCLR(3)	0->255 Green component
Integer DWNARWCLR (3)	array of 3 elements specifying reactive power animation arrow color(input).	
	DWNARWCLR(1)	0->255 Red component
	DWNARWCLR(2)	0->255 Blue component
	DWNARWCLR(3)	0->255 Green component

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Integer LABCLR (3)	array of 3 elements specifying label color(input).	
	LABCLR(1)	0->255 Red component
	LABCLR(2)	0->255 Blue component
	LABCLR(3)	0->255 Green component
Integer GRDSTYL	grid style (input).	
	GRDSTYL = 0	dotted
	GRDSTYL = 1	lined
Integer GRDPTRN	grid pattern (input).	
Real FLTVAL (2)	array of 2 elements specifying (input).	
	FLTVAL(1)	grid point size
	FLTVAL(2)	default line width in pixels
Real DBLVAL (10)	array of 10 elements specifying (input).	
	DBLVAL(1)	maximum zoom factor x 100, i.e. 10=1000% zoom
	DBLVAL(2)	minimum zoom factor
	DBLVAL(3)	zoom increment
	DBLVAL(4)	zoom at which to hide selection handles
	DBLVAL(5)	zoom at which to hide text
	DBLVAL(6)	zoom at which to hide grid
	DBLVAL(7)	multi-page print scaling
	DBLVAL(8)	pan distance
	DBLVAL(9)	grid spacing
	DBLVAL(10)	grid snap distance
Integer LOGVAL (7)	array of 7 elements specifying Each element has only two possible values. 0 = FALSE and 1 = TRUE. The elements are as follows(input).	
	LOGVAL(1)	no upside down text
	LOGVAL(2)	high quality printing
	LOGVAL(3)	display tooltips
	LOGVAL(4)	stipple grid
	LOGVAL(5)	hide unbound network items
	LOGVAL(6)	display filled busbars
	LOGVAL(7)	make branch connections perpendicular to branches
Integer LABLHIDE	label hide mode (input).	
	LABLHIDE = 0	show all labels and results
	LABLHIDE = 1	hide results
	LABLHIDE = 2	hide all labels and results
Integer PRNTMODE	printing mode (input).	
	PRNTMODE = 0	as seen on the screen, WYSIWYG
	PRNTMODE = 1	fit the whole diagram on the page
	PRNTMODE = 2	multi-page

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Character*260 IMGDIR	name of directory where image files for this diagram are stored (input).

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6.33. SETDIAGPROP_4

Use this API to specify basic properties in the active Slider diagram.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_SETDIAGPROP_4  BKGCLR(1)..BKGCLR(3) PRTCLR(1)..PRTCLR(3)
 GRDCLR(1)..GRDCLR(3) UPARWCLR(1)..UPARWCLR(3) DWNARWCLR(1)..DWNARWCLR(3)
 LABCLR(1)..LABCLR(3) GRDSTYL GRDPTRN FLTVAL(1)..FLTVAL(2)
 DBLVAL(1)..DBLVAL(11) LOGVAL(1)..LOGVAL(7) LABLHIDE PRNTMODE IMGDIR
```

Python command syntax:

```
setdiagprop_4(bkgclr, prtclr, grdclr, uparwclr, dwnarwclr, labclr,
 grdstyl, grdptrn, fltval, dblval, logval, lablhide, prntmode, imgdir)
```

Fortran command syntax:

```
CALL SETDIAGPROP_4(BKGCLR, PRTCLR, GRDCLR, UPARWCLR, DWNARWCLR, LABCLR,
 GRDSTYL, GRDPTRN, FLTVAL, DBLVAL, LOGVAL, LABLHIDE, PRNTMODE, IMGDIR)
```

Where:

Integer BKGCLR (3)	array of 3 elements specifying background color(input).	
	BKGCLR(1)	0->255 Red component
	BKGCLR(2)	0->255 Blue component
	BKGCLR(3)	0->255 Green component
Integer PRTCLR (3)	array of 3 elements specifying item connection port color(input).	
	PRTCLR(1)	0->255 Red component
	PRTCLR(2)	0->255 Blue component
	PRTCLR(3)	0->255 Green component
Integer GRDCLR (3)	array of 3 elements specifying on-diagram layout grid color(input).	
	GRDCLR(1)	0->255 Red component
	GRDCLR(2)	0->255 Blue component
	GRDCLR(3)	0->255 Green component
Integer UPARWCLR (3)	array of 3 elements specifying active power animation arrow color(input).	
	UPARWCLR(1)	0->255 Red component
	UPARWCLR(2)	0->255 Blue component
	UPARWCLR(3)	0->255 Green component
Integer DWNARWCLR (3)	array of 3 elements specifying reactive power animation arrow color(input).	
	DWNARWCLR(1)	0->255 Red component
	DWNARWCLR(2)	0->255 Blue component
	DWNARWCLR(3)	0->255 Green component
Integer LABCLR (3)	array of 3 elements specifying label color(input).	
	LABCLR(1)	0->255 Red component

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	LABCLR(2)	0->255 Blue component
	LABCLR(3)	0->255 Green component
Integer GRDSTYL	grid style (input).	
	GRDSTYL = 0	dotted
	GRDSTYL = 1	lined
Integer GRDPTRN	grid pattern (input).	
Real FLTVAL (2)	array of 2 elements specifying (input).	
	FLTVAL(1)	grid point size
	FLTVAL(2)	default line width in pixels
Real DBLVAL (11)	array of 11 elements specifying (input).	
	DBLVAL(1)	maximum zoom factor x 100, i.e. 10=1000% zoom
	DBLVAL(2)	minimum zoom factor
	DBLVAL(3)	zoom increment
	DBLVAL(4)	zoom at which to hide selection handles
	DBLVAL(5)	zoom at which to hide text
	DBLVAL(6)	zoom at which to hide grid
	DBLVAL(7)	multi-page print scaling
	DBLVAL(8)	pan distance
	DBLVAL(9)	grid spacing
	DBLVAL(10)	grid snap distance
	DBLVAL(11)	branch spacing
Integer LOGVAL (7)	array of 7 elements specifying Each element has only two possible values. 0 = FALSE and 1 = TRUE. The elements are as follows(input).	
	LOGVAL(1)	no upside down text
	LOGVAL(2)	high quality printing
	LOGVAL(3)	display tooltips
	LOGVAL(4)	stipple grid
	LOGVAL(5)	hide unbound network items
	LOGVAL(6)	display filled busbars
	LOGVAL(7)	make branch connections perpendicular to branches
Integer LABLHIDE	label hide mode (input).	
	LABLHIDE = 0	show all labels and results
	LABLHIDE = 1	hide results
	LABLHIDE = 2	hide all labels and results
Integer PRNTMODE	printing mode (input).	
	PRNTMODE = 0	as seen on the screen, WYSIWYG
	PRNTMODE = 1	fit the whole diagram on the page
	PRNTMODE = 2	multi-page
Character*260 IMGDIR	name of directory where image files for this diagram are stored (input).	

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6.34. SETDIAGRESACCC

This API routine is obsolete. It has been replaced by the API routine setdiagresaccc_2 .

Use this API to set the ACCC analysis annotation options in the active Slider diagram.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SETDIAGRESACCC  RATE CNTGOPT CONTOUR CNTROPT LOADPER FILE
```

Python command syntax:

```
setdiagresaccc(rate, cntgopt, contour, cntropt, loadper, file)
```

Fortran command syntax:

```
CALL SETDIAGRESACCC(RATE, CNTGOPT, CONTOUR, CNTROPT, LOADPER, FILE)
```

Where:

Integer RATE	branch rating set (input).	
	RATE = 1	RateA
	RATE = 2	RateB
	RATE = 3	RateC
Integer CNTGOPT	contingencies to view (input).	
	CNTGOPT = 1	all contingencies
	CNTGOPT = 2	contingencies causing overload violations
	CNTGOPT = 3	contingencies causing any violations
Integer CONTOUR	contour selection (input).	
	CONTOUR = 0	display data values as text
	CONTOUR = 1	display data values as a contour
Integer CNTROPT	contour value selection (input).	
	CNTROPT = 0	contour bus voltages
	CNTROPT = 1	contour % branch flows
Real LOADPER	loading percentage (input).	
Character*260 FILE	ACCC Contingency Solution data file (input; Not Allowed).	

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6.35. SETDIAGRESACCC_2

Use this API to set the ACCC analysis annotation options in the active Slider diagram

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_SETDIAGRESACCC_2  CNTGOPT SELCNTG CONTOUR CNTROPT
```

Python command syntax:

```
setdiagresaccc_2(cntgopt, selcntg, contour, cntropt)
```

Fortran command syntax:

```
CALL SETDIAGRESACCC_2(CNTGOPT, SELCNTG, CONTOUR, CNTROPT)
```

Where:

Integer SELCNTG	selected contingency (input).	
	SELCNTG = 0	none selected
	SELCNTG != 0	selected contingency from list determined by CNTGOPT
Integer CNTGOPT	contingency to view (input).	
	CNTGOPT = 1	all contingencies
	CNTGOPT = 2	contingencies causing overload violations
	CNTGOPT = 3	contingencies causing any violations
Integer CONTOUR	contour selection (input).	
	CONTOUR = 0	display data values as text
	CONTOUR = 1	display data values as a contour
Integer CNTROPT	contour value selection (input).	
	CNTROPT = 0	contour bus voltages
	CNTROPT = 1	contour % branch flows

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6.36. SETDIAGRESASCC

This API is used to set the ASCC short circuit analysis annotation options in the active Slider diagram.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SETDIAGRESASCC  ANNOT TYPE BUS BOE CTRIB FRM TO LAST ID STYLE WIDTH
 COLOR(1)..COLOR(3)
```

Python command syntax:

```
setdiagresascc(annot, type, bus, boe, ctrib, frm, to, last, id, style,
 width, color)
```

Fortran command syntax:

```
CALL SETDIAGRESASCC(ANNOT, TYPE, BUS, BOE, CTRIB, FRM, TO, LAST, ID,
 STYLE, WIDTH, COLOR)
```

Where:

Integer ANNOT	fault to display (input).	
	ANNOT = 0	fault currents (only bus faults)
	ANNOT = 1	three phase
	ANNOT = 2	line to Ground (LG)
	ANNOT = 3	line Line to Ground (LLG)
	ANNOT = 4	line to Line (LL)
Integer TYPE	fault display quantity (input).	
	TYPE = 1	Ia1
	TYPE = 2	Ia2
	TYPE = 3	Ia0
	TYPE = 4	IA
	TYPE = 5	IB
	TYPE = 6	IC
	TYPE = 7	3Ia0
Integer BUS	faulted bus (input).	
	BUS = 0	no bus faulted
	BUS > 0	bus to be faulted
Integer BOE	fault location (input).	
	BOE = 1	fault located at the faulted bus
	BOE = 2	fault located at the faulted bus with lines out-aged
	BOE = 3	fault located at the end of lines
Integer CTRIB	contributions at home bus for bus faults (input).	
	CTRIB = 0	contributions for selected fault

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	CTRIB = 1	contributions at home bus for all faulted buses for selected fault
Integer FRM	from bus if a line fault is applied (input).	
	FRM = 0	no line fault applied
	FRM > 0	from bus
Integer TO	to bus if a line fault is applied (input).	
	TO = 0	no line fault applied
	TO > 0	to bus
Integer LAST	third bus if a line fault is applied to a 3-winding transformer (input).	
	LAST = 0	no line fault applied
	LAST > 0	to bus
Character*2 ID	branch/transformer identifier (input).	
Integer STYLE	line style applied to the faulted bus diagram item (input).	
Integer WIDTH	line width applied to the faulted bus diagram item, >= 1 (input).	
Integer COLOR (3)	array of 3 elements specifying color for diagram items that are faulted(input).	
	COLOR(1)	0->255 Red component
	COLOR(2)	0->255 Blue component
	COLOR(3)	0->255 Green component

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6.37. SETDIAGRESDATA

This API routine is obsolete. It has been replaced by the API routine setdiagresdata_2 .

Use this API to specify the impedance data voltage and range checking annotation options in the active Slider Diagram.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESDATA  USEVLT VLNWDT(1)..VLNWDT(8) VLNSTY(1)..VLNSTY(8)
 VLTVAL(1)..VLTVAL(7) VLNCLRR(1)..VLNCLRR(8) VLNCLRG(1)..VLNCLRG(8)
 VLNCLRB(1)..VLNCLRB(8) OUTWDT OUTSTY OUTCLR(1)..OUTCLR(3) BUSANN BRNANN
 EQPANN BUSVLT
```

Python command syntax:

```
setdiagresdata(usevlt, vlnwdt, vlnsty, vltval, vlnclrr, vlnclrg, vlnclrb,
 outwdt, outsty, outclr, busann, brnann, eqpann, busvlt)
```

Fortran command syntax:

```
CALL SETDIAGRESDATA(USEVLT, VLNWDT, VLNSTY, VLTVAL, VLNCLRR, VLNCLRG,
 VLNCLRB, OUTWDT, OUTSTY, OUTCLR, BUSANN, BRNANN, EQPANN, BUSVLT)
```

Where:

Integer USEVLT	option to apply voltage level annotation to the Diagram elements (input).	
	USEVLT = 0	don't apply voltage level annotation
	USEVLT = 1	apply voltage level annotation
Integer VLNWDT (8)	array of 8 elements specifying used to set the width of Diagram elements that fall in the appropriate voltage range(input).	
	VLNWDT(1)	base voltage < VLTVAL(1)
	VLNWDT(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNWDT(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNWDT(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNWDT(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNWDT(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNWDT(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNWDT(8)	base voltage >= VLTVAL(7)
Integer VLNSTY (8)	array of 8 elements specifying used to set the line style of Diagram elements that fall in the appropriate voltage range(input).	
	VLNSTY(1)	base voltage < VLTVAL(1)
	VLNSTY(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNSTY(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNSTY(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNSTY(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNSTY(6)	VLTVAL(5) >= base voltage < VLTVAL(6)

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	VLNSTY(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNSTY(8)	base voltage >= VLTVAL(7)
Real VLTVAL (7)	used to set the voltage ranges used to assign annotation Diagram. Base voltage values should be in ascending order (input).	
Integer VLNCLRR (8)	array of 8 elements specifying used to set the Red color component of Diagram elements that fall in the appropriate voltage range, 0->255(input).	
	VLNCLRR(1)	base voltage < VLTVAL(1)
	VLNCLRR(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNCLRR(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNCLRR(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNCLRR(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNCLRR(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNCLRR(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNCLRR(8)	base voltage >= VLTVAL(7)
Integer VLNCLRG (8)	array of 8 elements specifying used to set Green color component of Diagram elements that fall in the appropriate voltage range, 0->255(input).	
	VLNCLRG(1)	base voltage < VLTVAL(1)
	VLNCLRG(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNCLRG(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNCLRG(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNCLRG(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNCLRG(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNCLRG(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNCLRG(8)	base voltage >= VLTVAL(7)
Integer VLNCLRB (8)	array of 8 elements specifying used to set Blue color component of Diagram elements that fall in the appropriate voltage range, 0->255(input).	
	VLNCLRB(1)	base voltage < VLTVAL(1)
	VLNCLRB(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNCLRB(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNCLRB(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNCLRB(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNCLRB(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNCLRB(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNCLRB(8)	base voltage >= VLTVAL(7)
Integer OUTWDT	line width applied to out of service Diagram items, >= 1 (input).	
Integer OUTSTY	line style applied to out of service Diagram items (input).	
Integer OUTCLR (3)	array of 3 elements specifying color of Diagram items that are out of service(input).	
	OUTCLR(1)	0->255 Red component
	OUTCLR(2)	0->255 Blue component

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	OUTCLR(3)	0->255 Green component
Integer BUSANN	bus ID annotation (input).	
	BUSANN = 0	number, no name
	BUSANN = 1	name, no number
	BUSANN = 2	number and name
	BUSANN = 3	no number, no name
Integer BRNANN	branch annotation (input).	
	BRNANN = 0	MW/MVAR flows
	BRNANN = 1	MVA
	BRNANN = 2	MVA/% rating
	BRNANN = 3	Amps/pu current
	BRNANN = 4	MW/%rating
	BRNANN = 5	Amps/% rating
	BRNANN = 6	None
Integer EQPANN	equipment annotation (input).	
	EQPANN = 0	MW/MVAR
	EQPANN = 2	none
Integer BUSVLT	bus voltage annotation (input).	
	BUSVLT = 0	magnitude (pu)
	BUSVLT = 1	magnitude (kV)
	BUSVLT = 2	angle (degrees)
	BUSVLT = 3	magnitude (pu) and angle (degrees)
	BUSVLT = 4	magnitude (pu) and magnitude (kV)
	BUSVLT = 5	magnitude (kV) and angle (degrees)
	BUSVLT = 7	magnitude (pu), magnitude (kV) and angle (degrees)
	BUSVLT = 8	none

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6.38. SETDIAGRESDATA_2

Use this API to specify the basic Power Flow annotation options in the active Slider Diagram.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SETDIAGRESDATA_2  BRNANN BUSANN
```

Python command syntax:

```
setdiagresdata_2(brnann, busann)
```

Fortran command syntax:

```
CALL SETDIAGRESDATA_2(BRNANN, BUSANN)
```

Where:

Integer BRNANN	branch annotation (input).	
	BRNANN = 6	display impedances
	BRNANN = 7	display ratings
	BRNANN = 8	display impedances & ratings
	BRNANN = 9	display nothing
Integer BUSANN	bus voltage annotation (input).	
	BUSANN = 0	don't display base kV
	BUSANN = 1	display base kV

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6.39. SETDIAGRESDYN

This API is used to set the dynamic simulation annotation options in the active Slider diagram.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SETDIAGRESDYN  MODE CONTOUR
```

Python command syntax:

```
setdiagresdyn(mode, contour)
```

Fortran command syntax:

```
CALL SETDIAGRESDYN(MODE, CONTOUR)
```

Where:

Integer MODE	specifies the simulation result/value to display (input).	
	MODE = 0	voltage violations as specified by SET_VOLTAGE_REC_CHECK and SET_VOLTAGE_DIP_CHECK
	MODE = 1	contour simulation quantities
Integer CONTOUR	specifies the dynamic quantity to contour (input).	
	CONTOUR = 0	bus voltage - kV

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6.40. SETDIAGRESGDIF

Use this API to specify the case comparison voltage and range checking annotation options in the active Slider diagram.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESGDIF  USEVLT VLNWDT(1)..VLNWDT(8) VLNSTY(1)..VLNSTY(8)
 VLTVAL(1)..VLTVAL(7) VLNCLRR(1)..VLNCLRR(8) VLNCLRG(1)..VLNCLRG(8)
 VLNCLRB(1)..VLNCLRB(8) OUTWDT OUTSTY OUTCLR(1)..OUTCLR(3) BUSANN
```

Python command syntax:

```
setdiagresgdif(usevlt, vlnwdt, vlnsty, vltval, vlnclrr, vlnclrg, vlnclrb,
 outwdt, outsty, outclr, busann)
```

Fortran command syntax:

```
CALL SETDIAGRESGDIF(USEVLT, VLNWDT, VLNSTY, VLTVAL, VLNCLRR, VLNCLRG,
 VLNCLRB, OUTWDT, OUTSTY, OUTCLR, BUSANN)
```

Where:

Integer USEVLT	flag to apply voltage level annotation to the diagram elements (input).	
	USEVLT = 0	don't apply voltage level annotation
	USEVLT = 1	apply voltage level annotation
Integer VLNWDT (8)	array of 8 elements specifying the width of diagram elements that fall in the appropriate voltage range(input).	
	VLNWDT(1)	base voltage < VLTVAL(1)
	VLNWDT(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNWDT(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNWDT(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNWDT(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNWDT(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNWDT(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNWDT(8)	base voltage >= VLTVAL(7)
Integer VLNSTY (8)	array of 8 elements specifying the line style of diagram elements that fall in the appropriate voltage range(input).	
	VLNSTY(1)	base voltage < VLTVAL(1)
	VLNSTY(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNSTY(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNSTY(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNSTY(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNSTY(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNSTY(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNSTY(8)	base voltage >= VLTVAL(7)

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Real VLTVAL (7)
the voltage ranges used to assign annotation diagram. Base voltage values should be in ascending order (input).

Integer VLNCLRR (8)
array of 8 elements specifying the Red color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRR(1)	base voltage < VLTVAL(1)
VLNCLRR(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRR(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRR(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRR(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRR(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRR(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRR(8)	base voltage >= VLTVAL(7)

Integer VLNCLRG (8)
array of 8 elements specifying the Green color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRG(1)	base voltage < VLTVAL(1)
VLNCLRG(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRG(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRG(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRG(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRG(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRG(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRG(8)	base voltage >= VLTVAL(7)

Integer VLNCLRB (8)
array of 8 elements specifying the Blue color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRB(1)	base voltage < VLTVAL(1)
VLNCLRB(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRB(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRB(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRB(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRB(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRB(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRB(8)	base voltage >= VLTVAL(7)

Integer OUTWDT
line width applied to out of service diagram items, >= 1 (input).

Integer OUTSTY
line style applied to out of service diagram items (input).

Integer OUTCLR (3)
array of 3 elements specifying color for diagram items that are out of service(input).

OUTCLR(1)	0->255 Red component
OUTCLR(2)	0->255 Blue component
OUTCLR(3)	0->255 Green component

Integer BUSANN
bus ID annotation (input).

BUSANN = 0	number, no name
BUSANN = 1	name, no number

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BUSANN = 2	number and name
BUSANN = 3	no number, no name

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6.41. SETDIAGRESIEC

Use this API to specify the IEC annotation options in the active Slider diagram.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SETDIAGRESIEC  ANNOT TYPE FLTBUS BOE CONTRIB FRM TO LAST ID STYLE
 WIDTH COLOR(1).. COLOR(3)
```

Python command syntax:

```
setdiagresiec(annot, type, fltbus, boe, contrib, frm, to, last, id, style,
 width, color)
```

Fortran command syntax:

```
CALL SETDIAGRESIEC(ANNOT, TYPE, FLTBUS, BOE, CONTRIB, FRM, TO, LAST, ID,
 STYLE, WIDTH, COLOR)
```

Where:

Integer ANNOT	specifies the fault to display (input).	
	ANNOT = 0	fault currents (only bus faults)
	ANNOT = 1	three phase
	ANNOT = 2	line to Ground (LG)
	ANNOT = 3	line Line to Ground (LLG)
	ANNOT = 4	line to Line (LL)
Integer TYPE	specifies the fault display quantity (input).	
	TYPE = 1	Ia1
	TYPE = 2	Ia2
	TYPE = 3	Ia0
	TYPE = 4	IA
	TYPE = 5	IB
	TYPE = 6	IC
	TYPE = 7	3Ia0
Integer FLTBUS	specifies the faulted bus (input).	
	FLTBUS = 0	no bus faulted
	FLTBUS > 0	bus to be faulted
Integer BOE	specifies the fault location (input).	
	BOE = 1	fault located at the faulted bus
	BOE = 2	fault located at the faulted bus with lines out-aged
	BOE = 3	fault located at the end of lines
Integer CONTRIB	specifies contributions at home bus for bus faults (input).	
	CONTRIB = 0	contributions for selected fault

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	CONTRIB = 1	contributions at home bus for all faulted buses for selected fault
Integer FRM	specifies the from bus if a line fault is applied (input).	
	FRM = 0	no line fault applied
	FRM > 0	from bus
Integer TO	specifies the to bus if a line fault is applied (input).	
	TO = 0	no line fault applied
	TO > 0	to bus
Integer LAST	specifies the third bus if a line fault is applied to a 3-winding transformer (input).	
	LAST = 0	no line fault applied
	LAST > 0	to bus
Character*2 ID	branch/transformer identifier (input).	
Integer STYLE	line style applied to the faulted bus diagram item (input).	
Integer WIDTH	line width applied to the faulted bus diagram item, >= 1 (input).	
Integer COLOR (3)	array of 3 elements specifying color for diagram items that are faulted(input).	
	COLOR(1)	0->255 Red component
	COLOR(2)	0->255 Blue component
	COLOR(3)	0->255 Green component

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6.42. SETDIAGRESOPF

Use this API to specify the OPF annotation options in the active Slider diagram.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_SETDIAGRESOPF  OPFBUSVOLTMAG OPFBUSPINJ OPFBUSQINJ OPFLOADSENS
 OPFFXSSENS OPFSWSSENS
```

Python command syntax:

```
setdiagresopf(opfbusvoltmag, opfbuspinj, opfbusqinj, opfloadsens,
 opffxssens, opfswssens)
```

Fortran command syntax:

```
CALL SETDIAGRESOPF(OPFBUSVOLTMAG, OPFBUSPINJ, OPFBUSQINJ, OPFLOADSENS,
 OPFFXSSENS, OPFSWSSENS)
```

Where:

Integer OPFBUSVOLTMAG	bus voltage magnitude flag (input).
Integer OPFBUSPINJ	bus P injection flag (input).
Integer OPFBUSQINJ	bus Q injection flag (input).
Integer OPFLOADSENS	load sensitivities flag (input).
Integer OPFFXSSENS	fixed shunt sensitivities flag (input).
Integer OPFSWSSENS	switched shunt sensitivities flag (input).

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6.43. SETDIAGRESPFLOWCHECK

This API routine is obsolete. It has been replaced by the API routine setdiagrespflowcheck_2 .

Use this API to specify the Power Flow voltage and range checking annotation options in the active Slider diagram.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESPFLOWCHECK  LOGARR(1)..LOGARR(4) VLNWDT(1)..VLNWDT(8)
 VLNSTY(1)..VLNSTY(8) VLTVAL(1)..VLTVAL(7) VLNCLRR(1)..VLNCLRR(8)
 VLNCLRG(1)..VLNCLRG(8) VLNCLRB(1)..VLNCLRB(8) LINLOD LLODARR(1)
 LLODARR(2) LLODCLR(1)..LLODCLR(3) BUSVLT(1) BUSVLT(2) BUSVLTWDT(1)
 BUSVLTWDT(2) BUSCLRR(1) BUSCLRR(2) BUSCLRG(1) BUSCLRG(2) BUSCLRB(1)
 BUSCLRB(2) OUTARR(1) OUTARR(2) OUTCLR(1)..OUTCLR(3) BRNRAT BARPERCENT
 BARCLRR(1) BARCLRR(2) BARCLRR(3) BARCLRR(4) BARCLRG(1) BARCLRG(2)
 BARCLRG(3) BARCLRG(4) BARCLRB(1) BARCLRB(2) BARCLRB(3) BARCLRB(4)
```

Python command syntax:

```
setdiagrespflowcheck(logarr, vlnwdt, vlnsty, vltval, vlnclrr, vlnclrg,
 vlnclrb, linlod, llodarr, llodclr, busvlt, busvltwdt, busclrr, busclrg,
 busclrb, outarr, outclr, brnrat, barpercent, barclrr, barclrg, barclrb)
```

Fortran command syntax:

```
CALL SETDIAGRESPFLOWCHECK(LOGARR, VLNWDT, VLNSTY, VLTVAL, VLNCLRR,
 VLNCLRG, VLNCLRB, LINLOD, LLODARR, LLODCLR, BUSVLT, BUSVLTWDT, BUSCLRR,
 BUSCLRG, BUSCLRB, OUTARR, OUTCLR, BRNRAT, BARPERCENT, BARCLRR, BARCLRG,
 BARCLRB)
```

Where:

Integer LOGARR (4)	array of 4 elements specifying flag to determine what annotation is applied. Each element has only two possible values. 0 = FALSE and 1 = TRUE(input).	
	LOGARR(1)	use Voltage level thresholds
	LOGARR(2)	use Line ratings
	LOGARR(3)	use Bus voltage
	LOGARR(4)	use out of service
Integer VLNWDT (8)	array of 8 elements specifying the width of diagram elements that fall in the appropriate voltage range(input).	
	VLNWDT(1)	base voltage < VLTVAL(1)
	VLNWDT(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNWDT(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNWDT(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNWDT(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNWDT(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNWDT(7)	VLTVAL(6) >= base voltage < VLTVAL(7)

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	VLNWDT(8)	base voltage >= VLTVAL(7)
Integer VLNSTY (8)	array of 8 elements specifying the line style of diagram elements that fall in the appropriate voltage range(input).	
	VLNSTY(1)	base voltage < VLTVAL(1)
	VLNSTY(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNSTY(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNSTY(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNSTY(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNSTY(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNSTY(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNSTY(8)	base voltage >= VLTVAL(7)
Real VLTVAL (7)	set the voltage ranges used to assign annotation diagram. Base voltage values should be in ascending order (input).	
Integer VLNCLRR (8)	array of 8 elements specifying the Red color component of diagram elements that fall in the appropriate voltage range, 0->255(input).	
	VLNCLRR(1)	base voltage < VLTVAL(1)
	VLNCLRR(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNCLRR(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNCLRR(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNCLRR(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNCLRR(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNCLRR(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNCLRR(8)	base voltage >= VLTVAL(7)
Integer VLNCLRG (8)	array of 8 elements specifying the Green color component of diagram elements that fall in the appropriate voltage range, 0->255(input).	
	VLNCLRG(1)	base voltage < VLTVAL(1)
	VLNCLRG(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNCLRG(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNCLRG(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNCLRG(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNCLRG(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNCLRG(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNCLRG(8)	base voltage >= VLTVAL(7)
Integer VLNCLRB (8)	array of 8 elements specifying the Blue color component of diagram elements that fall in the appropriate voltage range, 0->255(input).	
	VLNCLRB(1)	base voltage < VLTVAL(1)
	VLNCLRB(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNCLRB(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNCLRB(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNCLRB(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNCLRB(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNCLRB(7)	VLTVAL(6) >= base voltage < VLTVAL(7)

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	VLNCLRB(8)	base voltage >= VLTVAL(7)
Real LINLOD	% loading for line rating (input).	
Integer LLODARR (2)	array of 2 elements specifying the width and style of diagram items that exceed the % of line loading(input).	
	LLODARR(1)	Line width >= 1
	LLODARR(2)	Line style
Integer LLODCLR (3)	array of 3 elements specifying color for lines that exceed the % loading rating, LINLOD(input).	
	LLODCLR(1)	0->255 Red component
	LLODCLR(2)	0->255 Blue component
	LLODCLR(3)	0->255 Green component
Real BUSVLT (2)	array of 2 elements specifying the bus voltage check Min and Max(input).	
	BUSVLT(1)	Min. bus voltage
	BUSVLT(2)	Max. bus voltage
Integer BUSVLTWDT (2)	array of 2 elements specifying the bus voltage check width of diagram items at the Min and Max(input).	
	BUSVLTWDT(1)	Min. bus voltage width >= 1
	BUSVLTWDT(2)	Max. bus voltage width >= 1
Integer BUSCLRR (2)	array of 2 elements specifying the Red color component of diagram elements that fall in the appropriate bus voltage range, 0->255(input).	
	BUSCLRR(1)	base voltage < BUSVLTWDT(1)
	BUSCLRR(2)	BUSVLTWDT(1) >= base voltage < BUSVLTWDT (2)
Integer BUSCLRG (2)	array of 2 elements specifying the Green color component of diagram elements that fall in the appropriate bus voltage range, 0->255(input).	
	BUSCLRG(1)	base voltage < BUSVLTWDT (1)
	BUSCLRG(2)	BUSVLTWDT (1) >= base voltage < BUSVLTWDT (2)
Integer BUSCLRB (2)	array of 2 elements specifying the Blue color component of diagram elements that fall in the appropriate bus voltage range, 0->255(input).	
	BUSCLRB(1)	base voltage < BUSVLTWDT (1)
	BUSCLRB(2)	BUSVLTWDT(1) >= base voltage < BUSVLTWDT (2)
Integer OUTARR (2)	array of 2 elements specifying the width and style of diagram items that are out of service(input).	
	OUTARR(1)	Line width >= 1
	OUTARR(2)	Line style
Integer OUTCLR (3)	array of 3 elements specifying color for diagram items that are out of service(input).	

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	OUTCLR(1)	0->255 Red component
	OUTCLR(2)	0->255 Blue component
	OUTCLR(3)	0->255 Green component
Integer BRNRAT	branch rating set (input).	
	BRNRAT = 0	RateA
	BRNRAT = 1	RateB
	BRNRAT = 2	RateC
Integer BARPERCENT (4)	array of 4 elements specifying percent overloads at which to apply different color schemes for the line loading charts on the diagram(input).	
	BARPERCENT(1)	first overload voltage point
	BARPERCENT(2)	second overload voltage point
	BARPERCENT(3)	third overload voltage point
	BARPERCENT(4)	fourth overload voltage point
Integer BARCLRR (4)	array of 4 elements specifying the Red color component for the line loading charts on the diagram, 0->255(input).	
	BARCLRR(1)	first line loading chart red value
	BARCLRR(2)	second line loading chart red value
	BARCLRR(3)	third line loading chart red value
	BARCLRR(4)	fourth line loading chart red value
Integer BARCLRG (4)	array of 4 elements specifying the Green color component for the line loading charts on the diagram, 0->255(input).	
	BARCLRG(1)	first line loading chart green value
	BARCLRG(2)	second line loading chart green value
	BARCLRG(3)	third line loading chart green value
	BARCLRG(4)	fourth line loading chart green value
Integer BARCLRB (4)	array of 4 elements specifying the Blue color component for the line loading charts on the diagram, 0->255(input).	
	BARCLRB(1)	first line loading chart blue value
	BARCLRB(2)	second line loading chart blue value
	BARCLRB(3)	third line loading chart blue value
	BARCLRB(4)	fourth line loading chart blue value

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6.44. SETDIAGRESPFLOWCHECK_2

Use this API to specify the Power Flow voltage and range checking annotation options in the active Slider Diagram.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SETDIAGRESPFLOWCHECK_2  LOGARR(1) LOGARR(2) LINLOD LLODARR(1)
 LLODARR(2) LLODCLR(1)..LLODCLR(3) BUSVLT(1) BUSVLT(2) BUSVLTWDT(1)
 BUSVLTWDT(2) BUSCLRR(1) BUSCLRR(2) BUSCLRG(1) BUSCLRG(2) BUSCLRB(1)
 BUSCLRB(2) BRNRAT BARPERCENT BARCLRR(1) BARCLRR(2) BARCLRR(3) BARCLRR(4)
 BARCLRG(1) BARCLRG(2) BARCLRG(3) BARCLRG(4) BARCLRB(1) BARCLRB(2)
 BARCLRB(3) BARCLRB(4)
```

Python command syntax:

```
setdiagrespflowcheck_2(logarr, linlod, llodarr, llodclr, busvlt,
 busvltwdt, busclrr, busclrg, busclrb, brnrat, barpercent, barclrr,
 barclrg, barclrb)
```

Fortran command syntax:

```
CALL SETDIAGRESPFLOWCHECK_2(LOGARR, LINLOD, LLODARR, LLODCLR, BUSVLT,
 BUSVLTWDT, BUSCLRR, BUSCLRG, BUSCLRB, BRNRAT, BARPERCENT, BARCLRR,
 BARCLRG, BARCLRB)
```

Where:

Integer LOGARR (2)	array of 2 elements specifying flag to determine what annotation is applied. Each element has only two possible values. 0 = FALSE and 1 = TRUE(input).	
	LOGARR(1)	use Line ratings
	LOGARR(2)	use Bus voltage
Real LINLOD	% loading for line rating (input).	
Integer LLODARR (2)	array of 2 elements specifying the width and style of Diagram items that exceed the % of line loading(input).	
	LLODARR(1)	Line width >= 1
	LLODARR(2)	Line style
Integer LLODCLR (3)	array of 3 elements specifying the color for lines that exceed the % loading rating, LINLOD(input).	
	LLODCLR(1)	0->255 Red component
	LLODCLR(2)	0->255 Blue component
	LLODCLR(3)	0->255 Green component
Real BUSVLT (2)	array of 2 elements specifying the bus voltage check Min and Max(input).	
	BUSVLT(1)	Min. bus voltage
	BUSVLT(2)	Max. bus voltage

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Integer BUSVLTWDT (2) — array of 2 elements specifying the bus voltage check width of diagram items at the Min and Max(input).

BUSVLTWDT(1)	Min. bus voltage width >= 1
BUSVLTWDT(2)	Max. bus voltage width >= 1

Integer BUSCLRR (2) — array of 2 elements specifying the Red color component of Diagram elements that fall in the appropriate bus voltage range, 0->255(input).

BUSCLRR(1)	base voltage < BUSVLTWDT(1)
BUSCLRR(2)	BUSVLTWDT(1) >= base voltage < BUSVLTWDT (2)

Integer BUSCLRG (2) — array of 2 elements specifying the Green color component of Diagram elements that fall in the appropriate bus voltage range, 0->255(input).

BUSCLRG(1)	base voltage < BUSVLTWDT (1)
BUSCLRG(2)	BUSVLTWDT (1) >= base voltage < BUSVLTWDT (2)

Integer BUSCLRB (2) — array of 2 elements specifying the Blue color component of Diagram elements that fall in the appropriate bus voltage range, 0->255(input).

BUSCLRB(1)	base voltage < BUSVLTWDT (1)
BUSCLRB(2)	BUSVLTWDT(1) >= base voltage < BUSVLTWDT (2)

Integer BRNRAT — branch rating set (input).

BRNRAT = 0	RateA
BRNRAT = 1	RateB
BRNRAT = 2	RateC

Integer BARPERCENT (4) — array of 4 elements specifying percent overloads at which to apply different color schemes for the line loading charts on the Diagram(input).

BARPERCENT(1)	first overload voltage point
BARPERCENT(2)	second overload voltage point
BARPERCENT(3)	third overload voltage point
BARPERCENT(4)	fourth overload voltage point

Integer BARCLRR (4) — array of 4 elements specifying the Red color component for the line loading charts on the Diagram, 0->255(input).

BARCLRR(1)	first line loading chart red value
BARCLRR(2)	second line loading chart red value
BARCLRR(3)	third line loading chart red value
BARCLRR(4)	fourth line loading chart red value

Integer BARCLRG (4) — array of 4 elements specifying the Green color component for the line loading charts on the Diagram, 0->255(input).

BARCLRG(1)	first line loading chart green value
BARCLRG(2)	second line loading chart green value

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	BARCLRG(3)	third line loading chart green value
	BARCLRG(4)	fourth line loading chart green value
Integer BARCLRB (4)	array of 4 elements specifying the Blue color component for the line loading charts on the Diagram, 0->255(input).	
	BARCLRB(1)	first line loading chart blue value
	BARCLRB(2)	second line loading chart blue value
	BARCLRB(3)	third line loading chart blue value
	BARCLRB(4)	fourth line loading chart blue value

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6.45. SETDIAGRESPFLOWOPTIONS

This API routine is obsolete. It has been replaced by the API routine setdiagrespflowoptions_2 .

Use this API to specify the basic Power Flow annotation options in the active Slider diagram.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESPFLOWOPTIONS  BRNANN FLWOPT BUSANN BUSVLT EQPANN DIAGTTL
 PMVALINE PMVATRAN USEPGMSET
```

Python command syntax:

```
setdiagrespflowoptions(brnann, flwopt, busann, busvlt, eqpann, diagttl,
 pmvaline, pmvatran, usepgmset)
```

Fortran command syntax:

```
CALL SETDIAGRESPFLOWOPTIONS(BRNANN, FLWOPT, BUSANN, BUSVLT, EQPANN,
 DIAGTTL, PMVALINE, PMVATRAN, USEPGMSET)
```

Where:

Integer BRNANN	branch annotation (input).	
	BRNANN = 0	MW/MVAR flows
	BRNANN = 1	MVA
	BRNANN = 2	MVA/% rating
	BRNANN = 3	Amps/pu current
	BRNANN = 4	MW/%rating
	BRNANN = 5	Amps/% rating
	BRNANN = 6	None
Integer FLWOPT	flow direction (input).	
	FLWOPT = 0	arrows
	FLWOPT = 1	signs
Integer BUSANN	bus ID annotation (input).	
	BUSANN = 0	number, no name
	BUSANN = 1	name, no number
	BUSANN = 2	number and name
	BUSANN = 3	no number, no name
Integer BUSVLT	bus voltage annotation (input).	
	BUSVLT = 0	magnitude (pu)
	BUSVLT = 1	magnitude (kV)
	BUSVLT = 2	angle (degrees)
	BUSVLT = 3	magnitude (pu) and angle (degrees)
	BUSVLT = 4	magnitude (pu) and magnitude (kV)
	BUSVLT = 5	magnitude (kV) and angle (degrees)

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	BUSVLT = 7	magnitude (pu), magnitude (kV) and angle (degrees)
	BUSVLT = 8	none
Integer EQPANN	equipment annotation (input).	
	EQPANN = 0	MW/MVAR
	EQPANN = 1	MVA
	EQPANN = 2	none
Character*260 DIAGTTL	diagram title (input).	
Integer PMVALINE	loading basis for % loading on non-transformer branches (input).	
	PMVALINE = 0	current loading
	PMVALINE = 1	MVA loading
Integer PMVATRAN	loading basis for % loading on transformer branches (input).	
	PMVATRAN = 0	current loading
	PMVATRAN = 1	MVA loading
Integer USEPGMSET	flag for system or diagram settings for displaying MVATrans and MVA-Line values (input).	
	USEPGMSET = 0	use diagram settings
	USEPGMSET = 1	use System settings

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6.46. SETDIAGRESPFLOWOPTIONS_2

Use this API to specify the basic Power Flow annotation options in the active Slider diagram.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SETDIAGRESPFLOWOPTIONS_2  BRNANN FLWOPT BUSVLT PMVALINE PMVATRAN
 USEPGMSET DIAGTTL
```

Python command syntax:

```
setdiagrespflowoptions_2(brnann, flwopt, busvlt, pmvaline, pmvatran,
 usepgmset, diagttl)
```

Fortran command syntax:

```
CALL SETDIAGRESPFLOWOPTIONS_2(BRNANN, FLWOPT, BUSVLT, PMVALINE, PMVATRAN,
 USEPGMSET, DIAGTTL)
```

Where:

Integer BRNANN	branch annotation (input).	
	BRNANN = 0	MW/MVAR flows
	BRNANN = 1	MVA
	BRNANN = 2	MVA/% rating
	BRNANN = 3	Amps/pu current
	BRNANN = 4	MW/%rating
	BRNANN = 5	Amps/% rating
	BRNANN = 6	None
Integer FLWOPT	flow direction (input).	
	FLWOPT = 0	arrows
	FLWOPT = 1	signs
Integer BUSVLT	bus voltage annotation (input).	
	BUSVLT = 0	magnitude (pu)
	BUSVLT = 1	magnitude (kV)
	BUSVLT = 2	angle (degrees)
	BUSVLT = 3	magnitude (pu) and angle (degrees)
	BUSVLT = 4	magnitude (pu) and magnitude (kV)
	BUSVLT = 5	magnitude (kV) and angle (degrees)
	BUSVLT = 7	magnitude (pu), magnitude (kV) and angle (degrees)
	BUSVLT = 8	none
Integer PMVALINE	loading basis for % loading on non-transformer branches (input).	
	PMVALINE = 0	current loading
	PMVALINE = 1	MVA loading
Integer PMVATRAN	loading basis for % loading on transformer branches (input).	

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	PMVATRAN = 0	current loading
	PMVATRAN = 1	MVA loading
Integer USEPGMSET	flag for system or diagram settings for displaying MVATrans and MVA-Line values (input).	
	USEPGMSET = 0	use diagram settings
	USEPGMSET = 1	use System settings
Character*260 DIAGTTL	diagram title (input).	

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6.47. SETDIAGRESREL

This API routine is obsolete. It has been replaced by the API routine setdiagresrel_2 .

Use this API to specify the reliability annotation options in the active Slider diagram.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SETDIAGRESREL  OUTMODE MODE BUSMODE CNTBUS CNTBRN DETBUS DETBRN PRBBUS
 PRBBRN VOLTREC CNTLABEL CNTLIST
```

Python command syntax:

```
setdiagresrel(outmode, mode, busmode, cntbus, cntbrn, detbus, detbrn,
 prbbus, prbbrn, voltrec, cntlabel, cntlist)
```

Fortran command syntax:

```
CALL SETDIAGRESREL(OUTMODE, MODE, BUSMODE, CNTBUS, CNTBRN, DETBUS, DETBRN,
 PRBBUS, PRBBRN, VOLTREC, CNTLABEL, CNTLIST)
```

Where:

Integer OUTMODE	mode of results displayed (input).	
	OUTMODE = 1	post contingency stage
	OUTMODE = 2	post tripping stage
	OUTMODE = 3	post corrective actions stage
Integer MODE	type of reliability results displayed (input).	
	MODE = 0	contingency analysis results
	MODE = 1	deterministic reliability results
	MODE = 2	probabilistic reliability results
Integer BUSMODE	mode of bus reporting (input).	
	BUSMODE = 0	bus voltage violations
	BUSMODE = 1	bus load curtailments
Integer CNTBUS	If MODE = 0, is the option to specify whether Bus voltage is displayed (input).	
	CNTBUS = 0	don't display Bus voltage
	CNTBUS = 1	display Bus voltage
Integer CNTBRN	If MODE = 0, is the option to specify whether Branch flow loading is displayed (input).	
	CNTBRN = 0	don't display Branch flow loading
	CNTBRN = 1	display MVA loading
	CNTBRN = 2	display Percent (%)
	CNTBRN = 3	display both MVA and Percent (%)
Integer DETBUS	If MODE = 1, is the option to specify which bus voltage violations are displayed. Non-zero values may be summed (input).	

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	DETBUS = 0	don't display bus voltage violations
	DETBUS = 1	# of contingencies resulting in violations
	DETBUS = 2	contingency with maximum violation
	DETBUS = 4	average violations (AV)
	DETBUS = 8	maximum violation (MV)
Integer DETBRN	If MODE = 1, is the option to specify which branch flow overloading violations are displayed (input).	
	DETBRN = 0	don't display bus voltage violations
	DETBRN = 1	# of contingencies resulting in violations
	DETBRN = 2	Contingency with maximum loading
	DETBRN = 4	available capacity (AC)
	DETBRN = 6	maximum loading % (ML)
	DETBRN = 8	average violations (AV)
Integer PRBBUS	If MODE = 2, is the option to specify which bus voltage violations are displayed (input).	
	PRBBUS = 0	don't display bus voltage violations
	PRBBUS = 1	frequency (OC/Y)
	PRBBUS = 2	impact (VI)
	PRBBUS = 4	probability (H/Y)
	PRBBUS = 8	expected energy curtailed
Integer PRBBRN	If MODE = 2, is the option to specify which branch flow overloading violations are displayed (input).	
	PRBBRN = 0	don't display bus voltage violations
	PRBBRN = 1	frequency (OC/Y)s
	PRBBRN = 2	impact (FI)
	PRBBRN = 4	probability (H/Y)
Integer VOLTREC	index into the voltage limit record array for the current DFAX file (input).	
Character*32 CNTLABEL	contingency to display results for (input).	
Integer CNTLIST	what contingencies are listed for selection (input).	
	CNTLIST = 1	all contingencies are listed
	CNTLIST = 2	only those contingencies that cause overloading violations are listed
	CNTLIST = 3	only those contingencies that cause violations are listed

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6.48. SETDIAGRESREL_2

Use this API to specify the reliability annotation options in the active Slider diagram.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_SETDIAGRESREL_2  OUTMODE RELMODE BUSMODE BUSDETANN BRNDETANN BUSPRBANN
 BRNPRBANN VOLTREC
```

Python command syntax:

```
setdiagresrel_2(outmode, relmode, busmode, busdetann, brndetann,
 busprbann, brnprbann, voltrec)
```

Fortran command syntax:

```
CALL SETDIAGRESREL_2(OUTMODE, RELMODE, BUSMODE, BUSDETANN, BRNDETANN,
 BUSPRBANN, BRNPRBANN, VOLTREC)
```

Where:

Integer OUTMODE	mode of results displayed (input).	
	OUTMODE = 1	post contingency stage
	OUTMODE = 2	post tripping stage
	OUTMODE = 3	post corrective actions stage
Integer RELMODE	type of reliability results displayed (input).	
	RELMODE = 0	deterministic reliability results
	RELMODE = 1	probabilistic reliability results
Integer BUSMODE	mode of bus reporting (input).	
	BUSMODE = 0	bus voltage violations
	BUSMODE = 1	bus load curtailments
Integer BUSDETANN	If MODE = 1, is the option to specify which bus voltage violations are displayed (input).	
	BUSDETANN = 0	don't display Bus voltage violations
	BUSDETANN = BUSDETANN + 1	# of contingencies resulting in violations
	BUSDETANN = BUSDETANN + 2	contingency with maximum violations
	BUSDETANN = BUSDETANN + 4	average violations (AV)
	BUSDETANN = BUSDETANN + 8	maximum violations (MV)
Integer BRNDETANN	If MODE = 1, is the option to specify which branch flow overloading violations are displayed (input).	
	BRNDETANN = 0	don't display branch flow violations
	BRNDETANN = BRNDETANN + 1	# of contingencies resulting in violations

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	BRNDETANN = BRNDETANN + 2 contingency with maximum loading
	BRNDETANN = BRNDETANN + 4 available capacity (AC)
	BRNDETANN = BRNDETANN + 8 average violations (AV)
	BRNDETANN = BRNDETANN + 16 maximum loading % (ML)
Integer BUSPRBANN	If MODE = 2, is the option to specify which bus voltage violations are displayed (input).
	BUSPRBANN = 0 don't display Bus voltage violations
	BUSPRBANN = BUSPRBANN + 1 frequency (OC/Y)
	BUSPRBANN = BUSPRBANN + 2 impact (VI)
	BUSPRBANN = BUSPRBANN + 4 probability (H/Y)
	BUSPRBANN = BUSPRBANN + 8 expected energy curtailed
Integer BRNPRBANN	If MODE = 2, is the option to specify which branch flow overloading violations are displayed (input).
	BRNPRBANN = 0 don't display branch flow violations
	BRNPRBANN = BRNPRBANN + 1 Ffrequency (OC/Y)
	BRNPRBANN = BRNPRBANN + 2 impact (VI)
	BRNPRBANN = BRNPRBANN + 4 probability (H/Y)
Integer VOLTREC	index into the voltage limit record array for the current DFAX file (input).

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6.49. SETDIAGRESSCGR

Use this API to specify the short circuit analysis voltage and range checking annotation options in the active Slider diagram.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESSCGR  USEVLT VLNWDT(1)..VLNWDT(8) VLNSTY(1)..VLNSTY(8)
 VLTVAL(1)..VLTVAL(7) VLNCLRR(1)..VLNCLRR(8) VLNCLRG(1)..VLNCLRG(8)
 VLNCLRB(1)..VLNCLRB(8) OUTWDT OUTSTY OUTCLR(1)..OUTCLR(3) FLTWDT FLTSTY
 FLTCLR(1)..FLTCLR(3) BUSANN
```

Python command syntax:

```
setdiagresscgr(usevlt, vlnwdt, vlnsty, vltval, vlnclrr, vlnclrg, vlnclrb,
 outwdt, outsty, outclr, fltwdt, fltsty, fltclr, busann)
```

Fortran command syntax:

```
CALL SETDIAGRESSCGR(USEVLT, VLNWDT, VLNSTY, VLTVAL, VLNCLRR, VLNCLRG,
 VLNCLRB, OUTWDT, OUTSTY, OUTCLR, FLTWDT, FLTSTY, FLTCLR, BUSANN)
```

Where:

Integer USEVLT	flag to apply voltage level annotation to the diagram elements (input).	
	USEVLT = 0	don't apply voltage level annotation
	USEVLT = 1	apply voltage level annotation
Integer VLNWDT (8)	array of 8 elements specifying width of diagram elements that fall in the appropriate voltage range(input).	
	VLNWDT(1)	base voltage < VLTVAL(1)
	VLNWDT(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNWDT(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNWDT(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNWDT(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNWDT(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNWDT(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNWDT(8)	base voltage >= VLTVAL(7)
Integer VLNSTY (8)	array of 8 elements specifying line style of diagram elements that fall in the appropriate voltage range(input).	
	VLNSTY(1)	base voltage < VLTVAL(1)
	VLNSTY(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNSTY(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNSTY(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNSTY(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNSTY(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNSTY(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNSTY(8)	base voltage >= VLTVAL(7)

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Real VLTVAL (7)
voltage ranges used to assign annotation diagram. Base voltage values should be in ascending order (input).

Integer VLNCLRR (8)
array of 8 elements specifying set the Red color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRR(1)	base voltage < VLTVAL(1)
VLNCLRR(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRR(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRR(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRR(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRR(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRR(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRR(8)	base voltage >= VLTVAL(7)

Integer VLNCLRG (8)
array of 8 elements specifying set Green color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRG(1)	base voltage < VLTVAL(1)
VLNCLRG(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRG(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRG(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRG(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRG(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRG(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRG(8)	base voltage >= VLTVAL(7)

Integer VLNCLRB (8)
array of 8 elements specifying set Blue color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRB(1)	base voltage < VLTVAL(1)
VLNCLRB(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRB(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRB(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRB(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRB(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRB(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRB(8)	base voltage >= VLTVAL(7)

Integer OUTWDT
line width applied to out of service diagram items, >= 1 (input).

Integer OUTSTY
line style applied to out of service diagram items (input).

Integer OUTCLR (3)
array of 3 elements specifying color for diagram items that are out of service(input).

OUTCLR(1)	0->255 Red component
OUTCLR(2)	0->255 Blue component
OUTCLR(3)	0->255 Green component

Integer FLTWDT
line width applied to the faulted bus diagram item, >= 1 (input).

Integer FLTSTY
line style applied to the faulted bus diagram item (input).

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Integer FLTCLR (3)	array of 3 elements specifying color for Diagram items that are out of service(input).
	FLTCLR(1) 0->255 Red component
	FLTCLR(2) 0->255 Blue component
	FLTCLR(3) 0->255 Green component
Integer BUSANN	bus ID annotation (input).
	BUSANN = 0 number, no name
	BUSANN = 1 name, no number
	BUSANN = 2 number and name
	BUSANN = 3 no number, no name

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6.50. SETDIAGRESTYPEACC

Use this API to set the type of results displayed in the active Slider diagram to ACCC results. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 33.1.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEACC
```

Python command syntax:

```
setdiagrestypeacc()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEACC
```

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6.51. SETDIAGRESTYPEASCC

Use this API to set the type of results displayed in the active Slider diagram to ASCC fault calculation results. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEASCC
```

Python command syntax:

```
setdiagrestypeascc()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEASCC
```

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6.52. SETDIAGRESTYPEDATA

Use this API to set the type of results displayed in the active Slider diagram to impedance data values. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEDATA
```

Python command syntax:

```
setdiagrestypedata()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEDATA
```

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6.53. SETDIAGRESTYPEDYN

Use this API to set the type of results displayed in the active Slider diagram to Dynamics data results. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEDYN
```

Python command syntax:

```
setdiagrestypedyn()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEDYN
```

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6.54. SETDIAGRESTYPEGDIF

Use this API to set the type of results displayed in the active Slider diagram to the differences in values between the current network case and a supplied comparison case.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEGDIF  NAMEARG SFILE
```

Python command syntax:

```
setdiagrestypegdif(namearg, sfile)
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEGDIF(NAMEARG, SFILE)
```

Where:

Integer NAMEARG	flag indicating how to match up buses in the two cases (input). NameArg = 0 by bus number only NameArg = 1 by extended bus name only NameArg = 2 by both number and name.
Character*260 SFILE	name of an existing Saved case to compare the current case against (input).

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6.55. SETDIAGRESTYPEGIC

Use this API to set the type of results displayed in the active Slider diagram to GIC solution results. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 32.2.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEGIC
```

Python command syntax:

```
setdiagrestypegic()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEGIC
```

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6.56. SETDIAGRESTYPEIEC

Use this API to set the type of results displayed in the active Slider diagram to IEC Fault calculation results. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEIEC
```

Python command syntax:

```
setdiagrestypeiec()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEIEC
```

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6.57. SETDIAGRESTYPEOPF

Use this API to set the type of results displayed in the active Slider diagram to OPF solution results. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEOPF
```

Python command syntax:

```
setdiagrestypeopf()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEOPF
```

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6.58. SETDIAGRESTYPEPFLOW

Use this API to set the type of results displayed in the active Slider diagram to Power Flow results. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEPFLOW
```

Python command syntax:

```
setdiagrestypepflow()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEPFLOW
```

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6.59. SETDIAGRESTYPEREL

Use this API to set the type of results displayed in the active Slider diagram to reliability analysis results. Requires that current network elements are being displayed in the diagram view.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPEREL
```

Python command syntax:

```
setdiagrestyperel()
```

Fortran command syntax:

```
CALL SETDIAGRESTYPEREL
```

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6.60. SETDIAGRESTYPESCGR

Use this API to set the type of results displayed in the active Slider diagram to the results of a short circuit analysis.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETDIAGRESTYPESCGR  OPTIONS(1) .. OPTIONS(8)
```

Python command syntax:

```
setdiagrestypescgr(options)
```

Fortran command syntax:

```
CALL SETDIAGRESTYPESCGR(OPTIONS)
```

Where:

Integer OPTIONS (8)	array of 8 elements specifying analysis options(input).	
	OPTIONS(1)	faults
	OPTIONS(1) = 0	3 phase
	OPTIONS(1) = 1	3 phase and L-G
	OPTIONS(2)	L-G fault
	OPTIONS(2) = 0	A phase
	OPTIONS(2) = 1	zero sequence
	OPTIONS(3)	fault location
	OPTIONS(3) = 0	all
	OPTIONS(3) = 0	faulted bus number
	OPTIONS(4)	flat conditions
	OPTIONS(4) = 0	no
	OPTIONS(4) = 1	yes
	OPTIONS(5)	double precision
	OPTIONS(5) = 0	no
	OPTIONS(5) = 1	yes (ignored)
	OPTIONS(6)	dc line/FACTS device option
	OPTIONS(6) = 0	block
	OPTIONS(6) = 1	model as load
	OPTIONS(7)	zero sequence transformer impedance correction option
	OPTIONS(7) = 0	ignore
	OPTIONS(7) = 1	apply
	OPTIONS(8)	branch quantity option
	OPTIONS(8) = 0	currents
	OPTIONS(8) = 1	apparent impedances

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6.61. SETDIAGRESVRCS

Use this API to specify the system diagram defaults for Voltage coloring and style, Out of Service coloring and style etc.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SETDIAGRESVRCS  BUSANN EQPANN USEVCLR VLNWDT(1)..VLNWDT(8)
 VLNSTY(1)..VLNSTY(8) VLTVAL(1)..VLTVAL(7) VLNCLRR(1)..VLNCLRR(8)
 VLNCLRG(1)..VLNCLRG(8) VLNCLRB(1)..VLNCLRB(8) USEOOSCLR OOSWDT OOSSTY
 OUTCLR(1)..OUTCLR(3) BNDCLR(1)..BNDCLR(3) UNBNDCLR(1)..UNBNDCLR(3)
 TRANBARCHRT
```

Python command syntax:

```
setdiagresvrcs(busann, eqpann, usevclr, vlnwdt, vlnsty, vltval, vlnclrr,
 vlnclrg, vlnclrb, useoosclr, ooswdt, oossty, outclr, bndclr, unbndclr,
 tranbarchrt)
```

Fortran command syntax:

```
CALL SETDIAGRESVRCS(BUSANN, EQPANN, USEVCLR, VLNWDT, VLNSTY, VLTVAL,
 VLNCLRR, VLNCLRG, VLNCLRB, USEOOSCLR, OOSWDT, OOSSTY, OUTCLR, BNDCLR,
 UNBNDCLR, TRANBARCHRT)
```

Where:

Integer BUSANN	bus ID annotation (input).	
	BUSANN = 0	number, no name
	BUSANN = 1	name, no number
	BUSANN = 2	number and name
	BUSANN = 3	no number, no name
Integer EQPANN	equipment annotation (input).	
	EQPANN = 0	MW/MVAR
	EQPANN = 1	MVA
	EQPANN = 2	none
Integer USEVCLR	use voltage level coloring . The element has only two possible values. 0 = FALSE and 1 = TRUE (input).	
Integer VLNWDT (8)	array of 8 elements specifying width of diagram elements that fall in the appropriate voltage range(input).	
	VLNWDT(1)	base voltage < VLTVAL(1)
	VLNWDT(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
	VLNWDT(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
	VLNWDT(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
	VLNWDT(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
	VLNWDT(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
	VLNWDT(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
	VLNWDT(8)	base voltage >= VLTVAL(7)

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Integer VLNSTY (8) — array of 8 elements specifying line style of diagram elements that fall in the appropriate voltage range(input).

VLNSTY(1)	base voltage < VLTVAL(1)
VLNSTY(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNSTY(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNSTY(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNSTY(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNSTY(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNSTY(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNSTY(8)	base voltage >= VLTVAL(7)

Real VLTVAL (7) — voltage ranges used to assign annotation diagram. Base voltage values should be in ascending order (input).

Integer VLNCLRR (8) — array of 8 elements specifying the Red color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRR(1)	base voltage < VLTVAL(1)
VLNCLRR(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRR(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRR(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRR(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRR(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRR(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRR(8)	base voltage >= VLTVAL(7)

Integer VLNCLRG (8) — array of 8 elements specifying the Green color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRG(1)	base voltage < VLTVAL(1)
VLNCLRG(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRG(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRG(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRG(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRG(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRG(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRG(8)	base voltage >= VLTVAL(7)

Integer VLNCLRB (8) — array of 8 elements specifying the Blue color component of diagram elements that fall in the appropriate voltage range, 0->255(input).

VLNCLRB(1)	base voltage < VLTVAL(1)
VLNCLRB(2)	VLTVAL(1) >= base voltage < VLTVAL(2)
VLNCLRB(3)	VLTVAL(2) >= base voltage < VLTVAL(3)
VLNCLRB(4)	VLTVAL(3) >= base voltage < VLTVAL(4)
VLNCLRB(5)	VLTVAL(4) >= base voltage < VLTVAL(5)
VLNCLRB(6)	VLTVAL(5) >= base voltage < VLTVAL(6)
VLNCLRB(7)	VLTVAL(6) >= base voltage < VLTVAL(7)
VLNCLRB(8)	base voltage >= VLTVAL(7)

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Integer USEOOSCLR	Out-of-Service coloring. The element has only two possible values. 0 = FALSE and 1 = TRUE (input).
Integer OOSWDT	width of diagram items that are out of service. OOSWDT Line width >= 1 (input).
Integer OOSSTY	line style of diagram items that are out of service. OOSSTY Line style (input).
Integer OUTCLR (3)	array of 3 elements specifying color for diagram items that are out of service(input). OUTCLR(1) 0->255 Red component OUTCLR(2) 0->255 Blue component OUTCLR(3) 0->255 Green component
Integer BNDCLR (3)	array of 3 elements specifying bound item color(input). BNDCLR(1) 0->255 Red component BNDCLR(2) 0->255 Blue component BNDCLR(3) 0->255 Green component
Integer UNBNDCLR (3)	array of 3 elements specifying unbound item color(input). UNBNDCLR(1) 0->255 Red component UNBNDCLR(2) 0->255 Blue component UNBNDCLR(3) 0->255 Green component
Integer TRANBARCHRT	two-winding transformer bar chart display side. TRANBARCHRT = 0 High current loading side (input). TRANBARCHRT = 1 low current loading side TRANBARCHRT = 2 both sides

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6.62. SWITCHDIAGFILEDATASET

This API is used to rotate through values for the current element in the active data set on the active Slider diagram. Currently, this API is only used with ACCC results and is used for displaying the post contingency, post tripping and post corrective actions values.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SWITCHDIAGFILEDATASET
```

Python command syntax:

```
switchdiagfileset()
```

Fortran command syntax:

```
CALL SWITCHDIAGFILEDATASET
```

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6.63. UPDATEBUSLOCDIAGFILE

Use this API to update bus locations in the active Slider diagram in the GUI using the positions found in the current bus location file.

This API routine was first introduced in release 30.2.1.

Batch command syntax:

```
BAT_UPDATEBUSLOCDIAGFILE
```

Python command syntax:

```
updatebuslocdiagfile()
```

Fortran command syntax:

```
CALL UPDATEBUSLOCDIAGFILE
```

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Chapter 7
Single Element Data

This chapter defines API routines that are used to retrieve data for a single element (e.g., bus, branch, area, etc.) from the PSS®E engine and test on various program states.

These routines do not record themselves when called with recording enabled.

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7.1. ARDAT

Use this API to return area totals.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval  = ardat(iar, string)
```

Fortran command syntax:

```
CALL ARDAT(IAR, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL ARDAT(IAR, STRING, P, Q, IERR)
```

Where:

Integer IAR	Area number (input).	
Character STRING	String indicating the area total desired (input).	
	'LOAD'	Total owner load by bus owner assignment (net of load plus in-service distributed generation on load feeder)
	'LOADLD'	Total owner load by load owner assignment
	'LDGN'	Total distributed generation on load feeder by bus owner assignment
	'LDGNLD'	Total distributed generation on load feeder by load owner assignment
	'GEN'	Total owner generation
	'LOSS'	Total owner losses
	'INT'	Net area interchange
	'INDMAC'	Total owner induction machine powers by bus owner assignment
	'INDMACMC'	Total owner induction machine powers by machine owner assignment
	'INDGEN'	Total owner induction generator powers by bus owner assignment
	'INDGENMC'	Total owner induction generator powers by machine owner assignment
	'INDMOT'	Total owner induction motor powers by bus owner assignment
	'INDMOTMC'	Total owner induction motor powers by machine owner assignment
Real P	Real component of the desired complex power	
Real Q	Reactive component of the desired complex power	
Complex CMPVAL	Desired complex power (output).	
Integer IERR	error code (output).	

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IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
IERR = 1	Area number < 0 or > largest allowable area number; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 2	No in-service buses with in-service loads (for 'LOAD'), no in-service loads (for 'LOADLD'), no type 2 or type 3 buses (for 'GEN'), no branches (for 'LOSS'), or no ties (for 'INT') in area; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 3	Area not found; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 4	Bad 'STRING' value; 'P' and 'Q' or 'CMPVAL' unchanged

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7.2. AREDAT

Use this API to return real area data items.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval  = aredat(iar, string)
```

Fortran command syntax:

```
CALL AREDAT(IAR, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL AREDAT(IAR, STRING, RVAL, IERR)
```

Where:

Integer IAR	Area number (input).	
Character STRING	String indicating the area quantity desired (input).	
	'PDES'	Desired net interchange leaving area in MW
	'PTOL'	Interchange tolerance bandwidth in MW
	'O_PDES'	Same as 'PDES' in units determined by the power output option setting
	'O_PTOL'	Same as 'PTOL' in units determined by the power output option setting
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Area number <= 0 or > largest allowable area number; 'RVAL' unchanged
	IERR = 2	Area not found; 'RVAL' unchanged
	IERR = 3	Bad 'STRING' value; 'RVAL' unchanged

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7.3. AREINT

Use this API to return integer area data items.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival  = areint(iar, string)
```

Fortran command syntax:

```
CALL AREINT(IAR, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL AREINT(IAR, STRING, IVAL, IERR)
```

Where:

Integer IAR	Area number (input).	
Character STRING	String indicating the area quantity desired (input).	
	'SWING'	Area slack bus number
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Area number <= 0 or > largest allowable area number; 'IVAL' unchanged
	IERR = 2	Area not found; 'IVAL' unchanged
	IERR = 3	Bad 'STRING' value; 'IVAL' unchanged

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7.4. ARENAM

Use this API to return the area name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval  = arenam(iar)
```

Fortran command syntax:

```
CALL ARENAM(IAR, CVAL, IERR)
```

IPLAN command syntax:

```
CALL ARENAM(IAR, CVAL, IERR)
```

Where:

Integer IAR	Area number (input).	
Character*12 CVAL	Area name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Area number <= 0 or > largest allowable area number; 'CVAL' unchanged
	IERR = 2	Area not found; 'CVAL' unchanged

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7.5. ARENUM

Use this API to return the area number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, iar  = arenum(cval)
```

Fortran command syntax:

```
CALL ARENUM(IAR, CVAL, IERR)
```

IPLAN command syntax:

```
CALL ARENUM(IAR, CVAL, IERR)
```

Where:

Integer IAR	Area number (output).	
Character CVAL	Area name (input).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IAR' returned
	IERR = 1	Area with name 'CVAL' not found; 'IAR' unchanged
	IERR > 1	Multiple areas with the name 'CVAL' found, 'IAR' returned as lowest number of these

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7.6. AREUSE

Use this API to indicate if an area is in use.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ival = areuse(iar)
```

Fortran command syntax:

```
CALL AREUSE(IAR, IVAL)
```

IPLAN command syntax:

```
CALL AREUSE(IAR, IVAL)
```

Where:

Integer IAR	Area number (input).	
Integer IVAL	Area status (output).	
	IVAL = -1	Area number <= 0 or > largest allowable area number.
	IVAL = 0	If no equipment is assigned to area IAR.
	IVAL = 1	If any equipment is assigned to area IAR.

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7.7. ARITOJ

Use this API to return the interchange between two areas.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval  = aritoj(iar, jar)
```

Fortran command syntax:

```
CALL ARITOJ(IAR, JAR, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL ARITOJ(IAR, JAR, P, Q, IERR)
```

Where:

Integer IAR	From area number (input).	
Integer JAR	To area number (input).	
Real P	Real component of the complex interchange	
Real Q	Reactive component of the complex interchange	
Complex CMPVAL	Complex interchange (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Area number <= 0 or > largest allowable area number; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	No in-service branches between areas; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Area not found; 'P' and 'Q' or 'CMPVAL' unchanged

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7.8. BRNCUR

This API routine is obsolete. It has been replaced by the API routine BRNMSC .

Use this API to return branch flow values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = brncur(ibus,jbus,ickt)
```

Fortran command syntax:

```
CALL BRNCUR(IBUS, JBUS, ICKT, RVAL, IERR)
```

IPLAN command syntax:

```
CALL BRNCUR(IBUS, JBUS, ICKT, RVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*2 ICKT	Circuit or multisection line identifier (input).	
Real RVAL	Flow value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Branch not found; 'RVAL' unchanged
	IERR = 3	Branch out-of-service; 'RVAL' returned set to zero
	IERR = 4	No base kV at IBUS; 'RVAL' returned = per unit current

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7.9. BRNDAT

Use this API to return real branch parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = brndat(ibus,jbus,ickt,string)
```

Fortran command syntax:

```
CALL BRNDAT(IBUS, JBUS, ICKT, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL BRNDAT(IBUS, JBUS, ICKT, STRING, RVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*12 ICKT	Circuit identifier (input).	
Character STRING	String indicating the branch quantity desired (input).	
	'RATEn'	Rating from rating set "n" (n = 1 through 12)
	'RATEA'	First rating
	'RATEB'	Second rating
	'RATEC'	Third rating
	'RATE'	Default rating
	'LENGTH'	Line length
	'CHARG'	Total branch charging
	'CHARGZ'	Zero sequence charging
	'FRACT1'	Fraction of total ownership assigned to the first owner
	'FRACT2'	Fraction of total ownership assigned to the second owner
	'FRACT3'	Fraction of total ownership assigned to the third owner
	'FRACT4'	Fraction of total ownership assigned to the fourth owner
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Branch not found; 'RVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 4	'BRNDAT' invalid with multisection line identifier; 'RVAL' unchanged

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IERR = 5	Sequence data not in case; 'RVAL' unchanged
IERR = 6	For STRING = 'FRACTn', n > number of owners; 'RVAL' unchanged

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7.10. BRNDT2

Use this API to return complex positive and zero sequence parameters for non-transformer branches, and complex positive sequence parameters for transformer branches.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = brndt2(ibus,jbus,ickt,string)
```

Fortran command syntax:

```
CALL BRNDT2(IBUS, JBUS, ICKT, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL BRNDT2(IBUS, JBUS, ICKT, STRING, P, Q, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*12 ICKT	Circuit identifier (input).	
Character STRING	String indicating the branch quantity desired (input).	
	'RX'	Branch resistance and reactance in per unit
	'ISHNT'	Admittance of line shunt at IBUS end of the branch
	'JSHNT'	Admittance of line shunt at JBUS end of the branch
	'RXZ'	Zero sequence branch resistance and reactance
	'ISHNTZ'	Zero sequence admittance of line shunt at IBUS end of the branch
	'JSHNTZ'	Zero sequence admittance of line shunt at JBUS end of the branch
	'LOSSES'	Branch losses in MW and Mvar
	'O_LOSSES'	Branch losses in units determined by the power output option setting
Real P	Real component of the complex value indicated by STRING	
Real Q	Reactive component of the complex value indicated by STRING	
Complex CMPVAL	Complex value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Branch not found; 'P' and 'Q' or 'CMPVAL' unchanged

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IERR = 3	Invalid value of 'STRING'; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 4	'BRNDT2' invalid with multisection line identifier; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 5	Sequence data not in case; 'P' and 'Q' or 'CMPVAL' unchanged

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7.11. BRNFLO

Use this API to return the complex branch flow (P+jQ) as calculated at IBUS.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = brnflo(ibus,jbus,ickt)
```

Fortran command syntax:

```
CALL BRNFLO(IBUS, JBUS, ICKT, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL BRNFLO(IBUS, JBUS, ICKT, P, Q, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*2 ICKT	Circuit or multisection line identifier (input).	
Real P	Real component of the complex branch flow	
Real Q	Reactive component of the complex branch flow	
Complex CMPVAL	Complex branch flow (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Branch not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Branch is out-of-service; 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned

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7.12. BRNINT

Use this API to return integer branch parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = brnint(ibus,jbus,ickt,string)
```

Fortran command syntax:

```
CALL BRNINT(IBUS, JBUS, ICKT, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL BRNINT(IBUS, JBUS, ICKT, STRING, IVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*12 ICKT	Circuit or multisection line identifier (input).	
Character STRING	String indicating the branch quantity desired (input).	
	'STATUS'	Branch status
	'METER'	Bus number of metered end
	'NMETR'	Bus number of non-metered end
	'OWNERS'	Number of owners for this branch
	'OWN1'	Owner number of the first owner
	'OWN2'	Owner number of the second owner
	'OWN3'	Owner number of the third owner
	'OWN4'	Owner number of the fourth owner
	'STATION_I'	Bus IBUS substation number (returned as 0 if bus IBUS is not in a substation)
	'STATION_J'	Bus JBUS substation number (returned as 0 if bus JBUS is not in a substation)
	'SECTION_I'	Bus IBUS bus section number (returned as 0 if bus IBUS is not in a substation)
	'SECTION_J'	Bus JBUS bus section number (returned as 0 if bus JBUS is not in a substation)
	'NODE_I'	Number of the node to which the branch is connected at the bus IBUS end (returned as 0 if bus IBUS is not in a substation)
	'NODE_J'	Number of the node to which the branch is connected at the bus JBUS end (returned as 0 if bus JBUS is not in a substation)
	'SCTYPE'	MOV protection mode
Integer IVAL	Value indicated by STRING (output).	

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Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Branch not found; 'IVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	For 'STRING' = 'OWNn', n > number of owners; 'IVAL' unchanged

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7.13. BRNMSC

Use this API to return real branch flow values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = brnmsc(ibus,jbus,ickt,string)
```

Fortran command syntax:

```
CALL BRNMSC(IBUS, JBUS, ICKT, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL BRNMSC(IBUS, JBUS, ICKT, STRING, RVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character ICKT	Circuit or multisection line identifier (input).	
Character STRING	String indicating the branch quantity desired (input).	
	'MVA'	Branch MVA flow
	'O_MVA'	Same as 'MVA' in units determined by the power output option setting
	'AMPS'	Branch current in amps
	'PUCUR'	Branch current in pu
	'CURANG'	Angle of branch current in degrees
	'PCTRTA'	Percent current loading of RATEA
	'PCTRTB'	Percent current loading of RATEB
	'PCTRTC'	Percent current loading of RATEC
	'PCTMVA'	Percent MVA loading of RATEA
	'PCTMVB'	Percent MVA loading of RATEB
	'PCTMVC'	Percent MVA loading of RATEC
	'PCTCPA'	Percent current or MVA loading (according to the appropriate percent loading units program option setting) of RATEA
	'PCTCPB'	Percent current or MVA loading (according to the appropriate percent loading units program option setting) of RATEB
	'PCTCPC'	Percent current or MVA loading (according to the appropriate percent loading units program option setting) of RATEC
	'P'	Branch MW flow at the IBUS end of the branch
	'O_P'	Same as 'P' in units determined by the power output option setting

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	'Q'	Branch Mvar flow at the IBUS end of the branch
	'O_Q'	Same as 'Q' in units determined by the power output option setting
	'PLOS'	Branch active power losses in MW
	'O_PLOS'	Branch active power losses in units determined by the power output option setting
	'QLOS'	Branch reactive power losses in Mvar
	'O_QLOS'	Branch reactive power losses in units determined by the power output option setting
Real RVAL	Flow value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Branch not found; 'RVAL' unchanged
	IERR = 3	Branch out-of-service; 'RVAL' returned set to zero
	IERR = 4	No base kV at IBUS; 'RVAL' returned = per unit current
	IERR = 5	Invalid 'STRING'; 'RVAL' unchanged
	IERR = 6	Rating is zero; 'RVAL' returned set to zero

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7.14. BRNMVA

This API routine is obsolete. It has been replaced by the API routine BRNMSC .

Use this API to return return branch MVA.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = brnmva(ibus,jbus,ickt)
```

Fortran command syntax:

```
CALL BRNMVA(IBUS, JBUS, ICKT, RVAL, IERR)
```

IPLAN command syntax:

```
CALL BRNMVA(IBUS, JBUS, ICKT, RVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*2 ICKT	Circuit or multisection line identifier (input).	
Real RVAL	Flow value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Branch not found; 'RVAL' unchanged
	IERR = 3	Branch out-of-service; 'RVAL' returned set to zero

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7.15. BRNSTT

This API routine is obsolete. It has been replaced by the API routine BRNINT .

Return branch status.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = brnstt(ibus,jbus,ickt)
```

Fortran command syntax:

```
CALL BRNSTT(IBUS, JBUS, ICKT, IVAL, IERR)
```

IPLAN command syntax:

```
CALL BRNSTT(IBUS, JBUS, ICKT, IVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*2 ICKT	Circuit or multisection line identifier (input).	
Integer IVAL	Branch status (output).	
	IVAL = 0	out of service
	IVAL = 1	in-service
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Branch not found; 'IVAL' unchanged

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7.16. BSYSISDEF

Use this API to check whether a bus subsystem has been defined for a given subsystem ID. A return value of FALSE means that the given subsystem ID is operating in ALL buses mode, e.g., all buses are operated on.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
exist = bsysisdef(sid)
```

Fortran command syntax:

```
CALL SBSBDEF(SID,EXIST)
```

IPLAN command syntax:

```
CALL BSYSISDEF(SID,EXIST)
```

Where:

Integer SID	a valid bus subsystem ID. Valid bus subsystem IDs range from 0 to 11 (input).	
Logical EXIST	function return (output).	
	EXIST = False	no subsystem defined, operating in ALL buses mode.
	EXIST = True	subsystem is defined.

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7.17. BUSDAT

Use this API to return real bus values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = busdat(ibus ,string)
```

Fortran command syntax:

```
CALL BUSDAT(IBUS, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL BUSDAT(IBUS, STRING, RVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character STRING	String indicating the bus quantity desired (input).	
	'BASE'	Bus base voltage, in kV
	'PU'	Actual bus voltage magnitude, in pu
	'KV'	Actual bus voltage magnitude, in kV
	'ANGLE'	Bus voltage angle, in radians
	'ANGLED'	Bus voltage angle, in degrees
	'NVLMHI'	Normal high voltage limit, in pu
	'NVLMLO'	Normal low voltage limit, in pu
	'EVLMHI'	Emergency high voltage limit, in pu
	'EVLMLO'	Emergency low voltage limit, in pu
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Bad value of 'STRING'; 'RVAL' unchanged
	IERR = 3	Bus type code 4 or greater; 'RVAL' returned
	IERR = 4	No base voltage and 'STRING' = 'KV'; 'RVAL' unchanged

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7.18. BUSDT1

Use this API to return complex bus parameters as MVA. Values may be either nominal or actual.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = busdt1(ibus, string1, string2)
```

Fortran command syntax:

```
CALL BUSDT1(IBUS, STRING1, STRING2, RVAL, IERR)
```

IPLAN command syntax:

```
CALL BUSDT1(IBUS, STRING1, STRING2, RVAL, IERR)
```

Note:

When STRNG1 is one of the following: 'MVA', 'IL', 'YL', 'TOTAL', 'SC_MVA', 'SC_IL', 'SC_YL', 'SC_TOTAL', 'FX_MVA', 'FX_IL', 'FX_YL', 'FX_TOTAL', the values returned are the net load values (i.e., value includes the load plus contribution from any in-service distributed generation on the load feeder at that bus)

Where:

Integer IBUS	Bus number (input).	
Character STRING1	String indicating the bus quantity desired (input).	
	'MVA'	Total in-service constant MVA load
	'IL'	Total in-service constant current load
	'YL'	Total in-service constant admittance load
	'TOTAL'	Total in-service load
	'SC_MVA'	Total in-service scalable constant MVA load
	'SC_IL'	Total in-service scalable constant current load
	'SC_YL'	Total in-service scalable constant admittance load
	'SC_TOTAL'	Total in-service scalable load
	'FX_MVA'	Total in-service fixed constant MVA load
	'FX_IL'	Total in-service fixed constant current load
	'FX_YL'	Total in-service fixed constant admittance load
	'FX_TOTAL'	Total in-service fixed load
	'YS'	Total in-service fixed bus shunt
	'YSZERO'	Total nominal in-service zero sequence fixed bus shunt
	'YSZ'	Total nominal in-service zero sequence fixed bus shunt in per unit
	'YSW'	In-service switched shunt
	'YSWZ'	In-service zero sequence switched shunt

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	'SHUNTN'	Total in-service negative sequence load (pu nominal)
	'SHUNTZ'	Total in-service zero sequence load (pu nominal)
Character STRING2	String indicating the units to be used (ignored if STRING1 = 'YSZ' or 'YSWZ') (input).	
	'ACT'	Actual load or shunt (nominal shunt for STRING1 = 'YSZERO' or 'YSWZERO'). 'RVAL' returned in MVA
	'O_ACT'	Actual load or shunt (nominal shunt for STRING1 = 'YSZERO' or 'YSWZERO'). 'RVAL' in units determined by the power output option setting (units determined by power output options)
	'NOM'	Nominal load or shunt (at 1.0 pu voltage). 'RVAL' returned in MVA
	'O_NOM'	Nominal load or shunt (at 1.0 pu voltage). 'RVAL' in units determined by the power output option setting
Real RVAL	Magnitude of the complex value indicated by STRING1 in the units indicated by STRING2 (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Bad value of 'STRING1' or 'STRING2'; 'RVAL' unchanged
	IERR = 3	Bus type code 4 or greater; 'RVAL' returned
	IERR = 4	No loads at bus for 'STRING1' = 'MVA', 'IL', 'YL', or 'TOTAL'; 'RVAL' returned set to zero
	IERR = 5	No switched shunt at bus for 'STRING1' = 'YSW'; 'YSWZERO' or 'YSWZ'; 'RVAL' unchanged
	IERR = 6	Sequence data not in case for 'STRING1' = 'YSZERO', 'YSZ', 'YSWZERO', 'YSWZ', 'SHUNTN' or 'SHUNTZ'; 'RVAL' unchanged

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7.19. BUSDT2

Use this API to return complex bus parameters. Values may be either nominal or actual. Nominal values return negative values for reactors. Actual values show flow into reactor.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = busdt2(ibus, string1, string2)
```

Fortran command syntax:

```
CALL BUSDT2(IBUS, STRING1, STRING2, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL BUSDT2(IBUS, STRING1, STRING2, P, Q, IERR)
```

Note:

When STRNG1 is one of the following: 'MVA', 'IL', 'YL', 'TOTAL', 'SC_MVA', 'SC_IL', 'SC_YL', 'SC_TOTAL', 'FX_MVA', 'FX_IL', 'FX_YL', 'FX_TOTAL', the values returned are the net load values (i.e., value includes load plus the contribution from any in-service distributed generation on the load feeder at that bus)

Where:

Integer IBUS	Bus number (input).	
Character STRING1	String indicating the bus quantity desired (input).	
	'MVA'	Total in-service constant MVA load
	'IL'	Total in-service constant current load
	'YL'	Total in-service constant admittance load
	'TOTAL'	Total in-service load
	'SC_MVA'	Total in-service scalable constant MVA load
	'SC_IL'	Total in-service scalable constant current load
	'SC_YL'	Total in-service scalable constant admittance load
	'SC_TOTAL'	Total in-service scalable load
	'FX_MVA'	Total in-service fixed constant MVA load
	'FX_IL'	Total in-service fixed constant current load
	'FX_YL'	Total in-service fixed constant admittance load
	'FX_TOTAL'	Total in-service fixed load
	'YS'	Total in-service fixed bus shunt
	'YSZERO'	Total nominal in-service zero sequence fixed bus shunt
	'YSZ'	Total nominal in-service zero sequence fixed bus shunt in per unit
	'YSW'	In-service switched shunt

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	'YSWZERO'	Nominal in-service zero sequence switched shunt
	'YSWZ'	Nominal in-service zero sequence switched shunt in per unit
	'SHUNTN'	Total in-service negative sequence load (pu nominal)
	'SHUNTZ'	Total in-service zero sequence load (pu nominal)
Character STRING2	String indicating the units to be used (ignored if STRING1 = 'YSZ' or 'YSWZ') (input).	
	'ACT'	Actual load or shunt (nominal shunt for STRING1 = 'YSZERO' or 'YSWZERO'). 'P' 'Q' returned in MW Mvar
	'O_ACT'	Actual load or shunt (nominal shunt for STRING1 = 'YSZERO' or 'YSWZERO'). 'P' 'Q' in units determined by the power output option setting
	'NOM'	Nominal load (at 1.0 pu voltage). 'P' 'Q' returned in MW Mvar
	'O_NOM'	Nominal load (at 1.0 pu voltage). 'P' 'Q' in units determined by the power output option setting
Real P	Real component of the complex value indicated by STRING1 in the units indicated by STRING2 (output).	
Real Q	Reactive component of the complex value indicated by STRING1 in the units indicated by STRING2 (output).	
Complex CMPVAL	Complex value indicated by STRING1 in the units indicated by STRING2 (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Bad value of 'STRING1' or 'STRING2'; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Bus type code 4 or greater; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 4	No loads at bus for 'STRING1' = 'MVA', 'IL', 'YL', or 'TOTAL'; 'P' and 'Q' or 'CMPVAL' returned set to zero
	IERR = 5	No switched shunt at bus for 'STRING1' = 'YSW', 'YSWZERO' or 'YSWZ'; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 6	Sequence data not in case for 'STRING1' = 'YSZERO', 'YSZ', 'YSWZERO', 'YSWZ', 'SHUNTN' or 'SHUNTZ'; 'CMPVAL' unchanged

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7.20. BUSEXS

Use this API to check for the existence of a specified bus.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = busexs(ibus)
```

Fortran command syntax:

```
CALL BUSEXS(IBUS, IERR)
```

IPLAN command syntax:

```
CALL BUSEXS(IBUS, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Integer IERR	error code (output).	
	IERR = 0	Bus found
	IERR = 1	Bus not found

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7.21. BUSINT

Use this API to return integer bus parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = busint(ibus ,string)
```

Fortran command syntax:

```
CALL BUSINX(IBUS, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL BUSINT(IBUS, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character STRING	String indicating the bus quantity desired (input).	
	'STATION'	Bus IBUS substation number (returned as 0 if bus IBUS is not in a substation)
	'TYPE'	Bus type code
	'AREA'	Bus area number
	'ZONE'	Bus zone number
	'OWNER'	Bus owner number
	'DUMMY'	Returns 1 (one) if bus is a dummy bus of a multisection line else 0 (zero)
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Bad value of 'STRING'; 'IVAL' unchanged

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7.22. BUSMSM

Use this API to return complex bus mismatch.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = busmsm(ibus)
```

Fortran command syntax:

```
CALL BUSMSM(IBUS, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL BUSMSM(IBUS, P, Q, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Real P	Real component (MW) of the bus mismatch	
Real Q	Reactive component (Mvar) of the bus mismatch	
Complex CMPVAL	Complex bus mismatch (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Bus type code greater than 4; 'P' and 'Q' or 'CMPVAL' returned set to zero

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7.23. CCTMDLNAM_2WTD

Use this API to return model name of the CCT 2-winding transformer device model.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, model = cctmdlnam_2wtd(ibus, jbus, id)
```

Fortran command syntax:

```
CALL CCTMDLNAM_2WTD(IBUS, JBUS, ID, MODEL, IERR)
```

IPLAN command syntax:

```
CALL CCTMDLNAM_2WTD(IBUS, JBUS, ID, MODEL, IERR)
```

Where:

Integer IBUS	"From" Bus number of the 2-winding transformer (input).	
Integer JBUS	"To" Bus number of the 2-winding transformer (input).	
Character*2 ID	2-winding transformer circuit identifier (input).	
Character*16 MODEL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; MODEL name is returned
	IERR = 1	2-winding transformer not in case
	IERR = 2	No CCT models in case
	IERR = 3	No device model attached to the specified 2-winding transformer
	IERR = 4	Dynamics data not present in working memory
	IERR = 5	Specified branch is not a transformer

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7.24. CCTMDLNAM_3WTD

Use this API to return model name of the CCT 3-winding transformer device model.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, model = cctmdlnam_3wtd(ibus, jbus, kbus, id)
```

Fortran command syntax:

```
CALL CCTMDLNAM_3WTD(IBUS, JBUS, KBUS, ID, MODEL, IERR)
```

IPLAN command syntax:

```
CALL CCTMDLNAM_3WTD(IBUS, JBUS, KBUS, ID, MODEL, IERR)
```

Where:

Integer IBUS	"From" Bus number of the branch (input).	
Integer JBUS	"To" Bus number of the branch (input).	
Integer KBUS	"Third" Bus number of the branch (input).	
Character*2 ID	3-winding transformer branch circuit identifier (input).	
Character*16 MODEL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; MODEL name is returned
	IERR = 1	branch not in case;
	IERR = 2	No CCT models in case
	IERR = 3	No device model attached to the specified 3-winding transformer
	IERR = 4	Dynamics data not present in working memory
	IERR = 5	Specified branch is a transformer

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7.25. CCTMDLNAM_BRND

Use this API to return model name of the CCT branch device model.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, model = cctmdlnam_brnd(ibus, jbus, id)
```

Fortran command syntax:

```
CALL CCTMDLNAM_BRND(IBUS, JBUS, ID, MODEL, IERR)
```

IPLAN command syntax:

```
CALL CCTMDLNAM_BRND(IBUS, JBUS, ID, MODEL, IERR)
```

Where:

Integer IBUS	"From" Bus number of the branch (input).	
Integer JBUS	"To" Bus number of the branch (input).	
Character*2 ID	Branch circuit identifier (input).	
Character*16 MODEL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; MODEL name is returned
	IERR = 1	branch not in case;
	IERR = 2	No CCT models in case
	IERR = 3	No branch device model attached to the specified branch
	IERR = 4	Dynamics data not present in working memory
	IERR = 5	Specified branch is a transformer

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7.26. CCTMDLNAM_MSCO

Use this API to return model name of the CCT Miscellaneous other model.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, model = cctmdlnam_msco(mins)
```

Fortran command syntax:

```
CALL CCTMDLNAM_MSCO(MINS, MODEL, IERR)
```

IPLAN command syntax:

```
CALL CCTMDLNAM_MSCO(MINS, MODEL, IERR)
```

Where:

Integer MINS	Model instance number of the model (input).	
Character*16 MODEL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; MODEL name is returned
	IERR = 1	Invalid MINS value
	IERR = 2	No CCT models in case
	IERR = 3	No model attached for the specified MINS
	IERR = 4	Dynamics data not present in working memory

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7.27. CCTMIND_2DCO

Use this API to return starting array indices and status of CCT 2-terminal dc line other models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_2dco(name, model, string)
```

Fortran command syntax:

```
CALL CCTMIND_2DCO (NAME, MODEL, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_2DCO (NAME, MODEL, STRING, IVAL, IERR)
```

Where:

Character*12 NAME	Two-terminal dc line name (input).	
Character*16 MODEL	2-terminal dc other Model name (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect dc line status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	Model name specified in 'MODEL' not attached to this dc line; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged

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7.28. CCTMIND_2WTD

Use this API to return starting array indices and status of CCT 2-winding transformer device models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_2wtd(ibus, jbus, id, string)
```

Fortran command syntax:

```
CALL CCTMIND_2WTD(IBUS, JBUS, ID, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_2WTD(IBUS, JBUS, ID, STRING, IVAL, IERR)
```

Where:

Integer IBUS	"From" Bus number of the 2-winding transformer (input).	
Integer JBUS	"To" Bus number of the 2-winding transformer (input).	
Character*2 ID	2-winding transformer circuit identifier (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect 2-winding transformer status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	2-winding transformer not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	No device model attached to the specified 2-winding transformer; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged

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IERR = 7	Specified branch is not a transformer; 'IVAL' unchanged

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7.29. CCTMIND_3WTD

Use this API to return starting array indices and status of CCT 3-winding transformer device models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_3wtd(ibus, jbus, kbus, id, string)
```

Fortran command syntax:

```
CALL CCTMIND_3WTD(IBUS, JBUS, KBUS, ID, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_3WTD(IBUS, JBUS, KBUS, ID, STRING, IVAL, IERR)
```

Where:

Integer IBUS	"From" Bus number of the 3-winding transformer (input).	
Integer JBUS	"To" Bus number of the 3-winding transformer (input).	
Integer KBUS	"Third" Bus number of the 3-winding transformer (input).	
Character*2 ID	2-winding transformer circuit identifier (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect 3-winding transformer status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	4-winding transformer not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	No device model attached to the specified 3-winding transformer; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned

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IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged
IERR = 7	Specified branch is not a transformer; 'IVAL' unchanged

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7.30. CCTMIND_BRND

Use this API to return starting array indices and status of CCT branch device models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_brnd(ibus, jbus, id, string)
```

Fortran command syntax:

```
CALL CCTMIND_BRND(IBUS, JBUS, ID, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_BRND(IBUS, JBUS, ID, STRING, IVAL, IERR)
```

Where:

Integer IBUS	"From" Bus number of the branch (input).	
Integer JBUS	"To" Bus number of the branch (input).	
Character*2 ID	Branch circuit identifier (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect branch status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	branch not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	No device model attached to the specified branch; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged
	IERR = 7	Specified branch is a transformer; 'IVAL' unchanged

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7.31. CCTMIND_BRNO

Use this API to return starting array indices and status of CCT branch other models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_brno(ibus, jbus, id, model, string)
```

Fortran command syntax:

```
CALL CCTMIND_BRNO(IBUS, JBUS, ID, MODEL, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_BRNO(IBUS, JBUS, ID, MODEL, STRING, IVAL, IERR)
```

Where:

Integer IBUS	"From" Bus number of the branch (input).	
Integer JBUS	"To" Bus number of the branch (input).	
Character*2 ID	Branch circuit identifier (input).	
Character*16 MODEL	Branch other Model name (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect branch status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	branch not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	Model name specified in 'MODEL' not attached to the specified branch; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged

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IERR = 7	Branch is a transformer; 'IVAL' unchanged

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7.32. CCTMIND_BUSO

Use this API to return starting array indices and status of CCT Bus other models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_buso(ibus, model, string)
```

Fortran command syntax:

```
CALL CCTMIND_BUSO(IBUS, MODEL, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_BUSO(IBUS, MODEL, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*16 MODEL	Bus other Model name (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect bus status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	bus not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	Model name specified in 'MODEL' not attached to this bus; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged

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7.33. CCTMIND_MCNO

Use this API to return starting array indices and status of CCT machine other models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_mcno(ibus, id, model, string)
```

Fortran command syntax:

```
CALL CCTMIND_MCNO(IBUS, ID, MODEL, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_MCNO(IBUS, ID, MODEL, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Machine identifier (input).	
Character*16 MODEL	Machine other Model name (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect machine status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	machine not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	Model name specified in 'MODEL' not attached to this machine; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged

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7.34. CCTMIND_MCNP

Use this API to return starting array indices and status of CCT machine protection models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_mcnp(ibus, id, model, string)
```

Fortran command syntax:

```
CALL CCTMIND_MCNP(IBUS, ID, MODEL, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_MCNP(IBUS, ID, MODEL, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Machine identifier (input).	
Character*16 MODEL	Machine protection Model name (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect machine status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	machine not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	Model name specified in 'MODEL' not attached to this machine; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged

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7.35. CCTMIND_MSCO

Use this API to return starting array indices and status of CCT Miscellaneous other models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_msco(mins, string)
```

Fortran command syntax:

```
CALL CCTMIND_MSCO(MINS, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_MSCO(MINS, STRING, IVAL, IERR)
```

Where:

Integer MINS	Model instance number of the model (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	CON Starting CON index
	'STATE'	STATE Starting STATE index
	'VAR'	VAR Starting VAR index
	'ICON'	ICON Starting ICON index
	'NCON'	NCON Number of CON values used by the model
	'NSTATE'	NSTATE Number of STATE values used by the model
	'NVAR'	NVAR Number of VAR values used by the model
	'NICON'	NICON Number of ICON values used by the model
	'STATUS'	STATUS Bypassed flag: 0 = bypassed; 1 = not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Invalid MINS value; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	No model attached for the specified MINS; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged

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7.36. CCTMIND_SWSO

Use this API to return starting array indices and status of CCT Switched Shunt other models.

This API routine was first introduced in release 33.9.0.

Python command syntax:

```
ierr, ival = cctmind_swso(ibus, model, string)
```

Fortran command syntax:

```
CALL CCTMIND_SWSO(IBUS, MODEL, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL CCTMIND_SWSO(IBUS, MODEL, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number at which switched shunt is attached (input).	
Character*16 MODEL	Switched shunt other Model name (input).	
Character STRING	String indicating the model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect switched shunt status
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	bus not in case; 'IVAL' returned as 0
	IERR = 2	No CCT models in case
	IERR = 3	Model name specified in 'MODEL' not attached to this bus; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 6	Dynamics data not present in working memory; 'IVAL' unchanged
	IERR = 7	No switched shunt at the specified bus; 'IVAL' unchanged

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7.37. CHKTRE

Use this API to check for Type 4 (or greater) buses with in-service branches connected to them. If none are found, it then counts number of ac islands not containing a Type 3 (swing) bus. It may be inappropriate to call this subroutine with generators converted; in this case, there are no Type 3 buses, so there is normally one or more swingless islands.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ival = chktre()
```

Fortran command syntax:

```
CALL CHKTRE(IVAL)
```

IPLAN command syntax:

```
CALL CHKTRE(IVAL)
```

Where:

Integer IVAL	Return code (output).	
	IVAL = 0	None of the following found.
	IVAL = -1	No in-service buses remaining.
	IVAL = -2	There are Type 4 buses with in-service branches connected to them.
	IVAL > 0	Number of islands without a Type 3 bus.

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7.38. CHNVAL

Use this API to return the present value of the simulation variable assigned to a specified output channel.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = chnval(n)
```

Fortran command syntax:

```
CALL CHNVAL(N, RVAL, IERR)
```

IPLAN command syntax:

```
CALL CHNVAL(N, RVAL, IERR)
```

Where:

Integer N	Channel number (input).	
Real RVAL	Channel value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Invalid channel index; 'RVAL' unchanged
	IERR = 2	Invalid channel address; 'RVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'RVAL' unchanged

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7.39. DC2AUXMIND

Use this API to return the starting array indices and status of auxiliary signal model associated with 2-terminal dc line for the specified signal index.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = dc2auxmind(name, isgndx, string)
```

Fortran command syntax:

```
CALL DC2AUXMIND(NAME, ISGNDX, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DC2AUXMIND(NAME, ISGNDX, STRING, IVAL, IERR)
```

Where:

Character*12 NAME	Two-terminal dc line name (input).	
Integer ISGNDX	Auxiliary signal index (input).	
Character STRING	String indicating the 2-terminal dc line auxiliary signal model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. For a model to be called during simulations the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not in network but auxiliary signal model in model tables; 'IVAL' returned
	IERR = 2	dc line not in network and auxiliary signal model not in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables for the given auxiliary signal index; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned

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IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.40. DC2AUXMNAM

Use this API to return the auxiliary signal model name associated with a 2-terminal dc line for the specified auxiliary signal index.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = dc2auxmnam(name, isgndx)
```

Fortran command syntax:

```
CALL DC2AUXMNAM(NAME, ISGNDX, CVAL, IERR)
```

IPLAN command syntax:

```
CALL DC2AUXMNAM(NAME, ISGNDX, CVAL, IERR)
```

Where:

Character NAME	Two-terminal dc line name (input).	
Integer ISGNDX	Auxiliary signal index (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	dc line not in network but auxiliary signal model in model tables; 'CVAL' returned
	IERR = 2	dc line not in network and auxiliary signal model not in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables for the given auxiliary signal index; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.41. DC2DAT

This API routine is obsolete. It has been replaced by the API routine dc2dat_2 .

Use this API to return real two-terminal dc line quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = dc2dat(n, ibus, string)
```

Fortran command syntax:

```
CALL DC2DAT(N, IBUS, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL DC2DAT(N, IBUS, STRING, RVAL, IERR)
```

Where:

Integer N	Two-terminal dc line number (input).	
Integer IBUS	Converter bus number (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'DCCUR'	dc line current in amps (Line Quantity)
	'RDC'	dc line resistance in ohms (Line Quantity)
	'SETVAL'	Current (amps) or power (MW) demand (Line Quantity)
	'VSCHD'	Scheduled compounded dc voltage in kV (Line Quantity)
	'VCMOD'	Mode switch dc voltage in kV (Line Quantity)
	'RCOMP'	Compounding resistance in ohms (Line Quantity)
	'DELTI'	Margin in pu of desired dc power or current (Line Quantity)
	'DCVMIN'	Minimum compounded dc voltage in kV (Line Quantity)
	'PAC'	Apparent active power flow into the dc line in MW (Converter Quantity)
	'O_PAC'	Same as 'PAC' in units determined by the power output option setting (Converter Quantity)
	'QAC'	Apparent reactive power flow into the dc line in Mvar (Converter Quantity)
	'O_QAC'	Same as 'QAC' in units determined by the power output option setting (Converter Quantity)
	'MVA'	Apparent ac flow into the dc line in MVA (Converter Quantity)

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	'O_MVA'	Same as 'MVA' in units determined by the power output option setting (Converter Quantity)
	'ACAMPS'	ac current flow into the dc line in amps (Converter Quantity)
	'PUCUR'	ac current flow into the dc line in pu (Converter Quantity)
	'KVDC'	dc voltage in kV (Converter Quantity)
	'ANGLE'	Converter angle in degrees (Converter Quantity)
	'ANGMIN'	Minimum steady-state converter angle in degrees (Converter Quantity)
	'ANGMAX'	Nominal maximum converter angle in degrees (Converter Quantity)
	'RC'	Commutating transformer resistance per bridge in ohms (Converter Quantity)
	'XC'	Commutating transformer reactance per bridge in ohms (Converter Quantity)
	'EBASE'	Primary base ac voltage in kV (Converter Quantity)
	'TR'	Transformer ratio in pu (Converter Quantity)
	'TAP'	Tap setting in pu (Converter Quantity)
	'TPMX'	Maximum tap setting in pu (Converter Quantity)
	'TPMN'	Minimum tap setting in pu (Converter Quantity)
	'STP'	Tap step in pu (Converter Quantity)
	'XCAP'	Commutating capacitor reactance magnitude per bridge in ohms (Converter Quantity)
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	dc line not found; 'RVAL' unchanged
	IERR = 2	Bus not a converter bus of dc line 'N'; 'RVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 4	dc line blocked; 'RVAL' of 0.0 returned
	IERR = 5	No base kV at bus; 'RVAL' returned as the per unit current

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7.42. DC2DAT_2

This API is the second release of the API that returns real two-terminal dc line quantities.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, rval = dc2dat_2(name, ibus, string)
```

Fortran command syntax:

```
CALL DC2DAT_2(NAME, IBUS, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL DC2DAT_2(NAME, IBUS, STRING, RVAL, IERR)
```

Where:

Character NAME	Two-terminal dc line name (input).	
Integer IBUS	Converter bus number (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'DCCUR'	dc line current in amps (Line Quantity)
	'RDC'	dc line resistance in ohms (Line Quantity)
	'SETVAL'	Current (amps) or power (MW) demand (Line Quantity)
	'VSCHD'	Scheduled compounded dc voltage in kV (Line Quantity)
	'VCMOD'	Mode switch dc voltage in kV (Line Quantity)
	'RCOMP'	Compounding resistance in ohms (Line Quantity)
	'DELTI'	Margin in pu of desired dc power or current (Line Quantity)
	'DCVMIN'	Minimum compounded dc voltage in kV (Line Quantity)
	'PAC'	Apparent active power flow into the dc line in MW (Converter Quantity)
	'O_PAC'	Same as 'PAC' in units determined by the power output option setting (Converter Quantity)
	'QAC'	Apparent reactive power flow into the dc line in Mvar (Converter Quantity)
	'O_QAC'	Same as 'QAC' in units determined by the power output option setting (Converter Quantity)
	'MVA'	Apparent ac flow into the dc line in MVA (Converter Quantity)
	'O_MVA'	Same as 'MVA' in units determined by the power output option setting (Converter Quantity)

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	'ACAMPS'	ac current flow into the dc line in amps (Converter Quantity)
	'PUCUR'	ac current flow into the dc line in pu (Converter Quantity)
	'KVDC'	dc voltage in kV (Converter Quantity)
	'ANGLE'	Converter angle in degrees (Converter Quantity)
	'ANGMIN'	Minimum steady-state converter angle in degrees (Converter Quantity)
	'ANGMAX'	Nominal maximum converter angle in degrees (Converter Quantity)
	'RC'	Commutating transformer resistance per bridge in ohms (Converter Quantity)
	'XC'	Commutating transformer reactance per bridge in ohms (Converter Quantity)
	'EBASE'	Primary base ac voltage in kV (Converter Quantity)
	'TR'	Transformer ratio in pu (Converter Quantity)
	'TAP'	Tap setting in pu (Converter Quantity)
	'TPMX'	Maximum tap setting in pu (Converter Quantity)
	'TPMN'	Minimum tap setting in pu (Converter Quantity)
	'STP'	Tap step in pu (Converter Quantity)
	'XCAP'	Commutating capacitor reactance magnitude per bridge in ohms (Converter Quantity)
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	dc line not found; 'RVAL' unchanged
	IERR = 2	Bus not a converter bus of dc line 'NAME'; 'RVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 4	dc line blocked; 'RVAL' of 0.0 returned
	IERR = 5	No base kV at bus; 'RVAL' returned as the per unit current

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7.43. DC2INT

This API routine is obsolete. It has been replaced by the API routine dc2int_2 .

Use this API to return integer two-terminal dc line quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = dc2int(n, string)
```

Fortran command syntax:

```
CALL DC2INT(N, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DC2INT(N, STRING, IVAL, IERR)
```

Where:

Integer N	Two-terminal dc line number (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'MDC'	dc line control mode
	'RECT'	Bus number of rectifier end of dc line
	'INV'	Bus number of inverter end of dc line
	'METER'	Bus number of metered end of dc line
	'NBR'	Number of bridges in series rectifier end
	'NBI'	Number of bridges in series inverter end
	'ICR'	Rectifier firing angle measuring bus
	'ICI'	Inverter firing angle measuring bus
	'ANSTTR'	Firing angle status rectifier end: (-1) low; (0) regulating; (1) high
	'ANSTTI'	Firing angle status inverter end: (-1) low; (0) regulating (1) high
	'TPSTTR'	Tap status rectifier end: (-1) low; (0) regulating; (1) high; (-2) locked
	'TPSTTI'	Tap status inverter end: (-1) low; (0) regulating; (1) high; (-2) locked
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 2	dc line not found; 'IVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'IVAL' unchanged

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7.44. DC2INT_2

This API is the second release of the API that returns integer two-terminal dc line quantities.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = dc2int_2(name, string)
```

Fortran command syntax:

```
CALL DC2INT_2(NAME, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DC2INT_2(NAME, STRING, IVAL, IERR)
```

Where:

Character NAME	Two-terminal dc line name (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'MDC'	dc line control mode
	'RECT'	Bus number of rectifier end of dc line
	'INV'	Bus number of inverter end of dc line
	'METER'	Bus number of metered end of dc line
	'NBR'	Number of bridges in series rectifier end
	'NBI'	Number of bridges in series inverter end
	'ICR'	Rectifier firing angle measuring bus
	'ICI'	Inverter firing angle measuring bus
	'ANSTTR'	Firing angle status rectifier end: (-1) low; (0) regulating; (1) high
	'ANSTTI'	Firing angle status inverter end: (-1) low; (0) regulating (1) high
	'TPSTTR'	Tap status rectifier end: (-1) low; (0) regulating; (1) high; (-2) locked
	'TPSTTI'	Tap status inverter end: (-1) low; (0) regulating; (1) high; (-2) locked
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 2	dc line not found; 'IVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'IVAL' unchanged

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7.45. DC2MIND

Use this API to return 2-terminal dc line model starting array indices and status.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = dc2mind(name, string)
```

Fortran command syntax:

```
CALL DC2MIND(NAME, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DC2MIND(NAME, STRING, IVAL, IERR)
```

Where:

Character NAME	Two-terminal dc line name (input).	
Character STRING	String indicating the 2-terminal dc line model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect dc line status. For a model to be called during simulations the dc line must be in service the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not in network but in model tables; 'IVAL' returned
	IERR = 2	dc line not in network or in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.46. DC2MNAM

Use this API to return 2-terminal dc line model name.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = dc2mnam(name)
```

Fortran command syntax:

```
CALL DC2MNAM(NAME, CVAL, IERR)
```

IPLAN command syntax:

```
CALL DC2MNAM(NAME, CVAL, IERR)
```

Where:

Character NAME	Two-terminal dc line name (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	dc line not in network but in model tables; 'CVAL' returned
	IERR = 2	dc line not in network or in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.47. DCNAUXMIND

Use this API to return the starting array indices and status of auxiliary signal model associated with N-terminal dc line for the specified signal index.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = dcnauxmind(name, isgndx, string)
```

Fortran command syntax:

```
CALL DCNAUXMIND(NAME, ISGNDX, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNAUXMIND(NAME, ISGNDX, STRING, IVAL, IERR)
```

Where:

Character NAME	N-terminal dc line name (input).	
Integer ISGNDX	Auxiliary signal index (input).	
Character STRING	String indicating the N-terminal dc line auxiliary signal model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. For a model to be called during simulations the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not in network but auxiliary signal model in model tables; 'IVAL' returned
	IERR = 2	dc line not in network and auxiliary signal model not in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables for the given auxiliary signal index; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned

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IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.48. DCNAUXMNAM

Use this API to return the auxiliary signal model name associated with a N-terminal dc line for the specified auxiliary signal index.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = dcnauxmnam(name, isgndx)
```

Fortran command syntax:

```
CALL DCNAUXMNAM(NAME, ISGNDX, CVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNAUXMNAM(NAME, ISGNDX, CVAL, IERR)
```

Where:

Character NAME	N-terminal dc line name (input).	
Integer ISGNDX	Auxiliary signal index (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	dc line not in network but auxiliary signal model in model tables; 'CVAL' returned
	IERR = 2	dc line not in network and auxiliary signal model not in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables for the given auxiliary signal index; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.49. DCNCIN

This API routine is obsolete. It has been replaced by the API routine dcncin_2 .

Use this API to return integer multi-terminal dc line converter quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = dcncin(n, ibus, string)
```

Fortran command syntax:

```
CALL DCNCIN(N, IBUS, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNCIN(N, IBUS, STRING, IVAL, IERR)
```

Where:

Integer N	Multi-terminal dc line number (input).	
Integer IBUS	Converter bus number (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'ANSTT'	Firing angle status: (-1) low; (0) regulating; (1) high
	'TPSTT'	Tap status: (-1) low; (0) regulating; (1) high; (-2) locked
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not found; 'IVAL' unchanged
	IERR = 2	Bus not a converter bus of dc line 'N'; 'IVAL' unchanged
	IERR = 3	Invalid 'STRING'; 'IVAL' unchanged

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7.50. DCNCIN_2

This API is the second release of the API that returns integer multi-terminal dc line converter quantities.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = dcncin_2(name, ibus, string)
```

Fortran command syntax:

```
CALL DCNCIN_2(NAME, IBUS, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNCIN_2(NAME, IBUS, STRING, IVAL, IERR)
```

Where:

Character NAME	Multi-terminal dc line name (input).	
Integer IBUS	Converter bus number (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'ANSTT'	Firing angle status: (-1) low; (0) regulating; (1) high
	'TPSTT'	Tap status: (-1) low; (0) regulating; (1) high; (-2) locked
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not found; 'IVAL' unchanged
	IERR = 2	Bus not a converter bus of dc line 'NAME'; 'IVAL' unchanged
	IERR = 3	Invalid 'STRING'; 'IVAL' unchanged

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7.51. DCNDAT

This API routine is obsolete. It has been replaced by the API routine dcndat_2 .

Use this API to return real multi-terminal dc line quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = dcndat(n, ibus, string)
```

Fortran command syntax:

```
CALL DCNDAT(N, IBUS, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNDAT(N, IBUS, STRING, RVAL, IERR)
```

Where:

Integer N	Multi-terminal dc line number (input).	
Integer IBUS	Converter bus number (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'VCMOD'	Mode switch dc voltage in kV (Line Quantity)
	'PAC'	Active power into dc line at converter bus in MW (Converter Quantity)
	'O_PAC'	Same as 'PAC' in units determined by the power output option setting (Converter Quantity)
	'QAC'	Reactive power into dc line at converter bus in Mvar (Converter Quantity)
	'O_QAC'	Same as 'QAC' in units determined by the power output option setting (Converter Quantity)
	'KVDC'	dc voltage at converter bus end (Converter Quantity)
	'DCCUR'	dc current (Converter Quantity)
	'SETVAL'	Converter setpoint (Converter Quantity)
	'ANGLE'	Converter angle (Converter Quantity)
	'TAP'	Tap setting (Converter Quantity)
	'ANGMIN'	Minimum steady-state ALPHA or GAMMA angle in degrees (Converter Quantity)
	'ANGMAX'	Nominal maximum ALPHA or GAMMA angle in degrees (Converter Quantity)
	'RC'	Commutating resistance per bridge in ohms (Converter Quantity)
	'XC'	Commutating reactance per bridge in ohms (Converter Quantity)

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	'MVA'	Apparent ac flow into the dc line in MVA (Converter Quantity)
	'O_MVA'	Same as 'MVA' in units determined by the power output option setting (Converter Quantity)
	'ACAMPS'	ac current flow into the dc line in amps (Converter Quantity)
	'PUCUR'	ac current flow into the dc line in pu (Converter Quantity)
	'EBASE'	Primary base ac voltage in kV (Converter Quantity)
	'TR'	Transformer ratio in pu (Converter Quantity)
	'TPMX'	Maximum tap setting in pu (Converter Quantity)
	'TPMN'	Minimum tap setting in pu (Converter Quantity)
	'STP'	Tap step in pu (Converter Quantity)
	'DELTI'	Rectifier margin in pu of desired dc power or current (Converter Quantity)
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	dc line not found; 'RVAL' unchanged
	IERR = 2	Bus not a converter bus of dc line 'N'; 'RVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 4	dc line blocked; 'RVAL' of 0.0 returned
	IERR = 5	No base kV at bus; 'RVAL' returned as the per unit current

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7.52. DCNDAT_2

This API is the second release of the API that returns real multi-terminal dc line quantities.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, rval = dcndat_2(name, ibus, string)
```

Fortran command syntax:

```
CALL DCNDAT_2(NAME, IBUS, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNDAT_2(NAME, IBUS, STRING, RVAL, IERR)
```

Where:

Character NAME	Multi-terminal dc line name (input).	
Integer IBUS	Converter bus number (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'VCMOD'	Mode switch dc voltage in kV (Line Quantity)
	'PAC'	Active power into dc line at converter bus in MW (Converter Quantity)
	'O_PAC'	Same as 'PAC' in units determined by the power output option setting (Converter Quantity)
	'QAC'	Reactive power into dc line at converter bus in Mvar (Converter Quantity)
	'O_QAC'	Same as 'QAC' in units determined by the power output option setting (Converter Quantity)
	'KVDC'	dc voltage at converter bus end (Converter Quantity)
	'DCCUR'	dc current (Converter Quantity)
	'SETVAL'	Converter setpoint (Converter Quantity)
	'ANGLE'	Converter angle (Converter Quantity)
	'TAP'	Tap setting (Converter Quantity)
	'ANGMIN'	Minimum steady-state ALPHA or GAMMA angle in degrees (Converter Quantity)
	'ANGMAX'	Nominal maximum ALPHA or GAMMA angle in degrees (Converter Quantity)
	'RC'	Commutating resistance per bridge in ohms (Converter Quantity)
	'XC'	Commutating reactance per bridge in ohms (Converter Quantity)
	'MVA'	Apparent ac flow into the dc line in MVA (Converter Quantity)

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	'O_MVA'	Same as 'MVA' in units determined by the power output option setting (Converter Quantity)
	'ACAMPS'	ac current flow into the dc line in amps (Converter Quantity)
	'PUCUR'	ac current flow into the dc line in pu (Converter Quantity)
	'EBASE'	Primary base ac voltage in kV (Converter Quantity)
	'TR'	Transformer ratio in pu (Converter Quantity)
	'TPMX'	Maximum tap setting in pu (Converter Quantity)
	'TPMN'	Minimum tap setting in pu (Converter Quantity)
	'STP'	Tap step in pu (Converter Quantity)
	'DELTI'	Rectifier margin in pu of desired dc power or current (Converter Quantity)
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	dc line not found; 'RVAL' unchanged
	IERR = 2	Bus not a converter bus of dc line 'NAME'; 'RVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 4	dc line blocked; 'RVAL' of 0.0 returned
	IERR = 5	No base kV at bus; 'RVAL' returned as the per unit current

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7.53. DCNINT

This API routine is obsolete. It has been replaced by the API routine dcnint_2 .

Use this API to return integer multi-terminal dc line quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = dcnint(n, string)
```

Fortran command syntax:

```
CALL DCNINT(N, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNINT(N, STRING, IVAL, IERR)
```

Where:

Integer N	Multi-terminal dc line number (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'MDC'	Control mode
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 2	dc line not found; 'IVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'IVAL' unchanged

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

7.54. DCNINT_2

This API is the second release of the API that returns integer multi-terminal dc line quantities.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = dcnint_2(name, string)
```

Fortran command syntax:

```
CALL DCNINT_2(NAME, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNINT_2(NAME, STRING, IVAL, IERR)
```

Where:

Character NAME	Multi-terminal dc line name (input).	
Character STRING	String indicating the dc line quantity desired (input).	
	'MDC'	Control mode
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 2	dc line not found; 'IVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'IVAL' unchanged

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7.55. DCNMIND

Use this API to return N-terminal dc line model starting array indices and status.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = dcnmind(name, string)
```

Fortran command syntax:

```
CALL DCNMIND(NAME, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNMIND(NAME, STRING, IVAL, IERR)
```

Where:

Character NAME	N-terminal dc line name (input).	
Character STRING	String indicating the N-terminal dc line model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect dc line status. For a model to be called during simulations the dc line must be in service the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not in network but in model tables; 'IVAL' returned
	IERR = 2	dc line not in network or in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.56. DCNMNAM

Use this API to return N-terminal dc line model name.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = dcnmnam(name)
```

Fortran command syntax:

```
CALL DCNMNAM(NAME, CVAL, IERR)
```

IPLAN command syntax:

```
CALL DCNMNAM(NAME, CVAL, IERR)
```

Where:

Character NAME	N-terminal dc line name (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	dc line not in network but in model tables; 'CVAL' returned
	IERR = 2	dc line not in network or in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.57. DSCVAL

Use this API to return dynamics character array values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = dscval(string, indx)
```

Fortran command syntax:

```
CALL DSCVAL(STRING, INDX, CVAL, IERR)
```

IPLAN command syntax:

```
CALL DSCVAL(STRING, INDX, CVAL, IERR)
```

Where:

Character STRING	String indicating the dynamics quantity desired (input).	
	'ICON'	Character model parameters
Integer INDX	Index to the dynamics character array (input).	
Character*12 CVAL	Character dynamics array value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Invalid value of 'STRING'; 'CVAL' unchanged
	IERR = 2	'INDX' invalid; 'CVAL' unchanged
	IERR = 3	'INDX' points to integer icon; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.58. DSIVAL

Use this API to return dynamics integer array values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = dsival(string, indx)
```

Fortran command syntax:

```
CALL DSIVAL(STRING, INDX, IVAL, IERR)
```

IPLAN command syntax:

```
CALL DSIVAL(STRING, INDX, IVAL, IERR)
```

Where:

Character STRING	String indicating the dynamics quantity desired (input).	
	'ICON'	Integer model parameters
Integer INDX	Index to the dynamics integer array (input).	
Integer IVAL	Integer dynamics array value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 2	'INDX' invalid; 'IVAL' unchanged
	IERR = 3	'INDX' points to character icon; 'IVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.59. DSRVAL

Use this API to return dynamics real values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = dsrval(string, indx)
```

Fortran command syntax:

```
CALL DSRVAL(STRING, INDX, RVAL, IERR)
```

IPLAN command syntax:

```
CALL DSRVAL(STRING, INDX, RVAL, IERR)
```

Where:

Character STRING	String indicating the dynamics quantity desired (input).	
	'TIME'	Simulation time
	'DELT'	Simulation time step
	'STATE'	State variable values
	'CON'	Real model parameters
	'VAR'	Real model variable values
Integer INDX	Index to dynamics array Ignored for 'TIME' and 'DELT' (input).	
Real RVAL	Real dynamics quantity or array value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 2	'INDX' invalid; 'RVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'RVAL' unchanged

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7.60. FCDAUXMIND

Use this API to return the starting array indices and status of auxiliary signal model associated with FACTS device for the specified signal index.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = fcdauxmind(name, isgndx, string)
```

Fortran command syntax:

```
CALL FCDAUXMIND(NAME, ISGNDX, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL FCDAUXMIND(NAME, ISGNDX, STRING, IVAL, IERR)
```

Where:

Character NAME	FACTS device name (input).	
Integer ISGNDX	Auxiliary signal index (input).	
Character STRING	String indicating the FACTS device auxiliary signal model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. For a model to be called during simulations the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	FACTS device not in network but auxiliary signal model in model tables; 'IVAL' returned
	IERR = 2	FACTS device not in network and auxiliary signal model not in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables for the given auxiliary signal index; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged

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IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.61. FCDAUXMNAM

Use this API to return the auxiliary signal model name associated with a FACTS device for the specified auxiliary signal index.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = fcdauxmnam(name, isgndx)
```

Fortran command syntax:

```
CALL FCDAUXMNAM(NAME, ISGNDX, CVAL, IERR)
```

IPLAN command syntax:

```
CALL FCDAUXMNAM(NAME, ISGNDX, CVAL, IERR)
```

Where:

Character NAME	FACTS device name (input).	
Integer ISGNDX	Auxiliary signal index (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	FACTS device not in network but auxiliary signal model in model tables; 'CVAL' returned
	IERR = 2	FACTS device not in network and auxiliary signal model not in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables for the given auxiliary signal index; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.62. FCDDAT

This API routine is obsolete. It has been replaced by the API routine fcddat_2 .

Use this API to return FACTS device real quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = fcddat(n, string)
```

Fortran command syntax:

```
CALL FCDDAT(N, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL FCDDAT(N, STRING, RVAL, IERR)
```

Where:

Integer N	FACTS device number (input).	
Character STRING	String indicating the FACTS device quantity desired (input).	
	'PDES'	Desired active power flow at terminal end bus in MW
	'O_PDES'	Same as 'PDES' in units determined by the power output option setting
	'QDES'	Desired reactive power flow at terminal end bus in Mvar
	'O_QDES'	Same as 'QDES' in units determined by the power output option setting
	'VSET'	Voltage setpoint at sending end bus
	'SHMX'	Maximum shunt current at sending end bus in MVA at unity voltage
	'O_SHMX'	Same as 'SHMX' in units determined by the power output option setting
	'TRMX'	Maximum bridge real power transfer in MW
	'O_TRMX'	Same as 'TRMX' in units determined by the power output option setting
	'VTMN'	Minimum voltage at terminal end bus in pu
	'VTMX'	Maximum voltage at terminal end bus in pu
	'VSMX'	Maximum series voltage in pu
	'IMX'	Maximum series current in MVA at unity voltage
	'O_IMX'	Same as 'IMX' in units determined by the power output option setting
	'LINX'	Reactance of the dummy series element in pu
	'PSEND'	Active power into series element at sending end bus in MW

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'O_PSEND'	Same as 'PSEND' in units determined by the power output option setting
'QSEND'	Reactive power into series element at sending end bus in Mvar
'O_QSEND'	Same as 'QSEND' in units determined by the power output option setting
'PTERM'	Active power into series element at terminal end bus in MW
'O_PTERM'	Same as 'PTERM' in units determined by the power output option setting
'QTERM'	Reactive power into series element at terminal end bus in Mvar
'O_QTERM'	Same as 'QTERM' in units determined by the power output option setting
'PBRDG'	Active bridge power exchange in MW
'O_PBRDG'	Same as 'PBRDG' in units determined by the power output option setting
'QSHNT'	Reactive power into shunt element in Mvar
'O_QSHNT'	Same as 'QSHNT' in units determined by the power output option setting
'VSERES'	Series voltage magnitude
'MVASND'	Flow into series element at sending bus end in MVA
'O_MVASND'	Same as 'MVASND' in units determined by the power output option setting
'MVATRM'	Flow into series element at terminal bus end in MVA
'O_MVATRM'	Same as 'MVATRM' in units determined by the power output option setting
'MVASHN'	Flow into shunt element in MVA
'O_MVASHN'	Same as 'MVASHN' in units determined by the power output option setting
'IPUSND'	Current flow into series element at sending bus end in pu
'IPUTRM'	Current flow into series element at terminal bus end in pu
'IPUSHN'	Current flow into shunt element in pu
'AMPSND'	Current flow into series element at sending bus end in amps
'AMPTRM'	Current flow into series element at terminal bus end in amps
'AMPSHN'	Current flow into shunt element in amps
'PCTSND'	Percent series current loading at sending bus end
'PCTTRM'	Percent series current loading at terminal bus end

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	'PCTSHN'	Percent shunt current loading
	'PCTPBR'	Percent bridge real power loading
	'RMPCT'	Percent of contributed vars
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 2	FACTS device 'N' not found; 'RVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 4	No base kV at bus; 'RVAL' returned = per unit current
	IERR = 5	For quantities which are solution results, the FACTS device is blocked or out-of-service; "V" of 0.0 returned.

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7.63. FCDDAT_2

This API is the second release of the API that returns FACTS device real quantities.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, rval = fcddat_2(name, string)
```

Fortran command syntax:

```
CALL FCDDAT_2(NAME, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL FCDDAT_2(NAME, STRING, RVAL, IERR)
```

Where:

Character NAME	FACTS device name (input).	
Character STRING	String indicating the FACTS device quantity desired (input).	
	'PDES'	Desired active power flow at terminal end bus in MW
	'O_PDES'	Same as 'PDES' in units determined by the power output option setting
	'QDES'	Desired reactive power flow at terminal end bus in Mvar
	'O_QDES'	Same as 'QDES' in units determined by the power output option setting
	'VSET'	Voltage setpoint at sending end bus
	'SHMX'	Maximum shunt current at sending end bus in MVA at unity voltage
	'O_SHMX'	Same as 'SHMX' in units determined by the power output option setting
	'TRMX'	Maximum bridge real power transfer in MW
	'O_TRMX'	Same as 'TRMX' in units determined by the power output option setting
	'VTMN'	Minimum voltage at terminal end bus in pu
	'VTMX'	Maximum voltage at terminal end bus in pu
	'VSMX'	Maximum series voltage in pu
	'IMX'	Maximum series current in MVA at unity voltage
	'O_IMX'	Same as 'IMX' in units determined by the power output option setting
	'LINX'	Reactance of the dummy series element in pu
	'PSEND'	Active power into series element at sending end bus in MW
	'O_PSEND'	Same as 'PSEND' in units determined by the power output option setting

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'QSEND'	Reactive power into series element at sending end bus in Mvar
'O_QSEND'	Same as 'QSEND' in units determined by the power output option setting
'PTERM'	Active power into series element at terminal end bus in MW
'O_PTERM'	Same as 'PTERM' in units determined by the power output option setting
'QTERM'	Reactive power into series element at terminal end bus in Mvar
'O_QTERM'	Same as 'QTERM' in units determined by the power output option setting
'PBRDG'	Active bridge power exchange in MW
'O_PBRDG'	Same as 'PBRDG' in units determined by the power output option setting
'QSHNT'	Reactive power into shunt element in Mvar
'O_QSHNT'	Same as 'QSHNT' in units determined by the power output option setting
'VSERES'	Series voltage magnitude
'MVASND'	Flow into series element at sending bus end in MVA
'O_MVASND'	Same as 'MVASND' in units determined by the power output option setting
'MVATRM'	Flow into series element at terminal bus end in MVA
'O_MVATRM'	Same as 'MVATRM' in units determined by the power output option setting
'MVASHN'	Flow into shunt element in MVA
'O_MVASHN'	Same as 'MVASHN' in units determined by the power output option setting
'IPUSND'	Current flow into series element at sending bus end in pu
'IPUTRM'	Current flow into series element at terminal bus end in pu
'IPUSHN'	Current flow into shunt element in pu
'AMPSND'	Current flow into series element at sending bus end in amps
'AMPTRM'	Current flow into series element at terminal bus end in amps
'AMPSHN'	Current flow into shunt element in amps
'PCTSND'	Percent series current loading at sending bus end
'PCTTRM'	Percent series current loading at terminal bus end
'PCTSHN'	Percent shunt current loading
'PCTPBR'	Percent bridge real power loading

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	'RMPCT'	Percent of contributed vars
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 2	FACTS device 'NAME' not found; 'RVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 4	No base kV at bus; 'RVAL' returned = per unit current
	IERR = 5	For quantities which are solution results, the FACTS is blocked or out-of-service; "V" of 0.0 returned.

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7.64. FCDINT

This API routine is obsolete. It has been replaced by the API routine fcdint_2 .

Use this API to return FACTS device integer quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = fcdint(n, string)
```

Fortran command syntax:

```
CALL FCDINT(N, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL FCDINT(N, STRING, IVAL, IERR)
```

Where:

Integer N	FACTS device number (input).	
Character STRING	String indicating the FACTS device quantity desired (input).	
	'SEND'	Sending end bus number
	'TERM'	Terminal end bus number
	'REMOTE'	Bus number of remote bus or zero for local control
	'OWNER'	Owner number
	'MODE'	Control mode (negative for blocked during solution)
	'STATVT'	Terminal end voltage condition code returns: 0 Terminal end voltage not constrained. 1 Holding terminal end voltage high limit. -1 Holding terminal end voltage low limit
	'STATVS'	Series voltage limit code returns. 0 Series voltage not constrained. 1 Holding series voltage limit
	'STATBR'	Bridge active power transfer limit code returns. 0 Bridge real power transfer not constrained. 1 Holding bridge real power transfer limit
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 2	FACTS device 'N' not found; 'IVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'IVAL' unchanged

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7.65. FCDINT_2

This API is the second release of the API that returns FACTS device integer quantities.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = fcdint_2(name, string)
```

Fortran command syntax:

```
CALL FCDINT_2(NAME, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL FCDINT_2(NAME, STRING, IVAL, IERR)
```

Where:

Character NAME	FACTS device name (input).	
Character STRING	String indicating the FACTS device quantity desired (input).	
	'SEND'	Sending end bus number
	'TERM'	Terminal end bus number
	'REMOTE'	Bus number of remote bus or zero for local control
	'OWNER'	Owner number
	'MODE'	Control mode (negative for blocked during solution)
	'STATVT'	Terminal end voltage condition code returns: 0 Terminal end voltage not constrained. 1 Holding terminal end voltage high limit. -1 Holding terminal end voltage low limit
	'STATVS'	Series voltage limit code returns: 0 Series voltage not constrained. 1 Holding series voltage limit
	'STATBR'	Bridge active power transfer limit code returns: 0 Bridge real power transfer not constrained. 1 Holding bridge real power transfer limit
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 2	FACTS device 'NAME' not found; 'IVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'IVAL' unchanged

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7.66. FCDMIND

Use this API to return FACTS device model starting array indices and status.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = fcdmind(name, string)
```

Fortran command syntax:

```
CALL FCDMIND(NAME, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL FCDMIND(NAME, STRING, IVAL, IERR)
```

Where:

Character NAME	FACTS device name (input).	
Character STRING	String indicating the FACTS device model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect status of FACTS device. For a model to be called during simulations the FACTS device must be in service the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	FACTS device not in network but in model tables; 'IVAL' returned
	IERR = 2	FACTS device not in network or in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.67. FCDMNAM

Use this API to return FACTS device model name.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = fcdmnam(name)
```

Fortran command syntax:

```
CALL FCDMNAM(NAME, CVAL, IERR)
```

IPLAN command syntax:

```
CALL FCDMNAM(NAME, CVAL, IERR)
```

Where:

Character NAME	FACTS device name (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	FACTS device not in network but in model tables; 'CVAL' returned
	IERR = 2	FACTS device not in network or in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables; 'CVAL' unchanged

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7.68. FXSDT1

Use this API to return the magnitude of a specified fixed bus shunt.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, rval = fxsdt1(ibus, id ,string)
```

Fortran command syntax:

```
CALL FXSDT1(IBUS, ID, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL FXSDT1(IBUS, ID, STRING, RVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Shunt identifier (input).	
Character STRING	String indicating the units to be used (input).	
	'ACT'	Actual shunt. 'RVAL' returned in MVA
	'O_ACT'	Actual shunt. 'RVAL' in units determined by the power output option setting
	'NOM'	Nominal shunt (at 1.0 pu voltage). 'RVAL' returned in MVA
	'O_NOM'	Nominal shunt (at 1.0 pu voltage). 'RVAL' in units determined by the power output option setting
	'PQZERO'	Nominal zero sequence shunt (at 1.0 pu voltage). 'RVAL' returned in per unit
	'PQZ'	Nominal zero sequence shunt (at 1.0 pu voltage). 'RVAL' returned in MVA
	'O_PQZ'	Nominal zero sequence shunt (at 1.0 pu voltage). 'RVAL' returned in units determined by the power output option setting
Real RVAL	Magnitude of the complex shunt in the units indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Shunt not found; 'RVAL' unchanged
	IERR = 3	Bus type code is not 1, 2 or 3; 'RVAL' returned
	IERR = 4	Shunt out-of-service; 'RVAL' returned
	IERR = 5	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 6	Sequence data not in case for 'STRING1' = 'PQZERO', 'PQZ' or 'O_PQZ'; 'RVAL' unchanged

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7.69. FXSDT2

Use this API to return complex fixed bus shunt.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cmpval = fxsdt2(ibus, id ,string)
```

Fortran command syntax:

```
CALL FXSDT2(IBUS, ID, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL FXSDT2(IBUS, ID, STRING, P, Q, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Shunt identifier (input).	
Character STRING	String indicating the units to be used (input).	
	'ACT'	Actual shunt. 'CMPVAL' returned in MW Mvar
	'O_ACT'	Actual shunt. 'CMPVAL' in units determined by the power output option setting
	'NOM'	Nominal shunt (at 1.0 pu voltage). 'CMPVAL' returned in MW Mvar
	'O_NOM'	Nominal shunt (at 1.0 pu voltage). 'CMPVAL' in units determined by the power output option setting
	'PQZERO'	Nominal zero sequence shunt (at 1.0 pu voltage). 'CMPVAL' returned in per unit
	'PQZ'	Nominal zero sequence shunt (at 1.0 pu voltage). 'CMPVAL' returned in MW Mvar
	'O_PQZ'	Nominal zero sequence shunt (at 1.0 pu voltage). 'CMPVAL' returned in units determined by the power output option setting
Real P	Active power component of the complex shunt in the units indicated by STRING (output).	
Real Q	Reactive power component of the complex shunt in the units indicated by STRING (output).	
Complex CMPVAL	Complex shunt in the units indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Shunt not found; 'P' and 'Q' or 'CMPVAL' unchanged

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IERR = 3	Bus type code is not 1, 2 or 3; 'P' and 'Q' or 'CMPVAL' returned
IERR = 4	Shunt out-of-service; 'P' and 'Q' or 'CMPVAL' returned
IERR = 5	Invalid value of 'STRING'; 'P' and 'Q' or 'CMP-VAL' unchanged
IERR = 6	Sequence data not in case for 'STRING1' = 'PQZERO', 'PQZ' or 'O_PQZ'; 'P' and 'Q' or 'CMP-VAL' unchanged

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7.70. FXSINT

Use this API to return integer quantities of the specified fixed bus shunt.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = fxsint(ibus, id, string)
```

Fortran command syntax:

```
CALL FXSINT(IBUS, ID, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL FXSINT(IBUS, ID, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Shunt identifier (input).	
Character STRING	String indicating the shunt quantity desired (input).	
	'STATION'	Bus IBUS substation number (returned as 0 if bus IBUS is not in a substation)
	'SECTION'	Bus IBUS bus section number (returned as 0 if bus IBUS is not in a substation).
	'NODE'	Number of the node to which the fixed shunt is connected (returned as 0 if bus IBUS is not in a substation)
	'STATUS'	Shunt status; 1 (one) for in-service 0 (zero) for out-of-service
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Shunt not found; 'IVAL' unchanged
	IERR = 3	Bus type code is not 1, 2 or 3; 'IVAL' returned
	IERR = 4	Shunt out-of-service; 'IVAL' returned
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged

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7.71. GENCNV

Use this API to return a flag indicating whether generators are converted.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
icode = gencnv()
```

Fortran command syntax:

```
CALL GENCNV(ICODE)
```

IPLAN command syntax:

```
CALL GENCNV(ICODE)
```

Where:

Integer ICODE	0 if generators are converted, otherwise 1 Stand-alone simulator returns a 1 (output).

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7.72. GENDAT

Use this API to return plant total power output.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = gendat(ibus)
```

Fortran command syntax:

```
CALL GENDAT(IBUS, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL GENDAT(IBUS, P, Q, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Real P	Real component of the complex generation (output).	
Real Q	Reactive component of the complex generation (output).	
Complex CMPVAL	Complex generation (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Bus type code is not 2 or 3; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 3	No machines at bus; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 4	No online machines; 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned

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7.73. GENDT1

Returns total plant power output in MVA.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = gendt1(ibus)
```

Fortran command syntax:

```
CALL GENDT1(IBUS, RVAL, IERR)
```

IPLAN command syntax:

```
CALL GENDT1(IBUS, RVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Real RVAL	Magnitude of total plant generation (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Bus type code is not 2 or 3; 'RVAL' returned
	IERR = 3	No machines at bus; 'RVAL' unchanged
	IERR = 4	No online machines; 'RVAL' of 0.0 returned

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7.74. GET_MSTATE

Use this API to return the MSTATE value. MSTATE is meaningful in MODEs one through four and indicates the type of simulation being run by activities STRT/RUN, MSTR/MRUN, ESTR/ERUN, GSTR/GRUN or ASTR.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
n = get_mstate()
```

Fortran command syntax:

```
CALL GET_MSTATE(N)
```

IPLAN command syntax:

```
CALL GET_MSTATE(N)
```

Where:

Integer N	MSTATE value (output).	
	N = -1	No dynamics data is present.
	N = 0	Standard state-space dynamic simulation via activities STRT and RUN, or activity ASTR.
	N = 1	Excitation system response ratio test via activities ESTR and ERUN. Models other than generator and excitation system models are not called.
	N = 2	Excitation system open circuit step response test via activities ESTR and ERUN. Models other than generator and excitation system models are not called.
	N = 3	Governor response test via activities GSTR and GRUN. Models other than generator and turbine governor models are not called.
	N = 4	Extended term dynamic simulation via activities MSTR and MRUN.
	N = 5	Dynamics data is present but no initialization activity has been successfully executed.

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7.75. GNECHR

Use this API routine to return character quantities of a specified GNE device.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, cval = gnechr(name, string)
```

Fortran command syntax:

```
CALL GNECHR(NAME, STRING, CVAL, IERR)
```

IPLAN command syntax:

```
CALL GNECHR(NAME, STRING, CVAL, IERR)
```

Where:

Character NAME	GNE device name (input).	
Character STRING	String indicating the GNE device quantity desired (input).	
	'MODEL'	Name of the BOSL model used by this GNE device
	'CHARn'	nth two-character identifier required by this model; n must be between 1 and the number of character identifiers required by the model to calculate its inputs
Character*16 CVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	GNE device not found; 'CVAL' unchanged
	IERR = 2	Invalid value of 'STRING'; 'CVAL' unchanged
	IERR = 3	For 'STRING' value 'CHARn', n is not valid for this model; 'CVAL' unchanged

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7.76. GNEDAT

Use this API routine to return real quantities of a specified GNE device.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, rval = gnedat(name, string)
```

Fortran command syntax:

```
CALL GNEDAT(NAME, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL GNEDAT(NAME, STRING, RVAL, IERR)
```

Where:

Character NAME	GNE device name (input).	
Character STRING	String indicating the GNE device quantity desired (input).	
	'REALn'	nth real data item required by this model; n must be between 1 and the number of real data items used by the model
	'Pn'	active power in MW flowing into the GNE device at terminal n; n must be between 1 and the number of terminals used by the model
	'Qn'	reactive power in MVAR flowing into the GNE device at terminal n; n must be between 1 and the number of terminals used by the model
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	GNE device not found; 'RVAL' unchanged
	IERR = 2	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 3	For 'STRING' value 'REALn', n is not valid for this model; 'RVAL' unchanged

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7.77. GNEINT

Use this API routine to return integer quantities of a specified GNE device.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, ival = gneint(name, string)
```

Fortran command syntax:

```
CALL GNEINT(NAME, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL GNEINT(NAME, STRING, IVAL, IERR)
```

Where:

Character NAME	GNE device name (input).	
Character STRING	String indicating the GNE device quantity desired (input).	
	'NTERM'	Number of terminals used by this model (1 or 2)
	'BUS1'	Number of the first (or only) bus to which this model is connected
	'BUS2'	Number of the second bus to which this model is connected; 0 is returned for a 1 terminal model
	'STATUS'	Device status
	'OWNER'	Owner to which the device is assigned
	'NMETER'	Number of the non-metered end bus; used for GNE devices with more than one terminal
	'NREAL'	Number of floating point data items required by this model
	'NINTG'	Number of buses required by the model to calculate its inputs
	'NCHAR'	Number of number of two-character identifiers required by the model to calculate its inputs
	'INTGn'	nth bus number required by this model; n must be between 1 and the number of buses required by the model to calculate its inputs
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	GNE device not found; 'IVAL' unchanged
	IERR = 2	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 3	For 'STRING' value 'INTGn', n is not valid for this model; 'IVAL' unchanged

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7.78. INDDT1

Use this API routine to return real induction machine quantities.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, rval = inddt1(ibus, id, string)
```

Fortran command syntax:

```
CALL INDDT1(IBUS, ID, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL INDDT1(IBUS, ID, STRING, RVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Induction machine identifier (input).	
Character STRING	String indicating the induction machine quantity desired (input).	
	'MBASE'	Machine power base
	'RATEKV'	Machine rated voltage
	'PSET'	Power setpoint (positive for a motor)
	'RA'	Armature resistance
	'XA'	Armature leakage reactance
	'R1'	First cage resistance
	'X1'	First cage reactance
	'R2'	Second cage resistance
	'X2'	Second cage reactance
	'X3'	Third reactance
	'E1'	Main saturation parameter
	'SE1'	Main saturation parameter
	'E2'	Main saturation parameter
	'SE2'	Main saturation parameter
	'IA1'	Leakage saturation parameter
	'IA2'	Leakage saturation parameter
	'XAMULT'	Leakage saturation parameter
	'TRQA'	Load torque variation constant
	'TRQB'	Load torque variation constant
	'TRQD'	Load torque variation constant
	'TRQE'	Load torque variation constant
	'H'	Machine inertia constant
	'IRATIO'	Ratio of locked rotor current to rated current
	'ROVERX'	Ratio of Rpos to Xpos

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	'RZERO'	Machine zero sequence fault analysis resistance
	'XZERO'	Machine zero sequence fault analysis reactance
	'RGRND'	Machine zero sequence grounding resistance
	'XGRND'	Machine zero sequence grounding reactance
	'P'	Active power flowing from the bus to the machine in MW
	'O_P'	Same as 'P' in units determined by the power output option setting
	'Q'	Reactive power flowing from the bus to the machine in Mvar
	'O_Q'	Same as 'Q' in units determined by the power output option setting
	'MVA'	Complex power flowing from the bus to the machine in MVA
	'O_MVA'	Same as 'MVA' in units determined by the power output option setting
	'SLIP'	Machine slip
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Induction machine not found; 'RVAL' unchanged
	IERR = 3	Bus type code is not 2 or 3; 'RVAL' returned
	IERR = 4	Induction machine is out-of-service; 'RVAL' returned
	IERR = 5	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 6	Sequence data not in case (when STRING = 'IRATIO', 'ROVERX', 'RZERO', 'XZERO', 'RGRND' or 'XGRND')

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7.79. INDDT2

Use this API routine to return complex induction machine quantities.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, cmpval = inddt2(ibus, id, string)
```

Fortran command syntax:

```
CALL INDDT2(IBUS, ID, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL INDDT2(IBUS, ID, STRING, P, Q, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Induction machine identifier (input).	
Character STRING	String indicating the induction machine quantity desired (input).	
	'ZA'	Armature impedance (Ra + j Xa)
	'Z1'	First cage impedance (R1 + j X1)
	'Z2'	Second cage impedance (R2 + j X2)
	'ZZERO'	Machine zero sequence fault analysis impedance
	'ZGRND'	Machine zero sequence grounding impedance
	'PQ'	Actual power flowing from the bus to the machine in MW Mvar
	'O_PQ'	Same as 'PQ' in units determined by the power output option setting
Real P	Real component of the complex value indicated by STRING	
Real Q	Reactive component of the complex value indicated by STRING	
Complex CMPVAL	Complex value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Induction machine not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Bus type code is not 2 or 3; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 4	Induction machine is out-of-service; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 5	Invalid value of 'STRING'; 'P' and 'Q' or 'CMPVAL' unchanged

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IERR = 6	Sequence data not in case (when STRING = 'ZZERO' or ZGRND')

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7.80. INDINT

Use this API routine to return integer induction machine quantities.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, ival = indint(ibus, id, string)
```

Fortran command syntax:

```
CALL INDINT(IBUS, ID, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL INDINT(IBUS, ID, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Induction machine identifier (input).	
Character STRING	String indicating the induction machine quantity desired (input).	
	'STATUS'	Machine status; 1 (one) for in-service 0 (zero) for out-of-service
	'SCODE'	Machine standard type; 1 for NEMA 2 for IEC)
	'DCODE'	Machine design code; 0 for custom 1 through 5 for NEMA designs A through E
	'AREA'	Area number of its area
	'ZONE'	Zone number of its zone
	'OWNER'	Owner number of its owner
	'TCODE'	Type of mechanical torque variation; 1 for a simple power law 2 for the WECC model)
	'BCODE'	Base power code; 1 for mechanical power (MW) output of the machine 2 for apparent electrical power (MVA) drawn by the machine
	'PCODE'	Scheduled power code; 1 for mechanical power (MW) output of the machine 2 for electrical active power (MW) drawn by the machine
	'CZG'	Grounding impedance data input/outout (I/O) code (1 for per unit 2 for ohms)
	'CGR'	Grounding code (0 for an ungrounded machine 1 for a grounded machine)
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged

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IERR = 2	Induction machine not found; 'IVAL' unchanged
IERR = 3	Bus type code is not 2 or 3; 'IVAL' returned
IERR = 4	Induction machine is out-of-service, 'IVAL' returned
IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
IERR = 6	Sequence data not in case (when STRING = 'CZG' or 'CGR')

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7.81. INI2DC

Initializes the two-terminal dc line fetching routine 'NXT2DC' for retrieving two-terminal dc lines in dc line name alphabetical order.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr = ini2dc()
```

Fortran command syntax:

```
CALL INI2DC(IERR)
```

IPLAN command syntax:

```
CALL INI2DC(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	No error, 'NXT2DC' is properly initialized
	IERR = 1	No two-terminal dc lines in case; 'NXT2DC' may not be used

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7.82. INIBRN

Initializes the branch fetching routine 'NXTBRN' or 'NXTBRN3' for returning branches connected to IBUS. For multisection lines, 'NXTBRN' or 'NXTBRN3' returns the adjacent dummy bus as the to bus.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = inibrn(ibus, single)
```

Fortran command syntax:

```
CALL INIBRN(IBUS, SINGLE, IERR)
```

IPLAN command syntax:

```
CALL INIBRN(IBUS, SINGLE, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer SINGLE	If 1, 'NXTBRN' or 'NXTBRN3' will return only branches for which the from bus ('IBUS') number (or bus name) is lower than the to bus (or buses); if 2, then all branches connected to 'IBUS' are returned by 'NXTBRN' or 'NXTBRN3' (input).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'NXTBRN/NXTBRN3' is properly initialized
	IERR = 1	Bus not found; 'NXTBRN/NXTBRN3' may not be used
	IERR = 2	Invalid 'SINGLE' value; 'NXTBRN/NXTBRN3' may not be used

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7.83. INIBRX

Initializes the branch fetching routine, 'NXTBRN' or 'NXTBRN3' for returning branches connected to IBUS. For multisection lines, 'NXTBRN' or 'NXTBRN3' returns the other endpoint bus as the to bus.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = inibrx(ibus, single)
```

Fortran command syntax:

```
CALL INIBRX(IBUS, SINGLE, IERR)
```

IPLAN command syntax:

```
CALL INIBRX(IBUS, SINGLE, IERR)
```

Where:

Integer IBUS	From bus (input).	
Integer SINGLE	If 1, 'NXTBRN' or 'NXTBRN3' will return only branches for which the from bus ('IBUS') number (or bus name) is lower than the to bus (or buses); if 2, then all branches connected to 'IBUS' are returned by 'NXTBRN' or 'NXTBRN3' (input).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTBRN/NXTBRN3' properly initialized
	IERR = 1	Bus not found; 'NXTBRN/NXTBRN3' may not be used
	IERR = 2	Invalid 'SINGLE' value; 'NXTBRN/NXTBRN3' may not be used
	IERR = 3	'IBUS' is a multi-section line dummy bus; 'NXTBRN/NXTBRN3' may not be used

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7.84. INIBUS

Initializes the bus fetching routine 'NXTBUS' for retrieving buses in ascending numerical order. Dummy buses for multisection lines are included. Hidden star point buses of three-winding transformers are excluded.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = inibus(ibus)
```

Fortran command syntax:

```
CALL INIBUS(IBUS, IERR)
```

IPLAN command syntax:

```
CALL INIBUS(IBUS, IERR)
```

Where:

Integer IBUS	starting bus number, 0 for first bus in ordered list (input).	
Integer IERR	error code (output).	
	IERR = 0	no error; 'NXTBUS' is properly initialized
	IERR = 1	bus not found; 'NXTBUS' may not be used

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7.85. INIBUX

Initializes the bus fetching routine 'NXTBUS' for retrieving buses in ascending numerical order. Dummy buses of multisection lines are excluded. Hidden star point buses of three-winding transformers are excluded.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = inibux(ibus)
```

Fortran command syntax:

```
CALL INIBUX(IBUS, IERR)
```

IPLAN command syntax:

```
CALL INIBUX(IBUS, IERR)
```

Where:

Integer IBUS	Starting bus number, 0 for first bus in ordered list (input).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTBUS' properly initialized
	IERR = 1	Bus not found; 'NXTBUS' may not be used

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7.86. INIFAX

Initializes the FACTS device fetching routine 'NXTFAX' for retrieving FACTS devices in FACTS device name alphabetical order.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr = inifax()
```

Fortran command syntax:

```
CALL INIFAX(IERR)
```

IPLAN command syntax:

```
CALL INIFAX(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTFAX' is properly initialized
	IERR = 1	No FACTS devices in case; 'NXTFAX' may not be used

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7.87. INIFXS

Initializes the fixed bus shunt fetching routine 'NXTFXS' for returning fixed shunts attached to bus IBUS.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr = inifxs(ibus)
```

Fortran command syntax:

```
CALL INIFXS(IBUS, IERR)
```

IPLAN command syntax:

```
CALL INIFXS(IBUS, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'NXTFXS' is properly initialized
	IERR = 1	Bus not found; 'NXTFXS' may not be used
	IERR = 2	Bus has no fixed shunts; 'NXTFXS' may not be used

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7.88. INIIND

Initializes induction machine fetching routine 'NXTIND' for returning induction machines attached to IBUS.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr = iniind(ibus)
```

Fortran command syntax:

```
CALL INIIND(IBUS, IERR)
```

IPLAN command syntax:

```
CALL INIIND(IBUS, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'NXTIND' is properly initialized
	IERR = 1	Bus not found; 'NXTIND' may not be used
	IERR = 2	Bus has no induction machines connected to it; 'NXTIND' may not be used

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7.89. INILOD

Initializes load fetching routine 'NXTLOD' for returning loads attached to IBUS.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = inilod(ibus)
```

Fortran command syntax:

```
CALL INILOD(IBUS, IERR)
```

IPLAN command syntax:

```
CALL INILOD(IBUS, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'NXTLOD' is properly initialized
	IERR = 1	Bus not found; 'NXTLOD' may not be used
	IERR = 2	Bus has no load entries; 'NXTLOD' may not be used

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7.90. INIMAC

Initializes machine fetching routine 'NXTMAC' for returning machines attached to IBUS.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = inimac(ibus)
```

Fortran command syntax:

```
CALL INIMAC(IBUS, IERR)
```

IPLAN command syntax:

```
CALL INIMAC(IBUS, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'NXTMAC' is properly initialized
	IERR = 1	Bus not found; 'NXTMAC' may not be used
	IERR = 2	Bus not a generator bus; 'NXTMAC' may not be used

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7.91. INIMDC

Initializes the multi-terminal dc line fetching routine 'NXTMDC' for retrieving multi-terminal dc lines in dc line name alphabetical order.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr = inimdc()
```

Fortran command syntax:

```
CALL INIMDC(IERR)
```

IPLAN command syntax:

```
CALL INIMDC(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTMDC' is properly initialized
	IERR = 1	No multi-terminal dc lines in case; 'NXTMDC' may not be used

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7.92. INIMSL

Initializes the multi-section line member fetching routine 'NXTMSL' for retrieving lines that are sections of the specified multi-section line from IBUS to JBUS.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = inimsl(ibus, jbus, ickt)
```

Fortran command syntax:

```
CALL INIMSL(IBUS, JBUS, ICKT, IERR)
```

IPLAN command syntax:

```
CALL INIMSL(IBUS, JBUS, ICKT, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*2 ICKT	Multi-section line identifier (must begin with &) (input).	
Integer IERR	error code (output).	
	IERR = 0	No error; NXTMSL properly initialized
	IERR = 1	Bus not found; NXTMSL may not be used
	IERR = 2	& not the first character of 'ICKT'; NXTMSL may not be used
	IERR = 3	Multi-section line not found; NXTMSL may not be used

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7.93. ININAM

Initializes the bus fetching routine 'NXTBUS' for retrieving buses in ascending alphabetical order. Dummy buses for multisection lines are included. Hidden star point buses of three winding transformers are excluded.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = ininam(string)
```

Fortran command syntax:

```
CALL ININAM(STRING, IERR)
```

IPLAN command syntax:

```
CALL ININAM(STRING, IERR)
```

Where:

Character STRING	Extended bus name, blank for first bus in ordered list (input).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTBUS' is properly initialized
	IERR = 1	Bus not found; 'NXTBUS' may not be used

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7.94. ININAX

Initializes the bus fetching routine 'NXTBUS' for retrieving buses in ascending alphabetical order. Dummy buses of multisection lines are excluded. Hidden star point buses of three winding transformers are excluded.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = ininax(string)
```

Fortran command syntax:

```
CALL ININAX(STRING, IERR)
```

IPLAN command syntax:

```
CALL ININAX(STRING, IERR)
```

Where:

Character STRING	Extended bus name, blank for first bus in ordered list (input).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTBUS' properly initialized
	IERR = 1	Bus not found; 'NXTBUS' may not be used

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7.95. INITIE

Initializes the tie branch fetching routine 'NXTTIE' or 'NXTTIE3' for returning tie branches from area IAR. For multisection lines connected to a from area bus, the adjacent dummy bus is treated as the to bus. Dummy buses for multisection lines are included.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = initie(iar)
```

Fortran command syntax:

```
CALL INITIE(IAR, IERR)
```

IPLAN command syntax:

```
CALL INITIE(IAR, IERR)
```

Where:

Integer IAR	From area number (input).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTTIE' or 'NXTTIE3' is properly initialized
	IERR = 1	Area number <= 0 or > largest allowable area number; 'NXTTIE' or 'NXTTIE3' may not be used
	IERR = 2	No buses in area 'IAR'; 'NXTTIE' or 'NXTTIE3' may not be used

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7.96. INITIX

Initializes the tie branch fetching routine 'NXTTIE' or 'NXTTIE3' for returning tie branches from area IAR. For multisection lines connected to a from area bus, the other endpoint bus is treated as the to bus. Dummy buses of multisection lines are excluded.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = initix(iar)
```

Fortran command syntax:

```
CALL INITIX(IAR, IERR)
```

IPLAN command syntax:

```
CALL INITIX(IAR, IERR)
```

Where:

Integer IAR	From area number (input).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTTIE' or 'NXTTIE3' is properly initialized
	IERR = 1	Area number <= 0 or > largest allowable area number; 'NXTTIE' or 'NXTTIE3' may not be used
	IERR = 2	No buses in area 'IAR'; 'NXTTIE' or 'NXTTIE3' may not be used

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7.97. INIVSC

Initializes the VSC dc line fetching routine 'NXTVSC' for retrieving VSC dc lines in dc line name alphabetical order.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr = inivsc()
```

Fortran command syntax:

```
CALL INIVSC(IERR)
```

IPLAN command syntax:

```
CALL INIVSC(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	No error, 'NXTVSC' is properly initialized
	IERR = 1	No VSC dc lines in case; 'NXTVSC' may not be used

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7.98. ISLAND

Trips in-service branches connected to Type 4 (or greater) buses, and disconnects islands not containing a swing bus.
Note: Calling ISLAND with generators converted will disconnect all buses.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = island()
```

Fortran command syntax:

```
CALL ISLAND(IERR)
```

IPLAN command syntax:

```
CALL ISLAND(IERR)
```

Note:

Calling ISLAND with generators converted will disconnect all buses.

Where:

Integer IERR	error code (output).	
	IERR = 0	No errors.
	IERR = -1	No in-service buses remaining.
	IERR > 0	Number of islands tripped.

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7.99. ITERAT

Use this API to return the number of iterations used in the last solution attempt.

This API routine was first introduced in release 30.1.0.

Python command syntax:

```
ival = iterat()
```

Fortran command syntax:

```
CALL ITERAT(IVAL)
```

IPLAN command syntax:

```
CALL ITERAT(IVAL)
```

Where:

Integer IVAL	Number of iterations used in the last solution attempt (output).

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7.100. LMODIND

Returns load-related model starting array indices and status.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = lmodind(ibus, id ,string1, string2)
```

Fortran command syntax:

```
CALL LMODIND(IBUS, ID, STRING1, STRING2, IVAL, IERR)
```

IPLAN command syntax:

```
CALL LMODIND(IBUS, ID, STRING1, STRING2, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Load identifier (input).	
Character STRING1	String indicating the type of load model desired (input).	
	'CHARAC'	Load characteristic model
	'RELAY'	Load relay model
Character STRING2	String indicating the load model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'RESICON'	Starting reserved ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'NRESICON'	Number of reserved ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect load status. For a model to be called during simulations the load must be in-service bus type code must be 1 or 2 and the model's bypassed flag must be not bypassed
Integer IVAL	Load model array index or status (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Load not in network but in model tables; 'IVAL' returned

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IERR = 2	Load not in network or in model tables; 'IVAL' unchanged
IERR = 3	Invalid value of 'STRING1'; 'IVAL' unchanged
IERR = 4	No model of type 'STRING1' for this load; 'IVAL' unchanged
IERR = 5	Invalid value of 'STRING2'; 'IVAL' unchanged
IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.101. LMODNAM

Returns load-related model name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = lmodnam(ibus, id ,string)
```

Fortran command syntax:

```
CALL LMODNAM(IBUS, ID, STRING, CVAL, IERR)
```

IPLAN command syntax:

```
CALL LMODNAM(IBUS, ID, STRING, CVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Load identifier (input).	
Character STRING	String indicating the type of load model desired (input).	
	'CHARAC'	Load characteristic model
	'RELAY'	Load relay model
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Load not in network but in model tables; 'CVAL' returned
	IERR = 2	Load not in network or in model tables; 'CVAL' unchanged
	IERR = 3	Invalid value of 'STRING'; 'CVAL' unchanged
	IERR = 4	No model of type 'STRING' for this load; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.102. LODCNV

This obsolete API always returns a 0.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
icode = lodcnv()
```

Fortran command syntax:

```
CALL LODCNV(ICODE)
```

IPLAN command syntax:

```
CALL LODCNV(ICODE)
```

Where:

Integer ICODE	Returns a 0 (output).

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7.103. LODDT1

Returns load quantities in MVA.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = loddt1(ibus, id ,string1, string2)
```

Fortran command syntax:

```
CALL LODDT1(IBUS, ID, STRING1, STRING2, RVAL, IERR)
```

IPLAN command syntax:

```
CALL LODDT1(IBUS, ID, STRING1, STRING2, RVAL, IERR)
```

Note:

When STRNG1 is one of the following: 'MVA', 'IL', 'YL', 'TOTAL', the values returned are the net load values (i.e., value includes load plus contribution from any in-service distributed generation on the load feeder at that bus)

Where:

Integer IBUS	bus number (input).	
Character ID	load identifier (input).	
Character STRING1	string indicating the load characteristic desired (input).	
	'MVA'	constant MVA load
	'IL'	constant current load
	'YL'	constant admittance load
	'TOTAL'	total load
	'YNEG'	exceptional nominal negative sequence load
	'YZERO'	exceptional nominal zero sequence load
Character STRING2	string indicating the units to be used (input).	
	'ACT'	actual load (nominal load for STRING1 = 'YNEG' or 'YZERO'). 'P' 'Q' returned in MW Mvar
	'O_ACT'	actual load (nominal load for STRING1 = 'YNEG' or 'YZERO'). 'P' 'Q' in units determined by the power output option setting
	'NOM'	nominal load (at 1.0 pu voltage). 'RVAL' returned in MVA
	'O_NOM'	nominal load (at 1.0 pu voltage). 'RVAL' in units determined by the power output option setting
Real RVAL	magnitude of the complex value indicated by STRING1 in the units indicated by STRING2 (output).	
Integer IERR	error code (output).	

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IERR = 0	no error; 'RVAL' returned
IERR = 1	Bbus not found; 'RVAL' unchanged
IERR = 2	Load not found; 'RVAL' unchanged
IERR = 3	Bbus type code is not 1, 2 or 3; 'RVAL' returned
IERR = 4	Lload out-of-service; 'RVAL' returned
IERR = 5	Iinvalid value of 'STRING1' or 'STRING2'; 'RVAL' unchanged
IERR = 6	Ssequence data not in case for 'STRING1' = 'YNEG' or 'YZERO'; 'RVAL' unchanged

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7.104. LODDT2

Use this API to return complex load quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = loddt2(ibus, id ,string1, string2)
```

Fortran command syntax:

```
CALL LODDT2(IBUS, ID, STRING1, STRING2, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL LODDT2(IBUS, ID, STRING1, STRING2, P, Q, IERR)
```

Note:

When STRNG1 is one of the following: 'MVA', 'IL', 'YL', 'TOTAL', the values returned are the net load values (i.e., value includes load plus the contribution from any in-service distributed generation on the load feeder at that bus)

Where:

Integer IBUS	bus number (input).	
Character ID	load identifier (input).	
Character STRING1	string indicating the load characteristic desired (input).	
	'MVA'	constant MVA load
	'IL'	constant current load
	'YL'	constant admittance load
	'TOTAL'	total load
	'YNEG'	exceptional nominal negative sequence load
	'YZERO'	exceptional nominal zero sequence load
Character STRING2	String indicating the units to be used (input).	
	'ACT'	actual load (nominal load for STRING1 = 'YNEG' or 'YZERO'). 'P' 'Q' returned in MW Mvar
	'O_ACT'	actual load (nominal load for STRING1 = 'YNEG' or 'YZERO'). 'P' 'Q' in units determined by the power output option setting
	'NOM'	nominal load (at 1.0 pu voltage). 'P' 'Q' returned in MW Mvar
	'O_NOM'	nominal load (at 1.0 pu voltage). 'P' 'Q' in units determined by the power output option setting
Real P	real component of complex value indicated by STRING1 in the units indicated by STRING2	
Real Q	reactive component of complex value indicated by STRING1 in the units indicated by STRING2	

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Complex CMPVAL	complex value indicated by STRING1 in the units indicated by STRING2 (output).	
Integer IERR	error code (output).	
	IERR = 0	no error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	load not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	bus type code is not 1, 2 or 3; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 4	load out-of-service; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 5	invalid value of 'STRING1' or 'STRING2'; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 6	sequence data not in case for 'STRING1' = 'YNEG' or 'YZERO'; 'CMPVAL' unchanged

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7.105. LODIND

Use this API to return the load array index.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = lodind(ibus, id)
```

Fortran command syntax:

```
CALL LODIND(IBUS, ID, IVAL, IERR)
```

IPLAN command syntax:

```
CALL LODIND(IBUS, ID, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Load identifier (input).	
Integer IVAL	Load array index (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Load not found; 'IVAL' unchanged

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7.106. LODINT

Use this API to return integer load quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = lodint(ibus, id, string)
```

Fortran command syntax:

```
CALL LODINT(IBUS, ID, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL LODINT(IBUS, ID, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Load identifier (input).	
Character STRING	String indicating the load quantity desired (input).	
	'STATION'	Bus IBUS substation number (returned as 0 if bus IBUS is not in a substation)
	'SECTION'	Bus IBUS bus section number (returned as 0 if bus IBUS is not in a substation)
	'STATUS'	Load status; 1 (one) for in-service else 0 (zero)
	'AREA'	Load area number
	'ZONE'	Load zone number
	'OWNER'	Load owner number
	'SCALE'	Scalable load flag
	'CGR'	Load grounding code (0 for ungrounded 1 for grounded)
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Load not found; 'IVAL' unchanged
	IERR = 3	Bus type code is not 1, 2 or 3; 'IVAL' returned
	IERR = 4	Load out-of-service; 'IVAL' returned
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Sequence data not in case for 'STRING1' = 'CGR'; 'IVAL' unchanged

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7.107. MACDAT

Use this API to return real machine quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = macdat(ibus, id, string)
```

Fortran command syntax:

```
CALL MACDAT(IBUS, ID, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL MACDAT(IBUS, ID, STRING, RVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Machine identifier (input).	
Character STRING	String indicating the machine quantity desired (input).	
	'QMAX'	Maximum generator reactive output in Mvar
	'O_QMAX'	Same as 'QMAX' in units determined by the power output option setting
	'QMIN'	Minimum generator reactive output in Mvar
	'O_QMIN'	Same as 'QMIN' in units determined by the power output option setting
	'PMAX'	Maximum generator active output in MW
	'O_PMAX'	Same as 'PMAX' in units determined by the power output option setting
	'PMIN'	Minimum generator real active in MW
	'O_PMIN'	Same as 'PMIN' in units determined by the power output option setting
	'MBASE'	Total MVA base
	'MVA'	Machine loading in MVA
	'O_MVA'	Same as 'MVA' in units determined by the power output option setting
	'P'	Machine loading in MW
	'O_P'	Same as 'P' in units determined by the power output option setting
	'Q'	Machine loading in Mvar
	'O_Q'	Same as 'Q' in units determined by the power output option setting
	'PERCENT'	MVA loading of the machines as a percentage of machine base
	'GENTAP'	Step-up transformer off-nominal turns ratio
	'VSCHED'	Regulated voltage setpoint

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	'WPF'	'WPF Power factor used in setting reactive power limits for this wind machine when WMOD is 2 or 3
	'FRACT1'	Fraction of total ownership assigned to the first owner
	'FRACT2'	Fraction of total ownership assigned to the second owner
	'FRACT3'	Fraction of total ownership assigned to the third owner
	'FRACT4'	Fraction of total ownership assigned to the fourth owner
	'RMPCT'	Percentage of total MVAR required to regulate remote bus voltage
	'RPOS'	Positive sequence fault analysis machine resistance
	'XSUBTR'	Positive sequence fault analysis machine subtransient reactance
	'XTRANS'	Positive sequence fault analysis machine transient reactance
	'XSYNCH'	Positive sequence fault analysis machine synchronous reactance
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Machine not found; 'RVAL' unchanged
	IERR = 3	Bus type code is not 2 or 3; 'RVAL' returned
	IERR = 4	Machine off-line; 'RVAL' returned
	IERR = 5	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 6	For STRING = 'FRACTn', n > number of owners; 'RVAL' unchanged
	IERR = 7	For STRING = 'WPF', this machine is not a wind machine; 'RVAL' of 1.0 returned
	IERR = 8	Sequence data not in case for STRING = 'RPOS', 'XSUBTR', 'XTRANS', or 'XSYNCH'; 'RVAL' unchanged

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7.108. MACDT2

Use this API to return complex machine quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = macdt2(ibus, id, string)
```

Fortran command syntax:

```
CALL MACDT2(IBUS, ID, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL MACDT2(IBUS, ID, STRING, P, Q, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Machine identifier (input).	
Character STRING	String indicating the machine quantity desired (input).	
	'PQ'	Actual generator power output in MW/Mvar
	'O_PQ'	Same as 'PQ' in units determined by the power output option setting
	'ZSORCE'	Machine impedance
	'XTRAN'	Step-up transformer impedance
	'ZPOS'	Positive sequence fault analysis machine impedance (Rpos + j X)
	'ZNEG'	Negative sequence machine impedance
	'ZZERO'	Zero sequence machine impedance
	'ZGRND'	Zero sequence grounding impedance in per unit or ohms according to the machine's grounding impedance data input/output code
	'ZGRNDPU'	Zero sequence grounding impedance in per unit
Real P	Real component of the complex value indicated by STRING (output).	
Real Q	Reactive component of the complex value indicated by STRING (output).	
Complex CMPVAL	Complex value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Machine not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Bus type code is not 2 or 3; 'P' and 'Q' or 'CMPVAL' returned

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IERR = 4	Machine off-line; 'P' and 'Q' or 'CMPVAL' returned
IERR = 5	Invalid value of 'STRING'; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 6	Sequence data not in case for STRING = 'ZPOS', 'ZNEG', or 'ZZERO', 'ZGRND', or 'ZGRNDPU'; 'P' and 'Q' or 'CMPVAL' unchanged

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7.109. MACIND

Use this API to return the machine array index.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = macind(ibus, id)
```

Fortran command syntax:

```
CALL MACIND(IBUS, ID, IVAL, IERR)
```

IPLAN command syntax:

```
CALL MACIND(IBUS, ID, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Machine identifier (input).	
Integer IVAL	Machine array index (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Machine not found; 'IVAL' unchanged

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7.110. MACINT

Use this API to return integer machine quantities.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = macint(ibus, id, string)
```

Fortran command syntax:

```
CALL MACINT(IBUS, ID, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL MACINT(IBUS, ID, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Machine identifier (input).	
Character STRING	String indicating the machine quantity desired (input).	
	'STATION'	Bus IBUS substation number (returned as 0 if bus IBUS is not in a substation)
	'SECTION'	Bus IBUS bus section number (returned as 0 if bus IBUS is not in a substation)
	'STATUS'	Machine status; 1 (one) for in-service else 0 (zero)
	'IREG'	Bus number of remote regulated bus; zero for none
	'OWNERS'	Number of owners
	'WMOD'	Wind machine reactive power limits mode; 0 if this machine is not a wind machine
	'OWN1'	Owner number of the first owner
	'OWN2'	Owner number of the second owner
	'OWN3'	Owner number of the third owner
	'OWN4'	Owner number of the fourth owner
	'PERCENT'	MVA loading of the machines as a percentage of machine base
	'CZG'	Grounding impedance data input/output code
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Machine not found; 'IVAL' unchanged
	IERR = 3	Bus type code is not 2 or 3; 'IVAL' returned
	IERR = 4	Machine off-line, 'IVAL' returned

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IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
IERR = 6	For STRING = 'OWNn', n> number of owners; 'IVAL' unchanged
IERR = 7	Sequence data not in case for STRING = 'CZG'; 'IVAL' unchanged

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7.111. MACSTT

Use this API to return the machine status value.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = macstt(ibus, id)
```

Fortran command syntax:

```
CALL MACSTT(IBUS, ID, IVAL, IERR)
```

IPLAN command syntax:

```
CALL MACSTT(IBUS, ID, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Machine identifier (input).	
Integer IVAL	Machine status value; 0 for out of service, 1 for in service (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Machine not found; 'IVAL' unchanged

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7.112. MAXMSM

Use this API to return the complex bus mismatch at the bus with the largest MVA mismatch.

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, ibus, cmpval = maxmsm()
```

Fortran command syntax:

```
CALL MAXMSM(IBUS, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL MAXMSM(IBUS, P, Q, IERR)
```

Where:

Integer IBUS	Bus number of the bus with the largest MVA mismatch (input).	
Real P	Real component (MW) of the bus mismatch (output).	
Real Q	Reactive component (Mvar) of the bus mismatch (output).	
Complex CMPVAL	Complex bus mismatch (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IBUS', and 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	No in-service buses found; 'IBUS' of 0, and 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned

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7.113. MDLIND

Use this API to return plant-related model starting array indices and status.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = mdlind(ibus, id, string1, string2)
```

Fortran command syntax:

```
CALL MDLIND(IBUS, ID, STRING1, STRING2, IVAL, IERR)
```

IPLAN command syntax:

```
CALL MDLIND(IBUS, ID, STRING1, STRING2, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Machine identifier (input).	
Character STRING1	String indicating the type of plant-related model (input).	
	'GEN'	Generator model
	'COMP'	Compensator model
	'STAB'	Stabilizer model
	'EXC'	Excitation system model
	'GOV'	Turbine governor model
	'TLC'	Turbine Load Controller model
	'MINXL'	Minimum excitation limiter model
	'MAXXL'	Maximum excitation limiter model
Character STRING2	String indicating the plant-related model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect machine status. For a model to be called during simulations the machine must be in service the type code of the bus must be 2 and the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING Integer (output).	
Integer IERR	error code (output).	

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IERR = 0	No error; 'IVAL' returned
IERR = 1	Machine not in network but in model tables; 'IVAL' returned
IERR = 2	Machine not in network or in model tables; 'IVAL' unchanged
IERR = 3	Invalid value of 'STRING1', 'IVAL' unchanged
IERR = 4	No model of type 'STRING1' for this machine; 'IVAL' unchanged
IERR = 5	Invalid value of 'STRING2'; 'IVAL' unchanged
IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.114. MDLLIBCNT

Use this API to return the number of libraries in the list to be searched for dynamics library models.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
nstr = mdllibcnt()
```

Fortran command syntax:

```
CALL GETMODELLIBRARYCOUNT(NSTR)
```

IPLAN command syntax:

```
CALL MDLLIBCNT(NSTR)
```

Where:

Integer NSTR	number of libraries in the list (output).

7.115. MDLNAM

Use this API to return plant-related model name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = mdlnam(ibus, id, string)
```

Fortran command syntax:

```
CALL MDLNAM(IBUS, ID, STRING, CVAL, IERR)
```

IPLAN command syntax:

```
CALL MDLNAM(IBUS, ID, STRING, CVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Machine identifier (input).	
Character STRING	String indicating the type of plant-related model (input).	
	'GEN'	Generator model
	'COMP'	Compensator model
	'STAB'	Stabilizer model
	'EXC'	Excitation system model
	'GOV'	Turbine governor model
	'TLC'	Turbine Load Controller model
	'MINXL'	Minimum excitation limiter model
	'MAXXL'	Maximum excitation limiter model
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Machine not in network but in model tables; 'CVAL' returned
	IERR = 2	Machine not in network or in model tables; 'CVAL' unchanged
	IERR = 3	Invalid value of 'STRING', 'CVAL' unchanged
	IERR = 4	No model of type 'STRING' for this machine; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.116. MDLPGMCNT

Use this API to returns the number of model programs loaded from dynamics model libraries.

This API routine was first introduced in release 33.3.0.

Python command syntax:

```
nsiz = mdlpgmcnt()
```

Fortran command syntax:

```
CALL GETMODELPROGRAMCOUNT(NSIZ)
```

IPLAN command syntax:

```
CALL MDLPGMCNT(NSIZ)
```

Where:

Integer NSIZ	number of programs loaded (output).

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7.117. NATONO

Use this API to return the bus number for a specified 18-character extended bus name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ibus = natono(string)
```

Fortran command syntax:

```
CALL NATONO(STRING, IBUS, IERR)
```

IPLAN command syntax:

```
CALL NATONO(STRING, IBUS, IERR)
```

Where:

Character STRING	Extended bus name (input).	
Integer IBUS	Bus number (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IBUS' returned
	IERR = 1	Bus not found; 'IBUS' unchanged

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7.118. NOTONA

Use this API to return the bus 18-character extended bus name for a specified bus number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = notona(ibus)
```

Fortran command syntax:

```
CALL NOTONA(IBUS, CVAL, IERR)
```

IPLAN command syntax:

```
CALL NOTONA(IBUS, CVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*18 CVAL	Extended bus name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Bus not found; 'CVAL' unchanged
	IERR = 2	'CVAL' not long enough to contain entire extended bus name; 'CVAL' unchanged

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7.119. NXT2DC

Returns the next two-terminal dc line in dc line name alphabetical order. The routine INI2DC must be called first.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, string = nxt2dc()
```

Fortran command syntax:

```
CALL NXT2DC(STRING, IERR)
```

IPLAN command syntax:

```
CALL NXT2DC(STRING, IERR)
```

Where:

Character*12 STRING	Two-terminal dc line name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'STRING' returned
	IERR = 1	No more two-terminal dc lines; 'STRING' unchanged
	IERR = 2	'INI2DC' was not called; 'STRING' unchanged
	IERR = 3	'STRING' not long enough to contain entire dc line name; 'STRING' returned but contains a truncated dc line name

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7.120. NXTBRN

Returns the next branch connected to a bus, excluding three-winding transformers. Each time 'NXTBRN' is called it returns the next branch in either numeric or alphabetic ascending order, depending on the PSSE bus output option setting. 'INIBRN' or 'INIBRX' must first be called to establish the from bus number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, jbus, ickt = nxtbrn(ibus)
```

Fortran command syntax:

```
CALL NXTBRN(IBUS, JBUS, ICKT, IERR)
```

IPLAN command syntax:

```
CALL NXTBRN(IBUS, JBUS, ICKT, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (output).	
Character*2 ICKT	Circuit identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'JBUS' and 'ICKT' returned
	IERR = 1	No more branches from 'IBUS'; 'JBUS' is returned set to 0, and 'ICKT' is unchanged
	IERR = 2	'INIBRN' or 'INIBRX' was not called for bus 'IBUS'; 'JBUS', and 'ICKT' are unchanged

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7.121. NXTBRN3

Returns the next branch connected to a bus, including three-winding transformers. Each time 'NXTBRN3' is called it returns the next branch in either numeric or alphabetic ascending order, depending on the PSSE bus output option setting. 'INIBRN' or 'INIBRX' must first be called to establish the from bus number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, jbus, kbus, ickt = nxtbrn3(ibus)
```

Fortran command syntax:

```
CALL NXTBRN3(IBUS, JBUS, KBUS, ICKT, IERR)
```

IPLAN command syntax:

```
CALL NXTBRN3(IBUS, JBUS, KBUS, ICKT, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (output).	
Integer KBUS	Third bus number (output).	
Character*2 ICKT	Circuit identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'JBUS' and 'ICKT' returned
	IERR = 1	No more branches from 'IBUS'; 'JBUS' & 'KBUS' are returned set to 0, and 'ICKT' is unchanged
	IERR = 2	'INIBRN' or 'INIBRX' was not called for bus 'IBUS'; 'JBUS', 'KBUS', and 'ICKT' are unchanged

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7.122. NXTBUS

Returns the next bus in ordered sequence. One of the routines INIBUS, INIBUX, ININAM or ININAX must be called first. The sequence can also be affected by STAREA, STZONE, and STBSKV (more than one may be used; conditions are applied in an 'AND' fashion).

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ibus, cval = nxtbus()
```

Fortran command syntax:

```
CALL NXTBUS(IBUS, CVAL, IERR)
```

IPLAN command syntax:

```
CALL NXTBUS(IBUS, CVAL, IERR)
```

Where:

Integer IBUS	Bus number (output).	
Character*18 CVAL	Extended bus name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IBUS' and 'CVAL' returned
	IERR = 1	No more buses; 'IBUS' and 'CVAL' unchanged
	IERR = 2	'INIBUS', 'INIBUX','ININAM' or 'ININAX' was not called; 'IBUS' and 'CVAL' unchanged
	IERR = 3	'STRING' not long enough to contain entire extended bus name; 'IBUS' and 'CVAL' returned but 'CVAL' does not contain any part of the base voltage portion of extended bus name

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7.123. NXTFAX

Returns the next FACTS device in FACTS device name alphabetical order. The routine INIFAX must be called first.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, string = nxtfax()
```

Fortran command syntax:

```
CALL NXTFAX(STRING, IERR)
```

IPLAN command syntax:

```
CALL NXTFAX(STRING, IERR)
```

Where:

Character*12 STRING	FACTS device name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'STRING' returned
	IERR = 1	No more FACTS devices; 'STRING' unchanged
	IERR = 2	'INIFAX' was not called; 'STRING' unchanged
	IERR = 3	'STRING' not long enough to contain entire FACTS device name; 'STRING' returned but contains a truncated FACTS device name

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7.124. NXTFXS

Use this API to return the identifier of the next fixed shunt connected to a bus. Each time 'NXTFXS' is called it returns the next shunt identifier in ascending order. 'INIFXS' must first be called to establish the shunt bus number.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, id = nxtfxs(ibus)
```

Fortran command syntax:

```
CALL NXTFXS(IBUS, ID, IERR)
```

IPLAN command syntax:

```
CALL NXTFXS(IBUS, ID, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Shunt identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'ID' returned
	IERR = 1	No more fixed shunts at 'IBUS'; 'ID' unchanged
	IERR = 2	'INIFXS' was not called for bus 'IBUS'; 'ID' unchanged

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7.125. NXTIND

Use this API routine to return the identifier of the next induction machine connected to a bus. Each time 'NXTIND' is called it returns the next induction machine identifier in ascending order. 'INIIND' must first be called to establish the induction machine bus number.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, id = nxtind(ibus)
```

Fortran command syntax:

```
CALL NXTIND(IBUS, ID, IERR)
```

IPLAN command syntax:

```
CALL NXTIND(IBUS, ID, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Induction machine identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'ID' returned
	IERR = 1	No more induction machines at 'IBUS'; 'ID' unchanged
	IERR = 2	'INIIND' was not called for bus 'IBUS'; 'ID' unchanged

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7.126. NXTLOD

Use this API to return the identifier of the next load connected to a bus. Each time 'NXTLOD' is called it returns the next load identifier in ascending order. 'INILOD' must first be called to establish the load bus number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, id = nxtlod(ibus)
```

Fortran command syntax:

```
CALL NXTLOD(IBUS, ID, IERR)
```

IPLAN command syntax:

```
CALL NXTLOD(IBUS, ID, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Load identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'ID' returned
	IERR = 1	No more loads at 'IBUS'; 'ID' unchanged
	IERR = 2	'INILOD' was not called for bus 'IBUS'; 'ID' unchanged

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7.127. NXTMAC

Use this API to return the identifier of the next machine connected to a bus. Each time 'NXTMAC' is called it returns the next machine identifier in ascending order. 'INIMAC' must first be called to establish the plant bus number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, id = nxtmac(ibus)
```

Fortran command syntax:

```
CALL NXTMAC(IBUS, ID, IERR)
```

IPLAN command syntax:

```
CALL NXTMAC(IBUS, ID, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*2 ID	Machine identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'ID' returned
	IERR = 1	No more machines at 'IBUS'; 'ID' unchanged
	IERR = 2	'INIMAC' was not called for bus 'IBUS'; 'ID' unchanged

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7.128. NXTMDC

Returns the next multi-terminal dc line in dc line name alphabetical order. The routine INIMDC must be called first.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, string = nxtmdc()
```

Fortran command syntax:

```
CALL NXTMDC(STRING, IERR)
```

IPLAN command syntax:

```
CALL NXTMDC(STRING, IERR)
```

Where:

Character*12 STRING	Multi-terminal dc line name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'STRING' returned
	IERR = 1	No more multi-terminal dc lines; 'STRING' unchanged
	IERR = 2	'INIMDC' was not called; 'STRING' unchanged
	IERR = 3	'STRING' not long enough to contain entire dc line name; 'STRING' returned but contains a truncated dc line name

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7.129. NXTMSL

Use this API to return the next multi-section line member branch.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ibus, jbus, ickt = nxtmsl()
```

Fortran command syntax:

```
CALL NXTMSL(IBUS, JBUS, ICKT, IERR)
```

IPLAN command syntax:

```
CALL NXTMSL(IBUS, JBUS, ICKT, IERR)
```

Where:

Integer IBUS	From bus number (output).	
Integer JBUS	To bus number (output).	
Character*2 ICKT	Multi-section line identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IBUS', 'JBUS', and 'ICKT' returned
	IERR = 1	No more branches in multi-section line; 'IBUS', 'JBUS', and 'ICKT' unchanged
	IERR = 2	& not the first character of 'ICKT'; NXTMSL may not be used
	IERR = 3	INIMSL was not called; 'IBUS', 'JBUS', and 'ICKT' unchanged

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7.130. NXTTIE

Use this API to return the next tie branch from an area, excluding three-winding transformers. IBUS will be in the area IAR specified in the call to 'INITIE' or 'INITIX', which must be called first; JBUS will be in the other area.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ibus, jbus, ickt = nxttie()
```

Fortran command syntax:

```
CALL NXTTIE(IBUS, JBUS, ICKT, IERR)
```

IPLAN command syntax:

```
CALL NXTTIE(IBUS, JBUS, ICKT, IERR)
```

Where:

Integer IBUS	From bus number (output).	
Integer JBUS	To bus number (output).	
Character*2 ICKT	Circuit identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'IBUS', 'JBUS', and 'ICKT' returned
	IERR = 1	No more tie branches from area 'IAR'; 'IBUS' returned set to 0, and 'JBUS' and 'ICKT' unchanged
	IERR = 2	'INITIE' or 'INITIX' was not called; 'IBUS', 'JBUS' and 'ICKT' unchanged

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7.131. NXTTIE3

Use this API to return the next tie branch from an area, including three-winding transformers. IBUS will be in the area IAR specified in the call to 'INITIE' or 'INITIX', which must be called first; JBUS will be in the other area.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ibus, jbus, kbus, ickt = nxttie3()
```

Fortran command syntax:

```
CALL NXTTIE3(IBUS, JBUS, KBUS, ICKT, IERR)
```

IPLAN command syntax:

```
CALL NXTTIE3(IBUS, JBUS, KBUS, ICKT, IERR)
```

Where:

Integer IBUS	From bus number (output).	
Integer JBUS	To bus number (output).	
Integer KBUS	Third bus number (output).	
Character*2 ICKT	Circuit identifier (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'IBUS', 'JBUS', 'KBUS', and 'ICKT' returned
	IERR = 1	No more tie branches from area 'IAR'; 'IBUS' returned set to 0, and 'JBUS', 'KBUS', and 'ICKT' unchanged
	IERR = 2	'INITIE' or 'INITIX' was not called; 'IBUS', 'JBUS', 'KBUS', and 'ICKT' unchanged

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7.132. NXTVSC

Returns the next VSC dc line in dc line name alphabetical order. The routine INIVSC must be called first.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, string = nxtvsc()
```

Fortran command syntax:

```
CALL NXTVSC(STRING, IERR)
```

IPLAN command syntax:

```
CALL NXTVSC(STRING, IERR)
```

Where:

Character*12 STRING	VSC dc line name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'STRING' returned
	IERR = 1	No more VSC dc lines; 'STRING' unchanged
	IERR = 2	'INIVSC' was not called; 'STRING' unchanged
	IERR = 3	'STRING' not long enough to contain entire dc line name; 'STRING' returned but contains a truncated dc line name

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7.133. OKSTRT

Routine which may be called after activity STRT or MSTR to indicate the presence of INITIAL CONDITIONS SUSPECT errors (for activity STRT) or at least one MODEL NOT AVAILABLE IN MSTR/MRUN message (for MSTR).

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = okstrt()
```

Fortran command syntax:

```
CALL OKSTRT(IERR)
```

IPLAN command syntax:

```
CALL OKSTRT(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Simulation initialization errors were detected
	IERR = 9	Dynamics data not present in working memory

7.134. OPENUN

Connects an IPLAN unit number to the Fortran unit used by activity OPEN and returns that IPLAN unit number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
n = openun()
```

Fortran command syntax:

```
CALL OPENUN(N)
```

IPLAN command syntax:

```
CALL OPENUN(N)
```

Where:

Integer N	IPLAN unit number; zero if no unit is currently open by activity OPEN (output).

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7.135. ORDBUS

This API routine returns the bus number corresponding to the specified position in the ordered list (numeric or alphabetic) of non-hidden buses (i.e., star point buses and topological buses are excluded). Dummy buses of multi-section lines are included; hidden star point buses of three-winding transformers are not included; topological buses of buses represented by multiple bus sections are not included.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ibus = ordbus(iord)
```

Fortran command syntax:

```
CALL ORDBUS(IORD, IBUS, IERR)
```

IPLAN command syntax:

```
CALL ORDBUS(IORD, IBUS, IERR)
```

Where:

Integer IORD	Ordinal position; 1 through the number of non-hidden buses in the working case (input).	
Integer IBUS	Bus number (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IBUS' returned
	IERR = 1	'IORD' outside of range 1 through the number of non-hidden buses in the working case; 'IBUS' unchanged

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7.136. OWNDAT

Returns the data associated with an owner ('LOAD' gets loads at buses of owner 'IAR', 'LOADLD' gets loads assigned to owner 'IAR').

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = owndat(iar, string)
```

Fortran command syntax:

```
CALL OWDAT(IAR, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL OWNDAT(IAR, STRING, P, Q, IERR)
```

Where:

Integer IAR	Owner number (input).	
Character STRING	String indicating the owner total desired (input).	
	'LOAD'	Total owner load by bus owner assignment (net of load plus in-service distributed generation on load feeder)
	'LOADLD'	Total owner load by load owner assignment
	'LDGN'	Total distributed generation on load feeder by bus owner assignment
	'LDGNLD'	Total distributed generation on load feeder by load owner assignment
	'GEN'	Total owner generation
	'LOSS'	Total owner losses
	'INDMAC'	Total owner induction machine powers by bus owner assignment
	'INDMACMC'	Total owner induction machine powers by machine owner assignment
	'INDGEN'	Total owner induction generator powers by bus owner assignment
	'INDGENMC'	Total owner induction generator powers by machine owner assignment
	'INDMOT'	Total owner induction motor powers by bus owner assignment
	'INDMOTMC'	Total owner induction motor powers by machine owner assignment
Real P	Real component of complex power (output).	
Real Q	Reactive component of complex power (output).	
Complex CMPVAL	Complex power (output).	
Integer IERR	error code (output).	

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IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
IERR = 1	No in-service buses with in-service loads
IERR = 2	No in-service buses with in-service loads (for 'LOAD'), no in-service loads (for 'LOADLD'), no type 2 or type 3 buses (for 'GEN'), no in-service machines (for 'GENMC'), no in-service branches (for 'LOSS'), no in-service buses with in-service induction machines of the appropriate type (for 'INDMAC', 'INDGEN' and 'INDMOT'), or no inservice induction machines of the appropriate type (for 'INDMACMC', 'INDGENMC' and 'INDMOTMC') in owner; 'P' and 'Q' or 'CMPVAL' unchanged.
IERR = 3	Owner not found; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 4	Bad 'STRING' value; 'P' and 'Q' or 'CMPVAL' unchanged

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7.137. OWNNAM

Returns the owner name for a specified owner number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = ownnam(iar)
```

Fortran command syntax:

```
CALL OWNNAM(IAR, CVAL, IERR)
```

IPLAN command syntax:

```
CALL OWNNAM(IAR, CVAL, IERR)
```

Where:

Integer IAR	Owner number (input).	
Character*12 CVAL	Owner name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Owner number <= 0 or > largest allowable owner number; 'CVAL' unchanged
	IERR = 2	Owner not found; 'CVAL' unchanged

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7.138. OWNNUM

Returns the owner number for a specified owner name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, iar = ownnum(string)
```

Fortran command syntax:

```
CALL OWNNUM(IAR, STRING, IERR)
```

IPLAN command syntax:

```
CALL OWNNUM(IAR, STRING, IERR)
```

Where:

Integer IAR	Owner number (output).	
Character*12 STRING	Owner name (input).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IAR' returned
	IERR = 1	Owner with name 'STRING' not found; 'IAR' unchanged
	IERR > 1	Multiple owners with the name 'STRING' found; 'IAR' returned as the lowest numbered of these

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7.139. OWNUSE

Use this API to indicate whether an owner is in use.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ival = ownuse(iar)
```

Fortran command syntax:

```
CALL OWNUSE(IAR, IVAL)
```

IPLAN command syntax:

```
CALL OWNUSE(IAR, IVAL)
```

Where:

Integer IAR	Owner number (input).	
Integer IVAL	Owner status (output).	
	IVAL = -1	Owner number <= 0 or > largest allowable owner number.
	IVAL = 0	If no equipment is assigned to owner IAR.
	IVAL = 1	If any equipment is assigned to owner IAR.

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7.140. PRMDAT

Use this API routine to return real-valued solution parameters.

This API routine was first introduced in release 32.2.0.

Python command syntax:

```
ierr, rval = prmdat(string)
```

Fortran command syntax:

```
CALL PRMDAT(STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL PRMDAT(STRING, RVAL, IERR)
```

Where:

Character STRING	String indicating the solution parameter desired (input).	
	'ACCP'	Gauss-Seidel real component acceleration factor
	'ACCQ'	Gauss-Seidel imaginary component acceleration factor
	'ACCM'	special MSLV acceleration factor
	'TOL'	Gauss-Seidel convergence tolerance
	'ACCN'	Newton-Raphson acceleration factor
	'TOLN'	Newton_Raphson mismatch convergence tolerance
	'ACCTY'	TYSL acceleration factor
	'TOLTY'	TYSL convergence tolerance
	'BLOWUP'	blown-up voltage change threshold
	'PQBRAK'	constant power load characteristic breakpoint
	'THRSHZ'	zero impedance line threshold
	'ADJTHR'	automatic adjustment threshold tolerance
	'ACCTAP'	tap movement deceleration factor
	'TAPLIM'	maximum tap change per adjustment cycle
	'DVLIM'	largest | (delta V) / |V| | for Newton-Raphson solutions
	'NDVFCT'	non-divergent required improvement factor
	'VCTOLQ'	Newton-Raphson controlled bus reactive power mismatch convergence tolerance
	'VCTOLV'	Newton-Raphson controlled bus voltage error convergence tolerance
	'SWVBND'	percent of voltage controlling band mode switched shunts to be adjusted
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	

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IERR = 0	No error; 'RVAL' returned
IERR = 1	Invalid value of 'STRING'; 'RVAL' unchanged

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7.141. PRMINT

Use this API routine to return integer-valued solution parameters.

This API routine was first introduced in release 32.2.0.

Python command syntax:

```
ierr, ival = prmint(string)
```

Fortran command syntax:

```
CALL PRMINT(STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL PRMINT(STRING, IVAL, IERR)
```

Where:

Character STRING	String indicating the solution parameter desired (input).	
	'ITMX'	Gauss-Seidel maximum number of iterations
	'ITMXN'	Newton-Raphson maximum number of iterations
	'ITMXTY'	TYSL maximum number of iterations
	'MXTPSS'	maximum number of tap and/or switched shunt adjustment cycles that change data between iterations
	'MXSWIM'	maximum number of induction machine state switchings during a power flow solution
	'PFCODE'	last solution type performed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Invalid value of 'STRING'; 'IVAL' unchanged

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7.142. RMODIND

Returns the branch relay model starting array indices and status.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = rmodind(ibus, jbus, ckt, slot, string)
```

Fortran command syntax:

```
CALL RMODIND(IBUS, JBUS, CKT, SLOT, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL RMODIND(IBUS, JBUS, CKT, SLOT, STRING, IVAL, IERR)
```

Where:

Integer IBUS	from bus number (input).	
Integer JBUS	to bus number (input).	
Character CKT	circuit identifier (input).	
Integer SLOT	relay slot (1 or 2) (input).	
Character STRING	string indicating the relay model quantity desired (input).	
	'CON'	starting CON index
	'STATE'	starting STATE index
	'VAR'	starting VAR index
	'ICON'	starting ICON index
	'NCON'	number of CON values used by the model
	'NSTATE'	number of STATE values used by the model
	'NVAR'	number of VAR values used by the model
	'NICON'	number of ICON values used by the model
	'STATUS'	bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect branch status. For a model to be called during simulations the branch must be in-service and the model's bypassed flag must be not bypassed
Integer IVAL	branch model index or status (output).	
Integer IERR	error code (output).	
	IERR = 0	no error; 'IVAL' returned
	IERR = 1	branch not in network but in model tables; 'IVAL' returned
	IERR = 2	branch not in network or in model tables; 'IVAL' unchanged
	IERR = 3	invalid value of 'SLOT'; 'IVAL' unchanged
	IERR = 4	no SLOT 'SLOT' model for this branch; 'IVAL' unchanged

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IERR = 5	invalid value of 'STRING'; 'IVAL' unchanged
IERR = 6	model does not use any of this data type; 'IVAL' = 0 returned
IERR = 9	dynamics data not present in working memory; 'IVAL' unchanged

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7.143. RMODNAM

Returns the branch relay model name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = rmodnam(ibus, jbus, ickt, slot)
```

Fortran command syntax:

```
CALL RMODNAM(IBUS, JBUS, ICKT, SLOT, CVAL, IERR)
```

IPLAN command syntax:

```
CALL RMODNAM(IBUS, JBUS, ICKT, SLOT, CVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character*16 ICKT	Circuit identifier (input).	
Integer SLOT	Relay slot (1 or 2) (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Branch not in network but in model tables; 'CVAL' returned
	IERR = 2	Branch not in network or in model tables; 'CVAL' unchanged
	IERR = 3	Invalid value of 'SLOT'; 'CVAL' unchanged
	IERR = 4	No SLOT 'SLOT' model for this branch; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.144. SC3WND

Returns the three-winding transformer short circuit currents following activity SCMU.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ierr, cmpval = sc3wnd(ibus, jbus, kbus, ickt, string)
```

Fortran command syntax:

```
CALL SC3WND(IBUS, JBUS, KBUS, ICKT, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL SC3WND(IBUS, JBUS, KBUS, ICKT, STRING, P, Q, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Integer KBUS	Third bus number (input).	
Character ICKT	Circuit identifier (string (input).	
Character STRING	String indicating the short circuit current desired (input).	
	'IPOS'	Positive sequence
	'INEG'	Negative sequence
	'IZERO'	Zero sequence
	'IA'	A-phase
	'IB'	B-phase
	'IC'	C-phase
Real P	Real component of the complex current indicated by STRING (output).	
Real Q	Reactive component of the complex current indicated by STRING (output).	
Complex CMPVAL	Complex current indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	'SCINIT' not successfully called; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Branch value not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 4	Invalid 'STRING'; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 5	Branch out of service; 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned

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7.145. SCBRN2

Returns the branch short circuit currents following activity SCMU.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = scbrn2(ibus, jbus, ickt, string)
```

Fortran command syntax:

```
CALL SCBRN2(IBUS, JBUS, ICKT, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL SCBRN2(IBUS, JBUS, ICKT, STRING, P, Q, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character ICKT	Circuit identifier (input).	
Character STRING	String indicating the short circuit current desired (input).	
	'IPOS'	Positive sequence
	'INEG'	Negative sequence
	'IZERO'	Zero sequence
	'IA'	A-phase
	'IB'	B-phase
	'IC'	C-phase
Real P	Real component of the complex current indicated by STRING (output).	
Real Q	Reactive component of the complex current indicated by STRING (output).	
Complex CMPVAL	Complex current indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	'SCINIT' not successfully called; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Branch not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 4	Bad value of 'STRING1'; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 5	Branch out of service; 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned

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7.146. SCBUS2

Returns complex bus voltages and currents following activity SCMU.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = scbus2(ibus, string)
```

Fortran command syntax:

```
CALL SCBUS2(IBUS, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL SCBUS2(IBUS, STRING, P, Q, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character STRING	String indicating the quantity desired (input).	
	'VPOS'	Positive sequence voltage
	'VNEG'	Negative sequence voltage
	'VZERO'	Zero sequence voltage
	'VA'	A-phase voltage
	'VB'	B-phase voltage
	'VC'	C-phase voltage
	'SHUNTP'	Positive sequence shunt current
	'SHUNTN'	Negative sequence shunt current
	'SHUNTZ'	Zero sequence shunt current
	'SHUNTA'	A-phase shunt current
	'SHUNTB'	B-phase shunt current
	'SHUNTC'	C-phase shunt current
	'FAULTP'	Positive sequence fault current
	'FAULTN'	Negative sequence fault current
	'FAULTZ'	Zero sequence fault current
	'FAULTA'	A-phase fault current
	'FAULTB'	B-phase fault current
	'FAULTC'	C-phase fault current
Real P	Real component of the complex value indicated by STRING (output).	
Real Q	Reactive component of the complex value indicated by STRING (output).	
Complex CMPVAL	Complex value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	'SCINIT' not successfully called; 'P' and 'Q' or 'CMPVAL' unchanged

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IERR = 2	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 3	Bad value of 'STRING'; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 4	Bus disconnected; 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned
IERR = 5	Bus not faulted; 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned

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7.147. SCDONE

Use this API to restore the working case after 'SCINIT'.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = scdone()
```

Fortran command syntax:

```
CALL SCDONE(IERR)
```

IPLAN command syntax:

```
CALL SCDONE(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	'SCINIT' not successfully called

7.148. SCINIT

Use this API to initialize the short-circuit data retrieval APIs. It must be called when PSSE is at the power flow activity level and must follow each execution of activity SCMU. Can be followed by SCBUS2, SCBRN2, SC3WND and SCMAC2 calls. SCDONE must be called after executing these and before any other calls or PSSE activities can be executed.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = scinit()
```

Fortran command syntax:

```
CALL SCINIT(IERR)
```

IPLAN command syntax:

```
CALL SCINIT(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Sequence data not in case
	IERR = 2	Unable to access SCMU results; one of the following occurred: -No SCMU results file is open. -SCMU results are not compatible with the working case. -Error reading the SCMU results file

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7.149. SCMAC2

Use this API to return the machine short circuit currents following activity SCMU.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = scmac2(ibus, ickt, string)
```

Fortran command syntax:

```
CALL SCMAC2(IBUS, ICKT, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL SCMAC2(IBUS, ICKT, STRING, P, Q, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ICKT	Machine identifier (input).	
Character STRING	String indicating the short circuit current desired (input).	
	'IPOS'	Positive sequence
	'INEG'	Negative sequence
	'IZERO'	Zero sequence
	'IA'	A-phase
	'IB'	B-phase
	'IC'	C-phase
Real P	Real component of the complex current (output).	
Real Q	Reactive component of the complex current (output).	
Complex CMPVAL	Complex current (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	'SCINIT' not successfully called; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Machine not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 4	Bad value of 'STRING'; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 5	Machine out of service; 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned

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7.150. SELCTR

Use this API to return the activity selector indicator.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ival = selctr()
```

Fortran command syntax:

```
CALL SELCTR(IVAL)
```

IPLAN command syntax:

```
CALL SELCTR(IVAL)
```

Where:

Integer IVAL	Activity indicator (output).	
	IVAL = 0	Not at 'ACTIVITY'.
	IVAL = 1	At power flow activity selector.
	IVAL = 2	At dynamics activity selector.

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7.151. SFILES

Use this API to return the current Saved Case and Snapshot filenames.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
savfilnam, snpfilnam = sfiles()
```

Fortran command syntax:

```
CALL SFILES(SAVFILNAM, SNPFILNAM)
```

IPLAN command syntax:

```
CALL SFILES(SAVFILNAM, SNPFILNAM)
```

Where:

Character*260 SAVFILNAM	Current Saved Case File (output).
Character*260 SNPFILNAM	Current Snapshot File (output).

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7.152. SLMODIND

Use this API to return the subsystem load-related model starting array indices and status.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = slmodind(ctype, ibus, id, string1, string2)
```

Fortran command syntax:

```
CALL SLMODIND(CTYPE, ID, IBUS, STRING1, STRING2, IVAL, IERR)
```

IPLAN command syntax:

```
CALL SLMODIND(CTYPE, ID, IBUS, STRING1, STRING2, IVAL, IERR)
```

Where:

Character CTYPE	Subsystem type (input).	
	'BUS'	Subsystem specified by buses
	'AREA'	Subsystem specified by areas
	'ZONE'	Subsystem specified by zones
	'OWNER'	Subsystem specified by owners
	'ALL'	Subsystem of all loads
Integer IBUS	Number of bus, area, zone, or owner, depending on CTYPE. Not used if CTYPE='ALL' (input).	
Character ID	Load identifier (input).	
Character STRING1	String indicating the type of load model desired (input).	
	'CHARAC'	Load characteristic model
	'RELAY'	Load relay model
Character STRING2	String indicating the load model quantity desired (input).	
	'CON'	Starting CON index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect load status. For a model to be called during simulations the load must be in-service the bus type code must be 1 or 2 and the model's bypassed flag must be not bypassed
Integer IVAL	Load model index or status (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Invalid value of 'CTYPE'; 'IVAL' unchanged
	IERR = 2	Invalid value of 'STRING1'; 'IVAL' unchanged

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IERR = 3	No model of type 'STRING1' for this load; 'IVAL' unchanged
IERR = 4	Invalid value of 'STRING2'; 'IVAL' unchanged
IERR = 5	Model does not use any of this data type; 'IVAL' = 0 returned
IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.153. SLMODNAM

Use this API to return the subsystem load-related model name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = slmodnam(ctype, ibus, id, string)
```

Fortran command syntax:

```
CALL SLMODNAM(CTYPE, IBUS, ID, STRING, CVAL, IERR)
```

IPLAN command syntax:

```
CALL SLMODNAM(CTYPE, IBUS, ID, STRING, CVAL, IERR)
```

Where:

Character CTYPE	Subsystem type (input).	
	'BUS'	Subsystem specified by buses
	'AREA'	Subsystem specified by areas
	'ZONE'	Subsystem specified by zones
	'OWNER'	Subsystem specified by owners
	'ALL'	Subsystem of all loads
Integer IBUS	Number of bus, area, zone, or owner, depending on CTYPE. Not used if CTYPE='ALL' (input).	
Character ID	Load identifier (input).	
Character STRING	String indicating the type of load model desired (input).	
	'CHARAC'	Load characteristic model
	'RELAY'	Load relay model
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Invalid value of 'CTYPE'; 'CVAL' returned
	IERR = 2	Invalid value of 'STRING'; 'CVAL' unchanged
	IERR = 3	No model of type 'STRING' for this load; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.154. SOLSTR

Use this API to return a string indicating the result of the last solution attempt.

This API routine was first introduced in release 30.1.0.

Python command syntax:

```
desc = solstr()
```

Fortran command syntax:

```
CALL SOLSTR(DESC)
```

IPLAN command syntax:

```
CALL SOLSTR(DESC)
```

Where:

Character*48 DESC	string containing the results of the last solution attempt (output).

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7.155. SOLVED

Use this API to check whether the last solution attempt reached tolerance. This is not a test for whether the working case is a solved case.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ival = solved()
```

Fortran command syntax:

```
CALL SOLVED(IVAL)
```

IPLAN command syntax:

```
CALL SOLVED(IVAL)
```

Where:

Integer IVAL	Solution convergence indicator (output).	
	IVAL = 0	Met convergence tolerance.
	IVAL = 1	Iteration limit exceeded.
	IVAL = 2	Blown up (only when non-divergent option disabled).
	IVAL = 3	Terminated by non-divergent option.
	IVAL = 4	Terminated by console interrupt.
	IVAL = 5	Singular Jacobian matrix or voltage of 0.0 detected.
	IVAL = 6	Inertial power flow dispatch error (INLF).
	IVAL = 7	OPF solution met convergence tolerance (NOPF).
	IVAL = 9	Solution not attempted.

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7.156. SOLVED_CA

Use this API to check whether the last corrective action solution attempt reached tolerance.

This API routine was first introduced in release 33.3.0.

Python command syntax:

```
ival = solved_ca()
```

Fortran command syntax:

```
CALL SOLVED_CA(IVAL)
```

IPLAN command syntax:

```
CALL SOLVED_CA(IVAL)
```

Where:

Integer IVAL	Corrective action solution convergence indicator (output).	
	IVAL = 0	Removed all limit violations successfully
	IVAL = 1	Exceeded iteration limit (Number of AC load flows)
	IVAL = 2	Terminated with constraint violations
	IVAL = 3	Terminated because of insufficient controls
	IVAL = 4	Terminated because linear programming model cannot be setup
	IVAL = 5	Terminated because Linear Programming solution diverged
	IVAL = 6	Terminated because AC load flow exceeded iteration limit
	IVAL = 7	Terminated because AC load flow blown up (only when non-divergent option disabled)
	IVAL = 8	Terminated because AC load flow terminated by non-divergent option
	IVAL = 9	Terminated because AC load flow terminated by console interrupt
	IVAL = 10	Terminated because AC load flow terminated by singular jacobian matrix (FNSL) or voltage= 0.0 (FDNS)
	IVAL = 13	Terminated because AC load flow converged with stalled and/or tripped induction machines
	IVAL = 14	Solution not attempted

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7.157. STAREA

Use this API to restrict 'NXTBUS' to a single area. It must be called after INIBUS / INIBUX / ININAM / ININAX.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = starea(iar)
```

Fortran command syntax:

```
CALL STAREA(IAR, IERR)
```

IPLAN command syntax:

```
CALL STAREA(IAR, IERR)
```

Where:

Integer IAR	Area number (input).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid area number

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7.158. STBSKV

Use this API to restrict 'NXTBUS' to a single base voltage level. It must be called after INIBUS / INIBUX / ININAM / ININAX.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = stbskv(bskv)
```

Fortran command syntax:

```
CALL STBSKV(BSKV, IERR)
```

IPLAN command syntax:

```
CALL STBSKV(BSKV, IERR)
```

Where:

Real BSKV	Base kV level (input).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Negative base voltage

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7.159. STZONE

Use this API to restrict 'NXTBUS' to a single zone. It must be called after INIBUS / INIBUX / ININAM / ININAX.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr = stzone(iar)
```

Fortran command syntax:

```
CALL STZONE(IAR, IERR)
```

IPLAN command syntax:

```
CALL STZONE(IAR, IERR)
```

Where:

Integer IAR	Zone number (input).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Invalid zone number

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7.160. SWSBLK

Use this API to return the switched shunt number of steps and step size for block IBLK.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival, v = swsblk(ibus, iblk)
```

Fortran command syntax:

```
CALL SWSBLK(IBUS, IBLK, IVAL, V, IERR)
```

IPLAN command syntax:

```
CALL SWSBLK(IBUS, IBLK, IVAL, V, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Integer IBLK	Switched shunt block number (input).	
Integer IVAL	Number of steps in block 'IBLK' (output).	
Real V	Admittance in Mvar per step in block 'IBLK' (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' and 'V' returned
	IERR = 1	Bus not found; 'IVAL' and 'V' unchanged
	IERR = 2	No switched shunt at bus; 'IVAL' and 'V' unchanged
	IERR = 3	Bad value of 'IBLK'; 'IVAL' and 'V' unchanged
	IERR = 4	Bus type code greater than or equal to 4; 'IVAL' and 'V' returned
	IERR = 5	Status is 0; 'IVAL' and 'V' returned

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7.161. SWSBLZ

Use this API to return the switched shunt number of steps and zero sequence step size in per unit for block IBLK.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival, v = swsblz(ibus, iblk)
```

Fortran command syntax:

```
CALL SWSBLZ(IBUS, IBLK, IVAL, V, IERR)
```

IPLAN command syntax:

```
CALL SWSBLZ(IBUS, IBLK, IVAL, V, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Integer IBLK	Switched shunt block number (input).	
Integer IVAL	Number of steps in block 'IBLK' (output).	
Real V	Admittance in pu per step in block 'IBLK' (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' and 'V' returned
	IERR = 1	Bus not found; 'IVAL' and 'V' unchanged
	IERR = 2	No switched shunt at bus; 'IVAL' and 'V' unchanged
	IERR = 3	Bad value of 'IBLK'; 'IVAL' and 'V' unchanged
	IERR = 4	Bus type code greater than or equal to 4; 'IVAL' and 'V' returned
	IERR = 5	Status is 0; 'IVAL' and 'V' returned
	IERR = 6	Sequence data not in case; 'IVAL' and 'V' unchanged

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7.162. SWSBLZ2

Use this API to return the switched shunt number of steps and zero sequence step size in Mvar for block IBLK.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, ival, v = swsblz2(ibus, iblk)
```

Fortran command syntax:

```
CALL SWSBLZ2(IBUS, IBLK, IVAL, V, IERR)
```

IPLAN command syntax:

```
CALL SWSBLZ2(IBUS, IBLK, IVAL, V, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Integer IBLK	Switched shunt block number (input).	
Integer IVAL	Number of steps in block 'IBLK' (output).	
Real V	Admittance in Mvar per step in block 'IBLK' (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' and 'V' returned
	IERR = 1	Bus not found; 'IVAL' and 'V' unchanged
	IERR = 2	No switched shunt at bus; 'IVAL' and 'V' unchanged
	IERR = 3	Bad value of 'IBLK'; 'IVAL' and 'V' unchanged
	IERR = 4	Bus type code greater than or equal to 4; 'IVAL' and 'V' returned
	IERR = 5	Status is 0; 'IVAL' and 'V' returned
	IERR = 6	Sequence data not in case; 'IVAL' and 'V' unchanged

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7.163. SWSDAT

Use this API to return the switched shunt schedule voltage band.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, vhi, vlo = swsdat(ibus)
```

Fortran command syntax:

```
CALL SWSDAT(IBUS, VHI, VLO, IERR)
```

IPLAN command syntax:

```
CALL SWSDAT(IBUS, VHI, VLO, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Real VHI	Desired voltage upper limit (output).	
Real VLO	Desired voltage lower limit (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'VHI' and 'VLO' returned
	IERR = 1	Bus not found; 'VHI' and 'VLO' unchanged
	IERR = 2	No switched shunt at bus; 'VHI' and 'VLO' unchanged
	IERR = 4	Bus type code greater than or equal to 4; 'VHI' and 'VLO' returned
	IERR = 5	Status is 0; 'VHI' and 'VLO' returned

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7.164. SWSDT1

Use this API to return switched shunt real data.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ierr, rval = swsdt1(ibus, string)
```

Fortran command syntax:

```
CALL SWSDT1(IBUS, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL SWSDT1(IBUS, STRING, RVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character STRING	String indicating the switched shunt quantity desired (input).	
	'VSWHI'	Controlled quantity upper limit
	'VSWLO'	Controlled quantity lower limit
	'RMPCT'	Percentage of total Mvar required to regulate controlled bus voltage
	'BINIT'	Present admittance switched on in Mvar at unity voltage
	'O_BINIT'	Same as 'BINIT' units determined by power output option
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	No switched shunt at bus; 'RVAL' unchanged
	IERR = 3	Invalid 'STRING'; 'RVAL' unchanged
	IERR = 4	Bus type code greater than or equal to 4; 'RVAL' returned
	IERR = 5	Status is 0; 'RVAL' returned

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7.165. SWSINT

Use this API to return switched shunt integer parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = swsint(ibus, string)
```

Fortran command syntax:

```
CALL SWSINT(IBUS, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL SWSINT(IBUS, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character STRING	String indicating the switched shunt quantity desired (input).	
	'STATION'	Bus IBUS substation number (returned as 0 if bus IBUS is not in a substation)
	'SECTION'	Bus IBUS bus section number (returned as 0 if bus IBUS is not in a substation)
	'NODE'	Number of the node to which the switched shunt is connected (returned as 0 if bus IBUS is not in a substation)
	'MODE'	Control mode
	'STATUS'	Status flag
	'ADJM'	Adjustment method
	'REMOTE'	Bus number of regulated bus; zero for none
	'BLOCKS'	Number of blocks
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	No switched shunt at bus; 'IVAL' unchanged
	IERR = 3	Bad value of 'STRING'; 'IVAL' unchanged
	IERR = 4	Bus type code greater than or equal to 4; 'IVAL' returned
	IERR = 5	Status is 0; 'IVAL' returned

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7.166. SWSMIND

Use this API to return switched shunt model starting array indices and status.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = swsmind(ibus, string)
```

Fortran command syntax:

```
CALL SWSMIND(IBUS, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL SWSMIND(IBUS, STRING, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character STRING	String indicating the switched shunt model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect switched shunt status. For a model to be called during simulations the switched shunt must be in service the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Switched shunt not in network but in model tables; 'IVAL' returned
	IERR = 2	Switched shunt not in network or in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 7	Bus not found; 'IVAL' unchanged

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IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.167. SWSMNAM

Use this API to return switched shunt model name.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = swsmnam(ibus)
```

Fortran command syntax:

```
CALL SWSMNAM(IBUS, CVAL, IERR)
```

IPLAN command syntax:

```
CALL SWSMNAM(IBUS, CVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Switched shunt not in network but in model tables; 'CVAL' returned
	IERR = 2	Switched shunt not in network or in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables; 'CVAL' unchanged
	IERR = 7	Bus not found; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.168. SYSMSM

Use this API to return the total system MVA mismatch.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
mva = sysmsm()
```

Fortran command syntax:

```
CALL SYSMSM(MVA)
```

IPLAN command syntax:

```
CALL SYSMSM(MVA)
```

Where:

Real MVA	Total system MVA mismatch (output).

7.169. SYSMVA

Use this API to return the system base MVA.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
mvabas = sysmva()
```

Fortran command syntax:

```
CALL SYSMVA(MVABAS)
```

IPLAN command syntax:

```
CALL SYSMVA(MVABAS)
```

Where:

Real MVABAS	System base MVA (output).

7.170. SYSTOT

Use this API to return complex system-wide values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = systot(string)
```

Fortran command syntax:

```
CALL SYSTOT(STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL SYSTOT(STRING, P, Q, IERR)
```

Where:

Character STRING	String indicating the system total desired (input).	
	'LOAD'	Total system load
	'GEN'	Total system generation
	'LOSS'	Total system losses
	'INDMAC'	Total system induction machine powers
	'INDGEN'	Total system induction generator powers
	'INDMOT'	Total system induction motor powers
	'FACTSH'	Total of FACTS device shunt elements
	'BSSHNT'	Total of bus fixed and switched shunt elements
	'LNSHNT'	Total of line shunts and transformer magnetizing shunt elements
	'ALSHNT'	Total of FACTS device fixed switched line and transformer magnetizing shunt elements
Real P	Real component of the complex power (output).	
Real Q	Reactive component of the complex power (output).	
Complex CMPVAL	Complex power (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bad 'STRING' value; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	No in-service buses in case; 'P' and 'Q' or 'CMPVAL' unchanged

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7.171. TITLDT

Use this API to return the two line case title.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
line1, line2 = titldt()
```

Fortran command syntax:

```
CALL TITLDT(LINE1, LINE2)
```

IPLAN command syntax:

```
CALL TITLDT(LINE1, LINE2)
```

Where:

Character*60 LINE1	First 60-character line (output).
Character*60 LINE2	Second 60-character line (output).

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7.172. TOTBUS

Use this API to return the total number of buses in the working case. Dummy buses of multisection lines and star point buses of three-winding transformers are included.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
n = totbus()
```

Fortran command syntax:

```
CALL TOTBUS(N)
```

IPLAN command syntax:

```
CALL TOTBUS(N)
```

Where:

Integer N	Number of buses in the working case. If the working case is empty, then the number of buses returned is zero (output).

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7.173. TR3DAT

Use this API to return three-winding transformer real values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = tr3dat(ibus, jbus, kbus, ickt, string)
```

Fortran command syntax:

```
CALL TR3DAT(IBUS, JBUS, KBUS, ICKT, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL TR3DAT(IBUS, JBUS, KBUS, ICKT, STRING, RVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Integer KBUS	Third bus number (input).	
Character ICKT	Circuit identifier (input).	
Character STRING	String indicating the three-winding transformer quantity desired (input).	
	'FRACT1'	Fraction of total ownership assigned to the first owner
	'FRACT2'	Fraction of total ownership assigned to the second owner
	'FRACT3'	Fraction of total ownership assigned to the third owner
	'FRACT4'	Fraction of total ownership assigned to the fourth owner
	'VMSTAR'	Star bus voltage magnitude
	'ANSTAR'	Star bus voltage angle in degrees
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Three-winding transformer not found; 'RVAL' unchanged
	IERR = 3	Invalid 'STRING'; 'RVAL' unchanged
	IERR = 4	Error fetching transformer data; 'RVAL' unchanged
	IERR = 6	With STRING = FRACTn, n > number of owners; 'RVAL' unchanged

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7.174. TR3DT2

Use this API to return three-winding transformer complex values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = tr3dt2(ibus, jbus, kbus, ickt, string)
```

Fortran command syntax:

```
CALL TR3DT2(IBUS, JBUS, KBUS, ICKT, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL TR3DT2(IBUS, JBUS, KBUS, ICKT, STRING, P, Q, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Integer KBUS	Third bus number (input).	
Character ICKT	Circuit identifier (input).	
Character STRING	String indicating the three-winding transformer quantity desired (input).	
	'RX1-2'	Measured impedance in per unit between the winding 1 and 2 buses
	'RX2-3'	Measured impedance in per unit between the winding 2 and 3 buses
	'RX3-1'	Measured impedance in per unit between the winding 3 and 1 buses
	'YMAGNT'	Magnetizing admittance in pu
	'ZG1', 'ZGRND'	Zero-sequence winding 1 grounding impedance
	'Z01'	Zero-sequence winding 1 or winding 1 bus to winding 2 bus leakage impedance
	'ZG2'	Zero-sequence winding 2 grounding impedance
	'Z02'	Zero-sequence winding 2 or winding 2 bus to winding 3 bus leakage impedance
	'ZG3'	Zero-sequence winding 3 grounding impedance
	'Z03'	Zero-sequence winding 3 or winding 3 bus to winding 1 bus leakage impedance
	'ZNUTRL'	Zero-sequence common neutral grounding impedance
Real P	Real component of complex value indicated by STRING (output).	
Real Q	Reactive component of complex value indicated by STRING (output).	

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Complex CMPVAL	Complex value indicated by STRING (output).
Integer IERR	error code (output).

IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 2	Three-winding transformer not found; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 3	Invalid 'STRING'; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 4	Error fetching transformer data; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 5	Sequence data not in case for 'STRING' = 'ZGRND', 'ZG1', 'Z01', 'ZG2', ,'Z02', 'ZG3', 'Z03', or 'ZNUTRL'; 'P' and 'Q' or 'CMPVAL' unchanged

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7.175. TR3INT

Use this API to return three-winding transformer integer values.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = tr3int(ibus, jbus, kbus, ickt, string)
```

Fortran command syntax:

```
CALL TR3INT(IBUS, JBUS, KBUS, ICKT, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL TR3INT(IBUS, JBUS, KBUS, ICKT, STRING, IVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Integer KBUS	Third bus number (input).	
Character ICKT	Circuit identifier (input).	
Character STRING	String indicating the three-winding transformer quantity desired (input).	
	'STATUS'	Three-winding transformer status
	'NMETER'	Non-metered bus number
	'OWNERS'	Number of owners
	'OWN1'	Owner number of first owner
	'OWN2'	Owner number of second owner
	'OWN3'	Owner number of third owner
	'OWN4'	Owner number of fourth owner
	'CW'	Winding data I/O code
	'CZ'	Impedance data I/O code
	'CM'	Magnetizing admittance data I/O code
	'CZ0'	Leakage impedance data I/O code
	'CZG'	Grounding impedance data I/O code
	'CNXCOD'	Winding connection code
	'ZADCOD'	Impedance adjustment method code
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Three-winding transformer not found; 'IVAL' unchanged
	IERR = 3	Invalid 'STRING'; 'IVAL' unchanged

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IERR = 4	Error fetching transformer data; 'IVAL' unchanged
IERR = 5	Sequence data not in case for 'STRING' = 'CZ0', 'CZG' or 'CNXCOD'; 'IVAL' unchanged
IERR = 6	With 'STRING' = OWNn, n > number of owners; 'IVAL' unchanged

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7.176. TR3NAM

Use this API to return three-winding transformer name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = tr3nam(ibus, jbus, kbus, ickt)
```

Fortran command syntax:

```
CALL TR3NAM(IBUS, JBUS, KBUS, ICKT, CVAL, IERR)
```

IPLAN command syntax:

```
CALL TR3NAM(IBUS, JBUS, KBUS, ICKT, CVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Integer KBUS	Third bus number (input).	
Character ICKT	Circuit identifier (input).	
Character*40 CVAL	Transformer name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Bus not found; 'CVAL' unchanged
	IERR = 2	Three-winding transformer not found; 'CVAL' unchanged
	IERR = 4	Error fetching transformer data; 'CVAL' unchanged

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7.177. TRNDAT

This API routine is obsolete. It has been replaced by the API routine XFRDAT .

Return transformer Ratio and Angle.

This API routine was first introduced in release 30.3.3.

Python command syntax:

```
ierr, ratio, angle = trndat(ibus, jbus, ckt)
```

Fortran command syntax:

```
CALL TRNDAT(IBUS, JBUS, CKT, RATIO, ANGLE, IERR)
```

IPLAN command syntax:

```
CALL TRNDAT(IBUS, JBUS, CKT, RATIO, ANGLE, IERR)
```

Where:

Integer IBUS	from bus number (input).	
Integer JBUS	to bus number (input).	
Character*2 CKT	circuit identifier (input).	
Real RATIO	tap position (output).	
Real ANGLE	phase shift angle, in degrees (output).	
Integer IERR	error code (output).	
	IERR = 0	no error; 'RATIO' and 'ANGLE' returned
	IERR = 1	bus not found; 'RATIO' and 'ANGLE' unchanged
	IERR = 2	branch not found; 'RATIO' and 'ANGLE' unchanged
	IERR = 3	branch not a transformer; 'RATIO' and 'ANGLE' unchanged

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7.178. TRXDAT

Use this API to return the inter-area transfer MW.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ierr, rval = trxdat(iar, jar, id)
```

Fortran command syntax:

```
CALL TRXDAT(IAR, JAR, ID, RVAL, IERR)
```

IPLAN command syntax:

```
CALL TRXDAT(IAR, JAR, ID, RVAL, IERR)
```

Where:

Integer IAR	From area number (input).	
Integer JAR	To area number (input).	
Character*1 ID	Inter-area transfer identifier (input).	
Real RVAL	Value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Area number <= 0 or > largest allowable area number; 'RVAL' unchanged
	IERR = 2	Interarea transfer not found; 'RVAL' unchanged

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7.179. VSCAUXMIND

Use this API to return the starting array indices and status of auxiliary signal model associated with VSC dc line for the specified signal index.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = vscauxmind(name, isgndx, string)
```

Fortran command syntax:

```
CALL VSCAUXMIND(NAME, ISGNDX, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL VSCAUXMIND(NAME, ISGNDX, STRING, IVAL, IERR)
```

Where:

Character NAME	VSC dc line name (input).	
Integer ISGNDX	Auxiliary signal index (input).	
Character STRING	String indicating the VSC dc line auxiliary signal model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. For a model to be called during simulations the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not in network but auxiliary signal model in model tables; 'IVAL' returned
	IERR = 2	dc line not in network and auxiliary signal model not in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables for the given auxiliary signal index; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned

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IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.180. VSCAUXMNAM

Use this API to return the auxiliary signal model name associated with a VSC dc line for the specified auxiliary signal index.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = vscauxmnam(name, isgndx)
```

Fortran command syntax:

```
CALL VSCAUXMNAM(NAME, ISGNDX, CVAL, IERR)
```

IPLAN command syntax:

```
CALL VSCAUXMNAM(NAME, ISGNDX, CVAL, IERR)
```

Where:

Character NAME	VSC dc line name (input).	
Integer ISGNDX	Auxiliary signal index (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	dc line not in network but auxiliary signal model in model tables; 'CVAL' returned
	IERR = 2	dc line not in network and auxiliary signal model not in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables for the given auxiliary signal index; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.181. VSCCDT

Use this API to return VSC dc line real parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = vsccdt(id, ibus, string)
```

Fortran command syntax:

```
CALL VSCCDT(ID, IBUS, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL VSCCDT(ID, IBUS, STRING, RVAL, IERR)
```

Where:

Character ID	VSC dc line name (input).	
Integer IBUS	Converter bus number (input).	
Character STRING	String indicating the VSC dc line quantity desired (input).	
	'PAC'	Real power into dc line at converter bus in MW
	'O_PAC'	Same as 'PAC' in units determined by the power output option setting
	'QAC'	Reactive power into dc line at converter bus in MV
	'O_QAC'	Same as 'QAC' in units determined by the power output option setting
	'MVA'	Apparent ac flow into the dc line in MVA
	'O_MVA'	Same as 'MVA' in units determined by the power output option setting
	'PCTMVA'	ac loading percent of limit
	'ACAMPS'	ac current in amps
	'PUCUR'	ac current in pu
	'KVDC'	Voltage at controlling converter in kV
	'DCCUR'	dc current in amps
	'DCSET'	dc setpoint in pu
	'ACSET'	ac setpoint in pu
	'RDC'	dc line resistance in ohms
Real RVAL	Parameter value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	VSC dc line 'ID' not found; 'RVAL' unchanged
	IERR = 2	Invalid 'STRNG'; 'RVAL' unchanged
	IERR = 3	Bus 'IBUS' not a converter of this dc line; 'RVAL' unchanged

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IERR = 4	dc line blocked; 'RVAL' returned as 0.0
IERR = 5	No base kV at bus; 'RVAL' returned as the per unit current

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7.182. VSCCIN

Use this API to return VSC dc line converter integer parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = vsccin(id, ibus, string)
```

Fortran command syntax:

```
CALL VSCCIN(ID, IBUS, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL VSCCIN(ID, IBUS, STRING, IVAL, IERR)
```

Where:

Character ID	VSC dc line name (input).	
Integer IBUS	Converter bus number (input).	
Character STRING	String indicating the VSC dc line quantity desired (input).	
	'DCTYPE'	Converter dc control code
	'ACMODE'	Converter ac control mode
Integer IVAL	Parameter value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	VSC dc line 'ID' not found; 'IVAL' unchanged
	IERR = 2	Invalid 'STRING'; 'IVAL' unchanged
	IERR = 3	Bus 'IBUS' not a converter of this dc line; 'IVAL' unchanged

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7.183. VSCINT

Use this API to return VSC dc line integer parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = vscint(id, string)
```

Fortran command syntax:

```
CALL VSCINT(ID, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL VSCINT(ID, STRING, IVAL, IERR)
```

Where:

Character ID	VSC dc line name (input).	
Character STRING	String indicating the VSC dc line quantity desired (input).	
	'CONV1'	Converter 1 bus number
	'CONV2'	Converter 2 bus number
	'MDC'	Control mode
Integer IVAL	Parameter value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	VSC dc line 'ID' not found; 'IVAL' unchanged
	IERR = 2	Invalid 'STRING'; 'IVAL' unchanged

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7.184. VSCMIND

Use this API to return VSC dc line model starting array indices and status.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = vscmind(name, string)
```

Fortran command syntax:

```
CALL VSCMIND(NAME, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL VSCMIND(NAME, STRING, IVAL, IERR)
```

Where:

Character NAME	VSC dc line name (input).	
Character STRING	String indicating the VSC dc line model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect dc line status. For a model to be called during simulations the dc line must be in service the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	dc line not in network but in model tables; 'IVAL' returned
	IERR = 2	dc line not in network or in model tables; 'IVAL' unchanged
	IERR = 4	No model in model tables; 'IVAL' unchanged
	IERR = 5	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
	IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.185. VSCMNAM

Use this API to return VSC dc line model name.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = vscmnam(name)
```

Fortran command syntax:

```
CALL VSCMNAM(NAME, CVAL, IERR)
```

IPLAN command syntax:

```
CALL VSCMNAM(NAME, CVAL, IERR)
```

Where:

Character NAME	VSC dc line name (input).	
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	dc line not in network but in model tables; 'CVAL' returned
	IERR = 2	dc line not in network or in model tables; 'CVAL' unchanged
	IERR = 4	No model in model tables; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.186. WINDMIND

Use this API to return wind model starting array indices and status.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, ival = windmind(ibus, id, string1, string2)
```

Fortran command syntax:

```
CALL WINDMIND(IBUS, ID, STRING1, STRING2, IVAL, IERR)
```

IPLAN command syntax:

```
CALL WINDMIND(IBUS, ID, STRING1, STRING2, IVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Machine identifier (input).	
Character STRING1	String indicating the type of wind model (input).	
	'WGEN'	Wind Generator model
	'WELEC'	Wind Electrical model
	'WMECH'	Wind Mechanical model
	'WPICH'	Wind Pitch model
	'WAERO'	Wind Aerodynamic model
	'WGUST'	Wind Gust model
	'WAUX'	Wind Auxiliary Signal model
	'WTRQ'	Renewable Torque Control model.
Character STRING2	String indicating the plant-related model quantity desired (input).	
	'CON'	Starting CON index
	'STATE'	Starting STATE index
	'VAR'	Starting VAR index
	'ICON'	Starting ICON index
	'NCON'	Number of CON values used by the model
	'NSTATE'	Number of STATE values used by the model
	'NVAR'	Number of VAR values used by the model
	'NICON'	Number of ICON values used by the model
	'STATUS'	Bypassed flag: 0 = bypassed; 1 = not bypassed. Does not reflect machine status. For a model to be called during simulations the machine must be in service the type code of the bus must be 2 and the model's bypassed flag must be not bypassed
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	

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IERR = 0	No error; 'IVAL' returned
IERR = 1	Machine not in network but in model tables; 'IVAL' returned
IERR = 2	Machine not in network or in model tables; 'IVAL' unchanged
IERR = 3	Invalid value of 'STRING1', 'IVAL' unchanged
IERR = 4	No model of type 'STRING1' for this machine; 'IVAL' unchanged
IERR = 5	Invalid value of 'STRING2'; 'IVAL' unchanged
IERR = 6	Model does not use any of this data type; 'IVAL' = 0 returned
IERR = 9	Dynamics data not present in working memory; 'IVAL' unchanged

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7.187. WINDMNAM

Use this API to return wind model name.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, cval = windmnam(ibus, id, string)
```

Fortran command syntax:

```
CALL WINDMNAM(IBUS, ID, STRING, CVAL, IERR)
```

IPLAN command syntax:

```
CALL WINDMNAM(IBUS, ID, STRING, CVAL, IERR)
```

Where:

Integer IBUS	Bus number (input).	
Character ID	Machine identifier (input).	
Character STRING	String indicating the type of plant-related model (input).	
	'WGEN'	Wind Generator model
	'WELEC'	Wind Electrical model
	'WMECH'	Wind Mechanical model
	'WPICH'	Wind Pitch model
	'WAERO'	Wind Aerodynamic model
	'WGUST'	Wind Gust Model
	'WAUX'	Wind Auxiliary Signal model
	'WTRQ'	Renewable Torque Control model
Character*16 CVAL	Model name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'CVAL' returned
	IERR = 1	Machine not in network but in model tables; 'CVAL' returned
	IERR = 2	Machine not in network or in model tables; 'CVAL' unchanged
	IERR = 3	Invalid value of 'STRING', 'CVAL' unchanged
	IERR = 4	No model of type 'STRING' for this machine; 'CVAL' unchanged
	IERR = 9	Dynamics data not present in working memory; 'CVAL' unchanged

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7.188. WNDDAT

Use this API to return three-winding transformer real values for winding connected to bus IBUS.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = wnddat(ibus, jbus, kbus, ickt, string)
```

Fortran command syntax:

```
CALL WNDDAT(IBUS, JBUS, KBUS, ICKT, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL WNDDAT(IBUS, JBUS, KBUS, ICKT, STRING, RVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).
Integer JBUS	To bus number (input).
Integer KBUS	Third bus number (input).
Character ICKT	Circuit identifier (input).
Character STRING	String indicating the transformer winding quantity desired (input).

'RATEA'	Rating set A
'RATEB'	Rating set B
'RATEC'	Rating set C
'RATE'	Default rating set
'RATIO'	Winding tap ratio
'ANGLE'	Winding phase angle shift in degrees
'SBASE'	Winding MVA base
'NOMV'	Nominal winding voltage
'RMAX'	Ratio or angle high limit
'RMIN'	Ratio or angle low limit
'VMAX'	Controlled quantity high limit
'VMIN'	Controlled quantity low limit
'STEP'	Derived tap step
'CNXANG'	Winding connection angle in degrees
'MVA'	Winding MVA flow
'O_MVA'	Same as 'MVA' in units determined by the power output option setting
'AMPS'	Winding current in amps
'PUCUR'	Winding current in pu
'PCTRTA'	Percent current loading of RATEA
'PCTRTB'	Percent current loading of RATEB
'PCTRTC'	Percent current loading of RATEC
'PCTMVA'	Percent MVA loading of RATEA

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	'PCTMVB'	Percent MVA loading of RATEB
	'PCTMVC'	Percent MVA loading of RATEC
	'PCTCPA'	Percent current or MVA loading (according to the transformer percent loading units program option setting) of RATEA
	'PCTCPB'	Percent current or MVA loading (according to the transformer percent loading units program option setting) of RATEB
	'PCTCPC'	Percent current or MVA loading (according to the transformer percent loading units program option setting) of RATEC
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Three-winding transformer not found; 'RVAL' unchanged
	IERR = 3	Invalid 'STRING'; 'RVAL' unchanged
	IERR = 4	Error fetching transformer data; 'RVAL' unchanged
	IERR = 7	Branch out of service; 'RVAL' returned set to zero
	IERR = 8	No base kV at IBUS; 'RVAL' returned = per unit current
	IERR = 9	Rating is zero; 'RVAL' returned set to zero

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7.189. WNDDT2

Use this API to return three-winding transformer complex values for winding connected to bus IBUS.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = wnddt2(ibus, jbus, kbus, ickt, string)
```

Fortran command syntax:

```
CALL WNDDT2(IBUS, JBUS, KBUS, ICKT, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL WNDDT2(IBUS, JBUS, KBUS, ICKT, STRING, P, Q, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Integer KBUS	Third bus number (input).	
Character ICKT	Circuit identifier (input).	
Character STRING	String indicating the transformer winding quantity desired (input).	
	'RX'	Winding impedance in per unit
	'RXZ'	Zero-sequence winding impedance
	'COMPRX'	Compensating impedance
	'FLOW'	Complex flow in MW/Mvar
	'O_FLOW'	Same as 'FLOW' units determined by power output option
Real P	Real component of complex value indicated by STRING (output).	
Real Q	Reactive component of complex value indicated by STRING (output).	
Complex CMPVAL	Complex value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Bus not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	Three-winding transformer not found; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Invalid 'STRING'; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 4	Error fetching transformer data; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 5	Sequence data not in case for 'STRING' = 'RXZ'; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 7	Branch out of service; 'P' and 'Q' of 0.0 or 'CMPVAL' of (0.0,0.0) returned

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7.190. WNDINT

Use this API to return three-winding transformer integer values for winding connected to bus IBUS.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = wndint(ibus, jbus, kbus, ickt, string)
```

Fortran command syntax:

```
CALL WNDINT(IBUS, JBUS, KBUS, ICKT, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL WNDINT(IBUS, JBUS, KBUS, ICKT, STRING, IVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Integer KBUS	Third bus number (input).	
Character ICKT	Circuit identifier (input).	
Character STRING	String indicating the transformer winding quantity desired (input).	
	'STATUS'	Status
	'CNXCOD'	Winding connection code
	'WNDNUM'	Winding number
	'ICONT'	Controlled bus number
	'TABLE'	Transformer impedance correction table number (0 if none)
	'CNTRL'	Adjustment flag
	'CNTRL2'	Transformer adjustment code
	'NTPOSN'	Number of tap positions
	'TPSTT'	Transformer winding ratio limit code: (-1) low; (0) regulating; (1) high; (-2) locked
	'ANSTT'	Transformer winding angle limit code: (-1) low; (0) regulating; (1) high; (-2) locked
Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Three-winding transformer not found; 'IVAL' unchanged
	IERR = 3	Invalid 'STRING'; 'IVAL' unchanged
	IERR = 4	Error fetching transformer data; 'IVAL' unchanged

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IERR = 5	Sequence data not in case for 'STRING' = 'CNXCOD'; 'IVAL' unchanged

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7.191. XFRDAT

Use this API to return two-winding transformer real parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, rval = xfrdat(ibus, jbus, ickt, string)
```

Fortran command syntax:

```
CALL XFRDAT(IBUS, JBUS, ICKT, STRING, RVAL, IERR)
```

IPLAN command syntax:

```
CALL XFRDAT(IBUS, JBUS, ICKT, STRING, RVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character ICKT	Circuit identifier (input).	
Character STRING	String indicating the transformer quantity desired (input).	
	'RATIO'	Winding one off-nominal turns ratio
	'RATIO2'	Winding two off-nominal turns ratio
	'ANGLE'	Winding one-phase shift angle in degrees
	'RMAX'	Winding one ratio or angle upper limit
	'RMIN'	Winding one ratio or angle lower limit
	'VMAX'	Controlled quantity upper limit
	'VMIN'	Controlled quantity lower limit
	'STEP'	Winding one ratio or angle step width
	'CR'	Real component of load drop compensation impedance
	'CX'	Reactive component of load drop compensation impedance
	'CNXANG'	Winding connection angle in degrees
	'SBASE1'	Winding one base MVA
	'NOMV1'	Winding one nominal voltage in kV
	'NOMV2'	Winding two nominal voltage in kV
	'GMAGNT'	Conductance component of transformer magnetizing admittance in pu
	'BMAGNT'	Susceptance component of transformer magnetizing admittance in pu
	"RG1', 'RGRND"	Grounding resistance at the winding 1 bus
	"XG1', 'XGRND"	Grounding reactance at the winding 1 bus
	'R01'	Zero sequence leakage resistance connected to the winding 1 bus

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	'X01'	Zero sequence leakage reactance connected to the winding 1 bus
	"RG2', 'RGRND2"	Grounding resistance at the winding 2 bus
	"XG2', 'XGRND2"	Grounding reactance at the winding 1 bus
	'R02'	Zero sequence leakage resistance connected to the winding 2 bus
	'X02'	Zero sequence leakage reactance connected to the winding 2 bus
	'RNUTRL'	Common neutral grounding resistance
	'XNUTRL'	Common neutral grounding reactance
Real RVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'RVAL' returned
	IERR = 1	Bus not found; 'RVAL' unchanged
	IERR = 2	Branch not found; 'RVAL' unchanged
	IERR = 3	Branch not a transformer; 'RVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'RVAL' unchanged
	IERR = 5	Sequence data not in case for STRING = 'RGRND', 'XGRND', 'RGRND2', 'XGRND2', 'RG1', 'XG1', 'R01', 'X01', 'RG2', 'XG2', 'R02', 'X02', 'RNUTRL' or 'XNUTRL'; 'RVAL' unchanged

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7.192. XFRINT

Use this API to return two-winding transformer integer parameters.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, ival = xfrint(ibus, jbus, ickt, string)
```

Fortran command syntax:

```
CALL XFRINT(IBUS, JBUS, ICKT, STRING, IVAL, IERR)
```

IPLAN command syntax:

```
CALL XFRINT(IBUS, JBUS, ICKT, STRING, IVAL, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character ICKT	Circuit identifier (input).	
Character STRING	String indicating the transformer quantity desired (input).	
	'ICONT'	Controlled bus number
	'SIDCOD'	Controlled bus side code (negative if on the winding 1 side)
	'TAPPED'	Tapped side bus number
	'GCIBUS'	Bus IBUS old grounding code (derived from CNXCOD)
	'GCJBUS'	Bus JBUS old grounding code (derived from CNXCOD)
	'TABLE'	Transformer impedance correction table number (0 if none)
	'CNTRL'	Adjustment enable flag
	'CNTRL2'	Transformer adjustment code
	'NTPOSN'	Number of tap positions
	'CNXCOD'	Connection code
	'CW'	Winding data I/O code
	'CZ'	Impedance data I/O code
	'CM'	Magnetizing admittance I/O code
	'CZ0'	Leakage impedance data I/O code
	'CZG'	Grounding impedance data I/O code
	'TPSTT'	Transformer winding one ratio limit code: (-1) low; (0) regulating; (1) high; (-2) locked; (-3) controlling dc line
	'ANSTT'	Transformer winding one angle limit code: (-1) low; (0) regulating; (1) high; (-2) locked

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Integer IVAL	Value indicated by STRING (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'IVAL' returned
	IERR = 1	Bus not found; 'IVAL' unchanged
	IERR = 2	Branch not found; 'IVAL' unchanged
	IERR = 3	Branch not a transformer; 'IVAL' unchanged
	IERR = 4	Invalid value of 'STRING'; 'IVAL' unchanged
	IERR = 5	Sequence data not in case for STRING = 'CNX-COD', 'CZ0', 'CZG', 'GCIBUS' or 'GCJBUS'; IVAL unchanged
	IERR = 6	Connection code is 6 or greater for STRING = 'GCIBUS' or 'GCJBUS'; IVAL unchanged

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7.193. XFRNAM

Use this API to return two-winding transformer name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, string = xfrnam(ibus, jbus, ickt)
```

Fortran command syntax:

```
CALL XFRNAM(IBUS, JBUS, ICKT, STRING, IERR)
```

IPLAN command syntax:

```
CALL XFRNAM(IBUS, JBUS, ICKT, STRING, IERR)
```

Where:

Integer IBUS	From bus number (input).	
Integer JBUS	To bus number (input).	
Character ICKT	Circuit identifier (input).	
Character*40 STRING	Transformer name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'STRING' returned
	IERR = 1	Bus not found; 'STRING' unchanged
	IERR = 2	Transformer not found; 'STRING' unchanged
	IERR = 3	Branch not a transformer; 'STRING' unchanged

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7.194. ZNDAT

Use this API to return zone totals.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = zndat(iar, string)
```

Fortran command syntax:

```
CALL ZNDAT(IAR, STRING, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL ZNDAT(IAR, STRING, P, Q, IERR)
```

Where:

Integer IAR	Zone number (input).	
Character STRING	String indicating the zone total desired (input).	
	'LOAD'	Total zone load by bus zone assignment (net of load plus in-service distributed generation on load feeder)
	'LOADLD'	Total zone load by load zone assignment
	'LDGN'	Total distributed generation on load feeder by bus zone assignment
	'LDGNLD'	Total distributed generation on load feeder by load zone assignment
	'GEN'	Total zone generation
	'LOSS'	Total zone losses
	'INT'	Net zone interchange
	'INDMAC'	Total zone induction machine powers by bus zone assignment
	'INDMACMC'	Total zone induction machine powers by machine zone assignment
	'INDGEN'	Total zone induction generator powers by bus zone assignment
	'INDGENMC'	Total zone induction generator powers by machine zone assignment
	'INDMOT'	Total zone induction motor powers by bus zone assignment
	'INDMOTMC'	Total zone induction motor powers by machine zone assignment
Real P	Real component of the desired complex power (output).	
Real Q	Reactive component of the desired complex power (output).	
Complex CMPVAL	Desired complex power (output).	
Integer IERR	error code (output).	

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IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
IERR = 1	Zone number < 0 or > largest allowable zone number; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 2	No in-service buses with in-service loads (for 'LOAD'), no in-service loads (for 'LOADLD'), no type 2 or type 3 buses (for 'GEN'), no branches (for LOSS'), or no ties (for 'INT') in zone; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 3	Zone not found; 'P' and 'Q' or 'CMPVAL' unchanged
IERR = 4	Bad 'STRING' value; 'P' and 'Q' or 'CMPVAL' unchanged

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7.195. ZNITOJ

Use this API to return the interchange between two zones.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cmpval = znitoj(iar, jar)
```

Fortran command syntax:

```
CALL ZNITOJ(IAR, JAR, CMPVAL, IERR)
```

IPLAN command syntax:

```
CALL ZNITOJ(IAR, JAR, P, Q, IERR)
```

Where:

Integer IAR	From zone number (input).	
Integer JAR	To zone number (input).	
Real P	Real component of the complex interchange	
Real Q	Reactive component of the complex interchange	
Complex CMPVAL	Complex interchange (output).	
Integer IERR	error code (output).	
	IERR = 0	No error; 'P' and 'Q' or 'CMPVAL' returned
	IERR = 1	Zone number <= 0 or > largest allowable zone number; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 2	No in-service branches between zones; 'P' and 'Q' or 'CMPVAL' unchanged
	IERR = 3	Zone not found; 'P' and 'Q' or 'CMPVAL' unchanged

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7.196. ZONNAM

Use this API to return the zone name.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, cval = zonnam(iar)
```

Fortran command syntax:

```
CALL ZONNAM(IAR, CVAL, IERR)
```

IPLAN command syntax:

```
CALL ZONNAM(IAR, CVAL, IERR)
```

Where:

Integer IAR	Zone number (input).	
Character*12 CVAL	Zone name (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'CVAL' is returned
	IERR = 1	Zone number <= 0 or > largest allowable zone number; 'CVAL' is unchanged
	IERR = 2	Zone not found; 'CVAL' is unchanged

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7.197. ZONNUM

Use this API to return the zone number.

This API routine was first introduced in release 28.0.0.

Python command syntax:

```
ierr, iar = zonnum(string)
```

Fortran command syntax:

```
CALL ZONNUM(IAR, STRING, IERR)
```

IPLAN command syntax:

```
CALL ZONNUM(IAR, STRING, IERR)
```

Where:

Integer IAR	Zone number (integer (output).	
Character STRING	Zone name (string (input).	
Integer IERR	error code (output).	
	IERR = 0	No error, 'IAR' is returned
	IERR = 1	Zone with name 'STRING' not found, 'IAR' is unchanged
	IERR > 1	Multiple zones with the name 'STRING' found, 'IAR' returned as lowest number of these

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7.198. ZONUSE

Use this API to indicate whether a zone is in use.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ival = zonuse(iar)
```

Fortran command syntax:

```
CALL ZONUSE(IAR, IVAL)
```

IPLAN command syntax:

```
CALL ZONUSE(IAR, IVAL)
```

Where:

Integer IAR	Zone number (input).	
Integer IVAL	Zone status (output).	
	IVAL = -1	Zone number <= 0 or > largest allowable zone number.
	IVAL = 0	If no equipment is assigned to zone IAR.
	IVAL = 1	If any equipment is assigned to zone IAR.

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Chapter 8
Subsystem Data Retrieval

The subsystem data retrieval APIs are used to retrieve data from the PSS®E engine for elements (e.g., buses, branches, etc.) in a subsystem. One or more input data items to each API allow the user to restrict the elements for which data is returned to a subset of subsystem elements (e.g., to omit out-of-service elements).

These APIs are grouped into families with each family containing up to six APIs:

- An API that returns the number of array entries required to hold one set of subsystem data items returned by the data retrieval APIs in the family (e.g., GET_BUS_COUNT returns the number of buses in subsystem SID that meet the editing criterion indicated by FLAG).
- A data retrieval API that returns an array of one or more sets of integer data values.
- A data retrieval API that returns an array of one or more sets of real data values.
- A data retrieval API that returns an array of one or more sets of complex data values.
- A data retrieval API that returns an array of one or more sets of character data values.
- An API that returns information on the data types corresponding to specified STRING values, each of which may be specified to one or more of the data retrieval APIs in the family. STRING values that are valid input values for both the real and complex data retrieval APIs of a family are indicated as being of type complex.

Each of the data retrieval APIs returns an array of one or more columns containing subsystem data items. Input arguments specify the name of the array to be used, the number data items to be retrieved (which is also the minimum number of columns required in the array), and the number of elements in each column (which must be at least as large as the number returned by the first API listed above). Array elements beyond those required to accommodate the requested data items are left unchanged by the data retrieval APIs.

These routines do not record themselves when called with recording enabled.

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8.1. Character Data

Prior to calling any of the subsystem data retrieval APIs, space for the array that is to receive the set(s) of data items that are to be returned must be allocated. For character data items, the number of characters required for each array entry for the data corresponding to any of the STRING arguments of the character data retrieval APIs (i.e., the a*Char APIs) may be obtained from the GET_CHAR_LENGTH API.

8.1.1. GET_CHAR_LENGTH

Use this API to return the number of characters required to accommodate the data corresponding to the specified STRING entries, which can be for any of the a*Char data retrieval functions.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, nchars = get_char_length(string)
```

Fortran command syntax:

```
CALL GET_CHAR_LENGTH(NSTR, STRING, NCHARS, IERR)
```

Where:

Integer NSTR	number of elements in STRING and NCHARS (1 < NSTR < 50); no default allowed	
Character STRING (NSTR)	NSTR valid strings for the a*Char data retrieval functions to be used (input; no default allowed).	
Integer NCHARS (NSTR)	NSTR entries containing the number of characters of each element of the array associated with the corresponding STRING value (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1-50	Position of an element in the array STRING that contains a value that is not recognized by any of the a*Char API routines
	IERR = 99	Invalid NSTR value

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8.2. Bus Data

The bus data family of subsystem data retrieval APIs returns arrays of data for buses in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for all subsystem buses, or to restrict the array entries to data for in-service buses.

Data for subsystem buses is returned from all APIs in the bus data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API is referenced.

8.2.1. ABUSCHAR

Use this API to return an array of character values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = abuschar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_BUS_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service buses.
	FLAG = 2	for all buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus quantities desired (input; no default allowed).	
	'NAME'	Bus name (12 characters).
	'EXNAME'	Extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the GET_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of CARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	

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Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of CARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.2.2. ABUSCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the bus data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, buses = abuscount(sid, flag)
```

Fortran command syntax:

```
CALL GET_BUS_COUNT(SID, FLAG, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service buses.
	FLAG = 2	for all buses.
Integer BUSES	number of buses in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BUSES returned
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0
	IERR = 3	Invalid FLAG value; BUSES returned as 0

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8.2.3. ABUSCPLX

Use this API to return an array of complex values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = abuscplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_BUS_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service buses.
	FLAG = 2	for all buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus quantities desired (input; no default allowed).	
	'VOLTAGE'	Actual bus voltage, in pu, rectangular coordinates
	'SHUNTACT'	Actual total in-service fixed bus shunt, in MW and Mvar
	'O_SHUNTACT'	Actual total in-service fixed bus shunt, in units determined by the power output option setting
	'SHUNTNOM'	Nominal total in-service fixed bus shunt, in MW and Mvar at 1.0 pu voltage
	'O_SHUNTNOM'	Nominal total in-service fixed bus shunt, in units determined by the power output option setting at 1.0 pu voltage
	'SHUNTN'	Total in-service negative sequence load (pu nominal)
	'SHUNTZ'	Total in-service zero sequence load (pu nominal)
	'MISMATCH'	Bus mismatch, in MW and Mvar (0.0 if bus type code > 3)

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	'O_MISMATCH'	Bus mismatch, in units determined by the power output option setting (0.0 if bus type code > 3)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the GET_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of XARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value
	IERR = 7	Sequence data not in case (when STRING = 'SHUNTN' or 'SHUNTZ')

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8.2.4. ABUSINT

Use this API to return an array of integer values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = abusint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_BUS_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service buses.
	FLAG = 2	for all buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'TYPE'	Bus type code
	'AREA'	Bus area number
	'ZONE'	Bus zone number
	'OWNER'	Bus owner number
	'DUMMY'	Returns 1 if the bus is a dummy bus of a multisection line, or 0 if it is not
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the GET_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BUSES	number of elements in each column of IARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	

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IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of IARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.2.5. ABUSREAL

Use this API to return an array of real values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = abusreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_BUS_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service buses.
	FLAG = 2	for all buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus quantities desired (input; no default allowed).	
	'BASE'	Bus base voltage, in kV
	'PU'	Actual bus voltage magnitude, in pu
	'KV'	Actual bus voltage magnitude, in kV (in pu if base voltage is 0.0)
	'ANGLE'	Bus voltage phase angle, in radians
	'ANGLED'	Bus voltage phase angle, in degrees
	'NVLMHI'	Normal high voltage limit, in pu
	'NVLMLO'	Normal low voltage limit, in pu
	'EVLMHI'	Emergency high voltage limit, in pu
	'EVLMLO'	Emergency low voltage limit, in pu
	'MISMATCH'	Bus mismatch, in MVA (0.0 if bus type code > 3)
	'O_MISMATCH'	Bus mismatch, in units determined by the power output option setting (0.0 if bus type code > 3)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the GET_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	

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Integer BUSES	number of elements in each column of RARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.2.6. ABUSTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the bus data family (aBusInt, aBusReal, aBusCplx and aBusChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = abustypes(string)
```

Fortran command syntax:

```
CALL GET_BUS_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the bus data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the bus data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.3. Plant Bus Data

The plant bus data family of subsystem data retrieval APIs returns arrays of data for buses in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for all subsystem buses, or to restrict the array entries to data for a subset of buses according to their bus, plant, and/or machine configuration and statuses

Data for subsystem buses is returned from all APIs in the plant bus data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API is referenced.

8.3.1. AGENBUSCHAR

Use this API to return an array of character values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = agenbuschar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_PLANT_BUS_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service plant buses (type code 2 or 3) with at least one in-service machine.
	FLAG = 2	for only in-service plant buses (type code 2 or 3), including those with no in-service machines.
	FLAG = 3	for all in-service buses, including those that are not plant buses.
	FLAG = 4	for all plant buses, including those with no in-service machines.
	FLAG = 5	for all buses, including those that are not plant buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or plant quantities desired (input; no default allowed).	

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	'NAME'	Bus name (12 characters).
	'EXNAME'	Extended bus name (18 characters).
	'IREGNAME'	Bus name of regulated bus (12 characters).
	'IREGEXNAME'	Extended bus name of regulated bus (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the GET_PLANT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of CARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_PLANT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of CARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.3.2. AGENBUSCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the plant bus data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, buses = agenbuscount(sid, flag)
```

Fortran command syntax:

```
CALL GET_PLANT_BUS_COUNT(SID, FLAG, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service plant buses (type code 2 or 3) with at least one in-service machine.
	FLAG = 2	for only in-service plant buses (type code 2 or 3), including those with no in-service machines.
	FLAG = 3	for all in-service buses, including those that are not plant buses.
	FLAG = 4	for all plant buses, including those with no in-service machines.
	FLAG = 5	for all buses, including those that are not plant buses.
Integer BUSES	number of buses in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BUSES returned
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0
	IERR = 3	Invalid FLAG value; BUSES returned as 0

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8.3.3. AGENBUSCPLX

Use this API to return an array of complex values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = agenbuscplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_PLANT_BUS_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service plant buses (type code 2 or 3) with at least one in-service machine.
	FLAG = 2	for only in-service plant buses (type code 2 or 3), including those with no in-service machines.
	FLAG = 3	for all in-service buses, including those that are not plant buses.
	FLAG = 4	for all plant buses, including those with no in-service machines.
	FLAG = 5	for all buses, including those that are not plant buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or plant quantities desired (input; no default allowed).	
	'VOLTAGE'	Actual bus voltage, in pu, rectangular coordinates
	'PQGEN'	Power output of in-service machines at this plant (returns MW and Mvar)
	'MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (returns MW and Mvar)
	'O_PQGEN'	Power output of in-service machines at this plant (units determined by power output options)
	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)

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Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the GET_PLANT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Integer BUSES	number of elements in each column of XARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_PLANT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of XARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.3.4. AGENBUSINT

Use this API to return an array of integer values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = agenbusint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_PLANT_BUS_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service plant buses (type code 2 or 3) with at least one in-service machine.
	FLAG = 2	for only in-service plant buses (type code 2 or 3), including those with no in-service machines.
	FLAG = 3	for all in-service buses, including those that are not plant buses.
	FLAG = 4	for all plant buses, including those with no in-service machines.
	FLAG = 5	for all buses, including those that are not plant buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or plant quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'TYPE'	Bus type code
	'AREA'	Bus area number
	'ZONE'	Bus zone number
	'OWNER'	Bus owner number
	'DUMMY'	Returns 1 if the bus is a dummy bus of a multisection line or 0 if it is not
	'STATUS'	Inferred plant status. Returns 1 if the bus type code is 2 or 3 and at least one in-service machine is present or 0 otherwise

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	'IREG'	Bus number of the regulated bus
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the GET_PLANT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BUSES	number of buses in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of IARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.3.5. AGENBUSREAL

Use this API to return an array of real values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = agenbusreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_PLANT_BUS_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service plant buses (type code 2 or 3) with at least one in-service machine.
	FLAG = 2	for only in-service plant buses (type code 2 or 3), including those with no in-service machines.
	FLAG = 3	for all in-service buses, including those that are not plant buses.
	FLAG = 4	for all plant buses, including those with no in-service machines.
	FLAG = 5	for all buses, including those that are not plant buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or plant quantities desired (input; no default allowed).	
	'BASE'	Bus base voltage in kV
	'PU'	Actual bus voltage magnitude, in pu
	'KV'	Actual bus voltage magnitude, in kV (in pu if base voltage is 0.0)
	'ANGLE'	Bus voltage phase angle, in radians
	'ANGLED'	Bus voltage phase angle, in degrees
	'PERCENT'	MVA loading as a percentage of total plant MVA base, MBASE
	'IREGBASE'	Regulated bus base voltage, in kV
	'IREGPU'	Actual regulated bus voltage magnitude, in pu

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'IREGKV'	Actual regulated bus voltage magnitude, in kV (in pu if base voltage is 0.0)
'VSPU'	Regulated bus scheduled voltage magnitude, in pu
'VSKV'	Regulated bus scheduled voltage magnitude, in kV (in pu if base voltage is 0.0)
'RMPCT'	Percent of total Mvar to be contributed by this plant
'PGEN'	Active power output of in-service machines at this plant, in MW
'QGEN'	Reactive power output of in-service machines at this plant, in Mvar
'MVA'	|PGEN + j QGEN| of in-service machines at this plant, in MVA
'PMAX'	Active power upper limit of in-service machines at this plant, in MW
'PMIN'	Active power lower limit of in-service machines at this plant, in MW
'QMAX'	Reactive power upper limit of in-service machines at this plant, in Mvar
'QMIN'	Reactive power lower limit of in-service machines at this plant, in Mvar
'MISMATCH'	Bus mismatch, in MVA (0.0 if bus type code > 3)
'O_PGEN'	Active power output of in-service machines at this plant (units determined by power output options)
'O_QGEN'	Reactive power output of in-service machines at this plant (units determined by power output options)
'O_MVA'	|PGEN + j QGEN| of in-service machines at this plant (units determined by power output options)
'O_PMAX'	Active power upper limit of in-service machines at this plant (units determined by power output options)
'O_PMIN'	Active power lower limit of in-service machines at this plant (units determined by power output options)
'O_QMAX'	Reactive power upper limit of in-service machines at this plant (units determined by power output options)
'O_QMIN'	Reactive power lower limit of in-service machines at this plant (units determined by power output options)
'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)

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Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the GET_PLANT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BUSES	number of elements in each column of RARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_PLANT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.3.6. AGENBUSTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the plant bus data family (aGenBusInt, aGenBusReal, aGenBusCplx and aGenBusChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = agenbustypes(string)
```

Fortran command syntax:

```
CALL GET_PLANT_BUS_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the plant bus data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the plant bus data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.4. Machine Data

The machine data family of subsystem data retrieval APIs returns arrays of data for machines in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for all subsystem machines, or to restrict the array entries to data for a subset of machines according to their bus, plant, and/or machine configuration and statuses.

Data for subsystem machines is returned from all APIs in the machine data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API is referenced; multiple machines at the same bus are in ascending machine identifier order.

8.4.1. AMACHCHAR

Use this API to return an array of character values for subsystem machines.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = amachchar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_MACHINE_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,MACHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem machines to include (input; 1 by default).	
	FLAG = 1	for only in-service machines at in-service plants (type code 2 or 3).
	FLAG = 2	for all machines at in-service plants (type code 2 or 3).
	FLAG = 3	for only in-service machines, including those at Type 1 and 4 buses.
	FLAG = 4	for all machines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or machine quantities desired (input; no default allowed).	
	'ID'	Machine identifier (2 characters).
	'NAME'	Bus name (12 characters).

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	'EXNAME'	Extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of MACHS returned by the GET_MACHINE_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer MACHS	number of elements in each column of CARRAY that are set by this API. MACHS is identical to the value of MACHS returned by the GET_MACHINE_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of CARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.4.2. AMACHCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the machine data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, machs = amachcount(sid, flag)
```

Fortran command syntax:

```
CALL GET_MACHINE_COUNT(SID, FLAG, MACHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem machines to include (input; 1 by default).	
	FLAG = 1	for only in-service machines at in-service plants (type code 2 or 3).
	FLAG = 2	for all machines at in-service plants (type code 2 or 3).
	FLAG = 3	for only in-service machines, including those at type 1 and 4 buses.
	FLAG = 4	for all machines.
Integer MACHS	number of machines in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, MACHS returned
	IERR = 1	Working case is empty; MACHS returned as 0
	IERR = 2	Invalid SID value; MACHS returned as 0
	IERR = 3	Invalid FLAG value; MACHS returned as 0

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8.4.3. AMACHCPLX

Use this API to return an array of complex values for subsystem machines.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = amachcplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_MACHINE_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,MACHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem machines to include (input; 1 by default).	
	FLAG = 1	for only in-service machines at in-service plants (type code 2 or 3).
	FLAG = 2	for all machines at in-service plants (type code 2 or 3).
	FLAG = 3	for only in-service machines, including those at Type 1 and 4 buses.
	FLAG = 4	for all machines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or machine quantities desired (input; no default allowed).	
	'ZSORCE'	Machine impedance (ZR + j ZX)
	'XTRAN'	Step-up transformer impedance (RT + j XT)
	'ZPOS'	" Positive sequence fault analysis machine impedance (Rpos + j X"")"
	'ZNEG'	Negative sequence machine impedance (ZRNEG + j ZXNEG)
	'ZZERO'	Zero sequence machine impedance (RZERO + j XZERO)
	'ZGRND'	Grounding impedance. If the grounding impedance data I/O code is 2 and the bus base voltage is positive the impedance is returned in ohms; otherwise it is return in per unit.
	'ZGRNDPU'	Grounding impedance in per unit
	'PQGEN'	Power output (MW and Mvar)

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	'O_PQGEN'	Power output (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of MACHS returned by the GET_MACHINE_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer MACHS	number of elements in each column of XARRAY that are set by this API. MACHS is identical to the value of MACHS returned by the GET_MACHINE_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value
	IERR = 7	Sequence data not in case (when STRING = 'ZPOS', 'ZNEG', 'ZZERO', 'ZGRND' or 'ZGRNDPU')

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8.4.4. AMACHINT

Use this API to return an array of integer values for subsystem machines.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = amachint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_MACHINE_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,MACHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem machines to include (input; 1 by default).	
	FLAG = 1	for only in-service machines at in-service plants (type code 2 or 3).
	FLAG = 2	for all machines at in-service plants (type code 2 or 3).
	FLAG = 3	for only in-service machines, including those at Type 1 and 4 buses.
	FLAG = 4	for all machines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or machine quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'STATUS'	Machine status
	'WMOD'	Wind machine reactive power limits mode; 0 if the machine is not a wind machine
	'OWNERS'	Number of owners
	'OWN1'	First owner
	'OWN2'	Second owner
	'OWN3'	Third owner
	'OWN4'	Fourth owner
	'CZG'	Grounding impedance data input/outout (I/O) code (1 for per unit 2 for ohms)
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of MACHS returned by the	

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	GET_MACHINE_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer MACHS	number of elements in each column of IARRAY that are set by this API. MACHS is identical to the value of MACHS returned by the GET_MACHINE_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of IARRAY, is not large enough
IERR = 6	Invalid STRING value
IERR = 7	Sequence data not in case (when STRING = 'CZG')

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8.4.5. AMACHREAL

Use this API to return an array of real values for subsystem machines.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = amachreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_MACHINE_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,MACHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem machines to include (input; 1 by default).	
	FLAG = 1	for only in-service machines at in-service plants (type code 2 or 3).
	FLAG = 2	for all machines at in-service plants (type code 2 or 3).
	FLAG = 3	for only in-service machines, including those at Type 1 and 4 buses.
	FLAG = 4	for all machines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or machine quantities desired (input; no default allowed).	
	'FRACT1'	First owner fraction
	'FRACT2'	Second owner fraction
	'FRACT3'	Third owner fraction
	'FRACT4'	Fourth owner fraction
	'PERCENT'	MVA loading as a percentage of machine MVA base, MBASE
	'MBASE'	Machine MVA base
	'GENTAP'	Step-up transformer tap ratio
	'WPF'	Power factor used in setting reactive power limits for the wind machine when its WMOD is 2 or 3; 1.0 if the machine is not a wind machine
	'RPOS'	Positive sequence fault analysis machine resistance

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	'XSUBTR'	Positive sequence fault analysis machine subtransient reactance
	'XTRANS'	Positive sequence fault analysis machine transient reactance
	'XSYNCH'	Positive sequence fault analysis machine synchronous reactance
	'PGEN'	Active power output, in MW
	'QGEN'	Reactive power output, in Mvar
	'MVA'	|PGEN + j QGEN|, in MVA
	'PMAX'	Active power upper limit, in MW
	'PMIN'	Active power lower limit, in MW
	'QMAX'	Reactive power upper limit, in Mvar
	'QMIN'	Reactive power lower limit, in Mvar
	'O_PGEN'	Active power output (units determined by power output options)
	'O_QGEN'	Reactive power output (units determined by power output options)
	'O_MVA'	|PGEN + j QGEN| (units determined by power output options)
	'O_PMAX'	Active power upper limit (units determined by power output options)
	'O_PMIN'	Active power lower limit (units determined by power output options)
	'O_QMAX'	Reactive power upper limit (units determined by power output options)
	'O_QMIN'	Reactive power lower limit (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of MACHS returned by the GET_MACHINE_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer MACHS	number of elements in each column of RARRAY that are set by this API. MACHS is identical to the value of MACHS returned by the GET_MACHINE_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of RARRAY, is not large enough

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IERR = 6	Invalid STRING value
IERR = 7	Sequence data not in case (when STRING = 'RPOS', 'XSUBTR', 'XTRANS', or 'XSYNCH')

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8.4.6. AMACHTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the machine data family (aMachInt, aMachReal, aMachCplx and aMachChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = amachtypes(string)
```

Fortran command syntax:

```
CALL GET_MACHINE_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the machine data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the machine data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.5. Load Bus Data

The load bus data family of subsystem data retrieval APIs returns arrays of data for buses in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for all subsystem buses, or to restrict the array entries to data for a subset of buses according to their bus and/or load configuration and statuses.

Data for subsystem buses is returned from all APIs in the load bus data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API is referenced.

8.5.1. ALODBUSCHAR

Use this API to return an array of character values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = alodbuschar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_LOAD_BUS_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service load buses (type code 1, 2 or 3) with at least one in-service load.
	FLAG = 2	for only in-service load buses (type code 1, 2 or 3), including those with only out-of-service loads.
	FLAG = 3	for all in-service buses, including those that are not load buses.
	FLAG = 4	for all load buses, including those with only out-of-service loads.
	FLAG = 5	for all buses, including those that are not load buses.
	FLAG = 10	for only in-service load buses (type code 1, 2 or 3) with at least one in-service scalable load.

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	FLAG = 11	for only in-service load buses (type code 1, 2 or 3) with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 12	for all load buses with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 20	for only in-service load buses (type code 1, 2 or 3) with at least one in-service fixed load.
	FLAG = 21	for only in-service load buses (type code 1, 2 or 3) with at least one fixed load, including those with all such loads out-of-service.
	FLAG = 22	for all load buses with at least one fixed load, including those with all such loads out-of-service.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or load quantities desired (input; no default allowed).	
	'NAME'	Bus name (12 characters).
	'EXNAME'	Extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the GET_LOAD_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of CARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_LOAD_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of CARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.5.2. ALODBUSCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the load bus data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, buses = alodbuscount(sid, flag)
```

Fortran command syntax:

```
CALL GET_LOAD_BUS_COUNT(SID, FLAG, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service load buses (type code 1, 2 or 3) with at least one in-service load.
	FLAG = 2	for only in-service load buses (type code 1, 2 or 3), including those with only out-of-service loads.
	FLAG = 3	for all in-service buses, including those that are not load buses.
	FLAG = 4	for all load buses, including those with only out-of-service loads.
	FLAG = 5	for all buses, including those that are not load buses.
	FLAG = 10	for only in-service load buses (type code 1, 2 or 3) with at least one in-service scalable load.
	FLAG = 11	for only in-service load buses (type code 1, 2 or 3) with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 12	for all load buses with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 20	for only in-service load buses (type code 1, 2 or 3) with at least one in-service fixed load.
	FLAG = 21	for only in-service load buses (type code 1, 2 or 3) with at least one fixed load, including those with all such loads out-of-service.

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	FLAG = 22	for all load buses with at least one fixed load, including those with all such loads out-of-service.
Integer BUSES	number of buses in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BUSES returned
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0
	IERR = 3	Invalid FLAG value; BUSES returned as 0

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8.5.3. ALODBUSCPLX

Use this API to return an array of complex values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = alodbuscplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_LOAD_BUS_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service load buses (type code 1, 2 or 3) with at least one in-service load.
	FLAG = 2	for only in-service load buses (type code 1, 2 or 3), including those with only out-of-service loads.
	FLAG = 3	for all in-service buses, including those that are not load buses.
	FLAG = 4	for all load buses, including those with only out-of-service loads.
	FLAG = 5	for all buses, including those that are not load buses.
	FLAG = 10	for only in-service load buses (type code 1, 2 or 3) with at least one in-service scalable load.
	FLAG = 11	for only in-service load buses (type code 1, 2 or 3) with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 12	for all load buses with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 20	for only in-service load buses (type code 1, 2 or 3) with at least one in-service fixed load.
	FLAG = 21	for only in-service load buses (type code 1, 2 or 3) with at least one fixed load, including those with all such loads out-of-service.

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	FLAG = 22	for all load buses with at least one fixed load, including those with all such loads out-of-service.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or load quantities desired (input; no default allowed).	
	'VOLTAGE'	Actual bus voltage in pu rectangular coordinates
	'MVAACT'	Actual in-service constant MVA load
	'MVANOM'	Nominal in-service constant MVA load at 1.0 pu voltage
	'ILACT'	Actual in-service constant current load
	'ILNOM'	Nominal in-service constant current load at 1.0 pu voltage
	'YLACT'	Actual total in-service constant admittance load
	'YLNOM'	Nominal total in-service constant admittance load at 1.0 pu voltage
	'TOTALACT'	Actual in-service load
	'TOTALNOM'	Nominal in-service load at 1.0 pu voltage
	'MISMATCH'	Bus mismatch (0.0 if bus type code > 3)
	'O_MVAACT'	Actual in-service constant MVA load (units determined by power output options)
	'O_MVANOM'	Nominal in-service constant MVA load at 1.0 pu voltage (units determined by power output options)
	'O_ILACT'	Actual in-service constant current load (units determined by power output options)
	'O_ILNOM'	Nominal in-service constant current load at 1.0 pu voltage (units determined by power output options)
	'O_YLACT'	Actual total in-service constant admittance load (units determined by power output options)
	'O_YLNOM'	Nominal total in-service constant admittance load at 1.0 pu voltage (units determined by power output options)
	'O_TOTALACT'	Actual in-service load (units determined by power output options)
	'O_TOTALNOM'	Nominal in-service load at 1.0 pu voltage (units determined by power output options)
	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the	

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	GET_LOAD_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Integer BUSES	number of elements in each column of XARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_LOAD_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of XARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.5.4. ALODBUSINT

Use this API to return an array of integer values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = alodbusint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_LOAD_BUS_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service load buses (type code 1, 2 or 3) with at least one in-service load.
	FLAG = 2	for only in-service load buses (type code 1, 2 or 3), including those with only out-of-service loads.
	FLAG = 3	for all in-service buses, including those that are not load buses.
	FLAG = 4	for all load buses, including those with only out-of-service loads.
	FLAG = 5	for all buses, including those that are not load buses.
	FLAG = 10	for only in-service load buses (type code 1, 2 or 3) with at least one in-service scalable load.
	FLAG = 11	for only in-service load buses (type code 1, 2 or 3) with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 12	for all load buses with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 20	for only in-service load buses (type code 1, 2 or 3) with at least one in-service fixed load.
	FLAG = 21	for only in-service load buses (type code 1, 2 or 3) with at least one fixed load, including those with all such loads out-of-service.

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	FLAG = 22	for all load buses with at least one fixed load, including those with all such loads out-of-service.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or load quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'TYPE'	Bus type code
	'AREA'	Bus area number
	'ZONE'	Bus zone number
	'OWNER'	Bus owner number
	'DUMMY'	Returns 1 if the bus is a dummy bus of a multisection line, or 0 if it is not
	'STATUS'	Inferred load status. Returns 1 if the bus type code is 1, 2 or 3 and at least one in-service load is present or 0 otherwise
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the GET_LOAD_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of IARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_LOAD_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of IARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.5.5. ALODBUSREAL

Use this API to return an array of real values for subsystem buses.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = alodbusreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_LOAD_BUS_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service load buses (type code 1, 2 or 3) with at least one in-service load.
	FLAG = 2	for only in-service load buses (type code 1, 2 or 3), including those with only out-of-service loads.
	FLAG = 3	for all in-service buses, including those that are not load buses.
	FLAG = 4	for all load buses, including those with only out-of-service loads.
	FLAG = 5	for all buses, including those that are not load buses.
	FLAG = 10	for only in-service load buses (type code 1, 2 or 3) with at least one in-service scalable load.
	FLAG = 11	for only in-service load buses (type code 1, 2 or 3) with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 12	for all load buses with at least one scalable load, including those with all such loads out-of-service.
	FLAG = 20	for only in-service load buses (type code 1, 2 or 3) with at least one in-service fixed load.
	FLAG = 21	for only in-service load buses (type code 1, 2 or 3) with at least one fixed load, including those with all such loads out-of-service.

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	FLAG = 22	for all load buses with at least one fixed load, including those with all such loads out-of-service.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or load quantities desired (input; no default allowed).	
	'BASE'	Bus base voltage, in kV
	'PU'	Actual bus voltage magnitude, in pu
	'KV'	Actual bus voltage magnitude, in kV (in pu if base voltage is 0.0)
	'ANGLE'	Bus voltage phase angle, in radians
	'ANGLED'	Bus voltage phase angle, in degrees
	'MVAACT'	Actual in-service constant MVA load
	'MVANOM'	Nominal in-service constant MVA load at 1.0 pu voltage
	'ILACT'	Actual in-service constant current load
	'ILNOM'	Nominal in-service constant current load at 1.0 pu voltage
	'YLACT'	Actual total in-service constant admittance load
	'YLNOM'	Nominal total in-service constant admittance load at 1.0 pu voltage
	'TOTALACT'	Actual in-service load
	'TOTALNOM'	Nominal in-service load at 1.0 pu voltage
	'MISMATCH'	Bus mismatch (0.0 if bus type code > 3)
	'O_MVAACT'	Actual in-service constant MVA load (units determined by power output options)
	'O_MVANOM'	Nominal in-service constant MVA load at 1.0 pu voltage (units determined by power output options)
	'O_ILACT'	Actual in-service constant current load (units determined by power output options)
	'O_ILNOM'	Nominal in-service constant current load at 1.0 pu voltage (units determined by power output options)
	'O_YLACT'	Actual total in-service constant admittance load (units determined by power output options)
	'O_YLNOM'	Nominal total in-service constant admittance load at 1.0 pu voltage (units determined by power output options)
	'O_TOTALACT'	Actual in-service load (units determined by power output options)
	'O_TOTALNOM'	Nominal in-service load at 1.0 pu voltage (units determined by power output options)

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	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the GET_LOAD_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of RARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_LOAD_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Real RARRAY (NSTR*DIM)	(output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of RARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.5.6. ALODBUSTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the load bus data family (aLodBusInt, aLodBusReal, aLodBusCplx and aLodBusChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = alodbustypes(string)
```

Fortran command syntax:

```
CALL GET_LOAD_BUS_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the load bus data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the load bus data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.6. Load Data

The load data family of subsystem data retrieval APIs returns arrays of data for loads in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for all subsystem loads, or to restrict the array entries to data for a subset of loads according to their bus and/or load configuration and statuses.

Data for subsystem loads is returned from all APIs in the load data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API is referenced; multiple loads at the same bus are in ascending load identifier order.

8.6.1. ALOADCHAR

Use this API to return an array of character values for subsystem loads.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = aloadchar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_LOAD_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,LOADS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem loads to include (input; 1 by default).	
	FLAG = 1	for only in-service loads at in-service buses (type code < 4).
	FLAG = 2	for all loads at in-service buses (type code < 4).
	FLAG = 3	for only in-service loads, including those at type 4 buses.
	FLAG = 4	for all loads.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or load quantities desired (input; no default allowed).	
	'ID'	Load identifier (2 characters).
	'NAME'	Bus name (12 characters).
	'EXNAME'	Extended bus name (18 characters).

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Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of LOADS returned by the GET_LOAD_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer LOADS	number of elements in each column of CARRAY that are set by this API. LOADS is identical to the value of LOADS returned by the GET_LOAD_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of CARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.6.2. ALOADCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the load data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, loads = aloadcount(sid, flag)
```

Fortran command syntax:

```
CALL GET_LOAD_COUNT(SID, FLAG, LOADS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem loads to include (input; 1 by default).	
	FLAG = 1	for only in-service loads at in-service buses (type code < 4).
	FLAG = 2	for all loads at in-service buses (type code < 4).
	FLAG = 3	for only in-service loads, including those at type 4 buses.
	FLAG = 4	for all loads.
Integer LOADS	number of loads in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, LOADS returned
	IERR = 1	Working case is empty; LOADS returned as 0
	IERR = 2	Invalid SID value; LOADS returned as 0
	IERR = 3	Invalid FLAG value; LOADS returned as 0

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8.6.3. ALOADCPLX

Use this API to return an array of complex values for subsystem loads.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = aloadcplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_LOAD_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,LOADS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem loads to include (input; 1 by default).	
	FLAG = 1	for only in-service loads at in-service buses (type code < 4).
	FLAG = 2	for all loads at in-service buses (type code < 4).
	FLAG = 3	for only in-service loads, including those at type 4 buses.
	FLAG = 4	for all loads.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or load quantities desired. NOTE: In the description below, the "net" values is the load value minus any in-service distributed generation on that load feeder. (input; no default allowed).	
	'MVAACT'	Actual in-service constant MVA load (in MW and Mvar)
	'MVANOM'	Nominal in-service constant MVA load at 1.0 pu voltage (in MW and Mvar)
	'ILACT'	Actual in-service constant current load (in MW and Mvar)
	'ILNOM'	Nominal in-service constant current load at 1.0 pu voltage (in MW and Mvar)
	'YLACT'	Actual total in-service constant admittance load (in MW and Mvar)
	'YLNOM'	Nominal total in-service constant admittance load at 1.0 pu voltage (in MW and Mvar)

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	'TOTALACT'	Actual in-service load (in MW and Mvar)
	'TOTALNOM'	Nominal in-service load at 1.0 pu voltage (in MW and Mvar)
	'O_MVAACT'	Actual in-service constant MVA load (units determined by power output options)
	'O_MVANOM'	Nominal in-service constant MVA load at 1.0 pu voltage (units determined by power output options)
	'O_ILACT'	Actual in-service constant current load (units determined by power output options)
	'O_ILNOM'	Nominal in-service constant current load at 1.0 pu voltage (units determined by power output options)
	'O_YLACT'	Actual total in-service constant admittance load (units determined by power output options)
	'O_YLNOM'	Nominal total in-service constant admittance load at 1.0 pu voltage (units determined by power output options)
	'O_TOTALACT'	Actual in-service load (units determined by power output options)
	'O_TOTALNOM'	Nominal in-service load at 1.0 pu voltage (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of LOADS returned by the GET_LOAD_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer LOADS	number of elements in each column of XARRAY that are set by this API. LOADS is identical to the value of LOADS returned by the GET_LOAD_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.6.4. ALOADINT

Use this API to return an array of integer values for subsystem loads.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = aloadint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_LOAD_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,LOADS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem loads to include (input; 1 by default).	
	FLAG = 1	for only in-service loads at in-service buses (type code < 4).
	FLAG = 2	for all loads at in-service buses (type code < 4).
	FLAG = 3	for only in-service loads, including those at type 4 buses.
	FLAG = 4	for all loads.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or load quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'AREA'	Bus area number
	'ZONE'	Bus zone number
	'OWNER'	Bus owner number
	'STATUS'	Load status
	'SCALE'	Scalable load flag
	'CGR'	Grounding code (0 for an ungrounded load 1 for a grounded load)
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of LOADS returned by the GET_LOAD_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	

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Integer LOADS	number of elements in each column of IARRAY that are set by this API. LOADS is identical to the value of LOADS returned by the GET_LOAD_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of IARRAY, is not large enough
IERR = 6	Invalid STRING value
IERR = 7	Sequence data not in case (when STRING = 'CGR')

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8.6.5. ALOADREAL

Use this API to return an array of real values for subsystem loads.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = aloadreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_LOAD_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,LOADS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem loads to include (input; 1 by default).	
	FLAG = 1	for only in-service loads at in-service buses (type code < 4).
	FLAG = 2	for all loads at in-service buses (type code < 4).
	FLAG = 3	for only in-service loads, including those at type 4 buses.
	FLAG = 4	for all loads.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or load quantities desired. NOTE: In the description below, the "net" values is the load value minus any in-service distributed generation on that load feeder. (input; no default allowed).	
	'MVAACT'	Actual in-service constant MVA load, in MVA
	'MVANOM'	Nominal in-service constant MVA load at 1.0 pu voltage, in MVA
	'ILACT'	Actual in-service constant current load, in MVA
	'ILNOM'	Nominal in-service constant current load at 1.0 pu voltage, in MVA
	'YLACT'	Actual total in-service constant admittance load, in MVA
	'YLNOM'	Nominal total in-service constant admittance load at 1.0 pu voltage, in MVA
	'TOTALACT'	Actual in-service load, in MVA

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	'TOTALNOM'	Nominal in-service load at 1.0 pu voltage, in MVA
	'YNEG'	Nominal negative sequence total load at 1.0 pu voltage, in MVA
	'YZERO'	Nominal zero sequence total load at 1.0 pu voltage, in MVA
	'O_MVAACT'	Actual in-service constant MVA load (units determined by power output options)
	'O_MVANOM'	Nominal in-service constant MVA load at 1.0 pu voltage (units determined by power output options)
	'O_ILACT'	Actual in-service constant current load (units determined by power output options)
	'O_ILNOM'	Nominal in-service constant current load at 1.0 pu voltage (units determined by power output options)
	'O_YLACT'	Actual total in-service constant admittance load (units determined by power output options)
	'O_YLNOM'	Nominal total in-service constant admittance load at 1.0 pu voltage (units determined by power output options)
	'O_TOTALACT'	Actual in-service load (units determined by power output options)
	'O_TOTALNOM'	Nominal in-service load at 1.0 pu voltage (units determined by power output options)
	'O_YNEG'	Nominal negative sequence total load at 1.0 pu voltage (units determined by power output options)
	'O_YZERO'	Nominal zero sequence total load at 1.0 pu voltage (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of LOADS returned by the GET_LOAD_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer LOADS	number of elements in each column of RARRAY that are set by this API. LOADS is identical to the value of LOADS returned by the GET_LOAD_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value

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IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value
IERR = 7	Sequence data not in case (when STRING = 'YNEG', 'O_YNEG', 'YZERO' or 'O_YZERO')

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8.6.6. ALOADTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the load data family (aLoadInt, aLoadReal, aLoadCplx and aLoadChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = aloadtypes(string)
```

Fortran command syntax:

```
CALL GET_LOAD_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the load data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the load data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.7. Fixed Shunt Bus Data

The fixed shunt bus data family of subsystem data retrieval APIs returns arrays of data for buses in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for all subsystem buses, or to restrict the array entries to data for a subset of buses according to their bus and/or fixed shunt configuration and statuses.

Data for subsystem buses is returned from all APIs in the fixed shunt bus data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API is referenced.

8.7.1. AFXSHNTBUSCHAR

Use this API to return an array of character values for subsystem buses.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, carray = afxshntbuschar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SHUNT_BUS_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunt buses (type code 1, 2 or 3) with at least one in-service fixed shunt.
	FLAG = 2	for only in-service fixed shunt buses (type code 1, 2 or 3), including those with only out-of-service fixed shunts.
	FLAG = 3	for all in-service buses, including those that are not fixed shunt buses.
	FLAG = 4	for all fixed shunt buses, including those with only out-of-service fixed shunts.
	FLAG = 5	for all buses, including those that are not fixed shunt buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	

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Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or fixed shunt quantities desired (input; no default allowed). 'NAME' Bus name (12 characters). 'EXNAME' Extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the GET_SHUNT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Integer BUSES	number of elements in each column of CARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_SHUNT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer IERR	error code (output). IERR = 0 No error IERR = 1 Working case is empty IERR = 2 Invalid SID value IERR = 3 Invalid FLAG value IERR = 4 Invalid NSTR value IERR = 5 DIM, and hence the size of CARRAY, is not large enough IERR = 6 Invalid STRING value

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8.7.2. AFXSHNTBUSCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the fixed shunt bus data family.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, buses = afxshntbuscount(sid, flag)
```

Fortran command syntax:

```
CALL GET_SHUNT_BUS_COUNT(SID, FLAG, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunt buses (type code 1, 2 or 3) with at least one in-service fixed shunt.
	FLAG = 2	for only in-service fixed shunt buses (type code 1, 2 or 3), including those with only out-of-service fixed shunts.
	FLAG = 3	for all in-service buses, including those that are not fixed shunt buses.
	FLAG = 4	for all fixed shunt buses, including those with only out-of-service fixed shunts.
	FLAG = 5	for all buses, including those that are not fixed shunt buses.
Integer BUSES	number of buses in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BUSES returned
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0
	IERR = 3	Invalid FLAG value; BUSES returned as 0

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8.7.3. AFXSHNTBUSCPLX

Use this API to return an array of complex values for subsystem buses.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, xarray = afxshntbuscplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SHUNT_BUS_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunt buses (type code 1, 2 or 3) with at least one in-service fixed shunt.
	FLAG = 2	for only in-service fixed shunt buses (type code 1, 2 or 3), including those with only out-of-service fixed shunts.
	FLAG = 3	for all in-service buses, including those that are not fixed shunt buses.
	FLAG = 4	for all fixed shunt buses, including those with only out-of-service fixed shunts.
	FLAG = 5	for all buses, including those that are not fixed shunt buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or fixed shunt quantities desired (input; no default allowed).	
	'VOLTAGE'	Actual bus voltage, in pu, rectangular coordinates
	'PU_GBZERO'	Zero sequence total nominal in-service fixed bus shunt, in pu
	'SHUNTACT'	Total actual in-service fixed bus shunt (in MW and Mvar)
	'SHUNTNOM'	Total nominal in-service fixed bus shunt at 1.0 pu voltage (in MW and Mvar)
	'GBZERO'	Zero sequence total nominal in-service fixed bus shunt (in MW and Mvar)

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	'MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (in MW and Mvar)
	'O_SHUNTACT'	Total actual in-service fixed bus shunt (units determined by power output options)
	'O_SHUNTNOM'	Total nominal in-service fixed bus shunt at 1.0 pu voltage (units determined by power output options)
	'O_GBZERO'	Zero sequence total nominal in-service fixed bus shunt (units determined by power output options)
	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the GET_SHUNT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of XARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_SHUNT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value
	IERR = 7	Sequence data not in case; when STRING = 'PU_GBZERO', 'GBZERO' and 'O_GBZERO'

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8.7.4. AFXSHNTBUSINT

Use this API to return an array of integer values for subsystem buses.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, iarray = afxshntbusint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SHUNT_BUS_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunt buses (type code 1, 2 or 3) with at least one in-service fixed shunt.
	FLAG = 2	for only in-service fixed shunt buses (type code 1, 2 or 3), including those with only out-of-service fixed shunts.
	FLAG = 3	for all in-service buses, including those that are not fixed shunt buses.
	FLAG = 4	for all fixed shunt buses, including those with only out-of-service fixed shunts.
	FLAG = 5	for all buses, including those that are not fixed shunt buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or fixed shunt quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'TYPE'	Bus type code
	'AREA'	Bus area number
	'ZONE'	Bus zone number
	'OWNER'	Bus owner number
	'DUMMY'	Returns 1 if the bus is a dummy bus of a multisection line or 0 if it is not
	'STATUS'	Inferred shunt status. Returns 1 if the bus type code is 1 2 or 3 and at least one in-service fixed shunt is present or 0 otherwise

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Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the GET_SHUNT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Integer BUSES	number of elements in each column of IARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_SHUNT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of IARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.7.5. AFXSHNTBUSREAL

Use this API to return an array of real values for subsystem buses.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, rarray = afxshntbusreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SHUNT_BUS_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunt buses (type code 1, 2 or 3) with at least one in-service fixed shunt.
	FLAG = 2	for only in-service fixed shunt buses (type code 1, 2 or 3), including those with only out-of-service fixed shunts.
	FLAG = 3	for all in-service buses, including those that are not fixed shunt buses.
	FLAG = 4	for all fixed shunt buses, including those with only out-of-service fixed shunts.
	FLAG = 5	for all buses, including those that are not fixed shunt buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or fixed shunt quantities desired (input; no default allowed).	
	'BASE'	Bus base voltage, in kV
	'PU'	Actual bus voltage magnitude, in pu
	'KV'	Actual bus voltage magnitude, in kV (in pu if base voltage is 0.0)
	'ANGLE'	Bus voltage phase angle, in radians
	'ANGLED'	Bus voltage phase angle, in degrees
	'PU_GBZERO'	Zero sequence total nominal in-service fixed bus shunt, in pu
	'SHUNTACT'	Total actual in-service fixed bus shunt, in MVA

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	'SHUNTNOM'	Total nominal in-service fixed bus shunt at 1.0 pu voltage, in MVA
	'GBZERO'	Zero sequence total nominal in-service fixed bus shunt, in MVA
	'MISMATCH'	Bus mismatch (0.0 if bus type code > 3), in MVA
	'O_SHUNTACT'	Total actual in-service fixed bus shunt (units determined by power output options)
	'O_SHUNTNOM'	Total nominal in-service fixed bus shunt at 1.0 pu voltage (units determined by power output options)
	'O_GBZERO'	Zero sequence total nominal in-service fixed bus shunt (units determined by power output options)
	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the GET_SHUNT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of RARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_SHUNT_BUS_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of RARRAY, is not large enough
	IERR = 6	Invalid STRING value
	IERR = 7	Sequence data not in case; when STRING = 'PU_GBZERO', 'GBZERO' and 'O_GBZERO'

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8.7.6. AFXSHNTBUSTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the fixed shunt bus data family (aFxShntBusInt, aFxShntBusReal, aFxShntBusCplx and aFxShntBusChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = afxshntbustypes(string)
```

Fortran command syntax:

```
CALL GET_SHUNT_BUS_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the fixed shunt bus data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the fixed shunt bus data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.8. Fixed Shunt Data

The fixed shunt data family of subsystem data retrieval APIs returns arrays of data for fixed bus shunts in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for all subsystem fixed bus shunts, or to restrict the array entries to data for a subset of fixed bus shunts according to their bus and/or fixed shunt configuration and statuses.

Data for subsystem fixed bus shunts is returned from all APIs in the fixed shunt data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API is referenced; multiple fixed shunts at the same bus are in ascending shunt identifier order.

8.8.1. AFXSHUNTCHAR

Use this API to return an array of character values for subsystem fixed shunts.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, carray = afxshuntchar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SHUNT_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,SHUNTS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem fixed bus shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunts at in-service buses (type code < 4).
	FLAG = 2	for all fixed shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service fixed shunts, including those at type 4 buses.
	FLAG = 4	for all fixed bus shunts.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or fixed shunt quantities desired (input; no default allowed).	
	'ID'	Shunt identifier (2 characters).
	'NAME'	Bus name (12 characters).

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	'EXNAME'	Extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY DIM must be as least as large as the value of SHUNTS returned by the GET_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer SHUNTS	number of elements in each column of CARRAY that are set by this API. SHUNTS is identical to the value of SHUNTS returned by the GET_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of CARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.8.2. AFXSHUNTCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the fixed shunt data family.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, shunts = afxshuntcount(sid, flag)
```

Fortran command syntax:

```
CALL GET_SHUNT_COUNT(SID, FLAG, SHUNTS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem fixed bus shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunts at in-service buses (type code < 4).
	FLAG = 2	for all fixed shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service fixed shunts, including those at type 4 buses.
	FLAG = 4	for all fixed bus shunts.
Integer SHUNTS	number of fixed bus shunts in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, SHUNTS returned
	IERR = 1	Working case is empty; SHUNTS returned as 0
	IERR = 2	Invalid SID value; SHUNTS returned as 0
	IERR = 3	Invalid FLAG value; SHUNTS returned as 0

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8.8.3. AFXSHUNTCPLX

Use this API to return an array of complex values for subsystem fixed shunts.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, xarray = afxshuntcplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SHUNT_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,SHUNTS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem fixed bus shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunts at in-service buses (type code < 4).
	FLAG = 2	for all fixed shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service fixed shunts, including those at type 4 buses.
	FLAG = 4	for all fixed bus shunts.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or fixed shunt quantities desired (input; no default allowed).	
	'PU_GBZERO'	Zero sequence nominal fixed bus shunt, in per unit
	'SHUNTACT'	Actual in-service fixed bus shunt (in MW and Mvar)
	'SHUNTNOM'	Nominal fixed bus shunt at 1.0 pu voltage (in MW and Mvar)
	'GBZERO'	Zero sequence nominal fixed bus shunt (in MW and Mvar)
	'O_SHUNTACT'	Actual in-service fixed bus shunt (units determined by power output options)
	'O_SHUNTNOM'	Nominal fixed bus shunt at 1.0 pu voltage (units determined by power output options)
	'O_GBZERO'	Zero sequence nominal fixed bus shunt (units determined by power output options)

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Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of SHUNTS returned by the GET_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer SHUNTS	number of elements in each column of XARRAY that are set by this API. SHUNTS is identical to the value of SHUNTS returned by the GET_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of XARRAY, is not large enough
IERR = 6	Invalid STRING value
IERR = 7	Sequence data not in case; when STRING = 'PU_GBZERO', 'GBZERO' and 'O_GBZERO'

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8.8.4. AFXSHUNTINT

Use this API to return an array of integer values for subsystem fixed shunts.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, iarray = afxshuntint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SHUNT_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,SHUNTS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem fixed bus shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunts at in-service buses (type code < 4).
	FLAG = 2	for all fixed shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service fixed shunts, including those at type 4 buses.
	FLAG = 4	for all fixed bus shunts.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or fixed shunt quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'STATUS'	Shunt status
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of SHUNTS returned by the GET_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer SHUNTS	number of elements in each column of IARRAY that are set by this API. SHUNTS is identical to the value of SHUNTS returned by the GET_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	

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IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of IARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.8.5. AFXSHUNTREAL

Use this API to return an array of real values for subsystem fixed shunts.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
ierr, rarray = afxshuntreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SHUNT_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,SHUNTS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem fixed bus shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service fixed shunts at in-service buses (type code < 4).
	FLAG = 2	for all fixed shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service fixed shunts, including those at type 4 buses.
	FLAG = 4	for all fixed bus shunts.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or fixed shunt quantities desired (input; no default allowed).	
	'PU_GBZERO'	Zero sequence nominal fixed bus shunt, in pu
	'SHUNTACT'	Actual in-service fixed bus shunt, in MVA
	'SHUNTNOM'	Nominal fixed bus shunt at 1.0 pu voltage, in MVA
	'GBZERO'	Zero sequence nominal fixed bus shunt, in MVA
	'O_SHUNTACT'	Actual in-service fixed bus shunt (units determined by power output options)
	'O_SHUNTNOM'	Nominal fixed bus shunt at 1.0 pu voltage (units determined by power output options)
	'O_GBZERO'	Zero sequence nominal fixed bus shunt (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of SHUNTS returned by the	

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	GET_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer SHUNTS	number of elements in each column of RARRAY that are set by this API. SHUNTS is identical to the value of SHUNTS returned by the GET_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value
IERR = 7	Sequence data not in case; when STRING = 'PU_GBZERO', 'GBZERO' and 'O_GBZERO'

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8.8.6. AFXSHUNTTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the fixed shunt data family (aFxShuntInt, aFxShuntReal, aFxShuntCplx and aFxShuntChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = afxshunttypes(string)
```

Fortran command syntax:

```
CALL GET_SHUNT_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the fixed shunt data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the fixed shunt data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.9. Switched Shunt Data

The switched shunt data family of subsystem data retrieval APIs returns arrays of data for switched shunts in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for all subsystem switched shunts, or to restrict the array entries to data for switched shunts at in-service buses.

Data for subsystem switched shunts is returned from all APIs in the switched shunt data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API is referenced.

8.9.1. ASWSHCHAR

Use this API to return an array of character values for subsystem switched shunts.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = aswshchar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SWITCHED_SHUNT_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,SHNTS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem switched shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service switched shunts at in-service buses (type code < 4).
	FLAG = 2	for all switched shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service switched shunts, including those at type 4 buses.
	FLAG = 4	for all switched shunts.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or switched shunt quantities desired (input; no default allowed).	
	'VSCNAME'	VSC dc line name; non-blank name is returned only for those switched shunts for which the control mode, MODSW, is 4 (12 characters).

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	'FACTSNAME'	FACTS device name; non-blank name is returned only for those switched shunts for which the control mode, MODSW, is 6 (12 characters).
	'DEVICENAME'	VSC dc line name or FACTS device name; non-blank name is returned only for those switched shunts for which the control mode, MODSW, is 4 or 6 (12 characters).
	'NAME'	Bus name (12 characters).
	'EXNAME'	Extended bus name (18 characters).
	'IREGNAME'	Bus name of regulated bus (12 characters).
	'IREGEXNAME'	Extended bus name of regulated bus (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of SHNTS returned by the GET_SWITCHED_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer SHNTS	number of elements in each column of CARRAY that are set by this API. SHNTS is identical to the value of SHNTS returned by the GET_SWITCHED_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of CARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.9.2. ASWSHCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the switched shunt data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, shnts = aswshcount(sid, flag)
```

Fortran command syntax:

```
CALL GET_SWITCHED_SHUNT_COUNT(SID, FLAG, SHNTS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem switched shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service switched shunts at in-service buses (type code < 4).
	FLAG = 2	for all switched shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service switched shunts, including those at type 4 buses.
	FLAG = 4	for all switched shunts.
Integer SHNTS	number of switched shunts in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, SHNTS returned
	IERR = 1	Working case is empty; SHNTS returned as 0
	IERR = 2	Invalid SID value; SHNTS returned as 0
	IERR = 3	Invalid FLAG value; SHNTS returned as 0

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8.9.3. ASWSHCPLX

Use this API to return an array of complex values for subsystem switched shunts.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = aswshcplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SWITCHED_SHUNT_COMPLEX(SID, FLAG, NSTR, STRING, DIM, XARRAY,
 SHNTS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem switched shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service switched shunts at in-service buses (type code < 4).
	FLAG = 2	for all switched shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service switched shunts, including those at type 4 buses.
	FLAG = 4	for all switched shunts.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or switched shunt quantities desired (input; no default allowed).	
	'VOLTAGE'	Actual bus voltage in pu rectangular coordinates
	'YSWACT'	Present shunt switched on
	'MISMATCH'	Bus mismatch (0.0 if bus type code > 3)
	'O_YSWACT'	Present shunt switched on (units determined by power output options)
	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of SHNTS returned by the GET_SWITCHED_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	

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Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer SHNTS	number of elements in each column of XARRAY that are set by this API. SHNTS is identical to the value of SHNTS returned by the GET_SWITCHED_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of XARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.9.4. ASWSHINT

Use this API to return an array of integer values for subsystem switched shunts.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = aswshint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SWITCHED_SHUNT_INTEGER(SID, FLAG, NSTR, STRING, DIM, IARRAY,
 SHNTS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem switched shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service switched shunts at in-service buses (type code < 4).
	FLAG = 2	for all switched shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service switched shunts, including those at type 4 buses.
	FLAG = 4	for all switched shunts.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or switched shunt quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'TYPE'	Bus type code
	'AREA'	Bus area number
	'ZONE'	Bus zone number
	'OWNER'	Bus owner number
	'DUMMY'	Returns 1 if the bus is a dummy bus of a multisection line or 0 if it is not
	'MODE'	Switched shunt control mode
	'STATUS'	Switched shunt status flag
	'ADJMETHOD'	Switched shunt adjustment method
	'IREG'	Controlled bus
	'BLOCKS'	Number of blocks

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	'STEPSBLOCK1'	Number of steps in block 1
	'STEPSBLOCK2'	Number of steps in block 2
	'STEPSBLOCK3'	Number of steps in block 3
	'STEPSBLOCK4'	Number of steps in block 4
	'STEPSBLOCK5'	Number of steps in block 5
	'STEPSBLOCK6'	Number of steps in block 6
	'STEPSBLOCK7'	Number of steps in block 7
	'STEPSBLOCK8'	Number of steps in block 8
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of SHNTS returned by the GET_SWITCHED_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer SHNTS	number of elements in each column of IARRAY that are set by this API. SHNTS is identical to the value of SHNTS returned by the GET_SWITCHED_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of IARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.9.5. ASWSHREAL

Use this API to return an array of real values for subsystem switched shunts.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = aswshreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_SWITCHED_SHUNT_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,SHNTS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem switched shunts to include (input; 1 by default).	
	FLAG = 1	for only in-service switched shunts at in-service buses (type code < 4).
	FLAG = 2	for all switched shunts at in-service buses (type code < 4).
	FLAG = 3	for only in-service switched shunts, including those at type 4 buses.
	FLAG = 4	for all switched shunts.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or switched shunt quantities desired (input; no default allowed).	
	'BASE'	Bus base voltage in kV
	'PU'	Actual bus voltage magnitude, in pu
	'KV'	Actual bus voltage magnitude, in kV (in pu if base voltage is 0.0)
	'ANGLE'	Bus voltage phase angle, in radians
	'ANGLED'	Bus voltage phase angle, in degrees
	'VSWHI'	Controlled quantity upper limit
	'VSWLO'	Controlled quantity lower limit
	'RMPCT'	Percent of total Mvar to be contributed by this switched shunt
	'PU_BSWZERO'	Present zero sequence susceptance switched on, in pu
	'PUBZERBLOCK1'	Zero sequence pu step size in block 1
	'PUBZERBLOCK2'	Zero sequence pu step size in block 2

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'PUBZERBLOCK3'	Zero sequence pu step size in block 3
'PUBZERBLOCK4'	Zero sequence pu step size in block 4
'PUBZERBLOCK5'	Zero sequence pu step size in block 5
'PUBZERBLOCK6'	Zero sequence pu step size in block 6
'PUBZERBLOCK7'	Zero sequence pu step size in block 7
'PUBZERBLOCK8'	Zero sequence pu step size in block 8
'BSWNOM'	Present susceptance switched on (susceptance)
'BSWMAX'	Admittance with all reactors off, all capacitors on
'BSWMIN'	Admittance with all reactors on, all capacitors off
'BSWACT'	Present susceptance switched on (Mvar)
'BSWZERO'	Present zero sequence susceptance switched on
'BSTPBLOCK1'	Step size in block 1
'BSTPBLOCK2'	Step size in block 2
'BSTPBLOCK3'	Step size in block 3
'BSTPBLOCK4'	Step size in block 4
'BSTPBLOCK5'	Step size in block 5
'BSTPBLOCK6'	Step size in block 6
'BSTPBLOCK7'	Step size in block 7
'BSTPBLOCK8'	Step size in block 8
'BZERBLOCK1'	Zero sequence step size in block 1
'BZERBLOCK2'	Zero sequence step size in block 2
'BZERBLOCK3'	Zero sequence step size in block 3
'BZERBLOCK4'	Zero sequence step size in block 4
'BZERBLOCK5'	Zero sequence step size in block 5
'BZERBLOCK6'	Zero sequence step size in block 6
'BZERBLOCK7'	Zero sequence step size in block 7
'BZERBLOCK8'	Zero sequence step size in block 8
'MISMATCH'	Bus mismatch (0.0 if bus type code > 3)
'O_BSWNOM'	Present susceptance switched on (units determined by power output options)
'O_BSWMAX'	Admittance with all reactors off all capacitors on (units determined by power output options)
'O_BSWMIN'	Admittance with all reactors on all capacitors off (units determined by power output options)
'O_BSWACT'	Present susceptance switched on (units determined by power output options)
'O_BSWZERO'	Present zero sequence susceptance switched on (units determined by power output options)

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	'O_BSTPBLOCK1'	Step size in block 1 (units determined by power output options)
	'O_BSTPBLOCK2'	Step size in block 2 (units determined by power output options)
	'O_BSTPBLOCK3'	Step size in block 3 (units determined by power output options)
	'O_BSTPBLOCK4'	Step size in block 4 (units determined by power output options)
	'O_BSTPBLOCK5'	Step size in block 5 (units determined by power output options)
	'O_BSTPBLOCK6'	Step size in block 6 (units determined by power output options)
	'O_BSTPBLOCK7'	Step size in block 7 (units determined by power output options)
	'O_BSTPBLOCK8'	Step size in block 8 (units determined by power output options)
	'O_BZERBLOCK1'	Zero sequence step size in block 1 (units determined by power output options)
	'O_BZERBLOCK2'	Zero sequence step size in block 2 (units determined by power output options)
	'O_BZERBLOCK3'	Zero sequence step size in block 3 (units determined by power output options)
	'O_BZERBLOCK4'	Zero sequence step size in block 4 (units determined by power output options)
	'O_BZERBLOCK5'	Zero sequence step size in block 5 (units determined by power output options)
	'O_BZERBLOCK6'	Zero sequence step size in block 6 (units determined by power output options)
	'O_BZERBLOCK7'	Zero sequence step size in block 7 (units determined by power output options)
	'O_BZERBLOCK8'	Zero sequence step size in block 8 (units determined by power output options)
	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of SHNTS returned by the GET_SWITCHED_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer SHNTS	number of elements in each column of RARRAY that are set by this API. SHNTS is identical to the value of SHNTS returned by the GET_SWITCHED_SHUNT_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error

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IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value
IERR = 7	Sequence data not in case (when STRING = 'BSWZERO', 'PU_BSWZERO', 'O_BSWZERO', 'BZERBLOCKn', 'PUB-ZERBLOCKn', 'O_BZERBLOCKn')

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8.9.6. ASWSHTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the switched shunt data family (aSwshInt, aSwshReal, aSwshCplx and aSwshChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = aswshtypes(string)
```

Fortran command syntax:

```
CALL GET_SWITCHED_SHUNT_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the switched shunt data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the switched shunt data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.10. Branch Flow Data

The branch flow data family of subsystem data retrieval APIs returns arrays of loading data for ac branches in a specified subsystem. These APIs return loading data for non-transformer branches, two-winding transformers, and three-winding transformers.

One of the input items to each API allows the user to elect either to retrieve data for all subsystem branches, for subsystem tie branches, or for both subsystem and tie branches. Another input item allows the user to elect either to retrieve data for all such branches, or to restrict the array entries to data for in-service branches.

Data for subsystem branches is returned from all APIs in the branch flow data family in the same order. Unless otherwise restricted for tie branches, each branch appears in both directions (in all three directions for three-winding transformers) in the arrays returned by the APIs of the branch flow data family. Branches are ordered in ascending numerical or alphabetical order by from bus, according to the bus output option setting in effect at the time the API is referenced. For each from bus, branches are ordered in ascending order by to bus. Parallel branches are ordered in ascending circuit identifier order.

8.10.1. AFLOWCHAR

Use this API to return an array of character values for subsystem branches.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = aflowchar(sid, owner, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_BRANCH_FLOW_CHAR(SID, OWNER, TIES, FLAG, NSTR, STRING, DIM,
 CARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem branches only.
	TIES = 2	for the subsystem bus end of tie branches only.
	TIES = 3	for the non-subsystem bus end of tie branches only.

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	TIES = 4	for each end of tie branches only.
	TIES = 5	for each end of interior subsystem branches and the subsystem bus end of tie branches.
	TIES = 6	for each end of interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service branches.
	FLAG = 2	for all branches.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the branch quantities desired (input; no default allowed).	
	'ID'	Circuit identifier (2 characters).
	'FROMNAME'	From bus name (12 characters).
	'FROMEXNAME'	From bus extended bus name (18 characters).
	'TONAME'	To bus name (three-winding transformer name for a three-winding transformer winding) (12 characters).
	'TOEXNAME'	To bus extended bus name (three-winding transformer name and winding number for a three-winding transformer winding) (18 characters).
	'NMETERNAME'	Non-metered bus name (12 characters).
	'NMETEREXNAME'	Non-metered bus extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_BRANCH_FLOW_COUNT API when it is called with the same values specified for SID, OWNER, TIES and FLAG (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_BRANCH_FLOW_COUNT API when it is called with the same values specified for SID, OWNER, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid NSTR value

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IERR = 7	DIM, and hence the size of CARRAY, is not large enough
IERR = 8	Invalid STRING value

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8.10.2. AFLOWCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the branch flow data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, brnchs = aflowcount(sid, owner, ties, flag)
```

Fortran command syntax:

```
CALL GET_BRANCH_FLOW_COUNT(SID, OWNER, TIES, FLAG, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem branches only.
	TIES = 2	for the subsystem bus end of tie branches only.
	TIES = 3	for the non-subsystem bus end of tie branches only.
	TIES = 4	for each end of tie branches only.
	TIES = 5	for each end of interior subsystem branches and the subsystem bus end of tie branches.
	TIES = 6	for each end of interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service branches.
	FLAG = 2	for all branches.
Integer BRNCHS	number of array entries required for the subsystem indicated by SID and OWNER that meet the editing criteria indicated by TIES and FLAG (output).	
Integer IERR	error code (output).	

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IERR = 0	No error, BRNCHS returned
IERR = 1	Working case is empty; BRNCHS returned as 0
IERR = 2	Invalid SID value; BRNCHS returned as 0
IERR = 3	Invalid OWNER value; BRNCHS returned as 0
IERR = 4	Invalid TIES value; BRNCHS returned as 0
IERR = 5	Invalid FLAG value; BRNCHS returned as 0

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8.10.3. AFLOWCPLX

Use this API to return an array of complex values for subsystem branches.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = aflowcplx(sid, owner, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_BRANCH_FLOW_COMPLEX(SID, OWNER, TIES, FLAG, NSTR, STRING, DIM,
 XARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem branches only.
	TIES = 2	for the subsystem bus end of tie branches only.
	TIES = 3	for the non-subsystem bus end of tie branches only.
	TIES = 4	for each end of tie branches only.
	TIES = 5	for each end of interior subsystem branches and the subsystem bus end of tie branches.
	TIES = 6	for each end of interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service branches.
	FLAG = 2	for all branches.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the branch quantities desired (input; no default allowed).	

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	'PQ'	P + j Q flow at from bus end (in MW and Mvar)
	'PQLOSS'	Losses (in MW and Mvar)
	'O_PQ'	P + j Q flow at from bus end (units determined by power output options)
	'O_PQLOSS'	Losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_BRANCH_FLOW_COUNT API when it is called with the same values specified for SID, OWNER, TIES and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_BRANCH_FLOW_COUNT API when it is called with the same values specified for SID, OWNER, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of XARRAY, is not large enough
	IERR = 8	Invalid STRING value

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8.10.4. AFLOWINT

Use this API to return an array of integer values for subsystem branches.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = aflowint(sid, owner, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_BRANCH_FLOW_INTEGER(SID, OWNER, TIES, FLAG, NSTR, STRING, DIM,
 IARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem branches only.
	TIES = 2	for the subsystem bus end of tie branches only.
	TIES = 3	for the non-subsystem bus end of tie branches only.
	TIES = 4	for each end of tie branches only.
	TIES = 5	for each end of interior subsystem branches and the subsystem bus end of tie branches.
	TIES = 6	for each end of interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service branches.
	FLAG = 2	for all branches.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the branch quantities desired (input; no default allowed).	

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	'FROMNUMBER'	From bus number
	'TONUMBER'	To bus number (>10000000 for a three-winding transformer winding)
	'STATUS'	Branch status
	'NMETERNUMBER'	Non-metered end bus number
	'OWNERS'	Number of owners
	'OWN1'	First owner
	'OWN2'	Second owner
	'OWN3'	Third owner
	'OWN4'	Fourth owner
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_BRANCH_FLOW_COUNT API when it is called with the same values specified for SID, OWNER, TIES and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_BRANCH_FLOW_COUNT API when it is called with the same values specified for SID, OWNER, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of IARRAY, is not large enough
	IERR = 8	Invalid STRING value

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8.10.5. AFLOWREAL

Use this API to return an array of real values for subsystem branches.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = aflowreal(sid, owner, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_BRANCH_FLOW_REAL(SID, OWNER, TIES, FLAG, NSTR, STRING, DIM,
 RARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem branches only.
	TIES = 2	for the subsystem bus end of tie branches only.
	TIES = 3	for the non-subsystem bus end of tie branches only.
	TIES = 4	for each end of tie branches only.
	TIES = 5	for each end of interior subsystem branches and the subsystem bus end of tie branches.
	TIES = 6	for each end of interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service branches.
	FLAG = 2	for all branches.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the branch quantities desired (input; no default allowed).	

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'AMPS'	Branch current, in amps (0.0 if bus base voltage is 0.0)
'PUCUR'	Branch current, in pu
'PCTRATE'	Percent from bus current of default rating set
'PCTRATEA'	Percent from bus current of rating set A
'PCTRATEB'	Percent from bus current of rating set B
'PCTRATEC'	Percent from bus current of rating set C
'PCTMVARATE'	Percent from bus MVA of default rating set
'PCTMVARATEA'	Percent from bus MVA of rating set A
'PCTMVARATEB'	Percent from bus MVA of rating set B
'PCTMVARATEC'	Percent from bus MVA of rating set C
'PCTCORPRATE'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of default rating set
'PCTCORPRATEA'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set A
'PCTCORPRATEB'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set B
'PCTCORPRATEC'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set C
'FRACT1'	First owner fraction
'FRACT2'	Second owner fraction
'FRACT3'	Third owner fraction
'FRACT4'	Fourth owner fraction
'RATE'	Rating from default rating set
'RATEA'	Rating from rating set A
'RATEB'	Rating from rating set B
'RATEC'	Rating from rating set C
'P'	Active power flow, in MW
'Q'	Reactive power flow, in Mvar
'MVA'	|P + j Q|, in MVA
'PLOSS'	Active power losses, in MW
'QLOSS'	Reactive power losses, in Mvar
'O_P'	Active power flow (units determined by power output options)
'O_Q'	Reactive power flow (units determined by power output options)
'O_MVA'	|P + j Q| (units determined by power output options)
'O_PLOSS'	Active power losses (units determined by power output options)

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	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_BRANCH_FLOW_COUNT API when it is called with the same values specified for SID, OWNER, TIES and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_BRANCH_FLOW_COUNT API when it is called with the same values specified for SID, OWNER, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of RARRAY, is not large enough
	IERR = 8	Invalid STRING value

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8.10.6. AFLOWTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the branch flow data family (aFlowInt, aFlowReal, aFlowCplx and aFlowChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = aflowtypes(string)
```

Fortran command syntax:

```
CALL GET_BRANCH_FLOW_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the branch flow data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the branch flow data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.11. Branch Data

The branch data family of subsystem data retrieval APIs returns arrays of data for two-terminal ac branches in a specified subsystem. These APIs return data for non-transformer branches and/or two-winding transformers.

One of the input items to each API allows the user to elect either to retrieve data for all subsystem branches, for subsystem tie branches, or for both subsystem and tie branches. Another input item allows the user to elect to retrieve data for only non-transformer branches, only two-winding transformers, or both non-transformer branches and two-winding transformers, and either to retrieve data for all such branches, or to restrict the array entries to data for in-service branches.

Data for subsystem branches is returned from all APIs in the branch data family in the same order. One of the input items to each API allows the user to elect that either single entry arrays (each branch appears once) or double entry arrays (each branch appears in both directions) are to be returned. Branches are ordered in ascending numerical or alphabetical order by from bus, according to the bus output option setting in effect at the time the API is referenced. For each from bus, branches are ordered in ascending order by to bus. Parallel branches are ordered in ascending circuit identifier order.

8.11.1. ABRNCHAR

Use this API to return an array of character values for subsystem branches.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = abrnchar(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_BRANCH_CHAR(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 CARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem branches only.
	TIES = 2	for subsystem tie branches only.
	TIES = 3	for both interior subsystem branches and tie branches.

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Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default). FLAG = 1 for only in-service non-transformer branches. FLAG = 2 for all non-transformer branches. FLAG = 3 for only in-service non-transformer branches and two-winding transformers. FLAG = 4 for all non-transformer branches and two-winding transformers. FLAG = 5 for only in-service two-winding transformers. FLAG = 6 for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default). ENTRY = 1 for single entry (each branch once). ENTRY = 2 for double entry (each branch in both directions).
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).
Character STRING (NSTR)	NSTR of the following strings indicating the branch quantities desired (input; no default allowed). 'ID' Circuit identifier (2 characters). 'FROMNAME' From bus name (12 characters). 'FROMEXNAME' From bus extended bus name (18 characters). 'TONAME' To bus name (three-winding transformer name for a three-winding transformer winding) (12 characters). 'TOEXNAME' To bus extended bus name (three-winding transformer name and winding number for a three-winding transformer winding) (18 characters). 'METERNAME' Metered bus name (12 characters). 'METEREXNAME' Metered bus extended bus name (18 characters). 'NMETERNAME' Non-metered bus name (12 characters). 'NMETEREXNAME' Non-metered bus extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_BRANCH_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_BRANCH_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).

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Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of CARRAY, is not large enough
	IERR = 9	Invalid STRING value

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8.11.2. ABRNCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the branch data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, brnchs = abrncount(sid, owner, ties, flag, entry)
```

Fortran command syntax:

```
CALL GET_BRANCH_COUNT(SID, OWNER, TIES, FLAG, ENTRY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem branches only.
	TIES = 2	for subsystem tie branches only.
	TIES = 3	for both interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service non-transformer branches.
	FLAG = 2	for all non-transformer branches.
	FLAG = 3	for only in-service non-transformer branches and two-winding transformers.
	FLAG = 4	for all non-transformer branches and two-winding transformers.
	FLAG = 5	for only in-service two-winding transformers.
	FLAG = 6	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).

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Integer BRNCHS	number of array entries required for the subsystem indicated by SID and OWNER that meet the editing criteria indicated by TIES, FLAG and ENTRY (output).
Integer IERR	error code (output).

IERR = 0	No error, BRNCHS returned
IERR = 1	Working case is empty; BRNCHS returned as 0
IERR = 2	Invalid SID value; BRNCHS returned as 0
IERR = 3	Invalid OWNER value; BRNCHS returned as 0
IERR = 4	Invalid TIES value; BRNCHS returned as 0
IERR = 5	Invalid FLAG value; BRNCHS returned as 0
IERR = 6	Invalid ENTRY value; BRNCHS returned as 0

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8.11.3. ABRNCPLX

Use this API to return an array of complex values for subsystem branches.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = abrncplx(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_BRANCH_COMPLEX(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 XARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem branches only.
	TIES = 2	for subsystem tie branches only.
	TIES = 3	for both interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service non-transformer branches.
	FLAG = 2	for all non-transformer branches.
	FLAG = 3	for only in-service non-transformer branches and two-winding transformers.
	FLAG = 4	for all non-transformer branches and two-winding transformers.
	FLAG = 5	for only in-service two-winding transformers.
	FLAG = 6	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).

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Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).
Character STRING (NSTR)	NSTR of the following strings indicating the branch quantities desired (input; no default allowed).
	'RX' Branch impedance, in pu
	'FROMSHNT' Line shunt at from bus end, in pu
	'TOSHNT' Line shunt at to bus end, in pu
	'RXZERO' Zero sequence branch impedance, in pu
	'FROMSHNTZERO' Zero sequence line shunt at from bus end, in pu
	'TOSHNTZERO' Zero sequence line shunt at to bus end, in pu
	'PQ' P + j Q flow at from bus end (MW and Mvar)
	'PQLOSS' Losses (MW and Mvar)
	'O_PQ' P + j Q flow at from bus end (units determined by power output options)
	'O_PQLOSS' Losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_BRANCH_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_BRANCH_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).
Integer IERR	error code (output).
	IERR = 0 No error
	IERR = 1 Working case is empty
	IERR = 2 Invalid SID value
	IERR = 3 Invalid OWNER value
	IERR = 4 Invalid TIES value
	IERR = 5 Invalid FLAG value
	IERR = 6 Invalid ENTRY value
	IERR = 7 Invalid NSTR value
	IERR = 8 DIM, and hence the size of XARRAY, is not large enough
	IERR = 9 Invalid STRING value
	IERR = 10 Sequence data not in case (when STRING = 'RXZERO', 'FROMSHNTZERO' or 'TOSHNTZERO')

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8.11.4. ABRNINT

Use this API to return an array of integer values for subsystem branches.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = abrnint(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_BRANCH_INTEGER(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 IARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem branches only.
	TIES = 2	for subsystem tie branches only.
	TIES = 3	for both interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service non-transformer branches.
	FLAG = 2	for all non-transformer branches.
	FLAG = 3	for only in-service non-transformer branches and two-winding transformers.
	FLAG = 4	for all non-transformer branches and two-winding transformers.
	FLAG = 5	for only in-service two-winding transformers.
	FLAG = 6	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).

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Integer NSTR	the number of elements in STRING (1 < NSTR < 50) (input; no default allowed).
Character STRING (NSTR)	NSTR of the following strings indicating the branch quantities desired (input; no default allowed).

'FROMNUMBER'	From bus number
'TONUMBER'	To bus number
'STATUS'	Branch status
'METERNUMBER'	Metered end bus number
'NMETERNUMBER'	Non-metered end bus number
'OWNERS'	Number of owners
'OWN1'	First owner
'OWN2'	Second owner
'OWN3'	Third owner
'OWN4'	Fourth owner
'MOVTYPE'	System switching device type (returned as 0 if the branch is not a system switching device)

Integer DIM	the number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_BRANCH_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	the number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_BRANCH_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY. (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid OWNER value
IERR = 4	Invalid TIES value
IERR = 5	Invalid FLAG value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of IARRAY, is not large enough
IERR = 9	Invalid STRING value

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8.11.5. ABRNREAL

Use this API to return an array of real values for subsystem branches.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = abrnreal(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_BRANCH_REAL(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 RARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use branch ownership.
Integer TIES	a flag indicating which subsystem branches to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem branches only.
	TIES = 2	for subsystem tie branches only.
	TIES = 3	for both interior subsystem branches and tie branches.
Integer FLAG	a flag indicating which subsystem branches to include (input; 1 by default).	
	FLAG = 1	for only in-service non-transformer branches.
	FLAG = 2	for all non-transformer branches.
	FLAG = 3	for only in-service non-transformer branches and two-winding transformers.
	FLAG = 4	for all non-transformer branches and two-winding transformers.
	FLAG = 5	for only in-service two-winding transformers.
	FLAG = 6	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	

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Character STRING (NSTR)

NSTR of the following strings indicating the branch quantities desired (input; no default allowed).

'AMPS'	Branch current, in amps (0.0 if bus base voltage is 0.0)
'PUCUR'	Branch current, in pu
'PCTRATE'	Percent from bus current of default rating set
'PCTRATEA'	Percent from bus current of rating set A
'PCTRATEB'	Percent from bus current of rating set B
'PCTRATEC'	Percent from bus current of rating set C
'PCTRATE1'	Percent from bus current of rating set 1
'PCTRATE2'	Percent from bus current of rating set 2
'PCTRATE3'	Percent from bus current of rating set 3
'PCTRATE4'	Percent from bus current of rating set 4
'PCTRATE5'	Percent from bus current of rating set 5
'PCTRATE6'	Percent from bus current of rating set 6
'PCTRATE7'	Percent from bus current of rating set 7
'PCTRATE8'	Percent from bus current of rating set 8
'PCTRATE9'	Percent from bus current of rating set 9
'PCTRATE10'	Percent from bus current of rating set 10
'PCTRATE11'	Percent from bus current of rating set 11
'PCTRATE12'	Percent from bus current of rating set 12
'PCTMVARATE'	Percent from bus MVA of default rating set
'PCTMVARATEA'	Percent from bus MVA of rating set A
'PCTMVARATEB'	Percent from bus MVA of rating set B
'PCTMVARATEC'	Percent from bus MVA of rating set C
'PCTMVARATE1'	Percent from bus MVA of rating set 1
'PCTMVARATE2'	Percent from bus MVA of rating set 2
'PCTMVARATE3'	Percent from bus MVA of rating set 3
'PCTMVARATE4'	Percent from bus MVA of rating set 4
'PCTMVARATE5'	Percent from bus MVA of rating set 5
'PCTMVARATE6'	Percent from bus MVA of rating set 6
'PCTMVARATE7'	Percent from bus MVA of rating set 7
'PCTMVARATE8'	Percent from bus MVA of rating set 8
'PCTMVARATE9'	Percent from bus MVA of rating set 9
'PCTMVARATE10'	Percent from bus MVA of rating set 10
'PCTMVARATE11'	Percent from bus MVA of rating set 11
'PCTMVARATE12'	Percent from bus MVA of rating set 12
'PCTCORPRATE'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of default rating set
'PCTCORPRATEA'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set A

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'PCTCORPRATEB'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set B
'PCTCORPRATEC'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set C
'PCTCRPRATE'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of default rating set
'PCTCRPRATE1'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 1
'PCTCRPRATE2'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 2
'PCTCRPRATE3'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 3
'PCTCRPRATE4'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 4
'PCTCRPRATE5'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 5
'PCTCRPRATE6'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 6
'PCTCRPRATE7'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 7
'PCTCRPRATE8'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 8
'PCTCRPRATE9'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 9
'PCTCRPRATE10'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 10
'PCTCRPRATE11'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 11
'PCTCRPRATE12'	Percent from bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 12
'MAXPCTRATE'	Larger of percent from/to bus current of default rating set

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'MAXPCTRATEA'	Larger of percent from/to bus current of rating set A
'MAXPCTRATEB'	Larger of percent from/to bus current of rating set B
'MAXPCTRATEC'	Larger of percent from/to bus current of rating set C
'MAXPCTRATE1'	Larger of percent from/to bus current of rating set 1
'MAXPCTRATE2'	Larger of percent from/to bus current of rating set 2
'MAXPCTRATE3'	Larger of percent from/to bus current of rating set 3
'MAXPCTRATE4'	Larger of percent from/to bus current of rating set 4
'MAXPCTRATE5'	Larger of percent from/to bus current of rating set 5
'MAXPCTRATE6'	Larger of percent from/to bus current of rating set 6
'MAXPCTRATE7'	Larger of percent from/to bus current of rating set 7
'MAXPCTRATE8'	Larger of percent from/to bus current of rating set 8
'MAXPCTRATE9'	Larger of percent from/to bus current of rating set 9
'MAXPCTRATE10'	Larger of percent from/to bus current of rating set 10
'MAXPCTRATE11'	Larger of percent from/to bus current of rating set 11
'MAXPCTRATE12'	Larger of percent from/to bus current of rating set 12
'MXPCTMVARAT'	Larger of percent from/to bus MVA of default rating set
'MXPCTMVARATA'	Larger of percent from/to bus MVA of rating set A
'MXPCTMVARATB'	Larger of percent from/to bus MVA of rating set B
'MXPCTMVARATC'	Larger of percent from/to bus MVA of rating set C
'MXPCTPRATE'	Larger of percent from/to bus MVA of rating set C
'MXPCTPRATE1'	Larger of percent from/to bus MVA of rating set 1
'MXPCTPRATE2'	Larger of percent from/to bus MVA of rating set 2
'MXPCTPRATE3'	Larger of percent from/to bus MVA of rating set 3

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'MXPCTPRATE4'	Larger of percent from/to bus MVA of rating set 4
'MXPCTPRATE5'	Larger of percent from/to bus MVA of rating set 5
'MXPCTPRATE6'	Larger of percent from/to bus MVA of rating set 6
'MXPCTPRATE7'	Larger of percent from/to bus MVA of rating set 7
'MXPCTPRATE8'	Larger of percent from/to bus MVA of rating set 8
'MXPCTPRATE9'	Larger of percent from/to bus MVA of rating set 9
'MXPCTPRATE10'	Larger of percent from/to bus MVA of rating set 10
'MXPCTPRATE11'	Larger of percent from/to bus MVA of rating set 11
'MXPCTPRATE12'	Larger of percent from/to bus MVA of rating set 12
'MXPCTCRPRAT'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of default rating set
'MXPCTCRPRATA'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set A
'MXPCTCRPRATB'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set B
'MXPCTCRPRATC'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set C
'MXPCTCRPRA'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of default rating set
'MXPCTCRPRA1'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 1
'MXPCTCRPRA2'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 2
'MXPCTCRPRA3'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent

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	loading units program option setting) of rating set 3
'MXPCTCRPRA4'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 4
'MXPCTCRPRA5'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 5
'MXPCTCRPRA6'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 6
'MXPCTCRPRA7'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 7
'MXPCTCRPRA8'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 8
'MXPCTCRPRA9'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 9
'MXPCTCRPRA10'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 10
'MXPCTCRPRA11'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 11
'MXPCTCRPRA12'	Larger of percent from/to bus current or MVA loading (according to the appropriate percent loading units program option setting) of rating set 12
'FRACT1'	First owner fraction
'FRACT2'	Second owner fraction
'FRACT3'	Third owner fraction
'FRACT4'	Fourth owner fraction
'RATE'	Rating from default rating set
'RATEA'	Rating from rating set A
'RATEB'	Rating from rating set B
'RATEC'	Rating from rating set C
'RATE1'	Rating from rating set 1

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	'RATE2'	Rating from rating set 2
	'RATE3'	Rating from rating set 3
	'RATE4'	Rating from rating set 4
	'RATE5'	Rating from rating set 5
	'RATE6'	Rating from rating set 6
	'RATE7'	Rating from rating set 7
	'RATE8'	Rating from rating set 8
	'RATE9'	Rating from rating set 9
	'RATE10'	Rating from rating set 10
	'RATE11'	Rating from rating set 11
	'RATE12'	Rating from rating set 12
	'LENGTH'	Line length
	'CHARGING'	Total charging capacitance, in pu
	'CHARGINGZERO'	Zero sequence total charging capacitance, in pu
	'MOVIRATED'	MOV Rated Current for series capacitor branch, in kA
	'P'	Active power flow at from bus end, in MW
	'Q'	Reactive power flow at from bus end, in Mvar
	'MVA'	|P + j Q| at from bus end, in MVA
	'MAXMVA'	|P + j Q| at from bus or to bus end whichever is larger, in MVA
	'PLOSS'	Active power losses, in MW
	'QLOSS'	Reactive power losses, in Mvar
	'O_P'	Active power flow at from bus end (units determined by power output options)
	'O_Q'	Reactive power flow at from bus end (units determined by power output options)
	'O_MVA'	|P + j Q| at from bus end (units determined by power output options)
	'O_MAXMVA'	||P + j Q| at from bus or to bus end whichever is larger (units determined by power output options)
	'O_PLOSS'	Active power losses (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_BRANCH_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	

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Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_BRANCH_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid OWNER value
IERR = 4	Invalid TIES value
IERR = 5	Invalid FLAG value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of RARRAY, is not large enough
IERR = 9	Invalid STRING value
IERR = 10	Sequence data not in case (when STRING = 'CHARGINGZERO')

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8.11.6. ABRNTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the branch data family (aBrnInt, aBrnReal, aBrnCplx and aBrnChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = abrntypes(string)
```

Fortran command syntax:

```
CALL GET_BRANCH_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the branch data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the branch data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.12. Two-Winding Transformer Data

The two-winding transformer data family of subsystem data retrieval APIs returns arrays of data for two-winding transformers in a specified subsystem.

One of the input items to each API allows the user to elect either to retrieve data for all subsystem two-winding transformers, for subsystem tie two-winding transformers, or for both subsystem and tie two-winding transformers. Another input item allows the user to elect either to retrieve data for all such transformers, or to restrict the array entries to data for in-service transformers.

Data for subsystem two-winding transformers is returned from all APIs in the two-winding transformer data family in the same order. One of the input items to each API allows the user to elect that either single entry arrays (each transformer appears once) or double entry arrays (each transformer appears in both directions) are to be returned. Transformers are ordered in ascending numerical or alphabetical order by from bus, according to the bus output option setting in effect at the time the API is referenced. For each from bus, transformers are ordered in ascending order by to bus. Parallel transformers are ordered in ascending circuit identifier order.

8.12.1. ATRNCHAR

Use this API to return an array of character values for subsystem two-winding transformers.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = atrnchar(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_2WINDING_CHAR(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 CARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem two-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-winding transformers only.
	TIES = 2	for subsystem tie two-winding transformers only.

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	TIES = 3	for both interior subsystem and tie two-winding transformers.
Integer FLAG	a flag indicating which subsystem two-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service two-winding transformers.
	FLAG = 2	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	
	'ID'	Circuit identifier (2 characters).
	'FROMNAME'	From bus name (12 characters).
	'FROMEXNAME'	From bus extended bus name (18 characters).
	'TONAME'	To bus name (three-winding transformer name for a three-winding transformer winding) (12 characters).
	'TOEXNAME'	To bus extended bus name (three-winding transformer name and winding number for a three-winding transformer winding) (18 characters).
	'METERNAME'	Metered bus name (12 characters).
	'METEREXNAME'	Metered bus extended bus name (18 characters).
	'NMETERNAME'	Non-metered bus name (12 characters).
	'NMETEREXNAME'	Non-metered bus extended bus name (18 characters).
	'ICONTNAME'	Controlled bus name (12 characters).
	'ICONTEXNAME'	Controlled bus extended bus name (18 characters).
	'WIND1NAME'	Winding 1 bus name (12 characters).
	'WIND1EXNAME'	Winding 1 bus extended bus name (18 characters).
	'WIND2NAME'	Winding 2 bus name (12 characters).
	'WIND2EXNAME'	Winding 2 bus extended bus name (18 characters).
	'XFRNAME'	Transformer name (40 characters).
	'VECTORGROUP'	Vector group name (12 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_2WINDING_COUNT API when it is called with the same values	

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	specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_2WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid OWNER value
IERR = 4	Invalid TIES value
IERR = 5	Invalid FLAG value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of CARRAY, is not large enough
IERR = 9	Invalid STRING value

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8.12.2. ATRNCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the two-winding transformer data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, brnchs = atrncount(sid, owner, ties, flag, entry)
```

Fortran command syntax:

```
CALL GET_2WINDING_COUNT(SID,OWNER,TIES,FLAG,ENTRY,BRNCHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem two-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-winding transformers only.
	TIES = 2	for subsystem tie two-winding transformers only.
	TIES = 3	for both interior subsystem and tie two-winding transformers.
Integer FLAG	a flag indicating which subsystem two-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service two-winding transformers.
	FLAG = 2	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).
Integer BRNCHS	number of array entries required for the subsystem indicated by SID and OWNER that meet the editing criteria indicated by TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BRNCHS returned

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IERR = 1	Working case is empty; BRNCHS returned as 0
IERR = 2	Invalid SID value; BRNCHS returned as 0
IERR = 3	Invalid OWNER value; BRNCHS returned as 0
IERR = 4	Invalid TIES value; BRNCHS returned as 0
IERR = 5	Invalid FLAG value; BRNCHS returned as 0
IERR = 6	Invalid ENTRY value; BRNCHS returned as 0

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8.12.3. ATRNCPLX

Use this API to return an array of complex values for subsystem two-winding transformers.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = atrncplx(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_2WINDING_COMPLEX(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, XARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem two-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-winding transformers only.
	TIES = 2	for subsystem tie two-winding transformers only.
	TIES = 3	for both interior subsystem and tie two-winding transformers.
Integer FLAG	a flag indicating which subsystem two-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service two-winding transformers.
	FLAG = 2	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	
	'RXACT'	Actual transformer impedance in pu

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	'RXACTCZ'	Actual transformer impedance in units governed by CZ
	'RXNOM'	Nominal transformer impedance in pu
	'RXNOMCZ'	Nominal transformer impedance in units governed by CZ
	'YMAG'	Magnetizing admittance in pu
	'YMAGCM'	Magnetizing admittance in units governed by CM
	'COMPRX'	Load drop compensation impedance in pu
	'RXZERO'	Zero sequence branch impedance in pu
	'ZG1'	ZGRND' Zero sequence winding 1 grounding impedance in pu
	'ZG1CZG'	Zero sequence winding 1 grounding impedance in units governed by CZG
	'ZG2'	'ZGRND2' Zero sequence winding 2 grounding impedance in pu
	'ZG2CZG'	Zero sequence winding 2 grounding impedance in units governed by CZG
	'Z01'	Zero sequence leakage impedance connected to the winding 1 bus in pu
	'Z01CZ0'	Zero sequence leakage impedance connected to the winding 1 bus in units governed by CZ0
	'Z02'	Zero sequence leakage impedance connected to the winding 2 bus in pu
	'Z02CZ0'	Zero sequence leakage impedance connected to the winding 2 bus in units governed by CZ0
	'ZNUTRL'	Zero sequence common neutral grounding impedance in pu
	'ZNUTRLCZG'	Zero sequence common neutral grounding impedance in units governed by CZG
	'PQ'	P + j Q flow at from bus end
	'PQLOSS'	Losses (MW and Mvar)
	'O_PQ'	P + j Q flow at from bus end (units determined by power output options)
	'O_PQLOSS'	Losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_2WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	

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Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_2WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid OWNER value
IERR = 4	Invalid TIES value
IERR = 5	Invalid FLAG value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of XARRAY, is not large enough
IERR = 9	Invalid STRING value
IERR = 10	Sequence data not in case (when STRING = 'RXZERO', 'ZGRND', 'ZGRND2', 'ZG1', 'ZG1CZG', 'ZG2', 'ZG2CZG', 'Z01', 'Z01CZ0', 'Z02', 'Z02CZ0', 'ZNUTRL', or 'ZNUTRLCZG')

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8.12.4. ATRNINT

Use this API to return an array of integer values for subsystem two-winding transformers.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = atrnint(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_2WINDING_INTEGER(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, IARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem two-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-winding transformers only.
	TIES = 2	for subsystem tie two-winding transformers only.
	TIES = 3	for both interior subsystem and tie two-winding transformers.
Integer FLAG	a flag indicating which subsystem two-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service two-winding transformers.
	FLAG = 2	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	
	'FROMNUMBER'	From bus number

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	'TONUMBER'	To bus number
	'STATUS'	Transformer status
	'METERNUMBER'	Metered end bus number
	'NMETERNUMBER'	Non-metered end bus number
	'OWNERS'	Number of owners
	'OWN1'	First owner
	'OWN2'	Second owner
	'OWN3'	Third owner
	'OWN4'	Fourth owner
	'ICONTNUMBER'	Controlled bus number
	'WIND1NUMBER'	Winding 1 bus number
	'WIND2NUMBER'	Winding 2 bus number
	'TABLE'	Transformer impedance adjustment table number
	'CODE'	Transformer control mode for automatic adjustments
	'NTPOSN'	Number of tap positions
	'CW'	Winding data I/O code
	'CZ'	Impedance data I/O code
	'CM'	Magnetizing admittance I/O code
	'CZ0'	Leakage impedance data I/O code
	'CZG'	Grounding impedance data I/O code
	'CNXCOD'	Connection code
	'TPSTT'	Winding 1 ratio limit code: -1 for low, 0 for regulating, 1 for high, -2 for locked, -3 for controlling a dc line
	'ANSTT'	Winding 1 angle limit code: -1 for low, 0 for regulating, 1 for high, -2 for locked
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_2WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_2WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value

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IERR = 5	Invalid FLAG value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of IARRAY, is not large enough
IERR = 9	Invalid STRING value
IERR = 10	Sequence data not in case (when STRING = 'CZ0', 'CZG' or 'CNXCOD')

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8.12.5. ATRNREAL

Use this API to return an array of real values for subsystem two-winding transformers.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = atrnreal(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_2WINDING_REAL(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 RARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem two-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-winding transformers only.
	TIES = 2	for subsystem tie two-winding transformers only.
	TIES = 3	for both interior subsystem and tie two-winding transformers.
Integer FLAG	a flag indicating which subsystem two-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service two-winding transformers.
	FLAG = 2	for all two-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for single entry (each branch once).
	ENTRY = 2	for double entry (each branch in both directions).
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	
	'AMPS'	Branch current in amps (0.0 if bus base voltage is 0.0)

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'PUCUR'	Branch current in pu
'PCTRATE'	Percent from bus current of default rating set
'PCTRATEA'	Percent from bus current of rating set A
'PCTRATEB'	Percent from bus current of rating set B
'PCTRATEC'	Percent from bus current of rating set C
'PCTMVARATE'	Percent from bus MVA of default rating set
'PCTMVARATEA'	Percent from bus MVA of rating set A
'PCTMVARATEB'	Percent from bus MVA of rating set B
'PCTMVARATEC'	Percent from bus MVA of rating set C
'PCTCORPRATE'	Percent from bus current or MVA loading (according to the transformer percent loading units program option setting) of default rating set
'PCTCORPRATEA'	Percent from bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set A
'PCTCORPRATEB'	Percent from bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set B
'PCTCORPRATEC'	Percent from bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set C
'MAXPCTRATE'	Larger of percent from/to bus current of default rating set
'MAXPCTRATEA'	Larger of percent from/to bus current of rating set A
'MAXPCTRATEB'	Larger of percent from/to bus current of rating set B
'MAXPCTRATEC'	Larger of percent from/to bus current of rating set C
'MXPCTMVARAT'	Larger of percent from/to bus MVA of default rating set
'MXPCTMVARATA'	Larger of percent from/to bus MVA of rating set A
'MXPCTMVARATB'	Larger of percent from/to bus MVA of rating set B
'MXPCTMVARATC'	Larger of percent from/to bus MVA of rating set C
'MXPCTCRPRAT'	Larger of percent from/to bus current or MVA loading (according to the transformer percent loading units program option setting) of default rating set
'MXPCTCRPRATA'	Larger of percent from/to bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set A

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'MXPCTCRPRATB'	Larger of percent from/to bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set B
'MXPCTCRPRATC'	Larger of percent from/to bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set C
'FRACT1'	First owner fraction
'FRACT2'	Second owner fraction
'FRACT3'	Third owner fraction
'FRACT4'	Fourth owner fraction
'RATE'	Rating from default rating set
'RATEA'	Rating from rating set A
'RATEB'	Rating from rating set B
'RATEC'	Rating from rating set C
'RATIO'	Winding 1 tap ratio in pu
'RATIOCW'	Winding 1 tap ratio in units governed by CW
'RATIO2'	Winding 2 tap ratio in pu
'RATIO2CW'	Winding 2 tap ratio in units governed by CW
'ANGLE'	Winding 1 phase shift angle in degrees
'RMAX'	Winding 1 ratio or angle upper limit (in pu if ratio)
'RMAXCW'	Winding 1 ratio or angle upper limit (in units governed by CW if ratio)
'RMIN'	Winding 1 ratio or angle lower limit (in pu if ratio)
'RMINCW'	Winding 1 ratio or angle lower limit (in units governed by CW if ratio)
'VMAX'	Controlled quantity upper limit (in pu if voltage)
'VMAXKV'	Controlled quantity upper limit (in units governed by CW if voltage)
'VMIN'	Controlled quantity lower limit (in pu if voltage)
'VMINKV'	Controlled quantity lower limit (in units governed by CW if voltage)
'STEP'	Winding 1 ratio or angle step width (in pu if ratio)
'STEPCW'	Winding 1 ratio or angle step width (in units governed by CW if ratio)
'NOMV1'	Winding 1 nominal voltage in kv (0.0 for bus base voltage)
'NOMV2'	Winding 2 nominal voltage in kv (0.0 for bus base voltage)
'SBASE1'	Winding 1 base MVA

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	'P'	Active power flow at from bus end, in MW
	'Q'	Reactive power flow at from bus end, in Mvar
	'MVA'	|P + j Q| at from bus end, in MVA
	'MAXMVA'	|P + j Q| at from bus or to bus end, whichever is larger, in MVA
	'PLOSS'	Active power losses, in MW
	'QLOSS'	Reactive power losses, in Mvar
	'O_P'	Active power flow at from bus end (units determined by power output options)
	'O_Q'	Reactive power flow at from bus end (units determined by power output options)
	'O_MVA'	|P + j Q| at from bus end (units determined by power output options)
	'O_MAXMVA'	|P + j Q| at from bus or to bus end whichever is larger (units determined by power output options)
	'O_PLOSS'	Active power losses (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_2WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_2WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of RARRAY, is not large enough
	IERR = 9	Invalid STRING value

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8.12.6. ATRNTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the two-winding transformer data family (aTrnInt, aTrnReal, aTrnCplx and aTrnChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = atrntypes(string)
```

Fortran command syntax:

```
CALL GET_2WINDING_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the two-winding transformer data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the two-winding transformer data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.13. Three-Winding Transformer Data

The three-winding transformer data family of subsystem data retrieval APIs returns arrays of data for three-winding transformers in a specified subsystem.

One of the input items to each API allows the user to elect either to retrieve data for all subsystem three-winding transformers, for subsystem tie three-winding transformers, or for both subsystem and tie three-winding transformers. Another input item allows the user to elect either to retrieve data for all such transformers, or to restrict the array entries to data for in-service transformers.

Data for subsystem three-winding transformers is returned from all APIs in the three-winding transformer data family in the same order. One of the input items to each API allows the user to elect that transformers be ordered either by transformer name, or in ascending numerical or alphabetical order by winding 1 bus, according to the bus output option setting in effect at the time the API is referenced. For each winding 1 bus, transformers are ordered in ascending order by winding 2 bus, then winding 3 bus, then in ascending circuit identifier order.

8.13.1. ATR3CHAR

Use this API to return an array of character values for subsystem three-winding transformers.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = atr3char(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_3WINDING_CHAR(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 CARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.

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	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service three-winding transformers.
	FLAG = 2	for all three-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding 1 bus order.
	ENTRY = 2	for transformer name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	
	'ID'	Circuit identifier (2 characters).
	'WIND1NAME'	Winding 1 bus name (12 characters).
	'WIND1EXNAME'	Winding 1 bus extended bus name (18 characters).
	'WIND2NAME'	Winding 2 bus name (12 characters).
	'WIND2EXNAME'	Winding 2 bus extended bus name (18 characters).
	'WIND3NAME'	Winding 3 bus name (12 characters).
	'WIND3EXNAME'	Winding 3 bus extended bus name (18 characters).
	'NMETERNAME'	Non-metered bus name (12 characters).
	'NMETEREXNAME'	Non-metered bus extended bus name (18 characters).
	'XFRNAME'	Transformer name (40 characters).
	'VECTORGROUP'	Vector group name (12 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_3WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_3WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value

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IERR = 4	Invalid TIES value
IERR = 5	Invalid FLAG value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of CARRAY, is not large enough
IERR = 9	Invalid STRING value

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8.13.2. ATR3COUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the three-winding transformer data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, brnchs = atr3count(sid, owner, ties, flag, entry)
```

Fortran command syntax:

```
CALL GET_3WINDING_COUNT(SID,OWNER,TIES,FLAG,ENTRY,BRNCHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service three-winding transformers.
	FLAG = 2	for all three-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding 1 bus order.
	ENTRY = 2	for transformer name order.
Integer BRNCHS	number of array entries required for the subsystem indicated by SID and OWNER that meet the editing criteria indicated by TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BRNCHS returned

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IERR = 1	Working case is empty; BRNCHS returned as 0
IERR = 2	Invalid SID value; BRNCHS returned as 0
IERR = 3	Invalid OWNER value; BRNCHS returned as 0
IERR = 4	Invalid TIES value; BRNCHS returned as 0
IERR = 5	Invalid FLAG value; BRNCHS returned as 0
IERR = 6	Invalid ENTRY value; BRNCHS returned as 0

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8.13.3. ATR3CPLX

Use this API to return an array of complex values for subsystem three-winding transformers.

This API routine was first introduced in release 30.3.2.

Python command syntax:

```
ierr, xarray = atr3cplx(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_3WINDING_COMPLEX(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, XARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service three-winding transformers.
	FLAG = 2	for all three-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding 1 bus order.
	ENTRY = 2	for transformer name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	
	'RX1-2ACT'	Actual measured impedance between the winding 1 and winding 2 buses in pu

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'RX1-2ACTCZ'	Actual measured impedance between the winding 1 and winding 2 buses in units governed by CZ
'RX1-2NOM'	Nominal measured impedance between the winding 1 and winding 2 buses in pu
'RX1-2NOMCZ'	Nominal measured impedance between the winding 1 and winding 2 buses in units governed by CZ
'RX2-3ACT'	Actual measured impedance between the winding 2 and winding 3 buses in pu
'RX2-3ACTCZ'	Actual measured impedance between the winding 2 and winding 3 buses in units governed by CZ
'RX2-3NOM'	Nominal measured impedance between the winding 2 and winding 3 buses in pu
'RX2-3NOMCZ'	Nominal measured impedance between the winding 2 and winding 3 buses in units governed by CZ
'RX3-1ACT'	Actual measured impedance between the winding 3 and winding 1 buses in pu
'RX3-1ACTCZ'	Actual measured impedance between the winding 3 and winding 1 buses in units governed by CZ
'RX3-1NOM'	Nominal measured impedance between the winding 3 and winding 1 buses in pu
'RX3-1NOMCZ'	Nominal measured impedance between the winding 3 and winding 1 buses in units governed by CZ
'YMAG'	Magnetizing admittance in pu
'YMAGCM'	Magnetizing admittance in units governed by CM
'ZG1'	ZGRND' Zero sequence winding 1 grounding impedance in pu
'ZG1CZG'	Zero sequence winding 1 grounding impedance in units governed by CZG
'ZG2'	'ZGRND2' Zero sequence winding 2 grounding impedance in pu
'ZG2CZG'	Zero sequence winding 2 grounding impedance in units governed by CZG
'ZG3'	'ZGRND3' Zero sequence winding 3 grounding impedance in pu
'ZG3CZG'	Zero sequence winding 3 grounding impedance in units governed by CZG
'Z01'	Zero sequence winding 1 or bus 1 to bus 2 leakage impedance in pu
'Z01CZ0'	Zero sequence winding 1 or bus 1 to bus 2 leakage impedance in units governed by CZ0

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	'Z02'	Zero sequence winding 2 or bus 2 to bus 3 leakage impedance in pu
	'Z02CZ0'	Zero sequence winding 2 or bus 2 to bus 3 leakage impedance in units governed by CZ0
	'Z03'	Zero sequence winding 3 or bus 3 to bus 1 leakage impedance in pu
	'Z03CZ0'	Zero sequence winding 3 or bus 3 to bus 1 leakage impedance in units governed by CZ0
	'ZNUTRL'	Zero sequence common neutral grounding impedance in pu
	'ZNUTRLCZG'	Zero sequence common neutral grounding impedance in units governed by CZG
	'PQLOSS'	Losses (in MW and Mvar)
	'O_PQLOSS'	Losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_3WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_3WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of XARRAY, is not large enough
	IERR = 9	Invalid STRING value
	IERR = 10	Sequence data not in case (when STRING = 'ZGRND', 'ZG1', 'ZG1CZG', 'ZG2', 'ZG2CZG', 'ZG3', 'ZG3CZG', 'Z01', 'Z01CZ0', 'Z02', 'Z02CZ0', 'Z03', 'Z03CZ0', 'ZNUTRL', or 'ZNUTRLCZG')

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8.13.4. ATR3INT

Use this API to return an array of integer values for subsystem three-winding transformers.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = atr3int(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_3WINDING_INTEGER(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, IARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service three-winding transformers.
	FLAG = 2	for all three-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding 1 bus order.
	ENTRY = 2	for transformer name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	
	'WIND1NUMBER'	Winding 1 bus number

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	'WIND2NUMBER'	Winding 2 bus number
	'WIND3NUMBER'	Winding 3 bus number
	'STATUS'	Transformer status
	'NMETERNUMBER'	Non-metered end bus number
	'OWNERS'	Number of owners
	'OWN1'	First owner
	'OWN2'	Second owner
	'OWN3'	Third owner
	'OWN4'	Fourth owner
	'CW'	Winding data I/O code
	'CZ'	Impedance data I/O code
	'CM'	Magnetizing admittance I/O code
	'CZ0'	Leakage impedance data I/O code
	'CZG'	Grounding impedance data I/O code
	'CNXCOD'	Connection code

Integer DIM — number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_3WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).

Integer IARRAY (NSTR*DIM) — Is an array of at least NSTR columns, each of which contains DIM elements (output).

Integer BRNCHS — number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_3WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).

Integer IERR — error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid OWNER value
IERR = 4	Invalid TIES value
IERR = 5	Invalid FLAG value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of IARRAY, is not large enough
IERR = 9	Invalid STRING value
IERR = 10	Sequence data not in case (when STRING = 'CZ0', 'CZG' or 'CNXCOD')

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8.13.5. ATR3REAL

Use this API to return an array of real values for subsystem three-winding transformers.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = atr3real(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_3WINDING_REAL(SID, OWNER, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 RARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service three-winding transformers.
	FLAG = 2	for all three-winding transformers.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding 1 bus order.
	ENTRY = 2	for transformer name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	
	'FRACT1'	First owner fraction

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	'FRACT2'	Second owner fraction
	'FRACT3'	Third owner fraction
	'FRACT4'	Fourth owner fraction
	'VMSTAR'	Actual bus voltage magnitude at the star point bus, in pu
	'ANSTAR'	Bus voltage phase angle at the star point bus, in radians
	'PLOSS'	Active power losses, in MW
	'QLOSS'	Reactive power losses, in Mvar
	'O_PLOSS'	Active power losses (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_3WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_3WINDING_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of RARRAY, is not large enough
	IERR = 9	Invalid STRING value

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8.13.6. ATR3TYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the three-winding transformer data family (aTr3Int, aTr3Real, aTr3Cplx and aTr3Char).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = atr3types(string)
```

Fortran command syntax:

```
CALL GET_3WINDING_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the three-winding transformer data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the three-winding transformer data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.14. Three-Winding Transformer Winding Data

The three-winding transformer winding data family of subsystem data retrieval APIs returns arrays of winding data for three-winding transformers in a specified subsystem.

One of the input items to each API allows the user to elect either to retrieve data for all subsystem three-winding transformers, for subsystem tie three-winding transformers, or for both subsystem and tie three-winding transformers. Another input item allows the user to elect either to retrieve data for all such transformers, to restrict the array entries to data for all windings of in-service transformers, or to restrict the array entries to data for in-service windings of in-service transformers.

Data for subsystem three-winding transformers is returned from all APIs in the three-winding transformer winding data family in the same order. One of the input items to each API allows the user to elect that transformers be ordered either by transformer name with all windings of each transformer grouped together, or in ascending numerical or alphabetical order by winding bus, according to the bus output option setting in effect at the time the API is referenced.

8.14.1. AWNDCHAR

Use this API to return an array of character values for subsystem three-winding transformer windings.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, carray = awndchar(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_3WINDING_WINDINGS_CHAR(SID, OWNER, TIES, FLAG, ENTRY, NSTR,
 STRING, DIM, CARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.

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Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).
	FLAG = 1 — for only in-service windings of in-service three-winding transformers.
	FLAG = 2 — for all windings of in-service three-winding transformers.
	FLAG = 3 — for all three-winding transformer windings.
Integer ENTRY	a flag indicating organization (input; 1 by default).
	ENTRY = 1 — for winding bus order.
	ENTRY = 2 — for transformer name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).
	'ID' — Circuit identifier (2 characters).
	'WNDBUSNAME' — Name of the bus connected to this winding (12 characters).
	'WNDBUSEXNAME' — Extended bus name of the bus connected to this winding (18 characters).
	'OTHER1NAME' — Name of another bus connected to this winding (12 characters).
	'OTHER1EXNAME' — Extended bus name of another bus connected to this winding (18 characters).
	'OTHER2NAME' — Name of another bus connected to this winding (12 characters).
	'OTHER2EXNAME' — Extended bus name of another bus connected to this winding (18 characters).
	'WIND1NAME' — Winding 1 bus name (12 characters).
	'WIND1EXNAME' — Winding 1 bus extended bus name (18 characters).
	'WIND2NAME' — Winding 2 bus name (12 characters).
	'WIND2EXNAME' — Winding 2 bus extended bus name (18 characters).
	'WIND3NAME' — Winding 3 bus name (12 characters).
	'WIND3EXNAME' — Winding 3 bus extended bus name (18 characters).
	'NMETERNAME' — Non-metered bus name (12 characters).
	'NMETEREXNAME' — Non-metered bus extended bus name (18 characters).
	'XFRNAME' — Transformer name (40 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_3WINDING_WINDINGS_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).

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Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_3WINDING_WINDINGS_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of CARRAY, is not large enough
	IERR = 9	Invalid STRING value

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8.14.2. AWNDCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the three-winding transformer winding data family.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, brnchs = awndcount(sid, owner, ties, flag, entry)
```

Fortran command syntax:

```
CALL GET_3WINDING_WINDINGS_COUNT(SID,OWNER,TIES,FLAG,ENTRY,BRNCHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service windings of in-service three-winding transformers.
	FLAG = 2	for all windings of in-service three-winding transformers.
	FLAG = 3	for all three-winding transformer windings.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding 1 bus order.
	ENTRY = 2	for transformer name order.
Integer BRNCHS	number of array entries required for the subsystem indicated by SID and OWNER that meet the editing criteria indicated by TIES, FLAG and ENTRY (output).	

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Integer IERR	error code (output).	
	IERR = 0	No error, BRNCHS returned
	IERR = 1	Working case is empty; BRNCHS returned as 0
	IERR = 2	Invalid SID value; BRNCHS returned as 0
	IERR = 3	Invalid OWNER value; BRNCHS returned as 0
	IERR = 4	Invalid TIES value; BRNCHS returned as 0
	IERR = 5	Invalid FLAG value; BRNCHS returned as 0
	IERR = 6	Invalid ENTRY value; BRNCHS returned as 0

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8.14.3. AWNDCPLX

Use this API to return an array of complex values for subsystem three-winding transformer windings.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, xarray = awndcplx(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_3WINDING_WINDINGS_COMPLEX(SID, OWNER, TIES, FLAG, ENTRY, NSTR,
 STRING, DIM, XARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service windings of in-service three-winding transformers.
	FLAG = 2	for all windings of in-service three-winding transformers.
	FLAG = 3	for all three-winding transformer windings.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding bus order.
	ENTRY = 2	for transformer name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	

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	'RXACT'	Actual transformer impedance, in pu
	'RXNOM'	Nominal transformer impedance, in pu
	'COMPRX'	Load drop compensation impedance, in pu
	'RXZERO'	Zero sequence winding leakage impedance, in pu
	'ZGRND'	Zero sequence winding grounding impedance, in pu
	'PQ'	P + j Q flow at winding bus end, in MW and Mvar
	'PQLOSS'	Losses, in MW and Mvar
	'O_PQ'	P + j Q flow at winding bus end (units determined by power output options)
	'O_PQLOSS'	Losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_3WINDING_WINDINGS_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_3WINDING_WINDINGS_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of XARRAY, is not large enough
	IERR = 9	Invalid STRING value
	IERR = 10	Sequence data not in case (when STRING = 'RXZERO' or 'ZGRND')

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8.14.4. AWNDINT

Use this API to return an array of integer values for subsystem three-winding transformer windings.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, iarray = awndint(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_3WINDING_WINDINGS_INTEGER(SID, OWNER, TIES, FLAG, ENTRY, NSTR,
 STRING, DIM, IARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service windings of in-service three-winding transformers.
	FLAG = 2	for all windings of in-service three-winding transformers.
	FLAG = 3	for all three-winding transformer windings.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding 1 bus order.
	ENTRY = 2	for transformer name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).	

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	'WNDBUSNUMBER'	Number of the bus connected to this winding
	'OTHER1NUMBER'	Number of another bus connected to this transformer
	'OTHER2NUMBER'	Number of another bus connected to this transformer
	'WNDNUM'	Winding number
	'WIND1NUMBER'	Winding 1 bus number
	'WIND2NUMBER'	Winding 2 bus number
	'WIND3NUMBER'	Winding 3 bus number
	'NMETERNUMBER'	Non-metered end bus number
	'STATUS'	Winding status
	'OWNERS'	Number of owners
	'OWN1'	First owner
	'OWN2'	Second owner
	'OWN3'	Third owner
	'OWN4'	Fourth owner
	'ICONTNUMBER'	Controlled bus number
	'TABLE'	Transformer impedance adjustment table number
	'CODE'	Transformer control mode for automatic adjustments
	'NTPOSN'	Number of tap positions
	'CW'	Winding data I/O code
	'CZ'	Impedance data I/O code
	'CM'	Magnetizing admittance I/O code
	'CZ0'	Leakage impedance data I/O code
	'CZG'	Grounding impedance data I/O code
	'CNXCOD'	Connection code
	'TPSTT'	Winding 1 ratio limit code: -1 for low, 0 for regulating, 1 for high, -2 for locked
	'ANSTT'	Winding 1 angle limit code: -1 for low, 0 for regulating, 1 for high, -2 for locked
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_3WINDING_WINDINGS_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_3WINDING_WINDINGS_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	

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IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid OWNER value
IERR = 4	Invalid TIES value
IERR = 5	Invalid FLAG value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of IARRAY, is not large enough
IERR = 9	Invalid STRING value
IERR = 10	Sequence data not in case (when STRING = 'CZ0', 'CZG' or 'CNXCOD')

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8.14.5. AWNDREAL

Use this API to return an array of real values for subsystem three-winding transformer windings.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
ierr, rarray = awndreal(sid, owner, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_3WINDING_WINDINGS_REAL(SID, OWNER, TIES, FLAG, ENTRY, NSTR,
 STRING, DIM, RARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer OWNER	a flag indicating owner usage if ownership is a subsystem selection criterion (ignored if SID is negative) (input; 1 by default).	
	OWNER = 1	to use bus ownership.
	OWNER = 2	to use transformer ownership.
Integer TIES	a flag indicating which subsystem three-winding transformers to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem three-winding transformers only.
	TIES = 2	for subsystem tie three-winding transformers only.
	TIES = 3	for both interior subsystem and tie three-winding transformers.
Integer FLAG	a flag indicating which subsystem three-winding transformers to include (input; 1 by default).	
	FLAG = 1	for only in-service windings of in-service three-winding transformers.
	FLAG = 2	for all windings of in-service three-winding transformers.
	FLAG = 3	for all three-winding transformer windings.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for winding 1 bus order.
	ENTRY = 2	for transformer name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	

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Character STRING (NSTR)

NSTR of the following strings indicating the transformer quantities desired (input; no default allowed).

'AMPS'	Branch current in amps (0.0 if bus base voltage is 0.0)
'PUCUR'	Branch current in pu
'PCTRATE'	Percent winding bus current of default rating set
'PCTRATEA'	Percent winding bus current of rating set A
'PCTRATEB'	Percent winding bus current of rating set B
'PCTRATEC'	Percent winding bus current of rating set C
'PCTMVARATE'	Percent winding bus MVA of default rating set
'PCTMVARATEA'	Percent winding bus MVA of rating set A
'PCTMVARATEB'	Percent winding bus MVA of rating set B
'PCTMVARATEC'	Percent winding bus MVA of rating set C
'PCTCORPRATE'	Percent winding bus current or MVA loading (according to the transformer percent loading units program option setting) of default rating set
'PCTCORPRATEA'	Percent winding bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set A
'PCTCORPRATEB'	Percent winding bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set B
'PCTCORPRATEC'	Percent winding bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set C
'MAXPCTRATE'	Larger of percent winding/star point bus current of default rating set
'MAXPCTRATEA'	Larger of percent winding/star point bus current of rating set A
'MAXPCTRATEB'	Larger of percent winding/star point bus current of rating set B
'MAXPCTRATEC'	Larger of percent winding/star point bus current of rating set C
'MXPCTMVARAT'	Larger of percent winding/star point bus MVA of default rating set
'MXPCTMVARATA'	Larger of percent winding/star point bus MVA of rating set A
'MXPCTMVARATB'	Larger of percent winding/star point bus MVA of rating set B
'MXPCTMVARATC'	Larger of percent winding/star point bus MVA of rating set C
'MXPCTCRPRAT'	Larger of percent winding/star point bus current or MVA loading (according to the transformer percent loading units program option setting) of default rating set

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'MXPCTCRPRATA'	Larger of percent winding/star point bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set A
'MXPCTCRPRATB'	Larger of percent winding/star point bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set B
'MXPCTCRPRATC'	Larger of percent winding/star point bus current or MVA loading (according to the transformer percent loading units program option setting) of rating set C
'FRACT1'	First owner fraction
'FRACT2'	Second owner fraction
'FRACT3'	Third owner fraction
'FRACT4'	Fourth owner fraction
'RATE'	Rating from default rating set
'RATEA'	Rating from rating set A
'RATEB'	Rating from rating set B
'RATEC'	Rating from rating set C
'RATIO'	Winding tap ratio in pu
'RATIOCW'	Winding tap ratio in units governed by CW
'ANGLE'	Winding phase shift angle in degrees
'RMAX'	Winding ratio or angle upper limit (in pu if ratio)
'RMAXCW'	Winding ratio or angle upper limit (in units governed by CW if ratio)
'RMIN'	Winding ratio or angle lower limit (in pu if ratio)
'RMINCW'	Winding ratio or angle lower limit (in units governed by CW if ratio)
'VMAX'	Controlled quantity upper limit (in pu if voltage)
'VMAXKV'	Controlled quantity upper limit (in units governed by CW if voltage)
'VMIN'	Controlled quantity lower limit (in pu if voltage)
'VMINKV'	Controlled quantity lower limit (in units governed by CW if voltage)
'STEP'	Winding ratio or angle step width (in pu if ratio)
'STEPCW'	Winding ratio or angle step width (in units governed by CW if ratio)
'NOMV'	Winding nominal voltage in kv (0.0 for bus base voltage)
'SBASE'	Winding base MVA

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	'P'	Active power flow at winding bus end, in MW
	'Q'	Reactive power flow at winding bus end, in Mvar
	'MVA'	|P + j Q| at winding bus end, in MVA
	'MAXMVA'	|P + j Q| at winding bus or star point bus end whichever is larger, in MVA
	'PLOSS'	Active power losses, in MW
	'QLOSS'	Reactive power losses, in Mvar
	'O_P'	Active power flow at winding bus end (units determined by power output options)
	'O_Q'	Reactive power flow at winding bus end (units determined by power output options)
	'O_MVA'	|P + j Q| at winding bus end (units determined by power output options)
	'O_MAXMVA'	|P + j Q| at winding bus or star point bus end whichever is larger (units determined by power output options)
	'O_PLOSS'	Active power losses (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_3WINDING_WINDINGS_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_3WINDING_WINDINGS_COUNT API when it is called with the same values specified for SID, OWNER, TIES, FLAG and ENTRY (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid OWNER value
	IERR = 4	Invalid TIES value
	IERR = 5	Invalid FLAG value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of RARRAY, is not large enough
	IERR = 9	Invalid STRING value

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8.14.6. AWNDTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the three-winding transformer winding data family (aWndInt, aWndReal, aWndCplx and aWndChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = awndtypes(string)
```

Fortran command syntax:

```
CALL GET_3WINDING_WINDINGS_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the three-winding transformer winding data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the three-winding transformer winding data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.15. Two-Terminal dc Line Data

The two-terminal dc line data family of subsystem data retrieval APIs returns arrays of data for two-terminal dc lines in a specified subsystem.

One of the input items to each API allows the user to elect to retrieve data either for all subsystem two-terminal dc lines, for subsystem two-terminal dc lines that are tie branches, or for both subsystem and tie two-terminal dc lines. Another input item allows the user to elect to retrieve data for all such two-terminal dc lines, or to restrict the array entries to data for in-service two-terminal dc lines.

Data for subsystem two-terminal dc lines is returned from all APIs in the two-terminal dc line data family in dc line name alphabetical order.

8.15.1. A2TRMDCCHAR

Use this API to return an array of character values for subsystem two-terminal dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = a2trmdcchar(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_LINE_CHAR(SID, TIES, FLAG, NSTR, STRING, DIM, CARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-terminal dc lines only.
	TIES = 2	for subsystem tie two-terminal dc lines only.
	TIES = 3	for both interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	

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Character STRING (NSTR)	NSTR of the following strings indicating the two-terminal dc line quantities desired (input; no default allowed).	
	'DCNAME'	dc line name (12 characters).
	'FROMNAME'	Rectifier bus name (12 characters).
	'FROMEXNAME'	Rectifier bus extended bus name (18 characters).
	'TONAME'	Inverter bus name (12 characters).
	'TOEXNAME'	Inverter bus extended bus name (18 characters).
	'METERNAME'	Metered bus name (12 characters).
	'METEREXNAME'	Metered bus extended bus name (18 characters).
	'NMETERNAME'	Non-metered bus name (12 characters).
	'NMETEREXNAME'	Non-metered bus extended bus name (18 characters).
	'IFRNAME'	Rectifier firing angle measuring bus name (12 characters).
	'IFREXNAME'	Rectifier firing angle measuring bus extended bus name (18 characters).
	'IFINAME'	Inverter firing angle measuring bus name (12 characters).
	'IFIEXNAME'	Inverter firing angle measuring bus extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_TWO_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_TWO_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid NSTR value
	IERR = 6	DIM, and hence the size of CARRAY, is not large enough
	IERR = 7	Invalid STRING value

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8.15.2. A2TRMDCCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the two-terminal dc line data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, brnchs = a2trmdccount(sid, ties, flag)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_LINE_COUNT(SID, TIES, FLAG, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-terminal dc lines only.
	TIES = 2	for subsystem tie two-terminal dc lines only.
	TIES = 3	for both interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer BRNCHS	number of array entries required for the subsystem indicated by SID that meet the editing criteria indicated by TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BRNCHS returned
	IERR = 1	Working case is empty; BRNCHS returned as 0
	IERR = 2	Invalid SID value; BRNCHS returned as 0
	IERR = 3	Invalid TIES value; BRNCHS returned as 0
	IERR = 4	Invalid FLAG value; BRNCHS returned as 0

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8.15.3. A2TRMDCCPLX

Use this API to return an array of complex values for subsystem two-terminal dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = a2trmdccplx(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_LINE_COMPLEX(SID, TIES, FLAG, NSTR, STRING, DIM,
 XARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-terminal dc lines only.
	TIES = 2	for subsystem tie two-terminal dc lines only.
	TIES = 3	for both interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the two-terminal dc line quantities desired (input; no default allowed).	
	'PQAC_R'	Apparent ac power flow into the dc line at the rectifier (in MW and Mvar)
	'PQAC_I'	Apparent ac power flow into the dc line at the inverter (in MW and Mvar)
	'PQLOSS'	ac power losses (in MW and Mvar)
	'O_PQAC_R'	Apparent ac power flow into the dc line at the rectifier (units determined by power output options)
	'O_PQAC_I'	Apparent ac power flow into the dc line at the inverter (units determined by power output options)

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	'O_PQLOSS'	ac power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_TWO_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_TWO_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid NSTR value
	IERR = 6	DIM, and hence the size of XARRAY, is not large enough
	IERR = 7	Invalid STRING value

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8.15.4. A2TRMDCINT

Use this API to return an array of integer values for subsystem two-terminal dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = a2trmdcint(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_LINE_INTEGER(SID, TIES, FLAG, NSTR, STRING, DIM,
 IARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-terminal dc lines only.
	TIES = 2	for subsystem tie two-terminal dc lines only.
	TIES = 3	for both interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the two-terminal dc line quantities desired (input; no default allowed).	
	'FROMNUMBER'	Rectifier bus number
	'TONUMBER'	Inverter bus number
	'METERNUMBER'	Metered end bus number
	'NMETERNUMBER'	Non-metered end bus number
	'IFRNUMBER'	Rectifier firing angle measuring bus number
	'IFINUMBER'	Inverter extinction angle measuring bus number
	'MDC'	Control mode
	'NBR'	Rectifier number of bridges in series
	'NBI'	Inverter number of bridges in series

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	'ANSTT_R'	Rectifier angle limit code: -1 (at or below low limit), 0 (in range), or 1 (at or above high limit)
	'ANSTT_I'	Inverter angle limit code: -1 (at or below low limit), 0 (in range), or 1 (at or above high limit)
	'TPSTT_R'	Rectifier tap ratio limit code: -1 (at or below low limit), 0 (in range), 1 (at or above high limit) or -2 (locked)
	'TPSTT_I'	Inverter tap ratio limit code: -1 (at or below low limit), 0 (in range), 1 (at or above high limit) or -2 (locked)
	'CCCITMX'	Iteration limit for capacitor commutated converter solution
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_TWO_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_TWO_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid NSTR value
	IERR = 6	DIM, and hence the size of IARRAY, is not large enough
	IERR = 7	Invalid STRING value

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8.15.5. A2TRMDCREAL

Use this API to return an array of real values for subsystem two-terminal dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = a2trmdcreal(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_LINE_REAL(SID, TIES, FLAG, NSTR, STRING, DIM, RARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem two-terminal dc lines only.
	TIES = 2	for subsystem tie two-terminal dc lines only.
	TIES = 3	for both interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the two-terminal dc line quantities desired (input; no default allowed).	
	'DCCUR'	dc line current in amps
	'RDC'	dc line resistance in ohms
	'SETVAL'	Desired current (in amps) or power (in MW) demand
	'VSCHD'	Scheduled compounded dc voltage in kV
	'VCMOD'	Mode switch dc voltage in kV
	'RCOMP'	Compounding resistance in ohms
	'DELTI'	Margin in pu of desired dc power or current
	'DCVMIN'	Minimum compounded dc voltage in kV
	'ACAMPS_R'	ac current in amps at the rectifier
	'ACAMPS_I'	ac current in amps at the inverter

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'PUCUR_R'	ac current in pu at the rectifier
'PUCUR_I'	ac current in pu at the inverter
'KVDC_R'	Rectifier dc voltage in kV
'KVDC_I'	Inverter dc voltage in kV
'ANGLE_R'	Rectifier converter angle in degrees
'ANGLE_I'	Inverter converter angle in degrees
'ANGMIN_R'	Rectifier minimum steady-state converter angle in degrees
'ANGMIN_I'	Inverter minimum steady-state converter angle in degrees
'ANGMAX_R'	Rectifier nominal maximum converter angle in degrees
'ANGMAX_I'	Inverter nominal maximum converter angle in degrees
'RC_R'	Rectifier commutating transformer resistance per bridge in ohms
'RC_I'	Inverter commutating transformer resistance per bridge in ohms
'XC_R'	Rectifier commutating transformer reactance per bridge in ohms
'XC_I'	Inverter commutating transformer reactance per bridge in ohms
'EBASE_R'	Rectifier primary base ac voltage in kV
'EBASE_I'	Inverter primary base ac voltage in kV
'TR_R'	Rectifier transformer ratio in pu
'TR_I'	Inverter transformer ratio in pu
'TAP_R'	Rectifier tap setting in pu
'TAP_I'	Inverter tap setting in pu
'TPMX_R'	Rectifier maximum tap setting in pu
'TPMX_I'	Inverter maximum tap setting in pu
'TPMN_R'	Rectifier minimum tap setting in pu
'TPMN_I'	Inverter minimum tap setting in pu
'STP_R'	Rectifier tap step in pu
'STP_I'	Inverter tap step in pu
'XCAP_R'	Rectifier commutating capacitor reactance magnitude per bridge in pu
'XCAP_I'	Inverter commutating capacitor reactance magnitude per bridge in pu
'CCCACC'	Acceleration factor for capacitor commutated converter solution
'PAC_R'	Apparent active power flow into the dc line at the rectifier in MW
'PAC_I'	Apparent active power flow into the dc line at the inverter in MW

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	'QAC_R'	Apparent reactive power flow into the dc line at the rectifier in Mvar
	'QAC_I'	Apparent reactive power flow into the dc line at the inverter in Mvar
	'MVA_R'	Apparent ac power flow into the dc line at the rectifierin MVA
	'MVA_I'	Apparent ac power flow into the dc line at the inverter in MVA
	'PLOSS'	Active power losses in MW
	'QLOSS'	Reactive power losses in Mvar
	'O_PAC_R'	Apparent active power flow into the dc line at the rectifier (units determined by power output options)
	'O_PAC_I'	Apparent active power flow into the dc line at the inverter (units determined by power output options)
	'O_QAC_R'	Apparent reactive power flow into the dc line at the rectifier (units determined by power output options)
	'O_QAC_I'	Apparent reactive power flow into the dc line at the inverter (units determined by power output options)
	'O_MVA_R'	Apparent ac power flow into the dc line at the rectifier (units determined by power output options)
	'O_MVA_I'	Apparent ac power flow into the dc line at the inverter (units determined by power output options)
	'O_PLOSS'	Active power losses (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_TWO_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_TWO_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty

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IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid NSTR value
IERR = 6	DIM, and hence the size of RARRAY, is not large enough
IERR = 7	Invalid STRING value

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8.15.6. A2TRMDCTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the two-terminal dc line data family (a2TrmDcInt, a2TrmDcReal, a2TrmDcCplx and a2TrmDcChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = a2trmdctypes(string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_LINE_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the two-terminal dc line data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the two-terminal dc line data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.16. Two-Terminal dc Line Converter Data

The two-terminal dc line converter data family of subsystem data retrieval APIs returns arrays of data for two-terminal dc line converters in a specified subsystem.

One of the input items to each API allows the user to elect to retrieve data either for all subsystem two-terminal dc line converters, for subsystem two-terminal dc line converters of two-terminal dc lines that are tie branches, or for both subsystem and tie two-terminal dc line converters. Another input item allows the user to elect to retrieve data for all such two-terminal dc line converters, or to restrict the array entries to data for in-service two-terminal dc lines.

Data for subsystem converter buses is returned from all APIs in the two-terminal dc line converter data family in the same order. One of the input items to each API allows the user to elect that converter buses be ordered either in dc line name order with all converter buses of each two-terminal dc line grouped together, or in numerical or alphabetical order by converter bus, according to the bus output option setting in effect at the time the API is referenced.

8.16.1. A2TRMDCCONVCHAR

Use this API to return an array of character values for subsystem two-terminal dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = a2trmdcconvchar(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_CONV_CHAR(SID, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 CARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem two-terminal dc lines only.
	TIES = 2	for the subsystem bus end of tie two-terminal dc lines only.
	TIES = 3	for the non-subsystem bus end of tie two-terminal dc lines only.
	TIES = 4	for each end of tie two-terminal dc lines only.
	TIES = 5	for each end of interior subsystem two-terminal dc lines and the subsystem bus end of tie two-terminal dc lines.

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	TIES = 6	for each end of interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the two-terminal dc line converter quantities desired (input; no default allowed).	
	'DCNAME'	dc line name (12 characters).
	'FROMNAME'	From bus name (12 characters).
	'FROMEXNAME'	From bus extended bus name (18 characters).
	'TONAME'	To bus name (12 characters).
	'TOEXNAME'	To bus extended bus name (18 characters).
	'METERNAME'	Metered bus name (12 characters).
	'METEREXNAME'	Metered bus extended bus name (18 characters).
	'NMETERNAME'	Non-metered bus name (12 characters).
	'NMETEREXNAME'	Non-metered bus extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the GET_TWO_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of CARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_TWO_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid ENTRY value
	IERR = 6	Invalid NSTR value

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IERR = 7	DIM, and hence the size of CARRAY, is not large enough
IERR = 8	Invalid STRING value

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8.16.2. A2TRMDCCONVCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the two-terminal dc line converter data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, buses = a2trmdcconvcount(sid, ties, flag)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_CONV_COUNT(SID, TIES, FLAG, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem two-terminal dc lines only.
	TIES = 2	for the subsystem bus end of tie two-terminal dc lines only.
	TIES = 3	for the non-subsystem bus end of tie two-terminal dc lines only.
	TIES = 4	for each end of tie two-terminal dc lines only.
	TIES = 5	for each end of interior subsystem two-terminal dc lines and the subsystem bus end of tie two-terminal dc lines.
	TIES = 6	for each end of interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer BUSES	number of array entries required for the subsystem indicated by SID that meet the editing criteria indicated by TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BUSES returned
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0
	IERR = 3	Invalid TIES value; BUSES returned as 0

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IERR = 4	Invalid FLAG value; BUSES returned as 0

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8.16.3. A2TRMDCCONVCPLX

Use this API to return an array of complex values for subsystem two-terminal dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = a2trmdcconvcplx(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_CONV_COMPLEX(SID, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, XARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem two-terminal dc lines only.
	TIES = 2	for the subsystem bus end of tie two-terminal dc lines only.
	TIES = 3	for the non-subsystem bus end of tie two-terminal dc lines only.
	TIES = 4	for each end of tie two-terminal dc lines only.
	TIES = 5	for each end of interior subsystem two-terminal dc lines and the subsystem bus end of tie two-terminal dc lines.
	TIES = 6	for each end of interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the two-terminal dc line converter quantities desired (input; no default allowed).	

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	'PQAC'	Apparent ac power flow into the dc line at the from bus (in MW and Mvar)
	'PQLOSS'	ac power losses (in MW and Mvar)
	'O_PQAC'	Apparent ac power flow into the dc line at the from bus (units determined by power output options)
	'O_PQLOSS'	ac power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the GET_TWO_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of XARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_TWO_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid ENTRY value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of XARRAY, is not large enough
	IERR = 8	Invalid STRING value

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8.16.4. A2TRMDCCONVINT

Use this API to return an array of integer values for subsystem two-terminal dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = a2trmdcconvint(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_CONV_INTEGER(SID, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, IARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem two-terminal dc lines only.
	TIES = 2	for the subsystem bus end of tie two-terminal dc lines only.
	TIES = 3	for the non-subsystem bus end of tie two-terminal dc lines only.
	TIES = 4	for each end of tie two-terminal dc lines only.
	TIES = 5	for each end of interior subsystem two-terminal dc lines and the subsystem bus end of tie two-terminal dc lines.
	TIES = 6	for each end of interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the two-terminal dc line converter quantities desired (input; no default allowed).	

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	'FROMNUMBER'	From bus number
	'TONUMBER'	To bus number
	'METERNUMBER'	Metered end bus number
	'NMETERNUMBER'	Non-metered end bus number
	'CONVERTERTYP'	1 if from bus is the rectifier 2 if the inverter
	'MDC'	Control mode
	'ANSTT'	From bus converter angle limit code: -1 (at or below low limit), 0 (in range), or 1 (at or above high limit)
	'TPSTT'	From bus converter tap ratio limit code: -1 (at or below low limit), 0 (in range), 1 (at or above high limit) or -2 (locked)
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the GET_TWO_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of IARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_TWO_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid ENTRY value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of IARRAY, is not large enough
	IERR = 8	Invalid STRING value

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8.16.5. A2TRMDCCONVREAL

Use this API to return an array of real values for subsystem two-terminal dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = a2trmdcconvreal(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_CONV_REAL(SID, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 RARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of two-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem two-terminal dc lines only.
	TIES = 2	for the subsystem bus end of tie two-terminal dc lines only.
	TIES = 3	for the non-subsystem bus end of tie two-terminal dc lines only.
	TIES = 4	for each end of tie two-terminal dc lines only.
	TIES = 5	for each end of interior subsystem two-terminal dc lines and the subsystem bus end of tie two-terminal dc lines.
	TIES = 6	for each end of interior subsystem two-terminal dc lines and tie two-terminal dc lines.
Integer FLAG	a flag indicating which subsystem two-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service two-terminal dc lines.
	FLAG = 2	for all two-terminal dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the two-terminal dc line converter quantities desired (input; no default allowed).	
	'DCCUR'	dc line current in amps

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	'ACAMPS'	ac current in amps at the from bus
	'PUCUR'	ac current in pu at the from bus
	'KVDC'	From bus dc voltage in kV
	'ANGLE'	From bus converter angle in degrees
	'TAP'	From bus tap setting in pu
	'PAC'	Apparent active power flow into the dc line at the from bus
	'QAC'	Apparent reactive power flow into the dc line at the from bus
	'MVA'	Apparent ac power flow into the dc line at the from bus
	'PLOSS'	Active power losses
	'QLOSS'	Reactive power losses
	'O_PAC'	Apparent active power flow into the dc line at the from bus (units determined by power output options)
	'O_QAC'	Apparent reactive power flow into the dc line at the from bus (units determined by power output options)
	'O_MVA'	Apparent ac power flow into the dc line at the from bus (units determined by power output options)
	'O_PLOSS'	Active power losses (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the GET_TWO_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of RARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_TWO_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid ENTRY value
	IERR = 6	Invalid NSTR value

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IERR = 7	DIM, and hence the size of RARRAY, is not large enough
IERR = 8	Invalid STRING value

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8.16.6. A2TRMDCCONVTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the two-terminal dc line converter data family (a2TrmDcConvInt, a2TrmDcConvReal, a2TrmDcConvCplx and a2TrmDcConvChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = a2trmdcconvtypes(string)
```

Fortran command syntax:

```
CALL GET_TWO_TERM_DC_CONV_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the two-terminal dc line converter data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the two-terminal dc line converter data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.17. Multi-Terminal dc Line Data

The multi-terminal dc line data family of subsystem data retrieval APIs returns arrays of data for multi-terminal dc lines in a specified subsystem.

One of the input items to each API allows the user to elect to retrieve data for either all subsystem multi-terminal dc lines, for subsystem multi-terminal dc lines that are tie branches, or for both subsystem and tie multi-terminal dc lines. Another input item allows the user to elect to retrieve data for all such multi-terminal dc lines, or to restrict the array entries to data for in-service multi-terminal dc lines.

Data for subsystem multi-terminal dc lines is returned from all APIs in the multi-terminal dc line data family in dc line name alphabetical order.

8.17.1. AMULTITRMDCCHAR

Use this API to return an array of character values for subsystem multi-terminal dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = amultitrmdcchar(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_LINE_CHAR(SID, TIES, FLAG, NSTR, STRING, DIM,
 CARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem multi-terminal dc lines only.
	TIES = 2	for subsystem tie multi-terminal dc lines only.
	TIES = 3	for both interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	

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Character STRING (NSTR)	NSTR of the following strings indicating the multi-terminal dc line quantities desired (input; no default allowed).
	'DCNAME' dc line name (12 characters).
	'VCNPOSNAME' Positive pole voltage controlling converter bus name (12 characters).
	'VCNPOSEXNAME' Positive pole voltage controlling converter bus extended bus name (18 characters).
	'VCNNEGNAME' Negative pole voltage controlling converter bus name (12 characters).
	'VCNNEGEXNAME' Negative pole voltage controlling converter bus extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_MULTI_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_MULTI_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).
	IERR = 0 No error
	IERR = 1 Working case is empty
	IERR = 2 Invalid SID value
	IERR = 3 Invalid TIES value
	IERR = 4 Invalid FLAG value
	IERR = 5 Invalid NSTR value
	IERR = 6 DIM, and hence the size of CARRAY, is not large enough
	IERR = 7 Invalid STRING value

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8.17.2. AMULTITRMDCCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the multi-terminal dc line data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, brnchs = amultitrmdccount(sid, ties, flag)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_LINE_COUNT(SID, TIES, FLAG, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem multi-terminal dc lines only.
	TIES = 2	for subsystem tie multi-terminal dc lines only.
	TIES = 3	for both interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer BRNCHS	number of array entries required for the subsystem indicated by SID that meet the editing criteria indicated by TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BRNCHS returned
	IERR = 1	Working case is empty; BRNCHS returned as 0
	IERR = 2	Invalid SID value; BRNCHS returned as 0
	IERR = 3	Invalid TIES value; BRNCHS returned as 0
	IERR = 4	Invalid FLAG value; BRNCHS returned as 0

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8.17.3. AMULTITRMDCCPLX

Use this API to return an array of complex values for subsystem multi-terminal dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = amultitrmdccplx(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_LINE_COMPLEX(SID, TIES, FLAG, NSTR, STRING, DIM,
 XARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem multi-terminal dc lines only.
	TIES = 2	for subsystem tie multi-terminal dc lines only.
	TIES = 3	for both interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the multi-terminal dc line quantities desired (input; no default allowed).	
	'PQLOSS'	ac power losses (in MW and Mvar)
	'O_PQLOSS'	ac power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_MULTI_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	

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Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_MULTI_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid NSTR value
IERR = 6	DIM, and hence the size of XARRAY, is not large enough
IERR = 7	Invalid STRING value

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8.17.4. AMULTITRMDCINT

Use this API to return an array of integer values for subsystem multi-terminal dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = amultitrmdcint(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_LINE_INTEGER(SID, TIES, FLAG, NSTR, STRING, DIM,
 IARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem multi-terminal dc lines only.
	TIES = 2	for subsystem tie multi-terminal dc lines only.
	TIES = 3	for both interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the multi-terminal dc line quantities desired (input; no default allowed).	
	'CONVERTERS'	Number of converters
	'DCBUSES'	Number of dc buses
	'DCLINKS'	Number of dc links
	'VCNPOSNUMBER'	Positive pole voltage controlling converter bus number
	'VCNNEGNUMBER'	Negative pole voltage controlling converter bus number
	'MDC'	Control mode
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_MULTI_TERM_DC_LINE_COUNT API when it is called with the	

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	same values specified for SID, TIES and FLAG (input; no default allowed).
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_MULTI_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid NSTR value
IERR = 6	DIM, and hence the size of IARRAY, is not large enough
IERR = 7	Invalid STRING value

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8.17.5. AMULTITRMDCREAL

Use this API to return an array of real values for subsystem multi-terminal dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = amultitrmdcreal(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_LINE_REAL(SID, TIES, FLAG, NSTR, STRING, DIM,
 RARRAY, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem multi-terminal dc lines only.
	TIES = 2	for subsystem tie multi-terminal dc lines only.
	TIES = 3	for both interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the multi-terminal dc line quantities desired (input; no default allowed).	
	'VCMOD'	Mode switch dc voltage, in kV
	'PLOSS'	Active power losses, in MW
	'QLOSS'	Reactive power losses, in Mvar
	'O_PLOSS'	Active power losses (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_MULTI_TERM_DC_LINE_COUNT API when it is called with the	

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	same values specified for SID, TIES and FLAG (input; no default allowed).
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_MULTI_TERM_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid NSTR value
IERR = 6	DIM, and hence the size of RARRAY, is not large enough
IERR = 7	Invalid STRING value

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8.17.6. AMULTITRMDCTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the multi-terminal dc line data family (aMultiTrmDcInt, aMultiTrmDcReal, aMultiTrmDcCplx and aMultiTrmDcChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = amultitrmdctypes(string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_LINE_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the multi-terminal dc line data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the multi-terminal dc line data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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1996

8.18. Multi-Terminal dc Line Converter Data

The multi-terminal dc line converter data family of subsystem data retrieval APIs returns arrays of data for multi-terminal dc line converters in a specified subsystem.

One of the input items to each API allows the user to elect to retrieve data either for all subsystem multi-terminal dc line converters, for subsystem multi-terminal dc line converters of multi-terminal dc lines that are tie branches, or for both subsystem and tie multi-terminal dc line converters. Another input item allows the user to elect to retrieve data for all such multi-terminal dc line converters, or to restrict the array entries to data for in-service multi-terminal dc lines.

Data for subsystem converter buses is returned from all APIs in the multi-terminal dc line converter data family in the same order. One of the input items to each API allows the user to elect that converter buses be ordered either in dc line name order with all converter buses of each multi-terminal dc line grouped together, or in numerical or alphabetical order by converter bus, according to the bus output option setting in effect at the time the API is referenced.

8.18.1. AMULTITRMDCCONVCHAR

Use this API to return an array of character values for subsystem multi-terminal dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = amultitrmdcconvchar(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_CONV_CHAR(SID, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, CARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem multi-terminal dc lines only.
	TIES = 2	for the subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 3	for the non-subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 4	for each end of tie multi-terminal dc lines only.

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	TIES = 5	for each end of interior subsystem multi-terminal dc lines and the subsystem bus ends of tie multi-terminal dc lines.
	TIES = 6	for each end of interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the multi-terminal dc line converter quantities desired (input; no default allowed).	
	'DCNAME'	dc line name (12 characters).
	'NAME'	Converter bus name (12 characters).
	'EXNAME'	Converter bus extended bus name (18 characters).
	'DCBUSNAME'	Name of the dc bus connected to this converter bus (12 characters).
	'DCBUS2NAME'	Name of the optional second dc bus connected to this converter bus (12 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the GET_MULTI_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of CARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_MULTI_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid ENTRY value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of CARRAY, is not large enough

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IERR = 8	Invalid STRING value

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1999

8.18.2. AMULTITRMDCCONVCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the multi-terminal dc line converter data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, buses = amultitrmdcconvcount(sid, ties, flag)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_CONV_COUNT(SID, TIES, FLAG, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem multi-terminal dc lines only.
	TIES = 2	for the subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 3	for the non-subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 4	for each end of tie multi-terminal dc lines only.
	TIES = 5	for each end of interior subsystem multi-terminal dc lines and the subsystem bus ends of tie multi-terminal dc lines.
	TIES = 6	for each end of interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer BUSES	number of array entries required for the subsystem indicated by SID that meet the editing criteria indicated by TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BUSES returned
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0
	IERR = 3	Invalid TIES value; BUSES returned as 0

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IERR = 4	Invalid FLAG value; BUSES returned as 0

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8.18.3. AMULTITRMDCCONVCPLX

Use this API to return an array of complex values for subsystem multi-terminal dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = amultitrmdcconvcplx(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_CONV_COMPLEX(SID, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, XARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem multi-terminal dc lines only.
	TIES = 2	for the subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 3	for the non-subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 4	for each end of tie multi-terminal dc lines only.
	TIES = 5	for each end of interior subsystem multi-terminal dc lines and the subsystem bus ends of tie multi-terminal dc lines.
	TIES = 6	for each end of interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the multi-terminal dc line converter quantities desired (input; no default allowed).	

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	'PQAC'	Apparent ac power flow into the dc line at the from bus (in MW and Mvar)
	'PQLOSS'	Total dc line ac power losses (in MW and Mvar)
	'O_PQAC'	Apparent ac power flow into the dc line at the from bus (units determined by power output options)
	'O_PQLOSS'	Total dc line ac power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the GET_MULTI_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of XARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_MULTI_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid ENTRY value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of XARRAY, is not large enough
	IERR = 8	Invalid STRING value

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8.18.4. AMULTITRMDCCONVINT

Use this API to return an array of integer values for subsystem multi-terminal dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = amultitrmdcconvint(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_CONV_INTEGER(SID, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, IARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem multi-terminal dc lines only.
	TIES = 2	for the subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 3	for the non-subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 4	for each end of tie multi-terminal dc lines only.
	TIES = 5	for each end of interior subsystem multi-terminal dc lines and the subsystem bus ends of tie multi-terminal dc lines.
	TIES = 6	for each end of interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the multi-terminal dc line converter quantities desired (input; no default allowed).	

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2004

	'NUMBER'	Converter bus number
	'CONVERTERTYP'	1 for a voltage controlling converter, 3 for a rectifier, 4 for an inverter
	'POLE'	1 for a positive pole converter, -1 for a negative pole converter
	'DCBUSNUMBER'	Number of the dc bus connected to this converter bus
	'DCBUS2NUMBER'	Number of the optional second dc bus connected to this converter bus
	'NBRIDGES'	Number of bridges in series
	'ANSTT'	Converter angle limit code: -1 (at or below low limit), 0 (in range), or 1 (at or above high limit)
	'TPSTT'	Converter tap ratio limit code: -1 (at or below low limit), 0 (in range), 1 (at or above high limit), or -2 (locked)
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the GET_MULTI_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of IARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_MULTI_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid ENTRY value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of IARRAY, is not large enough
	IERR = 8	Invalid STRING value

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2005

8.18.5. AMULTITRMDCCONVREAL

Use this API to return an array of real values for subsystem multi-terminal dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = amultitrmdcconvreal(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_CONV_REAL(SID, TIES, FLAG, ENTRY, NSTR, STRING,
 DIM, RARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of multi-terminal dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem multi-terminal dc lines only.
	TIES = 2	for the subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 3	for the non-subsystem bus ends of tie multi-terminal dc lines only.
	TIES = 4	for each end of tie multi-terminal dc lines only.
	TIES = 5	for each end of interior subsystem multi-terminal dc lines and the subsystem bus ends of tie multi-terminal dc lines.
	TIES = 6	for each end of interior subsystem multi-terminal dc lines and tie multi-terminal dc lines.
Integer FLAG	a flag indicating which subsystem multi-terminal dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service multi-terminal dc lines.
	FLAG = 2	for all multi-terminal dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the multi-terminal dc line converter quantities desired (input; no default allowed).	

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2006

'DCCUR'	Converter dc current in amps
'SETVAL'	Voltage (kV) current (amps) or power (MW) setpoint
'ACAMPS'	ac current in amps at the converter bus
'PUCUR'	ac current in pu at the converter bus
'KVDC'	Converter bus dc voltage in kV
'ANGLE'	Converter angle in degrees
'ANGMIN'	Minimum steady-state converter angle in degrees
'ANGMAX'	Nominal maximum converter angle in degrees
'TAP'	Converter bus tap setting in pu
'TPMX'	Maximum tap setting in pu
'TPMN'	Minimum tap setting in pu
'STP'	Tap step in pu
'TR'	Transformer ratio in pu
'RC'	Commutating transformer resistance per bridge in ohms
'XC'	Commutating transformer reactance per bridge in ohms
'EBASE'	Primary base ac voltage in kV
'DELTI'	Rectifier margin in pu of desired dc power or current
'DCPF'	Converter participation factor
'PAC'	Apparent active power flow into the dc line at the converter bus in MW
'QAC'	Apparent reactive power flow into the dc line at the converter bus in Mvar
'MVA'	Apparent ac power flow into the dc line at the converter bus in MVA
'PLOSS'	Total dc line active power losses in MW
'QLOSS'	Total dc line reactive power losses in Mvar
'O_PAC'	Apparent active power flow into the dc line at the converter bus (units determined by power output options)
'O_QAC'	Apparent reactive power flow into the dc line at the converter bus (units determined by power output options)
'O_MVA'	Apparent ac power flow into the dc line at the converter bus (units determined by power output options)
'O_PLOSS'	Total dc line active power losses (units determined by power output options)
'O_QLOSS'	Total dc line reactive power losses (units determined by power output options)

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2007

Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the GET_MULTI_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).
Integer BUSES	number of elements in each column of RARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_MULTI_TERM_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid ENTRY value
IERR = 6	Invalid NSTR value
IERR = 7	DIM, and hence the size of RARRAY, is not large enough
IERR = 8	Invalid STRING value

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8.18.6. AMULTITRMDCCONVTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the two-terminal dc line converter data family (aMultiTrmDcConvInt, aMultiTrmDcConvReal, aMultiTrmDcConvCplx and aMultiTrmDcConvChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = amultitrmdcconvtypes(string)
```

Fortran command syntax:

```
CALL GET_MULTI_TERM_DC_CONV_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the two-terminal dc line converter data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the two-terminal dc line converter data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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2009

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2010

8.19. VSC dc Line Data

The VSC dc line data family of subsystem data retrieval APIs returns arrays of data for VSC dc lines in a specified subsystem.

VSC dc lines, for subsystem VSC dc lines that are tie branches, or for both subsystem and tie VSC dc lines. Another input item allows the user to elect to retrieve data for all such VSC dc lines, to restrict the array entries to data for in-service VSC dc lines with one or both converters in-service, or to restrict the array entries to data for in-service VSC dc lines with both converters in-service.

Data for subsystem VSC dc lines is returned from all APIs in the VSC dc line data family in dc line name alphabetical order.

8.19.1. AVSCDCCHAR

Use this API to return an array of character values for subsystem VSC dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = avscdcchar(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_LINE_CHAR(SID, TIES, FLAG, NSTR, STRING, DIM, CARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem VSC dc lines only.
	TIES = 2	for subsystem tie VSC dc lines only.
	TIES = 3	for both interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service VSC dc lines with one or both converters in-service.
	FLAG = 2	for only in-service VSC dc lines with both converters in-service.
	FLAG = 3	for all VSC dc lines.

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2011

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).
Character STRING (NSTR)	NSTR of the following strings indicating the VSC dc line quantities desired (input; no default allowed).
	'VSCNAME' dc line name (12 characters).
	'FROMNAME' Converter 1 bus name (12 characters).
	'FROMEXNAME' Converter 1 bus extended bus name (18 characters).
	'TONAME' Converter 2 bus name (12 characters).
	'TOEXNAME' Converter 2 bus extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_VSC_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_VSC_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).
	IERR = 0 No error
	IERR = 1 Working case is empty
	IERR = 2 Invalid SID value
	IERR = 3 Invalid TIES value
	IERR = 4 Invalid FLAG value
	IERR = 5 Invalid NSTR value
	IERR = 6 DIM, and hence the size of CARRAY, is not large enough
	IERR = 7 Invalid STRING value

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8.19.2. AVSCDCCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the VSC dc line data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, brnchs = avscdccount(sid, ties, flag)
```

Fortran command syntax:

```
CALL GET_VSC_DC_LINE_COUNT(SID, TIES, FLAG, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem VSC dc lines only.
	TIES = 2	for subsystem tie VSC dc lines only.
	TIES = 3	for both interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service VSC dc lines with one or both converters in-service.
	FLAG = 2	for only in-service VSC dc lines with both converters in-service.
	FLAG = 3	for all VSC dc lines.
Integer BRNCHS	number of array entries required for the subsystem indicated by SID that meet the editing criteria indicated by TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BRNCHS returned
	IERR = 1	Working case is empty; BRNCHS returned as 0
	IERR = 2	Invalid SID value; BRNCHS returned as 0
	IERR = 3	Invalid TIES value; BRNCHS returned as 0
	IERR = 4	Invalid FLAG value; BRNCHS returned as 0

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2013

8.19.3. AVSCDCCPLX

Use this API to return an array of complex values for subsystem VSC dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = avscdccplx(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_LINE_COMPLEX(SID, TIES, FLAG, NSTR, STRING, DIM, XARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem VSC dc lines only.
	TIES = 2	for subsystem tie VSC dc lines only.
	TIES = 3	for both interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service VSC dc lines with one or both converters in-service.
	FLAG = 2	for only in-service VSC dc lines with both converters in-service.
	FLAG = 3	for all VSC dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the VSC dc line quantities desired (input; no default allowed).	
	'PQLOSS'	ac power losses (in MW and Mvar)
	'O_PQLOSS'	ac power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_VSC_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	

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2014

Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_VSC_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid NSTR value
IERR = 6	DIM, and hence the size of XARRAY, is not large enough
IERR = 7	Invalid STRING value

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2015

8.19.4. AVSCDCINT

Use this API to return an array of integer values for subsystem VSC dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = avscdcint(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_LINE_INTEGER(SID, TIES, FLAG, NSTR, STRING, DIM, IARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem VSC dc lines only.
	TIES = 2	for subsystem tie VSC dc lines only.
	TIES = 3	for both interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service VSC dc lines with one or both converters in-service.
	FLAG = 2	for only in-service VSC dc lines with both converters in-service.
	FLAG = 3	for all VSC dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the VSC dc line quantities desired (input; no default allowed).	
	'FROMNUMBER'	Converter 1 bus number
	'TONUMBER'	Converter 2 bus number
	'MDC'	Control mode
	'OWNERS'	Number of owners
	'OWN1'	First owner
	'OWN2'	Second owner
	'OWN3'	Third owner
	'OWN4'	Fourth owner

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2016

Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_VSC_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_VSC_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid NSTR value
IERR = 6	DIM, and hence the size of IARRAY, is not large enough
IERR = 7	Invalid STRING value

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2017

8.19.5. AVSCDCREAL

Use this API to return an array of real values for subsystem VSC dc lines.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = avscdcreal(sid, ties, flag, string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_LINE_REAL(SID, TIES, FLAG, NSTR, STRING, DIM, RARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem VSC dc lines only.
	TIES = 2	for subsystem tie VSC dc lines only.
	TIES = 3	for both interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc lines to include (input; 1 by default).	
	FLAG = 1	for only in-service VSC dc lines with one or both converters in-service.
	FLAG = 2	for only in-service VSC dc lines with both converters in-service.
	FLAG = 3	for all VSC dc lines.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the VSC dc line quantities desired (input; no default allowed).	
	'DCCUR'	dc line current in amps
	'RDC'	dc line resistance in ohms
	'FRACT1'	First owner fraction
	'FRACT2'	Second owner fraction
	'FRACT3'	Third owner fraction
	'FRACT4'	Fourth owner fraction
	'PLOSS'	Active power losses, in MW
	'QLOSS'	Reactive power losses, in Mvar

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2018

	'O_PLOSS'	Active power losses (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_VSC_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_VSC_DC_LINE_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid NSTR value
	IERR = 6	DIM, and hence the size of RARRAY, is not large enough
	IERR = 7	Invalid STRING value

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8.19.6. AVSCDCTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the VSC dc line data family (aVscDcInt, aVscDcReal, aVscDcCplx and aVscDcChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = avscdctypes(string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_LINE_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the VSC dc line data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the VSC dc line data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.20. VSC dc Line Converter Data

The VSC dc line converter data family of subsystem data retrieval APIs returns arrays of data for VSC dc line converters in a specified subsystem.

One of the input items to each API allows the user to elect to retrieve data either for all subsystem VSC dc line converters, for subsystem VSC dc line converters of VSC dc lines that are tie branches, or for both subsystem and tie VSC dc line converters. Another input item allows the user to elect to retrieve data for all such VSC dc line converters, to restrict the array entries to data for all converters of in-service VSC dc lines, or to restrict the array entries to data for in-service converters of in-service VSC dc lines.

Data for subsystem converter buses is returned from all APIs in the VSC dc line converter data family in the same order. One of the input items to each API allows the user to elect that converter buses be ordered either in dc line name order with all converter buses of each VSC dc line grouped together, or in numerical or alphabetical order by converter bus, according to the bus output option setting in effect at the time the API is referenced.

8.20.1. AVSCDCCONVCHAR

Use this API to return an array of character values for subsystem VSC dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = avscdcconvchar(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_CONV_CHAR(SID, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 CARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem VSC dc lines only.
	TIES = 2	for the subsystem bus end of tie VSC dc lines only.
	TIES = 3	for the non-subsystem bus end of tie VSC dc lines only.
	TIES = 4	for each end of tie VSC dc lines only.
	TIES = 5	for each end of interior subsystem VSC dc lines and the subsystem bus end of tie VSC dc lines.

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	TIES = 6	for each end of interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service converters of in-service VSC dc lines.
	FLAG = 2	for all converters of in-service VSC dc lines.
	FLAG = 3	for all converters of all VSC dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the VSC dc line converter quantities desired (input; no default allowed).	
	'VSCNAME'	dc line name (12 characters).
	'FROMNAME'	From bus name (12 characters).
	'FROMEXNAME'	From bus extended bus name (18 characters).
	'TONAME'	To bus name (12 characters).
	'TOEXNAME'	To bus extended bus name (18 characters).
	'IREGNAME'	Bus name of the remote voltage controlled bus (12 characters).
	'IREGEXNAME'	Extended bus name of the remote voltage controlled bus (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the GET_VSC_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BUSES	number of elements in each column of CARRAY that are set by this API. DIM must be as least as large as the value of BUSES returned by the GET_VSC_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid ENTRY value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of CARRAY, is not large enough

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2023

IERR = 8	Invalid STRING value

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8.20.2. AVSCDCCONVCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the VSC dc line converter data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, buses = avscdcconvcount(sid, ties, flag)
```

Fortran command syntax:

```
CALL GET_VSC_DC_CONV_COUNT(SID, TIES, FLAG, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem VSC dc lines only.
	TIES = 2	for the subsystem bus end of tie VSC dc lines only.
	TIES = 3	for the non-subsystem bus end of tie VSC dc lines only.
	TIES = 4	for each end of tie VSC dc lines only.
	TIES = 5	for each end of interior subsystem VSC dc lines and the subsystem bus end of tie VSC dc lines.
	TIES = 6	for each end of interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service converters of in-service VSC dc lines.
	FLAG = 2	for all converters of in-service VSC dc lines.
	FLAG = 3	for all converters of all VSC dc lines.
Integer BUSES	number of array entries required for the subsystem indicated by SID that meet the editing criteria indicated by TIES and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BUSES returned
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0

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2025

IERR = 3	Invalid TIES value; BUSES returned as 0
IERR = 4	Invalid FLAG value; BUSES returned as 0

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2026

8.20.3. AVSCDCCONVCPLX

Use this API to return an array of complex values for subsystem VSC dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = avscdcconvcplx(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_CONV_COMPLEX(SID, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 XARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem VSC dc lines only.
	TIES = 2	for the subsystem bus end of tie VSC dc lines only.
	TIES = 3	for the non-subsystem bus end of tie VSC dc lines only.
	TIES = 4	for each end of tie VSC dc lines only.
	TIES = 5	for each end of interior subsystem VSC dc lines and the subsystem bus end of tie VSC dc lines.
	TIES = 6	for each end of interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service converters of in-service VSC dc lines.
	FLAG = 2	for all converters of in-service VSC dc lines.
	FLAG = 3	for all converters of all VSC dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	

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Character STRING (NSTR)	NSTR of the following strings indicating the VSC dc line converter quantities desired (input; no default allowed).
	'PQAC' Apparent ac power flow into the dc line at the from bus (in MW and Mvar)
	'PQLOSS' ac power losses (in MW and Mvar)
	'O_PQAC' Apparent ac power flow into the dc line at the from bus (units determined by power output options)
	'O_PQLOSS' ac power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the GET_VSC_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BUSES	number of elements in each column of XARRAY that are set by this API. DIM must be as least as large as the value of BUSES returned by the GET_VSC_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).
	IERR = 0 No error
	IERR = 1 Working case is empty
	IERR = 2 Invalid SID value
	IERR = 3 Invalid TIES value
	IERR = 4 Invalid FLAG value
	IERR = 5 Invalid ENTRY value
	IERR = 6 Invalid NSTR value
	IERR = 7 DIM, and hence the size of XARRAY, is not large enough
	IERR = 8 Invalid STRING value

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8.20.4. AVSCDCCONVINT

Use this API to return an array of integer values for subsystem VSC dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = avscdcconvint(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_CONV_INTEGER(SID, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 IARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem VSC dc lines only.
	TIES = 2	for the subsystem bus end of tie VSC dc lines only.
	TIES = 3	for the non-subsystem bus end of tie VSC dc lines only.
	TIES = 4	for each end of tie VSC dc lines only.
	TIES = 5	for each end of interior subsystem VSC dc lines and the subsystem bus end of tie VSC dc lines.
	TIES = 6	for each end of interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service converters of in-service VSC dc lines.
	FLAG = 2	for all converters of in-service VSC dc lines.
	FLAG = 3	for all converters of all VSC dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	

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Character STRING (NSTR)	NSTR of the following strings indicating the VSC dc line converter quantities desired (input; no default allowed).
	'FROMNUMBER' From bus number
	'TONUMBER' To bus number
	'IREG' Bus number of the remote voltage controlled bus
	'CONVERTER' 1 if the from bus is converter 1, 2 if converter 2
	'DCTYPE' Converter dc control code
	'ACMODE' Converter ac control mode
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the GET_VSC_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).
Integer BUSES	number of elements in each column of IARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_VSC_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer IERR	error code (output).
	IERR = 0 No error
	IERR = 1 Working case is empty
	IERR = 2 Invalid SID value
	IERR = 3 Invalid TIES value
	IERR = 4 Invalid FLAG value
	IERR = 5 Invalid ENTRY value
	IERR = 6 Invalid NSTR value
	IERR = 7 DIM, and hence the size of IARRAY, is not large enough
	IERR = 8 Invalid STRING value

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8.20.5. AVSCDCCONVREAL

Use this API to return an array of real values for subsystem VSC dc line converters.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = avscdcconvreal(sid, ties, flag, entry, string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_CONV_REAL(SID, TIES, FLAG, ENTRY, NSTR, STRING, DIM,
 RARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem converter buses of VSC dc lines to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem VSC dc lines only.
	TIES = 2	for the subsystem bus end of tie VSC dc lines only.
	TIES = 3	for the non-subsystem bus end of tie VSC dc lines only.
	TIES = 4	for each end of tie VSC dc lines only.
	TIES = 5	for each end of interior subsystem VSC dc lines and the subsystem bus end of tie VSC dc lines.
	TIES = 6	for each end of interior subsystem VSC dc lines and tie VSC dc lines.
Integer FLAG	a flag indicating which subsystem VSC dc line converters to include (input; 1 by default).	
	FLAG = 1	for only in-service converters of in-service VSC dc lines.
	FLAG = 2	for all converters of in-service VSC dc lines.
	FLAG = 3	for all converters of all VSC dc lines.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for converter bus order.
	ENTRY = 2	for dc line name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	

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Character STRING (NSTR)

NSTR of the following strings indicating the VSC dc line converter quantities desired (input; no default allowed).

'ACAMPS'	ac current in amps at the from bus
'PUCUR'	ac current in pu at the from bus.
'PCTMVA'	MVA as a percent of MVA rating (more restrictive of SMAX and IMAX) at the from bus
'KVDC'	From bus dc voltage in kV
'DCSET'	dc setpoint (kV or MW) at the from bus
'ACSET'	ac setpoint (power factor or pu voltage) at the from bus
'ALOSS'	From bus A loss coefficient in kW
'BLOSS'	From bus B loss coefficient in kW/amp
'MINLOSS'	From bus minimum converter losses in kW
'SMAX'	From bus converter MVA rating
'IMAX'	From bus converter current rating in amps
'PWF'	From bus power weighting factor fraction
'RMPCT'	Percent of total Mvar to be contributed by this converter
'PAC'	Apparent active power flow into the dc line at the from bus, in MW
'QAC'	Apparent reactive power flow into the dc line at the from bus, in Mvar
'MVA'	Apparent ac power flow into the dc line at the from bus, in MVA
'PLOSS'	Total active power losses, in MW
'QLOSS'	Total reactive power losses, in Mvar
'MAXQ'	Reactive power upper limit, in Mvar
'MINQ'	Reactive power lower limit, in Mvar
'O_PAC'	Apparent active power flow into the dc line at the from bus (units determined by power output options)
'O_QAC'	Apparent reactive power flow into the dc line at the from bus (units determined by power output options)
'O_MVA'	Apparent ac power flow into the dc line at the from bus (units determined by power output options)
'O_PLOSS'	Total active power losses (units determined by power output options)
'O_QLOSS'	Total reactive power losses (units determined by power output options)
'O_MAXQ'	Reactive power upper limit (units determined by power output options)
'O_MINQ'	Reactive power lower limit (units determined by power output options)

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Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the GET_VSC_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (input; no default allowed).
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BUSES	number of elements in each column of RARRAY that are set by this API. DIM must be as least as large as the value of BUSES returned by the GET_VSC_DC_CONV_COUNT API when it is called with the same values specified for SID, TIES and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid ENTRY value
IERR = 6	Invalid NSTR value
IERR = 7	DIM, and hence the size of RARRAY, is not large enough
IERR = 8	Invalid STRING value

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8.20.6. AVSCDCCONVTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the VSC dc line converter data family (aVscDcConvInt, aVscDcConvReal, aVscDcConvCplx and aVscDcConvChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = avscdcconvtypes(string)
```

Fortran command syntax:

```
CALL GET_VSC_DC_CONV_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the VSC dc line converter data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the VSC dc line converter data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.21. FACTS Device Data

The FACTS device data family of subsystem data retrieval APIs returns arrays of data for FACTS devices in a specified subsystem.

FACTS devices with a series element may be subsystem tie branches. One of the input items to each API allows the user to elect to retrieve data either for all subsystem FACTS devices, for subsystem FACTS devices that are tie branches, or for both subsystem and tie FACTS devices. Another input item allows the user to elect to retrieve data only for FACTS devices without a series element, only for FACTS devices with a series element, or all types of FACTS devices.

Another input item allows the user to elect to retrieve data for all such FACTS devices, or to restrict the array entries to data for in-service FACTS devices.

Data for subsystem FACTS devices is returned from all APIs in the FACTS device data family in ascending FACTS device name alphabetical order.

8.21.1. AFACTSCHAR

Use this API to return an array of character values for subsystem FACTS devices.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = afactschar(sid, ties, flag, fcttyp, string)
```

Fortran command syntax:

```
CALL GET_FACTS_CHAR(SID, TIES, FLAG, FCTTYP, NSTR, STRING, DIM, CARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem FACTS devices only.
	TIES = 2	for subsystem tie FACTS devices only.
	TIES = 3	for both interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.

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Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default). FCTTYP = 1 for only FACTS devices without a series element. FCTTYP = 2 for only FACTS devices with a series element. FCTTYP = 3 for both types of FACTS devices.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).
Character STRING (NSTR)	NSTR of the following strings indicating the FACTS device quantities desired (input; no default allowed). 'FACTSNAME' FACTS device name (12 characters). 'FROMNAME' Sending end bus name (12 characters). 'FROMEXNAME' Sending end bus extended bus name (18 characters). 'TONAME' Terminal end bus name (12 characters). 'TOEXNAME' Terminal end bus extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_FACTS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (input; no default allowed).
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer BRNCHS	number of elements in each column of CARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_FACTS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (output).
Integer IERR	error code (output). IERR = 0 No error IERR = 1 Working case is empty IERR = 2 Invalid SID value IERR = 3 Invalid TIES value IERR = 4 Invalid FLAG value IERR = 5 Invalid FCTTYP value IERR = 6 Invalid NSTR value IERR = 7 DIM, and hence the size of CARRAY, is not large enough IERR = 8 Invalid STRING value

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8.21.2. AFACTSCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the FACTS device data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, brnchs = afactscount(sid, ties, flag, fcttyp)
```

Fortran command syntax:

```
CALL GET_FACTS_COUNT(SID, TIES, FLAG, FCTTYP, BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem FACTS devices only.
	TIES = 2	for subsystem tie FACTS devices only.
	TIES = 3	for both interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.
Integer BRNCHS	number of array entries required for the subsystem indicated by SID that meet the editing criteria indicated by TIES, FLAG and FCTTYP (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BRNCHS returned
	IERR = 1	Working case is empty; BRNCHS returned as 0
	IERR = 2	Invalid SID value; BRNCHS returned as 0
	IERR = 3	Invalid TIES value; BRNCHS returned as 0

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IERR = 4	Invalid FLAG value; BRNCHS returned as 0
IERR = 5	Invalid FCTTYP value; BRNCHS returned as 0

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8.21.3. AFACTSCPLX

Use this API to return an array of complex values for subsystem FACTS devices.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = afactscplx(sid, ties, flag, fcttyp, string)
```

Fortran command syntax:

```
CALL GET_FACTS_COMPLEX(SID, TIES, FLAG, FCTTYP, NSTR, STRING, DIM, XARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem FACTS devices only.
	TIES = 2	for subsystem tie FACTS devices only.
	TIES = 3	for both interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the FACTS device quantities desired (input; no default allowed).	
	'PQDES'	Desired complex power flow at the terminal bus end (MW and Mvar)
	'PQSEND'	Complex power flow into the series element at the sending end bus (MW and Mvar)

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	'PQTERM'	Complex power flow into the series element at the terminal end bus (MW and Mvar)
	'PQSHNT'	Complex bridge (active) and shunt (reactive) power flow into the shunt element (MW and Mvar)
	'PQLOSS'	Losses (MW and Mvar)
	'O_PQDES'	Desired complex power flow at the terminal bus end (units determined by power output options)
	'O_PQSEND'	Complex power flow into the series element at the sending end bus (units determined by power output options)
	'O_PQTERM'	Complex power flow into the series element at the terminal end bus (units determined by power output options)
	'O_PQSHNT'	Complex bridge (active) and shunt (reactive) power flow into the shunt element (units determined by power output options)
	'O_PQLOSS'	Losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_FACTS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of XARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_FACTS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid FCTTYP value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of RARRAY, is not large enough
	IERR = 8	Invalid STRING value

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8.21.4. AFACTSINT

Use this API to return an array of integer values for subsystem FACTS devices.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = afactsint(sid, ties, flag, fcttyp, string)
```

Fortran command syntax:

```
CALL GET_FACTS_INTEGER(SID, TIES, FLAG, FCTTYP, NSTR, STRING, DIM, IARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem FACTS devices only.
	TIES = 2	for subsystem tie FACTS devices only.
	TIES = 3	for both interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the FACTS device quantities desired (input; no default allowed).	
	'SENDNUMBER'	Sending end bus number
	'TERMNUMBER'	Terminal end bus number
	'MODE'	Control mode (negative if blocked during solution)

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	'OWNER'	Owner number
	'VSREF'	Series voltage reference code
	'STATVT'	Terminal end voltage condition code: -1 (holding low limit), 0 (in range), or 1 (holding high limit)
	'STATVS'	Series voltage limit code: 0 (in range), or 1 (holding high limit)
	'STATBR'	Bridge active power transfer limit code: 0 (in range), or 1 (holding high limit)
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_FACTS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer BRNCHS	number of elements in each column of IARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_FACTS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid FCTTYP value
	IERR = 6	Invalid NSTR value
	IERR = 7	DIM, and hence the size of IARRAY, is not large enough
	IERR = 8	Invalid STRING value

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8.21.5. AFACTSREAL

Use this API to return an array of real values for subsystem FACTS devices.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = afactsreal(sid, ties, flag, fcttyp, string)
```

Fortran command syntax:

```
CALL GET_FACTS_REAL(SID, TIES, FLAG, FCTTYP, NSTR, STRING, DIM, RARRAY,
 BRNCHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for interior subsystem FACTS devices only.
	TIES = 2	for subsystem tie FACTS devices only.
	TIES = 3	for both interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the FACTS device quantities desired (input; no default allowed).	
	'VSET'	Voltage setpoint at the sending end bus in pu
	'VTMN'	Minimum voltage at the terminal end bus in pu
	'VTMX'	Maximum voltage at the terminal end bus in pu

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'VSMX'	Maximum series voltage magnitude in pu
'LINX'	Reactance of the dummy series element
'RMPCT'	Percentage of the total MVAR required to regulate the controlled bus voltage that are contributed by this device
'VSERES'	Series voltage magnitude in pu
'IPUSND'	Current flow into series element at sending end bus in pu. 'IPUTRM' Current flow into series element at terminal end bus in pu
'IPUSHN'	Current flow into shunt element in pu
'AMPSND'	Current flow into series element at sending end bus in amps
'AMPTRM'	Current flow into series element at terminal end bus in amps
'AMPSHN'	Current flow into shunt element in amps
'PCTSND'	Percent series current loading at sending bus end
'PCTTRM'	Percent series current loading at terminal bus end
'PCTSHN'	Percent shunt current loading
'PCTPBR'	Percent bridge active power loading
'PDES'	Desired active power flow at the terminal bus end, in MW
'QDES'	Desired reactive power flow at the terminal bus end, in Mvar
'SHMX'	Maximum shunt current at the sending end bus at 1.0 pu voltage, in MVA
'TRMX'	Maximum bridge active power transfer, in MVA
'IMX'	Maximum series current at 1.0 pu voltage, in MVA
'PSEND'	Active power flow into the series element at the sending end bus, in MW
'QSEND'	Reactive power flow into the series element at the sending end bus, in Mvar
'PTERM'	Active power flow into the series element at the terminal end bus, in MW
'QTERM'	Reactive power flow into the series element at the terminal end bus, in Mvar
'PBRDG'	Bridge active power exchange, in MW
'QSHNT'	Reactive power flow into the shunt element, in Mvar
'MVASND'	Power flow into the series element at the sending end bus, in MVA
'MVATRM'	Power flow into the series element at the terminal end bus, in MVA

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	'MVASHN'	Power flow into the shunt element, in MVA
	'QLOSS'	Reactive power losses, in Mvar
	'O_PDES'	Desired active power flow at the terminal bus end (units determined by power output options)
	'O_QDES'	Desired reactive power flow at the terminal bus end (units determined by power output options)
	'O_SHMX'	Maximum shunt current at the sending end bus at 1.0 pu voltage (units determined by power output options)
	'O_TRMX'	Maximum bridge active power transfer (units determined by power output options)
	'O_IMX'	Maximum series current at 1.0 pu voltage (units determined by power output options)
	'O_PSEND'	Active power flow into the series element at the sending end bus (units determined by power output options)
	'O_QSEND'	Reactive power flow into the series element at the sending end bus (units determined by power output options)
	'O_PTERM'	Active power flow into the series element at the terminal end bus (units determined by power output options)
	'O_QTERM'	Reactive power flow into the series element at the terminal end bus (units determined by power output options)
	'O_PBRDG'	Bridge active power exchange (units determined by power output options)
	'O_QSHNT'	Reactive power flow into the shunt element (units determined by power output options)
	'O_MVASND'	Power flow into the series element at the sending end bus (units determined by power output options)
	'O_MVATRM'	Power flow into the series element at the terminal end bus (units determined by power output options)
	'O_MVASHN'	Power flow into the shunt element (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BRNCHS returned by the GET_FACTS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	

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Integer BRNCHS	number of elements in each column of RARRAY that are set by this API. BRNCHS is identical to the value of BRNCHS returned by the GET_FACTS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid FCTTYP value
IERR = 6	Invalid NSTR value
IERR = 7	DIM, and hence the size of RARRAY, is not large enough
IERR = 8	Invalid STRING value

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8.21.6. AFACTSTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the FACTS device data family (aFactsInt, aFactsReal, aFactsCplx and aFactsChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = afactstypes(string)
```

Fortran command syntax:

```
CALL GET_FACTS_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the FACTS device data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the FACTS device data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.22. FACTS Device Bus Data

The FACTS device bus data family of subsystem data retrieval APIs returns arrays of data for buses that are in a specified subsystem and to which FACTS devices are connected.

One of the input items to each API allows the user to elect to retrieve data either for all subsystem FACTS device buses, for subsystem buses connected to FACTS devices with a series element that is a tie branch, or for both subsystem and tie FACTS device buses. Another input item allows the user to elect to retrieve data either for FACTS devices without a series element, for FACTS devices with a series element, or for both types of FACTS devices.

Another input item allows the user to elect to retrieve data for all such FACTS device buses, or to restrict the array entries to data for buses connected to in-service FACTS devices.

Data for subsystem buses is returned from all APIs in the FACTS device bus data family in the same order. One of the input items to each API allows the user to elect that FACTS device buses be ordered either in FACTS device name order with all buses of each FACTS device grouped together, or in numerical or alphabetical order by FACTS device bus, according to the bus output option setting in effect at the time the API is referenced.

8.22.1. AFACTSBUSCHAR

Use this API to return an array of character values for subsystem FACTS device buses.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = afactsbuschar(sid, ties, flag, fcttyp, entry, string)
```

Fortran command syntax:

```
CALL GET_FACTS_BUS_CHAR(SID, TIES, FLAG, FCTTYP, ENTRY, NSTR, STRING, DIM,
 CARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem buses connected to FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem FACTS devices only.
	TIES = 2	for the subsystem bus end of tie FACTS devices only.
	TIES = 3	for the non-subsystem bus end of tie FACTS devices only.
	TIES = 4	for each end of tie FACTS devices only.

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	TIES = 5	for each end of interior subsystem FACTS devices and the subsystem bus end of tie FACTS devices.
	TIES = 6	for each end of interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.
Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for bus order.
	ENTRY = 2	for FACTS device name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the FACTS device quantities desired (input; no default allowed).	
	'FACTSNAME'	FACTS device name (12 characters).
	'FROMNAME'	Bus name of this FACTS bus (12 characters).
	'FROMEXNAME'	Extended bus name of this FACTS bus (18 characters).
	'TONAME'	Bus name of the other end FACTS bus (12 characters).
	'TOEXNAME'	Extended bus name of the other end FACTS bus (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the GET_FACTS_BUS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (input; no default allowed).	
Integer BUSES	number of elements in each column of CARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_FACTS_BUS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (output).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value

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IERR = 3	Invalid TIES value
IERR = 4	Invalid FLAG value
IERR = 5	Invalid FCTTYP value
IERR = 6	Invalid ENTRY value
IERR = 7	Invalid NSTR value
IERR = 8	DIM, and hence the size of CARRAY, is not large enough
IERR = 9	Invalid STRING value

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8.22.2. AFACTSBUSCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the FACTS device bus data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, buses = afactsbuscount(sid, ties, flag, fcttyp)
```

Fortran command syntax:

```
CALL GET_FACTS_BUS_COUNT(SID, TIES, FLAG, FCTTYP, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem buses connected to FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem FACTS devices only.
	TIES = 2	for the subsystem bus end of tie FACTS devices only.
	TIES = 3	for the non-subsystem bus end of tie FACTS devices only.
	TIES = 4	for each end of tie FACTS devices only.
	TIES = 5	for each end of interior subsystem FACTS devices and the subsystem bus end of tie FACTS devices.
	TIES = 6	for each end of interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.

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Integer BUSES	number of array entries required for the subsystem indicated by SID that meet the editing criteria indicated by TIES, FLAG and FCTTYP (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, BUSES returned
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0
	IERR = 3	Invalid TIES value; BUSES returned as 0
	IERR = 4	Invalid FLAG value; BUSES returned as 0
	IERR = 5	Invalid FCTTYP value; BUSES returned as 0

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8.22.3. AFACTSBUSCPLX

Use this API to return an array of complex values for subsystem FACTS device buses.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = afactsbuscplx(sid, ties, flag, fcttyp, entry, string)
```

Fortran command syntax:

```
CALL GET_FACTS_BUS_COMPLEX(SID, TIES, FLAG, FCTTYP, ENTRY, NSTR, STRING,
 DIM, XARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem buses connected to FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem FACTS devices only.
	TIES = 2	for the subsystem bus end of tie FACTS devices only.
	TIES = 3	for the non-subsystem bus end of tie FACTS devices only.
	TIES = 4	for each end of tie FACTS devices only.
	TIES = 5	for each end of interior subsystem FACTS devices and the subsystem bus end of tie FACTS devices.
	TIES = 6	for each end of interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.

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Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for bus order.
	ENTRY = 2	for FACTS device name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the FACTS device quantities desired (input; no default allowed).	
	'PQFROM'	Complex power flow into the series element at the from bus end (MW and Mvar)
	'PQSHNT'	Complex bridge (active) and shunt (reactive) power flow into the shunt element (0.0 if the from bus is the terminal end bus) (MW and Mvar)
	'PQLOSS'	Losses (MW and Mvar)
	'O_PQFROM'	Complex power flow into the series element at the from bus end (units determined by power output options)
	'O_PQSHNT'	Complex bridge (active) and shunt (reactive) power flow into the shunt element (0.0 if the from bus is the terminal end bus) (units determined by power output options)
	'O_PQLOSS'	Losses (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the GET_FACTS_BUS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (input; no default allowed).	
Integer BUSES	number of elements in each column of XARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_FACTS_BUS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (output).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid FCTTYP value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of RARRAY, is not large enough
	IERR = 9	Invalid STRING value

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8.22.4. AFACTSBUSINT

Use this API to return an array of integer values for subsystem FACTS device buses.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = afactsbusint(sid, ties, flag, fcttyp, entry, string)
```

Fortran command syntax:

```
CALL GET_FACTS_BUS_INTEGER(SID, TIES, FLAG, FCTTYP, ENTRY, NSTR, STRING,
 DIM, IARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem buses connected to FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem FACTS devices only.
	TIES = 2	for the subsystem bus end of tie FACTS devices only.
	TIES = 3	for the non-subsystem bus end of tie FACTS devices only.
	TIES = 4	for each end of tie FACTS devices only.
	TIES = 5	for each end of interior subsystem FACTS devices and the subsystem bus end of tie FACTS devices.
	TIES = 6	for each end of interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.

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Integer ENTRY	a flag indicating organization (input; 1 by default).	
	ENTRY = 1	for bus order.
	ENTRY = 2	for FACTS device name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the FACTS device quantities desired (input; no default allowed).	
	'FROMNUMBER'	Bus number of this FACTS bus
	'TONUMBER'	Bus number of the other end FACTS bus
	'FACTSEND'	FACTS device end code of this FACTS bus: 1 (sending end bus) or 2 (terminal end bus)
	'MODE'	Control mode (negative if blocked during solution)
	'OWNER'	Owner number
	'VSREF'	Series voltage reference code
	'STATVT'	Terminal end voltage condition code: -1 (holding low limit), 0 (in range), or 1 (holding high limit)
	'STATVS'	Series voltage limit code: 0 (in range), or 1 (holding high limit)
	'STATBR'	Bridge active power transfer limit code: 0 (in range), or 1 (holding high limit)
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the GET_FACTS_BUS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (input; no default allowed).	
Integer BUSES	number of elements in each column of IARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_FACTS_BUS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (output).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid FCTTYP value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of IARRAY, is not large enough
	IERR = 9	Invalid STRING value

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8.22.5. AFACTSBUSREAL

Use this API to return an array of real values for subsystem FACTS device buses.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = afactsbusreal(sid, ties, flag, fcttyp, entry, string)
```

Fortran command syntax:

```
CALL GET_FACTS_BUS_REAL(SID, TIES, FLAG, FCTTYP, ENTRY, NSTR, STRING, DIM,
 RARRAY, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer TIES	a flag indicating which subsystem buses connected to FACTS devices with a series element to include (ignored if SID is negative) (input; 1 by default).	
	TIES = 1	for each end of interior subsystem FACTS devices only.
	TIES = 2	for the subsystem bus end of tie FACTS devices only.
	TIES = 3	for the non-subsystem bus end of tie FACTS devices only.
	TIES = 4	for each end of tie FACTS devices only.
	TIES = 5	for each end of interior subsystem FACTS devices and the subsystem bus end of tie FACTS devices.
	TIES = 6	for each end of interior subsystem FACTS devices and tie FACTS devices.
Integer FLAG	a flag indicating which subsystem FACTS devices to include (input; 1 by default).	
	FLAG = 1	for only in-service FACTS devices.
	FLAG = 2	for both in-service and out-of-service FACTS devices.
Integer FCTTYP	a flag indicating which types of FACTS devices to include (input; 1 by default).	
	FCTTYP = 1	for only FACTS devices without a series element.
	FCTTYP = 2	for only FACTS devices with a series element.
	FCTTYP = 3	for both types of FACTS devices.
Integer ENTRY	a flag indicating organization (input; 1 by default).	

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	ENTRY = 1	for bus order.
	ENTRY = 2	for FACTS device name order.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the FACTS device quantities desired (input; no default allowed).	
	'VSERES'	Series voltage magnitude in pu
	'IPUFROM'	Current flow into series element at the from bus end in pu
	'IPUSHN'	Current flow into shunt element in pu (0.0 if the from bus is the terminal end bus)
	'AMPSFROM'	Current flow into series element at the from bus end in amps
	'AMPSHN'	Current flow into shunt element in amps (0.0 if the from bus is the terminal end bus)
	'PCTFROM'	Percent series current loading at the from bus end
	'PCTSHN'	Percent shunt current loading (0.0 if the from bus is the terminal end bus)
	'PCTPBR'	Percent bridge active power loading (0.0 if the from bus is the terminal end bus)
	'PFROM'	Active power flow into the series element at the from bus end, in MW
	'QFROM'	Reactive power flow into the series element at the from bus end, in Mvar
	'PBRDG'	Bridge active power exchange (0.0 if the from bus is the terminal end bus), in MW
	'QSHNT'	Reactive power flow into the shunt element (0.0 if the from bus is the terminal end bus), in Mvar
	'MVAFROM'	Power flow into the series element at the from bus end, in MVA
	'MVASHN'	Power flow into the shunt element (0.0 if the from bus is the terminal end bus), in MVA
	'QLOSS'	Reactive power losses, in Mvar
	'O_PFROM'	Active power flow into the series element at the from bus end (units determined by power output options)
	'O_QFROM'	Reactive power flow into the series element at the from bus end (units determined by power output options)
	'O_PBRDG'	Bridge active power exchange (0.0 if the from bus is the terminal end bus) (units determined by power output options)

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	'O_QSHNT'	Reactive power flow into the shunt element (0.0 if the from bus is the terminal end bus) (units determined by power output options)
	'O_MVAFROM'	Power flow into the series element at the from bus end (units determined by power output options)
	'O_MVASHN'	Power flow into the shunt element (0.0 if the from bus is the terminal end bus) (units determined by power output options)
	'O_QLOSS'	Reactive power losses (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the GET_FACTS_BUS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (input; no default allowed).	
Integer BUSES	number of elements in each column of RARRAY that are set by this API. BUSES is identical to the value of BUSES returned by the GET_FACTS_BUS_COUNT API when it is called with the same values specified for SID, TIES, FLAG and FCTTYP (output).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid TIES value
	IERR = 4	Invalid FLAG value
	IERR = 5	Invalid FCTTYP value
	IERR = 6	Invalid ENTRY value
	IERR = 7	Invalid NSTR value
	IERR = 8	DIM, and hence the size of RARRAY, is not large enough
	IERR = 9	Invalid STRING value

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8.22.6. AFACTSBUSTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the FACTS device bus data family (aFactsBusInt, aFactsBusReal, aFactsBusCplx and aFactsBusChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = afactsbustypes(string)
```

Fortran command syntax:

```
CALL GET_FACTS_BUS_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the FACTS device bus data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the FACTS device bus data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.23. Area Data

The area data family of subsystem data retrieval APIs returns arrays of data for areas in a specified area subsystem. One of the input items to each API allows the user to elect either to retrieve data for areas with any ac buses, loads, induction machines, or dc buses assigned to them, or to restrict the array entries to data for areas with at least one ac bus assigned to them.

Data for subsystem areas is returned from all APIs in the area data family in the same order: ascending area number order.

8.23.1. AAREACHAR

Use this API to return an array of character values for subsystem areas.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = aareachar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_AREA_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,AREAS,IERR)
```

Where:

Integer SID	Defines the area subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all areas in the working case.
	SID >= 0	valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem areas to include (input; 1 by default).	
	FLAG = 1	for only subsystem areas with at least one ac bus assigned to them.
	FLAG = 2	for subsystem areas with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the area quantities desired (input; no default allowed).	
	'AREANAME'	Area name (12 characters).
	'NAME'	Bus name of the area swing bus (12 characters).
	'EXNAME'	Extended bus name of the area swing bus (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of AREAS returned by the	

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	GET_AREA_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer AREAS	number of elements in each column of CARRAY that are set by this API. AREAS is identical to the value of AREAS returned by the GET_AREA_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of CARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.23.2. AAREACOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the area data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, areas = aareacount(sid, flag)
```

Fortran command syntax:

```
CALL GET_AREA_COUNT(SID, FLAG, AREAS, IERR)
```

Where:

Integer SID	Defines the area subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all areas in the working case.
	SID >= 0	valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem areas to include (input; 1 by default).	
	FLAG = 1	for only subsystem areas with at least one ac bus assigned to them.
	FLAG = 2	for subsystem areas with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer AREAS	number of areas in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, AREAS returned
	IERR = 1	Working case is empty; AREAS returned as 0
	IERR = 2	Invalid SID value; AREAS returned as 0
	IERR = 3	Invalid FLAG value; AREAS returned as 0

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8.23.3. AAREACPLX

Use this API to return an array of complex values for subsystem areas.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = aareacplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_AREA_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,AREAS,IERR)
```

Where:

Integer SID	Defines the area subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all areas in the working case.
	SID >= 0	valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem areas to include (input; 1 by default).	
	FLAG = 1	for only subsystem areas with at least one ac bus assigned to them.
	FLAG = 2	for subsystem areas with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the area quantities desired (input; no default allowed).	
	'PQLOAD'	Total complex load on buses in the area (in MW and Mvar)
	'PQLOADLD'	Total complex load of loads in the area (in MW and Mvar)
	'PQGEN'	Total complex generation on buses in the area (in MW and Mvar)
	'PQLOSS'	Total complex losses assigned to the area (in MW and Mvar)
	'PQINT'	Total complex net interchange from the area (in MW and Mvar)
	'O_PQLOAD'	Total complex load on buses in the area (units determined by power output options)
	'O_PQLOADLD'	Total complex load of loads in the area (units determined by power output options)
	'O_PQGEN'	Total complex generation on buses in the area (units determined by power output options)

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	'O_PQLOSS'	Total complex losses assigned to the area (units determined by power output options)
	'O_PQINT'	Total complex net interchange from the area (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of AREAS returned by the GET_AREA_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer AREAS	number of elements in each column of XARRAY that are set by this API. AREAS is identical to the value of AREAS returned by the GET_AREA_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.23.4. AAREAINT

Use this API to return an array of integer values for subsystem areas.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = aareaint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_AREA_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,AREAS,IERR)
```

Where:

Integer SID	Defines the area subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all areas in the working case.
	SID >= 0	valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem areas to include (input; 1 by default).	
	FLAG = 1	for only subsystem areas with at least one ac bus assigned to them.
	FLAG = 2	for subsystem areas with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the area quantities desired (input; no default allowed).	
	'NUMBER'	Area number
	'SWING'	Number of the area swing bus used for area interchange control purposes
	'BUSES'	Number of ac buses assigned to the area
	'LOADS'	Number of loads assigned to the area
	'DCBUSES'	Number of dc buses assigned to the area
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of AREAS returned by the GET_AREA_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer AREAS	number of elements in each column of IARRAY that are set by this API. AREAS is identical to the value of AREAS returned by the GET_AREA_COUNT API when it is called with the same values specified for SID and FLAG (output).	

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Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of IARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.23.5. AAREAREAL

Use this API to return an array of real values for subsystem areas.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = aareareal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_AREA_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,AREAS,IERR)
```

Where:

Integer SID	Defines the area subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all areas in the working case.
	SID >= 0	valid area subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem areas to include (input; 1 by default).	
	FLAG = 1	for only subsystem areas with at least one ac bus assigned to them.
	FLAG = 2	for subsystem areas with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the area quantities desired (input; no default allowed).	
	'PDES'	Desired net active power interchange leaving the area, in MW
	'PTOL'	Interchange tolerance bandwidth, in MW
	'PLOAD'	Total active power load on buses in the area, in MW
	'QLOAD'	Total reactive power load on buses in the area, in Mvar
	'PLOADLD'	Total active power load of loads in the area, in MW
	'QLOADLD'	Total reactive power load of loads in the area, in Mvar
	'PGEN'	Total active power generation on buses in the area, in MW
	'QGEN'	Total reactive power generation on buses in the area, in Mvar
	'PLOSS'	Total active power losses assigned to the area, in MW

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	'QLOSS'	Total reactive power losses assigned to the area, in Mvar
	'PINT'	Total active power net interchange from the area, in MW
	'QINT'	Total reactive power net interchange from the area, in Mvar
	'O_PDES'	Desired net active power interchange leaving the area (units determined by power output options)
	'O_PTOL'	Interchange tolerance bandwidth (units determined by power output options)
	'O_PLOAD'	Total active power load on buses in the area (units determined by power output options)
	'O_QLOAD'	Total reactive power load on buses in the area (units determined by power output options)
	'O_PLOADLD'	Total active power load of loads in the area (units determined by power output options)
	'O_QLOADLD'	Total reactive power load of loads in the area (units determined by power output options)
	'O_PGEN'	Total active power generation on buses in the area (units determined by power output options)
	'O_QGEN'	Total reactive power generation on buses in the area (units determined by power output options)
	'O_PLOSS'	Total active power losses assigned to the area (units determined by power output options)
	'O_QLOSS'	Total reactive power losses assigned to the area (units determined by power output options)
	'O_PINT'	Total active power net interchange from the area (units determined by power output options)
	'O_QINT'	Total reactive power net interchange from the area (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of AREAS returned by the GET_AREA_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer AREAS	number of elements in each column of RARRAY that are set by this API. AREAS is identical to the value of AREAS returned by the GET_AREA_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	

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IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.23.6. AAREATYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the area data family (aAreaInt, aAreaReal, aAreaCplx and aAreaChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = aareatypes(string)
```

Fortran command syntax:

```
CALL GET_AREA_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the area data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the area data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.24. Owner Data

The owner data family of subsystem data retrieval APIs returns arrays of data for owners in a specified owner subsystem. One of the input items to each API allows the user to elect either to retrieve data for owners with any ac buses, loads, machines, induction machines, branches, FACTS devices, VSC dc lines, or dc buses assigned to them, or to restrict the array entries to data for owners with at least one ac bus assigned to them.

Data for subsystem owners is returned from all APIs in the owner data family in the same order: ascending owner number order.

8.24.1. AOWNERCHAR

Use this API to return an array of character values for subsystem owners.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = aownerchar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_OWNER_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,OWNERS,IERR)
```

Where:

Integer SID	Defines the owner subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all owners in the working case.
	SID >= 0	valid owner subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem owners to include (input; 1 by default).	
	FLAG = 1	for only subsystem owners with at least one ac bus assigned to them.
	FLAG = 2	for subsystem owners with any equipment (ac buses, loads, machines, ac branches, FACTS devices, VSC dc lines and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the owner quantities desired (input; no default allowed).	
	'OWNERNAME'	Owner name (12 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of OWNERS returned by the GET_OWNER_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	

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Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer OWNERS	number of elements in each column of CARRAY that are set by this API. OWNERS is identical to the value of OWNERS returned by the GET_OWNER_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of CARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.24.2. AOWNERCOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the owner data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, owners = aownercount(sid, flag)
```

Fortran command syntax:

```
CALL GET_OWNER_COUNT(SID, FLAG, OWNERS, IERR)
```

Where:

Integer SID	Defines the owner subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all owners in the working case.
	SID >= 0	valid owner subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem owners to include (input; 1 by default).	
	FLAG = 1	for only subsystem owners with at least one ac bus assigned to them.
	FLAG = 2	for subsystem owners with any equipment (ac buses, loads, machines, ac branches, FACTS devices, VSC dc lines and/or dc buses) assigned to them.
Integer OWNERS	number of owners in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, OWNERS returned
	IERR = 1	Working case is empty; OWNERS returned as 0
	IERR = 2	Invalid SID value; OWNERS returned as 0
	IERR = 3	Invalid FLAG value; OWNERS returned as 0

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8.24.3. AOWNERCPLX

Use this API to return an array of complex values for subsystem owners.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = aownercplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_OWNER_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,OWNERS,IERR)
```

Where:

Integer SID	Defines the owner subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all owners in the working case.
	SID >= 0	valid owner subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem owners to include (input; 1 by default).	
	FLAG = 1	for only subsystem owners with at least one ac bus assigned to them.
	FLAG = 2	for subsystem owners with any equipment (ac buses, loads, machines, ac branches, FACTS devices, VSC dc lines and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the owner quantities desired (input; no default allowed).	
	'PQLOAD'	Total complex load on the owner's buses (in MW and Mvar)
	'PQLOADLD'	Total complex load of the owner's loads (in MW and Mvar)
	'PQGEN'	Total complex generation on the owner's buses (in MW and Mvar)
	'PQGENMC'	Total complex generation on the owner's machines (in MW and Mvar)
	'PQLOSS'	Total complex losses assigned to the owner (in MW and Mvar)
	'O_PQLOAD'	Total complex load on the owner's buses (units determined by power output options)
	'O_PQLOADLD'	Total complex load of the owner's loads (units determined by power output options)

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	'O_PQGEN'	Total complex generation on the owner's buses (units determined by power output options)
	'O_PQGENMC'	Total complex generation on the owner's machines (units determined by power output options)
	'O_PQLOSS'	Total complex losses assigned to the owner (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of OWNERS returned by the GET_OWNER_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer OWNERS	number of elements in each column of XARRAY that are set by this API. OWNERS is identical to the value of OWNERS returned by the GET_OWNER_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.24.4. AOWNERINT

Use this API to return an array of integer values for subsystem owners.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = aownerint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_OWNER_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,OWNERS,IERR)
```

Where:

Integer SID	Defines the owner subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all owners in the working case.
	SID >= 0	valid owner subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem owners to include (input; 1 by default).	
	FLAG = 1	for only subsystem owners with at least one ac bus assigned to them.
	FLAG = 2	for subsystem owners with any equipment (ac buses, loads, machines, ac branches, FACTS devices, VSC dc lines and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the owner quantities desired (input; no default allowed).	
	'NUMBER'	Owner number
	'BUSES'	Number of ac buses assigned to the owner
	'LOADS'	Number of loads assigned to the owner
	'MACHINES'	Number of machine ownership specifications for the owner
	'BRANCHES'	Number of branch ownership specifications for the owner
	'FACTS'	Number of FACTS devices assigned to the owner
	'VSCDCLINES'	Number of VSC dc line ownership specifications for the owner
	'DCBUSES'	Number of dc buses assigned to the owner
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of OWNERS returned by the	

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	GET_OWNER_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).
Integer OWNERS	number of elements in each column of IARRAY that are set by this API. OWNERS is identical to the value of OWNERS returned by the GET_OWNER_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of IARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.24.5. AOWNERREAL

Use this API to return an array of real values for subsystem owners.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = aownerreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_OWNER_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,OWNERS,IERR)
```

Where:

Integer SID	Defines the owner subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all owners in the working case.
	SID >= 0	valid owner subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem owners to include (input; 1 by default).	
	FLAG = 1	for only subsystem owners with at least one ac bus assigned to them.
	FLAG = 2	for subsystem owners with any equipment (ac buses, loads, machines, ac branches, FACTS devices, VSC dc lines and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the owner quantities desired (input; no default allowed).	
	'PLOAD'	Total active power load on the owner's buses, in MW
	'QLOAD'	Total reactive power load on the owner's buses, in Mvar
	'PLOADLD'	Total active power load of the owner's loads, in MW
	'QLOADLD'	Total reactive power load of the owner's loads, in Mvar
	'PGEN'	Total active power generation on the owner's buses, in Mw
	'QGEN'	Total reactive power generation on the owner's buses, in Mvar
	'PGENMC'	Total active power generation on the owner's machines, in MW

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	'QGENMC'	Total reactive power generation on the owner's machines, in Mvar
	'PLOSS'	Total active power losses assigned to the owner, in MW
	'QLOSS'	Total reactive power losses assigned to the owner, in Mvar
	'O_PLOAD'	Total active power load on the owner's buses (units determined by power output options)
	'O_QLOAD'	Total reactive power load on the owner's buses (units determined by power output options)
	'O_PLOADLD'	Total active power load of the owner's loads (units determined by power output options)
	'O_QLOADLD'	Total reactive power load of the owner's loads (units determined by power output options)
	'O_PGEN'	Total active power generation on the owner's buses (units determined by power output options)
	'O_QGEN'	Total reactive power generation on the owner's buses (units determined by power output options)
	'O_PGENMC'	Total active power generation on the owner's machines (units determined by power output options)
	'O_QGENMC'	Total reactive power generation on the owner's machines (units determined by power output options)
	'O_PLOSS'	Total active power losses assigned to the owner (units determined by power output options)
	'O_QLOSS'	Total reactive power losses assigned to the owner (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of OWNERS returned by the GET_OWNER_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer OWNERS	number of elements in each column of RARRAY that are set by this API. OWNERS is identical to the value of OWNERS returned by the GET_OWNER_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value

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IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.24.6. AOWNERTYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the owner data family (aOwnerInt, aOwnerReal, aOwnerCplx and aOwnerChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = aownertypes(string)
```

Fortran command syntax:

```
CALL GET_OWNER_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the owner data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the owner data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.25. Zone Data

The zone data family of subsystem data retrieval APIs returns arrays of data for zones in a specified zone subsystem. One of the input items to each API allows the user to elect either to retrieve data for zones with any ac buses, loads, induction machines, or dc buses assigned to them, or to restrict the array entries to data for zones with at least one ac bus assigned to them.

Data for subsystem zones is returned from all APIs in the zone data family in the same order: ascending zone number order.

8.25.1. AZONECHAR

Use this API to return an array of character values for subsystem zones.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, carray = azonechar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_ZONE_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,ZONES,IERR)
```

Where:

Integer SID	Defines the zone subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all zones in the working case.
	SID >= 0	valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem zones to include (input; 1 by default).	
	FLAG = 1	for only subsystem zones with at least one ac bus assigned to them.
	FLAG = 2	for subsystem zones with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the zone quantities desired (input; no default allowed).	
	'ZONENAME'	Zone name (12 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of ZONES returned by the GET_ZONE_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	

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Integer ZONES	number of elements in each column of CARRAY that are set by this API. ZONES is identical to the value of ZONES returned by the GET_ZONE_COUNT API when it is called with the same values specified for SID and FLAG (output).
Integer IERR	error code (output).

IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of CARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.25.2. AZONECOUNT

Use this API to return the number of array entries required to accommodate the data to be returned by the remaining members of the zone data family.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, zones = azonecount(sid, flag)
```

Fortran command syntax:

```
CALL GET_ZONE_COUNT(SID, FLAG, ZONES, IERR)
```

Where:

Integer SID	Defines the zone subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all zones in the working case.
	SID >= 0	valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem zones to include (input; 1 by default).	
	FLAG = 1	for only subsystem zones with at least one ac bus assigned to them.
	FLAG = 2	for subsystem zones with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer ZONES	number of zones in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error, ZONES returned
	IERR = 1	Working case is empty; ZONES returned as 0
	IERR = 2	Invalid SID value; ZONES returned as 0
	IERR = 3	Invalid FLAG value; ZONES returned as 0

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8.25.3. AZONECPLX

Use this API to return an array of complex values for subsystem zones.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, xarray = azonecplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_ZONE_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,ZONES,IERR)
```

Where:

Integer SID	Defines the zone subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all zones in the working case.
	SID >= 0	valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem zones to include (input; 1 by default).	
	FLAG = 1	for only subsystem zones with at least one ac bus assigned to them.
	FLAG = 2	for subsystem zones with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the zone quantities desired (input; no default allowed).	
	'PQLOAD'	Total complex load on buses in the zone (in MW and Mvar)
	'PQLOADLD'	Total complex load of loads in the zone (in MW and Mvar)
	'PQGEN'	Total complex generation on buses in the zone (in MW and Mvar)
	'PQLOSS'	Total complex losses assigned to the zone (in MW and Mvar)
	'PQINT'	Total complex net interchange from the zone (in MW and Mvar)
	'O_PQLOAD'	Total complex load on buses in the zone (units determined by power output options)
	'O_PQLOADLD'	Total complex load of loads in the zone (units determined by power output options)
	'O_PQGEN'	Total complex generation on buses in the zone (units determined by power output options)

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	'O_PQLOSS'	Total complex losses assigned to the zone (units determined by power output options)
	'O_PQINT'	Total complex net interchange from the zone (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of ZONES returned by the GET_ZONE_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer ZONES	number of elements in each column of XARRAY that are set by this API. ZONES is identical to the value of ZONES returned by the GET_ZONE_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.25.4. AZONEINT

Use this API to return an array of integer values for subsystem zones.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, iarray = azoneint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_ZONE_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,ZONES,IERR)
```

Where:

Integer SID	Defines the zone subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all zones in the working case.
	SID >= 0	valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem zones to include (input; 1 by default).	
	FLAG = 1	for only subsystem zones with at least one ac bus assigned to them.
	FLAG = 2	for subsystem zones with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the zone quantities desired (input; no default allowed).	
	'NUMBER'	Zone number
	'BUSES'	Number of ac buses assigned to the zone
	'LOADS'	Number of loads assigned to the zone
	'DCBUSES'	Number of dc buses assigned to the zone
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of ZONES returned by the GET_ZONE_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer ZONES	number of elements in each column of IARRAY that are set by this API. ZONES is identical to the value of ZONES returned by the GET_ZONE_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	

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IERR = 0	No error
IERR = 1	Working case is empty
IERR = 2	Invalid SID value
IERR = 3	Invalid FLAG value
IERR = 4	Invalid NSTR value
IERR = 5	DIM, and hence the size of IARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.25.5. AZONEREAL

Use this API to return an array of real values for subsystem zones.

This API routine was first introduced in release 30.3.0.

Python command syntax:

```
ierr, rarray = azonereal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_ZONE_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,ZONES,IERR)
```

Where:

Integer SID	Defines the zone subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API to assume a subsystem containing all zones in the working case.
	SID >= 0	valid zone subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem zones to include (input; 1 by default).	
	FLAG = 1	for only subsystem zones with at least one ac bus assigned to them.
	FLAG = 2	for subsystem zones with any equipment (ac buses, loads, and/or dc buses) assigned to them.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the zone quantities desired (input; no default allowed).	
	'PLOAD'	Total active power load on buses in the zone, in MW
	'QLOAD'	Total reactive power load on buses in the zone, in Mvar
	'PLOADLD'	Total active power load of loads in the zone, in MW
	'QLOADLD'	Total reactive power load of loads in the zone, in Mvar
	'PGEN'	Total active power generation on buses in the zone, in MW
	'QGEN'	Total reactive power generation on buses in the zone, in Mvar
	'PLOSS'	Total active power losses assigned to the zone, in MW
	'QLOSS'	Total reactive power losses assigned to the zone, in Mvar

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	'PINT'	Total active power net interchange from the zone, in MW
	'QINT'	Total reactive power net interchange from the zone, in Mvar
	'O_PLOAD'	Total active power load on buses in the zone (units determined by power output options)
	'O_QLOAD'	Total reactive power load on buses in the zone (units determined by power output options)
	'O_PLOADLD'	Total active power load of loads in the zone (units determined by power output options)
	'O_QLOADLD'	Total reactive power load of loads in the zone (units determined by power output options)
	'O_PGEN'	Total active power generation on buses in the zone (units determined by power output options)
	'O_QGEN'	Total reactive power generation on buses in the zone (units determined by power output options)
	'O_PLOSS'	Total active power losses assigned to the zone (units determined by power output options)
	'O_QLOSS'	Total reactive power losses assigned to the zone (units determined by power output options)
	'O_PINT'	Total active power net interchange from the zone (units determined by power output options)
	'O_QINT'	Total reactive power net interchange from the zone (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of ZONES returned by the GET_ZONE_COUNT API when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer ZONES	number of elements in each column of RARRAY that are set by this API. ZONES is identical to the value of ZONES returned by the GET_ZONE_COUNT API when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value

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IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.25.6. AZONETYPES

Use this API to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the zone data family (aZoneInt, aZoneReal, aZoneCplx and aZoneChar).

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
ierr, types = azonetypes(string)
```

Fortran command syntax:

```
CALL GET_ZONE_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the zone data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the zone data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.26. Induction Machine Bus Data

The induction machine bus data family of subsystem data retrieval API routines returns arrays of data for buses in a specified subsystem. One of the input items to each API routine allows the user to elect either to retrieve data for all subsystem buses, or to restrict the array entries to data for a subset of buses according to their bus and/or induction machine configuration and statuses.

Data for subsystem buses is returned from all API routines in the induction machine bus data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API routine is referenced.

8.26.1. AINDMACBUSCHAR

Use this API routine to return an array of character values for subsystem buses.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, carray = aindmacbuschar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_BUS_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction machine.
	FLAG = 2	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 3	for all in-service subsystem buses (type code 1, 2 or 3), including those that have no induction machines at the bus.
	FLAG = 4	for all subsystem buses with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 5	for all subsystem buses, including those that have no induction machines at the bus.

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	FLAG = 10	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction generator.
	FLAG = 11	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction generator, even if all induction generators at the bus are out-of-service.
	FLAG = 12	for all subsystem buses with at least one induction generator, even if all induction generators at the bus are out-of-service.
	FLAG = 20	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction motor.
	FLAG = 21	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction motor, even if all induction motors at the bus are out-of-service.
	FLAG = 22	for all subsystem buses with at least one induction motor, even if all induction motors at the bus are out-of-service.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or induction machine quantities desired (input; no default allowed).	
	'NAME'	Bus name (12 characters).
	'EXNAME'	Extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of BUSES returned by the aIndMacBusCount API routine when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of CARRAY that are set by this API routine. BUSES is identical to the value of BUSES returned by the aIndMacBusCount API routine when it is called with the same values specified for SID and FLAG (output).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of CARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.26.2. AINDMACBUSCOUNT

Use this API routine to obtain the number of array entries required to accommodate the data to be returned by the remaining members of the induction machine bus data family.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, buses = aindmacbuscount(sid, flag)
```

Fortran command syntax:

```
CALL GET_IND_MAC_BUS_COUNT(SID, FLAG, BUSES, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction machine.
	FLAG = 2	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 3	for all in-service subsystem buses (type code 1, 2 or 3), including those that have no induction machines at the bus.
	FLAG = 4	for all subsystem buses with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 5	for all subsystem buses, including those that have no induction machines at the bus.
	FLAG = 10	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction generator.
	FLAG = 11	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction generator, even if all induction generators at the bus are out-of-service.
	FLAG = 12	for all subsystem buses with at least one induction generator, even if all induction generators at the bus are out-of-service.

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	FLAG = 20	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction motor.
	FLAG = 21	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction motor, even if all induction motors at the bus are out-of-service.
	FLAG = 22	for all subsystem buses with at least one induction motor, even if all induction motors at the bus are out-of-service.
Integer BUSES	number of buses in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty; BUSES returned as 0
	IERR = 2	Invalid SID value; BUSES returned as 0
	IERR = 3	Invalid FLAG value; BUSES returned as 0

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8.26.3. AINDMACBUSCPLX

Use this API routine to return an array of complex values for subsystem buses.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, xarray = aindmacbuscplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_BUS_COMPLEX(SID, FLAG, NSTR, STRING, DIM, XARRAY, BUSES,
 IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction machine.
	FLAG = 2	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 3	for all in-service subsystem buses (type code 1, 2 or 3), including those that have no induction machines at the bus.
	FLAG = 4	for all subsystem buses with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 5	for all subsystem buses, including those that have no induction machines at the bus.
	FLAG = 10	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction generator.
	FLAG = 11	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction generator, even if all induction generators at the bus are out-of-service.
	FLAG = 12	for all subsystem buses with at least one induction generator, even if all induction generators at the bus are out-of-service.

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	FLAG = 20	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction motor.
	FLAG = 21	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction motor, even if all induction motors at the bus are out-of-service.
	FLAG = 22	for all subsystem buses with at least one induction motor, even if all induction motors at the bus are out-of-service.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or induction machine quantities desired (input; no default allowed).	
	'VOLTAGE'	Actual bus voltage, in pu, rectangular coordinates
	'TOTAL'	Flow from the bus to the in-service induction machines at the bus (in MW and Mvar)
	'MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (in MW and Mvar)
	'O_TOTAL'	Flow from the bus to the in-service induction machines at the bus (units determined by power output options)
	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of BUSES returned by the aIndMacBusCount API routine when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of XARRAY that are set by this API routine. BUSES is identical to the value of BUSES returned by the aIndMacBusCount API routine when it is called with the same values specified for SID and FLAG (output).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.26.4. AINDMACBUSINT

Use this API routine to return an array of integer values for subsystem buses.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, iarray = aindmacbusint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_BUS_INTEGER(SID, FLAG, NSTR, STRING, DIM, IARRAY, BUSES,
 IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction machine.
	FLAG = 2	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 3	for all in-service subsystem buses (type code 1, 2 or 3), including those that have no induction machines at the bus.
	FLAG = 4	for all subsystem buses with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 5	for all subsystem buses, including those that have no induction machines at the bus.
	FLAG = 10	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction generator.
	FLAG = 11	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction generator, even if all induction generators at the bus are out-of-service.
	FLAG = 12	for all subsystem buses with at least one induction generator, even if all induction generators at the bus are out-of-service.

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	FLAG = 20	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction motor.
	FLAG = 21	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction motor, even if all induction motors at the bus are out-of-service.
	FLAG = 22	for all subsystem buses with at least one induction motor, even if all induction motors at the bus are out-of-service.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or induction machine quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'TYPE'	Bus type code
	'AREA'	Bus area number
	'ZONE'	Bus zone number
	'OWNER'	Bus owner number
	'DUMMY'	Returns 1 if the bus is a dummy bus of a multisection line, or 0 if it is not
	'STATUS'	Inferred induction machine status. Returns 1 if the bus type code is 1, 2 or 3, and at least one in-service induction machine is present or 0 otherwise
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of BUSES returned by the aIndMacBusCount API routine when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of IARRAY that are set by this API routine. BUSES is identical to the value of BUSES returned by the aIndMacBusCount API routine when it is called with the same values specified for SID and FLAG (output).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of IARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.26.5. AINDMACBUSREAL

Use this API routine to return an array of real values for subsystem buses.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, rarray = aindmacbusreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_BUS_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,BUSES,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem buses to include (input; 1 by default).	
	FLAG = 1	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction machine.
	FLAG = 2	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 3	for all in-service subsystem buses (type code 1, 2 or 3), including those that have no induction machines at the bus.
	FLAG = 4	for all subsystem buses with at least one induction machine, even if all induction machines at the bus are out-of-service.
	FLAG = 5	for all subsystem buses, including those that have no induction machines at the bus.
	FLAG = 10	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction generator.
	FLAG = 11	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction generator, even if all induction generators at the bus are out-of-service.
	FLAG = 12	for all subsystem buses with at least one induction generator, even if all induction generators at the bus are out-of-service.

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	FLAG = 20	for all in-service subsystem buses (type code 1, 2 or 3) with at least one in-service induction motor.
	FLAG = 21	for all in-service subsystem buses (type code 1, 2 or 3) with at least one induction motor, even if all induction motors at the bus are out-of-service.
	FLAG = 22	for all subsystem buses with at least one induction motor, even if all induction motors at the bus are out-of-service.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or induction machine quantities desired (input; no default allowed).	
	'BASE'	Bus base voltage, in kV
	'PU'	Actual bus voltage magnitude, in pu
	'KV'	Actual bus voltage magnitude, in kV (in pu if base voltage is 0.0)
	'ANGLE'	Bus voltage phase angle, in radians
	'ANGLED'	Bus voltage phase angle, in degrees
	'TOTAL'	Flow from the bus to the in-service induction machines at the bus, in MVA
	'MISMATCH'	Bus mismatch (0.0 if bus type code > 3), in MVA
	'O_TOTAL'	Flow from the bus to the in-service induction machines at the bus (units determined by power output options)
	'O_MISMATCH'	Bus mismatch (0.0 if bus type code > 3) (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of BUSES returned by the aIndMacBusCount API routine when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer BUSES	number of elements in each column of RARRAY that are set by this API routine. BUSES is identical to the value of BUSES returned by the aIndMacBusCount API routine when it is called with the same values specified for SID and FLAG (output).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value

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IERR = 5	DIM, and hence the size of RARRAY, is not large enough
IERR = 6	Invalid STRING value

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8.26.6. AINDMACBUSTYPES

Use this API routine to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the induction machine bus data family (aIndMacBusInt, aIndMacBusReal, aIndMacBusCplx and aIndMacBusChar).

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, types = aindmacbustypes(string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_BUS_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the induction machine bus data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the induction machine bus data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.27. Induction Machine Data

The induction machine data family of subsystem data retrieval API routines returns arrays of data for induction machines in a specified subsystem. One of the input items to each API routine allows the user to elect either to retrieve data for all subsystem induction machines, or to restrict the array entries to data for a subset of induction machines according to their bus and/or induction machine configuration and statuses.

Data for subsystem induction machines is returned from all API routines in the induction machine data family in the same order: ascending bus number order or alphabetic extended bus name order, according to the bus output option setting in effect at the time the API routine is referenced; multiple induction machines at the same bus are in ascending induction machine identifier order.

8.27.1. AINDMACCHAR

Use this API routine to return an array of character values for subsystem induction machines.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, carray = aindmacchar(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_CHAR(SID,FLAG,NSTR,STRING,DIM,CARRAY,MACHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem induction machines to include (input; 1 by default).	
	FLAG = 1	for all in-service induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 2	for all induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 3	for all in-service induction machines at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 4	for all induction machines at all subsystem buses.
	FLAG = 11	for all in-service induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 12	all induction generators at in-service subsystem buses (type code 1, 2 or 3).

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	FLAG = 13	for all in-service induction generators at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 14	for all induction generators at all subsystem buses.
	FLAG = 21	for all in-service induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 22	all induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 23	for all in-service induction motors at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 24	for all induction motors at all subsystem buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or induction machine quantities desired (input; no default allowed).	
	'ID'	Induction machine identifier (2 characters).
	'NAME'	Bus name (12 characters).
	'EXNAME'	Extended bus name (18 characters).
Integer DIM	number of elements in each column of the array CARRAY. DIM must be as least as large as the value of MACHS returned by the aIndMacCount API routine when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Character CARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer MACHS	number of elements in each column of CARRAY that are set by this API routine. MACHS is identical to the value of MACHS returned by the aIndMacCount API routine when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of CARRAY, is not large enough
	IERR = 6	Invalid STRING value

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8.27.2. AINDMACCOUNT

Use this API routine to obtain the number of array entries required to accommodate the data to be returned by the remaining members of the induction machine data family.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, machs = aindmaccount(sid, flag)
```

Fortran command syntax:

```
CALL GET_IND_MAC_COUNT(SID, FLAG, MACHS, IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem induction machines to include (input; 1 by default).	
	FLAG = 1	for all in-service induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 2	for all induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 3	for all in-service induction machines at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 4	for all induction machines at all subsystem buses.
	FLAG = 11	for all in-service induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 12	all induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 13	for all in-service induction generators at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 14	for all induction generators at all subsystem buses.
	FLAG = 21	for all in-service induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 22	all induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 23	for all in-service induction motors at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).

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	FLAG = 24	for all induction motors at all subsystem buses.
Integer MACHS	number of induction machines in the subsystem indicated by SID that meet the editing criteria indicated by FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty; MACHS returned as 0
	IERR = 2	Invalid SID value; MACHS returned as 0
	IERR = 3	Invalid FLAG value; MACHS returned as 0

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8.27.3. AINDMACCPLX

Use this API routine to return an array of complex values for subsystem induction machines.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, xarray = aindmaccplx(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_COMPLEX(SID,FLAG,NSTR,STRING,DIM,XARRAY,MACHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem induction machines to include (input; 1 by default).	
	FLAG = 1	for all in-service induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 2	for all induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 3	for all in-service induction machines at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 4	for all induction machines at all subsystem buses.
	FLAG = 11	for all in-service induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 12	all induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 13	for all in-service induction generators at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 14	for all induction generators at all subsystem buses.
	FLAG = 21	for all in-service induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 22	all induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 23	for all in-service induction motors at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).

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	FLAG = 24	for all induction motors at all subsystem buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or induction machine quantities desired (input; no default allowed).	
	'ZA'	(RA + j XA)
	'Z1'	(R1 + j X1)
	'Z2'	(R2 + j X2)
	'ZZERO'	Machine zero sequence fault analysis impedance
	'ZGROUND'	Machine zero sequence grounding impedance
	'PQ'	Complex power flow from the bus to the machine (in MW and Mvar)
	'O_PQ'	Complex power flow from the bus to the machine (units determined by power output options)
Integer DIM	number of elements in each column of the array XARRAY. DIM must be as least as large as the value of MACHS returned by the aIndMacCount API routine when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Complex XARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer MACHS	number of elements in each column of XARRAY that are set by this API routine. MACHS is identical to the value of MACHS returned by the aIndMacCount API routine when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of XARRAY, is not large enough
	IERR = 6	Invalid STRING value
	IERR = 7	Sequence data not in case (when STRING = 'ZZERO' or 'ZGROUND')

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8.27.4. AINDMACINT

Use this API routine to return an array of integer values for subsystem induction machines.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, iarray = aindmacint(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_INTEGER(SID,FLAG,NSTR,STRING,DIM,IARRAY,MACHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem induction machines to include (input; 1 by default).	
	FLAG = 1	for all in-service induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 2	for all induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 3	for all in-service induction machines at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 4	for all induction machines at all subsystem buses.
	FLAG = 11	for all in-service induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 12	all induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 13	for all in-service induction generators at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 14	for all induction generators at all subsystem buses.
	FLAG = 21	for all in-service induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 22	all induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 23	for all in-service induction motors at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).

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	FLAG = 24	for all induction motors at all subsystem buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or induction machine quantities desired (input; no default allowed).	
	'NUMBER'	Bus number
	'AREA'	Machine area assignment
	'ZONE'	Machine zone assignment
	'OWNER'	Machine owner assignment
	'STATUS'	Machine status
	'STANDARDCODE'	Machine standard code (1 for NEMA, 2 for IEC)
	'DESIGNCODE'	Machine design code (0 through 5)
	'PSETCODE'	Power setpoint code (1 for mechanical, 2 for electrical)
	'BASECODE'	Machine base power code (1 for mechanical, 2 for electrical)
	'TRQVARCODE'	Mechanical torque variation code (1 for simple power law, 2 for WECC model)
	'CZG'	Grounding impedance data input/outout (I/O) code (1 for per unit, 2 for ohms)
	'CGR'	Grounding code (0 for an ungrounded machine, 1 for a grounded machine)
Integer DIM	number of elements in each column of the array IARRAY. DIM must be as least as large as the value of MACHS returned by the aIndMacCount API routine when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Integer IARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer MACHS	number of elements in each column of IARRAY that are set by this API routine. MACHS is identical to the value of MACHS returned by the aIndMacCount API routine when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of IARRAY, is not large enough
	IERR = 6	Invalid STRING value
	IERR = 7	Sequence data not in case (when STRING = 'CZG' or 'CGR')

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8.27.5. AINDMACREAL

Use this API routine to return an array of real values for subsystem induction machines.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, rarray = aindmacreal(sid, flag, string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_REAL(SID,FLAG,NSTR,STRING,DIM,RARRAY,MACHS,IERR)
```

Where:

Integer SID	Defines the bus subsystem to be used (input; -1 by default).	
	SID < 0	to instruct the API routine to assume a subsystem containing all buses in the working case.
	SID >= 0	valid bus subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined.
Integer FLAG	a flag indicating which subsystem induction machines to include (input; 1 by default).	
	FLAG = 1	for all in-service induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 2	for all induction machines at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 3	for all in-service induction machines at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 4	for all induction machines at all subsystem buses.
	FLAG = 11	for all in-service induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 12	all induction generators at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 13	for all in-service induction generators at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).
	FLAG = 14	for all induction generators at all subsystem buses.
	FLAG = 21	for all in-service induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 22	all induction motors at in-service subsystem buses (type code 1, 2 or 3).
	FLAG = 23	for all in-service induction motors at subsystem buses, even if the bus to which it is connected is out-of-service (type code 4).

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	FLAG = 24	for all induction motors at all subsystem buses.
Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR of the following strings indicating the bus and/or induction machine quantities desired (input; no default allowed).	
	'MBASE'	Machine base power
	'RATEKV'	Machine rated voltage
	'PSETPOINT'	Power setpoint (positive for a motor)
	'RA'	Armature resistance
	'XA'	Armature leakage reactance
	'XM'	Magnetizing reactance
	'R1'	First cage resistance
	'X1'	First cage reactance
	'R2'	Second cage resistance
	'X2'	Second cage reactance
	'X3'	Third reactance
	'E1'	First saturation voltage
	'SE1'	SE(E1)
	'E2'	Second saturation voltage
	'SE2'	SE(E2)
	'IA1'	Starting stator current for applying stator leakage reactance saturation
	'IA2'	Stator current at which the stator leakage reactance is fully saturated
	'XAMULT'	Xa multiplier at full leakage reactance saturation
	'TRQA'	Load torque variation constant A for WECC model
	'TRQB'	Load torque variation constant B for WECC model
	'TRQD'	Load torque variation constant D for both models
	'TRQE'	Load torque variation constant E for WECC model
	'SLIP'	Slip
	'H'	Machine inertia
	'ILKDOVERIRAT'	Ratio of locked rotor current to rated current
	'ROVERX'	Ratio of Rpos to Xpos
	'RZERO'	Machine zero sequence fault analysis resistance
	'XZERO'	Machine zero sequence fault analysis reactance
	'RGROUND'	Machine zero sequence grounding resistance
	'XGROUND'	Machine zero sequence grounding reactance

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	'P'	Active power flow from the bus to the machine, in MW
	'Q'	Reactive power flow from the bus to the machine, in Mvar
	'MVA'	MVA flow from the bus to the machine, in MVA
	'O_P'	Active power flow from the bus to the machine (units determined by power output options)
	'O_Q'	Reactive power flow from the bus to the machine (units determined by power output options)
	'O_MVA'	MVA flow from the bus to the machine (units determined by power output options)
Integer DIM	number of elements in each column of the array RARRAY. DIM must be as least as large as the value of MACHS returned by the aIndMacCount API routine when it is called with the same values specified for SID and FLAG (input; no default allowed).	
Real RARRAY (NSTR*DIM)	Is an array of at least NSTR columns, each of which contains DIM elements (output).	
Integer MACHS	number of elements in each column of RARRAY that are set by this API routine. MACHS is identical to the value of MACHS returned by the aIndMacCount API routine when it is called with the same values specified for SID and FLAG (output).	
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = 1	Working case is empty
	IERR = 2	Invalid SID value
	IERR = 3	Invalid FLAG value
	IERR = 4	Invalid NSTR value
	IERR = 5	DIM, and hence the size of RARRAY, is not large enough
	IERR = 6	Invalid STRING value
	IERR = 7	Sequence data not in case (when STRING = 'ILKDOVERIRAT', 'ROVERX', 'RZERO', 'XZERO', 'XZERO', 'RGROUND' or 'XGROUND')

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8.27.6. AINDMACTYPES

Use this API routine to return an array of character values indicating the data types corresponding to a set of specified STRING values that are valid input values to any of the data retrieval routines of the induction machine data family (aIndMacInt, aIndMacReal, aIndMacCplx and aIndMacChar).

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, types = aindmactypes(string)
```

Fortran command syntax:

```
CALL GET_IND_MAC_TYPES(NSTR,STRING,TYPES,IERR)
```

Where:

Integer NSTR	number of elements in STRING (1 < NSTR < 50) (input; no default allowed).	
Character STRING (NSTR)	NSTR strings that are valid STRING values for any of the data retrieval routines of the induction machine data family (input; no default allowed).	
Character*1 TYPES (NSTR)	each of which contains one of the following characters indicating the data type of the data associated with the corresponding element of STRING (output).	
	'I'	Integer.
	'R'	Real.
	'X'	Complex.
	'C'	Character.
	'?'	Invalid STRING value.
Integer IERR	error code (output).	
	IERR = 0	No error
	IERR = n	Position of the first element in STRING that contains a value that is not a valid STRING value for any of the data retrieval routines in the induction machine data family (1 < n < NSTR)
	IERR = 99	Invalid NSTR value

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8.28. Substation Data

The substation data family of subsystem data retrieval APIs returns arrays of data for substations in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for substations with a least one subsystem bus, or to restrict the array entries to data for substations with all of their buses in the subsystem.

Data for subsystem substations is returned from all APIs in the substation data family in ascending substation number order.

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8.29. Substation Node Data

The substation node data family of subsystem data retrieval APIs returns arrays of data for substation nodes in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for substations with a least one subsystem bus, or to restrict the array entries to data for substations with all of their buses in the subsystem.

Data for subsystem substation nodes is returned from all APIs in the substation node data family in ascending substation number order, and in ascending node number order for each substation.

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8.30. Substation Switching Device Data

The substation switching device data family of subsystem data retrieval APIs returns arrays of data for substation switching devices in a specified subsystem. One of the input items to each API allows the user to select either to retrieve data for substations with a least one subsystem bus, or to restrict the array entries to data for substations with all of their buses in the subsystem.

Data for subsystem substation switching devices is returned from all APIs in the substation switching device data family in ascending substation number order. For the nodes connected by each switching device, the node with the smaller node number is returned as the "from" node, and the node with the larger node number is returned as the "to" node. Station switching devices for a substation are returned in ascending "from" node number order, and in ascending "to" node number order for each "from" node.

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8.31. Substation Terminal Data

The substation terminal data family of subsystem data retrieval APIs returns arrays of data for substation terminals in a specified subsystem. One of the input items to each API allows the user to elect either to retrieve data for substations with a least one subsystem bus, or to restrict the array entries to data for substations with all of their buses in the subsystem.

Data for subsystem substation terminals is returned from all APIs in the substation terminal data family in ascending substation number order, and in ascending bus number order for each substation.

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8.32. Zero Sequence Mutuals Data

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Chapter 9
Graphical Analysis Output

This chapter defines API routines that interact with displays and files associated with the graphical analysis capabilities of the PSS®E GUI. They will elicit no action if a GUI is not present when invoked.

These routines are not recorded when recording is enabled.

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9.1. POLY_PRINT

Use this API to print the results of activity POLY from a *.pol results file.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_POLY_PRINT  NPLOT  PLOTARRAY(1)..PLOTARRAY(NPLOT)  BOUNDVAL  PRTPARAMS
  RESFILE
```

Python command syntax:

```
poly_print(nplot, plotarray, boundval, prtparams, resfile)
```

Fortran command syntax:

```
CALL POLY_PRINT(NPLOT, PLOTARRAY, BOUNDVAL, PRTPARAMS, RESFILE)
```

Where:

Integer NPLOT	number of plots (input).
Integer PLOTARRAY (NPLOT)	an array of the numbers of plots, selected for printing. The base case is marked by 0, then there are all the contingencies, beginning from 1, and the last number means all the cases together. The array consists of the set of these numbers. The size of the array is equal to NPLOT (input).
Real BOUNDVAL	boundary value for the plot area. It is equal to maximum opposing system delta interchange used for calculations (input).
Integer PRTPARAMS	print (or not) the set of parameters used for calculations (input). PRTPARAMS = 0 no printing of parameters. PRTPARAMS = 1 the set of parameters will be printed.
Character*260 RESFILE	name of the existing POLY results file (*.pol) file (input).

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9.2. PV_PRINT

Use this API to print the results of PV analysis from the *.pv results file

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_PV_PRINT  RESFILE  RESTYPE  NCONTING  CONTINGARRAY(1)..
 CONTINGARRAY(NCONTING)  NOUTPUTS  OUTPUTSARRAY(1)..
 OUTPUTSARRAY(NOUTPUTS)  INTERFACE  ENFORCESCALE  USELINESTYLES
 NUMBERLINES  PRTPARAMS
```

Python command syntax:

```
pv_print(resfile, restype, nconting, contingarray, noutputs, outputsarray,
 interface, enforcescale, uselinestyles, numberlines, prtparams)
```

Fortran command syntax:

```
CALL PV_PRINT(RESFILE, RESTYPE, NCONTING, CONTINGARRAY, NOUTPUTS,
 OUTPUTSARRAY, INTERFACE, ENFORCESCALE, USELINESTYLES, NUMBERLINES,
 PRTPARAMS)
```

Where:

Character*260 RESFILE	name of the existing *.pv file (input).
Integer RESTYPE	type of the results (input). RESTYPE = 0 bus voltage RESTYPE = 1 generator MW RESTYPE = 2 branch MVA flow RESTYPE = 3 interface MW flow RESTYPE = 4 bus voltage against interface flow
Integer NCONTING	number of specified contingencies (input).
Integer CONTINGARRAY (NCONTING)	an array of the numbers of specified contingencies. The base case is marked by 0. The size of the array is equal to NCONTING (input).
Integer NOUTPUTS	number of specified outputs (input).
Integer OUTPUTSARRAY (NOUTPUTS)	an array of the numbers of specified outputs. The count starts from 0. The size of the array is equal to NOUTPUTS (input).
Integer INTERFACE	number of specified interface. Valid only in case of bus voltage against interface flow (RESTYPE = 4) (input).
Integer ENFORCESCALE	allows to print the results with the different scales (input). ENFORCESCALE = 0 the scales will be not changed according with the results; this allows to print the results from different files in the same scales. ENFORCESCALE = 1 the maximum and minimum values for the used scales will be taken from the file of results, and the printing will be done according with these values.

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Integer USELINESTYLES	use (or not) different line styles for plotting (input).	
	USELINESTYLES = 0	all the lines will be solid
	USELINESTYLES = 1	five different (standard) styles of lines will be used: Solid, Dash-Dash, Dot-Dot, Dash-Dot, Dash-Dot-Dot.
Integer NUMBERLINES	put the numbers (or not) on the graphs (input).	
	NUMBERLINES = 0	no numbering of the graphs.
	NUMBERLINES = 1	each graph will have its number for better recognition
Integer PRTPARAMS	print (or not) the set of parameters used for calculations (input).	
	PRTPARAMS = 0	no printing of parameters
	PRTPARAMS = 1	the set of parameters will be printed

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9.3. QV_PRINT

Use this API to print the results of QV analysis from a *.qv resultsfile.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_QV_PRINT  RESFILE  NCONTING  CONTINGARRAY(1).. CONTINGARRAY(NCONTING)
 ENFORCESCALE  USELINESTYLES  NUMBERLINES  PRTPARAMS
```

Python command syntax:

```
qv_print(resfile, nconting, contingarray, enforcescale, uselinestyles,
 numberlines, prtparams)
```

Fortran command syntax:

```
CALL QV_PRINT(RESFILE, NCONTING, CONTINGARRAY, ENFORCESCALE,
 USELINESTYLES, NUMBERLINES, PRTPARAMS)
```

Where:

Character*260 RESFILE	name of the existing *.qv file (input).	
Integer NCONTING	number of specified contingencies (input).	
Integer CONTINGARRAY (NCONTING)	array of the numbers of specified contingencies. The base case is marked by 0. The size of the array is equal to NCONTING (input).	
Integer ENFORCESCALE	allows to print the results with the different scales (input).	
	ENFORCESCALE = 0	the scales will be not changed according with the results; this allows to print the results from different files in the same scales
	ENFORCESCALE = 1	the maximum and minimum values for the used scales will be taken from the file of results, and the printing will be done according with these values
Integer USELINESTYLES	use (or not) different line styles for plotting (input).	
	USELINESTYLES = 0	all the lines will be solid
	USELINESTYLES = 1	five different (standard) styles of lines will be used: Solid, Dash-Dash, Dot-Dot, Dash-Dot, Dash-Dot-Dot
Integer NUMBERLINES	put the numbers (or not) on the graphs (input).	
	NUMBERLINES = 0	no numbering of the graphs
	NUMBERLINES = 1	each graph will have its number for better recognition
Integer PRTPARAMS	print (or not) the set of parameters used for calculations (input).	
	PRTPARAMS = 0	no printing of parameters
	PRTPARAMS = 1	the set of parameters will be printed

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9.4. SETFULLVIEWGRAPHAREA

Use this API to specify the parameters of the scales used for plotting the results of PV and QV analysis.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETFULLVIEWGRAPHAREA AREA VAXISSHOW VAXISSTYLE VAXISCLR(1)..
 VAXISCLR(3) VGRIDSHOW VGRIDSTYLE VGRIDCLR(1).. VGRIDCLR(3)
 HAXISSHOW HAXISSTYLE HAXISCLR(1).. HAXISCLR(3) HGRIDSHOW HGRIDSTYLE
  HGRIDCLR(1).. HGRIDCLR(3) BORDERSHOW BORDERSTYLE BORDERCLR(1)..
 BORDERCLR(3) SETAREACLR AREACLR(1).. AREACLR(3)
```

Python command syntax:

```
setfullviewgrapharea(area, vaxisshow, vaxisstyle, vaxisclr, vgridshow,
 vgridstyle, vgridclr, haxisshow, haxisstyle, haxisclr, hgridshow,
 hgridstyle, hgridclr, bordershow, borderstyle, borderclr, setareaclr,
 areaclr)
```

Fortran command syntax:

```
CALL SETFULLVIEWGRAPHAREA(AREA, VAXISSHOW, VAXISSTYLE, VAXISCLR,
 VGRIDSHOW, VGRIDSTYLE, VGRIDCLR, HAXISSHOW, HAXISSTYLE, HAXISCLR,
 HGRIDSHOW, HGRIDSTYLE, HGRIDCLR, BORDERSHOW, BORDERSTYLE, BORDERCLR,
 SETAREACLR, AREACLR)
```

Where:

Integer AREA	number of the plotting area (input).	
	AREA = 0	QV analysis, area for Q(V) plot
	AREA = 1	QV analysis, area for mismatch plot
	AREA = 2	PV analysis, area for bus voltage
	AREA = 3	PV analysis, area for generator MW
	AREA = 4	PV analysis, area for branch MVA flow
	AREA = 5	PV analysis, area for interface MW flow
	AREA = 6	PV analysis, area for bus voltage against interface flow.
Integer VAXISSHOW	show (or not) the vertical axis (input).	
Integer VAXISSTYLE	style of the vertical axis. Valid only if VAXISSHOW=1 (input).	
	VAXISSTYLE = 0	solid line
	VAXISSTYLE = 1	dash - dash - dash line
	VAXISSTYLE = 2	dot - dot - dot line
	VAXISSTYLE = 3	dash - dot line
	VAXISSTYLE = 4	dash - dot - dot line
Integer VAXISCLR (3)	array of 3 elements specifying color of the vertical axis. Valid only if VAXISSHOW=1(input).	
	VAXISCLR(1)	Red component; the value between 0 and 255

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	VAXISCLR(2)	Green component; the value between 0 and 255
	VAXISCLR(3)	Blue component; the value between 0 and 255
Integer VGRIDSHOW	show (or not) the vertical grid (input).	
Integer VGRIDSTYLE	style of the vertical grid. Valid only if VGRIDSHOW =1 (input).	
	VGRIDSTYLE = 0	solid line
	VGRIDSTYLE = 1	dash - dash - dash line
	VGRIDSTYLE = 2	dot - dot - dot line
	VGRIDSTYLE = 3	dash - dot line
	VGRIDSTYLE = 4	dash - dot - dot line
Integer VGRIDCLR (3)	array of 3 elements specifying color of the vertical grid. Valid only if VGRIDSHOW =1(input).	
	VGRIDCLR(1)	Red component; the value between 0 and 255
	VGRIDCLR(2)	Green component; the value between 0 and 255
	VGRIDCLR(3)	Blue component; the value between 0 and 255
Integer HAXISSHOW	show (or not) the horizontal axis (input).	
Integer HAXISSTYLE	style of the horizontal axis. Valid only if HAXISSHOW=1 (input).	
	HAXISSTYLE = 0	solid line
	HAXISSTYLE = 1	dash - dash - dash line
	HAXISSTYLE = 2	dot - dot - dot line
	HAXISSTYLE = 3	dash - dot line
	HAXISSTYLE = 4	dash - dot - dot line
Integer HAXISCLR (3)	array of 3 elements specifying color of the horizontal axis. Valid only if HAXISSHOW=1(input).	
	HAXISCLR(1)	Red component; the value between 0 and 255
	HAXISCLR(2)	Green component; the value between 0 and 255
	HAXISCLR(3)	Blue component; the value between 0 and 255
Integer HGRIDSHOW	show (or not) the horizontal grid (input).	
Integer HGRIDSTYLE	style of the horizontal grid. Valid only if HGRIDSHOW =1 (input).	
	HGRIDSTYLE = 0	solid line
	HGRIDSTYLE = 1	dash - dash - dash line
	HGRIDSTYLE = 2	dot - dot - dot line
	HGRIDSTYLE = 3	dash - dot line
	HGRIDSTYLE = 4	dash - dot - dot line
Integer HGRIDCLR (3)	array of 3 elements specifying color of the horizontal grid. Valid only if HGRIDSHOW =1(input).	

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	HGRIDCLR(1)	Red component; the value between 0 and 255
	HGRIDCLR(2)	Green component; the value between 0 and 255
	HGRIDCLR(3)	Blue component; the value between 0 and 255
Integer BORDERSHOW	show (or not) the border around the plot (input).	
Integer BORDERSTYLE	style of the border. Valid only if BORDERSHOW =1 (input).	
	BORDERSTYLE = 0	solid line
	BORDERSTYLE = 1	dash - dash - dash line
	BORDERSTYLE = 2	dot - dot - dot line
	BORDERSTYLE = 3	dash - dot line
	BORDERSTYLE = 4	dash - dot - dot line
Integer BORDERCLR (3)	array of 3 elements specifying color of the border. Valid only if BORDERSHOW =1(input).	
	BORDERCLR(1)	Red component; the value between 0 and 255
	BORDERCLR(2)	Green component; the value between 0 and 255
	BORDERCLR(3)	Blue component; the value between 0 and 255
Integer SETAREACLR	set (or not) the back color of the plotting area (input).	
Integer AREACLR (3)	array of 3 elements specifying color of the plotting area. Valid only if SETAREACLR =1(input).	
	AREACLR(1)	Red component; the value between 0 and 255
	AREACLR(2)	Green component; the value between 0 and 255
	AREACLR(3)	Blue component; the value between 0 and 255

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9.5. SETFULLVIEWSCALE

Use this API to specify the parameters of the scales used for plotting the results of PV and QV analysis.

This API routine was first introduced in release 30.0.0.

Batch command syntax:

```
BAT_SETFULLVIEWSCALE SCALE LTVAL RBVAL ADJUST(1) ADJUST(2) AXISPLACE
 AXISVALUE GRIDCALCTYPE GRIDSTEP GRIDMINLINES SHOWNUM (1).. SHOWNUM(3)
 LTPOS INPOS RBPOS LTPLACES INPLACES RBPLACES
```

Python command syntax:

```
setfullviewscale(scale, ltval, rbval, adjust, axisplace, axisvalue,
 gridcalctype, gridstep, gridminlines, shownum, ltpos, inpos, rbpos,
 ltplaces, inplaces, rbplaces)
```

Fortran command syntax:

```
CALL SETFULLVIEWSCALE(SCALE, LTVAL, RBVAL, ADJUST, AXISPLACE, AXISVALUE,
 GRIDCALCTYPE, GRIDSTEP, GRIDMINLINES, SHOWNUM, LTPOS, INPOS, RBPOS,
 LTPLACES, INPLACES, RBPLACES)
```

Where:

Integer SCALE	number of the scale (input).	
	SCALE = 0	QV analysis, horizontal scale for Q(V) plot
	SCALE = 1	QV analysis, vertical scale for Q(V) plot
	SCALE = 2	QV analysis, vertical scale for mismatch plot
	SCALE = 3	PV analysis, horizontal scale for incremental transfer
	SCALE = 4	PV analysis, horizontal scale for interface MW flow (bus voltage against interface flow)
	SCALE = 5	PV analysis, vertical scale for bus voltage
	SCALE = 6	PV analysis, vertical scale for generator MW
	SCALE = 7	PV analysis, vertical scale for branch MVA flow
	SCALE = 8	PV analysis, vertical scale for interface MW flow
	SCALE = 9	PV analysis, vertical scale for bus voltage (bus voltage against interface flow)
Real LTVAL	left value (for horizontal scale) or top value (for vertical scale) (input).	
Real RBVAL	right value (for horizontal scale) or bottom value (for vertical scale) (input).	
Integer ADJUST (2)	array of 2 elements specifying (input).	
	ADJUST(1)	to adjust (or not) the left (top) value of the scale

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	ADJUST(2)	to adjust (or not) the right (bottom) value of the scale
Integer AXISPLACE	place of the axis (input).	
	AXISPLACE = 0	on the left (top) border
	AXISPLACE = 1	the axis place is set in AXISVALUE
	AXISPLACE = 2	on the right (bottom) border
	AXISPLACE = 3	undefined, the exact value will be calculated
Real AXISVALUE	place of the axis. Valid only in case AXISPLACE = 1 (input).	
Integer GRIDCALCTYPE	way the grid is calculated (input).	
	GRIDCALCTYPE = 0	grid step is set in GRIDSTEP
	GRIDCALCTYPE = 1	grid is defined by minimum grid lines set in GRIDMINLINES
Real GRIDSTEP	tep of the grid. Valid only in case GRIDCALCTYPE = 0 (input).	
Integer GRIDMINLINES	Minimum number of the grid lines. Valid only in case GRIDCALCTYPE = 1 (input).	
Integer SHOWNUM (3)	array of 3 elements specifying (input).	
	SHOWNUM(1)	to show (or not) the left (top) value of the scale
	SHOWNUM(2)	to show (or not) all the intermediate values for the grid
	SHOWNUM(3)	to show (or not) the right (bottom) value of the scale
Integer LTPOS	place of the left (top) value relative to the left (top) border (input).	
	LTPOS = 0	to the left of the border (above the border)
	LTPOS = 1	centered to the border
	LTPOS = 2	to the right of the border (below the border)
Integer INPOS	place of the intermediate values relative to the intermediate grid lines (input).	
	INPOS = 0	to the left of the line (above the line)
	INPOS = 1	centered to the line
	INPOS = 2	to the right of the line (below the line)
Integer RBPOS	place of the right (bottom) value relative to the right (bottom) border (input).	
	RBPOS = 0	to the left of the border (above the border)
	RBPOS = 1	centered to the border
	RBPOS = 2	to the right of the border (below the border)
Integer LTPLACES	number of digits to the right of the decimal point for the left (top) value (input).	
Integer INPLACES	number of digits to the right of the decimal point for intermediate values (input).	
Integer RBPLACES	number of digits to the right of the decimal point for the right (bottom) value (input).	

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Chapter 10
Scenarios

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10.1. CLOSESCENARIOFILE

This API routine is obsolete. It has been replaced by the API routine SCENARIOCLOSE .

Use this API to close the currently open Scenario. If a file in the active group has was specified to "Load Last", this file will be opened prior to the closing of the Scenario.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CLOSESCENARIOFILE
```

Python command syntax:

```
closescenariofile()
```

Fortran command syntax:

```
CALL CLOSESCENARIOFILE()
```

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10.2. NEWSCENARIOFILE

This API routine is obsolete. It has been replaced by the API routine SCENARIONEW .

Use this API to create a new Scenario. At least one group must be added to the Scenario through SCENARIOGROUPADD before files may be added to it. If adding files through file tracking is also desired, a group must become the active group which can be done by calling SCENARIOGROUPOPEN.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_NEWSCENARIOFILE  SFILE
```

Python command syntax:

```
newscenariofile(sfile)
```

Fortran command syntax:

```
CALL NEWSCENARIOFILE(SFILE)
```

Where:

Character*260 SFILE	full file path of the new scenario file (input; blank by default. The name of the Scenario will be determined by the file name portion of the full file path. When the default is used no name is given to the Scenario and a full file name will be required upon saving. The root path of the Scenario will be determined by the path portion of the full file path. When the default is used the root path will default to the current working directory).

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10.3. OPENSCENARIOFILE

This API routine is obsolete. It has been replaced by the API routine SCENARIOOPEN .

Use this API to open a previously created Scenario.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_OPENSCENARIOFILE  SFILE
```

Python command syntax:

```
openscenariofile(sfile)
```

Fortran command syntax:

```
CALL OPENSCENARIOFILE(SFILE)
```

Where:

Character*260 SFILE	full file path of the scenario file to open (input; no default allowed).

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10.4. SAVESCENARIOFILE

This API routine is obsolete. It has been replaced by the API routine SCENARIOSAVE .

Use this API to save the currently open Scenario.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SAVESCENARIOFILE  SFILE
```

Python command syntax:

```
savescenariofile(sfile)
```

Fortran command syntax:

```
CALL SAVESCENARIOFILE (SFILE)
```

Where:

Character*260 SFILE	full file path to save the scenario file to (input; unchanged by default. When the default is used the Scenario will be saved to the last file-name given in a SCENARIONEW, SCENARIOOPEN, or SCENARIOSAVE operation, if a call to SCENARIONEW with default arguments was the previous call, the Scenario will not be saved due to no previous file-name being available).

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10.5. SCENARIOCLOSE

Use this API to close the currently open Scenario. If a file in the active group has was specified to "Load Last", this file will be opened prior to the closing of the Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOCLOSE
```

Python command syntax:

```
ierr = scenarioclose()
```

Fortran command syntax:

```
CALL SCENARIOCLOSE(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory

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10.6. SCENARIOCOPYFILE

Use this API to copy a file from one group to another group.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOCOPYFILE  SRCGROUP SFILE DSTGROUP
```

Python command syntax:

```
ierr = scenariocopyfile(srcgroup, sfile, dstgroup)
```

Fortran command syntax:

```
CALL SCENARIOCOPYFILE(SRCGROUP, SFILE, DSTGROUP, IERR)
```

Where:

Character*72 SRCGROUP	name of the group containing the file to be copied (input; no default allowed).	
Character*260 SFILE	name of the file to be copied (input; no default allowed).	
Character*72 DSTGROUP	name of the group the file will be copied to (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	invalid file name
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.7. SCENARIOFILEADD

Use this API to add a file to the specified group.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOFILEADD  SGROUP SFILE STYPE LOAD SARGS
```

Python command syntax:

```
ierr = scenariofileadd(sgroup, sfile, stype, load, sargs)
```

Fortran command syntax:

```
CALL SCENARIOFILEADD(SGROUP, SFILE, STYPE, LOAD, SARGS, IERR)
```

Where:

Character*72 SGROUP	name of the group to add a file to (input; no default allowed).	
Character*260 SFILE	name of the file to add (input; no default allowed).	
Character*8 STYPE	file type identifier for the specified file (input; blank by default).	
Integer LOAD	file's position in the load order of its group. A load order of 0 specifies that the file is not loaded (input; 0 by default).	
Character*260 SARGS	argument string to be used for the file when automatically opening the file (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	invalid file name
	IERR = 3	specified group not found

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10.8. SCENARIOFILECOUNT

Use this API to retrieve the number of files in a specified group in the Scenario.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, nfiles = scenariofilecount(sgroup)
```

Fortran command syntax:

```
CALL SCENARIOFILECOUNT(SGROUP, NFILES, IERR)
```

Where:

Character*72 SGROUP	group to retrieve the file count of input (input; ignored).	
Integer NFILES	number of the files in the specified group (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found

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10.9. SCENARIOFILEGETATTRS

Use this API to retrieve the attributes of the specified file.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, stype, load, sargs = scenariofilegetattrs(sgroup, sfile)
```

Fortran command syntax:

```
CALL SCENARIOFILEGETATTRS(SGROUP, SFILE, STYPE, LOAD, SARGS, IERR)
```

Where:

Character*72 SGROUP	group that contains the specified file (input; no default allowed).	
Character*260 SFILE	name of the file whose attributes are to be retrieved (input; no default allowed).	
Character*8 STYPE	file type identifier for the specified file (output).	
Integer LOAD	Denotes the file's position in the load order of its group. A load order of 0 specifies that the file is not loaded (output).	
Character*260 SARGS	argument string to be used for the file when automatically opening the file (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.10. SCENARIOFILEGETCOMMENTS

Use this API to retrieve the comments for the specified file.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, comments = scenariofilegetcomments(sgroup, sfile)
```

Fortran command syntax:

```
CALL SCENARIOFILEGETCOMMENTS(SGROUP, SFILE, COMMENTS, IERR)
```

Where:

Character*72 SGROUP	group that contains the specified file (input; no default allowed).	
Character*260 SFILE	name of the file whose attributes are to be retrieved (input; no default allowed).	
Character COMMENTS	the comment string for the specified file. Use SCENARIOFILELENGTHCOMMENTS to get required buffer size, if needed. (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.11. SCENARIOFILELENGTHCOMMENTS

Use this API to determine the buffer size needed to hold the comments for the specified file.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, length = scenariofilelengthcomments(sgroup, sfile)
```

Fortran command syntax:

```
CALL SCENARIOFILELENGTHCOMMENTS(SGROUP, SFILE, LENGTH, IERR)
```

Where:

Character*72 SGROUP	group that contains the specified file (input; no default allowed).	
Character*260 SFILE	name of the file whose comment length is to be returned (input; no default allowed).	
Integer LENGTH	length of the Scenario comment string (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.12. SCENARIOFILELIST

Use this API to retrieve the list of files in the specified group.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, sfiles = scenariofilelist(sgroup)
```

Fortran command syntax:

```
CALL SCENARIOFILELIST(SGROUP, NFILES, SFILES, IERR)
```

Where:

Character*72 SGROUP	group to retrieve the file list of (input; no default allowed).	
Integer NFILES	max number of files to list (input; no default allowed).	
Character*260 SFILES (NFILES)	list of files in the specified group (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found

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10.13. SCENARIOFILEREMOVE

Use this API to remove a file from the specified group.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOFILEREMOVE  SGROUP SFILE
```

Python command syntax:

```
ierr = scenariofileremove(sgroup, sfile)
```

Fortran command syntax:

```
CALL SCENARIOFILEREMOVE(SGROUP, SFILE, IERR)
```

Where:

Character*72 SGROUP	name of the group to remove a file from (input; no default allowed).	
Character*260 SFILE	name of the file to be removed (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	invalid file name
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.14. SCENARIOFILERENAME

Use this API to rename one of the files in the Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOFILERENAME  SGROUP SFILE SNEWNAME
```

Python command syntax:

```
ierr = scenariofilerename(sgroup, sfile, snewname)
```

Fortran command syntax:

```
CALL SCENARIOFILERENAME(SGROUP, SFILE, SNEWNAME, IERR)
```

Where:

Character*72 SGROUP	group that contains the specified file (input; no default allowed).	
Character*260 SFILE	name of the file whose name is to be changed (input; no default allowed).	
Character*260 SNEWNAME	new name to be given to the specified file (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	invalid file name
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.15. SCENARIOFILESETATTRS

Use this API to modify the attributes of the specified file.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOFILESETATTRS  SGROUP SFILE STYPE LOAD SARGS
```

Python command syntax:

```
ierr = scenariofilesetattrs(sgroup, sfile, stype, load, sargs)
```

Fortran command syntax:

```
CALL SCENARIOFILESETATTRS(SGROUP, SFILE, STYPE, LOAD, SARGS, IERR)
```

Where:

Character*72 SGROUP	group that contains the specified file (input; no default allowed).	
Character*260 SFILE	name of the file whose attributes are to be modified (input; no default allowed).	
Character*8 STYPE	new file type identifier for the specified file (input; ignored).	
Integer LOAD	file's new position in the load order of its group. A load order of 0 specifies that the file is not loaded (input; ignored).	
Character*260 SARGS	argument string to be used for the file when automatically opening the file (input; ignored).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.16. SCENARIOFILESETCOMMENTS

Use this API to define or modify the comments for the specified file.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOFILESETCOMMENTS  SGROUP SFILE NCOMMENTBLOCKS COMMENTS APPEND
```

Python command syntax:

```
ierr = scenariofilesetcomments(sgroup, sfile, ncommentblocks, comments,
 append)
```

Fortran command syntax:

```
CALL SCENARIOFILESETCOMMENTS(SGROUP, SFILE, NCOMMENTBLOCKS, COMMENTS,
 APPEND, IERR)
```

Where:

Character*72 SGROUP	group that contains the specified file (input; no default allowed).	
Character*260 SFILE	name of the file whose attributes are to be modified (input; no default allowed).	
Integer NCOMMENTBLOCKS	number of 72 character strings that make up the comment. In python, this number can be set to 1 and the full comment string given as the first comment block (input; 0 by default).	
Character*72 COMMENTS (NCOMMENTBLOCKS)	comment string to be defined for or added to the comments for the specified file (input; unchanged).	
Logical APPEND	flag for appending comments. If TRUE, COMMENTS is appended to the current comments. If FALSE, the current comments are overwritten by COMMENTS (input; FALSE).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.17. SCENARIOGETATTRS

Use this API to retrieve the attributes of the Scenario.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, rootpath, startupgroup, creationdate, createdby, changedate,
 changedby, name, version, versionpsse = scenariogetattrs()
```

Fortran command syntax:

```
CALL SCENARIOGETATTRS(ROOTPATH, STARTUPGROUP, CREATIONDATE, CREATEDBY,
 CHANGEDATE, CHANGEDBY, NAME, VERSION, VERSIONPSSE, IERR)
```

Where:

Character*260 ROOTPATH	root path of the Scenario (output).	
Character*72 STARTUPGROUP	name of the startup group (output).	
Character*23 CREATIONDATE	date of creation (output).	
Character*256 CREATEDBY	username of the creator (output).	
Character*23 CHANGEDATE	last date of modification (output).	
Character*256 CHANGEDBY	username of the last modifier (output).	
Character*260 NAME	name of the Scenario (output).	
Character*14 VERSION	version of Scenario the file was saved under (output).	
Character*14 VERSIONPSSE	version of PSSE the file was saved under (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory

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10.18. SCENARIOGETCOMMENTS

Use this API to retrieve the Scenario comments.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, comments = scenariogetcomments()
```

Fortran command syntax:

```
CALL SCENARIOGETCOMMENTS(COMMENTS, IERR)
```

Where:

Character COMMENTS	the comment string for the Scenario. Use SCENARIOLENGTHCOMMENTS to get required buffer size, if needed. (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory

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10.19. SCENARIOGETOPT

Use this API to retrieve the Scenario options. Note: Scenario settings are not attached to the Scenario file. They are user preferences that are saved on the user's machine.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, trackopt, uziprootopt, uziplclpopt, uziplclopt, uziprmtpopt,
 uziprmtopt = scenariogetopt()
```

Fortran command syntax:

```
CALL SCENARIOGETOPT(TRACKOPT, UZIPROOTOPT, UZIPLCLPOPT, UZIPLCLOPT,
 UZIPRMTPOPT, UZIPRMTOPT, IERR)
```

Where:

Integer TRACKOPT	setting for file tracking (output).	
	TRACKOPT = 0	tracking is turned off.
	TRACKOPT = 1	the user will be prompted as soon as possible when a new file is being tracked
	TRACKOPT = 2	the user will be prompted about tracked files when the current active group is being closed
	TRACKOPT = 3	tracked files will be added to the current active group immediately.
Integer UZIPROOTOPT	setting for conflict resolution for root files (output).	
	UZIPROOTOPT = 0	skip the file.
	UZIPROOTOPT = 1	prompt the user for action.
	UZIPROOTOPT = 2	overwrite the file on disk.
Integer UZIPLCLPOPT	setting for placement of local files (output).	
	UZIPLCLPOPT = 0	do not unzip.
	UZIPLCLPOPT = 1	unzip to the root path.
	UZIPLCLPOPT = 2	unzip to the same path the file was at before the file was zipped.
	UZIPLCLPOPT = 3	unzip to the same path the file was at before, relative to the root path.
Integer UZIPLCLOPT	setting for conflict resolution for local files (output).	
	UZIPLCLOPT = 0	skip the file.
	UZIPLCLOPT = 1	prompt the user for action.
	UZIPLCLOPT = 2	overwrite the file on disk.
Integer UZIPRMTPOPT	setting for placement of remote files Note: Cannot choose to unzip remote files by relative path (output).	
	UZIPRMTPOPT = 0	do not unzip.
	UZIPRMTPOPT = 1	unzip to the root path.
	UZIPRMTPOPT = 2	unzip to the same path the file was at before the file was zipped.

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Integer UZIPRMTOPT	setting for conflict resolution for remote files (output).	
	UZIPRMTOPT = 0	skip the file.
	UZIPRMTOPT = 1	prompt the user for action.
	UZIPRMTOPT = 2	overwrite the file on disk.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred

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10.20. SCENARIOGROUPADD

Use this API to add a group to the Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOGROUPADD  SGROUP
```

Python command syntax:

```
ierr = scenariogroupadd(sgroup)
```

Fortran command syntax:

```
CALL SCENARIOGROUPADD(SGROUP, IERR)
```

Where:

Character*72 SGROUP	name of the new group to add to the Scenario (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	invalid group name

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10.21. SCENARIOGROUPCOUNT

Use this API to retrieve the number of groups in the Scenario.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, ngroups = scenariogroupcount()
```

Fortran command syntax:

```
CALL SCENARIOGROUPCOUNT(NGROUPS, IERR)
```

Where:

Integer NGROUPS	number of the group in the Scenario (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory

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10.22. SCENARIOGROUPGETCOMMENTS

Use this API to retrieve the comments for the specified group.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, comments = scenariogroupgetcomments(sgroup)
```

Fortran command syntax:

```
CALL SCENARIOGROUPGETCOMMENTS(SGROUP, COMMENTS, IERR)
```

Where:

Character*72 SGROUP	name of the group whose comments are to be returned (input; no default allowed).	
Character COMMENTS	the comment string for the specified group. Use SCENARIOGROUPLENGTHCOMMENTS to get required buffer size, if needed. (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found

10.23. SCENARIOGROUPLENGTHCOMMENTS

Use this API to determine the buffer size needed to hold the comments for the specified group.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, length = scenariogrouplengthcomments(sgroup)
```

Fortran command syntax:

```
CALL SCENARIOGROUPLENGTHCOMMENTS(SGROUP, LENGTH, IERR)
```

Where:

Character*72 SGROUP	name of the group whose comment length is to be returned (input; no default allowed).	
Integer LENGTH	length of the comment string for the specified group (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found

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10.24. SCENARIOGROUPLIST

Use this API to retrieve the list of groups in the Scenario.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, sgroups = scenariogrouplist()
```

Fortran command syntax:

```
CALL SCENARIOGROUPLIST(NGROUPS, SGROUPS, IERR)
```

Where:

Integer NGROUPS	max number of groups to list (input; no default allowed).	
Character*72 SGROUPS (NGROUPS)	list of groups in the Scenario (output; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory

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10.25. SCENARIOGROUPOPEN

Use this API to add a group to the Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOGROUPOPEN  SGROUP
```

Python command syntax:

```
ierr = scenariogroupopen(sgroup)
```

Fortran command syntax:

```
CALL SCENARIOGROUPOPEN(SGROUP, IERR)
```

Where:

Character*72 SGROUP	name of the group to open (input; ignored by default. When default is used the startup group will be opened if one is defined in the Scenario. If the name of a group in the Scenario is given, that group will be opened. If the string is blank or does not match any of the group names in the Scenario no group is opened).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found

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10.26. SCENARIOGROUPREMOVE

Use this API to remove a group from the Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOGROUPREMOVE  SGROUP
```

Python command syntax:

```
ierr = scenariogroupremove (sgroup)
```

Fortran command syntax:

```
CALL SCENARIOGROUPREMOVE (SGROUP, IERR)
```

Where:

Character*72 SGROUP	name of the group to be removed from the Scenario (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found

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10.27. SCENARIOGROUPRENAME

Use this API to rename one of the groups in the Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOGROUPRENAME  SGROUP SNEWNAME
```

Python command syntax:

```
ierr = scenariogrouprename(sgroup, snewname)
```

Fortran command syntax:

```
CALL SCENARIOGROUPRENAME(SGROUP, SNEWNAME, IERR)
```

Where:

Character*72 SGROUP	name of the group to be renamed (input; no default allowed).	
Character*72 SNEWNAME	new name to be given to the specified group (input; no default allowed).	
Integer IERR	error code (output).	
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	invalid group name
	IERR = 3	specified group not found

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10.28. SCENARIOGROUPSETCOMMENTS

Use this API to define or modify the comments for the specified group.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOGROUPSETCOMMENTS  SGROUP NCOMMENTBLOCKS COMMENTS APPEND
```

Python command syntax:

```
ierr = scenariogroupsetcomments(sgroup, ncommentblocks, comments, append)
```

Fortran command syntax:

```
CALL SCENARIOGROUPSETCOMMENTS(SGROUP, NCOMMENTBLOCKS, COMMENTS, APPEND,
 IERR)
```

Where:

Character*72 SGROUP	name of the group whose comments are to be modified (input; no default allowed).	
Integer NCOMMENTBLOCKS	number of 72 character strings that make up the comment. In python, this number can be set to 1 and the full comment string given as the first comment block (input; 0 by default).	
Character*72 COMMENTS (NCOMMENTBLOCKS)	comment string to be defined for or added to the comments for the specified group (input; unchanged).	
Logical APPEND	flag for appending comments. If TRUE, COMMENTS is appended to the current comments. If FALSE, the current comments are overwritten by COMMENTS (input; FALSE).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found

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10.29. SCENARIOISDIRTY

Use this API to determine if there are any unsaved changes in the current Scenario.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
isdirty = scenarioisdirty()
```

Fortran command syntax:

```
CALL SCENARIOISDIRTY(ISDIRTY)
```

Where:

Logical ISDIRTY	TRUE when the Scenario in memory currently contains unsaved changes (output).

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10.30. SCENARIOISVALID

Use this API to determine if there is a valid Scenario in memory.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
isvalid = scenarioisvalid()
```

Fortran command syntax:

```
CALL SCENARIOISVALID(ISVALID)
```

Where:

Logical ISVALID	TRUE when a valid Scenario is in memory, currently returns FALSE when in Zip Mode (output).

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10.31. SCENARIOLENGTHCOMMENTS

Use this API to determine the buffer size needed to hold the current Scenario comments.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, length = scenariolengthcomments()
```

Fortran command syntax:

```
CALL SCENARIOLENGTHCOMMENTS(LENGTH, IERR)
```

Where:

Integer LENGTH	length of the Scenario comment string (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory

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10.32. SCENARIOMODELGROUP

Use this API to create a new group that is identical to another group.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOMODELGROUP  SGROUP SNEWNAME
```

Python command syntax:

```
ierr = scenariomodelgroup(sgroup, snewname)
```

Fortran command syntax:

```
CALL SCENARIOMODELGROUP(SGROUP, SNEWNAME, IERR)
```

Where:

Character*72 SGROUP	name of the group to be copied (input; no default allowed).	
Character*72 SNEWNAME	name of the new group (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	invalid file name
	IERR = 3	specified group not found

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10.33. SCENARIOMOVEFILE

Use this API to move a file from one group to another group.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOMOVEFILE  SRCGROUP SFILE DSTGROUP
```

Python command syntax:

```
ierr = scenariomovefile(srcgroup, sfile, dstgroup)
```

Fortran command syntax:

```
CALL SCENARIOMOVEFILE(SRCGROUP, SFILE, DSTGROUP, IERR)
```

Where:

Character*72 SRCGROUP	name of the group containing the file to be moved (input; no default allowed).	
Character*260 SFILE	name of the file to be moved (input; no default allowed).	
Character*72 DSTGROUP	name of the group the file will be moved to (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	invalid file name
	IERR = 3	specified group not found
	IERR = 4	specified file not found

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10.34. SCENARIONEW

Use this API to create a new Scenario. At least one group must be added to the Scenario through SCENARIOGROUPADD before files may be added to it. If adding files through file tracking is also desired, a group must become the active group which can be done by calling SCENARIOGROUPOPEN.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIONEW  SFILE
```

Python command syntax:

```
ierr = scenarionew(sfile)
```

Fortran command syntax:

```
CALL SCENARIONEW(SFILE, IERR)
```

Where:

Character*260 SFILE	full file path of the new scenario file (input; blank by default. The name of the Scenario will be determined by the file name portion of the full file path. When the default is used no name is given to the Scenario and a full file name will be required upon saving. The root path of the Scenario will be determined by the path portion of the full file path. When the default is used the root path will default to the current working directory).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred

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10.35. SCENARIOOPEN

Use this API to open a previously created Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOOPEN  SFILE SGROUP
```

Python command syntax:

```
ierr = scenarioopen(sfile, sgroup)
```

Fortran command syntax:

```
CALL SCENARIOOPEN(SFILE, SGROUP, IERR)
```

Where:

Character*260 SFILE	full file path of the scenario file to open (input; no default allowed).	
Character*72 SGROUP	name of the group to open when the scenario is loaded. When the default is used the startup group will be opened if one is defined in the Scenario. If the name of a group in the Scenario is given, that group will be opened. If the string is blank or does not match any of the group names in the Scenario, no group is opened. (input; blank by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 2	file path is invalid

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10.36. SCENARIOSAVE

Use this API to save the currently open Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOSAVE  SFILE
```

Python command syntax:

```
ierr = scenariosave(sfile)
```

Fortran command syntax:

```
CALL SCENARIOSAVE(SFILE, IERR)
```

Where:

Character*260 SFILE	full file path to save the scenario file to (input; unchanged by default. When the default is used the Scenario will be saved to the last file-name given in a SCENARIONEW, SCENARIOOPEN, or SCENARIOSAVE operation, if a call to SCENARIONEW with default arguments was the previous call, the Scenario will not be saved due to no previous file-name being available).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 2	file path is invalid

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10.37. SCENARIOSETATTRS

Use this API to modify the root path and startup group attributes of the Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOSETATTRS  ROOTPATH STARTUPGROUP
```

Python command syntax:

```
ierr = scenariosetattrs(rootpath, startupgroup)
```

Fortran command syntax:

```
CALL SCENARIOSETATTRS(ROOTPATH, STARTUPGROUP, IERR)
```

Where:

Character*260 ROOTPATH	directory to be used as the root path of the Scenario (input; unchanged).	
Character*72 STARTUPGROUP	name of the group to be designated as the startup group (input; unchanged).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory
	IERR = 3	specified group not found

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10.38. SCENARIOSETCOMMENTS

Use this API to define or modify the Scenario comments.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOSETCOMMENTS  NCOMMENTBLOCKS COMMENTS APPEND
```

Python command syntax:

```
ierr = scenariosetcomments(ncommentblocks, comments, append)
```

Fortran command syntax:

```
CALL SCENARIOSETCOMMENTS(NCOMMENTBLOCKS, COMMENTS, APPEND, IERR)
```

Where:

Integer NCOMMENTBLOCKS	number of 72 character strings that make up the comment. In python, this number can be set to 1 and the full comment string given as the first comment block (input; 0 by default).	
Character*72 COMMENTS (NCOMMENTBLOCKS)	comment string to be defined for or added to the comments for the Scenario (input; unchanged).	
Logical APPEND	flag for appending comments. If TRUE, COMMENTS is appended to the current Scenario comments. If FALSE, the current Scenario comments are overwritten by COMMENTS (input; FALSE).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory

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10.39. SCENARIOSETOPT

Use this API to modify the Scenario options. Note: Scenario settings are not attached to the Scenario file. They are user preferences that are saved on the user's machine.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOSETOPT  TRACKOPT UZIPROOTOPT UZIPLCLPCOPT UZIPLCLOPT
 UZIPRMTPCOPT UZIPRMTOPT
```

Python command syntax:

```
ierr = scenariosetopt(trackopt, uziprootopt, uziplclpcopt, uziplclopt,
 uziprmtpcopt, uziprmtopt)
```

Fortran command syntax:

```
CALL SCENARIOSETOPT(TRACKOPT, UZIPROOTOPT, UZIPLCLPCOPT, UZIPLCLOPT,
 UZIPRMTPCOPT, UZIPRMTOPT, IERR)
```

Where:

Integer TRACKOPT	setting for file tracking (input; unchanged).	
	TRACKOPT = 0	The user will not be prompted about tracked files. If there are files that are currently being tracked the user will be prompted as soon as possible for those files.
	TRACKOPT = 1	The user will be prompted as soon as possible when a new file is being tracked
	TRACKOPT = 2	The user will be prompted about tracked files when the current active group is being closed
	TRACKOPT = 3	Tracked files will be added to the current active group immediately. If there are files that are currently being tracked the user will be prompted as soon as possible for those files.
Integer UZIPROOTOPT	setting for conflict resolution for root files (input; unchanged).	
	UZIPROOTOPT = 0	Skip the file.
	UZIPROOTOPT = 1	Prompt the user for action.
	UZIPROOTOPT = 2	Overwrite the file on disk.
Integer UZIPLCLPCOPT	setting for placement of local files (input; unchanged).	
	UZIPLCLPCOPT = 0	Do not unzip.
	UZIPLCLPCOPT = 1	Unzip to the root path.
	UZIPLCLPCOPT = 2	Unzip to the same path the file was at before the file was zipped.
	UZIPLCLPCOPT = 3	Unzip to the same path the file was at before, relative to the root path.
Integer UZIPLCLOPT	setting for conflict resolution for local files (input; unchanged).	
	UZIPLCLOPT = 0	Skip the file.

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	UZIPLCLOPT = 1	Prompt the user for action.
	UZIPLCLOPT = 2	Overwrite the file on disk.
Integer UZIPRMTPCOPT	setting for placement of remote files (input; unchanged by default). Note: Cannot choose to unzip remote files by relative path).	
	UZIPRMTPCOPT = 0	Do not unzip.
	UZIPRMTPCOPT = 1	Unzip to the root path.
	UZIPRMTPCOPT = 2	Unzip to the same path the file was at before the file was zipped.
Integer UZIPRMTOPT	setting for conflict resolution for remote files (input; unchanged).	
	UZIPRMTOPT = 0	Skip the file.
	UZIPRMTOPT = 1	Prompt the user for action.
	UZIPRMTOPT = 2	Overwrite the file on disk.
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred

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10.40. SCENARIOTRACKFILEADD

Use this API to add the files currently being tracked.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOTRACKFILEADD  SGROUP NFILES INDEXES DISCARD
```

Python command syntax:

```
ierr = scenariotrackfileadd(sgroup, nfiles, indexes, discard)
```

Fortran command syntax:

```
CALL SCENARIOTRACKFILEADD(SGROUP, NFILES, INDEXES, DISCARD, IERR)
```

Where:

Character*72 SGROUP	group to add tracked file to (input; current active group).	
Integer NFILES	number of files currently being tracked to add (input; no default allowed).	
Integer INDEXES (NFILES)	indexes of the files to add based on the list from SCENARIOTRACKFILEADD (input; no default allowed).	
Logical DISCARD	a flag to specify whether or not to discard the files being tracked that were not added (input; FALSE).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred

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10.41. SCENARIOTRACKFILELIST

Use this API to retrieve the list of files being tracked.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, sfiles, stypes = scenariotrackfilelist()
```

Fortran command syntax:

```
CALL SCENARIOTRACKFILELIST(NFILES, SFILES, STYPES, IERR)
```

Where:

Integer NFILES	max number of files currently being tracked to return (input; no default allowed).	
Character*260 SFILES (NFILES)	list of files currently being tracked (output).	
Character*8 STYPES (NFILES)	list of file type identifiers for each file being tracked (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred

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10.42. SCENARIOTRACKFILENUM

Use this API to retrieve the current number of files being tracked.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, nfiles = scenariotrackfilenum()
```

Fortran command syntax:

```
CALL SCENARIOTRACKFILENUM(NFILES, IERR)
```

Where:

Integer NFILES	number of files currently being tracked (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred

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10.43. SCENARIOUNZIP

Use this API to unzip a zipped Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOUNZIP SFILE
```

Python command syntax:

```
ierr = scenariounzip(sfile)
```

Fortran command syntax:

```
CALL SCENARIOUNZIP(SFILE, IERR)
```

Where:

Character*260 SFILE	full file path of the scenario archive to unzip (input; no default allowed).	
Integer IERR	error code (output).	
	IERR = 0	no conflicts occurred
	IERR > 0	conflicts occurred

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10.44. SCENARIOUNZIPPROBLEMS

Use this API to retrieve the problems that occurred during the last unzip attempt.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
ierr, indexes, problems, paths, types = scenariounzipproblems()
```

Fortran command syntax:

```
CALL SCENARIOUNZIPPROBLEMS(ERRCNT, INDEXES, PROBLEMS, PATHS, TYPES, IERR)
```

Where:

Integer ERRCNT	return value from either SCENARIOUNZIP or SCEBARIOUNZIPSOLUTIONS (input; no default allowed).	
Integer INDEXES (ERRCNT)	internal problem index which will be used in SCENARIOUNZIPSOLUTIONS (output).	
Integer PROBLEMS (ERRCNT)	problems that were encountered (output).	
	PROBLEMS = 0	file already exists.
	PROBLEMS = 1	path does not exist.
	PROBLEMS = 2	a file with the specified name does not exist in the archive.
	PROBLEMS = 3	a directory with the specified file name already exists.
Character*260 PATHS (ERRCNT)	full file paths of all files that had problems being extracted (output).	
Integer TYPES (ERRCNT)	file types associated with each file (output).	
Integer IERR	error code (output).	
	IERR = 0	no conflicts occurred
	IERR > 0	conflicts occurred

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10.45. SCENARIOUNZIPSOLUTIONS

Use this API to submit potential solutions to the problems that occurred during the last unzip attempt.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOUNZIPSOLUTIONS  NSOLUTIONS INDEXES SOLUTIONS PATHS
```

Python command syntax:

```
ierr = scenariounzipsolutions(nsolutions, indexes, solutions, paths)
```

Fortran command syntax:

```
CALL SCENARIOUNZIPSOLUTIONS(NSOLUTIONS, INDEXES, SOLUTIONS, PATHS, IERR)
```

Where:

Integer NSOLUTIONS	number of solutions being submitted (input; no default allowed).	
Integer INDEXES (NSOLUTIONS)	internal problem indexes which were obtained from SCENARIOUNZIPPROBLEMS (input; no default allowed).	
Integer SOLUTIONS (NSOLUTIONS)	solutions to attempt Note: These solutions do not apply to all problems. For example if the file is missing from the archive only a solution of 0 will solve the problem (input; no default allowed).	
	SOLUTIONS = 0	skip the file.
	SOLUTIONS = 1	retry to unzip, possibly with a different file name.
	SOLUTIONS = 2	to overwrite the current item on disk.
Character*260 PATHS (NSOLUTIONS)	used to attempt to unzip a file to a new path or under a new name All name changes will be reflected in the Scenario. If no name is specified for a particular problem the name will not change (input).	
Integer IERR	error code (output).	
	IERR = 0	no conflicts occurred
	IERR > 0	conflicts occurred

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10.46. SCENARIOZIP

Use this API to zip up the current Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOZIP  SFILE
```

Python command syntax:

```
ierr = scenariozip(sfile)
```

Fortran command syntax:

```
CALL SCENARIOZIP(SFILE, IERR)
```

Where:

Character*260 SFILE	full file path of the file to zip the scenario to (input; unchanged by default). When the default is used the name of the zip will be the same name as the Scenario but with the ZIP extension instead of the PSSXML extension. If a different name is given the Scenario's name and the Scenario PSSXML file name will be changed to reflect the new name).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred
	IERR = 1	no Scenario in memory

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10.47. SCENARIOZIPACTIVE

Use this API to determine if the Scenario is currently in Zip Mode.

This API routine was first introduced in release 33.0.0.

Python command syntax:

```
active = scenariozipactive()
```

Fortran command syntax:

```
CALL SCENARIOZIPACTIVE(ACTIVE)
```

Where:

Logical ACTIVE	TRUE when the Scenario is in Zip Mode (output).

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10.48. SCENARIOZIPOPEN

Use this API to open a zipped Scenario.

This API routine was first introduced in release 33.0.0.

Batch command syntax:

```
BAT_SCENARIOZIPOPEN  SFILE SGROUP
```

Python command syntax:

```
ierr = scenariozipopen(sfile, sgroup)
```

Fortran command syntax:

```
CALL SCENARIOZIPOPEN(SFILE, SGROUP, IERR)
```

Where:

Character*260 SFILE	full file path of the scenario file to open (input; no default allowed).	
Character*72 SGROUP	Reserved for future use (input; ignored).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred
	IERR > 0	fatal error occurred

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10.49. SCENARIOZIPSAVE

This API routine was first introduced in release 33.2.0.

Batch command syntax:

```
BAT_SCENARIOZIPSAVE  SFILE
```

Python command syntax:

```
ierr = scenariozipsave(sfile)
```

Fortran command syntax:

```
CALL SCENARIOZIPSAVE(SFILE, IERR)
```

Where:

Character*260 SFILE	full file path of the scenario file to save to (input; no default allowed).
Integer IERR	error code (output).

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Chapter 11
Event Studies

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

11.1. CLOSEEVENTSTUDYFILE

Use this API to close an open Event Study file.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_CLOSEEVENTSTUDYFILE
```

Python command syntax:

```
closeeventstudyfile()
```

Fortran command syntax:

```
CALL CLOSEEVENTSTUDYFILE
```

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11.2. OPENEVENTSTUDYFILE

Use this API to open a new or existing event study file in the GUI.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_OPENEVENTSTUDYFILE  SFILE
```

Python command syntax:

```
openeventstudyfile(sfile)
```

Fortran command syntax:

```
CALL OPENEVENTSTUDYFILE(SFILE)
```

Where:

Character*260 SFILE	name of the new or existing event study file (input).

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11.3. RUNDYNAMICEVENTSTUDY

Use this API to run a dynamic study using the active event study in the open event study file.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_RUNDYNAMICEVENTSTUDY
```

Python command syntax:

```
rundynamiceventstudy()
```

Fortran command syntax:

```
CALL RUNDYNAMICEVENTSTUDY
```

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11.4. RUNPOWERFLOWEVENTSTUDY

Use this API to run a power flow study using the active event study in the open event study file.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_RUNPOWERFLOWEVENTSTUDY
```

Python command syntax:

```
runpowerfloweventstudy()
```

Fortran command syntax:

```
CALL RUNPOWERFLOWEVENTSTUDY
```

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11.5. SAVEEVENTSTUDYFILE

Use this API to save existing event studies in an event study file.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SAVEEVENTSTUDYFILE  SFILE
```

Python command syntax:

```
saveeventstudyfile(sfile)
```

Fortran command syntax:

```
CALL SAVEEVENTSTUDYFILE(SFILE)
```

Where:

Character*260 SFILE	name of the destination event study file (input).

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11.6. SETACTIVEEVENTSTUDY

Use this API to select an Event Study in an Event file as the active Event Study for running dynamic or power flow event studies.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_SETACTIVEEVENTSTUDY  SFILE
```

Python command syntax:

```
setactiveeventstudy(sfile)
```

Fortran command syntax:

```
CALL SETACTIVEEVENTSTUDY(SFILE)
```

Where:

Character*260 SFILE	name of an event study in the open event study file (input).

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Chapter 12
Measurement Interface

This chapter describes the API routines used in PSS®E Measurement Interface.

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12.1. HDAIMPORTDATA

Use this API to import historical data from an OPC HDA database.

This API routine was first introduced in release 33.9.0.

Batch command syntax:

```
BAT_HDAIMPORTDATA  HOST SERVER MAPFILE TIMEFROM(1)..TIMEFROM(6)
 TIMETO(1)..TIMETO(6) AGGREGATE SID ATTRIBUTECOUNT
 ATTRIBUTES(1)..ATTRIBUTES(ATTRIBUTECOUNT)
```

Python command syntax:

```
ierr = hdaimportdata(host, server, mapfile, timefrom, timeto, aggregate,
 sid, attributecount, attributes)
```

Fortran command syntax:

```
CALL HDAIMPORTDATA(HOST, SERVER, MAPFILE, TIMEFROM, TIMETO, AGGREGATE,
 SID, ATTRIBUTECOUNT, ATTRIBUTES, IERR)
```

Where:

Character*260 HOST	name of the address of the machine hosting the server (input).
Character*260 SERVER	name of the OPC HDA compliant server (input).
Character*260 MAPFILE	name of the excel file mapping network items to HDA tags (input).
Integer TIMEFROM	time from which to start gathering data. The order is Year, Month, Day, Hour, Minute, Second. If all zeros specified, then asnapshot will be taken from the time specified by TIMETO and AGGREGATE will be ignored. (input).
Integer TIMETO	time to gather data to. The order is Year, Month, Day, Hour, Minute, Second (input).
Integer AGGREGATE	ID of the aggregate to use when gathering data. Use the help function in the python module psspihist to retrieve aggregate IDs from a server. (input).
Integer SID	subsystem to gather data for and apply data to (input).
Integer ATTRIBUTECOUNT	number of attributes to gather data for (input).
Character*61 ATTRIBUTES	list of attributes to gather. There are specified by network element type followed by result type, e.g. LOAD-PL. (input).
Integer IERR	error code (output).

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Chapter 13
Multiple Element Data

All material contained in this documentation is proprietary to Siemens Industry Inc., Siemens Power Technologies International

13.1. GETMODELLIBRARIES

Return the list of libraries that are searched for dynamically (i.e. run-time) loaded dynamics model programs.
In particular this will be the list of libraries searched for user-written dynamics model programs.

This API routine was first introduced in release 31.0.0.

Python command syntax:

```
names = getmodellibraries()
```

Fortran command syntax:

```
CALL GETMODELLIBRARIES(NSTR, NAMES, NRET)
```

Where:

Integer NSTR	declared size of NAMES (input).
Character*260 NAMES (NSTR)	names of all libraries in the search list (output).
Integer NRET	number of libraries returned (output).

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13.2. GETMODELPROGRAMS

Return the list dynamically (i.e. run-time) loaded dynamics model programs currently in use.
In particular this will include all user-written dynamics model programs.

This API routine was first introduced in release 33.3.0.

Python command syntax:

```
progs,libs,args = getmodelprograms()
```

Fortran command syntax:

```
CALL GETMODELPROGRAMS(NSIZ,PROGS,LIBS,ARGS,PCNT)
```

Where:

Integer NSIZ	declared size of PROGS and LIBS (input).
Character*260 PROGS (NSIZ)	(output).
Character*260 LIBS (NSIZ)	names of the libraries each program has been loaded from; the array index of program can be used to identify the correcponding library in this array. (output).
Integer ARGS (NSIZ)	number of arguments for each program; the array index of program can be used to identify the correcponding entry in this array. (output).
Integer PCNT	number of programs returned (input).

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13.3. TREEDAT

Use this API to return Swing Bus and Island Data of Working Case Network.

This API routine was first introduced in release 33.5.1.

Python command syntax:

```
treeobj = treedat(sizeislands)
```

Fortran command syntax:

```
CALL TREEDAT(NBUS, SIZEISLANDS, NSWINGBUS, NISLANDS, NB_ISLANDS,
 NP_ISLANDS, BUSNUMXX, BUSNAMEXX, BUSKVXX, BUSAREAXX, GENPQXX, LOADXX,
 SHUNTXX, IERR)
```

Where:

Integer NBUS	number of buses in the case (input; no default allowed).
Integer SIZEISLANDS	number of (possible) islands (input; 20 by default).
Integer NSWINGBUS	number of swing buses in the case (output).
Integer NISLANDS	number of actual islands (output).
Integer NB_ISLANDS (SIZEISLANDS)	number of buses in each island (output).
Integer NP_ISLANDS (SIZEISLANDS)	number of plant buses in each island (output).
Integer BUSNUMXX (NBUS)	bus number array, composed from swing + islanded buses, (1:n_swbus) -- swing buses, (n_swbus+1,n_swbus+nb_islands(1)) -- buses in island 1 and so on (output).
Character*12 BUSNAMEXX (NBUS)	bus name array, composed same way as BUSNUMXX (output).
Real BUSKVXX (NBUS)	bus base kV array, composed same way as BUSNUMXX (output).
Integer BUSAREAXX (NBUS)	bus area array, composed same way as BUSNUMXX (output).
Real GENPQXX (SIZEISLANDS*4)	pgen, qgen, qt, qb values for each island (output).
Complex LOADXX (SIZEISLANDS*3)	(PQ, IL, YL) loads for each island (output).
Complex SHUNTXX (SIZEISLANDS)	shunts for each island (output).
Integer TREEOBJ	Returned dictionary object. (output).
Integer IERR	error code (output).
	IERR = 0 — no error occurred
	IERR = 1 — Warning: All islands data is not fetched. Re-run the activity with argument sizeislands value greater than provided value
	IERR = 2 — Error: Working case is empty
	IERR = 3 — Error: No sizeislands input value provided
	IERR = 4 — Error: Invalid sizeislands input value, must be greater than 0

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The returned object 'robj' contains the following attributes:

robj['ierr']: Error Code
= 0, No error
= 1, All islands data is not fetched. Re-run the activity
with higher argument sizeislands value.
= 2, Working case is empty.
= 3, No sizeislands input value provided.
= 4, Invalid sizeislands input value, must be greater than 0.

Following values are not returned for robj['ierr']=2,3,4.

robj['nswingbus']: Number of Swing Buses
robj['nislands']: Number of Islands
robj['island_nbus']: [] of size nislands, Number of buses in each island
robj['island_nplants']: [] of size nislands, Number of plant buses in each island

robj['swing_busnum']: [] of size nswingbus, Swing Bus Numbers
robj['swing_busname']: [] of size nswingbus, Swing Bus Names
robj['swing_buskv']: [] of size nswingbus, Swing Bus Base kV
robj['swing_busarea']: [] of size nswingbus, Swing Bus Area

robj['island_busnum']: [] of size nislands, each item is [] of Island Bus Numbers
robj['island_busname']: [] of size nislands, each item is [] of Island Bus Names
robj['island_buskv']: [] of size nislands, each item is [] of Island Bus Base kV
robj['island_busarea']: [] of size nislands, each item is [] of Island Bus Area
robj['island_plants']: [] of size nislands, each item is [] of Island Plant Bus Numbers

robj['island_pgen']: [] of size nislands, Total Pgen of each Island (MVA)
robj['island_qgen']: [] of size nislands, Total Qgen of each Island (MVAR)
robj['island_qt']: [] of size nislands, Total QT of each Island (maximum Q limit) (MVAR)
robj['island_qb']: [] of size nislands, Total QB of each Island (minimum Q limit) (MVAR)

robj['island_load_pq']: [] of size nislands, Total Constant MVA (PL+jQL) loads of each Island (MVA)
robj['island_load_il']: [] of size nislands, Total Constant Current (IP+jIQ) loads of each Island (MVA)
robj['island_load_yl']: [] of size nislands, Total Constant admittance (YP+jYQ) loads of each Island (MVA)

robj['island_shunt'] --> [] of size nislands, Total Bus Shunts (G+jB) of each Island (MVA)

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Chapter 14
Non-Engineering

This chapter defines API routines that provide functions not associated with PSS®E data, analysis, or graphics capabilities.

Most of these routines do not record themselves when called with recording enabled. Those that have a batch format documented can record themselves.

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14.1. ALERT

Use this API to send output to the alert device.
Note: A Fortran program that calls ALERTS must also include a USE statement for KEYPTI.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
alert(string)
```

Fortran command syntax:

```
CALL ALERTS(STRING)

CALL ALERTS(STRING-ARRAY)
```

Where:

Character*220 STRING	string to be written (input).
Character*220 STRING-ARRAY	array of strings to be written

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14.2. BEGINREPORT

Use this API to create a new report tab in the GUI.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
beginreport()
```

Fortran command syntax:

```
CALL RPTBEG()
```

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14.3. FILEIN

Get input from file (Fortran unit number).

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ierr, string, flags = filein(lunit)
```

Where:

Integer LUNIT	Fortran logical unit number of the file (input).
Character*512 STRING	buffer read from file (output).
Integer ILEN	declared length (input).
Logical FLAGS	the first element of flags is set to true if an end-of-file is encountered; the second element of flags is set to true if the file is closed after the read attempt (output).
Integer IERR	error code (output). IERR = 0 no error occurred.

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14.4. GETBATDEFAULTS

Retrieve "no input" value for integers and reals.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
idef,rdef = getbatdefaults()
```

Fortran command syntax:

```
CALL GETBATDEFAULTS(IDEF, RDEF)
```

Where:

Integer IDEF	default integer value used in the batch command processor (output).
Real RDEF	default real value used in the batch command processor (output).

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14.5. GETDEFAULTCHAR

Retrieve "no input" value for characters (i.e., strings) other than filenames. Specify this value to indicate that "no input" is intended.

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
cdef = getdefaultchar()
```

Fortran command syntax:

```
CALL GETDEFAULTCHAR(CDEF)
```

Where:

Character*1 CDEF	"no character value (output).

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14.6. GETDEFAULTINT

Retrieve "no input" value for integers. Specify this value to indicate that "no input" is intended.

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
idef = getdefaultint()
```

Fortran command syntax:

```
CALL GETDEFAULTINT(IDEF)
```

Where:

Integer IDEF	"no integer value (output).

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14.7. GETDEFAULTREAL

Retrieve "no input" value for reals (floats). Specify this value to indicate that "no input" is intended.

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
rdef = getdefaultreal()
```

Fortran command syntax:

```
CALL GETDEFAULTREAL(RDEF)
```

Where:

Real RDEF	"no real value (output).

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14.8. GETLOADEDMODULES

Get list of all modules currently loaded.

This API routine was first introduced in release 32.0.1.

Python command syntax:

```
m = getloadedmodules()
```

Where:

Character M	list of loaded libraries (output).

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14.9. GETMODFUNCLIST

Get list of external callable methods given a module name. Module is imported in process.

This API routine was first introduced in release 32.0.0.

Python command syntax:

```
c = getmodfunclist(mod)
```

Where:

Character MOD	module name (input).
Character C	list of callable methods (output).

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14.10. IOFLUSH

Force all files in PSSE to flush buffers to disk, if possible.

This API routine was first introduced in release 30.2.1.

Python command syntax:

```
ioflush()
```

Fortran command syntax:

```
CALL IOFLSH()
```

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14.11. PAGEREPORT

Use this API to place a report separator on the active report tab of the GUI.

This API routine was first introduced in release 30.2.0.

Python command syntax:

```
pagereport()
```

Fortran command syntax:

```
CALL RPTPGBK()
```

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14.12. PROGRESS

Send output to progress device.
Note: A Fortran program that calls PROGRESS must also include a USE statement for KEYPTI.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
progress(string)
```

Fortran command syntax:

```
CALL PROGRESS(STRING)
```

Where:

Character*220 STRING	string to be written (input).
Character*220 STRING-ARRAY	array of strings to be written

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14.13. PROMPT

Send output to prompt device. Prompts are stored until the next terminal input request. Up to 36 buffers of up to 220 characters can be stored.
Note: A Fortran program that calls PROMPTS must also include a USE statement for KEYPTI.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
prompt(string)
```

Fortran command syntax:

```
CALL PROMPTS(STRING)

CALL PROMPTS(STRING-ARRAY)
```

Where:

Character*220 STRING	string to be written (input).
Character*220 STRING-ARRAY	array of strings to be written

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14.14. PSSEINIT

Initialize PSSE.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ierr = psseinit(buses)
```

Fortran command syntax:

```
CALL PSSEINIT(BUSES, IERR)
```

Where:

Integer BUSES	requested bus size; zero for default size level (input).
Integer IERR	error code (output).

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14.15. PSSEVERSION

Get PSSE version information.

This API routine was first introduced in release 30.1.0.

Python command syntax:

```
name,major,minor,update,date,stat = psseversion()
```

Fortran command syntax:

```
CALL REVPSSE(NAME, MAJOR, MINOR, UPDATE, DATE, STAT)
```

Where:

Character*32 NAME	product name (output).
Integer MAJOR	major release number (output).
Integer MINOR	minor release number (output).
Integer UPDATE	update release number (output).
Character*30 DATE	release date (output).
Character*132 STAT	any special release information (output).

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14.16. REFRESHGUI

Refresh the graphical user interface.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
refreshgui()
```

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14.17. REPORT

Send output to report device.
Note: A Fortran program that calls REPORTS must also include a USE statement for KEYPTI.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
report(string)
```

Fortran command syntax:

```
CALL REPORTS(STRING)
```

Where:

Character*220 STRING	string to be written (input).
Character*220 STRING-ARRAY	array of strings to be written

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14.18. RUNIPLANFILE

Execute an IPLAN program.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
runiplanfile(iplname)
```

Fortran command syntax:

```
CALL EXECRUN(IPLNAME)
```

Where:

Character*260 IPLNAME	the filename; the filename can be followed by arguments as part of a single string, in which case the filename must be delimited (e.g. enclosed in quotes). The -arg option can be specified this way as well. (input).

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14.19. RUNRSPNSFILE

Run a response file. OPerates by replacing standard input with a file.
Then begin and continue reading from terminal until standard input is no longer redirected using PSSE command line techniques (allowing interaction with IPLAN programs, for example).
Pass any results to Command Line Interface (CLI). The CLI can recognize batch commands and Line Mode commands and direct them to the proper interpreters.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
runrspnsfile(rspname)
```

Where:

Character*260 RSPNAME	the filename; the filename can be followed by arguments as part of a single string, in which case the filename must be delimited (e.g. enclosed in quotes). (input).

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14.20. SHOWLOADEDMODULES

Display list of all modules currenly loaded.

This API routine was first introduced in release 32.0.2.

Python command syntax:

```
showloadedmodules()
```

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14.21. STARTRECORDING

Start recording program operation.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ierr = startrecording(iopt,string)
```

Fortran command syntax:

```
CALL STARTRECORDING(IOPT, STRING, IERR)
```

Where:

Integer IOPT	type of format to be recorded (input).	
	IOPT = 0	record in batch (BAT_) format.
	IOPT = 1	record in python format.
Character*260 STRING	filename (input).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred.
	IERR = 1	error closing file.
	IERR = 2	not recording.

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14.22. STOPRECORDING

Stop recording program operation.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ierr = stoprecording()
```

Fortran command syntax:

```
CALL STOPRECORDING(IERR)
```

Where:

Integer IERR	error code (output).	
	IERR = 0	no error occurred.
	IERR = 1	error closing file.
	IERR = 2	not recording.

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14.23. USERIN

Get input from interactive device.

This API routine was first introduced in release 30.0.0.

Python command syntax:

```
ierr,string = userin()
```

Where:

Character*512 STRING	buffer input by user (output).	
Integer IERR	error code (output).	
	IERR = 0	no error occurred.

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Chapter 15
PSSPLOT Module

This chapter defines API routines that are used to manipulate plots in the interface. They will elicit no action if the GUI is not present when invoked.

Please note that these routines are members of the pssplot Python extension module, not psspy.

These routines will be recorded when recording is enabled, and the Python recording will reference each of these functions within the pssplot namespace.

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15.1. DELETESELPLOT

Use this API to delete the selected plot on the active page in the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_DELETESELPLOT
```

Python command syntax:

```
deleteselplot()
```

Fortran command syntax:

```
CALL DELETESELPLOT
```

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15.2. DRAGDROPPLOTDATA

Use this API to load a Channel Output File into the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_DRAGDROPPLOTDATA  SFILENAME SCHANNELNAME
```

Python command syntax:

```
dragdropplotdata(sfilename, schannelname)
```

Fortran command syntax:

```
CALL DRAGDROPPLOTDATA(SFILENAME, SCHANNELNAME)
```

Where:

Character*260 SFILENAME	Channel File name (input).
Character*260 SCHANNELNAME	Channel to be dragged to the plot (input).

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15.3. EXPORTPLOT

Use this API to export the selected plot on the active page in the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_EXPORTPLOT SFILENAME NTYPE
```

Python command syntax:

```
exportplot(sfilename, ntype)
```

Fortran command syntax:

```
CALL EXPORTPLOT(SFILENAME, NTYPE)
```

Where:

Character*260 SFILENAME	filename (input).	
Integer NTYPE	graphics format for the exported file (input).	
	NTYPE = 1	Metafile export
	NTYPE = 2	Bitmap export
	NTYPE = 3	JPEG export
	NTYPE = 4	PDF export

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15.4. INSERTPAGE

Use this API to insert a page into the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_INSERTPAGE
```

Python command syntax:

```
insertpage()
```

Fortran command syntax:

```
CALL INSERTPAGE
```

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15.5. INSERTPLOT

Use this API to add an empty plot on the active page in the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_INSERTPLOT
```

Python command syntax:

```
insertplot()
```

Fortran command syntax:

```
CALL INSERTPLOT
```

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15.6. NEWPLOTBOOK

Use this API to create a new plotting book.

This API routine was first introduced in release 32.0.0.

Batch command syntax:

```
BAT_NEWPLOTBOOK
```

Python command syntax:

```
newplotbook()
```

Fortran command syntax:

```
CALL NEWPLOTSHEET
```

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15.7. OPENCHANDATAFILE

Use this API to load a Channel Output File into the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_OPENCHANDATAFILE  SFILENAME
```

Python command syntax:

```
openchandatafile(sfilename)
```

Fortran command syntax:

```
CALL OPENCHANDATAFILE(SFILENAME)
```

Where:

Character*260 SFILENAME	name of the channel file to be opened (input).

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15.8. PRINTSELPLOT

Use this API to print the selected plot on the active page in the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_PRINTSELPLOT SFILENAME NORIENT NINDEX
```

Python command syntax:

```
printselplot(sfilename, norient, nindex)
```

Fortran command syntax:

```
CALL PRINTSELPLOT(SFILENAME, NORIENT, NINDEX)
```

Where:

Character*260 SFILENAME	name of the file to print (input).
Integer NORIENT	orientation of the paper (input). NORIENT = 0 Portrait NORIENT = 1 Landscape
Integer NINDEX	index for the printer you wish to use; This index refers to the list of printers defined on your system; default = 1 (input).

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15.9. SETSELECTEDPAGE

Use this API to set the active page in the plotting book

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SETSELECTEDPAGE    NPAGENUM
```

Python command syntax:

```
setselectedpage(npagenum)
```

Fortran command syntax:

```
CALL SETSELECTEDPAGE(NPAGENUM)
```

Where:

Integer NPAGENUM	page number in the plotting book (input).

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15.10. SETSELECTEDPLOT

Use this API to set the selected plot on the active page in the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SETSELECTEDPLOT    NPLOTNUM
```

Python command syntax:

```
setselectedplot(nplotnum)
```

Fortran command syntax:

```
CALL SETSELECTEDPLOT(NPLOTNUM)
```

Where:

Integer NPLOTNUM	plot number on the selected page (input).

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15.11. SHOWPLOTEDITOR

Use this API to display the editor for the selected plot in the plotting book.

This API routine was first introduced in release 31.0.0.

Batch command syntax:

```
BAT_SHOWPLOTEDITOR
```

Python command syntax:

```
showploteditor()
```

Fortran command syntax:

```
CALL SHOWPLOTEDITOR
```

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Chapter 16
PSSRDB Module

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16.1. DVRMATTACHANDLAUNCHWORKBOOK

Use this API to create an ACCC or Dynamics visualization workbook and launch the data visualizer.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_DVRMATTACHANDLAUNCHWORKBOOK SQLSERV SQLDBASE DVRMTMPWRKBK SQLSCRIPT
 DVRMWRKBK ATCHWRKBK LAUNCHDVRM
```

Python command syntax:

```
ierr = dvrmattachandlaunchworkbook(sqlserv, sqldbase, dvrmtmpwrkbk,
 sqlscript, dvrmwrkbk, atchwrkbk, launchdvrm)
```

Fortran command syntax:

```
CALL DVRMATTACHANDLAUNCHWORKBOOK(SQLSERV, SQLDBASE, DVRMTMPWRKBK,
 SQLSCRIPT, DVRMWRKBK, ATCHWRKBK, LAUNCHDVRM, IERR)
```

Where:

Character*260 SQLSERV	name of the host machine where the database to use is located (input).
Character*260 SQLDBASE	name of the results database to use (input).
Character*260 DVRMTMPWRKBK	name of the DVRM Template Workbook to attach to SQLSERV and SQLDBASE (input).
Character*260 SQLSCRIPT	name of the SQL script to be run against SQLDBASE. This script is usually used to optimize the performance of the database with the DVRM Workbook. (input).
Character*260 DVRMWRKBK	name of the DVRM Workbook to use to launch the DVRM Visualizer. This workbook has been modified to attach to a data source. (input).
Integer ATCHWRKBK	flag used to indicate whether the DVRM Template Workbook identified by DVRMTMPWRKBK is to be "attached" to the identified SQLSERV and SQLDBASE. (input; 0 by default). ATCHWRKBK = 0 — do not create an "attached" DVRM workbook ATCHWRKBK = 1 — create an "attached" DVRM workbook from the DVRM template workbook
Integer LAUNCHDVRM	used to indicate "launch" the DVRM Visualizer with the "attached" DVRM Workbook identified by DVRMWRKBK (input; 0 by default). LAUNCHDVRM = 0 — do not launch the DVRM visualizer LAUNCHDVRM = 1 — launch the DVRM visualizer
Integer IERR	error code (output). IERR = 0 — no error occured IERR = 1 — no SQL Server specified IERR = 2 — no SQL Database specified IERR = 3 — no template workbook specified IERR = 4 — no attached workbook specified

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16.2. DVRMCREATEACCDATABASE

Use this API to create a DVRM ACC results database.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_DVRMCREATEACCDATABASE NFILES ACFILES(1)..ACFILE(NFILES) SQLSERV
 SQLDBASE THRESHOLD
```

Python command syntax:

```
ierr = dvrmcreateaccdatabase(nfiles, acfiles, sqlserv, sqldbase,
 threshold)
```

Fortran command syntax:

```
CALL DVRMCREATEACCDATABASE(NFILES, ACFILES, SQLSERV, SQLDBASE, THRESHOLD,
 IERR)
```

Where:

Integer NFILES	number of ACC results files to process into database (input; no default allowed).	
Character*260 ACFILES (NFILES)	array of NFILES names of .ACC result files (input; no default allowed).	
Character*260 SQLSERV	name of the host machine where the database will be created (input; no default allowed).	
Character*260 SQLDBASE	name to give the results database (input; no default allowed).	
Integer THRESHOLD	threshold value used to filter out overloads in the ACCC results database. Any elements overloaded by less than this value will not be added to the ACCC results database. (input; 90 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occured
	IERR = 1	no ACFILES specified
	IERR = 2	no SQL Server specified
	IERR = 3	no SQL Database specified
	IERR = 4	error creating the SQL Database

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16.3. DVRMCREATEDYNDATABASE

Use this API to create a DVRM Dynamic results database.

This API routine was first introduced in release 33.8.0.

Batch command syntax:

```
BAT_DVRMCREATEACCDATABASE NFILES OTXFILES(1)..OTXFILE(NFILES) SQLSERV
 SQLDBASE THRESHOLD
```

Python command syntax:

```
ierr = dvrmcreatedyndatabase(nfiles, otxfiles, sqlserv, sqldbase,
 threshold)
```

Fortran command syntax:

```
CALL DVRMCREATEDYNDATABASE(NFILES, OTXFILES, SQLSERV, SQLDBASE, THRESHOLD,
 IERR)
```

Where:

Integer NFILES	number of .OUTX results files to process into database (input; no default allowed).	
Character*260 OTXFILES (NFILES)	array of NFILES names of .OUTX result files (input; no default allowed).	
Character*260 SQLSERV	name of the host machine where the database will be created (input; no default allowed).	
Character*260 SQLDBASE	name to give the results database (input; no default allowed).	
Integer THRESHOLD	cutoff threshold for records to place into database. (input; 90 by default).	
Integer IERR	error code (output).	
	IERR = 0	no error occured
	IERR = 1	no OTXFILES specified
	IERR = 2	no SQL Server specified
	IERR = 3	no SQL Database specified
	IERR = 4	error creating the SQL Database

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Chapter 17
PSSARRAYS Module

The pssarrays module provides Python functions to retrieve PSS®E solution results in arrays. These arrays can then be used in a Python script for post-processing, to create customized reports for export to a text file or an Excel spreadsheet, etc.

These functions do not record themselves when called while recording is enabled in PSS®E.

Please note that these routines are members of the pssarrays Python extension module, not psspy. The syntax on the following pages assumes that the module has been imported by executing:

```
import pssarrays
```

17.1. ACCC_SUMMARY

This function returns AC Contingency monitored element labels, contingency labels etc.

Python syntax:

```
rlst = pssarrays.accc_summary(accfile)
```

where:

string	accfile	Is the name of Contingency Solution Output file (input, no default allowed), maximum string length=260

Returned object 'rlst' contains the following attributes:

integer	rlst.ierr	Is the error code =0 no error occurred =4 error opening accfile =5 error reading accfile =6 prerequisite requirements for function not met
integer	rlst.acccsize.nmline	Is the number of monitored branches
integer	rlst.acccsize.ninter	Is the number of monitored interfaces
integer	rlst.acccsize.ncase	Is the number of contingencies + 1 (for base case)
integer	rlst.acccsize.nmvbus	Is the number of voltage monitored buses
integer	rlst.acccsize.nmvrec	Is the number of voltage monitored records
integer	rlst.acccsize.nmvbusrec	Is the number of voltage monitored bus records
integer	rlst.acccsize.nbus	Is the number of buses in the case
integer	rlst.acccsize. ncntlshed	Is the number of load sheds due to dispatch and contingency
integer	rlst.acccsize. ntrplshed	Is the number of load sheds due to tripping
integer	rlst.acccsize. ncactlshed	Is the number of load sheds due to corrective actions
integer	rlst.acccsize. ncactgdisp	Is the number of generation dispatched due to corrective actions
integer	rlst.acccsize. ncactphsftr	Is the number of phase shifter changed due to corrective actions
string	rlst.casetitle.line1	Is the case short title line 1, maximum string length=60
string	rlst.casetitle.line2	Is the case short title line 2, maximum string length=60
string	rlst.file.acc	Is the name of contingency output (.acc) file, maximum string length=260
string	rlst.file.sav	Is the name of saved case (.sav) file, maximum string length=260

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string	rlst.file.dfx	Is the name of distribution factor data (.dfx) file, maximum string length=260
string	rlst.file.sub	Is the name of subsystem definition data (.sub) file, maximum string length=260
string	rlst.file.mon	Is the name of monitored element data (.mon) file, maximum string length=260
string	rlst.file.con	Is the name of contingency description data (.con) file, maximum string length=260
string	rlst.file.thr	Is the name of load throwover data (.thr) file, maximum string length=260
string	rlst.file.inl	Is the name of unit inertia and governor data (.inl) file, maximum string length=260
string	rlst.file.trp	Is the name of tripping element data (.trp) file, maximum string length=260
string	rlst.melement	Is a list of length (nmline+ninter) of monitored branch and interface names, maximum string length=54
float	rlst.rating.a	Is a list of length (nmline+ninter) of monitored element rating A
float	rlst.rating.b	Is a list of length (nmline+ninter) of monitored element rating B
float	rlst.rating.c	Is a list of length (nmline+ninter) of monitored element rating C
string	rlst.mvbuslabel	Is a list of length (nmvbusrec) of monitored voltage bus label, maximum string length=25
string	rlst.mvreclabel	Is a list of length (nmvbusrec) of monitored voltage record label, maximum string length=20
float	rlst.mvrecmax	Is a list of length (nmvbusrec) of monitored voltage bus maximum
float	rlst.mvrecmin	Is a list of length (nmvbusrec) of monitored voltage bus minimum
string	rlst.mvrectype	Is a list of length (nmvbusrec) of monitored voltage record type (RANGE / DEVIATION), maximum string length=9
string	rlst.colabel	Is a list of length (ncase) of contingency labels, maximum string length=12
string	rlst.busname	Is a list of length (nbus) of extended bus names, maximum string length=25

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17.2. ACCC_SOLUTION

This function returns ACCC post-contingency, post-tripping or post-corrective action solution monitored MVA flows, ampere flows and bus voltages for one contingency.

Python syntax:

```
rlst = pssarrays.accc_solution(accfile, colabel,
stype, busmsm, sysmsm)
```

where:

string	accfile	Is the name of Contingency Solution Output file (input, no default allowed), maximum string length=260
string	colabel	Is the name of contingency label to get ACCC solution for, only one label allowed (input, no default allowed), maximum string length=12
string	stype	Is the name of solution type to get (input, *contingency* by default), allowed values: *contingency*, *con*, *tripping*, *trp*, *caction*, *contingency action*, *cor*
float	busmsm	Is the value of bus mismatch tolerance in MVA (input, 0.5 MVA by default)
float	sysmsm	Is the value of system mismatch tolerance in MVA (input, 5.0 MVA by default)

Returned object 'rlst' contains the following attributes:

integer	rlst.ierr	Is the error code =0 no error occurred =4 error opening accfile =5 error reading accfile =6 prerequisite requirements for function not met
string	rlst.codesc	Is the list of length (up to 16) of contingency events description, maximum string length=152
boolean	rlst.cnvflag	Is the convergence status (True when converged)
string	rlst.cnvcond	Is the convergence condition description, maximum string length=34
integer	rlst.island	Is the number of islands
float	rlst.mvaworst	Is the worst MVA mismatch
float	rlst.mvatotal	Is the total MVA mismatch
float	rlst.volts	Is a list of length (nmvbusrec) of monitored bus voltage in pu
float	rlst.mvaflow	Is a list of length (nmline+ninter) of monitored branch and interface MVA flow

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float	rlst.ampflow	Is a list of length (nmline) of monitored branch ampere flow
string	rlst.lshedbus	Is a list of extended bus names of the load shed buses, maximum string length=25, list length: = ncntlshed, for post-contingency solution = ntrplshed, for post-tripping solution = ncactlshed, for post-corrective action solution
float	rlst.loadshed	Is a list of loads shed in MW, list length: = ncntlshed, for post-contingency solution = ntrplshed, for post-tripping solution = 2 columns of ncactlshed, for post-corrective action solution, 1st column gives initial load in MW, and 2nd column gives load shed in MW.
string	rlst.gdispbus	Is a list of length (ncactgdisp) of extended bus names of the generation dispatch buses, maximum string length=25, available for post-corrective action solution only.
float	rlst.gendisp	Is a list of generation dispatches in MW, available for post-corrective action solution only, list length = 2 columns of ncactgdisp, 1st column gives initial generation in MW, and 2nd column gives generation dispatch in MW.
string	rlst.phsftr	Is a list of length (ncactphsftr) of phase shifter names, maximum string length=54, available for post-corrective action solution only.
float	rlst.phsftrang	Is a list of phase shifter angles in degrees, available for post-corrective action solution only, list length = 2 columns of ncactphsftr, 1st column gives initial angle in degrees, and 2nd column gives new angle in degrees.

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17.3. ACCC_SUMMARY_REPORT

This function generates text report of lists returned by accc_summary.

Python syntax:

```
ierr = pssarrays.accc_summary_report(accfile,
rptfile)
```

where:

string	accfile	Is the name of Contingency Solution Output file (input, no default allowed), maximum string length=260
string	rptfile	Is the name of report file (input, PSS®E Report by default), maximum string length=260

returns:

integer	rlst.ierr	Is the error code =0 no error occurred =4 error opening accfile =5 error reading accfile =6 prerequisite requirements for function not met

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17.4. SENSITIVITY_FLOW_TO_MW

This API is used to calculate sensitivity factors of a branch flow to MW power at generator buses and MW power at load buses.

The documentation for this API is available by running the following commands in either PSS®E CLI or any Python Interpreter:

```
import pssarrays

  help(pssarrays.sensitivity_flow_to_mw)
```

For error codes details please refer to Section 1.291, SENSITIVITY_FLOWS.

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17.5. ACCC_SOLUTION_REPORT

This function generates text report of lists returned by accc_solution.

Python syntax:

```
ierr = pssarrays.accc_solution_report(accfile,
colabels, stype, busmsm, sysmsm, rating, rptfile)
```

where:

string	accfile	Is the name of Contingency Solution Output file (input, no default allowed), maximum string length=260
string	colabels	Is the list of contingency label names (input, all contingencies in accfile by default), maximum string length=12
string	stype	Is the name of solution type to get (input, *contingency* by default allowed), allowed values: *contingency*, *con*, *tripping*, *trp*, *caction*, *contingency action*, *cor*
float	busmsm	Is the value of bus mismatch tolerance in MVA (input, 0.5 MVA by default)
float	sysmsm	Is the value of system mismatch tolerance in MVA (input, 5.0 MVA by default)
string	rating	Is the name of rating ('a' or 'b' or 'c') to use for calculating percentage loading (input, 'a' by default), maximum string length=1
string	rptfile	Is the name of report file (input, PSS®E Report by default), maximum string length=260

returns:

integer	rlst.ierr	Is the error code =0 no error occurred =4 error opening accfile =5 error reading accfile =6 prerequisite requirements for function not met

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17.6. ACCC_VIOLATIONS_REPORT

This function generates text report of monitored flow and bus voltage violations of ACCC post-contingency, post-tripping or post-corrective action solution. For each monitored element flow and bus voltage, only the worst violation and contingency which results that violation are reported. That is each element violations is reported once and that violation is the worst violation.

Python syntax:

```
ierr = pssarrays.accc_violations_report
(accfile,stype,busmsm,sysmsm,rating,flowlimit,rptfile)
```

where:

string	accfile	Is the name of Contingency Solution Output file (input, no default allowed), maximum string length=260
string	stype	Is the name of solution type (input, *contingency* by default allowed), allowed values: *contingency*, *con*, *tripping*, *trp*, *caction*, *contingency action*, *cor*
float	busmsm	Is the value of bus mismatch tolerance in MVA (input, 0.5 MVA by default)
float	sysmsm	Is the value of system mismatch tolerance in MVA (input, 5.0 MVA by default)
string	rating	Is the name of rating ('a' or 'b' or 'c') to use for calculating percentage loading (input, 'a' by default), maximum string length=1
float	flowlimit	Is the value of percent of flow rating (input, 100% by default)
string	rptfile	Is the name of report file (input, PSS®E Report by default), maximum string length=260

returns:

integer	rlst.ierr	Is the error code =0 no error occurred =4 error opening accfile =5 error reading accfile =6 prerequisite requirements for function not met

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17.7. ASCC_CURRENTS

This function returns short circuit currents and Thevenin impedance in three-phase ac systems from API. Results are returned for bus faults only. Results for line out and line end faults are not returned.

Python syntax:

```
rlst = pssarrays.ascc_currents
  (sid,all,flt3ph,fltlg,fltllg,fltll,linout,linend,voltop,genxop,
  tpunty,dcload,zcorec,lnchrg,shntop,loadop,machpq,volts,relfil,fcdfil,
  scfile)
```

int	sid	Is a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default).
int	all	Is the *all buses*or selected subsystem flag (input; 1 by default) = 1 process all buses = 0 process only buses in subsystem SID
int	flt3ph	Is to report three phase faults (0 by default). = 0 omit = 1 include
int	fltlg	Is to report line-to-ground faults (0 by default). = 0 omit = 1 include
int	fltllg	Is to report line-line-to-ground faults (0 by default). = 0 omit = 1 include
int	fltll	Is to report line-to-line faults (0 by default). = 0 omit = 1 include
int	linout	Is to report line-out faults (0 by default). = 0 omit = 1 include
int	linend	Is to report line-end faults (0 by default). = 0 omit = 1 include
int	voltop	Is the bus voltage option (0 by default).

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		= 0 use bus voltages from power flow solution = 1 set all bus voltages at specified value and at 0 deg = 2 set faulted bus voltage at specified value and at 0 deg
int	genxop	Is the synchronous machine reactance option, ignored if generators are already “converted” (0 by default). = 0 use subtransient reactance = 1 use transient reactance = 2 use synchronous reactance
int	tpunty	Is the transformer tap tatios and phase angle option (0 by default) = 0 leave tap ratios and phase shift angles unchanged = 1 set tap ratios to 1.0 pu and phase shift angles to 0 deg = 2 set tap ratios to 1.0 pu and phase shift angles unchanged = 3 set tap ratios unchanged and phase shift angles to 0 deg
int	dcload	Is the dc line and FACTS device option (0 by default) = 0 blocked = 1 represent as load
int	zcorec	Is the zero sequence transformer impedance correction option (0 by default) = 0 ignore = 1 apply
int	lnchrg	Is the line charging option (0 by default) = 0 leave unchanged = 1 set to 0.0 in the positive and negative sequences = 2 set to 0.0 in all sequences
int	shntop	Is the line shunts, fixed shunts, switched shunts and transformer magnetizing admittance option (0 by default)

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		= 0 leave unchanged = 1 set to 0.0 in the positive and negative sequences = 2 set to 0.0 in all sequences
int	loadop	Is the load option (0 by default) = 0 leave unchanged = 1 set to 0.0 in the positive and negative sequences = 2 set to 0.0 in all sequences
int	machpq	Is the synchronous and asynchronous machines power output option (0 by default) = 0 use real and reactive power outputs from power flow solution = 1 set real and reactive power outputs to 0.0
int	volts	Is the user specified bus voltage value in pu (1.0 by default) This is used only when voltop = 1 or 2.
str	relfil	Is the Relay Output Data (.rel) file; blank for none (input; blank by default); maximum string length=260
str	fcdfil	Is the name of fault control (.fcd) file; blank for none (input; blank by default); maximum string length=260.
str	scfile	Is the ASCC_3 Results (.sc) file; blank for none (output; blank by default); maximum string length=260.

Returned object 'rlst' contains the following attributes:

Index i is the ith faulted bus.

int	rlst.ierr	Is the error code = 0, no error occurred = 1, invalid SID value or subsystem SID is not defined = 2, invalid ALL value = 3, invalid STATUS value = 4, line end fault invalid; bus table is full = 5, error opening RELFIL = 6, error opening FCDFIL

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		= 7, prerequisite requirements for function not met = 8, no faults specified = 9, error opening SCFILE = 10, SCFILE error = 11, bus SC results not available. = 12, branch SC results not available. = 13, three winding transformer SC results not available. = 14, incorrect SCFILE. = 15, SCFILE not allowed.
str	rlst.scunit	Is the units of returned fault currents = pu = physical (r.m.s. amperes)
str	rlst.scfmt	Is the coordinates of returned fault currents = rectangular = polar
int	rlst.fltbus	Is a list of faulted bus numbers
complex	rlst.flt3ph[i].ia1	Is three phase fault, positive sequence current = I"k
complex	rlst.flt3ph[i].ia2	Is three phase fault, negative sequence current
complex	rlst.flt3ph[i].ia0	Is three phase fault, zero sequence current
complex	rlst.flt3ph[i].ia	Is three phase fault, Phase A current
complex	rlst.flt3ph[i].ib	Is three phase fault, Phase B current
complex	rlst.flt3ph[i].ic	Is three phase fault, Phase C current
complex	rlst.fltlg[i].ia1	Is line to ground fault, positive sequence current
complex	rlst.fltlg[i].ia2	Is line to ground fault, negative sequence current
complex	rlst.fltlg[i].ia0	Is line to ground fault, zero sequence current = I"k/3
complex	rlst.fltlg[i].ia	Is line to ground fault, Phase A current
complex	rlst.fltlg[i].ib	Is line to ground fault, Phase B current
complex	rlst.fltlg[i].ic	Is line to ground fault, Phase C current
complex	rlst.fltllg[i].ia1	Is line line to ground fault, positive sequence current
complex	rlst.fltllg[i].ia2	Is line line to ground fault, negative sequence current
complex	rlst.fltllg[i].ia0	Is line line to ground fault, zero sequence current = I"k/3
complex	rlst.fltllg[i].ia	Is line line to ground fault, Phase A current
complex	rlst.fltllg[i].ib	Is line line to ground fault, Phase B current
complex	rlst.fltllg[i].ic	Is line line to ground fault, Phase C current

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complex	rlst.fltll[i].ia1	Is line to line fault, positive sequence current
complex	rlst.fltll[i].ia2	Is line to line fault, negative sequence current
complex	rlst.fltll[i].ia0	Is line to line fault, zero sequence current
complex	rlst.fltll[i].ia	Is line to line fault, Phase A current
complex	rlst.fltll[i].ib	Is line to line fault, Phase B current = I"k
complex	rlst.fltll[i].ic	Is line to line fault, Phase C current = -I"k
complex	rlst.thevz[i].z1	Is Thevenin impedance, positive sequence, in PU or OHM as set by SCUNIT
complex	rlst.thevz[i].z2	Is Thevenin impedance, negative sequence, in PU or OHM as set by SCUNIT
complex	rlst.thevz[i].z0	Is Thevenin impedance, zero sequence, in PU or OHM as set by SCUNIT
complex	rlst.thevzpu[i].z1	Is PU Thevenin impedance, positive sequence
complex	rlst.thevzpu[i].z2	Is PU Thevenin impedance, negative sequence
complex	rlst.thevzpu[i].z0	Is PU Thevenin impedance, zero sequence
complex	rlst.maxflt[i].ia1	Is maximum fault current, positive sequence current
complex	rlst.maxflt[i].ia2	Is maximum fault current, negative sequence current
complex	rlst.maxflt[i].ia0	Is maximum fault current, zero sequence current
complex	rlst.maxflt[i].ia	Is maximum fault current, Phase A current
complex	rlst.maxflt[i].ib	Is maximum fault current, Phase B current
complex	rlst.maxflt[i].ic	Is maximum fault current, Phase C current
str	rlst.maxfltdsc[i]	Is description of fault condition for maximum fault current

Note: Maximum fault current is the largest fault current among analyzed faults at a bus.

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17.8. IECS_CURRENTS

This function returns short circuit currents and Thevenin impedance in three-phase ac systems according to IEC 60909 standard (IECS_3 API). Results are returned for bus faults only. Results for line out and line end faults are not returned.

Python syntax:

```
rlst = pssarrays.iecs_currents(sid, all, flt3ph, fltlg, fltllg, fltll,
 fltloc, linout, linend, tpunty, lnchrg, shntop, dcload, zcorec, optnftrc,
 loadop, genxop, brktime, vfactorc, iecfil, fcdfil, scfile)
```

where:

int	sid	Is a valid subsystem identifier. Valid subsystem identifiers range from 0 to 11. Subsystem SID must have been previously defined (input; 0 by default)
int	all	Is the *all buses* or specified subsystem flag (input; 1 by default). = 1 process all buses. = 0 process only buses in subsystem SID.
int	flt3ph	Is to report three phase faults, (0 by default). = 0 omit = 1 include
int	fltlg	Is to report line-to-ground faults (0 by default). = 0 omit = 1 include
int	fltllg	Is to report line-line-to-ground faults, (0 by default). = 0 omit = 1 include
int	fltll	Is to report line-to-line faults, (0 by default). = 0 omit = 1 include
int	fltloc	Is the location of fault (1 by default) =0 fault at network bus =1 fault at LV bus of Power Station Unit (PSU) =2 fault at auxiliary transformer (connected to PSU) LV bus
int	linout	Is to report line-out faults (0 by default).

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		= 0 omit = 1 include
int	linend	Is to report line-end faults (0 by default). = 0 omit = 1 include
int	tpunty	Is the transformer tap ratios and phase angle option (0 by default) = 0 leave tap ratios and phase shift angels unchanged = 1 set tap ratios to 1.0 pu and phase shift angels to 0 deg = 2 set tap ratios to 1.0 pu and phase shift angels unchanged = 3 set tap ratios unchanged and phase shift angels to 0 deg
int	lnchrg	Is the line charging option (1 by default). = 0 leave unchanged = 1 set to 0.0 in the positive and negative sequences = 2 set to 0.0 in all sequences
int	shntop	Is the line shunts, fixed shunts, switched shunts and transformer magnetizing admittance option (1 by default). = 0 leave unchanged = 1 set to 0.0 in the positive and negative sequences = 2 set to 0.0 in all sequences
int	dcload	Is the dc line and FACTS device option (0 by default) = 0 blocked = 1 represent as load
int	zcorec	Is the zero sequence transformer impedance correction option (0 by default) = 0 ignore = 1 apply
int	optnftrc	Is the option for Voltage Factor C

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		= 0, Voltage Factor C for maximum fault currents (default) = 1, Voltage Factor C for minimum fault currents = 2, Voltage Factor C as specified by 'vfactorc' value
int	loadop	Is the load option (1 by default) = 0 leave unchanged = 1 set to 0.0 in the positive and negative sequences = 2 set to 0.0 in all sequences
int	genxop	Is the synchronous machine reactance option, ignored if generators are already “converted” (0 by default). = 0 use subtransient reactance = 1 use transient reactance = 2 use synchronous reactance
float	brktime	Is the breaker contact parting time in seconds (5/BaseFrequency by default)
float	vfactorc	Is the user specified Voltage Factor C This is used only when optnftrc=3.
str	iecfil	Is the name of IEC data (.iec) file; blank for none (input; blank by default); maximum string length=260
str	fcdfil	Is the name of fault control (.fcd) file; blank for none (input; blank by default); maximum string length=260
str	scfile	Is the IECS_3 Results (.sc) file; blank for none (output; blank by default); maximum string length=260.

Returned object ’rlst’ contains the following attributes::

Index i is the ith faulted bus.

int	rlst.ierr	Is the error code = 0 no error occurred. = 1 invalid sid value or subsystem sid is not defined. = 2 invalid all value. = 3 invalid STATUS value. = 4 invalid brktime value. = 5 no faults specified.

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		= 6 error opening iecfile. = 7 error opening fcdfile. = 8 prerequisite requirements for API are not met. = 9 error opening SCFILE = 10 SCFILE error = 11 bus SC results not available. = 12 branch SC results not available. = 13 three winding transformer SC results not available. = 14 incorrect SCFILE. = 15 SCFILE not allowed.
str	rlst.scunit	Is the units of returned fault currents = pu = physical (r.m.s. amperes)
str	rlst.scfmt	Is the coordinates of returned fault currents = rectangular = polar
int	rlst.fltbus	Is a list of faulted bus numbers
complex	rlst.flt3ph[i].ia1	Is three phase fault, positive sequence current = I"k
complex	rlst.flt3ph[i].ia2	Is three phase fault, negative sequence current
complex	rlst.flt3ph[i].ia0	Is three phase fault, zero sequence current
complex	rlst.flt3ph[i].ia	Is three phase fault, Phase A current
complex	rlst.flt3ph[i].ib	Is three phase fault, Phase B current
complex	rlst.flt3ph[i].ic	Is three phase fault, Phase C current
complex	rlst.flt3ph[i].ipb	Is three phase fault, peak current - Method B, ip(B)
complex	rlst.flt3ph[i].ipc	Is three phase fault, peak current - Method C, ip(C)
complex	rlst.flt3ph[i].idc	Is three phase fault, dc component of asymmetrical breaking current, idc
complex	rlst.flt3ph[i].ibsym	Is three phase fault, symmetrical breaking current (r.m.s.), ib(sym)
complex	rlst.flt3ph[i].ibuns	Is three phase fault, symmetrical breaking current (r.m.s.), ib(uns)
complex	rlst.fltlg[i].ia1	Is line to ground fault, positive sequence current
complex	rlst.fltlg[i].ia2	Is line to ground fault, negative sequence current
complex	rlst.fltlg[i].ia0	Is line to ground fault, zero sequence current = I”k/3
complex	rlst.fltlg[i].ia	Is line to ground fault, Phase A current

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complex	rlst.fltlg[i].ib	Is line to ground fault, Phase B current
complex	rlst.fltlg[i].ic	Is line to ground fault, Phase C current
complex	rlst.fltlg[i].ipb	Is line to ground fault, peak current - Method B, ip(B)
complex	rlst.fltlg[i].ipc	Is line to ground fault, peak current - Method C, ip(C)
complex	rlst.fltlg[i].idc	Is line to ground fault dc component of asymmetrical breaking current
complex	rlst.fltlg[i].ibsym	Is line to ground fault, symmetrical breaking current (r.m.s.)
complex	rlst.fltlg[i].ibuns	Is line to ground fault, asymmetrical breaking current (r.m.s.)
complex	rlst.fltllg[i].ia1	Is line line to ground fault, positive sequence current
complex	rlst.fltllg[i].ia2	Is line line to ground fault, negative sequence current
complex	rlst.fltllg[i].ia0	Is line line to ground fault, zero sequence current = I"k / 3
complex	rlst.fltllg[i].ia	Is line line to ground fault, Phase A Current
complex	rlst.fltllg[i].ib	Is line line to ground fault, Phase B Current
complex	rlst.fltllg[i].ic	Is line line to ground fault, Phase C Current
complex	rlst.fltllg[i].ipb	Is line line to ground fault, peak current - Method B, ip(B)
complex	rlst.fltllg[i].ipc	Is line line to ground fault, peak current - Method C, ip(C)
complex	rlst.fltllg[i].idc	Is line line to ground fault dc component of asymmetrical breaking current, idc
complex	rlst.fltllg[i].ibsym	Is line line to ground fault, symmetrical breaking current (r.m.s.), ib(sym)
complex	rlst.fltllg[i].ibuns	Is line line to ground fault, asymmetrical breaking current (r.m.s.), ib(uns)
complex	rlst.fltll[i].ia1	Is line to line fault, positive sequence current
complex	rlst.fltll[i].ia2	Is line to line fault, negative sequence current
complex	rlst.fltll[i].ia0	Is line to line fault, zero sequence current
complex	rlst.fltll[i].ia	Is line to line fault, Phase A current
complex	rlst.fltll[i].ib	Is line to line fault, Phase B current = I”k
complex	rlst.fltll[i].ic	Is line to line fault, Phase C current = -I”k
complex	rlst.fltll[i].ipb	Is line to line fault, peak current - Method B, ip(B)
complex	rlst.fltll[i].ipc	Is line to line fault, peak current - Method C, ip(C)
complex	rlst.fltll[i].idc	Is line to line fault, dc component of asymmetrical breaking current, idc
complex	rlst.fltll[i].ibsym	Is line to line fault, symmetrical breaking current (r.m.s.), ib(sym)
complex	rlst.fltll[i].ibuns	Is line to line fault, asymmetrical breaking current (r.m.s.), ib(uns)
complex	rlst.thevz[i].z1	Is PU Thevenin impedance, positive sequence

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complex	rlst.thevzpu[i].z2	Is PU Thevenin impedance, negative sequence
complex	thevzpu[i].z0	Is PU Thevenin impedance, zero sequence
complex	rlst.maxflt[i].ia1	Is maximum fault current, positive sequence current
complex	rlst.maxflt[i].ia2	Is maximum fault current, negative sequence current
complex	rlst.maxflt[i].ia0	Is maximum fault current, zero sequence current
complex	rlst.maxflt[i].ia	Is maximum fault current, Phase A current
complex	rlst.maxflt[i].ib	Is maximum fault current, Phase B current
complex	rlst.maxflt[i].ic	Is maximum fault current, Phase C current
str	rlst.maxfltdsc[i]	Is description of fault condition for maximum fault current

Note: Maximum fault current is the largest fault current among analyzed faults at a bus.

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17.9. OTDF_FACTORS

This function returns the OTDF factors array. The PSS®E case must be opened prior to execution of this function. Also the dfxfile provided must correspond to the PSS®E case opened.

Python syntax:

```
rlst = pssarrays.otdf_factors(dfxfile)
```

where:

string	dfxfile	Is the name of distribution factor .dfx file (input, no default allowed), maximum string length=260

Returned object 'rlst' contains the following attributes:

integer	rlst.ierr	Is the error code =0 no error occurred =1 no Distribution Factor Data File specified =2 generators are converted =3 buses in island(s) without a swing bus; use activity TREE =4 error opening file DFXFILE =5 file DFXFILE is not in the form of a PSS®E-25 or later DFAX file; run DFAX =6 file does not contain distribution factors; run DFAX with distribution factor option enabled =7 no monitored branches or interfaces defined =8 prerequisite requirements for API are not met =9 requires opened SAV file, for retrieving OTDF factors
integer	rlst.size.nmline	Is the number of monitored branches
integer	rlst.size.ninter	Is the number of monitored interfaces
integer	rlst.size.ncase	Is the number of contingencies + 1 (for base case)
string	rlst.casetitle.line1	Is the case short title line 1, maximum string length=60
string	rlst.casetitle.line2	Is the case short title line 2, maximum string length=60
string	rlst.file.sav	Is the name of saved case (.sav) file, maximum string length=260
string	rlst.file.dfx	Is the name of distribution factor data (.dfx) file, maximum string length=260

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string	rlst.file.sub	Is the name of subsystem definition data (.sub) file, maximum string length=260
string	rlst.file.mon	Is the name of monitored element data (.mon) file, maximum string length=260
string	rlst.file.con	Is the name of contingency description data (.con) file, maximum string length=260
string	rlst.melement	Is a list of length (nmline+ninter) of monitored branch and interface names, maximum string length=54
string	rlst.colabel	Is a list of length (ncase) of contingency labels, maximum string length=12
string	rlst.codesc	Is a list of length (ncase) of contingency description, maximum string length=152
float	rlst.factor	Is a list of list OTDF factors, list of ncase elements with each element is a list of nmline+ninter values

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17.10. PV_SUMMARY

This functions returns PV analysis summary.

Python syntax:

```
rlst = pssarrays.pv_summary(pvfile)
```

where:

string	pvfile	PV output filename (.pv), (input, no default allowed), maximum string length=260

Returned object 'rlst' contains the following attributes:

int	rlst.ierr	error code: =0 no error =1 file not PV type =4 requires SAV file
int	rlst.pvsize.ncase	number of contingencies + 1 (for base case)
int	rlst.pvsize.nmline	number of monitored branches
int	rlst.pvsize.ninter	number of monitored interfaces
int	rlst.pvsize.nmvbus	number of voltage monitored buses
int	rlst.pvsize.nmvrec	number of voltage monitored records
int	rlst.pvsize.nmgnbus	number of monitored plant (generator) buses
int	rlst.pvsize.nmldbus	number of monitored load buses
int	rlst.pvsize.nmxtrns	maximum number of MW transfer changes
int	rlst.options	PV solution options (same as in API manual)
float	rlst.values	PV solution values (same as in API manual)
string	rlst.brnflowunits.xfrcur	PV solution transformer branch flow units ('mbrnamp' in pv_solution return list)
string	rlst.brnflowunits.nxfrcr	PV solution non-transformer branch flow units ('mbrnamp' in pv_solution return list)
string	rlst.casetitle.line1	short title line 1
string	rlst.casetitle.line2	short title line 2
string	rlst.file.pv	contingency output (.acc) filename
string	rlst.file.sav	saved case (.sav) filename
string	rlst.file.ecd	economic dispatch data (.ecd) filename
string	rlst.file.thr	load throwover data (.thr) filename
string	rlst.file.dfx	distribution factor data (.dfx) filename
string	rlst.file.sub	subsystem definition data (.sub) filename
string	rlst.file.mon	monitored element data (.mon) filename
string	rlst.file.con	contingency description data (.con) filename
string	rlst.srcsink	source and sink subsystem names

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string	rlst.mbranch	monitored branch names
float	rlst.mbrnrating.a	rating A
float	rlst.mbrnrating.b	rating B
float	rlst.mbrnrating.c	rating C
string	rlst.minterface	monitored interface names
float	rlst.mitfrating	selected rating of monitored interface
string	rlst.mgenbus	monitored plant (generator) bus label
string	rlst.mloadbus	monitored load bus label
string	rlst.mvbuslabel	monitored voltage bus label
string	rlst.mvreclabel	monitored voltage record label
float	rlst.mvrecmax	monitored voltage bus maximum
float	rlst.mvrecmin	monitored voltage bus minimum
string	rlst.mvrectype	monitored voltage record type (range/deviation)
string	rlst.colabel	contingency labels
string	rlst.codesc	contingency description
float	rlst.maxmw	maximum MW transfer

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17.11. PV_SUMMARY_REPORT

This function creates PV analysis summary text report.

Python syntax:

```
ierr = pv_summary_report(pvfile,rptfile)
```

where:

string	pvfile	PV output filename (.pv), (input, no default allowed), maximum string length=260
string	rptfile	Is the name of report file (input, PSS®E Report by default), maximum string length=260

Returns:

int	rlst.ierr	error code: =0 no error =1 file not PV type =4 requires SAV file

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17.12. PV_SOLUTION

This functions returns PV analysis monitored flows and bus voltages for one contingency.

Python syntax:

```
rlst = pssarrays.pv_solution(pvfile,colabel)
```

where:

string	pvfile	PV output filename (.pv), (input, no default allowed), maximum string length=260
string	colabel	Is the name of contingency label to get PV solution for, only one label allowed (input, no default allowed), maximum string length=12

Returned object 'rlst' contains the following attributes:

int	rlst.ierr	error code: =0 no error =1 file not PV type =4 requires SAV file
int	rlst.island	number of islands
float	rlst.mwtransfer	MW transactions, list of size [ntrns]
boolean	rlst.cnvflag	convergence flag (True when converged), list of size [ntrns]
string	rlist.cnvcond	convergence condition
float	rlst.mvaworst	largest bus MVA mismatch, list of size [ntrns]
float	rlst.mvatotal	total system MVA mismatch, list of size [ntrns]
float	rlst.volts	monitored bus voltage (pu), list of size [nmvbus][ntrns]
float	rlst.mgenmw	monitored plant MW, list of size [nmgnbus][ntrns]
float	rlst.mgenmvar	monitored plant MVAR, list of size [nmgnbus][ntrns]
float	rlst.mloadmw	monitored load MW, list of size [nmldbus][ntrns]
float	rlst.mloadmvar	monitored load MVAR, list of size [nmldbus][ntrns]
float	rlst.mbrnmva	monitored branch MVA flow (MVA), list of size [nmline][ntrns]
float	rlst.mbrnamp	monitored branch flow, AMPS expressed in MVA or MVA, list of size [nmline][ntrns]. The brnflowunits variable in pv_summary determine the units. Use this value to calculate branch loadings.
float	rlst.mitfmw	monitored interface MW flow (MW), list of size [ninter][ntrns]

where

ntrans =number of MW transfer changes

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nmvbus =number of voltage monitored buses

nmgnbus =number of monitored plant (generator) buses

nmldbus =number of monitored load buses

nmline =number of monitored branches

ninter =number of monitored interfaces

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17.13. PV_SOLUTION_REPORT

This function creates PV analysis solution text report.

Python syntax:

```
ierr = pv_solution_report(pvfile,colabels,rptfile)
```

where:

string	pvfile	PV output filename (.pv), (input, no default allowed), maximum string length=260
string	colabels	Is the list of contingency label names (input, all contingencies in accfile by default), maximum string length=12
string	rptfile	Is the name of report file (input, PSS®E Report by default), maximum string length=260

Returns:

int	rlst.ierr	error code: =0 no error =1 file not PV type =4 requires SAV file

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17.14. QV_SOLUTION

This functions returns QV analysis monitored flows and bus voltages for one contingency.

Python syntax:

```
rlst = pssarrays.qv_solution(qvfile,colabel)
```

where:

string	qvfile	QV output filename (.qv), (input, no default allowed), maximum string length=260
string	colabel	Is the name of contingency label to get QV solution for, only one label allowed (input, no default allowed), maximum string length=12

Returned object 'rlst' contains the following attributes:

int	rlst.ierr	error code: =0 no error =2 file not QV type =5 requires SAV file
int	rlst.island	number of islands
float	rlst.vsetpoint	voltage setpoints, list of size [nvstp]
boolean	rlst.cnvflag	convergence flag (True when converged), list of size [nvstp]
string	rlst.convcond	convergence condtion
float	rlst.mvaworst	largest bus MVA mismatch, list of size [nvstp]
float	rlst.mvatotal	total system MVA mismatch, list of size [nvstp]
float	rlst.volts	monitored bus voltage (pu), list of size [nmvbus][nvstp]
float	rlst.mgenmvar	monitored plant Mvar, list of size [nmgnbus][nvstp]

where

ntrans = number of voltage setpoint changes

nmvbus = number of voltage monitored buses

nmgnbus = number of monitored plant (generator) buses

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17.15. QV_SOLUTION_REPORT

This function creates QV analysis solution text report.

Python syntax:

```
ierr = qv_solution_report(qvfile,colabels,rptfile)
```

where:

string	qvfile	QV output filename (.qv), (input, no default allowed), maximum string length=260
string	colabels	Is the list of contingency label names (input, all contingencies in accfile by default), maximum string length=12
string	rptfile	Is the name of report file (input, PSS®E Report by default), maximum string length=260

Returned list (accessed by attributes as):

int	rlst.ierr	error code: =0 no error =2 file not QV type =5 requires SAV file

17.16. QV_SUMMARY

This functions returns QV analysis summary.

Python syntax:

```
rlst = pssarrays.qv_summary(qvfile)
```

where:

string	qvfile	QV output filename (.qv), (input, no default allowed), maximum string length=260

Returned object 'rlst' contains the following attributes:

int	rlst.ierr	error code: =0 no error =2 file not QV type, =5 requires SAV file
int	rlst.qvbus	QV analysis bus name
int	rlst.qvsize.ncase	number of contingencies + 1 (for base case)
int	rlst.qvsize.nmvbus	number of voltage monitored buses
int	rlst.qvsize.nmvrec	number of voltage monitored records
int	rlst.qvsize.nmgnbus	number of monitored plant (generator) buses
int	rlst.qvsize.nmxvstp	maximum number of voltage setpoint changes
int	rlst.options	QV solution options (same as in API manual)
string	rlst.values	QV solution values (same as in API manual)
string	rlst.casetitle.line1	short title line 1
string	rlst.casetitle.line2	short title line 2
string	rlst.file.qv	contingency output (.acc) filename
string	rlst.file.sav	saved case (.sav) filename
string	rlst.file.thr	load throwover data (.thr) filename
string	rlst.file.dfx	distribution factor data (.dfx) filename
string	rlst.file.sub	subsystem definition data (.sub) filename
string	rlst.file.mon	monitored element data (.mon) filename
string	rlst.file.con	contingency description data (.con) filename
string	rlst.mgenbus	monitored plant (generator) bus label
string	rlst.mvbuslabel	monitored voltage bus label
string	rlst.mvreclabel	monitored voltage record label
float	rlst.mvrecmax	monitored voltage bus maximum
float	rlst.mvrecmin	monitored voltage bus minimum
string	rlst.mvrectype	monitored voltage record type (range/deviation)
string	rlst.colabel	contingency labels
string	rlst.codesc	contingency description

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float	rlst.minvstp	minimum voltage setpoint
float	rlst.maxvstp	maximum voltage setpoint
float	rlst.minmvar	minimum MVAR change
float	rlst.maxmvar	maximum MVAR change
float	rlst.maxmsm	maximum MVAR mismatch

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17.17. QV_SUMMARY_REPORT

This function creates QV analysis summary text report.

Python syntax:

```
ierr = qv_summary_report(qvfile,rptfile)
```

where:

string	qvfile	QV output filename (.qv), (input, no default allowed), maximum string length=260
string	rptfile	Is the name of report file (input, PSS®E Report by default), maximum string length=260

Returned list (accessed by attributes as):

int	rlst.ierr	error code: =0 no error =2 file not QV type =5 requires SAV file

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Chapter 18
PSSEXCEL Module

The pssexcel module provides Python functions to export PSS®E data or solution results to Excel spreadsheets.

These functions do not record themselves when called while recording is enabled in PSS®E.

Please note that these routines are members of the pssexcel Python extension module, not psspy. The syntax on the following pages assumes that the module has been imported by executing:

```
import pssexcel
```

18.1. ACCC

This function exports ACCC analysis results to Excel Spreadsheet.

Python syntax:

```
pssexcel.accc(accfile, string, colabel='',
stype='contingency', busmsm=0.5, sysmsm=5.0, rating='a', namesplit=True,
xlsfile='', sheet='', overwritesheet=True, show=True, ratecon='b',
 baseflowvio=True,
basevoltvio=True, flowlimit=100.0, flowchange=0.0, voltchange=0.0)
```

where:

string	accfile	ACCC analysis output filename (.acc), no default allowed
string	string	Name or list of names indicating which results to export, no default allowed. Allowed strings: 's' or 'summary' ACCC Analysis Summary 'e' or 'events' Contingency Events Description 'b' or 'branch' Monitored Branch Flow (MVA) 'i' or 'interface' Monitored Interface Flow (MW) 'v' or 'voltage' Monitored Bus Voltage 'l' or 'load' Loads Shed (MW) 'g' or 'generator' Generator Dispatch (MW) 'p' or 'phase shifter' Phase Shifter Angle

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		Example: string =
		
		`'v`
		
		' or
		
		`['s','v','g','l','b','i']`

string	colabel	Contingency label or list of contingency labels for which the solution is exported, default - all contingencies, not required when exporting only summary
string	stype	Is the name of solution type to get (input, contingency by default), allowed values: contingency, con, tripping, trp, caction, contingency action, cor
float	busmsm	Is the value of bus mismatch tolerance in MVA (input, 0.5 MVA by default)
float	sysmsm	Is the value of system mismatch tolerance in MVA (input, 5.0 MVA by default)
string	rating	s the value of rating to use to calculate base case percentage overload, default 'a', allowed values:'a', 'b', or 'c'
string	stype	Is the name of solution type to get (input, *contingency* by default allowed), allowed values: *contingency*, *con*, *tripping*, *trp*, *caction*, *contingency action*, *cor*
float	busmsm	Is the value of bus mismatch tolerance in MVA (input, 0.5 MVA by default)
float	sysmsm	Is the value of system mismatch tolerance in MVA (input, 5.0 MVA by default)
string	rating	Rating to use to calculate percentage overload. Default 'a', allowed values: 'a', 'b', or 'c'.
boolean	namesplit	Split extended bus names, default True = True bus names split into three values: number name and bus voltage (three columns in worksheet) = False bus names are kept as single value string (one column in worksheet)
string	xlsfile	Excel workbook name, default Book#.xls

string	sheet	Excel worksheet name, default (blank string), depending on string input, worksheets created are:
		
		`Summary`
		
		`ACCC`
		`Analysis Summary`
		
		`Contingency Events`
		
		`Events Description`

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		Branch Flow Monitored Branch Flow (MVA) Interface Flow Monitored Interface Flow (MW) Bus Voltage Monitored Bus Voltage Load Shed Loads Shed (MW) Generator Dispatch Generator Dispatch (MW) Phase Shifter Angle Phase Shifter Angle (degrees)
		All worksheet names are prefixed with sheet name, when provided.
boolean	overwritesheet	Overwrite worksheets flag, default True = True existing worksheets are overwritten = False existing worksheets are copied and their names appended with (#) where # is next sequence number.
boolean	show	Show or Hide Excel Spreadsheet flag, default True = True open and show Excel Spreadsheet = False do not show Excel Spreadsheet (create and save)
string	ratecon	Is the value of rating to use to calculate Contingency Case percentage overload default 'b', allowed values:'a', 'b', or 'c'

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boolean	baseflowvio	Exclude elements with base case loading violations from contingency reports default True, allowed values: True or False
boolean	basevoltvio	Exclude buses with base case voltage range violations from contingency reports default True, allowed values: True or False
float	flowlimit	Percent of flow rating, default=100 for 100% Show monitored elements with flow above flowlimit value. flowlimit=0 will export flows for all elements.
float	flowchange	Minimum contingency case flow change in MVA for overload reports default 0 MVA, allowed values: Any number
float	voltchange	Minimum contingency case voltage change in pu for voltage range violations default 0 pu, allowed values: Any number

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18.2. iec_data_file

This function exports synchronous machines, asynchronous machines and transformer data from PSS®E Saved Case required to assemble IEC fault data input file as an Excel® spreadsheet. IEC data file (.iec) is used by IECS API. The PSS ®E Saved case must be opened prior to execution of this function.

It creates three worksheets 'Machines', 'Transformers' and 'IEC Data'.

This file is used as a convenient way to create an .iec file by copy/paste generator and transformer records to form a power station unit, modify the generator, motor and transformer data as required by IECS API. After appropriate modifications, save IEC Data worksheet as comma separated file with extension .iec. This file then can be used as *Fault data input file* to the IECS API.

Python syntax:

```
pssexcel.iec_data_file(xlsfile, sheet, overwritesheet,
show)
```

where:

string	xlsfile	Is the name of excel workbook filename (input, default savefile prefix and extension .iec)

string	sheet	Excel worksheet name, default (blank string), worksheets created are:
		
		`Machines`
		
		`Saved case machines data`
		
		`Transformers`
		
		`Saved case two and three windings transformers data`
		
		`IEC Data`
		
		`Data required by IECS API`
		
		All worksheet names are prefixed with sheet name, when provided.

boolean	overwritesheet	Overwrite worksheets flag, default True = True existing worksheets are overwritten = False existing worksheets are copied and their names appended with (#), where # is next sequence number.
boolean	show	Show or Hide Excel Spreadsheet flag, default True = True open and show Excel Spreadsheet

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= False do not show Excel Spreadsheet (create and save)

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18.3. pv

This function exports PV solution results to Excel Spreadsheet.

Python syntax:

```
pssexcel.pv(pvfile, string, colabel, namesplit,
xlsfile, sheet, overwritesheet, show)
```

where:

string	pvfile	PV analysis output filename (.pv), no default allowed
string	string	Name or list of names indicating which results to export, no default allowed. Allowed strings:

```
's' or 'summary'

PV
Solution Summary

'v' or 'voltage'

Monitored
Bus Voltage

'm' or 'mismatch'

Largest and Total Mismatch

'g' or 'generator'

Monitored Plants MW and MVAR

'l' or 'load'

Monitored
Loads MW and MVAR

'b' or 'branch'

Monitored
Branch Flow (MVA)

'i' or 'interface'

Monitored Interface Flow (MW)
```

Example:

```
string = 'v' or

string =
 ['s','m','v','g','l','b','i']
```

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string	colabel	Contingency label or list of contingency labels for which the solution is exported, default - all contingencies, not required when exporting only summary
boolean	namesplit	Split extended bus names, default True = True bus names split into three values: number name and bus voltage (three columns in worksheet) = False bus names are kept as single value string (one column in worksheet)
string	xlsfile	Excel workbook name, default Book#.xls

string	sheet	Excel worksheet name, default ” (blank string), depending on string input, worksheets created are:
		
		`Summary`
		
		`PV`
		`Solution Summary`
		
		`Bus Voltage`
		
		`Monitored`
		`Bus Voltage`
		
		`Mismatch`
		
		`Largest`
		`and Total Mismatch`
		
		`Generator Dispatch`
		
		`Monitored Plants MW and MVAR`
		
		`Bus Load`
		
		`Monitored`
		`Loads MW and MVAR`
		
		`Branch Flow`
		
		`Monitored`
		`Branch Flow (MVA)`
		
		`Interface Flow`
		
		`Monitored`
		`Interface Flow (MW)`
		
		All worksheet names are prefixed with sheet name, when provided.

boolean	overwritesheet	Overwrite worksheets flag, default True = True existing worksheets are overwritten

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		= False existing worksheets are copied and their names appended with (#) where # is next sequence number.
boolean	show	Show or Hide Excel Spreadsheet flag, default True = True open and show Excel Spreadsheet = False do not show Excel Spreadsheet (create and save)

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18.4. qv

This function exports QV solution results to Excel Spreadsheet.

Python syntax:

```
pssexcel.qv(qvfile, string, colabel, namesplit,
xlsfile, sheet, overwritesheet, show)
```

where:

string	qvfile	QV analysis output filename (.qv), no default allowed

string	string	Name or list of names indicating which results to export, no default allowed. Allowed strings:
		`'s' or 'summary'`
		`QV Solution Summary`
		`'v' or 'voltage'`
		`Monitored Bus Voltage`
		`'m' or 'mismatch'`
		`Largest and Total Mismatch`
		`'g' or 'generator'`
		`Monitored Plants MW and MVAR`
		Example:
		`string = 'v' or`
		`string = ['s','m','v','g']`

string	colabel	Contingency label or list of contingency labels for which the solution is exported, default - all contingencies, not required when exporting only summary
boolean	namesplit	Split extended bus names, default True = True bus names split into three values: number name and bus voltage (three columns in worksheet) = False bus names are kept as single value string (one column in worksheet)
string	xlsfile	Excel workbook name, default Book#.xls
string	sheet	Excel worksheet name, default '' (blank string), depending on string input, worksheets created are:

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		Summary QV Solution Summary Bus Voltage Monitored Bus Voltage Mismatch Largest and Total Mismatch Generator Dispatch Monitored Plants MVAR All worksheet names are prefixed with sheet name, when provided.
boolean	overwritesheet	Overwrite worksheets flag, default True = True existing worksheets are overwritten = False existing worksheets are copied and their names appended with (#) where # is next sequence number.
boolean	show	Show or Hide Excel Spreadsheet flag, default True = True open and show Excel Spreadsheet = False do not show Excel Spreadsheet (create and save)

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Chapter 19
EXCELPY Module

The excelpy module provides Python functions to interface with Excel. Use this module to create, populate and format Excel workbooks from Python.

These functions do not record themselves when called while recording is enabled in PSS®E.

Please note that these routines are members of the excelpy Python extension module, not psspy. The syntax on the following pages assumes that the module has been imported by executing:

```
import excelpy
```

Following are the definition of arguments used by various functions defined in this module.

string or int	address	tuple of integers/strings. Allowed address when specifying cell: 'adrstr' or ('adrstr') or (row,'clnstr') or (row, column) Example: Cell address referencing 2nd row, column A, can be provided as- 'a2' or ('a2') or (2,'a') or (2,1) range: ('adrstr1','adrstr2') or (toprow, 'leftcolumnstr', bottomrow, 'rightcolumnstr') or (toprow, leftcolumn, bottomrow, rightcolumn) or ('adrstr1', (bottomrow, rightcolumn)) or ('adrstr1', (bottomrow, 'rightcolumnstr')) ((toprow, leftcolumn), 'adrstr2') or ((toprow, 'leftcolumnstr'), 'adrstr2')

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		where row, column, toprow, leftcolumn, bottomrow, rightcolumn are integer values, adrstr, adrstr1, adrstr2 are cell address string values, and clnstr, leftcolumnstr, rightcolumnstr are column name string values. Note: Column and address refernce string values are case-insensitive.
string	alignv	alignment value for cell, range, rows or columns. Allowed values when specifying horizontal alignment: 'left','right','h_center' vertical alignment: 'top','bottom','v_center'
string	borderv	border name. Allowed border names when formatting: cell: 'diagonaldown', 'diagnalup', 'top', 'bottom', 'left', 'right', 'none', 'outline' range: 'diagonaldown', 'diagonalup', 'top', 'bottom', 'left', 'right', 'inside', 'hinside', 'vinside', 'none', 'outline'
string or int	borderColor	string, color name; or integer, color index. Allowed color names: 'black', 'white', 'red', 'green', 'blue', 'yellow', 'magenta', 'cyan', 'brown', 'dgreen', 'dblue', 'orange'

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		Allowed ColorIndex: `1 to 56 as per ColorIndex property in Excel Visual Basic Reference.`
string	borderStyle	border style name. Allowed names: `'continuous', 'dash','dashdot', 'dashdotdot', 'dot', 'double', 'none', 'slantdashdot'`
string	borderWeight	border weight name. Allowed names: `'hairline', 'medium', 'thick', 'thin'`

string	fontStyle	font style name, specified as style name or tuple of style names.
		Allowed font style names:
		`'regular', 'bold', 'italic', 'underline'`
		Example:
		`(Italic,Bold) or`
		`'bold'`

string or int	fontColor	string, color name; or integer, color index. This is same as color in 'borderColor'.
string	numberFormat	number format Example: 0.000 for three decimal places
int	leftCol	left column number
string	newSheet	name of the new worksheet
string	oldSheet	name of the worksheet used as reference for worksheet operations
boolean	overwritesheet	overwrite if worksheet exists, Allowed values: = True existing worksheet will be overwritten. = False existing sheet copied and its name appended with (#), where # is next sequence number.

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		Used when creating new worksheets or re-naming existing worksheets.
string	sheetspreadsheet	name of the excel worksheet. If sheet is not provided, active worksheet used.
int	topRow	top row number
boolean	wrapText	wrap text or not, Allowed values: True or False
string	xlsfile	name of the Excel workbook

Default Workbook Style

The excel workbook is created with following default font style:

```
Black, Regular, Courier New, size 10, no wrapText,
General number format
```

How to Use?

Create xls file object as below and apply various methods defined here.

(1) When used to create new Excel files or add worksheets to existing Excel files:

testxls = excelpy.workbook()

testxls = excelpy.workbook(r"c:\working dir\ex1.xls")

testxls = excelpy.workbook(r"c:\working dir\ex1.xls", "MySheet") or

testxls = excelpy.workbook(r"c:\working dir\ex1.xls", "MySheet", False) or

testxls = excelpy.workbook(r"c:\working dir\ex1.xls", "MySheet", False, 'w') or

testxls = excelpy.workbook(xlsfile=r"c:\working dir\ex1.xls", sheet="MySheet",

overwritesheet=False, mode='w')

(2) When used to read existing Excel files:

testxls = excelpy.workbook(r"c:\working dir\ex1.xls", mode='r')

Note: When using Excel 2007 and later, create file in either 'xlsx' or 'xls' format.

Excel 2003 and earlier - default save format 'xls'.

Excel 2007 and later - default save format 'xlsx'.

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19.1. Methods

The following methods are members of the excelpy class that can be instantiated using the workbook functions defined in Section 19.2, "Workbook Functions". In all these methods, when sheet=None or not provided, active worksheet is used.

align(address, alignv, sheet=None)

Align Cell or Range.

cell, address = (row,col)

range, address = (topRow,leftCol,bottomRow,rightCol)

align_columns(address, alignv, sheet=None)

Align one or more columns.

One column, address = (row,col)

Multiple columns, address = (topRow,leftCol,topRow,rightCol)

align_rows(address, alignv, sheet=None)

Align one or more rows.

One row, address = (row,col)

Multiple rows, address = (topRow,leftCol,bottomRow,leftCol)

autofit_columns(address, sheet=None)

Autofit column or columns.

One column, address = (row,col)

Multiple columns, address = (topRow,leftCol,topRow,rightCol)

autofit_rows(address, sheet=None)

Autofit row or rows.

One row, address = (row,col)

Multiple rows, address = (topRow,leftCol,bottomRow,leftCol)

border(address, borderv, borderStyle='continuous', borderWeight='hairline', borderColor='black', sheet=None)

Format Cell or Range borders.

cell, address = (row,col)

range, address = (topRow,leftCol,bottomRow,rightCol)

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close()

Close active Excel workbook and quit Excel application.

delete_columns(address, howmany=1, sheet=None)

Delete columns to the right of specified row.

row address = (row,col)

Number of columns deleted is provided by integer value 'howmany'.

delete_rows(address, howmany=1, sheet=None)

Delete rows below specified row.

row address = (row,col)

Number of rows deleted is provided by integer value 'howmany'.

fontaddress, fontStyle=None, fontName=None, fontSize=None, fontColor=None, wrapText=False, numberFormat=None, sheet=None)

Set cell or range font properties: style, name, size, color, wrap, number format

Cell, address = (row,col)

Range, address = (topRow,leftCol,bottomRow,rightCol)

font_color(address, color, sheet=None)

Set cell or range font color.

Cell, address = (row,col)

Range, address = (topRow,leftCol,bottomRow,rightCol)

font_sheet(fontStyle=('Regular',), fontName='Courier

New', fontSize=10, fontColor='black', wrapText=False, numberFormat=None, sheet=None)

Set font properties for Excel sheet: style, name, size, color, wrap, number format

freezepanes(address, sheet=None)

Freeze worksheet panes.

address = (row,1) to freeze worksheet rows above 'row'

address = (1, col) to freeze worksheet columns left of 'col'

address = (row,col to freeze worksheet rows above 'row' and columns

left of 'col'

height(address, height, sheet=None)

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Set height of one row or multiple rows. Height is provided in points.

One row, address = (row,col)

Multiple rows, address = (topRow,leftCol,bottomRow,rightCol)

hide()

Hide opened Excel workbook.

merge(address, sheet=None)

Merge cells in the range.

page_footer(left=None, center=None, right=None, sheet=None)

Set page left, center, and right footers. Footer input value is a string.

page_format(orientation='portrait', left=1.0, right=1.0, top=1.0, bottom=1.0, header=0.5, footer=0.5, sheet=None)

Set page format: orientation, margins (given in inches).

```
   Orientation = 'portrait'  or 'p'  or 1
               = 'landscape' or 'l'  or 2
     Defaults:
     orientation = Portrait
     margins: left   = 1.0 inch
              right  = 1.0 inch
              top    = 1.0 inch
              bottom = 1.0 inch
              header = 0.5 inch
              footer = 0.5 inch
```

page_header(left=None, center=None, right=None, sheet=None)

Set page left, center, and right headers. Header input value is a string.

page_landscape(sheet=None)

Set page orientation to landscape.

page_margin(left=1.0, right=1.0, top=1.0, bottom=1.0, header=0.5, footer=0.5, sheet=None)

Set page margins. The margin value is given in inches.

page_portrait(sheet=None)

Set page orientation to portrait.

save(xlsfile=None)

Save active Excel workbook.

xlsfile = save(xlsfile)

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Returns name of the saved Excel file.

set_active_sheet(sheet)

Set 'sheet' as active worksheet.

set_cell(address, value, fontStyle=None, fontName=None, fontSize=None, fontColor=None, wrapText=False, numberFormat=None, sheet=None)

Set value to one cell.

set_range(topRow, leftCol, data, transpose=False, fontStyle=None, fontName=None, fontSize=None, fontColor=None, wrapText=False, numberFormat=None, sheet=None)

Set data to cells in the range.

bottomRow, rightCol = set_range(topRow,leftCol, data, ...)

Boolean, transpose: Allowed values: True or False

Excel methods write data row by row. If data in Python list of lists, or tuple of tuples or any combination of these, is not stored row by row, transpose that data. This functions works same as matrix transpose.

Returns:

bottomRow: Bottom row number after writing input data to worksheet.

rightCol : Right column number after writing input data to worksheet.

show( )

Show opened Excel workbook.

show_alerts(display)

Show or suppresses all POP-UP windows, like File Overwrite Yes/No/Cancel, Merge Overwrite.

Boolean, display: Allowed values: True or False, 1 or 0

= True or 1, show alerts

= False or 0, do not show alerts. Data will be overwritten.

transpose_data(data, transpose=True)

Transpose data.

Excel methods write data row by row. If data in Python list of lists, or tuple of tuples or any combination of these, is not stored row by row, transpose that data. This functions works same as matrix transpose.

nrows, nclns, datat = set_range(data, transpose=True)

data : any type of data to be tranposed

transpose: boolean, allowed values: True or False, True by default

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Returns:

nrows - number of rows in transposed data

nclns - number of columns in transposed data

datat - transposed data

width(address, width, sheet=None)

Set width of one column or multiple columns. Width is provided in points.

One column, address = (row,col)

Multiple columns, address = (topRow,leftCol,topRow,rightCol)

worksheet_add_after(newSheet=None, oldSheet=None, overwritesheet=True)spreadsheet

Add a newSheet after the oldSheet or active sheet. If newSheet name is not provided, Sheet# worksheet is added. If oldSheet name is not provided, worksheet is added at the end. Sheet# is a next available sheet number in the active workbook.

worksheet_add_before(newSheet=None, oldSheet=None, overwritesheet=True)

Add a newSheet before the oldSheet or active sheet. If newSheet name is not provided, Sheet# worksheet is added. If oldSheet name is not provided, worksheet is added at the end. Sheet# is a next available sheet number in the active workbook.spreadsheet

worksheet_add_begin(sheet=None, overwritesheet=True)

Add a sheet at the beginning. If sheet name is not provided, Sheet# worksheet is added. Sheet# is a next available sheet number in the active workbook.spreadsheet

worksheet_add_end(sheet=None, overwritesheet=True)spreadsheet

Add a sheet at the end. If sheet name is not provided, Sheet# worksheet is added. Sheet# is a next available sheet number in the active workbook.

worksheet_delete(sheet)

Delete a sheet from active workbook.

worksheet_left_columns2repeat(columns, sheet=None)spreadsheet

Repeat 'columns' at the left of each page of the worksheet.

'columns' is specified as tuple or list of column numbers,

e.g., columns = (1,3) to repeat columns 1 to 3 at the left or

columns = 1 to repeat column 1 at the left.

worksheet_rename(newSheet, oldSheet=None, overwritesheet=True)spreadsheet

Rename Excel 'oldSheet' or active sheet with newSheet name.

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overwritesheet = True, contents of oldSheet are deleted, so newSheet is blank worksheet

= False, contents of oldSheet are written to newSheet.\

worksheet_size_violation(topRow, leftCol, data=None, transpose=False)

Get worksheet size violations and extents.

Inputs:

data - List or List of Lists values to be written transpose - Logical

Excel methods write data row by row. If data in Python list of lists, or tuple of tuples or any combination of these, is not stored that way, transpose that data. This functions same as matrix transpose.

Returns:

Returned tuple contains:

row_violation = rtple[0], True when top/bottom row exceeds maximum allowable rows

col_violation = rtple[2], True when left/right column exceeds maximum allowable columns

topRow = rtple[3], top row number

leftCol = rtple[4], left column number

bottomRow = rtple[5], bottom row number

rightCol = rtple[6], right column number

wshtmaxrows = rtple[7], = 65536, maximum number of rows allowed in a worksheet

wshtmaxcols = rtple[8], = 256, maximum number of columns allowed in a worksheet

worksheet_top_rows2repeat(rows, sheet=None)spreadsheet

Repeat 'rows' at the top of each page of the worksheet.

'rows' is specified as tuple or list of row numbers,

e.g., rows = (1,3) to repeat rows 1 to 3 at the top or

rows = 1 to repeat row 1 at the top.

wraptext(address, sheet=None)

Wrap Text inside the cell or range.

Cell, address = (row,col)

Range, address = (topRow,leftCol,topRow,rightCol)

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19.2. Workbook Functions

Use this function to create an Excel workbook object that will possess the methods defined in Section 19.1, "Methods".

Python syntax:

```
excelpy.workbook(xlsfile=None, sheet=None,
overwritesheet=True)
```

Examples:

```
testxls = excelpy.workbook()

testxls = excelpy.workbook("c:\working dir\test1.xls")

testxls = excelpy.workbook("c:\working dir\test1.xls",
"MySheet")

testxls = excelpy.workbook("c:\working dir\test1.xls",
"MySheet",
                           False)

testxls = excelpy.workbook(xlsfile="c:\working
dir\test1.xls",
                           sheet="MySheet", overwritesheet=False)
```

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Chapter 20
GICDATA Module

The gicdata module provides Python functions to create the GIC data file required in GIC analysis.

These functions do not record themselves when called while recording is enabled in PSS®E.

Please note that these routines are members of the gicdata Python extension module, not psspy. The syntax on the following pages assumes that the module has been imported by executing:

```
import gicdata
```

GIC (Geomagnetically Induced Current) Analysis requires an additional GIC data file (See Section 7.2, GIC Data File Contents in the *Program Operation Manual*) that contains:

- Substation data
- Bus Substation data and
- Transformer data

Functions in the gicdata module help to create GIC data file applicable to a specific Power Flow Network Case for an entire network or just desired study subsystem.

Creating GIC data using the gicdata module is a two step process.

Step 1

Create GIC data file template using functions:

- template_excel(..)
- template_py(..)

This will create structure of GIC data that is required.

Step 2

Edit/Update the GIC data file template created in Step (1) to provide:

- Substation data
- Bus Substation data
- Transformer data

For transformers, vector group information must be provided. If power flow Saved Case has vector group information, GIC data file template will have this information as well. If PSSE Saved Case does not have vector group information, this needs to be provided. The GIC data file then can default values for all other transformer record data items.

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Once GIC data file template is updated, create GIC data file using functions:

- excel2gicfile(..)
- py2gicfile(..)

This GIC data file can be used in GIC Analysis.

Argument Definitions:

string savfile- PSSE Saved Case file name

string pyfile- GIC data file formatted in Python structures

This module defines Python structures and saves in this file (Step 1 above).

User only edits/updates values.

string excelfile- GIC data file formatted in Excel worksheets

This module creates this Excel file and worksheets in it (Step 1 above).

User only edits/updates values.

string gicfile- GIC data file name (output) that can be used in GIC Analysis.

The following optional arguments are used to define subsystem, if GIC study is to be performed for a sub-system.

real basekv- List of two real values: minimum voltage and maximum voltage,

For example [minVolt, maxVolt], default = []

integer areas- List of areas, default = []

integer buses- List of buses, default = []

integer owners- List of owners, default = []

integer zones- List of zones, default = []

integer tielevels- Number of intertie levels, default = 0

Functions:

check_gicfile(gicfile0, gicfile='')

Check GIC data file 'gicfile0' and create new GIC data file 'gicfile'.

Input gicfile0 is read and processed to do following:

- Add quotes (if already not there) around character data items likes "SUBNAM"
- Remove tabs and such non-ASCII characters

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- Check if same geographical location is provided for different substations and if found remove that substation from 'substation' record groups and replace that substation in 'bus subsation' record group.

Output 'gicfile' created only if it is specified.

excel2gicfile(excelfile, gicfile)

Create GIC data file (.gic) from GIC data template in Excel.

py2gicfile(pyfile, gicfile)

Create GIC data file (.gic) from GIC data template in Python structures.

template_excel(savfile, excelfile, basekv=[], areas=[], buses=[], owners=[], zones=[], tielevels=0)

Create GIC data file template in Excel Spreadsheet.

template_py(savfile, pyfile, basekv=[], areas=[], buses=[], owners=[], zones=[], tielevels=0)

Create GIC data file template in Python structures.

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Chapter 21
PFCONTROL Module

This chapter describes Python functions available in the PSS®E Remedial Action Scheme (RAS) and Advanced Contingency module.

These functions allow the user to program advanced RAS scenarios and events not supported by the existing SUB/MON/CON and TRP files to be executed in conjunction with the Power Flow Customization Interface (PCI) embedded in the PSS®E engine.

These functions do not record themselves when called while recording is enabled in PSS®E.

Please note that these routines are members of the pfcontrol Python extension module, not psspy. The syntax on the following pages assumes that the module has been imported by executing:

```
import pfcontrol
```

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21.1. PFDSCNBUS

This API is used to disconnect a bus in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr, typcode = pfdscnbus(iocode, frombus)
```

Python syntax:

```
ierr = pfdscnbus(frombus)
```

where:

Integer IOCODE	Is the action code (input). The values are as follows. 0 by default.
	IOCODE = 0 disconnect the bus
	IOCODE = 1 return the type code in typcode
Integer FROMBUS	Is the bus number (input).
Integer TYPCODE	Is the bus type code (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 invalid IOCODE value.
	IERR = 6 bus is already disconnected (JCODE > 3).

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21.2. PFBRCHINT

This API is used to get/set specific integer quantities on a branch in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pfbrchint(frombus, tobus, id, namval, ival)

ierr, ival = pfbrchint(frombus, tobus, id, namval)
```

where:

Integer FROMBUS	Is the branch from bus number (input).
Integer TOBUS	Is the branch to bus number (input).
Character ID*2	Is the branch Id (input).
Character NAMVAL*6	Is the value to set or return (input). The values are as follows.
	STATUS the branch status
Integer IVAL	Is the integer value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 branch not found.
	IERR = 6 multi-section line reporting option is disabled.
	IERR = 7 invalid NAMVAL string.
	IERR = 8 invalid IOCODE value.
	IERR = 9 at least one of the endpoint buses is not energized.

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21.3. PFSUBSWDINT

This API is used to get/set specific integer quantities on a substation switching device in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pfsubswdint(statnum, fromnode, tonode, id, namval, ival)

ierr, ival = pfsubswdint(statnum, fromnode, tonode, id, namval)
```

where:

Integer STATNUM	Is the substation switching device substation number (input).
Integer FROMNODE	Is the substation switching device from node number (input).
Integer TONODE	Is the substation switching device to node number (input).
Character ID*2	Is the substation switching device Id (input).
Character NAMVAL*6	Is the value to set or return (input). The values are as follows.
	STATUS the substation switching device status
Integer IVAL	Is the integer value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 substation STATNUM not found.
	IERR = 5 substation from node not found in substation STATNUM.
	IERR = 6 substation to node not found in substation STATNUM.
	IERR = 7 switching device not found in substation STATNUM.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 at least one of the nodes is out of service
	IERR = 11 attempting to open a stuck breaker.

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21.4. PFFXSINT

This API is used to get/set specific integer quantities on a fixed shunt in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pffxsint(frombus, id, namval, ival)

ierr, ival = pffxsint(frombus, id, namval)
```

where:

Integer FROMBUS	Is the fixed shunt from bus number (input).
Character ID*2	Is the fixed shunt Id (input).
Character NAMVAL*6	Is the value to set or return (input). The values are as follows.
	STATUS the fixed shunt status
Integer IVAL	Is the integer value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 fixed shunt not found.
	IERR = 6 bus is out of service.
	IERR = 7 shunt is out of service.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 fatal error from SHUNT_API

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21.5. PFFXSREAL

This API is used to get/set specific real quantities on a fixed shunt in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pffxsreal(frombus, id, namval, rval)

ierr, rval = pffxsreal(frombus, id, namval)
```

where:

Integer FROMBUS	Is the fixed shunt from bus number (input).
Character ID*2	Is the fixed shunt Id (input).
Character NAMVAL*8	Is the value to set or return (input). The values are as follows.
	PNOMINAL the nominal active power in MW
	QNOMINAL the nominal reactive power in MVAR
Integer RVAL	Is the real value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 fixed shunt not found.
	IERR = 6 bus is out of service.
	IERR = 7 shunt is out of service.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 fatal error from SHUNT_API

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21.6. PFSWSINT

This API is used to get/set specific integer quantities on a switched shunt in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pfswsint(frombus, namval, ival)

ierr, ival = pfswsint(frombus, namval)
```

where:

Integer FROMBUS	Is the switched shunt from bus number (input).
Character NAMVAL*6	Is the value to set or return (input). The values are as follows.
	STATUS the switched shunt status
Integer IVAL	Is the integer value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 switched shunt not found.
	IERR = 6 bus is out of service.
	IERR = 7 shunt is out of service.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 fatal error from SWITCHED_SHUNT_API_3

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21.7. PFSWSREAL

This API is used to get/set specific real quantities on a switched shunt in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pfswsreal(frombus, namval, rval)

ierr, rval = pfswsreal(frombus, namval)
```

where:

Integer FROMBUS	Is the switched shunt from bus number (input).
Character NAMVAL*5	Is the value to set or return (input). The values are as follows.
	QINIT the present switched shunt admittance in MVAR
	VMAX the desired voltage upper limit
	VMIN the desired voltage lower limit
Real RVAL	Is the real value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 switched shunt not found.
	IERR = 6 bus is out of service.
	IERR = 7 shunt is out of service.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 fatal error from SWITCHED_SHUNT_API_3

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21.8. PFLODINT

This API is used to get/set specific integer quantities on a load in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax to set an integer quantity:

```
ierr = pflodint(frombus, id, namval, ival)

ierr, ival = pflodint(frombus, id, namval)
```

where:

Integer FROMBUS	Is the load from bus number (input).
Character ID*2	Is the load Id (input).
Character NAMVAL*6	Is the value to set or return (input). The values are as follows.
	STATUS the load status
Integer IVAL	Is the integer value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 load not found.
	IERR = 6 bus is out of service.
	IERR = 7 load is out of service.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 fatal error from LOAD_API_5

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21.9. PFLODREAL

This API is used to get/set specific real quantities on a load in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pflodreal(frombus, id, namval, rval)

ierr, rval = pflodreal(frombus, id, namval)
```

where:

Integer FROMBUS	Is the load from bus number (input).
Character ID*2	Is the load Id (input).
Character NAMVAL*5	Is the value to set or return (input). The values are as follows.
	PLOAD the nominal active constant power load in MW
	QLOAD the nominal reactive constant power load in MVAR
Integer RVAL	Is the real value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 load not found.
	IERR = 6 bus is out of service.
	IERR = 7 load is out of service.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 fatal error from LOAD_API_5

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21.10. PFMACINT

This API is used to get/set specific integer quantities on a machine in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pfmacint(frombus, id, namval, ival)

ierr, ival = pfmacint(frombus, id, namval)
```

where:

Integer FROMBUS	machine from bus number (input).
Character ID*2	Is the machine Id (input).
Character NAMVAL*6	Is the value to set or return (input). The values are as follows.
	STATUS the machine status
	Is the integer value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 machine not found.
	IERR = 6 bus is out of service.
	IERR = 7 machine is out of service.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 fatal error from MACHINE_API_2

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21.11. PFMACREAL

This API is used to get/set specific real quantities on a machine in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pfmacreal(frombus, id, namval, rval)

ierr, rval = pfmacreal(frombus, id, namval)
```

where:

Integer FROMBUS	Is the machine from bus number (input).
Character ID*2	Is the machine Id (input).
Character NAMVAL*4	Is the value to set or return (input). The values are as follows.
	PGEN the active power generation in MW
	QMAX the reactive power high limit in MVAR
	QMIN the reactive power low limit in MVAR
Integer RVAL	Is the real value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 machine not found.
	IERR = 6 bus is out of service.
	IERR = 7 machine is out of service.
	IERR = 8 invalid NAMVAL string.
	IERR = 9 invalid IOCODE value.
	IERR = 10 fatal error from MACHINE_API_2

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21.12. PFPLNTREAL

This API is used to get/set specific real quantities on a plant in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pfplntreal(frombus, section, namval, rval)

ierr, rval = pfplntreal(frombus, section, namval)
```

where:

Integer FROMBUS	Is the plant from bus number (input).
Integer SECTION	Is the plant bus section number (input).
	SECTION = 0 If bus FROMBUS is not in a substation or if FROMBYS is not represented by multiple bus sections or if the same change applies to all plants at all bus sections
	SECTION != 0 plant bus section to apply change to
Character NAMVAL*4	Is the value to set or return (input). The values are as follows.
	VSET the desired voltage setpoint
Integer RVAL	Is the real value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from PCI automation function
	IERR = 2 not called from allowable PCI automation function.
	IERR = 3 working case is empty.
	IERR = 4 bus not found.
	IERR = 5 invalid node number.
	IERR = 6 bus is not part of a substation.
	IERR = 7 node is not found in this substation.
	IERR = 8 no plant present on this bus section.
	IERR = 9 bus is out of service.
	IERR = 10 plant is out of service.
	IERR = 11 invalid NAMVAL string.
	IERR = 12 invalid IOCODE value.

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IERR = 13 fatal error from PLANT_BUS_SECTION_CHNG.

IERR = 14 no in-service plant found on any bus section

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21.13. PFCNTLVSC

This API is used to get/set specific real quantities on a VSC DC line in a PCI automation function. Its use is only supported in the context of a call from a PCI automation function.

Python syntax:

```
ierr = pfcntlvsc(name, cnvbus, namval, rval)

ierr, rval = pfcntlvsc(name, cnvbus, namval)
```

where:

Character NAME*12	Is the VSC DC line name (input).
Integer CNVBUS	Is the VSC DC line converter bus number (input).
Character NAMVAL*5	Is the value to set or return (input). The values are as follows.
	PGEN the active power generation in MW
	QMAX the reactive power high limit in MVAR
	QMIN the reactive power low limit in MVAR
Integer RVAL	Is the real value to set if specified (input) else returned (output).
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 not called from allowable PCI automation function.
	IERR = 2 working case is empty.
	IERR = 3 VSC DC line not found
	IERR = 4 converter bus not found.
	IERR = 5 bus is not a converter bus of this VSC DC line.
	IERR = 6 invalid NAMVAL string.
	IERR = 7 invalid IOCODE value.
	IERR = 8 for getting 'QSET', VSQMAX .NE. VSQMIN.
	IERR = 9 for setting 'QMAX' or 'QMIN', called by Pre_Iteration or Post_Iteration.
	IERR = 10 fatal error from VSC_DC_LINE_CONVERTER_API_2.

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Chapter 22
PSSACCSS Module

This chapter describes Python functions available in the PSS Â®E Remedial Action Scheme (RAS) and Advanced Contingency module.

These functions allow the user to read an ACC file directly using Python calls. In some situations these APIs are faster than the APIs in the pssarrays module since they deal with smaller ammounts of data.

These functions do not record themselves when called while recording is enabled in PSS®E.

Please note that these routines are members of the pssaccss Python extension module, not psspy. The syntax on the following pages assumes that the module has been imported by executing:

```
import pssaccss
```

22.1. ACCC_FILE_INIT

This API is used to initialize an ACCC file for direct access to the contingency data contained within.

Python syntax:

```
ierr = accc_file_init(rfile, acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_INITIALIZE(RFILE, ACCCSIZE, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping. ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions. ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions. ACCCSIZE (14) Name Out option. ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name. ACCCSIZE (14) = -1 Undecided.

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	ACCCSIZE (15) ACCC file type.
	ACCCSIZE (15) = 0 ACCC.
	ACCCSIZE (15) = 1 N-1-1.
	ACCCSIZE (16) Number of Areas.
	ACCCSIZE (17) Number of Zones.
	ACCCSIZE (18) Number of Owners.
	ACCCSIZE (19) Number of Voltage levels.
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 1 error opening RFILE.
	IERR = 10 prerequisite requirements for API are not met.

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22.2. ACCC_FILE_SIZE

This API is used to get the number of elements contained within and ACCC file so that storage may be allocated for retrieving items (if desired).

Python syntax:

```
ierr, acccsize = accc_file_size(rfile)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_SIZE(RFILE, ACCCSIZE, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Output). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping. ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions. ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions. ACCCSIZE (14) Name Out option. ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name.

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	ACCCSIZE (14) = -1 Undecided.
	ACCCSIZE (15) ACCC file type.
	ACCCSIZE (15) = 0 ACCC.
	ACCCSIZE (15) = 1 N-1-1.
	ACCCSIZE (16) Number of Areas.
	ACCCSIZE (17) Number of Zones.
	ACCCSIZE (18) Number of Owners.
	ACCCSIZE (19) Number of Voltage levels.
Integer IERR	Is the error code (output).
	IERR = 0 no error occurred.
	IERR = 4 error opening RFILE.
	IERR = 5 error reading RFILE.
	IERR = 10 prerequisite requirements for API are not met.

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22.3. ACCC_FILE_GET_SUM_VOLT

This API is used to return information about monitored voltage records and limits contained in the contingency data file.

Python syntax:

```
ierr, mvrec_labelxx, mvrec_limxx, mvrec_typexx, mvbus_seqxx,
 vbaselmxx, vemergnxx, mvrec_ivbxx, mvrnd_labelxx, mvrnd_limxx,
 mvrnd_typexx, mvnod_seqxx, vnbaselmxx, vnemergnxx, mvrnd_ivbxx =
 accc_file_get_sum_volt(rfile, acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_SUMMARY_VOLTAGES(RFILE, ACCCSIZE, MVREC_LABELxx,
 MVREC_LIMxx, MVREC_TYPExx, MVBUS_SEQxx, VBASELMxx, VEMERGNxx,
 MVREC_IVBxx, MVRND_LABELxx, MVRND_LIMxx, MVRND_TYPExx, MVNOD_SEQxx,
 VNBASELMxx, VNEMERGNxx, MVRND_IVBxx, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping. ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions. ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions.

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	ACCCSIZE (14) Name Out option. ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name. ACCCSIZE (14) = -1 Undecided. ACCCSIZE (15) ACCC file type. ACCCSIZE (15) = 0 ACCC. ACCCSIZE (15) = 1 N-1-1. ACCCSIZE (16) Number of Areas. ACCCSIZE (17) Number of Zones. ACCCSIZE (18) Number of Owners. ACCCSIZE (19) Number of Voltage levels.
Character MVREC_LABELxx*34 (ACCCSIZE(5))	Is an array containing monitored voltage record label for all monitored voltage records in the file (output).
Real MVREC_LIMxx (2*ACCCSIZE(5))	Is an array containing monitored voltage record limits for all monitored voltage records in the file (output).
Character MVREC_TYPExx*9 (ACCCSIZE(5))	Is an array containing monitored voltage record type for all monitored voltage records in the file (output). Either 'DEVIATION', 'RANGE' or 'LIMIT'
Integer MVBUS_SEQxx (ACCCSIZE(7))	Is an array containing monitored bus sequence numbers for all buses in the monitored bus list in the file (output).
Real VBASELMxx (2*ACCCSIZE(7))	Is an array containing monitored bus normal limits for all buses in the monitored bus list in the file (output).
Real VEMERGNxx (2*ACCCSIZE(7))	Is an array containing monitored bus emergency limits for all buses in the monitored bus list in the file (output).
Integer MVREC_IVBxx (ACCCSIZE(7)* ACCCSIZE(5))	Is an array containing flags to indicate if the bus is monitored in a record, indexed by the bus sequence number and monitor record in the file (output).
Character MVRND_LABELxx*34 (ACCCSIZE(23))	Monitored voltage record label
Real MVRND_LIMxx (ACCCSIZE(23))	Monitored voltage record limits
Character MVRND_TYPExx*9 (ACCCSIZE(23))	Monitor voltage record TYPE
Integer MVNOD_SEQxx (ACCCSIZE(21))	Contains the number in monitored buses list, indexed by the bus sequence number in the bus table
Real VNBASELMxx (ACCCSIZE(21))	The normal voltage limits, indexed by the bus sequence in the bus table
Real VNEMERGNxx (ACCCSIZE(21))	The emergency voltage limits, indexed by the bus sequence in the bus table

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Integer MVRND_IVBxx (ACCCSIZE(21)* ACCCSIZE(23))	Contains a flag to indicate if the bus is monitored in a record, indexed by the bus sequence number and monitor record
Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = 1-4 error in internal parsing routines. IERR = 5 RFILE does not match the one called in ACCC_FILE_SIZE. IERR = 6 error in internal ordering of buses. IERR = 10 prerequisite requirements for API are not met.

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22.4. ACCC_FILE_CLOSE

This API is used to close an open ACCC file.

Python syntax:

```
ierr = accc_file_close()
```

Fortran syntax:

```
CALL ACCC_FILE_CLOSE(IERR)
```

where:

Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = X error closing open ACCC file. IERR = 10 prerequisite requirements for API are not met.

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22.5. ACCC_FILE_SUMMARY

This API is used to get a summary of the data contained within an ACCC file.

Python syntax:

```
shrttitle, filenames, cktlabels, mon_ratexx, mvbus_labelxx, mvrev_labelxx,
 mvrec_maxxx, mvrec_minxx, mvrec_typexx, lblcntxx, busnamaccxx, addcntxx,
 addtrpxx, addcorxx, mvrec_ivbxx, ierr = accc_file_summary(rfile,
 acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_SUMMARY(RFILE, ACCCSIZE, SHRTTITLE, FILENAMES,
 CKTLABELS, MON_RATExx, MVBUS_LABELxx, MVREV_LABELxx, MVREC_MAXxx,
 MVREC_MINxx, MVREC_TYPExx, LBLCNTxx, BUSNAMACCxx, ADDCNTxx, ADDTRPxx,
 ADDCORxx, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; No default allowed).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping. ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions. ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions. ACCCSIZE (14) Name Out option.

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	ACCCSIZE (14) = 0 Number.
	ACCCSIZE (14) = 1 Name.
	ACCCSIZE (14) = -1 Undecided.
	ACCCSIZE (15) ACCC file type.
	ACCCSIZE (15) = 0 ACCC.
	ACCCSIZE (15) = 1 N-1-1.
	ACCCSIZE (16) Number of Areas.
	ACCCSIZE (17) Number of Zones.
	ACCCSIZE (18) Number of Owners.
	ACCCSIZE (19) Number of Voltage levels.
Character SHRTTITLE*60(2)	Is an array of two elements that contain the two short title lines of the case contained within the ACCC file. SHRTTITLE (1) First short title line.
	SHRTTITLE (2) Second short title line.
Character FILENAMES*260(9)	Is an array of nine elements that contain the file names of all files used in the case contained within the ACCC file. FILENAMES (1) ACC file name.
	FILENAMES (2) SAV file name.
	FILENAMES (3) DFX file name.
	FILENAMES (4) SUB file name.
	FILENAMES (5) MON file name.
	FILENAMES (6) CON file name.
	FILENAMES (7) THR file name.
	FILENAMES (8) INL file name.
	FILENAMES (9) TRP file name.
Character CKTLABELS*62 (ACCCSIZE(1)* ACCCSIZE(2))	Is an array that contains the labels of all monitored lines contained within the ACCC file. Multi-section line labels are not stored.
Float MON_RATExx ((ACCCSIZE(1)+ ACCCSIZE(2))* ACCCSIZE(20))	Is an array that contains the Monintored Line Flow ratings.
Character MVBUS_LABELxx*29 (ACCCSIZE(6))	Is an array containing the Monitored Bus labels.
Character MVREV_LABELxx*34 (ACCCSIZE(6))	Is an array containing the Monitored Bus record labels.

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Real MVREC_MAXxx (ACCCSIZE(6))	Is an array containing the Monitored Bus voltage record maximum.
Real MVREC_MINxx (ACCCSIZE(6))	Is an array containing the Monitored Bus voltage record minimum.
Character MVREC_TYPExx*9 (ACCCSIZE(6))	Is an array containing the Monitored Bus voltage record type (Range, Deviation, Limit).
Charcter LBLCNTxx*152 (ACCCSIZE(3))	Is an array containing the Contingency labels.
Character BUSNAMACCxx*29 (ACCCSIZE(7))	Is an array containing the extended Bus names.
Long Integer ADDCNTxx (ACCCSIZE(3))	Is an array containing ACCC Contingency solution result addresses. Used as input to other APIs.
Long Integer ADDTRPxx (ACCCSIZE(3))	Is an array containing ACCC Tripping solution result addresses. Used as input to other APIs.
Long Integer ADDCORxx (ACCCSIZE(3))	Is an array containing ACCC Corrective Action solution result addresses. Used as input to other APIs.
Integer MVREC_IVBxx (ACCCSIZE(4)* ACCCSIZE(5))	Is an array containing flags used to indicate if the Bus is monitored in a record (indexed by the bus sequence number and monitored bus record).
Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = 4 error opening RFILE. IERR = 5 RFILE does not match the one called in ACCC_FILE_SIZE. IERR = 6 error in fetching contingency strings. IERR = 10 prerequisite requirements for API are not met.

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22.6. ACCC_FILE_SUM_AREAS

This API is used to return information about areas, owners and zone information contained in the contingency data file.

Python syntax:

```
ierr, bsarea_outxxx, bszone_outxx, bsowne_outxx, bsvolt_outxx,
 nuarea_outxx, nuzone_outxx, nuowne_outxx, nuvolt_outxx, naarea_outxx,
 nazone_outxx, naowne_outxx = accc_file_sum_areas(rfile, acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_SUMMARY_AREAS(RFILE, ACCCSIZE, BSAREA_OUTxxx,
 BSZONE_OUTxx, BSOWNE_OUTxx, BSVOLT_OUTxx, NUAREA_OUTxx, NUZONE_OUTxx,
 NUOWNE_OUTxx, NUVOLT_OUTxx, NAAREA_OUTxx, NAZONE_OUTxx, NAOWNE_OUTxx,
 NUVOLT_OUTxx, NAAREA_OUTxx, NAZONE_OUTxx, NAOWNE_OUTxx, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows.
	ACCCSIZE (1) Number of monitored lines.
	ACCCSIZE (2) Number of interfaces.
	ACCCSIZE (3) Number of Contingencies + Base Case.
	ACCCSIZE (4) Number of voltage buses.
	ACCCSIZE (5) Number of voltage records.
	ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)).
	ACCCSIZE (7) Number of buses in ACCC case.
	ACCCSIZE (8) Acc file version number.
	ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency.
	ACCCSIZE (10) Max # of LOAD SHEDS due to tripping.
	ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions.
	ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions.
	ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions.
	ACCCSIZE (14) Name Out option.

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	ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name. ACCCSIZE (14) = -1 Undecided. ACCCSIZE (15) ACCC file type. ACCCSIZE (15) = 0 ACCC. ACCCSIZE (15) = 1 N-1-1. ACCCSIZE (16) Number of Areas. ACCCSIZE (17) Number of Zones. ACCCSIZE (18) Number of Owners. ACCCSIZE (19) Number of Voltage levels.
Integer BSAREA_OUTxxx (ACCCSIZE(7))	Is an array containing all the Area numbers for the buses in the file (output).
Integer , BSZONE_OUTxxx (ACCCSIZE(7))	Is an array containing all the Zone numbers for the buses in the file (output).
Integer BSOWNE_OUTxxx (ACCCSIZE(7))	Is an array containing all the Owner numbers for the buses in the file (output).
Real BSVOLT_OUTxxx (ACCCSIZE(7))	Is an array containing all the Voltages for the buses in the file (output).
Integer NUAREA_OUTxxx (ACCCSIZE(16))	Is an array containing all the Area numbers in the file (output).
Integer NUZONE_OUTxxx (ACCCSIZE(17))	Is an array containing all the Zone numbers in the file (output).
Integer NUOWNE_OUTxxx (ACCCSIZE(18))	Is an array containing all the Owner numbers in the file (output).
Real NUOWNE_OUTxxx (ACCCSIZE(19))	Is an array containing all the Voltage levels numbers in the file (output).
Character NAAREA_OUTxxx*12 (ACCCSIZE(16))	Is an array containing all the Area names in the file (output).
Character NAZONE_OUTxxx*12 (ACCCSIZE(17))	Is an array containing all the Zone names in the file (output).
Character NAOWNE_OUTxxx*12 (ACCCSIZE(18))	Is an array containing all the Owner names in the file (output).
Integer IERR	Is the error code (output). IERR = -1 No results IERR = 0 no error occurred. IERR = 5 RFILE does not match the one called in ACCC_FILE_SIZE. IERR = 6 error in fetching contingency strings.

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IERR = 10 prerequisite requirements for API are not met.

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22.7. ACCC_FILE_CNTG_VOLT

This API is used to return voltages for a specified contingency and contingency type for all monitored buses in the contingency data file.

Python syntax:

```
ierr, mv_vmag_outxx = accc_file_cntg_volt(rfile, getwhat, cntpos,
 acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_CONTINGENCY_VOLTAGE(RFILE, GETWHAT, CNTPOS, ACCCSIZE,
 MV_VMAG_OUTxx, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer GETWHAT	Is the voltages to return for the specified contingency (Input). The values are as follows. GETWHAT = 1 Contingency solution voltages GETWHAT = 2 Tripping solution voltages GETWHAT = 3 Corrective action solution voltages
Long Integer CNTPOS	Is the contingency position in the Contingency Solution Output file from which to read the desired voltages (Input).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping. ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions.

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	ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions. ACCCSIZE (14) Name Out option. ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name. ACCCSIZE (14) = -1 Undecided. ACCCSIZE (15) ACCC file type. ACCCSIZE (15) = 0 ACCC. ACCCSIZE (15) = 1 N-1-1. ACCCSIZE (16) Number of Areas. ACCCSIZE (17) Number of Zones. ACCCSIZE (18) Number of Owners. ACCCSIZE (19) Number of Voltage levels.
Real MV_VMAG_OUTxx (ACCCSIZE(4))	Is an array containing all the desired voltages for the desired contingency (output).
Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = 4 error opening RFILE. IERR = 5 error reading RFILE. IERR = 6 error in fetching contingency strings. IERR = 10 prerequisite requirements for API are not met.

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22.8. ACCC_FILE_CNTG_FLOW

This API is used to return flows for a specified contingency and contingency type for all monitored lines in the contingency data file.

Python syntax:

```
ierr, acflow_outxx, rat_num_outxx = accc_file_cntg_flow(rfile, getwhat,
 cntpos, acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_CONTINGENCY_FLOW(RFILE, GETWHAT, CNTPOS, ACCCSIZE,
 ACFLOW_OUTxx, RAT_NUM_OUTxx, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer GETWHAT	Is the flows to return for the specified contingency (Input). The values are as follows. GETWHAT = 1 Contingency solution flows GETWHAT = 2 Tripping solution flows GETWHAT = 3 Corrective action solution flows
Long Integer CNTPOS	Is the contingency position in the Contingency Solution Output file from which to read the desired flows (Input).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping. ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions.

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	ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions. ACCCSIZE (14) Name Out option. ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name. ACCCSIZE (14) = -1 Undecided. ACCCSIZE (15) ACCC file type. ACCCSIZE (15) = 0 ACCC. ACCCSIZE (15) = 1 N-1-1. ACCCSIZE (16) Number of Areas. ACCCSIZE (17) Number of Zones. ACCCSIZE (18) Number of Owners. ACCCSIZE (19) Number of Voltage levels.
Real ACFLOW_OUTxx ((ACCCSIZE(1)+ ACCCSIZE(2))	Is an array containing all the desired MVA flows for the desired contingency (output).
Real RAT_NUM_OUTxx (ACCCSIZE(1))	Is an array containing all the desired Ampere flows for the desired contingency (output).
Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = 4 error opening RFILE. IERR = 5 error reading RFILE. IERR = 6 error in fetching contingency strings. IERR = 10 prerequisite requirements for API are not met.

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22.9. ACCC_FILE_CNTG_VOLT_FLOW

This API is used to return voltages and flows for a specified contingency and contingency type for all monitored buses and lines in the contingency data file.

Python syntax:

```
ierr, mv_vmag_outxx, acflow_outxx, rat_num_outxx =
 accc_file_cntg_volt_flow(rfile, getwhat, cntpos, acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_CONTINGENCY_VOLT_FLOW(RFILE, GETWHAT, CNTPOS, ACCCSIZE,
 MV_VMAG_OUTxx, ACFLOW_OUTxx, RAT_NUM_OUTxx,  IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer GETWHAT	Is the flows to return for the specified contingency (Input). The values are as follows. GETWHAT = 1 Contingency solution voltages and flows GETWHAT = 2 Tripping solution voltages and flows GETWHAT = 3 Corrective action solution voltages and flows
Long Integer CNTPOS	Is the contingency position in the Contingency Solution Output file from which to read the desired flows (Input).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping. ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions.

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	ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions. ACCCSIZE (14) Name Out option. ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name. ACCCSIZE (14) = -1 Undecided. ACCCSIZE (15) ACCC file type. ACCCSIZE (15) = 0 ACCC. ACCCSIZE (15) = 1 N-1-1. ACCCSIZE (16) Number of Areas. ACCCSIZE (17) Number of Zones. ACCCSIZE (18) Number of Owners. ACCCSIZE (19) Number of Voltage levels.
Real MV_VMAG_OUTxx (ACCCSIZE(4))	Is an array containing all the desired voltages for the desired contingency (output).
Real ACFLOW_OUTxx (ACCCSIZE(1) +ACCCSIZE(2))	Is an array containing all the desired MVA flows for the desired contingency (output).
Real RAT_NUM_OUTxx (ACCCSIZE(1))	Is an array containing all the desired Ampere flows for the desired contingency (output).
Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = 5 error reading RFILE. IERR = 6 error in fetching contingency strings. IERR = 10 prerequisite requirements for API are not met.

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22.10. ACCC_FILE_CNTG_DSCR

This API is used to return voltages and flows for a specified contingency and contingency type for all monitored buses and lines in the contingency data file.

Python syntax:

```
ierr, num_island, case_worst, case_total, cnv_cond, cnt_desc =
 accc_file_cntg_dscr(rfile, getwhat, cntpos, acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_CONTINGENCY_DESCRIPTION(RFILE, GETWHAT, CNTPOS,
 ACCCSIZE, NUM_ISLAND, CASE_WORST, CASE_TOTAL, CNV_COND, CNT_DESC, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer GETWHAT	Is the flows to return for the specified contingency (Input). The values are as follows. GETWHAT = 1 Contingency solution voltages and flows GETWHAT = 2 Tripping solution voltages and flows GETWHAT = 3 Corrective action solution voltages and flows
Long Integer CNTPOS	Is the contingency position in the Contingency Solution Output file from which to read the desired flows (Input).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping. ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions.

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	ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions. ACCCSIZE (14) Name Out option. ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name. ACCCSIZE (14) = -1 Undecided. ACCCSIZE (15) ACCC file type. ACCCSIZE (15) = 0 ACCC. ACCCSIZE (15) = 1 N-1-1. ACCCSIZE (16) Number of Areas. ACCCSIZE (17) Number of Zones. ACCCSIZE (18) Number of Owners. ACCCSIZE (19) Number of Voltage levels.
Integer NUM_ISLAND	Is the number of islanded buses in the desired contingency (Output).
Real CASE_WORST	Is the worst case in the desired contingency (Output).
Real CASE_TOTAL	Is the case total in the desired contingency (Output).
Character CNV_COND*34	Is the convergence solution text for the desired contingency (Output).
Character CNT_DESC*160 (32)	Is the description of every contingency in the specified contingency group (Output).
Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = 5 error reading RFILE. IERR = 6 error in fetching contingency strings. IERR = 10 prerequisite requirements for API are not met.

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22.11. ACCC_FILE_CNTG_LOAD

This API is used to return voltages and flows for a specified contingency and contingency type for all monitored buses and lines in the contingency data file.

Python syntax:

```
ierr, num_loadshed, lshed_bus_outxx, lshed_outxx, num_gendisp,
 gdisp_bus_outxx, disp_outxx, num_phsshft, pshft_bus_outxx, pshft_outxx =
 accc_file_cntg_load(rfile, getwhat, cntpos, acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_CONTINGENCY_LOAD(RFILE, GETWHAT, CNTPOS, ACCCSIZE,
 NUM_LOADSHED, LSHED_BUS_OUTxx, LSHED_OUTxx, NUM_GENDISP, GDISP_BUS_OUTxx,
 GDISP_OUTxx, NUM_PHSSHFT, PSHFT_BUS_OUTxx, PSHFT_OUTxx, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer GETWHAT	Is the flows to return for the specified contingency (Input). The values are as follows. GETWHAT = 1 Contingency solution loads GETWHAT = 2 Tripping solution loads GETWHAT = 3 Corrective action solution voltages and flows
Long Integer CNTPOS	Is the contingency position in the Contingency Solution Output file from which to read the desired flows (Input).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows. ACCCSIZE (1) Number of monitored lines. ACCCSIZE (2) Number of interfaces. ACCCSIZE (3) Number of Contingencies + Base Case. ACCCSIZE (4) Number of voltage buses. ACCCSIZE (5) Number of voltage records. ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)). ACCCSIZE (7) Number of buses in ACCC case. ACCCSIZE (8) Acc file version number. ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency. ACCCSIZE (10) Max # of LOAD SHEDS due to tripping.

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	ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions. ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions. ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions. ACCCSIZE (14) Name Out option. ACCCSIZE (14) = 0 Number. ACCCSIZE (14) = 1 Name. ACCCSIZE (14) = -1 Undecided. ACCCSIZE (15) ACCC file type. ACCCSIZE (15) = 0 ACCC. ACCCSIZE (15) = 1 N-1-1. ACCCSIZE (16) Number of Areas. ACCCSIZE (17) Number of Zones. ACCCSIZE (18) Number of Owners. ACCCSIZE (19) Number of Voltage levels.
Integer NUM_LOADSHED	Is the number of loads shed for the desired contingency (Output).
Real LSHED_BUS_OUTxx (ACCCSIZE(11))	Is an array containing all the load shedding buses for the desired contingency (Output).
Real LSHED_OUTxx (2*ACCCSIZE(11))	Is an array containing all the load shed at the load shedding buses for the desired contingency (Output).
Integer NUM_GENDISP	Is the number of Generators dispatched for the desired contingency (Output).
Real GDISP_BUS_OUTxx (ACCCSIZE(12))	Is an array containing all the generation dispatching buses for the desired contingency (Output).
Real GDISP_OUTxx (2*ACCCSIZE(12))	Is an array containing all the generation dispatch at the generation dispatching buses for the desired contingency (Output).
Integer NUM_PHSSHFT	Is the number of phase shifters for the desired contingency (Output).
Character PSHFT_BUS_OUTxx*33 (ACCCSIZE(13))	Is an array containing all the phase shifting buses for the desired contingency (Output).
Real PSHFT_OUTxx (2*ACCCSIZE(13))	Is an array containing all the phase shifts for the phase shifting buses for the desired contingency (Output).
Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = 5 error reading RFILE.

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IERR = 6 error in fetching contingency strings.

IERR = 10 prerequisite requirements for API are not met.

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22.12. ACCC_FILE_SOLUTION

This API is used to return voltages and flows for a specified contingency and contingency type for all monitored buses and lines in the contingency data file.

Python syntax:

```
ierr, cnt_desc, cnv_flg, cnv_cond, num_island, case_worst, case_total,
 mv_vmag_outxx, acflow_outxx, rat_num_outxx, lshed_bus_outxx, lshed_outxx
 = accc_file_solution(rfile, getwhat, cntpos, bus_msm, sys_msm, acccsize)
```

or:

```
ierr, cnt_desc, cnv_flg, cnv_cond, num_island, case_worst, case_total,
 mv_vmag_outxx, acflow_outxx, rat_num_outxx, lshed_bus_outxx, lshed_outxx,
 gdispbusxx, gdispxx, phsftnamexx, pfsftangxx = accc_file_solution(rfile,
 getwhat, cntpos, bus_msm, sys_msm, acccsize)
```

Fortran syntax:

```
CALL ACCC_FILE_GET_1CNTGNCY_SOLUTION(RFILE, GETWHAT, CNTPOS, BUS_MSM,
 SYS_MSM, ACCCSIZE, CNT_DESC, CNV_FLG, CNV_COND, NUM_ISLAND, CASE_WORST,
 CASE_TOTAL, MV_VMAG_OUTxx, ACFLOW_OUTxx, RAT_NUM_OUTxx, LSHED_BUS_OUTxx,
 LSHED_OUTxx, IERR)
```

or:

```
CALL ACCC_FILE_GET_1TRIPPING_SOLUTION(RFILE, GETWHAT, CNTPOS, BUS_MSM,
 SYS_MSM, ACCCSIZE, CNT_DESC, CNV_FLG, CNV_COND, NUM_ISLAND, CASE_WORST,
 CASE_TOTAL, MV_VMAG_OUTxx, ACFLOW_OUTxx, RAT_NUM_OUTxx, LSHED_BUS_OUTxx,
 LSHED_OUTxx, IERR)
```

or:

```
CALL ACCC_FILE_GET_CACTION_SOLUTION(RFILE, GETWHAT, CNTPOS, BUS_MSM,
 SYS_MSM, ACCCSIZE, CNT_DESC, CNV_FLG, CNV_COND, NUM_ISLAND, CASE_WORST,
 CASE_TOTAL, MV_VMAG_OUTxx, ACFLOW_OUTxx, RAT_NUM_OUTxx, LSHED_BUS_OUTxx,
 LSHED_OUTxx, GDISPBUSxx, GDISPxx, PHSFTNAMExx, PFSFTANGxx, IERR)
```

where:

Character RFILE*260	Is the Contingency Solution Output File (input; no default allowed).
Integer GETWHAT	Is the flows to return for the specified contingency (Input). The values are as follows. GETWHAT = 1 Contingency solution loads GETWHAT = 2 Tripping solution loads GETWHAT = 3 Corrective action solution voltages and flows
Long Integer CNTPOS	Is the contingency position in the Contingency Solution Output file from which to read the desired flows (Input).
Integer BUS_MSM	Is the bus mismatch value to use when reporting contingencies from the contingency file (Input).

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Integer SYS_MSM	Is the system mismatch value to use when reporting contingencies from the contingency file (Input).
Integer ACCCSIZE(19)	Is an array containing the number of the various elements in the ACCC file (Input). The values are as follows.

ACCCSIZE (1) Number of monitored lines.

ACCCSIZE (2) Number of interfaces.

ACCCSIZE (3) Number of Contingencies + Base Case.

ACCCSIZE (4) Number of voltage buses.

ACCCSIZE (5) Number of voltage records.

ACCCSIZE (6) number monitored bus records (could be more than ACCCSIZE(4)).

ACCCSIZE (7) Number of buses in ACCC case.

ACCCSIZE (8) Acc file version number.

ACCCSIZE (9) Max # of LOAD SHEDS due to dispatch and contingency.

ACCCSIZE (10) Max # of LOAD SHEDS due to tripping.

ACCCSIZE (11) Max # of LOAD SHEDS due to corrective actions.

ACCCSIZE (12) Max # of GENERATION DISPACHED due to corrective actions.

ACCCSIZE (13) max # of PHASE SHIFTER changed due to corrective actions.

ACCCSIZE (14) Name Out option.

ACCCSIZE (14) = 0 Number.

ACCCSIZE (14) = 1 Name.

ACCCSIZE (14) = -1 Undecided.

ACCCSIZE (15) ACCC file type.

ACCCSIZE (15) = 0 ACCC.

ACCCSIZE (15) = 1 N-1-1.

ACCCSIZE (16) Number of Areas.

ACCCSIZE (17) Number of Zones.

ACCCSIZE (18) Number of Owners.

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	ACCCSIZE (19) Number of Voltage levels.
Character CNT_DESC*160 (32)	Is the description of every contingency in the specified contingency group (Output).
Logical CNV_FLAG	Is a logical flag indicating whether the specified contingency converged or not (Output).
Character CNV_COND*34	Is the convergence solution text for the desired contingency (Output).
Integer NUM_ISLAND	Is the number of islanded buses in the desired contingency (Output).
Real CASE_WORST	Is the worst case in the desired contingency (Output).
Real CASE_TOTAL	Is the case total in the desired contingency (Output).
Real MV_VMAG_OUTxx * ACCCSIZE(4)	Is an array containing all the desired voltages for the desired contingency (output).
Real ACFLOW_OUTxx (ACCCSIZE(1)+ ACCCSIZE(2))	Is an array containing all the desired MVA flows for the desired contingency (output).
Real RAT_NUM_OUTxx (ACCCSIZE(1))	Is an array containing all the desired Ampere flows for the desired contingency (output).
Real LSHED_BUS_OUTxx (ACCCSIZE(11))	Is an array containing all the load shedding buses for the desired contingency (Output).
Real LSHED_OUTxx (2*ACCCSIZE(11))	Is an array containing all the load shed at the load shedding buses for the desired contingency (Output).
Integer GDISPBUS (ACCCSIZE(11))	Is an array containing all the generation dispatch buses for the desired contingency (Output).
Real GDISPxx (ACCCSIZE(11))	Is an array containing all the generation dispatched at the generation dispatch buses for the desired contingency (Output).
Character PHSFTNAMExx*62 (ACCCSIZE(11))	Is an array containing all the Phase shifter names (Output).
Real PHSFTANGxx (ACCCSIZE(11))	Is an array containing all the Phase shifter angles (Output).
Integer IERR	Is the error code (output). IERR = 0 no error occurred. IERR = 4 error opening RFILE. IERR = 5 error reading RFILE. IERR = 6 error in fetching contingency strings. IERR = 10 prerequisite requirements for API are not met.

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Chapter 23
Other Python Modules

This chapter identifies other Python modules that are delivered as part of the PSS®E API.

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23.1. Modules Supplied with a GUI

These modules can be imported or run by themselves. They display a GUI that presents their functionality.

createusrdllw – Create a user dll for dynamics simulation. Combines the traditional compile and cload4 processes in a more user-friendly way.

excelexportw – Provides tools to export psse results to Excel, e.g., pv, qv, and acc files.

plot2wordw – Provides tools to create Word documents from plots.

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23.2. Modules that Define Functions

Each of these modules contains help information about the functions that they define. Import the module and run the help function on them for more information.

bsntools – Bus renaming tools. Functions to modify bus names by area. Requires psspy.

caspy – Create an object to directly returns arrays of data from a saved case file. Encapsulates the functionality of the Saved Case Data Extraction routines (USRCAS).

dyntools – Tools for processing channel output files, including export to Excel.

pssgrpg – Provides functions intended to be used to supply values to report labels on 1-line diagrams.

pssppe – Convert csv file to a dictionary of tuples.

pssras – Define a Remedial Action Scheme. Used with the Powerflow Customization Interface.

redirect – Tools for directing PSS®E output to Python, and vice versa.

sliderPy – Provides functions to directly manipulate diagram elements.

wordpy – Provides Python functions to interface with Microsoft Word.

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